Institutional Quantitative Finance Reference Catalogue

Foundations, methods, markets and research resources

13 October 2026

Companion: curated learning guide

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How to use the reference catalogue

This catalogue is a reference work for institutional quantitative research. Its four parts separate foundations and research methods; portfolio, pricing and trading methods; markets and instruments; and data, systems and research resources. Chapters bring foundations, implementation references and recent research together by subject. A work or explicitly identified version has one primary home; links connect its other applications.

For a selected sequence, use the separate learning guide. Its overview routes and sequenced specialist pathways specify required readings, prerequisites, outcomes and replication checkpoints. Specialist readings connect contract mechanics, valuation, evidence, data and implementation without a fixed reading-count cap. *Core* means required for one of those routes; catalogue inclusion alone is not a priority ranking.

Each entry separates **difficulty**, **resource type** and **publication/version status**. Intro assumes mathematical maturity but no specialism; Intermediate bridges introductory and specialist material; Advanced assumes graduate methods or market knowledge; Research denotes specialist work. “Not specified” is used where the inherited entry gives no defensible level. Type describes the resource, and status records the selected edition, date, draft or living documentation. An unspecified date is not a claim of currency. Paired works and editions retain their separate names and version qualifications within the same primary entry.

Default annotations describe contribution and applicability. Longer qualifications are available through an entry’s *Extended note* link in the companion HTML file. Keep that file and both PDFs together when sharing them. Entries retain searchable identifiers, and chapter/section links support browsing without a prescribed reading order. Version-record links expose publication dates, selected revisions, verification scope, version relationships and a review policy for every entry. Inherited metadata is labelled separately from dated source inspections, including the 12 October 2026 crypto-roadmap enhancement; unknown fields remain explicit. This is not an exhaustive external-source or legal-currentness clearance.

Underlying assets, legal instruments, pooled vehicles and strategies describe different layers. A wrapper, label or region does not establish an independent source of diversification. Read contract, cash-flow, data, financing and implementation references together. Working papers, bounded case studies and restricted data retain their limitations. Unresolved leads and intentional exclusions remain in the editorial records, outside the recommended readings.

Part I

Foundations and research methods

Chapter 1

Mathematics, probability and statistics

Calculus and linear algebra support introductory probability; proof-based analysis supports measure theory and graduate inference. Different texts offer alternative treatments, not a compulsory sequence.

1.1 Probability and inference

Joseph K. Blitzstein and Jessica Hwang, *Introduction to Probability*, 2nd ed. (2019)

Difficulty: Intro **Type:** Book / monograph

Status/version: 2nd ed. (2019)

Calculus-based probability and conditional reasoning; the Harvard Stat 110 hub includes the free second-edition text and companion lectures.

Version record

Sheldon Ross, *A First Course in Probability*, 9th ed. (2014)

Difficulty: Intro **Type:** Book / monograph

Status/version: 9th ed. (2014)

Calculus-based discrete and continuous probability with extensive examples and problems; the ninth edition remains useful (2014) as a historical selection. The tenth Global edition is identified by Pearson as copyright 2019, ISBN 9781292269207; exercise numbering differs.

Version record

Jim Pitman, *Probability*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Mathematical arguments and examples that develop probabilistic intuition.

Version record

Charles M. Grinstead and J. Laurie Snell, *Introduction to Probability*

Difficulty: Intro **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Probability with proofs and exercises; an online-text alternative.

Version record

George Casella and Roger L. Berger, *Statistical Inference*, 2nd ed. (2002)

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: 2nd ed. (2002)

Sufficiency, estimation, testing and likelihood theory from calculus and probability. The second edition is the main reference; the earlier first edition is an alternative.

Version record

Morris H. DeGroot and Mark J. Schervish, *Probability and Statistics*

Difficulty: Intro **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Likelihood, frequentist inference and Bayesian reasoning in an integrated probability/statistics text.

Version record

Larry Wasserman, *All of Statistics: A Concise Course in Statistical Inference*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Springer Texts in Statistics, first edition; copyright 2004. Hardcover released 4 December 2003; later softcover and eBook format dates are not new editions.

Compact overview of inference, bootstrap and statistical learning; shorter proofs than specialist texts.

Version record

William Feller, *An Introduction to Probability Theory and Its Applications, Volume I*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Wiley, third edition, 1968 (ISBN 9780471257080).

Classic discrete probability: random walks, generating functions, renewal theory and Markov chains with unusually rich examples; third edition (1968).

Version record

Geoffrey Grimmett and David Stirzaker, *Probability and Random Processes*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Oxford University Press, fourth edition, 2020 (ISBN 9780198847595).

Probability through Markov chains, martingales, diffusions and stationary processes; fourth edition (2020) with the companion exercise book already in the catalogue.

Version record

Dimitri P. Bertsekas and John N. Tsitsiklis, *Introduction to Probability*

Difficulty: Intro **Type:** Book / monograph

Status/version: Athena Scientific, second edition, 2008.

Engineering-style probability with inference and limit theorems; second edition (2008), the text for MIT 6.041.

Version record

John A. Rice, *Mathematical Statistics and Data Analysis*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Thomson Brooks/Cole, third edition, 2007 (ISBN 9780534399429).

Statistics text that pairs theory with data analysis: estimation, testing, bootstrap and linear models; third edition (2007).

Version record

1.2 Measure theory and graduate probability

H. L. Royden and P. M. Fitzpatrick, *Real Analysis*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

General real analysis, with measure and integration as foundations for probability.

Version record

Elias M. Stein and Rami Shakarchi, *Real Analysis: Measure Theory, Integration, and Hilbert Spaces*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Measure theory, integration and Hilbert spaces within a broader analysis treatment.

Version record

Sheldon Axler, *Measure, Integration & Real Analysis*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Author-hosted open-access analysis text with prerequisites and a concluding probability-measures chapter.

Version record

Sidney I. Resnick, *A Probability Path* (1999)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 1999. Version distinctions remain in the annotation.

Measure and integration developed in probabilistic context, followed by distributions and limit theory.

Version record

David Williams, *Probability with Martingales* (1991)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 1991. Version distinctions remain in the annotation.

Conditional expectation, filtrations, stopping times and martingales for stochastic modelling and pricing.

Version record

David Pollard, *A User's Guide to Measure Theoretic Probability*
Difficulty: Advanced **Type:** Book / monograph
Status/version: Edition/revision and publication date not established for the selected resource(s).
Concise measure-theoretic probability connecting formal definitions with probabilistic arguments.
Version record

Jeffrey S. Rosenthal, *A First Look at Rigorous Probability Theory*
Difficulty: Advanced **Type:** Book / monograph
Status/version: 2nd ed. (2006); author errata and selected solutions linked.
Short introduction to sigma-fields, integration and limit laws; second edition (2006), with author errata and selected solutions.
Version record

Marek Capiński and Peter Ekkehard Kopp, *Measure, Integral and Probability*
Difficulty: Advanced **Type:** Book / monograph
Status/version: 2nd ed. (2004)
Measure and integration, Radon–Nikodym theory, martingales and mathematical finance; second edition (2004).
Version record

Jean Jacod and Philip Protter, *Probability Essentials*
Difficulty: Advanced **Type:** Book / monograph
Status/version: Edition/revision and publication date not established for the selected resource(s).
Concise graduate probability with measure, integration and exercises; linked second edition.
Version record

Rick Durrett, *Probability: Theory and Examples*, 5th ed. (2019)
Difficulty: Advanced **Type:** Book / monograph
Status/version: 5th ed. (2019)
Limit theorems, random walks, Markov chains and martingales, with proofs and examples.
Version record

Patrick Billingsley, *Probability and Measure*
Difficulty: Advanced **Type:** Book / monograph
Status/version: Edition/revision and publication date not established for the selected resource(s).
Detailed reference for measure, integration and convergence.
Version record

Achim Klenke, *Probability Theory: A Comprehensive Course*
Difficulty: Advanced **Type:** Book / monograph
Status/version: Linked 2nd ed. (2014); edition distinctions retained in annotation.
Broad graduate probability reference; the linked second edition remains useful (2014). The third edition (2020; e-book 30 October, print 31 October) adds exercises and section summaries. Measure-theoretic foundations through stochastic integration support asset-price modelling.
Version record

Olav Kallenberg, *Foundations of Modern Probability*
Difficulty: Research **Type:** Book / monograph
Status/version: Linked 3rd ed. (2021); 2nd edition also retained.
Convergence, conditioning, point processes and random measures. Linked third edition (2021); the second edition is also retained.
Version record

1.3 Theoretical, Bayesian and nonparametric statistics

1.3.1 Classical, Bayesian and decision-theoretic inference

Robert W. Keener, *Theoretical Statistics: Topics for a Core Course*
Difficulty: Advanced **Type:** Book / monograph
Status/version: Edition/revision and publication date not established for the selected resource(s).
Decision theory, estimation, testing and large-sample theory; assumes measure-theoretic probability.
Version record

E. L. Lehmann and George Casella, *Theory of Point Estimation*, 2nd ed. (1998)

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: 2nd ed. (1998)

Estimation, unbiasedness, equivariance, minimaxity and optimality.

Version record

E. L. Lehmann and Joseph P. Romano, *Testing Statistical Hypotheses*, 3rd ed. (2005)

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: 3rd ed. (2005)

Exact and asymptotic hypothesis testing, power and optimality. The third edition remains useful (2005); select the fourth edition (Springer, 2022; e-book 22 June, hardcover 24 June) for updated coverage. Graduate probability is prerequisite; mathematical error control does not by itself handle a financial researcher's unrecorded selection process.

Version record

A. W. van der Vaart, *Asymptotic Statistics* (1998)

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: Selected citation year(s): 1998. Version distinctions remain in the annotation.

M-estimation, asymptotic efficiency, delta methods, bootstrap and empirical processes.

Version record

Thomas S. Ferguson, *A Course in Large Sample Theory* (1996)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 1996. Version distinctions remain in the annotation.

Complementary derivations of large-sample methods, including Bayesian asymptotics.

Version record

Jun Shao, *Mathematical Statistics*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Rigorous mathematical statistics and a substantial problem collection.

Version record

Andrew Gelman et al., *Bayesian Data Analysis*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Applied Bayesian model construction, computation and data analysis.

Version record

José M. Bernardo and Adrian F. M. Smith, *Bayesian Theory*

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Bayesian foundations and decision theory.

Version record

Peter M. Lee, *Bayesian Statistics: An Introduction*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Bayesian reasoning and its mathematical foundations.

Version record

Peter D. Hoff, *A First Course in Bayesian Statistical Methods*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Bayesian derivations and practical statistical inference.

Version record

A. W. van der Vaart and Jon A. Wellner, *Weak Convergence and Empirical Processes: With Applications to Statistics*

Difficulty: Research **Type:** Book / monograph

Status/version: 2nd ed. (2023); historical 1st ed. (1996) also identified.

Select the second edition (Springer, 2023; e-book 11 July, hardcover 12 July), with expanded preservation results and statistical applications; the 1996 first edition remains useful historical groundwork. Graduate analysis and probability are prerequisites. Supports asymptotic arguments for estimators and resampling, with dependence assumptions checked separately for financial panels.

Version record

James O. Berger, *Statistical Decision Theory and Bayesian Analysis*, 2nd ed. (1985)

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: 2nd ed. (1985)

Minimax, invariant, Bayesian and sequential decisions; James O. Berger is distinct from Casella coauthor Roger L. Berger.

Version record

Larry Wasserman, *All of Nonparametric Statistics*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Density estimation, smoothing and nonparametric inference.

Version record

1.3.2 Computational and modern inference

Bradley Efron and Trevor Hastie, *Computer Age Statistical Inference: Algorithms, Evidence, and Data Science*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2016); DOI 10.1017/cbo9781316576533.

Connects classical frequentist and Bayesian inference to the bootstrap, empirical Bayes, false-discovery rates, the lasso and post-selection inference. Useful bridge from textbook inference to the multiple-testing problems in factor and strategy research; the authors host a free PDF.

Version record

Bradley Efron and Robert J. Tibshirani, *An Introduction to the Bootstrap*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Chapman and Hall/CRC (1994); DOI 10.1201/9780429246593.

Standard reference for bootstrap standard errors, confidence intervals and bias correction. Independent resampling is the starting point only; dependent financial data need the block and stationary variants listed in the validation chapter.

Version record

Christian P. Robert and George Casella, *Monte Carlo Statistical Methods*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer New York (2004); DOI 10.1007/978-1-4757-4145-2.

Simulation-based inference: importance sampling, Markov chain Monte Carlo, convergence diagnostics and stochastic optimisation. Supports Bayesian estimation of latent-volatility, regime and hierarchical models.

Version record

Bradley Efron, *Large-Scale Inference: Empirical Bayes Methods for Estimation, Testing, and Prediction*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2010); DOI 10.1017/cbo9780511761362.

Empirical Bayes and local false-discovery rates for thousands of simultaneous tests; directly relevant to screening many signals or funds.

Version record

David M. Blei, Alp Kucukelbir and Jon D. McAuliffe, "Variational Inference: A Review for Statisticians"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of the American Statistical Association, 112(518), 859–877 (2017); DOI 10.1080/01621459.2017.1285773.

Mean-field and stochastic variational inference as a scalable alternative to MCMC, with its approximation biases.

Version record

Michael Betancourt, "A Conceptual Introduction to Hamiltonian Monte Carlo"

Difficulty: Advanced **Type:** Research paper

Status/version: arXiv 1701.02434, January 2017 (later revisions).

Geometric explanation of HMC and its diagnostics; background for Stan-based Bayesian finance models.

Extended note. Version record

1.3.3 Bayesian methods in finance

Doron Avramov and Guofu Zhou, "Bayesian Portfolio Analysis"

Difficulty: Advanced **Type:** Survey

Status/version: Annual Review of Financial Economics, 2(1), 25–47 (2010); DOI 10.1146/annurev-financial-120209-133947.

Review of Bayesian portfolio analysis. Priors can carry information about events, macro conditions, asset-pricing theories and the forces driving security returns, and the Bayesian framework accounts for parameter and model uncertainty that standard estimates ignore. Covers i.i.d. and predictable returns and applications from single assets and equity portfolios to mutual and hedge funds.

Version record

Philippe Jorion, "Bayes-Stein Estimation for Portfolio Analysis"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial and Quantitative Analysis, 21(3), 279 (1986); DOI 10.2307/2331042.

Treats expected-return estimation for a portfolio as simultaneous estimation of normal means under the investor's loss, where Stein showed the sample mean is inadmissible. Proposes a simple empirical Bayes (Bayes–Stein) estimator that shrinks means toward a common value; simulations show significant gains in portfolio selection.

Version record

Shmuel Kandel and Robert F. Stambaugh, "On the Predictability of Stock Returns: An Asset-Allocation Perspective"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 51(2), 385–424 (1996); DOI 10.1111/j.1540-6261.1996.tb02689.x.

Judges monthly stock-return predictability from the perspective of a risk-averse Bayesian investor allocating between stocks and cash. A regression that looks weak by the usual statistical measures can still move the portfolio decision substantially, even when prior beliefs are weighted against predictability: statistical and economic significance differ.

Version record

Nicholas Barberis, "Investing for the Long Run when Returns Are Predictable"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 55(1), 225–264 (2000); DOI 10.1111/0022-1082.00205.

Long-horizon portfolio choice when returns are predictable, with explicit estimation risk. Even after accounting for parameter uncertainty, predictability leads long-horizon investors to hold substantially more stocks; but because the evidence is statistically weak, an investor who ignores estimation risk can over-allocate to stocks by a sizeable amount.

Version record

Ľuboš Pástor, "Portfolio Selection and Asset Pricing Models"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 55(1), 179–223 (2000); DOI 10.1111/0022-1082.00204.

Uses an asset-pricing model as an informative prior for portfolio selection, with the degree of belief in the model as a parameter. Applied to home bias and to value and size effects: US investors' belief in the domestic CAPM must be very strong to justify observed home bias, and the same prior still gives large, stable positions in the Fama–French book-to-market portfolio.

Version record

Ľuboš Pástor and Robert F. Stambaugh, "Comparing asset pricing models: an investment perspective"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 56(3), 335–381 (2000); DOI 10.1016/s0304-405x(00)00044-1.

Compares risk-based and characteristic-based pricing models by the portfolios that investors centred on each would hold. Dogmatic beliefs and unconstrained positions give economically large differences; modest uncertainty about the models, and realistic position-size constraints, shrink the differences and sometimes remove them.

Version record

Christopher S. Jones and Jay Shanken, "Mutual fund performance with learning across funds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 78(3), 507–552 (2005); DOI 10.1016/j.jfineco.2004.08.009.

Lets beliefs about one fund's alpha depend on the alphas of other funds. With independent priors the largest posterior alpha grows without bound as the number of funds rises, even when no manager has skill; with prior dependence each alpha is shrunk toward an aggregate estimate. Also shows that restricting estimation to surviving funds biases the average alpha upward.

Version record

Bradley Efron, "Tweedie's Formula and Selection Bias"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of the American Statistical Association, 106(496), 1602–1614 (2011); DOI 10.1198/jasa.2011.tm11181.

When many estimates are observed, the largest ones overstate their true values (selection bias, or regression to the mean). Tweedie's formula gives a simple empirical-Bayes correction; the paper examines its merits and limits and its link to James–Stein estimation. Directly applicable to the best of many backtested strategies or funds.

Version record

Eric Jacquier, Nicholas G. Polson and Peter E. Rossi, "Bayesian Analysis of Stochastic Volatility Models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 12(4), 371–389 (1994); DOI 10.1080/07350015.1994.10524553.

Develops a Markov chain Monte Carlo sampler for stochastic-volatility models with autoregressive log variance, giving exact finite-sample posterior inference, the filtered and smoothed volatility path, and predictive densities that include parameter uncertainty. On stock and exchange-rate data and in sampling experiments, the Bayes estimators beat method-of-moments and quasi-maximum-likelihood estimators.

Version record

Sangjoon Kim, Neil Shephard and Siddhartha Chib, "Stochastic Volatility: Likelihood Inference and Comparison with ARCH Models"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 65(3), 361–393 (1998); DOI 10.1111/1467-937X.00050.

Likelihood-based MCMC framework for stochastic volatility that samples all latent volatilities at once through an offset mixture approximation with importance reweighting. Adds simulation-based filtering, likelihood evaluation and model diagnostics, and compares stochastic-volatility and GARCH models with non-nested likelihood ratios and Bayes factors.

Version record

Jun Tu and Guofu Zhou, "Incorporating Economic Objectives into Bayesian Priors: Portfolio Choice under Parameter Uncertainty"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 45(4), 959–986 (2010); DOI 10.1017/s0022109010000335.

Places priors on the solution of the portfolio problem rather than on the primitive return parameters. On the Fama–French 25 portfolios and three factors (1965–2004), objective-based priors change certainty-equivalent returns by more than 10% a year in many cases and can beat both alternative priors and strong classical strategies out of sample.

Version record

Siddhartha Chib, Xiaming Zeng and Lingxiao Zhao, "On Comparing Asset Pricing Models"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(1), 551–577 (2020); DOI 10.1111/jofi.12854.

Shows that the Barillas–Shanken Bayesian comparison of factor models is unsound because the Jeffreys priors on nuisance parameters violate a change-of-variable property across models; simulations confirm it performs poorly. Derives a class of improper priors that gives valid marginal likelihoods and reliable comparisons of which factors are risk factors.

Version record

Andrew Gelman, Aki Vehtari, Daniel Simpson et al., "Bayesian Workflow"

Difficulty: Advanced **Type:** Research preprint

Status/version: arXiv preprint 2011.01808 (2020); DOI 10.48550/arXiv.2011.01808.

Describes applied Bayesian analysis as a workflow rather than a single fit: iterative model building, prior and posterior predictive checking, validation, troubleshooting computation, understanding and comparing many models. Worked examples; a practical companion to Bayesian Data Analysis. Preprint.

Version record

Eric Jacquier and Nicholas Polson, "Bayesian Methods In Finance"

Difficulty: Advanced **Type:** Survey

Status/version: In J. Geweke, G. Koop and H. van Dijk (eds.), The Oxford Handbook of Bayesian Econometrics, Oxford University Press, 438–512 (2011); DOI 10.1093/oxfordhb/9780199559084.013.0010.

Handbook survey of Bayesian methods across finance: return predictability and its economic relevance for allocation, MCMC and particle-filtering algorithms and their role in stochastic-volatility modelling, Bayesian contributions to empirical asset pricing, shrinkage and the use of theory in priors, and filtering with parameter learning.

Version record

Michael Johannes and Nicholas Polson, "MCMC Methods for Continuous-Time Financial Econometrics"

Difficulty: Research **Type:** Book chapter

Status/version: In Y. Aït-Sahalia and L. P. Hansen (eds.), Handbook of Financial Econometrics, Vol. 2: Applications, North-Holland, 1–72 (2010); DOI 10.1016/B978-0-444-53548-1.50003-9.

Tutorial on MCMC for continuous-time asset-pricing models: the Clifford–Hammersley theorem, Gibbs and Metropolis–Hastings samplers and their convergence, worked algorithms for equity-price, option-pricing, term-structure and regime-switching models, and sequential inference for parameters and states.

Version record

Gael M. Martin, David T. Frazier, Worapree Maneesoonthorn et al., "Bayesian forecasting in economics and finance: A modern review"

Difficulty: Advanced **Type:** Survey

Status/version: International Journal of Forecasting, 40(2), 811–839 (2024); DOI 10.1016/j.ijforecast.2023.05.002.

Review of Bayesian forecasting in economics and finance: how uncertainty about model, parameters and latent states enters the forecast distribution by integration or averaging, and how modern Bayesian computation makes forecasts feasible for large and complex problems. Historical context with computational detail.

Version record

1.4 High-dimensional statistics and random matrices

Large panels and many signals require non-asymptotic and random-matrix tools. Read these before covariance shrinkage, factor-count selection and high-dimensional portfolio tests.

Martin J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2019); DOI 10.1017/9781108627771.

Concentration inequalities, sparse regression, covariance estimation and matrix completion when parameters outnumber observations. Supplies the theory behind shrinkage, lasso screening and large-panel factor estimation.

Version record

Roman Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, second edition, February 2026 (ISBN 9781009490672); first edition 2018 (DOI 10.1017/9781108231596); author PDF available.

Concentration of measure, random vectors and matrices with data-science applications; the 2026 second edition adds a refresher chapter and about 200 exercises. Probability prerequisite for high-dimensional covariance and portfolio estimation.

Version record

Marc Potters and Jean-Philippe Bouchaud, *A First Course in Random Matrix Theory: for Physicists, Engineers and Data Scientists*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2020); DOI 10.1017/9781108768900.

Random-matrix spectra, free probability and eigenvalue cleaning, with chapters aimed at sample covariance matrices. Explains why sample eigenvalues are biased and how rotationally invariant estimators correct them.

Version record

Laurent Laloux, Pierre Cizeau, Jean-Philippe Bouchaud and Marc Potters, "Noise Dressing of Financial Correlation Matrices"

Difficulty: Research **Type:** Research paper

Status/version: Physical Review Letters, 83(7), 1467–1470 (1999); DOI 10.1103/physrevlett.83.1467.

Early demonstration that most eigenvalues of an empirical stock correlation matrix lie inside the Marchenko–Pastur noise band. Historical starting point for eigenvalue clipping; later shrinkage estimators are better calibrated.

Version record

Joël Bun, Jean-Philippe Bouchaud and Marc Potters, "Cleaning large correlation matrices: Tools from Random Matrix Theory"

Difficulty: Research **Type:** Research paper

Status/version: Physics Reports, 666, 1–109 (2017); DOI 10.1016/j.physrep.2016.10.005.

Survey of rotationally invariant and nonlinear-shrinkage estimators, with out-of-sample portfolio-risk comparisons. Read with Ledoit–Wolf to compare the physics and statistics routes to the same estimation problem.

Version record

Jianqing Fan, Yuan Liao and Martina Mincheva, "Large Covariance Estimation by Thresholding Principal Orthogonal Complements"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Royal Statistical Society Series B: Statistical Methodology, 75(4), 603–680 (2013); DOI 10.1111/rssb.12016.

POET: a factor-plus-sparse-residual covariance estimator with convergence rates in high dimensions. Relevant when a statistical factor model is combined with residual-covariance regularisation.

Version record

Peter Bühlmann and Sara van de Geer, *Statistics for High-Dimensional Data: Methods, Theory and Applications*

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (2011); DOI 10.1007/978-3-642-20192-9.

Lasso, group lasso, boosting and stability-oriented methods with oracle inequalities. A more applied-statistics complement to Wainwright for sparse signal screening.

Version record

Zhidong Bai and Jack W. Silverstein, *Spectral Analysis of Large Dimensional Random Matrices*

Difficulty: Research **Type:** Book / monograph

Status/version: Springer New York (2010); DOI 10.1007/978-1-4419-0661-8.

Rigorous limiting spectral distributions, including Marchenko–Pastur and sample-covariance results; second edition (2010).

Version record

Olivier Ledoit and Michael Wolf, "A well-conditioned estimator for large-dimensional covariance matrices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Multivariate Analysis, 88(2), 365–411 (2004); DOI 10.1016/s0047-259x(03)00096-4.

Optimal linear shrinkage toward the identity with a consistent estimator of the intensity; the statistical companion to the portfolio paper.

Version record

Trevor Hastie, Robert Tibshirani and Martin Wainwright, *Statistical Learning with Sparsity: The Lasso and Generalizations*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Chapman and Hall/CRC (2015); DOI 10.1201/b18401.

Lasso variants, graphical lasso, sparse PCA and post-selection inference, with free author PDF.

Version record

Peter J. Bickel and Elizaveta Levina, "Covariance regularization by thresholding"

Difficulty: Research **Type:** Research paper

Status/version: The Annals of Statistics, 36(6), 2577–2604 (2008); DOI 10.1214/08-aos600.

Hard-thresholding covariance estimators consistent in high dimensions under sparsity.

Version record

Jianqing Fan, Yingying Fan and Jinchi Lv, "High dimensional covariance matrix estimation using a factor model"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 147(1), 186–197 (2008); DOI 10.1016/j.jeconom.2008.09.017.

Factor-model covariance estimators and their advantage for precision-matrix and portfolio-risk estimation.

Version record

1.5 Heavy tails, extremes and dependence

Tail estimation, extreme-value theory and copulas underpin stress calibration, tail-risk hedging, catastrophe pricing and default dependence. Small tail samples make model uncertainty part of the result.

Rama Cont, "Empirical properties of asset returns: stylized facts and statistical issues"

Difficulty: Advanced **Type:** Research paper

Status/version: Quantitative Finance, 1(2), 223–236 (2001); DOI 10.1080/713665670.

Concise catalogue of heavy tails, volatility clustering, leverage effects, aggregational Gaussianity and absence of linear autocorrelation. A checklist for whether a model or simulator reproduces basic return behaviour.

Version record

Benoit Mandelbrot, "The Variation of Certain Speculative Prices"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Business, 36(4), 394 (1963); DOI 10.1086/294632.

Historical source for non-Gaussian, heavy-tailed price changes using cotton prices. Its infinite-variance stable model is not the modern consensus, but the paper frames the tail-risk question.

Version record

Paul Embrechts, Claudia Klüppelberg and Thomas Mikosch, *Modelling Extremal Events for Insurance and Finance*

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (1997); DOI 10.1007/978-3-642-33483-2.

Extreme-value theory, peaks-over-threshold, tail-index estimation and ruin theory. Reference text for tail quantiles, catastrophe losses and stress calibration where the sample contains few extremes.

Version record

Harry Joe, *Dependence Modeling with Copulas*

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: Chapman and Hall/CRC (2014); DOI 10.1201/b17116.

Copula families, tail dependence, vine constructions and inference. Use for joint-default, multi-asset tail and insurance-loss dependence; the Gaussian-copula critique in the credit section shows the limits of a misapplied model.

Extended note. Version record

Roger B. Nelsen, *An Introduction to Copulas*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer New York (2006); DOI 10.1007/0-387-28678-0.

Shorter mathematical introduction to copula construction, Sklar theorem and measures of association before vine and estimation methods.

Version record

Nassim Nicholas Taleb, *Statistical Consequences of Fat Tails: Real World Preasymptotics, Epistemology, and Applications*

Difficulty: Advanced **Type:** Book / monograph

Status/version: STEM Academic Press, Technical Incerto collection, 2020; arXiv 2001.10488.

Argues that slow convergence of sample moments under fat tails undermines naive estimates of means, correlations and Sharpe ratios. Polemical in places; useful as a stress test of estimation assumptions rather than a calibrated model. The author distributes an arXiv version.

Extended note. Version record

Alexander J. McNeil and Rüdiger Frey, "Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 7(3-4), 271–300 (2000); DOI 10.1016/s0927-5398(00)00012-8.

GARCH-filtered EVT for conditional VaR and ES; a standard approach for daily tail forecasting.

Version record

Bruce M. Hill, "A Simple General Approach to Inference About the Tail of a Distribution"

Difficulty: Advanced **Type:** Research paper

Status/version: The Annals of Statistics, 3(5) (1975); DOI 10.1214/aos/1176343247.

The Hill tail-index estimator; read with threshold-sensitivity diagnostics.

Version record

François Longin and Bruno Solnik, "Extreme Correlation of International Equity Markets"
Difficulty: Research **Type:** Research paper
Status/version: The Journal of Finance, 56(2), 649–676 (2001); DOI 10.1111/0022-1082.00340.
EVT-based evidence that correlation rises in bear markets but not in bull markets; foundation for asymmetric dependence.
Version record

Andrew J. Patton, "Modelling Asymmetric Exchange Rate Dependence"
Difficulty: Research **Type:** Research paper
Status/version: International Economic Review, 47(2), 527–556 (2006); DOI 10.1111/j.1468-2354.2006.00387.x.
Time-varying conditional copulas for asymmetric dependence; methodological template for dynamic copula models.
Version record

Kjersti Aas, Claudia Czado, Arnoldo Frigessi and Henrik Bakken, "Pair-copula constructions of multiple dependence"
Difficulty: Research **Type:** Research paper
Status/version: Insurance: Mathematics and Economics, 44(2), 182–198 (2009); DOI 10.1016/j.insmatheco.2007.02.001.
Vine (pair-copula) constructions for flexible high-dimensional dependence with inference procedures.
Version record

Xavier Gabaix, "Power Laws in Economics and Finance"
Difficulty: Advanced **Type:** Research paper
Status/version: Annual Review of Economics, 1(1), 255–294 (2009); DOI 10.1146/annurev.economics.050708.142940.
Survey of power laws in returns, volume, firm size and city size, with mechanisms that generate them.
Version record

Ethan Ratliff-Crain, Colin M. Van Oort, Matthew T. K. Koehler and Brian F. Tivnan, "Revisiting Cont's stylized facts for modern stock markets"
Difficulty: Research **Type:** Research paper
Status/version: Quantitative Finance, 25(9), 1343–1373 (2025); DOI 10.1080/14697688.2025.2530637.
Re-tests Cont's eleven stylised facts on modern markets to check which still hold.
Version record

1.6 Stochastic processes and calculus

Graduate probability supports these texts. See also Shreve, Volumes I and II, Karatzas–Shreve and Section 7.1.

Mark S. Joshi, *The Concepts and Practice of Mathematical Finance*
Difficulty: Advanced **Type:** Book / monograph
Status/version: Edition/revision and publication date not established for the selected resource(s).
Risk-neutral pricing, Black–Scholes and exotic derivatives; chapter numbering depends on edition.
Version record

Martin Baxter and Andrew Rennie, *Financial Calculus: An Introduction to Derivative Pricing*
Difficulty: Advanced **Type:** Book / monograph
Status/version: Edition/revision and publication date not established for the selected resource(s).
Compact martingale-pricing introduction, including continuous-time measure changes.
Version record

Bernt Øksendal, *Stochastic Differential Equations: An Introduction with Applications*
Difficulty: Advanced **Type:** Book / monograph
Status/version: Linked 6th ed. (2003)
Stochastic integration, differential equations and applications; linked sixth edition (2003).
Version record

Steven E. Shreve, *Stochastic Calculus for Finance*, Volumes I and II
Difficulty: Advanced **Type:** Book / monograph
Status/version: 2004
Volume I, *The Binomial Asset Pricing Model*, covers discrete-time pricing; Volume II, *Continuous-Time Models*, covers Brownian motion, Itô calculus, valuation and hedging. The Volume II publisher record and author errata complement both volumes. Crypto jumps, funding and venue frictions require extensions.
Applications: All assets, including crypto; derivatives, hedging, risk.
Version record

Hans Föllmer and Alexander Schied, ***Stochastic Finance: An Introduction in Discrete Time***

Difficulty: Advanced **Type:** Book / monograph

Status/version: De Gruyter; 2025-08-15 (publisher lists ebook and paperback). Fifth, revised and expanded edition. DOI 10.1515/9783111045283.

A rigorous discrete-time mathematical-finance text on arbitrage, preferences, risk measures, incomplete markets, contingent claims and dynamic hedging. The fifth edition adds current material on law-invariant risk measures, relative entropy and universal portfolios. It complements continuous-time stochastic-calculus texts; it does not replace continuous-time pricing or instrument-specific market conventions.

Extended note. Version record

Ioannis Karatzas and Steven E. Shreve, ***Brownian Motion and Stochastic Calculus***

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2nd ed., 1991

Deep stochastic-calculus reference for Itô/Stratonovich calculus, martingale representation, Girsanov and stochastic differential equations. Use it to check filtration, measure-change and path-dependence assumptions behind continuous-time desk models.

Applications: Derivatives, rates; stochastic models. Version record

J. Michael Steele, ***Stochastic Calculus and Financial Applications***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer New York (2001); DOI 10.1007/978-1-4684-9305-4.

Brownian motion, martingales, Ito calculus and Girsanov theory with intuitive arguments and finance examples. A bridge between Shreve and the measure-theoretic texts.

Version record

Philip E. Protter, ***Stochastic Integration and Differential Equations***

Difficulty: Research **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (2005); DOI 10.1007/978-3-662-10061-5.

Semimartingale approach to stochastic integration, covering jumps and general SDEs. Reference for jump-diffusion and Levy-driven models beyond continuous Brownian settings.

Version record

Peter E. Kloeden and Eckhard Platen, ***Numerical Solution of Stochastic Differential Equations***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (1992); DOI 10.1007/978-3-662-12616-5.

Euler, Milstein and higher-order schemes with strong and weak convergence theory. Use before discretising volatility or short-rate SDEs in Monte Carlo pricing.

Version record

Daniel Revuz and Marc Yor, ***Continuous Martingales and Brownian Motion***

Difficulty: Research **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (1999); DOI 10.1007/978-3-662-06400-9.

Graduate reference on Brownian motion and continuous martingales: local times, excursions, Girsanov, time changes and Bessel processes. The place to go when Karatzas–Shreve runs out.

Version record

L. C. G. Rogers and David Williams, ***Diffusions, Markov Processes and Martingales, Volume 1: Foundations***

Difficulty: Research **Type:** Book / monograph

Status/version: Cambridge University Press, edition 2 (2000); DOI 10.1017/cbo9781107590120.

Two-volume classic (Volume 2 covers Ito calculus) on Markov processes, martingales and diffusions, written with probabilistic intuition and many finance-relevant examples.

Version record

Jean Jacod and Albert N. Shiryaev, ***Limit Theorems for Stochastic Processes***

Difficulty: Research **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (2003); DOI 10.1007/978-3-662-05265-5.

Semimartingale characteristics and functional limit theorems; the technical foundation behind high-frequency asymptotics and jump-process inference.

Version record

Jean-François Le Gall, *Brownian Motion, Martingales, and Stochastic Calculus*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer International Publishing (2016); DOI 10.1007/978-3-319-31089-3.

Concise modern graduate text from Brownian motion and martingales to stochastic integration, SDEs and local times; a compact alternative to the longer references.

Version record

David Applebaum, *Lévy Processes and Stochastic Calculus*

Difficulty: Research **Type:** Book / monograph

Status/version: Cambridge University Press, edition 2 (2009); DOI 10.1017/cbo9780511809781.

Levy processes, Levy–Ito decomposition and stochastic calculus with jumps, with chapters on exponential Levy models for finance (second edition, 2009).

Version record

Freddy Delbaen and Walter Schachermayer, *The Mathematics of Arbitrage*

Difficulty: Research **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (2006); DOI 10.1007/978-3-540-31299-4.

The fundamental theorem of asset pricing in full generality (no free lunch with vanishing risk for semimartingales); the rigorous core behind risk-neutral pricing.

Version record

Grigorios A. Pavliotis, *Stochastic Processes and Applications: Diffusion Processes, the Fokker-Planck and Langevin Equations*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer New York (2014); DOI 10.1007/978-1-4939-1323-7.

Applied treatment of diffusions, Fokker–Planck equations, Ornstein–Uhlenbeck and Langevin dynamics, ergodicity and parameter estimation for SDEs.

Version record

Stewart N. Ethier and Thomas G. Kurtz, *Markov Processes: Characterization and Convergence*

Difficulty: Research **Type:** Book / monograph

Status/version: Wiley, edition 1 (1986); DOI 10.1002/9780470316658.

Martingale-problem characterisation of Markov processes and weak convergence; used for diffusion limits of order books, queues and Markov-chain approximations.

Version record

1.7 Stochastic control, BSDEs and mean-field games

Dynamic programming, viscosity solutions and backward SDEs connect portfolio choice, execution, market making and nonlinear pricing. Mean-field games describe many interacting traders.

Robert C. Merton, "Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case"

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Economics and Statistics, 51(3), 247 (1969); DOI 10.2307/1926560.

Continuous-time consumption and portfolio choice solved by dynamic programming. The benchmark Merton fraction underlies dynamic allocation, Kelly comparisons and transaction-cost extensions.

Version record

Robert C. Merton, "Optimum consumption and portfolio rules in a continuous-time model"

Difficulty: Advanced/Research **Type:** Research paper

Status/version: Journal of Economic Theory, 3(4), 373–413 (1971); DOI 10.1016/0022-0531(71)90038-x.

Extends the 1969 problem to general utility and stochastic investment opportunities, introducing intertemporal hedging demands.

Version record

Huyên Pham, *Continuous-time Stochastic Control and Optimization with Financial Applications*

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (2009); DOI 10.1007/978-3-540-89500-8.

Dynamic programming, viscosity solutions, BSDEs and optimal stopping, applied to portfolio choice, hedging and real options.

Version record

Wendell H. Fleming and H. Mete Soner, ***Controlled Markov Processes and Viscosity Solutions***

Difficulty: Research **Type:** Book / monograph

Status/version: Springer, second edition, 2006; DOI 10.1007/0-387-31071-1.

Graduate reference for Hamilton–Jacobi–Bellman equations and viscosity solutions underlying optimal execution, market-making and consumption problems.

Version record

Nizar Touzi, ***Optimal Stochastic Control, Stochastic Target Problems, and Backward SDE***

Difficulty: Research **Type:** Book / monograph

Status/version: Springer New York (2013); DOI 10.1007/978-1-4614-4286-8.

Concise lecture-based treatment of control, super-hedging as a stochastic target problem and backward SDEs.

Version record

Nicole El Karoui, Shige Peng and Marie Claire Quenez, "Backward Stochastic Differential Equations in Finance"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 7(1), 1–71 (1997); DOI 10.1111/1467-9965.00022.

Foundational survey connecting BSDEs to hedging, recursive utility and nonlinear pricing; background for funding-adjusted valuation and deep BSDE solvers.

Version record

René Carmona and François Delarue, ***Probabilistic Theory of Mean Field Games with Applications I***

Difficulty: Research **Type:** Book / monograph

Status/version: Springer International Publishing (2018); DOI 10.1007/978-3-319-58920-6.

Probabilistic approach to mean-field games and controls, with applications to systemic risk and crowded trading. Volume II treats common noise and master equations.

Version record

Pierre Cardaliaguet and Charles-Albert Lehalle, "Mean field game of controls and an application to trade crowding"

Difficulty: Research **Type:** Research paper

Status/version: Mathematics and Financial Economics, 12(3), 335–363 (2018); DOI 10.1007/s11579-017-0206-z.

Optimal liquidation when many traders affect a common price, solved as a mean-field game of controls. Formalises crowding in execution schedules.

Version record

Dimitri P. Bertsekas, ***Dynamic Programming and Optimal Control***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Athena Scientific; Volume I fourth edition 2017, Volume II fourth edition 2012.

Discrete-time dynamic programming, approximate DP and rollout methods; a bridge between control theory and reinforcement learning.

Version record

Jiongmin Yong and Xun Yu Zhou, ***Stochastic Controls: Hamiltonian Systems and HJB Equations***

Difficulty: Research **Type:** Book / monograph

Status/version: Springer New York (1999); DOI 10.1007/978-1-4612-1466-3.

Links the stochastic maximum principle and dynamic programming, with linear-quadratic control and forward-backward SDEs; the standard reference for mean-variance control problems.

Version record

Bernt Øksendal and Agnès Sulem, ***Applied Stochastic Control of Jump Diffusions***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer International Publishing (2019); DOI 10.1007/978-3-030-02781-0.

Optimal stopping, singular and impulse control for jump diffusions, with portfolio, transaction-cost and harvesting examples (third edition, 2019).

Version record

Jean-Michel Lasry and Pierre-Louis Lions, "Mean field games"

Difficulty: Research **Type:** Research paper

Status/version: Japanese Journal of Mathematics, 2(1), 229–260 (2007); DOI 10.1007/s11537-007-0657-8.

Founding paper of mean-field game theory: coupled HJB and Fokker–Planck equations for the Nash equilibrium of many small interacting agents.

Version record

Minyi Huang, Roland P. Malhamé and Peter E. Caines, "Large population stochastic dynamic games: closed-loop McKean-Vlasov systems and the Nash certainty equivalence principle"

Difficulty: Research **Type:** Research paper

Status/version: Communications in Information and Systems, 6(3), 221–252 (2006); DOI 10.4310/cis.2006.v6.n3.a5.

The independent engineering origin of mean-field games (Nash certainty equivalence), parallel to Lasry–Lions.

Version record

Étienne Pardoux and Shige Peng, "Adapted solution of a backward stochastic differential equation"

Difficulty: Research **Type:** Research paper

Status/version: Systems & Control Letters, 14(1), 55–61 (1990); DOI 10.1016/0167-6911(90)90082-6.

Existence and uniqueness for nonlinear BSDEs: the starting point of the BSDE approach to hedging, recursive utility and nonlinear pricing.

Version record

Jianfeng Zhang, *Backward Stochastic Differential Equations: From Linear to Fully Nonlinear Theory*

Difficulty: Research **Type:** Book / monograph

Status/version: Springer New York (2017); DOI 10.1007/978-1-4939-7256-2.

Modern BSDE text covering numerics, path-dependent PDEs and second-order BSDEs, with financial applications.

Version record

René Carmona, *Lectures on BSDEs, Stochastic Control, and Stochastic Differential Games with Financial Applications*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Society for Industrial and Applied Mathematics (2016); DOI 10.1137/1.9781611974249.

Course-length bridge from stochastic control to BSDEs and mean-field games, worked through finance examples.

Version record

Harold J. Kushner and Paul Dupuis, *Numerical Methods for Stochastic Control Problems in Continuous Time*

Difficulty: Research **Type:** Book / monograph

Status/version: Springer New York (2001); DOI 10.1007/978-1-4613-0007-6.

Markov-chain approximation methods with convergence proofs: the standard numerical toolkit for continuous-time control.

Version record

Goran Peskir and Albert Shiryaev, *Optimal Stopping and Free-Boundary Problems*

Difficulty: Research **Type:** Book / monograph

Status/version: Birkhäuser Basel (2006); DOI 10.1007/978-3-7643-7390-0.

Optimal stopping via martingale and Markov methods and the free-boundary characterisation; American options, sequential testing and quickest detection.

Version record

Ioannis Karatzas and Steven E. Shreve, *Methods of Mathematical Finance*

Difficulty: Research **Type:** Book / monograph

Status/version: Springer New York (1998); DOI 10.1007/b98840.

Martingale and duality methods for optimal consumption and investment, incomplete markets and constraints; the continuous-time portfolio-choice reference.

Version record

1.8 Optimisation and numerical methods

Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press first print edition, 8 March 2004; author-hosted book and later digital format linked. The 2023 lecture-slide update is not a book revision.

The mathematical and computational foundation for constrained portfolio construction, execution, calibration and risk budgeting. Its duality and sensitivity analysis chapters are essential for diagnosing why a production optimiser chooses a position, not merely obtaining a numerical solution.

Applications: All assets; portfolio, execution, calibration.

Version record

Jorge Nocedal and Stephen J. Wright, *Numerical Optimization*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2nd ed., 2006

Newton and quasi-Newton methods, line searches, trust regions and constrained optimisation. Complements Boyd–Vandenberghe with numerical stability and ill-conditioning issues that matter when portfolio, calibration and ML routines reach production.

Applications: All assets; optimisation, calibration. Version record

Aharon Ben-Tal, Laurent El Ghaoui and Arkadi Nemirovski, *Robust Optimization*

Difficulty: Research **Type:** Book / monograph

Status/version: Princeton University Press (2009); DOI 10.1515/9781400831050.

Uncertainty sets, tractable robust counterparts and chance constraints. Theory behind robust mean-variance and robust ALM formulations.

Version record

Manfred Gilli, Dietmar Maringer and Enrico Schumann, *Numerical Methods and Optimization in Finance*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Elsevier (2019); DOI 10.1016/c2017-0-01621-x.

Finite differences, Monte Carlo, simulation and heuristic optimisation (threshold accepting, differential evolution) for non-convex portfolio and calibration problems, with code.

Version record

Mykel J. Kochenderfer and Tim A. Wheeler, *Algorithms for Optimization*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: MIT Press, 2019 (ISBN 9780262039420); open-access online edition.

Derivative-based, stochastic, population and surrogate optimisation algorithms with compact Julia implementations; useful for calibration and hyperparameter search. MIT Press provides an open-access edition.

Version record

Dimitris Bertsimas and John N. Tsitsiklis, *Introduction to Linear Optimization*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Athena Scientific, 1997.

Simplex, duality, sensitivity and network flows. Background for cash-flow matching, index tracking and integer constraints on lot sizes.

Version record

Gene H. Golub and Charles F. Van Loan, *Matrix Computations*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Johns Hopkins University Press, fourth edition, 2013 (ISBN 9781421407944).

Reference for factorizations, eigenvalue and least-squares algorithms and their numerical stability; fourth edition (2013).

Version record

Lloyd N. Trefethen and David Bau III, *Numerical Linear Algebra*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: SIAM, twenty-fifth anniversary (second) edition, 2022.

Lecture-based introduction to QR, SVD, conditioning and iterative methods; second edition (SIAM 2022).

Version record

Dimitris Bertsimas and Melvyn Sim, "The Price of Robustness"

Difficulty: Research **Type:** Research paper

Status/version: Operations Research, 52(1), 35–53 (2004); DOI 10.1287/opre.1030.0065.

Budgeted uncertainty sets that keep robust linear programs linear and control conservatism.

Version record

Bartolomeo Stellato, Goran Banjac, Paul Goulart, Alberto Bemporad and Stephen Boyd, "OSQP: an operator splitting solver for quadratic programs"

Difficulty: Advanced **Type:** Research paper

Status/version: 2018 UKACC 12th International Conference on Control (CONTROL), 339–339 (2018); DOI 10.1109/control.2018.8516834.

ADMM-based QP solver used for large portfolio and control problems; warm starts suit rolling rebalancing.

Version record

Akshay Agrawal, Brandon Amos, Shane Barratt, Stephen Boyd, Steven Diamond and J. Zico Kolter, "Differentiable Convex Optimization Layers"

Difficulty: Research **Type:** Research paper

Status/version: NeurIPS 2019 proceedings; arXiv 1910.12430.

cvxpylayers: differentiate through convex programs to train end-to-end decision-focused models such as portfolio layers.

Extended note. Version record

Alexander Shapiro, Darinka Dentcheva and Andrzej Ruszczyński, *Lectures on Stochastic Programming: Modeling and Theory*

Difficulty: Research **Type:** Book / monograph

Status/version: SIAM, third edition, 2021.

Two-stage and multistage stochastic programs, risk-averse optimisation and sample-average approximation; third edition (SIAM 2021).

Version record

1.9 Combinatorics, decisions and game theory

Optional references for combinatorial reasoning, estimation and strategic decisions.

Frederick Mosteller, *Fifty Challenging Problems in Probability with Solutions*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Worked probability problems for practising conditional reasoning, event definitions and solution checks. A useful exercise companion to the Blitzstein–Hwang / Harvard Stat 110 route.

Version record

Geoffrey Grimmett and David Stirzaker, *One Thousand Exercises in Probability*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

An advanced exercise collection for probability proofs and stochastic-process topics. The linked author record identifies the 2001 volume. Pair with the Williams stochastic-process reading as appropriate.

Version record

David Patrick, *Introduction to Counting & Probability*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

A worked foundational route through counting and probability, useful before the more advanced combinatorics readings below.

Version record

Chen Chuan-Chong and Koh Khee-Meng, *Principles and Techniques in Combinatorics* (1992); John Riordan, *Introduction to Combinatorial Analysis*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 1992. Version distinctions remain in the annotation.

Substantive extensions into combinatorial proofs, recurrences, generating functions and inclusion–exclusion. The links provide bibliographic catalogue records; a Riordan edition is not specified.

Version record

Lawrence Weinstein and John A. Adam, *Guesstimation: Solving the World's Problems on the Back of a Cocktail Napkin* (2008); Douglas Hubbard, *How to Measure Anything*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2008. Version distinctions remain in the annotation.

Optional estimation and measurement practice: decompose a quantity into measurable factors, track units and expose assumptions that dominate the estimate. The Hubbard edition is unspecified.

Version record

Avinash Dixit and Barry Nalebuff, *The Art of Strategy*; *Thinking Strategically*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Accessible strategic-reasoning readings for payoff comparisons, backward induction and opponent responses. Editions are unspecified.

Version record

Martin J. Osborne, *An Introduction to Game Theory*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

A formal route through equilibrium concepts, mixed strategies and zero-sum games. The author's course page identifies the Oxford University Press 2004 text.

Version record

Thomas S. Ferguson, *A Course in Game Theory* (World Scientific, 2020)

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2020. Version distinctions remain in the annotation.

A formal game-theory reference, identified under its published title on the author's UCLA page. The author page does not establish current free access to the complete text.

Version record

Daniel Kahneman, *Thinking, Fast and Slow*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Optional adjacent reading on human judgement and biases, rather than a mathematics or game-theory text. For Kelly-related quantitative decision material, use the Kelly paper and MacLean–Thorp–Ziemba collection.

Version record

Thomas M. Cover and Joy A. Thomas, *Elements of Information Theory*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2005); DOI 10.1002/047174882x.

Entropy, mutual information and the information-theoretic treatment of growth-optimal portfolios and universal portfolios. Complements Kelly and Thorp.

Version record

Chapter 2

Asset pricing, econometrics and forecasting

Connect return restrictions and economic mechanisms to estimators and forecasting evidence. Recent foundation models follow the statistical forecasting references they extend.

2.1 Asset-pricing foundations

2.1.1 Theory texts

John H. Cochrane, *Asset Pricing*, revised edition

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press revised edition, February 2005; the author page identifies this selected edition and separately labels draft material for a possible next edition.

The most efficient bridge from stochastic discount factors and Euler equations to cross-sectional and time-series return restrictions. Institutional researchers should use it to distinguish compensation for risk from mispricing or a statistical predictor that merely correlates with realised returns, and to communicate that distinction to portfolio managers and risk committees.

Applications: All assets; factor investing, macro, derivatives. Version record

Darrell Duffie, *Dynamic Asset Pricing Theory*

Difficulty: Research **Type:** Book / monograph

Status/version: Princeton University Press, third edition, 2001 (ISBN 9780691090221).

Graduate treatment of state prices, equivalent martingale measures, term-structure and credit models in discrete and continuous time. The third edition (2001) remains the standard theory reference behind Cochrane.

Version record

Kerry E. Back, *Asset Pricing and Portfolio Choice Theory*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Oxford University Press (2017); DOI 10.1093/acprof:oso/9780190241148.001.0001.

Second edition (2017) covering single-period and dynamic portfolio choice, SDFs, consumption-based pricing, heterogeneous beliefs and production. Problem-driven alternative to Cochrane and Duffie.

Version record

John Y. Campbell, *Financial Decisions and Markets: A Course in Asset Pricing*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press, 2018 (ISBN 9780691160801).

Integrated graduate course on static portfolio choice, the SDF, the equity premium, present-value relations, intertemporal models and behavioural finance. Strong on linking theory to empirical regularities.

Version record

Jonathan E. Ingersoll Jr., *Theory of Financial Decision Making*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Rowman & Littlefield, 1987 (ISBN 9780847673599).

Rigorous classic on utility, portfolio choice, arbitrage, CAPM/APT and continuous-time pricing (1987).

Version record

Costis Skiadas, *Asset Pricing Theory*

Difficulty: Research **Type:** Book / monograph

Status/version: Princeton University Press (2009); DOI 10.1515/9781400830145.

Rigorous treatment of arbitrage, optimality and equilibrium with recursive utility in discrete and continuous time.

Version record

George Pennacchi, *Theory of Asset Pricing*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Pearson/Addison-Wesley, 2008 (ISBN 9780321127204).

Accessible first-year PhD text covering single-period, consumption-based and continuous-time models and term structure.

Version record

Claus Munk, *Financial Asset Pricing Theory*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Oxford University Press (2013); DOI 10.1093/acprof:oso/9780199585496.001.0001.

Graduate text unifying SDF, consumption-based, factor and term-structure models with continuous-time tools.

Version record

2.1.2 Factor models and empirical tests

Andrew Ang, Asset Management: A Systematic Approach to Factor Investing

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Unifies macro and style risks across equities, bonds, credit and alternatives. Study value, momentum, carry and liquidity, mean–variance estimation problems, illiquidity and hedge-fund exposures. Analyse portfolios by shared economic risks as well as asset-class labels.

Version record

Eugene F. Fama and James D. MacBeth, "Risk, Return, and Equilibrium: Empirical Tests"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Political Economy, 81(3), 607–636 (1973); DOI 10.1086/260061.

Introduces cross-sectional regressions averaged over time; the Fama–MacBeth procedure remains the default for characteristic and beta pricing tests. Pair with Shanken for errors-in-variables corrections.

Version record

Michael R. Gibbons, Stephen A. Ross and Jay Shanken, "A Test of the Efficiency of a Given Portfolio"

Difficulty: Advanced **Type:** Research paper

Status/version: Econometrica, 57(5), 1121 (1989); DOI 10.2307/1913625.

The GRS finite-sample test that factor portfolios span test assets, interpreted as a Sharpe-ratio comparison. Predecessor of the high-dimensional test in the validation chapter.

Version record

Lars Peter Hansen, "Large Sample Properties of Generalized Method of Moments Estimators"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 50(4), 1029 (1982); DOI 10.2307/1912775.

Foundational GMM paper: estimation and over-identification tests from moment conditions such as Euler equations and SDF restrictions.

Version record

Lars Peter Hansen and Ravi Jagannathan, "Implications of Security Market Data for Models of Dynamic Economies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 99(2), 225–262 (1991); DOI 10.1086/261749.

Hansen–Jagannathan volatility bounds on admissible SDFs; a quick diagnostic for whether a candidate pricing kernel is volatile enough to price observed Sharpe ratios.

Version record

Jay Shanken, "On the Estimation of Beta-Pricing Models"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 5(1), 1–33 (1992); DOI 10.1093/rfs/5.1.1.

Errors-in-variables correction for two-pass cross-sectional regressions; essential when estimated betas enter second-stage risk-premium tests.

Version record

Eugene F. Fama and Kenneth R. French, "A five-factor asset pricing model"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 116(1), 1–22 (2015); DOI 10.1016/j.jfineco.2014.10.010.

Adds profitability and investment to market, size and value; makes HML largely redundant in the sample. Benchmark model for anomaly alphas, with factor data in the French library.

Version record

Kewei Hou, Chen Xue and Lu Zhang, "Digesting Anomalies: An Investment Approach"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 28(3), 650–705 (2015); DOI 10.1093/rfs/hhu068.

Investment-based q-factor model (market, size, investment, profitability) motivated by production theory; a competitor to Fama–French factors in anomaly tests.

Version record

Bryan T. Kelly, Seth Pruitt and Yinan Su, "Characteristics are covariances: A unified model of risk and return"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 134(3), 501–524 (2019); DOI 10.1016/j.jfineco.2019.05.001.

Instrumented PCA: latent factors with loadings that are linear in observable characteristics. Separates whether characteristics proxy for risk exposure or mispricing.

Version record

Martin Lettau and Markus Pelger, "Factors That Fit the Time Series and Cross-Section of Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(5), 2274–2325 (2020); DOI 10.1093/rfs/hhaa020.

Risk-premium PCA adds a pricing-error penalty to PCA so that weak but priced factors are recovered.

Version record

Serhiy Kozak, Stefan Nagel and Shrihari Santosh, "Shrinking the cross-section"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 135(2), 271–292 (2020); DOI 10.1016/j.jfineco.2019.06.008.

Bayesian-motivated shrinkage of SDF coefficients on many characteristic portfolios; finds characteristics-sparse SDFs perform poorly while PC-sparse ones do well.

Version record

Guanhao Feng, Stefano Giglio and Dacheng Xiu, "Taming the Factor Zoo: A Test of New Factors"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(3), 1327–1370 (2020); DOI 10.1111/jofi.12883.

Double-selection lasso to evaluate whether a new factor adds pricing information beyond a large set of existing factors.

Version record

Stefano Giglio and Dacheng Xiu, "Asset Pricing with Omitted Factors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 129(7), 1947–1990 (2021); DOI 10.1086/714090.

Three-pass estimator that recovers risk premia of an observable factor even when the model omits other priced factors.

Version record

Svetlana Bryzgalova, Jiantao Huang and Christian Julliard, "Bayesian Solutions for the Factor Zoo: We Just Ran Two Quadrillion Models"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 78(1), 487–557 (2023); DOI 10.1111/jofi.13197.

Bayesian model averaging over about 2.25 quadrillion linear factor models, robust to weak factors; an R implementation (BayesianFactorZoo) is on CRAN.

Version record

Turan G. Bali, Robert F. Engle and Scott Murray, *Empirical Asset Pricing: The Cross Section of Stock Returns*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, 2016 (ISBN 9781118095041).

Practical text on portfolio sorts, Fama–MacBeth regressions, data screens and the main characteristics (beta, size, value, momentum, liquidity, skewness). Good first replication manual.

Version record

John H. Cochrane, "Presidential Address: Discount Rates"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 66(4), 1047–1108 (2011); DOI 10.1111/j.1540-6261.2011.01671.x.

Reframes asset pricing around time-varying discount rates and the multidimensional cross-section; source of the factor-zoo question.

Version record

Alexander Dickerson, Christian Julliard and Philippe Mueller, "The co-pricing factor zoo"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 182, 104295 (2026); DOI 10.1016/j.jfineco.2026.104295.

Joint pricing of bonds and stocks across 18 quadrillion models: equity and nontradable factors explain corporate-bond premia once Treasury term-structure risk is controlled, making most bond factors redundant.

Version record

Yacine Aït-Sahalia, Jean Jacod and Dacheng Xiu, "Continuous-Time Fama-MacBeth Regressions"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(12), 3542–3579 (2025); DOI 10.1093/rfs/hhaf072.

Continuous-time Fama–MacBeth regressions with high-frequency data, separating risk premia on continuous moves and jumps of different sizes.

Version record

Marie-Claude Beaulieu, Jean-Marie Dufour and Lynda Khalaf, "Identification-robust and simultaneous inference in multifactor asset pricing models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 248, 105915 (2025); DOI 10.1016/j.jeconom.2024.105915.

Identification-robust and simultaneous inference for multifactor asset-pricing models.

Extended note. Version record

Jiantao Huang and Ran Shi, "Model uncertainty in the cross-section of stock returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 256, 106066 (2026); DOI 10.1016/j.jeconom.2025.106066.

Model uncertainty in cross-sectional return models.

Extended note. Version record

Thuy Duong Dang, Fabian Hollstein and Marcel Prokopczuk, "Factor pricing across asset classes"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 87, 101688 (2026); DOI 10.1016/j.jempfin.2026.101688.

Factor pricing across asset classes.

Extended note. Version record

2.1.3 Return predictability and expected returns

Tobias Moskowitz, Yao Hua Ooi and Lasse Heje Pedersen, "Time Series Momentum"

Difficulty: Research **Type:** Research paper

Status/version: 2012

Own-return continuation across liquid futures, with volatility scaling and exposure implications. Historical crisis behaviour depends on trend reversals, rolls and implementation costs.

Applications: Futures, FX forwards; trend, CTA. Version record

Ralph Koijen, Tobias Moskowitz, Lasse Heje Pedersen and Evert Vrugt, "Carry"

Difficulty: Research **Type:** Research paper

Status/version: 2018

Provides a common definition and empirical study of carry across asset classes. Its main practical contribution is to expose the shared economic and tail-risk structure beneath strategies that are often managed in separate asset-class silos.

Applications: FX, bonds, commodities, equities, options; carry. Version record

Brian Hurst, Yao Hua Ooi and Lasse Heje Pedersen, "A Century of Evidence on Trend-Following Investing"

Difficulty: Advanced **Type:** Research paper

Status/version: 2017

Extends trend evidence over a much longer history and across many futures markets. It is useful for regime and crisis analysis and for communicating implementation-aware evidence to non-quant stakeholders, though the oldest data require particular scrutiny concerning tradability, costs and contract construction.

Applications: Futures, FX; trend, crisis risk mitigation. Version record

Antti Ilmanen, *Expected Returns*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley first publication 21 March 2011; selected Expected Returns is distinct from Ilmanen's 2022 Investing Amid Low Expected Returns.

Cross-asset evidence on value, momentum, carry and liquidity rewards, including bond carry, rolldown and curve strategies. Distinguishes economic risk premia from implementation-dependent returns.

Applications: Multi-asset; risk premia, macro allocation. Version record

Ivo Welch and Amit Goyal, "A Comprehensive Look at the Empirical Performance of Equity Premium Prediction" (2008)

Difficulty: Research **Type:** Research paper

Status/version: Selected DOI-linked version; print 2008-07; online 2007-03-17.

The source's nested return-predictability reading. Compare economic predictors with historical-mean benchmarks using genuinely out-of-sample forecasts. Its historical failures motivate scepticism about in-sample explanatory power, not a proof that every future predictor must fail.

Version record

John Y. Campbell and Robert J. Shiller, "The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 1(3), 195–228 (1988); DOI 10.1093/rfs/1.3.195.

Log-linear present-value identity linking valuation ratios to expected returns and dividend growth; the basis of most predictive-regression interpretation.

Version record

John H. Cochrane, "The Dog That Did Not Bark: A Defense of Return Predictability"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 21(4), 1533–1575 (2008); DOI 10.1093/rfs/hhm046.

Uses the present-value identity to argue that absent dividend-growth predictability strengthens the case for return predictability.

Version record

John Y. Campbell and Samuel B. Thompson, "Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 21(4), 1509–1531 (2008); DOI 10.1093/rfs/hhm055.

Shows that simple economic restrictions (sign constraints, non-negative premium) improve out-of-sample predictive regressions; a direct response to Welch–Goyal.

Version record

David E. Rapach, Jack K. Strauss and Guofu Zhou, "Out-of-Sample Equity Premium Prediction: Combination Forecasts and Links to the Real Economy"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 23(2), 821–862 (2010); DOI 10.1093/rfs/hhp063.

Forecast combination across individual predictors yields stable out-of-sample gains; a template for shrinkage by averaging.

Version record

Amit Goyal, Ivo Welch and Athanasse Zafirov, "A Comprehensive 2022 Look at the Empirical Performance of Equity Premium Prediction"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 37(11), 3490–3557 (2024); DOI 10.1093/rfs/hhae044.

Updates Welch–Goyal (2008) with newer predictors and data through 2021; most variables still fail out of sample. The current reference vintage for equity-premium prediction claims.

Version record

Ian Martin, "What is the Expected Return on the Market?"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 132(1), 367–433 (2017); DOI 10.1093/qje/qjw034.

Derives a lower bound on the equity premium from index option prices (SVIX); a forward-looking alternative to historical averages.

Version record

Elroy Dimson, Paul Marsh and Mike Staunton, ***Triumph of the Optimists: 101 Years of Global Investment Returns***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press (2002); DOI 10.1515/9781400829477.

Long-run international return history across equities, bonds, bills and currencies; survivorship-aware base rates. Updated annually in the UBS Global Investment Returns Yearbook.

Version record

Nusret Cakici, Christian Fieberg, Tobias Neumaier, Thorsten Poddig and Adam Zaremba, "Pockets of Predictability: A Replication"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(6), 3771–3790 (2025); DOI 10.1111/jofi.13484.

Replication finding that the "pockets of predictability" kernel regressions used in-sample information; correcting the code to one-sided kernels removes much of the reported out-of-sample predictability. A model replication-audit reading.

Version record

Jules van Binsbergen, Sophia Hua, Jonas Peeters and Jessica Wachter, "Is the United States a Lucky Survivor? A Hierarchical Bayesian Approach"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(4), 2355–2388 (2025); DOI 10.1111/jofi.13452.

International cross-learning about crash risk suggests survivorship bias explains about one-third of the US equity premium over the past century; caution against extrapolating US history.

Version record

Ricardo Delao, Xiao Han and Sean Myers, "The return of return dominance: Decomposing the cross-section of prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 169, 104059 (2025); DOI 10.1016/j.jfineco.2025.104059.

About 75% of cross-sectional dispersion in P/E ratios reflects differences in future returns and 25% future earnings growth, reconciling evidence on prices and profitability.

Version record

Benjamin Golez and Peter Koudijs, "Equity duration and predictability"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 172, 104114 (2025); DOI 10.1016/j.jfineco.2025.104114.

Since 1945 expected returns dominate equity price variation; rising equity duration explains why because discount-rate variation matters more for distant payouts.

Version record

R. David McLean, Jeffrey Pontiff and Christopher Reilly, "Taking sides on return predictability"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 173, 104158 (2025); DOI 10.1016/j.jfineco.2025.104158.

Which investor types trade with or against return predictors (anomalies), and what that implies for mispricing persistence.

Extended note. Version record

Magnus Dahlquist and Markus Ibert, "Institutions' return expectations across assets and time"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 175, 104188 (2026); DOI 10.1016/j.jfineco.2025.104188.

Capital-market assumptions of asset managers, consultants, advisors, pension funds and forecasters: subjective premia track objective real-time premia and are countercyclical, with large cross-sectional dispersion.

Version record

Kai Li, Yingying Li, Changlei Lyu and Jialin Yu, "How to Dominate the Historical Average"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(10), 3086–3116 (2025); DOI 10.1093/rfs/hhaf010.

Equity-premium forecasts with a conservative constant slope that has lower bias than the historical mean at equal variance, dominating it out of sample.

Version record

Emanuel Moench and Tobias Stein, "Equity Premium Predictability over the Business Cycle"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(3), 1216–1246 (2026); DOI 10.1017/s0022109025102093.

Equity returns follow a V-shape around recessions; term-spread probit models mistime recessions and predictability concentrates around business-cycle turns.

Version record

2.1.4 Intermediaries, demand and behavioural finance

Ralph S. J. Koijen and Motohiro Yogo, A Demand System Approach to Asset Pricing

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy 127(4), 1475–1515, August 2019; published online 19 June 2019; DOI 10.1086/701683.

Models heterogeneous institutional and household demand from portfolio holdings and asset characteristics, using an instrumental-variable approach to address the price–demand endogeneity. The empirical application is US equities: inferred demand curves depend on holdings coverage, instruments, and investor aggregation, and are not a direct trade-level impact or capacity measure for every asset class.

Extended note. Version record

Valentin Haddad and Tyler Muir, "Market Macrostructure: Institutions and Asset Prices"; NBER WP 33434

Difficulty: Research **Type:** Research paper

Status/version: 2025

Annual Review of Financial Economics 17, 133–150 (November 2025; online 18 July); NBER working paper January 2025. A model and selective review connecting passive ownership, central-bank asset purchases and levered intermediaries to asset prices. [8]

Applications: Equities, bonds, macro; flows, intermediary constraints. Version record

Lasse Heje Pedersen, *Efficiently Inefficient: How Smart Money Invests and Market Prices Are Determined* (2015)

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2015. Version distinctions remain in the annotation.

Academic/practitioner bridge on competition, trading frictions and limits to arbitrage, including global macro and managed futures.

Version record

Jean-Philippe Bouchaud, "The Inelastic Market Hypothesis: A Microstructural Interpretation"; 2022 journal article

Difficulty: Research **Type:** Research paper

Status/version: 2021 preprint

Microstructural interpretation of inelastic markets, including the 2010–2011 CFM random-trade campaign and short-term impact comparison.

Version record

Zhiguo He and Arvind Krishnamurthy, "Intermediary Asset Pricing"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 103(2), 732–770 (2013); DOI 10.1257/aer.103.2.732.

Risk premia rise when specialist intermediaries are capital constrained; theoretical basis for funding-liquidity and crisis dynamics.

Version record

Zhiguo He, Bryan Kelly and Asaf Manela, "Intermediary asset pricing: New evidence from many asset classes"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 126(1), 1–35 (2017); DOI 10.1016/j.jfineco.2017.08.002.

Primary-dealer capital-ratio factor prices returns across equities, bonds, options, CDS, commodities and FX; factor data are published by the authors.

Version record

Tobias Adrian, Erkko Etula and Tyler Muir, "Financial Intermediaries and the Cross-Section of Asset Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(6), 2557–2596 (2014); DOI 10.1111/jofi.12189.

Broker-dealer leverage as a pricing factor; compare its procyclical sign with the He–Kelly–Manela capital ratio.

Version record

Xavier Gabaix and Ralph S. J. Koijen, "In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 28967 (2021); DOI 10.3386/w28967. Working-paper record; no journal version established in this check.

Estimates that a dollar of flow into equities raises aggregate market value by about five dollars; links fund mandates and flows to prices. Status and later versions should be checked before citing figures.

Extended note. Version record

Andrei Shleifer and Robert W. Vishny, "The Limits of Arbitrage"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 52(1), 35–55 (1997); DOI 10.1111/j.1540-6261.1997.tb03807.x.

Agency-constrained arbitrageurs lose capital when mispricing widens, so arbitrage is weakest when most needed; foundation for crowding and drawdown analysis.

Version record

Nicholas Barberis and Richard Thaler, "A Survey of Behavioral Finance"

Difficulty: Advanced **Type:** Research paper

Status/version: Handbook of the Economics of Finance, 1053–1128 (2003); DOI 10.1016/s1574-0102(03)01027-6.

Handbook survey of limits to arbitrage and psychology-based explanations for the equity premium, volatility, predictability and the cross-section.

Version record

Malcolm Baker and Jeffrey Wurgler, "Investor Sentiment and the Cross-Section of Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 61(4), 1645–1680 (2006); DOI 10.1111/j.1540-6261.2006.00885.x.

Sentiment index predicting returns of hard-to-value and hard-to-arbitrage stocks; widely used conditioning variable.

Version record

Pedro Bordalo, Nicola Gennaioli, Rafael La Porta and Andrei Shleifer, "Diagnostic Expectations and Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 74(6), 2839–2874 (2019); DOI 10.1111/jofi.12833.

Overreaction in analyst long-term growth expectations explains return reversals for high-expectation stocks; a disciplined behavioural expectations model.

Version record

Tim de Silva and David Thesmar, "Noise in Expectations: Evidence from Analyst Forecasts"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 37(5), 1494–1537 (2024); DOI 10.1093/rfs/hhad091.

Separates noise from bias in analyst forecasts and shows that noise explains a large part of apparent expectation errors; 2025 RFS Brennan runner-up.

Version record

Sebastien Betermier, Laurent E. Calvet, Samuli Knüpfer and Jens Soerlie Kvaerner, "Investor Factors"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(5), 2789–2830 (2025); DOI 10.1111/jofi.13474.

Extracts pricing factors from investor holdings: a market factor plus a long-short portfolio from holdings sorted by investor age or wealth explains Norwegian portfolio holdings and returns.

Version record

Carter Davis, Mahyar Kargar and Jiacui Li, "Why do portfolio choice models predict inelastic demand?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 172, 104096 (2025); DOI 10.1016/j.jfineco.2025.104096.

Decomposes demand elasticity into price pass-through and unspanned returns to explain why optimisers imply far higher elasticities than estimated.

Version record

Philippe van der Beck, "Short versus long-run demand elasticities in asset pricing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 184, 104337 (2026); DOI 10.1016/j.jfineco.2026.104337.

Short- versus long-run demand elasticities in asset pricing and implications for flow-driven price effects.

Extended note. Version record

Johannes Ruf and Yueying Sun, "Mandate Models and the Inelastic Market Hypothesis"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, mafi.70052 (2026); DOI 10.1111/mafi.70052.

Mathematical analysis of mandate models behind the inelastic markets hypothesis.

Version record

2.2 Econometrics and time series

Regression and inference are prerequisites. Financial forecasting references, including Tsay, Hamilton and Hyndman–Athanasopoulos, appear in Section 2.2. Physical-measure volatility forecasts and implied-surface pricing answer different questions; see Section 7.1 for pricing and hedging references.

2.2.1 Econometrics texts and large panels

George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel and Greta M. Ljung, *Time Series Analysis: Forecasting and Control*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 5th ed. (2015)

ARIMA modelling, identification, forecasting and control; fifth edition (2015).

Version record

John Y. Campbell, Andrew W. Lo and A. Craig MacKinlay, *The Econometrics of Financial Markets*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 1997 print book; DOI metadata also lists a 2019 online edition/date; contents equivalence not established.

Empirical asset pricing, predictability, GMM and financial time-series econometrics (1997).

Version record

Chris Brooks, *Introductory Econometrics for Finance*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Selected DOI-linked version; print 2019-03-28; online 2019-03-18.

Financial regression, time series, volatility, panels and limited dependent variables; linked fourth edition (2019), with companion data/code.

Version record

Jeffrey M. Wooldridge, *Introductory Econometrics: A Modern Approach*, 8th International edition; prior seventh-edition record

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Cengage International Edition, eighth edition, 28 April 2025 paperback listing; regional/format identity retained.

Regression, panel data and limited dependent variables. Retain the separately linked seventh edition for older coursework; the eighth International edition is copyright 2025 (this paperback listing: 28 April 2025). Its contents identify Chapters 13–15 for panels/instruments and 19 for causal inference. Use after basic regression; textbook identification assumptions still need a credible financial research design. Formats and courseware dates differ.

Version record

William H. Greene, *Econometric Analysis*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Matrix-based econometrics; eighth-edition author companion includes errata and supporting material.

Version record

Ruey S. Tsay, Analysis of Financial Time Series

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Financial ARIMA, ARCH/GARCH, regime-switching and multivariate models; physical-volatility forecasting includes asymmetric and stochastic-volatility specifications.

Version record

Shuping Shi, Xiaohu Wang and Tao Zeng (eds.), ***Financial Econometrics: Theory and Applications***

Difficulty: Advanced **Type:** Book / edited research collection

Status/version: Cambridge University Press; 2025-02-20 online; publisher lists 2025-02-27 as the other format date. First edition; edited research collection in Themes in Modern Econometrics. DOI 10.1017/9781108910095.

An advanced, research-oriented collection on current financial-econometric methods. Its chapters connect low-frequency trend, bubble and factor-regression questions with high-frequency continuous-time inference and Bayesian volatility/model selection. Use it to deepen selected methods after a standard time-series course, not as a first econometrics text.

Extended note. Version record

James D. Hamilton, Time Series Analysis

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Graduate econometric foundations for stationarity, ARMA/ARIMA, cointegration, state-space methods and Kalman filtering. Use the mathematical treatment to check identification, persistence and model assumptions behind financial forecasts.

Version record

Rob J. Hyndman and George Athanasopoulos, Forecasting: Principles and Practice

Difficulty: Intro **Type:** Book / monograph

Status/version: 3rd ed. (2021); living online text.

Freely available third edition (2021). The author/publisher page reports online revision 28 September 2026 versus print revision 31 May 2021. Begin with exponential smoothing and ARIMA, graphical diagnostics, benchmarks and forecast-accuracy evaluation before finance-specific targets and dependence-aware validation. The third-edition examples use tsibble/fable; preserve code and package versions when comparing older forecast-package examples.

Version record

Bruce E. Hansen, ***Econometrics***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press, 2022 (ISBN 9780691235899); author-hosted draft available.

Comprehensive modern graduate econometrics text (Princeton 2022) covering regression, asymptotics, bootstrap, GMM, panel, IV and time series with free author drafts.

Extended note. Version record

Fumio Hayashi, ***Econometrics***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press, 2000 (ISBN 9780691010182).

GMM-centred graduate text with clear treatment of time-series asymptotics, HAC inference and empirical finance examples.

Version record

Graham Elliott and Allan Timmermann, ***Economic Forecasting***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press, 2016 (ISBN 9780691140131).

Loss functions, forecast combination, model instability, density and interval forecasts, and forecast evaluation with finance examples. The most directly relevant forecasting text for return prediction.

Version record

Helmut Lütkepohl, ***New Introduction to Multiple Time Series Analysis***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer Berlin Heidelberg (2005); DOI 10.1007/978-3-540-27752-1.

VAR, VECM, cointegration and structural VAR estimation; background for cointegration-based pairs trading and macro-finance VARs.

Version record

Lutz Kilian and Helmut Lütkepohl, ***Structural Vector Autoregressive Analysis***

Difficulty: Research **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2017); DOI 10.1017/9781108164818.

Identification by short-run, long-run, sign and external-instrument restrictions; relevant to oil-shock and monetary-policy-shock studies.

Version record

Jianqing Fan and Qiwei Yao, *The Elements of Financial Econometrics*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press (2017); DOI 10.1017/9781108120616.

Compact text on return predictability, GARCH, multivariate volatility, high-dimensional covariance and factor pricing with statistical emphasis.

Version record

James H. Stock and Mark W. Watson, "Forecasting Using Principal Components From a Large Number of Predictors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the American Statistical Association, 97(460), 1167–1179 (2002); DOI 10.1198/016214502388618960.

Diffusion-index forecasting: extract principal components from many predictors and forecast with a few factors; a baseline for macro and return forecasting with large panels.

Version record

Jushan Bai and Serena Ng, "Determining the Number of Factors in Approximate Factor Models"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 70(1), 191–221 (2002); DOI 10.1111/1468-0262.00273.

Information criteria for choosing the number of factors in large panels; standard tool before statistical factor models or PCA risk models.

Version record

2.2.2 Volatility and high-frequency econometrics

Ser-Huang Poon and Clive W. J. Granger, "Forecasting Volatility in Financial Markets: A Review" (2003)

Difficulty: Advanced **Type:** Research paper

Status/version: Selected citation year(s): 2003. Version distinctions remain in the annotation.

Survey of historical-volatility, conditional-variance and option-implied forecasting approaches and their evaluation. Use its bibliography and model comparisons to motivate strong simple baselines; relative accuracy depends on horizon, data and loss function.

Version record

Robert F. Engle, "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation" (1982)

Difficulty: Advanced **Type:** Research paper

Status/version: Selected citation year(s): 1982. Version distinctions remain in the annotation.

Original ARCH framework for time-varying conditional variance. Work through volatility clustering, estimation and forecast diagnostics before using conditional variance for risk scaling, volatility targeting or option-strategy research.

Version record

Tim Bollerslev, "Generalized Autoregressive Conditional Heteroskedasticity" (1986)

Difficulty: Advanced **Type:** Research paper

Status/version: Selected citation year(s): 1986. Version distinctions remain in the annotation.

GARCH extends ARCH with lagged conditional variance. Study persistence, stationarity restrictions and multi-step forecasts, then compare asymmetric variants in Tsay. A statistical variance forecast is distinct from an option-implied risk-neutral volatility surface.

Version record

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold and Paul Labys, "Modeling and Forecasting Realized Volatility" (2003)

Difficulty: Research **Type:** Research paper

Status/version: Selected DOI-linked version; print 2003-03.

High-frequency returns, realised-volatility models, density forecasts and risk quantiles; sampling and microstructure noise matter.

Version record

Yacine Aït-Sahalia and Jean Jacod, *High-Frequency Financial Econometrics*

Difficulty: Research **Type:** Book / monograph

Status/version: Princeton University Press, edition 1 (2014); DOI 10.23943/princeton/9780691161433.001.0001.

Rigorous treatment of realised measures, jumps, microstructure noise and asynchronous sampling for intraday data.

Version record

Fulvio Corsi, "A Simple Approximate Long-Memory Model of Realized Volatility"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Econometrics, 7(2), 174–196 (2008); DOI 10.1093/jjfinec/nbp001.

HAR-RV: daily, weekly and monthly realised-volatility components in a linear regression; a hard-to-beat volatility forecasting benchmark.

Version record

Robert Engle, "Dynamic Conditional Correlation"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 20(3), 339–350 (2002); DOI 10.1198/073500102288618487.

DCC-GARCH for time-varying correlations in large systems; widely used in risk models and hedge-ratio estimation.

Version record

Peter R. Hansen and Asger Lunde, "A forecast comparison of volatility models: does anything beat a GARCH(1,1)?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 20(7), 873–889 (2005); DOI 10.1002/jae.800.

Compares 330 ARCH-type models using SPA tests; a lesson in benchmark choice and data-snooping-aware model comparison.

Version record

Andrew J. Patton, "Volatility forecast comparison using imperfect volatility proxies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 160(1), 246–256 (2011); DOI 10.1016/j.jeconom.2010.03.034.

Identifies loss functions (MSE, QLIKE) that rank volatility forecasts consistently when the true variance is replaced by a noisy proxy.

Version record

Ole E. Barndorff-Nielsen and Neil Shephard, "Econometric Analysis of Realized Volatility and its Use in Estimating Stochastic Volatility Models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Royal Statistical Society Series B: Statistical Methodology, 64(2), 253–280 (2002); DOI 10.1111/1467-9868.00336.

Asymptotic theory for realised variance and its use in fitting stochastic-volatility models.

Version record

Tim Bollerslev, Andrew J. Patton and Rogier Quaedvlieg, "Exploiting the errors: A simple approach for improved volatility forecasting"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 192(1), 1–18 (2016); DOI 10.1016/j.jeconom.2015.10.007.

HARQ adjusts HAR coefficients for measurement error in realised variance using realised quarticity.

Version record

Daniel B. Nelson, "Conditional Heteroskedasticity in Asset Returns: A New Approach"

Difficulty: Advanced **Type:** Research paper

Status/version: Econometrica, 59(2), 347 (1991); DOI 10.2307/2938260.

EGARCH: asymmetric volatility response to negative shocks without parameter positivity constraints.

Version record

Lawrence R. Glosten, Ravi Jagannathan and David E. Runkle, "On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 48(5), 1779–1801 (1993); DOI 10.1111/j.1540-6261.1993.tb05128.x.

GJR-GARCH threshold model for the leverage effect and evidence on the risk-return relation.

Version record

Jia Li, Peter C. B. Phillips, Shuping Shi and Jun Yu, "Weak Identification of Long Memory with Implications for Volatility Modeling"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(10), 3117–3148 (2025); DOI 10.1093/rfs/hhaf022.

Long-memory models with weak dynamics and rough models with near-unit roots are near-observationally equivalent, so standard tests struggle to tell them apart; relevant to the rough-volatility debate.

Version record

Chen Liu, Minh-Ngoc Tran, Chao Wang, Richard Gerlach and Robert Kohn, "Neural-Network Volatility Forecasting"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 24(4), nbag019 (2026); DOI 10.1093/jjfinec/nbag019.

Across 10,000+ stocks, sample-size scaling improves neural volatility forecasts more than model size or architecture choice.

Version record

Colin Bowers and Chris Heaton, "Empirical Evaluation of Competing High-Frequency Estimators of Quadratic Variation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 23(3), nbaf007 (2025); DOI 10.1093/jjfinec/nbaf007.

Tests for differences in bias, variance and MSE among competing realised-variance estimators.

Version record

2.3 Forecasting models and benchmarks

Subject context and qualifications. Subject context and qualifications.

2.3.1 Forecast evaluation, scoring and combination

Tilmann Gneiting and Adrian E. Raftery, "Strictly Proper Scoring Rules, Prediction, and Estimation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the American Statistical Association, 102(477), 359–378 (2007); DOI 10.1198/016214506000001437.

Proper scoring rules (log score, CRPS, interval score) for probabilistic forecasts; the evaluation basis for density, quantile and tail forecasts.

Version record

Tilmann Gneiting, "Making and Evaluating Point Forecasts"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the American Statistical Association, 106(494), 746–762 (2011); DOI 10.1198/jasa.2011.r10138.

A forecast should be evaluated with a loss consistent for the functional it targets (mean, quantile, expectile); explains why some risk measures are hard to backtest.

Version record

Allan Timmermann, "Forecast Combinations"

Difficulty: Advanced **Type:** Research paper

Status/version: Handbook of Economic Forecasting, 135–196 (2006); DOI 10.1016/s1574-0706(05)01004-9.

Handbook chapter on why simple averages often beat estimated optimal weights; guidance on combining model forecasts and signals.

Version record

Spyros Makridakis, Evangelos Spiliotis and Vassilios Assimakopoulos, "The M4 Competition: 100,000 time series and 61 forecasting methods"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Forecasting, 36(1), 54–74 (2020); DOI 10.1016/j.ijforecast.2019.04.014.

Large open forecasting competition showing combinations and hybrid statistical/ML methods dominating; open data for benchmarking.

Version record

Spyros Makridakis, Evangelos Spiliotis and Vassilios Assimakopoulos, "M5 accuracy competition: Results, findings, and conclusions"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Forecasting, 38(4), 1346–1364 (2022); DOI 10.1016/j.ijforecast.2021.11.013.

Hierarchical retail-sales competition where gradient-boosted trees won; lessons on feature engineering and cross-learning across related series.

Version record

J. M. Bates and C. W. J. Granger, "The Combination of Forecasts"

Difficulty: Advanced **Type:** Research paper

Status/version: OR, 20(4), 451 (1969); DOI 10.2307/3008764.

The origin of forecast combination: weighting competing forecasts by inverse error variance often beats the best single model.

Version record

Tilmann Gneiting and Matthias Katzfuss, "Probabilistic Forecasting"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Statistics and Its Application, 1(1), 125–151 (2014); DOI 10.1146/annurev-statistics-062713-085831.

Review of probabilistic forecasting: calibration and sharpness, proper scoring rules and evaluation of predictive distributions.

Version record

Francis X. Diebold, Todd A. Gunther and Anthony S. Tay, "Evaluating Density Forecasts with Applications to Financial Risk Management"

Difficulty: Advanced **Type:** Research paper

Status/version: International Economic Review, 39(4), 863 (1998); DOI 10.2307/2527342.

Probability-integral-transform tests for density forecasts, applied to risk management; the basis for distributional backtests.

Version record

Rob J. Hyndman and Anne B. Koehler, "Another look at measures of forecast accuracy"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Forecasting, 22(4), 679–688 (2006); DOI 10.1016/j.ijforecast.2006.03.001.

Compares accuracy measures and introduces the mean absolute scaled error (MASE), avoiding the pathologies of percentage errors.

Version record

Fotios Petropoulos et al., "Forecasting: theory and practice"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Forecasting, 38(3), 705–871 (2022); DOI 10.1016/j.ijforecast.2021.11.001.

Encyclopaedic review of forecasting methods and practice, from statistical models and combinations to judgement and evaluation, with an extensive bibliography.

Version record

Xiaoqian Wang, Rob J. Hyndman, Feng Li and Yanfei Kang, "Forecast combinations: An over 50-year review"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Forecasting, 39(4), 1518–1547 (2023); DOI 10.1016/j.ijforecast.2022.11.005.

Survey of point and probabilistic forecast combination since Bates–Granger, including the "forecast combination puzzle".

Version record

Clive W. J. Granger and M. Hashem Pesaran, "Economic and statistical measures of forecast accuracy"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Forecasting, 19(7), 537–560 (2000); DOI 10.1002/1099-131x(200012)19:73.0.co;2-g.

Argues forecasts should be judged by the decisions they support; links accuracy to economic loss, which is how trading signals should be evaluated.

Version record

M. Hashem Pesaran and Allan Timmermann, "A Simple Nonparametric Test of Predictive Performance"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 10(4), 461–465 (1992); DOI 10.1080/07350015.1992.10509922.

The standard test of directional (sign) forecasting accuracy, widely used for market-timing signals.

Version record

2.3.2 Deep and foundation forecasting models

Bryan Lim et al., "Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting"

Difficulty: Research **Type:** Research paper

Status/version: 2021

Combines variable selection, recurrent processing and attention for multi-horizon probabilistic forecasts with known and observed covariates. For finance, its greatest value is structured handling of heterogeneous features and horizons, not evidence that attention alone predicts returns.

Applications: Macro, volatility, liquidity, risk forecasts. Version record

Yuqi Nie et al., "A Time Series Is Worth 64 Words: Long-term Forecasting with Transformers"

Difficulty: Research **Type:** Research paper

Status/version: 2023

PatchTST tokenises local temporal patches rather than individual observations, improving efficiency and receptive field. It is a strong architecture to test for curves, volatility and market-state forecasts, while cross-sectional return prediction still requires finance-specific targets and validation.

Applications: Multi-asset forecasting, volatility, curves. Version record

Yong Liu et al., "iTransformer: Inverted Transformers Are Effective for Time Series Forecasting" and official code

Difficulty: Research **Type:** Research paper

Status/version: 2024

Inverts the conventional representation so that variables become tokens and temporal observations become features. This is conceptually attractive for panels of related assets or curve points, although asset entry/exit and unstable cross-sectional identities remain difficult. [19]

Applications: Multi-asset panels, curves, cross-sectional signals. Version record

Abdul Fatir Ansari et al., "Chronos: Learning the Language of Time Series" and code

Difficulty: Research **Type:** Research paper

Status/version: 2024

Quantises time-series values into tokens and trains language-model architectures for probabilistic zero-shot forecasting. It is useful as a pretrained benchmark and scenario forecaster, but finance teams must test performance under transformations, structural breaks and distributional tails absent from general benchmarks. [20]

Applications: Macro, risk, volatility, demand and liquidity forecasts. Version record

Abhimanyu Das et al., "A Decoder-only Foundation Model for Time-Series Forecasting" and TimesFM code

Difficulty: Research **Type:** Report / research note

Status/version: 2024–2026

TimesFM established a large pretrained univariate forecasting model and has since evolved to TimesFM 2.5, with longer context, quantile forecasts and covariate support. It should be used as a zero-/few-shot baseline, not presumed to contain tradeable financial information merely because it performs well on generic forecasting datasets. [21]

Applications: Macro, volatility, liquidity, curves, scenarios. Version record

Yicheng Wang and Sandro Claudio Lera, "Meta-Learning for Return Prediction in Shifting Market Regimes"; journal version

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets 79 (June 2026), 101042; earlier working paper retained.

Conditions return forecasts on recent feature–return relationships with a Transformer-based Financial Prior-data Fitted Network (FinPFN). The journal abstract describes daily Chinese A-shares and monthly U.S. equities. Assess forward information availability, model selection and turnover before treating reported forecasting gains as implementable returns. The 2024 working paper and 2026 journal version remain separately identified. [22]

Applications: Equities; adaptive ML, regime change. Version record

Taha Aksu, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Silvio Savarese, Caiming Xiong and Doyen Sahoo, GIFT-Eval: A Benchmark for General Time Series Forecasting Model Evaluation; official benchmark and change log

Difficulty: Research **Type:** Research paper

Status/version: Preprint first posted 14 October 2024; v2 11 November 2024; evolving benchmark repository

Compares statistical, deep and foundation models across frequencies, horizons and univariate or multivariate tasks, with an explicitly separated pretraining collection. Aggregate scores depend on dataset composition, loss and model access to training data. Freeze the benchmark commit and evaluation manifest, then add point-in-time financial holdouts, stress regimes and economic losses; a clean benchmark split does not prove a third-party model's training corpus was uncontaminated.

Extended note. Version record

Md Asif Bin Syed, Md Younus Ahamed and Azmine Toushik Wasi, Position: Time-Series Foundation Models Require Explicit Domain-Level Benchmarks

Difficulty: Research **Type:** Research paper

Status/version: ICML 2026 position paper, PMLR 306, 172221–172247; proceedings volume published 29 September 2026

Argues that pooled forecasting leaderboards can hide domain-specific failures and illustrates the problem with comparisons across financial and nonfinancial datasets. It provides a recent counterweight to model-release claims: sampling frequency, missingness, stochastic dynamics and concept drift change what a forecast benchmark measures. This position paper motivates domain-aware evaluation; its benchmark comparisons do not establish tradable financial performance or a preferred model for every desk.

Extended note. Version record

Iñaki Aldasoro, Peter Hördahl, Andreas Schrimpf and Xingyu Sonya Zhu, Predicting financial market stress with machine learning

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1250, published 17 March 2025; PDF title page dated February 2025. No journal version established.

Constructs market-specific condition indicators for US Treasury, FX and money markets, then uses quantile random forests to forecast their distributions and tail stress. Provides a finance-native alternative to generic time-series benchmarks for risk and liquidity forecasts.

Extended note. Version record

Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo, Unified Training of Universal Time Series Forecasting Transformers; Official model code

Difficulty: Research **Type:** Research paper

Status/version: ICML 2024, Proceedings of Machine Learning Research 235, 53140–53164, July 2024.

Introduces Moirai, a pretrained forecasting transformer designed for mixed frequencies, variable-dimensional multivariate series, and varying distributions across a large cross-domain archive. It offers a useful comparison with the catalogue's Chronos and TimesFM entries when testing zero-shot forecasts for curves, macro releases, volumes, and volatility. Check whether each target series appeared in pretraining before claiming zero-shot validity.

Extended note. Version record

Abdul Fatir Ansari et al., Chronos-2: From Univariate to Universal Forecasting; Official model repository

Difficulty: Research **Type:** Research paper

Status/version: Amazon Science arXiv preprint first posted 17 October 2025; official repository announced model release 20 October 2025; no peer-reviewed venue established in inspected record.

Extends the catalogue's original Chronos reading from mostly univariate forecasts to multivariate and covariate-informed zero-shot tasks using group attention and synthetic multivariate pretraining. It gives a concrete updated baseline for macro panels, yield curves, liquidity, and volatility where related series and known covariates matter. Pretraining and generic benchmark results do not validate excess-return forecasts or net strategy performance.

Extended note. Version record

Bryan Lim and Stefan Zohren, "Time-series forecasting with deep learning: a survey"

Difficulty: Advanced **Type:** Research paper

Status/version: Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 379(2194), 20200209 (2021); DOI 10.1098/rsta.2020.0209.

Survey of encoder designs, hybrid statistical-deep models and interpretability for multi-horizon forecasting.

Version record

Yu Shi, Zongliang Fu, Shuo Chen, Bohan Zhao, Wei Xu, Changshui Zhang and Jian Li, "Kronos: A Foundation Model for the Language of Financial Markets"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2508.02739, first posted 2 August 2025; NeurIPS 2025 workshop presentation; code and weights released under MIT licence.

Tokenises OHLCV candlesticks and pre-trains a decoder-only transformer on more than 12 billion K-line records from 45 exchanges; reports zero-shot forecasting, volatility and synthetic-data gains. Author claims; check leakage across venues and dates before relying on reported improvements.

Extended note. Version record

Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister and Frank Hutter, "Accurate predictions on small data with a tabular foundation model"

Difficulty: Research **Type:** Research paper

Status/version: Nature, 637(8045), 319–326 (2025); DOI 10.1038/s41586-024-08328-6.

TabPFN: a transformer pre-trained on synthetic tabular tasks that performs in-context prediction on small datasets. Relevant to small-sample panels; financial time ordering and non-stationarity are not part of its training design.

Version record

2.4 Regimes and adaptation

Research reliability references appear in Section 5.1; see Narang for an overview of quantitative trading systems.

Amara Mulliner, Campbell R. Harvey, Chao Xia, Ed Fang and Otto van Hemert, *Regimes, Systematic Models and the Power of Prediction*; *AIMA Journal* 142, 29–31 (Q2 2025)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2025. Version distinctions remain in the annotation.

Short practitioner treatment of historical-state similarity for timing six equity factors. The AIMA version credits all five authors; conclusions depend on variable selection and point-in-time availability.

Version record

Amara Mulliner, Campbell R. Harvey, Chao Xia, Ed Fang and Otto van Hemert, *Regimes*, *Journal of Portfolio Management* 52(4), 6–25 (February 2026); author copy

Difficulty: Not specified **Type:** Research paper

Status/version: Selected DOI-linked version; print 2026-01-31; online 2025-12-19.

Full research article, distinct from the shorter practitioner treatment. Uses distance across seven economic state variables to compare similar regimes and dissimilar "anti-regimes" for six equity factors over 1985–2024. Reported timing results require scrutiny of calibration, variable selection and available-at-the-time inputs.

Version record

QuantStart / Michael Halls-Moore: HMM regime detection in QSTrader; QuantInsti HMM tutorial

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Edition/revision and publication date not established for the selected resource(s).

Hands-on state estimation and conditional strategy allocation. Fit states only on past data and use filtered probabilities available at the decision time. Full-sample smoothed states introduce hindsight. Evaluate transaction costs, switching delay, rare-state sample size and an unconditional baseline.

Applications: Supplementary tutorials.

Version record

Two Sigma, "A Machine Learning Approach to Regime Modeling" (2021)

Difficulty: Advanced **Type:** Practitioner / commentary

Status/version: Selected citation year(s): 2021. Version distinctions remain in the annotation.

Gaussian mixture models and factor-based state variables provide a practitioner example of regime monitoring and portfolio-risk framing. Compare state-conditioned constraints with unconditional baselines and estimate states using information available at the decision time.

Applications: industry.

Version record

James D. Hamilton, "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle"

Difficulty: Advanced **Type:** Research paper

Status/version: Econometrica, 57(2), 357 (1989); DOI 10.2307/1912559.

Markov-switching autoregression and filtering of latent regimes; the base model for regime-switching allocation.

Version record

Andrew Ang and Allan Timmermann, "Regime Changes and Financial Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 4(1), 313–337 (2012); DOI 10.1146/annurev-financial-110311-101808.

Review of regime-switching models for returns, volatility, term structure and portfolio choice, with warnings on in-sample regime fitting.

Version record

Jushan Bai and Pierre Perron, "Computation and analysis of multiple structural change models"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Applied Econometrics, 18(1), 1–22 (2003); DOI 10.1002/jae.659.

Practical algorithms and tests for multiple structural breaks; a check before assuming a stable signal or factor premium.

Version record

M. Hashem Pesaran and Allan Timmermann, "Selection of estimation window in the presence of breaks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 137(1), 134–161 (2007); DOI 10.1016/j.jeconom.2006.03.010.

Trade-off between bias from pre-break data and variance from short windows; guidance on rolling versus expanding estimation.

Version record

Peter Nystrup, Petter N. Kolm and Erik Lindström, "Feature selection in jump models"

Difficulty: Research **Type:** Research paper

Status/version: Expert Systems with Applications, 184, 115558 (2021); DOI 10.1016/j.eswa.2021.115558.

Statistical jump models with feature selection for persistent regimes; more stable regime assignments than HMMs in their tests.

Version record

Yizhan Shu, Chenyu Yu and John M. Mulvey, "Downside risk reduction using regime-switching signals: a statistical jump model approach"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Asset Management, 25(5), 493–507 (2024); DOI 10.1057/s41260-024-00376-x.

Applies jump-model regimes to equity-index risk reduction with out-of-sample evaluation; read for the implementation details and the backtest controls.

Version record

Yu Bai, Bin Peng, Shuping Shi and Wenying Yao, "Optimal Bandwidth Selection for Forecasting under Parameter Instability"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 24(2), nbag001 (2026); DOI 10.1093/jjfinec/nbag001.

Bandwidth selection for local estimators under parameter instability, chosen by expected end-of-sample loss.

Version record

D. C. Oliveira, D. Sandfelder, A. Fujita, X. Dong and M. Cucuringu, "Tactical asset allocation with macroeconomic regime detection"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(9), 1465–1491 (2026); DOI 10.1080/14697688.2026.2659195.

Macroeconomic regime classification and forecasting with ML integrated into tactical allocation.

Version record

Eric Bouyé and Jérôme Teiletche, "Regime-Based Strategic Asset Allocation"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 81(4), 84–102 (2025); DOI 10.1080/0015198x.2025.2558354.

Strategic allocation with macro regimes modelled as mixtures: analytical results for mean-variance and risk-budgeting portfolios and regime-based construction rules that beat asset-based portfolios empirically.

Version record

2.5 Event studies and long-run abnormal returns

Design and inference for short- and long-horizon event studies: normal-return models, event-induced variance, cross-correlation of clustered events, characteristic-matched benchmarks and calendar-time tests. The Kothari–Warner survey and the Lyon–Barber–Tsai and Fama (1998) papers are in Section 10.4.

Stephen J. Brown and Jerold B. Warner, "Using daily stock returns"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 14(1), 3–31 (1985); DOI 10.1016/0304-405x(85)90042-x.

Examines how the properties of daily returns affect event-study methods. Daily data present few difficulties: standard market-model procedures are usually well specified even when daily features are ignored, though allowing for autocorrelation and event-induced variance can help, and tests that ignore cross-sectional dependence can be well specified and more powerful. Extends the authors' 1980 monthly-data study.

Version record

Ekkehart Boehmer, Jim Musumeci and Annette B. Poulsen, "Event-study methodology under conditions of event-induced variance"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 30(2), 253–272 (1991); DOI 10.1016/0304-405x(91)90032-f.

Simulates events with stochastic effects and finds that even small event-induced increases in variance make the common tests reject a true null of zero average abnormal return too often. A simple adjustment to the cross-sectional test, standardising each abnormal return by its estimation-period standard deviation, restores correct size without losing power.

Version record

James W. Kolari and Seppo Pynnönen, "Event Study Testing with Cross-sectional Correlation of Abnormal Returns"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 23(11), 3996–4025 (2010); DOI 10.1093/rfs/hhq072.

When event dates cluster, even low cross-correlation among abnormal returns makes standard tests over-reject. Proposes a correction to the Boehmer–Musumeci–Poulsen statistic that accounts for cross-correlation, works for multi-day cumulative returns and beats the portfolio approach, which is less powerful.

Version record

Brad M. Barber and John D. Lyon, "Detecting long-run abnormal stock returns: The empirical power and specification of test statistics"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 43(3), 341–372 (1997); DOI 10.1016/s0304-405x(96)00890-2.

Studies the power and size of tests for one- to five-year abnormal returns. Abnormal returns measured against a reference portfolio such as a market index give misspecified tests, for three identified reasons; matching each sample firm to a control firm of similar size and book-to-market yields well-specified tests in nearly all sampling situations.

Version record

Kent Daniel, Mark Grinblatt, Sheridan Titman and Russ Wermers, "Measuring Mutual Fund Performance with Characteristic-Based Benchmarks"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 52(3), 1035–1058 (1997); DOI 10.1111/j.1540-6261.1997.tb02724.x.

Builds characteristic-matched benchmarks from 125 passive portfolios sorted on size, book-to-market and prior-year return, and defines characteristic-selectivity and characteristic-timing measures. On holdings of more than 2,500 equity funds (1975–1994), funds, especially aggressive-growth funds, show some selectivity but no characteristic timing. The DGTW benchmark is now standard for abnormal returns of holdings and event samples.

Version record

Mark L. Mitchell and Erik Stafford, "Managerial Decisions and Long-Term Stock Price Performance"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Business, 73(3), 287–329 (2000); DOI 10.1086/209645.

Re-examines long-run abnormal returns after acquisitions, equity issues and repurchases and finds little reliable evidence. Cross-sectional dependence of event-firm returns inflates test statistics, and once the benchmark model's known mispricing is accounted for, the remaining abnormal performance is small and concentrated in small stocks. The case for calendar-time portfolio tests.

Version record

Eugene F. Fama, Lawrence Fisher, Michael C. Jensen and Richard Roll, "The Adjustment of Stock Prices to New Information"

Difficulty: Advanced **Type:** Research paper

Status/version: International Economic Review, 10(1), 1–21 (1969); DOI 10.2307/2525569.

The first modern event study. Asks how quickly common-stock prices adjust to the information implicit in a stock split, rather than inferring efficiency from the independence of successive price changes, and in doing so proposes the event-study methodology for measuring the effect of an event on security prices.

Version record

Stephen J. Brown and Jerold B. Warner, "Measuring security price performance"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 8(3), 205–258 (1980); DOI 10.1016/0304-405x(80)90002-1.

Introduces abnormal performance into actual monthly stock returns to compare event-study methods. A simple market-model method performs well under a wide variety of conditions; in some situations even simpler methods that do not adjust for market-wide factors or risk do no worse; and misuse of any method can produce false inferences about abnormal performance. The monthly-data companion to the 1985 daily-data study.

Version record

James W. Kolari and Seppo Pynnonen, "Nonparametric rank tests for event studies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 18(5), 953–971 (2011); DOI 10.1016/j.jempfin.2011.08.003.

Rank tests dominate parametric tests for single-day abnormal returns because returns are not normal, but have been hard to apply to cumulative abnormal returns. Proposes a generalised rank (GRANK) procedure for single-day and cumulative returns, derives its asymptotic distribution and shows in CRSP simulations that it beats earlier rank tests of CARs and is robust to serial correlation and event-induced volatility.

Version record

Charles J. Corrado, "Event studies: A methodology review"

Difficulty: Advanced **Type:** Survey

Status/version: Accounting & Finance, 51(1), 207–234 (2011); DOI 10.1111/j.1467-629x.2010.00375.x.

Review of short-horizon event-study methodology by the author of the standard nonparametric rank test: the fundamental design topics, the variations in method and their relative merits, with new perspectives on open questions. Notes how event studies have spread from accounting and finance to economics, history, law, management, marketing and political science.

Version record

Tim Loughran and Jay R. Ritter, "Uniformly least powerful tests of market efficiency"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 55(3), 361–389 (2000); DOI 10.1016/s0304-405x(99)00054-9.

Counter-argument to Fama (1998): because methods weight returns differently, long-run abnormal returns should differ across methods in predictable ways. Identifies three problems that give value-weighted three-factor time-series regressions low power after managerial actions, illustrates them with the new-issues puzzle and simulations, and argues that multifactor models as used cannot test market efficiency.

Version record

Paul Goldsmith-Pinkham and Tianshu Lyu, "Causal Inference in Financial Event Studies"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2511.15123 (2025); DOI 10.48550/arXiv.2511.15123.

Shows that when the factor model is misspecified, as it almost always is, standard event-study estimators of abnormal returns are inconsistent, with the worst bias in volatile periods, at long horizons and when event timing correlates with market conditions. Proposes synthetic-control replicating portfolios; revisiting four applications, some established findings may reflect misspecification. Traditional methods remain reliable for short windows with random timing. Preprint (November 2025).

Version record

Kirill Borusyak, Xavier Jaravel and Jann Spiess, "Revisiting Event-Study Designs: Robust and Efficient Estimation"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 91(6), 3253–3285 (2024); DOI 10.1093/restud/rdae007.

Framework for difference-in-differences with staggered adoption and heterogeneous effects: conventional regression estimators are biased without strong homogeneity restrictions. Derives the efficient imputation estimator, inference tools and tests of identifying assumptions, applying with time-varying controls, triple differences and some non-binary treatments. The panel event-study design, as distinct from return-based event studies.

Version record

A. Craig MacKinlay, "Event Studies in Economics and Finance"

Difficulty: Intro **Type:** Survey

Status/version: Journal of Economic Literature, 35(1), 13–39 (1997); no DOI registered.

The standard introduction to event studies: measuring an event's effect on firm value from security prices around it, on the premise that rational markets reflect the effect immediately. Describes the procedure step by step, including its potential complications, with a worked example.

Version record

2.6 Standard errors, clustering and weak instruments

Inference for regressions on financial panels and time series: heteroskedasticity- and autocorrelation-robust covariance estimators, one-way, two-way and design-based clustering, predictive-regression bias, and weak instruments. Newey–West (1987) and nonstandard errors are in Section 5.2.

Halbert White, "A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity"

Difficulty: Advanced **Type:** Research paper

Status/version: Econometrica, 48(4), 817 (1980); DOI 10.2307/1912934.

The heteroskedasticity-consistent covariance estimator for linear regression, valid without a model for the form of heteroskedasticity, and a direct test that compares it with the usual estimator. The starting point for every robust and clustered standard error used in finance.

Version record

Donald W. K. Andrews, "Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 59(3), 817 (1991); DOI 10.2307/2938229.

Makes heteroskedasticity- and autocorrelation-consistent covariance estimation operational: compares kernels by asymptotic truncated mean squared error, derives optimal kernels and bandwidths, and introduces data-dependent automatic bandwidths, with Monte Carlo evidence on finite-sample behaviour. Read with Newey–West before choosing a lag length by habit.

Version record

Mitchell A. Petersen, "Estimating Standard Errors in Finance Panel Data Sets: Comparing Approaches"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 22(1), 435–480 (2009); DOI 10.1093/rfs/hhn053.

In finance panels residuals can be correlated across firms or across time, and OLS standard errors are then biased. Compares the methods used in the corporate-finance and asset-pricing literatures, explains when they agree and when they diverge, and gives guidance on which to use. The standard reference for panel standard errors in finance.

Version record

Samuel B. Thompson, "Simple formulas for standard errors that cluster by both firm and time"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 99(1), 1–10 (2011); DOI 10.1016/j.jfineco.2010.08.016.

Standard errors robust to correlation across firms and across time at once: the two-way covariance estimator equals the firm-clustered estimator plus the time-clustered estimator minus the heteroskedasticity-robust OLS estimator, so any package with a clustering command can compute it.

Version record

A. Colin Cameron, Jonah B. Gelbach and Douglas L. Miller, "Robust Inference With Multiway Clustering"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 29(2), 238–249 (2011); DOI 10.1198/jbes.2010.07136.

Variance estimator for OLS and for nonlinear estimators (logit, probit, GMM) with two-way or multiway non-nested clustering, extending the one-way sandwich estimator under similar weak assumptions. Monte Carlo evidence includes a two-dimensional placebo-law extension of Bertrand, Duflo and Mullainathan.

Version record

A. Colin Cameron and Douglas L. Miller, "A Practitioner's Guide to Cluster-Robust Inference"

Difficulty: Advanced **Type:** Survey

Status/version: Journal of Human Resources, 50(2), 317–372 (2015); DOI 10.3368/jhr.50.2.317.

Practitioner's guide to cluster-robust inference when errors are correlated within clusters: default standard errors can greatly overstate precision. Covers the basic method and the complications met in practice: cluster-specific fixed effects, few clusters, multiway clustering and estimators other than OLS.

Version record

John C. Driscoll and Aart C. Kraay, "Consistent Covariance Matrix Estimation with Spatially Dependent Panel Data"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 80(4), 549–560 (1998); DOI 10.1162/003465398557825.

Covariance estimator robust to very general cross-sectional (spatial) and temporal dependence in panels as the time dimension becomes large, from a simple extension of common nonparametric covariance estimators. Monte Carlo and empirical examples; suited to macro-finance panels with long time series.

Version record

Ian D. Gow, Gaizka Ormazabal and Daniel J. Taylor, "Correcting for Cross-Sectional and Time-Series Dependence in Accounting Research"

Difficulty: Advanced **Type:** Research paper

Status/version: The Accounting Review, 85(2), 483–512 (2010); DOI 10.2308/accr.2010.85.2.483.

Evaluates the corrections for cross-sectional and time-series dependence used in accounting research and finds that none in common use is robust to both; the Z2 statistic and Newey–West-adjusted Fama–MacBeth standard errors do not correct for both, contrary to claims. Well-specified two-way corrections overturn several published findings on the cost of capital, cost of debt and conservatism.

Version record

Alberto Abadie, Susan Athey, Guido W. Imbens and Jeffrey M. Wooldridge, "When Should You Adjust Standard Errors for Clustering?"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 138(1), 1–35 (2023); DOI 10.1093/qje/qjac038.

Asks why and when to cluster. Adds a design component (variation from treatment assignment) to the usual sampling component. When the sample contains a non-negligible fraction of the population's clusters, as with all states or all firms, conventional clustered standard errors can be severely inflated; proposes corrected variance estimators.

Version record

Marianne Bertrand, Esther Duflo and Sendhil Mullainathan, "How Much Should We Trust Differences-In-Differences Estimates?"

Difficulty: Advanced **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 119(1), 249–275 (2004); DOI 10.1162/003355304772839588.

Random placebo laws in state-level wage data are found significant at the 5% level up to 45% of the time when difference-in-differences standard errors ignore serial correlation. Parametric corrections do poorly; block bootstrap, clustered variance estimators and collapsing to pre and post periods work, depending on the number of groups.

Version record

Robert F. Stambaugh, "Predictive regressions"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 54(3), 375–421 (1999); DOI 10.1016/s0304-405x(99)00041-0.

When returns are regressed on a lagged persistent regressor such as the dividend yield, the disturbance is correlated with the regressor's innovation and OLS has finite-sample bias. Derives the exact properties and Bayesian posteriors under different priors on the regressor's autocorrelation and initial observation, and shows that asset allocations are sensitive to these choices.

Version record

Robert J. Hodrick, "Dividend Yields and Expected Stock Returns: Alternative Procedures for Inference and Measurement"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 5(3), 357–386 (1992); DOI 10.1093/rfs/5.3.351.

Compares inference procedures for long-horizon return regressions on dividend yields. Monte Carlo evidence shows the Hansen–Hodrick standard errors are biased at long horizons, while estimators derived under the null, a reformulated one-period regression and VAR-implied long-horizon statistics perform better.

Version record

Wei Jiang, "Have Instrumental Variables Brought Us Closer to the Truth"

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Corporate Finance Studies, 6(2), 127–140 (2017); DOI 10.1093/rcfs/cfx015.

Surveys 255 instrumental-variable papers in the top three finance journals: IV estimates exceed the uninstrumented estimates in about 80% of studies, regardless of the expected sign of the bias, and are on average nine times larger. Offers several explanations for these implausibly large estimates and proposes best practices for identification.

Version record

Isaiah Andrews, James H. Stock and Liyang Sun, "Weak Instruments in Instrumental Variables Regression: Theory and Practice"

Difficulty: Research **Type:** Survey

Status/version: Annual Review of Economics, 11(1), 727–753 (2019); DOI 10.1146/annurev-economics-080218-025643.

Review of weak instruments in linear IV with emphasis on heteroskedastic, serially correlated and clustered data, where many classical results do not apply. A survey of AER papers from 2014 to 2018 shows weak instruments remain a practical problem, and the authors set out simple steps for handling them, drawing on the literature on detecting weak instruments and robust confidence sets.

Version record

James H. Stock and Motohiro Yogo, "Testing for Weak Instruments in Linear IV Regression"

Difficulty: Research **Type:** Book chapter

Status/version: In D. W. K. Andrews and J. H. Stock (eds.), Identification and Inference for Econometric Models, Cambridge University Press, 80–108 (2005); DOI 10.1017/CBO9780511614491.006.

Defines weak instruments by the maximum IV bias or Wald-test size distortion they cause and tabulates critical values for the first-stage F (or Cragg–Donald) statistic with one or more endogenous regressors. The source of the rule-of-thumb thresholds still quoted in finance papers.

Version record

David S. Lee, Justin McCrary, Marcelo J. Moreira and Jack Porter, "Valid t-ratio Inference for IV"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 112(10), 3260–3290 (2022); DOI 10.1257/aer.20211063.

Single-instrument t-ratio inference can be badly distorted. Introduces the tF critical value, a standard-error adjustment that is a smooth function of the first-stage F; in a quarter of the specifications from 61 AER papers, corrected standard errors are at least 49% (5% level) and 136% (1% level) larger than conventional 2SLS ones.

Version record

Alwyn Young, "Consistency without Inference: Instrumental Variables in Practical Application"

Difficulty: Research **Type:** Research paper

Status/version: European Economic Review, 147, 104112 (2022); DOI 10.1016/j.euroecorev.2022.104112.

Counter-evidence on IV practice from 1,309 regressions in 30 AEA-journal papers: non-i.i.d. errors and highly leveraged regressions distort size and power and raise IV's bias relative to OLS; first-stage F pre-tests are largely uninformative; and IV rarely rejects the OLS estimate even when the two differ substantially. Uses Monte Carlo, the jackknife and the bootstrap.

Version record

Chapter 3

Machine learning and alternative data

Separate predictive modelling from asset-pricing claims. Compare flexible models with simple baselines, and track dependence, missingness, data availability and model selection.

3.1 Statistical learning foundations

Shai Shalev-Shwartz and Shai Ben-David, *Understanding Machine Learning: From Theory to Algorithms*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected DOI-linked version; print 2014-05-19; online 2014-07-05.

PAC learning, generalisation and VC dimension; independent-sample guarantees need care with dependent market data.

Version record

Michael Ludkovski and Jimmy Risk, *Gaussian Process Models for Quantitative Finance*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer Cham; 2025-03-06 ebook; 2025-03-07 softcover. First edition; SpringerBriefs in Quantitative Finance. DOI 10.1007/978-3-031-80874-6.

Introduces Gaussian-process methods for option sensitivities, optimal stopping, yield and forward curves, and stochastic control. The companion notebooks connect kernels and constrained regression to pricing and uncertainty estimates. Check approximation error, extrapolation, constraints and kernel sensitivity against a numerical benchmark before using a surrogate for valuation or risk.

Extended note. Version record

Marcos López de Prado, *Advances in Financial Machine Learning*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2018

A useful catalogue of finance-specific data sampling (including volume bars), triple-barrier labels, meta-labels, fractional differentiation, feature importance and clustering, purged / embargoed cross-validation (including combinatorial splits), and backtest diagnostics. It should be read critically and checked against original statistical literature; it is not a replacement for probability, econometrics or asset-pricing foundations.

Applications: Multi-asset; ML, backtesting, sizing. Version record

Trevor Hastie, Robert Tibshirani and Jerome Friedman, *The Elements of Statistical Learning*, 2nd ed.

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2009

Statistical learning reference for supervised methods, regularisation, model complexity and resampling. Supplies the more demanding companion to ISL and a statistical-learning foundation; connect its validation principles to dependent financial samples.

Applications: All assets; ML, model selection. Version record

Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, *An Introduction to Statistical Learning*, 2nd ed.

Difficulty: Intro **Type:** Book / monograph

Status/version: 2021

Accessible statistical-learning refresher on regularisation, resampling, classification and model selection. Use to align quant and engineering vocabulary and diagnose leakage, poor validation and unstable pipelines before moving to ESL or finance-specific ML papers.

Applications: All assets; ML foundations. Version record

Murphy, *Probabilistic Machine Learning: An Introduction* (2022)

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2022. Version distinctions remain in the annotation.

Probabilistic prediction, calibration, uncertainty and distribution shift; a foundation for interpreting model outputs.

Version record

Goodfellow, Bengio and Courville, *Deep Learning* (2016)

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2016. Version distinctions remain in the annotation.

Feedforward networks, optimisation and regularisation; mathematical foundations for evaluating neural architectures.

Version record

Bishop, *Pattern Recognition and Machine Learning*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Probabilistic models, model selection and approximate inference.

Version record

Hao Fei, *A Beginner's Guide to AI Research: Insights from a Multimodal AI Perspective*

Difficulty: Intro **Type:** Course / learning resource

Status/version: Introductory lecture slides, 4 September 2025

Introductory lecture slides on problem formulation and experimental design; multimodal AI context rather than financial validation.

Version record

Leo Breiman, "Random Forests"

Difficulty: Advanced **Type:** Research paper

Status/version: Machine Learning, 45(1), 5–32 (2001); DOI 10.1023/a:1010933404324.

Bagged decision trees with random feature selection; robust nonlinear baseline for tabular return and default prediction.

Version record

Tianqi Chen and Carlos Guestrin, "XGBoost: A Scalable Tree Boosting System"

Difficulty: Advanced **Type:** Research paper

Status/version: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794 (2016); DOI 10.1145/2939672.2939785.

Regularised gradient boosting with sparsity-aware splits; the reference gradient-boosted-tree implementation for tabular finance data.

Version record

Léo Grinsztajn, Edouard Oyallon and Gaël Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?"

Difficulty: Research **Type:** Research paper

Status/version: NeurIPS 2022 Datasets and Benchmarks Track; arXiv 2207.08815.

Benchmark showing tree ensembles beat deep nets on medium-size tabular data because of irregular target functions and uninformative features. NeurIPS 2022 Datasets and Benchmarks.

Extended note. Version record

Simon J. D. Prince, *Understanding Deep Learning*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: MIT Press, 2023; author-hosted free PDF and notebooks.

Clear modern deep-learning text (MIT Press 2023) with free PDF and notebooks, covering transformers, diffusion models and reinforcement learning.

Version record

Christopher M. Bishop and Hugh Bishop, *Deep Learning: Foundations and Concepts*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer International Publishing (2024); DOI 10.1007/978-3-031-45468-4.

Probabilistic foundations of neural networks through transformers, graph networks and diffusion models; successor in spirit to Bishop (2006).

Version record

Kevin P. Murphy, *Probabilistic Machine Learning: Advanced Topics*

Difficulty: Research **Type:** Book / monograph

Status/version: MIT Press, 2023; author-hosted draft.

Advanced companion covering inference, generative models, causality, decision making and distribution shift (MIT Press 2023; free draft).

Extended note. Version record

Anastasios N. Angelopoulos and Stephen Bates, "Conformal Prediction: A Gentle Introduction"

Difficulty: Advanced **Type:** Research paper

Status/version: Foundations and Trends® in Machine Learning, 16(4), 494–591 (2023); DOI 10.1561/2200000101.

Practitioner introduction to split conformal prediction, conditional coverage and distribution shift, with code examples.

Version record

Sylvain Arlot and Alain Celisse, "A survey of cross-validation procedures for model selection"

Difficulty: Advanced **Type:** Survey

Status/version: Statistics Surveys, 4(none), 40–79 (2010); DOI 10.1214/09-ss054.

Survey of cross-validation for risk estimation and model selection that separates rigorous theoretical results from empirical claims and ends with guidelines for choosing a procedure given the problem's features. Read before assuming k-fold results transfer to dependent data.

Version record

Christoph Bergmeir, Rob J. Hyndman and Bonsoo Koo, "A note on the validity of cross-validation for evaluating autoregressive time series prediction"

Difficulty: Advanced **Type:** Research paper

Status/version: Computational Statistics & Data Analysis, 120, 70–83 (2018); DOI 10.1016/j.csda.2017.11.003.

Shows that standard k-fold cross-validation is valid for purely autoregressive forecasting models whose errors are uncorrelated, as when a flexible ML model nests the right specification. Theory, simulation and a real example show k-fold then performs favourably against out-of-sample evaluation and non-dependent cross-validation. The conditions matter: they fail when errors stay autocorrelated.

Version record

Vitor Cerqueira, Luis Torgo and Igor Mozetič, "Evaluating time series forecasting models: an empirical study on performance estimation methods"

Difficulty: Advanced **Type:** Research paper

Status/version: Machine Learning, 109(11), 1997–2028 (2020); DOI 10.1007/s10994-020-05910-7.

Compares cross-validation variants, holdout and prequential evaluation on 174 real and three synthetic time series. Blocked cross-validation suits stationary series; for non-stationary series the most accurate estimates come from out-of-sample methods, particularly a holdout repeated over several test periods. Most financial series fall in the second case.

Version record

3.2 Financial learning and alpha research

Read with Gu–Kelly–Xiu, Nagel and the factor-model references. These papers connect statistical learning limits and economic restrictions to dynamic portfolio objectives; the complexity critique supplies an explicit counterpoint to the random-feature result.

3.2.1 Surveys and texts

Marcos López de Prado, Machine Learning for Asset Managers (2020)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 2020. Version distinctions remain in the annotation.

Statistical learning for asset-manager research, feature assessment and portfolio decisions; distinct from Advances in Financial Machine Learning.

Version record

Stefan Nagel, *Machine Learning in Asset Pricing*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2021

Gives a rigorous asset-pricing interpretation of regularisation, trees, neural networks, latent factors and text methods. It is the preferred conceptual bridge for quants who know machine learning but need to understand financial identification, portfolio tests and economic restrictions. [3]

Applications: Equities, cross-asset factors; ML.

Version record

Stefan Jansen, *Machine Learning for Trading*, 3rd ed. (2026); historical second-edition book/code (2020)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Packt, third edition, 24 July 2026; historical second-edition code retained.

The current edition extends the research workflow through selection-aware validation, causal methods, modern neural models, deployment and monitoring. Use the author chapter map for the selected edition. Chapter 4 of the retained 2020 repository belongs to the second edition; chapter numbers and examples must not be silently transferred to the third. Contents were inspected; examples were not executed.

Applications: reproducible practice. Extended note.

Version record

Bryan Kelly and Dacheng Xiu, "Financial Machine Learning"

Difficulty: Advanced/Research **Type:** Research paper

Status/version: Foundations and Trends® in Finance, 13(3-4), 205–363 (2023); DOI 10.1561/500000064.

Monograph-length survey of return prediction, factor models, optimal portfolios, text and complexity in finance; the most complete academic map of the field.

Version record

Stefano Giglio, Bryan Kelly and Dacheng Xiu, "Factor Models, Machine Learning, and Asset Pricing"

Difficulty: Research **Type:** Research paper

Status/version: Annual Review of Financial Economics, 14(1), 337–368 (2022); DOI 10.1146/annurev-financial-101521-104735.

Annual Review survey linking PCA, IPCA, RP-PCA and SDF estimation to machine-learning methods.

Version record

Matthew F. Dixon, Igor Halperin and Paul Bilokon, *Machine Learning in Finance: From Theory to Practice*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer International Publishing (2020); DOI 10.1007/978-3-030-41068-1.

Bayesian learning, neural networks, sequence models and reinforcement learning for pricing, hedging and allocation, with exercises.

Version record

Guillaume Coqueret and Tony Guida, *Machine Learning for Factor Investing*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Chapman & Hall/CRC, 2023 Python version; DOI 10.1201/9781003121596. The original R-based edition (2020) has a free online text.

Data preparation, labels, penalised regression, trees, neural nets and backtesting on a characteristics panel; R and Python versions with online text.

Version record

Agostino Capponi and Charles-Albert Lehalle (eds.), *Machine Learning and Data Sciences for Financial Markets: A Guide to Contemporary Practices*

Difficulty: Advanced/Research **Type:** Book / edited research collection

Status/version: Cambridge University Press, edition 1 (2023); DOI 10.1017/9781009028943.

Sixty-plus contributors on robo-advice, price formation, deep hedging, portfolio construction, nowcasting, alternative data and algorithmic ethics. Chapters vary in maturity.

Version record

Ronen Israel, Bryan T. Kelly and Tobias J. Moskowitz, "Can Machines "Learn" Finance?"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: SSRN working paper 3624052 (2020); DOI 10.2139/ssrn.3624052. Working-paper record; no journal version established in this check.

Why ML in return prediction differs from image recognition: small signal-to-noise, non-stationarity and adversarial markets; realistic expectations.

Extended note. Version record

Marcos López de Prado, *Causal Factor Investing: Can Factor Investing Become Scientific?*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2023); DOI 10.1017/9781009397315.

Cambridge Element arguing that factor models require causal graphs and that associational specifications produce false discoveries; read critically beside the causal-inference chapter.

Version record

Zihao Zhang and Stefan Zohren, *Deep Learning in Quantitative Trading*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2025); DOI 10.1017/9781009707091.

Cambridge Element covering deep sequence models, order-book prediction, momentum networks and portfolio optimisation with code-oriented examples. Previously shortlisted; added now as a compact, finance-specific deep-learning sequence.

Version record

Andrew Green, *Deep Learning in Quantitative Finance*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, first edition, March 2026 (ISBN 9781119685241); retailer listings differ on date and page count.

Wiley (2026) practitioner text on networks, sequence models, generative models and reinforcement learning applied to derivative valuation, PDE/BSDE solvers, volatility models and hedging, with Python notebooks. Listings disagree on page count and date; confirm the edition before citing.

Extended note. Version record

Marcos López de Prado and Vincent Zoonekynd, "Correcting the Factor Mirage: A Research Protocol for Causal Factor Investing"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(3), 10–47 (2025); DOI 10.3905/jpm.2025.1.794.

The "factor mirage": standard specification practice (notably controlling for colliders) can make factor strategies lose money even when premia are correctly signed; proposes a causal research protocol for factor investing.

Version record

3.2.2 Return prediction and ML asset pricing

Bryan Kelly, Semyon Malamud and Kangying Zhou, "The Virtue of Complexity in Return Prediction", and Shihao Gu, Bryan Kelly and Dacheng Xiu, "Empirical Asset Pricing via Machine Learning"

Difficulty: Research **Type:** Research paper

Status/version: 2020–2024

These works explain why flexible nonlinear models can improve conditional return estimation when regularised and evaluated cross-sectionally. Their most important institutional contribution is comparative benchmarking across model classes rather than a claim that neural networks universally dominate. [2]

Applications: Equities; ML alpha, asset pricing. Version record

J. B. Heaton, N. G. Polson and J. H. Witte, "Deep Learning for Finance: Deep Portfolios" (2017)

Difficulty: Research **Type:** Research paper

Status/version: Selected DOI-linked version; print 2017-01; online 2016-10-07.

Early application of deep representation learning to financial prediction and portfolios. Reproduce against linear, shrinkage and tree baselines; evaluate stability across time, economic constraints and costs. Complements the Gu–Kelly–Xiu and Nagel readings.

Applications: academic. Version record

Rui Da, Stefan Nagel, and Dacheng Xiu, The Statistical Limit of Arbitrage; Author research page and revision status

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33070, issued October 2024; author's research page reports a June 2025 revision; no journal publication established as of 4 October 2026.

Models arbitrageurs who must learn sparse, weak alphas from finite data and shows that even optimal statistical procedures can leave genuine pricing errors unexploited. It supplies a useful economic distinction between an infeasible perfect-information Sharpe ratio and the return a research team can realistically estimate and trade. It is a theoretical and empirical working-paper framework, not a direct estimate of any desk's attainable portfolio.

Extended note. Version record

Luyang Chen, Markus Pelger, and Jason Zhu, Deep Learning in Asset Pricing

Difficulty: Research **Type:** Research paper

Status/version: Management Science 70(2), 714–750, February 2024 issue; published online 20 February 2023.

Builds a conditional asset-pricing model in which a neural network is disciplined by no-arbitrage moments, adversarial test assets, and macro state extraction. It connects flexible prediction to pricing restrictions rather than treating improved cross-sectional return forecasts as a complete economic explanation. Report model sensitivity, point-in-time macro inputs, and net portfolio results separately; published out-of-sample gains do not certify executable alpha.

Extended note. Version record

Bryan T. Kelly, Boris Kuznetsov, Semyon Malamud and Teng Andrea Xu, Artificial Intelligence Asset Pricing Models

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33351, issued January 2025; NBER lists revision June 2026. No journal publication established.

Places a transformer in the stochastic discount factor to share conditional pricing information across assets; a linear-transformer surrogate decomposes the learned pricing mechanisms. Read as a modern conditional asset-pricing architecture and compare against simpler factor and autoencoder baselines.

Extended note. Version record

Theis Ingerslev Jensen, Bryan Kelly, Semyon Malamud, and Lasse Heje Pedersen, Machine Learning and the Implementable Efficient Frontier

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 39(10), 3035–3078, October 2026 issue; published online 15 March 2026 and typeset 6 April 2026.

Integrates trading costs into dynamic portfolio choice and learns portfolio weights against an economic objective, yielding an implementable frontier defined by out-of-sample net return at each risk level. The paper shows how cost-blind ML can overselect fast-decaying, small-stock characteristics even when prediction metrics look strong. Its empirical model still assumes a particular cost function and stationary information process; desk-specific execution, borrow, and regime stress remain necessary.

Extended note. Version record

Stefan Nagel, Seemingly Virtuous Complexity in Return Prediction

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 34104, August 2025; author lists a July 2025 revision; no journal publication established as of 4 October 2026.

Analyzes the random-Fourier-feature setting behind the catalogue's Kelly–Malamud–Zhou complexity reading. When features vastly outnumber a short training window, the forecast reduces to a similarity-weighted average of recent returns, effectively a volatility-timed momentum rule; artificial reversal data exposes that inductive bias. Reproduce the simple implied rule and test alternative return regimes before attributing performance to learned economic structure; the critique targets the stated short-window setting, not all nonlinear ML.

Extended note. Version record

Rob Lehnherr, Manan Mehta, and Stefan Nagel, Optimal Factor Timing in a High-Dimensional Setting

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal 81(2), 51–66, 2025; published online 27 March 2025.

Combines many factor-return predictors and equity factors while shrinking the resulting timing portfolio toward skepticism about large gains. It uses macro variables and spreads between each factor's long and short legs, then reports out-of-sample evidence including large-cap factors. Validate release lags, revisions, trading frequency, and net turnover for each predictor; the reported historical gains are conditional on the chosen factor set and calibration.

Extended note. Version record

Joachim Freyberger, Andreas Neuhierl and Michael Weber, "Dissecting Characteristics Nonparametrically"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(5), 2326–2377 (2020); DOI 10.1093/rfs/hhz123.

Adaptive group lasso on nonparametric characteristic functions selects a small set of characteristics with incremental information.

Version record

Bryan Kelly, Semyon Malamud and Lasse Heje Pedersen, "Principal Portfolios"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 78(1), 347–387 (2023); DOI 10.1111/jofi.13199.

Uses the full matrix of cross-predictability between signals and assets to form principal portfolios; decomposes into alpha and beta components.

Version record

Doron Avramov, Si Cheng and Lior Metzker, "Machine Learning vs. Economic Restrictions: Evidence from Stock Return Predictability"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 69(5), 2587–2619 (2023); DOI 10.1287/mnsc.2022.4449.

ML return predictability concentrates in microcaps, distressed and illiquid stocks; profits shrink under economic restrictions and costs.

Version record

Markus Leippold, Qian Wang and Wenyu Zhou, "Machine learning in the Chinese stock market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 145(2), 64–82 (2022); DOI 10.1016/j.jfineco.2021.08.017.

Replicates ML return prediction in China, where retail dominance, liquidity and state ownership change which predictors work.

Version record

Daniele Bianchi, Matthias Büchner and Andrea Tamoni, "Bond Risk Premiums with Machine Learning"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 34(2), 1046–1089 (2021); DOI 10.1093/rfs/hhaa062.

Neural networks on yields and macro data predict Treasury excess returns; illustrates ML return prediction outside equities.

Version record

Yijie Wang, Hao Gao, Campbell R. Harvey, Yan Liu and Xinyuan Tao, "Machine Learning Meets Markowitz"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 34861, February 2026; no journal publication established.

Argues against the predict-then-optimise split and trains ML models on portfolio objectives directly. Working paper; compare after costs with the implementable-frontier evidence.

Extended note. Version record

Sebastian Bell, Ali Kakhbod, Martin Lettau and Abdolreza Nazemi, "AlphaGlass: Interpretable Characteristic-Based Portfolio Choice"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 35186, May 2026; no journal publication established.

Interpretable framework mapping stock characteristics into additive signals with sparse interactions, then optimising long-short portfolio objectives directly. From the authors of the glass-box corporate-bond paper; working paper, so compare after costs with the implementable-frontier evidence.

Extended note. Version record

Antoine Didisheim, Shikun (Barry) Ke, Bryan T. Kelly and Semyon Malamud, "APT or "AIPT"? The Surprising Dominance of Large Factor Models"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33012 (2024); DOI 10.3386/w33012. Working-paper record; no journal version established in this check.

Argues that SDFs built from very many factors outperform sparse factor models out of sample; part of the complexity debate with Nagel.

Extended note. Version record

Jingwen Jiang, Bryan Kelly and Dacheng Xiu, "(Re-)Imag(in)ing Price Trends"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 78(6), 3193–3249 (2023); DOI 10.1111/jofi.13268.

CNNs trained on price-chart images predict returns and transfer across horizons and markets; a careful image-based signal study.

Version record

Svetlana Bryzgalova, Markus Pelger and Jason Zhu, "Forest through the Trees: Building Cross-Sections of Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(5), 2447–2506 (2025); DOI 10.1111/jofi.13477.

AP-Trees: decision-tree portfolio splits chosen to span the SDF give interpretable test assets and factors that beat conventional double sorts as building blocks.

Version record

Lin William Cong, Guanhao Feng, Jingyu He and Xin He, "Growing the efficient frontier on panel trees"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 167, 104024 (2025); DOI 10.1016/j.jfineco.2025.104024.

P-Trees grow interpretable panel trees under mean-variance efficiency to build test assets that expand the efficient frontier, with alphas unexplained by benchmark models.

Version record

Bin Li, Alberto G. Rossi, Xuemin (Sterling) Yan and Lingling Zheng, "Machine learning from a "Universe" of signals: The role of feature engineering"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 172, 104138 (2025); DOI 10.1016/j.jfineco.2025.104138.

ML on a large universe of mechanically generated signals: how much performance comes from feature engineering versus model choice.

Extended note. Version record

Yingguang Zhang, Yandi Zhu and Juhani T. Linnainmaa, "Man versus Machine Learning Revisited"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(12), 3768–3790 (2025); DOI 10.1093/rfs/hhaf066.

Attributes the large alpha of a random-forest analyst-error strategy (van Binsbergen, Han and Lopez-Lira 2023) to look-ahead bias; removing it cuts the alpha sharply. RFS has issued an expression of concern for the original article.

Extended note. Version record

Rohit Allena, "Confident Risk Premiums and Investments Using Machine Learning Uncertainties"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(5), 1463–1505 (2026); DOI 10.1093/rfs/hhaf087.

Ex-ante confidence intervals for risk-premium forecasts from linear and ML models; strategies trading only precisely forecast stocks improve performance.

Version record

Darwin Choi, Wenxi Jiang and Chao Zhang, "Alpha Go Everywhere: Machine Learning and International Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Asset Pricing Studies, 15(3-4), 288–331 (2025); DOI 10.1093/rapstu/raaf005.

Neural networks trained market by market predict international returns better than models trained on US data; ML gains are market-specific.

Version record

Joachim Grammig, Constantin Hanenberg, Christian Schlag and Jantje Sönksen, "Diverging Roads: Theory-Based vs. Machine Learning-Implied Stock Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 23(2), nbaf005 (2025); DOI 10.1093/jjfinec/nbaf005.

Compares theory-based and ML equity risk-premium estimates by horizon and combines them in hybrid strategies.

Version record

Yang Bai and Kuntara Pukthuanthong, "Machine Learning Classification and Portfolio Construction: Does the Loss Function Matter?"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 1–17 (2026); DOI 10.1080/0015198x.2026.2726133.

Matched ML models trained as classifiers beat regression models in portfolio construction (value-weighted Sharpe 2.08 versus 1.39 for a stacking ensemble), with spanning tests favouring classification.

Version record

Iro Tasitsiomi, "Effective Model Complexity in Equity-Premium Forecasting"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial Data Science, 8(3), 10–37 (2026); DOI 10.3905/jfds.2026.011.

Revisits the "virtue of complexity" with Goyal–Welch predictors: effective complexity depends on feature geometry, kernel bandwidth, spectrum and regularisation, not raw feature count. A useful check on random-feature return forecasting.

Version record

Tingting Cheng, Shan Jiang, Albert Bo Zhao and Junyi Zhao, "Is machine learning a necessity? A regression-based approach for stock return prediction"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 81, 101598 (2025); DOI 10.1016/j.jempfin.2025.101598.

A regression-based alternative to machine learning for stock return prediction.

Extended note. Version record

3.2.3 Feature selection and interpretability

Meinshausen and Bühlmann (2010), "Stability Selection"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2010. Version distinctions remain in the annotation.

Subsampling-based selection frequency and false-discovery perspective.

Version record

Lundberg and Lee (2017), "A Unified Approach to Interpreting Model Predictions"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2017. Version distinctions remain in the annotation.

SHAP framework and limitations of attribution.

Version record

Tibshirani (1996), LASSO

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 1996. Version distinctions remain in the annotation.

Regularized variable selection and shrinkage.

Version record

Emmanuel Candès, Yingying Fan, Lucas Janson and Jinchi Lv, "Panning for Gold: Model-X Knockoffs for High-Dimensional Controlled Variable Selection"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Royal Statistical Society Series B: Statistical Methodology, 80(3), 551–577 (2018); DOI 10.1111/rssb.12265.

Model-X knockoffs control the false-discovery rate of selected features without p-values for each variable.

Version record

Scott M. Lundberg et al., "From local explanations to global understanding with explainable AI for trees"

Difficulty: Advanced **Type:** Research paper

Status/version: Nature Machine Intelligence, 2(1), 56–67 (2020); DOI 10.1038/s42256-019-0138-9.

TreeSHAP exact Shapley values for tree ensembles, interaction values and global summaries.

Version record

Aaron Fisher, Cynthia Rudin and Francesca Dominici, "All Models are Wrong, but Many are Useful: Learning a Variable's Importance by Studying an Entire Class of Prediction Models Simultaneously"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Machine Learning Research 20(177), 1–81 (2019).

Model reliance and model-class reliance: importance ranges across the Rashomon set rather than one fitted model.

Version record

Yonghwan Jo and Yong Hwi Kim, "Rethinking Variable Importance in Machine Learning: An Economic Perspective on Empirical Asset Pricing"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 82(2), 92–135 (2026); DOI 10.1080/0015198x.2026.2621646.

Economic rather than statistical variable importance for ML portfolios over 94 predictors: outside microcaps, low-risk and liquidity-volatility signals matter most and short-term reversal fades, because ML tilts towards microcaps.

Version record

Jerome H. Friedman, "Greedy function approximation: A gradient boosting machine"

Difficulty: Advanced **Type:** Research paper

Status/version: The Annals of Statistics, 29(5) (2001); DOI 10.1214/aos/1013203451.

Introduces gradient boosting and partial-dependence plots; the foundation for the tree ensembles used in return prediction.

Version record

Daniel W. Apley and Jingyu Zhu, "Visualizing the Effects of Predictor Variables in Black Box Supervised Learning Models"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of the Royal Statistical Society Series B: Statistical Methodology, 82(4), 1059–1086 (2020); DOI 10.1111/rssb.12377.

Accumulated local effects (ALE) plots: an alternative to partial dependence that remains valid with correlated features, as financial characteristics usually are.

Version record

Marco Tulio Ribeiro, Sameer Singh and Carlos Guestrin, ""Why Should I Trust You?": Explaining the Predictions of Any Classifier"

Difficulty: Advanced **Type:** Research paper

Status/version: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1135–1144 (2016); DOI 10.1145/2939672.2939778.

LIME: local surrogate explanations for any classifier; read alongside SHAP for model-agnostic interpretation.

Version record

Hui Zou and Trevor Hastie, "Regularization and variable selection via the elastic net"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of the Royal Statistical Society Series B: Statistical Methodology, 67(2), 301–320 (2005); DOI 10.1111/j.1467-9868.2005.00503.x.

Elastic net combines lasso and ridge penalties to select groups of correlated predictors, the default penalised regression in many factor-zoo studies.

Version record

Rina Foygel Barber and Emmanuel J. Candès, "Controlling the false discovery rate via knockoffs"

Difficulty: Research **Type:** Research paper

Status/version: The Annals of Statistics, 43(5), 2055–2085 (2015); DOI 10.1214/15-aos1337.

Original fixed-X knockoff filter giving exact finite-sample FDR control in variable selection; predecessor of model-X knockoffs.

Version record

Carolin Strobl, Anne-Laure Boulesteix, Achim Zeileis and Torsten Hothorn, "Bias in random forest variable importance measures: Illustrations, sources and a solution"

Difficulty: Advanced **Type:** Research paper

Status/version: BMC Bioinformatics, 8(1), 25 (2007); DOI 10.1186/1471-2105-8-25.

Shows impurity-based random-forest importance is biased towards variables with many split points; motivates permutation and conditional importance.

Version record

Cynthia Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead"

Difficulty: Advanced **Type:** Research paper

Status/version: Nature Machine Intelligence, 1(5), 206–215 (2019); DOI 10.1038/s42256-019-0048-x.

Argues for inherently interpretable models over post-hoc explanations in high-stakes settings; a useful counterpoint for model-governance discussions.

Version record

3.3 Graph and relational models

Subject context and qualifications.

Fuli Feng et al., "Temporal Relational Ranking for Stock Prediction" and code

Difficulty: Research **Type:** Research paper

Status/version: 2019

Incorporates firm relationships into a temporal ranking model rather than forecasting stocks independently. It is an early finance-native graph approach and a useful baseline for sector, ownership, supply-chain and knowledge-graph signals. [24]

Applications: Equities; graph ML, cross-sectional ranking. Version record

Wentao Xu et al., "HIST: A Graph-based Framework for Stock Trend Forecasting via Mining Concept-Oriented Shared Information"

Difficulty: Research **Type:** Research paper

Status/version: 2022

Separates predefined concepts from hidden latent concepts shared across companies. Its institutional appeal is that it attempts to represent changing economic groupings beyond static sector classifications, though hidden concepts require stability and interpretability checks.

Applications: Equities; graph ML, latent factors. Version record

Tong Li, Zhaoyang Liu, Yanyan Shen, Xue Wang, Haokun Chen and Sen Huang, "MASTER: Market-Guided Stock Transformer for Stock Price Forecasting" and official code

Difficulty: Research **Type:** Research paper

Status/version: 2024

Combines cross-sectional attention with market-state guidance for stock ranking. It is a useful recent benchmark for dynamic stock interactions, but reported gains should be stress-tested against universe leakage, China-specific microstructure and realistic portfolio constraints. [25]

Applications: Equities; transformers, cross-sectional alpha. Version record

Uddin et al., "Attention-based Dynamic Graph Neural Network for Asset Pricing"

Difficulty: Research **Type:** Research paper

Status/version: 2023

Learns time-varying asset relationships rather than imposing a fixed graph. The framework is relevant to covariance, contagion and cross-sectional prediction, especially where economic linkages change faster than conventional classifications. [26]

Applications: Equities, multi-asset; graph factors, risk. Version record

Xavier Gabaix, Ralph S. J. Koijen, Robert J. Richmond and Motohiro Yogo, Asset Embeddings

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33651, issued April 2025; no journal publication established from inspected records.

Uses institutional holdings co-occurrence to learn asset and investor representations, drawing on recommender systems, Word2Vec and transformer methods. The paper evaluates relative valuations, stock comovement and portfolio decisions, and discusses crowded trades and stress testing.

Extended note. Version record

Microsoft Research, Temporal Routing Adaptor and Optimal Transport

Difficulty: Research **Type:** Research paper

Status/version: 2021

TRA routes observations to specialised predictors and uses optimal transport to avoid degenerate routing. It is an important concept-drift model available within Qlib and is particularly relevant when multiple market regimes coexist rather than switch cleanly. [27]

Applications: Equities; adaptive ML, regime mixtures. Version record

Lauren Cohen and Andrea Frazzini, "Economic Links and Predictable Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 63(4), 1977–2011 (2008); DOI 10.1111/j.1540-6261.2008.01379.x.

Customer returns predict supplier returns because of limited attention; the canonical economic-link signal underlying graph features.

Version record

Charles M. C. Lee, Stephen Teng Sun, Rongfei Wang and Ran Zhang, "Technological links and predictable returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 132(3), 76–96 (2019); DOI 10.1016/j.jfineco.2018.11.008.

Patent-based technological proximity predicts returns of linked firms; a second economically motivated graph.

Version record

Francis X. Diebold and Kamil Yılmaz, "On the network topology of variance decompositions: Measuring the connectedness of financial firms"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 182(1), 119–134 (2014); DOI 10.1016/j.jeconom.2014.04.012.

Variance-decomposition networks for volatility connectedness; widely used spillover measures.

Version record

Thomas N. Kipf and Max Welling, "Semi-Supervised Classification with Graph Convolutional Networks"

Difficulty: Advanced **Type:** Research paper

Status/version: ICLR 2017; arXiv 1609.02907.

The GCN layer underlying most graph-based stock models; ICLR 2017.

Extended note. Version record

Lior Menzly and Oguzhan Ozbas, "Market Segmentation and Cross-predictability of Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 65(4), 1555–1580 (2010); DOI 10.1111/j.1540-6261.2010.01578.x.

Supplier–customer industry returns cross-predict, more strongly with less analyst and institutional coverage.

Version record

Bernard Herskovic, Bryan Kelly, Hanno Lustig and Stijn Van Nieuwerburgh, "Firm Volatility in Granular Networks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 128(11), 4097–4162 (2020); DOI 10.1086/710345.

Customer–supplier network concentration explains common variation in firm idiosyncratic volatility.

Version record

Jian Feng, Xiaolin Huo, Xin Liu, Yifei Mao and Hong Xiang, "Economic links from bonds and cross-stock return predictability"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 171, 104110 (2025); DOI 10.1016/j.jfineco.2025.104110.

Economic links inferred from corporate-bond markets predict cross-stock returns.

Extended note. Version record

3.4 Alternative data and information value

Subject context and qualifications. Subject context and qualifications.

Alexander Denev and Saeed Amen, *The Book of Alternative Data* (2020)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley first edition, August 2020, 416 pages; ISBN 978-1-119-60179-1. Publisher contents and author identities inspected.

Taxonomy, sourcing, cleaning and assessing non-traditional data. Research uses include consumer behaviour, weather, text, web activity, job postings and satellite imagery. Evaluate entity mapping, historical coverage, changing definitions, data cost and incremental information beyond conventional factors.

Applications: practitioner. Version record

Tony Guida, ed., *Big Data and Machine Learning in Quantitative Investment* (2018 online / 2019 copyright)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected DOI-linked version; print 2018-11-30; online 2018-12-12.

Practitioner case studies on alternative-data integration, macro signals, consumer receipts, gradient boosting, social media, NLP, energy-event detection and deep/RL methods. Separate the PDF's illustrative credit-card and search-trend examples from a claim about the exact contents of any particular chapter.

Applications: case studies. Version record

Johan Bollen, Huina Mao and Xiaojun Zeng, "Twitter Mood Predicts the Stock Market" (2011 journal version)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2011. Version distinctions remain in the annotation.

Historical example of turning aggregate social text into predictors. Useful for replication, feature-selection and sample-stability exercises; its reported result is not evidence of durable, executable social-media alpha. Compare with Tetlock and current text baselines.

Applications: academic. Version record

Tobias Preis, Helen Susannah Moat and H. Eugene Stanley, "Quantifying Trading Behavior in Financial Markets Using Google Trends" (2013)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2013. Version distinctions remain in the annotation.

Attention-based search-volume example. Replication should preserve historical query vintages, normalisation windows, keyword selection and release timing. Test the entire keyword-selection procedure out of sample, including fees and alternative explanations.

Applications: academic. Version record

J.P. Morgan, "Trading Insights: All About Alternative Data"

Difficulty: Not specified **Type:** Research paper

Status/version: Edition/revision and publication date not established for the selected resource(s).

Practitioner discussion of dataset selection, adoption and implementation obstacles. Optional context alongside Denev–Amen and the AIMA survey; test examples independently with procurement costs and capacity assumptions.

Version record

AIMA and SS&C, *Casting the Net: How Hedge Funds Are Using Alternative Data* (2020)

Difficulty: Not specified **Type:** Report / research note

Status/version: Selected citation year(s): 2020. Version distinctions remain in the annotation.

Industry survey and interviews about alternative-data adoption, uses and implementation challenges. The report describes 100 responding managers with roughly US$720 billion under management; this is an adoption survey, not a test of returns or proof that any dataset produces alpha.

Applications: Industry survey. Version record

RavenPack, "Event Trading Using Market Response" (22 July 2011)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2011. Version distinctions remain in the annotation.

Vendor research on news-event trading and market response; the linked title identifies the retained resource.

Applications: supplementary. Version record

David Leinweber and Jacob Sisk, "Event-Driven Trading and the 'New News"', *Journal of Portfolio Management* 38(1), 110–124 (2011); SSRN record

Difficulty: Research **Type:** Research paper

Status/version: Selected DOI-linked version; print 2011-10-31; online 2011-10-31.

Distinct news-analysis paper using US event studies and portfolio simulations for 2006–2009, with a 2010 holdout. Its historical reported results are not independently replicated here; news timestamps, availability and trading costs matter.

Version record

Svetlana Bryzgalova, Sven Lerner, Martin Lettau, and Markus Pelger, Missing Financial Data

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 38(3), 803–882, March 2025 issue; published online 2 July 2024.

Shows that missing firm fundamentals are widespread and structured rather than isolated random holes; the paper proposes an imputation approach using both time and cross-sectional dependence. Use it to define explicit missing-data diagnostics and compare imputed, missing-indicator, and observed-only strategies. Any imputation used for live forecasts must be fitted using only information available at the decision date.

Extended note. Version record

Joachim Freyberger, Bjoern Hoeppner, Andreas Neuhierl, and Michael Weber, Missing Data in Asset Pricing Panels

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 38(3), 760–802, March 2025 issue; published online 27 January 2024.

Develops conditional-mean imputation and weighted least squares inside a GMM framework so asset-pricing panels can retain return observations with partly missing predictors and still support valid inference under the paper's assumptions. It complements Bryzgalova et al., which focuses on the structure and reconstruction of missing fundamentals. Historical panel validity does not by itself guarantee that the imputed feature existed at a live decision time.

Extended note. Version record

Christina Zhu, Big Data as a Governance Mechanism

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 32(5), 2021–2061, May 2019; published online 4 April 2019.

Studies the introduction of consumer-transaction and satellite data to asset managers and finds evidence of greater price informativeness and altered managerial behavior at covered firms. It supplies a concrete empirical alternative-data case beyond a vendor taxonomy and shows why information value can erode as adoption grows. The paper's covered firms, 2014–2016 data, and costly licensed feeds constrain transfer to current markets.

Extended note. Version record

Saketh Aleti and Tim Bollerslev, News and Asset Pricing: A High-Frequency Anatomy of the SDF

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 38(3), 712–759, March 2025 issue; first published online 16 May 2024.

Matches real-time newswire categories to an intraday estimated stochastic discount factor, separating which news moves priced risk from news that merely attracts attention. Monetary-policy and finance stories dominate average SDF variation in its sample, with changing importance across time and factor premia. A priced-news association is not a ready long–short trading rule: feed latency, surprise measurement, overlapping releases, and executable prices must be rebuilt for a desk.

Extended note. Version record

Maryam Farboodi, Dhruv Singal, Laura Veldkamp, and Venky Venkateswaran, Valuing Financial Data

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 38(3), 938–980, March 2025 issue; first published online 22 July 2024.

Develops a sufficient-statistics approach to the private value of financial data using equilibrium return moments and other investors' information demand. It clarifies why a dataset's value depends on the investor's portfolio, competitors, information type, and market liquidity, even when the raw feed is identical. The model is an economic valuation framework; actual license fees, historical reconstruction costs, and marginal execution value still require an institution-specific pilot.

Extended note. Version record

Abhiroop Mukherjee, George Panayotov and Janghoon Shon, "Eye in the sky: Private satellites and government macro data"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 141(1), 234–254 (2021); DOI 10.1016/j.jfineco.2021.03.002.

Satellite oil-storage data reduce the information content of official inventory releases; an alternative-data study with a clear information-timing design.

Version record

T. Clifton Green, Ruoyan Huang, Quan Wen and Dexin Zhou, "Crowdsourced employer reviews and stock returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 134(1), 236–251 (2019); DOI 10.1016/j.jfineco.2019.03.012.

Changes in Glassdoor ratings predict returns and earnings surprises; illustrates panel construction from crowd-sourced data.

Version record

Jiekun Huang, "The customer knows best: The investment value of consumer opinions"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 128(1), 164–182 (2018); DOI 10.1016/j.jfineco.2018.02.001.

Abnormal Amazon customer ratings predict returns and earnings surprises.

Version record

Marko Kolanovic and Rajesh T. Krishnamachari, "Big Data and AI Strategies: Machine Learning and Alternative Data Approach to Investing"

Difficulty: Advanced **Type:** Report / research note

Status/version: J.P. Morgan Quantitative and Derivatives Strategy, May 2017; distribution and access vary.

Influential 2017 J.P. Morgan practitioner report classifying alternative-data sources and ML methods; dated vendor landscape, still a useful taxonomy.

Version record

Paul Lieberman, Atanas Mihov, Andy Naranjo and Mihail Velikov, "Show me the receipts: B2B payment timeliness and expected returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 172, 104108 (2025); DOI 10.1016/j.jfineco.2025.104108.

B2B payment timeliness data predict expected stock returns; an alternative-data signal tied to firm cash flows.

Extended note. Version record

Yuan Xue, Bohui Zhang and Xiaofeng Zhao, "Nowcasting Firms' Operating Activities from Satellite Data on Thermal Infrared Radiation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(3), 1073–1111 (2026); DOI 10.1017/s0022109025102147.

Thermal infrared radiation from 28,236 factories of Chinese listed firms tracks operations in real time and forecasts fundamentals; a physically grounded alternative-data source.

Version record

Alexandre d'Aspremont, Simon Ben Arous, Jean-Charles Bricongne, Benjamin Lietti and Baptiste Meunier, "Satellites turn "concrete": Tracking cement with satellite data and neural networks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 249, 105923 (2025); DOI 10.1016/j.jeconom.2024.105923.

Satellite infrared images of about 1,000 cement plants, processed with physics and neural networks, build a real-time activity index that beats benchmarks for nowcasting cement and construction output.

Version record

Ashby Monk, Marcel Prins and Dane Rook, "Rethinking Alternative Data in Institutional Investment"

Difficulty: Intro **Type:** Practitioner article

Status/version: The Journal of Financial Data Science, 1(1), 14–31 (2019); DOI 10.3905/jfds.2019.1.1.014.

Argues that institutional investors should rethink alternative data's value proposition, how they characterise datasets and how they access them, choosing datasets that fit their own resources. Favours defensive, defensible strategies and partnerships over racing to exploit short-lived signals.

Version record

Kenneth Froot, Namho Kang, Gideon Ozik and Ronnie Sadka, "What do measures of real-time corporate sales say about earnings surprises and post-announcement returns?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 125(1), 143–162 (2017); DOI 10.1016/j.jfineco.2017.04.008.

Builds real-time proxies of retail sales from sources including about 50 million mobile devices. The within-quarter measure predicts sales growth, revenue and earnings surprises, with average excess announcement returns of 3.4%; the post-quarter measure relates negatively to announcement returns and positively to later returns, as managers guide pessimistically when holding good private news.

Version record

Zsolt Katona, Marcus O. Painter, Panos N. Patatoukas and Jean Zeng, "On the Capital Market Consequences of Big Data: Evidence from Outer Space"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(2), 551–579 (2025); DOI 10.1017/s0022109023001448.

Uses the start of satellite coverage of major retailers to study unequal access to big data. Investors with the data traded profitably, especially ahead of bad-news reports; coverage brought more informed short selling, less informed individual buying and lower liquidity around reports, so unequal access raised information asymmetry without immediately improving price discovery.

Version record

Jung Koo Kang, Lorien Stice-Lawrence and Yu Ting Forester Wong, "The Firm Next Door: Using Satellite Images to Study Local Information Advantage"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Accounting Research, 59(2), 713–750 (2021); DOI 10.1111/1475-679x.12360.

Satellite car counts at 92,668 stores of 71 US retailers serve as a timely store-level performance measure. Institutional investors trade on the performance of nearby stores, profitably on average, but some rely on local signals even when they predict poorly; industry expertise and short-term incentives reduce that overreliance.

Version record

Ilia D. Dichev and Jingyi Qian, "The benefits of transaction-level data: The case of NielsenIQ scanner data"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Accounting and Economics, 74(1), 101495 (2022); DOI 10.1016/j.jacceco.2022.101495.

NielsenIQ retail scanner data (2006–2018), aggregated to manufacturer-quarter consumer purchases, strongly predict GAAP revenue. Analysts' revenue forecasts have predictable errors, and hedge portfolios on abnormal consumer purchases earn about 14–19% a year depending on specification: transaction-level data give an information edge over reported revenue.

Version record

Hailiang Chen, Prabuddha De, Yu (Jeffrey) Hu and Byoung-Hyoun Hwang, "Wisdom of Crowds: The Value of Stock Opinions Transmitted Through Social Media"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(5), 1367–1403 (2014); DOI 10.1093/rfs/hhu001.

Textual analysis of articles on a leading investor social-media platform, and of the readers' comments on them. The views in both predict future stock returns and earnings surprises, evidence that crowd opinion carries value-relevant information.

Version record

Jillian Grennan and Roni Michaely, "FinTechs and the Market for Financial Analysis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 56(6), 1877–1907 (2021); DOI 10.1017/s0022109020000721.

Describes the hundreds of equity-market-intelligence FinTechs that aggregate traditional and non-traditional sources (Twitter, blogs) and use AI to make recommendations. They substitute for traditional information providers, and the non-traditional data they synthesise contain valuable crowd information linked to informational efficiency.

Version record

Olivier Dessaint, Thierry Foucault and Laurent Frésard, "Does Alternative Data Improve Financial Forecasting? The Horizon Effect"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 79(3), 2237–2287 (2024); DOI 10.1111/jofi.13323.

Counter-evidence on the benefits of alternative data: because such data make short-term information cheaper, forecasters shift attention from the long term to the short term, so long-horizon forecasts become less informative while short-horizon ones improve. Confirmed with equity analysts' forecasts at several horizons and their exposure to social-media data.

Version record

Maryam Farboodi and Laura Veldkamp, "Long-Run Growth of Financial Data Technology"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 110(8), 2485–2523 (2020); DOI 10.1257/aer.20171349.

Model in which traders choose between learning about future asset values and learning about others' demands as data processing improves. Unbiased technical change can shift the whole market toward extracting information from order flow, yet both kinds of data keep being processed; the results revise common views on price informativeness and on whether liquidity-providing trades make markets more resilient.

Version record

Juliane Begenau, Maryam Farboodi and Laura Veldkamp, "Big data in finance and the growth of large firms"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 97, 71–87 (2018); DOI 10.1016/j.jmoneco.2018.05.013.

Hypothesis that big data in finance lowers the cost of capital more for large firms, which generate more data and attract a growing share of data processing as technology improves, letting them invest more cheaply and grow larger. Links alternative-data coverage to firm size.

Version record

Simona Abis and Laura Veldkamp, "The Changing Economics of Knowledge Production"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 37(1), 89–118 (2023); DOI 10.1093/rfs/hhad059.

Estimates investment firms' data stocks and knowledge production functions from hiring and wage data. Big-data technology changes how data and labour combine; the structural estimates predict a 5% fall in the labour share of income in investment management, comparable to estimates for the industrial revolution.

Version record

3.5 Text and language models

Subject context and qualifications.

3.5.1 Text as data

Tim Loughran and Bill McDonald, When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 66(1), 35–65, February 2011; first published online 6 January 2011.

Demonstrates that generic sentiment dictionaries misread finance vocabulary and builds finance-specific word lists using 10-K filings. It is the essential transparent baseline before claiming that FinBERT or an LLM extracts incremental signal from reports, conference calls, or news. Tone associations with market outcomes are neither causal effects nor proof of a durable post-cost strategy.

Extended note. Version record

Dogu Araci, FinBERT: Financial Sentiment Analysis with Pre-trained Language Models

Difficulty: Research **Type:** Research paper

Status/version: arXiv preprint first posted 27 August 2019; no journal/proceedings status established from the inspected primary record.

Adapts BERT to financial sentiment when labeled data are scarce, providing a contextual baseline between Loughran–McDonald dictionaries and generative LLMs. Its benchmark gains concern sentiment classification on the evaluated datasets, not returns or incremental portfolio value. For a news alpha study, test target-entity labels, event and publication timestamps, domain shifts, calibration, and the cost of updating the model. Compare against the exact dictionary and base-model versions available at the research date.

Version record

Matthew Gentzkow, Bryan Kelly and Matt Taddy, "Text as Data"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Literature, 57(3), 535–574 (2019); DOI 10.1257/jel.20181020.

Survey of text representation, dictionary, topic and supervised methods for economics; the conceptual entry point before LLM features.

Version record

Zheng Tracy Ke, Bryan T. Kelly and Dacheng Xiu, "Predicting Returns with Text Data"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 3388293 (2019); DOI 10.2139/ssrn.3388293. Working-paper record; no journal version established in this check.

SESTM: supervised sentiment extraction screened for return-relevant words; transparent baseline against which LLM sentiment can be compared.

Extended note. Version record

Leland Bybee, Bryan Kelly, Asaf Manela and Dacheng Xiu, "Business News and Business Cycles"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 79(5), 3105–3147 (2024); DOI 10.1111/jofi.13377.

Topic model of Wall Street Journal text tracks economic attention and adds explanatory power for macro and asset-price fluctuations.

Version record

Justin Grimmer, Margaret E. Roberts and Brandon M. Stewart, *Text as Data: A New Framework for Machine Learning and the Social Sciences*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press, 2022 (ISBN 9780691207551).

Research-design framework for discovery, measurement, prediction and causal inference with text.

Version record

Tim Loughran and Bill McDonald, "Textual Analysis in Accounting and Finance: A Survey"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Accounting Research, 54(4), 1187–1230 (2016); DOI 10.1111/1475-679x.12123.

Survey of dictionary, readability and document-similarity methods in finance, with practical pitfalls in parsing filings.

Version record

Paul C. Tetlock, Maytal Saar-Tsechansky and Sofus Macskassy, "More Than Words: Quantifying Language to Measure Firms' Fundamentals"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 63(3), 1437–1467 (2008); DOI 10.1111/j.1540-6261.2008.01362.x.

Negative words in firm-specific news predict earnings and returns, with underreaction; a canonical news-text signal.

Version record

Narasimhan Jegadeesh and Di Wu, "Word power: A new approach for content analysis"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 110(3), 712–729 (2013); DOI 10.1016/j.jfineco.2013.08.018.

Estimates word weights from market reactions instead of fixed dictionaries, improving tone measures for 10-K text.

Version record

Lauren Cohen, Christopher Malloy and Quoc Nguyen, "Lazy Prices"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 75(3), 1371–1415 (2020); DOI 10.1111/jofi.12885.

Year-on-year changes in 10-K/10-Q language predict negative future returns; investors are slow to notice textual changes.

Version record

Gerard Hoberg and Gordon Phillips, "Text-Based Network Industries and Endogenous Product Differentiation"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Political Economy, 124(5), 1423–1465 (2016); DOI 10.1086/688176.

Product-description similarity defines time-varying text-based industries (TNIC); widely used for peers, competition and relative-value signals.

Version record

Scott R. Baker, Nicholas Bloom and Steven J. Davis, "Measuring Economic Policy Uncertainty"

Difficulty: Advanced **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 131(4), 1593–1636 (2016); DOI 10.1093/qje/qjw024.

Newspaper-based economic policy uncertainty index and its links to volatility, investment and employment.

Version record

Asaf Manela and Alan Moreira, "News implied volatility and disaster concerns"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 123(1), 137–162 (2017); DOI 10.1016/j.jfineco.2016.01.032.

Text of Wall Street Journal front pages reconstructs implied volatility back to 1890 (NVIX) and reveals disaster-concern premia.

Version record

Tarek A. Hassan, Stephan Hollander, Laurence van Lent and Ahmed Tahoun, "Firm-Level Political Risk: Measurement and Effects"

Difficulty: Advanced **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 134(4), 2135–2202 (2019); DOI 10.1093/qje/qjz021.

Earnings-call transcripts measure firm-level political risk; a template for building firm-level risk exposures from conversational text.

Version record

Diego Garcia, "Sentiment during Recessions"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 68(3), 1267–1300 (2013); DOI 10.1111/jofi.12027.

A century of New York Times financial columns: news sentiment predicts returns, mostly in recessions.

Version record

Feng Li, "The Information Content of Forward-Looking Statements in Corporate Filings–A Naive Bayesian Machine Learning Approach"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Accounting Research, 48(5), 1049–1102 (2010); DOI 10.1111/j.1475-679x.2010.00382.x.

Early supervised-learning classification of MD&A tone that predicts future earnings; a bridge from dictionaries to machine learning.

Version record

Lin William Cong, Tengyuan Liang, Xiao Zhang and Wu Zhu, "Textual Factors: A Scalable, Interpretable, and Data-Driven Approach to Analyzing Unstructured Information"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 71(12), 10727–10739 (2025); DOI 10.1287/mnsc.2020.01180.

Textual factors: word embeddings clustered by locality-sensitive hashing and refined by topic modelling into interpretable factors for downstream regressions. Applications include macro forecasting from news, interpreting factor models from filings and measuring technology breakthroughs from patents; a statistical package is distributed.

Version record

3.5.2 Language models for finance

Shijie Wu et al., BloombergGPT: A Large Language Model for Finance

Difficulty: Research **Type:** Research paper

Status/version: arXiv preprint first posted 30 March 2023; no peer-reviewed venue established from the inspected primary record.

Describes a finance-specific 50-billion-parameter language model trained on both proprietary financial text and general corpora, with evaluation on financial and general NLP tasks. It is useful for understanding domain pretraining, benchmark selection, and the limits of comparing proprietary text models with open baselines. Use a point-in-time model/data cutoff and explicit downstream economic tests before financial forecasting claims.

Extended note. Version record

Tian Guo and Emmanuel Hauptmann, Fine-Tuning Large Language Models for Stock Return Prediction Using Newsflow

Difficulty: Research **Type:** Research paper

Status/version: EMNLP 2024 Industry Track, 1028–1045, November 2024; Association for Computational Linguistics proceedings.

Compares encoder and decoder language-model embeddings fine-tuned from timestamped news into numerical return forecasts, then ranks stocks in long-only and long–short portfolios. The paper reports that aggregated token representations can beat sentiment-based rankings in its tested universes, with model-class differences across universe sizes. The retrieved paper text did not reveal an explicit transaction-cost or turnover analysis.

Extended note. Version record

Jens Ludwig, Sendhil Mullainathan, and Ashesh Rambachan, Large Language Models: An Applied Econometric Framework; NBER working-paper history

Difficulty: Not specified **Type:** Research paper

Status/version: Annual Review of Economics 18, 283–316, August 2026 volume; first published as Review in Advance 6 April 2026. Earlier NBER Working Paper 33344 issued January 2025 and revised December 2025.

Separates using an LLM to predict future outcomes from using it to measure a textual construct for downstream inference. For prediction, it requires no training leakage into the evaluated sample; for measurement, a human-labeled validation sample is needed to quantify model/prompt errors and produce defensible inference. Merely prompting a model to 'ignore future information' does not remove knowledge already learned during pretraining.

Extended note. Version record

Bryan T. Kelly, Semyon Malamud, Johannes Schwab and Teng Andrea Xu, Scaling Point-in-Time Language Models; Author revision record

Difficulty: Research **Type:** Research paper

Status/version: Author preprint dated 30 April 2026; NBER Working Paper 35247 issued May 2026. SSRN mirror of the NBER version reports revision 29 July 2026; the original author preprint has a different abstract, so cite the selected NBER version. No journal publication established.

Builds monthly language-model checkpoints trained only on text available by each cutoff from 2013–2024, with released pipeline. The current NBER abstract also describes an economic evaluation of portfolios built from time-valid text embeddings. This is a concrete protocol for avoiding retrospective LLM training contamination.

Extended note. Version record

Ties de Kok, ChatGPT for Textual Analysis? How to Use Generative LLMs in Accounting Research

Difficulty: Research **Type:** Research paper

Status/version: Management Science 71(9), 7888–7906, September 2025 issue; published online 13 January 2025.

Frames generative LLMs as measurement tools for accounting text and emphasizes construct validity, model choice, prompt design, and validation against alternative coding methods. The ability to code text at scale does not make the resulting variable causal or a profitable signal.

Extended note. Version record

Michael Krumdick, Rik Koncel-Kedziorski, Viet Dac Lai, Varshini Reddy, Charles Lovering, and Chris Tanner, BizBench: A Quantitative Reasoning Benchmark for Business and Finance

Difficulty: Research **Type:** Research paper

Status/version: ACL 2024 Long Papers, 8309–8332, August 2024; Association for Computational Linguistics proceedings.

Separates financial document reading, numerical extraction, concept knowledge, and program synthesis across eight quantitative tasks, exposing errors that a general chat benchmark can hide. Use it when selecting an LLM for audited financial-statement or analyst workflows and add firm-specific held-out documents with known answers. The paper's own limitations note possible dataset errors and incomplete coverage of finance, including private markets and languages beyond English.

Extended note. Version record

Alejandro Lopez-Lira and Yuehua Tang, "Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 184, 104335 (2026); DOI 10.1016/j.jfineco.2026.104335.

Evidence that LLM headline scores predict next-day returns, concentrated in small stocks and after bad news; now published in the Journal of Financial Economics (2026). Check model training cutoffs against sample dates.

Version record

Yifei Chen, Bryan T. Kelly and Dacheng Xiu, "Expected Returns and Large Language Models"

Difficulty: Research **Type:** Research paper

Status/version: Working paper circulated from 2022–2023; presented in the AEA 2026 programme; SSRN record not retrievable in this pass; no journal publication established.

Compares embeddings from many language models for return prediction across 16 international markets; larger contextual models help most outside the US.

Extended note. Version record

Alex G. Kim, Maximilian Muhn and Valeri V. Nikolaev, "Financial Statement Analysis with Large Language Models"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4835311 (2024); DOI 10.2139/ssrn.4835311. Working-paper record; no journal version established in this check.

Reports that an LLM given anonymised statements predicts earnings direction comparably to analysts and specialised models; anonymisation reduces but does not eliminate memorisation risk.

Extended note. Version record

Paul Glasserman and Caden Lin, "Assessing Look-Ahead Bias in Stock Return Predictions Generated by GPT Sentiment Analysis"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Financial Data Science, 6(1), 25–42 (2024); DOI 10.3905/jfds.2023.1.143.

Separates look-ahead from distraction effects by anonymising firm names; anonymised headlines can perform better in-sample.

Version record

Songrun He, Linying Lv, Asaf Manela and Jimmy Wu, "Chronologically Consistent Large Language Models"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2502.21206, first draft February 2025; revised July 2025 (v3); no journal publication established.

ChronoBERT and ChronoGPT are trained only on text available at each vintage; in a news-return task their Sharpe ratios are similar to a larger model, suggesting modest look-ahead in that setting. A follow-up instruction-tuned family appeared in October 2025.

Extended note. Version record

Suproteem K. Sarkar and Keyon Vafa, "Lookahead Bias in Pretrained Language Models"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4754678 (2024); DOI 10.2139/ssrn.4754678. Working-paper record; no journal version established in this check.

Shows that pretrained models can encode post-period information and proposes tests; complements Levy and the point-in-time models.

Extended note. Version record

Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer and Bertie Vidgen, "FinanceBench: A New Benchmark for Financial Question Answering"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2311.11944, November 2023; open sample dataset.

Open-book question answering over public filings; documents retrieval and hallucination failures in document-grounded financial analysis.

Extended note. Version record

Zhiyu Chen et al., "FinQA: A Dataset of Numerical Reasoning over Financial Data"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, 3697–3711 (2021); DOI 10.18653/v1/2021.emnlp-main.300.

Expert-annotated numerical reasoning over earnings reports with executable programs; a benchmark for arithmetic reliability.

Version record

Andrea L. Eisfeldt, Gregor Schubert, Miao Ben Zhang and Bledi Taska, "Generative AI and Firm Values"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, jofi.70089 (2026); DOI 10.1111/jofi.70089.

Firm workforce exposure to generative AI: an "Artificial-Minus-Human" portfolio earned about 5% in the two weeks after ChatGPT's release, larger for firms with substitutable labour.

Version record

Jinfei Sheng, Zheng Sun, Baozhong Yang and Alan L. Zhang, "Generative AI and Asset Management"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag050 (2026); DOI 10.1093/rfs/hhag050.

Hedge funds that adopted generative AI after ChatGPT earned 2–4% higher annualised abnormal returns in a difference-in-differences design; non-hedge funds did not.

Version record

Jang Ho Kim, Jaekyun Park, Jungwoo Shin and Geumil Bae, "Enhancing Portfolio Construction with Correlation Estimates from Large Language Models"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial Data Science, 8(1), 31–47 (2026); DOI 10.3905/jfds.2025.1.209.

Feeds LLM-derived asset-relationship information into a modified Gerber-statistic correlation estimate for portfolio optimisation, reporting more efficient portfolios.

Version record

3.5.3 Agents and LLM strategy evaluation

Weixian Waylon Li, Hyeonjun Kim, Mihai Cucuringu and Tiejun Ma, "Can LLM-based Financial Investing Strategies Outperform the Market in Long Run?"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2505.07078, first posted May 2025; later revisions note KDD 2026 acceptance; code public.

FINSABER backtests LLM timing strategies over two decades and 100+ symbols with survivorship, look-ahead and snooping controls; reported advantages largely disappear and strategies mis-time bull and bear markets. Accepted to KDD 2026 per the authors.

Extended note. Version record

Ralph S. J. Koijen and Bradford Levy, "Assessing the Benefits of Optimized Agentic AI Systems for Asset Pricing"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 35431, July 2026; previously shortlisted on 12 October 2026; no journal publication established.

Real-time benchmark testing how well AI systems explain same-day stock returns around earnings announcements using only information available at release; optimised agentic systems raise explained variation from about 8% to nearly 20%. Explanation of contemporaneous returns is distinct from forecasting tradable returns.

Extended note. Version record

Yijia Xiao, Edward Sun, Di Luo and Wei Wang, "TradingAgents: Multi-Agents LLM Financial Trading Framework"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2412.20138, December 2024; open-source code.

Multi-agent analyst/trader/risk-manager architecture. Short evaluation windows and limited universes make it a design reference rather than performance evidence; contrast with FINSABER.

Extended note. Version record

Authors as listed on arXiv (StockBench), "StockBench: Can LLM Agents Trade Stocks Profitably In Real-world Markets?"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2510.02209, October 2025; revised March 2026.

Contamination-free benchmark (test period March–July 2025, updated) with daily decisions from prices, fundamentals and news; most frontier LLM agents fail to beat equal-weight buy-and-hold.

Extended note. Version record

Authors as listed on arXiv (LiveTradeBench, UIUC), "LiveTradeBench: Seeking Real-World Alpha with Large Language Models"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2511.03628, November 2025.

Live streaming evaluation of 21 LLMs over 50 days allocating across US stocks and Polymarket contracts; general benchmark scores do not predict trading outcomes.

Extended note. Version record

3.6 Generative models and scenarios

Subject context and qualifications.

Jinsung Yoon, Daniel Jarrett and Mihaela van der Schaar, "Time-series Generative Adversarial Networks"

Difficulty: Research **Type:** Research paper

Status/version: 2019

TimeGAN combines supervised temporal dynamics with adversarial distribution matching. It is a useful baseline for synthetic market paths, but matching marginal distributions and discriminator scores is insufficient without tail, dependence and downstream-policy tests.

Applications: Multi-asset scenarios, simulator training. Version record

Tianlin Xu et al., "COT-GAN: Generating Sequential Data via Causal Optimal Transport"

Difficulty: Research **Type:** Research paper

Status/version: 2020

Uses causal optimal-transport distances designed for sequential data. The method is attractive for financial paths because it evaluates conditional temporal structure rather than treating observations as exchangeable samples.

Applications: Scenarios, volatility and market simulation. Version record

Hao Ni, Lukasz Szpruch, Marc Sabate-Vidales, Baoren Xiao, Magnus Wiese and Shujian Liao, "Sig-Wasserstein GANs for Time Series Generation"; preprint

Difficulty: Research **Type:** Research paper

Status/version: 2021

ICAIF 2021 paper using expected path signatures to compare generated and observed time-series laws. Relevant to path-dependent risk and hedging; signature matching requires tests of tails, dependence and downstream performance. Distinct from the conditional extension below.

Applications: Multi-asset paths, derivatives, risk. Version record

Shujian Liao, Hao Ni, Marc Sabate-Vidales, Lukasz Szpruch, Magnus Wiese and Baoren Xiao, "Sig-Wasserstein GANs for conditional time series generation"; preprint

Difficulty: Research **Type:** Research paper

Status/version: 2024

Conditional path generation using signature-based distribution comparisons. *Mathematical Finance* 34(2), 622–670 (2024; online 2023); preprint first posted 2020, revised October 2023. Complements the ICAIF paper for conditioning scenarios on observed histories.

Applications: Conditional scenarios, derivatives, risk. Version record

Kashif Rasul et al., "Autoregressive Denoising Diffusion Models for Multivariate Probabilistic Time Series Forecasting"

Difficulty: Research **Type:** Research paper

Status/version: 2021

TimeGrad introduced diffusion-style probabilistic modelling for multivariate forecasts. It provides a foundation for conditional scenario generation, though high-dimensional cross-asset tails and long-horizon dependence remain demanding.

Applications: Multi-asset risk, macro and demand forecasting. Version record

Tomonori Takahashi and Takayuki Mizuno, "Generation of synthetic financial time series by diffusion models"; 2024 arXiv preprint

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 25(10), 1507–1516 (2025; online 5 August 2025); DOI 10.1080/14697688.2025.2528697. Originally a 2024 arXiv working paper.

Applies diffusion models specifically to synthetic financial returns and examines stylised facts. It is a useful finance-native benchmark, but the decisive test is whether generated paths improve stress, hedging or policy learning on held-out real regimes.

Applications: Synthetic returns, stress testing. Version record

Gihun Kim, Sun-Yong Choi and Yeoneung Kim, "A diffusion-based generative model for financial time series via geometric Brownian motion"

Difficulty: Research **Type:** Research paper

Status/version: Preprint v1, 25 July 2025

Incorporates price-proportional noise into a score-based diffusion process and reports improved reproduction of heavy tails, volatility clustering and leverage effects. The author lists it as submitted; it does not replace historically grounded scenario and extreme-value models. [33]

Applications: Equity scenarios, risk simulation. Version record

Giuseppe Masi, Andrea Coletta and Novella Bartolini, "High-Quality Synthetic Financial Time-Series using a GAN-Diffusion Framework"

Difficulty: Research **Type:** Research paper

Status/version: Preprint v1, 26 May 2026

CoMeTS-GAN jointly generates correlated stocks' mid-prices and volumes; its critic guides diffusion sampling to improve cross-stock correlations. This preprint informs multi-asset simulator design but requires independent validation under stress and downstream execution tasks. [34]

Applications: Equities; multivariate simulation, execution research. Version record

Magnus Wiese, Robert Knobloch, Ralf Korn and Peter Kretschmer, "Quant GANs: deep generation of financial time series"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 20(9), 1419–1440 (2020); DOI 10.1080/14697688.2020.1730426.

Temporal-convolution GAN reproducing heavy tails and volatility clustering in equity index returns; benchmark for scenario generators.

Version record

Hans Buehler, Blanka Horvath, Terry Lyons, Imanol Perez Arribas and Ben Wood, "A Data-driven Market Simulator for Small Data Environments"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2006.14498, June 2020.

Signature-based conditional VAE for generating realistic paths from little data; motivated by deep-hedging training.

Extended note. Version record

Terry Lyons and Andrew D. McLeod, "Signature Methods in Machine Learning"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2206.14674, June 2022 (later revisions).

Survey of path signatures, kernels and expected signatures for sequential data; theory behind signature trading, simulators and volatility models elsewhere in the catalogue.

Extended note. Version record

Ilya Chevyrev and Andrey Kormilitzin, "A Primer on the Signature Method in Machine Learning"

Difficulty: Advanced **Type:** Research paper

Status/version: arXiv 1603.03788, March 2016.

Accessible introduction to path signatures and feature extraction with worked examples.

Extended note. Version record

Adil Rengim Cetingoz and Charles-Albert Lehalle, "Synthetic data for portfolios: a throw of the dice will never abolish chance"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(8), 1311–1342 (2026); DOI 10.1080/14697688.2026.2673155.

Assesses when generative synthetic data help portfolio decisions and why adoption lags interest; a validation lens on generated scenarios.

Version record

Beatrice Acciaio, Stephan Eckstein and Songyan Hou, "Time-Causal VAE: Robust Financial Time Series Generator"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 17(1), 245–279 (2026); DOI 10.1137/24m1711650.

Time-causal VAE for synthetic financial time series whose loss bounds the causal Wasserstein distance, so generated paths preserve values of dynamic optimisation problems.

Version record

David Hirnschall, Paul Krühner and Kurt Hornik, "The log-signature-based time series Wasserstein generative adversarial network"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Computational Finance (2026); DOI 10.21314/jcf.2026.003.

A Wasserstein GAN using log-signature features to generate financial time series.

Extended note. Version record

3.7 Reinforcement learning and control in finance

Reinforcement learning applies to execution, market making, hedging and allocation only when the simulator, reward and evaluation reflect the market. Read the control foundations and the simulation-validation readings alongside these surveys.

Sutton and Barto, *Reinforcement Learning*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Value functions, policies, exploration and sequential decisions; a foundation for execution and hedging applications.

Version record

Ben Hambly, Renyuan Xu and Huining Yang, "Recent advances in reinforcement learning in finance"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 33(3), 437–503 (2023); DOI 10.1111/mafi.12382.

Survey of RL for execution, market making, portfolio choice, hedging and robo-advice, with links to stochastic control.

Version record

Petter N. Kolm and Gordon Ritter, "Modern Perspectives on Reinforcement Learning in Finance"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper 3449401 (2019); DOI 10.2139/ssrn.3449401. Working-paper record; no journal version established in this check.

Short practitioner-oriented review mapping trading problems to Markov decision processes and model-free RL.

Extended note. Version record

Haoran Wang, Thaleia Zariphopoulou and Xun Yu Zhou, "Reinforcement Learning in Continuous Time and Space: A Stochastic Control Approach"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Machine Learning Research 21(198), 1–34 (2020).

Entropy-regularised exploratory control with Gaussian optimal policies for linear-quadratic problems; continuous-time foundation for RL portfolio choice.

Version record

Arthur Charpentier, Romuald Élie and Carenini Remlinger, "Reinforcement Learning in Economics and Finance"

Difficulty: Advanced **Type:** Research paper

Status/version: Computational Economics, 62(1), 425–462 (2023); DOI 10.1007/s10614-021-10119-4.

Survey connecting RL to dynamic programming, bandits, games and financial applications.

Version record

John Moody and Matthew Saffell, "Learning to trade via direct reinforcement"

Difficulty: Advanced **Type:** Research paper

Status/version: IEEE Transactions on Neural Networks, 12(4), 875–889 (2001); DOI 10.1109/72.935097.

Recurrent direct reinforcement optimising differential Sharpe ratio with costs; early trading RL template.

Version record

Igor Halperin, "QLBS: Q-Learner in the Black-Scholes(-Merton) Worlds"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 3087076 (2017); DOI 10.2139/ssrn.3087076. Working-paper record; no journal version established in this check.

Discrete-time Q-learning formulation of option pricing and hedging that recovers Black–Scholes in the limit.

Extended note. Version record

Petter N. Kolm and Gordon Ritter, "Dynamic Replication and Hedging: A Reinforcement Learning Approach"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Financial Data Science, 1(1), 159–171 (2019); DOI 10.3905/jfds.2019.1.1.159.

RL hedging of options with transaction costs and a mean-variance reward.

Version record

Brian Ning, Franco Ho Ting Lin and Sebastian Jaimungal, "Double Deep Q-Learning for Optimal Execution"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 28(4), 361–380 (2021); DOI 10.1080/1350486x.2022.2077783.

Model-free DDQN execution compared with TWAP on market data; illustrates benchmark and simulator choices.

Version record

Xiao-Yang Liu et al., "FinRL: Deep Reinforcement Learning Framework to Automate Trading in Quantitative Finance"

Difficulty: Advanced **Type:** Research paper

Status/version: Proceedings of the Second ACM International Conference on AI in Finance, 1–9 (2021); DOI 10.1145/3490354.3494366.

Open-source RL environments and baselines for trading; useful for experiments, not evidence of tradable performance.

Version record

Xuefeng Gao, Lingfei Li and Xun Yu Zhou, "Reinforcement Learning for Jump-Diffusions, With Financial Applications"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, mafi.70027 (2026); DOI 10.1111/mafi.70027.

Continuous-time entropy-regularised RL for jump-diffusion control problems with financial applications.

Version record

Patrick Cheridito and Moritz Weiss, "Reinforcement learning for trade execution with market and limit orders"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(6), 833–853 (2026); DOI 10.1080/14697688.2026.2631116.

RL framework allocating market and limit orders for optimal execution in a limit order book.

Version record

Enrique Nicolas Corimayo and Enrique Gabriel Baquela, "Applications of Reinforcement Learning in Derivatives Pricing: A Systematic Literature Review"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial Data Science, 7(3), 44–85 (2025); DOI 10.3905/jfds.2025.1.197.

PRISMA-style systematic review of reinforcement learning for derivatives pricing and hedging.

Version record

Ayush Jha, "Reinforcement Learning for Strategic Asset Allocation: Why the Objective Function Dominates the Agent Architecture"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial Data Science, jfds.2026.012 (2026); DOI 10.3905/jfds.2026.012.

Decomposes RL strategic allocation into agent, objective and rebalancing frequency on a 12-ETF 60/40 universe; the objective function dominates performance, not the RL algorithm.

Version record

Alessio Brini and Petter N. Kolm, "On Deep Reinforcement Learning for Dynamic Trading with PPO: Challenges and Future Directions"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial Data Science, 8(2), 31–52 (2026); DOI 10.3905/jfds.2026.001.

Benchmarks PPO against a closed-form optimal trading policy with time-varying alpha and impact; how the continuous action space is clipped and rescaled decides convergence and bias.

Version record

Boyu Wang, Xuefeng Gao and Lingfei Li, "Reinforcement learning for continuous-time optimal execution: actor–critic algorithm and error analysis"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 30(2), 597–655 (2026); DOI 10.1007/s00780-026-00589-5.

Entropy-regularised continuous-time actor–critic RL for Almgren–Chriss execution with closed-form Gaussian policies, a recalibration step and finite-time error bounds.

Version record

Chapter 4

Causal inference and experimental design

Start with the estimand and identification assumptions. Platform interference, adaptive policies and trading feedback require more than a conventional independent-unit experiment.

4.1 Identification and causal learning

Subject context and qualifications.

4.1.1 Research designs: DiD, synthetic control, RD and IV

Guido Imbens and Donald Rubin, ***Causal Inference for Statistics, Social, and Biomedical Sciences***

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2015

The rigorous foundation for potential outcomes, randomisation, compliance, confounding, heterogeneous treatment effects and inference. Understand why controlled interventions identify effects and what assumptions replace randomisation when it is unavailable; adjustment alone does not eliminate latent confounding. Read before causal forests or double machine learning on financial policies, client pricing and execution.

Applications: Execution, pricing, client experiments, policy research. Version record

Angrist and Pischke, ***Mostly Harmless Econometrics***

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Design-based identification, instruments and empirical causal reasoning; assess assumptions before interpreting predictive sensitivity as a treatment effect.

Version record

Scott Cunningham, ***Causal Inference: The Mixtape***

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Yale University Press, 2021; free online edition.

Accessible design-based text (Yale University Press 2021; free online) on DAGs, matching, RD, IV, panel, DiD and synthetic control with Stata/R code.

Version record

Nick Huntington-Klein, ***The Effect: An Introduction to Research Design and Causality***

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Chapman & Hall/CRC, second edition, 9 July 2025 (ISBN 9781032580227); free online edition; first edition 2021.

Research-design-first causal inference text; the second edition (2025) adds partial identification and updates methods, with R, Stata and Python code.

Version record

Alberto Abadie, "Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Literature, 59(2), 391–425 (2021); DOI 10.1257/jel.20191450.

Guidance on when synthetic controls are credible: pre-treatment fit, donor pools and inference.

Version record

Brantly Callaway and Pedro H. C. Sant'Anna, "Difference-in-Differences with multiple time periods"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 225(2), 200–230 (2021); DOI 10.1016/j.jeconom.2020.12.001.

Group-time average treatment effects for staggered adoption; avoids the negative-weight problem of two-way fixed effects.

Version record

Andrew Goodman-Bacon, "Difference-in-differences with variation in treatment timing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 225(2), 254–277 (2021); DOI 10.1016/j.jeconom.2021.03.014.

Decomposes the two-way fixed-effects DiD estimator into 2x2 comparisons and shows why staggered timing biases it.

Version record

Jonathan Roth, Pedro H. C. Sant'Anna, Alyssa Bilinski and John Poe, "What's trending in difference-in-differences? A synthesis of the recent econometrics literature"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Econometrics, 235(2), 2218–2244 (2023); DOI 10.1016/j.jeconom.2023.03.008.

Synthesis of staggered timing, parallel-trends violations, sensitivity analysis and inference; practical checklist for event studies of regulation.

Version record

Dmitry Arkhangelsky, Susan Athey, David A. Hirshberg, Guido W. Imbens and Stefan Wager, "Synthetic Difference-in-Differences"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 111(12), 4088–4118 (2021); DOI 10.1257/aer.20190159.

Combines unit and time weights from synthetic control with DiD; robust when parallel trends hold only approximately.

Version record

Matias D. Cattaneo, Nicolás Idrobo and Rocío Titiunik, *A Practical Introduction to Regression Discontinuity Designs: Foundations*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2019); DOI 10.1017/9781108684606.

Cambridge Element on sharp RD estimation, bandwidth choice, robust inference and falsification, with software.

Version record

Michael R. Roberts and Toni M. Whited, "Endogeneity in Empirical Corporate Finance"

Difficulty: Advanced **Type:** Research paper

Status/version: Handbook of the Economics of Finance, 493–572 (2013); DOI 10.1016/b978-0-44-453594-8.00007-0.

Handbook chapter on IV, DiD, RD and matching in corporate-finance data; finance-specific sources of endogeneity.

Version record

Judea Pearl, *Causality: Models, Reasoning, and Inference*

Difficulty: Research **Type:** Book / monograph

Status/version: Cambridge University Press, edition 2 (2009); DOI 10.1017/cbo9780511803161.

Structural causal models, do-calculus and backdoor/frontdoor criteria; foundation for graph-based identification.

Version record

Stefan Wager, *Causal Inference: A Statistical Learning Approach*

Difficulty: Research **Type:** Book / monograph

Status/version: Author-hosted draft (Stanford); no published edition established in this check.

Graduate text on potential outcomes, doubly robust estimation, heterogeneous effects and policy learning; check the publisher record for the final edition date.

Extended note. Version record

Kjell G. Nyborg and Jiri Woschitz, "Robust difference-in-differences analysis when there is a term structure"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 170, 104081 (2025); DOI 10.1016/j.jfineco.2025.104081.

Standard DiD is predisposed to misspecification for variables with a term structure (yields, spreads); proposes robust specifications for maturity heterogeneity.

Version record

Jonathan B. Cohn, Travis L. Johnson, Zack Liu and Malcolm I. Wardlaw, "Past is prologue: Inference from the cross section of returns around an event"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 180, 104278 (2026); DOI 10.1016/j.jfineco.2026.104278.

Studies inference from the cross-section of returns around events, relevant to event-study design.

Extended note. Version record

Susan Athey and Guido W. Imbens, "The State of Applied Econometrics: Causality and Policy Evaluation"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 31(2), 3–32 (2017); DOI 10.1257/jep.31.2.3.

Overview of modern identification strategies (RD, synthetic control, DiD) and supplementary analyses for robustness.

Version record

Liyang Sun and Sarah Abraham, "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 225(2), 175–199 (2021); DOI 10.1016/j.jeconom.2020.09.006.

Shows two-way fixed-effects event-study leads and lags are contaminated under staggered adoption and proposes an interaction-weighted estimator.

Version record

Clément de Chaisemartin and Xavier D'Haultfœuille, "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 110(9), 2964–2996 (2020); DOI 10.1257/aer.20181169.

TWFE estimates can put negative weights on some treatment effects; proposes a robust alternative estimator.

Version record

Andrew C. Baker, David F. Larcker and Charles C. Y. Wang, "How much should we trust staggered difference-in-differences estimates?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 144(2), 370–395 (2022); DOI 10.1016/j.jfineco.2022.01.004.

Finance-specific demonstration that staggered DiD designs common in corporate finance can produce wrong-signed estimates, with recommended fixes.

Version record

4.1.2 Causal ML, policy learning and platform experiments

Susan Athey and Stefan Wager, Policy Learning With Observational Data

Difficulty: Research **Type:** Research paper

Status/version: Econometrica 89(1), 133–161, January 2021; first published online 15 January 2021.

Uses doubly robust value estimates to learn constrained treatment-assignment rules from observational data when an identification strategy is available. Separate estimation of heterogeneous effects from the choice of action under costs, budget, and capacity constraints. Historical trade or quote logs need overlap, defensible treatment assignment assumptions, and an interference assessment before the learned policy's value can be interpreted causally.

Extended note. Version record

Victor Chernozhukov et al., "Double/Debiased Machine Learning for Treatment and Structural Parameters"

Difficulty: Research **Type:** Research paper

Status/version: 2018

Uses orthogonal scores and cross-fitting so that flexible nuisance models can estimate confounding relationships without dominating inference on the target treatment effect. It is valuable for high-dimensional policy evaluation, provided the identification assumptions are economically defensible.

Applications: Execution, market design, macro and corporate finance. Version record

Victor Chernozhukov, Christian Hansen, Nathan Kallus, Martin Spindler and Vasilis Syrgkanis, *Applied Causal Inference Powered by ML and AI*; Fixed original arXiv version

Difficulty: Advanced **Type:** Book / online manuscript

Status/version: Online book; original arXiv posting 4 March 2024. Author website reports an update on 25 September 2026. That website date is not an independently verified PDF revision or a new print edition; arXiv still identifies v1 from 2024.

A worked learning bridge from causal identification to double/debiased machine learning, with author-provided Python and R exercises. Use it to formulate a treatment, outcome and information set before estimating financial policy or execution effects. It supplements the original DML paper with a coherent textbook sequence; predictive accuracy and flexible controls alone do not establish causality.

Extended note. Version record

Victor Chernozhukov, Carlos Cinelli, Whitney K. Newey, Amit Sharma and Vasilis Syrgkanis, Long Story Short: Omitted Variable Bias in Causal Machine Learning; Journal DOI

Difficulty: Research **Type:** Research paper

Status/version: The Review of Economics and Statistics, 2026, as identified by the authors' arXiv journal reference; original preprint 26 December 2021. Selected arXiv v6, 21 September 2026. Author's May 2026 MIT CV describes the journal article as online early; issue, pagination and full-text equivalence not independently established.

Adds sensitivity analysis for omitted confounders to modern causal ML. Bounds on omitted-variable bias translate assumptions about unobserved explanatory power into uncertainty about treatment, derivative and policy effects. Use after DML estimation to ask how strong missing information would have to be to change an execution or allocation-policy conclusion.

Extended note. Version record

Susan Athey, Julie Tibshirani and Stefan Wager, "Generalized Random Forests"

Difficulty: Research **Type:** Research paper

Status/version: 2019

Extends forests to local moment estimation, including heterogeneous treatment effects and instrumental-variable settings. In institutional finance it can identify which clients, instruments or liquidity states respond differently to an intervention.

Applications: Pricing, execution, client routing, heterogeneous effects. Version record

Kevin Webster and Nicholas Westray, "Getting More for Less: Better A/B Testing via Causal Regularization"

Difficulty: Research **Type:** Research paper

Status/version: 2022 working paper; 2023 article

Combines plentiful observational trading data with scarce interventional data to address alpha/impact prediction bias. A realistic order-simulator study illustrates the bias/variance and experiment-cost trade-off. Randomisation supports causal comparisons; it does not by itself make every fitted impact model unbiased. Read the 2022 working paper alongside the October 2023 Risk article; lending-pricing applicability is an extension, not a reported deployment.

Applications: Execution algorithms, market impact, A/B testing. Version record

Ramesh Johari, Hannah Li, Inessa Liskovich and Gabriel Y. Weintraub, "Experimental Design in Two-Sided Platforms: An Analysis of Bias"; preprint

Difficulty: Research **Type:** Research paper

Status/version: 2022

Compares customer-side, listing-side and two-sided randomisation in a marketplace model; bias depends on supply/demand balance. One participant's treatment can change availability for others, motivating lending, internal-crossing, OTC-quoting and AMM applications. Distinguish two-sided assignment from stratification and examine residual spillovers; a chosen grouping does not guarantee identification of a full-rollout effect.

Applications: Two-sided markets, pricing, liquidity. Version record

Yuhang Wu, Zeyu Zheng, Guangyu Zhang, Zuohua Zhang and Chu Wang, "Non-Stationary A/B Tests"

Difficulty: Research **Type:** Research paper

Status/version: 2022

Time as a continuous covariate, temporal post-stratification and time-grouped randomisation for non-stationary treatment effects. CUPED and switchbacks are distinct methods in the lending bibliography.

Applications: Execution, quoting, client pricing. Version record

Emanuele Olivetti, Vincent Zoonekynd, Marcos López de Prado, Guido W. Imbens and Miguel A. Hernán, "Evidence from ADIA Lab's Causal Discovery Challenge"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial Data Science, 8(3), 61–76 (2026); DOI 10.3905/jfds.2026.010.

Results of the ADIA Lab causal discovery challenge (47,000 synthetic datasets, 3,343 submissions): supervised learning beat constraint-based causal discovery at classifying confounders, mediators and colliders.

Version record

Stefan Wager and Susan Athey, "Estimation and Inference of Heterogeneous Treatment Effects using Random Forests"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the American Statistical Association, 113(523), 1228–1242 (2018); DOI 10.1080/01621459.2017.1319839.

Causal forests with asymptotic normality and valid confidence intervals for heterogeneous treatment effects.

Version record

Sören R. Künzel, Jasjeet S. Sekhon, Peter J. Bickel and Bin Yu, "Metalearners for estimating heterogeneous treatment effects using machine learning"

Difficulty: Advanced **Type:** Research paper

Status/version: Proceedings of the National Academy of Sciences, 116(10), 4156–4165 (2019); DOI 10.1073/pnas.1804597116.

S-, T- and X-learners for conditional treatment effects with any ML base learner; a practical toolkit for uplift-style analyses.

Version record

X. Nie and S. Wager, "Quasi-oracle estimation of heterogeneous treatment effects"

Difficulty: Research **Type:** Research paper

Status/version: Biometrika, 108(2), 299–319 (2021); DOI 10.1093/biomet/asaa076.

The R-learner: residual-on-residual loss that makes heterogeneous-effect estimation robust to nuisance-model errors.

Version record

Susan Athey and Guido W. Imbens, "Machine Learning Methods That Economists Should Know About"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Economics, 11(1), 685–725 (2019); DOI 10.1146/annurev-economics-080217-053433.

Survey of supervised learning, causal ML, bandits and matrix completion for empirical economists.

Version record

A. Belloni, V. Chernozhukov and C. Hansen, "Inference on Treatment Effects after Selection among High-Dimensional Controls"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Economic Studies, 81(2), 608–650 (2014); DOI 10.1093/restud/rdt044.

Post-double-selection lasso: select controls that predict either treatment or outcome to keep inference valid with many candidate controls.

Version record

Jonas Peters, Dominik Janzing and Bernhard Schölkopf, *Elements of Causal Inference: Foundations and Learning Algorithms*

Difficulty: Advanced **Type:** Book / monograph

Status/version: MIT Press, 2017 (ISBN 9780262037310); open-access PDF from the publisher.

Structural causal models, identifiability and causal discovery from data, with an open-access PDF; background for causal factor-investing arguments.

Version record

Peter Spirtes, Clark Glymour and Richard Scheines, *Causation, Prediction, and Search*

Difficulty: Research **Type:** Book / monograph

Status/version: The MIT Press, edition 2 (2001); DOI 10.7551/mitpress/1754.001.0001.

Graphical causal discovery (PC and related algorithms) under faithfulness; the constraint-based methods benchmarked in the ADIA causal challenge.

Version record

4.2 Experimental design and evaluation

Subject context and qualifications.

Douglas C. Montgomery, *Design and Analysis of Experiments*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Wiley, 10th ed., 2019

Randomisation, blocking, factorial designs, power and experimental planning; the tenth edition (2019) is the selected reference.

Version record

Alan S. Gerber and Donald P. Green, *Field Experiments: Design, Analysis, and Interpretation*

Difficulty: Advanced **Type:** Book / monograph

Status/version: W. W. Norton, 2012

Field randomisation, clustered designs, stratification, compliance, ethics and treatment-effect analysis in operational settings. Translate these questions to clients and securities whose behaviour and interactions complicate controlled pricing tests. Use alongside Imbens–Rubin for potential outcomes, identification and the limits of adjustment when randomisation is unavailable.

Version record

Ron Kohavi, Diane Tang and Ya Xu, *Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, 2020

Practical experiment design, power, bias/variance, metrics, invariant checks, platform operation and common pitfalls. Adapt the overall evaluation criterion and guardrail approach to quote acceptance, lending revenue and risk. Pair early monitoring with the longitudinal sequential-testing reading below; technology-company experience must be adapted to lending interactions and delayed outcomes.

Version record

David Holtz, Felipe Lobel, Ruben Lobel, Inessa Liskovich and Sinan Aral, "Reducing Interference Bias in Online Marketplace Experiments Using Cluster Randomization: Evidence from a Pricing Meta-experiment on Airbnb"

Difficulty: Research **Type:** Research paper

Status/version: *Management Science* 71(1), 390–406; online 2024, issue 2025

Meta-experiment comparing individual and cluster randomisation for an Airbnb platform-fee increase. Relevant to spillovers among substitutes; arbitrary clusters need not remove interference.

Version record

Sascha Füllbrunn and Ernan Haruvy (eds.), *Handbook of Experimental Finance*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edward Elgar, 2022

Broader experimental-finance collection for comparing laboratory and field approaches, financial behaviour and finance-specific research questions. Read after the core pricing-experiment sequence to broaden methodological judgement; it is a contextual collection, not a securities-lending implementation manual.

Version record

Totte Harinen and Bonnie Li, "Using Causal Inference to Improve the Uber User Experience"

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Engineering article; 19 June 2019

Practitioner discussion of pre-experiment covariate adjustment/CUPED, incomplete treatment uptake, heterogeneous effects and observational alternatives. Connects methods to an internal experiment platform. Use pre-treatment information carefully; covariate adjustment cannot by itself repair arbitrary allocation failures or establish causality. The source's abbreviated title and 2020 date are corrected.

Version record

Sebastian Ankargren, Mattias Frånberg and Mårten Schultzberg, "Bringing Sequential Testing to Experiments with Longitudinal Data (Part 1): The Peeking Problem 2.0"

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Engineering article; 18 July 2023

Sequential-testing complications with repeated observations and longitudinal outcomes; ordinary repeated fixed-horizon p-values do not justify early stopping.

Version record

Peter M. Aronow and Cyrus Samii, "Estimating Average Causal Effects Under General Interference, with Application to a Social Network Experiment"

Difficulty: Not specified **Type:** Research paper

Status/version: *Annals of Applied Statistics* 11(4), 1912–1947, 2017

Design-based causal estimation under specified interference. Shared inventory or collateral can connect lending units; clustered standard errors alone do not repair misspecified exposures.

Version record

Iavor Bojinov, David Simchi-Levi and Jinglong Zhao, "Design and Analysis of Switchback Experiments"

Difficulty: Not specified **Type:** Research paper

Status/version: *Management Science* 69(7), 3759–3777; online 2022, issue 2023

Policy randomisation over time with explicit carryover assumptions; outstanding loans and borrower relationships can persist beyond a time block.

Version record

Alex Deng, Ya Xu, Ron Kohavi and Toby Walker, "Improving the Sensitivity of Online Controlled Experiments by Utilizing Pre-Experiment Data"

Difficulty: Not specified **Type:** Research paper

Status/version: WSDM 2013

Original pre-experiment-covariate adjustment reference. Precision gains depend on predictive covariates; post-treatment utilisation is not a valid pre-treatment adjustment.

Version record

Aleksander Fabijan et al., "Diagnosing Sample Ratio Mismatch in Online Controlled Experiments: A Taxonomy and Rules of Thumb for Practitioners"

Difficulty: Not specified **Type:** Research paper

Status/version: KDD 2019

Taxonomy for diagnosing assignment and recorded-sample mismatches. Accepted-loan counts can legitimately differ when acceptance is treatment-affected.

Version record

Steven R. Howard, Aaditya Ramdas, Jon McAuliffe and Jasjeet Sekhon, "Time-uniform, Nonparametric, Nonasymptotic Confidence Sequences"; author preprint

Difficulty: Advanced **Type:** Research paper

Status/version: *Annals of Statistics* 49(2), 1055–1080, 2021

Time-uniform uncertainty under stated process and moment assumptions; not valid for arbitrary dependence or heavy tails without suitable conditions.

Version record

Miguel A. Hernán and James M. Robins, *Causal Inference: What If*

Difficulty: Not specified **Type:** Book / monograph

Status/version: author's recommended citation 2020; living online text

Selection, longitudinal causal inference and time-varying interventions. Loan termination differs from administrative censoring; observation and identification assumptions remain necessary.

Version record

Susan Athey, Raj Chetty, Guido W. Imbens and Hyunseung Kang, "The Surrogate Index: Combining Short-Term Proxies to Estimate Long-Term Treatment Effects More Rapidly and Precisely"; NBER version

Difficulty: Advanced **Type:** Research paper

Status/version: *Review of Economic Studies* 93(4), 2284–2312; online 2025, issue 2026

Conditions for inferring delayed effects from short-term proxies and historical data. Early utilisation does not automatically identify lifetime income or retention.

Version record

Miroslav Dudík, John Langford and Lihong Li, "Doubly Robust Policy Evaluation and Learning"

Difficulty: Advanced **Type:** Research paper

Status/version: ICML 2011

Off-policy evaluation combining reward models and action propensities. Requires support and identification; cannot create evidence for unoffered prices or resolve shared-inventory effects.

Version record

Vitor Hadad, David A. Hirshberg, Ruohan Zhan, Stefan Wager and Susan Athey, "Confidence Intervals for Policy Evaluation in Adaptive Experiments"; open article

Difficulty: Advanced **Type:** Research paper

Status/version: *PNAS* 118(15), e2014602118, 2021

Inference with adaptive treatment probabilities under stated asymptotic assumptions. Reward maximisation and effect estimation remain different objectives.

Version record

Nicholas Larsen, Jonathan Stallrich, Srijan Sengupta, Alex Deng, Ron Kohavi and Nathaniel T. Stevens, "Statistical Challenges in Online Controlled Experiments: A Review of A/B Testing Methodology"

Difficulty: Advanced **Type:** Survey

Status/version: The American Statistician, 78(2), 135–149 (2024); DOI 10.1080/00031305.2023.2257237.

Review of the challenges that large-scale online controlled experiments raise and that need new statistical methods. Places current A/B-testing methods within their statistical lineages with illustrative applications, and discusses the practice and culture of online experimentation.

Version record

Ramesh Johari, Pete Koomen, Leonid Pekelis and David Walsh, "Always Valid Inference: Continuous Monitoring of A/B Tests"

Difficulty: Research **Type:** Research paper

Status/version: Operations Research, 70(3), 1806–1821 (2022); DOI 10.1287/opre.2021.2135.

Fixed-horizon p-values are invalid when users monitor a test continuously and stop when it looks significant. Defines always-valid p-values and confidence intervals that permit stopping at any time, trade off sample size and power efficiently, and extend to multiple-testing control; deployed on a commercial A/B platform for hundreds of thousands of experiments.

Version record

Chapter 5

Research validation and reproducibility

Read design, selection and replication together. A significant forecast, an implementable strategy and a reproducible experiment are different claims.

5.1 Research design and reporting

These resources address research design, empirical evaluation, reporting and research-to-production reliability. Controlled-experiment books appear in Section 4.2; software documentation appears in Section 21.3.

John Schulman, "An Opinionated Guide to ML Research" (source: 2020)

Difficulty: Research **Type:** Practitioner / commentary

Status/version: Selected citation year(s): 2020. Version distinctions remain in the annotation.

Research judgement and idea-driven versus goal-driven inquiry. Optional essays: Richard Hamming, You and Your Research; Michael Nielsen, Principles of Effective Research, Part I (8 July 2004), and the collected technical note 0404 (27 July 2004; hosted copy). The note is a later, broader text.

Applications: essay. Version record

Jackson Kamiri and Geoffrey Mariga, "Research Methods in Machine Learning: A Content Analysis"

Difficulty: Not specified **Type:** Research paper

Status/version: 10(2), 2021; methodological survey

Methodological content analysis of 100 IEEE-journal articles published since 2019; a bounded survey of research designs.

Version record

Official NeurIPS Paper Checklist; Joelle Pineau et al., *Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program)*

Difficulty: Not specified **Type:** Report / research note

Status/version: JMLR 22(164), 2021

Reporting of assumptions, baselines, hyperparameters, code/data access and limitations. The JMLR report concerns the 2019 programme; the 2021 checklist announcement provides historical context. A checklist is a disclosure aid, not proof of reproducibility.

Version record

Paul Cottrell, "Research Methods in Quantitative Finance"

Difficulty: Not specified **Type:** Research paper

Status/version: SSRN working-paper lead, source: 2016

Identified SSRN working paper on quantitative-finance research designs; full text was unavailable, so detailed scope remains unconfirmed.

Version record

Taylan Mavruk, *Quantitative Research Methods in Corporate Finance: Exemplified by Stata, Python, and R*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Cambridge University Press, 2025

Empirical corporate-finance research, panels, event studies and endogeneity, with Stata and accompanying Python/R code; published 20 March 2025.

Version record

Adrian R. Bell, Chris Brooks and Marcel Prokopczuk (eds.), *Handbook of Research Methods and Applications in Empirical Finance* (2013); Keith Cuthbertson and Dirk Nitzsche, *Quantitative Financial Economics: Stocks, Bonds and Foreign Exchange* (2nd ed., 2004)

Difficulty: Not specified **Type:** Book / monograph

Status/version: 2013 handbook; Quantitative Financial Economics, 2nd ed. (2004).

Complementary empirical-finance methods and financial-economics texts. The handbook first appeared in 2013 (2014 paperback); the textbook reading is the second edition (2004).

Version record

Randle Rabe, ***Best Research Practices for Your Quantitative Finance Research Group***

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Hudson & Thames, January 2022

Practitioner guidance on literature review, bias control, reproducibility, collaboration and research reporting; January 2022 edition.

Version record

Kevin J. Davey, ***Building Winning Algorithmic Trading Systems: A Trader's Journey From Data Mining to Monte Carlo Simulation to Live Trading*** **(2014)**

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2014. Version distinctions remain in the annotation.

Practitioner treatment of strategy evaluation, Monte Carlo analysis and feedback from live trading; outcomes are not guaranteed.

Applications: Practitioner book; live-feedback.

Version record

Rob Arnott, Campbell R. Harvey and Harry Markowitz, "A Backtesting Protocol in the Era of Machine Learning"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: The Journal of Financial Data Science, 1(1), 64–74 (2019); DOI 10.3905/jfds.2019.1.064.

Seven-point protocol covering research motivation, multiple testing, sample choice, cross-validation, model dynamics, complexity and research culture. A compact pre-registration checklist.

Version record

Campbell R. Harvey, "Presidential Address: The Scientific Outlook in Financial Economics"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 72(4), 1399–1440 (2017); DOI 10.1111/jofi.12530.

Explains p-hacking and publication incentives in finance and proposes a Bayesianised p-value (minimum Bayes factor) for reporting.

Version record

Sayash Kapoor and Arvind Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science"

Difficulty: Advanced **Type:** Research paper

Status/version: Patterns, 4(9), 100804 (2023); DOI 10.1016/j.patter.2023.100804.

Taxonomy of eight leakage types across 294 affected papers in 17 fields, with model-info sheets for preventing them; directly applicable to feature construction in return prediction.

Version record

Geir Kjetil Sandve, Anton Nekrutenko, James Taylor and Eivind Hovig, "Ten Simple Rules for Reproducible Computational Research"

Difficulty: Intro **Type:** Research paper

Status/version: PLoS Computational Biology, 9(10), e1003285 (2013); DOI 10.1371/journal.pcbi.1003285.

Short practical rules on recording provenance, versions, seeds and intermediate results.

Version record

David H. Bailey, Jonathan M. Borwein, Marcos López de Prado and Qiji Jim Zhu, "Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting on Out-of-Sample Performance"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: Notices of the American Mathematical Society, 61(5), 458 (2014); DOI 10.1090/noti1105.

Shows how quickly a high in-sample Sharpe ratio can be found by trying configurations, and estimates the minimum backtest length.

Version record

Howard Peter Henson and Jacques Francois Joubert, "The Evolution of Walk-Forward Backtesting"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(4), 44–65 (2026); DOI 10.3905/jpm.2025.1.802.

How to build a walk-forward backtester: the professional systematic workflow, then vectorised, loop-based and event-driven architectures, ending with a functional-reactive design. Engineering complement to the statistical backtesting readings.

Version record

5.2 Selection, forecast comparison and uncertainty

These extensions complement White, Hansen, deflated Sharpe ratios, causal identification and purged validation. Match the guarantee to the sampling process, dependence assumptions and decision being made; uncertainty intervals do not directly validate a trading rule.

5.2.1 Selection bias, multiple testing and look-ahead

David H. Bailey et al., "The Probability of Backtest Overfitting"

Difficulty: Research **Type:** Research paper

Status/version: David H. Bailey, Jonathan Borwein, Marcos Lopez de Prado and Qiji Jim Zhu; 2017 journal article, with the earlier 2014–15 working manuscript retained as a distinct version.

Model-selection overfitting across backtest splits. Versions include 2014 working material, the February 2015 manuscript, and the 2017 journal article; dependence and split design matter.

Applications: All assets and strategies; validation. Version record

David H. Bailey and Marcos López de Prado, "The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality" (author copy)

Difficulty: Research **Type:** Research paper

Status/version: 2014

Adjusts apparent performance for selection bias, non-normal returns and the number of trials. It is a useful governance statistic, although its assumptions and definition of an independent trial must be examined rather than mechanically accepted.

Applications: All assets and strategies; model selection. Version record

Campbell Harvey, Yan Liu and Heqing Zhu, "...and the Cross-Section of Expected Returns"

Difficulty: Research **Type:** Research paper

Status/version: 2016

Recalibrates significance thresholds in response to the proliferation of published factors. Institutional researchers should treat it as a minimum standard for false-discovery awareness rather than as a universal t-statistic cutoff.

Applications: Equities, factors, alternative data. Version record

White (2000), "A Reality Check for Data Snooping"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2000. Version distinctions remain in the annotation.

Model-search bias and benchmark superiority. *Econometrica* 68, 1097–1126 (2000).

Version record

Hansen (2005), "A Test for Superior Predictive Ability"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2005. Version distinctions remain in the annotation.

A test for superior predictive ability when comparing many candidate models. *Journal of Business & Economic Statistics* (2005).

Version record

Sullivan, Timmermann and White, *Data-Snooping, Technical Trading Rule Performance, and the Bootstrap* (1999)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 1999. Version distinctions remain in the annotation.

Trading-rule data snooping and bootstrap evaluation. *Journal of Finance* 54, 1647–1691 (1999).

Version record

Campbell R. Harvey and Yan Liu, "Backtesting'' (2015)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2015. Version distinctions remain in the annotation.

Selection bias from testing many strategies, adjusted Sharpe ratios and return hurdles; author-hosted code accompanies the paper.

Version record

Campbell R. Harvey and Yan Liu, "Evaluating Trading Strategies", *Journal of Portfolio Management* 40(5), 108–118 (2014)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2014. Version distinctions remain in the annotation.

Multiple testing and the inflation of apparent strategy significance; distinct from the authors' 2015 Backtesting paper.

Version record

Stefano Giglio, Yuan Liao, and Dacheng Xiu, Thousands of Alpha Tests

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 34(7), 3456–3496, July 2021; published online 24 September 2020.

Constructs a multiple-testing procedure for large families of alpha tests in linear asset-pricing models, allowing latent factors and missing data and using a wild bootstrap for finite-sample inference. Its asymptotic and bootstrap guarantees depend on the model and sampling assumptions; they do not solve hidden researcher degrees of freedom or execution costs.

Extended note. Version record

Albert J. Menkveld et al. (FINance Crowd Analysis Project), Nonstandard Errors

Difficulty: Research **Type:** Contract / specification / guidance

Status/version: The Journal of Finance 79(3), 2339–2390, June 2024; published online 17 April 2024.

Shows that analyst choices create an evidence-generation uncertainty beyond ordinary sampling error: 164 teams analyzed the same finance data and produced materially different estimates. Its peer-review stages reduced variation, suggesting practical value in challenge sessions. The experiment measures variation in a particular multi-team design, not a universal adjustment that can be added mechanically to any strategy's standard error.

Extended note. Version record

Bradford Levy, Caution Ahead: Numerical Reasoning and Look-Ahead Bias in AI Models

Difficulty: Research **Type:** Research paper

Status/version: Journal of Accounting Research 64(3), 1139–1188, 2026; first published online 6 May 2026. Open-access version of record.

Tests LLM arithmetic and financial-statement reasoning, then compares time-vintaged EDGAR-trained models and commercial-model outputs to diagnose memorization and post-training look-ahead. Offers operational validation checks, including tool-assisted calculations and truly forward evaluation.

Extended note. Version record

Yoav Benjamini and Yosef Hochberg, "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of the Royal Statistical Society Series B: Statistical Methodology, 57(1), 289–300 (1995); DOI 10.1111/j.2517-6161.1995.tb02031.x.

The Benjamini–Hochberg FDR procedure; standard alternative to family-wise control when screening many signals or funds.

Version record

Joseph P. Romano and Michael Wolf, "Stepwise Multiple Testing as Formalized Data Snooping"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 73(4), 1237–1282 (2005); DOI 10.1111/j.1468-0262.2005.00615.x.

Bootstrap stepdown procedure that improves on White Reality Check by identifying which strategies beat the benchmark while controlling family-wise error.

Version record

Campbell R. Harvey and Yan Liu, "Lucky factors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 141(2), 413–435 (2021); DOI 10.1016/j.jfineco.2021.04.014.

Bootstrap-based sequential factor selection that adjusts for the number of candidate factors tried.

Version record

Tarun Chordia, Amit Goyal and Alessio Saretto, "Anomalies and False Rejections"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(5), 2134–2179 (2020); DOI 10.1093/rfs/hhaa018.

Data-mines about two million accounting-based signals to calibrate multiple-testing thresholds; most survive conventional cut-offs but few survive adjusted ones.

Version record

Laurent Barras, Olivier Scaillet and Russ Wermers, "False Discoveries in Mutual Fund Performance: Measuring Luck in Estimated Alphas"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 65(1), 179–216 (2010); DOI 10.1111/j.1540-6261.2009.01527.x.

Estimates the proportions of skilled, unskilled and zero-alpha funds using the false-discovery rate; transferable to strategy screening.

Version record

Andrew Y. Chen and Tom Zimmermann, "Publication Bias in Asset Pricing Research"

Difficulty: Research **Type:** Research paper

Status/version: Oxford Research Encyclopedia of Economics and Finance (2023); DOI 10.1093/acrefore/9780190625979.013.888.

Estimates publication bias using the open-source anomaly data and finds that bias-adjusted returns remain mostly positive; a counter-weight to replication-crisis narratives.

Version record

Jefferson Duarte, Christopher S. Jones, Mehdi Khorram, Haitao Mo and Junbo L. Wang, "Too Good to Be True: Look-Ahead Bias in Empirical Options Research"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag061 (2026); DOI 10.1093/rfs/hhag061.

Many high-Sharpe options strategies in published research reflect look-ahead bias from using future information to filter noisy quotes; a direct validation warning for options backtests.

Version record

Nabil Bouamara, Sébastien Laurent and Shuping Shi, "A Stepwise Cauchy Combination Test for Multiple Testing Problems with Financial Applications"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 23(5), nbaf020 (2025); DOI 10.1093/jjfinec/nbaf020.

Stepwise Cauchy combination test for dependent test statistics and sparse signals, with strong family-wise error control and higher power than conventional procedures.

Version record

Joel M. Vanden, "Low Power of Alpha Tests When Factors Are Constructed From Sorted Portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 24(3), nbag013 (2026); DOI 10.1093/jjfinec/nbag013.

Alpha tests have low power when factors are built from the same sorted portfolios used as test assets.

Version record

Marcos López de Prado and Emilio Porcu, "The Deflated Sharpe Ratio: A Unified Framework for Search-Adjusted Performance Inference"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, jpm.2026.070 (2026); DOI 10.3905/jpm.2026.070.

Unifies the Deflated Sharpe Ratio variants (location-only DSR-L, location-and-dispersion DSR-LS and related forms) as search-adjusted significance tests that judge the winning strategy against what the search alone could have produced.

Version record

5.2.2 Sharpe-ratio and performance inference

Mikhail Chernov, Bryan T. Kelly, Semyon Malamud and Johannes Schwab, A Test of the Efficiency of a Given Portfolio in High Dimensions

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33565, issued March 2025; NBER revision September 2025. CEPR Discussion Paper 19999 published 5 March 2025 and revised 8 September 2025. No journal publication established.

Extends the Gibbons–Ross–Shanken portfolio-efficiency test to many more test assets than observations. Random-matrix corrections address bias caused by regularizing a singular or ill-conditioned return covariance matrix; simulations and factor-portfolio applications support the method.

Extended note. Version record

Andrew W. Lo, "The Statistics of Sharpe Ratios"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 58(4), 36–52 (2002); DOI 10.2469/faj.v58.n4.2453.

Standard errors of Sharpe ratios under IID and serially correlated returns, and the bias in annualising by square-root-of-time.

Version record

Oliver Ledoit and Michael Wolf, "Robust performance hypothesis testing with the Sharpe ratio"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 15(5), 850–859 (2008); DOI 10.1016/j.jempfin.2008.03.002.

HAC and studentised bootstrap tests for the difference between two Sharpe ratios; replaces the Jobson–Korkie test for fat-tailed, dependent returns.

Version record

David H. Bailey and Marcos López de Prado, "The Sharpe Ratio Efficient Frontier"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Risk, 15(2), 3–44 (2012); DOI 10.21314/jor.2012.255.

Probabilistic Sharpe ratio and minimum track-record length accounting for skewness and kurtosis.

Version record

Dimitris N. Politis and Joseph P. Romano, "The Stationary Bootstrap"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the American Statistical Association, 89(428), 1303–1313 (1994); DOI 10.1080/01621459.1994.10476870.

Random-block-length bootstrap preserving stationarity; standard resampling for dependent return series in Sharpe and Reality-Check tests.

Version record

Marcos López de Prado, Alexander Lipton and Vincent Zoonekynd, "Sharpe Ratio Inference: A New Standard for Decision Making and Reporting"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(6), 6–50 (2026); DOI 10.3905/jpm.2026.1.837.

Proposes a reporting standard for Sharpe-ratio inference built around five recurring errors: point estimates without significance, i.i.d.-Normal assumptions, ignoring power and minimum track-record length, reading p-values as posterior probabilities, and no multiple-testing correction. A checklist for any performance memo.

Version record

J. D. Jobson and Bob M. Korkie, "Performance Hypothesis Testing with the Sharpe and Treynor Measures"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 36(4), 889–908 (1981); DOI 10.1111/j.1540-6261.1981.tb04891.x.

The original test for the difference of two Sharpe ratios, later corrected by Memmel and made robust by Ledoit and Wolf.

Version record

J. D. Opdyke, "Comparing Sharpe ratios: So where are the p-values?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Asset Management, 8(5), 308–336 (2007); DOI 10.1057/palgrave.jam.2250084.

Asymptotic distribution of the Sharpe ratio and of Sharpe-ratio differences under non-Normal, serially correlated returns.

Version record

Steven E. Pav, *The Sharpe Ratio: Statistics and Applications*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Chapman and Hall/CRC, edition 1 (2021); DOI 10.1201/9781003181057.

Book-length treatment of Sharpe-ratio sampling distributions, tests, power, portfolio extensions and overfitting.

Version record

William Goetzmann, Jonathan Ingersoll, Matthew Spiegel and Ivo Welch, "Portfolio Performance Manipulation and Manipulation-proof Performance Measures"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 20(5), 1503–1546 (2007); DOI 10.1093/rfs/hhm025.

Shows Sharpe ratios and alphas can be gamed with option-like payoffs and derives a manipulation-proof performance measure.

Version record

5.2.3 Forecast comparison and predictive inference

Francis X. Diebold and Roberto S. Mariano, "Comparing Predictive Accuracy'', *Journal of Business & Economic Statistics* 13(3), 253–263 (1995)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected DOI-linked version; print 1995-07.

Paired forecast-loss comparisons with serial-dependence assumptions; an explicit reading choice for the original author/method lead.

Version record

Whitney K. Newey and Kenneth D. West, "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix'' (1986 technical working paper; 1987 journal article)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 1986, 1987. Version distinctions remain in the annotation.

HAC covariance estimation with a positive semi-definite construction; explicit reading choice for the original method lead.

Version record

Peter R. Hansen, Asger Lunde, and James M. Nason, The Model Confidence Set

Difficulty: Advanced **Type:** Research paper

Status/version: Econometrica 79(2), 453–497, March 2011 issue; first published online 14 February 2011.

Provides a set of forecasting models that can include the best model at a chosen confidence level, making uncertainty across close competitors explicit instead of crowning the smallest observed loss. This extends the catalogue's Diebold–Mariano pairwise test and Hansen's superior-predictive-ability test into a practical model-comparison workflow. The confidence set is conditional on the candidate family and evaluation sample; repeated tuning against it invalidates its interpretation.

Extended note. Version record

Isaac Gibbs and Emmanuel Candès, Adaptive Conformal Inference Under Distribution Shift

Difficulty: Research **Type:** Research paper

Status/version: NeurIPS 2021, Advances in Neural Information Processing Systems 34.

Adapts conformal prediction-set calibration online as the data-generating distribution shifts, with a long-run coverage-frequency guarantee under the paper's update scheme. It is useful for monitoring volatility, liquidity, or return-prediction intervals after a regime change when static holdout calibration decays. Long-run average coverage is not the same as conditional coverage during a crash, so risk limits need separate tail and path stress.

Extended note. Version record

Rina Foygel Barber and Ashwin Pananjady, Predictive inference for time series: why is split conformal effective despite temporal dependence?; January 2026 preprint revision

Difficulty: Research **Type:** Research paper

Status/version: ALT 2026, PMLR 313, 1–24; conference 23–26 February 2026. Original preprint 2 October 2025.

Examines why split-conformal intervals can work with temporally dependent data and predictors that use past observations. Coverage-loss bounds use a switch coefficient and are sharp over stationary beta-mixing processes. Read beside adaptive conformal inference to separate serial dependence from distribution shift; the results do not supply unconditional crisis coverage for arbitrary financial processes.

Extended note. Version record

Todd E. Clark and Kenneth D. West, "Approximately normal tests for equal predictive accuracy in nested models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 138(1), 291–311 (2007); DOI 10.1016/j.jeconom.2006.05.023.

Adjusts MSE comparisons for nested models, where the larger model is penalised by estimation noise; the standard test for return predictors against the historical mean.

Version record

Raffaella Giacomini and Halbert White, "Tests of Conditional Predictive Ability"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 74(6), 1545–1578 (2006); DOI 10.1111/j.1468-0262.2006.00718.x.

Tests forecasting methods (model plus estimation window) conditionally on information, allowing nested models and rolling estimation.

Version record

Sílvia Gonçalves, Michael W. McCracken and Yongxu Yao, "Bootstrapping out-of-sample predictability tests with real-time data"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 247, 105916 (2025); DOI 10.1016/j.jeconom.2024.105916.

Block bootstrap for West-type out-of-sample predictability tests when forecasts use revised real-time data.

Version record

Hyeong Jin Hyun and Xiao Wang, "Neural Conformal Inference for jump diffusion processes"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 251, 106061 (2025); DOI 10.1016/j.jeconom.2025.106061.

Neural conformal prediction for jump-diffusion processes.

Extended note. Version record

Giuseppe Buccheri, Roberto Renò and Giorgio Vocalelli, "Taking advantage of biased proxies for forecast evaluation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 251, 106068 (2025); DOI 10.1016/j.jeconom.2025.106068.

Optimally combining possibly biased proxies (for example realised-volatility measures) to rank forecasts reliably, preserving loss robustness and Diebold–Mariano testing.

Version record

5.3 Replication and implementability

Read the papers as a comparison of definitions, samples and statistical designs. Apparently conflicting replication findings can reflect weighting, microcaps, missing data and different null hypotheses; net implementability is a further question.

Kewei Hou, Chen Xue, and Lu Zhang, Replicating Anomalies

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 33(5), 2019–2133, May 2020; first published online 10 December 2018.

Rebuilds a library of 452 reported cross-sectional anomalies using common portfolio conventions, NYSE breakpoints, and value-weighted returns; most fail even a single-test hurdle under those choices. This is a diagnostic for microcap dependence, inconsistent original implementations, and the difference between a published return spread and an investable portfolio. Pair with Chen–Zimmermann and Jensen–Kelly–Pedersen, which apply different replication definitions and reach more favorable conclusions.

Extended note. Version record

Andrew Y. Chen and Tom Zimmermann, Open Source Cross-Sectional Asset Pricing; Author project code

Difficulty: Research **Type:** Research paper

Status/version: Critical Finance Review 11(2), 207–264, 2022; publisher record dated 3 May 2022.

Provides code and data for 319 stock-return characteristics and compares reproduced statistics with the original articles, finding high fidelity for characteristics originally presented as clearly significant. It is a practical reference for building a versioned factor library: retain the original definition, reporting lag, universe, breakpoints, and portfolio weighting before comparing a standardized variant. Reproduction of published statistics does not establish persistence, uniqueness, capacity, or net alpha.

Extended note. Version record

Theis Ingerslev Jensen, Bryan Kelly, and Lasse Heje Pedersen, Is There a Replication Crisis in Finance?; Project data and documentation

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 78(5), 2465–2518, October 2023 issue; first published online 26 May 2023.

Uses a hierarchical Bayesian model and a harmonized global factor dataset to argue that many published equity factors replicate, cluster into a smaller number of themes, and retain evidence outside the original U.S. samples. The global dataset is valuable for external-validity tests, but a replicated gross factor premium is not an institution-specific tradable edge after financing, turnover, borrow, and crowding.

Extended note. Version record

Robert Novy-Marx and Mihail Velikov, A Taxonomy of Anomalies and Their Trading Costs

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 29(1), 104–147, January 2016; first published online 10 November 2015.

Measures how trading costs change a broad set of anomaly portfolios and shows why turnover and size are first-order economic filters. A buy/hold spread that is stricter for opening than retaining positions improves many lower-turnover signals, while high-turnover signals seldom survive in the paper's tests. Historical cost estimates are not a substitute for present venue, borrow, queue, and market-impact models.

Extended note. Version record

R. David McLean and Jeffrey Pontiff, Does Academic Research Destroy Stock Return Predictability?

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 71(1), 5–32, February 2016; first published online 13 October 2015.

Tracks 97 published cross-sectional predictors beyond their original samples and after publication, distinguishing ordinary out-of-sample decay from additional postpublication erosion. The results motivate an explicit discovery-date clock, data-vintage reconstruction, and forward-only tests when promoting a literature signal into a live queue. The paper's average historical decline is not a universal haircut for every signal or market.

Extended note. Version record

Benjamin Marrow and Stefan Nagel, Real-Time Discovery and Tracking of Return-Based Anomalies; Coauthor research listing

Difficulty: Research **Type:** Research paper

Status/version: University of Chicago author-hosted working paper, October 2024; no journal acceptance established as of 4 October 2026.

Reconstructs what a Bayesian researcher could infer about momentum and reversals at each historical date using Gaussian-process prediction and empirical-Bayes shrinkage over many return lags. Its real-time posterior separates an ex-ante discovery from a pattern selected after observing decades of future data, and suggests several return-based anomalies weaken around publication. This remains a working-paper result; turnover, costs, and other markets need separate validation.

Extended note. Version record

Campbell R. Harvey and Yan Liu, "False (and Missed) Discoveries in Financial Economics"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(5), 2503–2553 (2020); DOI 10.1111/jofi.12951.

Double-bootstrap method balancing type I and type II errors when evaluating many strategies or funds.

Version record

Andrew Y. Chen, "The Limits of p-Hacking: Some Thought Experiments"

Difficulty: Research **Type:** Research paper

Status/version: Federal Reserve Finance and Economics Discussion Series 2019.016, 2019.0(16) (2019); DOI 10.17016/feds.2019.016. Working-paper record; no journal version established in this check.

Argues that p-hacking cannot plausibly explain most published anomaly t-statistics.

Version record

Heiko Jacobs and Sebastian Müller, "Anomalies across the globe: Once public, no longer existent?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 135(1), 213–230 (2020); DOI 10.1016/j.jfineco.2019.06.004.

Post-publication decline of anomalies is a US phenomenon; international anomalies decay little.

Version record

Andrew Detzel, Robert Novy-Marx and Mihail Velikov, "Model Comparison with Transaction Costs"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 78(3), 1743–1775 (2023); DOI 10.1111/jofi.13225.

Factor-model rankings change once factor trading costs are included.

Version record

Andrew Y. Chen and Mihail Velikov, "Zeroing in on the Expected Returns of Anomalies"

Difficulty: Research **Type:** Research paper

Status/version: Federal Reserve FEDS 2020-039 (May 2020); later journal version not established in this check.

After costs and publication effects, expected anomaly returns are close to zero for recent periods.

Version record

Ivo Welch, "Reproducing, Extending, Updating, Replicating, Reexamining, and Reconciling"

Difficulty: Advanced **Type:** Research paper

Status/version: Critical Finance Review, 8(1-2), 301–304 (2019); DOI 10.1561/104.00000082.

Typology of replication activities and incentives in finance research.

Version record

Dmitriy Muravyev, Neil D. Pearson and Joshua M. Pollet, "Anomalies and Their Short-Sale Costs"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(6), 3639–3694 (2025); DOI 10.1111/jofi.13501.

Across 162 anomalies, stock-borrow fees eliminate long-short abnormal returns (0.14% per month before short-sale costs); the long leg remains but is weaker. Essential implementability counter-evidence.

Extended note. Version record

Nolan Alexander and Frank J. Fabozzi, "Measuring Strategy-Decay Risk: Minimum Regime Performance and the Durability of Systematic Investing"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(4), 198–219 (2026); DOI 10.3905/jpm.2025.1.807.

Minimum regime performance, the worst realised risk-adjusted return across historical regimes, as a lower-bound measure of strategy durability and decay risk, applied to a broad set of factor strategies.

Version record

5.4 Reproducible code, data pipelines and ML systems

Computational reproducibility, research-software practice, data validation and documentation, and the engineering of machine-learning systems. Replication of published results is in Section 5.3; model governance and production controls in Chapter 22.

Christophe Pérignon, Olivier Akmansoy, Christophe Hurlin et al., "Computational Reproducibility in Finance: Evidence from 1,000 Tests"

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Financial Studies, 37(11), 3558–3593 (2024); DOI 10.1093/rfs/hhae029.

Re-runs the code behind more than 1,000 answers to six finance research questions from 168 teams: the same code on the same raw data reproduces the results exactly only 52% of the time. Reproducibility is higher with better coding skills and more effort, lower for technical questions, complex code and tail results; researchers are overconfident about their own reproducibility. Gives guidelines for researchers and journals.

Version record

Garret Christensen and Edward Miguel, "Transparency, Reproducibility, and the Credibility of Economics Research"

Difficulty: Advanced **Type:** Survey

Status/version: Journal of Economic Literature, 56(3), 920–980 (2018); DOI 10.1257/jel.20171350.

Surveys evidence that publication bias, failure to replicate and specification searching remain widespread in economics, and the remedies: better research design, registration and pre-analysis plans, disclosure standards and open data and code. Separates practices with emerging consensus from those still disputed.

Version record

Lars Vilhuber, "Reproducibility and Replicability in Economics"

Difficulty: Intro **Type:** Survey

Status/version: Harvard Data Science Review, 2(4) (2020); DOI 10.1162/99608f92.4f6b9e67.

History and current state of reproducibility and replicability in economics by the AEA's data editor: code and data sharing, working papers, public and restricted data, proprietary software and the rise of journal verification of replication packages. Useful for finance research built on licensed vendor data.

Version record

Greg Wilson, D. A. Aruliah, C. Titus Brown et al., "Best Practices for Scientific Computing"

Difficulty: Intro **Type:** Practitioner article

Status/version: PLoS Biology, 12(1), e1001745 (2014); DOI 10.1371/journal.pbio.1001745.

Best practices for scientific software that rest on research and experience: write programs for people, let the computer do the work, make incremental changes, don't repeat yourself, plan for mistakes, optimise only after the code works, document design and purpose, and collaborate. A checklist for research code; the section headings were read on the article page.

Version record

Roger D. Peng, "Reproducible Research in Computational Science"

Difficulty: Intro **Type:** Practitioner article

Status/version: Science, 334(6060), 1226–1227 (2011); DOI 10.1126/science.1213847.

Short Science perspective proposing reproducibility, rerunning the authors' data and code, as the minimum standard for judging computational claims when full independent replication is not possible.

Version record

Joao Felipe Pimentel, Leonardo Murta, Vanessa Braganholo and Juliana Freire, "A Large-Scale Study About Quality and Reproducibility of Jupyter Notebooks"

Difficulty: Advanced **Type:** Research paper

Status/version: IEEE/ACM International Conference on Mining Software Repositories (MSR 2019), 507–517 (2019); DOI 10.1109/MSR.2019.00077.

Studies 1.4 million Jupyter notebooks on GitHub for the characteristics that affect reproducibility, and proposes best practices to raise the reproducibility rate. Evidence for checking notebook-based research by rerunning it from a clean state before trusting it.

Version record

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt et al., "Array programming with NumPy"

Difficulty: Intro **Type:** Research paper

Status/version: Nature, 585(7825), 357–362 (2020); DOI 10.1038/s41586-020-2649-2.

Reviews the array-programming concepts behind NumPy, the foundation of the scientific Python stack, and its role as an interoperability layer for other array libraries. Background for vectorised research code and its numerical conventions.

Version record

Wes McKinney, "Data Structures for Statistical Computing in Python"

Difficulty: Intro **Type:** Research paper

Status/version: Proceedings of the 9th Python in Science Conference (SciPy 2010), 56–61 (2010); DOI 10.25080/Majora-92bf1922-00a.

Introduces pandas for the data sets common in finance and statistics, discussing its design issues with examples and comparisons with R. Read for the origin of the data structures most research pipelines now use.

Version record

Sebastian Schelter, Dustin Lange, Philipp Schmidt, Meltem Celikel, Felix Biessmann and Andreas Grafberger, "Automating large-scale data quality verification"

Difficulty: Advanced **Type:** Research paper

Status/version: Proceedings of the VLDB Endowment, 11(12), 1781–1794 (2018); DOI 10.14778/3229863.3229867.

System for unit tests on data: a declarative API combines standard quality constraints (completeness, uniqueness, ranges) with custom checks, runs incrementally on growing data sets, and uses ML to suggest constraints and flag anomalies in data-quality time series. A template for validating vendor files before they reach a model.

Version record

Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III and Kate Crawford, "Datasheets for datasets"

Difficulty: Intro **Type:** Research paper

Status/version: Communications of the ACM, 64(12), 86–92 (2021); DOI 10.1145/3458723.

Proposes that every dataset ship with a datasheet documenting its motivation, composition, collection process and recommended uses, to improve communication between data creators and users. A structure for documenting vendor and alternative data sets.

Version record

Margaret Mitchell, Simone Wu, Andrew Zaldivar et al., "Model Cards for Model Reporting"

Difficulty: Intro **Type:** Research paper

Status/version: Conference on Fairness, Accountability, and Transparency (FAT* 2019), 220–229 (2019); DOI 10.1145/3287560.3287596.

Model cards: short documents released with a trained model that report its intended use, evaluation procedure and benchmarked performance across relevant conditions and groups, and where it should not be used. Complements model-risk documentation.

Version record

Eric Breck, Shanqing Cai, Eric Nielsen, Michael Salib and D. Sculley, "The ML test score: A rubric for ML production readiness and technical debt reduction"

Difficulty: Advanced **Type:** Research paper

Status/version: IEEE International Conference on Big Data (Big Data 2017), 1123–1132 (2017); DOI 10.1109/BigData.2017.8258038.

Twenty-eight specific tests and monitoring needs for production ML systems, drawn from experience with many such systems, set out as a road map for production readiness and for paying down ML technical debt; testing matters because a model's prediction behaviour is hard to specify in advance.

Version record

Saleema Amershi, Andrew Begel, Christian Bird et al., "Software Engineering for Machine Learning: A Case Study"

Difficulty: Advanced **Type:** Research paper

Status/version: IEEE/ACM International Conference on Software Engineering, SEIP track (ICSE-SEIP 2019), 291–300 (2019); DOI 10.1109/ICSE-SEIP.2019.00042.

Case study of Microsoft teams building AI applications around a nine-stage ML workflow. Three differences from ordinary software stand out: data discovery, management and versioning are harder; model customisation and reuse need different skills; and models are entangled components with non-monotonic errors.

Version record

Andrei Paleyes, Raoul-Gabriel Urma and Neil D. Lawrence, "Challenges in Deploying Machine Learning: A Survey of Case Studies"

Difficulty: Advanced **Type:** Survey

Status/version: ACM Computing Surveys, 55(6), 1–29 (2023); DOI 10.1145/3533378.

Survey of published case studies of ML deployment, mapping the practical problems found to the stages of the deployment workflow and showing that practitioners face issues at every stage.

Version record

Christophe Pérignon, Kamel Gadouche, Christophe Hurlin, Roxane Silberman and Eric Debonnel, "Certify reproducibility with confidential data"

Difficulty: Intro **Type:** Practitioner article

Status/version: Science, 365(6449), 127–128 (2019); DOI 10.1126/science.aaw2825.

Policy proposal for research on confidential microdata that referees cannot access: an agency accredited by the data producers certifies that the researcher's code run on the restricted data produces the reported results, giving a reproducibility signal without releasing the data.

Version record

Part II

Portfolio, pricing and trading methods

Chapter 6

Portfolio construction, risk and ALM

Move from objectives and estimation to constraints, costs and path dependence. Liability-relative risk adds payment timing and collateral needs to asset-only allocation.

6.1 Portfolio foundations and estimation

Section 6.3 adds the signature and RKHS readings for policies that depend on price and signal history.

6.1.1 Mean-variance theory and estimation error

Richard C. Grinold and Ronald N. Kahn, *Active Portfolio Management*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 1999

The canonical treatment of forecasts, information coefficients, breadth, risk models and constrained active portfolios. The Fundamental Law is best understood as an organising approximation whose assumptions fail when bets, signals and transaction costs are correlated.

Applications: Equities, multi-asset; factor portfolios, signal combination. Version record

Fischer Black and Robert Litterman, "Global Portfolio Optimization"

Difficulty: Advanced **Type:** Research paper

Status/version: 1992

Introduces a disciplined Bayesian-style mechanism for blending equilibrium returns with uncertain views. The paper is foundational for combining discretionary, macro and quantitative forecasts without allowing noisy expected returns to generate implausibly extreme portfolios.

Applications: Multi-asset; allocation, signal blending. Version record

Victor DeMiguel, Lorenzo Garlappi and Raman Uppal, "Optimal Versus Naive Diversification"

Difficulty: Advanced **Type:** Research paper

Status/version: 2009

Demonstrates how estimation error can overwhelm theoretically optimal portfolio rules. It is a necessary corrective to papers that report only in-sample utility or Sharpe improvements from increasingly elaborate optimisers.

Applications: Multi-asset; portfolio construction. Version record

Oliver Hellum, Theis I. Jensen, Bryan T. Kelly and Semyon Malamud, Complex Modern Portfolio Theory

Difficulty: Research **Type:** Research paper

Status/version: SSRN author version dated 19 March 2026, posted 20 March 2026; NBER Working Paper 35246 issued May 2026, NBER page lists June 2026 revision. SSRN separately lists a later upload for its NBER copy; cite a fixed version. No journal publication established.

Revisits mean–variance portfolio estimation when asset count exceeds training observations. The authors report that portfolio performance can improve beyond the N>T threshold across several universes and regularizers, making it a useful counterpoint to simple diversification and shrinkage readings.

Extended note. Version record

Harry Markowitz, "Portfolio Selection"

Difficulty: Intro **Type:** Research paper

Status/version: 1952

Original mean–variance portfolio selection and efficient frontier. Derive the optimisation problem and then make estimation error, constraints and input stability explicit; the institutional lesson is not to optimise noisy raw expected returns mechanically.

Applications: Multi-asset; portfolio construction. Version record

Ludwig B. Chincarini and Daehwan Kim, Quantitative Equity Portfolio Management: An Active Approach to Portfolio Construction and Management

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

End-to-end equity process from the source's "Alpha Mojo" signal research to risk models, optimisation, trading costs and performance attribution. Compare screening with ranking, normalise signals with z-scores, and trace value, momentum and quality anomalies to original references. Distinct from the Qian–Hua–Sorensen book already listed.

Version record

Guangliang He and Robert Litterman, The Intuition Behind Black-Litterman Model Portfolios (1999)

Difficulty: Advanced **Type:** Research paper

Status/version: Selected citation year(s): 1999. Version distinctions remain in the annotation.

Equilibrium weights, investor views and confidence in Black–Litterman allocation; the linked paper carries a 1999 copyright, distinct from its SSRN posting date.

Version record

Frank J. Fabozzi, Petter N. Kolm, Dessislava A. Pachamanova and Sergio M. Focardi, Robust Portfolio Optimization and Management

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Estimation uncertainty, robust allocation and alternatives to plug-in mean–variance optimisation; uncertainty sets and tail-risk criteria are distinct modelling choices.

Version record

Daniel P. Palomar, *Portfolio Optimization: Theory and Application*; Author companion

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press; 2025-06-12 ebook; 2025-06-17 print (Cambridge lists both format dates). First edition. DOI 10.1017/9781009428095.

A computationally oriented graduate reference connecting heavy-tailed financial-data models and graph estimation to implementable portfolio algorithms. It consolidates mean-variance, tail/drawdown, risk-parity, robust, index-tracking, pairs-trading and deep-learning portfolios with Python/R examples. Use alongside the institution's own constraints, trading-cost model and point-in-time data; the published numerical examples do not validate a particular mandate or strategy.

Extended note. Version record

Gérard Cornuéjols and Reha Tütüncü, Optimization Methods in Finance

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Linear, quadratic, integer and stochastic programming for cash-flow matching, portfolios, index tracking and decisions under uncertainty. This is the two-author edition; later editions add Javier Peña.

Version record

Dimitris Bertsimas, David B. Brown and Constantine Caramanis, "Theory and Applications of Robust Optimization" (2011)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2011. Version distinctions remain in the annotation.

Uncertainty sets, computational tractability and adaptable multistage decisions; a survey of robust optimisation.

Version record

Richard C. Grinold, "The Fundamental Law of Active Management" (1989)

Difficulty: Advanced **Type:** Research paper

Status/version: Selected DOI-linked version; print 1989-04-30; online 1989-04-30.

Original paper linking information ratio, skill and independent breadth. Study it separately from the Grinold–Kahn book: adding correlated signals does not create the independent bets assumed by the approximation. Connect forecast combination to risk, implementation and net performance.

Version record

Edward Qian, Ronald Hua and Eric Sorensen, *Quantitative Equity Portfolio Management: Modern Techniques and Applications* (2007)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 2007. Version distinctions remain in the annotation.

Connects value, growth, quality and momentum definitions to return forecasts, risk models, Fama–MacBeth regressions, portfolio constraints, attribution, turnover and execution. Useful for sector-aware accounting signals and controlled long–short or enhanced-index portfolios.

Applications: practitioner. Version record

Michael Isichenko, *Quantitative Portfolio Management: The Art and Science of Statistical Arbitrage* (2021)

Difficulty: Advanced/Research **Type:** Book / monograph

Status/version: Selected citation year(s): 2021. Version distinctions remain in the annotation.

Links financial data, return forecasts, forecast combination, risk models and cost-aware portfolio optimisation. Corrects the PDF's attribution to "Girault & Bouchaud, 2016"; this is the publisher-identified book with the supplied title.

Version record

Richard O. Michaud, "The Markowitz Optimization Enigma: Is "Optimized" Optimal?"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 45(1), 31–42 (1989); DOI 10.2469/faj.v45.n1.31.

Classic statement that optimisers act as error maximisers; motivates resampling, shrinkage and constraints.

Version record

Vijay K. Chopra and William T. Ziemba, "The Effect of Errors in Means, Variances, and Covariances on Optimal Portfolio Choice"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 19(2), 6–11 (1993); DOI 10.3905/jpm.1993.409440.

Quantifies that errors in means matter roughly ten times more than errors in variances and much more than covariances.

Version record

Michael J. Best and Robert R. Grauer, "On the Sensitivity of Mean-Variance-Efficient Portfolios to Changes in Asset Means: Some Analytical and Computational Results"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 4(2), 315–342 (1991); DOI 10.1093/rfs/4.2.315.

Analytical sensitivity of efficient weights to mean perturbations; small changes can swing weights dramatically.

Version record

Raymond Kan and Guofu Zhou, "Optimal Portfolio Choice with Parameter Uncertainty"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 42(3), 621–656 (2007); DOI 10.1017/s0022109000004129.

Derives the expected out-of-sample utility loss from estimated mean-variance inputs and optimal three-fund combinations.

Version record

Victor DeMiguel, Lorenzo Garlappi, Francisco J. Nogales and Raman Uppal, "A Generalized Approach to Portfolio Optimization: Improving Performance by Constraining Portfolio Norms"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 55(5), 798–812 (2009); DOI 10.1287/mnsc.1080.0986.

Norm-constrained minimum-variance portfolios nest shrinkage and short-sale constraints and improve out-of-sample Sharpe ratios.

Version record

Michael W. Brandt, Pedro Santa-Clara and Rossen Valkanov, "Parametric Portfolio Policies: Exploiting Characteristics in the Cross-Section of Equity Returns"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(9), 3411–3447 (2009); DOI 10.1093/rfs/hhp003.

Optimises weights as a linear function of characteristics, bypassing return and covariance estimation for large universes.

Version record

Petter N. Kolm, Reha Tütüncü and Frank J. Fabozzi, "60 Years of portfolio optimization: Practical challenges and current trends"

Difficulty: Advanced **Type:** Research paper

Status/version: European Journal of Operational Research, 234(2), 356–371 (2014); DOI 10.1016/j.ejor.2013.10.060.

Survey of practical optimisation: estimation error, constraints, transaction costs, robust and multi-period formulations.

Version record

Stephen Boyd, Kasper Johansson, Ronald Kahn, Philipp Schiele and Thomas Schmelzer, "Markowitz Portfolio Construction at Seventy"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Portfolio Management 50(8), 117–160 (July 2024); arXiv 2401.05080.

Modern convex Markowitz formulation with costs, leverage and robustness to forecast uncertainty, with open-source code; current practical reference for the single-period problem.

Extended note. Version record

Giuseppe A. Paleologo, *The Elements of Quantitative Investing*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley Finance, 22 April 2025 (ISBN 9781394265459).

Practitioner text (Wiley 2025) spanning return statistics, factor models, backtesting, portfolio construction, risk management and hedging from a fundamental-quant platform perspective.

Version record

Giuseppe A. Paleologo, ***Advanced Portfolio Management: A Quant's Guide for Fundamental Investors***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, 2021 (ISBN 9781119789796).

Factor-model-based risk decomposition, sizing and hedging for discretionary long-short managers.

Version record

Miquel Noguer Alonso, Julian Antolin Camarena and Alberto Bueno Guerrero, ***Quantitative Portfolio Optimization: Advanced Techniques and Applications***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley Finance, January 2025 (ISBN 9781394281312).

Wiley (2025) survey of mean-variance, Black–Litterman, risk parity, hierarchical risk parity, factor and robust optimisation with code examples.

Version record

Richard C. Grinold and Ronald N. Kahn, ***Advances in Active Portfolio Management: New Developments in Quantitative Investing***

Difficulty: Advanced **Type:** Book / monograph

Status/version: McGraw Hill, 2019 (ISBN 9781260453713).

Sequel to Active Portfolio Management covering dynamic portfolio management, expected-return research, smart beta and ESG.

Version record

Roger Clarke, Harindra de Silva and Steven Thorley, "Portfolio Constraints and the Fundamental Law of Active Management"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 58(5), 48–66 (2002); DOI 10.2469/faj.v58.n5.2468.

Introduces the transfer coefficient: constraints reduce realised information ratios below the fundamental-law prediction.

Version record

Bryan Kelly, Semyon Malamud, Mo Pourmohammadi and Fabio Trojani, "Universal Portfolio Shrinkage"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag084 (2026); DOI 10.1093/rfs/hhag084.

Universal portfolio shrinkage approximator (UPSA): nonlinear shrinkage built from closed-form basis portfolios that reweights principal components to maximise out-of-sample portfolio performance.

Version record

Raymond Kan and Nathan Lassance, "Optimal Portfolio Choice with Fat Tails and Parameter Uncertainty"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(8), 3753–3790 (2025); DOI 10.1017/s0022109025101622.

Fat tails worsen out-of-sample mean-variance and GMV performance relative to normal-based theory; new combination rules account for both tails and parameter uncertainty.

Version record

Shingo Goto and Toru Yamada, "Selection versus diversification in noisy alpha environments"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 189, 107726 (2026); DOI 10.1016/j.jbankfin.2026.107726.

Combining many weak signals beats selecting only the most significant ones; out-of-sample information ratios peak at p-value cutoffs of 5–10%, looser than false-discovery control suggests.

Version record

6.1.2 Covariance estimation and factor risk models

Olivier Ledoit and Michael Wolf, "Analytical Nonlinear Shrinkage of Large-Dimensional Covariance Matrices"

Difficulty: Research **Type:** Research paper

Status/version: 2020

Provides a modern large-dimensional covariance estimator with direct relevance to factor-neutral portfolios and risk forecasts. It should be studied alongside eigenvalue cleaning and factor models rather than treating sample covariance as a reliable input.

Applications: Equities, multi-asset; risk, optimisation.

Version record

Olivier Ledoit and Michael Wolf, "Honey, I Shrunk the Sample Covariance Matrix"

Difficulty: Advanced **Type:** Research paper

Status/version: 2003 manuscript / 2004 article

Linear shrinkage of the sample covariance for portfolio use, addressing noisy inputs and unstable optimisation. This is a separate foundational reading from the 2020 nonlinear-shrinkage paper; compare estimation and out-of-sample portfolio behaviour.

Applications: Multi-asset; covariance, optimisation. Version record

Gregory Connor, Lisa R. Goldberg and Robert A. Korajczyk, *Portfolio Risk Analysis*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press (2010); DOI 10.1515/9781400835294.

Fundamental, statistical and macro factor risk models, estimation, risk decomposition and model evaluation from authors with commercial risk-model experience.

Version record

Olivier Ledoit and Michael Wolf, "The Power of (Non-)Linear Shrinking: A Review and Guide to Covariance Matrix Estimation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 20(1), 187–218 (2022); DOI 10.1093/jjfinec/nbaa007.

Practitioner-oriented review comparing linear, nonlinear and factor-based shrinkage with guidance on when each is appropriate.

Version record

Robert F. Engle, Olivier Ledoit and Michael Wolf, "Large Dynamic Covariance Matrices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 37(2), 363–375 (2019); DOI 10.1080/07350015.2017.1345683.

Combines DCC dynamics with nonlinear shrinkage for large universes; improves minimum-variance portfolio risk.

Version record

Jose Menchero, D. J. Orr and Jun Wang, "The Barra US Equity Model (USE4): Methodology Notes"

Difficulty: Advanced **Type:** Report / research note

Status/version: MSCI Barra methodology notes, August 2011; access may require MSCI client credentials.

Commercial risk-model methodology: style and industry factors, eigenfactor risk adjustment, volatility regime adjustment and specific-risk Bayesian shrinkage.

Version record

Gianluca De Nard and Damjan Kostovic, "Learning the Shrinkage Intensity: A Data-Driven Approach for Risk-Optimized Portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 24(2), nbag002 (2026); DOI 10.1093/jjfinec/nbag002.

Learns a state-dependent shrinkage intensity by contextual bandits for linear and nonlinear covariance shrinkage, improving risk-optimised portfolios.

Version record

M. Sipke Dom, Clint Howard, Maarten Jansen and Harald Lohre, "Beyond GMV: the relevance of covariance matrix estimation for risk-based portfolio construction"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 25(3), 403–419 (2025); DOI 10.1080/14697688.2025.2468268.

Covariance-estimator choice matters for constrained, implementable risk-based portfolios, not only unconstrained GMV.

Version record

Jorge Guijarro-Ordonez, Misha van Beek and Amandeep Dhaliwal, "A Responsive Variance and Covariance Matrix Forecaster for Multiple Horizons and Asset Classes"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, jpm.2026.072 (2026); DOI 10.3905/jpm.2026.072.

A responsive variance and covariance forecaster designed for multiple horizons and asset classes.

Extended note. Version record

Lisa R. Goldberg, Hubeyb Gurdogan and Alec Kercheval, "Portfolio optimisation via strategy-specific eigenvector shrinkage"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 29(3), 665–706 (2025); DOI 10.1007/s00780-025-00566-4.

Generalised James–Stein eigenvector shrinkage towards constraint-gradient subspaces reduces excess volatility of constrained minimum-variance portfolios.

Version record

Richard Luger, "Regularizing stock return covariance matrices via multiple testing of correlations"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 248, 105753 (2025); DOI 10.1016/j.jeconom.2024.105753.

Regularises large stock covariance matrices by multiple testing of pairwise correlations (FWER or FDP control, robust to heavy tails and GARCH effects) followed by shrinkage to positive definiteness.

Version record

6.1.3 Risk-based, hierarchical and growth-optimal allocation

John L. Kelly Jr., "A New Interpretation of Information Rate" (1956)

Difficulty: Advanced **Type:** Research paper

Status/version: Selected citation year(s): 1956. Version distinctions remain in the annotation.

Original capital-growth sizing argument based on maximising expected logarithmic wealth. Derive the assumptions before translating it to leverage or position sizes. Estimation error, drawdowns and financing constraints can dominate asymptotic growth; Kelly optimisation is not a guarantee against ruin.

Version record

Leonard C. MacLean, Edward O. Thorp and William T. Ziemba, eds., The Kelly Capital Growth Investment Criterion: Theory and Practice

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Retains the nested "MacLean et al." modern-treatment recommendation. Read capital-growth and fractional-sizing discussions alongside Kelly, portfolio constraints and security/drawdown objectives; distinguish growth optimality from short-horizon risk tolerance.

Version record

Sébastien Maillard, Thierry Roncalli and Jérôme Teïletche, "The Properties of Equally Weighted Risk Contribution Portfolios"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 36(4), 60–70 (2010); DOI 10.3905/jpm.2010.36.4.060.

Defines equal-risk-contribution portfolios and shows their volatility lies between minimum-variance and equal-weight portfolios.

Version record

Thierry Roncalli, *Introduction to Risk Parity and Budgeting*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Chapman and Hall/CRC (2016); DOI 10.1201/b15151.

Risk budgeting theory, algorithms and applications to equities, bonds, alternatives and multi-asset portfolios.

Version record

Marcos López de Prado, "Building Diversified Portfolios that Outperform Out of Sample"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 42(4), 59–69 (2016); DOI 10.3905/jpm.2016.42.4.059.

Hierarchical risk parity: cluster assets, quasi-diagonalise the covariance matrix and allocate by recursive bisection without inversion. Results depend on universe and backtest design.

Version record

Yves Choueifaty and Yves Coignard, "Toward Maximum Diversification"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 35(1), 40–51 (2008); DOI 10.3905/jpm.2008.35.1.40.

Maximum diversification ratio portfolios as a risk-based alternative to market-cap and minimum-variance weights.

Version record

Attilio Meucci, "Fully Flexible Views: Theory and Practice"

Difficulty: Advanced **Type:** Research paper

Status/version: Risk 21(10), 97–102 (October 2008); SSRN author version.

Entropy pooling: blend arbitrary views on scenarios with a prior distribution; generalises Black–Litterman beyond normal returns.

Extended note. Version record

Denis Chaves, Jason Hsu, Feifei Li and Omid Shakernia, "Risk Parity Portfolio vs. Other Asset Allocation Heuristic Portfolios"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Investing, 20(1), 108–118 (2011); DOI 10.3905/joi.2011.20.1.108.

Risk parity compared with equal weight, minimum variance and 60/40 across asset classes; risk parity diversifies better but depends on the asset set.

Version record

Thomas M. Cover, "Universal Portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 1(1), 1–29 (1991); DOI 10.1111/j.1467-9965.1991.tb00002.x.

Online portfolio selection that asymptotically matches the best constant-rebalanced portfolio in hindsight; the root of online-learning portfolio methods.

Version record

Enzo Busseti, Ernest K. Ryu and Stephen Boyd, "Risk-Constrained Kelly Gambling"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Investing, 25(3), 118–134 (2016); DOI 10.3905/joi.2016.25.3.118.

Convex formulation of growth-optimal betting with a drawdown-risk constraint, a practical fractional-Kelly alternative.

Version record

6.2 Costs, constraints and dynamic trading

Read after the optimisation foundations. The selection connects forecasts to positions, trading schedules and net portfolio objectives; evaluate capital, turnover, financing and risk constraints jointly.

Ravi Jagannathan and Tongshu Ma; correction by Ravi Jagannathan, Tongshu Ma, and Jiaqi Zhang, Risk Reduction in Large Portfolios: Why Imposing the Wrong Constraints Helps; A Note on 'Risk Reduction in Large Portfolios: Why Imposing the Wrong Constraints Helps'; 2019 published correction

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance 58(4), 1651–1683, August 2003, DOI 10.1111/1540-6261.00580; correction in The Journal of Finance 74(5), 2689–2696, October 2019, DOI 10.1111/jofi.12824.

Shows why nonnegative weight constraints can reduce risk from estimated covariances even when the true optimal portfolio contains shorts, giving a rigorous counterpoint to unrestricted plug-in mean–variance optimization and the 1/N benchmark. Read the 2019 note with the article because it corrects an error in the proof of Proposition 2; it does not erase the empirical result.

Extended note. Version record

Stephen Boyd, Enzo Busseti, Steven Diamond, Ronald N. Kahn, Kwangmoo Koh, Peter Nystrup, and Jan Speth, Multi-Period Trading via Convex Optimization

Difficulty: Advanced **Type:** Research paper

Status/version: Foundations and Trends in Optimization 3(1), 1–76, August 2017, DOI 10.1561/2400000023; Stanford author page hosts final article, slides, and code.

Builds a coherent single- and multiperiod trading objective that balances expected return, risk, transaction costs, holding costs, and convex constraints, with only the first planned trade executed before replanning. The paper explicitly leaves the forecast problem outside scope; production use also requires calibrated impact, borrow, liquidity, and hard operational constraints rather than treating solver output as executable orders.

Extended note. Version record

Nicolae Gârleanu and Lasse Heje Pedersen, Dynamic Trading with Predictable Returns and Transaction Costs; author manuscript

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 68(6), 2309–2340, December 2013; first published online 26 July 2013; DOI 10.1111/jofi.12080.

Derives a costly-trading policy for several correlated assets and predictors with different decay rates: aim toward expected future targets and adjust only partway today. The closed-form policy depends on the assumed return process and cost specification; estimated alpha decay, futures roll, funding, and stress-state impact must be checked before applying its reported net advantage.

Extended note. Version record

Pierre Collin-Dufresne, Kent D. Daniel, and Mehmet Sağlam, Optimal Dynamic Asset Allocation with Transaction Costs: The Role of Hedging Demands

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33058, issued October 2024, DOI 10.3386/w33058; no journal publication asserted.

Extends costly dynamic allocation to non-myopic investors whose future investment opportunities change, making both the aim portfolio and trading speed depend on intertemporal hedging demand. Read after Campbell–Viceira and Gârleanu–Pedersen to separate long-horizon opportunity hedges from short-horizon signal exploitation. The paper illustrates its mechanism with simplified cases and a

daily retail-imbalance strategy, so a pension or macro portfolio would need its own horizon, opportunity-state, transaction-cost, and liability calibration before adopting the rule.

Version record

Ruslan Goyenko, Bryan T. Kelly, Tobias J. Moskowitz, Yinan Su, and Chao Zhang, Trading Volume Alpha; July 2026 author revision

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33037, October 2024; separate author manuscript dated 10 July 2026, uploaded 11 July 2026.

Treats predictability of individual-stock volume as an input to portfolio optimization, trading off tracking error against expected performance after trading cost instead of optimizing gross alpha alone. The empirical result concerns stocks, and volume is only one component of execution cost; an institutional implementation must calibrate spread, impact, urgency, participation, and stressed depth for its own instruments. The July 2026 author abstract emphasizes an asymmetric, fund-size-dependent economic training loss; compare versions before transferring numerical claims.

Extended note. Version record

John Y. Campbell and Luis M. Viceira, *Strategic Asset Allocation: Portfolio Choice for Long-Term Investors* (2002)

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected DOI-linked version; print 2002-01-03; online 2003-11-01.

Intertemporal portfolio choice with changing investment opportunities, rates and inflation; supports economic reasoning rather than prescribing a tactical rule.

Version record

M. H. A. Davis and A. R. Norman, "Portfolio Selection with Transaction Costs"

Difficulty: Research **Type:** Research paper

Status/version: Mathematics of Operations Research, 15(4), 676–713 (1990); DOI 10.1287/moor.15.4.676.

No-trade region for Merton portfolios with proportional costs; foundation of band-rebalancing rules.

Version record

Ciamac C. Moallemi and Mehmet Sağlam, "Dynamic Portfolio Choice with Linear Rebalancing Rules"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 52(3), 1247–1278 (2017); DOI 10.1017/s0022109017000345.

Tractable near-optimal linear rebalancing policies with costs and constraints.

Version record

Victor DeMiguel, Francisco J. Nogales and Raman Uppal, "Stock Return Serial Dependence and Out-of-Sample Portfolio Performance"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(4), 1031–1073 (2014); DOI 10.1093/rfs/hhu002.

Exploiting VAR-type serial dependence in portfolios survives moderate costs out of sample.

Version record

Gordon Ritter, "Explicit methods for portfolio optimization with realistic costs and constraints"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(9), 1363–1370 (2026); DOI 10.1080/14697688.2026.2707213.

Coordinate-descent mean-variance optimisation with borrow costs, spreads and square-root impact, each step solved in closed form.

Version record

Jörgen Blomvall, Jonas Ekblom and John R. Birge, "Approximate dynamic programming for high-dimensional portfolio choice with transaction costs"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(9), 1371–1396 (2026); DOI 10.1080/14697688.2026.2708220.

Approximate dynamic programming for high-dimensional portfolio choice with transaction costs.

Extended note. Version record

6.3 Path-dependent portfolios and control

Owen Futter, Blanka Horvath and Magnus Wiese, *Signature Trading: A Path-Dependent Extension of the Mean-Variance Framework with Exogenous Signals* (2023 preprint)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2023. Version distinctions remain in the annotation.

Linear policies in truncated price/signal signatures extend mean–variance allocation to path history. Signature depth and moment estimation drive complexity; endogenous drawdown control is not a hard maximum-drawdown constraint.

Applications: graduate probability, statistics and optimisation. Version record

Owen Futter, Nicola Muça Cirone and Blanka Horvath, *Kernel Learning for Mean-Variance Trading Strategies* (arXiv v1, 14 July 2025; also listed under preprints in the coauthor's CV, inspected 4 October 2026)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2025, 2026. Version distinctions remain in the annotation.

History-dependent policies in an RKHS, including signature and neural embeddings, with a closed-form mean–variance solution in the stated setting. Reported gains depend on feature choice, regularisation and estimation.

Applications: graduate probability, statistics and optimisation. Version record

Andrea Macrì, Sebastian Jaimungal and Fabrizio Lillo, *Deep Reinforcement Learning for Optimal Trading with Partial Information* (arXiv v1, 31 October 2025)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2025. Version distinctions remain in the annotation.

Recurrent RL for a partially observed regime-switching signal; posterior-regime features outperform the other tested state representations in the authors' examples, not universally.

Applications: graduate probability, statistics and optimisation. Version record

Yahui Bai, Yuhe Gao, Runzhe Wan, Sheng Zhang and Rui Song, *A Review of Reinforcement Learning in Financial Applications* (2024, v1; especially Section 6.1.4)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2024. Version distinctions remain in the annotation.

Survey of financial RL, including history, latent information and state design (Section 6.1.4). Longer memory can raise noise and data needs; the corrected linked authors/title are retained.

Applications: graduate probability, statistics and optimisation. Version record

Sanford J. Grossman and Zhongquan Zhou, *Optimal Investment Strategies for Controlling Drawdowns* (1993)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 1993. Version distinctions remain in the annotation.

Continuous-time wealth-floor control tied to the running maximum; state augmentation connects drawdown constraints to risky investment. Model guarantees need not survive discrete trading, gaps or costs.

Applications: graduate probability, statistics and optimisation. Version record

Alexei Chekhlov, Stanislav Uryasev and Michael Zabarankin, *Portfolio Optimization with Drawdown Constraints* (linked manuscript dated 13 January 2003)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2003. Version distinctions remain in the annotation.

Conditional drawdown-at-risk allocation using sampled paths and linear programming. CDaR differs from terminal-loss CVaR; this 2003 manuscript is distinct from the 2005 paper Drawdown Measure in Portfolio Optimization.

Applications: graduate probability, statistics and optimisation. Version record

Kexin Chen, Mei Choi Chiu and Hoi Ying Wong, *Time-Consistent Mean-Variance Pairs-Trading Under Regime-Switching Cointegration* (2019)

Difficulty: Research **Type:** Research paper

Status/version: Selected DOI-linked version; print 2019-01; online 2019-06-27.

Time-consistent mean–variance pairs trading under regime-switching cointegration. Regime switching does not by itself solve hidden-state filtering; the institutional record confirms the 2019 SIAM identity.

Applications: graduate probability, statistics and optimisation. Version record

Jakša Cvitanić and Ioannis Karatzas, "On portfolio optimization under drawdown constraints"

Difficulty: Research **Type:** Research paper

Status/version: The IMA Volumes in Mathematics and its Applications, 35–45 (1995); DOI 10.1007/978-1-4757-2435-6_3.

Continuous-time generalisation of Grossman–Zhou drawdown-constrained investment.

Version record

Enzo Busseti, Ernest K. Ryu and Stephen Boyd, "Risk-Constrained Kelly Gambling"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Investing, 25(3), 118–134 (2016); DOI 10.3905/joi.2016.25.3.118.

Convex formulation of Kelly betting with a drawdown-probability constraint.

Version record

Edward O. Thorp, "The Kelly Criterion in Blackjack, Sports Betting, and the Stock Market"

Difficulty: Advanced **Type:** Research paper

Status/version: Handbook of Asset and Liability Management, 385–428 (2008); DOI 10.1016/b978-044453248-0.50015-0.

Practitioner exposition of growth-optimal betting, fractional Kelly and estimation risk.

Version record

Pascal J. Maenhout, Hao Xing and Anne G. Balter, "Model Ambiguity versus Model Misspecification in Dynamic Portfolio Choice"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(3), 1741–1795 (2026); DOI 10.1111/jofi.70027.

Dynamic portfolio choice under model ambiguity versus misspecification: risk-averse investors fear return persistence, risk-tolerant investors fear mean reversion, changing robust allocations.

Version record

Samir Varma, "The False Promise of Drawdown Rules: New Evidence and a Better Framework"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(1), 145–161 (2025); DOI 10.3905/jpm.2025.1.765.

Fixed drawdown stop rules on ETFs and long-short portfolios (1993–2022) often amplify rather than cut losses; proposes a coherent drawdown-adjusted performance metric that treats gains and losses symmetrically.

Version record

Christa Cuchiero and Janka Möller, "Signature Methods in Stochastic Portfolio Theory"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 16(4), 1239–1303 (2025); DOI 10.1137/24m1700223.

Signature portfolios in stochastic portfolio theory: linear functionals of signatures of ranked market weights are universal approximators of path-dependent portfolios and can approximate growth-optimal portfolios.

Version record

6.4 Risk measures and budgeting

For running-maximum wealth constraints and conditional drawdown-at-risk, see Grossman–Zhou and Chekhlov–Uryasev–Zabarankin in Section 6.3. Risk contributions and leverage: Qian and Clarke–de Silva–Thorley; drawdown constraints: Section 6.3.

6.4.1 VaR, expected shortfall and backtesting

Alexander McNeil, Rüdiger Frey and Paul Embrechts, *Quantitative Risk Management*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton revised edition, 2015, ISBN 978-0-691-16627-8; 2005 first edition remains a separate historical version.

The standard reference for heavy tails, dependence, copulas, extreme-value theory and market/credit risk. It remains indispensable when Gaussian covariance and volatility forecasts are inadequate for leveraged or derivatives portfolios.

Applications: Multi-asset, credit, derivatives; tail risk. Version record

R. Tyrrell Rockafellar and Stanislav Uryasev, "Optimization of Conditional Value-at-Risk"

Difficulty: Advanced **Type:** Research paper

Status/version: 2000 article

Scenario-based tail-loss optimisation using a tractable auxiliary-variable formulation. Supports expected-shortfall constraints in portfolio construction. CVaR of one-period losses is not itself a path-dependent maximum-drawdown measure.

Applications: Multi-asset; tail risk, optimisation. Version record

Philippe Jorion, Value at Risk: The New Benchmark for Managing Financial Risk

Difficulty: Intro **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Parametric VaR, historical simulation, stress testing and communication of market, credit and operational risk. Use as foundational risk vocabulary while examining tail dependence, model limitations and the information lost by reporting a single quantile.

Version record

John C. Hull, *Risk Management and Financial Institutions*, 6th ed. (Wiley, February 2023)

Difficulty: Intro **Type:** Book / monograph

Status/version: 6th ed. (Wiley, February 2023)

Connects market, credit and liquidity risk to enterprise limits, capital and institutional governance. Regulatory discussions belong to the chosen edition's historical setting; consult applicable current primary rules for an actual capital calculation.

Version record

Philippe Artzner, Freddy Delbaen, Jean-Marc Eber and David Heath, "Coherent Measures of Risk" (1999)

Difficulty: Advanced **Type:** Research paper

Status/version: Selected DOI-linked version; print 1999-07; online 2001-12-25.

Monotonicity, translation invariance, positive homogeneity and subadditivity define coherence. Understand why VaR can fail subadditivity and how tail-loss measures address different questions. Pair with Rockafellar–Uryasev for portfolio implementation; this article is not a current regulatory rulebook.

Version record

Peter F. Christoffersen, *Elements of Financial Risk Management*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Academic Press, second edition, 2012 (ISBN 9780123744487).

Second edition (2012): volatility, correlation and tail models, VaR/ES estimation and backtesting with spreadsheet exercises.

Version record

Jón Daníelsson, *Financial Risk Forecasting: The Theory and Practice of Forecasting Market Risk with Implementation in R and Matlab*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2012); DOI 10.1002/9781119205869.

Practical VaR/ES forecasting, backtesting and model risk with code; author site maintains Python/R/Julia versions.

Version record

Paul H. Kupiec, "Techniques for Verifying the Accuracy of Risk Measurement Models"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Derivatives, 3(2), 73–84 (1995); DOI 10.3905/jod.1995.407942.

Proportion-of-failures test for VaR exception counts; still the regulatory starting point.

Version record

Peter F. Christoffersen, "Evaluating Interval Forecasts"

Difficulty: Advanced **Type:** Research paper

Status/version: International Economic Review, 39(4), 841 (1998); DOI 10.2307/2527341.

Independence and conditional-coverage tests for VaR exceptions; catches clustered violations that count tests miss.

Version record

Carlo Acerbi and Dirk Tasche, "On the coherence of expected shortfall"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Banking & Finance, 26(7), 1487–1503 (2002); DOI 10.1016/s0378-4266(02)00283-2.

Defines expected shortfall consistently for general distributions and proves coherence; the measure later adopted by FRTB.

Version record

Tobias Fissler and Johanna F. Ziegel, "Higher order elicitability and Osband's principle"

Difficulty: Research **Type:** Research paper

Status/version: The Annals of Statistics, 44(4), 1680–1707 (2016); DOI 10.1214/16-aos1439.

Shows that VaR and ES are jointly elicitable, enabling consistent scoring functions for comparing ES forecasts.

Version record

Basel Committee on Banking Supervision, "Minimum capital requirements for market risk"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: BCBS d457, January 2019; incorporated in the Basel Framework (MAR); national implementation dates vary.

Fundamental Review of the Trading Book (January 2019 standard): expected shortfall at 97.5%, liquidity horizons, desk-level P&L attribution and backtesting. Jurisdictional implementation dates differ.

Extended note. Version record

David Barrera, Stéphane Crépey, Emmanuel Gobet, Hoang Dong Nguyen and Bouazza Saadeddine, "Statistical Learning of Value-at-Risk and Expected Shortfall"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 36(1), 156–179 (2026); DOI 10.1111/mafi.70000.

Non-asymptotic learning guarantees for conditional VaR and ES under heavy tails using a two-step approach.

Version record

Tobias Fissler, Fangda Liu, Ruodu Wang and Linxiao Wei, "Elicitability and Identifiability of Tail Risk Measures"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 36(3), 451–464 (2026); DOI 10.1111/mafi.70016.

Joint identifiability and elicitability of tail risk measures (VaR, ES, range VaR), enabling consistent backtesting and forecast comparison.

Version record

Hubert Dichtl, Wolfgang Drobetz, Tizian Otto and Tatjana Puhan, "Bootstrapping and Bias: The Economic Costs of Misjudging Downside Risk"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, jpm.2026.063 (2026); DOI 10.3905/jpm.2026.063.

Compares bootstrap methods for estimating the distribution of maximum drawdown for stock/bond mixes; the stationary bootstrap is most precise, and biased estimates have quantifiable economic costs in risk-budgeting decisions.

Version record

6.4.2 Tail risk, hedging and stress testing

Vineer Bhansali, *Tail Risk Hedging: Creating Robust Portfolios for Volatile Markets*

Difficulty: Advanced **Type:** Book / monograph

Status/version: McGraw Hill, 2014 (ISBN 9780071791755).

Practitioner treatment of option-based and indirect tail hedges, their carry cost and monetisation rules.

Version record

Campbell R. Harvey, Edward Hoyle, Sandy Rattray, Matthew Sargaison, Dan Taylor and Otto Van Hemert, "The Best of Strategies for the Worst of Times: Can Portfolios Be Crisis Proofed?"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 45(5), 7–28 (2019); DOI 10.3905/jpm.2019.45.5.007.

Compares puts, Treasuries, gold, credit protection, trend following and quality factors as crisis hedges over drawdowns since 1985.

Version record

Bryan Kelly and Hao Jiang, "Tail Risk and Asset Prices"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(10), 2841–2871 (2014); DOI 10.1093/rfs/hhu039.

Panel-based estimate of time-varying common tail risk that predicts market returns and is priced in the cross-section.

Version record

Riccardo Rebonato, *Coherent Stress Testing: A Bayesian Approach to the Analysis of Financial Stress*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2012); DOI 10.1002/9781118374719.

Bayesian-network construction of stress scenarios with explicit causal links and probabilities, avoiding unconditioned historical replays.

Version record

Roni Israelov, "Pathetic Protection: The Elusive Benefits of Protective Puts"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Alternative Investments, 21(3), 6–33 (2018); DOI 10.3905/jai.2018.1.066.

Protective puts reduce drawdowns less efficiently than lower equity exposure once their cost is counted.

Version record

Antti Ilmanen, "Do Financial Markets Reward Buying or Selling Insurance and Lottery Tickets?"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 69(2) (2013); DOI 10.2469/faj.v69.n2.13.

Across markets, selling insurance-like payoffs earns a premium while buying lottery-like assets is penalised. Read with the 2013 Taleb comment and Ilmanen reply in the same journal (FAJ 69(2)) on tail-hedging critiques.

Extended note. Version record

Cecilia Parlatore and Thomas Philippon, "Designing Stress Scenarios"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(2), 833–873 (2025); DOI 10.1111/jofi.13422.

Optimal design of supervisory stress scenarios when a principal learns agents' unknown exposures from reported scenario losses; a Kalman-filter learning problem with implications for scenario choice and capital requirements.

Version record

Hervé Zumbach and Gilles Zumbach, "A Quantitative Approach to Historical Stress Tests"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 51(9), 234–255 (2025); DOI 10.3905/jpm.2025.1.742.

An objective method to identify and classify historical stress episodes (crash, crisis, rally, recovery) using an LM-ARCH volatility definition, so historical stress tests can be found exhaustively and compared quantitatively.

Version record

Guido Baltussen, Martin Martens and Lodewijk van der Linden, "The Best Defensive Strategies: Two Centuries of Evidence"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 82(1), 6–34 (2026); DOI 10.1080/0015198x.2025.2602270.

More than 220 years of evidence: low-risk, quality and value equity factors protect well in drawdowns, while gold and puts are less cost-effective; defensive absolute-return portfolios and trend following are the most effective, complementary hedges.

Version record

Min Cao and Thomas Conlon, "Tail Risk Hedging: The Superiority of the Naïve Hedging Strategy"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 45(8), 977–1005 (2025); DOI 10.1002/fut.22602.

Across asset classes the naive one-for-one futures hedge is hard to beat for tail-risk hedging, especially in low-growth periods, because optimised hedge ratios carry estimation error.

Version record

6.4.3 Systemic risk and fire sales

Tobias Adrian and Markus K. Brunnermeier, "CoVaR"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 106(7), 1705–1741 (2016); DOI 10.1257/aer.20120555.

Measures a firm's contribution to system tail risk as the change in system VaR conditional on the firm's distress.

Version record

Viral V. Acharya, Lasse H. Pedersen, Thomas Philippon and Matthew Richardson, "Measuring Systemic Risk"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 30(1), 2–47 (2017); DOI 10.1093/rfs/hhw088.

Systemic expected shortfall and marginal expected shortfall linking undercapitalisation in a crisis to market data.

Version record

Christian Brownlees and Robert F. Engle, "SRISK: A Conditional Capital Shortfall Measure of Systemic Risk"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 30(1), 48–79 (2017); DOI 10.1093/rfs/hhw060.

Expected capital shortfall in a market decline; published by the NYU V-Lab for global financials.

Version record

Monica Billio, Mila Getmansky, Andrew W. Lo and Loriana Pelizzon, "Econometric measures of connectedness and systemic risk in the finance and insurance sectors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 104(3), 535–559 (2012); DOI 10.1016/j.jfineco.2011.12.010.

Principal-components and Granger-causality networks across hedge funds, banks, brokers and insurers.

Version record

Larry Eisenberg and Thomas H. Noe, "Systemic Risk in Financial Systems"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 47(2), 236–249 (2001); DOI 10.1287/mnsc.47.2.236.9835.

Clearing-payment vector for interbank obligations; foundation for network contagion and default-cascade models.

Version record

Robin Greenwood, Augustin Landier and David Thesmar, "Vulnerable banks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 115(3), 471–485 (2015); DOI 10.1016/j.jfineco.2014.11.006.

Fire-sale spillovers when banks delever by selling common assets; a tractable stress-test model of indirect contagion.

Version record

Rama Cont and Eric Schaanning, "Fire sales, indirect contagion and systemic stress testing"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 2955646 (2017); DOI 10.2139/ssrn.2955646. Working-paper record; no journal version established in this check.

Price-mediated contagion through overlapping portfolios in stress tests; shows losses can exceed direct exposures.

Extended note. Version record

Paymon Khorrami and Fernando Mendo, "Dynamic Self-Fulfilling Fire Sales"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(5), 3087–3126 (2026); DOI 10.1111/jofi.70069.

In a Brunnermeier–Sannikov economy with fully hedgeable fundamentals, fire sales are inherently self-fulfilling coordination failures rather than consequences of fundamental shocks.

Version record

Sulkhan Chavleishvili and Manfred Kremer, "CISS of death: measuring financial crises in real time"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 29(3), 685–710 (2025); DOI 10.1093/rof/rfaf013.

The ECB Composite Indicator of Systemic Stress: a real-time, look-ahead-free financial-stress index that weights market stress indicators by their time-varying cross-correlations.

Version record

Lukas Gonon, Thilo Meyer-Brandis and Niklas Weber, "Computing Systemic Risk Measures with Graph Neural Networks"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 17(2), 565–619 (2026); DOI 10.1137/24m1697402.

Systemic risk measures with random allocations on Eisenberg–Noe networks, computed with permutation-equivariant graph neural networks.

Version record

6.4.4 Risk budgeting and volatility targeting

Attilio Meucci, *Risk and Asset Allocation*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2005

Estimation, simulation, dependence, allocation and scenario-based risk. Entropy pooling belongs to Meucci's broader work and is not a verified chapter attribution to this 2005 book.

Applications: Multi-asset; portfolio construction, risk. Version record

Wai Lee, *Risk-Based Asset Allocation: A New Answer to an Old Question?*, *Journal of Portfolio Management* 37(4), 11–28 (Summer 2011)

Difficulty: Research **Type:** Research paper

Status/version: Selected DOI-linked version; print 2011-07-31; online 2011-07-31.

Compares equal-weight, minimum-variance, maximum-diversification and risk-parity allocations through mean–variance assumptions. Risk-only allocation rules do not guarantee better returns; complements Qian's risk-budgeting perspective.

Version record

Ray Dalio, Principles

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Optional management and diversification perspective; no fixed number of return streams guarantees diversification.

Applications: Supplementary / conceptual. Version record

Bernd Scherer, *Portfolio Construction and Risk Budgeting*

Difficulty: Advanced **Type:** Book / monograph

Status/version: The linked publisher edition is dated 2015.

Portfolio estimation, diversification, risk parity, views, scenario analysis and transaction costs. The linked publisher edition is dated 2015. The PDF's "Bertrand & Lapin, 2002" attribution is incorrect; its intended allocation and risk-budgeting topics are retained here.

Applications: practitioner. Version record

Bridgewater, *The All Weather Story* (2012); Bob Prince, IPE interview (2011)

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Selected citation year(s): 2012, 2011. Version distinctions remain in the annotation.

Optional practitioner context on growth/inflation exposure balance and leverage. Economic-driver balance, equal dollar allocation and equal risk contribution are different constructions.

Version record

Edward Qian, *Risk Parity Portfolios: Efficient Portfolios Through True Diversification* (September 2005); Roger Clarke, Harindra de Silva and Steven Thorley, *Risk Parity, Maximum Diversification, and Minimum Variance: An Analytic Perspective* (2013)

Difficulty: Not specified **Type:** Mixed resources

Status/version: Selected citation year(s): 2005, 2013. Version distinctions remain in the annotation.

Distinct risk-based allocation references comparing contributions, covariance assumptions and leverage; risk balancing does not guarantee resilience in every regime.

Version record

Alan Moreira and Tyler Muir, *Volatility-Managed Portfolios*, *Journal of Finance* (2017)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected DOI-linked version; print 2017-08; online 2017-05-15.

Conditional risk scaling of factor portfolios; inverse-variance management and constant-volatility targeting are distinct rules.

Version record

Ray Dalio, *Engineering Targeted Returns and Risks* (August 2011)

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Selected citation year(s): 2011. Version distinctions remain in the annotation.

Practitioner framework separating alpha and beta and considering economic-environment exposures.

Version record

Campbell R. Harvey, Edward Hoyle, Russell Korgaonkar, Sandy Rattray, Matthew Sargaison and Otto Van Hemert, "The Impact of Volatility Targeting"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 45(1), 14–33 (2018); DOI 10.3905/jpm.2018.45.1.014.

Volatility targeting reduces left-tail events and improves Sharpe ratios for risk assets, but not for bonds or currencies.

Version record

Scott Cederburg, Michael S. O'Doherty, Feifei Wang and Xuemin (Sterling) Yan, "On the performance of volatility-managed portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 138(1), 95–117 (2020); DOI 10.1016/j.jfineco.2020.04.015.

Counter-evidence to Moreira–Muir: across 103 strategies, volatility-managed portfolios do not systematically beat unmanaged ones out of sample.

Extended note. Version record

Benjamin T. Hood and Cameron Raughtigan, "Volatility Targeting Is Trendy: How Trend Following Explains Alpha in Volatility-Managed Strategies"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(1), 100–121 (2025); DOI 10.3905/jpm.2025.1.764.

Volatility-targeting alpha in equities is mostly an implicit trend-following loading created by the leverage effect; in bonds, currencies and commodities, where the leverage effect is absent, the link disappears.

Version record

Alessio de Longis, David Happersberger, Scott Hixon, Han Liang and Ingmar Nolte, "Regime-Aware Risk Parity: Conditioning the Covariance Matrix on Macroeconomic and Stock Market Regimes"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(5), 51–76 (2026); DOI 10.3905/jpm.2026.1.818.

Regime-conditional covariance matrices built from macroeconomic data and the VIX term structure give risk-parity portfolios tighter ex post risk control and better out-of-sample performance across regimes.

Version record

Tim Bollerslev, Benjamin Hood, John Huss and Lasse Heje Pedersen, "Risk Everywhere: Modeling and Managing Volatility"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 31(7), 2729–2773 (2018); DOI 10.1093/rfs/hhy041.

High-frequency data for more than fifty commodities, currencies, equity indices and fixed-income instruments show strong similarities in realised-volatility patterns within and across asset classes. Panel realised-volatility models that exploit them forecast risk better out of sample, a utility-based evaluation shows economic gains, and the paper assesses the trading costs and speed of implementing each risk model.

Version record

Dion Bongaerts, Xiaowei Kang and Mathijs van Dijk, "Conditional Volatility Targeting"

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal, 76(4), 54–71 (2020); DOI 10.1080/0015198x.2020.1790853.

Conventional volatility targeting does not consistently improve performance in global equity markets and can deepen drawdowns. Changing exposure only in extreme high- and low-volatility states raises Sharpe ratios and reduces drawdowns and tail risk, with low turnover and leverage, in major equity markets and momentum factors, and can be applied to tactical multi-asset allocation.

Version record

Victor DeMiguel, Alberto Martín-Utrera and Raman Uppal, "A Multifactor Perspective on Volatility-Managed Portfolios"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 79(6), 3859–3891 (2024); DOI 10.1111/jofi.13395.

Single-factor volatility-managed portfolios fail out of sample (Cederburg et al.) and after trading costs (Barroso and Detzel). A conditional multifactor portfolio that scales exposures with volatility beats its unconditional counterpart out of sample and net of costs, and factor risk prices generally fall when market volatility rises.

Version record

6.5 Performance measurement, attribution and manager evaluation

Separate return calculation, attribution, risk adjustment and skill inference. Smoothed returns, survivorship and luck can make an ordinary record look exceptional.

William F. Sharpe, "The Sharpe Ratio"

Difficulty: Intro **Type:** Research paper

Status/version: The Journal of Portfolio Management, 21(1), 49–58 (1994); DOI 10.3905/jpm.1994.409501.

Sharpe's own restatement of the reward-to-variability ratio and its proper use relative to a benchmark.

Version record

Gary P. Brinson, L. Randolph Hood and Gilbert L. Beebower, "Determinants of Portfolio Performance"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 42(4), 39–44 (1986); DOI 10.2469/faj.v42.n4.39.

Allocation, selection and interaction attribution of pension-fund returns; foundation of Brinson attribution.

Version record

William F. Sharpe, "Asset allocation: Management style and performance measurement"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 18(2), 7–19 (1992); DOI 10.3905/jpm.1992.409394.

Returns-based style analysis with constrained regression on asset-class indices.

Version record

Mark M. Carhart, "On Persistence in Mutual Fund Performance"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 52(1), 57–82 (1997); DOI 10.1111/j.1540-6261.1997.tb03808.x.

Four-factor model with momentum; shows most persistence in fund returns is explained by common factors and costs.

Version record

Eugene F. Fama and Kenneth R. French, "Luck versus Skill in the Cross-Section of Mutual Fund Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 65(5), 1915–1947 (2010); DOI 10.1111/j.1540-6261.2010.01598.x.

Bootstrap simulation of fund alphas under zero true alpha to separate skill from luck.

Version record

Jonathan B. Berk and Richard C. Green, "Mutual Fund Flows and Performance in Rational Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 112(6), 1269–1295 (2004); DOI 10.1086/424739.

Decreasing returns to scale and competitive flows explain lack of persistence even with skilled managers.

Version record

Jonathan B. Berk and Jules H. van Binsbergen, "Measuring skill in the mutual fund industry"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 118(1), 1–20 (2015); DOI 10.1016/j.jfineco.2015.05.002.

Measures skill as value added in dollars rather than percentage alpha; skill persists for up to ten years.

Version record

Mila Getmansky, Andrew W. Lo and Igor Makarov, "An econometric model of serial correlation and illiquidity in hedge fund returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 74(3), 529–609 (2004); DOI 10.1016/j.jfineco.2004.04.001.

Return smoothing creates serial correlation that understates volatility and inflates Sharpe ratios; provides an unsmoothing correction for hedge funds and private assets.

Version record

Vikas Agarwal and Narayan Y. Naik, "Risks and Portfolio Decisions Involving Hedge Funds"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 17(1), 63–98 (2004); DOI 10.1093/rfs/hhg044.

Option-based risk factors capture the nonlinear, short-put-like payoffs of many hedge-fund strategies.

Version record

Rajesh K. Aggarwal and Philippe Jorion, "Hidden Survivorship in Hedge Fund Returns"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 66(2) (2010); DOI 10.2469/faj.v66.n2.test.

Survivorship bias hidden in multi-fund families and database merges inflates reported hedge-fund returns.

Version record

Carl R. Bacon, *Practical Portfolio Performance Measurement and Attribution*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2023); DOI 10.1002/9781119831976.

Return calculation, risk-adjusted measures, arithmetic and geometric attribution, fixed-income attribution and GIPS context. Check for the latest edition.

Version record

CFA Institute, "Global Investment Performance Standards (GIPS) for Firms, 2020 edition"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: GIPS 2020 standards, effective 1 January 2020; guidance statements and Q&A updated separately.

Voluntary standard for composite construction, return calculation, pooled-fund reporting and disclosure; relevant to presenting institutional track records.

Extended note. Version record

James Brugler, Marta Khomyn and Tālis J. Putniņš, "Benchmarking benchmarks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 168, 104018 (2025); DOI 10.1016/j.jfineco.2025.104018.

State-space separation of information and noise in reference rates: alternative risk-free rates are less noisy than LIBOR in most currencies, but USD SOFR is noisier.

Version record

Feng Jiao, Sergei Sarkissian and David Schumacher, "Liquidity picking and fund performance"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 170, 104085 (2025); DOI 10.1016/j.jfineco.2025.104085.

Skilled mutual funds trade cross-listed firms in their most liquid venue, consistent with informed trading.

Version record

Shiyang Huang, Xu Lu, Yang Song and Hong Xiang, "Remeasuring Scale in Active Management"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag037 (2026); DOI 10.1093/rfs/hhag037.

Including institutional twin vehicles shows active-fund scale is understated by at least 65%, changing estimates of returns to scale.

Version record

Yong Chen, Sophia Zhengzi Li, Yushan Tang and Guofu Zhou, "Anomalies as New Hedge Fund Factors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(8), 3660–3693 (2025); DOI 10.1017/s0022109025101270.

A nine-factor hedge-fund model including five anomaly factors outperforms existing hedge-fund factor models in and out of sample.

Version record

Bruce J. Feibel, "Rethinking Performance Attribution in the Age of Systematic Investing and AI"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, jpm.2026.069 (2026); DOI 10.3905/jpm.2026.069.

How to supplement Brinson attribution with exposure, factor, risk, decision and model-explainability analysis for portfolios using derivatives, systematic rules and AI, and how to report what attribution cannot establish.

Version record

Christos Antoniadis and Spyros Skouras, "A Reassessment of Hedge Fund Returns Using Daily Return Data"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 82(2), 159–185 (2026); DOI 10.1080/0015198x.2026.2640981.

Daily hedge-fund returns reveal more factor exposures, more time variation and lower alphas than monthly data suggest, including a commodity-put factor and liquidity-timing effects.

Version record

6.6 Climate risk and sustainable investing

Climate exposure, ESG ratings and green premia are measured with disputed data. Read equilibrium theory, empirical premia, counter-evidence and scenario design together; disclosure rules and scenario vintages change.

Stefano Giglio, Bryan Kelly and Johannes Stroebel, "Climate Finance"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 13(1), 15–36 (2021); DOI 10.1146/annurev-financial-102620-103311.

Survey of how climate risk enters macro-finance models and is priced in real estate, equities and fixed income, with hedging-portfolio approaches.

Version record

Patrick Bolton and Marcin Kacperczyk, "Do investors care about carbon risk?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 142(2), 517–549 (2021); DOI 10.1016/j.jfineco.2021.05.008.

Finds higher returns for firms with higher emission levels and changes, interpreted as a carbon-risk premium.

Version record

Jitendra Aswani, Aneesh Raghunandan and Shivaram Rajgopal, "Are Carbon Emissions Associated with Stock Returns?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 28(1), 75–106 (2024); DOI 10.1093/rof/rfad013.

Counter-evidence: the carbon premium is driven by vendor-estimated emissions and scaling choices and disappears for disclosed, unscaled emissions.

Extended note. Version record

Ľuboš Pástor, Robert F. Stambaugh and Lucian A. Taylor, "Sustainable investing in equilibrium"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 142(2), 550–571 (2021); DOI 10.1016/j.jfineco.2020.12.011.

Green assets have lower expected returns in equilibrium but can outperform when climate concerns rise unexpectedly.

Version record

Ľuboš Pástor, Robert F. Stambaugh and Lucian A. Taylor, "Dissecting green returns"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2021); Journal of Financial Economics publication (2022) reported by the authors, DOI not established in this check.

Attributes recent green outperformance to unanticipated increases in climate concern rather than higher expected returns.

Extended note. Version record

Robert F. Engle, Stefano Giglio, Bryan Kelly, Heebum Lee and Johannes Stroebel, "Hedging Climate Change News"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(3), 1184–1216 (2020); DOI 10.1093/rfs/hhz072.

Builds climate-news indices from text and mimicking portfolios that hedge climate-news innovations out of sample.

Version record

Florian Berg, Julian F. Kölbel and Roberto Rigobon, "Aggregate Confusion: The Divergence of ESG Ratings"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Finance, 26(6), 1315–1344 (2022); DOI 10.1093/rof/rfac033.

Decomposes ESG rating divergence into scope, measurement and weight; rater effects mean ratings are not interchangeable signals.

Version record

Philipp Krueger, Zacharias Sautner and Laura T. Starks, "The Importance of Climate Risks for Institutional Investors"

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Financial Studies, 33(3), 1067–1111 (2020); DOI 10.1093/rfs/hhz137.

Survey evidence on how institutions perceive and manage regulatory and physical climate risks.

Version record

Zacharias Sautner, Laurence van Lent, Grigory Vilkov and Ruishen Zhang, "Firm-Level Climate Change Exposure"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 78(3), 1449–1498 (2023); DOI 10.1111/jofi.13219.

Earnings-call text measures of opportunity, regulatory and physical climate exposure, with public data.

Version record

Peter Tankov and Ruixun Zhang (eds.), *Handbook of Quantitative Sustainable Finance*

Difficulty: Advanced/Research **Type:** Book / edited research collection

Status/version: Chapman & Hall/CRC Financial Mathematics Series, 12 December 2025; ISBN 9781032627922 (print), 9781032636252 (ebook).

CRC (December 2025) edited volume on quantitative climate and ESG modelling, scenarios, pricing and portfolio construction. Chapter quality and maturity vary.

Version record

Riccardo Rebonato, *Climate Scenarios for Financial Modelling*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge Elements in Quantitative Finance; Cambridge page listed expected online publication 10 August 2026; retailer lists paperback (ISBN 9781047777872) dated 30 September 2026.

Cambridge Element arguing that current climate scenarios underweight outcome uncertainty and physical risk and lack the probabilities needed for valuation; shows a mechanically negative climate risk premium under specific conditions.

Extended note. Version record

Network for Greening the Financial System, "NGFS Climate Scenarios for central banks and supervisors"

Difficulty: Advanced **Type:** Data / specification

Status/version: Phase V (November 2024); later phases may supersede; scenario vintage must be recorded.

Orderly, disorderly and hot-house scenario narratives with transition and physical-risk variables used in supervisory climate stress tests; vintages change between phases.

Version record

International Sustainability Standards Board, "IFRS S2 Climate-related Disclosures"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Issued June 2023; effective for periods from 1 January 2024 where adopted.

Disclosure standard for climate risks and opportunities, including scenario analysis and Scope 1–3 emissions; adoption is jurisdiction-specific.

Extended note. Version record

Mathias S. Kruttli, Brigitte Roth Tran and Sumudu W. Watugala, "Pricing Poseidon: Extreme Weather Uncertainty and Firm Return Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(2), 783–832 (2025); DOI 10.1111/jofi.13416.

Hurricanes raise implied volatility for exposed firms before and after landfall; volatility-premium dynamics suggest investors underestimate extreme-weather uncertainty.

Version record

Philippe van der Beck, "Flow-Driven ESG Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, jofi.70083 (2026); DOI 10.1111/jofi.70083.

Recent ESG outperformance is largely price impact from flows into ESG portfolios; estimated demand elasticities imply each dollar of flow raises ESG prices substantially. Distinguish realised from expected returns.

Version record

Ehsan Azarmsa and Joel Shapiro, "The Market for ESG Ratings"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, jofi.70078 (2026); DOI 10.1111/jofi.70078.

Model of competing ESG raters: specialisation and aggregation incentives explain rating divergence and its welfare effects.

Version record

Lasse Heje Pedersen, "Carbon Pricing versus Green Finance"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(2), 561–602 (2026); DOI 10.1111/jofi.70022.

Unified model: green finance should not be used when carbon prices reflect social cost, but can partially substitute when carbon prices are too low; quantifies the trade-offs.

Version record

Gianpaolo Parise and Mirco Rubin, "Green Window Dressing"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(6), 3555–3588 (2025); DOI 10.1111/jofi.13499.

Mutual funds time ESG trades around disclosure dates to inflate sustainability ratings; post-disclosure reversals reveal window dressing.

Version record

Hyeyoon Jung, Robert F. Engle and Richard Berner, "CRISK: Measuring the climate risk exposure of the financial system"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 171, 104076 (2025); DOI 10.1016/j.jfineco.2025.104076.

CRISK: a measure of financial institutions' expected capital shortfall under climate stress, extending systemic-risk capital-shortfall methods to climate risk.

Extended note. Version record

Andrea Macrina and Peter Tankov, "Climate and Nature Risk in Mathematical Finance"

Difficulty: Research **Type:** Report / research note

Status/version: Mathematical Finance, 36(4), 623–625 (2026); DOI 10.1111/mafi.70059.

Special-issue editorial mapping how stochastic modelling, control, games and equilibrium contribute to climate and nature risk.

Version record

Joop Huij, Dries Laurs, Philip Stork and Remco C. J. Zwinkels, "Carbon Beta: A Market-Based Measure of Climate Transition Risk Exposure"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 82(3), 111–137 (2026); DOI 10.1080/0015198x.2026.2652222.

Carbon beta: a market-based measure of a firm's climate transition-risk exposure estimated from returns rather than disclosed emissions.

Extended note. Version record

Tobias Bauckloh and Victor Beyer, "Non-standard errors in carbon premia"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 189, 107727 (2026); DOI 10.1016/j.jbankfin.2026.107727.

Across 100,000+ portfolio construction paths, the carbon premium averages -0.16% per month with a non-standard error of 0.26%; proxy, weighting and double-sorting choices drive the dispersion.

Version record

Abraham Lioui and Sanjay Misra, "Which carbon factor?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 188, 107699 (2026); DOI 10.1016/j.jbankfin.2026.107699.

Compares candidate carbon factors.

Extended note. Version record

6.7 Asset–liability management

Subject context and qualifications.

William F. Sharpe and Lawrence G. Tint, Liabilities—A New Approach

Difficulty: Advanced **Type:** Research paper

Status/version: *Journal of Portfolio Management* 16(2), 5–10, Winter 1990; version-of-record DOI, subscription access may apply.

Places the liability in portfolio choice and asks what risk is measured relative to the obligation rather than only relative to cash. Define benefit cash flows and a liability valuation basis before estimating surplus covariance. Its early inflation-linked-security setting is not a modern pension rule or a full stochastic cash-flow model; benefits, contribution options and collateral needs need separate treatment.

Extended note. Version record

Roy P. M. M. Hoevenaars, Roderick D. J. Molenaar, Peter C. Schotman and Tom B. M. Steenkamp, Strategic Asset Allocation with Liabilities: Beyond Stocks and Bonds

Difficulty: Research **Type:** Research paper

Status/version: *Journal of Economic Dynamics and Control* 32(9), 2939–2970, September 2008; published article, author-university record.

Models inflation and real-rate risk in pension liabilities and estimates intertemporal covariance among liabilities, bonds, equities and alternatives. It separates the global minimum-variance portfolio from a liability hedge, then shows why asset-only and surplus-optimal allocations differ. Read after Sharpe–Tint and the long-horizon allocation texts. Estimated VAR dynamics and pre-2008 market data are model and sample dependent; real-world liquidity, leverage and collateral may make a statistical hedge infeasible.

Extended note. Version record

Robert Novy-Marx and Joshua D. Rauh, Policy Options for State Pension Systems and Their Impact on Plan Liabilities

Difficulty: Research **Type:** Research paper

Status/version: *Journal of Pension Economics and Finance* 10(2), 173–194, 2011; NBER WP 16453 gives abstract and published-version record.

Revalues US state pension cash flows under different benefit definitions and municipal or Treasury discount curves, showing large sensitivity to both the promised-benefit perimeter and discount basis. For public plans, retain ABO versus broader projected-benefit definitions, COLAs, mortality and fiscal support in every scenario. Numerical estimates concern June 2009 and cannot be used as current deficits; Treasury discounting also imports Treasury liquidity and inflation-premium features.

Extended note. Version record

The Pensions Regulator, Using Leveraged Liability-Driven Investment

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: UK regulator guidance replacing its 2022 statements; living page reviewed 4 October 2026.

Turns liability-rate hedging into collateral operations: eligible assets, operational and market-stress buffers, dealing cycles, replenishment and trustee governance. The page's illustrative buffers are regulatory guidance for leveraged LDI, not a prescribed international hedge ratio; investor account, counterparties and eligible collateral still govern calls.

Extended note. Version record

Gabor Pinter, Emil Siriwardane and Danny Walker, Fire Sales of Safe Assets

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1233, 6 December 2024; public PDF. An author reports a forthcoming *Journal of Financial Economics* version; no journal issue citation verified by 4 October 2026.

Uses UK gilt trades around the September–October 2022 LDI disruption to identify price pressure from forced selling and frictions transferring cash from solvent pension schemes to pooled LDI vehicles. Stress intragroup and pooled-fund transfer time, market depth and variation margin together. The paper's episode-specific price effects are not a standing gilt haircut or evidence that every LDI scheme was insolvent.

Extended note. Version record

Bank of England, System-Wide Exploratory Scenario Exercise: Final Report

Difficulty: Advanced **Type:** Report / research note

Status/version: November 2024 final report; public central-bank scenario, Annex 4 on DB pensions and LDI.

Traces the feedback from rates, repo and derivatives margin through LDI recapitalization, gilt sales, banks' intermediation capacity and FX hedges. Build one dated cash ladder across all demands and identify who can mobilize assets within five days. Participant actions are responses to a designed scenario, not a forecast; the report also documents improved buffers since 2022 and should not be read as unchanged crisis conditions.

Extended note. Version record

IFRS Foundation, IFRS 17, Insurance Contracts, especially B74–B78

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Issued standard text, 2022 hosted PDF; check current amendments before regulatory or accounting implementation.

Requires rates appropriate to the timing, currency, inflation and liquidity characteristics of insurance cash flows; asset return premiums enter only where the liability itself varies with those returns. Use for an insurer's asset–liability key-rate and inflation map, with guarantee and policyholder-option scenarios. IFRS reporting and economic hedging are related but not identical, and this 2022 PDF is not certification of every later amendment or local capital regime.

Extended note. Version record

European Insurance and Occupational Pensions Authority, Updated Risk-Free-Rate Technical Documentation and calculation examples

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Released 26 May 2026; revised method scheduled to apply 30 January 2027, not yet operative at this cut-off.

Explains forthcoming Solvency II curve interpolation, extrapolation, risk-corrected spreads and volatility adjustment, with spreadsheet examples. Compare a long insurer cash-flow vector under the currently published monthly curve and the prospective method to isolate methodology risk from bond-market movement. For 4 October 2026 use the contemporaneously applicable EIOPA monthly curve and methodology; the 2027 document is forward preparation. Local insurer portfolio and matching-adjustment eligibility need separate review.

Extended note. Version record

Jeffrey R. Brown, Stephen G. Dimmock, Jun-Koo Kang and Scott J. Weisbenner, How University Endowments Respond to Financial Market Shocks: Evidence and Implications

Difficulty: Research **Type:** Research paper

Status/version: *American Economic Review* 104(3), 931–962, March 2014; published article, free publisher PDF.

Finds asymmetric payout behavior after negative endowment shocks and links cuts to university operations. Read after private-fund pacing: an endowment's liability is a spending and mission policy that can adjust, but reducing distributions has real consequences. Model spending smoothing, restricted funds and emergency operating needs alongside portfolio liquidity. Historical university behavior and policy discretion do not define every foundation's legal payout or a necessary future response.

Extended note. Version record

International Monetary Fund, Sovereign Asset-Liability Management: Guidance for Resource-Rich Economies

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: IMF Policy Paper, 11 June 2014; final institutional guidance.

Separates stabilization, intergenerational savings, development, pension-reserve and reserve-investment purposes. The same asset mix cannot serve all: fiscal commodity shocks may require short-notice withdrawals while distant pension liabilities permit longer investments. Policy guidance is not a sovereign's legal mandate or proof that an illiquid investment is accessible during a budget shock; country budgets, currency needs and political decisions matter.

Extended note. Version record

Ding Ding, Xiang Fang, Bryan Hardy and Karen K. Lewis, Global Pension Asset Allocations and Debt Markets

Difficulty: Research **Type:** Research paper

Status/version: BIS Paper 172, 3 June 2026; public 23-page research paper.

Documents global shifts away from direct fixed income toward mutual funds and alternatives and studies their association with local government yields. Read after the ALM foundation as evidence that actual pension portfolios and bond-market demand have changed, then decompose a plan's debt exposure through pooled and private holdings. Classification across countries and fund wrappers can obscure duration, leverage, credit and liquidity; reconstruct those exposures for a total-fund stress.

Extended note. Version record

David F. Swensen, *Pioneering Portfolio Management: An Unconventional Approach to Institutional Investment*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Free Press, fully revised and updated edition, 2009 (ISBN 9781416544692).

Yale endowment model: equity orientation, illiquidity tolerance, manager selection and rebalancing; revised edition 2009. Read with evidence on endowment returns.

Version record

Itay Goldstein, Hao Jiang and David T. Ng, "Investor flows and fragility in corporate bond funds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 126(3), 592–613 (2017); DOI 10.1016/j.jfineco.2016.11.007.

Concave flow-performance in illiquid bond funds creates run-like fragility; motivates swing pricing and liquidity buffers.

Version record

Dunhong Jin, Marcin Kacperczyk, Bige Kahraman and Felix Suntheim, "Swing Pricing and Fragility in Open-End Mutual Funds"

Difficulty: Research **Type:** Research paper

Status/version: IMF Working Paper 19/227 (2019) record; later journal publication not established in this check.

UK evidence that swing pricing reduces outflows during stress by passing liquidation costs to redeeming investors.

Extended note. Version record

Qi Chen, Itay Goldstein and Wei Jiang, "Payoff complementarities and financial fragility: Evidence from mutual fund outflows"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 97(2), 239–262 (2010); DOI 10.1016/j.jfineco.2010.03.016.

Funds holding illiquid assets show stronger outflow sensitivity to bad performance, consistent with strategic complementarities.

Version record

Yiming Ma, Kairong Xiao and Yao Zeng, "Bank Debt, Mutual Fund Equity, and Swing Pricing in Liquidity Provision"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhaf105 (2025); DOI 10.1093/rfs/hhaf105.

Mutual funds issuing demandable equity provide material liquidity insurance; swing pricing affects that liquidity provision. Bond funds supply about 12.5% of bank-provided liquidity.

Version record

Kristy A. E. Jansen, Sven Klingler, Angelo Ranaldo and Patty Duijm, "Pension Liquidity Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag068 (2026); DOI 10.1093/rfs/hhag068.

Dutch pension funds with weaker funding ratios use interest-rate swaps more aggressively, exposing them to margin calls that can exceed 6% of assets.

Version record

Ziang Li, "Long Rates, Life Insurers, and Credit Spreads"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag079 (2026); DOI 10.1093/rfs/hhag079.

With negative duration gaps, higher long rates raise life insurers' equity and risk-bearing capacity, lowering credit spreads.

Version record

Redouane Elkamhi and Jacky S. H. Lee, "Total Portfolio Approach"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 51(8), 109–127 (2025); DOI 10.3905/jpm.2025.1.728.

Defines the total portfolio approach as a governance mindset rather than a strategy, contrasts it with strategic asset allocation, and sets out its theoretical basis and implementation toolkits for asset owners.

Version record

Chapter 7

Derivatives valuation, volatility and hedging

Read valuation assumptions alongside surface constraints, numerical approximations and hedge outcomes. Recent learning methods complement model and market-convention checks.

7.1 Pricing and numerical foundations

Related references: Sinclair, Gatheral, Heston, Bergomi and rough volatility.

7.1.1 Textbooks, practice and market data

Tomas Björk, *Arbitrage Theory in Continuous Time*, fourth edition

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2019

A rigorous yet coherent route through martingales, change of measure, term-structure models, incomplete markets and optimal control. It is a stronger theoretical foundation for derivatives research than collections of product-specific formulas. [9]

Applications: All derivatives, rates, credit. Version record

Jim Gatheral, *The Volatility Surface*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2006

The standard practitioner text on implied volatility, local volatility, stochastic volatility, jumps and surface parameterisations. It remains essential for understanding what an arbitrage-free surface must encode before applying machine learning to calibration or interpolation. [10]

Applications: Equity, FX and commodity options; volatility. Version record

Lorenzo Bergomi, *Stochastic Volatility Modeling*

Difficulty: Research **Type:** Book / monograph

Status/version: Routledge/Chapman & Hall first edition, 5 January 2016; publisher identifies a single Lorenzo Bergomi monograph.

Forward-variance-centred treatment of smile dynamics, forward-start options, cliquets, volatility swaps and exotic hedging. This is a single monograph.

Applications: Equity derivatives, volatility products, exotics. Version record

Rama Cont and Peter Tankov, *Financial Modelling with Jump Processes*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2004

The standard reference for Lévy processes, jump models, calibration and incomplete-market hedging. It supplies the mathematical tools needed to model gaps, event risk and short-maturity smile behaviour that continuous diffusions miss.

Applications: Options, credit, event risk. Version record

Cboe DataShop, Option Quote Intervals; Option EOD Summary; data conventions FAQ

Difficulty: Advanced **Type:** Data / specification

Status/version: Living purchasable OPRA option-quote products; product fields and conventions inspected 4 October 2026.

Provides a specified route to historical SPX option NBBO bid/ask and size snapshots by series, timestamp, strike, expiry and type, with one-minute or custom intervals and EOD alternatives. Cboe advertises historical purchases from January 2012 on the current product pages and downloadable samples. For SPX, underlying index bid/ask and active underlying price require a Cboe Global Indices Feed license; the option quote purchase by itself does not supply a matched underlying quote. The FAQ distinguishes quote snapshots from interval trades and notes AM-expiry zeros.

Extended note. Version record

Marcus Overhaus, Ana Bermúdez, Hans Buehler, Andrew Ferraris, Christopher Jordinson and Aziz Lamnouar, *Equity Hybrid Derivatives*

Difficulty: Research **Type:** Book / monograph

Status/version: 2007

A desk-level treatment of hybrid equity–rates–FX products, local/stochastic volatility, correlation and numerical implementation. Despite its age, it remains one of the few public references organised around the actual modelling demands of structured-equity books.

Applications: Equity hybrids, convertibles, structured products. Version record

John C. Hull, *Options, Futures, and Other Derivatives*, 11th Global ed. (published 2021; copyright 2022)

Difficulty: Intro **Type:** Book / monograph

Status/version: Selected 11th Global edition paperback published 5 July 2021, copyright 2022; ISBN 9781292410654. Preserve Global-edition pagination and matching author errata.

Derivatives mechanics, Black–Scholes, Greeks, implied versus historical volatility and basic exotics. This explicitly selected Global edition has different pagination from the US edition; use the matching author errata.

Version record

Sheldon Natenberg, *Option Volatility & Pricing: Advanced Trading Strategies and Techniques*, 2nd ed. (published 2014; copyright 2015)

Difficulty: Intro **Type:** Book / monograph

Status/version: 2nd ed.; published 2014, copyright 2015.

Market-maker introduction to option valuation, volatility, Greeks, spreads, straddles and dynamic hedging. Builds the practical intuition needed to interpret model values and manage option-position risks.

Version record

Nassim Nicholas Taleb, Dynamic Hedging (1997)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 1997. Version distinctions remain in the annotation.

Option-portfolio rehedging under smile changes, liquidity limits, leverage and discontinuous moves; fitted Greeks are incomplete protection against tail risk.

Version record

Lawrence G. McMillan, *Options as a Strategic Investment*, 5th ed. (Prentice Hall Press, August 2012)

Difficulty: Intro **Type:** Book / monograph

Status/version: 5th ed. (Prentice Hall Press, August 2012)

Supplementary encyclopaedia of payoff structures, including calls, puts, spreads, straddles, iron condors and ratio spreads. Use payoff diagrams and scenario analysis to communicate with traders and structurers; a catalogue of structures is not evidence of a systematic edge.

Version record

Emanuel Derman and Michael B. Miller, The Volatility Smile (2016)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 2016. Version distinctions remain in the annotation.

Connects the limitations of Black–Scholes to observed smiles and skews, local volatility, stochastic volatility and jumps. Compare models fitted to the same prices by their dynamic hedging and extrapolation behaviour, especially for structured derivatives.

Version record

***Frank J. Fabozzi, ed., Handbook of Structured Financial Products (September 1998)*; book record / preview**

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 1998. Version distinctions remain in the annotation.

Collected practitioner reference on securitisation, including asset-backed and mortgage-backed instruments and collateralised loan/bond obligations. Retained for foundational cash-flow engineering and historical context; conventions and market practices require an edition-aware update.

Version record

Paul Wilmott, *Paul Wilmott on Quantitative Finance*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, second edition, 2006 (three-volume set; ISBN 9780470018705); the former Wiley product page no longer resolves.

Second edition (2006, three volumes): PDE-centred derivatives modelling, numerical methods and practitioner commentary on model risk.

Version record

7.1.2 Classic pricing papers

Fischer Black and Myron Scholes, "The Pricing of Options and Corporate Liabilities"

Difficulty: Advanced **Type:** Research paper

Status/version: 1973

Replication and no-arbitrage lead to the option-pricing PDE and its valuation representation. Use the original model to locate failure points: discrete rehedging, costs, jumps, stochastic volatility and smile risk.

Applications: Options; pricing, hedging. Version record

Steven L. Heston, "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options"

Difficulty: Advanced **Type:** Research paper

Status/version: 1993

Characteristic functions make stochastic-volatility pricing computationally tractable. Connect smile/skew dynamics to parameter estimation and calibration rather than treating the stochastic differential equations as a complete implementation.

Applications: Equity, rates, FX options; calibration. Version record

Darrell Duffie, Jun Pan and Kenneth Singleton, "Transform Analysis and Asset Pricing for Affine Jump-Diffusions"; author PDF

Difficulty: Research **Type:** Research paper

Status/version: 2000

Transform methods for affine jump-diffusions support pricing with stochastic volatility, jumps and interest rates. Connect model specifications to tractable conditional transforms and Fourier valuation. *Econometrica* 68(6), 1343–1376 (2000); preceded by NBER Working Paper 7105 (1999).

Applications: Options, credit, affine models. Version record

Peter Carr and Dilip Madan, "Option Valuation Using the Fast Fourier Transform"

Difficulty: Advanced **Type:** Research paper

Status/version: 1999

Fast Fourier methods convert a characteristic function into efficient option prices across strikes. Study damping, integration and numerical error when building calibration routines; efficient pricing must remain consistent with the model and payoff conventions.

Applications: Options; numerical pricing. Version record

Robert C. Merton, *Theory of Rational Option Pricing*, *Bell Journal of Economics and Management Science* 4(1), 141–183 (1973)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 1973. Version distinctions remain in the annotation.

No-arbitrage option-pricing foundations; distinct from the Black–Scholes paper.

Version record

J. Michael Harrison and David M. Kreps, "Martingales and arbitrage in multiperiod securities markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Economic Theory, 20(3), 381–408 (1979); DOI 10.1016/0022-0531(79)90043-7.

Links absence of arbitrage to equivalent martingale measures; the starting point of risk-neutral pricing theory.

Version record

J. Michael Harrison and Stanley R. Pliska, "Martingales and stochastic integrals in the theory of continuous trading"

Difficulty: Research **Type:** Research paper

Status/version: Stochastic Processes and their Applications, 11(3), 215–260 (1981); DOI 10.1016/0304-4149(81)90026-0.

Continuous-time formulation of self-financing strategies, completeness and martingale pricing.

Version record

John C. Cox, Stephen A. Ross and Mark Rubinstein, "Option pricing: A simplified approach"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 7(3), 229–263 (1979); DOI 10.1016/0304-405x(79)90015-1.

Binomial tree pricing and its convergence to Black–Scholes; remains the standard teaching model and early-exercise workhorse.

Version record

Douglas T. Breeden and Robert H. Litzenberger, "Prices of State-Contingent Claims Implicit in Option Prices"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Business, 51(4), 621 (1978); DOI 10.1086/296025.

Second strike-derivative of call prices recovers the risk-neutral density; basis for implied-density and variance-swap replication.

Version record

Robert C. Merton, "Option pricing when underlying stock returns are discontinuous"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 3(1-2), 125–144 (1976); DOI 10.1016/0304-405x(76)90022-2.

Jump-diffusion option pricing with diversifiable jump risk; the original jump model behind short-dated smile explanations.

Version record

Bruno Dupire, "Pricing with a Smile"

Difficulty: Advanced **Type:** Research paper

Status/version: Risk 7(1), 18–20 (January 1994); not openly hosted, and the former Risk.net article URL no longer resolves.

Local-volatility function implied by the full call surface (the Dupire equation). Published in Risk (January 1994); access is through Risk.net or reprints.

Version record

Patrick S. Hagan, Deep Kumar, Andrew S. Lesniewski and Diana E. Woodward, "Managing Smile Risk"

Difficulty: Advanced **Type:** Research paper

Status/version: Wilmott Magazine, September 2002, 84–108.

SABR model and its implied-volatility asymptotic formula; market standard for swaption and cap smiles, with known low-strike arbitrage issues.

Version record

David S. Bates, "Jumps and Stochastic Volatility: Exchange Rate Processes Implicit in Deutsche Mark Options"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 9(1), 69–107 (1996); DOI 10.1093/rfs/9.1.69.

Stochastic volatility with jumps (SVJ) fitted to currency options; the standard Heston-plus-jumps extension.

Version record

S. G. Kou, "A Jump-Diffusion Model for Option Pricing"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 48(8), 1086–1101 (2002); DOI 10.1287/mnsc.48.8.1086.166.

Proposes a double-exponential jump-diffusion that captures fat, asymmetric return tails and the volatility smile while remaining simple enough for analytical prices of calls, puts, interest-rate derivatives and path-dependent options.

Version record

Peter Carr, Hélyette Geman, Dilip B. Madan and Marc Yor, "The Fine Structure of Asset Returns: An Empirical Investigation"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Business, 75(2), 305–333 (2002); DOI 10.1086/338705.

Introduces the CGMY process, whose jump component can have finite or infinite activity and variation. Time-series and option evidence suggest that, under both the statistical and the risk-neutral measure, index returns are pure-jump processes of infinite activity and finite variation with no diffusion component; individual stocks may retain one in their statistical dynamics.

Version record

7.1.3 Model specification and empirical tests

Gurdip Bakshi, Charles Cao and Zhiwu Chen, "Empirical Performance of Alternative Option Pricing Models"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 52(5), 2003–2049 (1997); DOI 10.1111/j.1540-6261.1997.tb02749.x.

Derives an option model in which volatility, interest rates and jumps are all stochastic, then compares nested models on S&P 500 options by the consistency of implied parameters with time-series data, out-of-sample pricing and hedging. Stochastic volatility and jumps both matter for pricing and internal consistency, but for hedging, stochastic volatility alone does best.

Version record

Bernard Dumas, Jeff Fleming and Robert E. Whaley, "Implied Volatility Functions: Empirical Tests"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 53(6), 2059–2106 (1998); DOI 10.1111/0022-1082.00083.

Tests the deterministic-volatility-function (local-volatility) model of Derman–Kani, Dupire and Rubinstein on S&P 500 options from June 1988 to December 1993. Its predictive and hedging performance is no better than an ad hoc rule that smooths Black–Scholes implied volatilities across strikes and maturities, so an exact fit to today's surface does not establish stable dynamics.

Version record

David S. Bates, "Post-'87 Crash Fears in the S&P 500 Futures Option Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 94(1–2), 181–238 (2000); DOI 10.1016/s0304-4076(99)00021-4.

Post-crash implied distributions are strongly negatively skewed. Nesting stochastic volatility with time-varying negative-mean jumps and fitting both to S&P 500 futures options over 1988–1993, finds that stochastic volatility alone matches the skew only with implausible parameters, such as very high volatility of volatility, whereas adding jumps gives plausible estimates. Evidence is also presented against volatility following a diffusion.

Version record

Bjørn Eraker, Michael Johannes and Nicholas Polson, "The Impact of Jumps in Volatility and Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 58(3), 1269–1300 (2003); DOI 10.1111/1540-6261.00566.

Estimates stochastic-volatility models with jumps in returns and in volatility by likelihood-based methods on S&P 500 and Nasdaq 100 returns, recovering spot volatility, jump times and sizes through the 1987, 1997 and 1998 stress episodes. Finds strong evidence for both kinds of jump and shows how they, and estimation risk, affect option prices.

Version record

Mark Broadie, Mikhail Chernov and Michael Johannes, "Model Specification and Risk Premia: Evidence from Futures Options"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 62(3), 1453–1490 (2007); DOI 10.1111/j.1540-6261.2007.01241.x.

Uses S&P 500 futures options from 1987 to 2003 to test for jumps in volatility, with strong time-series evidence, and in prices, with strong cross-sectional evidence, and to estimate diffusive and jump risk premia. Jump risk premia are economically and statistically significant and important for understanding option returns.

Version record

Peter Christoffersen, Steven Heston and Kris Jacobs, "The Shape and Term Structure of the Index Option Smirk: Why Multifactor Stochastic Volatility Models Work So Well"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 55(12), 1914–1932 (2009); DOI 10.1287/mnsc.1090.1065.

The level and slope of the index option smirk move largely independently, which a one-factor stochastic-volatility model cannot reproduce. A two-factor model whose factors have different correlations with returns generates stochastic correlation and a more flexible term structure, improving on Heston by 24% in sample and 23% out of sample.

Version record

Wim Schoutens, Erwin Simons and Jurgen Tistaert, "A Perfect Calibration! Now What?"

Difficulty: Advanced **Type:** Research paper

Status/version: Wilmott, 2004(2), 66–78 (2004); DOI 10.1002/wilm.42820040216.

Calibrates Heston (with and without jumps), Barndorff-Nielsen–Shephard and time-changed Lévy models to the same equity surface; all reproduce vanilla prices almost equally well. Their prices for a range of exotic options nevertheless differ substantially because the models' path behaviour differs, so a good vanilla fit does not settle exotic prices.

Version record

7.1.4 Rough and path-dependent volatility

Jim Gatheral, Thibault Jaisson and Mathieu Rosenbaum, "Volatility Is Rough"

Difficulty: Research **Type:** Research paper

Status/version: 2018

First circulated as a 2014 preprint and published in 2018. Documents a fractional scaling structure in high-frequency volatility and introduces rough volatility as a parsimonious explanation. It changed the research agenda for volatility forecasting and option-surface dynamics, but implementation requires care over estimation noise and discretisation. [11]

Applications: Equity and index volatility, options.

Version record

Julien Guyon and Jordan Lekeufack, Volatility Is (Mostly) Path-Dependent

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 23(9), 1221–1258, 2023; published online 19 July 2023; DOI 10.1080/14697688.2023.2221281.

Fits index implied and realised volatility using weighted histories of returns and squared returns, then builds a path-dependent continuous-time volatility model with exponential-kernel Markovian approximations. It gives a concrete alternative to interpreting every persistent volatility pattern as fractional roughness. Study the historical-versus-risk-neutral parameter distinction, train/test split and joint SPX/VIX smile fit before comparing it with rough Bergomi or rough Heston. Publisher abstract claims high explanatory power in its equity-index samples; it does not establish universal performance across assets, volatility regimes or tradable hedges after costs.

Extended note.

Version record

Eduardo Abi Jaber and Shaun (Xiaoyuan) Li, Volatility Models in Practice: Rough, Path-Dependent, or Markovian?

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance 35(4), 796–817, October 2025; first published online 6 May 2025; DOI 10.1111/mafi.12463.

Compares rough, path-dependent and Markovian models against SPX option smiles using matched parameter budgets. The publisher abstract reports that the tested rough specifications miss the global at-the-money skew shape and underperform a one-factor Markovian benchmark over one week to three months; a non-rough path-dependent and a two-factor Markovian model fit one-week to three-year smiles better in the study. Use it to design matched calibration and hedging comparisons after the PDV and rough-volatility foundations. These sample-specific option-price results do not prove that every rough model fails or that lower fit error yields better hedges.

Extended note.

Version record

Omar El Euch and Mathieu Rosenbaum, "The Characteristic Function of Rough Heston Models"

Difficulty: Research **Type:** Research paper

Status/version: 2019

Makes rough Heston computationally tractable through fractional Riccati equations. It is the natural next step after the empirical rough-volatility paper for researchers interested in calibration, pricing and surface dynamics.

Applications: Options, volatility surfaces. Version record

Christian Bayer, Peter Friz, Jim Gatheral and Mathieu Rosenbaum, eds., *Rough Volatility*

Difficulty: Research **Type:** Book / monograph

Status/version: 2023

A specialist collection covering theory, numerics, calibration and empirical implications of rough volatility. It is preferable to relying on isolated preprints once a desk is seriously considering rough models for production. [12]

Applications: Options, volatility, numerical methods. Version record

Eduardo Abi Jaber and Louis-Amand Gérard, "Signature Volatility Models: Pricing and Hedging with Fourier"

Difficulty: Research **Type:** Research paper

Status/version: 2025

Uses path signatures to construct flexible non-Markovian volatility models while retaining Fourier-based pricing machinery. This is a significant recent bridge between rough/path-dependent modelling and computationally useful derivative pricing. [13]

Applications: Options, volatility surfaces, path-dependent products. Version record

Hervé Andrès, Alexandre Boumezoued and Benjamin Jourdain, "The implied volatility surface (also) is path-dependent"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(7), 1017–1047 (2026); DOI 10.1080/14697688.2026.2637739.

Joint simulation of the implied-volatility surface and the underlying with path-dependent dynamics, extending Guyon–Lekeufack.

Version record

Ranieri Dugo, Giacomo Giorgio and Paolo Pigato, "Multivariate rough volatility"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 1–28 (2026); DOI 10.1080/14697688.2026.2638519.

Multivariate fractional OU model for joint log-volatilities, a multivariate rough FSV model.

Version record

Xiyue Han and Alexander Schied, "On the Rate of Convergence of Estimating the Hurst Parameter of Rough Stochastic Volatility Models"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 16(4), 1336–1349 (2025); DOI 10.1137/25m1750767.

Consistency and almost-sure convergence rates for a scale-invariant Hurst estimator applied to rough stochastic volatility from discrete observations.

Version record

7.1.5 Numerical pricing: Monte Carlo, PDE, Fourier and AAD

Paul Glasserman, *Monte Carlo Methods in Financial Engineering*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer first edition; hardcover published 7 August 2003, eBook published 9 March 2013.

The definitive reference for simulation, variance reduction, discretisation-error control, Greeks, American options and rare-event methods. Its techniques remain central to exotics, XVA, counterparty exposure and scenario engines even when neural approximators are introduced.

Applications: Derivatives, structured products, risk. Version record

Francis A. Longstaff and Eduardo S. Schwartz, "Valuing American Options by Simulation: A Simple Least-Squares Approach"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 14(1), 113–147 (2001); DOI 10.1093/rfs/14.1.113.

Least-squares Monte Carlo for early exercise; standard method for Bermudans, callable structures and exposure simulation.

Version record

Fang Fang and Cornelis W. Oosterlee, "A Novel Pricing Method for European Options Based on Fourier-Cosine Series Expansions"

Difficulty: Advanced **Type:** Research paper

Status/version: SIAM Journal on Scientific Computing, 31(2), 826–848 (2009); DOI 10.1137/080718061.

COS method: fast, exponentially convergent Fourier-cosine pricing for models with known characteristic functions.

Version record

Leif Andersen, "Simple and efficient simulation of the Heston stochastic volatility model"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Computational Finance, 11(3), 1–42 (2008); DOI 10.21314/jcf.2008.189.

Quadratic-exponential scheme that avoids negative variance and bias of naive Euler discretisation in Heston.

Version record

Roger Lord and Christian Kahl, "Complex Logarithms in Heston-Like Models"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 20(4), 671–694 (2010); DOI 10.1111/j.1467-9965.2010.00416.x.

If the complex logarithm in a Heston-type characteristic function is restricted to its principal branch, as most software does, the function can become discontinuous and Fourier prices completely wrong. Proves that a formulation exists in which the principal branch is always correct, and shows how to avoid the discontinuities in the Variance Gamma and Schöbel–Zhu models and in Broadie–Kaya exact Heston simulation.

Version record

Michael B. Giles, "Multilevel Monte Carlo Path Simulation"

Difficulty: Research **Type:** Research paper

Status/version: Operations Research, 56(3), 607–617 (2008); DOI 10.1287/opre.1070.0496.

Multilevel estimators combining coarse and fine discretisations to cut Monte Carlo cost for SDE expectations.

Version record

Michael Giles and Paul Glasserman, "Smoking adjoints: fast Monte Carlo Greeks"

Difficulty: Advanced **Type:** Research paper

Status/version: Risk 19(1), 88–92 (January 2006).

Adjoint (reverse-mode) pathwise sensitivities for LIBOR-market-model Greeks; the paper that introduced AAD to finance practice. Published in Risk (January 2006).

Version record

Antoine Savine, *Modern Computational Finance: AAD and Parallel Simulations*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, 2018 (ISBN 9781119539452); public companion code.

Builds a production-style parallel Monte Carlo library and algorithmic adjoint differentiation in C++; book code is public.

Version record

Luca Capriotti, "Fast Greeks by algorithmic differentiation"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Computational Finance, 14(3), 3–35 (2011); DOI 10.21314/jcf.2011.234.

Shows adjoint algorithmic differentiation computes all Monte Carlo Greeks at a small multiple of the pricing cost.

Version record

Daniel J. Duffy, *Finite Difference Methods in Financial Engineering: A Partial Differential Equation Approach*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2006); DOI 10.1002/9781118673447.

Finite-difference schemes, stability and boundary conditions for option PDEs, including Crank–Nicolson pitfalls and ADI methods.

Version record

Brian Huge and Antoine Savine, "Differential Machine Learning"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2005.02347 (May 2020); summary article in Risk (October 2020).

Trains neural pricing approximators on pathwise differentials as well as values, improving accuracy for risk and exposure calculations.

Extended note. Version record

Daniel A. Cline, "Efficient upper bounds for American options: regression-based duals from backward primals"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(7), 1071–1089 (2026); DOI 10.1080/14697688.2026.2647986.

Regression-based dual upper bounds for American and Bermudan options from primal paths, cheaply tightening price intervals.

Version record

Nicola F. Zaugg and Lech A. Grzelak, "Lifted Heston Model: Efficient Monte Carlo Simulation with Large Time Steps"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 17(3), 1279–1307 (2026); DOI 10.1137/25m1815614.

An implicit integrated-variance scheme for simulating the lifted Heston model with large time steps.

Version record

Christian Bayer, Luca Pelizzari and John Schoenmakers, "Primal and dual optimal stopping with signatures"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 29(4), 981–1014 (2025); DOI 10.1007/s00780-025-00570-8.

Signature-based primal (Longstaff–Schwartz) and dual (martingale) methods for American options in non-Markovian models such as rough volatility.

Version record

7.2 Surfaces, rough volatility and calibration

Read Gatheral and Bergomi first. Progress from static arbitrage and tail constraints to stochastic dynamics, efficient numerical pricing and learned calibration; each step answers a different validation question.

7.2.1 Surfaces and calibration

Roger W. Lee, The Moment Formula for Implied Volatility at Extreme Strikes; author manuscript

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance 14(3), 469–480 (2004); first published 18 June 2004.

Derives model-independent links between the asymptotic slope of total implied variance and finite positive or inverse moments of the underlying distribution. It gives a disciplined boundary check for wing extrapolation when listed strikes are sparse, and helps distinguish a surface that fits observed quotes from one with defensible tail behaviour. The formula constrains asymptotic slopes; it does not identify a unique extrapolated price or settle finite-strike liquidity and event-risk assumptions.

Version record

Jim Gatheral; Antoine Jacquier, Arbitrage-free SVI volatility surfaces

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 14(1), 59–71 (2014 issue); accepted 7 June 2013; online 11 September 2013.

Constructs a tractable SVI implied-volatility surface and calibration conditions that rule out static butterfly and calendar arbitrage. A finite quote grid, bid–ask spreads and extreme-wing extrapolation still require explicit checks; a static-arbitrage-free fit does not validate its future smile dynamics or hedge performance.

Extended note. Version record

Christian Bayer; Peter Friz; Jim Gatheral, Pricing under rough volatility; publisher erratum

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 16(6), 887–904 (2016); online 18 November 2015. Publisher records a 2016 erratum correcting Figure 8, not the stated findings.

Introduces the rough Bergomi pricing route for claims on the underlying and integrated variance, linking rough forward-variance dynamics to equity-index skew and variance-swap term structure. It gives a concrete model to read after the catalogue's empirical rough-volatility and Bergomi references, then compare with rough Heston's transform approach. Read the corrected version because the publisher records a figure erratum.

Extended note. Version record

Ryan McCrickerd; Mikko S. Pakkanen, Turbocharging Monte Carlo pricing for the rough Bergomi model; author preprint

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 18(11), 1877–1886 (2018); online 30 April 2018; arXiv first posted 8 August 2017.

Combines variance-reduction techniques to reduce the runtime-adjusted noise of rough Bergomi implied-volatility estimates, directly addressing calibration by simulation. Pair it with Bayer–Friz–Gatheral and Glasserman to audit discretisation bias, Monte Carlo error, confidence targets and hardware-adjusted costs before using a numerical pricer in daily calibration.

Extended note. Version record

Christian Bayer; Simon Breneis, Efficient option pricing in the rough Heston model using weak simulation schemes; author manuscript

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 24(9), 1247–1261 (2024); accepted 27 July 2024; online 2 September 2024; arXiv first posted 6 October 2023.

Builds a weak simulation method for rough Heston from a low-dimensional Markovian approximation, with numerical experiments on vanilla, path-dependent and Bermudan payoffs. The reported second-order weak convergence and linear time-step cost apply to the tested scheme; factor truncation, square-root boundary behaviour and exercise-regression bias remain separate validation tasks.

Extended note. Version record

Fabio Baschetti; Giacomo Bormetti; Pietro Rossi, Deep calibration with random grids; University of Bologna accepted manuscript

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 24(9), 1263–1285 (2024); accepted 7 March 2024; online 12 April 2024; author accepted manuscript released by Bologna in 2025.

Trains a neural calibration map on random strike–maturity grids, combining gridwise robustness with pointwise fitting so a desk need not impose one fixed quote lattice. Its speed and fit are conditional on simulator coverage and training parameter ranges; one must validate out-of-domain quotes, calibration stability, Greeks and arbitrage constraints independently before substituting it for a trusted pricer.

Extended note. Version record

Milena Vuletić; Rama Cont, VolGAN: A Generative Model for Arbitrage-Free Implied Volatility Surfaces; Oxford version of record; author code

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance 31(4), 203–238, in a 2024-labelled issue; accepted 19 February 2025, first online 6 March 2025. Oxford cites the article as 2025.

Learns joint scenarios for the SPX implied-volatility surface and underlying, aiming to retain cross-strike dependence, VIX behaviour, non-Gaussian moves and time-varying correlations. Its no-arbitrage construction and reported hedge comparison should be checked on the generated quote grid and out-of-sample regimes; neither neural scenario realism nor a delta/delta-vega comparison establishes executable hedges after liquidity, transaction costs and stressed correlations.

Extended note. Version record

Pascal Letourneau and Lars Stentoft, "Efficient Pricing and Model Calibration With Large Panels of Options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 23(5), nbaf019 (2025); DOI 10.1093/jjfinec/nbaf019.

Prices large option panels through time with a single simulation, enabling calibration of models with early exercise and time-varying volatility.

Version record

Karim Moussa, "Arbitrage filtering of option prices: a simple real-time approach"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(3), 325–340 (2026); DOI 10.1080/14697688.2025.2574907.

Real-time arbitrage filter adjusting option quotes that violate no-arbitrage conditions using more liquid quotes, without discarding data.

Version record

Christa Cuchiero, Eva Flonner and Kevin Kurt, "Robust financial calibration: a Bayesian approach for neural stochastic differential equations"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Computational Finance (2025); DOI 10.21314/jcf.2025.009.

Bayesian approach to robust calibration of neural stochastic differential equation models.

Extended note. Version record

7.2.2 Model risk, robust and model-free pricing

Rama Cont, "Model uncertainty and its impact on the pricing of derivative instruments"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 16(3), 519–547 (2006); DOI 10.1111/j.1467-9965.2006.00281.x.

Coherent and convex measures of model risk based on the range of prices across calibrated models.

Version record

Mathias Beiglböck, Pierre Henry-Labordère and Friedrich Penkner, "Model-independent bounds for option prices—a mass transport approach"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 17(3), 477–501 (2013); DOI 10.1007/s00780-013-0205-8.

Martingale optimal transport: price bounds for exotic options consistent with observed vanilla prices.

Version record

Pierre Henry-Labordère, *Model-free Hedging: A Martingale Optimal Transport Viewpoint*

Difficulty: Research **Type:** Book / monograph

Status/version: Chapman and Hall/CRC, edition 1 (2017); DOI 10.1201/9781315161747.

Monograph on robust bounds and hedges for exotics via martingale optimal transport, with numerical methods.

Version record

Julien Guyon and Pierre Henry-Labordère, *Nonlinear Option Pricing*

Difficulty: Research **Type:** Book / monograph

Status/version: Chapman and Hall/CRC (2013); DOI 10.1201/b16332.

Uncertain volatility, local stochastic volatility calibration, BSDEs and particle methods for nonlinear pricing problems.

Version record

Rama Cont and Purba Das, "Rough Volatility: Fact or Artefact?"

Difficulty: Research **Type:** Research paper

Status/version: Sankhya B, 86(1), 191–223 (2024); DOI 10.1007/s13571-024-00322-2.

Model-free roughness estimator shows realised-volatility series look rough even when spot volatility is smooth, attributing apparent roughness to estimation error. Read against the rough-volatility entries.

Version record

Blanka Horvath, Aitor Muguruza and Mehdi Tomas, "Deep learning volatility: a deep neural network perspective on pricing and calibration in (rough) volatility models"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 21(1), 11–27 (2021); DOI 10.1080/14697688.2020.1817974.

Learns the pricing map offline so calibration of rough and other expensive models becomes near-instant.

Version record

Julien Guyon, "The Joint S&P 500/VIX Smile Calibration Puzzle Solved"

Difficulty: Research **Type:** Research paper

Status/version: Risk, April 2020; SSRN 3397382.

Constructs a model that exactly fits SPX and VIX smiles jointly using a dispersion-constrained martingale optimal transport. Published in Risk (April 2020).

Extended note. Version record

Jim Gatheral, Paul Jusselin and Mathieu Rosenbaum, "The quadratic rough Heston model and the joint S&P 500/VIX smile calibration problem"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2001.01789 (January 2020); Risk, May 2020.

Rough model with Zumbach effect that fits SPX and VIX smiles jointly with few parameters.

Extended note. Version record

Ferenc Horvath, "Arbitrage-based recovery"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 163, 103969 (2025); DOI 10.1016/j.jfineco.2024.103969.

Recovery theorem based on no-arbitrage that needs Arrow–Debreu prices for only one maturity; tested on 26 years of S&P 500 options.

Version record

Zixing Chen, Yihan Qi, Shanlan Que, Julian Sester and Xiao Zhang, "Empirical analysis of the model-free valuation approach: hedging gaps, conservatism, and trading opportunities"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 1–17 (2026); DOI 10.1080/14697688.2026.2694470.

Empirical hedging gaps of model-free super-hedges on S&P 500 constituents (2018–2022) and their conservatism.

Version record

Julien Guyon, "Dispersion-Constrained Martingale Schrödinger Bridges: Joint Entropic Calibration of Stochastic Volatility Models to S&P 500 and VIX Smiles"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 16(3), 834–874 (2025); DOI 10.1137/22m1511242.

Continuous-time dispersion-constrained martingale Schrödinger bridges that jointly and exactly calibrate SPX smiles, VIX futures and VIX smiles by minimum entropy relative to a reference SV model.

Version record

7.3 Delta hedging, transaction costs and smile dynamics

An option model is finally judged by the hedge it implies. These readings show what happens to a delta hedge when volatility is misspecified, rebalancing is discrete and trading is costly, how to attribute option profit and loss to its Greeks, and which delta to use when the smile moves with the underlying. The practitioner texts by Natenberg and Taleb are in Section 7.1; learned and scenario-based hedges follow in Section 7.4.

Nicole El Karoui, Monique Jeanblanc-Picqué and Steven E. Shreve, "Robustness of the Black and Scholes Formula"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 8(2), 93–126 (1998); DOI 10.1111/1467-9965.00047.

If a hedger's assumed volatility, a function of time and the stock price, dominates the true stochastic volatility, the resulting price dominates the true price and the hedge super-replicates almost surely; the reverse holds when it is dominated. The comparison can fail when the assumed volatility is not a function of time and price. The results cover European and American options and give a bound and hedge for Asian options.

Version record

Peter Carr and Liuren Wu, "Option Profit and Loss Attribution and Pricing: A New Framework"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(4), 2271–2316 (2020); DOI 10.1111/jofi.12894.

Uses the Black–Merton–Scholes formula to attribute an option's short-horizon profit and loss to moves in the underlying and in the option's own implied volatility. Requiring no dynamic arbitrage then links an option's fair implied volatility to the underlying's volatility, corrected for the implied volatility's expected drift, its variance and its covariance with the underlying return.

Version record

Stephen Figlewski, "Options Arbitrage in Imperfect Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 44(5), 1289–1311 (1989); DOI 10.1111/j.1540-6261.1989.tb02654.x.

Simulates the standard option arbitrage of hedging against the underlying when volatility is uncertain, trading is costly and indivisible, and rebalancing is discrete. In a market such as stock index options the arbitrage carries so much risk and cost that it sets only very wide bounds on option prices, which matters for price formation and for tests of valuation models.

Version record

Hayne E. Leland, "Option Pricing and Replication with Transactions Costs"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 40(5), 1283–1301 (1985); DOI 10.1111/j.1540-6261.1985.tb02383.x.

With transaction costs, discrete Black–Scholes delta hedging leaves errors that are correlated with the market and do not vanish as rebalancing becomes more frequent. A modified replicating strategy, depending on the cost level and the rebalancing interval, gives errors that are uncorrelated with the market and tend to zero, and yields the cost of replication and bounds on option prices.

Version record

A. Elizabeth Whalley and Paul Wilmott, "An Asymptotic Analysis of an Optimal Hedging Model for Option Pricing with Transaction Costs"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 7(3), 307–324 (1997); DOI 10.1111/1467-9965.00034.

The utility-based hedging model of Hodges–Neuberger and Davis–Panas–Zariphopoulou leads to a slow three-dimensional free-boundary problem. An asymptotic analysis for small transaction costs reduces it to a two-variable diffusion equation and gives a simple analytical hedging band around delta that depends on the option's gamma.

Version record

Stéphane Crépey, "Delta-Hedging Vega Risk?"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 4(5), 559–579 (2004); DOI 10.1080/14697680400000038.

Compares delta hedging with the Black–Scholes implied delta and with the local-volatility delta recalibrated daily to the smile. In negatively skewed markets, where slow rallies and fast sell-offs are more likely than the reverse, the local delta hedges better on average; the result is checked on simulated and real equity-index data, with transaction costs, and discussed for barrier options.

Version record

Carol Alexander and Leonardo M. Nogueira, "Model-Free Hedge Ratios and Scale-Invariant Models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 31(6), 1839–1861 (2007); DOI 10.1016/j.jbankfin.2006.11.011.

Most processes used to price options are scale invariant, and under them the price hedge ratios of any claim with a homogeneous payoff, including path-dependent ones, are model free. Those ratios are minimum-variance only when price–volatility correlation is zero; otherwise, and in scale-invariant local-volatility models, they hedge less efficiently than minimum-variance ratios.

Version record

John Hull and Alan White, "Optimal Delta Hedging for Options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 82, 180–190 (2017); DOI 10.1016/j.jbankfin.2017.05.006.

Because volatility tends to move with the underlying, the usual delta does not minimise the variance of a hedged position. Estimates an empirical minimum-variance delta and shows that on S&P 500 options it beats stochastic-volatility models recalibrated daily for each maturity, with further results for S&P 100, Dow Jones, single-stock, and commodity and interest-rate ETF options.

Version record

7.4 Hedging with learning and scenarios

Subject context and qualifications. These readings connect learned market dynamics to specific hedge decisions. Compare with delta and delta-vega baselines, then examine the traded instruments, transaction costs and stress regimes behind any claimed improvement.

Hans Buehler, Lukas Gonon, Josef Teichmann and Ben Wood, "Deep Hedging"

Difficulty: Research **Type:** Research paper

Status/version: 2019

Recasts hedging under transaction costs, liquidity constraints and general risk measures as a neural optimisation problem. Its institutional importance lies less in "AI pricing" than in jointly learning nonlinear hedges under realistic frictions and constraints. [14]

Applications: Listed and OTC derivatives; hedging, RL. Version record

Hans Buehler, Phillip Murray and Ben Wood, "Deep Bellman Hedging"; SSRN draft

Difficulty: Research **Type:** Research paper

Status/version: Preprint 3 July 2022; v4 25 June 2024

Actor–critic hedging from historical transitions, with frictions and no fixed terminal-maturity boundary. Reuses a policy across portfolios and market states near the training data; historical coverage limits generalisation. The arXiv v4 manuscript is internally dated April 2024. [15]

Applications: Options, OTC books; hedging, RL. Version record

Oskari Mikkilä and Juho Kanniainen, "Empirical Deep Hedging"

Difficulty: Research **Type:** Research paper

Status/version: 2023; online 2022

Evaluates deep hedging using intraday equity and option data rather than only idealised simulations. It is valuable for studying whether learned policies survive real volatility surfaces, discrete rebalancing and trading costs. [16]

Applications: Equity options; empirical hedging. Version record

Eva Lütkebohmert, Thorsten Schmidt and Julian Sester, "Robust Deep Hedging"; preprint

Difficulty: Research **Type:** Research paper

Status/version: 2022; preprint 2021

Extends deep hedging to ambiguity about the data-generating model. This is directly relevant to model risk because hedging policies trained on one simulator can fail sharply when volatility, jumps or dependence are misspecified. [28]

Applications: Options, multi-asset derivatives; robust hedging. Version record

Phillip Murray, Ben Wood, Hans Buehler, Magnus Wiese and Mikko S. Pakkanen, "Deep Hedging: Continuous Reinforcement Learning for Hedging of General Portfolios across Multiple Risk Aversions"

Difficulty: Research **Type:** Research paper

Status/version: 2022 preprint

Trains continuous-control policies across a range of risk preferences rather than retraining for a single objective. It is useful for desks that must quote or hedge under changing risk limits, capital charges and inventory conditions. [29]

Applications: Options; RL hedging, risk control. Version record

Supriyo Ghosh, Laura Wynter, Shiau Hong Lim and Duc Thien Nguyen, "Neural-progressive hedging: Enforcing constraints in reinforcement learning with stochastic programming"

Difficulty: Research **Type:** Research paper

Status/version: 2022

Uses a transition model and stochastic programming to adjust an RL policy for constraints, including CVaR. UAI 2022, PMLR 180, 707–717; experiments include portfolio liquidity and bike sharing. Derivatives applications are a research extension, not a demonstrated deployment. [30]

Applications: Constrained policies, portfolios, CVaR. Version record

Yuming Ma, "Deep Hedging to Manage Tail Risk"

Difficulty: Research **Type:** Research paper

Status/version: Preprint v1, 27 June 2025

Optimises CVaR-style tail-risk hedges using crisis-era bootstrap simulators and explicit frictions. It is a recent working paper rather than settled evidence, but it identifies the right institutional question: whether learned hedges retain value under tail scenarios and stressed liquidity. [32]

Applications: Multi-asset overlays, options; tail hedging. Version record

Rama Cont; Milena Vuletić, Data-driven hedging with generative models; Oxford version of record

Difficulty: Research **Type:** Research paper

Status/version: Annals of Operations Research, version of record online 16 October 2025; accepted 18 September 2025; DOI 10.1007/s10479-025-06867-3. No volume/pages asserted from consulted record.

Uses a conditional generative market model to simulate states from current conditions and chooses option-portfolio hedges by minimising risk over those scenarios, including the possibility of transaction costs and impact. Institutional use hinges on conditional tail fidelity, available hedge instruments, execution assumptions and stability under regime change; scenario-model misspecification can be mistaken for hedge improvement.

Extended note. Version record

Sebastian Egebjerg, Deep hedging 0DTE options

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Stability, vol. 84, article 101535, June 2026.

Tests deep reinforcement-learning hedges for same-day SPX options using high-frequency option data and actual transaction costs, with Black–Scholes delta hedging and regime robustness as central comparisons. Reported outperformance is context-specific: the underlying high-frequency SPX dataset is proprietary, so external replication, latency and fill assumptions, and stress beyond the studied periods need separate scrutiny before use in an options book.

Extended note. Version record

James M. Hutchinson, Andrew W. Lo and Tomaso Poggio, "A Nonparametric Approach to Pricing and Hedging Derivative Securities Via Learning Networks"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 49(3), 851–889 (1994); DOI 10.1111/j.1540-6261.1994.tb00081.x.

Early demonstration that neural networks learn option pricing and hedging maps from data.

Version record

Johannes Ruf and Weiguan Wang, "Neural networks for option pricing and hedging: a literature review"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Computational Finance (2020); DOI 10.21314/jcf.2020.390.

Reviews 150+ papers with emphasis on data leakage, benchmark choice and hedging evaluation.

Version record

Sebastian Becker, Patrick Cheridito and Arnulf Jentzen, "Deep Optimal Stopping"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Machine Learning Research 20(74), 1–25 (2019).

Neural stopping decisions for high-dimensional Bermudan options with lower and upper price bounds.

Version record

Jiequn Han, Arnulf Jentzen and Weinan E, "Solving high-dimensional partial differential equations using deep learning"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the National Academy of Sciences, 115(34), 8505–8510 (2018); DOI 10.1073/pnas.1718942115.

Deep BSDE method for high-dimensional semilinear PDEs, including nonlinear pricing with default risk.

Version record

Justin Sirignano and Konstantinos Spiliopoulos, "DGM: A deep learning algorithm for solving partial differential equations"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Computational Physics, 375, 1339–1364 (2018); DOI 10.1016/j.jcp.2018.08.029.

Mesh-free deep Galerkin method for high-dimensional PDEs, demonstrated on American options.

Version record

Hui Chen, Antoine Didisheim and Simon Scheidegger, "Deep surrogates for finance: With an application to option pricing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 177, 104222 (2026); DOI 10.1016/j.jfineco.2025.104222.

Deep surrogates replace expensive structural model solutions with trained networks, applied to option pricing and estimation.

Extended note. Version record

Zhonghao Xian, Xing Yan, Cheuk Hang Leung and Qi Wu, "Risk-neutral generative networks"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(6), 961–980 (2026); DOI 10.1080/14697688.2026.2679119.

Generative networks for risk-neutral densities and option prices across maturities with no-arbitrage conditions imposed.

Version record

Lei Fan and Justin Sirignano, "Machine learning methods for pricing financial derivatives"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(6), 931–959 (2026); DOI 10.1080/14697688.2026.2623901.

Neural-SDE models with neural drift and volatility calibrated to derivative prices, with training methods for pricing and hedging.

Version record

Deqing Jiang, Justin Sirignano and Samuel N. Cohen, "Global Convergence of Deep Galerkin and PINN Methods for Solving Partial Differential Equations"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 17(2), 620–645 (2026); DOI 10.1137/24m1701502.

Global convergence results for Deep Galerkin and PINN training for high-dimensional PDEs from option pricing, mean-field and portfolio problems.

Version record

John Armstrong and Andrei Ionescu, "Gamma hedging and rough paths"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 29(4), 933–979 (2025); DOI 10.1007/s00780-025-00576-2.

Pathwise result: discrete gamma hedging replicates European (and some exotic) options for price signals of finite p-variation with $p<3$, without assuming a probabilistic model.

Version record

7.5 Variance, correlation, VIX and dividends

Separate the payoff being traded from the risk premium being estimated. Move from variance replication to index-versus-single-name correlation, volatility futures and dividend strips, using methodology documents and explicit counter-evidence to test implementation. Single-stock option returns, and short-dated and retail option trading, have their own subsections; option-implied densities and tail risk follow in Section 7.6.

7.5.1 Variance and correlation risk premia

Oleg Bondarenko, "Why Are Put Options So Expensive?" (2014)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2014. Version distinctions remain in the annotation.

The historical put-pricing puzzle and limits of standard explanations of option returns; crash-risk compensation alone does not exhaust the paper.

Version record

George M. Constantinides, Jens Carsten Jackwerth and Alexi Savov, "The Puzzle of Index Option Returns"

Difficulty: Research **Type:** Research paper

Status/version: Review of Asset Pricing Studies, 3(2), 229–257 (2013); DOI 10.1093/rapstu/rat004.

Builds S&P 500 call and put portfolios rebalanced daily to a fixed maturity, moneyness and unit market beta; their monthly returns are close to normal, so linear factor models can be tested. A single crisis-related factor (price jumps, volatility jumps or liquidity) alongside the market is not rejected, even with factor premia constrained to their equity values, and the alphas of short-dated out-of-the-money puts become insignificant.

Version record

Peter Carr and Liuren Wu, Variance Risk Premiums

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies 22(3), 1311–1341, March 2009 issue; published online 10 April 2008.

Connects option-implied variance to realised variance and studies the compensation for bearing variance risk. Use it to design a variance-swap replication and to separate a volatility forecast from the premium embedded in option prices. Finite strike coverage, discrete sampling, jumps, transaction costs and margin change practical replication; an attractive average short-variance return must be assessed together with drawdowns and funding needs.

Extended note. Version record

Joost Driessen, Pascal J. Maenhout and Grigory Vilkov, The Price of Correlation Risk: Evidence from Equity Options

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance 64(3), 1377–1406, 2009.

Uses index and constituent options to investigate the pricing of correlation risk. It gives dispersion research an economic framework beyond comparing index implied volatility with a weighted average of single-name volatilities. The empirical findings depend on the available option panel, and the paper discusses trading frictions: correlation exposure, skew exposure and realised trading P&L are related but distinct objects.

Extended note. Version record

Adrian Buss and Grigory Vilkov, "Measuring Equity Risk with Option-Implied Correlations"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 25(10), 3113–3140 (2012); DOI 10.1093/rfs/hhs087.

Estimates option-implied correlations and from them forward-looking factor betas. Implied market betas give a monotonically increasing risk–return relation with a slope close to the market excess return, which rolling-window betas do not show, because they predict realised betas better with smaller and less systematic errors.

Version record

Oleg Bondarenko and Carole Bernard, "Option-Implied Dependence and Correlation Risk Premium"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 59(7), 3139–3189 (2024); DOI 10.1017/s0022109023000960.

Recovers, without a model, a joint risk-neutral distribution consistent with options on several assets and on their weighted index, and applies it to the nine industry sectors of the S&P 500. Implied dependence is highly asymmetric and time-varying: the risk premium on correlation in falling markets is strongly negative, while that on correlation in rising markets is positive.

Version record

Peter Carr and Dilip Madan, "Towards a theory of volatility trading"

Difficulty: Advanced **Type:** Research paper

Status/version: Option Pricing, Interest Rates and Risk Management, 458–476 (None); DOI 10.1017/cbo9780511569708.013.

Static replication of variance and other volatility payoffs with a strip of options; basis of variance-swap pricing.

Version record

Kresimir Demeterfi, Emanuel Derman, Michael Kamal and Joseph Zou, "A Guide to Volatility and Variance Swaps"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Derivatives, 6(4), 9–32 (1999); DOI 10.3905/jod.1999.319129.

Practitioner derivation of variance-swap replication, skew effects and discrete-strike errors (Goldman Sachs note published in J. Derivatives).

Version record

Gurdip Bakshi and Nikunj Kapadia, "Delta-Hedged Gains and the Negative Market Volatility Risk Premium"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 16(2), 527–566 (2003); DOI 10.1093/rfs/hhg002.

Delta-hedged index option losses imply a negative price of volatility risk.

Version record

Gurdip Bakshi, Nikunj Kapadia and Dilip Madan, "Stock Return Characteristics, Skew Laws, and the Differential Pricing of Individual Equity Options"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 16(1), 101–143 (2003); DOI 10.1093/rfs/16.1.0101.

Model-free risk-neutral skewness and kurtosis from option prices; widely used moment estimators.

Version record

Tim Bollerslev, George Tauchen and Hao Zhou, "Expected Stock Returns and Variance Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(11), 4463–4492 (2009); DOI 10.1093/rfs/hhp008.

Implied minus realised variance predicts quarterly equity returns.

Version record

Walid Khalfallah and Nikolaos Tessaromatis, "Volatility Carry Strategies Across Global Asset Classes"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 1–22 (2026); DOI 10.1080/0015198x.2026.2726134.

Cross-sectional volatility-carry strategies in volatility swaps and forward volatility agreements across equities, FX, rates and commodities earn strong Sharpe ratios; after costs, swap-based implementations are most robust.

Version record

7.5.2 Option returns, demand and dealer hedging

Joshua D. Coval and Tyler Shumway, "Expected Option Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 56(3), 983–1009 (2001); DOI 10.1111/0022-1082.00352.

Option returns are too low relative to a CAPM-type benchmark, suggesting priced volatility risk.

Version record

Nicolae Gârleanu, Lasse Heje Pedersen and Allen M. Poteshman, "Demand-Based Option Pricing"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(10), 4259–4299 (2009); DOI 10.1093/rfs/hhp005.

Dealers who cannot hedge perfectly charge for end-user demand, explaining index-option expensiveness and skew.

Version record

Dmitriy Muravyev, "Order Flow and Expected Option Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 71(2), 673–708 (2016); DOI 10.1111/jofi.12380.

Inventory risk of market makers is priced: option order-flow imbalance predicts returns.

Version record

Dmitriy Muravyev and Neil D. Pearson, "Options Trading Costs Are Lower than You Think"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(11), 4973–5014 (2020); DOI 10.1093/rfs/hhaa010.

Accounting for execution timing, effective spreads on options are much smaller than quoted spreads suggest.

Version record

Peter Christoffersen, Ruslan Goyenko, Kris Jacobs and Mehdi Karoui, "Illiquidity Premia in the Equity Options Market"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 31(3), 811–851 (2018); DOI 10.1093/rfs/hhx113.

Option illiquidity predicts higher delta-hedged returns, consistent with compensation for market-maker risk.

Version record

Guido Baltussen, Zhi Da, Sten Lammers and Martin Martens, "Hedging demand and market intraday momentum"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 142(1), 377–403 (2021); DOI 10.1016/j.jfineco.2021.04.029.

Negative gamma positioning by option dealers and leveraged ETFs predicts end-of-day momentum across markets.

Version record

Sophie X. Ni, Neil D. Pearson, Allen M. Poteshman and Joshua White, "Does Option Trading Have a Pervasive Impact on Underlying Stock Prices?"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 34(4), 1952–1986 (2021); DOI 10.1093/rfs/hhaa082.

Delta rebalancing by option market makers affects underlying volatility around expirations.

Version record

Benjamin Anderegg, Florian Ulmann and Didier Sornette, "The Impact of Option Hedging on the Spot Market Volatility"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 124, 102627 (2022); DOI 10.1016/j.jimonfin.2022.102627.

Models how the net delta hedging of aggregated FX option market makers and takers feeds into spot volatility, which rises when market makers are short gamma. Reconstructing market makers' gamma from trade-repository data for October 2017 to June 2018, finds it negative and significant: a gamma exposure of about $-1{,}000$ billion U.S. dollars raises volatility by an absolute 0.7 percentage points in EURUSD and 0.9 in USDJPY.

Version record

Steven L. Heston, Christopher S. Jones, Mehdi Khorram, Shuaiqi Li and Haitao Mo, "The Variance Premium and Seasonal Momentum in Option Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag057 (2026); DOI 10.1093/rfs/hhag057.

Model-free option portfolios isolate the variance premium and reveal quarterly seasonal continuation in realised and implied variance.

Version record

7.5.3 Single-stock option returns and option factors

Patrick Dennis and Stewart Mayhew, "Risk-Neutral Skewness: Evidence from Stock Options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 37(3), 471–493 (2002); DOI 10.2307/3594989.

Risk-neutral skewness of CBOE stock options is more negative for high-beta stocks, in periods of high market volatility and when index option skew is more negative; size and trading volume also help, but leverage and the put–call ratio do not. Firm-specific factors explain more of the variation than systematic ones.

Version record

Peter Christoffersen, Mathieu Fournier and Kris Jacobs, "The Factor Structure in Equity Options"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 31(2), 595–637 (2018); DOI 10.1093/rfs/hhx089.

The first principal components of equity implied-volatility levels, skews and term structures explain much of their cross-sectional variation and are highly correlated with the corresponding S&P 500 index option measures. A valuation model built on this structure predicts that high-beta firms have higher implied volatilities, steeper moneyness slopes and term structures that covary more with the market, as the data confirm.

Version record

Amit Goyal and Alessio Saretto, "Cross-Section of Option Returns and Volatility"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 94(2), 310–326 (2009); DOI 10.1016/j.jfineco.2009.01.001.

Sorts stocks on the difference between historical realised volatility and at-the-money implied volatility. A zero-cost option strategy long the portfolio with a large positive difference and short the one with a large negative difference earns an economically and statistically significant monthly return that is robust across market conditions, risk characteristics, industries and option liquidity and is not explained by standard risk factors.

Version record

Jie Cao and Bing Han, "Cross Section of Option Returns and Idiosyncratic Stock Volatility"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 108(1), 231–249 (2013); DOI 10.1016/j.jfineco.2012.11.010.

Delta-hedged equity option returns fall monotonically as the underlying's idiosyncratic volatility rises, unexplained by standard risk factors or known anomalies. Consistent with constrained intermediaries, dealers charge more for options on high-idiosyncratic-volatility stocks; controlling for limits to arbitrage cuts the relation by about 40%.

Version record

Turan G. Bali and Scott Murray, "Does Risk-Neutral Skewness Predict the Cross-Section of Equity Option Portfolio Returns?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 48(4), 1145–1171 (2013); DOI 10.1017/s0022109013000410.

Builds skewness assets from a long and a short option position and the underlying stock, removing exposure to the stock price (delta) and to implied volatility (vega). Their returns are strongly negatively related to risk-neutral skewness, consistent with a preference for positive skewness, and standard equity and option factors do not explain them.

Version record

Aurelio Vasquez, "Equity Volatility Term Structures and the Cross Section of Option Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 52(6), 2727–2754 (2017); DOI 10.1017/s002210901700076x.

The slope of a stock's implied-volatility term structure predicts its option returns: straddle portfolios with high slopes outperform those with low slopes by an economically and statistically significant amount that traditional factors, higher-order option factors and jump risk do not explain.

Version record

Sophie X. Ni, Jun Pan and Allen M. Poteshman, "Volatility Information Trading in the Option Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 63(3), 1059–1091 (2008); DOI 10.1111/j.1540-6261.2008.01352.x.

Non-market-maker net demand for volatility, built from individual equity option volume, predicts the underlying's realised volatility and raises option prices. The price impact rises by 40% as information asymmetry about volatility intensifies before earnings announcements and returns to normal once the uncertainty is resolved.

Version record

Andrew Dubinsky, Michael Johannes, Andreas Kaeck and Norman J. Seeger, "Option Pricing of Earnings Announcement Risks"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 32(2), 646–687 (2019); DOI 10.1093/rfs/hhy060.

Introduces reduced-form models and estimators that separate the price uncertainty around earnings announcements from ordinary daily volatility in option prices. The anticipated announcement uncertainty is large, varies over time and is informative about future return volatility, and the paper quantifies its effect on option pricing models.

Version record

Chao Gao, Yuhang Xing and Xiaoyan Zhang, "Anticipating Uncertainty: Straddles around Earnings Announcements"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 53(6), 2587–2617 (2018); DOI 10.1017/s0022109018000285.

Single-stock straddles generally earn negative returns, but at-the-money straddles held from three days before an earnings announcement to the announcement earn a significant 3.34%. Returns are higher for small, volatile, high-kurtosis firms with volatile past surprises and higher trading costs, suggesting that investors underestimate announcement uncertainty.

Version record

Lykourgos Alexiou, Amit Goyal, Alexandros Kostakis and Leonidas Rompolis, "Pricing event risk: evidence from concave implied volatility curves"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 29(4), 963–1007 (2025); DOI 10.1093/rof/rfaf016.

Concave short-term implied-volatility curves before earnings announcements signal bimodal outcomes; straddle and calendar-spread returns show investors pay a large premium to hedge announcement gamma risk.

Version record

Xintong (Eunice) Zhan, Bing Han, Jie Cao and Qing Tong, "Option Return Predictability"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 35(3), 1394–1442 (2022); DOI 10.1093/rfs/hhab067.

Expected returns to writing delta-hedged calls fall with stock price, profit margin and profitability and rise with cash holdings, cash-flow variance, share issuance, external financing, distress risk and analyst-forecast dispersion. The strategies have annual Sharpe ratios above two and remain profitable after transaction costs; two option factors explain them, equity factors do not.

Version record

Turan G. Bali, Heiner Beckmeyer, Mathis Mörke and Florian Weigert, "Option Return Predictability with Machine Learning and Big Data"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 36(9), 3548–3602 (2023); DOI 10.1093/rfs/hhad017.

With over 12 million equity option observations from 1996 to 2020, nonlinear machine-learning models substantially improve out-of-sample prediction of option returns and give long–short profits after transaction costs. Option characteristics are the strongest standalone predictors, stock characteristics add substantially, and the evidence points to informational frictions and option mispricing.

Version record

Amit Goyal and Alessio Saretto, "Can Equity Option Returns Be Explained by a Factor Model? IPCA Says Yes"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(6), 1783–1821 (2025); DOI 10.1093/rfs/hhae087.

Many delta-hedged equity option strategies earn large average returns, over 80 basis points a month across 46 long–short strategies built on published signals. An instrumented principal components (IPCA) factor model brings their average alpha close to zero even before transaction costs and can serve as a benchmark for evaluating other option portfolios.

Version record

Alex Horenstein, Aurelio Vasquez and Xiao Xiao, "Common Factors in Equity Option Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(3), 835–874 (2026); DOI 10.1093/rfs/hhaf060.

Delta-hedged equity option returns have a low-dimensional factor structure; a four-factor characteristic model performs comparably to a latent model.

Version record

Niclas Käfer, Mathis Mörke and Tobias Wiest, "Option Factor Momentum"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(1), 32–60 (2026); DOI 10.1017/s0022109025000225.

Time-series and cross-sectional momentum across 28 equity-option factors, not explained by prominent option factor models.

Version record

Niclas Käfer, Mathis Mörke, Florian Weigert and Tobias Wiest, "A Bayesian Stochastic Discount Factor for the Cross-Section of Individual Equity Options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(4), 1632–1659 (2026); DOI 10.1017/s0022109025102251.

Bayesian model-averaged SDF for single-stock options outperforms reduced-form benchmarks in pricing option anomalies.

Version record

Liuren Wu and Yuzhao Zhang, "Common Pricing of Decentralized Risk: A Linear Option Pricing Model"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(6), 1822–1867 (2025); DOI 10.1093/rfs/hhaf016.

Top-down linear option pricing that prices each risk source in proportion to decentralised risk estimates, outperforming bottom-up dynamic models.

Version record

Alexei Boulatov, Assaf Eisdorfer, Amit Goyal and Alexei Zhdanov, "Cheap Options Are Expensive"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Asset Pricing Studies, 16(2), 283–326 (2026); DOI 10.1093/rapstu/raag001.

Retail demand makes options on low-price stocks expensive: delta-hedged returns underperform high-price stocks' options, confirmed by splits and mini-index options.

Version record

Dmitriy Muravyev, Neil D. Pearson and Joshua M. Pollet, "Is There a Risk Premium in the Stock Lending Market? Evidence from Equity Options"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 77(3), 1787–1828 (2022); DOI 10.1111/jofi.13129.

Option prices embed the present value of uncertain stock-borrowing fees, so the gap between option-implied and realised fees estimates any premium for borrow-fee risk. The premium is small, and borrow-fee risk does not predict fee-adjusted short-selling returns.

Version record

Dmitriy Muravyev, Neil D. Pearson and Joshua M. Pollet, "Why does options market information predict stock returns?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 172, 104153 (2025); DOI 10.1016/j.jfineco.2025.104153.

Option-implied volatility spreads predict stock returns because they embed stock-borrow fees; the predictability is consistent with known fee effects.

Version record

7.5.4 VIX futures, volatility products and dividend claims

Robert E. Whaley, "Understanding the VIX"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 35(3), 98–105 (2009); DOI 10.3905/jpm.2009.35.3.098.

Explains what the Cboe Volatility Index measures, its history and purpose, common misreadings of it, and its place among indicators of where markets stand relative to recent history.

Version record

Cboe Global Indices, Cboe Volatility Index Methodology

Difficulty: Advanced **Type:** Data / specification

Status/version: Version 6.0; last revised 26 February 2026.

The authoritative operational companion for VIX research: identifies the option inputs, maturity handling, calculation and dissemination conventions, and the distinction between the spot index and derivatives settlement. Reproduce a worked calculation and separately model the special opening quotation used at expiry. An index value is a calculated benchmark; it is not the execution price of a variance swap, a futures contract or a volatility ETP.

Extended note. Version record

Javier Mencía and Enrique Sentana, "Valuation of VIX Derivatives"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 108(2), 367–391 (2013); DOI 10.1016/j.jfineco.2012.12.003.

Tests VIX futures and option valuation models before, during and after the 2008–2009 crisis. Restrictive mean reversion and heteroskedasticity cause large crisis-period errors, so the authors add a time-varying central tendency, jumps and stochastic volatility; a log-VIX process with central tendency and stochastic volatility prices VIX derivatives reliably, and a significant risk premium shifts the long-run volatility level.

Version record

Ing-Haw Cheng, The VIX Premium

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies 32(1), 180–227, January 2019; first online 2018.

Studies the premium in VIX futures relative to a statistical expectation of future VIX and links its behaviour to hedging demand and trader positions. Read it after variance risk premia to distinguish a futures premium, a realised-variance payoff and the convexity effects of volatility. A steep futures curve is not sufficient evidence of an exploitable premium or a stable rolling strategy.

Extended note. Version record

Bjørn Eraker and Yue Wu, "Explaining the Negative Returns to Volatility Claims: An Equilibrium Approach"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 125(1), 72–98 (2017); DOI 10.1016/j.jfineco.2017.04.007.

Documents a large negative return premium on VIX futures and VIX exchange-traded notes: a constant-maturity one-month VIX futures position lost about 30% a year over 2006–2013. An equilibrium model in which rising volatility lowers stock prices explains these negative returns and other features of VIX futures and ETN returns.

Version record

Carol Alexander, Dimitris Korovilas and Julia Kapraun, "Diversification with Volatility Products"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 65, 213–235 (2016); DOI 10.1016/j.jimonfin.2016.03.002.

Derives the expected returns at which equity and bond investors would see ex ante diversification gains from adding volatility products. In U.S. and European data the gains often appear optimal ex ante but are almost never realised because of high roll and transaction costs; volatility futures diversified effectively only during the banking crisis.

Version record

Patrick Augustin, Ing-Haw Cheng and Ludovic Van den Bergen, "Volmageddon and the Failure of Short Volatility Products"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 77(3), 35–51 (2021); DOI 10.1080/0015198x.2021.1913040.

Describes the 5 February 2018 collapse of short-volatility exchange-traded products and how hedge and leverage rebalancing in a concentrated, volatile market drove it. The episode carries risk-management lessons reminiscent of the losses on portfolio insurance.

Version record

John M. Griffin and Amin Shams, "Manipulation in the VIX?"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 31(4), 1377–1417 (2018); DOI 10.1093/rfs/hhx085.

At VIX settlement, SPX option volume spikes only in the out-of-the-money options used to compute the VIX, and more for options with a larger and discontinuous influence on it. Tests using puts and calls, open interest around settlement and a European volatility contract with a different settlement procedure are inconsistent with hedging or coordinated liquidity trading but consistent with manipulation, which shows the importance of settlement design.

Version record

Christine Bangsgaard and Thomas Kokholm, "The stock market impact of volatility hedging: Evidence from end-of-day trading by VIX ETPs"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 180, 107556 (2025); DOI 10.1016/j.jbankfin.2025.107556.

End-of-day VIX ETP rebalancing demand predicts SPX futures returns in the direction of dealers' hedges, followed by reversal: hedging flow moving prices without information.

Version record

Jules van Binsbergen, Michael Brandt and Ralph Koijen, On the Timing and Pricing of Dividends

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review 102(4), 1596–1618, June 2012.

Extracts prices of short-horizon dividend claims and studies the term structure of equity risk. It adds dividend futures and strips to the equity-derivatives route, linking near-term cash-flow claims to the value of a long-lived index. The reported short-horizon premium is a measurement-sensitive empirical result; read the subsequent comment alongside it before turning the term structure into a strategy or a calibration target.

Extended note. Version record

Florian Schulz, On the Timing and Pricing of Dividends: Comment

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review 106(10), 3185–3223, October 2016.

Re-examines dividend-strip returns and shows how market frictions and measurement choices affect the inferred equity-premium term structure. Replication must distinguish the return on a theoretical claim from the return an investor could actually realise.

Extended note. Version record

Jules van Binsbergen, Wouter Hueskes, Ralph Koijen and Evert Vrugt, "Equity Yields"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 110(3), 503–519 (2013); DOI 10.1016/j.jfineco.2013.08.017.

Uses traded dividend claims with maturities up to ten years in the U.S., Europe and Japan to build equity yields, analogous to bond yields, and splits them into expected dividend growth and risk premia by maturity. Both vary strongly over time, especially at short maturities, and equity yields help predict dividend growth and other measures of economic growth such as consumption.

Version record

Jac Kragt, Frank de Jong and Joost Driessen, "The Dividend Term Structure"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 55(3), 829–867 (2020); DOI 10.1017/s002210901900036x.

Fits Euro Stoxx 50 dividend futures with a two-factor model of discounted risk-adjusted dividend growth: one factor mean-reverts within a year, the other at business-cycle horizons. Estimated on dividend futures alone, the model explains a large part of daily stock market returns.

Version record

Ravi Bansal, Shane Miller, Dongho Song and Amir Yaron, "The Term Structure of Equity Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 142(3), 1209–1228 (2021); DOI 10.1016/j.jfineco.2021.05.043.

In U.S., European and Japanese dividend strips over 2004–2017, the term structure of dividend risk premia slopes up in expansions and down in recessions. A regime-switching consumption model matches this pattern and implies an upward-sloping unconditional term structure; a downward average slope can arise when a short sample over-represents recessions.

Version record

Niels Joachim Gormsen, "Time Variation of the Equity Term Structure"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 76(4), 1959–1999 (2021); DOI 10.1111/jofi.13020.

The slope of the term structure of one-period expected returns on dividend claims is countercyclical. Long-run-risk and habit models fit the cyclicality but not the average downward slope, while models built to match the downward slope miss the cyclicality; a model with two priced factors reconciles both.

Version record

Maxim Ulrich, Stephan Florig and Ralph Seehuber, "A Model-Free Term Structure of U.S. Dividend Premiums"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 36(3), 1289–1318 (2023); DOI 10.1093/rfs/hhac035.

Combines analysts' multi-horizon dividend forecasts for individual firms with option-implied S&P 500 dividend growth and risk-free rates to estimate an ex ante, model-free term structure of the dividend risk premium over 2004–2021.

Version record

7.5.5 Short-dated options and retail option trading

Torben G. Andersen, Nicola Fusari and Viktor Todorov, "Short-Term Market Risks Implied by Weekly Options"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 72(3), 1335–1386 (2017); DOI 10.1111/jofi.12486.

Weekly S&P 500 options isolate volatility and jump risks because they are largely insensitive to shifts in the longer-run economic environment. A semi-nonparametric approach uncovers variation in negative jump-tail risk that market volatility does not span and that helps predict future equity returns.

Version record

Viktor Todorov and Yang Zhang, "Intraday volatility patterns from short-dated options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 254, 105732 (2026); DOI 10.1016/j.jeconom.2024.105732.

Recovers intraday volatility patterns from short-dated options.

Extended note. Version record

Federico M. Bandi, Nicola Fusari and Roberto Renò, "0DTE Option Pricing"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4503344 (2023); DOI 10.2139/ssrn.4503344. Working-paper record; no journal version established in this check.

Pricing framework for same-day options with intraday volatility and jump components.

Extended note. Version record

Carsten H. Chong and Viktor Todorov, "Do Equity and Options Markets Agree about Volatility?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, jofi.70070 (2026); DOI 10.1111/jofi.70070.

Same-day-expiry options impose tight pricing-kernel restrictions on small, frequent price moves; violations create pseudo-arbitrage between equity and options markets.

Version record

Caio Almeida, Gustavo Freire and Rodrigo Hizmeri, "0DTE Asset Pricing"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4701401 (2024); DOI 10.2139/ssrn.4701401. Working-paper record; no journal version established in this check.

Zero-days-to-expiry (0DTE) options, about half of S&P 500 option volume, show investors demanding more compensation for positive than for negative market returns: the 0DTE variance premium is high, driven mainly by upside risk, and negatively predicts market returns. Most 0DTEs violate price bounds implied by risk aversion; a strategy exploiting the violations was highly profitable until 2022 but faded once daily expiries were available.

Extended note. Version record

Chukwuma Dim, Bjørn Eraker and Grigory Vilkov, "0DTEs: Trading, Gamma Risk and Volatility Propagation"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4692190 (2024); DOI 10.2139/ssrn.4692190. Working-paper record; no journal version established in this check.

Finds no evidence that zero-days-to-expiry option trading destabilises S&P 500 volatility through dealer gamma; check the revision and publication status.

Extended note. Version record

Jonathan Brogaard, Jaehee Han and Peter Y. Won, "Does 0DTE Options Trading Increase Volatility?"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4426358 (2023); DOI 10.2139/ssrn.4426358. Working-paper record; no journal version established in this check.

Monthly volume of index 0DTE options rose from 0.08 million contracts in January 2011 to 34.4 million in August 2023, 48% of index option trading. Using the staggered introduction of weekly index options as an instrument, a one-standard-deviation rise in 0DTE trading raises volatility by 9.1% of its mean; the effect survives controls for market makers' gamma hedging and is driven mainly by speculative retail investors.

Extended note. Version record

Grigory Vilkov, "0DTE Trading Rules"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4641356 (2023); DOI 10.2139/ssrn.4641356. Working-paper record; no journal version established in this check. Revision with data to 2 February 2026.

Studies realised payoffs of S&P 500 0DTE options and standard multi-leg structures from September 2016 to February 2026. A positive same-day variance premium exists but is small and hard to monetise after realistic frictions; P&L is wide, state-dependent and tail-dominated, loading more on direction and skewness than on realised variance. Conditional rules set at 10:00 ET under a strict out-of-sample protocol nonetheless deliver meaningful net performance for some strategies and baskets.

Extended note. Version record

Josef Lakonishok, Inmoo Lee, Neil D. Pearson and Allen M. Poteshman, "Option Market Activity"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 20(3), 813–857 (2007); DOI 10.1093/rfs/hhl025.

Describes purchased and written open interest and opening buy and sell volume by investor class. Straddle and strangle volatility trades are a small share of volume, much call writing is part of covered calls, and during the late-1990s bubble the least sophisticated investors sharply increased call purchases on growth but not value stocks.

Version record

Rob Bauer, Mathijs Cosemans and Piet Eichholtz, "Option Trading and Individual Investor Performance"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 33(4), 731–746 (2009); DOI 10.1016/j.jbankfin.2008.11.005.

Most individual investors lose heavily on options, far more than on their equity trades, because of poor market timing driven by overreaction to past returns, compounded by high trading costs. Gambling and entertainment, not hedging, appear to motivate most option trading, and performance persists across traders.

Version record

Gregory W. Eaton, T. Clifton Green, Brian S. Roseman and Yanbin Wu, "Retail option traders and the implied volatility surface"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 177, 104238 (2026); DOI 10.1016/j.jfineco.2026.104238.

Retail option trading shapes the implied volatility surface, linking retail demand to option prices.

Extended note. Version record

Tim de Silva, Kevin Smith and Eric C. So, "Losing is Optional: Retail Option Trading and Expected Announcement Volatility"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4050165 (2023); DOI 10.2139/ssrn.4050165. Working-paper record; no journal version established in this check. Version read: 8 June 2023 (first draft January 2022); an earlier SSRN title ends "Earnings Announcement Volatility".

Retail investors buy options heavily before earnings announcements, especially when expected abnormal volatility is high, and trade options rather than stocks more as anticipated volatility rises. They overpay relative to realised volatility, pay large bid–ask spreads and react slowly to announcements, losing 5–9% on average and 10–14% around high-expected-volatility announcements.

Extended note. Version record

Heiner Beckmeyer, Nicole Branger and Leander Gayda, "Retail Traders Love 0DTE Options... But Should They?"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 4404704 (2023); DOI 10.2139/ssrn.4404704. Working-paper record; no journal version established in this check.

Identifies retail S&P 500 option trades through the Automated Price Improvement Mechanism and finds that more than 75% of them involve 0DTE contracts. Despite lower effective spreads from price improvement, retail investors lose substantially; the paper traces the losses to their positioning, both in aggregate and trade by trade.

Extended note. Version record

7.5.6 Volatility trading practice

Euan Sinclair, *Volatility Trading*, 2nd ed. (2013)

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2nd ed. (2013)

Practical volatility forecasting, option valuation, hedging, sizing and risk. Use alongside Section 7.1 to study earnings straddles, event jumps and implied-versus-realised volatility. Event payoffs, executable option quotes and hedging costs must be modelled explicitly.

Applications: practitioner. Version record

Colin Bennett, *Trading Volatility: Trading Volatility, Correlation, Term Structure and Skew*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Self-published (CreateSpace), 2014.

Practitioner guide to variance swaps, dispersion, skew trading and volatility term structure from a sell-side strategist; self-published 2014.

Version record

Euan Sinclair, *Positional Option Trading: An Advanced Guide*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, 2020 (ISBN 9781119583516).

Edge identification, sizing and hedging for multi-day option positions, with emphasis on variance-premium harvesting risks.

Version record

Riccardo Rebonato, *Volatility and Correlation: The Perfect Hedger and the Fox*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2004); DOI 10.1002/9781118673539.

Practitioner and modeller view of smile dynamics, correlation and hedging for rates and equity options; second edition (2004).

Version record

Peter Carr and Roger Lee, "Volatility Derivatives"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 1(1), 319–339 (2009); DOI 10.1146/annurev.financial.050808.114304.

Review of variance and volatility swaps, VIX-style replication and model-free hedging of volatility exposure.

Version record

Sébastien Bossu, *Advanced Equity Derivatives: Volatility and Correlation*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2014); DOI 10.1002/9781118835364.

Practitioner text on variance and volatility swaps, dispersion, correlation products and their replication.

Version record

Euan Sinclair, *Option Trading: Pricing and Volatility Strategies and Techniques*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2012); DOI 10.1002/9781119198673.

Practical option trading: volatility forecasting, hedging choices, trade sizing and the economics of selling and buying volatility.

Version record

Roni Israelov and Lars N. Nielsen, "Covered Calls Uncovered"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 71(6), 44–57 (2015); DOI 10.2469/faj.v71.n6.1.

Decomposes covered-call returns into equity, short-volatility and implicit equity-timing exposures; the timing component is uncompensated.

Version record

Robert E. Whaley, "Return and Risk of CBOE Buy Write Monthly Index"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Derivatives, 10(2), 35–42 (2002); DOI 10.3905/jod.2002.319194.

The BXM buy-write benchmark earned equity-like returns with lower volatility over 1988–2001, an early systematic short-volatility reference.

Version record

7.6 Option-implied densities, recovery and tail risk

Option prices across strikes reveal the market's risk-neutral distribution. These readings extract it, extend it into the tails, try to separate beliefs from risk aversion, and test what it forecasts; jump and tail premia are measured from the same quotes. The Breeden–Litzenberger density result is in Section 7.1, the Bakshi–Kapadia–Madan moment estimators in Section 7.5, and Horvath's arbitrage-based recovery with model-free pricing in Section 7.2.

7.6.1 Risk-neutral densities and implied moments

Peter Christoffersen, Kris Jacobs and Bo Young Chang, "Forecasting with Option-Implied Information"

Difficulty: Advanced **Type:** Book chapter

Status/version: Chapter in *Handbook of Economic Forecasting* (Elsevier), 581–656 (2013); the DOI is in the title link and version record.

Surveys how to extract option-implied volatility, skewness, kurtosis and densities and use them in forecasting, including implied covariance, correlation and beta and cross-sectional equity forecasts. Discusses how to adjust implied quantities for risk premia before using them in forecasting regressions.

Version record

Jens Carsten Jackwerth and Mark Rubinstein, "Recovering Probability Distributions from Option Prices"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 51(5), 1611–1631 (1996); DOI 10.1111/j.1540-6261.1996.tb05219.x.

Recovers risk-neutral distributions of the S&P 500 from European options by nonparametric optimisation constrained to fit option and index prices, and proposes a fast maximum-smoothness criterion. Since the 1987 crash, a three (four) standard deviation fall over a year, about -36% (-46%), is about 10 (100) times more likely than under lognormality.

Version record

Yacine Aït-Sahalia and Andrew W. Lo, "Nonparametric Estimation of State-Price Densities Implicit in Financial Asset Prices"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 53(2), 499–547 (1998); DOI 10.1111/0022-1082.215228.

Constructs a nonparametric estimator of the state-price density implicit in option prices, derives its asymptotic sampling theory and applies it to S&P 500 options. The estimate is arbitrage-free, captures negative skewness, excess kurtosis and smiles, and can price new, complex or illiquid claims.

Version record

Mark Britten-Jones and Anthony Neuberger, "Option Prices, Implied Price Processes, and Stochastic Volatility"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 55(2), 839–866 (2000); DOI 10.1111/0022-1082.00228.

Characterises all continuous price processes consistent with current option prices, extending local-volatility results to stochastic volatility. The characterisation gives a volatility forecast that needs only current option prices, and a fast way to fit arbitrary volatility processes exactly to today's prices and price exotics on lattices.

Version record

George J. Jiang and Yisong S. Tian, "The Model-Free Implied Volatility and Its Information Content"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 18(4), 1305–1342 (2005); DOI 10.1093/rfs/hhi027.

Extends the Britten-Jones–Neuberger model-free implied variance to price processes with jumps and shows how to compute it from option quotes. On S&P 500 options it subsumes the information in Black–Scholes implied volatility and past realised volatility and forecasts future realised volatility more efficiently.

Version record

Stephen Figlewski, "Estimating the Implied Risk-Neutral Density for the US Market Portfolio"

Difficulty: Research **Type:** Book chapter

Status/version: Chapter in *Volatility and Time Series Econometrics* (Oxford University Press), 323–353 (2010); the DOI is in the title link and version record.

Sets out a practical procedure for complete, well-behaved risk-neutral densities: smooth implied volatilities, incorporate the bid–ask spread in the fit, and complete the tails with generalised extreme value distributions. Applies it to S&P 500 options from 1996 to 2008.

Version record

Thomas R. Bollinger, William R. Melick and Charles P. Thomas, "Principled Pasting: Attaching Tails to Risk-Neutral Probability Density Functions Recovered from Option Prices"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 23(12), 1751–1768 (2023); DOI 10.1080/14697688.2023.2272677.

When tails are attached to a curve-fitted risk-neutral density, requires them also to price the options at the attachment strikes, not only to match the density and distribution there. The resulting densities price options better both inside and outside the fitting set; put–call parity complicates which options count as out of sample.

Version record

7.6.2 Risk aversion, recovery and expected returns

Jens Carsten Jackwerth, "Recovering Risk Aversion from Option Prices and Realized Returns"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 13(2), 433–451 (2000); DOI 10.1093/rfs/13.2.433.

Combines S&P 500 option-implied and realised-return distributions to derive implied risk-aversion functions. Before the 1987 crash they are positive and decreasing in wealth; afterwards they are partly negative and partly increasing, which the paper attributes to option mispricing, and simulated strategies exploiting it earn excess returns after costs, further crashes and downside hedges.

Version record

Robert R. Bliss and Nikolaos Panigirtzoglou, "Option-Implied Risk Aversion Estimates"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 59(1), 407–446 (2004); DOI 10.1111/j.1540-6261.2004.00637.x.

Adjusts risk-neutral densities from FTSE 100 and S&P 500 options with power and exponential utility to estimate the representative investor's relative risk aversion by horizon. The estimates are reasonable, consistent across utility functions and markets, decline with horizon and are lower when market volatility is high.

Version record

Steve Ross, "The Recovery Theorem"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 70(2), 615–648 (2015); DOI 10.1111/jofi.12092.

State prices are the product of the pricing kernel and natural probabilities. The Recovery Theorem separates the two from state prices alone, recovering the market's return forecast, risk aversion, risk premium and probability of a catastrophe, and allowing model-free tests of market efficiency.

Version record

Jaroslav Borovička, Lars Peter Hansen and José A. Scheinkman, "Misspecified Recovery"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 71(6), 2493–2544 (2016); DOI 10.1111/jofi.12404.

Uses Perron–Frobenius theory to isolate a positive martingale component of the stochastic discount factor. The probability measure that recovery delivers absorbs long-term risk adjustments, so when the martingale is non-degenerate, as in many structural models, treating it as investors' beliefs distorts inference about risk–return trade-offs.

Version record

Ian W. R. Martin and Christian Wagner, "What Is the Expected Return on a Stock?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 74(4), 1887–1929 (2019); DOI 10.1111/jofi.12778.

Derives a stock's expected return from the risk-neutral variance of the market and the stock's excess risk-neutral variance relative to the average stock, all computable from index and stock options with no free parameters. The formula performs well in and out of sample and implies more variation in expected returns, over time and across stocks, than previously acknowledged.

Version record

7.6.3 Jump and tail risk premia

Jun Pan, "The Jump-Risk Premia Implicit in Options: Evidence from an Integrated Time-Series Study"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 63(1), 3–50 (2002); DOI 10.1016/s0304-405x(01)00088-5.

Estimates an arbitrage-free model with stochastic volatility and jumps jointly from the S&P 500 index and near-the-money short-dated options. The jump-risk premia move quickly with market volatility, are larger in turbulent markets and help explain the volatility smirk in the cross-section of options.

Extended note. Version record

Tim Bollerslev and Viktor Todorov, "Tails, Fears, and Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 66(6), 2165–2211 (2011); DOI 10.1111/j.1540-6261.2011.01695.x.

Compensation for rare events accounts for a large share of the average equity and variance risk premia. Combining extreme-value approximations of jump tails estimated from high-frequency data with short-dated out-of-the-money options and model-free implied variation measures, builds an Investor Fears index that shows large, time-varying compensation for disaster fears.

Version record

Torben G. Andersen, Nicola Fusari and Viktor Todorov, "The Risk Premia Embedded in Index Options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 117(3), 558–584 (2015); DOI 10.1016/j.jfineco.2015.06.005.

Using time series of index option surfaces, finds priced left-tail risk that market volatility and its components cannot span and introduces a tail factor. The factor adds nothing to forecasts of future volatility or jumps but is critical for predicting equity and variance risk premia, which react far more persistently after crises than the risks themselves.

Version record

Bryan Kelly, Ľuboš Pástor and Pietro Veronesi, "The Price of Political Uncertainty: Theory and Evidence from the Option Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 71(5), 2417–2480 (2016); DOI 10.1111/jofi.12406.

Isolates political uncertainty with national elections and global summits. Options whose lives span these events are more expensive, as a model of government policy choice predicts, because they protect against price, variance and tail risks; the protection is worth more in a weak economy and amid high political uncertainty, and the effects spill over across countries.

Version record

7.7 Convertibles, exotics and structured products

Subject context and qualifications.

7.7.1 Convertible bonds and convertible arbitrage

Jan De Spiegeleer and Wim Schoutens, The Handbook of Convertible Bonds: Pricing, Strategies and Risk Management

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2011 copyright; Wiley online edition 2 January 2012.

Introduces a hybrid market where equity optionality, issuer credit, contractual calls and puts, dividends and stock borrow interact. Use it to build a term-sheet checklist and an attribution separating equity delta, volatility, credit spread, rates and financing. Historical market examples require current confirmations, corporate-action adjustments and financing inputs before they can inform live relative value.

Extended note. Version record

Elie Ayache, Peter A. Forsyth and Kenneth R. Vetzal, Valuation of Convertible Bonds With Credit Risk; Author manuscript

Difficulty: Research **Type:** Research paper

Status/version: Journal of Derivatives 11(1), 9–29, Fall 2003; linked author manuscript dated April 2003.

Develops convertible valuation with explicit treatment of credit events and the associated stock-price behaviour. It is a technical companion to the handbook for examining how default, recovery and contractual exercise features enter a numerical pricing problem. Callable and puttable features require careful exercise logic and numerical convergence; apparently similar bond/equity decompositions can produce inconsistent values when their default assumptions differ.

Extended note. Version record

Michael J. Brennan and Eduardo S. Schwartz, "Convertible Bonds: Valuation and Optimal Strategies for Call and Conversion"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 32(5), 1699–1715 (1977); DOI 10.1111/j.1540-6261.1977.tb03364.x.

Contingent-claims valuation of convertibles with optimal call and conversion policies.

Version record

Kostas Tsiveriotis and Chris Fernandes, "Valuing Convertible Bonds with Credit Risk"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Fixed Income, 8(2), 95–102 (1998); DOI 10.3905/jfi.1998.408243.

Split of convertible value into cash-only and equity components discounted at different rates; industry-standard PDE approach.

Version record

The Southern Company, Second Supplemental Indenture: Series 2026B 3.50% Convertible Senior Notes due 2029; issuance report

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Executed supplemental indenture dated 6 August 2026; SEC Exhibit 4.3.

Connects equity-credit valuation to conditional conversion, VWAP averaging, settlement election and event adjustments. Read Sections 1.02, 2.01–2.03 and 2.07 alongside the incorporated base indenture. The 40-trading-day VWAP observation period followed by settlement creates path dependence beyond a terminal equity option; the notes have no issuer optional redemption. Build a term-sheet state machine before attempting a hedge. This US issuer example does not represent all convertible structures, and market calibration requires separate bond quotes, stock borrow and volatility inputs.

Extended note. Version record

Darwin Choi, Mila Getmansky, Brian Henderson and Heather Tookes, "Convertible bond arbitrageurs as suppliers of capital"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 23(6), 2492–2522 (2010); DOI 10.1093/rfs/hhq003.

Convertible arbitrage funds supply capital to issuers; the 2008 short-sale ban disrupted issuance and pricing.

Version record

Stephen J. Brown, Bruce D. Grundy, Craig M. Lewis and Patrick Verwijmeren, "Convertibles and Hedge Funds as Distributors of Equity Exposure"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 25(10), 3077–3112 (2012); DOI 10.1093/rfs/hhs088.

Hedge funds buying convertibles and shorting stock act as distributors of the issuer's equity exposure; issuance-day price pressure is short-sale driven.

Version record

Vikas Agarwal, William H. Fung, Yee Cheng Loon and Narayan Y. Naik, "Risk and return in convertible arbitrage: Evidence from the convertible bond market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 18(2), 175–194 (2011); DOI 10.1016/j.jempfin.2010.11.008.

Builds convertible-arbitrage risk factors from bond-level data and explains hedge-fund convertible-arbitrage returns.

Version record

Mark Mitchell, Lasse Heje Pedersen and Todd Pulvino, "Slow Moving Capital"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 97(2), 215–220 (2007); DOI 10.1257/aer.97.2.215.

Convertible-bond and merger arbitrage after capital shocks (2005 redemptions, 2008): prices recover slowly because arbitrage capital is slow.

Version record

Manuel Ammann, Axel Kind and Christian Wilde, "Are convertible bonds underpriced? An analysis of the French market"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Banking & Finance, 27(4), 635–653 (2003); DOI 10.1016/s0378-4266(01)00256-4.

Model prices exceed market prices of French convertibles by a few percent, consistent with the underpricing that funds convertible arbitrage.

Version record

7.7.2 Exotic options and static hedging

Mohamed Bouzoubaa and Adel Osseiran, *Exotic Options and Hybrids: A Guide to Structuring, Pricing and Trading*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2012); DOI 10.1002/9781119206965.

Structuring and risk of barriers, cliquets, autocallables, baskets, correlation and hybrid products from a desk perspective.

Version record

Mark Broadie, Paul Glasserman and Steven Kou, "A Continuity Correction for Discrete Barrier Options"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 7(4), 325–349 (1997); DOI 10.1111/1467-9965.00035.

Most barrier contracts are monitored on discrete dates, for which closed-form prices are lacking. Shifting the barrier away from the underlying by a factor $\exp(\beta\sigma\sqrt{\Delta t})$, with $\beta \approx 0.5826$, lets continuous-monitoring formulas price them with remarkable accuracy; the correction is justified theoretically and numerically.

Version record

Emanuel Derman, Deniz Ergener and Iraj Kani, "Static Options Replication"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Derivatives, 2(4), 78–95 (1995); DOI 10.3905/jod.1995.407927.

Constructs a fixed-weight portfolio of standard options with several strikes and maturities that replicates a target option over a range of stock prices and times before expiry without rebalancing. The method gives static hedges for exotic options, a way to build exotic payoffs from standard options, and a valuation check against more liquid quotes.

Version record

Peter Carr, Katrina Ellis and Vishal Gupta, "Static Hedging of Exotic Options"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 53(3), 1165–1190 (1998); DOI 10.1111/0022-1082.00048.

Builds static hedges for several exotic options from standard options, using a relationship between European puts and calls with different strikes. The method allows for constant volatility or for volatility smiles and frowns.

Version record

7.7.3 Autocallables and retail structured products

Tristan Guillaume, "Autocallable Structured Products"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Derivatives, 22(3), 73–94 (2015); DOI 10.3905/jod.2015.22.3.073.

Gives a general analytical valuation of autocallable structured products with contingent and memory coupons, partial principal protection with conversion into the stock, upside participation and best-of or worst-of features, the terms issuers use to raise yield for income-seeking investors. The available abstract is the journal's editorial summary.

Version record

Yeda Cui, Lingfei Li and Gongqiu Zhang, Pricing and hedging autocallable products by Markov chain approximation

Difficulty: Research **Type:** Research paper

Status/version: Review of Derivatives Research 27, 259–303, 2024; published 4 October 2024.

Develops continuous-time Markov-chain approximations for pricing autocallables and investigates their hedging. It supplies a concrete implementation route for products with coupon, call and barrier discontinuities under the paper's model classes. Use it to compare grid accuracy, runtime and hedge performance with Monte Carlo or PDE benchmarks. Results for the studied state spaces and contracts do not automatically extend to a multi-asset worst-of book.

Extended note. Version record

Sun-Joong Yoon, "Structured Products Markets and Implied Volatility Distortion"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Derivatives and Quantitative Studies, 22(3), 433–464 (2014); DOI 10.1108/jdqs-03-2014-b0003.

Most Korean equity-linked securities (ELS) are step-down autocallables that leave their issuers long vega, which issuers offset by selling volatility through warrants and listed or over-the-counter options. Finds that growth in ELS issuance pushes Korean implied volatility down and narrows the spread between implied and realised volatility, and proposes listed VKOSPI derivatives to balance volatility supply and demand.

Version record

Martin Wallmeier and Martin Diethelm, "Market Pricing of Exotic Structured Products: The Case of Multi-Asset Barrier Reverse Convertibles in Switzerland"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Derivatives, 17(2), 59–72 (2009); DOI 10.3905/jod.2009.17.2.059.

Values Swiss multi-asset barrier reverse convertibles, which pay a fixed coupon and repay par or shares of the worst-performing of usually three stocks if any hits its barrier, with a multinomial tree. Market prices exceed model values on average, more so for less commonly used underlyings and for products in low-interest-rate currencies.

Version record

Carole Bernard, Phelim P. Boyle and William Gornall, "Locally Capped Investment Products and the Retail Investor"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Derivatives, 18(4), 72–88 (2011); DOI 10.3905/jod.2011.18.4.072.

Values retail products that pay a guaranteed minimum plus capped index returns summed over subperiods (locally capped, globally floored). They are overpriced by about 6.5% on average, and issuers' marketing relies on cherry-picked scenarios in which the product beats the alternatives. The available abstract is the journal's editorial summary.

Version record

Brian J. Henderson and Neil D. Pearson, "The dark side of financial innovation: A case study of the pricing of a retail financial product"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 100(2), 227–247 (2011); DOI 10.1016/j.jfineco.2010.12.006.

Retail structured notes (SPARQS) were priced almost 8% above fair value at issue; expected returns were below the risk-free rate.

Version record

Claire Célérier and Boris Vallée, "Catering to Investors Through Security Design: Headline Rate and Complexity"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 132(3), 1469–1508 (2017); DOI 10.1093/qje/qjx007.

European retail structured products became more complex as rates fell, to offer attractive headline rates; complexity raised issuer margins.

Version record

Mark Egan, "Brokers versus Retail Investors: Conflicting Interests and Dominated Products"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 74(3), 1217–1260 (2019); DOI 10.1111/jofi.12763.

Using retail convertible-bond data, finds that cheap and expensive otherwise-identical products coexist and that brokers earn about twice the fees for selling the dominated ones. An estimated broker-intermediated search model shows that search costs are compounded when brokers steer clients toward high-fee, inferior products.

Version record

Petra Vokata, "Engineering lemons"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 142(2), 737–755 (2021); DOI 10.1016/j.jfineco.2021.04.035.

US yield-enhancing structured products (reverse convertibles, autocallables) earn returns well below fair value, with losses concentrated in high-coupon products.

Version record

Cliff Asness, Jeffrey Cao, Antti Ilmanen and Dan Villalon, "Rebuffed: An Empirical Review of Buffer Funds"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 51(10), 120–142 (2025); DOI 10.3905/jpm.2025.1.757.

Empirical and theoretical review of buffer (defined-outcome) funds: like earlier downside-protection products they do not deliver a free lunch relative to simple equity/cash mixes.

Version record

Chapter 8

Microstructure, market making and execution

Follow information, inventory and market impact through order placement, dealer quoting, execution and simulation. Fill probabilities and counterfactual outcomes require separate evidence.

8.1 Market structure and liquidity

Section 8.2 adds signature-based execution, nonlinear transient impact and the Graham Capital execution-path stress note, and supplements the Cartea–Jaimungal–Penalva reading.

8.1.1 Theory, texts and dealer models

Maureen O'Hara, *Market Microstructure Theory*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 1995

The foundational theoretical treatment of asymmetric information, dealer inventories, price formation and market design. It supplies the economic models needed to interpret order-book statistics rather than treating imbalance features as unexplained predictors.

Applications: Exchange-traded and dealer markets; market making. Version record

Larry Harris, *Trading and Exchanges*

Difficulty: Intro **Type:** Book / monograph

Status/version: Oxford University Press publication date 24 October 2002; the 2003 date is a common citation year for the selected book, not a verified new edition.

Orders, venues, dealer behaviour and liquidity provision. The Oxford record dates publication to October 2002; 2003 is the commonly cited edition year.

Applications: Equities, futures, options, FX; execution. Version record

Joel Hasbrouck, *Empirical Market Microstructure*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Oxford University Press, 4 January 2007; print ISBN 9780195301649.

Develops econometric methods for trades, quotes, liquidity, price impact, price discovery and information shares, including applications to TAQ and limit-order-book data. It should be read before using high-frequency data because sampling, bid–ask bounce and asynchronous markets can create convincing but spurious signals.

Applications: Equities, futures, FX; HFT, price discovery. Version record

Álvaro Cartea, Sebastian Jaimungal and José Penalva, *Algorithmic and High-Frequency Trading*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2015

Market making, optimal execution, inventory, signals, order flow and stochastic control across traditional and crypto markets. Author resources include code, data and errata; richer state representations raise computational demands.

Applications: Multi-asset; HFT, execution, market making. Extended note. Version record

Albert S. Kyle, "Continuous Auctions and Insider Trading"

Difficulty: Advanced **Type:** Research paper

Status/version: 1985

Canonical informed-trading model connecting order flow, information revelation and price impact. Use it to develop intuition for information leakage and impact; richer transient-impact models require additional assumptions.

Applications: All markets; information, impact. Version record

Lawrence R. Glosten and Paul R. Milgrom, "Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders"

Difficulty: Advanced **Type:** Research paper

Status/version: 1985

Adverse selection produces spreads even for risk-neutral market makers. Provides the information-risk foundation for spread decomposition, toxic order flow and market-making research.

Applications: Market making; adverse selection. Version record

Foucault, Pagano and Röell, *Market Liquidity: Theory, Evidence, and Policy*, 2nd ed. (2023)

Difficulty: Not specified **Type:** Book / monograph

Status/version: 2nd ed. (2023)

Institutions, liquidity measurement, order flow, depth, adverse selection, inventory and market design.

Version record

Gábor Pintér, Semih Üslü and Jean-Charles Wijnandts, Comparing Search and Intermediation Frictions Across Markets; Earlier Bank of England version

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1283; earlier Bank of England Staff Working Paper 974 dated 2022-04-14 and updated 2025-05; no journal version established. Issued 2025-08-08.

Places gilt and corporate-bond liquidity on comparable transaction-level footing by decomposing search and intermediation frictions; useful for understanding why the same investor can face materially different OTC execution costs across sectors.

Extended note. Version record

Ho and Stoll dealer inventory models

Difficulty: Not specified **Type:** Reference resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Dealer inventory risk and quote adjustment. This is an author/topic lead; the supplied bibliography does not identify a particular Ho–Stoll paper.

Version record

Bouchaud et al., *Trades, Quotes and Prices*

Difficulty: Not specified **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Modern empirical and statistical view of microstructure and impact.

Version record

Darrell Duffie, *Dark Markets: Asset Pricing and Information Transmission in Over-the-Counter Markets*

Difficulty: Research **Type:** Book / monograph

Status/version: Princeton University Press (2012); DOI 10.1515/9781400840519.

Search-based theory of OTC pricing, bargaining and information percolation (Princeton Lectures in Finance).

Version record

Darrell Duffie, Nicolae Gârleanu and Lasse Heje Pedersen, "Over-the-Counter Markets"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 73(6), 1815–1847 (2005); DOI 10.1111/j.1468-0262.2005.00639.x.

Search-and-bargaining model of OTC spreads and prices; foundation for the dealer-market literature.

Version record

Ibrahim Ekren, Brad Mostowski and Gordan Žitković, "Kyle's model with stochastic liquidity"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 29(4), 1195–1231 (2025); DOI 10.1007/s00780-025-00574-4.

Kyle's insider-trading model extended to stochastic liquidity.

Extended note. Version record

8.1.2 Liquidity measures and information

Yakov Amihud, "Illiquidity and stock returns: cross-section and time-series effects"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Markets, 5(1), 31–56 (2002); DOI 10.1016/s1386-4181(01)00024-6.

Absolute return per dollar volume as a low-frequency price-impact proxy; the most used illiquidity measure.

Version record

Richard Roll, "A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 39(4), 1127 (1984); DOI 10.2307/2327617.

Infers effective spread from negative serial covariance of price changes.

Version record

Ľuboš Pástor and Robert F. Stambaugh, "Liquidity Risk and Expected Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 111(3), 642–685 (2003); DOI 10.1086/374184.

Aggregate liquidity factor based on return reversal after volume; stocks with high liquidity betas earn higher returns.

Version record

Yakov Amihud and Haim Mendelson, "Asset pricing and the bid-ask spread"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 17(2), 223–249 (1986); DOI 10.1016/0304-405x(86)90065-6.

Expected returns increase with spread, with clientele effects by holding period.

Version record

David Easley, Nicholas M. Kiefer, Maureen O'Hara and Joseph B. Paperman, "Liquidity, Information, and Infrequently Traded Stocks"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 51(4), 1405–1436 (1996); DOI 10.1111/j.1540-6261.1996.tb04074.x.

Probability of informed trading (PIN) estimated from buy and sell order arrivals.

Version record

David Easley, Marcos M. López de Prado and Maureen O'Hara, "Flow Toxicity and Liquidity in a High-frequency World"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 25(5), 1457–1493 (2012); DOI 10.1093/rfs/hhs053.

Volume-synchronised PIN as a real-time toxicity indicator; read with Andersen–Bondarenko for the critique.

Extended note. Version record

Torben G. Andersen and Oleg Bondarenko, "VPIN and the flash crash"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 17, 1–46 (2014); DOI 10.1016/j.finmar.2013.05.005.

Shows VPIN's apparent predictive power for the 2010 flash crash depends on trade classification and is largely mechanical.

Version record

Craig W. Holden and Stacey Jacobsen, "Liquidity Measurement Problems in Fast, Competitive Markets: Expensive and Cheap Solutions"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 69(4), 1747–1785 (2014); DOI 10.1111/jofi.12127.

Monthly TAQ and withdrawn-quote filters distort spread measures; prescribes daily TAQ interpolation fixes.

Version record

Yashar H. Barardehi, Peter Dixon and Qiyu Liu, "Detecting Informed Trading Risk from Undercutting Activity"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(4), 2109–2164 (2026); DOI 10.1111/jofi.70047.

A simple trade-and-quote measure of informed-trading risk based on reduced undercutting by liquidity providers; an alternative to PIN/VPIN-style measures.

Version record

Yuehao Dai, Chao Shi and Ruixun Zhang, "Estimating market liquidity from daily data: Marrying microstructure models and machine learning"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 79, 101019 (2026); DOI 10.1016/j.finmar.2025.101019.

Combines microstructure models with machine learning to estimate liquidity from daily data.

Extended note. Version record

8.1.3 Impact, execution and quoting foundations

Robert Almgren and Neil Chriss, "Optimal Execution of Portfolio Transactions"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Risk article, 2001; publisher DOI 10.21314/JOR.2001.041.

Establishes the canonical trade-off between expected impact cost and execution risk. Although its linear–quadratic assumptions are stylised, it remains the benchmark against which more elaborate stochastic-control and RL execution policies should be measured.

Applications: Equities, futures, FX; optimal execution. Version record

Anna A. Obizhaeva and Jiang Wang, "Optimal Trading Strategy and Supply/Demand Dynamics"; author PDF

Difficulty: Research **Type:** Research paper

Status/version: 2013

Models a resilient limit-order book in which consumed liquidity replenishes over time. It gives a structural basis for execution schedules and transient impact beyond the static Almgren–Chriss framework.

Applications: Exchange-traded assets; execution, impact. Version record

Jim Gatheral, "No-Dynamic-Arbitrage and Market Impact"

Difficulty: Research **Type:** Research paper

Status/version: 2010

Derives restrictions on impact models required to prevent price-manipulation strategies. It is a critical model-risk reference whenever temporary, transient or nonlinear impact functions are embedded in execution simulators.

Applications: Equities, futures; impact, execution. Version record

Rama Cont, Arseniy Kukanov and Sasha Stoikov, "The Price Impact of Order Book Events"; preprint

Difficulty: Research **Type:** Research paper

Status/version: 2014; online 2013

Shows that order-flow imbalance constructed from submissions, cancellations and trades explains short-horizon price changes better than trade volume alone. It is a foundational feature-engineering paper for limit-order-book research.

Applications: Equities, futures, crypto; HFT, order flow. Version record

Sasha Stoikov, The micro-price: a high-frequency estimator of future prices; Author working paper and sample code

Difficulty: Advanced **Type:** Research paper

Status/version: Quantitative Finance 18(12), 1959-1966 (2018); published online 3 September 2018. Published journal article; author working paper was revised 3 June 2020.

Defines a micro-price that adjusts the midpoint for bid-ask spread and book imbalance, then tests its short-horizon predictive value against midpoint and weighted-midpoint baselines. Read after order-flow imbalance and queue-reactive models. Estimate the transition probabilities only from information available at each quote time; align venue timestamps, ticks and cancellations before testing forecasts. A conditional price estimate is not a directly executable edge: evaluate signal decay, fill selection, fees and latency separately. The published evidence concerns historical high-frequency data, so recalibrate on the intended venue.

Extended note. Version record

Andrea Frazzini, Ronen Israel and Tobias Moskowitz, "Trading Costs"

Difficulty: Research **Type:** Research paper

Status/version: 2018

Estimates realised costs across a large institutional trading history and studies their relation to size, volatility and participation. It is essential for converting academic factor returns into credible capacity and net-alpha estimates.

Applications: Equities; factor implementation, TCA. Version record

Yuriy Nevmyvaka, Yi Feng and Michael Kearns, "Reinforcement Learning for Optimized Trade Execution"

Difficulty: Research **Type:** Research paper

Status/version: 2006

A seminal empirical application of RL to execution using limit-order data. It remains an important baseline because it formulates a bounded sequential problem with measurable costs rather than attempting unconstrained price prediction.

Applications: Equities; RL, execution. Version record

Álvaro Cartea, Fayçal Drissi and Pierre Osselin, "Bandits for Algorithmic Trading with Signals"

Difficulty: Research **Type:** Research paper

Status/version: 2023; revised 2026

SSRN revision 16 July 2026 lists *SIAM Journal on Financial Mathematics*, forthcoming. Connects contextual bandits, predictive signals and trading decisions while accounting for costs. It is a useful recent alternative to full RL when the main decision is adaptive action selection rather than long-horizon state control. [17]

Applications: Equities, futures; adaptive execution and trading. Version record

Marco Avellaneda and Sasha Stoikov, "High-Frequency Trading in a Limit Order Book"

Difficulty: Advanced **Type:** Research paper

Status/version: Quantitative Finance 8(3), 217–224, April 2008; published online 28 March 2008.

Stochastic-control market-making model with inventory- and order-flow-sensitive quotes. Use as a benchmark and make its stylised arrival, fill and inventory assumptions explicit in a simulator.

Applications: LOB markets; market making. Version record

8.1.4 High-frequency trading, fragmentation and market design

Citadel Securities, "Exchange Best Practices for Reducing Operational Risk (Market Lens)"

Difficulty: Advanced **Type:** Practitioner / commentary

Status/version: 2021

Practitioner venue-risk document connecting exchange operations to execution monitoring, kill switches and venue certification. Use it to identify operational assumptions missing from research models; it is a firm perspective.

Applications: Venues; operational risk, execution. Version record

Citadel Securities, "Enhancing Competition and Innovation in U.S. Financial Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: 2025

Market-structure white paper on competition and innovation. Read for proposals that could alter liquidity, routing and execution economics; distinguish the firm's policy advocacy from enacted requirements or neutral empirical consensus.

Applications: US markets; structure, execution. Version record

Citadel Securities, *Market Internals*

Difficulty: Advanced **Type:** Practitioner / commentary

Status/version: Ongoing; 2026 example

Recurring commentary on liquidity, dispersion, positioning and structural change, including a February 2026 example. A changing portal is not a frozen historical dataset.

Applications: Multi-asset; market-state monitoring. Version record

Eric Budish, Peter Cramton and John Shim, "The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 130(4), 1547–1621 (2015); DOI 10.1093/qje/qjv027.

Continuous limit order books create mechanical arbitrage rents from latency; proposes frequent batch auctions.

Version record

Terrence Hendershott, Charles M. Jones and Albert J. Menkveld, "Does Algorithmic Trading Improve Liquidity?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 66(1), 1–33 (2011); DOI 10.1111/j.1540-6261.2010.01624.x.

Uses NYSE autoquote as an instrument and finds algorithmic trading narrowed spreads and improved price informativeness.

Version record

Jonathan Brogaard, Terrence Hendershott and Ryan Riordan, "High-Frequency Trading and Price Discovery"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(8), 2267–2306 (2014); DOI 10.1093/rfs/hhu032.

HFTs trade in the direction of permanent price changes and against transitory errors overall.

Version record

Albert J. Menkveld, "High frequency trading and the new market makers"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 16(4), 712–740 (2013); DOI 10.1016/j.finmar.2013.06.006.

Case study of one HFT market maker across Chi-X and Euronext: inventory, position-taking and cross-venue activity.

Version record

Albert J. Menkveld, "The Economics of High-Frequency Trading: Taking Stock"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 8(1), 1–24 (2016); DOI 10.1146/annurev-financial-121415-033010.

Annual Review survey of HFT theory and evidence on liquidity, price discovery and welfare.

Version record

Maureen O'Hara and Mao Ye, "Is market fragmentation harming market quality?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 100(3), 459–474 (2011); DOI 10.1016/j.jfineco.2011.02.006.

More fragmented US stocks show lower costs and faster execution, consistent with a virtual single market under Reg NMS.

Version record

Haoxiang Zhu, "Do Dark Pools Harm Price Discovery?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(3), 747–789 (2014); DOI 10.1093/rfs/hht078.

Dark pools attract uninformed flow, which can concentrate informed trading on lit venues and improve price discovery.

Version record

Matthew Baron, Jonathan Brogaard, Björn Hagströmer and Andrei Kirilenko, "Risk and Return in High-Frequency Trading"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 54(3), 993–1024 (2019); DOI 10.1017/s0022109018001096.

HFT profits are concentrated in the fastest firms and in aggressive strategies; entry and latency competition shape returns.

Version record

James Brugler and Carole Comerton-Forde, "Differential access to dark markets and execution outcomes"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 171, 104086 (2025); DOI 10.1016/j.jfineco.2025.104086.

Dark pools that restrict HFT access show less information leakage, adverse selection and post-trade imbalance; closures provide quasi-experimental evidence.

Version record

Nittai K. Bergman, Ohad Kadan, Roni Michaely and Pamela C. Moulton, "Do proprietary traders provide liquidity?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 184, 104325 (2026); DOI 10.1016/j.jfineco.2026.104325.

Asks whether proprietary trading firms provide or consume liquidity.

Extended note. Version record

Jean-Edouard Colliard, Thierry Foucault and Stefano Lovo, "Algorithmic Pricing and Liquidity in Securities Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag010 (2026); DOI 10.1093/rfs/hhag010.

Q-learning algorithmic market makers adapt to adverse selection but fail to learn competitive pricing, leading to wider spreads; implications for AI in liquidity provision.

Version record

Yi-Tsung Lee, Roberto Riccó and Kai Wang, "Frequent batch auctions vs. continuous trading: Evidence from Taiwan"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 101082 (2026); DOI 10.1016/j.finmar.2026.101082.

Evidence comparing frequent batch auctions with continuous trading from Taiwan's market-design change.

Extended note. Version record

Yashar H. Barardehi, Peter Dixon, Qiyu Liu and Ariel Lohr, "When does the tick size help or harm market quality? Evidence from the Tick Size Pilot"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 78, 101024 (2026); DOI 10.1016/j.finmar.2025.101024.

Tick Size Pilot evidence: a 5-cent tick harmed liquidity for narrow-spread stocks but improved it for wider-spread stocks, through pricing fidelity versus undercutting.

Version record

Sergio Pulido, Mathieu Rosenbaum and Emmanouil Sfendourakis, "Understanding the worst-kept secret of high-frequency trading"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 30(2), 329–396 (2026); DOI 10.1007/s00780-026-00585-9.

Explains why order-book volume imbalance predicts price moves: in a market-making model with an efficient price and uncertainty zones, imbalance arises as the optimal quoting response.

Version record

Bruno Biais, Thierry Foucault and Sophie Moinas, "Equilibrium fast trading"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 116(2), 292–313 (2015); DOI 10.1016/j.jfineco.2015.03.004.

Fast connections help investors find quotes in fragmented markets but let fast traders learn information first, imposing adverse selection on slow traders. Institutions overinvest in speed in equilibrium; welfare is highest with one market where fast and slow traders coexist and a Pigovian tax on speed.

Version record

Thierry Foucault, Roman Kozhan and Wing Wah Tham, "Toxic Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 30(4), 1053–1094 (2017); DOI 10.1093/rfs/hhw103.

Short-lived arbitrage opportunities arise when quotes lag news, exposing dealers to stale-quote risk. With triangular-arbitrage data, illiquidity is higher on days when toxic opportunities are more frequent and arbitrageurs relatively faster: the efficiency gain from fast arbitrage comes with more adverse selection.

Version record

Jean-Edouard Colliard and Thierry Foucault, "Trading Fees and Efficiency in Limit Order Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 25(11), 3389–3421 (2012); DOI 10.1093/rfs/hhs089.

Lower trading fees raise gains per trade but can lead investors to post limit orders that execute less often, so investors may end up worse off. The model gives testable predictions for how total fees and their split between liquidity suppliers and demanders affect fill rates and spreads.

Version record

Katya Malinova and Andreas Park, "Subsidizing Liquidity: The Impact of Make/Take Fees on Market Quality"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 70(2), 509–536 (2015); DOI 10.1111/jofi.12230.

Uses a Toronto Stock Exchange fee change. Quotes adjust so that liquidity demanders' costs are unchanged once fees are counted, but narrower posted spreads lead traders, especially retail, to use aggressive orders more, and adverse-selection costs fall.

Version record

Robert Battalio, Shane A. Corwin and Robert Jennings, "Can Brokers Have It All? On the Relation between Make-Take Fees and Limit Order Execution Quality"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 71(5), 2193–2238 (2016); DOI 10.1111/jofi.12422.

Identifies retail brokers that appear to sell market orders and send limit orders to the venues paying the largest rebates. Limit-order execution quality falls as rebates rise, so routing to maximise rebates does not maximise customers' fills.

Version record

8.1.5 Retail flow, auctions and execution quality

Thomas J. Boulton, Thomas D. Shohfi and Michael Walz, "How Does Payment for Order Flow Influence Markets? Evidence from Robinhood Crypto Token Introductions"

Difficulty: Research **Type:** Research paper

Status/version: January 2025 draft

Three-author DERA draft, January 2025 (first draft June 2024). The SSRN revision, 17 October 2025, adds Benjamin Miller. Studies token introductions and market quality; versions differ in reported fee comparisons. Crypto findings do not establish equity/option effects or an SEC policy determination.

Applications: Crypto; PFOF, routing, market quality. Version record

Ekkehart Boehmer, Charles M. Jones, Xiaoyan Zhang and Xinran Zhang, "Tracking Retail Investor Activity"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 76(5), 2249–2305 (2021); DOI 10.1111/jofi.13033.

Identifies wholesaler-internalised retail trades from sub-penny price improvement in TAQ; retail order imbalance predicts returns.

Version record

Brad M. Barber, Xing Huang, Philippe Jorion, Terrance Odean and Christopher Schwarz, "A (Sub)penny for Your Thoughts: Tracking Retail Investor Activity in TAQ"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 79(4), 2403–2427 (2024); DOI 10.1111/jofi.13334.

Experimental trades show the BJZZ algorithm misclassifies many retail trades and mis-signs others; proposes a quote-midpoint refinement.

Version record

Christopher Schwarz, Brad M. Barber, Xing Huang, Philippe Jorion and Terrance Odean, "The 'Actual Retail Price' of Equity Trades"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(5), 2507–2541 (2025); DOI 10.1111/jofi.13467.

Simultaneous experimental trades across brokers reveal large dispersion in execution prices for identical retail orders; Journal of Finance (2025) and a 2025 Dimensional distinguished paper.

Version record

Vincent Bogousslavsky and Dmitriy Muravyev, "Who trades at the close? Implications for price discovery and liquidity"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 66, 100852 (2023); DOI 10.1016/j.finmar.2023.100852.

Closing auctions now account for a large share of volume, driven by index and passive funds; closing prices show more noise.

Version record

Thomas Ernst, Chester Spatt and Jian Sun, "Would Order-By-Order Auctions Be Competitive?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(4), 1879–1927 (2025); DOI 10.1111/jofi.13449.

Models broker routing versus order-by-order auctions for retail orders: auctions improve allocative efficiency but a winner's curse can reduce retail welfare; relevant to the SEC order-competition proposal.

Version record

Anne Haubo Dyhrberg, Andriy Shkilko and Ingrid M. Werner, "The retail execution quality landscape"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 168, 104051 (2025); DOI 10.1016/j.jfineco.2025.104051.

Wholesaler executions save retail investors money; the largest wholesalers offer the lowest costs through scale, and entry raises costs by reducing incumbents' scale.

Version record

Thomas Ernst and Chester Spatt, "Payment for Order Flow and Option Internalization"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhaf108 (2026); DOI 10.1093/rfs/hhaf108.

Option wholesalers internalise retail option flow through auctions and the book; DMM allocation rights create advantages in internalising order flow.

Version record

Hans Degryse and Nikolaos Karagiannis, "Priority Rules, Internalization, and Payment for Order Flow"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Asset Pricing Studies, 15(3-4), 217–246 (2025); DOI 10.1093/rapstu/raaf004.

Theory of broker internalisation and PFOF: effects on order choice, dealer participation and investor welfare.

Version record

Yashar H. Barardehi, Dan Bernhardt, Zhi Da and Mitch Warachka, "Institutional Liquidity Costs, Internalized Retail Trade Imbalances, and the Cross Section of Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(8), 3826–3865 (2025); DOI 10.1017/s0022109025000043.

Wholesalers use internalised retail flow to supply liquidity to institutions when liquidity is scarce; retail imbalances relate to institutional costs and returns.

Version record

Umut Çetin and Albina Danilova, "Order Routing and Market Quality: Who Benefits From Internalization?"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 36(2), 397–421 (2026); DOI 10.1111/mafi.70014.

Compares market designs with and without retail internalisation to assess who benefits from wholesaler order routing.

Version record

Martin ŠmÍd and AleŠ AntonÌn KubĚna, "Distribution of price and volume in a call auction"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(4), 525–540 (2026); DOI 10.1080/14697688.2026.2636565.

Unconditional distribution of clearing price and volume in a call auction with general supply and demand curves.

Version record

Jan Harren, "Institutional trading costs, Robinhood activity, and expected returns"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(8), 1253–1268 (2026); DOI 10.1080/14697688.2026.2691814.

Decomposes permanent price impact into retail and institutional components; institutional impact predicts returns.

Version record

Robert P. Bartlett, Justin McCrary and Maureen O'Hara, "The Market Inside the Market: Odd-Lot Quotes"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 38(3), 661–711 (2025); DOI 10.1093/rfs/hhad074.

Odd-lot quotes routinely improve on the NBBO, contribute to price discovery and are most useful to traders with proprietary data feeds; a model built on them predicts short-run prices. The SEC's round-lot redefinition reduces, but does not remove, better-priced odd-lot quotes inside the NBBO.

Version record

Michael J. Barclay and Terrence Hendershott, "Price Discovery and Trading After Hours"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 16(4), 1041–1073 (2003); DOI 10.1093/rfs/hhg030.

After-hours trading produces significant but inefficient price discovery: individual trades carry more information than during the day, and pre-open price changes are larger and less noisy than post-close ones. A baseline for judging overnight sessions.

Version record

8.1.6 Fractional shares, odd lots and contract size

Trading in units smaller than a round lot or a whole share: fractional trading at retail brokers and how it is reported, odd lots in the consolidated data, the nominal share price, and smaller option and futures contracts. Tokenised fractional claims are in Tokenised securities and real-world assets.

Robert P. Bartlett, Justin McCrary and Maureen O'Hara, "Tiny trades, big questions: Fractional shares"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 157, 103836 (2024); DOI 10.1016/j.jfineco.2024.103836.

Identifies fractional-share trades on the consolidated tape by the execution latency of the two largest fractional broker-dealers (Robinhood and DriveWealth). High-priced stocks, meme stocks, IPOs, SPACs and popular retail names have many tiny trades, largely the fractional legs of dollar-based orders; the measure predicts liquidity and volatility and proxies app-based retail trading. Reporting rules then rounded fractions up, inflating volume and censoring samples.

Version record

Robert Bartlett, Justin McCrary and Maureen O'Hara, "A Fractional Solution to a Stock Market Mystery"

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal, 81(3), 5–26 (2025); DOI 10.1080/0015198x.2025.2489924.

Why did Berkshire Hathaway A volume jump from about 375 shares a day to over 20,000 from February 2021 and fall ninefold in June 2024? The volume was phantom: fractional retail trades reported as whole shares. It distorted the relation between the A and B shares, left arbitrage unexploited, raised trading costs and invited manipulation; a reporting change removed most of it. Reporting rules become a limit to arbitrage.

Version record

Zhi Da, Vivian W. Fang and Wenwei Lin, "Fractional Trading"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(3), 623–660 (2025); DOI 10.1093/rfs/hhae012.

Fractional trading, introduced by US brokers from late 2019, lets capital-constrained retail investors buy high-priced stocks. One-share off-exchange trades surge in stocks priced above USD 100 relative to cheaper stocks; when coordinated around attention-grabbing events these tiny trades exert large price pressure and can fuel meme-stock frenzies and bubbles in high-priced stocks.

Version record

William C. Weld, Roni Michaely, Richard H. Thaler and Shlomo Benartzi, "The Nominal Share Price Puzzle"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 23(2), 121–142 (2009); DOI 10.1257/jep.23.2.121.

Average nominal share prices on the NYSE have stayed near USD 35 since the Great Depression, kept there by splits, even though consumer prices rose more than tenfold and trading ten USD 35 shares cost far more in commissions than one USD 350 share. Signalling and optimal-trading-range explanations fail; custom and norms fit best. The background to why fractional trading matters only where firms stop splitting.

Version record

Maureen O'Hara, Chen Yao and Mao Ye, "What's Not There: Odd Lots and Market Data"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(5), 2199–2236 (2014); DOI 10.1111/jofi.12185.

Odd-lot trades were then missing from the consolidated tape and TAQ even though algorithms and high-frequency traders use them heavily: they are a median 24 per cent of trades and over 60 per cent in some stocks, and contribute 35 per cent of price discovery. Omitting them distorts order-imbalance and sentiment measures.

Version record

Robert Battalio, Shane A. Corwin and Robert Jennings, "Unrecognized Odd Lot Liquidity Supply: A Hidden Trading Cost for High Priced Stocks"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Trading, 12(1), 35–41 (2016); DOI 10.3905/jot.2017.12.1.035.

Odd-lot limit orders are not protected quotes, so trades, especially off-exchange trades benchmarked to the protected quote, can execute at worse prices than displayed odd lots. In ten high-priced stocks over one week in 2015, odd lots bettered the protected quote 37 per cent of the time and about 55,000 trades were disimproved, costing USD 554,675. The cost follows from firms not splitting high-priced shares.

Version record

Alexander Kurov and Dennis J. Lasser, "Price Dynamics in the Regular and E-Mini Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 39(2), 365–384 (2004); DOI 10.1017/s0022109000003112.

Transactions with trader-type codes for the regular and E-mini S&P 500 and Nasdaq-100 futures: price discovery begins in the smaller electronic E-mini contracts, and exchange locals' trades are more informative than customers', apparently because locals see pit order flow. Shows that a smaller contract can lead the full-size one.

Version record

U.S. Securities and Exchange Commission, Order Granting Accelerated Approval . . . to List and Trade Option Contracts Overlying 10 Shares of Certain Securities (Release 34-67948); MIAX, rule change deleting mini-options (Release 34-101191)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC order of 28 September 2012, 77 FR 60735 (4 October 2012); MIAX filing SR-MIAX-2024-38, 89 FR 79985 (1 October 2024).

The closest US precedent for fractional options. NYSE Arca and ISE were allowed to list 'mini options' on 10 shares of SPY, AAPL, GLD, GOOG and AMZN, all priced above USD 100 and heavily traded, so that investors holding odd lots could hedge high-priced stocks. Trading began in 2013; MIAX's 2024 filing records that mini options never gained significant market acceptance, were all delisted several years earlier and would not be relisted.

Version record

NYSE Research (Intercontinental Exchange), Fractional Reporting: Early Findings

Difficulty: Advanced **Type:** Report / research note

Status/version: NYSE Research factsheet, March 2026.

First look at fractional trades after FINRA's trade-reporting facilities began, on 23 February 2026, to require decimal quantities (truncated at six decimals) instead of rounding sub-share trades up to one share. In 2025 just under 19 per cent of TRF-reported trades were one-share prints, far more in high-priced stocks; in the first two weeks of decimal reporting 9.41 per cent of TRF trades had a fractional component, over 95 per cent of them below one share. The data change ends the phantom-volume problem Bartlett, McCrary and O'Hara describe.

Version record

Board of the International Organization of Securities Commissions (IOSCO), Neo-Brokers: Final Report (FR/18/25)

Difficulty: Advanced **Type:** Report / research note

Status/version: IOSCO Final Report FR/18/25, November 2025.

App-based, execution-only brokers that rely on engaging interfaces, nudges, gamification, copy trading and social media. IOSCO recommends that neo-brokers act honestly and fairly, disclose all direct and indirect fees, avoid 'zero cost' claims where clients pay other firms' charges, and address ancillary services, non-commission revenue and IT resilience. IOSCO's retail-conduct work lists fractional asset trading alongside finfluencers, copy trading and digital engagement practices.

Version record

8.1.7 Market-structure rules, 2024–2026

U.S. Securities and Exchange Commission, Disclosure of Order Execution Information (Release 34-99679); compliance-date extension (Release 34-104147)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final rule adopted 6 March 2024; compliance date extended from 14 December 2025 to 1 August 2026.

Amends the monthly Rule 605 execution-quality reports: broker-dealers with larger customer bases now report alongside market centres, covered orders include certain orders outside regular hours and stop orders, and order-size and order-type categories change, including fractional-share orders. Use the reports as data on retail execution quality from August 2026.

Extended note. Version record

U.S. Securities and Exchange Commission, Regulation NMS: Minimum Pricing Increments, Access Fees, and Transparency of Better Priced Orders (Release 34-101070); exemptive order on compliance dates (Press Release 2025-130)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final rule adopted September 2024, effective 9 December 2024; tick-size and access-fee compliance postponed to the first business day of November 2026.

Sets a smaller minimum quoting increment for certain NMS stocks, lowers exchange access-fee caps, and adds a revised round-lot definition and odd-lot information to consolidated data. The tick and fee provisions were stayed pending judicial review; after the D.C. Circuit denied review, an October 2025 order moved their compliance to November 2026. Check current status before modelling queue or fee effects.

Extended note. Version record

U.S. Securities and Exchange Commission, Withdrawal of Proposed Regulatory Actions (Release 34-103247)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Notice published in the Federal Register on 17 June 2025.

Formally withdraws fourteen rule proposals from 2020–2023, among them the proposed Order Competition Rule (88 FR 128) and proposed Regulation Best Execution (88 FR 5440). The Commission states it will issue new proposals if it acts in these areas; treat research on those designs as policy analysis, not current rules.

Extended note. Version record

U.S. Securities and Exchange Commission, order approving Nasdaq's 23-hour trading day (Release 34-105199); New York Stock Exchange, Extended-Hours Trading FAQs (version 4.0)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC order dated 10 April 2026; NYSE FAQ updated August 2026.

The SEC granted accelerated approval for Nasdaq to trade NMS stocks 23 hours a day, five days a week. NYSE Arca plans an overnight session from 9 p.m. to 4 a.m. Eastern time starting 6 December 2026, subject to SEC approvals, SIP availability and DTCC modernisation. Overnight sessions move liquidity, price discovery and execution-quality measurement outside regular hours.

Extended note. Version record

European Parliament and Council, Regulation (EU) 2024/791 amending Regulation (EU) No 600/2014 (MiFIR review); ESMA, Consolidated Tape Providers

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Regulation of 28 February 2024, published in the Official Journal on 8 March 2024; ESMA tape-provider selections 2025–2026.

Amends MiFIR to enhance data transparency, remove obstacles to consolidated tapes, optimise trading obligations and prohibit receiving payment for order flow. ESMA has selected tape providers for bonds (Ediphy, fairCT), shares and ETFs (EuroCTP) and OTC derivatives (Etrading Software); check authorisation and start dates before using a tape as a best-execution or TCA benchmark.

Extended note. Version record

8.2 Execution methods and implementation

8.2.1 Price impact: empirical laws and models

Gianbiagio Curato, Jim Gatheral and Fabrizio Lillo, *Optimal Execution with Nonlinear Transient Market Impact* (2014 preprint; 2016 online publication; 2017 journal issue)

Difficulty: Research **Type:** Research paper

Status/version: Selected DOI-linked version; print 2017-01-02; online 2016-05-24.

Nonlinear transient impact links execution costs to past trades. Negative expected costs in some specifications signal undesirable manipulation incentives; spread costs or convex impact can mitigate them.

Applications: graduate probability, statistics and optimisation. Version record

Robert Almgren, Chee Thum, Emmanuel Hauptmann and Hong Li, "Direct Estimation of Equity Market Impact"

Difficulty: Advanced **Type:** Research paper

Status/version: Risk 18(7), 58–62 (July 2005); author PDF.

Estimates temporary and permanent impact from Citigroup US equity orders; impact scales with a power near 0.6 of participation. Published in Risk (July 2005).

Version record

Bence Tóth, Yves Lempérière, Cyril Deremble, Joachim de Lataillade, Julien Kockelkoren and Jean-Philippe Bouchaud, "Anomalous Price Impact and the Critical Nature of Liquidity in Financial Markets"

Difficulty: Research **Type:** Research paper

Status/version: Physical Review X, 1(2), 021006 (2011); DOI 10.1103/physrevx.1.021006.

Square-root impact law across CFM metaorders and a latent-liquidity explanation.

Version record

Yuki Sato and Kiyoshi Kanazawa, "Strict Universality of the Square-Root Law in Price Impact across Stocks: A Complete Survey of the Tokyo Stock Exchange"

Difficulty: Research **Type:** Research paper

Status/version: Physical Review Letters, 135(25), 257401 (2025); DOI 10.1103/65jz-81kv.

Eight years of account-level Tokyo Stock Exchange data: impact exponent equals one half within error for stocks and traders, rejecting two non-universal models. Physical Review Letters 135, 257401 (15 December 2025).

Version record

Albert S. Kyle and Anna A. Obizhaeva, "Market Microstructure Invariance: Empirical Hypotheses"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 84(4), 1345–1404 (2016); DOI 10.3982/ecta10486.

Invariance predicts how bet sizes, transaction costs and impact scale with trading activity; tested on portfolio-transition data.

Version record

Jean-Philippe Bouchaud, Yuval Gefen, Marc Potters and Matteo Wyart, "Fluctuations and response in financial markets: the subtle nature of random price changes"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 4(2), 176–190 (2004); DOI 10.1080/14697680400000022.

Propagator model: transient impact decaying in time reconciles long-memory order flow with diffusive prices.

Version record

Aurélien Alfonsi, Antje Fruth and Alexander Schied, "Optimal execution strategies in limit order books with general shape functions"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 10(2), 143–157 (2010); DOI 10.1080/14697680802595700.

Optimal execution with resilient order books and transient impact; shows how book shape changes the schedule.

Version record

Elia Zarinelli, Michele Treccani, J. Doyne Farmer and Fabrizio Lillo, "Beyond the square root: Evidence for logarithmic dependence of market impact on size and participation rate"

Difficulty: Research **Type:** Research paper

Status/version: Market Microstructure and Liquidity, 01(02), 1550004 (2015); DOI 10.1142/s2382626615500045.

Large broker metaorder sample suggests impact is better described by a logarithm than a square root at large sizes.

Version record

Kevin T. Webster, *Handbook of Price Impact Modeling*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Chapman and Hall/CRC, edition 1 (2023); DOI 10.1201/9781003316923.

Unified treatment of propagator, Obizhaeva–Wang and nonlinear impact models, alpha-impact interaction, and statistical biases in fitting impact (selection, prediction and causal issues). CRC (May 2023).

Version record

Guillaume Maitrier, Grégoire Loeper, Kiyoshi Kanazawa and Jean-Philippe Bouchaud, "The 'double' square-root law: evidence for the mechanical origin of market impact using Tokyo stock exchange data"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(4), 491–503 (2026); DOI 10.1080/14697688.2026.2615106.

Account-level Tokyo data support a mechanical origin of the square-root impact law ("double" square-root law).

Version record

M. Naviglio, G. Bormetti, F. Campigli, G. Rodikov and F. Lillo, "Why is the estimation of metaorder impact with public market data so challenging?"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(4), 505–523 (2026); DOI 10.1080/14697688.2026.2628221.

Explains why public-data models give linear rather than concave metaorder price paths and why impact estimation from public data is hard.

Version record

Guillaume Maitrier and Jean-Philippe Bouchaud, "The subtle interplay between square-root impact, order imbalance & volatility: a unifying framework"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(6), 855–877 (2026); DOI 10.1080/14697688.2026.2641766.

Reconciles square-root metaorder impact, linear order-imbalance impact and random-walk prices in one framework.

Version record

8.2.2 Optimal execution and TCA

Andrea Frazzini, Ronen Israel and Tobias Moskowitz, *Trading Costs of Asset Pricing Anomalies* (2012 working-paper record)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2012. Version distinctions remain in the annotation.

Implementation costs of equity anomalies; distinct from the authors' 2018 Trading Costs record. Equity evidence does not establish futures or FX capacity.

Version record

Barry Johnson, *Algorithmic Trading and DMA: An Introduction to Direct Access Trading Strategies* (2010)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 2010. Version distinctions remain in the annotation.

Practical guide to direct market access, order types, execution algorithms and transaction-cost analysis. Connects intraday mean reversion and liquidity provision to order handling and implementation. Retained from the source; venue conventions require current documentation.

Applications: practitioner.

Version record

Charles-Albert Lehalle and Sophie Laruelle, eds., *Market Microstructure in Practice*, 2nd ed. (2018)

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2nd ed. (2018)

Order books, fragmented markets, liquidity measurement and optimal trading. Extends the source's abbreviated "Laruelle et al." attribution. Pair with Harris, Hasbrouck and Cartea–Jaimungal–Penalva when turning order-flow and bid–ask effects into testable signals.

Applications: practitioner.

Version record

Michael Durbin, *All About High-Frequency Trading* (2010)

Difficulty: Intro **Type:** Book / monograph

Status/version: Selected citation year(s): 2010. Version distinctions remain in the annotation.

Accessible introduction to HFT operations and strategy families, including cross-market arbitrage and liquidity provision. Useful orientation before the mathematical microstructure readings; historical technology and market descriptions are not current specifications.

Applications: supplementary.

Version record

Irene Aldridge, *High-Frequency Trading: A Practical Guide to Algorithmic Strategies and Trading Systems*, 2nd ed. (2013)

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: 2nd ed. (2013)

Practitioner overview of high-frequency data, strategy design, backtesting and implementation. Use the short-horizon examples to investigate latency, turnover and cost sensitivity; realistic queue and fill models need independent validation.

Applications: supplementary.

Version record

Olivier Guéant, ***The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Chapman & Hall/CRC, first edition, 2016.

Mathematical treatment of liquidity-constrained trading, impact, execution schedules and market-making inventory. Connect execution speed, timing risk and cost; read after Harris/Lehalle and alongside Almgren–Chriss and Cartea–Jaimungal–Penalva.

Version record

Robert Kissell and Morton Glantz, Optimal Trading Strategies: Quantitative Approaches for Managing Market Impact and Trading Risk (2003)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 2003. Version distinctions remain in the annotation.

Practical impact estimation, pre-trade costs, trading risk and transaction-cost analysis. Connect participation-of-volume (POV) and VWAP schedules to urgency, capacity and opportunity cost. Morton Glantz is restored from the author's bibliography.

Version record

Robert Kissell, The Science of Algorithmic Trading and Portfolio Management (2013)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 2013. Version distinctions remain in the annotation.

Algorithm selection, market-impact models and portfolio implementation; distinguishes pre-trade estimates from realised shortfall.

Version record

Dimitris Bertsimas and Andrew W. Lo, "Optimal Control of Execution Costs" (1998)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 1998. Version distinctions remain in the annotation.

Dynamic execution policy for a fixed block over a finite horizon with market-state-dependent impact. Compare its expected-cost objective with Almgren–Chriss's explicit cost/risk trade-off; retain both frameworks rather than calling the earlier one universally superseded.

Version record

Jasdeep Kalsi, Terry Lyons and Imanol Perez Arribas, ***Optimal Execution with Rough Path Signatures*** **(2019 preprint; SIAM Journal on Financial Mathematics 11(2), 470–493, 2020)**

Difficulty: Research **Type:** Research paper

Status/version: Selected DOI-linked version; print 2020-01; online 2020-04-27.

Signature-based execution policies for path-dependent prices and trading-speed impact; richer history increases approximation and computational demands.

Applications: graduate probability, statistics and optimisation. Version record

Erik Forseth, Nastja Bethke, Chuck Rogers and Ed Tricker (Graham Capital Management), ***Putting March 2023 In Perspective: Path-Dependency in Execution in Volatile Times*** **(March 2023 research note)**

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2023. Version distinctions remain in the annotation.

March 2023 Treasury-futures liquidation illustration of execution-path sensitivity. Ex-post daily high/low fill bounds are not attainable strategy returns; the note also compares VWAP windows and longer history.

Applications: execution stress. Version record

Raja Velu, Maxence Hardy and Daniel Nehren, ***Algorithmic Trading and Quantitative Strategies***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Chapman & Hall/CRC, 2020 release; 2021 copyright

Market microstructure, TCA, execution and financial performance measurement; a bridge from experimental design to trading outcomes.

Version record

M. Sotiropoulos and A. Battle, ***Extended Transaction Cost Analysis (TCA)*****, Deutsche Bank; cited in Webster–Westray's Risk article**

Difficulty: Not specified **Type:** Research paper

Status/version: 2017 note, cited in October 2023

Historical TCA note cited indirectly by Webster–Westray; the original was not obtained. Illustrates the cost/power trade-off in live trading experiments.

Version record

Clearpool-sponsored "Best Ex Requires the Best Tools", *Institutional Investor*

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: 26 February 2019

Historical sponsored account of routing comparisons and real-time TCA; not an independently validated experiment or claim of present product availability.

Version record

André F. Perold, "The implementation shortfall: Paper versus reality"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 14(3), 4–9 (1988); DOI 10.3905/jpm.1988.409150.

Defines implementation shortfall as paper-portfolio return minus realised return; the basis of modern TCA.

Version record

Robert Kissell, *Algorithmic Trading Methods: Applications Using Advanced Statistics, Optimization, and Machine Learning Techniques*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Elsevier (2021); DOI 10.1016/c2017-0-03456-0.

Second edition (Academic Press 2021): impact models, TCA, algorithm design, portfolio trading and machine-learning cost estimation.

Version record

Robert Engle, Robert Ferstenberg and Jeffrey Russell, "Measuring and Modeling Execution Cost and Risk"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 38(2), 14–28 (2012); DOI 10.3905/jpm.2012.38.2.014.

Estimates execution cost and its risk from Morgan Stanley algorithmic orders, giving a liquidation-risk frontier.

Version record

Emmanuel Bacry, Adrian Iuga, Matthieu Lasnier and Charles-Albert Lehalle, "Market Impacts and the Life Cycle of Investors Orders"

Difficulty: Research **Type:** Research paper

Status/version: Market Microstructure and Liquidity, 01(02), 1550009 (2015); DOI 10.1142/s2382626615500094.

Large broker metaorder sample showing concave impact during execution and partial reversion afterwards.

Version record

Eduardo Abi Jaber and Eyal Neuman, "Optimal Liquidation With Signals: The General Propagator Case"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 35(4), 841–866 (2025); DOI 10.1111/mafi.12465.

Optimal liquidation with Volterra-propagator transient impact, temporary impact and a predictive signal, solved explicitly.

Version record

Eduardo Abi Jaber, Eyal Neuman and Sturmius Tuschmann, "Optimal Portfolio Choice With Cross-Impact Propagators"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, mafi.70025 (2026); DOI 10.1111/mafi.70025.

Portfolio choice with matrix-valued Volterra cross-impact propagators and temporary impact.

Version record

Steven Campbell and Marcel Nutz, "Optimal Execution Among N Traders With Transient Price Impact"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, mafi.70037 (2026); DOI 10.1111/mafi.70037.

N-player optimal execution games with Obizhaeva–Wang transient impact: regularisation restores equilibrium existence.

Version record

Peter Bank, Álvaro Cartea and Laura Körber, "Optimal execution and speculation with trade signals"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 30(4), 1269–1321 (2026); DOI 10.1007/s00780-026-00602-x.

Price impact driven purely by mutually exciting limit and market order flow; an execution and speculation problem with short-term order-flow signals.

Version record

8.3 Queues, placement and market making

Use Harris, Hasbrouck, Cartea and Almgren–Chriss as foundations. The papers below move from queue dynamics and routing to inventory control and dealer markets, with explicit assumptions about fills and participant response.

8.3.1 Queues, placement and point processes

Rama Cont; Adrien de Larrard, Price Dynamics in a Markovian Limit Order Market; author preprint

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics 4(1), 1–25 (2013); online 29 January 2013.

Models limit and market orders plus cancellations as a queueing system so price changes emerge when book queues deplete. It is a useful theoretical baseline before queue-reactive simulators, passive fill models or learned execution policies. The small Markovian state omits long-memory order flow, venue latency, hidden liquidity and strategic responses to one's own orders, so it is not sufficient as a production fill simulator.

Extended note. Version record

Weibing Huang; Charles-Albert Lehalle; Mathieu Rosenbaum, Simulating and Analyzing Order Book Data: The Queue-Reactive Model

Difficulty: Research **Type:** Research paper

Status/version: Journal of the American Statistical Association 110(509), 107–122 (2015); accepted author version posted 20 November 2014; publisher online 22 April 2015.

Estimates state-dependent order-arrival and cancellation intensities during intervals with a fixed reference price, then stitches those intervals into a full book simulator. It can supply a transparent baseline for ABIDES or JAX-LOB calibration. The conditional Markov assumption and exponential event timing miss clustered latency races and a trader's longer-lived impact; reproduce both book-state statistics and policy costs before trusting simulated fills.

Extended note. Version record

Rama Cont; Arseniy Kukanov, Optimal order placement in limit order markets; Oxford author manuscript

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 17(1), 21–39 (2017 issue); accepted 28 April 2016; online 17 June 2016.

Formulates the immediate choice of limit versus market orders and allocation across fragmented exchanges as a convex order-placement problem. The single-venue result is explicit; the multi-venue case uses a stochastic algorithm. Fill probabilities and venue-response assumptions need re-estimation for each symbol and market regime; the convex optimum is only as credible as those inputs and the selected objective.

Extended note. Version record

Matteo Aquilina; Eric Budish; Peter O'Neill, Quantifying the High-Frequency Trading 'Arms Race'

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics 137(1), 493–564 (February 2022); online 10 September 2021.

Uses exchange message data, including failed trade and cancel attempts, to identify latency-arbitrage races invisible in ordinary successful-order-book records. Its race-frequency, volume and cost estimates are specific to the studied UK venue/data and methodology; they should inform stress cases, not be copied as a universal spread or latency surcharge for every exchange or asset.

Extended note. Version record

Rama Cont, Sasha Stoikov and Rishi Talreja, "A Stochastic Model for Order Book Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: Operations Research, 58(3), 549–563 (2010); DOI 10.1287/opre.1090.0780.

Continuous-time Markov queueing model of the order book with Laplace-transform probabilities for price moves and fills.

Version record

Ciamac C. Moallemi and Kai Yuan, "A Model for Queue Position Valuation in a Limit Order Book"

Difficulty: Research **Type:** Research paper

Status/version: Author working paper (2016 version).

Values queue priority in pro-rata versus price-time markets; quantifies the benefit of joining early.

Extended note. Version record

Charles-Albert Lehalle and Othmane Mounjid, "Limit order strategic placement with adverse selection risk and the role of latency"

Difficulty: Research **Type:** Research paper

Status/version: Market Microstructure and Liquidity, 03(01), 1750009 (2017); DOI 10.1142/s2382626617500095.

Optimal placement and cancellation using order-book imbalance, with explicit latency effects.

Version record

Emmanuel Bacry, Iacopo Mastromatteo and Jean-François Muzy, "Hawkes Processes in Finance"

Difficulty: Research **Type:** Research paper

Status/version: Market Microstructure and Liquidity, 01(01), 1550005 (2015); DOI 10.1142/s2382626615500057.

Review of self- and mutually-exciting point processes for order flow, volatility, impact and endogeneity.

Version record

Frédéric Abergel, Marouane Anane, Anirban Chakraborti, Aymen Jedidi and Ioane Muni Toke, *Limit Order Books*

Difficulty: Research **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2016); DOI 10.1017/cbo9781316683040.

Empirical facts, queueing and Hawkes models, and agent-based simulation of limit order books (Cambridge 2016).

Version record

Álvaro Arroyo, Álvaro Cartea, Fernando Moreno-Pino and Stefan Zohren, "Deep attentive survival analysis in limit order books: estimating fill probabilities with convolutional-transformers"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 24(1), 35–57 (2024); DOI 10.1080/14697688.2023.2286351.

Survival-analysis model for limit-order fill times with proper scoring; directly relevant to placement and market-making simulators.

Version record

Bidisha Chakrabarty, Terrence Hendershott, Samarpan Nawn and Roberto Pascual, "Order Exposure in High-Frequency Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(1), 61–98 (2026); DOI 10.1017/s0022109025101282.

Hidden-order use by algorithmic versus non-algorithmic traders and how it varies with volatility, adverse selection and tick size.

Version record

Corey Garriott, Vincent van Kervel and Marius Zoican, "Queuing and inventories in limit order markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 75, 100982 (2025); DOI 10.1016/j.finmar.2025.100982.

Queue position affects adverse selection and limits inventory risk sharing in limit-order markets, crowding out liquidity after inventory shocks.

Version record

Rama Cont, Pierre Degond and Lifan Xuan, "A Mathematical Framework for Modeling Order Book Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 16(1), 123–166 (2025); DOI 10.1137/22m1541538.

A general limit-order-book framework splitting dynamics into an order-flow point process and a market-clearing "mass transport" operator; nests many LOB models and their scaling limits.

Version record

8.3.2 Market making, inventory and learning

Álvaro Cartea; Ryan Donnelly; Sebastian Jaimungal, Enhancing trading strategies with order book signals; Oxford accepted manuscript and metadata

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance 25(1), 1–35 (2018); accepted 25 January 2018; publication date 23 February 2018.

Uses Nasdaq order-book volume imbalance to predict the direction of the next market order and its immediate price response, then embeds the signal in a stochastic-control limit-order strategy with inventory penalties. Its 11-stock 2014 calibration/test split is narrow; queue priority, latency, venue fees and feature decay must be rebuilt for any later asset or production execution setting.

Extended note. Version record

Philippe Bergault; Olivier Guéant, Size matters for OTC market makers: General results and dimensionality reduction techniques; author preprint

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance 31(1), 279–322 (January 2021 issue); first published online 23 August 2020.

Extends inventory-aware OTC quoting to many correlated assets and different RFQ trade sizes, then uses a factor representation to reduce the numerical dimension. It informs bond, credit or FX quote ladders where inventory can hedge across instruments and client sizes alter win probabilities. The model still requires observed RFQ response curves, cross-asset covariance stability, hedging costs and client-flow segmentation before a quote can be treated as executable.

Extended note. Version record

Philippe Bergault, David Evangelista, Olivier Guéant and Douglas Vieira, Closed-form Approximations in Multi-asset Market Making; Author preprint, arXiv v5

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Mathematical Finance 28(2), 101-142 (2021 issue); published online 10 March 2022. Published article; arXiv v5 revised 26 September 2022.

Derives tractable approximations to value functions and optimal quotes in several correlated-asset extensions of Avellaneda-Stoikov. Read the finite-horizon and ergodic cases as practical baselines when a full numerical control solution is too expensive. Compare proxy quotes with a small exact solver, then stress covariance shifts, asymmetric client flow and estimation error. The approximations can initialize a learned controller or supply a simple quoting rule; neither use proves that the model's Poisson fill assumptions or frictionless rebalancing match a crypto venue.

Extended note. Version record

Alexander Barzykin, Philippe Bergault and Olivier Guéant, Algorithmic market making in dealer markets with hedging and market impact; Author preprint, arXiv v6

Difficulty: Advanced **Type:** Research paper

Status/version: Mathematical Finance 33(1), 41-79 (January 2023). Published journal article; arXiv v6 revised 11 December 2022.

Adds external hedging to an inventory-aware dealer quotation model: the dealer can internalize client flow by changing quotes or trade in another liquidity pool, paying execution costs and market impact. Derive the inventory region in which waiting for offsetting flow is preferable to hedging, then estimate its width from client RFQ response curves and hedge costs. The paper illustrates the model with USDCNH spot parameters, so crypto use requires new fill, impact, collateral and cross-venue latency inputs. Compare the policy with simple threshold hedges before relying on it.

Extended note. Version record

Bastien Baldacci; Philippe Bergault; Olivier Guéant, Algorithmic market making for options

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 21(1), 85–97 (2021); accepted 1 May 2020; online 4 August 2020.

Reduces an option-market-maker's high-dimensional portfolio-control problem by approximating inventory with vega exposure, yielding a tractable equation under Heston-style volatility. This joins two catalogue strands usually read separately: stochastic-volatility surface risk and inventory-sensitive quote control. The paper's numerical examples establish tractability under its model, not robust live options quoting with discrete strikes, complex fills and hedging latency.

Extended note. Version record

Winston Wei Dou, Itay Goldstein and Yan Ji, "AI-Powered Trading, Algorithmic Collusion, and Price Efficiency"

Difficulty: Research **Type:** Research paper

Status/version: 2025

NBER WP 34054 (July 2025); remains a working paper on the author's research page. A model and simulations show RL speculators sustaining collusive profits without communication; this is a market-design and conduct-risk result in the studied environments, not evidence of actual-market collusion. [31]

Applications: Market making, multi-agent trading, regulation. Version record

Olivier Guéant, Charles-Albert Lehalle and Joaquin Fernandez-Tapia, "Dealing with the inventory risk: a solution to the market making problem"

Difficulty: Research **Type:** Research paper

Status/version: Mathematics and Financial Economics, 7(4), 477–507 (2013); DOI 10.1007/s11579-012-0087-0.

Closed-form-like solution of the Avellaneda–Stoikov problem with inventory bounds via a linear ODE system.

Version record

Álvaro Cartea, Sebastian Jaimungal and Jason Ricci, "Buy Low, Sell High: A High Frequency Trading Perspective"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 5(1), 415–444 (2014); DOI 10.1137/130911196.

Market making with adverse selection, short-term alpha and order-flow feedback in a stochastic control setting.

Version record

Thomas Spooner, John Fearnley, Rahul Savani and Andreas Koukorinis, "Market Making via Reinforcement Learning"

Difficulty: Research **Type:** Research paper

Status/version: AAMAS 2018 proceedings; arXiv 1804.04216.

Temporal-difference agents with reward shaping for inventory risk, tested on historical order-book replay.

Extended note. Version record

Bruno Gašperov, Stjepan Begušić, Petra Posedel Šimović and Zvonko Kostanjčar, "Reinforcement Learning Approaches to Optimal Market Making"

Difficulty: Advanced **Type:** Research paper

Status/version: Mathematics, 9(21), 2689 (2021); DOI 10.3390/math9212689.

Systematic review of RL market-making papers, data, reward designs and evaluation weaknesses.

Version record

Marcel Nutz, Kevin Webster and Long Zhao, "Unwinding Stochastic Order Flow: When to Warehouse Trades"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 36(3), 500–527 (2026); DOI 10.1111/mafi.70019.

Central risk book problem: when to warehouse and internalise stochastic order flow versus route to market, minimising costs.

Version record

Álvaro Cartea, Gerardo Duran-Martin and Leandro Sánchez-Betancourt, "Detecting toxic flow"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(4), 541–561 (2026); DOI 10.1080/14697688.2026.2619539.

Methods for detecting toxic order flow, relevant to internalisation and quoting decisions.

Extended note. Version record

Sudip Gupta, "Tacit Coordination in AI-Driven Liquidity Provision: A Practitioner's Diagnostic Tool Kit"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial Data Science, 8(3), 203–216 (2026); DOI 10.3905/jfds.2026.015.

Q-learning market makers that never communicate still converge to wider spreads and higher dealer profits; a diagnostic for algorithmic tacit coordination in liquidity provision.

Version record

Álvaro Cartea and Leandro Sánchez-Betancourt, "Brokers and Informed Traders: Dealing with Toxic Flow and Extracting Trading Signals"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 16(2), 243–270 (2025); DOI 10.1137/24m1660243.

Optimal broker strategy with toxic informed flow and noise flow: losses to informed traders pay for a learnable signal that drives internalise/externalise and speculation decisions.

Version record

Robert Boyce, Martin Herdegen and Leandro Sánchez-Betancourt, "Market Making with Exogenous Competition"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 16(2), 692–706 (2025); DOI 10.1137/24m1679811.

Market making against competitors who quote by an inventory rule of thumb; approximate closed-form linear-quadratic solution when fill probabilities depend on relative depth.

Version record

Paulin Aubert, Etienne Chevalier and Vathana Ly Vath, "Option Market Making with Hedging-Induced Market Impact"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 33(1), 1–45 (2026); DOI 10.1080/1350486x.2026.2671725.

Option market making when delta hedging moves the underlying: Cox-process option flow, permanent and transient impact, and the manipulation risks this feedback creates.

Version record

Terrence Hendershott and Albert J. Menkveld, "Price pressures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 114(3), 405–423 (2014); DOI 10.1016/j.jfineco.2014.08.001.

NYSE intermediary data show price pressures averaging 0.49% with a half-life of 0.92 days as risk-averse intermediaries use prices to mean-revert inventory. A dynamic inventory model identifies their risk aversion and puts the social cost at about 10% of the cost of immediacy.

Version record

8.4 Order-book learning

Zihao Zhang, Stefan Zohren and Stephen Roberts, "DeepLOB: Deep Convolutional Neural Networks for Limit Order Books"

Difficulty: Research **Type:** Research paper

Status/version: 2019

Combines convolutional and recurrent architectures for limit-order-book prediction and became a standard benchmark in financial deep learning. Institutional replication must control for overlapping labels, event-time leakage, venue-specific sequencing and whether forecast horizons exceed achievable latency.

Applications: Equities, futures, crypto; HFT prediction. Version record

Yichi Zhang, Mihai Cucuringu, Alexander Y. Shestopaloff and Stefan Zohren, "ClusterLOB: Enhancing Trading Strategies by Clustering Orders in Limit Order Books"; preprint

Difficulty: Research **Type:** Research paper

Status/version: 2026; preprint 2025, v3 10 May

Clusters MBO orders into inferred behavioural groups and constructs cluster-specific order-flow imbalance using NASDAQ stocks. Trading results require independent testing across venues, latency assumptions and fees. *Quantitative Finance* 26(8), 1285–1309; online 16 June 2026, August issue. [18]

Applications: Equities; MBO analytics, HFT. Version record

Sirignano, "Deep Learning for Limit Order Books''

Difficulty: Not specified **Type:** Research paper

Status/version: Edition/revision and publication date not established for the selected resource(s).

Spatial architecture and tail performance in order-book modelling; connect network design to the geometry of market data. Justin A. Sirignano's published version is *Quantitative Finance* 19(4), 549–570 (2019; online 23 November 2018). The 2016 preprint; publisher abstract/metadata checked, with live execution performance unverified.

Version record

Zhang, Zohren and Roberts, *BDLOB: Bayesian Deep Convolutional Neural Networks for Limit Order Books* (2018)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2018. Version distinctions remain in the annotation.

Bayesian dropout and predictive uncertainty for limit-order-book models; study uncertainty calibration before translating forecasts into position sizes. The arXiv record identifies the Third Workshop on Bayesian Deep Learning (NeurIPS 2018); this is workshop status, not a main-conference publication.

Version record

Justin Sirignano and Rama Cont, "Universal features of price formation in financial markets: perspectives from deep learning"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 19(9), 1449–1459 (2019); DOI 10.1080/14697688.2019.1622295.

A single LSTM trained on pooled order-book data across stocks outperforms stock-specific models, suggesting universal price-formation features.

Version record

Petter N. Kolm, Jeremy Turiel and Nicholas Westray, "Deep order flow imbalance: Extracting alpha at multiple horizons from the limit order book"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 33(4), 1044–1081 (2023); DOI 10.1111/mafi.12413.

Stationary order-flow features outperform raw order-book inputs for multi-horizon return forecasting in deep networks.

Version record

Rama Cont, Mihai Cucuringu and Chao Zhang, "Cross-impact of order flow imbalance in equity markets"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 23(10), 1373–1393 (2023); DOI 10.1080/14697688.2023.2236159.

Multi-level OFI integrated across book depth explains contemporaneous returns; cross-asset OFI adds forecasting power at short horizons.

Version record

Adamantios Ntakaris, Martin Magris, Juho Kanniainen, Moncef Gabbouj and Alexandros Iosifidis, "Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Forecasting, 37(8), 852–866 (2018); DOI 10.1002/for.2543.

FI-2010 public LOB benchmark; widely used but small and dated, so results transfer poorly to current markets.

Version record

Matteo Prata et al., "LOB-Based Deep Learning Models for Stock Price Trend Prediction: A Benchmark Study (LOBCAST)"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2308.01915, August 2023; later journal version to be confirmed.

Re-evaluates fifteen deep LOB models in an open framework; all degrade sharply on new data.

Extended note. Version record

Antonio Briola, Silvia Bartolucci and Tomaso Aste, "Deep limit order book forecasting: a microstructural guide"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 25(7), 1101–1131 (2025); DOI 10.1080/14697688.2025.2522911.

LOBFrame code base; argues standard ML metrics mislead and evaluates forecasts by the probability of completing profitable transactions.

Version record

Jiwon Jung and Kiseop Lee, "Attention-based reading, highlighting, and forecasting of the limit order book"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 25(7), 1015–1027 (2025); DOI 10.1080/14697688.2025.2522914.

Attention models that forecast the multi-level order book, not only the mid-price.

Version record

8.5 Simulation and counterfactual validation

Read after the queueing and impact models, alongside ABIDES and JAX-LOB. Matching unconditional price statistics, reproducing order events and predicting responses to an intervention are distinct levels of evidence.

Nelson Vadori et al., Towards multi-agent reinforcement learning-driven over-the-counter market simulations

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance 34(2), 262–347 (2024); first published online 20 September 2023.

Models a dealer OTC market as interacting liquidity providers and takers whose policies learn quoting, hedging, execution and market-share trade-offs. Shared-policy learning and a calibration procedure make it an institutional complement to exchange-book simulators such as ABIDES and JAX-LOB, especially for FX-style bilateral flow. Before using it for policy selection, benchmark against observed RFQ flow, tiering, rejection rates, hedging venues and regime-specific client response.

Extended note. Version record

Peer Nagy et al., LOB-Bench: Benchmarking Generative AI for Finance - an Application to Limit Order Book Data; benchmark project

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, 45437–45460 (2025).

Provides a reproducible evaluation suite for generated LOBSTER-format message-by-order data, testing conditional and unconditional spreads, volumes, imbalance, inter-arrival times, discriminator scores and event-specific price responses. The reported model ranking is tied to its data and chosen metrics; an institution must add closed-loop replay validity, participant response, queue/fill economics and held-out strategy outcomes for its target venue.

Extended note. Version record

Junjie Li; Yang Liu; Weiqing Liu; Shikai Fang; Lewen Wang; Chang Xu; Jiang Bian, MarS: a Financial Market Simulation Engine Powered by Generative Foundation Model; official author code; preprint

Difficulty: Research **Type:** Research paper

Status/version: International Conference on Learning Representations (ICLR) 2025 conference paper; initial arXiv preprint 4 September 2024.

Introduces an order-level generative market model and interactive simulator intended to respond to participant actions while allowing controlled market scenarios. The conference paper reports scaling and market-impact demonstrations; a production research team still needs exchange-rule fidelity, valid cancellation references, closed-loop stability, regime holdouts and downstream execution comparisons against simpler simulators before treating generated P&L as credible.

Extended note. Version record

Patrick Noble; Mathieu Rosenbaum; Saad Souilmi, Bridging the Reality Gap in Limit Order Book Simulation

Difficulty: Research **Type:** Research paper

Status/version: arXiv v1 submitted 25 March 2026; manuscript internally dated 27 March 2026. No journal/proceedings status established in consulted source.

Builds an interactive simulator for large-tick order books by projecting the state to spread and imbalance, fitting non-Poisson event timing, and adding decaying signed-flow feedback for transient impact and reversion. Read beside ABIDES/JAX-LOB and LOB-Bench to design calibration and adversarial execution tests. Its reported stock examples depend on tick regime, measured round-trip latency, impact kernel and chosen strategy; replication on target symbols and venues is essential.

Extended note. Version record

Junoh Kang; Kiseop Lee; Bohyung Han, ReLOBGen: Replayable Limit Order Book Message Generation

Difficulty: Research **Type:** Research paper

Status/version: arXiv v1 submitted 26 September 2026; no reviewed conference or journal venue established from consulted primary sources.

Makes generated LOB messages executable against the current resting book by choosing eligible referenced orders and masking invalid event fields, avoiding post-hoc correction or rejection. This addresses a concrete failure mode in closed-loop generative simulators: a cancellation or execution that refers to an absent order cannot produce a valid fill path. Read as an implementation check on MarS-like generators.

Extended note. Version record

David Byrd, Maria Hybinette and Tucker Hybinette Balch, "ABIDES: Towards High-Fidelity Multi-Agent Market Simulation"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the 2020 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation, 11–22 (2020); DOI 10.1145/3384441.3395986.

Discrete-event agent-based simulator with nanosecond timestamps and exchange messaging, the basis of ABIDES-JPMC.

Version record

Andrea Coletta, Matteo Prata, Michele Conti, Emanuele Mercanti, Novella Bartolini, Aymeric Moulin, Svitlana Vyetrenko and Tucker Balch, "Towards Realistic Market Simulations: a Generative Adversarial Networks Approach"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the Second ACM International Conference on AI in Finance, 1–9 (2021); DOI 10.1145/3490354.3494411.

Conditional GAN agents trained on order-book data inside ABIDES to produce reactive background flow.

Version record

Robert L. Axtell and J. Doyne Farmer, "Agent-Based Modeling in Economics and Finance: Past, Present, and Future"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Literature, 63(1), 197–287 (2025); DOI 10.1257/jel.20221319.

Comprehensive JEL review of agent-based models, with finance achievements in volatility clustering, impact, systemic risk and housing.

Version record

Svitlana Vyetrenko, David Byrd, Nick Petosa, Mahmoud Mahfouz, Danial Dervovic, Manuela Veloso and Tucker Balch, "Get Real: Realism Metrics for Robust Limit Order Book Market Simulations"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the First ACM International Conference on AI in Finance, 1–8 (2020); DOI 10.1145/3383455.3422561.

Stylised-fact metrics for judging simulated LOB realism; companion to ABIDES.

Version record

Luca Lalor and Anatoliy Swishchuk, "Event-Based Limit Order Book Simulation under a Neural Hawkes Process: Application in Market-Making"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 32(2), 128–155 (2025); DOI 10.1080/1350486x.2025.2548448.

Event-based LOB simulator with twelve event types driven by a neural (LSTM) Hawkes process, reproducing high-frequency stylised facts.

Version record

8.6 Fixed-income and OTC electronic trading

Dealer markets differ from limit order books: requests for quote, portfolio trading, transparency and balance-sheet capacity determine costs. Read with the funding and Treasury-market readings.

Hendrik Bessembinder, Chester Spatt and Kumar Venkataraman, "A Survey of the Microstructure of Fixed-Income Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 55(1), 1–45 (2020); DOI 10.1017/s0022109019000231.

Survey of dealer intermediation, transparency, electronic trading and regulation across Treasuries, corporates, munis and securitised products.

Version record

Maureen O'Hara and Xing (Alex) Zhou, "The electronic evolution of corporate bond dealers"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 140(2), 368–390 (2021); DOI 10.1016/j.jfineco.2021.01.001.

Electronic RFQ trading lowers costs for small trades; dealers adapt by changing inventory behaviour.

Version record

Terrence Hendershott and Ananth Madhavan, "Click or Call? Auction versus Search in the Over-the-Counter Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 70(1), 419–447 (2015); DOI 10.1111/jofi.12164.

Electronic multi-dealer auctions versus voice search in corporate bonds; auctions dominate for liquid, smaller trades.

Version record

Amy K. Edwards, Lawrence E. Harris and Michael S. Piwowar, "Corporate Bond Market Transaction Costs and Transparency"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 62(3), 1421–1451 (2007); DOI 10.1111/j.1540-6261.2007.01240.x.

TRACE-based transaction costs decline with trade size and with post-trade transparency.

Version record

Jessica Li, Maureen O'Hara, Andreas C. Rapp and Xing (Alex) Zhou, "Bond Market Illiquidity: Is Portfolio Trading the Solution?"

Difficulty: Research **Type:** Research paper

Status/version: SSRN 4495516, first posted 2023; revised 2025; no journal publication established.

Portfolio trading generally lowers costs for risky and illiquid bonds but becomes costly for liquidity providers in stress. Working paper; check for later revisions.

Extended note. Version record

Jian Li and Haiyue Yu, "Investor Composition and the Liquidity Component in the U.S. Corporate Bond Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(2), 871–922 (2026); DOI 10.1111/jofi.70024.

Short-horizon holders such as mutual funds and ETFs increase secondary trading and amplify frictions in prices. Journal of Finance 81(2), 871–922 (April 2026); confirm author names from the publisher record.

Version record

Terrence Hendershott, Dan Li, Dmitry Livdan and Norman Schürhoff, "Relationship Trading in Over-the-Counter Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(2), 683–734 (2020); DOI 10.1111/jofi.12864.

Insurers' dealer networks: size and number of relationships shape execution costs in corporate bonds.

Version record

Marco Di Maggio, Amir Kermani and Zhaogang Song, "The value of trading relations in turbulent times"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 124(2), 266–284 (2017); DOI 10.1016/j.jfineco.2017.01.003.

Dealer relationships provide liquidity in stress; peripheral dealers face higher costs.

Version record

Hendrik Bessembinder, Stacey Jacobsen, William Maxwell and Kumar Venkataraman, "Capital Commitment and Illiquidity in Corporate Bonds"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 73(4), 1615–1661 (2018); DOI 10.1111/jofi.12694.

Dealers committed less capital after the crisis; block-trade costs and inventory changed accordingly.

Version record

Michael A. Goldstein and Edith S. Hotchkiss, "Providing liquidity in an illiquid market: Dealer behavior in US corporate bonds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 135(1), 16–40 (2020); DOI 10.1016/j.jfineco.2019.05.014.

Dealers prefer offsetting trades within the day in infrequently traded bonds; matched trades lower costs.

Version record

Maureen O'Hara and Xing (Alex) Zhou, "Anatomy of a liquidity crisis: Corporate bonds in the COVID-19 crisis"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2020); Journal of Financial Economics version (2021) reported by the authors, DOI not established in this check.

Dealer balance-sheet constraints and investor selling drove the March 2020 corporate bond liquidity collapse.

Extended note. Version record

Mahyar Kargar, Benjamin Lester, David Lindsay, Shuo Liu, Pierre-Olivier Weill and Diego Zúñiga, "Corporate Bond Liquidity during the COVID-19 Crisis"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 34(11), 5352–5401 (2021); DOI 10.1093/rfs/hhab063.

Measures the price of immediacy in March 2020 and the effect of Fed facilities.

Version record

Lynn Riggs, Esen Onur, David Reiffen and Haoxiang Zhu, "Swap trading after Dodd-Frank: Evidence from index CDS"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 137(3), 857–886 (2020); DOI 10.1016/j.jfineco.2020.03.008.

RFQ behaviour on swap execution facilities: customers contact few dealers; dealers respond strategically.

Version record

Tomy Lee and Chaojun Wang, "Regulating Over-the-Counter Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(4), 1929–1962 (2025); DOI 10.1111/jofi.13461.

OTC dealers cream-skim uninformed flow from exchanges; restricting this induces "cheap substitution" of traders, with ambiguous effects on costs and volumes. Theory for OTC regulation debates.

Version record

Terrence Hendershott, Dan Li, Dmitry Livdan, Norman Schürhoff and Kumar Venkataraman, "Quote Competition in Corporate Bonds"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(4), 2165–2216 (2026); DOI 10.1111/jofi.70048.

Indicative dealer quotes in corporate bonds lower costs and raise volume; better quotes attract flow, but competition is imperfect and relationships still matter.

Version record

Mahyar Kargar, Benjamin Lester, Sébastien Plante and Pierre-Olivier Weill, "Sequential Search for Corporate Bonds"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(5), 3035–3086 (2026); DOI 10.1111/jofi.70068.

Platform request data reveal how long corporate-bond customers search, how many dealers they contact and what search costs, beyond transaction records.

Version record

Stacey Jacobsen and Kumar Venkataraman, "Receiving investors in the block market for corporate bonds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 170, 104061 (2025); DOI 10.1016/j.jfineco.2025.104061.

Studies which investors receive dealer-intermediated block trades in corporate bonds and how block-market access affects pricing.

Extended note. Version record

Batchimeg Sambalaibat, "Heterogeneous clienteles and dealer networks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 174, 104185 (2025); DOI 10.1016/j.jfineco.2025.104185.

Links heterogeneous client types to the structure of dealer networks in OTC markets.

Extended note. Version record

Viet-Dung Doan, "Exchange-traded funds and transparency in over-the-counter markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 29(4), 1043–1065 (2025); DOI 10.1093/rof/rfaf019.

Daily ETF holdings disclosure acts as price transparency in municipal bonds: ETF-held bonds carry 5–9 bp lower retail dealer markups.

Version record

Geir Høidal Bjønnes, Neophytos Kathitziotis and Carol Osler, "Who Pays the Most to Trade? Cross-client Dispersion in OTC Liquidity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, rfag042 (2026); DOI 10.1093/rof/rfag042.

A dealing bank's complete OTC FX record shows spreads under 1 bp for hedge funds and over 50 bp for individuals and small firms, driven mainly by clients' execution efficiency.

Version record

8.7 Dealer market making in bonds, swaps and FX

Bond, swap and currency dealers quote on request or stream prices to clients, then decide whether to hold the resulting inventory, offset it against later client flow or hedge it with other dealers. These readings cover dealer quoting models, evidence on dealer inventory, balance-sheet capacity and networks, and FX practices such as internalisation, aggregators and last look. The previous section covers the same markets from the client's side.

8.7.1 Bond and swap dealers: RFQs, inventory and capacity

Jean-David Fermanian, Olivier Guéant and Jiang Pu, "The Behavior of Dealers and Clients on the European Corporate Bond Market: The Case of Multi-Dealer-to-Client Platforms"

Difficulty: Research **Type:** Research paper

Status/version: Market Microstructure and Liquidity, 2(3–4), 1750004 (2016); DOI 10.1142/S2382626617500046.

Uses a large proprietary set of requests for quotes sent through Bloomberg's multi-dealer-to-client platform to one of the main liquidity providers in European corporate bonds. Models the RFQ process and the competition among the dealers asked, and uses the model to infer how dealers and clients behave on these platforms.

Version record

Olivier Guéant and Jiang Pu, "Mid-Price Estimation for European Corporate Bonds: A Particle Filtering Approach"

Difficulty: Research **Type:** Research paper

Status/version: Market Microstructure and Liquidity, 4(1–2), 1950005 (2018 issue; published online 2019); DOI 10.1142/S2382626619500059.

Estimates the mid-price of illiquid European corporate bonds in real time from the information a dealer actually has: its own dealer-to-dealer and dealer-to-client trades, the outcomes of the RFQs it answered, and composite prices such as Bloomberg CBBT. The Bayesian method uses particle filtering (sequential Monte Carlo).

Version record

Philippe Bergault and Olivier Guéant, "Liquidity Dynamics in RFQ Markets and Impact on Pricing"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2309.04216, posted 8 September 2023; version 3 revised 19 June 2024.

Asks how to value securities in RFQ-based OTC markets when transaction prices are scarce. Extends the limit-order-book micro-price to RFQ markets, models liquidity imbalance with bidimensional Markov-modulated Poisson processes, and introduces a Fair Transfer Price that can be used when the market is illiquid or one-sided.

Extended note. Version record

Tobias Adrian, Nina Boyarchenko and Or Shachar, "Dealer balance sheets and bond liquidity provision"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 89, 92–109 (2017); DOI 10.1016/j.jmoneco.2017.03.011.

Builds dealer-level liquidity measures for U.S. corporate bonds from uncapped transaction data that identify each dealer. Over the full sample, bonds are more liquid when the institutions trading them are more levered, earn more trading revenue and rely less on repo funding. After January 2014, as post-crisis rules took effect, bonds traded by more vulnerable institutions or those with greater liquidity mismatch are less liquid.

Version record

Jack Bao, Maureen O'Hara and Xing (Alex) Zhou, "The Volcker Rule and corporate bond market making in times of stress"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 130(1), 95–113 (2018); DOI 10.1016/j.jfineco.2018.06.001.

Treats rating downgrades, which force sales of corporate bonds, as stress events. After the Volcker Rule, stressed bonds became less liquid: dealers subject to the rule cut their market making and unaffected dealers replaced only part of it. Volcker-affected dealers that were not constrained by Basel III or CCAR also changed behaviour, which points to the Volcker Rule rather than the other regulations.

Version record

Jens Dick-Nielsen and Marco Rossi, "The Cost of Immediacy for Corporate Bonds"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 32(1), 1–41 (2019); DOI 10.1093/rfs/hhy080.

Uses index exclusions, when uninformed index trackers demand immediacy, as a natural experiment. The cost of immediacy in corporate bonds more than doubled after 2008, and its supply became more elastic to price. Dealers revert to target inventory faster than before the crisis, consistent with rules that favour smaller inventories; the paper also compares bank and nonbank dealers.

Version record

David Cimon and Corey Garriott, "Banking regulation and market making"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 109, 105653 (2019); DOI 10.1016/j.jbankfin.2019.105653.

Models market makers under Basel III-style liquidity and capital constraints and a Volcker-style position limit. Regulation makes market makers refuse principal positions, yet bid-ask spreads can narrow because new market makers enter, and assets carry a liquidity premium. The costs of regulation show up in principal positions and asset prices, not in bid-ask spreads.

Version record

Jaewon Choi, Yesol Huh and Sean Seunghun Shin, "Customer Liquidity Provision: Implications for Corporate Bond Transaction Costs"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 70(1), 187–206 (2024); DOI 10.1287/mnsc.2022.4646.

Shows that customers increasingly provide liquidity in corporate bonds after the post-2008 banking rules, so bid-ask measures that assume dealers always provide it understate the cost of dealer liquidity. In large trades where dealers use inventory, customers pay spreads 40–60 percent wider than before the crisis. Customers' balance-sheet capacity and their dealer relationships determine who provides liquidity.

Version record

Dan Li and Norman Schürhoff, "Dealer Networks"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 74(1), 91–144 (2019); DOI 10.1111/jofi.12728.

Maps the interdealer network of the U.S. municipal bond market from regulatory data: 10 to 30 hub dealers and over 2,000 peripheral broker-dealers, with bonds flowing from the periphery to the core and partly back. Central dealers charge investors up to twice the round-trip markup of peripheral dealers but provide immediacy, matching buyers and sellers more directly, especially in stress. Investors trade off execution cost against speed.

Version record

Péter Kondor and Gábor Pintér, "Clients' Connections: Measuring the Role of Private Information in Decentralized Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 77(1), 505–544 (2022); DOI 10.1111/jofi.13087.

Proposes the number of dealers a client trades with in a period, its connections, as a measure of private information. In U.K. government bond trade data, clients with more connections trade ahead of future price moves, and market-wide connections help explain yields. The evidence suggests dealers pass information from informed clients to their affiliates.

Version record

Jason Allen and Milena Wittwer, "Centralizing Over-the-Counter Markets?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 131(12), 3310–3351 (2023); DOI 10.1086/725361.

Uses trade-level data from the Canadian government bond market to ask whether bilateral trading should move to a central platform. Intermediaries charge investors a markup. A structural model finds that many investors would not use the platform even if they could, because it is costly, competition for investors is low and they value dealer relationships; centralisation can lower welfare unless competition is strong.

Version record

Pierre Collin-Dufresne, Benjamin Junge and Anders B. Trolle, "Market Structure and Transaction Costs of Index CDSs"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(5), 2719–2763 (2020); DOI 10.1111/jofi.12953.

Finds that post-Dodd-Frank index CDS trading remains two-tiered. Dealer-to-client trades cost more than interdealer trades, but the gap is explained by the larger, mostly permanent price impact of client trades. Interdealer trades are mostly liquidity-motivated and use low-cost, low-immediacy protocols; dealer-to-client trades are nonanonymous, almost always improve on executable interdealer quotes, and dealers appear to price them on their expected impact.

Version record

Evangelos Benos, Richard Payne and Michalis Vasios, "Centralized Trading, Transparency, and Interest Rate Swap Market Liquidity: Evidence from the Implementation of the Dodd–Frank Act"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 55(1), 159–192 (2020); DOI 10.1017/S0022109018001527.

Uses proprietary interest rate swap transactions to measure the effect of the Dodd-Frank trade mandate. Swaps with the most centralised trading saw liquidity measures improve by 12 to 19 percent against a control group, driven by more competition among dealers. Interdealer trading in euro swaps moved from the United States to Europe, consistent with dealers avoiding the mandate to keep market power.

Version record

Gino Cenedese, Angelo Ranaldo and Michalis Vasios, "OTC premia"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 136(1), 86–105 (2020); DOI 10.1016/j.jfineco.2019.09.010.

Studies interest rate swap prices with transaction- and identity-level data. Prices differ persistently across clients, consistent with dealers passing regulatory costs on through valuation adjustments (xVA): a client pays a higher fixed rate for a swap that is not centrally cleared, and less when it posts initial margin or is more creditworthy. Dealers pay no such premium to each other.

Version record

8.7.2 FX dealers: inventory, internalisation and last look

Richard K. Lyons, "Tests of microstructural hypotheses in the foreign exchange market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 39(2–3), 321–351 (1995); DOI 10.1016/0304-405X(95)00832-Y.

Tests the two classic dealer channels, information and inventory control, on a three-part transactions dataset from the spot FX market. Trades move quoted prices through both channels, supporting both strands of microstructure theory. Because FX has no insiders, the information effect implies a broader view of private information than equity studies usually take.

Version record

Geir Høidal Bjønnes and Dagfinn Rime, "Dealer behavior and trading systems in foreign exchange markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 75(3), 571–605 (2005); DOI 10.1016/j.jfineco.2004.08.001.

Studies the complete transactions of four interbank spot FX dealers. Incoming trades carry information, and trade direction matters more than size. All four dealers control inventory strongly, but through their trades rather than by shading their own prices as in Lyons (1995); the authors suggest electronic brokers gave dealers more ways to trade. Trading styles, especially inventory control, differ across the dealers.

Version record

Angelo Ranaldo and Fabricius Somogyi, "Asymmetric information risk in FX markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 140(2), 391–411 (2021); DOI 10.1016/j.jfineco.2020.12.007.

Measures the information content of FX trades with an order-flow dataset that separates groups of market participants across many currency pairs. Information advantages vary across participant groups, over time and across pairs, consistent with asymmetric information and a fragmented OTC market. A strategy based on permanent price impact earns high returns after risk, transaction costs and common FX factors.

Version record

Joel Hasbrouck and Richard M. Levich, "Network structure and pricing in the FX market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 141(2), 705–729 (2021); DOI 10.1016/j.jfineco.2021.04.013.

Builds the trading network from CLS Bank settlement data spanning currency pairs, participants and execution platforms. The more central counterparty earns a higher return on a trade, and the gap widens as the two parties' centralities diverge, a pervasive centrality premium. Losses that central participants incur in supplying liquidity partly offset it.

Version record

Harald Hau, Peter Hoffmann, Sam Langfield and Yannick Timmer, "Discriminatory Pricing of Over-the-Counter Derivatives"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 67(11), 6660–6677 (2021); DOI 10.1287/mnsc.2020.3787.

Uses regulatory data to measure price discrimination against nonfinancial clients in FX derivatives. Transaction costs fall with client sophistication: the median client pays 10.9 pips more than blue-chip companies, against an average effective spread of 6.9 pips. The discrimination disappears when clients trade electronically on multidealer platforms; less sophisticated clients also pay more with their relationship bank and in fast markets, but only in bilaterally negotiated trades.

Version record

Roel Oomen, "Execution in an aggregator"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 17(3), 383–404 (2017); DOI 10.1080/14697688.2016.1201589.

Models execution in an aggregator, which consolidates quotes from several liquidity providers into one book and is widely used in spot FX. Derives the inside spread, last-look reject rate, effective spread and the providers' market shares and revenues. Aggregation gives the provider that wins a request a winner's curse that grows as more providers join; when the trader splits orders, one provider that hedges quickly can push all of them to hedge externally, raising impact costs.

Version record

Roel Oomen, "Last look"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 17(7), 1057–1070 (2017); DOI 10.1080/14697688.2016.1262545.

Treats last look as the step in which a liquidity provider accepts, re-prices or rejects a deal request after the market may have moved. Compares symmetric and asymmetric last-look designs and protocols that do or do not pass price improvement and slippage to the trader. The design and protocol set the trader's execution risk but need not change effective costs; when the trader adversely selects providers, the protocol becomes the main driver of execution risk.

Version record

Álvaro Cartea, Sebastian Jaimungal and Jamie Walton, "Foreign exchange markets with Last Look"

Difficulty: Research **Type:** Research paper

Status/version: Mathematics and Financial Economics, 13(1), 1–30 (2019); DOI 10.1007/s11579-018-0218-3.

Models spot FX spreads set by risk-neutral brokers who recover losses to latency arbitrageurs from other traders. With one venue, a last-look option to reject loss-making trades lowers quoted spreads. With two venues, traders settle into an equilibrium in which both venues or only one survive, and a last-look venue can quote wider spreads than one without it.

Version record

Alexander Barzykin, Philippe Bergault and Olivier Guéant, "Dealing with multi-currency inventory risk in FX cash markets"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2207.04100, posted 8 July 2022; version 4 revised 30 October 2023.

Gives FX cash dealers a framework to maximise expected profit while controlling inventory risk across many currency pairs. Dealers skew prices to attract or divert client flow and hedge with other dealers in the dealer-to-dealer market; approximation techniques make the framework scale to any number of pairs.

Extended note. Version record

Alexander Barzykin, Robert Boyce and Eyal Neuman, "FX Market Making with Internal Liquidity"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2512.04603, posted 4 December 2025; also SSRN 5859484.

Studies a market maker that offers clients passive access to an internal liquidity pool. Acting as principal, it can fill resting orders as part of its risk management or adjust prices to its external OTC clients to encourage matching flow. Derives the optimal multi-objective strategy and draws qualitative lessons, including for liquidity providers who expect a certain execution speed.

Extended note. Version record

Alexander Barzykin, "Dynamic Slippage Control and Rejection Feedback in Spot FX Market Making"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 6372499 (2026); DOI 10.2139/ssrn.6372499.

Extends Avellaneda–Stoikov quoting to a spot FX dealer that streams size-dependent quotes on a ladder, with adverse selection from price moves during a latency window. The dealer chooses a slippage tolerance, beyond which it rejects the trade, jointly with its quotes, so the acceptance decision works like an option written on the latency move. Past rejections lower client flow through a reputation score, and an approximation makes the controls computable.

Extended note. Version record

Alexander Barzykin, Philippe Bergault and Olivier Guéant, "Market Making in Spot Precious Metals"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2404.15478, posted 23 April 2024; version 5 revised 18 January 2026.

Spot precious-metals dealers face liquidity that sits mainly in futures. Models the exchange-for-physical spread between futures and spot as a nested Ornstein–Uhlenbeck process, then optimises a market maker's expected P&L against inventory risk across spot and futures, with a fast approximation to the HJB solution for near-real-time use.

Extended note. Version record

8.8 Options, futures and ETF execution

Execution differs by instrument: option dealers hedge discretely and price volatility risk, futures trade under exchange matching and settlement rules, and ETFs and index funds concentrate trading around creations, reconstitutions and the close.

8.8.1 Options market structure and market making

Robert Battalio, Brian Hatch and Robert Jennings, "Toward a National Market System for U.S. Exchange-listed Equity Options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 59(2), 933–962 (2004); DOI 10.1111/j.1540-6261.2004.00653.x.

Asks whether multiply listed equity options deliver efficient trades and best execution as the market moved toward a national market system. Between June 2000, when quotes were often ignored, and January 2002, the market came markedly closer to a single national market.

Version record

Mel Jameson and William Wilhelm, "Market Making in the Options Markets and the Costs of Discrete Hedge Rebalancing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 47(2), 765–779 (1992); DOI 10.1111/j.1540-6261.1992.tb04409.x.

Shows that option market makers face inventory risks specific to options: the inability to rebalance hedges continuously and uncertainty about the underlying's volatility each explain a significant share of quoted CBOE spreads.

Version record

Sahn-Wook Huh, Hao Lin and Antonio S. Mello, "Options market makers' hedging and informed trading: Theory and evidence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 23, 26–58 (2015); DOI 10.1016/j.finmar.2015.01.001.

Models option market makers who hedge against informed trading. Hedging widens spreads in both the option and the stock, more so in options and when hedging with the underlying rather than other options, and it changes how informed traders split their orders; the empirical tests support the main prediction.

Version record

Bjørn Eraker and Daniela Osterrieder, "Market Maker Inventory, Bid–Ask Spreads, and the Computation of Option Implied Risk Measures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 21(5), 1820–1851 (2023); DOI 10.1093/jjfinec/nbac025.

Asking prices react more than bids to shocks, so option spreads are skewed, and a market-making model matches these asymmetries. Correcting quotes for them changes option-implied risk measures; fear and risk premia around some events are overstated when computed from raw quotes.

Version record

Svetlana Bryzgalova, Anna Pavlova and Taisiya Sikorskaya, "Retail Trading in Options and the Rise of the Big Three Wholesalers"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 78(6), 3465–3514 (2023); DOI 10.1111/jofi.13285.

Retail trading in U.S. options grew to over 60% of market volume, facilitated by payment for order flow, nearly 90% of which comes from three wholesalers. A transaction-level measure of retail flow shows a preference for cheap weekly options with average spreads of 12.6%, and retail investors lose money on average.

Version record

8.8.2 Futures matching, rolls and settlement

Joel Hasbrouck, "Intraday Price Formation in U.S. Equity Index Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 58(6), 2375–2400 (2003); DOI 10.1046/j.1540-6261.2003.00609.x.

Measures price discovery across floor-traded index futures, E-mini futures, ETFs and sector ETFs. For the S&P 500 and Nasdaq-100 most price discovery occurs in the E-mini; for the S&P 400 it is shared between the regular futures and the ETF.

Version record

CME Group, CME Globex Matching Algorithms (Client Systems Wiki)

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living documentation; page last updated 11 February 2025.

Describes how Globex allocates an incoming order across resting orders: time priority (FIFO), pro-rata allocation with rounding and FIFO remainders, and top-order priority where a product supports it. Queue value, order sizing and fill probability depend on the product's algorithm, so check it before modelling passive futures execution.

Version record

Angelo Aspris, Sean Foley, Drew Harris and Peter O'Neill, "Time Pro-rata Matching: Evidence of a Change in LIFFE STIR Futures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 35(6), 522–541 (2015); DOI 10.1002/fut.21708.

Studies LIFFE's 2007 move from pure pro-rata to time pro-rata allocation in short-term interest-rate futures. Removing pure pro rata reduced displayed depth but improved execution quality, consistent with pro rata encouraging traders to oversize orders to win allocation.

Version record

Hendrik Bessembinder, Allen Carrion, Laura Tuttle and Kumar Venkataraman, "Liquidity, resiliency and market quality around predictable trades: Theory and evidence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 121(1), 142–166 (2016); DOI 10.1016/j.jfineco.2016.02.011.

Extends strategic-trading theory around a predictable liquidation to include market resiliency, then studies the large, predictable futures roll of an ETF. Roll dates show narrower spreads, more depth and better resiliency, with more accounts supplying liquidity and no evidence of systematic predation.

Version record

Yiqun Mou, "Limits to Arbitrage and Commodity Index Investment: Front-Running the Goldman Roll"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 1716841 (2010); DOI 10.2139/ssrn.1716841.

Working paper on trading ahead of the predictable monthly roll of a major commodity index and the limits to arbitrage around it. Read with the evidence on predictable roll trades before assuming a roll is costly or free.

Extended note. Version record

Craig Pirrong, "Strategic trading and manipulation in trade at settlement contracts"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 43(5), 615–634 (2023); DOI 10.1002/fut.22401.

Trade-at-settlement contracts let uninformed traders transact at the settlement price, but large intermediaries with concentrated TAS positions can profit from strategic or manipulative trading into the settlement window, causing excess and partly permanent price moves.

Version record

8.8.3 ETFs, index events and closing auctions

Martin Lettau and Ananth Madhavan, "Exchange-Traded Funds 101 for Economists"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 32(1), 135–154 (2018); DOI 10.1257/jep.32.1.135.

Describes how ETFs are structured and organised, how they differ from mutual funds, and how creation and redemption work, before turning to their growth across indexes, bonds and smart-beta strategies.

Version record

Travis Box, Ryan Davis, Richard Evans and Andrew Lynch, "Intraday arbitrage between ETFs and their underlying portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 141(3), 1078–1095 (2021); DOI 10.1016/j.jfineco.2021.04.023.

Tests minute by minute whether ETF price shocks pass to the underlying through arbitrage and finds little such transmission: ETF returns do not lead portfolio prices, and gaps opened by order imbalances in the underlying close mainly through ETF quote adjustments rather than arbitrage trading.

Version record

Andrei Shleifer, "Do Demand Curves for Stocks Slope Down?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 41(3), 579–590 (1986); DOI 10.1111/j.1540-6261.1986.tb04518.x.

Stocks added to the S&P 500 from 1976 earned significant abnormal returns at announcement that did not reverse within ten days and rose with index-fund buying, consistent with downward-sloping demand curves rather than certification of quality.

Version record

Honghui Chen, Gregory Noronha and Vijay Singal, "The Price Response to S&P 500 Index Additions and Deletions: Evidence of Asymmetry and a New Explanation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 59(4), 1901–1930 (2004); DOI 10.1111/j.1540-6261.2004.00683.x.

Added firms gain permanently in price, but deleted firms show no permanent decline. Changes in investor awareness explain the asymmetry, which symmetric price-pressure or demand-curve accounts cannot.

Version record

Robin Greenwood, "Short- and long-term demand curves for stocks: theory and evidence on the dynamics of arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 75(3), 607–649 (2005); DOI 10.1016/j.jfineco.2004.03.007.

Theory and evidence on how arbitrageurs absorb an index-driven demand shock over short and long horizons.

Extended note. Version record

Ananth Madhavan, "The Russell Reconstitution Effect"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 59(4), 51–64 (2003); DOI 10.2469/faj.v59.n4.2545.

Annual Russell reconstitutions from 1996 to 2002 brought significant returns from both temporary price pressure and permanent index-membership effects. They cost index funds that trade on the reconstitution date and reward liquidity providers, who bear high trading costs, price risk and intraday volatility.

Version record

Antti Petajisto, "The index premium and its hidden cost for index funds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 18(2), 271–288 (2011); DOI 10.1016/j.jempfin.2010.10.002.

From 1990 to 2005, S&P 500 and Russell 2000 additions rose 8.8% and 4.7% between announcement and effective date, and deletions fell 15.1% and 4.6%. The resulting index turnover cost to index funds is at least 21–28 bp a year for the S&P 500 and 38–77 bp for the Russell 2000.

Version record

Narasimhan Jegadeesh and Yanbin Wu, "Closing auctions: Nasdaq versus NYSE"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 143(3), 1120–1139 (2022); DOI 10.1016/j.jfineco.2021.12.003.

Closing auctions reached about 10% of volume by 2019. The NYSE auction offers more depth than Nasdaq's; in both, the temporary price impact takes three to five days to dissipate, and strategies trading its reversal are profitable.

Version record

Amit Goyal, Narasimhan Jegadeesh and Yanbin Wu, "Price Impact in Closing Auctions, Opening Auctions, and Continuous Markets: A Benchmark"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(6), 2620–2655 (2026); DOI 10.1017/s0022109026102592.

Price impact is lower in closing auctions than in the continuous market for all stocks except Nasdaq microcaps, and opening auctions are illiquid. Placing anomaly-strategy trades in the cheaper mechanism puts annual costs at 17–41 bp, or 9–21 bp in closing auctions excluding microcaps.

Version record

Chapter 9

Funding, collateral and securities lending

Funding, collateral and borrow connect valuation with feasible positions. Securities lending has its own supply, recall, fee and settlement mechanisms. CFDs and total return swaps deliver the same exposure synthetically, priced by their financing, borrow and margin terms.

9.1 Funding liquidity, demand and crowded trades

These readings explain how positions, intermediary capacity and common demand can change prices and the cost of exiting. Connect them to risk budgeting and transaction-cost research before interpreting a factor return as scalable alpha.

9.1.1 Funding liquidity, leverage and runs

Markus K. Brunnermeier and Lasse Heje Pedersen, Market Liquidity and Funding Liquidity

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies 22(6), 2201–2238, June 2009; published online 26 November 2008.

Formalizes the feedback between traders' funding and securities' market liquidity: falling prices and tighter margins can force sales that further damage depth and prices. The spiral occurs under model conditions rather than every selloff; translate it into scenarios with collateral eligibility, haircut and margin shocks, liquidity horizons, dealer capacity, and correlated portfolio liquidation.

Extended note. Version record

Lucy Cordes and Sebastian Infante, Repo Rate Sensitivity to Treasury Issuance and Quantitative Tightening

Difficulty: Advanced **Type:** Report / research note

Status/version: FEDS Notes, 12 February 2025; table, figures, public data and code corrected 1 July 2026.

Estimates how the Treasury tri-party repo spread to ON RRP reacts to changes in bill and coupon supply as quantitative tightening changes the level of market liquidity. Use the corrected July 2026 exhibits and preserve publication and revision dates; the estimated sensitivities vary over regimes, and average repo spreads do not substitute for security-specific specials, bilateral haircuts, or a fund's actual funding terms.

Extended note. Version record

Adam Copeland and Owen Engbretson, Repo and the Liquidity Risk Premium

Difficulty: Research **Type:** Research paper

Status/version: Federal Reserve Bank of New York Staff Report 1189; working paper, no journal version established. Issued 2026-03.

Connects dealer liquidity buffers, reserve supply and administered rates to repo rate dispersion. This gives the funding section a quantified mechanism distinct from Treasury supply and QT effects on repo spreads.

Extended note. Version record

Sarah Breeden, Bank of England, Gilt edged resilience: strengthening liquidity provision in the repo market; September 2025 discussion paper

Difficulty: Advanced **Type:** Practitioner / commentary

Status/version: 17 July 2026; policy commentary following the September 2025 discussion paper.

Connects the cash gilt market to repo intermediation, leveraged non-bank demand and stress liquidity. Read it with the discussion paper to assess central clearing, minimum haircuts and the capacity of dealers to supply financing under stress. The discussion concerns potential reforms and their trade-offs, not a statement that every proposed rule is in force; scenario results also depend on participants' assumed behaviour.

Extended note. Version record

Gary Gorton and Andrew Metrick, "Securitized banking and the run on repo"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 104(3), 425–451 (2012); DOI 10.1016/j.jfineco.2011.03.016.

The 2007–08 crisis as a run on repo through rising haircuts on securitised collateral.

Version record

Adam Copeland, Antoine Martin and Michael Walker, "Repo Runs: Evidence from the Tri-Party Repo Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(6), 2343–2380 (2014); DOI 10.1111/jofi.12205.

Tri-party repo haircuts were stable in 2008; runs occurred through quantity withdrawal, especially for Lehman.

Version record

Tobias Adrian and Hyun Song Shin, "Liquidity and leverage"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Intermediation, 19(3), 418–437 (2010); DOI 10.1016/j.jfi.2008.12.002.

Procyclical broker-dealer leverage and its link to aggregate liquidity and risk premia.

Version record

Nicolae Gârleanu and Lasse Heje Pedersen, "Margin-based Asset Pricing and Deviations from the Law of One Price"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 24(6), 1980–2022 (2011); DOI 10.1093/rfs/hhr027.

Margin constraints create a funding-liquidity premium and explain bond–CDS and CIP-type basis deviations.

Version record

Zoltan Pozsar, Tobias Adrian, Adam Ashcraft and Hayley Boesky, "Shadow Banking"

Difficulty: Advanced **Type:** Report / research note

Status/version: Federal Reserve Bank of New York Staff Report 458, July 2010, revised February 2012; FRBNY Economic Policy Review 2013.

Map of the shadow-banking credit intermediation chain and its funding sources, revised 2012.

Version record

Moorad Choudhry, *The Repo Handbook*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Elsevier (2010); DOI 10.1016/c2009-0-16714-4.

Repo mechanics, tri-party, margining, documentation (GMRA) and the use of repo in funding and relative value; second edition 2010.

Version record

Adam Copeland and Antoine Martin, "Repo over the Financial Crisis"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(2), 911–936 (2025); DOI 10.1111/jofi.13406.

New data on bilateral and tri-party repo through 2007–08: bilateral activity fell much more, and much of the decline was in Treasury-backed repo.

Version record

Sebastian Infante and Zack Saravay, "Treasury Reuse and the Demand for Safe Assets"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhaf094 (2025); DOI 10.1093/rfs/hhaf094.

Collateral reuse of Treasuries lets intermediaries reallocate safety benefits and alleviate safe-asset scarcity.

Version record

9.1.2 Crowded trades and arbitrage capital

Carter Davis, The Elasticity of Quantitative Investment

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 38(10), 2845–2886, October 2025; published online 21 July 2025, corrected/typeset 16 August 2025; DOI 10.1093/rfs/hhaf047.

Tests the demand elasticity implied by thirteen modern cross-sectional statistical-arbitrage models and finds surprisingly inelastic demand because trading against price changes is difficult. The models and application concern equity characteristics, not a universal hard capacity number; financing, market impact, mandate limits, and the response of other arbitrageurs must be measured separately before converting elasticity into trading limits.

Extended note. Version record

Amir E. Khandani and Andrew W. Lo, What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data; published DOI

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets 14(1), 1–46, February 2011, DOI 10.1016/j.finmar.2010.07.005; MIT hosts the author's final manuscript issued August 2010. Distinct from their shorter 2007 Journal of Investment Management article with a similar title.

Combines long–short equity factor portfolios with transaction-level simulations to reconstruct the August 2007 quant dislocation and examine coordinated deleveraging and reduced market-making risk capital. The simulations identify patterns consistent with unwinds, but do not prove one unique initiating trade or guarantee that the event's magnitudes transfer to today's markets.

Extended note. Version record

Matteo Aquilina, Marco Jacopo Lombardi, Andreas Schrimpf, and Vladyslav Sushko, The Market Turbulence and Carry Trade Unwind of August 2024

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Bulletin 90, 27 August 2024; official eight-page policy analysis, not a peer-reviewed journal article.

Dissects the July–August 2024 reversal of yen-funded FX carry and related leveraged equity positions, emphasizing how margin and volatility feedback amplified a modest macro surprise. BIS position-size estimates are rough because forwards, swaps, and off-balance-sheet financing are only partly visible; the episode should inform stress design rather than serve as a stable expected-return or position-size formula.

Extended note. Version record

Mark Mitchell and Todd Pulvino, "Arbitrage crashes and the speed of capital"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 104(3), 469–490 (2012); DOI 10.1016/j.jfineco.2011.09.002.

Prime-broker funding withdrawal in 2008 forced arbitrageurs out of convertible and merger arbitrage, widening spreads.

Version record

Lasse Heje Pedersen, "When Everyone Runs for the Exit"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Central Banking 5(4), 177–199 (December 2009).

Liquidity spirals, crowding and the 2007 quant event explained through funding constraints.

Version record

Paul, Weiss, Rifkind, Wharton & Garrison LLP for the Credit Suisse Special Committee, "Credit Suisse Group Special Committee of the Board of Directors Report on Archegos Capital Management"

Difficulty: Advanced **Type:** Report / research note

Status/version: Report dated 29 July 2021; hosting may have moved after the UBS merger.

Case study of total-return-swap financing, margin settings, concentration and risk-control failure in a prime-brokerage relationship.

Version record

Emil N. Siriwardane, Adi Sunderam and Jonathan Wallen, "Segmented Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(5), 2543–2590 (2025); DOI 10.1111/jofi.13469.

Thirty-two arbitrage spreads across equity, fixed income and FX have low average pairwise correlation (22%), inconsistent with single-constraint intermediary models; segmentation of funding and balance sheets drives arbitrage dynamics.

Version record

Xi Dong, Namho Kang and Joel Peress, "Fast and Slow Arbitrage: The Predictive Power of (Persistent) Capital Flows for Factor Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(10), 2936–2987 (2025); DOI 10.1093/rfs/hhaf036.

Persistent aggregate flows to hedge and mutual funds predict monthly factor returns out of sample (R2 up to 6.6%), reflecting active managers' capital constraints.

Version record

Albert S. Kyle, Anna A. Obizhaeva and Yajun Wang, "Trading in Crowded Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(1), 137–175 (2026); DOI 10.1017/s0022109025101683.

Traders who underestimate crowding trade too aggressively and earn less than expected; equilibrium effects of misperceived crowdedness.

Version record

9.1.3 Treasury basis, relative value and dealer capacity

Mathias S. Kruttli, Phillip J. Monin, Lubomir Petrasek, and Sumudu W. Watugala, LTCM Redux? Hedge Fund Treasury Trading, Funding Fragility, and Risk Constraints; Federal Reserve author publication record

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics 169, article 104017, July 2025; DOI 10.1016/j.jfineco.2025.104017.

Uses granular regulatory fund and creditor data around March 2020 to show Treasury arbitrageurs cut positions and held more cash while average repo financing remained broadly stable; internal VaR constraints and precautionary liquidity were more explanatory than a simple repo-credit shutoff. Evidence is episode- and filing-specific, so institutional scenario design should stress both external financing and internally imposed limits, client concentration, margin calls, and same-day liquidity needs.

Extended note. Version record

Daniel Barth and R. Jay Kahn, Hedge Funds and the Treasury Cash-Futures Basis Trade

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics 155, article 103823, October 2025; DOI 10.1016/j.jmoneco.2025.103823; supersedes the 2021 OFR working-paper title Hedge Funds and the Treasury Cash-Futures Disconnect.

Models hedge funds as repo-financed warehouses connecting cash Treasuries, futures demand, and constrained dealers, then uses regulatory and market data to explain basis-trade quantities and the March 2020 unwind. Estimates from sample-specific supervisory data are not a guaranteed arbitrage return; stress tests must include funding, delivery, and liquidation paths through convergence.

Extended note. Version record

Vladyslav Sushko and Karamfil Todorov, Sizing Up Hedge Funds' Relative Value Trades in US Treasuries and Interest Rate Swaps

Difficulty: Advanced **Type:** Research paper

Status/version: Box B in BIS Quarterly Review, December 2025, pp. 57–59; official research box, not a peer-reviewed journal article.

Separates cash–futures basis from the cash-Treasury versus pay-fixed-swap spread trade and argues that the latter drove much recent growth in hedge-fund repo leverage and April 2025 Treasury stress. The swap-trade amount is an upper bound inferred from aggregate residual exposures, because Treasuries also serve as collateral and liquidity; do not compare it directly with fund-level estimates as though both measure identical positions.

Extended note. Version record

Phillip J. Monin, Decomposing Hedge Funds' U.S. Treasury Exposures

Difficulty: Research **Type:** Research paper

Status/version: FEDS Notes, 22 June 2026, DOI 10.17016/2380-7172.4082; Federal Reserve staff analysis, not an official Board position or peer-reviewed journal article.

Decomposes large hedge funds' September 2025 Treasury longs into estimated cash–futures basis, swap-spread, maturity-matched, curve, and liquidity uses, rather than treating all cash bonds or all short futures as basis trades. Form PF lacks trade-level position matching, so these are strategy proxies with meaningful classification and duration-conversion uncertainty, not exact exposures or a mechanically reconcilable update of BIS aggregate bounds.

Extended note. Version record

Darrell Duffie, "Post-Crisis Bank Regulations and Financial Market Liquidity"

Difficulty: Advanced **Type:** Report / research note

Status/version: Banca d'Italia Baffi Lecture, March 2018; later book Fragmenting Markets (De Gruyter 2022).

Baffi Lecture on how leverage, liquidity and capital rules raised the balance-sheet cost of market making in Treasuries, repo and CIP arbitrage.

Version record

Darrell Duffie, Michael J. Fleming, Frank M. Keane, Claire Nelson, Or Shachar and Peter Van Tassel, "Dealer Capacity and US Treasury Market Functionality"

Difficulty: Research **Type:** Report / research note

Status/version: Federal Reserve Bank of New York Staff Report 1070, August 2023 (later revisions possible).

Measures dealer balance-sheet utilisation and shows functionality deteriorates nonlinearly when capacity binds.

Version record

Nellie Liang and Haoxiang Zhu, "Clearing the Path for Treasury Market Resilience"

Difficulty: Advanced **Type:** Report / research note

Status/version: Hutchins Center Working Paper 103, Brookings, July 2025; updated February 2026.

Estimates dealer netting benefits from cleared repo (up to about USD 1.3 trillion if all primary-dealer Treasury repo were cleared, December 2025 data) and discusses SLR design. Check exact title on the Brookings page.

Extended note. Version record

Andreas Schrimpf, Hyun Song Shin and Vladyslav Sushko, "Leverage and margin spirals in fixed income markets during the Covid-19 crisis"

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Bulletin No. 2, April 2020.

How the March 2020 Treasury dash for cash interacted with leveraged basis trades and margin increases.

Version record

Annette Vissing-Jorgensen, "The Treasury Market in Spring 2020 and the Response of the Federal Reserve"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 124, 19–47 (2021); DOI 10.1016/j.jmoneco.2021.10.007.

Decomposes March 2020 Treasury selling by foreign official, mutual fund and hedge-fund holders and assesses Fed purchases.

Version record

Matthias Fleckenstein and Francis A. Longstaff, "Renting Balance Sheet Space: Intermediary Balance Sheet Rental Costs and the Valuation of Derivatives"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(11), 5051–5091 (2020); DOI 10.1093/rfs/hhaa033.

Futures–cash and swap–Treasury deviations imply a rental cost of dealer balance sheet that varies with regulation and stress.

Version record

Sven Klingler and Suresh Sundaresan, "An Explanation of Negative Swap Spreads: Demand for Duration from Underfunded Pension Plans"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 74(2), 675–710 (2019); DOI 10.1111/jofi.12750.

Underfunded pension plans' demand for long-duration swaps, with limited dealer balance sheet, explains negative 30-year swap spreads.

Version record

Nina Boyarchenko, Thomas M. Eisenbach, Pooja Gupta, Or Shachar and Peter Van Tassel, "Bank-Intermediated Arbitrage"

Difficulty: Advanced **Type:** Report / research note

Status/version: SSRN working paper 3200041 (2018); DOI 10.2139/ssrn.3200041. Working-paper record; no journal version established in this check.

How post-crisis leverage rules raised the balance-sheet cost of CIP, Treasury–swap and futures-basis arbitrage for banks.

Extended note. Version record

9.2 Collateral and valuation adjustments

Read with the retained historical Gregory edition, the fifth-edition update below and versioned SIMM methodology. Contractual discounting, default close-out, funding, collateral and initial margin are separate inputs; faster numerical methods do not remove those choices.

Jon Gregory, *The xVA Challenge: Counterparty Credit Risk, Funding, Collateral and Capital*, 3rd ed.

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected historical third edition (2015); Wiley released a materially updated fifth edition in July 2026.

Desk-facing account of counterparty credit, funding, collateral and capital adjustments. Use the source's CVA/FVA/KVA/MVA agenda to explain why model values and desk P&L diverge; margin methodology is a separate, versioned input. The catalogue also provides the fifth-edition companion (July 2026) for current FRTB-CVA, margin, capital, compression and computational coverage; compare editions before selecting a desk implementation text.

Applications: OTC derivatives; funding, collateral, risk. Version record

ISDA, *ISDA SIMM Methodology*, version 2.8+2506; version 2.8+2512 release

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: 2.8+2512 published 12 June 2026, effective 11 July; governance update 18 September 2026. Historical 2.8+2506 retained.

Retains the October 2025 release for historical comparison and adds the June 2026 recalibration using data through 31 December 2025. Version 2.8+2512 became effective 11 July 2026 (calculations from 10 July close; first exchange 13 July). The 18 September 2026 governance update explicitly reflects at-least-semiannual recalibration; this governance date is distinct from a methodology's effective date.

Applications: OTC derivatives; initial margin. Extended note. Version record

Vladimir Piterbarg, Funding beyond discounting: collateral agreements and derivatives pricing

Difficulty: Research **Type:** Research paper

Status/version: Risk technical article, 16 March 2010; original article distinct from later Landmarks retrospective.

Shows that collateral and funding agreements alter discounting, forwards and implied volatilities relative to a single risk-free-curve valuation. This supplies the contractual pricing step missing between the catalogue's rates-model monographs and Gregory's broad xVA desk reference. The stylised agreement must be replaced by actual CSA and netting terms; quoted model value is not automatically a transferable close-out price.

Extended note. Version record

Masaaki Fujii; Yasufumi Shimada; Akihiko Takahashi, A Market Model of Interest Rates With Dynamic Basis Spreads in the Presence of Collateral and Multiple Currencies

Difficulty: Research **Type:** Research paper

Status/version: Wilmott 2011, issue 54, 61–73; first published online 20 September 2011 (issue listed July 2011).

Extends a rates market model to dynamic tenor and cross-currency basis under collateralisation, making basis exposures visible in values and risk sensitivities. It adds a concrete multi-curve, multi-currency bridge to Brigo–Mercurio and Andersen–Piterbarg, whose original formulations need post-crisis collateral conventions. Collateral currency choice and legal CSA terms remain product-specific inputs.

Extended note. Version record

Christoph Burgard; Mats Kjaer, Partial differential equation representations of derivatives with bilateral counterparty risk and funding costs; publisher addendum; original paper PDF

Difficulty: Research **Type:** Research paper

Status/version: Journal of Credit Risk 7(3), September 2011; publisher addendum corrects notation in earlier versions without changing results.

Derives PDE and Feynman–Kac representations for bilateral counterparty default and funding costs, decomposing a derivative's total value into a clean price and valuation adjustments under stated hedging and funding assumptions. It is the analytical counterpart to Gregory's xVA overview and Glasserman's exposure simulation route. Close-out, own-default treatment, recoveries and desk funding policy must be matched to the actual trade and institution.

Extended note. Version record

Kristoffer Andersson; Alessandro Gnoatto, Multi-Layer Deep xVA: Structural Credit Models, Measure Changes and Convergence Analysis

Difficulty: Research **Type:** Research paper

Status/version: arXiv preprint v1 20 February 2025; v2 21 February 2025. No journal/proceedings status established from sources reviewed.

Represents portfolio-wide clean value, collateral, initial margin, credit/debit, margin and funding adjustments as layered coupled BSDE problems. Its deep solver uses quantile regression for margin over the risk period and measure changes to sample rare defaults, targeting the nested-simulation cost that limits large xVA books. Benchmark against simpler exposure engines and independent sensitivities.

Extended note. Version record

Jon Gregory, The xVA Challenge: A Valuation Adjustment Framework for Modern Derivatives Markets

Difficulty: Advanced **Type:** Book / monograph

Status/version: 5th ed., July 2026; ISBN 9781394354450.

The current-edition companion to the historical third-edition entry, extending the reading route across counterparty exposure, collateral and valuation adjustments. The publisher identifies updated coverage of FRTB-CVA, collateral valuation, margin and capital adjustments, compression, and computational techniques including AAD. Edition recency does not establish a jurisdiction's effective regulatory treatment; verify local rules and the actual CSA, close-out and funding assumptions for a portfolio.

Extended note. Version record

Andrew Green, *XVA: Credit, Funding and Capital Valuation Adjustments*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2015); DOI 10.1002/9781119161233.

CVA, DVA, FVA, KVA and MVA modelling with Monte Carlo exposure simulation and accounting context.

Version record

Damiano Brigo, Massimo Morini and Andrea Pallavicini, *Counterparty Credit Risk, Collateral and Funding: With Pricing Cases for All Asset Classes*

Difficulty: Research **Type:** Book / monograph

Status/version: Wiley, edition 1 (2013); DOI 10.1002/9781118818589.

Arbitrage-free CVA/DVA/FVA with wrong-way risk and collateral, with asset-class case studies.

Version record

Darrell Duffie and Haoxiang Zhu, "Does a Central Clearing Counterparty Reduce Counterparty Risk?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Asset Pricing Studies, 1(1), 74–95 (2011); DOI 10.1093/rapstu/rar001.

Clearing one asset class through a CCP can increase total exposure by sacrificing bilateral netting across classes.

Version record

Rama Cont, "Central clearing and risk transformation"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper 2955647 (2017); DOI 10.2139/ssrn.2955647. Working-paper record; no journal version established in this check.

CCPs transform counterparty risk into liquidity and margin risk; implications for stress testing and default waterfalls.

Extended note. Version record

Basel Committee on Banking Supervision and International Organization of Securities Commissions, "Margin requirements for non-centrally cleared derivatives"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: BCBS/IOSCO, revised April 2020 (d499); earlier 2015 version superseded.

Global framework for initial and variation margin on uncleared derivatives with phase-in thresholds; context for ISDA SIMM.

Extended note. Version record

Leif Andersen, Michael Pykhtin and Alexander Sokol, "Rethinking the margin period of risk"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Credit Risk, 13(1) (2017); DOI 10.21314/jcr.2016.218.

Detailed timeline of collateral disputes and settlement during default that produces exposure spikes missed by simple MPOR models.

Extended note. Version record

T. van der Zwaard, L. A. Grzelak and C. W. Oosterlee, "On the Hull-White model with volatility smile for Valuation Adjustments"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 25(10), 1535–1555 (2025); DOI 10.1080/14697688.2025.2565270.

Hull–White extension that captures market smile for xVA exposure calculations.

Version record

Michael B. Giles, Abdul-Lateef Haji-Ali and Jonathan Spence, "Efficient risk estimation for the credit valuation adjustment"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Stochastics, 30(4), 983–1021 (2026); DOI 10.1007/s00780-026-00601-y.

Efficient estimation of risk measures on credit valuation adjustment.

Extended note. Version record

Lokman Abbas Turki, Stéphane Crépey, Botao Li and Bouazza Saadeddine, "An explicit scheme for pathwise cross valuation adjustment computations"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Computational Finance (2025); DOI 10.21314/jcf.2025.003.

An explicit scheme for pathwise XVA computation.

Extended note. Version record

9.3 Securities-lending economics and lifecycle

Historical papers, association guidance and protocol documentation supply distinct product perspectives. Living pages and deployment-specific documents require their relevant versions.

Darrell Duffie, Nicolae Gârleanu and Lasse Heje Pedersen, "Securities Lending, Shorting, and Pricing"

Difficulty: Not specified **Type:** Research paper

Status/version: *Journal of Financial Economics* 66(2–3), 307–339, 2002

Search frictions, bargaining and lending availability connect borrow fees to asset prices. Provides the economic motivation for measuring inventory and outside options when experimenting with prices. Its model is a foundation for mechanism design and hypotheses, not proof that a quoted-fee increase improves portfolio income across asset classes.

Version record

Gene D'Avolio, "The Market for Borrowing Stock"

Difficulty: Not specified **Type:** Research paper

Status/version: *Journal of Financial Economics* 66(2–3), 271–306, 2002

Empirical anatomy of lending supply, borrow costs, specialness and recall risk using a US intermediary's April 2000–September 2001 data. Helps select covariates and understand selection into observed loans. Preserve its historical US-equity scope when designing modern global securities or token-lending studies.

Version record

CPSS and IOSCO Technical Committee, *Securities lending transactions: market development and implications*, CPSS Publications No. 32

Difficulty: Not specified **Type:** Report / research note

Status/version: Historical technical reference; July 1999

Joint survey of lending mechanisms, collateral, settlement and counterparty/operational risk. Historical institutional context for lifecycle and inventory research; neither a current rulebook nor evidence from a lending-price experiment. CPSS is the committee's name at publication.

Version record

International Securities Lending Association, *Securities Lending Best Practice Handbook*; downloadable compilation

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Living ISLA handbook hub in migration; linked 29-page compilation filename dated 31-03-2025, with individual practices separately dated and some under review; checked 4 October 2026.

Operational companion on securities-loan lifecycle, collateral, margin, settlement and related practices. The ISLA hub states that the handbook is being migrated; the linked compilation contains separately dated, sometimes under-review practices and says the governing

GMSLA or equivalent agreement prevails. Pin the selected practice and agreement rather than reading the PDF filename as a single revision date.

Version record

ICMA European Repo and Collateral Council, Repo FAQ 13: "What is the difference between repo and securities lending?"

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Living ICMA ERCC Repo FAQ 13; page has no publication or revision date displayed; checked 4 October 2026.

Clarifies the typical cash-financing motivation of repo versus security borrowing, including special repo, cash/non-cash collateral and fee/rebate conventions. Use to define whose revenue is measured and the relevant counterfactual funding cost. Open repo termination and securities-loan recall are not interchangeable contractual rights.

Version record

Federal Reserve Bank of New York, Reference Rates: TGCR, BGCR and SOFR; Additional Information about Reference Rates

Difficulty: Advanced **Type:** Data / benchmark methodology

Status/version: Living official benchmark pages, checked 4 October 2026; daily observations and revisions are separately dated.

Defines the transaction pools behind Treasury general collateral repo, broad general collateral repo and SOFR: TGCR uses tri-party Treasury general collateral, BGCR adds FICC GCF, and SOFR adds filtered FICC delivery-versus-payment repo. The published volume-weighted medians, percentiles and volumes support a dated funding benchmark; they do not quote a specific security loan, bilateral haircut or dealer funding line.

Extended note. Version record

LBMA, LPPM and LPMCL; edited by Jonathan Spall, *A Guide to the Loco London Precious Metals Market*, Chapter 9: "Lending and Borrowing Metal"

Difficulty: Not specified **Type:** Book chapter

Status/version: Loco London guide copyright October 2017; Chapter 9 live HTML checked 4 October 2026, without chapter-level revision date.

Metal leases/deposits, interest conventions and the relation between metal and currency financing in forwards/swaps. Explains principal credit exposure and why an outright metal lease differs from an exchange of metal and cash. Builds a commodity-lending module grounded in precious metals; do not extrapolate its conventions to every commodity or treat its numerical examples as current quotes.

Version record

Emilio Frangella and Lasse Herskind, *Aave V3 Technical Paper*; Aave Interest Rate Strategy documentation; Aave V3 Pool contract interface

Difficulty: Not specified **Type:** Protocol paper / documentation

Status/version: paper 27 January 2022; living docs accessed 2026-10-03 Selected companion documents inspected 12 October 2026; earlier source verification retained separately.

Pooled lending with a kinked utilisation-rate curve and collateral/liquidation feedback. The linked historical core repository was archived on 21 October 2025; use the separately linked origin migration notes and deployment-specific documentation for later versions. The V3.2 migration notes remove stable-rate logic and change the rate-strategy return interface. Pin chain, block, contracts and parameters; the 2022 paper does not identify later deployments. Bilateral token lending can have different pricing and risk mechanics. For liquidation execution, use the current Pool interface for account health, reserve configuration, liquidationCall and flashLoan. Pin chain, block, oracle and market parameters; a health factor below one does not establish an executable net gain.

Extended note. Version record

Kolasinski, Reed and Ringgenberg (2013), "A Multiple Lender Approach to Understanding Supply and Search in the Equity Lending Market"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected DOI-linked version; print 2013-04; online 2013-03-07.

Lending supply, search frictions and fee dispersion; a reference for modelling nonlinear fees and conditional demand.

Version record

Henderson, Jostova and Philipov, "Strategic Behavior by Equity Lenders''

Difficulty: Not specified **Type:** Research paper

Status/version: Edition/revision and publication date not established for the selected resource(s).

Strategic pricing by equity lenders and the distinction between intrinsic fees and demand response. Working-paper revision: 30 August 2025; online publication: 9 September 2025; journal issue: *Management Science* 72(5), 4359–4384 (May 2026).

Version record

Chague et al., "Well-Connected Short-Sellers Pay Lower Loan Fees: A Market-Wide Analysis"

Difficulty: Not specified **Type:** Research paper

Status/version: Edition/revision and publication date not established for the selected resource(s).

Borrower search costs, network position and fee dispersion for the same stock and day.

Version record

Duong et al. (2017), "The Information Value of Stock Lending Fees: Are Lenders Price Takers?"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2017. Version distinctions remain in the annotation.

Evidence on lender information and the information content of stock-lending fees.

Version record

Taisiya Sikorskaya, "Institutional Investor Mandates, Securities Lending, and Short-Selling Constraints''

Difficulty: Not specified **Type:** Research paper

Status/version: Edition/revision and publication date not established for the selected resource(s).

Institutional mandates, endogenous lending supply and short-selling constraints. The January 2026 working-paper recommendation is distinct from the later record: *Journal of Financial Economics* 184, article 104349 (October 2026). The SSRN link preserves working-paper access; its recorded revision is 13 August 2026.

Version record

Pan Yiming (also listed as Yiming Pan in a conference author field), *Lendable Inventory Concentration, Borrow Fragility, and Equity Pricing Anomalies*; differently titled May 2026 conference draft

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper: written 9 June, posted 3 July, revised 24 July 2026; separate conference PDF dated 31 May 2026.

Using IHS Markit securities-lending data, the SSRN abstract connects concentrated lendable inventory with unstable borrowing conditions, delayed bad-news incorporation and larger fee-adjusted anomaly spreads driven by short candidates. The Paris meeting lists the same VGSF researcher as both Yiming Pan (author field) and Pan Yiming (presenter field); its single-author PDF has a Pan Yiming byline and an earlier, different title. Preserve the SSRN title and sole-author attribution. Exact draft-to-SSRN text equivalence and formal given/family-name fields remain unverified.

Version record

Spencer Andrews, Brian J. Henderson and Adam V. Reed, "ETF Lending Premia and the Limits of Create-to-Lend Arbitrage" (revised June 2026)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2026. Version distinctions remain in the annotation.

ETF create-to-lend constraints, ownership concentration and lending premia. Preliminary working paper, first posted 20 December 2023, revised 26 June 2026.

Version record

George Nurisso, "I Don't Recall: Information and the Governance Tradeoffs in Securities Lending" (May 2026)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2026. Version distinctions remain in the annotation.

Information and governance trade-offs in securities lending and recalls. Preliminary theoretical working paper, posted 18 May 2026.

Version record

Atmaz, Basak and Ruan (2024), "Dynamic Equilibrium with Costly Short-Selling and Lending Market"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2024. Version distinctions remain in the annotation.

Dynamic links among disagreement, lending fees, short interest, volatility and returns. *Review of Financial Studies* 37(2), 444–506; online August 2023, issue February 2024.

Version record

Hong Kong SFC, *Report on the Thematic Review of Prime Services and Related Equity Derivatives Activities in Hong Kong*

Difficulty: Not specified **Type:** Report / research note

Status/version: Regulator report; 10 June 2019

Historical prime-services and securities-lending context in Hong Kong; a regulatory review rather than a controlled pricing study or current cross-jurisdictional rulebook.

Version record

Danqi Hu, Charles M. Jones, Xiaoyan Zhang and Xinran Zhang, "When do short sellers trade? Evidence from intraday data and implications for informed trading models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 172, 104148 (2025); DOI 10.1016/j.jfineco.2025.104148.

Intraday short-sale data show when short sellers trade and what that implies for informed-trading models.

Extended note. Version record

Amit Goyal, Adam V. Reed, Esad Smajlbegovic and Amar Soebhag, "Stealthy shorts: Informed liquidity supply"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 172, 104155 (2025); DOI 10.1016/j.jfineco.2025.104155.

The most informed short sellers supply rather than demand liquidity; transaction-level decomposition of short volume.

Version record

George C. Nurisso, "Learning by lending securities"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 178, 104237 (2026); DOI 10.1016/j.jfineco.2026.104237.

Studies what securities lenders learn from their lending activity; read with the same author's securities-lending governance paper already in the catalogue.

Extended note. Version record

Pekka Honkanen, "Securities Lending and Trading by Active and Passive Funds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(3), 1272–1309 (2025); DOI 10.1017/s0022109024000966.

Active funds use information from securities-lending demand to rebalance away from borrowed stocks; passive funds do not, with family spillovers.

Version record

Bastian von Beschwitz, Pekka Honkanen and Daniel Schmidt, "Passive ownership and short selling"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 29(4), 1137–1188 (2025); DOI 10.1093/rof/rfaf023.

Using the Russell cutoff, passive ownership raises lendable supply and short interest without changing fees; more reliable passive lending attracts informed short sellers and improves price efficiency.

Version record

Stefan Greppmair and Stephan Jank, "Collateral scarcity and market functioning: insights from the Eurosystem securities lending facilities"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 30(5), 1703–1738 (2026); DOI 10.1093/rof/rfag004.

The Eurosystem's 2020 securities-lending fee cut as a natural experiment: lending use surged, repo activity expanded through the collateral multiplier, and repo and cash-market quality improved.

Version record

9.4 CFDs, total return swaps and synthetic exposure

A contract for difference or equity total return swap delivers the price change of a share or index without ownership. Its cost sits in the financing spread, dividend and borrow pass-through, and margin and counterparty terms. The same contract raises disclosure and tax questions for institutions, and leverage and conduct questions when sold to retail clients.

9.4.1 Contracts, pricing and financing

Christine Brown, Jonathan Dark and Kevin Davis, "Exchange traded contracts for difference: Design, pricing, and effects"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Futures Markets, 30(12), 1108–1149 (2010); DOI 10.1002/fut.20475.

Analyses CFD contract design and pricing, and tests theoretical pricing relations on trades and quotes in the exchange-traded equity CFDs introduced by the Australian Securities Exchange. Draws lessons for designing new derivative contracts. The paper notes that the larger UK over-the-counter CFD market had no published academic study.

Version record

F. Douglas Foster, Adrian D. Lee and Wai-Man Liu, "CFDs, forwards, futures and the cost-of-carry"

Difficulty: Advanced **Type:** Research paper

Status/version: Pacific-Basin Finance Journal, 54, 183–198 (2019); DOI 10.1016/j.pacfin.2018.05.004.

Derives cost-of-carry parity relations between CFDs, forwards and futures with stochastic interest rates and uncertain dividends. On the S&P/ASX 200, listed index CFDs generate cash flows similar to listed futures once an open-interest charge is ignored, so a CFD can stand in for a forward.

Version record

Todd M. Hazelkorn, Tobias J. Moskowitz and Kaushik Vasudevan, "Beyond Basis Basics: Liquidity Demand and Deviations from the Law of One Price"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 78(1), 301–345 (2023); DOI 10.1111/jofi.13198.

Reads the futures–cash basis of global equity indices as liquidity demand for equity exposure. The basis comoves with dealer and investor futures positions and negatively predicts returns, implying an annual premium of 5 to 6 percent for supplying index exposure. A benchmark for the implied financing spread of synthetic equity.

Version record

Jules H. van Binsbergen, William F. Diamond and Marco Grotteria, "Risk-free interest rates"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 143(1), 1–29 (2022); DOI 10.1016/j.jfineco.2021.06.012.

Infers risk-free rates from option prices at minute frequency without a model of risk. The convenience yield on Treasuries is about 40 basis points, larger below three months and four times larger in the 2008 crisis. Gives a financing benchmark against which to judge a CFD or swap funding spread.

Version record

Eurex, Index Total Return Futures: a listed solution for implied equity repo trading

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living exchange product page; inspected 13 October 2026.

Exchange-listed futures that replicate an index total return swap: the buyer receives index price changes and distributions and pays overnight funding plus a traded spread, which is the implied equity repo rate. Cleared and portfolio-margined with other equity index derivatives, they are a listed alternative to bilateral swaps. Check current contract specifications and funding-rate conventions before use.

Version record

9.4.2 Prime brokerage, disclosure and tax

Antoine Bouveret and Martin Haferkorn, European Securities and Markets Authority, Leverage and derivatives – the case of Archegos

Difficulty: Advanced **Type:** Report / research note

Status/version: ESMA TRV Risk Analysis, ESMA50-165-2096, May 2022; DOI 10.2856/613372 (EU Publications Office).

Uses EMIR trade-repository data to reconstruct Archegos's total return swap positions with EU counterparties, showing the steep rise in concentrated exposure in February and March 2021 before a default that cost banks more than $10 billion. By spreading similar positions across dealers, the family office built exposures that no single counterparty or regulator saw. Shows how derivatives reporting can monitor leverage and concentration.

Version record

George O. Aragon and Philip E. Strahan, "Hedge funds as liquidity providers: Evidence from the Lehman bankruptcy"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 103(3), 570–587 (2012); DOI 10.1016/j.jfineco.2011.10.004.

Hedge funds using Lehman as prime broker lost funding liquidity after its September 2008 bankruptcy, and the stocks they held became less liquid, especially ex ante illiquid stocks; no similar effect followed the Bear Stearns failure. Evidence that a prime broker's failure reaches the market liquidity of its clients' positions.

Version record

Henry T. C. Hu and Bernard Black, "Hedge funds, insiders, and the decoupling of economic and voting ownership: Empty voting and hidden (morphable) ownership"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Corporate Finance, 13(2–3), 343–367 (2007); DOI 10.1016/j.jcorpfin.2007.02.004.

Shows how equity swaps, stock lending and other derivatives let hedge funds and insiders separate votes from economic ownership, often without disclosure: empty voting holds more votes than economic exposure, and hidden (morphable) ownership holds undisclosed economic exposure that can be turned into votes. The conceptual basis for disclosure rules on cash-settled derivatives.

Version record

U.S. Securities and Exchange Commission, Modernization of Beneficial Ownership Reporting: Final Rule

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Release 33-11253, adopted October 2023; 88 FR 76896 (7 November 2023); effective 5 February 2024.

Shortens Schedule 13D and 13G filing deadlines and, instead of adopting the proposed rule, gives guidance on when holders of cash-settled derivatives may already be beneficial owners of the reference shares. Item 6 of Schedule 13D now requires disclosure of cash-settled derivatives, including security-based swaps, that reference the issuer's securities.

Version record

U.S. Securities and Exchange Commission, Withdrawal of Proposed Regulatory Actions

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Release 33-11377, 12 June 2025; 90 FR 25531 (17 June 2025).

Withdraws fourteen proposed rules, among them proposed Rule 10B-1, which would have required public reporting of large security-based swap positions after Archegos. The Commission said it does not intend to adopt the withdrawn proposals and that any future action would need a new proposal.

Version record

Internal Revenue Service, Notice 2026-61: Extension of the Phase-in Period for the Enforcement and Administration of Section 871(m)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: IRS Notice 2026-61, issued September 2026.

Extends by two years the phase-in of the U.S. dividend-equivalent withholding rules for non-U.S. holders of equity derivatives. The rules for non-delta-one transactions will not apply to those issued through 2028, while delta-one transactions, such as equity swaps and CFDs on U.S. shares, remain covered under a good-faith enforcement standard through 2028; the anti-abuse rule continues to apply.

Version record

Thiess Buettner, Carolin Holzmann, Felix Kreidl and Hendrik Scholz, "Withholding-tax non-compliance: the case of cum-ex stock-market transactions"

Difficulty: Research **Type:** Research paper

Status/version: International Tax and Public Finance, 27(6), 1425–1452 (2020); DOI 10.1007/s10797-020-09602-9.

Models cum-ex trades around German ex-dividend dates, which produced illegitimate withholding-tax refunds, and shows they pay for both sides only with collusive fraud. From 2009 to 2015, trading in suitable stocks rose just before ex-dates until refunds were reformed, without price effects. A case study of how dividend-related derivative and stock trades can exploit tax mechanics.

Version record

9.4.3 Retail CFDs: regulation, brokers and outcomes

European Securities and Markets Authority, ESMA agrees to prohibit binary options and restrict CFDs to protect retail investors

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: ESMA statement, 27 March 2018; temporary EU product intervention measures.

The EU-wide restrictions on retail CFDs: leverage caps from 30:1 for major currency pairs to 2:1 for cryptocurrencies, close-out when margin falls to 50 percent of the requirement, negative balance protection per account, a ban on trading incentives and a standardised warning giving the provider's share of losing retail accounts. National evidence cited 74 to 89 percent of retail accounts losing money. The EU measures were temporary; check the national rules now in force.

Version record

Financial Conduct Authority, PS19/18: Restricting contract for difference products sold to retail clients

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: FCA policy statement, July 2019; rules in force 1 August 2019 for CFDs and 1 September 2019 for CFD-like options.

The UK's permanent rules for retail CFDs and CFD-like options: leverage limits of 30:1 to 2:1 by underlying, close-out at 50 percent of required margin, negative balance protection, a ban on inducements and a standardised warning with the provider's percentage of losing accounts.

Extended note. Version record

Australian Securities and Investments Commission, ASIC's CFD product intervention order extended for five years (22-082MR) and Report 724

Difficulty: Advanced **Type:** Report / research note

Status/version: ASIC media release 22-082MR, 2022, with Report 724; order in force from 29 March 2021, extended to 23 May 2027.

Extends Australia's CFD product intervention order, which limits leverage to between 30:1 and 2:1 and standardises margin close-out, negative balance protection and an inducements ban. Report 724 finds that in the order's first six months, aggregate retail net losses fell 91 percent, from $372 million to $33 million a quarter, across more than 60 issuers. It is a before–after comparison, not a controlled estimate.

Version record

Brad M. Barber, Yi-Tsung Lee, Yu-Jane Liu and Terrance Odean, "Just How Much Do Individual Investors Lose by Trading?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(2), 609–632 (2009); DOI 10.1093/rfs/hhn046.

Uses the complete trading history of all investors in Taiwan. Individuals' aggregate portfolio loses 3.8 percentage points a year, about 2.2 percent of GDP, almost entirely through aggressive orders, while institutions gain. A full-market benchmark for the size of retail trading losses.

Version record

Adrian D. Lee and Shan Choy, "Contracts for dummies? The performance of investors in contracts for difference"

Difficulty: Research **Type:** Research paper

Status/version: Accounting & Finance, 54(3), 965–997 (2014); DOI 10.1111/acfi.12018.

Measures after-cost returns of investors in ASX-listed share CFDs. Market-order trades earn small positive daily returns but negative returns over one month to one year because of financing costs, and net selling suggests use for shorting. Liquidity demanders get favourable execution. Exchange-listed CFDs differ from the over-the-counter retail CFDs that most providers sell.

Version record

Petar Peshev, "Analyzing CFD Retail Investors' Performance in a Post MiFID II Environment"

Difficulty: Research **Type:** Research paper

Status/version: Economic Studies (Bulgarian Academy of Sciences), 30(4), 53–73 (2021). No DOI established.

Studies retail CFD clients of EU-regulated firms after MiFID II. More than 80 percent of the sample lose money, and low equity and high used margin-to-equity ratios predict negative holding-period returns. MiFID II and the CFD sales restriction changed returns and margin use, possibly not as regulators intended. Sample details are limited in the abstract; read the paper before generalising.

Version record

Rawley Heimer and Alp Simsek, "Should retail investors' leverage be limited?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 132(3), 1–21 (2019); DOI 10.1016/j.jfineco.2018.10.017.

Evaluates the 2010 U.S. leverage cap on retail foreign-exchange trading with a difference-in-differences design. Volume fell 23 percent, high-leverage traders' losses fell 40 percent and brokers' operating capital fell 25 percent, with no change in relative bid–ask prices. The closest causal evidence on leverage caps from a related retail leveraged market.

Version record

Rawley Z. Heimer and Alex Imas, "Biased by Choice: How Financial Constraints Can Reduce Financial Mistakes"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 35(4), 1643–1681 (2022); DOI 10.1093/rfs/hhab073.

Shows that restricting retail leverage improved trading performance: by raising the cost of postponing losses, the constraint improved market timing and reduced the disposition effect. Two experiments replicate the mechanism.

Version record

M. Butz and R. Oomen, "Internalisation by electronic FX spot dealers"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 19(1), 35–56 (2019); DOI 10.1080/14697688.2018.1504167.

Models a dealer's choice between internalising client trades, warehousing the risk until offsetting flow arrives, and hedging them in the market. Calibrated to BIS survey data, a tier-one FX dealer takes several minutes to internalise in liquid currencies and tens of minutes in emerging-market currencies; internalisation costs less for dealers that hold more risk. The same choice underlies how a CFD provider hedges client flow.

Version record

Maira Andrade, Daniel Costa, Leonardo Weiss-Cohen, Jamie Torrance and Philip Newall, "Trading is a losing game: an audit of deceptive choice architecture in demo-mode Contract for Difference (CFD) trading apps"

Difficulty: Advanced **Type:** Research paper

Status/version: Behavioural Public Policy, 1–23 (2025), first published online; DOI 10.1017/bpp.2024.62.

Audits the demo modes of the 14 most popular UK CFD trading apps. 31.6 percent of risk warnings did not meet the regulator's standards, only 35.7 percent of apps showed warnings in their main tabs, and educational material encouraged hope of winning rather than acknowledging luck. Mandated loss disclosures can be blunted by app design.

Version record

Philipp Chapkovski, Mariana Khapko and Marius Zoican, "Trading Gamification and Investor Behavior"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 72(1), 32–56 (2026); DOI 10.1287/mnsc.2022.02650.

Randomised online experiment on trading-app gamification. Hedonic features such as confetti and badges raise trading volume by about 5 percent, but self-selection of less financially literate traders explains most of the gap between platforms, and price-trend notifications reinforce mistakes for traders with wrong beliefs.

Version record

Part III

Markets and instruments

Chapter 10

Equities

Combine cash-flow valuation, factor evidence, short-selling and other trader signals, corporate events, statistical arbitrage and market structure, then combine the signals into long–short portfolios. Research validation and cost-aware portfolio methods are cross-cutting prerequisites.

10.1 Factors, statistical arbitrage and events

Subject context and qualifications.

10.1.1 Factor premia and their risks

Eugene Fama and Kenneth French, "Common Risk Factors in the Returns on Stocks and Bonds"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics 33(1), 3–56, February 1993; DOI 10.1016/0304-405X(93)90023-5.

Establishes the empirical factor-model template that still underlies institutional equity risk and performance attribution. Use it to reason about factor correlation, redundancy and interpretation. The paper is also a reminder that an explanatory factor model is not automatically an investable strategy after shorting, turnover and implementation constraints.

Applications: Equities, bonds; factor investing. Version record

Narasimhan Jegadeesh and Sheridan Titman, "Returns to Buying Winners and Selling Losers"

Difficulty: Advanced **Type:** Research paper

Status/version: 1993

The foundational cross-sectional momentum paper. Reproduction should incorporate delistings, point-in-time universes, skip periods, sector effects, trading costs and the severe episodic crash behaviour of momentum portfolios.

Applications: Equities; momentum. Version record

Andrea Frazzini and Lasse Heje Pedersen, "Betting Against Beta"

Difficulty: Research **Type:** Research paper

Status/version: 2014

Shows how leverage constraints can produce a low-beta premium across asset classes. The institutional lesson is that financing conditions, beta estimation, leverage, derivatives implementation and crowding are integral to the strategy's economics.

Applications: Equities, bonds, FX, commodities; defensive factors. Version record

Clifford Asness, Tobias Moskowitz and Lasse Heje Pedersen, "Value and Momentum Everywhere" (author copy)

Difficulty: Research **Type:** Research paper

Status/version: 2013

Value and momentum across equities, indices, currencies, government bonds and commodities, including common variation and negative correlation. Cross-asset ranking differs from an asset's own time-series trend. [1]

Applications: Multi-asset; value, momentum, risk premia. Version record

Clifford Asness, Andrea Frazzini and Lasse Pedersen, "Quality Minus Junk"

Difficulty: Research **Type:** Research paper

Status/version: 2019

Defines a broad quality factor using profitability, growth, safety and payout characteristics. The paper is valuable for its explicit construction choices and for analysing interactions between quality, valuation, leverage and defensive equity exposures.

Applications: Equities; quality, factor portfolios. Version record

Stephen Penman and Peter F. Pope, Financial Statement Analysis for Value Investing

Difficulty: Advanced **Type:** Book / monograph

Status/version: Columbia Business School Publishing, April 2025; publisher hardcover ISBN 9780231215688; 432 pages.

Builds a fundamental-equity route from balance sheet, income statement, leverage, profitability and growth to value and credit risk. Read it beside factor papers to understand how accounting definitions create and sometimes distort value, quality and investment signals. Work through a company filing, reconcile earnings with cash flow, estimate reinvestment and financing needs, and examine whether a low multiple reflects accounting treatment or economic risk. The publisher exposes contents and an excerpt, but the full book was not reviewed; its case examples are foundations for analysis, not point-in-time evidence of an investable strategy.

Extended note. Version record

Robert Novy-Marx, "The other side of value: The gross profitability premium"
Difficulty: Advanced **Type:** Research paper
Status/version: Journal of Financial Economics, 108(1), 1–28 (2013); DOI 10.1016/j.jfineco.2013.01.003.
Gross profits to assets predicts returns as strongly as book-to-market and hedges value; foundation of profitability factors.
Version record

Richard G. Sloan, "Do Stock Prices Fully Reflect Information in Accruals and Cash Flows about Future Earnings?"
Difficulty: Advanced **Type:** Research paper
Status/version: The Accounting Review, 71(3), 289–315 (1996); DOI 10.2308/tar-9608042309.
Accrual component of earnings is less persistent than cash flows, and prices overweight it; origin of the accrual anomaly.
Version record

Joseph D. Piotroski, "Value Investing: The Use of Historical Financial Statement Information to Separate Winners from Losers"
Difficulty: Advanced **Type:** Research paper
Status/version: Journal of Accounting Research, 38, 1 (2000); DOI 10.2307/2672906.
Nine-signal F-score improves value-stock returns; a transparent accounting-quality screen.
Version record

Kent Daniel and Tobias J. Moskowitz, "Momentum crashes"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 122(2), 221–247 (2016); DOI 10.1016/j.jfineco.2015.12.002.
Momentum crashes in panic states after market declines, when losers rebound; dynamic weighting mitigates.
Version record

Pedro Barroso and Pedro Santa-Clara, "Momentum has its moments"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 116(1), 111–120 (2015); DOI 10.1016/j.jfineco.2014.11.010.
Scaling momentum by its own realised volatility nearly removes crash risk and raises Sharpe ratios.
Version record

Clifford Asness, Andrea Frazzini, Ronen Israel and Tobias Moskowitz, "Fact, Fiction, and Momentum Investing"
Difficulty: Intro/Advanced **Type:** Research paper
Status/version: The Journal of Portfolio Management, 40(5), 75–92 (2014); DOI 10.3905/jpm.2014.40.5.075.
Addresses common objections to momentum (size, costs, taxes, persistence) with long-sample evidence.
Version record

Sina Ehsani and Juhani T. Linnainmaa, "Factor Momentum and the Momentum Factor"
Difficulty: Research **Type:** Research paper
Status/version: The Journal of Finance, 77(3), 1877–1919 (2022); DOI 10.1111/jofi.13131.
Factor autocorrelation explains much of stock momentum; factor momentum is a distinct timing signal.
Version record

Andrew Ang, Robert J. Hodrick, Yuhang Xing and Xiaoyan Zhang, "The Cross-Section of Volatility and Expected Returns"
Difficulty: Research **Type:** Research paper
Status/version: The Journal of Finance, 61(1), 259–299 (2006); DOI 10.1111/j.1540-6261.2006.00836.x.
High idiosyncratic volatility stocks earn abnormally low returns; part of the low-risk anomaly family.
Version record

Malcolm Baker, Brendan Bradley and Jeffrey Wurgler, "Benchmarks as Limits to Arbitrage: Understanding the Low-Volatility Anomaly"
Difficulty: Advanced **Type:** Research paper
Status/version: Financial Analysts Journal, 67(1), 40–54 (2011); DOI 10.2469/faj.v67.n1.4.
Benchmarked institutional mandates discourage exploiting low-volatility stocks; behavioural demand for lottery stocks contributes.
Version record

Ronen Israel, Kristoffer Laursen and Scott Richardson, "Is (Systematic) Value Investing Dead?"
Difficulty: Advanced **Type:** Research paper
Status/version: The Journal of Portfolio Management, 47(2), 38–62 (2020); DOI 10.3905/jpm.2020.1.194.
Decomposes the 2018–2020 value drawdown into valuation spread widening versus fundamental deterioration.
Version record

Antti S. Ilmanen, Ronen Israel, Tobias J. Moskowitz, Ashwin K. Thapar and Franklin Wang, "Factor Premia and Factor Timing: A Century of Evidence"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper (2019) under this title; later journal version not established in this check.

Value, momentum, carry and defensive premia across asset classes over a century; little predictability from macro states.

Extended note. Version record

Pedro Barroso, Andrew Detzel and Paulo Maio, "The volatility puzzle of the beta anomaly"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 165, 103994 (2025); DOI 10.1016/j.jfineco.2025.103994.

Studies how volatility relates to the beta anomaly; read with betting-against-beta and low-volatility evidence.

Extended note. Version record

Geert Bekaert, Mikael Bergbrant and Haimanot Kassa, "Expected idiosyncratic volatility"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 167, 104023 (2025); DOI 10.1016/j.jfineco.2025.104023.

Separates expected from realised idiosyncratic volatility and revisits its relation to returns.

Extended note. Version record

Daniele Bianchi, Andrea De Polis and Ivan Petrella, "Time-Varying Skewness and Momentum Crashes"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Asset Pricing Studies, raag010 (2026); DOI 10.1093/rapstu/raag010.

Time-varying skewness in momentum returns deepens in crashes; a volatility–skewness crash indicator predicts left-tail momentum outcomes.

Version record

Xavier Gérard and Laura Jehl, "The Many Facets of Stock Momentum: Distinguishing Factor and Stock Components"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 82(1), 103–127 (2026); DOI 10.1080/0015198x.2025.2562790.

Separates stock-specific from factor momentum using earnings-announcement returns during formation; the stock-specific component predicts returns without long-run reversal in the US, Europe and Japan.

Version record

Ludwig Chincarini, Renato Lazo-Paz and Fabio Moneta, "Crowded spaces and anomalies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 182, 107579 (2026); DOI 10.1016/j.jbankfin.2025.107579.

Anomaly returns come mainly from the most crowded long-leg and least crowded short-leg stocks and persist after publication, consistent with crowding-related crash risk as a limit to arbitrage.

Version record

Guido Baltussen, Laurens Swinkels and Pim van Vliet, "Global factor premiums"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 142(3), 1128–1154 (2021); DOI 10.1016/j.jfineco.2021.06.030.

Replication and new-sample evidence over more than 200 years for 24 global factor premia across asset classes; most premia are present out of sample after allowing for p-hacking.

Version record

Clifford Asness, Andrea Frazzini, Ronen Israel, Tobias J. Moskowitz and Lasse Heje Pedersen, "Size Matters, If You Control Your Junk"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 129(3), 479–509 (2018); DOI 10.1016/j.jfineco.2018.05.006.

The size premium has been criticised as weak, unstable, concentrated in microcaps and in January, and absent outside the United States. Controlling for quality, or its inverse, junk, restores a significant size premium that is stable over time, not concentrated in microcaps and present in 30 industries and 24 international equity markets.

Version record

David Blitz, Joop Huij and Martin Martens, "Residual Momentum"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 18(3), 506–521 (2011); DOI 10.1016/j.jempfin.2011.01.003.

Ranks stocks on residual returns from a factor model rather than on total returns. Residual momentum earns risk-adjusted profits about twice those of total-return momentum, and the usual explanations for momentum do not account for it.

Version record

10.1.2 Statistical arbitrage and relative value

Evan Gatev, William Goetzmann and K. Geert Rouwenhorst, "Pairs Trading"

Difficulty: Advanced **Type:** Research paper

Status/version: 2006

The canonical empirical study of distance-based relative-value trading. It is most useful as a research-design benchmark; institutional stat arb generally requires richer factor residualisation, borrow modelling, dynamic relationships and portfolio-level execution.

Applications: Equities, ETFs; stat arb, mean reversion. Version record

Marco Avellaneda and Jeong-Hyun Lee, "Statistical Arbitrage in the U.S. Equities Market"

Difficulty: Research **Type:** Research paper

Status/version: 2010

Develops PCA- and ETF-factor residual strategies for large equity universes. The paper links factor-model residuals, mean reversion and portfolio construction in a form that is much closer to institutional stat arb than isolated pairs trading.

Applications: Equities; stat arb, residual mean reversion. Version record

Andrew Pole, *Statistical Arbitrage: Algorithmic Trading Insights and Techniques* (2007)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected DOI-linked version; print 2012-01-02; online 2015-10-02.

Practitioner treatment of equity mean reversion, portfolio construction, data processing, risk and implementation. Read alongside Gatev and Avellaneda–Lee to compare pairs, baskets and residual signals; test stability and borrow availability rather than assuming that dollar neutrality removes systematic risk.

Applications: practitioner. Version record

Ganapathy Vidyamurthy, *Pairs Trading: Quantitative Methods and Analysis* (2004)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Selected citation year(s): 2004. Version distinctions remain in the annotation.

Companion on cointegration, spread construction, pair selection and execution. Distinguish distance-based selection from Engle–Granger cointegration; estimate hedge ratios and stationarity only in the training window and test sensitivity to structural breaks and multiple pair searches.

Applications: methods. Version record

Christopher Krauss, "Statistical Arbitrage Pairs Trading Strategies: Review and Outlook"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Surveys, 31(2), 513–545 (2017); DOI 10.1111/joes.12153.

Taxonomy of distance, cointegration, time-series, stochastic-control and ML pairs methods with evaluation shortcomings.

Version record

Robert J. Elliott, John van der Hoek and William P. Malcolm, "Pairs trading"

Difficulty: Advanced **Type:** Research paper

Status/version: Quantitative Finance, 5(3), 271–276 (2005); DOI 10.1080/14697680500149370.

Mean-reverting Gaussian spread model with Kalman filtering and threshold rules.

Version record

Binh Do and Robert Faff, "Does Simple Pairs Trading Still Work?"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 66(4), 83–95 (2010); DOI 10.2469/faj.v66.n4.1.

Distance-method pairs profits declined after 2002 but rebound in turbulent markets; a decay study with costs.

Version record

Jorge Guijarro-Ordonez, Markus Pelger and Greg Zanotti, "Deep Learning Statistical Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 72(9), 7502–7549 (2026); DOI 10.1287/mnsc.2022.03132.

Builds residual portfolios from factor models, extracts time-series signals with convolutional transformers and optimises trading jointly; now published in Management Science. Reported Sharpe ratios should be judged after realistic costs and capacity.

Version record

Andrew W. Lo and A. Craig MacKinlay, "When Are Contrarian Profits Due to Stock Market Overreaction?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 3(2), 175–205 (1990); DOI 10.1093/rfs/3.2.175.

Decomposes contrarian profits into autocorrelation and lead–lag cross-autocorrelation components.

Version record

Stefan Nagel, "Evaporating Liquidity"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 25(7), 2005–2039 (2012); DOI 10.1093/rfs/hhs066.

Short-term reversal returns proxy for liquidity-provision returns, which rise sharply when VIX is high.

Version record

10.1.3 Events, flows and calendar effects

Ray Ball and Philip Brown, "An Empirical Evaluation of Accounting Income Numbers", together with Victor Bernard and Jacob Thomas, "Post-Earnings-Announcement Drift: Delayed Price Response or Risk Premium?"

Difficulty: Advanced **Type:** Research paper

Status/version: 1968 /1989

These papers provide the empirical foundation for earnings-event signals and delayed information incorporation. Modern implementations must use exact timestamps, announcement-session classification, restatement-safe fundamentals and realistic overnight execution assumptions.

Applications: Equities; event-driven, earnings. Version record

Robin Greenwood and Marco Sammon, The Disappearing Index Effect

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 80(2), 657–698, April 2025; DOI 10.1111/jofi.13410; Harvard open manuscript record.

Re-examines the price response to S&P 500 additions and deletions as passive ownership grew. The paper reports that abnormal addition returns fell from roughly 7.4% in the 1990s to 0.3% over the prior decade, with a similar decline for deletions and other index families. Use it to test anticipation, announcement and effective-date windows rather than assuming an index rebalance creates a stable arbitrage. A reconstruction needs point-in-time constituents, announcement timestamps, float adjustments, fund flows and executable closing prices. Historical averages do not establish the payoff to a current index-event strategy.

Extended note. Version record

Mark Mitchell and Todd Pulvino, "Characteristics of Risk and Return in Risk Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: 2001

The canonical empirical treatment of merger arbitrage and its nonlinear downside when deals fail. A modern desk should extend it with competing bids, antitrust probabilities, financing conditions, borrow, options and portfolio crowding.

Applications: Equities, options; merger arbitrage. Version record

Paul Tetlock, "Giving Content to Investor Sentiment"

Difficulty: Research **Type:** Research paper

Status/version: 2007

A seminal demonstration that textual media content contains information about market activity and reversals. It remains a useful baseline for evaluating whether modern language models add information beyond transparent dictionaries, timestamps and standard market controls.

Applications: Equities; NLP, news, event-driven. Version record

David Ikenberry, Josef Lakonishok and Theo Vermaelen, "Market underreaction to open market share repurchases"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 39(2-3), 181–208 (1995); DOI 10.1016/0304-405x(95)00826-z.

Long-run abnormal returns after repurchase announcements, concentrated in value stocks.

Version record

Alon Brav, Wei Jiang, Frank Partnoy and Randall Thomas, "Hedge Fund Activism, Corporate Governance, and Firm Performance"
Difficulty: Research **Type:** Research paper
Status/version: The Journal of Finance, 63(4), 1729–1775 (2008); DOI 10.1111/j.1540-6261.2008.01373.x.
13D filings by activist hedge funds generate announcement returns and operational improvements.
Version record

Dong Lou, Christopher Polk and Spyros Skouras, "A tug of war: Overnight versus intraday expected returns"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 134(1), 192–213 (2019); DOI 10.1016/j.jfineco.2019.03.011.
Momentum earns overnight while other anomalies earn intraday, consistent with clientele effects.
Version record

Steven L. Heston and Ronnie Sadka, "Seasonality in the cross-section of stock returns"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 87(2), 418–445 (2008); DOI 10.1016/j.jfineco.2007.02.003.
Stocks tend to repeat relative performance in the same calendar month for decades.
Version record

Lei Gao, Yufeng Han, Sophia Zhengzi Li and Guofu Zhou, "Market intraday momentum"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 129(2), 394–414 (2018); DOI 10.1016/j.jfineco.2018.05.009.
First half-hour return predicts last half-hour return of the S&P 500 ETF, strongest on volatile days.
Version record

Joshua Coval and Erik Stafford, "Asset fire sales (and purchases) in equity markets"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 86(2), 479–512 (2007); DOI 10.1016/j.jfineco.2006.09.007.
Flow-driven mutual fund sales create price pressure that later reverses; a template for flow-based signals.
Version record

Dong Lou, "A Flow-Based Explanation for Return Predictability"
Difficulty: Research **Type:** Research paper
Status/version: Review of Financial Studies, 25(12), 3457–3489 (2012); DOI 10.1093/rfs/hhs103.
Expected flow-induced trading predicts returns and partly explains momentum and fund-performance persistence.
Version record

Ekkehart Boehmer, Charles M. Jones and Xiaoyan Zhang, "Which Shorts Are Informed?"
Difficulty: Research **Type:** Research paper
Status/version: The Journal of Finance, 63(2), 491–527 (2008); DOI 10.1111/j.1540-6261.2008.01324.x.
Institutional non-program short sales predict negative returns; uses NYSE system order data.
Version record

David E. Rapach, Matthew C. Ringgenberg and Guofu Zhou, "Short interest and aggregate stock returns"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 121(1), 46–65 (2016); DOI 10.1016/j.jfineco.2016.03.004.
Aggregate short interest is a strong predictor of market returns, related to cash-flow news.
Version record

Yizhang Li, Stanislav Sokolinski and Andrea Tamoni, "Which investors drive anomaly returns and how?"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 179, 104257 (2026); DOI 10.1016/j.jfineco.2026.104257.
Identifies which investor types drive anomaly returns and through which trading channels.
Extended note. Version record

10.1.4 Alpha research practice

Igor Tulchinsky, ed., *Finding Alphas: A Quantitative Approach to Building Trading Strategies*, 2nd ed. (2019)

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2nd ed. (2019)

WorldQuant practitioner chapters on hypothesis generation, alpha design, correlation, testing and combining many small forecasts. Includes equity, futures, forwards, intraday and event-driven perspectives. Use the examples to build a documented hypothesis queue, with separate validation and implementation decisions.

Applications: industry.

Version record

Zura Kakushadze, "101 Formulaic Alphas" (2016)

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2016. Version distinctions remain in the annotation.

A concrete catalogue of formulaic equity signals and their common operators. Reimplement ranks, lags, rolling statistics and neutralisation with point-in-time data; examine correlation, decay, turnover and net returns. The paper describes historical production alphas; publication does not establish a present tradable edge.

Version record

Ernest P. Chan, *Quantitative Trading: How to Build Your Own Algorithmic Trading Business*; *Algorithmic Trading: Winning Strategies and Their Rationale* (2013)

Difficulty: Intro **Type:** Book / monograph

Status/version: Selected DOI-linked version; print 2012-01-02; online 2015-10-03.

Restored supplementary introductions to idea generation, mean reversion, trend following, out-of-sample tests, trading frictions and operational pitfalls, with MATLAB/Python-oriented examples. The first title carries a 2009 copyright in the linked Wiley record. Rebuild historical examples with current data and explicit fill, borrow and funding assumptions.

Applications: supplementary. Extended note.

Version record

Zura Kakushadze and Juan Andrés Serur, *151 Trading Strategies*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Springer International Publishing (2018); DOI 10.1007/978-3-030-02792-6.

Catalogue of strategy definitions across asset classes with formulas and references; a map of ideas, not performance evidence.

Version record

Ernest P. Chan, *Machine Trading: Deploying Computer Algorithms to Conquer the Markets*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Wiley, 2017 (ISBN 9781119219606).

Practitioner treatment of factor models, ML, options strategies and intraday trading with code (Wiley 2017).

Version record

Frank J. Fabozzi, Sergio M. Focardi and Petter N. Kolm, *Quantitative Equity Investing: Techniques and Strategies*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, 2010 (Frank J. Fabozzi Series).

End-to-end quantitative equity process: factor and signal construction, forecasting, risk models, portfolio construction and trading costs.

Version record

Ronald N. Kahn, "The Future of Investment Management"

Difficulty: Advanced **Type:** Report / research note

Status/version: CFA Institute Research Foundation, 2018.

CFA Institute Research Foundation monograph on the history and future of active, smart-beta and quantitative management from a leading practitioner.

Version record

Zura Kakushadze and Igor Tulchinsky, "Performance v. turnover: a story by 4,000 alphas"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Investment Strategies, 5(2), 75–89 (2016); DOI 10.21314/jois.2016.066.

Empirical regularities across 4,000 real-life production alphas: return scales with volatility and turnover in a simple power law.

Version record

10.2 Valuation and ETFs

Cash-flow valuation complements the factor and event-driven readings. ETFs add a vehicle layer whose creation, redemption and price-discovery mechanisms vary with the underlying market; read the contrasting equity and bond evidence together.

10.2.1 Valuation and accounting

Aswath Damodaran, Investment Valuation: Tools and Techniques for Determining the Value of Any Asset

Difficulty: Intro **Type:** Book / monograph

Status/version: 4th ed.; selected hardcover ISBN 9781394254606, publisher listing February 2025.

Adds cash-flow valuation to the catalogue's strong factor-investing coverage. Work through financial statements, financing costs, terminal value and equity value per share, then compare discounted cash flows, relative valuation and real options for mature firms, start-ups, distressed businesses and property. Valuation uncertainty remains sensitive to accounting choices, reinvestment and terminal assumptions; a point estimate is neither a timing signal nor an executable portfolio.

Extended note. Version record

Tim Koller, Marc Goedhart and David Wessels (McKinsey & Company), *Valuation: Measuring and Managing the Value of Companies*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Wiley, eighth edition, May 2025 (hardcover ISBN 9781394279418); seventh edition 2020 retained as prior version.

Standard practitioner DCF and economic-profit text with ROIC/growth decomposition, cost of capital and market-implied expectations; eighth edition (2025).

Version record

James A. Ohlson, "Earnings, Book Values, and Dividends in Equity Valuation"

Difficulty: Research **Type:** Research paper

Status/version: Contemporary Accounting Research, 11(2), 661–687 (1995); DOI 10.1111/j.1911-3846.1995.tb00461.x.

Residual-income valuation linking price to book value and abnormal earnings; theoretical basis for accounting-based value signals.

Version record

Richard Frankel and Charles M. C. Lee, "Accounting valuation, market expectation, and cross-sectional stock returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Accounting and Economics, 25(3), 283–319 (1998); DOI 10.1016/s0165-4101(98)00026-3.

Analyst-based residual-income value-to-price predicts returns better than book-to-market.

Version record

Andrea L. Eisfeldt, Edward Kim and Dimitris Papanikolaou, "Intangible Value"

Difficulty: Research **Type:** Research paper

Status/version: Critical Finance Review, 11(2), 299–332 (2022); DOI 10.1561/104.00000113.

Adds capitalised intangible capital to book equity; an intangible-adjusted HML performs better in recent decades.

Version record

Gerald A. Feltham and James A. Ohlson, "Valuation and Clean Surplus Accounting for Operating and Financial Activities"

Difficulty: Research **Type:** Research paper

Status/version: Contemporary Accounting Research, 11(2), 689–731 (1995); DOI 10.1111/j.1911-3846.1995.tb00462.x.

Residual-income valuation separating operating and financial activities; the theoretical basis of accounting-based value signals.

Version record

Charles M. C. Lee, James Myers and Bhaskaran Swaminathan, "What is the Intrinsic Value of the Dow?"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 54(5), 1693–1741 (1999); DOI 10.1111/0022-1082.00164.

Residual-income intrinsic values predict Dow index returns better than price-to-book or earnings yield.

Version record

William R. Gebhardt, Charles M. C. Lee and Bhaskaran Swaminathan, "Toward an Implied Cost of Capital"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Accounting Research, 39(1), 135–176 (2001); DOI 10.1111/1475-679x.00007.

Implied cost of capital from analyst forecasts and residual income; a forward-looking expected-return proxy used across asset pricing.

Version record

Scott Richardson, İrem Tuna and Peter Wysocki, "Accounting anomalies and fundamental analysis: A review of recent research advances"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Accounting and Economics, 50(2-3), 410–454 (2010); DOI 10.1016/j.jacceco.2010.09.008.

Review of accounting-based anomalies (accruals, earnings quality, fundamentals) and how to judge whether they are risk or mispricing.

Version record

Eugene F. Fama and Kenneth R. French, "Profitability, investment and average returns"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 82(3), 491–518 (2006); DOI 10.1016/j.jfineco.2005.09.009.

Uses the valuation identity to predict that profitability and investment forecast returns, the bridge from accounting to the five-factor model.

Version record

Ray Ball, Joseph Gerakos, Juhani T. Linnainmaa and Valeri Nikolaev, "Accruals, cash flows, and operating profitability in the cross section of stock returns"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 121(1), 28–45 (2016); DOI 10.1016/j.jfineco.2016.03.002.

Cash-based operating profitability subsumes the accrual anomaly and outperforms accrual-based profitability measures.

Version record

Andrey Golubov and Theodosia Konstantinidi, "Where Is the Risk in Value? Evidence from a Market-to-Book Decomposition"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 74(6), 3135–3186 (2019); DOI 10.1111/jofi.12836.

Splits market-to-book into market-to-value and value-to-book; the value premium comes from the market-to-value (mispricing-like) component.

Version record

Ryan H. Peters and Lucian A. Taylor, "Intangible capital and the investment-q relation"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 123(2), 251–272 (2017); DOI 10.1016/j.jfineco.2016.03.011.

Constructs firm-level intangible capital from capitalised R&D and SG&A; the standard intangible-adjusted book value input.

Version record

Robert D. Arnott, Campbell R. Harvey, Vitali Kalesnik and Juhani T. Linnainmaa, "Reports of Value's Death May Be Greatly Exaggerated"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 77(1), 44–67 (2021); DOI 10.1080/0015198x.2020.1842704.

Value's 2007–2020 drawdown is mostly revaluation (cheap got cheaper) plus intangibles mismeasurement, not a structural break.

Version record

Stephen H. Penman, *Financial Statement Analysis and Security Valuation*

Difficulty: Advanced **Type:** Book / monograph

Status/version: McGraw-Hill Education, fifth edition, 2013.

Standard text on accounting-based valuation: reformulated statements, residual earnings and growth, and fundamental-analysis forecasting.

Version record

Aswath Damodaran, *Narrative and Numbers: The Value of Stories in Business*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Columbia University Press (2017); DOI 10.7312/damo18048.

Links qualitative business narratives to valuation inputs and shows how to test narratives against numbers.

Version record

Partha S. Mohanram, "Separating Winners from Losers among Low Book-to-Market Stocks using Financial Statement Analysis"

Difficulty: Research **Type:** Research paper

Status/version: Review of Accounting Studies, 10(2–3), 133–170 (2005); DOI 10.1007/s11142-005-1526-4.

Builds a G-score for growth (low book-to-market) firms from earnings and cash-flow profitability, the stability of earnings and growth, and the intensity of R&D, capital expenditure and advertising. Buying high-score and shorting low-score firms earns excess returns after controls for size, book-to-market, momentum and accruals, and analysts are more often positively surprised by high-score firms.

Version record

Messod D. Beneish, "The Detection of Earnings Manipulation"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 55(5), 24–36 (1999); DOI 10.2469/faj.v55.n5.2296.

Profiles firms that manipulated earnings and estimates a model from financial-statement variables that capture distortions and the incentives to manipulate. It identifies about half of the manipulators before public discovery. Use it as a screen: a high score still needs a check on whether the distortion has another structural cause.

Version record

U-Wen Kok, Jason Ribando and Richard Sloan, "Facts about Formulaic Value Investing"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 73(2), 81–99 (2017); DOI 10.2469/faj.v73.n2.2.

Quantitative value strategies built from ratios such as book-to-price or earnings-to-price make no attempt to estimate intrinsic value. The authors find little compelling evidence that these strategies outperform in US equities, and show that they systematically select firms with temporarily inflated accounting numbers.

Version record

Kewei Hou, Mathijs A. van Dijk and Yinglei Zhang, "The Implied Cost of Capital: A New Approach"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Accounting and Economics, 53(3), 504–526 (2012); DOI 10.1016/j.jacceco.2011.12.001.

Estimates the implied cost of capital for a large 1968–2008 sample from earnings forecasts made by a cross-sectional model rather than by analysts. The model forecasts beat analysts on coverage, bias and earnings response, and the resulting implied cost of capital is a more reliable proxy for expected returns.

Version record

Eric C. So, "A New Approach to Predicting Analyst Forecast Errors: Do Investors Overweight Analyst Forecasts?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 108(3), 615–640 (2013); DOI 10.1016/j.jfineco.2013.02.002.

Shows that the usual methods for predicting analyst forecast errors are biased and replaces them with characteristic forecasts, which map current firm characteristics into future earnings. Comparing these with analyst forecasts predicts forecast errors and revisions. Prices do not fully reflect the predictable errors, so investors appear to overweight analyst forecasts, and sorting on predicted errors earns abnormal returns.

Version record

Söhnke M. Bartram and Mark Grinblatt, "Agnostic Fundamental Analysis Works"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 128(1), 125–147 (2018); DOI 10.1016/j.jfineco.2016.11.008.

Estimates each firm's peer-implied fair value by regressing market values on reported accounting items, a deliberately finance-agnostic method chosen to limit data snooping. Buying firms priced below and shorting firms priced above their peer-implied values earns risk-adjusted returns of up to 10% a year in both large and small stocks; the gap closes over about 34 months.

Version record

Kai Cao and Haifeng You, "Fundamental Analysis via Machine Learning"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 80(2), 74–98 (2024); DOI 10.1080/0015198x.2024.2313692.

Uses machine learning to forecast corporate earnings. The forecasts are more accurate than leading statistical models and the analyst consensus, apparently because they find new predictors and nonlinear relationships. The new information they contain predicts stock returns: the most and least favourable quintiles differ by about 34–77 basis points a month on a risk-adjusted basis.

Version record

10.2.2 ETFs, indexing and passive ownership

Ananth N. Madhavan, Exchange-Traded Funds and the New Dynamics of Investing

Difficulty: Advanced **Type:** Book / monograph

Status/version: Oxford University Press, published 1 August 2016; print ISBN 9780190279394.

A coherent entry to ETF economics, index investing, factor exposures and the links between funds and their underlying markets. Use it before studying ETF premiums, creation and redemption, execution quality, and the choice between a fund and direct securities. ETF shares, underlying holdings and published NAVs can have different liquidity and valuation times, so a displayed premium is not automatically an arbitrage opportunity.

Extended note. Version record

Itzhak Ben-David, Francesco Franzoni and Rabih Moussawi, Do ETFs Increase Volatility?

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance 73(6), 2471–2535, December 2018 issue; first published online 22 September 2018.

Studies how ETF ownership can transmit trading shocks to constituent equities through the arbitrage mechanism. Read alongside ETF mechanics to distinguish price discovery, temporary price pressure and changes in fundamental risk. The empirical design and equity-market setting matter: ownership is endogenous, measured volatility depends on horizon, and the result should not be transferred mechanically to bond ETFs or interpreted as proof that every passive flow destabilises prices.

Extended note. Version record

Anna Agapova, Margarita Kaprielyan and Nikanor Volkov, ETFs and the price volatility of underlying bonds

Difficulty: Research **Type:** Research paper

Status/version: Financial Review 60(3), 667–700, 2025; online 16 January 2025.

Examines corporate-bond ETF ownership and flows using 2010–2022 observations and finds an association between ownership and lower underlying bond volatility. Use the comparison to formulate competing liquidity-buffer and price-pressure hypotheses. The estimates are conditional on the sample and identification choices; they do not establish that all ETF structures remain stabilising in an unprecedented redemption shock.

Extended note. Version record

Itzhak Ben-David, Francesco Franzoni and Rabih Moussawi, "Exchange-Traded Funds"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 9(1), 169–189 (2017); DOI 10.1146/annurev-financial-110716-032538.

Annual Review of ETF structure, arbitrage mechanism, liquidity, and effects on underlying securities.

Version record

Antti Petajisto, "Inefficiencies in the Pricing of Exchange-Traded Funds"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 73(1), 24–54 (2017); DOI 10.2469/faj.v73.n1.7.

Measures ETF premiums and discounts relative to NAV and peer ETFs, larger for illiquid underlying assets.

Version record

David C. Brown, Shaun William Davies and Matthew C. Ringgenberg, "ETF Arbitrage, Non-Fundamental Demand, and Return Predictability"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 25(4), 937–972 (2021); DOI 10.1093/rof/rfaa027.

ETF creation and redemption flows signal non-fundamental demand and predict reversals.

Version record

Kevin Pan and Yao Zeng, "ETF Arbitrage Under Liquidity Mismatch"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 3723406 (2017); DOI 10.2139/ssrn.3723406. Working-paper record; no journal version established in this check.

Authorised participants' dual role as dealers in illiquid bond markets limits ETF arbitrage when bonds are illiquid.

Extended note. Version record

Alex Chinco and Marco Sammon, "The passive ownership share is double what you think it is"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 157, 103860 (2024); DOI 10.1016/j.jfineco.2024.103860.

Counts index-linked holdings of active funds and institutions; passive-linked ownership is roughly twice the index-fund share.

Version record

Valentin Haddad, Paul Huebner and Erik Loualiche, "How Competitive Is the Stock Market? Theory, Evidence from Portfolios, and Implications for the Rise of Passive Investing"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 115(3), 975–1018 (2025); DOI 10.1257/aer.20230505.

Institutions' demand elasticities respond strategically to others; rise of passive reduces aggregate elasticity less than naive estimates.

Version record

Richard B. Evans, Rabih Moussawi, Michael S. Pagano and John Sedunov, "Operational shorting and ETF liquidity provision"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 180, 104241 (2026); DOI 10.1016/j.jfineco.2026.104241.

Studies operational shorting by ETF liquidity providers and its role in ETF liquidity provision.

Extended note. Version record

Rabih Moussawi, Ke Shen and Raisa Velthuis, "The Role of Taxes in the Rise of ETFs"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(10), 2988–3039 (2025); DOI 10.1093/rfs/hhaf044.

ETF tax efficiency via in-kind redemptions and heartbeat trades is central to ETF growth.

Version record

Hao Jiang, Dimitri Vayanos and Lu Zheng, "Passive Investing and the Rise of Mega-Firms"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(12), 3461–3496 (2025); DOI 10.1093/rfs/hhaf085.

Passive flows disproportionately raise prices of the largest firms, especially those in noise-trader demand; aggregate prices can rise even from active-to-passive switching.

Version record

Marco Sammon and John J. Shim, "Who Clears the Market When Passive Investors Trade?"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(10), 3330–3377 (2026); DOI 10.1093/rfs/hhag032.

Firms, not other institutions, are the main sellers accommodating index-fund buying, responding one-for-one to exogenous passive demand.

Version record

10.3 Short selling and informed-trader signals

Short sellers, option traders, insiders and retail investors leave records that can be used as stock signals: short interest and loan fees, option volume and implied-volatility spreads, insider filings and retail order flow. These readings separate information from short-sale constraints, borrow costs and attention, and record the rules that govern short-sale data. Section 9.3 covers securities-lending mechanics and fees.

10.3.1 Short interest, lending fees and short-selling rules

Paul Asquith, Parag A. Pathak and Jay R. Ritter, "Short Interest, Institutional Ownership, and Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 78(2), 243–276 (2005); DOI 10.1016/j.jfineco.2005.01.001.

Treats short interest as a proxy for shorting demand and institutional ownership as a proxy for lendable supply. Stocks with high short interest and low institutional ownership underperform by 215 basis points a month equal-weighted in 1988–2002, but by an insignificant 39 basis points value-weighted. For most stocks the two levels make binding short-sale constraints unlikely.

Version record

Lauren Cohen, Karl B. Diether and Christopher J. Malloy, "Supply and Demand Shifts in the Shorting Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 62(5), 2061–2096 (2007); DOI 10.1111/j.1540-6261.2007.01269.x.

Uses proprietary loan fees and quantities from a large institutional lender to separate shifts in shorting demand from shifts in lending supply. An increase in shorting demand predicts an abnormal return of -2.98% the following month, and more so where public information is scarce, so the lending market reveals private information.

Version record

Karl B. Diether, Kuan-Hui Lee and Ingrid M. Werner, "Short-Sale Strategies and Return Predictability"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(2), 575–607 (2009); DOI 10.1093/rfs/hhn047.

Uses the short-sale data the SEC required for 2005: short sales are 24% of NYSE and 31% of Nasdaq share volume. Short sellers trade more after positive returns and correctly predict negative abnormal returns, consistent with trading against short-term overreaction. A strategy based on daily short selling is profitable in the sample.

Version record

Joseph E. Engelberg, Adam V. Reed and Matthew C. Ringgenberg, "How Are Shorts Informed? Short Sellers, News, and Information Processing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 105(2), 260–278 (2012); DOI 10.1016/j.jfineco.2012.03.001.

Matches short sales with news releases. The negative relation between short sales and future returns is twice as large on news days and four times as large on days with negative news. The most informed short sales come from clients rather than market makers, and there is little evidence that short sellers anticipate news: their edge is mainly skill in processing public information.

Version record

Joseph E. Engelberg, Adam V. Reed and Matthew C. Ringgenberg, "Short-Selling Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 73(2), 755–786 (2018); DOI 10.1111/jofi.12601.

Short sellers face the risk that loan fees rise or that loans are recalled. Stocks with more of this short-selling risk have lower returns, less price efficiency and less short selling, so the risk itself limits arbitrage.

Version record

Itamar Drechsler and Qingyi Freda Drechsler, "The Shorting Premium and Asset Pricing Anomalies"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 20282, July 2014; DOI 10.3386/w20282. Working-paper record; no journal version established in this check.

A portfolio long cheap-to-short and short expensive-to-short stocks earns 1.43% a month gross and 0.91% net of fees. Eight well-known anomalies largely disappear among the 80% of stocks with low fees and are amplified among high-fee stocks. The authors attribute the premium to concentrated short risk borne by the few investors who do most of the shorting.

Extended note. Version record

Robert F. Stambaugh, Jianfeng Yu and Yu Yuan, "The Short of It: Investor Sentiment and Anomalies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 104(2), 288–302 (2012); DOI 10.1016/j.jfineco.2011.12.001.

Combines market-wide sentiment with short-sale impediments, which make overpricing more common than underpricing. Across a broad set of anomalies, long–short profits are higher after high sentiment, the extra profit comes from the short leg, and sentiment is unrelated to long-leg returns.

Version record

Amy K. Edwards, Adam V. Reed and Pedro A. C. Saffi, "A Survey of Short-Selling Regulations"

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Asset Pricing Studies, 14(4), 613–639 (2024); DOI 10.1093/rapstu/raae011.

Organises short-selling regulation into trading restrictions, securities-lending rules and disclosure requirements, and reviews the academic evidence and data on 45 regulations from 1896 to 2021. Draws conclusions on which approaches work and on their wider effects on markets. Read it before using short-selling data across jurisdictions or regimes.

Version record

Lorien Stice-Lawrence, Yu Ting Forester Wong and Wuyang Zhao, "Short Squeezes After Short-Selling Attacks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Accounting Research, 63(3), 1187–1236 (2025); DOI 10.1111/1475-679x.12595.

Positive returns after activist short attacks tend to reverse fully and come with heavy short covering, higher volatility, squeeze-related news and lending-market disruption. About 15% of attacks are squeezed. Squeeze risk rises with the short seller's visibility, falls with the credibility of its evidence, and is often triggered by insider purchases, share recalls, retail trading or firm disclosures.

Version record

Antonio Gargano, Juan Sotes-Paladino and Patrick Verwijmeren, "Short of Capital: Stock Market Implications of Short Sellers' Losses"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 71(10), 8347–8370 (2025); DOI 10.1287/mnsc.2023.01356.

Using US equity-lending data, short sellers' mark-to-market losses, but not their gains, reduce later shorting of a stock, more so for large losses and for stocks with high margin requirements. Loss-driven shorting does not change price efficiency or return predictability, suggesting the constraint binds mainly on uninformed short sellers.

Version record

U.S. Securities and Exchange Commission, Order Granting Temporary Exemptive Relief from Compliance with Rule 13f-2 and Form SHO, and from Certain Aspects of Rule 10c-1a (Release 34-104303)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Exemptive order dated 3 December 2025; Form SHO reporting deferred to 2 January 2028.

Rule 13f-2, adopted in October 2023, requires institutional managers above set thresholds to report monthly short positions and activity on Form SHO, which the SEC will publish in aggregate by security; Rule 10c-1a requires securities loans to be reported to FINRA. After the Fifth Circuit remanded both rules without vacatur on 25 August 2025, this order defers Form SHO reporting until 2 January 2028, so the first reports cover January 2028, and defers Rule 10c-1a loan reporting to 28 September 2028 and public dissemination to 29 March 2029. Check the current status before planning research on these data.

Extended note. Version record

10.3.2 Option, insider and retail-flow signals

Jun Pan and Allen M. Poteshman, "The Information in Option Volume for Future Stock Prices"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 19(3), 871–908 (2006); DOI 10.1093/rfs/hhj024.

Builds put–call ratios from option volume initiated by buyers opening new positions. Stocks with low ratios beat stocks with high ratios by more than 40 basis points the next day and more than 1% the next week. The predictability comes from option traders' non-public information rather than from inefficiency, and is greater where informed traders concentrate and where options offer more leverage.

Version record

Martijn Cremers and David Weinbaum, "Deviations from Put-Call Parity and Stock Return Predictability"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 45(2), 335–367 (2010); DOI 10.1017/s002210901000013x.

Measures deviations from put–call parity as the implied-volatility gap between matched calls and puts. Stocks with relatively expensive calls beat stocks with relatively expensive puts by 50 basis points a week, and lending rebate rates show that the result is not driven by hard-to-borrow stocks. Predictability is greater when option liquidity is high and stock liquidity low, and it declines over the sample.

Version record

Yuhang Xing, Xiaoyan Zhang and Rui Zhao, "What Does the Individual Option Volatility Smirk Tell Us About Future Equity Returns?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 45(3), 641–662 (2010); DOI 10.1017/s0022109010000220.

Stocks with the steepest option volatility smirks underperform those with the flattest by 10.9% a year on a risk-adjusted basis, for at least six months, and have the worst earnings shocks the following quarter. Consistent with informed traders buying out-of-the-money puts on bad news.

Version record

Travis L. Johnson and Eric C. So, "The Option to Stock Volume Ratio and Future Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 106(2), 262–286 (2012); DOI 10.1016/j.jfineco.2012.05.008.

In a model with equity short-sale costs, high option volume relative to stock volume signals bad news even when trade direction is unobserved. Firms in the lowest decile of the option-to-stock volume ratio beat the highest decile by 1.47% a month risk-adjusted. The signal is stronger when short-sale costs are high or option leverage is low, and it predicts firm-specific earnings news.

Version record

Byeong-Je An, Andrew Ang, Turan G. Bali and Nusret Cakici, "The Joint Cross Section of Stocks and Options"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(5), 2279–2337 (2014); DOI 10.1111/jofi.12181.

Stocks whose call (put) implied volatilities rose most over the past month have higher (lower) future returns; the decile spread is about 1% a month and persists for up to six months. Past stock returns also predict changes in option implied volatility. Consistent with rational models of informed trading.

Version record

Andreas Neuhierl, Xiaoxiao Tang, Rasmus T. Varneskov and Guofu Zhou, "Do Option Characteristics Predict the Underlying Stock Returns in the Cross-Section?"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 72(4), 2742–2760 (2026); DOI 10.1287/mnsc.2024.04720.

Tests a large set of option characteristics against a large set of firm characteristics with portfolio sorts and high-dimensional methods. Only a few option characteristics add predictive power, and the strongest are associated with mispricing, future tail returns and short-selling costs.

Version record

Josef Lakonishok and Inmoo Lee, "Are Insider Trades Informative?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 14(1), 79–111 (2001); DOI 10.1093/rfs/14.1.79.

Covers insider trading in all NYSE, AMEX and Nasdaq firms in 1975–1995. Prices barely move when insiders trade or report their trades. Insiders are contrarian in aggregate but predict market moves better than simple contrarian rules, and they predict cross-sectional returns mainly in smaller firms. Purchases carry the information; sales do not.

Version record

Lauren Cohen, Christopher Malloy and Lukasz Pomorski, "Decoding Inside Information"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 67(3), 1009–1043 (2012); DOI 10.1111/j.1540-6261.2012.01740.x.

Separates predictable routine insider trades, which carry no information, from opportunistic ones. Trading only on opportunistic insiders earns value-weighted abnormal returns of 82 basis points a month. The most informed are local, non-executive insiders in geographically concentrated, poorly governed firms; they are more likely to face SEC enforcement and trade less after enforcement waves.

Version record

Hans K. Hvide and Kasper Meisner Nielsen, "Flying below the Radar: Insider Trading by Executives below the Top"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 181, 104282 (2026); DOI 10.1016/j.jfineco.2026.104282.

Uses Norwegian administrative register data on executives below the top, whose trades are not publicly reported. Their purchases of own-company stock earn abnormal returns of about 50–100 basis points over one month, while their purchases of other stocks earn negative returns, so the gains reflect information rather than investing ability.

Version record

Brad M. Barber and Terrance Odean, "All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 21(2), 785–818 (2008); DOI 10.1093/rfs/hhm079.

Individual investors are net buyers of attention-grabbing stocks: stocks in the news, with abnormal trading volume or with extreme one-day returns. Attention limits the choice set when buying among thousands of stocks, but not when selling, because investors mostly sell stocks they already own.

Version record

Zhi Da, Joseph Engelberg and Pengjie Gao, "In Search of Attention"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 66(5), 1461–1499 (2011); DOI 10.1111/j.1540-6261.2011.01679.x.

Measures investor attention with Google's Search Volume Index for Russell 3000 stocks in 2004–2008. The measure is timelier than other attention proxies and probably captures retail attention. A rise predicts higher prices over the next two weeks and a reversal within the year, and it helps explain the first-day returns and long-run underperformance of IPOs.

Version record

Eric K. Kelley and Paul C. Tetlock, "How Wise Are Crowds? Insights from Retail Orders and Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 68(3), 1229–1265 (2013); DOI 10.1111/jofi.12028.

Using retail order data that address selection bias, both aggressive (market) and passive (limit) net buying predict monthly returns without reversal. Only aggressive orders predict firm news, including earnings surprises. Passive net buying follows negative returns, consistent with liquidity provision. Retail trading contributes to market efficiency.

Version record

Brad M. Barber, Xing Huang, Terrance Odean and Christopher Schwarz, "Attention-Induced Trading and Returns: Evidence from Robinhood Users"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 77(6), 3141–3190 (2022); DOI 10.1111/jofi.13183.

Robinhood users engage in more attention-induced trading than other retail investors, partly because of the app's design; outages disproportionately reduce trading in high-attention stocks. Intense buying by Robinhood users forecasts negative returns: the top stocks bought each day average -4.7% abnormal returns over the next 20 days.

Version record

10.4 Corporate events and merger arbitrage

Event studies measure how prices respond to earnings announcements, analyst recommendations, equity issuance and takeover bids, and how much of the response is delayed. Long-horizon results depend heavily on method, so the methods readings come first. Merger arbitrage and other equity relative-value trades depend on deal completion, short-sale constraints and the supply of arbitrage capital.

10.4.1 Event-study methods and earnings announcements

S. P. Kothari and Jerold B. Warner, "Econometrics of Event Studies"

Difficulty: Advanced **Type:** Research paper

Status/version: Handbook of Empirical Corporate Finance, Volume 1 (Elsevier, 2007), 3–36; DOI 10.1016/B978-0-444-53265-7.50015-9.

Surveys event-study methods. Short-horizon methods are reliable; long-horizon methods have improved but remain seriously limited. Shows that the properties of the tests vary by calendar period and with sample characteristics such as volatility, so check them for non-random samples before trusting an abnormal return.

Version record

John D. Lyon, Brad M. Barber and Chih-Ling Tsai, "Improved Methods for Tests of Long-Run Abnormal Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 54(1), 165–201 (1999); DOI 10.1111/0022-1082.00101.

Two approaches give well-specified tests in random samples: buy-and-hold abnormal returns against carefully built reference portfolios, with skewness-adjusted or bootstrapped inference, and mean monthly returns on calendar-time portfolios. Both are misspecified in many non-random samples, so long-run abnormal-return analysis is treacherous.

Version record

Eugene F. Fama, "Market Efficiency, Long-Term Returns, and Behavioral Finance"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 49(3), 283–306 (1998); DOI 10.1016/s0304-405x(98)00026-9.

Argues that long-term return anomalies do not overturn market efficiency. Apparent overreaction is about as common as underreaction, post-event continuation about as common as reversal, and many anomalies are sensitive to how they are measured and disappear with reasonable changes in method.

Version record

Pavel Savor and Mungo Wilson, "Earnings Announcements and Systematic Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 71(1), 83–138 (2016); DOI 10.1111/jofi.12361.

Firms scheduled to report earnings earn an annualised abnormal return of 9.9%. The authors explain it as risk: investors learn about non-announcing firms from announcers, so announcers' covariance with market cash-flow news spikes. Announcer returns forecast aggregate earnings, early announcers earn more than late ones, and exposure to announcement risk is priced.

Version record

Owen Lamont and Andrea Frazzini, "The Earnings Announcement Premium and Trading Volume"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 13090, May 2007; DOI 10.3386/w13090. Working-paper record; no journal version established in this check.

Stock prices rise on average around scheduled earnings announcements. The premium is large, robust and highest for stocks with high past announcement-period volume, and it coincides with imputed small-investor buying, consistent with investors buying when an announcement catches their attention.

Extended note. Version record

Stefano DellaVigna and Joshua M. Pollet, "Investor Inattention and Friday Earnings Announcements"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 64(2), 709–749 (2009); DOI 10.1111/j.1540-6261.2009.01447.x.

Earnings announced on Fridays get a 15% smaller immediate response, a 70% larger delayed response and 8% lower trading volume than those on other weekdays, and trading the extra Friday drift earns abnormal returns. Supports limited-attention explanations of post-earnings-announcement drift.

Version record

David Hirshleifer, Sonya Seongyeon Lim and Siew Hong Teoh, "Driven to Distraction: Extraneous Events and Underreaction to Earnings News"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 64(5), 2289–2325 (2009); DOI 10.1111/j.1540-6261.2009.01501.x.

When many other firms announce earnings on the same day, the immediate price and volume reaction to a firm's earnings surprise is weaker and post-announcement drift is stronger. A strategy based on this distraction effect earns substantial alphas; news from unrelated industries and large surprises distract most.

Version record

Charles Martineau, "Rest in Peace Post-Earnings Announcement Drift"

Difficulty: Research **Type:** Research paper

Status/version: Critical Finance Review, 11(3–4), 613–646 (2022); DOI 10.1561/104.00000122.

Finds that prices now fully reflect earnings surprises on the announcement date: the drift has been absent for large stocks since 2006 and has disappeared only recently for microcaps. Recommends how researchers should test for drift and suggests where to study price formation after earnings news instead.

Version record

10.4.2 Analysts, issuance and corporate actions

Kent L. Womack, "Do Brokerage Analysts' Recommendations Have Investment Value?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 51(1), 137–167 (1996); DOI 10.1111/j.1540-6261.1996.tb05205.x.

New buy and sell recommendations from major US brokers move prices when issued, although few coincide with public news. The reaction is incomplete: drift after buy recommendations is modest (+2.4%) and short-lived, but after sell recommendations it is larger (-9.1%) and lasts six months.

Version record

Narasimhan Jegadeesh, Joonghyuk Kim, Susan D. Krische and Charles M. C. Lee, "Analyzing the Analysts: When Do Recommendations Add Value?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 59(3), 1083–1124 (2004); DOI 10.1111/j.1540-6261.2004.00657.x.

Sell-side analysts favour glamour stocks: positive momentum, high growth, high volume and relatively expensive. The level of the consensus recommendation adds value only among value and positive-momentum stocks and hurts elsewhere, but the quarterly change in consensus is a robust predictor that is largely independent of other signals.

Version record

Tim Loughran and Jay R. Ritter, "The New Issues Puzzle"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 50(1), 23–51 (1995); DOI 10.1111/j.1540-6261.1995.tb05166.x.

Firms that issued stock in 1970–1990, through an IPO or a seasoned offering, were poor long-run investments: average returns over the next five years were only 5% a year for IPOs and 7% for seasoned issuers. Book-to-market accounts for only a modest part of the shortfall.

Version record

Jeffrey Pontiff and Artemiza Woodgate, "Share Issuance and Cross-sectional Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 63(2), 921–945 (2008); DOI 10.1111/j.1540-6261.2008.01335.x.

Since 1970, net share issuance predicts the cross-section of stock returns more significantly than size, book-to-market or momentum individually, even after excluding the repurchase, seasoned-offering and stock-merger samples behind earlier studies. Before 1970 it has no significant predictive power for most holding periods.

Version record

Kent Daniel and Sheridan Titman, "Market Reactions to Tangible and Intangible Information"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 61(4), 1605–1643 (2006); DOI 10.1111/j.1540-6261.2006.00884.x.

Future returns are unrelated to past accounting performance but strongly negatively related to the intangible return, the part of past returns orthogonal to that performance. Book-to-market predicts returns because it proxies for the intangible return, and a composite equity-issuance measure also predicts returns independently.

Version record

10.4.3 Merger arbitrage and equity relative value

Malcolm Baker and Serkan Savaşoglu, "Limited Arbitrage in Mergers and Acquisitions"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 64(1), 91–115 (2002); DOI 10.1016/s0304-405x(02)00072-7.

A diversified portfolio of risk-arbitrage positions earned abnormal returns of 0.6–0.9% a month in 1981–1996. In the authors' model, completion risk and position size limit arbitrageurs' risk-bearing capacity. Returns rise with an ex ante measure of completion risk and with target size, and the paper also relates them to the supply of arbitrage capital.

Version record

Mark Mitchell, Todd Pulvino and Erik Stafford, "Price Pressure around Mergers"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 59(1), 31–63 (2004); DOI 10.1111/j.1540-6261.2004.00626.x.

Examines professional trading around 2,130 mergers announced in 1994–2000. Uninformed shifts in demand cause short-lived price pressure. Nearly half of the negative announcement return of acquirers in stock-financed mergers reflects short selling by merger arbitrageurs, so earlier estimates of acquirers' wealth losses are biased downward.

Version record

Gaurav Jetley and Xinyu Ji, "The Shrinking Merger Arbitrage Spread: Reasons and Implications"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 66(2), 54–68 (2010); DOI 10.2469/faj.v66.n2.3.

The merger-arbitrage spread fell by more than 400 basis points after 2002, in line with lower returns and higher inflows at merger-arbitrage hedge funds. More trading in target stocks after announcement, lower transaction costs and changed deal risk explain part of the decline, so some of it is probably permanent.

Version record

Stefano Giglio and Kelly Shue, "No News Is News: Do Markets Underreact to Nothing?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(12), 3389–3440 (2014); DOI 10.1093/rfs/hhu052.

In the year after a merger announcement, the passage of time without news changes the probability that the deal completes. Variation in the hazard rate of completion strongly predicts returns, consistent with underreaction to the passage of time rather than with risk or frictions.

Version record

Wei Jiang, Tao Li and Danqing Mei, "Influencing Control: Jawboning in Risk Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 73(6), 2635–2675 (2018); DOI 10.1111/jofi.12721.

Activist arbitrageurs campaign publicly to improve the terms of announced deals, targeting low premiums and deals with managerial conflicts such as going-private transactions and large CEO payouts. They make it less likely that the target is sold to the announced bidder, raise premiums, and earn more than passive arbitrageurs.

Version record

Charles Cao, Bradley A. Goldie, Bing Liang and Lubomir Petrasek, "What Is the Nature of Hedge Fund Manager Skills? Evidence from the Risk-Arbitrage Strategy"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 51(3), 929–957 (2016); DOI 10.1017/s0022109016000387.

Using portfolio holdings, hedge funds beat a naive risk-arbitrage portfolio by 3.7% a year on a risk-adjusted basis, while other institutional arbitrageurs do not. The advantage comes from managing downside risk, not from predicting or influencing deal outcomes.

Version record

Mark Mitchell, Todd Pulvino and Erik Stafford, "Limited Arbitrage in Equity Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 57(2), 551–584 (2002); DOI 10.1111/1540-6261.00434.

Studies 82 cases in which a company's market value is less than the value of its stake in a subsidiary. In 30% of them the link between parent and subsidiary is severed before the gap closes, and a specialised arbitrageur would earn 50% more if convergence were smooth. Uncertainty about the return distribution and its risks limits arbitrage.

Version record

Owen A. Lamont and Richard H. Thaler, "Can the Market Add and Subtract? Mispricing in Tech Stock Carve-outs"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 111(2), 227–268 (2003); DOI 10.1086/367683.

In 1998–2000 technology carve-outs, such as 3Com and Palm, the parent traded below the value of the subsidiary shares it was due to distribute. Short-sale constraints kept the overpriced subsidiary expensive or impossible to short; option prices imply shorting costs high enough to remove exploitable arbitrage.

Version record

Kenneth A. Froot and Emil M. Dabora, "How Are Stock Prices Affected by the Location of Trade?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 53(2), 189–216 (1999); DOI 10.1016/s0304-405x(99)00020-3.

Studies large Siamese-twin companies that pool their cash flows but whose shares trade in different markets. A twin's relative price rises with the market where it trades most. Of the explanations tested, only tax-induced investor heterogeneity explains some of the facts; country-specific sentiment and market segmentation remain candidates.

Version record

Minmo Gahng, Jay R. Ritter and Donghang Zhang, "SPACs"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 36(9), 3463–3501 (2023); DOI 10.1093/rfs/hhad019.

Going public by merging with a special purpose acquisition company costs much more than a traditional IPO. SPAC IPO investors and sponsors earned high annualised returns, though the authors expect recent deals to disappoint, while public investors in the merged companies earned very low market-adjusted returns on an equal-weighted basis, cushioned by high redemptions in the worst deals.

Version record

10.5 Signal combination and long–short portfolio construction

These readings turn equity signals into a portfolio: scaling scores into return forecasts, combining many signals, aligning the alpha with the risk model and constraints, choosing sector neutrality and the size of the short side, and netting out trading costs. Section 6.1 covers general portfolio theory, covariance estimation and factor risk models.

Richard C. Grinold, "Alpha Is Volatility Times IC Times Score"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 20(4), 9–16 (1994); DOI 10.3905/jpm.1994.409482.

Practitioner rule for turning a standardised signal score into a return forecast by scaling it with residual volatility and the information coefficient (IC). No abstract could be retrieved, so this annotation rests on the title, author and venue; read the paper before relying on details.

Extended note. Version record

Zura Kakushadze, "Combining Alphas via Bounded Regression"

Difficulty: Research **Type:** Research paper

Status/version: Risks, 3(4), 474–490 (2015); DOI 10.3390/risks3040474.

Gives an algorithm and source code for weighting many alpha streams when the history is too short for a sample covariance matrix. Regression on principal components can produce poorly diversified or turnover-skewed weights; bounding the weights inside the regression corrects this. The method also applies to stock portfolio construction.

Version record

Sebastián Ceria, Anureet Saxena and Robert A. Stubbs, "Factor Alignment Problems and Quantitative Portfolio Management"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 38(2), 29–43 (2012); DOI 10.3905/jpm.2012.38.2.029.

When the expected-return model, the risk model and the constraints are built from different factors, the optimiser magnifies the mismatch. The authors survey common sources of these factor alignment problems, document their effect on the ex post performance of optimised portfolios, and propose augmented risk models as a remedy.

Version record

Shaun Fitzgibbons, Jacques Friedman, Lukasz Pomorski and Laura Serban, "Long-Only Style Investing: Don't Just Mix, Integrate"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Investing, 26(4), 153–164 (2017); DOI 10.3905/joi.2017.26.4.153.

Compares a mix of stand-alone style indexes with a portfolio that integrates the styles at stock level. Integration substantially improves returns and the information ratio by avoiding stocks with offsetting style exposures and including stocks with balanced positive exposures.

Version record

Sina Ehsani, Campbell R. Harvey and Feifei Li, "Is Sector Neutrality in Factor Investing a Mistake?"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 79(3), 95–117 (2023); DOI 10.1080/0015198x.2023.2196931.

Splits a characteristic's predictive power into a component across industries and one within industries. Shows analytically and empirically that the average long–short investor gains from hedging out sector bets, whereas a long-only investor should, on average, not sector-neutralise.

Version record

Kent Daniel, Lira Mota, Simon Rottke and Tano Santos, "The Cross-Section of Risk and Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(5), 1927–1979 (2020); DOI 10.1093/rfs/hhaa021.

Portfolios sorted on characteristics carry unpriced as well as priced risk. Removing the unpriced part with covariances estimated from past returns raises the squared Sharpe ratio of the optimal combination of the five Fama–French characteristic portfolios from 1.17 to 2.13.

Version record

Victor DeMiguel, Alberto Martín-Utrera, Francisco J. Nogales and Raman Uppal, "A Transaction-Cost Perspective on the Multitude of Firm Characteristics"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(5), 2180–2222 (2020); DOI 10.1093/rfs/hhz085.

Transaction costs raise the number of characteristics that are jointly significant for an optimal portfolio from 6 to 15, because the trades needed to rebalance different characteristics often cancel out. Costs therefore give an economic reason to combine more characteristics than standard factor models use.

Version record

Ronen Israel and Tobias J. Moskowitz, "The Role of Shorting, Firm Size, and Time on Market Anomalies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 108(2), 275–301 (2013); DOI 10.1016/j.jfineco.2012.11.005.

Over 86 years of US data, long positions supply almost all of size profits, 60% of value and half of momentum. Shorting matters less for momentum and more for value as firm size falls; value is weak among the largest stocks, while momentum shows no reliable relation with size. Changes in trading costs and institutional ownership have little effect on these returns.

Version record

Andrea Frazzini, Ronen Israel and Tobias J. Moskowitz, "Trading Costs"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 3229719, 2018; DOI 10.2139/ssrn.3229719. Working-paper record; no journal version established in this check.

Uses $1.7 trillion of live trade executions by a large institutional manager across 21 developed equity markets over 19 years. Measured costs are an order of magnitude smaller than earlier studies suggest, and a price-impact model calibrated to these trades describes independent broker costs and index-fund trading costs out of sample better than models from the literature.

Extended note. Version record

Lasse Heje Pedersen, Abhilash Babu and Ari Levine, "Enhanced Portfolio Optimization"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 77(2), 124–151 (2021); DOI 10.1080/0015198x.2020.1854543.

Identifies the portfolios that make standard mean–variance optimisation fail in practice and proposes enhanced portfolio optimisation (EPO), a simple way to correct them. Applied to industry momentum and time-series momentum across equities and global asset classes, EPO earns significant alpha beyond the market, the 1/N portfolio and standard factors.

Version record

Chapter 11

Rates, inflation and cash instruments

Cash-flow timing, curves, inflation indexation and contract conventions belong together. Monetary policy and debt issuance connect the curve to the central bank and the Treasury. Sovereign and municipal credit also require the credit and regional-market chapters.

11.1 Fixed-income foundations and term premia

11.1.1 Texts, curve construction and relative value

Bruce Tuckman and Angel Serrat, *Fixed Income Securities: Tools for Today's Markets*, fourth edition

Difficulty: Intro **Type:** Book / monograph

Status/version: Fourth edition, Wiley, first published online 7 September 2022; print ISBN 9781119835554.

The most practical institutional introduction to government bonds, swaps, futures, repo, curve risk and relative value. The fourth edition incorporates post-LIBOR benchmarks and the modern electronic and collateralised rates environment. [4]

Applications: Rates, swaps, futures; relative value, hedging.

Version record

Frank J. Fabozzi, ed., with the assistance of Francesco A. Fabozzi and Steven V. Mann, *The Handbook of Fixed Income Securities*, 9th ed. (McGraw Hill, June 2021)

Difficulty: Advanced **Type:** Book / monograph

Status/version: 9th ed. (McGraw Hill, June 2021)

Selected ninth edition. Covers MBS/CMOs, mortgage valuation, embedded options and term-structure models, alongside fixed-income risk and portfolio methods. Earlier editions have different contributors and market conventions.

Version record

Patrick S. Hagan and Graeme West, "Interpolation Methods for Curve Construction"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Mathematical Finance, 13(2), 89–129 (2006); DOI 10.1080/13504860500396032.

Compares interpolation schemes for yield curves on positivity of forwards, locality and hedge stability; recommends monotone-convex.

Version record

Ferdinando M. Ametrano and Marco Bianchetti, "Everything You Always Wanted to Know About Multiple Interest Rate Curve Bootstrapping but Were Afraid to Ask"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper 2219548 (2013); DOI 10.2139/ssrn.2219548. Working-paper record; no journal version established in this check.

Practical multi-curve bootstrapping with OIS discounting, tenor basis, turn-of-year effects and interpolation choices; QuantLib-based.

Extended note.

Version record

Marc Henrard, *Interest Rate Modelling in the Multi-Curve Framework: Foundations, Evolution and Implementation*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Palgrave Macmillan UK (2014); DOI 10.1057/9781137374660.

Multi-curve pricing, convexity, collateral and implementation from an OpenGamma perspective.

Version record

Charles R. Nelson and Andrew F. Siegel, "Parsimonious Modeling of Yield Curves"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Business, 60(4), 473 (1987); DOI 10.1086/296409.

Three-component exponential yield-curve form used by central banks and as the basis of dynamic Nelson–Siegel models.

Version record

Francis X. Diebold and Canlin Li, "Forecasting the term structure of government bond yields"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 130(2), 337–364 (2006); DOI 10.1016/j.jeconom.2005.03.005.

Dynamic Nelson–Siegel with level, slope and curvature factors forecast with AR models.

Version record

Refet S. Gürkaynak, Brian Sack and Jonathan H. Wright, "The U.S. Treasury yield curve: 1961 to the present"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Monetary Economics, 54(8), 2291–2304 (2007); DOI 10.1016/j.jmoneco.2007.06.029.

Documents the Federal Reserve daily Svensson zero-coupon curve and data; the standard public US curve dataset.

Version record

Yan Liu and Jing Cynthia Wu, "Reconstructing the yield curve"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 142(3), 1395–1425 (2021); DOI 10.1016/j.jfineco.2021.05.059.

Kernel-smoothing zero-coupon curve with smaller pricing errors than GSW; data published by the authors.

Version record

Robert Litterman and José Scheinkman, "Common Factors Affecting Bond Returns"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: The Journal of Fixed Income, 1(1), 54–61 (1991); DOI 10.3905/jfi.1991.692347.

Level, steepness and curvature explain almost all Treasury return variation; basis of PCA hedging.

Version record

Galen D. Burghardt, Terrence M. Belton, Morton Lane and John Papa, *The Treasury Bond Basis: An in-Depth Analysis for Hedgers, Speculators, and Arbitrageurs*

Difficulty: Advanced **Type:** Book / monograph

Status/version: McGraw-Hill, third edition, 2005 (ISBN 9780071456104).

Cheapest-to-deliver, delivery options, implied repo and basis trading in Treasury futures; third edition 2005.

Version record

Doug Huggins and Christian Schaller, *Fixed Income Relative Value Analysis: A Practitioner's Guide to the Theory, Tools, and Trades*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, second edition, 2024 (ISBN 9781394189083); first edition 2013.

Practitioner treatment of curve, swap-spread, futures-basis, cross-currency and volatility relative-value trades; the second edition adds the SOFR transition, sovereign risk, capital constraints and multivariate mean-reversion tools, with companion spreadsheets.

Version record

Riccardo Rebonato, *Bond Pricing and Yield-Curve Modelling: A Structural Approach*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2018); DOI 10.1017/9781316694169.

Expectations, risk premia and convexity in affine term-structure models with an investor's focus on what drives yields.

Version record

Alexander Kemarsky, Wouter van der Helm and Vladimir Piterbarg, "Fast calculation of cheapest-to-deliver curves"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Computational Finance (2026); DOI 10.21314/jcf.2025.015.

Fast analytic approximation of cheapest-to-deliver discounting curves for multi-currency CSAs, improving speed and accuracy over earlier approximations.

Version record

11.1.2 Term-structure models and premia

Damiano Brigo and Fabio Mercurio, *Interest Rate Models: Theory and Practice*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer second edition; hardcover published 4 August 2006, eBook 26 September 2007.

A core reference for short-rate models, Heath–Jarrow–Morton, LIBOR market models, calibration and interest-rate derivatives. Some conventions pre-date the full multi-curve transition, so the mathematical models should be supplemented with contemporary collateral and fallback documentation. [5]

Applications: Rates, OTC derivatives, structured products.

Version record

Leif Andersen and Vladimir Piterbarg, *Interest Rate Modeling*

Difficulty: Research **Type:** Book / monograph

Status/version: 2010

A three-volume practitioner-research treatment of curve construction, model dynamics, calibration, numerical implementation and complex rates products. It is one of the closest public substitutes for an institutional rates-derivatives modelling manual.

Applications: Rates, inflation, hybrids, exotics.

Version record

Oldrich Vasicek, "An Equilibrium Characterization of the Term Structure"

Difficulty: Advanced **Type:** Research paper

Status/version: 1977

Foundational affine short-rate model linking interest-rate dynamics to bond prices and yields. Work through the expectation/PDE representation as a first-principles check on term-structure behaviour before more flexible rates models.

Applications: Rates; term structure. Version record

David Heath, Robert Jarrow and Andrew Morton, "Bond Pricing and the Term Structure of Interest Rates: A New Methodology for Contingent Claims Valuation"

Difficulty: Advanced **Type:** Research paper

Status/version: Econometrica 60(1), 77–105, January 1992; DOI 10.2307/2951677.

Models forward-rate volatility and derives the no-arbitrage drift restrictions. Study the original framework alongside Brigo–Mercurio to understand the conditions behind arbitrage-free term-structure dynamics and contingent-claim valuation.

Applications: Rates; derivatives pricing. Version record

Pierre-Olivier Gourinchas, Walker Ray, and Dimitri Vayanos, A Preferred-Habitat Model of Term Premia, Exchange Rates, and Monetary Policy Spillovers

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review 115(11), 3788–3824, November 2025; DOI 10.1257/aer.20220379.

Links currency and bond risk premia in a two-country preferred-habitat model with segmented investor clienteles and capital-constrained arbitrageurs. An institution can use it to organize duration, FX hedge, and cross-border demand scenarios, but the equilibrium calibration does not supply a directly tradeable timing rule and requires market-specific flows and funding assumptions.

Extended note. Version record

Tobias Adrian, Richard K. Crump, and Emanuel Moench, Pricing the Term Structure with Linear Regressions; New York Fed ACM term-premium data notes

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics 110(1), 110–138, October 2013; New York Fed Staff Report 340 issued August 2008, revised April 2013.

Estimates a no-arbitrage yield-curve model through a computationally convenient three-step linear-regression method, separating fitted expected short-rate and term-premium components. The premium is latent and depends on factor choice, historical sample, zero-coupon curve inputs, and revisions; a daily estimate should not be mistaken for a directly observed or guaranteed future bond return.

Extended note. Version record

Michael T. Kiley, Why Have Long-term Treasury Yields Fallen Since the 1980s? Expected Short Rates and Term Premiums in (Quasi-) Real Time

Difficulty: Research **Type:** Research paper

Status/version: Federal Reserve Finance and Economics Discussion Series 2024-054, July 2024; page last updated 31 January 2025; DOI 10.17016/FEDS.2024.054. Staff working paper, not official Board view.

Constructs a quasi-real-time alternative to full-sample Treasury yield decompositions and finds that their long historical decline can be attributed more to expected short rates and less to term premia than standard decompositions suggest. The result is a model comparison, not an observed decomposition; changing sample endpoints, survey inputs, and maturity choice can shift the inferred premium and should be versioned in backtests.

Extended note. Version record

Eugene F. Fama and Robert R. Bliss, *The Information in Long-Maturity Forward Rates*, *American Economic Review* (1987); John Y. Campbell and Robert J. Shiller, *Yield Spreads and Interest Rate Movements: A Bird's Eye View* (1991)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 1987, 1991. Version distinctions remain in the annotation.

Forward-rate information, term premia and subsequent interest-rate movements.

Version record

John C. Cox, Jonathan E. Ingersoll Jr. and Stephen A. Ross, "A Theory of the Term Structure of Interest Rates"

Difficulty: Advanced **Type:** Research paper

Status/version: Econometrica, 53(2), 385 (1985); DOI 10.2307/1911242.

Square-root short-rate model with non-negative rates and closed-form bond prices.

Version record

John Hull and Alan White, "Pricing Interest-Rate-Derivative Securities"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 3(4), 573–592 (1990); DOI 10.1093/rfs/3.4.573.

Extended Vasicek model fitting the initial curve; widely used for Bermudan swaptions and xVA.

Version record

Darrell Duffie and Rui Kan, "A Yield-Factor Model of Interest Rates"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 6(4), 379–406 (1996); DOI 10.1111/j.1467-9965.1996.tb00123.x.

Multi-factor affine term-structure framework with yields as factors.

Version record

Qiang Dai and Kenneth J. Singleton, "Specification Analysis of Affine Term Structure Models"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 55(5), 1943–1978 (2000); DOI 10.1111/0022-1082.00278.

Classifies admissible affine models and tests three-factor specifications.

Version record

Gregory R. Duffee, "Term Premia and Interest Rate Forecasts in Affine Models"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 57(1), 405–443 (2002); DOI 10.1111/1540-6261.00426.

Essentially affine market prices of risk allow time-varying premia and improve forecasts.

Version record

Scott Joslin, Kenneth J. Singleton and Haoxiang Zhu, "A New Perspective on Gaussian Dynamic Term Structure Models"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 24(3), 926–970 (2011); DOI 10.1093/rfs/hhq128.

Canonical Gaussian DTSM parameterisation that separates risk-neutral and physical dynamics for fast estimation.

Version record

John H. Cochrane and Monika Piazzesi, "Bond Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 95(1), 138–160 (2005); DOI 10.1257/0002828053828581.

A single tent-shaped combination of forward rates predicts bond excess returns across maturities.

Version record

Dimitri Vayanos and Jean-Luc Vila, "A Preferred-Habitat Model of the Term Structure of Interest Rates"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 89(1), 77–112 (2021); DOI 10.3982/ecta17440.

Arbitrageurs and maturity-specific demand determine term premia; framework for QE and pension demand effects.

Version record

Arvind Krishnamurthy and Annette Vissing-Jorgensen, "The Aggregate Demand for Treasury Debt"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 120(2), 233–267 (2012); DOI 10.1086/666526.

Treasury supply affects corporate–Treasury spreads through a convenience yield for safety and liquidity.

Version record

Jordan Brooks and Tobias J. Moskowitz, "Yield Curve Premia"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 2956411 (2017); DOI 10.2139/ssrn.2956411. Working-paper record; no journal version established in this check.

Value, momentum and carry signals price government-bond returns across countries better than affine factors alone.

Extended note. Version record

Otto Randl, Giorgia Simion and Josef Zechner, "Pricing and constructing international government bond portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 173, 104152 (2025); DOI 10.1016/j.jfineco.2025.104152.

SDF for currency-hedged developed-market government bonds; the mean-variance-efficient portfolio doubles average Sharpe ratios and its price of risk peaks in crises.

Version record

Andrés Schneider, "Financial Intermediaries and the Yield Curve"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhaf102 (2026); DOI 10.1093/rfs/hhaf102.

General-equilibrium model where binding intermediary constraints shape yield-curve dynamics through savers' marginal utility.

Version record

Damir Filipovic, Markus Pelger and Ye Ye, "Shrinking the term structure"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, rfag033 (2026); DOI 10.1093/rof/rfag033.

Smoothness-based factors for the full cross-section of Treasury returns: four investable factors explain time variation and risk premia, and generalise PCA level-slope-curvature shapes.

Version record

Xin Gao, Guanglian Hu, Bingxin Li and Rui Liu, "Risk premiums in the U.S. Treasury futures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 186, 107669 (2026); DOI 10.1016/j.jbankfin.2026.107669.

Risk premiums in US Treasury futures.

Extended note. Version record

Refet S. Gürkaynak and Jonathan H. Wright, "Macroeconomics and the Term Structure"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Literature, 50(2), 331–367 (2012); DOI 10.1257/jel.50.2.331.

Surveys the term structure where macroeconomics and finance meet: why the expectations hypothesis fails, how time-varying risk premia account for the anomalies, and how policymakers read expectations from long rates. Inflation uncertainty is a leading candidate driver of the premia. Use it to frame the model choices in the curve and policy readings.

Version record

Jens H. E. Christensen, Francis X. Diebold and Glenn D. Rudebusch, "The affine arbitrage-free class of Nelson–Siegel term structure models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 164(1), 4–20 (2011); DOI 10.1016/j.jeconom.2011.02.011.

Derives arbitrage-free Nelson–Siegel (AFNS) models as restricted versions of the canonical three-factor affine model, keeping the level, slope and curvature loadings while restoring no-arbitrage. The restriction eases estimation and can improve forecasts; compare it directly with dynamic Nelson–Siegel and an unrestricted Gaussian model.

Version record

Don H. Kim and Jonathan H. Wright, "An Arbitrage-Free Three-Factor Term Structure Model and the Recent Behavior of Long-Term Yields and Distant-Horizon Forward Rates"

Difficulty: Research **Type:** Research paper

Status/version: Federal Reserve Finance and Economics Discussion Series 2005-33, August 2005; DOI 10.17016/FEDS.2005.33. Staff working paper.

Documents the three-factor arbitrage-free model behind the Federal Reserve Board staff's Kim–Wright term-premium series, fitted to Treasury yields from 1990. It attributes much of the 2004–05 fall in long yields and distant forwards to lower term premia, mostly real term premia. Compare it with the ACM regression estimates: the two series differ by model, inputs and sample.

Extended note. Version record

11.1.3 Bond risk premia and rates strategies

Sydney C. Ludvigson and Serena Ng, "Macro Factors in Bond Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(12), 5027–5067 (2009); DOI 10.1093/rfs/hhp081.

Extracts real and inflation factors from a large macroeconomic panel and shows that they forecast excess Treasury returns beyond forward rates and yield spreads, which standard affine models rule out. Read it with the real-time-data and small-sample critiques before treating the macro factors as tradable.

Version record

Gregory R. Duffee, "Information in (and not in) the Term Structure"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 24(9), 2895–2934 (2011); DOI 10.1093/rfs/hhr033.

Shows that almost half the variation in bond risk premia cannot be detected from the cross-section of yields. A filtered hidden factor forecasts both short rates and excess returns and moves against economic activity, yet macroeconomic variables explain little of it. A direct challenge to models that use yields as their only factors.

Version record

Scott Joslin, Marcel Priebsch and Kenneth J. Singleton, "Risk Premiums in Dynamic Term Structure Models with Unspanned Macro Risks"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(3), 1197–1233 (2014); DOI 10.1111/jofi.12131.

Builds an arbitrage-free term-structure model in which output and inflation risks are not spanned by the yield curve but move the market prices of level, slope and curvature risk. Between 1985 and 2007 these risks account for much of the variation in forward term premia. The standard framework for unspanned macro risk; compare it with the Bauer–Hamilton robustness tests.

Version record

Anna Cieslak and Pavol Povala, "Expected Returns in Treasury Bonds"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 28(10), 2859–2901 (2015); DOI 10.1093/rfs/hhv032.

Decomposes Treasury yields into persistent inflation expectations and maturity-specific rate cycles, then extracts a risk-premium factor that forecasts excess returns in and out of sample and subsumes the forward-rate combination of Cochrane and Piazzesi. Check that the inflation trend can be built from data available at each forecast date.

Version record

Michael D. Bauer and James D. Hamilton, "Robust Bond Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 31(2), 399–448 (2018); DOI 10.1093/rfs/hhx096.

Shows that standard tests of bond-return predictability suffer large size distortions when regressors are persistent and correlated with lagged returns. With robust tests, only the level and slope of the curve predict excess returns, and the evidence for unspanned macro risk does not survive. Apply its bootstrap before accepting a new bond-premium predictor.

Version record

Eric Ghysels, Casidhe Horan and Emanuel Moench, "Forecasting through the Rearview Mirror: Data Revisions and Bond Return Predictability"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 31(2), 678–714 (2018); DOI 10.1093/rfs/hhx098.

Re-runs macro-factor forecasts of bond returns on real-time data vintages instead of final revised data. Predictive power and the implied countercyclicality of term premia fall substantially. A data-timing control for any macroeconomic bond signal.

Version record

Daniel L. Thornton and Giorgio Valente, "Out-of-Sample Predictions of Bond Excess Returns and Forward Rates: An Asset Allocation Perspective"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 25(10), 3141–3168 (2012); DOI 10.1093/rfs/hhs069.

Evaluates forward-rate forecasts of bond excess returns in a dynamic asset-allocation exercise. The models add no systematic economic value over a no-predictability benchmark, and their relative performance deteriorates over time. The out-of-sample counterweight to the in-sample predictability evidence.

Version record

Antonio Gargano, Davide Pettenuzzo and Allan Timmermann, "Bond Return Predictability: Economic Value and Links to the Macroeconomy"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 65(2), 508–540 (2019); DOI 10.1287/mnsc.2017.2829.

Reconciles the statistical and economic evidence on bond predictability. A model with a forward spread, a forward-rate factor and a macro factor, allowing for volatility dynamics and parameter uncertainty, improves out-of-sample accuracy and risk-adjusted portfolio returns. Its forecasts are strongly negatively correlated with survey forecasts of short rates.

Version record

John Y. Campbell, Adi Sunderam and Luis M. Viceira, "Inflation Bets or Deflation Hedges? The Changing Risks of Nominal Bonds"

Difficulty: Research **Type:** Research paper

Status/version: Critical Finance Review, 6(2), 263–301 (2017); DOI 10.1561/104.00000043.

Documents how the covariance between Treasury and stock returns has changed since 1953: high in the early 1980s and negative in the 2000s, especially in downturns. A term-structure model with a time-varying nominal–real covariance state fits the change; a high covariance implies a higher term premium and a more concave curve. Use it to judge whether bonds hedge equity risk in a given inflation regime.

Version record

John Y. Campbell, Carolin Pflueger and Luis M. Viceira, "Macroeconomic Drivers of Bond and Equity Risks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 128(8), 3148–3185 (2020); DOI 10.1086/707766.

Explains why the stock-market exposure of Treasury bonds turned from positive to negative after 2001 with a habit-formation model of time-varying risk premia. A change in the comovement of inflation and the output gap drives the switch, but only when risk premia respond endogenously. Links the stock–bond correlation to the monetary-policy regime.

Version record

11.1.4 Treasury market structure and functioning

Michael J. Fleming, Bruce Mizrach and Giang Nguyen, "The Microstructure of a U.S. Treasury ECN: The BrokerTec Platform"

Difficulty: Research **Type:** Research paper

Status/version: 2009

Staff Report 381 (July 2009; revised March 2017), published in *Journal of Financial Markets* 40, 2–22 (2018). Empirical evidence on electronic interdealer Treasury trading, depth, spreads and price discovery. Rates quants should read it together with newer Federal Reserve and New York Fed work on principal trading firms and liquidity fragility.

Applications: Treasuries; HFT, execution, liquidity. Version record

Oliver Ashtari-Tafti, Rodrigo Guimaraes, Gábor Pintér and Jean-Charles Wijnandts, The Liquidity State Dependence of Monetary Policy Transmission

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1289; earlier three-author drafts circulated in 2023; no journal version established. Issued 2025-09-09.

Shows why monetary-policy shocks can have different long-rate effects depending on Treasury-market intermediation, linking a market-liquidity state variable to real-term-premium transmission.

Extended note. Version record

Alain Chaboud, Ellen Correia Golay, Michael J. Fleming, Yesol Huh, Frank M. Keane, and Or Shachar, Liquidity and Trading Dynamics in the Off-the-Run U.S. Treasury Market

Difficulty: Research **Type:** Research paper

Status/version: Federal Reserve Bank of New York Staff Report 1170, November 2025, DOI 10.59576/sr.1170; working report, with no journal publication asserted.

Measures trading activity and liquidity in seasoned Treasuries, which are most of outstanding supply but differ materially from the on-the-run benchmark during stress. Some underlying regulatory transaction data are restricted, and market-structure proposals are prospective; cost and liquidation assumptions still need contemporaneous instrument-level validation.

Extended note. Version record

Michael J. Fleming, Weiling Liu and Giang Nguyen, Intraday Price Pressure and Order Flow Around U.S. Treasury Auctions

Difficulty: Research **Type:** Research paper

Status/version: Federal Reserve Bank of New York Staff Report 1188, revised July 2026; working paper, no journal version established. Issued 2026-03. Issue date 2026-03; selected revision 2026-07.

A long-horizon empirical account of auction-linked inventory pressure and order flow: yields rise ahead of auctions and reverse afterward, with the size of that pattern conditioned by dealer capacity and demand. Gives the auction-data entries a concrete price-formation mechanism.

Extended note. Version record

Darrell Duffie, "Still the World's Safe Haven? Redesigning the U.S. Treasury Market After the COVID-19 Crisis"

Difficulty: Advanced **Type:** Report / research note

Status/version: Hutchins Center Working Paper 62, Brookings, June 2020.

Diagnoses the March 2020 dash for cash and proposes central clearing, all-to-all trading and a standing repo facility.

Extended note. Version record

U.S. Treasury, Federal Reserve Board, Federal Reserve Bank of New York, SEC and CFTC staff, "Recent Disruptions and Potential Reforms in the U.S. Treasury Market: A Staff Progress Report"

Difficulty: Advanced **Type:** Report / research note

Status/version: Inter-Agency Working Group on Treasury Market Surveillance, November 2021.

Inter-agency account of the March 2020 disruption and reform workstreams: clearing, transparency, dealer capacity and leverage.

Version record

Zhiguo He, Stefan Nagel and Zhaogang Song, "Treasury inconvenience yields during the COVID-19 crisis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 143(1), 57–79 (2022); DOI 10.1016/j.jfineco.2021.05.044.

Treasuries traded at a discount to swaps and OIS in March 2020 as dealer balance sheets were saturated.

Version record

Wenxin Du, Benjamin Hébert and Wenhao Li, "Intermediary balance sheets and the treasury yield curve"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 150(3), 103722 (2023); DOI 10.1016/j.jfineco.2023.103722.

Swap spreads and Treasury basis reflect a shift from Treasury scarcity to dealer balance-sheet constraint after 2008.

Version record

Urban Jermann, "Negative Swap Spreads and Limited Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(1), 212–238 (2020); DOI 10.1093/rfs/hhz030.

Model of negative swap spreads when arbitrageurs face holding costs for Treasuries.

Version record

William Diamond and Peter Van Tassel, "Risk-Free Rates and Convenience Yields around the World"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(4), 2063–2108 (2026); DOI 10.1111/jofi.70045.

Infers risk-free rates from index options to estimate safe-asset convenience yields in ten G11 currencies; convenience yields rise with interest-rate levels and in crises.

Version record

Viral V. Acharya and Toomas Laarits, "When Do Treasuries Earn the Convenience Yield?—A Hedging Perspective"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, jofi.70085 (2026); DOI 10.1111/jofi.70085.

Treasury convenience yields are high when Treasuries hedge equity downturns (low stock–bond covariance), a hedging perspective on safe-asset value.

Version record

Dong Lou, Gabor Pinter, Semih Üslü and Danny Walker, "Yield drifts when issuance comes before macro news"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 165, 103993 (2025); DOI 10.1016/j.jfineco.2025.103993.

Studies Treasury yield drifts when auction issuance is scheduled before macroeconomic news releases.

Extended note. Version record

Lena Boneva, Jakub Kastl and Filip Zikes, "Dealer balance sheets and bidding behavior in the Bank of England's QE reverse auctions"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 174, 104182 (2025); DOI 10.1016/j.jfineco.2025.104182.

Dealer balance-sheet constraints affect bidding in Bank of England QE reverse auctions, linking intermediation capacity to QE prices.

Extended note. Version record

Dan Li, Lubomir Petrasek and Mary Tian, "Risk-averse dealers in a risk-free market—The role of trading desk risk limits"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 181, 104290 (2026); DOI 10.1016/j.jfineco.2026.104290.

Desk-level VaR limits constrain US dealers' Treasury intermediation in stress; dealers reduce positions as they approach internal limits.

Version record

Thomas M. Eisenbach and Gregory Phelan, "Fragility of Safe Asset Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(5), 1310–1361 (2026); DOI 10.1093/rfs/hhaf064.

Strategic interaction between safety and liquidity investors with constrained dealers explains March 2020 safe-asset crashes.

Version record

Falk Bräuning and Hillary Stein, "The Effect of Primary Dealer Constraints on Intermediation in the Treasury Market"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag043 (2026); DOI 10.1093/rfs/hhag043.

Confidential microdata show SLR and desk-VaR constraint shocks reduce primary-dealer Treasury positions and market liquidity.

Version record

Michael J. Fleming and Eli M. Remolona, "Price Formation and Liquidity in the U.S. Treasury Market: The Response to Public Information"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 54(5), 1901–1915 (1999); DOI 10.1111/0022-1082.00172.

Finds a two-stage response to macroeconomic announcements in the interdealer Treasury market: an almost instantaneous price change with lower volume and sharply wider spreads, then a prolonged period of high volume, persistent volatility and moderately wide spreads as traders reconcile private views.

Version record

Michael W. Brandt and Kenneth A. Kavajecz, "Price Discovery in the U.S. Treasury Market: The Impact of Orderflow and Liquidity on the Yield Curve"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 59(6), 2623–2654 (2004); DOI 10.1111/j.1540-6261.2004.00711.x.

Order-flow imbalances account for up to 26 percent of day-to-day yield variation on days without major announcements. The effect is permanent and strongest when liquidity is low, so price discovery through trading, not only public news, moves the curve.

Version record

Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation and Office of the Comptroller of the Currency, Enhanced Supplementary Leverage Ratio Standards: Final Rule; Federal Register notice

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final rule issued 25 November 2025 and effective 1 April 2026, with optional adoption from 1 January 2026; OCC Bulletin 2025-41, Federal Register document 2025-21626.

Recalibrates the enhanced supplementary leverage ratio for U.S. global systemically important bank holding companies and their depository institutions. The fixed add-on becomes a buffer tied to the method 1 GSIB surcharge, capped at one percentage point for the bank subsidiaries, and the eSLR leaves the well-capitalised definition. The agencies aim to reduce disincentives to low-risk intermediation such as Treasury market making; whether dealer capacity expands is an empirical question.

Version record

11.2 Monetary policy, asset purchases and the yield curve

Policy expectations, term premia and central-bank balance sheets move the curve together. Measure surprises in narrow windows, adjust futures-implied paths for risk premia, and treat purchase or runoff effects as conditional on market functioning and on which securities change hands.

Kenneth N. Kuttner, "Monetary policy surprises and interest rates: Evidence from the Fed funds futures market"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Monetary Economics, 47(3), 523–544 (2001); DOI 10.1016/S0304-3932(01)00055-1.

Uses federal funds futures to split target-rate changes into anticipated and surprise components. Bill, note and bond yields barely respond to anticipated changes but respond strongly to surprises. The original futures-based policy-shock measure behind later event studies.

Version record

Monika Piazzesi and Eric T. Swanson, "Futures prices as risk-adjusted forecasts of monetary policy"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 55(4), 677–691 (2008); DOI 10.1016/j.jmoneco.2008.04.003.

Documents positive, countercyclical excess returns on federal funds and eurodollar futures, predictable from employment growth, Treasury yield spreads and corporate bond spreads. Ignoring these premia biases futures-implied policy paths and some shock measures; adjust before reading a futures strip as the expected path.

Version record

Samuel G. Hanson and Jeremy C. Stein, "Monetary policy and long-term real rates"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 115(3), 429–448 (2015); DOI 10.1016/j.jfineco.2014.11.001.

A 100 basis-point rise in the two-year yield on an FOMC day is associated with a 42 basis-point rise in the ten-year forward real rate, more than sticky-price models allow. The response appears to reflect term premia, consistent with demand from yield-oriented investors.

Version record

Samuel G. Hanson, David O. Lucca and Jonathan H. Wright, "Rate-Amplifying Demand and the Excess Sensitivity of Long-Term Rates"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 136(3), 1719–1781 (2021); DOI 10.1093/qje/qjab011.

Since 2000, long-term rates have become more sensitive to daily and monthly moves in short rates while the low-frequency link has weakened. Short-rate changes temporarily move term premia because investor demand for long bonds shifts in a rate-amplifying direction and arbitrage absorbs it slowly. Read it with the mortgage-convexity evidence.

Version record

Eric T. Swanson, "Measuring the effects of federal reserve forward guidance and asset purchases on financial markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 118, 32–53 (2021); DOI 10.1016/j.jmoneco.2020.09.003.

Extends the Gürkaynak–Sack–Swanson factor method to separate forward guidance from large-scale asset purchases during the 2009–15 zero lower bound. Forward guidance moved short-term yields more but its effects decayed within months; purchases moved longer-term Treasury and corporate yields and rate uncertainty more persistently.

Version record

Joseph Gagnon, Matthew Raskin, Julie Remache and Brian Sack, "The Financial Market Effects of the Federal Reserve's Large-Scale Asset Purchases"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Central Banking, 7(1), March 2011. No DOI.

Federal Reserve staff account of how the 2008–10 purchases of agency debt, agency MBS and Treasuries were carried out, with event-study evidence that they lowered longer-term rates meaningfully and durably, including on securities not purchased. Most of the decline is attributed to lower risk and term premia rather than expected short rates. Announcement windows cannot fully separate purchase news from other information released the same day.

Version record

Arvind Krishnamurthy and Annette Vissing-Jorgensen, "The Effects of Quantitative Easing on Interest Rates: Channels and Implications for Policy"

Difficulty: Research **Type:** Research paper

Status/version: Brookings Papers on Economic Activity, 2011(2), 215–287; DOI 10.1353/eca.2011.0019.

Event study of QE1 and QE2 across asset classes. Signalling, safe-asset demand and inflation channels operate in both programmes; MBS-prepayment and default-risk channels appear in QE1 only. Effects depend on which assets are bought, so Treasury yields alone are an incomplete policy target.

Version record

Stefania D'Amico and Thomas B. King, "Flow and stock effects of large-scale treasury purchases: Evidence on the importance of local supply"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 108(2), 425–448 (2013); DOI 10.1016/j.jfineco.2012.11.007.

Uses security-level prices and quantities from the Federal Reserve's 2009 purchase of $300 billion of Treasuries to document a local-supply effect: yields in a maturity sector responded more to purchases in that sector. Estimates a persistent stock effect of about 30 basis points over the programme, plus smaller flow effects around operations.

Version record

Jing Cynthia Wu and Fan Dora Xia, "Measuring the Macroeconomic Impact of Monetary Policy at the Zero Lower Bound"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 48(2–3), 253–291 (2016); DOI 10.1111/jmcb.12300.

Makes a shadow-rate term-structure model tractable near the zero lower bound and uses the shadow rate to summarise unconventional policy and its macroeconomic effects. The shadow-rate series depends on the model and its yield inputs; treat it as a model summary, not an observed policy rate.

Version record

Kathryn Holston, Thomas Laubach and John C. Williams, "Measuring the natural rate of interest: International trends and determinants"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 108, S59–S75 (2017); DOI 10.1016/j.jinteco.2017.01.004.

Applies the Laubach–Williams model to the United States, Canada, the euro area and the United Kingdom and finds large declines in trend growth and the natural rate over 25 years, with substantial comovement across the four. Natural-rate estimates are imprecise and change as data are revised.

Version record

Michael D. Bauer and Glenn D. Rudebusch, "Interest Rates under Falling Stars"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 110(5), 1316–1354 (2020); DOI 10.1257/aer.20171822.

Adds time-varying trend inflation and equilibrium real rates to an arbitrage-free term-structure model. Accounting for these trends changes term-premium estimates, improves out-of-sample yield forecasts and matters for predicting excess bond returns.

Version record

Carlo Altavilla, Luca Brugnolini, Refet S. Gürkaynak, Roberto Motto and Giuseppe Ragusa, "Measuring euro area monetary policy"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 108, 162–179 (2019); DOI 10.1016/j.jmoneco.2019.08.016.

Builds the euro-area event-study measures from tick data around ECB decisions and press conferences. Three factors explain yield-curve moves in each window; a QE factor dominates recently, and forward-guidance and QE effects persist at long maturities. The associated database is the standard source of euro-area policy surprises.

Version record

Wenxin Du, Kristin Forbes and Matthew N. Luzzetti, "Quantitative Tightening Around the Globe: What Have We Learned?"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 32321, April 2024; DOI 10.3386/w32321.

Compares quantitative tightening in seven advanced economies. QT announcements raise government bond yields and steepen curves, active sales more than passive runoff at long maturities, with limited effects on most other market prices; implementation has coincided with modestly higher overnight funding spreads.

Extended note. Version record

Board of Governors of the Federal Reserve System, Implementation Note issued October 29, 2025; Roberto Perli, Eric LeSueur and Linsey Molloy, Federal Reserve Bank of New York, The Implementation of Reserve Management Purchases to Maintain Ample Reserves

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: FOMC implementation note, 29 October 2025; New York Fed Teller Window article, 31 March 2026. Dated operating settings.

The directive that ended balance-sheet runoff on 1 December 2025: maturing Treasuries are rolled over at auction and agency principal is reinvested in Treasury bills. The New York Fed article explains the reserve-management purchases of bills that began in mid-December 2025 at about $40 billion a month and the expected slower pace after the April 2026 tax season. Later implementation notes supersede these settings.

Version record

11.3 Government debt issuance, auctions and demand

Auction design, the issuance calendar, maturity choice and investor demand set the supply side of government debt. Read the dated Treasury documents with the auction and demand evidence; announcement and auction effects depend on dealer capacity and the holder base.

Kenneth D. Garbade, *Treasury Debt Management under the Rubric of Regular and Predictable Issuance: 1983–2012*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Federal Reserve Bank of New York monograph, 2015.

A program-management history of how the Treasury ran regular and predictable issuance from 1983 to 2012, adapting offering sizes, maturities and the auction calendar to changing financing needs. Institutional background for auction cycles, reopenings and new-issue sizes.

Version record

U.S. Department of the Treasury, Quarterly Refunding

Difficulty: Advanced **Type:** Data / specification

Status/version: Living official hub for refunding statements, Treasury Borrowing Advisory Committee materials and archives; inspected 13 October 2026.

Treasury's hub for quarterly refunding statements, Treasury Borrowing Advisory Committee charges and minutes, primary-dealer information and archives. Use the dated statements to identify announced coupon sizes, bill-share guidance and buyback parameters at each decision date; later statements revise guidance.

Version record

Ali Hortaçsu, Jakub Kastl and Allen Zhang, "Bid Shading and Bidder Surplus in the US Treasury Auction System"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 108(1), 147–169 (2018); DOI 10.1257/aer.20160675.

Estimates a structural bidding model on uniform-price Treasury bill and note auctions from July 2009 to October 2013. Primary dealers bid higher yields than direct and indirect bidders because their ability to shade bids exceeds their higher willingness to pay. Bidder surplus averages about three basis points, with efficiency losses of about two.

Version record

Dong Lou, Hongjun Yan and Jinfan Zhang, "Anticipated and Repeated Shocks in Liquid Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 26(8), 1891–1912 (2013); DOI 10.1093/rfs/hht034.

Secondary-market Treasury prices fall in the days before auctions and recover shortly afterwards, although auction dates and sizes are announced in advance. The pattern is linked to dealers' limited risk-bearing capacity and slow end-investor capital, implying a hidden issuance cost of 9–18 basis points of auction size.

Version record

Robin Greenwood and Dimitri Vayanos, "Bond Supply and Excess Bond Returns"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(3), 663–713 (2014); DOI 10.1093/rfs/hht133.

Tests a term-structure model in which risk-averse arbitrageurs absorb supply shocks: the maturity-weighted debt-to-GDP ratio predicts higher yields and excess returns, more so for long bonds and after arbitrageurs have lost money. The empirical basis for supply channels in debt management and QE.

Version record

Robin Greenwood, Samuel G. Hanson and Jeremy C. Stein, "A Comparative-Advantage Approach to Government Debt Maturity"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 70(4), 1683–1722 (2015); DOI 10.1111/jofi.12253.

Models the government's maturity choice when short-term riskless debt earns a monetary premium but must be refinanced more often. If private money-like debt creates negative externalities, the government should tilt issuance shorter and partly crowd it out. A framework for the bill-share question.

Version record

Wenxin Du, Joanne Im and Jesse Schreger, "The U.S. Treasury Premium"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 112, 167–181 (2018); DOI 10.1016/j.jinteco.2018.01.001.

Measures the convenience premium of U.S. Treasuries over other developed-market government bonds as a covered-interest-parity deviation between bond yields. The five-year premium averaged about 21 basis points before the 2008 crisis, rose to 90 during it and averaged about -8 afterwards, while the short-maturity premium stayed positive.

Version record

Zhengyang Jiang, Hanno Lustig, Stijn Van Nieuwerburgh and Mindy Z. Xiaolan, "The U.S. Public Debt Valuation Puzzle"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 92(4), 1309–1347 (2024); DOI 10.3982/ecta20497.

Compares the market value of U.S. federal debt with the present value of fiscal surpluses in a realistic asset-pricing model. Because surpluses are risky, the gap is large: Treasury investors do not appear to impose the government budget constraint, and Treasuries may be overpriced.

Version record

Walker Ray, Michael Droste and Yuriy Gorodnichenko, "Unbundling Quantitative Easing: Taking a Cue from Treasury Auctions"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 132(9), 3115–3172 (2024); DOI 10.1086/729581.

Uses high-frequency identification around Treasury auctions to isolate demand shocks that work only through preferred habitat. Finds large localised yield-curve effects when markets are disrupted; a New Keynesian model with preferred habitat reproduces the financial effects of QE and implies modest macroeconomic stimulus. A replication package is available.

Version record

Rashad Ahmed and Iñaki Aldasoro, Stablecoins and safe asset prices

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1270, first published 28 May 2025; later revised with data to March 2026. Working paper.

Estimates how flows into dollar stablecoins affect Treasury bill yields, using daily data, local projections and an instrument built from idiosyncratic token growth. Inflows lower three-month bill yields by a few basis points within days, with limited effects at longer maturities and larger effects in Treasury market stress. Reported magnitudes differ between versions; cite the version read.

Extended note. Version record

11.4 Modern rates and contract conventions

Subject context and qualifications.

Andrei Lyashenko and Fabio Mercurio, Looking Forward to Backward-Looking Rates: A Modeling Framework for Term Rates Replacing LIBOR

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper written 6 February 2019, posted 5 March 2019, last revised 13 February 2020; no journal version asserted.

Defines forward-looking term objects from overnight risk-free rates and shows how their dynamics can be modeled for derivatives whose payoff depends on a rate realized over an accrual period. It is a bridge from the list's general interest-rate texts to the post-LIBOR payoff timing problem. Its 2019 contract context predates later SOFR standardization; recalibrate curves, observation shifts, payment delays, and collateral conventions before implementation.

Extended note. Version record

Claudio Fontana, Short Communication: Caplet Pricing in Affine Models for Alternative Risk-Free Rates; Author institution accepted manuscript

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics 14(1), SC1–SC16, March 2023; DOI 10.1137/22M1513691; supersedes 2022 preprint

Derives tractable caplet and floorlet valuation for alternative overnight rates under an affine short-rate process, explicitly separating forward-looking and backward-looking payoffs. The difference matters because a cap on a compounded coupon acquires fixing exposure through the accrual period, unlike an old LIBOR caplet fixed at its start. Model assumptions and Fourier numerics still need validation against the live cap surface and contract conventions.

Extended note. Version record

Marco Bianchetti and Mattia Carlicchi, Interest Rates After the Credit Crunch: Multiple-Curve Vanilla Derivatives and SABR

Difficulty: Advanced **Type:** Research paper

Status/version: 2011 arXiv preprint; historical transition study, no later journal publication asserted

Develops no-arbitrage vanilla pricing with separate discount and tenor curves and extends SABR smile treatment to that multi-curve setting, with empirical comparisons from the post-crisis transition. Read it as historical evidence for why a single curve became inadequate, not as a current SOFR calibration recipe: IBOR basis, collateral and quote conventions have changed, and SABR fits require explicit static-arbitrage and extrapolation checks.

Extended note. Version record

CME Group, Pricing and Hedging USD SOFR Interest Rate Swaps with SOFR Futures

Difficulty: Advanced **Type:** Research paper

Status/version: CME practitioner article, 2025; exchange source, not independent academic research

Works from exchange-traded SOFR futures to a USD swap curve, futures-strip hedge ratios, and the convexity adjustment between futures and forwards. This adds contract-level implementation to the already listed New York Fed SOFR fixing methodology. Exchange examples are indicative; margin timing, curve interpolation, collateral, and liquidity differ for a specific portfolio.

Extended note. Version record

CME Group, The Treasury Futures Delivery Process, Eighth Edition; CME conversion-factor detail

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: CME exchange manual, eighth edition available by 2024; use current rulebook for any live delivery

Explains the deliverable basket, conversion factors, invoice amount, notice, and the long and short positions' obligations in U.S. Treasury futures delivery. The manual's examples should be recomputed for the target expiry and eligible CUSIPs; changing basket composition or the optimal delivery date can overturn a static hedge ratio.

Extended note. Version record

ICE Benchmark Administration, About the ICE Swap Rate and Inflation Swap Benchmarks; IBA benchmark hub

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: IBA benchmark explainer available in 2026; living benchmark and tenor coverage

Identifies ICE Swap Rate as the administered reference used for cash-settled swaptions and selected swap close-outs, and lists benchmark families linked to SOFR, SONIA, €STR, and EURIBOR. The explainer establishes benchmark scope, not a full swaption pricing model; confirm the applicable methodology, licence, fallback, and trade confirmation for valuation or backtesting.

Extended note. Version record

International Swaps and Derivatives Association, 2021 ISDA Interest Rate Derivatives Definitions; ISDA 2021 Definitions InfoHub

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: First version published 11 June 2021; controlled definitions with subsequent updates

Provides the market's contractual framework for floating-rate options, compounding, dates, payments, and fallbacks in interest-rate derivatives. It belongs beside an RFR caplet model because a mathematically correct coupon can still disagree with the trade if

observation periods, business-day treatment, or payment delay differ. The public InfoHub verifies the release and scope; the controlled definitions may require licensed access and must be pinned by version.

Extended note. Version record

Alan Brace, Dariusz Gatarek and Marek Musiela, "The Market Model of Interest Rate Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 7(2), 127–155 (1997); DOI 10.1111/1467-9965.00028.

LIBOR market model consistent with Black cap pricing; forward-rate framework adapted to backward-looking RFRs.

Version record

Anders B. Trolle and Eduardo S. Schwartz, "A General Stochastic Volatility Model for the Pricing of Interest Rate Derivatives"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(5), 2007–2057 (2009); DOI 10.1093/rfs/hhn040.

HJM model with unspanned stochastic volatility fitted jointly to caps and swaptions.

Version record

Fabio Mercurio, "A Simple Multi-Curve Model for Pricing SOFR Futures and Other Derivatives"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper 3225872 (2018); DOI 10.2139/ssrn.3225872. Working-paper record; no journal version established in this check.

Short-rate approach to SOFR futures convexity and spreads to fed funds.

Extended note. Version record

Jacob Bjerre Skov and David Skovmand, "Dynamic term structure models for SOFR futures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 41(10), 1520–1544 (2021); DOI 10.1002/fut.22246.

Estimates Gaussian DTSMs on SOFR futures, with policy-meeting jumps and convexity adjustments.

Version record

Sven Klingler and Olav Syrstad, "The SOFR discount"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 164, 103989 (2025); DOI 10.1016/j.jfineco.2024.103989.

SOFR-linked floating-rate notes price at lower spreads than comparable LIBOR notes, evidence of a benefit from the benchmark transition.

Version record

Matthew Bickersteth, Yining Ding and Marek Rutkowski, "Pricing and Hedging of SOFR Derivatives"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 36(1), 180–202 (2026); DOI 10.1111/mafi.70004.

Pricing and hedging framework for SOFR-based derivatives after the LIBOR transition.

Version record

Yining Ding, Ruyi Liu and Marek Rutkowski, "Cross-Currency Basis Swaps Referencing Backward-Looking Rates"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 17(2), 491–544 (2026); DOI 10.1137/24m1701538.

Pricing and hedging collateralised cross-currency basis swaps when both legs reference backward-looking RFRs (compounded SOFR versus averaged AONIA).

Version record

Teemu Pennanen and Waleed Taoum, "Statistical Modeling of SOFR Term Structure"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 32(4), 253–288 (2025); DOI 10.1080/1350486x.2026.2620091.

Statistical (incomplete-market) SOFR term-structure model with macro drivers of policy-rate jumps, for risk management and simulation.

Version record

Andreas Schrimpf and Vladyslav Sushko, "Beyond LIBOR: a primer on the new benchmark rates"

Difficulty: Intermediate **Type:** Report / research note

Status/version: BIS Quarterly Review, March 2019 (published 5 March 2019).

Primer on the move from interbank offered rates to transaction-based overnight risk-free rates: how the new benchmarks are built, why they omit banks' marginal term funding costs, and why cash markets face the hardest transition. Explains why some jurisdictions pair risk-free rates with credit-sensitive term benchmarks.

Version record

Francis A. Longstaff, Pedro Santa-Clara and Eduardo S. Schwartz, "The Relative Valuation of Caps and Swaptions: Theory and Empirical Evidence"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 56(6), 2067–2109 (2001); DOI 10.1111/0022-1082.00399.

Uses a string market model to back out the correlation matrix implied by swaptions and compare it with cap prices. Swaption prices are generated by four factors, implied correlations are below historical ones, long-dated swaptions appear mispriced, and caps periodically deviate from swaption-implied values, notably in the 1998 hedge-fund crisis.

Version record

Pierre Collin-Dufresne and Robert S. Goldstein, "Do Bonds Span the Fixed Income Markets? Theory and Evidence for Unspanned Stochastic Volatility"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 57(4), 1685–1730 (2002); DOI 10.1111/1540-6261.00475.

Swap rates explain little of the returns on at-the-money straddles, so volatility risk is not spanned by bonds. Gives conditions under which affine models exhibit unspanned stochastic volatility and shows that derivatives, not bonds alone, are then needed to identify the model.

Version record

Torben G. Andersen and Luca Benzoni, "Do Bonds Span Volatility Risk in the U.S. Treasury Market? A Specification Test for Affine Term Structure Models"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 65(2), 603–653 (2010); DOI 10.1111/j.1540-6261.2009.01546.x.

Uses model-free realised yield volatility from intraday data to test term-structure models. The yield curve does not span realised volatility, so broad affine, quadratic-Gaussian and jump-diffusion classes fail, and Treasury volatility risk cannot be hedged with Treasuries alone.

Version record

Patrick S. Hagan, Deep Kumar, Andrew Lesniewski and Diana Woodward, "Arbitrage-Free SABR"

Difficulty: Advanced **Type:** Research paper

Status/version: Wilmott, 2014(69), 60–75; DOI 10.1002/wilm.10290.

The explicit SABR implied-volatility expansion can imply arbitrage at low strikes. Reduces SABR to an effective one-dimensional forward equation for the density and solves it numerically; prices match the standard formulas except in a boundary layer at very low rates, where they stay arbitrage-free.

Version record

Samuel G. Hanson, "Mortgage convexity"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 113(2), 270–299 (2014); DOI 10.1016/j.jfineco.2014.05.002.

Refinancing-driven changes in MBS duration shift the amount of interest-rate risk that bond investors must hold. With limited short-run risk tolerance, bond risk premia are high when aggregate MBS duration is high, which can make long rates look excessively sensitive to short rates and explains how convexity hedging amplifies rate moves.

Version record

Hoyong Choi, Philippe Mueller and Andrea Vedolin, "Bond Variance Risk Premiums"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 21(3), 987–1022 (2017); DOI 10.1093/rof/rfw072.

Shows that Treasury realised variance can be replicated with a static position in Treasury futures options and dynamic futures trading, then builds variance swaps across tenors. Implied variance term structures are linked to economic activity, and selling Treasury variance swaps earned a Sharpe ratio near two in the sample; treat that as a crash-exposed premium.

Version record

11.5 Inflation and real yields

Start with linker cash flows and index conventions, then the joint nominal/real pricing model, then breakeven decomposition and empirical evidence. Inflation expectations, inflation risk premia, liquidity and index basis are separate exposures.

Jessica James, Michael Leister and Christoph Rieger, Inflation-Linked Bonds and Derivatives: Investing, Hedging and Valuation Principles for Practitioners

Difficulty: Advanced **Type:** Book / monograph

Status/version: De Gruyter book, 2023; eBook and hardback published 30 January 2023; DOI 10.1515/9783110787429

Dedicated, compact treatment of inflation indices, CPI linkage, breakevens, deflation floors, inflation curves, carry, seasonality, risk premia, and inflation options. It fills the large gap between a chapter in the broad rates texts and direct TIPS or linker implementation.

The publisher provides a detailed contents page, but individual gated chapters were not reviewed; use instrument terms and national index rules for production cash flows.

Extended note. Version record

Robert A. Jarrow and Yildiray Yildirim, Pricing Treasury Inflation Protected Securities and Related Derivatives Using an HJM Model

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis 38(2), 337–358, June 2003; selected journal article

Constructs nominal and real zero-coupon curves and an arbitrage-free HJM framework linked through CPI, then applies it to TIPS hedging and an inflation-index option. It supplies the modeling foundation for joint nominal-rate, real-rate, and index exposures that the dedicated practitioner book organizes. The paper's historical hedging exercise is a methodological example, not evidence of present-day model accuracy.

Extended note. Version record

Stefania D'Amico, Don H. Kim and Min Wei, Tips from TIPS: The Informational Content of Treasury Inflation-Protected Security Prices; Fed discussion of the published decomposition

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis 53(1), 395–436, February 2018; select journal version over Fed working papers

Uses no-arbitrage term-structure models to show that TIPS yields contain a liquidity premium, so nominal-minus-real breakevens combine expected inflation, inflation risk premium, and TIPS-specific liquidity. Re-estimate with chosen sample, surveys, and curve inputs and check behavior during stress; model components are latent rather than directly observed. The early TIPS and crisis-period estimates are informative about mechanisms, not fixed adjustments to apply to 2026 breakevens.

Extended note. Version record

Robert A. Jarrow and Yildiray Yildirim, Inflation-Adjusted Bonds, Swaps, and Derivatives

Difficulty: Advanced/Research **Type:** Research paper

Status/version: Annual Review of Financial Economics 15, 449–471, November 2023; online 20 June 2023; DOI 10.1146/annurev-financial-110921-110855

Reviews the valuation and empirical literature for indexed bonds, inflation swaps, and options, connecting real-versus-nominal HJM modeling to estimates of expectations and inflation risk premia. Read after the 2003 model to see which questions have accumulated evidence and which remain thin, especially the efficiency of inflation-derivative markets. The authors expressly identify limited comparative evidence on derivative valuation models.

Extended note. Version record

U.S. Department of the Treasury, TreasuryDirect, TIPS/CPI Data and Daily Index Ratios; Treasury TIPS investor guide

Difficulty: Advanced **Type:** Data / specification

Status/version: Living Treasury instrument and data page; inspected 4 October 2026

Publishes reference CPI and daily index ratios used to adjust TIPS principal and coupon cash flows. It is the necessary source when turning an inflation model into a security-level calculation, especially around settlement dates, index lag, interpolation, and the maturity principal floor. Treasury mechanics establish contractual payments, while market prices still contain real-rate, inflation-risk, liquidity, and supply-demand components.

Extended note. Version record

U.S. Department of the Treasury, Interest Rate Statistics and Daily Interest Rate XML Feed

Difficulty: Advanced **Type:** Data / specification

Status/version: Living nominal and TIPS par-yield data and XML interface; methodology and availability inspected 4 October 2026.

Treasury publishes daily nominal and real constant-maturity par-yield series with a documented XML query path. The real curve is fitted from indicative bid-side secondary-market quotations on recently auctioned TIPS, not observed transaction prices, executable security-level quotes or a complete curve from individual bonds. Its history begins in 2003, with a 1 December 2008 TIPS knot-method break and a 6 December 2021 Treasury spline-method break.

Extended note. Version record

U.S. Department of the Treasury, TreasuryDirect, Historical Auction Search and About Auctions

Difficulty: Advanced **Type:** Data / specification

Status/version: Living auction-result query and download formats; inspected 4 October 2026.

TreasuryDirect offers CUSIP/date-filtered auction results in CSV, JSON, TSV and XML, including TIPS auctions since their 1997 introduction. A successful auction has an awarded price corresponding to the high accepted yield, providing a real primary-market price for an identified security and auction date. These sparse primary auctions are not a daily secondary-market bid–ask panel.

Extended note. Version record

United Kingdom Debt Management Office, Index-linked Gilts: Index Ratio Data and Formulae; DMO price and accrued-interest examples

Difficulty: Advanced **Type:** Data / specification

Status/version: Living DMO data/specification page; inspected 4 October 2026

Provides the official UK linker index-ratio series and a route to calculation formulae for the three-month-lag gilt design. Pair it with the U.S. TIPS page to stop a cross-market inflation strategy from silently equating UK RPI-linked cash flows with U.S. CPI-U TIPS. The DMO series verifies mechanics; comparison of real yields still requires currency hedging, tax, liquidity, and index-basis assumptions.

Extended note. Version record

Matthias Fleckenstein, Francis A. Longstaff and Hanno Lustig, "The TIPS-Treasury Bond Puzzle"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(5), 2151–2197 (2014); DOI 10.1111/jofi.12032.

Synthetic Treasuries from TIPS plus inflation swaps were cheaper than nominal Treasuries; a persistent arbitrage limited by capital.

Version record

Fabio Mercurio, "Pricing inflation-indexed derivatives"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 5(3), 289–302 (2005); DOI 10.1080/14697680500148851.

Market-model pricing of inflation-indexed swaps, caps and floors.

Version record

Jeroen Kerkhof, "Inflation Derivatives Explained: Markets, Products, and Pricing"

Difficulty: Advanced **Type:** Report / research note

Status/version: Lehman Brothers Fixed Income Quantitative Research, July 2005; available through secondary hosts.

Widely circulated Lehman Brothers primer on inflation swaps, seasonality, indexation lags and option pricing (July 2005).

Version record

Carolin Pflueger, "Back to the 1980s or not? The drivers of inflation and real risks in Treasury bonds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 167, 104027 (2025); DOI 10.1016/j.jfineco.2025.104027.

Asks whether recent inflation and real-rate risks in Treasuries resemble the 1980s, decomposing drivers of bond risk and stock–bond correlation.

Extended note. Version record

Xiang Fang, Yang Liu and Nikolai Roussanov, "Getting to the Core: Inflation Risks Within and Across Asset Classes"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(3), 702–743 (2026); DOI 10.1093/rfs/hhaf050.

Stocks have negative core-inflation betas but positive energy betas; real assets mostly hedge energy rather than core inflation, and only core inflation carries a priced (negative) premium.

Version record

Refet S. Gürkaynak, Brian Sack and Jonathan H. Wright, "The TIPS Yield Curve and Inflation Compensation"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics, 2(1), 70–92 (2010); DOI 10.1257/mac.2.1.70.

Documents the smoothed TIPS curve fitted daily at the Federal Reserve Board and published with the nominal curve, and computes inflation compensation from the two. Shows that breakevens are not pure inflation expectations because of inflation-risk and TIPS-liquidity premia, and gives a first estimate of each.

Version record

Joseph Haubrich, George Pennacchi and Peter Ritchken, "Inflation Expectations, Real Rates, and Risk Premia: Evidence from Inflation Swaps"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 25(5), 1588–1629 (2012); DOI 10.1093/rfs/hhs003.

Estimates a nominal and real term-structure model whose volatility factors drive only risk premia, using Treasury yields, survey inflation forecasts and inflation swap rates. Separates short- and long-horizon real and inflation risk premia and finds TIPS undervalued before 2004 and during the 2008 crisis.

Version record

Michael Abrahams, Tobias Adrian, Richard K. Crump, Emanuel Moench and Rui Yu, "Decomposing real and nominal yield curves"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 84, 182–200 (2016); DOI 10.1016/j.jmoneco.2016.10.006.

Decomposes nominal and TIPS yields into expected real short rates, expected inflation, and real, inflation and liquidity premia with an affine model. LSAP announcements lowered yields mainly through real term premia, which also explain the response of long real forwards to policy surprises.

Version record

Martin M. Andreasen, Jens H. E. Christensen and Simon Riddell, "The TIPS Liquidity Premium"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 25(6), 1639–1675 (2021); DOI 10.1093/rof/rfab018.

Identifies TIPS liquidity risk from individual bond prices by modelling how seasoned TIPS migrate into buy-and-hold portfolios. Finds a sizeable, countercyclical liquidity premium; accounting for it improves inflation forecasts and the fit to survey expectations.

Version record

Matthias Fleckenstein, Francis A. Longstaff and Hanno Lustig, "Deflation Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 30(8), 2719–2760 (2017); DOI 10.1093/rfs/hhx021.

Extracts the market distribution of inflation from inflation swaps and options. Expected inflation is about 2.5 percent over 30 years, yet markets put substantial weight on extended deflation, priced like other tail risks and correlated with financial-market and consumer-confidence outcomes. Relevant to the TIPS deflation floor.

Version record

Yuriy Kitsul and Jonathan H. Wright, "The economics of options-implied inflation probability density functions"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 110(3), 696–711 (2013); DOI 10.1016/j.jfineco.2013.08.013.

Builds inflation probability densities from U.S. inflation caps and floors and studies their response to macroeconomic news. The implied densities put more mass on deflation and high inflation than time-series models, giving a U-shaped empirical pricing kernel. The densities rest on over-the-counter dealer quotes in a young market.

Version record

UK Statistics Authority and HM Treasury, Response to the joint consultation on reforming the methodology of the Retail Prices Index

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Published 25 November 2020; change scheduled for February 2030.

Confirms that RPI will take on the methods and data of CPIH, a change the Authority says it can make in February 2030, after the last specific index-linked gilt matures. Long RPI-linked gilts, swaps and liabilities must price the switch; aspects of the reform were under judicial review when the response was published.

Version record

11.6 Sovereign and municipal debt

Distinguish local-currency from hard-currency claims, domestic from foreign law, and sovereign from subsovereign obligations. Debt sustainability and recovery processes provide economic scenarios; security-level disclosure, tax treatment and liquidity determine the actual investment.

11.6.1 Sovereign default, restructuring and local-currency debt

Wenxin Du and Jesse Schreger, Local Currency Sovereign Risk

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 71(3), 1027–1070, June 2016; online 3 February 2016; DOI 10.1111/jofi.12389

Constructs a local-currency sovereign credit spread against a synthetic local risk-free rate from cross-currency swaps, then compares its level and global-risk sensitivity with foreign-currency sovereign spreads. It is essential for separating currency and default exposures when comparing domestic bonds with hard-currency Eurobonds. Hedge availability, collateral, and law of issuance must be checked for each country and tenor.

Extended note. Version record

Stefan Avdjiev, Bryan Hardy and Maximilian Jager, The Evolving Nexus: Sovereigns, Banks and NBFIs

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1369; working paper, no journal version established. Issued 2026-07-20.

Documents a changing path for sovereign stress transmission through nonbank investors and their bank financing links, extending the reading list's mostly direct bank-sovereign and sovereign-debt treatment.

Extended note. Version record

Juan J. Cruces and Christoph Trebesch, Sovereign Defaults: The Price of Haircuts

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics 5(3), 85–117, July 2013; DOI 10.1257/mac.5.3.85

Builds a historical investor-loss database for 180 sovereign restructurings in 68 countries and connects larger present-value haircuts with higher subsequent spreads and longer market exclusion. It supplies recovery and reputation evidence absent from a standard sovereign yield decomposition. The sample ends in 2010, and associations do not imply an invariant causal penalty; legal terms, official-sector treatment, and creditor mix vary across later episodes.

Extended note. Version record

International Monetary Fund, Staff Guidance Note on the Sovereign Risk and Debt Sustainability Framework for Market Access Countries; IMF framework and template hub

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: IMF Policy Paper 2022/039, 8 August 2022; DOI 10.5089/9798400216862.007

Sets out the IMF framework for near-term sovereign stress signals, debt fan charts, gross financing needs, and judgment in market-access country analysis. The framework is designed for surveillance and program analysis, not security pricing or trading signals. Country staff assumptions, contingent liabilities, official financing, and domestic debt coverage can differ from an investor's information set and must be documented separately.

Extended note. Version record

Global Sovereign Debt Roundtable co-chairs, Sovereign Debt Restructuring: A Playbook for Country Authorities; IMF release confirming update

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Updated 15 April 2026 GSDR co-chairs document; first version April 2025; non-binding guidance

Maps the practical sequence and concepts of a sovereign debt restructuring for authorities, including coordination with different creditor groups. It is a current process companion to empirical haircut and sovereign-spread papers: timing, perimeter, comparability, and negotiation can matter as much as a model's terminal recovery assumption. The document explicitly is non-binding and is not a contract, legal opinion, prediction of creditor participation, or promise of a particular recovery.

Extended note. Version record

World Bank, International Debt Statistics; World Bank access and archive guidance

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living annual external-debt stock and flow database; methodology and vintages required

Provides annual external debt stocks, flows, debt-service measures, and creditor composition for reporting low- and middle-income countries. It helps a sovereign analyst reconcile marketable bonds with wider public and publicly guaranteed external obligations before estimating funding needs or restructuring perimeter. External-debt aggregates do not by themselves cover all local-currency domestic debt or bond-level legal clauses.

Extended note. Version record

Francis A. Longstaff, Jun Pan, Lasse H. Pedersen and Kenneth J. Singleton, "How Sovereign Is Sovereign Credit Risk?"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics, 3(2), 75–103 (2011); DOI 10.1257/mac.3.2.75.

Sovereign CDS spreads are driven largely by global factors (US equity, volatility, risk premia) rather than local fundamentals.

Version record

Jun Pan and Kenneth J. Singleton, "Default and Recovery Implicit in the Term Structure of Sovereign CDS Spreads"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 63(5), 2345–2384 (2008); DOI 10.1111/j.1540-6261.2008.01399.x.

Separates default intensity, recovery and risk premia from sovereign CDS term structures.

Version record

Robert F. Dittmar, Alex Hsu, Guillaume Roussellet and Peter Simasek, "Default Risk and the Pricing of U.S. Sovereign Bonds"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 81(2), 829–869 (2026); DOI 10.1111/jofi.70014.

US default risk affects relative pricing of nominal Treasuries and TIPS: hedged breakevens rise with US default risk through inflation-swap premia and yield shocks.

Version record

Cristina Arellano, "Default Risk and Income Fluctuations in Emerging Economies"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 98(3), 690–712 (2008); DOI 10.1257/aer.98.3.690.

Small open-economy model in which default probabilities and spreads depend on repayment incentives. Default is more likely in recessions, when repaying noncontingent debt is most costly, and the model matches Argentine interest-rate and consumption volatility. The workhorse quantitative model of sovereign default.

Version record

Josefin Meyer, Carmen M. Reinhart and Christoph Trebesch, "Sovereign Bonds Since Waterloo"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 137(3), 1615–1680 (2022); DOI 10.1093/qje/qjac007.

Compiles about 266,000 monthly prices of foreign-currency government bonds traded in London and New York from 1815 to 2016 for up to 91 countries. Real ex post returns average more than 6 percent a year including defaults and wars, 3–4 percentage points above U.S. or U.K. government bonds, mainly because of high coupons. The sample covers bonds listed in those two markets.

Version record

Benjamin Hébert and Jesse Schreger, "The Costs of Sovereign Default: Evidence from Argentina"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 107(10), 3119–3145 (2017); DOI 10.1257/aer.20151667.

Identifies the cost of default from legal rulings in NML Capital v. Republic of Argentina that moved default probabilities. A 10 percent increase in default probability lowers Argentine equity values by about 6 percent and depreciates the exchange rate by about 1 percent.

Version record

Julian Schumacher, Christoph Trebesch and Henrik Enderlein, "Sovereign defaults in court"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 131, 103388 (2021); DOI 10.1016/j.jinteco.2020.103388.

Builds a four-decade dataset of creditor lawsuits against defaulting sovereigns. Litigation has become more common, legal changes have strengthened creditors, and disputes in U.S. and U.K. courts can cut governments off from international capital markets. Holdout and litigation risk therefore belong in restructuring scenarios.

Version record

Wenxin Du, Carolin E. Pflueger and Jesse Schreger, "Sovereign Debt Portfolios, Bond Risks, and the Credibility of Monetary Policy"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(6), 3097–3138 (2020); DOI 10.1111/jofi.12965.

Governments whose local-currency debt offers greater hedging benefits borrow more in foreign currency. Risk-averse lenders and a lack of monetary commitment explain the pattern: expected countercyclical inflation raises the risk premium on local debt and discourages local-currency borrowing.

Version record

Boris Hofmann, Ilhyock Shim and Hyun Song Shin, "Bond Risk Premia and The Exchange Rate"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 52(S2), 497–520 (2020); DOI 10.1111/jmcb.12760.

In emerging markets, currency appreciation compresses sovereign spreads, even on local-currency bonds, through a lower credit risk premium. The bilateral U.S. dollar rate matters, not the trade-weighted rate, because global investors evaluate returns in dollars.

Version record

11.6.2 Euro-area, UK and Japanese government bonds

Alessandro Beber, Michael W. Brandt and Kenneth A. Kavajecz, "Flight-to-Quality or Flight-to-Liquidity? Evidence from the Euro-Area Bond Market"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(3), 925–957 (2009); DOI 10.1093/rfs/hhm088.

Exploits the negative cross-country correlation between credit quality and liquidity in euro-area government bonds. Credit quality explains most of the spreads, liquidity matters for low-risk issuers and in uncertain markets, and large flows chase liquidity rather than credit quality.

Version record

Paul De Grauwe and Yuemei Ji, "Self-fulfilling crises in the Eurozone: An empirical test"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of International Money and Finance, 34, 15–36 (2013); DOI 10.1016/j.jimonfin.2012.11.003.

Finds that a significant part of the 2010–11 rise in peripheral euro-area spreads was disconnected from debt and fiscal fundamentals and consistent with self-fulfilling negative sentiment, a fragility not seen in stand-alone countries with their own central banks.

Version record

Viral Acharya, Itamar Drechsler and Philipp Schnabl, "A Pyrrhic Victory? Bank Bailouts and Sovereign Credit Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(6), 2689–2739 (2014); DOI 10.1111/jofi.12206.

Models and documents the sovereign–bank loop: bailouts raised European sovereign CDS in 2008, and later sovereign CDS changes explain bank CDS changes after aggregate and bank-level controls, through government guarantees and bank holdings of sovereign bonds.

Version record

Loriana Pelizzon, Marti G. Subrahmanyam, Davide Tomio and Jun Uno, "Sovereign credit risk, liquidity, and European Central Bank intervention: Deus ex machina?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 122(1), 86–115 (2016); DOI 10.1016/j.jfineco.2016.06.001.

Studies the Italian government bond market in the euro crisis. Credit risk drives liquidity: a 10 percent change in the CDS spread leads to a 13 percent change in the bid–ask spread, more strongly above 500 basis points. The ECB's long-term refinancing operations weakened the link, showing the role of market makers' funding liquidity.

Version record

Arvind Krishnamurthy, Stefan Nagel and Annette Vissing-Jorgensen, "ECB Policies Involving Government Bond Purchases: Impact and Channels"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 22(1), 1–44 (2018); DOI 10.1093/rof/rfx053.

Kalman-filter-augmented event study of the SMP, OMT and LTROs. For Italy, Spain and Portugal, the SMP and OMT lowered yields considerably: default risk accounts for about 37 percent of the decline, redenomination risk for 13 percent and market segmentation for 50 percent.

Version record

Ralph S. J. Koijen, François Koulischer, Benoît Nguyen and Motohiro Yogo, "Inspecting the mechanism of quantitative easing in the euro area"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 140(1), 1–20 (2021); DOI 10.1016/j.jfineco.2020.11.006.

Uses security-level holdings of all euro-area investors to trace rebalancing under QE from March 2015 to December 2017: investors outside the euro area accommodated most Eurosystem purchases. An instrumented demand system implies government bond yields fell by 65 basis points on average, from 38 to 83 across countries.

Version record

European Central Bank, The Transmission Protection Instrument

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: ECB press release, 21 July 2022.

Sets out the instrument the Governing Council can activate against unwarranted, disorderly market moves that threaten monetary transmission: purchases of public-sector securities with one to ten years to maturity, not restricted ex ante, for jurisdictions meeting four cumulative criteria on EU fiscal rules, macroeconomic imbalances, debt sustainability and sound policies. Activation and eligibility are judged at the time.

Version record

Gabor Pinter, An anatomy of the 2022 gilt market crisis

Difficulty: Research **Type:** Research paper

Status/version: Bank of England Staff Working Paper 1019, 31 March 2023. Staff working paper.

Uses transaction-level gilt, repo and interest-rate swap data for September and October 2022. LDI, pension and insurance firms with larger repo and swap exposures sold more gilts, hedge funds were paid to supply liquidity, transaction costs rose most for small trades and investors outside the LDI sector, and price dispersion more than doubled within days, consistent with illiquidity spreading across segments.

Extended note. Version record

Bank of Japan, Statement on Monetary Policy (June 2025), with the plan for the reduction of JGB purchases

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Bank of Japan statement, 17 June 2025; interim assessment scheduled for the June 2026 meeting.

Sets the path of reduced JGB purchases: cuts of about 400 billion yen a quarter through January–March 2026, then about 200 billion yen a quarter, reaching about 2 trillion yen a month by January–March 2027. The Bank may increase purchases, use fixed-rate operations or modify the plan if long rates rise rapidly, and it eased securities-lending conditions for more issues.

Version record

11.6.3 Municipal bonds

Municipal Securities Rulemaking Board, Sources of Repayment; MSRB guide to bond-specific federal and state tax treatment

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Living MSRB investor guide; inspected 4 October 2026

Distinguishes a general obligation issuer's taxing pledge from revenue bonds paid by a specified system, facility, tax, or obligor, and explains why the actual source and priority can vary under state law. Use the guide to frame questions, then read the issue's official statement on EMMA for pledge limits, covenants, call rights, and remedies.

Extended note. Version record

Municipal Securities Rulemaking Board, EMMA: Electronic Municipal Market Access; MSRB coverage and caveats

Difficulty: Advanced **Type:** Data / specification

Status/version: Living official municipal disclosure and trade-data portal; inspected 4 October 2026

Provides official statements, continuing disclosures, advance-refunding records, and reported secondary trades for U.S. municipal issues. Preserve filing and trade timestamps, subsequent corrections, and CUSIP mapping in any backtest. EMMA is a disclosure and observation system, not a guarantee that every offering has a full statement or that a reported print is executable in size.

Extended note. Version record

Lawrence E. Harris and Michael S. Piwowar, Secondary Trading Costs in the Municipal Bond Market

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 61(3), 1361–1397, June 2006; DOI 10.1111/j.1540-6261.2006.00875.x

Estimates bond-level trading costs across a large one-year U.S. municipal transaction sample and finds higher costs for smaller trades, riskier or more complex issues, and older securities. It counters the temptation to compare a municipal yield with a Treasury yield while ignoring fragmented execution and tax-specific investor demand. Its historical estimates predate later transparency and electronic protocols and should not be pasted into today's executable cost model.

Extended note. Version record

Robert Novy-Marx and Joshua D. Rauh, Fiscal Imbalances and Borrowing Costs: Evidence from State Investment Losses

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Economic Policy 4(2), 182–213, May 2012; DOI 10.1257/pol.4.2.182

Uses unequal 2008 state pension-investment losses to test whether municipal investors charge for fiscal pressure beyond ratings and standard fiscal controls. The authors find that spreads rose most for lower-credit-quality states, providing a concrete channel from contingent pension liabilities to borrowing costs. The shock identifies one historical setting; its estimated basis-point response is not a current pricing rule or a default probability.

Extended note. Version record

Michael Schwert, "Municipal Bond Liquidity and Default Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 72(4), 1683–1722 (2017); DOI 10.1111/jofi.12511.

Most of the municipal yield spread compensates default risk rather than illiquidity.

Version record

Andrew Ang, Vineer Bhansali and Yuhang Xing, "Taxes on Tax-Exempt Bonds"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 65(2), 565–601 (2010); DOI 10.1111/j.1540-6261.2009.01545.x.

Discount munis carry hidden capital-gains tax effects that are priced; relevant for muni relative value.

Version record

Richard C. Green, Burton Hollifield and Norman Schürhoff, "Financial Intermediation and the Costs of Trading in an Opaque Market"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 20(2), 275–314 (2007); DOI 10.1093/rfs/hhl012.

Estimates a bargaining model on dealer–customer municipal bond trades from a period when trade data were released only with a lag. Dealers earn lower average markups on larger trades and exercise substantial market power, which falls with trade size and rises with the complexity of the trade.

Version record

Marcus Painter, "An inconvenient cost: The effects of climate change on municipal bonds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 135(2), 468–482 (2020); DOI 10.1016/j.jfineco.2019.06.006.

Counties more exposed to climate change pay higher underwriting fees and initial yields on long-term, but not short-term, municipal bonds. The gap is concentrated in lower-rated bonds and widened after the 2006 Stern Review, pointing to investor attention.

Version record

Paul Goldsmith-Pinkham, Matthew T. Gustafson, Ryan C. Lewis and Michael Schwert, "Sea-Level Rise Exposure and Municipal Bond Yields"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 36(11), 4588–4635 (2023); DOI 10.1093/rfs/hhad041.

Municipal bond markets began pricing sea-level-rise exposure in 2013, more for long maturities and not only through near-term flood risk. A structural credit model separates asset-value and uncertainty effects; the premium follows a different trend from house prices.

Version record

11.7 Cash and money markets

Short maturity does not remove rollover, liquidity or counterparty risk. Combine funding-market economics with fund holdings and instrument definitions; use the earlier repo and securities-lending readings for secured financing and collateral specialness.

Matteo Aquilina, Andreas Schrimpf and Karamfil Todorov, CP and CDs Markets: A Primer

Difficulty: Advanced **Type:** Research paper

Status/version: BIS Quarterly Review special feature, published 18 September 2023.

Explains issuance and investor structure of commercial paper and negotiable certificates of deposit across major currencies, and documents the reduced but still important role of money-market funds in absorbing short-term paper. It clarifies that CP and CDs rarely trade before maturity, so liquidity analysis cannot assume continuous bond-market price discovery. The cross-jurisdictional aggregates are historical and imperfect; issuer credit, programme support, and live placement capacity require instrument-level evidence.

Extended note. Version record

Iñaki Aldasoro, Sebastian Doerr and Haonan Zhou, Liquidity Regulation and Bank Funding Costs

Difficulty: Research **Type:** Contract / specification / guidance

Status/version: BIS Working Paper 1352, 27 May 2026; working paper, not peer-reviewed journal article

Uses money-fund holdings of bank certificates of deposit, commercial paper, asset-backed commercial paper, and repo to study how liquidity regulation affects banks' funding cost and maturity structure. It adds causal research to the CP/CD primer and separates an issuer's regulatory balance-sheet choice from generic market-level cash rates. The source's granular sample is February 2011–December 2015; historical estimates and institution-specific regulation should not be treated as current CP quotes.

Extended note. Version record

Marco Cipriani and Gabriele La Spada, Sophisticated and Unsophisticated Runs

Difficulty: Research **Type:** Research paper

Status/version: New York Fed Staff Report 956, December 2020, revised April 2024; selected version remains a staff report.

Separates institutional and retail prime money-market fund behavior during March 2020, finding different roles for fundamentals, information spillovers, and the availability of safer alternatives. It makes fund-run stress more precise than a uniform redemption shock and links holdings of CP and CDs to the ability of issuers to refinance. Staff-report revision and any later publication should be checked before treating the working draft as final.

Extended note. Version record

U.S. Securities and Exchange Commission, Form N-MFP Data Sets; SEC series and class identification file

Difficulty: Advanced **Type:** Data / specification

Status/version: Living monthly as-filed data; available through August 2026 at cutoff; N-MFP3 introduced into extracts July 2025

Provides flat-file extracts of monthly money-market fund filings, including fund and portfolio-holding information, so a researcher can measure exposure to CP, CDs, repo, liquidity buckets, and issuer concentration. It is the public data complement to the prime-fund run paper. Never substitute a present download for a historical decision-time information set.

Extended note. Version record

Federal Reserve Board, Commercial Paper Rates and Outstanding Summary; Fed download and revision notices

Difficulty: Advanced **Type:** Data / specification

Status/version: Living Federal Reserve statistical release and methodology; 1 October 2026 release available at cutoff

Defines the Fed's commercial-paper rate indexes and outstanding categories for financial, nonfinancial, and asset-backed issuers, including the DTCC transaction basis used for rate estimation. It gives cash desks a consistent series for funding conditions and cross-category stress without confusing CP supply with fund demand. Match maturity and issuer category before comparison.

Extended note. Version record

U.S. Securities and Exchange Commission, Money Market Fund Reforms; Full final-rule PDF

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC final rule adopted July 2023; use current rule text and compliance dates for 2026 operations

Documents the increase in daily and weekly liquid-asset minimums, removal of redemption gates and the old fee trigger link, and the liquidity-fee framework for covered funds. Read the rule alongside N-MFP observations rather than assuming the 2020 run regime still applies unchanged. The final-rule page is an official historical rulemaking record; an operational policy must check effective amendments, fund category, and current prospectus language.

Extended note. Version record

Gara Afonso and Ricardo Lagos, "Trade Dynamics in the Market for Federal Funds"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 83(1), 263–313 (2015); DOI 10.3982/ecta10586.

Search-and-bargaining model of the fed funds market with intraday dynamics.

Version record

Marcin Kacperczyk and Philipp Schnabl, "How Safe Are Money Market Funds?"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 128(3), 1073–1122 (2013); DOI 10.1093/qje/qjt010.

Prime funds reached for yield in 2007–08, with risk-taking linked to sponsor incentives.

Version record

Darrell Duffie and Arvind Krishnamurthy, "Passthrough Efficiency in the Fed's New Monetary Policy Setting"

Difficulty: Advanced **Type:** Report / research note

Status/version: Federal Reserve Bank of Kansas City, Jackson Hole symposium paper, 2016.

How frictions and balance-sheet costs weaken pass-through from policy rates to money-market rates (Jackson Hole 2016).

Version record

Lorenzo Casavecchia, Chanyuan Ge, C. Wei Li and Ashish Tiwari, "Prime Time for Prime Funds: Floating NAV, Intraday Redemptions, and Liquidity Risk during Crises"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Asset Pricing Studies, 16(3), 327–377 (2026); DOI 10.1093/rapstu/raag006.

Multi-strike prime money funds with intraday NAVs hold safer assets yet face larger crisis outflows; evidence on post-2016 MMF design.

Version record

Marcia Stigum and Anthony Crescenzi, *Stigum's Money Market*

Difficulty: Advanced **Type:** Book / monograph

Status/version: McGraw-Hill, fourth edition, 2007.

Classic institutional guide to the money market: fed funds, repo, Treasury bills, commercial paper, CDs and dealer operations (fourth edition, 2007).

Version record

Darrell Duffie, "Special Repo Rates"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 51(2), 493–526 (1996); DOI 10.1111/j.1540-6261.1996.tb02692.x.

Why specific collateral trades at special repo rates and how specialness links to on-the-run premia and supply.

Version record

Gara Afonso, Anna Kovner and Antoinette Schoar, "Stressed, Not Frozen: The Federal Funds Market in the Financial Crisis"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 66(4), 1109–1139 (2011); DOI 10.1111/j.1540-6261.2011.01670.x.

After Lehman the fed funds market became more sensitive to borrower risk but did not freeze.

Version record

Arvind Krishnamurthy, Stefan Nagel and Dmitry Orlov, "Sizing Up Repo"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 69(6), 2381–2417 (2014); DOI 10.1111/jofi.12168.

Money-fund and securities-lender repo data show the shadow-banking repo run in 2007–08 was concentrated in private-label collateral and dealers.

Version record

Lawrence Schmidt, Allan Timmermann and Russ Wermers, "Runs on Money Market Mutual Funds"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review, 106(9), 2625–2657 (2016); DOI 10.1257/aer.20140678.

Daily share-class flows in the September 2008 run: institutional investors ran first and redemptions show strategic complementarities.

Version record

Daniel Covitz, Nellie Liang and Gustavo A. Suarez, "The Evolution of a Financial Crisis: Collapse of the Asset-Backed Commercial Paper Market"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 68(3), 815–848 (2013); DOI 10.1111/jofi.12023.

Program-level evidence of runs on ABCP in 2007, driven by sponsor and collateral risk.

Version record

Viral V. Acharya, Philipp Schnabl and Gustavo Suarez, "Securitization without risk transfer"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 107(3), 515–536 (2013); DOI 10.1016/j.jfineco.2012.09.004.

ABCP conduits with bank guarantees left risk on bank balance sheets while avoiding capital charges; losses returned to sponsors in 2007.

Version record

Marcin Kacperczyk and Philipp Schnabl, "When Safe Proved Risky: Commercial Paper during the Financial Crisis of 2007–2009"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 24(1), 29–50 (2010); DOI 10.1257/jep.24.1.29.

Commercial-paper market structure, the 2007 ABCP collapse and the 2008 money-fund run, written as an accessible overview.

Version record

Chapter 12

Credit and securitisation

Distinguish issuer and collateral cash flows, recourse, recovery, tranching and optionality. Legal and reporting resources complement pricing models and empirical evidence.

12.1 Credit models and market liquidity

12.1.1 Structural and reduced-form models

Darrell Duffie and Kenneth Singleton, ***Credit Risk: Pricing, Measurement, and Management***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press, 2003; author site also supplies table corrections.

Develops the reduced-form framework for defaultable claims and credit derivatives. It is essential for understanding hazard rates, recovery conventions, risk-neutral versus physical default probabilities and the interaction of credit and interest-rate risk. [6]

Applications: Corporate bonds, CDS, credit derivatives. Version record

David Lando, ***Credit Risk Modeling***

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2004

Provides rigorous structural and intensity-based credit modelling with a strong stochastic-process foundation. It is particularly useful for researchers working on transition risk, incomplete information and portfolio credit models.

Applications: Credit, corporate bonds, counterparty risk. Version record

Dominic O'Kane, ***Modelling Single-name and Multi-name Credit Derivatives***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Book copyright 2008; Wiley online edition first published 2 January 2012.

An implementation-focused account of CDS curves, tranche models, copulas and credit correlation. Although written before several post-crisis market reforms, its pricing mechanics remain important for understanding structured credit and model-risk history.

Applications: CDS, indices, tranches, structured credit. Version record

Alexander Lipton and Andrew Rennie, eds., ***The Oxford Handbook of Credit Derivatives***

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2011

Oxford University Press, 27 January 2011. Collected treatments of default/recovery, single- and multi-name credit models, counterparty risk, non-Gaussian tails and securitisation. Compare dependence assumptions and model limitations; the historical models do not replace current contractual terms, funding or hedging-liquidity analysis.

Applications: Structured credit, OTC derivatives. Version record

Robert C. Merton, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 29(2), 449 (1974); DOI 10.2307/2978814.

Equity as a call on firm assets; the structural model underlying distance-to-default and credit–equity linkages.

Version record

Fischer Black and John C. Cox, "Valuing Corporate Securities: Some Effects of Bond Indenture Provisions"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 31(2), 351–367 (1976); DOI 10.1111/j.1540-6261.1976.tb01891.x.

First-passage default with safety covenants; the barrier extension of Merton.

Version record

Hayne E. Leland, "Corporate Debt Value, Bond Covenants, and Optimal Capital Structure"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 49(4), 1213–1252 (1994); DOI 10.1111/j.1540-6261.1994.tb02452.x.

Endogenous default and optimal leverage with taxes and bankruptcy costs.

Version record

Robert A. Jarrow and Stuart M. Turnbull, "Pricing Derivatives on Financial Securities Subject to Credit Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 50(1), 53–85 (1995); DOI 10.1111/j.1540-6261.1995.tb05167.x.

Reduced-form default as a Poisson event; foundation of intensity-based credit pricing.

Version record

Darrell Duffie and Kenneth J. Singleton, "Modeling Term Structures of Defaultable Bonds"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 12(4), 687–720 (1999); DOI 10.1093/rfs/12.4.687.

Default-adjusted short-rate discounting under recovery of market value.

Version record

Pierre Collin-Dufresne, Robert S. Goldstein and J. Spencer Martin, "The Determinants of Credit Spread Changes"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 56(6), 2177–2207 (2001); DOI 10.1111/0022-1082.00402.

Structural variables explain only a quarter of spread changes; a common factor dominates the residuals.

Version record

Jing-Zhi Huang and Ming Huang, "How Much of the Corporate-Treasury Yield Spread Is Due to Credit Risk?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Asset Pricing Studies, 2(2), 153–202 (2012); DOI 10.1093/rapstu/ras011.

Calibrated structural models explain a small share of investment-grade spreads: the credit spread puzzle.

Version record

Simon Gilchrist and Egon Zakrajšek, "Credit Spreads and Business Cycle Fluctuations"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 102(4), 1692–1720 (2012); DOI 10.1257/aer.102.4.1692.

Excess bond premium from bond-level spreads predicts economic activity; Fed publishes updates.

Version record

Francis A. Longstaff, Sanjay Mithal and Eric Neis, "Corporate Yield Spreads: Default Risk or Liquidity? New Evidence from the Credit Default Swap Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 60(5), 2213–2253 (2005); DOI 10.1111/j.1540-6261.2005.00797.x.

CDS-implied default component explains most of the spread; the remainder relates to liquidity.

Version record

Jennie Bai and Pierre Collin-Dufresne, "The CDS-Bond Basis"

Difficulty: Research **Type:** Research paper

Status/version: Financial Management, 48(2), 417–439 (2019); DOI 10.1111/fima.12252.

Negative CDS-bond bases in 2008 linked to funding cost, collateral quality and counterparty risk.

Version record

Bryan Kelly, Gerardo Manzo and Diogo Palhares, "Credit-Implied Volatility"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 81(2), 89–116 (2025); DOI 10.1080/0015198x.2025.2473251.

Credit-implied volatility: a volatility measure inferred from credit-market prices.

Extended note. Version record

12.1.2 CDS contracts and standard models

ISDA and S&P Global, ISDA CDS Standard Model, source code and test grids

Difficulty: Advanced **Type:** Data / specification

Status/version: Production version 1.8.3, released and effective 7 April 2021; official download and documentation pages checked 4 October 2026.

Supplies the standard single-name CDS valuation implementation and published IBOR/RFR test grids. Use it after O'Kane to reconcile trade dates, accrual, recovery, discount factors, hazard calibration and the conversion between quoted running spread and upfront fee. Reproduce a documented example and compare an independent implementation at multiple maturities and recovery assumptions. The production code version is 2021, while curve inputs and contractual documents have separate revision histories. A matching model output does not prove that a trade confirmation, settlement price or observed bond-CDS basis is correct.

Extended note. Version record

ISDA and S&P Global, RFR Interest Rate Curve Specification–6 Currencies

Difficulty: Advanced **Type:** Data / specification

Status/version: Official RFR curve specification dated 15 June 2026; documentation index checked 4 October 2026; detailed linked document not independently inspected.

Updates the rate-curve input layer used with the CDS Standard Model and makes the RFR transition a concrete calibration task. Match each quoted tenor, instrument type, day count, currency and interpolation rule to the dated specification before constructing discount factors. Reprice an otherwise unchanged CDS under the chosen RFR curve and a historical IBOR setup to isolate rate-input effects from spread, recovery and contract changes. The documentation index verifies a six-currency 15 June 2026 revision; detailed conventions require reading the linked specification and relevant market data before live implementation.

Extended note. Version record

International Swaps and Derivatives Association, 2014 Credit Derivatives Definitions; 2019 NTCE Supplement; Physical Settlement Matrix and Confirmation

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: 2014 Definitions (subscription text); NTCE Supplement published 15 July 2019; ISDA matrix/confirmation page lists 6 October 2025 as most recent version at 4 October 2026 check.

Connects CDS model outputs to the actual credit-event, obligation, delivery and settlement terms that determine a protection payoff. Use the 2014 Definitions as the contract framework, then inspect the 2019 narrowly tailored credit-event amendment and the dated transaction-type matrix before modelling a reference entity. Reconstruct a hypothetical failure-to-pay or restructuring scenario, tracing the credit-event condition to deliverable obligations, auction or physical settlement and payment timing. The 2014 Definitions require a subscription; the public ISDA descriptions and supplement establish scope but do not replace a signed confirmation or legal review.

Extended note. Version record

Mark Paddrik and Stathis Tompaidis, "Intermediation networks and derivative market liquidity: evidence from credit default swap markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, rfag023 (2026); DOI 10.1093/rof/rfag023.

Supervisory single-name CDS data reconstruct dealer networks; Shapley-value connectivity measures predict volumes, inventory management, execution costs and spreads.

Version record

John Hull and Alan White, "Valuing Credit Default Swaps I: No Counterparty Default Risk"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Derivatives, 8(1), 29–40 (2000); DOI 10.3905/jod.2000.319115.

Standard reduced-form CDS valuation by bootstrapping default probabilities from bond prices, the model family behind the ISDA standard.

Version record

Darrell Duffie, "Credit Swap Valuation"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 55(1), 73–87 (1999); DOI 10.2469/faj.v55.n1.2243.

Accessible derivation of CDS spreads from par floating-rate spreads and hazard rates, with the CDS-bond basis intuition.

Version record

Patrick Augustin, Marti G. Subrahmanyam, Dragon Yongjun Tang and Sarah Qian Wang, "Credit Default Swaps: A Survey"

Difficulty: Advanced **Type:** Research paper

Status/version: Foundations and Trends® in Finance, 9(1-2), 1–196 (2014); DOI 10.1561/0500000040.

Comprehensive survey of CDS market structure, pricing, the basis, informational role and real effects.

Version record

Martin Oehmke and Adam Zawadowski, "The Anatomy of the CDS Market"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 30(1), 80–119 (2017); DOI 10.1093/rfs/hhw068.

Position-level DTCC data show CDS used for hedging and for liquidity-motivated speculation in place of illiquid bonds.

Version record

Patrick Bolton and Martin Oehmke, "Credit Default Swaps and the Empty Creditor Problem"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 24(8), 2617–2655 (2011); DOI 10.1093/rfs/hhr002.

Creditors hedged with CDS can be tougher in renegotiation, raising bankruptcy risk; relevant for restructuring and credit-event analysis.

Version record

Navneet Arora, Priyank Gandhi and Francis A. Longstaff, "Counterparty credit risk and the credit default swap market"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 103(2), 280–293 (2012); DOI 10.1016/j.jfineco.2011.10.001.

Dealer-quote data show counterparty risk is priced in CDS but the effect is small given collateralisation.

Version record

Emil N. Siriwardane, "Limited Investment Capital and Credit Spreads"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 74(5), 2303–2347 (2019); DOI 10.1111/jofi.12777.

CDS sellers' capital shocks move spreads: concentrated dealer and fund positions make CDS prices sensitive to intermediary capital.

Version record

Fan Yu, "How Profitable Is Capital Structure Arbitrage?"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 62(5), 47–62 (2006); DOI 10.2469/faj.v62.n5.4282.

Equity-implied CDS spreads versus market spreads: capital-structure arbitrage has hedge-fund-like returns with occasional large losses.

Version record

Nikunj Kapadia and Xiaoling Pu, "Limited arbitrage between equity and credit markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 105(3), 542–564 (2012); DOI 10.1016/j.jfineco.2011.10.014.

Short-horizon equity and CDS price changes are often discordant, consistent with limits to capital-structure arbitrage.

Version record

12.1.3 Credit portfolio models and default prediction

Janet M. Tavakoli, Structured Finance and Collateralized Debt Obligations, 2nd ed. (2008)

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2nd ed. (2008)

CDO collateral, securitisation, tranches, credit derivatives and synthetic structures, with attention to ratings, leverage and discrepancies between model assumptions and contractual reality. Source historical crisis context retained; distinguish the author's commentary from independent validation of every prediction.

Version record

Moorad Choudhry, Structured Credit Products: Credit Derivatives and Synthetic Securitisation, 2nd ed. (2010)

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2nd ed. (2010)

Practical building blocks of credit risk transfer: CDS, CDOs, credit-linked notes and synthetic securitisation. Connect default dependence, tranche pricing and cash-flow waterfalls. The source's "latest developments" description belongs to this historical edition.

Version record

David X. Li, "On Default Correlation: A Copula Function Approach" (2000)

Difficulty: Advanced **Type:** Research paper

Status/version: Selected citation year(s): 2000. Version distinctions remain in the annotation.

Read the original default-time dependence model before its crisis commentary. Examine marginal default distributions, copula and correlation assumptions, tranche sensitivity and stress breakdowns. The 1999 working paper and 2000 journal article are versions of the same work; a convenient dependence model does not establish stable joint tails.

Version record

Felix Salmon, "Recipe for Disaster: The Formula That Killed Wall Street" (Wired, 2009)

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Selected citation year(s): 2009. Version distinctions remain in the annotation.

Journalistic companion to Li on model adoption, correlation assumptions and crisis exposure; distinct from an empirical causal account of the financial crisis.

Applications: Supplementary / journalism.

Version record

Oldrich Vasicek, "Loan portfolio value"

Difficulty: Advanced **Type:** Research paper

Status/version: Risk 15(12), 160–162 (December 2002); earlier KMV working papers 1987/1991.

Single-factor Gaussian limiting loss distribution behind Basel IRB formulas. Published in Risk (December 2002).

Extended note. Version record

Michael B. Gordy, "A risk-factor model foundation for ratings-based bank capital rules"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Intermediation, 12(3), 199–232 (2003); DOI 10.1016/s1042-9573(03)00040-8.

Asymptotic single-risk-factor conditions under which portfolio-invariant capital charges are valid.

Version record

Greg M. Gupton, Christopher C. Finger and Mickey Bhatia, "CreditMetrics: Technical Document"

Difficulty: Advanced **Type:** Report / research note

Status/version: J.P. Morgan, April 1997; later RiskMetrics Group/MSCI hosting.

Rating-migration portfolio credit VaR with asset-correlation simulation (J.P. Morgan 1997).

Version record

Tyler Shumway, "Forecasting Bankruptcy More Accurately: A Simple Hazard Model"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Business, 74(1), 101–124 (2001); DOI 10.1086/209665.

Discrete-time hazard model using market and accounting variables outperforms static scoring.

Version record

John Y. Campbell, Jens Hilscher and Jan Szilagyi, "In Search of Distress Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 63(6), 2899–2939 (2008); DOI 10.1111/j.1540-6261.2008.01416.x.

Reduced-form failure model; distressed stocks earn anomalously low returns.

Version record

Darrell Duffie, Leandro Saita and Ke Wang, "Multi-period corporate default prediction with stochastic covariates"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 83(3), 635–665 (2007); DOI 10.1016/j.jfineco.2005.10.011.

Term structure of default probabilities from distance-to-default and macro covariates.

Version record

Sanjiv R. Das, Darrell Duffie, Nikunj Kapadia and Leandro Saita, "Common Failings: How Corporate Defaults Are Correlated"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 62(1), 93–117 (2007); DOI 10.1111/j.1540-6261.2007.01202.x.

Defaults cluster beyond what conditional intensities imply, rejecting doubly-stochastic correlation models.

Version record

Edward I. Altman, "Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: The Journal of Finance, 23(4), 589–609 (1968); DOI 10.1111/j.1540-6261.1968.tb00843.x.

Z-score discriminant model; historical baseline for default prediction.

Version record

12.1.4 Corporate bond factors and liquidity

Bryan Kelly, Diogo Palhares and Seth Pruitt, Modeling Corporate Bond Returns

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 78(4), 1967–2008, 2023; first published online 24 April 2023; DOI 10.1111/jofi.13233.

Proposes a conditional five-factor model for corporate-bond returns with time-varying loadings and reports stronger fit and out-of-sample portfolio performance than earlier bond models. Read it before the 2026 glass-box ML paper to establish a transparent competing factor benchmark and to separate common credit exposure from bond-level characteristics. Reproduce factor construction and portfolio tests only with a clearly defined TRACE/ICE universe, return alignment and rebalancing clock. The publisher identifies data access through the authors' AQR connection and supplies supporting information and replication code; published Sharpe ratios are not executable net returns.

Extended note. Version record

Alexander Dickerson, Philippe Mueller and Cesare Robotti, Priced Risk in Corporate Bonds; author manuscript

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics 150(2), article 103707, November 2023; author manuscript dated 14 June 2023. Later arXiv upload in 2026 is not the publication date.

Challenges incremental pricing claims for corporate-bond factors using portfolio- and bond-level tests, finding the bond market factor difficult to improve on, with traded liquidity a marginal exception. Its author manuscript traces a lead/lag construction error in previously published bond-factor data and examines sensitivity to return truncation, weighting and robust inference. Read against Kelly–Palhares–Pruitt and subsequent ML results as competing, method-dependent evidence. The authors' public factor portal and versioned replication package give a route to corrected BBW series and published-table checks. Rebuild factor dates before comparing models; a large in-sample cross-sectional fit does not establish a tradable premium after sparse trading, costs or data corrections. The original Bai–Bali–Wen article (Journal of Financial Economics 131(3), 619–642, 2019; DOI 10.1016/j.jfineco.2018.08.002) has been retracted: see the December 2023 retraction notice. Cite corrected series as the authors' replication output, not as support for the retracted article's findings.

Extended note. Version record

Alexander Dickerson, Philippe Mueller and Cesare Robotti, Open Source Bond Asset Pricing; Priced Risk in Corporate Bonds replication package

Difficulty: Advanced **Type:** Data / code / research

Status/version: Author-maintained factor/data portal; Mendeley replication package version 2 published 10 August 2023; inspected 4 October 2026.

Provides a concrete public route to the corrected Bai–Bali–Wen bond-factor series and reproducible tables for Dickerson–Mueller–Robotti. The author site lists processed corporate-bond panels, factor downloads and a TRACE cleaning pipeline; the versioned Mendeley package describes Matlab/Python scripts for published tables. These released series support factor comparison without first reconstructing all raw trades, but are not Kelly–Palhares–Pruitt IPCA factors or a substitute for entitled transaction-level TRACE. The original Bai–Bali–Wen article (Journal of Financial Economics 131(3), 619–642, 2019; DOI 10.1016/j.jfineco.2018.08.002) has been retracted: see the December 2023 retraction notice. Cite corrected series as the authors' replication output, not as support for the retracted article's findings.

Extended note. Version record

Sebastian Bell, Ali Kakhbod, Martin Lettau and Abdolreza Nazemi, "Glass Box Machine Learning and Corporate Bond Returns"

Difficulty: Research **Type:** Research paper

Status/version: 2026 journal; WP 2024–25

Applies interpretable machine learning to the sparse, illiquid corporate-bond cross-section rather than importing equity methods unchanged. The published version is *Journal of Financial Economics* 181, 104294 (July 2026); retain NBER 33320 (December 2024; revised July 2025) for version history. Interpretable forecasts do not establish causality or executable net returns; publisher metadata and accessible sections checked, with full replication unperformed. [7]

Applications: Corporate bonds; ML, cross-sectional alpha. Extended note. Version record

Jack Bao, Jun Pan, and Jiang Wang, The Illiquidity of Corporate Bonds

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 66(3), 911–946, June 2011; first published online 23 May 2011; DOI 10.1111/j.1540-6261.2011.01655.x.

Uses 2003–2009 corporate-bond transactions to show that illiquidity exceeds simple bid–ask measures and explains substantial high-grade spread variation and bond-level spread differences. The historical TRACE sample, trade-size selection, and changing post-crisis market structure mean its coefficients should not be pasted into current executable spread or risk models.

Extended note. Version record

Andrea L. Eisfeldt, Bernard Herskovic, and Shuo Liu, Interdealer Price Dispersion and Intermediary Capacity

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 32998, issued September 2024, DOI 10.3386/w32998; no journal publication asserted.

Connects dispersion of interdealer corporate-bond prices to misallocated credit risk and limited dealer-sector capacity, then relates dispersion to yield spreads, returns, and bond–fair-value basis changes. Interdealer dispersion is an inferred capacity indicator, not a quoted execution cost or independent causal proof; dealer network access, selection of traded bonds, and inventory states matter for implementation.

Extended note. Version record

Naz Koont, Yiming Ma, Lubos Pastor, and Yao Zeng, Steering a Ship in Illiquid Waters: Active Management of Passive Funds

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies 38(10), 2887–2935, October 2025; published online 19 May 2025, corrected/typeset 2 July 2025; DOI 10.1093/rfs/hhaf034.

Shows how corporate-bond ETFs choose creation and redemption baskets to balance index tracking against liquidity transformation, so a nominally passive vehicle makes active bond-level allocation decisions. The paper finds basket inclusion usually improves liquidity but can worsen it during large creation/redemption imbalances; results should not be interpreted as an always-liquid ETF hedge or as a uniform effect across funds and bonds.

Extended note. Version record

Ronen Israel, Diogo Palhares and Scott Richardson, "Common Factors in Corporate Bond Returns"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper (2015); later journal publication not established in this check.

Carry, quality, momentum and value in corporate bonds with transaction-cost-aware construction.

Extended note. Version record

Patrick Houweling and Jeroen van Zundert, "Factor Investing in the Corporate Bond Market"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 73(2), 100–115 (2017); DOI 10.2469/faj.v73.n2.1.

Size, low-risk, value and momentum factors in IG and HY corporate bonds with long-only implementation.

Version record

Antonio Coppola, "In Safe Hands: The Financial and Real Impact of Investor Composition over the Credit Cycle"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(8), 2275–2325 (2025); DOI 10.1093/rfs/hhaf017.

Bonds held mostly by insurers rather than mutual funds suffer milder crisis losses; investor-composition elasticities affect prices and firm investment. RFS Brennan Award 2026.

Version record

Lorenzo Bretscher, Lukas Schmid, Ishita Sen and Varun Sharma, "Institutional Corporate Bond Pricing"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(3), 605–660 (2026); DOI 10.1093/rfs/hhaf067.

Demand-system model of corporate bond pricing estimated on holdings: inelastic insurers in investment grade, elastic mutual funds in high yield.

Version record

Maureen O'Hara, Andreas C. Rapp and Xing (Alex) Zhou, "Bond Market Resiliency: The Role of Insurers"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(5), 1362–1410 (2026); DOI 10.1093/rfs/hhaf097.

Insurers bought corporate bonds that mutual funds fire-sold in March 2020, especially insurers with stable funding and dealer relationships.

Version record

Lev Dynkin, Jay Hyman, Simon Polbennikov, Matt Steinaway, Matt Nest and Arkady Ho, "The Rise in Systematic Credit Investing"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 51(10), 67–84 (2025); DOI 10.3905/jpm.2025.1.756.

Why systematic investing came late to credit and what changed (liquidity, transparency, electronic protocols), with a case study of a rules-based value, momentum and sentiment credit portfolio with low tracking error.

Version record

12.2 Leveraged and distressed credit

Subject context and qualifications.

Edward I. Altman, Edith Hotchkiss and Wei Wang, Corporate Financial Distress, Restructuring, and Bankruptcy: Analyze Leveraged Finance, Distressed Debt, and Bankruptcy, Fourth Edition

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley fourth edition first published 27 February 2019; DOI 10.1002/9781119541929

Adds an integrated institutional treatment of high-yield bonds, leveraged finance, distress prediction, bankruptcy process, creditor recovery, and restructuring outcomes that the list's default-intensity texts do not supply. It is the appropriate foundation before calibrating a distressed bond's spread or comparing bonds with loans across the capital structure. Wiley exposes the contents and chapter summaries; gated book chapters were not represented as fully reviewed.

Extended note. Version record

Viral V. Acharya, Sreedhar T. Bharath and Anand Srinivasan, Does Industry-wide Distress Affect Defaulted Firms? Evidence from Creditor Recoveries; Author-hosted manuscript

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics 85(3), 787–821, September 2007; DOI 10.1016/j.jfineco.2006.05.011

Shows that creditor recoveries in a historical U.S. default sample fall when the debtor's whole industry is distressed, with effects tied to asset specificity, industry concentration, and claim seniority. It challenges fixed recovery percentages in high-yield and loan stress tests: default probability and loss severity can rise together when likely buyers of collateral are also constrained. Use its mechanism to design

sector-wide liquidation scenarios, while recognizing the 1982–1999 sample, present-value recovery measurement, and bankruptcy process differ from today's loan and bond mix.

Extended note. Version record

Victoria Ivashina and Boris Vallée, Weak Credit Covenants

Difficulty: Research **Type:** Research paper

Status/version: Management Science 71(12), 9998–10021, 2025; NBER WP 27316 issued June 2020; selected journal version

Uses 1,240 credit agreements to quantify how carve-outs and deductibles weaken negative covenants even when a headline covenant is present, and connects the measures to the J.Crew restructuring price reaction. Issuance spread and event evidence do not reveal the payoff of every negotiated transaction; apply the contract definitions and amendment history to each facility before estimating recovery or priming exposure.

Extended note. Version record

Robert Prilmeier and René M. Stulz, The Cov-lite Liquidity Advantage, Regulatory Pressures, and the Evolution of the Leveraged Loan Market

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 35617, August 2026; no journal publication asserted

Links the growth of loans without maintenance covenants after the financial crisis to regulatory pressure on bank retention and to a liquidity advantage that can lower loan spreads. Compare its issuance evidence with the contract-level weak-covenant study before attributing spread differences to protection alone. It is a recent working paper, and the observed historical relation is not a guarantee of secondary liquidity under stress.

Extended note. Version record

S&P Dow Jones Indices, S&P UBS Leveraged Loan Index and Methodology; 2024 index-family launch and name history

Difficulty: Advanced **Type:** Data / specification

Status/version: Living index and methodology page; current naming follows 4 December 2024 S&P–UBS launch

Defines the investable, market-value-weighted USD leveraged-loan benchmark and links its methodology and constituent data. It is needed to interpret performance, default rates, discount margins, repricings, and attribution in an asset class where facilities are private and lifecycle events are less transparent than listed bonds. The public index page does not provide all licensed loan-level terms or executable bids for a portfolio of a different size.

Extended note. Version record

FDIC, Federal Reserve Board and Office of the Comptroller of the Currency, 2025 Shared National Credit Program Report

Difficulty: Advanced **Type:** Report / research note

Status/version: Joint supervisory report released 12 January 2026; loans originated by 30 June 2025

Gives a dated supervisory view of large syndicated credits, including leveraged borrowers and non-pass classifications, with bank and nonbank ownership distinctions. It is a useful external check on leveraged-loan risk selection and the denominator behind claims about credit deterioration. It covers shared commitments above the program threshold, not the whole private-credit or high-yield bond market, and classifications are supervisory judgments rather than traded prices.

Extended note. Version record

ICE Swap Trade, Markit CDX North American High Yield Index Options Contract Specifications

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: ICE OTC contract specification dated 15 June 2026; verify live series and rule changes

Lists the traded CDX.NA.HY option's series, roll, tenor, currency, quotation, and notional rules. It extends the list's generic multi-name-credit models to an actual high-yield index hedge, where a series roll or option expiry can create basis against cash bonds and leveraged loans. Exchange terms alone cannot establish cash-bond hedge effectiveness, and the selected 2026 PDF must be superseded for later trades.

Extended note. Version record

Stuart C. Gilson, *Creating Value through Corporate Restructuring: Case Studies in Bankruptcies, Buyouts, and Breakups*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, second edition, 2010 (ISBN 9780470502143); the former Wiley product page no longer resolves.

Harvard case studies of Chapter 11, exchange offers, distressed investing and spin-offs; second edition 2010.

Version record

Stephen G. Moyer, *Distressed Debt Analysis: Strategies for Speculative Investors*

Difficulty: Advanced **Type:** Book / monograph

Status/version: J. Ross Publishing, 2004 (ISBN 9781932159189).

Practitioner text on capital-structure analysis, valuation in bankruptcy, plan negotiation and recovery estimation.

Version record

Edward I. Altman, "Measuring Corporate Bond Mortality and Performance"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 44(4), 909 (1989); DOI 10.2307/2328616.

Introduces cohort mortality-rate tables for high-yield bonds, the basis of rating-agency default studies and loss-adjusted spread analysis.

Version record

Edward I. Altman, Brooks Brady, Andrea Resti and Andrea Sironi, "The Link between Default and Recovery Rates: Theory, Empirical Evidence, and Implications"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Business, 78(6), 2203–2228 (2005); DOI 10.1086/497044.

Recovery rates fall when default rates rise, so credit-loss models that assume independence understate tail risk.

Version record

Rainer Jankowitsch, Florian Nagler and Marti G. Subrahmanyam, "The determinants of recovery rates in the US corporate bond market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 114(1), 155–177 (2014); DOI 10.1016/j.jfineco.2014.06.001.

TRACE trades around default: recoveries depend on bond characteristics, firm fundamentals, macro conditions and liquidity.

Version record

Gregor Andrade and Steven N. Kaplan, "How Costly is Financial (Not Economic) Distress? Evidence from Highly Leveraged Transactions that Became Distressed"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 53(5), 1443–1493 (1998); DOI 10.1111/0022-1082.00062.

Distressed LBOs that were economically sound lost 10–20% of value to financial distress; a benchmark estimate of distress costs.

Version record

Edith S. Hotchkiss and Robert M. Mooradian, "Vulture investors and the market for control of distressed firms"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 43(3), 401–432 (1997); DOI 10.1016/s0304-405x(96)00900-2.

Distressed-debt investors who gain control improve post-restructuring performance; the governance role of vulture funds.

Version record

Wei Jiang, Kai Li and Wei Wang, "Hedge Funds and Chapter 11"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 67(2), 513–560 (2012); DOI 10.1111/j.1540-6261.2012.01724.x.

Hedge funds in Chapter 11 influence outcomes: their presence raises emergence probability and junior-claim recoveries.

Version record

Victoria Ivashina and Zheng Sun, "Institutional demand pressure and the cost of corporate loans"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 99(3), 500–522 (2011); DOI 10.1016/j.jfineco.2010.10.009.

Institutional (CLO and fund) demand for leveraged loans lowers spreads at syndication, an early demand-system result for loans.

Version record

Amir Kermani and Yueran Ma, "Asset Specificity of Nonfinancial Firms"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 138(1), 205–264 (2022); DOI 10.1093/qje/qjac030.

Hand-collected liquidation values show low and industry-specific recoveries of physical assets, explaining differences in recovery rates.

Version record

Michael Schwert, "Does Borrowing from Banks Cost More than Borrowing from the Market?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(2), 905–947 (2020); DOI 10.1111/jofi.12849.

Bank loans carry a premium over equivalent bonds after adjusting for seniority and covenants, consistent with bank market power.

Version record

12.3 Covered bonds

Subject context and qualifications.

European Union, Directive (EU) 2019/2162 on covered bonds

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Official Journal text, 2019; read Articles 4–6 and 10–17 alongside applicable national implementing law

Defines dual recourse, bankruptcy remoteness, eligible cover assets, segregation, coverage and liquidity buffers. Article 17 confines extendable maturity to objective triggers and specifies its effect on claims and priority. Prerequisite: distinguish the issuer's unsecured estate from the cover-pool estate and map asset collections, overcollateralisation and extension dates. The Directive establishes an EU framework, not identical national collateral eligibility, supervisory practice or a quoted fair spread.

Extended note. Version record

Bank for International Settlements, The Covered Bond Market

Difficulty: Not specified **Type:** Report / research note

Status/version: BIS Quarterly Review, September 2007; public PDF

Explains how issuer credit, cover-pool quality, legal protection and liquidity enter the same instrument and surveys the historic European market. Build a spread decomposition against senior bank debt and government bonds, then test the residual for issue size, market-making and eligible-collateral effects. The sample predates the EU covered-bond Directive, post-crisis bank resolution and today's market, so its observed spreads are a conceptual starting point, not calibration targets.

Extended note. Version record

Chris Kenyon, Pricing Covered Bonds

Difficulty: Research **Type:** Research paper

Status/version: C.R.E.D.I.T. 2009 proceedings paper; author-posted SSRN version revised August 2014

Models a cover pool whose assets are replaced after specified credit events until the issuer's liquidity is exhausted, then adds issuer default in a factor-copula pricing framework. Prerequisite: portfolio loss dependence, credit-event triggers and risk-neutral valuation. The model analyses hypothetical entity-level pools and idealised replacement rules; it does not deliver deal-specific mortgage cash flows, soft-bullet extension timing or a current spread calibration.

Extended note. Version record

Dirk Tasche, Fitting a Distribution to Value-at-Risk and Expected Shortfall, with an Application to Covered Bonds; author preprint

Difficulty: Research **Type:** Research paper

Status/version: Journal of Credit Risk 12(2), 77–111, June 2016; public author preprint

Compares one- and two-asset structural loss models that separate the cover pool from the issuer's remaining estate and make the unsecured residual claim explicit. Shows that plausible cover-pool and issuer PD/LGD inputs need not identify a unique two-asset calibration. Prerequisite: joint asset-value distributions and expected shortfall. Its one-period loss model is not a traded-price term structure, stochastic cover-pool replenishment or a soft-bullet cash-flow engine.

Extended note. Version record

Marcel Prokopczuk, Jan B. Siewert and Volker Vonhoff, Credit Risk in Covered Bonds

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance 21, 102–120, March 2013; publisher abstract, gated full text

Uses hand-collected German cover-pool disclosures and bond spreads to separate issuer, collateral and liquidity effects. It rejects the shortcut of interpreting the full spread to Bunds as a liquidity premium: cover-pool credit quality also mattered, especially in the financial and sovereign crises. The data and estimated coefficients are historical German Pfandbrief evidence, not a current cross-country spread model; full derivations were not accessible in the publisher preview.

Extended note. Version record

S&P Global Ratings, Credit Rating Model: Covered Bond Monitor; Methodology for Rating Covered Bonds

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Model and criteria dated 25 July 2025; official criteria page

Quantifies whether cover-pool cash flows, overcollateralisation and refinancing under stressed spreads can cover liabilities after issuer failure, including maturity mismatch and extension features. Prerequisite: loan prepayments, default and recovery, swap counterparty and legal-maturity data. The public model description and criteria are rating tools, not executable code or a market fair-value curve; the detailed model and loan tape are proprietary, and some criteria content may require access.

Extended note. Version record

CCF Société de Financement de l'Habitat, Base Prospectus, Soft Bullet Covered Bonds provisions

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Issuer programme prospectus dated 16 December 2024; read Condition 7(a), risk factors and issue-specific final terms

Specifies a trigger-dependent move from scheduled to extended legal maturity, continued interest on unpaid principal and no holder acceleration solely from extension. Prerequisite: discounting delayed principal and distinguishing payment default from an authorised extension. Create hard-bullet and triggered soft-bullet cash-flow scenarios, then test duration and liquidity under stressed refinancing. Final terms and later programme amendments control the live trade.

Extended note. Version record

Toni Ahnert, Covered Bonds as a Source of Funding for Banks' Mortgage Portfolios

Difficulty: Advanced **Type:** Report / research note

Status/version: Bank of Canada Financial System Review, June 2018; public PDF

Connects dynamic cover-pool replenishment and overcollateralisation to cheaper mortgage funding, while tracing the cost imposed on residual unsecured creditors as assets become encumbered. Prerequisite: understand bank capital structure and the mechanics of dual recourse. The discussion is anchored in Canadian rules and historical evidence; it does not supply a universal optimal issuance share or a current Canadian programme valuation.

Extended note. Version record

Australian Prudential Regulation Authority, APS 121 Covered Bonds

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Prudential standard effective 1 January 2023; official current standards page as checked 4 October 2026

Shows how Australian cover-pool maintenance, prudential supervision and restrictions on issuing banks differ from the EU framework. Read the actual standard and the deal's programme documents after the EU Directive; jurisdiction is a model input, not a footnote. APRA rules govern Australian authorised deposit-taking institutions; they do not replace an issue's contract or resolve investor recovery in another country.

Extended note. Version record

European Banking Authority, Advice on the Review of the EU Covered Bond Framework

Difficulty: Not specified **Type:** Report / research note

Status/version: Advice published 23 September 2025; press release links the full advice

Assesses implementation of the EU framework and recommends more harmonisation, investor transparency and a third-country equivalence regime. It also considers green cover pools and a possible SME dual-recourse-like instrument. Prerequisite: distinguish Directive provisions from proposals. It is supervisory advice, not enacted EU law; the proposed changes and SME concept should not be coded as current covered-bond rights.

Extended note. Version record

12.4 Bank capital, hybrids and preferred shares

Subject context and qualifications.

Basel Committee on Banking Supervision, Basel Framework, CAP10 Definition of Capital

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Basel Framework online chapter, consulted 4 October 2026

Sets the regulatory distinction: Additional Tier 1 is perpetual and absorbs losses through contractual conversion or write-down, with fully discretionary non-cumulative distributions; Tier 2 is gone-concern capital with a minimum original maturity and declining recognition near maturity. Prerequisite: common-equity capital ratios and bank resolution hierarchy. Basel eligibility is a standard for supervisors, not a substitute for local implementation, resolution powers or the security's governing documents.

Extended note. Version record

European Banking Authority, Monitoring of AT1, T2 and TLAC/MREL Instruments: Update

Difficulty: Advanced **Type:** Report / research note

Status/version: EBA supervisory report, June 2024; public PDF

Reviews issuance terms, calls, triggers and loss-absorption language against EU capital and resolution requirements. It clarifies why a first-call date is not maturity and why capital recognition and investor payoff require different analyses. Read after CAP10 and before modeling callable spread duration. This report is supervisory monitoring at a particular date, not a pricing model or a ruling on every outstanding instrument; verify subsequent law and issue terms.

Extended note. Version record

Swiss Financial Market Supervisory Authority, FINMA Provides Information about the Write-down of Credit Suisse AT1 Capital Instruments

Difficulty: Not specified **Type:** Report / research note

Status/version: Official release, 23 March 2023

Documents FINMA's stated contractual and emergency-law basis for the Credit Suisse AT1 write-down and distinguishes the treatment of Tier 2. Prerequisite: compare the actual prospectus trigger with the jurisdiction's extraordinary support and resolution powers. The Swiss Federal Administrative Court later revoked FINMA's order in a 1 October 2025 partial decision; FINMA appealed. Read the following court entry before treating FINMA's 2023 account as the current legal conclusion.

Extended note. Version record

Swiss Federal Administrative Court, AT1 partial judgment release; judgment; 22 October stay order; FINMA appeal notice and decision register

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: B-2334/2023, 1 October 2025; release 14 October; official texts/register inspected through 4 October 2026.

The partial judgment annulled FINMA's decree for lack of a contractual viability event and adequate legal basis, reserving whether/how the write-down should be reversed. The 22 October order stayed the other administrative-court AT1 cases pending the Federal Supreme Court's outcome; FINMA's register records an appeal. The administrative stay is not itself a Supreme Court ruling on enforceability; the 2025 legal victory does not establish restored claims.

Extended note. Version record

José Manuel Corcuera, Jan De Spiegeleer, José Fajardo, Henrik Jönsson, Wim Schoutens and Arturo Valdivia, Close Form Pricing Formulas for Coupon Cancellable CoCos

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance 42, 339–351, May 2014; publisher abstract, gated article

Derives a first-passage valuation for conversion-to-equity CoCos with a separate coupon-cancellation barrier. It shows why a single default spread cannot represent both lost coupons and principal/equity conversion. Prerequisite: barrier options, stock-process calibration and contractual capital triggers. The paper substitutes observable market-price barriers for accounting-capital events and does not cover every write-down contract; the publisher's abstract and preview, not full derivations, were accessible.

Extended note. Version record

Axel Kind, Philippe Oster and Franziska J. Peter, The Determinants of Banks' AT1 CoCo Spreads

Difficulty: Research **Type:** Research paper

Status/version: European Financial Management 28(2), 567–604, March 2022; online 29 April 2021

Studies euro-area AT1 spreads with regulatory, instrument and market variables and finds the maximum-distributable-amount framework economically relevant to prices. Prerequisite: understand the difference between a contractual capital-ratio conversion point and an earlier restriction on distributions. The sample is euro-area and period-specific, with publisher abstract and metadata public; full data and model estimates may require journal access.

Extended note. Version record

Christian Koziol and Philipp Roßmann, Contingent Convertible Bonds: Optimal Call Strategy and the Impact of Refinancing

Difficulty: Research **Type:** Research paper

Status/version: Journal of Corporate Finance 77, article 102277, December 2022; publisher article and abstract

Models the issuer's call decision within its whole capital structure and shows that refinancing with equity, senior or junior debt changes the optimal call boundary. It tests the effect on 79 European CoCos. Prerequisite: callable debt, refinancing constraints and equity-value maximisation. Its empirical securities are European listed-bank CoCos observed as of March 2019; the structural assumptions and sample do not forecast every post-2023 call.

Extended note. Version record

S&P Global Ratings, Hybrid Capital: Methodology and Assumptions

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Criteria issued 13 October 2025; public page republished 11 December 2025 for a nonmaterial correction

Provides a cross-sector framework for how subordination, coupon deferral, conversion or write-down, permanency and issuer intent affect hybrid equity content and issue ratings. Read after cash-flow seniority and call-option basics. It makes corporate hybrids comparable to, but distinct from, bank regulatory capital. Rating-agency equity content is a proprietary credit opinion, not an accounting conclusion, legal priority opinion or fair-value model; inspect the live criteria and issue documentation.

Extended note. Version record

Ameren Corporation, Junior Subordinated Notes due 2057: Final Prospectus Supplement

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final Rule 424(b)(2) prospectus supplement dated 8 September 2026.

A corporate-hybrid case linking ranking, coupon reset, cumulative deferral and event calls. The initial 6.450% coupon resets from March 2032 to five-year Treasury plus 1.868%, subject to the initial-rate floor. Permitted deferrals can last ten years and compound semiannually; they differ from permanent coupon cancellation. Read the subordination, redemption and deferral clauses together. Model cash timing and issuer choice separately from rating-agency equity content; the first call window is not promised repayment. This prospectus does not establish a live quote or realised investor return.

Extended note. Version record

IFRS Foundation, IAS 32 Financial Instruments: Presentation

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Standard hub, consulted 4 October 2026; full issued text may require access

Classifies a preferred instrument by its contractual obligation to deliver cash or another financial asset, rather than by its name. Redemption obligations and issuer discretion over dividends can therefore place superficially similar preferred shares on different sides of the liability/equity boundary. IAS 32 classification neither determines bank regulatory capital nor guarantees that a preferred holder ranks as the issuer reports in a non-IFRS jurisdiction.

Extended note. Version record

David Emanuel, A Theoretical Model for Valuing Preferred Stock

Difficulty: Not specified **Type:** Research paper

Status/version: The Journal of Finance 38(4), 1133–1155, September 1983; publisher abstract, full text gated

Uses contingent-claim valuation to price preferred stock when dividends can be omitted, deriving distinct results for cumulative and non-cumulative rights. It explains why neither a riskless perpetuity nor a simple risky bond fully captures preferred dividends. The model is a theoretical baseline from 1983, not a modern bank-resolution or tax calibration; the publisher abstract and metadata were verified, while full derivations may need access.

Extended note. Version record

Mark J. Flannery, "Contingent Capital Instruments for Large Financial Institutions: A Review of the Literature"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 6(1), 225–240 (2014); DOI 10.1146/annurev-financial-110613-034331.

Review of CoCo design (triggers, conversion terms, death spirals) and evidence, written by the proponent of the idea.

Version record

Suresh Sundaresan and Zhenyu Wang, "On the Design of Contingent Capital with a Market Trigger"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 70(2), 881–920 (2015); DOI 10.1111/jofi.12134.

Market-price-triggered CoCos may admit no or multiple equilibrium prices unless conversion avoids wealth transfers; key design caution.

Version record

Paul Glasserman and Behzad Nouri, "Contingent Capital with a Capital-Ratio Trigger"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 58(10), 1816–1833 (2012); DOI 10.1287/mnsc.1120.1520.

Valuation of CoCos that convert gradually on a book capital-ratio trigger, with closed-form results under a structural model.

Version record

Stefan Avdjiev, Bilyana Bogdanova, Patrick Bolton, Wei Jiang and Anastasia Kartasheva, "CoCo issuance and bank fragility"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 138(3), 593–613 (2020); DOI 10.1016/j.jfineco.2020.06.008.

CoCo issuance lowers issuers' CDS spreads, more so for equity-conversion and high-trigger designs.

Version record

Jarl G. Kallberg, Crocker H. Liu and Sriram Villupuram, "Preferred stock: Some insights into capital structure"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Corporate Finance, 21, 77–86 (2013); DOI 10.1016/j.jcorpfin.2013.01.005.

Who issues preferred stock and why: preferred as a hybrid between debt and equity in capital structure.

Version record

12.5 Sukuk

Subject context and qualifications.

Islamic Financial Services Board, IFSB-19: Guiding Principles on Disclosure Requirements for Islamic Capital Market Products

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Standard issued April 2017; public PDF in migrated IFSB archive

Requires investors to examine structure, underlying assets, ownership, payment sources, recourse, default and enforcement terms, and Sharia governance. Prerequisite: distinguish beneficial ownership and economic exposure from a claim on the obligor under a purchase undertaking. IFSB-19 is disclosure guidance, not proof that an asset transfer is enforceable in insolvency or that a court and Sharia board will interpret the documents identically.

Extended note. Version record

International Monetary Fund staff, Islamic Finance, Consumer Protection, and Financial Stability

Difficulty: Advanced **Type:** Research paper

Status/version: IMF Working Paper 15/107, 2015; public abstract and paper

Separates structures with a substantive true sale and asset recourse from asset-based certificates where an obligor's purchase undertaking supplies the economic repayment. It discusses the East Cameron and Investment Dar defaults as different enforcement and restructuring paths. The case outcomes depend on the governing contracts and courts; IMF staff research is not a legal opinion on a live sukuk, and asset ownership and enforceability require transaction-specific counsel.

Extended note. Version record

Sajjad Zaheer and Sweder van Wijnbergen, Sukuk Defaults: On Distress Resolution in Islamic Finance

Difficulty: Research **Type:** Research paper

Status/version: Qualitative Research in Financial Markets 17(2), 292–311, 2025; final journal article in University of Amsterdam repository

Reconstructs the East Cameron, Investment Dar and Saad/Golden Belt defaults from offering documents and subsequent resolution paths, exposing differences between stated ownership, enforceable title, purchase undertakings and cross-border recourse. Prerequisite: the asset-backed versus asset-based distinction and a legal-entity cash-flow map. Three historical, selected defaults cannot estimate population default frequencies or current recovery rates; some underlying offering and court materials require independent transaction-level verification.

Extended note. Version record

Yusuf Ayturk, Mehmet Asutay and Ercument Aksak, What Explains Corporate Sukuk Primary Market Spreads?; author-university record

Difficulty: Research **Type:** Research paper

Status/version: Research in International Business and Finance 40, 141–149, April 2017; accepted manuscript on Durham record

Studies 63 international corporate sukuk issues from 2004–2015. Credit rating and maturity help explain issue spreads, while scholar reputation and broad sukuk contract type are not significant in this small sample. Prerequisite: cross-sectional identification and consistent primary-market spread benchmarks. The sample is small, historical and subject to issuer and structure selection; insignificance of a broad type dummy does not make transaction ownership or default recourse irrelevant.

Extended note. Version record

Nawaf Almaskati, Sukuk versus Bonds: New Evidence from the Primary Market

Difficulty: Research **Type:** Research paper

Status/version: Borsa Istanbul Review 22(5), 1033–1038, September 2022; open-access journal article

Uses propensity-score matching on primary issues from 2000–2021 and estimates lower sukuk coupons than comparable conventional bonds, with estimates varying by matching method. Prerequisite: distinguish new-issue coupon from secondary yield and check issuer, currency, tenor and time balance. The paper proposes demand from Islamic investors as an explanation, not a separately identified causal channel; matched observables cannot remove every contract or liquidity difference.

Extended note. Version record

Islamic Financial Services Board, IFSB-23: Revised Capital Adequacy Standard for Institutions Offering Islamic Financial Services (Banking Segment), especially §6

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Adopted December 2021; IFSB recommends national implementation from January 2023

Updates the prudential treatment of sukuk holdings and exposures as originator, issuer, servicer or credit-enhancement provider, including true sale, risk transfer, liquidity facilities and retained securitisation positions. IFSB-23 revises earlier IFSB guidance but is a standard for Islamic banks, not a universal investor risk weight or evidence of actual national adoption; check the applicable local rules.

Extended note. Version record

Islamic Financial Services Board, Islamic Financial Services Industry Stability Report 2025

Difficulty: Not specified **Type:** Report / research note

Status/version: Annual report, May 2025; public PDF

Documents the recent mix of wakalah, hybrid and other sukuk structures and discusses trading constraints for debt-based components. Prerequisite: understand how the composition of tangible assets versus receivables can affect Sharia tradability under relevant interpretations. The report aggregates heterogeneous markets and external data; its structure shares and liquidity observations are not security-level bid depth or a universal Sharia ruling.

Extended note. Version record

Christophe J. Godlewski, Rima Turk-Ariss and Laurent Weill, "Sukuk vs. conventional bonds: A stock market perspective"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Comparative Economics, 41(3), 745–761 (2013); DOI 10.1016/j.jce.2013.02.006.

Event study: Malaysian stock prices react negatively to sukuk issuance announcements but not to conventional bond issues.

Version record

Paul-Olivier Klein and Laurent Weill, "Why do companies issue sukuk?"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Economics, 31(1), 26–33 (2016); DOI 10.1016/j.rfe.2016.05.003.

Issuer-level determinants of choosing sukuk over conventional bonds.

Version record

Saad Azmat, Michael Skully and Kym Brown, "The (little) difference that makes all the difference between Islamic and conventional bonds"

Difficulty: Advanced **Type:** Research paper

Status/version: Pacific-Basin Finance Journal, 42, 46–59 (2017); DOI 10.1016/j.pacfin.2015.12.010.

Contrasts asset-based and asset-backed sukuk with conventional bonds: most sukuk are economically close to bonds, which shapes their credit analysis.

Version record

Houcem Smaoui and Mohsin Khawaja, "The Determinants of Sukuk Market Development"

Difficulty: Advanced **Type:** Research paper

Status/version: Emerging Markets Finance and Trade, 53(7), 1501–1518 (2017); DOI 10.1080/1540496x.2016.1224175.

Cross-country macroeconomic and institutional drivers of sukuk market development.

Version record

Andrea Paltrinieri, M. Kabir Hassan, Salman Bahoo and Ashraf Khan, "A bibliometric review of sukuk literature"

Difficulty: Advanced **Type:** Research paper

Status/version: International Review of Economics & Finance, 86, 897–918 (2023); DOI 10.1016/j.iref.2019.04.004.

Bibliometric map of sukuk research by theme, a starting list for further reading.

Version record

Accounting and Auditing Organization for Islamic Financial Institutions (AAOIFI), "Shari'ah Standards, including Standard No. 17 (Investment Sukuk)"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: AAOIFI Shari'ah Standards (consolidated edition); access via AAOIFI.

The standard-setter's definitions of sukuk types and permissible structures; read with IFSB capital standards for regulatory treatment.

Extended note. Version record

12.6 Synthetic risk transfer

Subject context and qualifications.

Basel Committee on Banking Supervision, Synthetic Risk Transfers

Difficulty: Not specified **Type:** Report / research note

Status/version: BCBS report, February 2026; public PDF

Surveys synthetic protection on loans that remain with the originating bank, tranche design, capital relief, protection sellers and supervisory concerns. Prerequisite: portfolio loss distributions, credit derivatives and securitisation capital treatment. Use the report to build a reference-pool-to-tranche bridge and a checklist for retained risk, counterparty exposure and capital recognition. It is a cross-jurisdictional supervisory survey, not an individual bank's approved relief, complete transaction database or model for a quoted tranche premium.

Version record

Bank for International Settlements, The Rise and Risks of Synthetic Risk Transfers

Difficulty: Advanced **Type:** Research paper

Status/version: BIS Quarterly Review, March 2026; public article

Compares direct bank-issued credit-linked notes, SPV-issued collateralised notes and unfunded guarantees or credit default swaps. Its transaction diagrams expose collateral location, premium and collateral return, bank-provided investor financing and potential recycling of risk into the banking system. The article distinguishes the broad EU/UK regulatory phrase "significant risk transfer" from the narrower synthetic mechanism. Its market aggregates and illustrative structures do not resolve the economics of a private deal.

Extended note. Version record

Board of Governors of the Federal Reserve System, Regulation Q Frequently Asked Questions: Credit-Linked Notes and Synthetic Securitisation

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official supervisory FAQ, 28 September 2023; consult current page and later rules

Explains how collateralised SPV credit-linked notes and direct bank-issued notes can differ for US securitisation capital treatment; relief for a direct issue is not automatic and may require case-specific supervisory action. Prerequisite: separate credit-risk transfer from the regulatory recognition of that transfer. The FAQ is US-specific guidance at its publication date, not a cross-border approval or a guarantee of relief for a proposed structure.

Extended note. Version record

Alex Osberghaus and Glenn Schepens, Synthetic, but How Much Risk Transfer?

Difficulty: Research **Type:** Research paper

Status/version: ECB Working Paper 3210, March 2026; selected ECB version, public PDF

Uses transaction and loan-level data to study which loans enter SRT reference portfolios, how issuing banks redeploy capital and how monitoring changes after transfer. It tests the claim that regulatory capital relief equals complete economic risk transfer. The empirical identification and confidential data cover the observed market and period; the working paper is not a universal causal parameter for future issuers or a public deal-level loss model.

Extended note. Version record

Fabio Cortes, Fernando Fernandez Dionis, Yi Li, Andres Ramirez and Mengxue Zhang (authors as listed by IMF), "Recycling Risk: Synthetic Risk Transfers"

Difficulty: Advanced **Type:** Report / research note

Status/version: IMF Working Paper 2025/200; verify author list on the IMF page.

IMF review of SRT growth (over USD 1 trillion securitised since 2016), common structures, investor base and regulatory treatment across jurisdictions.

Extended note. Version record

European Banking Authority, "Report on significant risk transfer in securitisation"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: EBA/Rep/2020/32, November 2020.

Supervisory assessment of SRT tests, structural features and recommendations for consistent capital relief (November 2020).

Extended note. Version record

Darrell Duffie, "Innovations in credit risk transfer: implications for financial stability"

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Working Papers No. 255, July 2008.

BIS overview of how CDS, CDOs and loan sales transfer bank credit risk and where incentive and stability problems arise.

Extended note. Version record

Franklin Allen and Elena Carletti, "Credit risk transfer and contagion"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 53(1), 89–111 (2006); DOI 10.1016/j.jmoneco.2005.10.004.

Credit risk transfer from banks to insurers can create contagion through shared asset markets.

Version record

Gary B. Gorton and George G. Pennacchi, "Banks and loan sales: Marketing nonmarketable assets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 35(3), 389–411 (1995); DOI 10.1016/0304-3932(95)01199-x.

Implicit guarantees and retained shares let banks sell loans despite moral hazard; the incentive logic behind risk-retention rules in SRT.

Version record

Christine A. Parlour and Andrew Winton, "Laying off credit risk: Loan sales versus credit default swaps"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 107(1), 25–45 (2013); DOI 10.1016/j.jfineco.2012.08.004.

When banks prefer CDS (keeping control rights) to loan sales, and the monitoring consequences.

Version record

Bernadette A. Minton, René Stulz and Rohan Williamson, "How Much Do Banks Use Credit Derivatives to Hedge Loans?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Services Research, 35(1), 1–31 (2009); DOI 10.1007/s10693-008-0046-3.

Few US banks used credit derivatives to hedge loans before 2008, and mostly for large investment-grade exposures.

Version record

Joshua Coval, Jakub Jurek and Erik Stafford, "The Economics of Structured Finance"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 23(1), 3–25 (2009); DOI 10.1257/jep.23.1.3.

Tranching concentrates systematic risk and amplifies parameter error; senior tranches are economic catastrophe bonds. Core intuition for SRT tranche risk.

Version record

12.7 Sustainable and transition debt

Subject context and qualifications.

International Capital Market Association, Green Bond Principles, Social Bond Principles and Sustainability Bond Guidelines

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: ICMA voluntary guidance; selected GBP and SBP June 2025, SBG June 2021

Separates environmental, social and combined eligible use of proceeds and the four process components: proceeds, project evaluation, management of proceeds and reporting. Prerequisite: distinguish an issuer's general credit from earmarked project allocation and impact metrics. These voluntary principles do not by themselves ring-fence collateral, increase bondholder seniority or specify a monetary remedy; enforceable obligations are in the issue documents and applicable law.

Extended note. Version record

International Capital Market Association, Sustainability-Linked Bond Principles

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: ICMA voluntary guidance, June 2024; public PDF

Defines the KPI, sustainability performance target, bond-characteristic change, reporting and external-verification components of an issuer-level sustainability-linked bond. Unlike a use-of-proceeds bond, proceeds may fund general corporate purposes. A labelled bond can meet the voluntary process framework while offering little financial penalty, and the principles alone do not establish that the target will be met or cause emissions reductions.

Extended note. Version record

International Capital Market Association, Climate Transition Bond Guidelines and Climate Transition Finance Handbook

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Selected November 2025 guidance; June 2026 FAQ linked from ICMA's climate-transition hub

Introduces a standalone use-of-proceeds climate-transition bond label and pairs project safeguards with entity-level transition strategy and disclosure. It addresses financing in high-emitting sectors that a simple green-project screen may miss. Prerequisite: sector decarbonisation pathways and carbon-lock-in risk. The guidelines are voluntary and evolving; a self-described "transition" bond need not conform, and an eligible project does not establish the issuer's net emissions path.

Extended note. Version record

European Union, Regulation (EU) 2023/2631 on European Green Bonds

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official Journal regulation; European Green Bond Standard applies from 21 December 2024

Makes the European Green Bond designation conditional on EU Taxonomy-related allocation, disclosure templates and external review, with defined flexibility and supervision. Read after the ICMA principles so a regulated EuGB label is not confused with a voluntary green-bond label. Rules and delegated measures evolve; the regulation governs use of its designation, not all bonds marketed as green or the issuer's entire environmental footprint.

Extended note. Version record

Caroline Flammer, Corporate Green Bonds; author-hosted journal PDF

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics 142(2), 499–516, 2021; selected journal version

Studies corporate green issuance, investor response and subsequent environmental outcomes. In its matched same-issuer bond comparison, the estimated green-bond yield difference is not statistically significant, even while some issuer outcomes improve. Its sample, issuer-selection controls and time period matter; observed corporate changes do not prove that every financed project was additional or that today's primary-market premium is zero.

Extended note. Version record

Pauline Lam and Jeffrey Wurgler, Green Bonds: New Label, Same Projects

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 32960, September 2024; selected working-paper version

Classifies the novelty of projects financed by US corporate and municipal green bonds and examines whether pricing and ownership distinguish more additional projects. It makes environmental additionality a separate empirical outcome from compliant allocation. The estimated small share of newly novel features depends on its US sample and classification method; it is not a global impact estimate or a conclusion that all refinancing lacks benefit.

Extended note. Version record

Peter Feldhütter, Kristoffer Halskov and Arthur Krebbers, Pricing of Sustainability-Linked Bonds

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics 162, article 103944, 2024; public journal PDF

Values the bond's sustainability-linked payoff and estimates a small pure label premium after accounting for the contractual coupon step-up, while extracting market-implied target-miss probabilities. Prerequisite: discount contingent cash flows and distinguish a step-up's present value from a spread concession. The sample and target-probability inference depend on model and disclosure quality; correct pricing does not establish that targets are ambitious or drive real decarbonisation.

Extended note. Version record

Fabio Fornari, Daniele Pianeselli and Andrea Zaghini, Environmental Score and Bond Pricing: It Better Be Good, It Better Be Green

Difficulty: Research **Type:** Research paper

Status/version: ECB Working Paper 3176, 22 January 2026; official author page links paper

Matches new green issues to conventional bonds and finds a baseline green-label yield premium of about 16 basis points, larger for issuers with high environmental scores and stronger project credentials. Prerequisite: distinguish primary-issue matching from secondary-market OAS and examine environmental-score construction. This working-paper estimate is conditional on its sample, matching and proprietary score inputs; it need not agree numerically with Flammer's different sample or identify causal environmental impact.

Extended note. Version record

Olivier David Zerbib, "The effect of pro-environmental preferences on bond prices: Evidence from green bonds"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 98, 39–60 (2019); DOI 10.1016/j.jbankfin.2018.10.012.

Matched green and conventional bonds show a small negative yield premium (greenium).

Version record

David F. Larcker and Edward M. Watts, "Where's the greenium?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Accounting and Economics, 69(2-3), 101312 (2020); DOI 10.1016/j.jacceco.2020.101312.

Counter-evidence: in municipal bonds from the same issuer, the greenium is essentially zero.

Extended note. Version record

Malcolm Baker, Daniel Bergstresser, George Serafeim and Jeffrey Wurgler, "The Pricing and Ownership of US Green Bonds"

Difficulty: Research **Type:** Research paper

Status/version: Annual Review of Financial Economics, 14(1), 415–437 (2022); DOI 10.1146/annurev-financial-111620-014802.

Green municipal and corporate bonds price at a premium and are held more concentrated by green-preferring investors.

Version record

Dragon Yongjun Tang and Yupu Zhang, "Do shareholders benefit from green bonds?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Corporate Finance, 61, 101427 (2020); DOI 10.1016/j.jcorpfin.2018.12.001.

Stock prices react positively to green bond announcements, with institutional ownership and liquidity effects.

Version record

Stefania D'Amico, Johannes Klausmann and N. Aaron Pancost, "The benchmark greenium"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 176, 104217 (2026); DOI 10.1016/j.jfineco.2025.104217.

Estimates a benchmark greenium from closely matched green and conventional bonds.

Extended note. Version record

12.8 Mortgage valuation and securitisation

Read collateral behaviour, servicing incentives and deal waterfalls together. Residential prepayment, commercial-property refinancing, consumer-loan selection and CLO reinvestment generate different cash-flow risks. OAS and tranche spreads remain conditional on the selected model and documents. Subject context and qualifications.

James Vickery and Joshua Wright, TBA Trading and Liquidity in the Agency MBS Market; 2013 full EPR issue

Difficulty: Advanced **Type:** Research paper

Status/version: Federal Reserve Bank of New York Economic Policy Review 19(1), 1–18, May 2013; predecessor Staff Report 468 first August 2010, revised 2014. Cite the 2013 EPR article as the selected version.

Explains the agency MBS to-be-announced convention, where a forward trade specifies eligible pool characteristics but leaves exact deliverables to later allocation, and estimates a material liquidity benefit around the crisis. The estimated 2009–2010 liquidity benefit is historical and the government guarantee alone is not the full explanation; prepayment, delivery eligibility, dollar-roll and settlement details still require separate modelling.

Extended note. Version record

Dave Na, Ellie Newman, and Bernd Schlusche, The Evolution of the Federal Reserve's Agency MBS Holdings

Difficulty: Advanced **Type:** Report / research note

Status/version: FEDS Notes, 20 September 2024, DOI 10.17016/2380-7172.3612; Fed staff scenario analysis, not official Board forecast.

Documents how the coupon and age profile of the Federal Reserve's agency MBS portfolio interacts with borrowers' refinancing option, housing turnover, and alternative rate paths to drive principal payments. The scenarios describe SOMA holdings, not every investor's MBS portfolio or a priced option-adjusted spread; projecting cash flows still requires loan/pool composition, borrower behavior, and revised rate assumptions.

Extended note. Version record

Eduardo S. Schwartz and Walter N. Torous, Prepayment and the Valuation of Mortgage-Backed Securities

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 44(2), 375–392, June 1989; DOI 10.1111/j.1540-6261.1989.tb05062.x

Builds a mortgage security valuation model around an estimated, state-dependent prepayment probability rather than an assumption that every borrower exercises a refinancing option optimally. That distinction is central to scenario cash flows, effective duration, and option-adjusted spread. Its aggregate GNMA evidence is historical and cannot calibrate today's UMBS, servicing, borrower-credit, or specified-pool behavior without substantial redesign and backtesting.

Extended note. Version record

Matthew Dyer, Valuing and Analyzing Mortgage-Backed and Asset-Backed Securities

Difficulty: Advanced **Type:** Book chapter

Status/version: Chapter 26 in Debt Markets and Investments, Oxford University Press, 16 October 2019; DOI 10.1093/oso/9780190877439.003.0026

Lays out the operational sequence from prepayment and default forecasts to monthly asset-backed cash flows, discounting, zero-volatility spreads, and Monte Carlo option-adjusted spreads. It is a concise implementation companion to the broad Fabozzi material already present, especially when comparing a pass-through's yield with a callable cash-flow scenario. Publisher abstract and metadata were reviewed; the gated chapter's derivations were not.

Extended note. Version record

Nina Boyarchenko, Andreas Fuster and David O. Lucca, Understanding Mortgage Spreads; Fed staff-report version, revised June 2018

Difficulty: Research **Type:** Report / research note

Status/version: The Review of Financial Studies 32(10), 3799–3850, October 2019; DOI 10.1093/rfs/hhz004

Uses stripped MBS prices to identify non-interest-rate prepayment risk and investigate why mortgage spreads vary across coupons and time. It helps a portfolio desk distinguish the coupon smile associated with prepayment exposure from broader spread movements that a simple OAS comparison might label uniformly as borrower optionality. Reconcile quoted strips, TBA deliverability, and funding before trading its decomposition.

Extended note. Version record

Jiakai Chen, Haoyang Liu, David Rubio, Asani Sarkar and Zhaogang Song, MBS Market Dysfunctions in the Time of COVID-19; Related New York Fed Staff Report 933, revised September 2026

Difficulty: Advanced **Type:** Report / research note

Status/version: New York Fed research exposition, 17 July 2020; underlying Staff Report 933 revised September 2026

Documents the breakdown in the normal relationship between specified-pool cash MBS and forward TBA prices during March 2020, despite dealers increasing liquidity provision. Use it to test whether a specified-pool hedge can be sold or funded when the benchmark TBA remains liquid. The post is a research summary of one crisis episode; obtain the linked staff report for model details and avoid extrapolating its measured dislocation mechanically.

Extended note. Version record

Federal Housing Finance Agency, Prepayment Monitoring Report

Difficulty: Advanced **Type:** Report / research note

Status/version: Quarterly official series; latest selected report is second-quarter 2026, released 30 September 2026

Tracks Fannie Mae and Freddie Mac prepayment alignment for cohorts of TBA-eligible UMBS, holding coupon, maturity, and loan-origination year comparable. It offers an observable check on whether an agency assumption or pool-selection backtest is confounded by issuer-specific speed differences. The public cohort sample and ex post speeds are not a complete pool-level forecast; specified pools and loan characteristics require separate treatment, and published vintages should be retained.

Extended note. Version record

Fannie Mae, Fannie Mae Single-Family Loan Performance Data; Fannie Mae Data Dynamics access route

Difficulty: Advanced **Type:** Data / specification

Status/version: Living loan-level dataset; 31 July 2026 refresh adds first-quarter 2026 acquisition and performance data

Supplies acquisition and monthly performance records for a defined subset of Fannie Mae fixed-rate loans, with fields useful for prepayment, delinquency, modification, and credit models. It gives an institution a reproducible public starting point for hazard estimation instead of using only aggregate CPR assumptions. Preserve download vintages, loan identifiers, definition files, and publication lags before any historical validation.

Extended note. Version record

Tomasz Piskorski, Amit Seru and Vikrant Vig, Securitization and Distressed Loan Renegotiation: Evidence from the Subprime Mortgage Crisis; Author-hosted journal PDF

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics 97(3), 369–397, September 2010; selected journal article

Studies how securitization status affects foreclosure versus renegotiation after serious delinquency in 2005–2006 mortgages, showing why contractual servicing incentives can change RMBS loss timing and severity even for otherwise similar collateral. The evidence belongs to a subprime crisis vintage and does not imply the same treatment effect under current pooling agreements, servicing standards, or borrower relief programs.

Extended note. Version record

Peimin Chen, Igor Kozhanov, Peng Liu and Chunchi Wu, Commercial Mortgage-Backed Security Pricing with Real Estate Liquidity Risk

Difficulty: Research **Type:** Research paper

Status/version: Real Estate Economics 49(S2), 490–525, September 2021; first online 19 August 2019; DOI 10.1111/1540-6229.12297

Adds property-market liquidity frictions to a structural commercial-mortgage model and finds that deteriorating sale liquidity changes default sensitivities and the subordination needed for structured claims. Stress debt-service coverage, property values, occupancy, and workout timing jointly. The empirical model does not reproduce every conduit waterfall or special-servicer action; validate deal documents and property type before translating its coefficients to tranche prices.

Extended note. Version record

Elizabeth Klee and Chaehee Shin, Post-crisis Signals in Securitization: Evidence from Auto ABS

Difficulty: Research **Type:** Research paper

Status/version: Federal Reserve FEDS 2020-042, May 2020; working paper, DOI 10.17016/FEDS.2020.042

Uses granular regulatory data to show that auto-loan originators' decision to hold higher-quality loans longer can signal private information, affect new ABS pricing, and predict later performance. The result comes from auto ABS issuance and should not be assumed for credit-card master trusts or student-loan structures; the regulatory microdata used in the study are not fully public.

Extended note. Version record

Larry Cordell, Michael R. Roberts and Michael Schwert, CLO Performance; Earlier NBER working-paper version

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 78(3), 1235–1278, June 2023; online 21 March 2023; DOI 10.1111/jofi.13224

Measures CLO debt and equity outcomes and relates equity returns to the gap between leveraged-loan collateral pricing and rated CLO liabilities, while emphasizing long-term funding and reinvestment options. Rebuild overcollateralization tests, reinvestment periods,

optional calls, and manager trading for any actual deal; average historical returns are not a substitute for indenture-specific cash-flow projections or stress losses.

Extended note. Version record

U.S. Securities and Exchange Commission, Asset-Backed Securities Disclosure and Registration, Regulation AB II; SEC adoption summary

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC final rule Release 33-9638, 2014; consult subsequent amendments for current compliance

Specifies standardized, tagged asset-level disclosure for covered public securitizations, including residential and commercial mortgages, auto loans and leases, and debt securities. Map asset records to bonds, servicing periods, corrections, and deal waterfalls before calculating default or prepayment rates. Coverage depends on offering type and asset class; the rule is a data standard, not a guarantee of complete loan history or current market value.

Extended note. Version record

Basel Committee on Banking Supervision, Basel Framework, CRE40 Securitisation: General Provisions

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Online Basel Framework, consulted 4 October 2026; especially CRE40.12–.13, .24–.28 and .34

Defines tranche exposure, excess spread, revolving-pool early amortisation, clean-up calls and the structural information a bank needs before applying the capital framework. Prerequisite: be able to distinguish asset defaults from liability priority and an investor note's legal maturity from expected amortisation. Basel is a prudential standard, not an issue-specific waterfall or an investor valuation model; national implementation and deal documents control.

Extended note. Version record

Benjamin J. Keys, Tanmoy Mukherjee, Amit Seru and Vikrant Vig, Did Securitization Lead to Lax Screening? Evidence from Subprime Loans

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics 125(1), 307–362, February 2010; author-hosted journal PDF

Uses a credit-score threshold affecting ease of securitisation to identify weaker screening and subsequent default among comparable US subprime loans. It adds the origination-incentive layer to the list's servicing and prepayment papers. Prerequisite: regression discontinuity and borrower soft-information selection. The causal design belongs to a historical subprime segment and threshold; it should not be mechanically applied to current qualified mortgages, agency guarantees or all RMBS collateral.

Extended note. Version record

Freddie Mac, Single-Family Credit Risk Transfer Handbook and Reference Pool Disclosure File Layouts

Difficulty: Advanced **Type:** Data / specification

Status/version: Handbook September 2025; file layout version 4.2 effective July 2026

Connects a reference pool of guaranteed residential mortgages to STACR note and insurance loss allocations; the July 2026 layout specifies loan-level and supplemental disclosures needed to build a vintage panel. Prerequisite: separate the underlying guaranteed MBS cash flow from the reference-pool credit-risk transfer. Deal memoranda and data access conditions govern each transaction.

Extended note. Version record

Rithm Capital, 2025 Form 10-K; filing PDF

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Year ended 31 December 2025; filed 19 February 2026. Selected full risk discussion, Note 5 and financing tables inspected; historical annual snapshot.

Connects borrower delinquency and foreclosure duration to advances, reimbursement priority and funding. Non-agency advances generally rank ahead of trust note payments; recoverability may depend on loan/property proceeds, pool collections and servicer compliance. Note 5 distinguishes principal/interest, escrow and foreclosure advances, provisions and write-offs. Prerequisite: servicing cash flows versus credit-loss allocation. This US issuer portfolio does not replace a particular CMBS PSA or monthly collateral tape.

Extended note. Version record

Andrew Davidson and Alexander Levin, *Mortgage Valuation Models: Embedded Options, Risk, and Uncertainty*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Oxford University Press (2014); DOI 10.1093/acprof:oso/9780199998166.001.0001.

Prepayment and default modelling, OAS, risk-neutral prepayment adjustments and credit-sensitive MBS valuation from practitioner modellers.

Version record

Frank J. Fabozzi (ed.), *The Handbook of Mortgage-Backed Securities*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Oxford University Press (2016); DOI 10.1093/acprof:oso/9780198785774.001.0001.

Seventh edition (Oxford 2016) covering agency and non-agency products, prepayment models, CMOs and valuation.

Version record

Andreas Fuster, David O. Lucca and James Vickery, "Mortgage-Backed Securities"

Difficulty: Advanced **Type:** Report / research note

Status/version: Federal Reserve Bank of New York Staff Report 1001, January 2022; also a handbook chapter.

Survey of agency MBS market structure, prepayment, pricing, TBA and policy, with references to the research literature.

Version record

Chris Downing, Richard Stanton and Nancy Wallace, "An Empirical Test of a Two-Factor Mortgage Valuation Model: How Much Do House Prices Matter?"

Difficulty: Research **Type:** Research paper

Status/version: Real Estate Economics, 33(4), 681–710 (2005); DOI 10.1111/j.1540-6229.2005.00135.x.

Joint prepayment and default with house prices as a state variable in MBS valuation.

Version record

Zhiguo He and Zhaogang Song, "Agency MBS as Safe Assets"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(2), 387–426 (2026); DOI 10.1093/rfs/hhaf052.

New agency MBS carry a convenience premium over half the long-term Treasury premium, varying with mortgage rates through prepayment.

Version record

12.9 CMBS and workouts

Subject context and qualifications.

CRE Finance Council, Investor Reporting Package, Version 8.4

Difficulty: Not specified **Type:** Report / research note

Status/version: Industry reporting standard adopted 13 March 2024; public version and field specifications

Standardises loan, property, bond, modification and special-servicing reporting so a CMBS analyst can connect occupancy, debt-service coverage, appraisals and realised losses to the monthly bond statement. Prerequisite: distinguish mortgage loan, whole loan, companion note and trust certificate identifiers. The IRP improves comparability but cannot supply every pooling-and-servicing covenant, verify a property's current sale value or guarantee that all trustee data are publicly accessible.

Extended note. Version record

BBCMS Mortgage Trust 2025-C39, SEC-filed structural and collateral term sheet

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Preliminary free-writing prospectus dated 3 December 2025; preserved alongside the final 424B2, 1 December PSA and 2025 10-K. Final document identities, selected 10-K text, the March 2026 Exhibit 99.1 indexed cash-flow rows and the March 2026 Form 10-D inspected; final prospectus/PSA, full exhibit and ABS-EE XML retrieval remain incomplete

Specifies a liquidated-loan proceeds order after advances and expenses, treatment of interest not advanced because of appraisal reductions, and special-servicer sale and whole-loan constraints. The indexed March Exhibit 99.1 reports certificate interest of $4,122,095.20, principal of $248,015.80, and total fees of $10,952.80. The referenced ABS-EE tape and full exhibit still require reconciliation before a worked waterfall can be validated. This is one deal's term sheet; abbreviated disclosure and later definitive documents take precedence, and its percentages or servicer rights are not universal CMBS rules.

Extended note. Version record

Sheridan Titman and Sergey Tsyplakov, "Originator Performance, CMBS Structures, and the Risk of Commercial Mortgages"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 23(9), 3558–3594 (2010); DOI 10.1093/rfs/hhq055.

Loans from originators with poor stock performance have higher spreads and default rates; incentives in conduit CMBS.

Version record

Richard Stanton and Nancy Wallace, "CMBS Subordination, Ratings Inflation, and Regulatory-Capital Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: Financial Management, 47(1), 175–201 (2018); DOI 10.1111/fima.12183.

Subordination levels fell while risk rose, consistent with regulatory-capital-driven ratings inflation in CMBS.

Version record

Andra C. Ghent and Rossen Valkanov, "Comparing Securitized and Balance Sheet Loans: Size Matters"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 62(10), 2784–2803 (2016); DOI 10.1287/mnsc.2015.2260.

Large commercial loans are more likely securitised; selection affects comparisons of CMBS and portfolio-loan performance.

Version record

Sean J. Flynn Jr., Andra C. Ghent and Alexei Tchistyi, "Informational Efficiency in Securitization after Dodd-Frank"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(11), 5131–5172 (2020); DOI 10.1093/rfs/hhaa003.

After risk-retention rules, CMBS prices incorporate more loan-level information.

Version record

Xudong An, Yongheng Deng and Stuart A. Gabriel, "Asymmetric information, adverse selection, and the pricing of CMBS"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 100(2), 304–325 (2011); DOI 10.1016/j.jfineco.2010.12.002.

Portfolio lenders retain better loans; conduit CMBS loans priced for adverse selection.

Version record

Andra C. Ghent, Walter N. Torous and Rossen I. Valkanov, "Commercial Real Estate as an Asset Class"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 11(1), 153–171 (2019); DOI 10.1146/annurev-financial-110118-123121.

Annual Review of CRE return measurement, illiquidity, leverage and the role of debt markets.

Version record

Matteo Crosignani and Saketh Prazad, "Extend-and-Pretend in the U.S. CRE Market"

Difficulty: Research **Type:** Report / research note

Status/version: Federal Reserve Bank of New York Staff Report 1130, October 2024 (later revisions possible).

Argues banks extended impaired CRE mortgages to avoid capital write-downs, crowding out new CRE lending since 2022. Read with Glancy (2026) for the opposing interpretation.

Version record

David Glancy, "Pretend or Amend? On Evergreening in CRE"

Difficulty: Research **Type:** Report / research note

Status/version: Federal Reserve Board FEDS working paper, May 2026.

Counter-evidence: CRE loan extensions mostly address temporary payment frictions, and banks tightened extension terms after 2023.

Extended note. Version record

Brent W. Ambrose and Anthony B. Sanders, "Commercial Mortgage-Backed Securities: Prepayment and Default"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Real Estate Finance and Economics, 26(2-3), 179–196 (2003); DOI 10.1023/a:1022978708728.

Loan-level CMBS evidence on prepayment and default and the role of prepayment penalties and lockouts.

Version record

Lamont Black, John Krainer and Joseph Nichols, "From Origination to Renegotiation: A Comparison of Portfolio and Securitized Commercial Real Estate Loans"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Real Estate Finance and Economics, 55(1), 1–31 (2017); DOI 10.1007/s11146-016-9548-1.

Securitised CRE loans are harder to modify and resolve than portfolio loans, affecting workout outcomes and loss severities.

Version record

David Glancy, John R. Krainer, Robert J. Kurtzman and Joseph B. Nichols, "Intermediary Segmentation in the Commercial Real Estate Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 54(7), 2029–2080 (2022); DOI 10.1111/jmcb.12889.

Banks, life insurers and CMBS lend to different CRE borrowers; segmentation shapes how credit shocks transmit across lenders.

Version record

John M. Griffin and Alex Priest, "Is COVID Revealing a Virus in CMBS 2.0?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 78(4), 2233–2276 (2023); DOI 10.1111/jofi.13228.

Post-2010 CMBS underwriting overstated income; loans with larger income overstatement defaulted more during COVID-19.

Version record

12.10 Consumer ABS

Subject context and qualifications.

Volkswagen Auto Loan Enhanced Trust 2025-2, SEC-filed prospectus

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final Rule 424(b)(5) prospectus filed 2025; issuer document on EDGAR

Shows how monthly auto-loan collections pass through servicer advances, fees, note interest, principal, reserve top-up and residual certificates, with overcollateralisation and a yield-supplement amount. Prerequisite: distinguish gross receivable yield, losses, prepayments, servicing cost and liability interest. The prospectus is a single prime programme's contract, not a population estimate for subprime auto ABS or leases; trustee reports are required to validate realised performance.

Extended note. Version record

Board of Governors of the Federal Reserve System, Report to Congress on Risk Retention

Difficulty: Not specified **Type:** Report / research note

Status/version: Official report, October 2010; historical crisis analysis

Explains revolving credit-card trusts, seller's interest, excess spread, reserve accounts and early amortisation, including the funding shock to the originator when principal collections stop purchasing new receivables. Prerequisite: distinguish a revolving trust from a static amortising auto pool. The crisis evidence is historical; the report predates current card balances, trust documents and later risk-retention implementation.

Extended note. Version record

Citibank Credit Card Issuance Trust, Class 2025-A1 Notes Prospectus

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final prospectus dated 18 June 2025, filed 23 June 2025; issuer document on EDGAR

Details the master-trust collateral certificate, depositor/seller interest, allocation across series, excess spread and events redirecting principal collections. Prerequisite: read definitions of investor percentage, expected principal date and legal maturity before treating a card ABS as bullet debt. A Citibank Citiseries issue illustrates one architecture; other master trusts and card vintages can use different triggers, subordination and collateral definitions.

Extended note. Version record

Consumer Financial Protection Bureau, Private Education Loan Ombudsman Annual Report

Difficulty: Not specified **Type:** Report / research note

Status/version: Official annual report, January 2026; public PDF

Separates private education loans from federal student-loan programmes and documents borrower, servicer and owner problems that can affect delinquency and recovery timing. Prerequisite: distinguish in-school, grace, repayment, deferment and forbearance states. Use the report to build plausible transition and servicing-stress scenarios for private-loan ABS rather than transplanting a federally guaranteed loss assumption. Complaint data are selected and policy-focused; they do not estimate a pool's unconditional default probability or provide a security's waterfall.

Version record

SLM Student Loan Trust 2005-7, Class A-5 Remarketing Free-Writing Prospectus

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Preliminary July 2025 remarketing document for a 2005 trust; issuer filing on EDGAR

Exposes FFELP consolidation-loan guarantees, deferment and forbearance, prepayment, reset-rate and failed-remarketing extension risks in a seasoned trust. Prerequisite: distinguish the governmental guarantee on eligible loans from a guarantee of ABS notes, which the filing expressly disclaims. This is a preliminary remarketing of legacy FFELP collateral, not a newly originated 2025 pool; final terms and subsequent trust reporting are necessary for an actual trade.

Extended note. Version record

Efraim Benmelech, Ralf R. Meisenzahl and Rodney Ramcharan, "The Real Effects of Liquidity During the Financial Crisis: Evidence from Automobiles"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 132(1), 317–365 (2017); DOI 10.1093/qje/qjw031.

Collapse of ABS funding for captive auto lenders reduced car sales in 2008–09; shows the funding role of auto ABS.

Version record

Manuel Adelino, Antoinette Schoar and Felipe Severino, "Dynamics of Housing Debt in the Recent Boom and Great Recession"

Difficulty: Research **Type:** Research paper

Status/version: NBER Macroeconomics Annual, 32, 265–311 (2018); DOI 10.1086/696054.

Middle- and high-income borrowers drove mortgage growth and defaults; challenges the subprime-only narrative relevant to consumer-credit securitisation.

Version record

William Adams, Liran Einav and Jonathan Levin, "Liquidity Constraints and Imperfect Information in Subprime Lending"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review, 99(1), 49–84 (2009); DOI 10.1257/aer.99.1.49.

Subprime auto-lending data show adverse selection and moral hazard in down-payment and loan-size choices, informing auto-ABS collateral risk.

Version record

Liran Einav, Mark Jenkins and Jonathan Levin, "Contract Pricing in Consumer Credit Markets"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 80(4), 1387–1432 (2012); DOI 10.3982/ecta7677.

Structural model of subprime auto-loan pricing with risk-based contract terms.

Version record

Sumit Agarwal, Souphala Chomsisengphet, Neale Mahoney and Johannes Stroebel, "Regulating Consumer Financial Products: Evidence from Credit Cards"

Difficulty: Advanced **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 130(1), 111–164 (2015); DOI 10.1093/qje/qju037.

The CARD Act cut hidden card fees without raising interest charges; useful for card-ABS yield and excess-spread analysis.

Version record

Amir E. Khandani, Adlar J. Kim and Andrew W. Lo, "Consumer credit-risk models via machine-learning algorithms"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Banking & Finance, 34(11), 2767–2787 (2010); DOI 10.1016/j.jbankfin.2010.06.001.

Machine-learning forecasts of card delinquency from transaction data, with cost–benefit evaluation of credit-line cuts.

Version record

Tobias Berg, Valentin Burg, Ana Gombović and Manju Puri, "On the Rise of FinTechs: Credit Scoring Using Digital Footprints"

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Financial Studies, 33(7), 2845–2897 (2020); DOI 10.1093/rfs/hhz099.

Simple digital-footprint variables predict consumer default as well as credit bureau scores and complement them.

Version record

Andreas Fuster, Paul Goldsmith-Pinkham, Tarun Ramadorai and Ansgar Walther, "Predictably Unequal? The Effects of Machine Learning on Credit Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 77(1), 5–47 (2022); DOI 10.1111/jofi.13090.

More flexible ML credit models raise average accuracy but change outcomes unequally across groups; fairness and model-risk implications.

Version record

12.11 CLOs and specialist collateral

Subject context and qualifications.

Carlyle Direct Lending CLO 2024-1, Executed Indenture

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Executed 29 October 2024; issuer filing on EDGAR

Specifies interest and principal proceeds, reinvestment constraints, overcollateralisation and interest-diversion tests, collateral haircuts, maturity and default remedies for a direct-lending CLO. Prerequisite: distinguish par from market-value tests and separate the loan's interest receipts from principal and trading proceeds. This is a middle-market/direct-lending deal, not the exact covenant set of a broadly syndicated-loan CLO; private collateral data can limit independent calibration.

Extended note. Version record

Arbor Realty Trust, First-Quarter 2026 Form 10-Q, CLO Compliance Tests

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Filed 8 May 2026; April 2026 determination dates

Reports actual overcollateralisation and interest-coverage results for several CRE-loan CLOs, making the link from indenture definitions to periodic surveillance observable. Prerequisite: recompute each ratio using the deal's collateral valuation, debt balances and treatment of troubled loans. These CRE CLO ratios are not directly transferable to Carlyle's direct-lending indenture or broadly syndicated-loan CLOs; a 10-Q is not a full asset tape or deal-level forecast.

Extended note. Version record

S&P Global Ratings, Global Aircraft ABS: Methodology and Assumptions

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Criteria dated 10 March 2023; republished 26 August 2025 for nonmaterial changes

Frames aircraft and engine lease cash flows around lessee credit, lease expiry, re-lease downtime, maintenance, repossession and residual value. Prerequisite: understand how asset-specific recovery and remarketing differ from consumer-loan charge-offs. Rating criteria are a transparent scenario framework, not a market-clearing price or assurance of access to the aircraft appraisals and servicing data needed for an investor model.

Extended note. Version record

Efraim Benmelech, Jennifer Dlugosz and Victoria Ivashina, "Securitization without adverse selection: The case of CLOs"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 106(1), 91–113 (2012); DOI 10.1016/j.jfineco.2012.05.006.

Securitised loans in CLOs do not underperform comparable loans, unlike subprime mortgages.

Version record

Redouane Elkamhi and Yoshio Nozawa, "Fire-sale risk in the leveraged loan market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 146(3), 1120–1147 (2022); DOI 10.1016/j.jfineco.2022.05.003.

CLO overcollateralisation constraints force sales of downgraded loans, creating price pressure.

Version record

Maria Loumioti and Florin P. Vasvari, "Portfolio performance manipulation in collateralized loan obligations"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Accounting and Economics, 67(2-3), 438–462 (2019); DOI 10.1016/j.jacceco.2018.09.003.

CLO managers near covenant thresholds trade to meet compliance tests; shows why test mechanics matter.

Version record

Stavros Peristiani and João A. C. Santos, "CLO trading and collateral manager bank affiliation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Intermediation, 39, 47–58 (2019); DOI 10.1016/j.jfi.2018.06.003.

Bank-affiliated CLO managers sell loans arranged by their bank well before default, suggesting an information advantage.

Version record

Efraim Benmelech and Jennifer Dlugosz, "The alchemy of CDO credit ratings"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Monetary Economics, 56(5), 617–634 (2009); DOI 10.1016/j.jmoneco.2009.04.007.

How CLO and CDO tranches achieved AAA ratings on low-rated collateral and why those ratings failed; read with CLO rating methodologies.

Version record

John M. Griffin and Jordan Nickerson, "Are CLO Collateral and Tranche Ratings Disconnected?"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 36(6), 2319–2360 (2023); DOI 10.1093/rfs/hhac078.

CLO tranche ratings appear too generous relative to the underlying loan risk once collateral ratings and correlation assumptions are scrutinised.

Version record

Vitaly M. Bord and João A. C. Santos, "Does Securitization of Corporate Loans Lead to Riskier Lending?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 47(2-3), 415–444 (2015); DOI 10.1111/jmcb.12181.

Loans sold to CLOs at origination underperform similar loans, consistent with weaker screening.

Version record

Taylor D. Nadauld and Michael S. Weisbach, "Did securitization affect the cost of corporate debt?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 105(2), 332–352 (2012); DOI 10.1016/j.jfineco.2012.03.002.

Loans eligible for CLO securitisation carried lower spreads, showing how CLO demand feeds back into leveraged-loan pricing.

Version record

Rustom M. Irani, Rajkamal Iyer, Ralf R. Meisenzahl and José-Luis Peydró, "The Rise of Shadow Banking: Evidence from Capital Regulation"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 34(5), 2181–2235 (2021); DOI 10.1093/rfs/hhaa106.

Capital-constrained banks sell syndicated-loan shares to nonbanks such as CLOs and funds; the loans then become more fragile in stress.

Version record

Roberto Liebscher and Thomas Mählmann, "Are Professional Investment Managers Skilled? Evidence from Syndicated Loan Portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 63(6), 1892–1918 (2017); DOI 10.1287/mnsc.2015.2412.

CLO managers show persistent skill in loan selection and trading; evidence for manager due diligence.

Version record

Chapter 13

FX, macro and regional markets

Connect cross-asset premia, macro regimes and news, exchange-rate determination, currency premia and hedging to regional access, settlement and data. A country label is not a substitute for an instrument and contract specification.

13.1 Macro and cross-asset research

Related core works: Ilmanen, Drobny, Gliner, Soros, Reinhart–Rogoff, Value and Momentum Everywhere, Time Series Momentum, A Century of Evidence, and Carry.

13.1.1 Macro books, history and discretionary context

Greg Gliner, Global Macro Trading: Profiting in a New World Economy

Difficulty: Intro **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Source foundational macro handbook: connect economic releases, central-bank policy, currencies and rates to trade construction and risk management. Its short applied chapters provide a reference alongside the more formal asset-pricing and systematic-macro readings.

Version record

Steven Drobny, Inside the House of Money: Top Hedge Fund Traders on Profiting in the Global Markets

Difficulty: Intro **Type:** Book / monograph

Status/version: Discretionary macro interviews and scenario judgement; revised 2013 edition.

Discretionary macro interviews and scenario judgement; revised 2013 edition. Context for formulating hypotheses rather than empirical validation of systematic rules.

Version record

Carmen M. Reinhart and Kenneth S. Rogoff, This Time Is Different: Eight Centuries of Financial Folly

Difficulty: Intro **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Historical sovereign defaults, banking crises, debt and credit cycles. Place contemporary indicators in a long-run setting and compare crisis, deleveraging and recovery episodes; historical parallels are inputs to scenarios rather than deterministic timing rules.

Version record

George Soros, The Alchemy of Finance

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Conceptual account of reflexivity between beliefs, prices, policy and financing. Useful for hypotheses and stress scenarios; a discretionary and philosophical account rather than empirical validation.

Version record

Andrew W. Lo, Adaptive Markets: Financial Evolution at the Speed of Thought

Difficulty: Intro **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Links changing market efficiency to adaptation, competition and behaviour. Use its regime perspective when examining unstable correlations, factor premia and strategy decay; the hypothesis does not by itself specify a profitable regime-switching rule.

Version record

Jack D. Schwager, *Market Wizards: Interviews with Top Traders* (1989; Wiley reissue 2012); *Hedge Fund Market Wizards: How Winning Traders Win* (2012)

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 1989, 2012. Version distinctions remain in the annotation.

Distinct books in the Market Wizards series, offering practitioner case histories on risk judgement and adaptation. Later series titles and revised editions are separate selections. Anecdotal accounts support hypothesis formation; they do not validate systematic rules.

Version record

Ray Dalio, *Principles for Navigating Big Debt Crises* (2018); *How the Economic Machine Works* (2013)

Difficulty: Not specified **Type:** Mixed resources

Status/version: Selected citation year(s): 2018, 2013. Version distinctions remain in the annotation.

Optional book and explanatory presentation on debt cycles, deleveraging and policy responses; practitioner frameworks for macro scenarios.

Version record

Antti Ilmanen, *Investing Amid Low Expected Returns: Making the Most When Markets Offer the Least*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Wiley, 2022 (ISBN 9781119860198).

Successor to Expected Returns: long-run return expectations, illiquidity, factor premia, alternatives and implementation discipline.

Version record

Dirk Willer and Alex Saunders, *Trading Global Macro Markets*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, 30 April 2026 (hardback ISBN 9781394273751); listings differ on page count.

Practitioner text on trading macro regimes across rates, FX, equities and emerging markets from a sell-side head of macro strategy; confirm the final edition before citing page references.

Extended note. Version record

13.1.2 Cross-asset style premia and factor timing

Value, momentum, carry and defensive premia measured across equity indices, bonds, currencies and commodities, and the evidence on timing them. Core cross-asset papers, among them Value and Momentum Everywhere, Carry and Betting Against Beta, are in Sections 2.1 and 10.1.

Martin Lettau, Matteo Maggiori and Michael Weber, "Conditional risk premia in currency markets and other asset classes"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 114(2), 197–225 (2014); DOI 10.1016/j.jfineco.2014.07.001.

A downside-risk CAPM prices the cross-section of currency returns: the market-beta gap between high- and low-interest-rate currencies is larger when market returns are bad, which is also when the price of risk is high. The same model jointly prices equities, equity index options, commodities, sovereign bonds and currencies, which models built for a single asset class fail to do.

Version record

Clifford S. Asness, John M. Liew and Ross L. Stevens, "Parallels Between the Cross-Sectional Predictability of Stock and Country Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Portfolio Management, 23(3), 79–87 (1997); DOI 10.3905/jpm.1997.409606.

Country-level book-to-market, market equity and one-year momentum help explain the cross-section of expected equity-index returns, as their stock-level counterparts do within countries. The January seasonal in size and the pattern of profits around portfolio formation also carry over from stocks to countries.

Version record

Sanjeev Bhojraj and Bhaskaran Swaminathan, "Macromomentum: Returns Predictability in International Equity Indices"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Business, 79(1), 429–451 (2006); DOI 10.1086/497416.

Country equity indices show momentum in the year after portfolio formation and reversal over the following two years. Positive currency momentum predicts low future index returns, weakening momentum and strengthening reversal in dollar-denominated returns; the authors read the pattern as consistent with behavioural theories in which momentum is followed by reversal.

Version record

Aleksi Pitkäjärvi, Matti Suominen and Lauri Vaittinen, "Cross-asset signals and time series momentum"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 136(1), 63–85 (2020); DOI 10.1016/j.jfineco.2019.02.011.

In a broad international sample, past bond-market returns predict equity-market returns positively and past equity returns predict bond returns negatively. A diversified cross-asset time-series momentum portfolio built on these signals has a Sharpe ratio more than 40% higher than standard time-series momentum; the authors attribute both effects to slow-moving capital.

Version record

Tarun Gupta and Bryan Kelly, "Factor Momentum Everywhere"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Portfolio Management, 45(3), 13–36 (2019); DOI 10.3905/jpm.2019.45.3.013.

Across 65 characteristic-based equity factors around the world, each factor can be timed with its own recent performance. A combined time-series factor momentum portfolio earns an annual Sharpe ratio of 0.84 and adds to stock, industry momentum and value strategies, suggesting that momentum stems largely from persistence in common factor returns.

Version record

Nick Baltas, "The Impact of Crowding in Alternative Risk Premia Investing"

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal, 75(3), 89–104 (2019); DOI 10.1080/0015198x.2019.1600955.

A framework, based on each strategy's mechanics, for how crowding affects alternative risk premia. Divergence premia such as momentum tend to underperform after crowded periods, whereas convergence premia such as value tend to outperform as investor flows into them grow.

Version record

Clifford Asness, Swati Chandra, Antti Ilmanen and Ronen Israel, "Contrarian Factor Timing is Deceptively Difficult"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Portfolio Management, 43(5), 72–87 (2017); DOI 10.3905/jpm.2017.43.5.072.

Value spreads seem to predict style returns, but mainly because value timing adds value exposure. In a multi-style portfolio that already holds value, contrarian timing adds that exposure intermittently and less efficiently than a risk-targeted strategic allocation and can reduce diversification; results stay weak at longer horizons and at valuation extremes.

Version record

Valentin Haddad, Serhiy Kozak and Shrihari Santosh, "Factor Timing"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(5), 1980–2018 (2020); DOI 10.1093/rfs/hhaa017.

The optimal factor-timing portfolio is equivalent to the stochastic discount factor; the paper estimates both with restrictions on expected-return dynamics. Market-neutral equity factors are strongly and robustly predictable, timing them improves on static factor investing, and the implied discount factor is more volatile and cyclically different from estimates that ignore predictability.

Version record

13.1.3 Trend following and managed futures

Robert Carver, Systematic Trading

Difficulty: Intro **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Practical progression from individual trend, carry and mean-reversion rules to a combined portfolio. The source emphasis on simple robust parameters, forecast/risk scaling, position sizing, turnover and correlation between strategies.

Version record

Alex Greyserman and Kathryn Kaminski, *Trend Following with Managed Futures: The Search for Crisis Alpha* (2014); Vineer Bhansali, Josh Davis, Matt Dorsten and Graham Rennison, *Carry and Trend in Lots of Places* (2015)

Difficulty: Not specified **Type:** Mixed resources

Status/version: Selected DOI-linked version; print 2014-08-25; online 2014-12-05.

Managed-futures history and carry/trend diversification. Crisis behaviour is conditional on trends, reversals, correlations and costs.

Version record

Brian Hurst, Yao Hua Ooi and Lasse Heje Pedersen, *Understanding Managed Futures* (2010); Clifford Asness, Antti Ilmanen, Ronen Israel and Tobias Moskowitz, *Investing with Style*, *Journal of Investment Management* (2015)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2010, 2015. Version distinctions remain in the annotation.

Managed-futures intuition and a distinct multi-style account of value, momentum, carry and defensive exposures.

Version record

Man Group, *Trend Following and Drawdowns: Is This Time Different?*; Campbell R. Harvey, Sandy Rattray, Andrew Sinclair and Otto Van Hemert, *Man vs. Machine: Comparing Discretionary and Systematic Hedge Fund Performance* (2017)

Difficulty: Not specified **Type:** Mixed resources

Status/version: Selected citation year(s): 2017. Version distinctions remain in the annotation.

Drawdown context and a separate 1996–2014 hedge-fund comparison adjusting for volatility and factors; its scope extends beyond macro.

Version record

William Fung and David A. Hsieh, *The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers* (2001); *Asset-Based Style Factors for Hedge Funds* (2002); *Hedge Fund Benchmarks: A Risk-Based Approach* (2004)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected DOI-linked version; print 2001-04; online 2001-04-01.

Three distinct papers on option-like payoffs, nonlinear exposures and asset-based benchmarks. A lookback-straddle representation does not imply literal ownership of protective options.

Version record

Yves Lempérière, Cyril Deremble, Philip Seager, Marc Potters and Jean-Philippe Bouchaud, "Two centuries of trend following"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Investment Strategies, 3(3), 41–61 (2014); DOI 10.21314/jois.2014.043.

Trend signals across asset classes since 1800 with stable t-statistics; a long-sample robustness check.

Version record

Nick Baltas and Robert Kosowski, "Demystifying Time-Series Momentum Strategies: Volatility Estimators, Trading Rules and Pairwise Correlations"

Difficulty: Advanced **Type:** Research paper

Status/version: Market Momentum, 30–67 (2020); DOI 10.1002/9781119599364.ch3.

Effects of volatility estimators, signal definitions and correlation adjustments on trend-following turnover and performance.

Version record

Abby Y. Kim, Yiuman Tse and John K. Wald, "Time series momentum and volatility scaling"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 30, 103–124 (2016); DOI 10.1016/j.finmar.2016.05.003.

Argues much of TSMOM performance comes from volatility scaling rather than the trend signal.

Version record

Dashan Huang, Jiangyuan Li, Liyao Wang and Guofu Zhou, "Time series momentum: Is it there?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 135(3), 774–794 (2020); DOI 10.1016/j.jfineco.2019.08.004.

Pooled regressions overstate predictability; asset-by-asset evidence for time-series momentum is weak. Counter-evidence to the trend literature.

Extended note. Version record

Ari Levine and Lasse Heje Pedersen, "Which Trend Is Your Friend?"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 72(3), 51–66 (2016); DOI 10.2469/faj.v72.n3.3.

Shows that moving-average crossovers and time-series momentum are equivalent filters of past returns.

Version record

Abhilash Babu, Ari Levine, Yao Hua Ooi, Lasse Heje Pedersen and Erik Stamelos, "Trends Everywhere"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper 3386035 (2018); DOI 10.2139/ssrn.3386035. Working-paper record; no journal version established in this check.

Trend following in 58 alternative markets beyond traditional futures, with lower correlation to standard trend.

Extended note. Version record

Robert Carver, *Advanced Futures Trading Strategies: 30 Fully Tested Strategies for Multiple Trading Styles and Time Frames*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Harriman House, 18 April 2023 (ISBN 9780857199683).

Systematic futures strategies with explicit position sizing, forecast scaling, carry, trend, relative value and cost accounting; companion open-source code (pysystemtrade).

Version record

Christian L. Goulding, Campbell R. Harvey and Michele G. Mazzoleni, "Momentum turning points"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 149(3), 378–406 (2023); DOI 10.1016/j.jfineco.2023.05.007.

Slow and fast time-series momentum signals classify four stock-market phases: bull, correction, bear and rebound. Bear phases concentrate in high-risk states yet predict negative expected returns; blending slow and fast momentum into intermediate-speed portfolios turns the information in these phases into positive unconditional alpha, which the authors decompose.

Version record

Geetesh Bhardwaj, Gary B. Gorton and K. Geert Rouwenhorst, "Fooling Some of the People All of the Time: The Inefficient Performance and Persistence of Commodity Trading Advisors"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(11), 3099–3132 (2014); DOI 10.1093/rfs/hhu040.

Commodity trading advisors (CTAs) returned 85 basis points a year over Treasury bills to investors in 1994–2007, insignificantly different from zero, against estimated gross excess returns of 5.4%, so fees absorbed most of the performance. Even before fees CTAs show no alpha relative to simple futures strategies in the public domain, and voluntary reporting biases the databases.

Version record

13.1.4 Cross-asset allocation and macro data

Campbell R. Harvey, *The Real Term Structure and Consumption Growth* (1988); Richard Meese and Kenneth Rogoff, *Empirical Exchange Rate Models of the Seventies: Do They Fit Out of Sample?* (1983)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected DOI-linked version; print 1983-02.

Consumption-growth forecasting from the term structure, and a separate random-walk benchmark for FX forecasts. The latter result is specific to its tested models, currencies and horizons.

Version record

Macrosynergy, *Macroeconomic Data and Systematic Trading Strategies* (2024)

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Page displays 23 April 2024; later revision date unestablished.

Macro-quantamental signals and data problems: revisions, release delays, short histories and heterogeneous reporting.

Version record

David Blitz and Pim van Vliet, *Global Tactical Cross-Asset Allocation: Applying Value and Momentum Across Asset Classes* (2008)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2008. Version distinctions remain in the annotation.

Cross-asset value/momentum allocation and reported out-of-sample evidence; implementation depends on instruments, financing and costs.

Version record

Chris Brightman and Shane Shepherd, *Systematic Global Macro* (December 2016)

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Selected citation year(s): 2016. Version distinctions remain in the annotation.

Practitioner account of twelve liquid strategies across stocks, bonds, commodities and currencies. The historical landing page may redirect; reported performance requires reconstruction.

Version record

Clifford S. Asness, Andrea Frazzini and Lasse H. Pedersen, "Leverage Aversion and Risk Parity"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 68(1), 47–59 (2012); DOI 10.2469/faj.v68.n1.1.

Leverage-constrained investors overweight risky assets, so levered risk parity can earn higher risk-adjusted returns.

Version record

Robert M. Anderson, Stephen W. Bianchi and Lisa R. Goldberg, "Will My Risk Parity Strategy Outperform?"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 68(6), 75–93 (2012); DOI 10.2469/faj.v68.n6.7.

Counter-evidence: risk-parity outperformance is sensitive to the period, financing cost and transaction costs.

Extended note. Version record

Mikhail Chernov, Lars A. Lochstoer and Dongho Song, "The Real Channel for Nominal Bond-Stock Puzzles"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, jofi.70092 (2026); DOI 10.1111/jofi.70092.

Consumption dynamics change when the bond–stock correlation switches sign; regime learning about permanent versus transitory shocks explains nominal bond–stock puzzles.

Version record

Michael Stamos, "Forecasting Stock–Bond Correlation"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 51(5), 134–142 (2025); DOI 10.3905/jpm.2025.1.677.

Forecasting models for the stock–bond correlation, a key input for multi-asset risk and hedging decisions.

Extended note. Version record

13.2 Macro regimes, inflation and the stock–bond correlation

How asset classes and styles behave across growth, inflation and volatility regimes, how to model regime changes for allocation, and what drives the sign of the stock–bond correlation. Regime-switching econometrics is in Section 2.4; inflation-linked bonds and real yields are in Section 11.5.

13.2.1 Regime models and macro sensitivities

Andrew Ang and Geert Bekaert, "International Asset Allocation With Regime Shifts"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 15(4), 1137–1187 (2002); DOI 10.1093/rfs/15.4.1137.

Solves a U.S. investor's dynamic international portfolio problem when returns follow a regime-switching process in which correlations and volatilities rise in bad times. International diversification remains valuable and currency hedging adds further benefit; ignoring regimes costs little for all-equity portfolios but more when a conditionally risk-free asset can be held.

Version record

Massimo Guidolin and Allan Timmermann, "Asset allocation under multivariate regime switching"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Economic Dynamics and Control, 31(11), 3503–3544 (2007); DOI 10.1016/j.jedc.2006.12.004.

Four regimes (crash, slow growth, bull and recovery) are needed to capture the joint distribution of stock and bond returns, and optimal allocations vary considerably across them as investors update state probabilities. Horizon effects on the stock allocation reverse between crash and bull states; ignoring regimes can be costly even after allowing for parameter uncertainty, and out-of-sample forecasts confirm the regimes' economic importance.

Version record

Mark Kritzman, Sébastien Page and David Turkington, "Regime Shifts: Implications for Dynamic Strategies"

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal, 68(3), 22–39 (2012); DOI 10.2469/faj.v68.n3.3.

Applies Markov-switching models to forecast regimes in market turbulence, inflation and economic growth, and shifts asset allocation with them. In backtests the dynamic process outperformed static allocation, especially for investors seeking to avoid large losses.

Version record

Antti Ilmanen, Thomas Maloney and Adrienne Ross, "Exploring Macroeconomic Sensitivities: How Investments Respond to Different Economic Environments"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Portfolio Management, 40(3), 87–99 (2014); DOI 10.3905/jpm.2014.40.3.087.

Measures how traditional asset classes and dynamic style strategies perform across growth, inflation, real-yield, volatility and illiquidity environments. Macroeconomic factors are not directly investable; style premia carry meaningfully less macro exposure than asset classes, and diversification reduces a portfolio's macro exposures.

Version record

13.2.2 Inflation and the stock–bond correlation

Geert Bekaert and Xiaozheng Wang, "Inflation risk and the inflation risk premium"

Difficulty: Research **Type:** Research paper

Status/version: Economic Policy, 25(64), 755–806 (2010); DOI 10.1111/j.1468-0327.2010.00253.x.

Estimates inflation betas of standard bond and diversified equity indices for more than 45 countries and finds them poor inflation hedges; adding Treasury bills, foreign bonds, real estate and gold helps but leaves inflation risk hard to hedge. Reviews term-structure estimates of the inflation risk premium, which most studies find sizeable and time-varying, and the case for inflation-linked bonds such as TIPS.

Version record

Henry Neville, Teun Draaisma, Ben Funnell, Campbell R. Harvey and Otto Van Hemert, "The Best Strategies for Inflationary Times"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Portfolio Management, 47(8), 8–37 (2021); DOI 10.3905/jpm.2021.1.274.

Passive and active strategies in the United States, the United Kingdom and Japan over 95 years of inflation surges. Unexpected inflation hurts bonds and equities, local inflation most; commodities earn positive returns in surges, with wide variation across the complex; among active strategies trend following gives the most reliable protection, and some equity factors such as quality help.

Version record

Robert Connolly, Chris Stivers and Licheng Sun, "Stock Market Uncertainty and the Stock-Bond Return Relation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 40(1), 161–194 (2005); DOI 10.1017/s0022109000001782.

Links daily stock–Treasury comovement to stock-market uncertainty measured by index-option implied volatility and detrended turnover. Higher uncertainty predicts a lower future stock–bond correlation, and bonds outperform stocks on days when implied volatility jumps or turnover is unexpectedly high, so diversification benefits rise with uncertainty.

Version record

Lieven Baele, Geert Bekaert and Koen Inghelbrecht, "The Determinants of Stock and Bond Return Comovements"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 23(6), 2374–2428 (2010); DOI 10.1093/rfs/hhq014.

A dynamic factor model with regime-switching macro state variables, risk aversion, macro uncertainty and liquidity proxies for stock–bond comovement. Macro fundamentals explain little of the stock–bond correlation and liquidity proxies matter more; macro factors fit bond volatility, the variance premium is critical for stock volatility, and the model fails mainly in fitting covariances.

Version record

Alfie Brixton, Jordan Brooks, Pete Hecht, Antti Ilmanen, Thomas Maloney and Nicholas McQuinn, "A Changing Stock–Bond Correlation: Drivers and Implications"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Portfolio Management, 49(4), 64–80 (2023); DOI 10.3905/jpm.2023.1.459.

The stock–bond correlation was consistently negative from 2000 to 2020 but positive in the 1970s, 1980s and 1990s, and higher inflation uncertainty could bring that back. A simple macro model of its drivers, with international evidence, shows how a positive correlation would raise portfolio risk or force allocation changes that lower expected returns, and considers alternatives as replacement diversifiers.

Version record

Roderick Molenaar, Edouard Sénéchal, Laurens Swinkels and Zhenping Wang, "Empirical Evidence on the Stock–Bond Correlation"

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal, 80(3), 17–36 (2024); DOI 10.1080/0015198x.2024.2317333.

History shows abrupt shifts in the stock–bond correlation after long stable periods. Inflation, real rates and government creditworthiness explain much of the correlation, and increases in it coincide with higher multi-asset portfolio risk and higher bond risk premia.

Version record

Christopher S. Jones and Sungjune Pyun, "Consumption Growth Persistence and the Stock–Bond Correlation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(2), 810–838 (2025); DOI 10.1017/s002210902400019x.

A consumption model in which the correlation between shocks to consumption and to expected future consumption growth varies over time. Time variation in this consumption-growth persistence drives the stock–bond correlation and its link to the power of bond yields to predict stock returns; the authors suggest asset-price fluctuations are the main source of the variation.

Version record

13.3 Macro news, announcements and nowcasting

How prices respond to scheduled macro releases and central-bank decisions, how much return is earned around them, and how to track the economy in real time. Policy-surprise construction and the yield-curve effects of policy are in Section 11.2; real-time data vintages are in Section 20.2.

13.3.1 Announcement effects, policy news and premia

Pavel Savor and Mungo Wilson, "How Much Do Investors Care About Macroeconomic Risk? Evidence from Scheduled Economic Announcements"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 48(2), 343–375 (2013); DOI 10.1017/s002210901300015x.

Equity and bond returns are much higher on scheduled announcement days, consistent with priced macro risk.

Version record

David O. Lucca and Emanuel Moench, "The Pre-FOMC Announcement Drift"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 70(1), 329–371 (2015); DOI 10.1111/jofi.12196.

Large S&P 500 returns in the 24 hours before scheduled FOMC announcements; later samples show weakening.

Version record

Refet S. Gürkaynak, Brian Sack and Eric T. Swanson, "Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements"

Difficulty: Research **Type:** Research paper

Status/version: FEDS 2004-66; journal version International Journal of Central Banking 1(1), 55–93 (May 2005).

Two-factor high-frequency decomposition of FOMC surprises into target and path factors.

Version record

Emi Nakamura and Jón Steinsson, "High-Frequency Identification of Monetary Non-Neutrality: The Information Effect"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 133(3), 1283–1330 (2018); DOI 10.1093/qje/qjy004.

Monetary surprises move real rates and expected growth, implying an information effect of policy announcements.

Version record

Michael D. Bauer and Eric T. Swanson, "A Reassessment of Monetary Policy Surprises and High-Frequency Identification"

Difficulty: Research **Type:** Research paper

Status/version: NBER Macroeconomics Annual, 37, 87–155 (2023); DOI 10.1086/723574.

Surprises are predictable from public data; orthogonalised surprises reduce the apparent information effect.

Version record

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold and Clara Vega, "Micro Effects of Macro Announcements: Real-Time Price Discovery in Foreign Exchange"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 93(1), 38–62 (2003); DOI 10.1257/000282803321455151.

High-frequency FX responses to US announcement surprises, with asymmetric reaction to bad news.

Version record

Benjamin Golez and Ben Matthies, "Fed information effects: Evidence from the equity term structure"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 165, 103988 (2025); DOI 10.1016/j.jfineco.2024.103988.

Uses the equity term structure (dividend strips/futures) to assess Fed information effects of monetary-policy announcements.

Extended note. Version record

Francesco Bianchi, Sydney C. Ludvigson and Sai Ma, "What Hundreds of Economic News Events Say About Belief Overreaction in the Stock Market"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(10), 3191–3239 (2026); DOI 10.1093/rfs/hhag062.

Structural estimation with ML across hundreds of economic news events finds systematic investor overreaction, often dampening market responses.

Version record

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold and Clara Vega, "Real-time price discovery in global stock, bond and foreign exchange markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 73(2), 251–277 (2007); DOI 10.1016/j.jinteco.2007.02.004.

High-frequency futures data show that U.S. macroeconomic news causes conditional mean jumps in U.S., German and British stock, bond and currency markets. Equity responses depend on the stage of the business cycle, which explains the low average stock–bond correlation; conditioning on the state of the economy, equities and currencies respond as strongly as bonds, and cross-market links remain after controlling for news.

Version record

John H. Boyd, Jian Hu and Ravi Jagannathan, "The Stock Market's Reaction to Unemployment News: Why Bad News Is Usually Good for Stocks"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 60(2), 649–672 (2005); DOI 10.1111/j.1540-6261.2005.00742.x.

News of rising unemployment is on average good for stocks in expansions and bad in contractions. The news bundles information about future interest rates, the equity risk premium and corporate earnings; interest-rate information dominates in expansions and dividend information in contractions.

Version record

Vadim Elenev, Tzuo-Hann Law, Dongho Song and Amir Yaron, "Fearing the Fed: How wall street reads main street"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 153, 103790 (2024); DOI 10.1016/j.jfineco.2024.103790.

The stock market's sensitivity to major macro announcements is countercyclical: prices react more to good or bad surprises when the economy is below potential and policy easing is expected. Regressions and a no-arbitrage model attribute the pattern to procyclical monetary-policy expectations and a countercyclical price of risk.

Version record

Thomas Gilbert, Chiara Scotti, Georg Strasser and Clara Vega, "Is the intrinsic value of a macroeconomic news announcement related to its asset price impact?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 92, 78–95 (2017); DOI 10.1016/j.jmoneco.2017.09.008.

Asset prices respond very differently to different macro releases. A measure of each announcement's intrinsic value, its ability to nowcast GDP growth, inflation and the federal funds target, explains a significant share of the variation in Treasury-yield responses; timeliness and revision premia matter most.

Version record

Hengjie Ai and Ravi Bansal, "Risk Preferences and the Macroeconomic Announcement Premium"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 86(4), 1383–1430 (2018); DOI 10.3982/ecta14607.

Stock returns around scheduled macro announcements such as the employment report and FOMC statements account for 55% of the market equity premium. A characterisation theorem shows that a nonnegative announcement premium identifies a significant departure from time-separable expected utility, supporting a broad class of non-expected-utility preferences, and gives conditions for a pre-announcement drift.

Version record

Anna Cieslak, Adair Morse and Annette Vissing-Jorgensen, "Stock Returns over the FOMC Cycle"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 74(5), 2201–2248 (2019); DOI 10.1111/jofi.12818.

Since 1994 the U.S. equity premium has been earned entirely in even weeks of the FOMC cycle, counting from the last meeting. Intermeeting target changes, fed funds futures and Board meetings tie this causally to unexpectedly accommodating Fed policy, with evidence of systematic informal communication by Fed officials with the media and the financial sector.

Version record

Ben S. Bernanke and Kenneth N. Kuttner, "What Explains the Stock Market's Reaction to Federal Reserve Policy?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 60(3), 1221–1257 (2005); DOI 10.1111/j.1540-6261.2005.00760.x.

An unanticipated 25 basis point cut in the federal funds target is associated on average with about a 1% rise in broad stock indices. A Campbell–Ammer decomposition attributes most of the response to changes in expected excess returns.

Version record

Michael D. Bauer and Eric T. Swanson, "An Alternative Explanation for the 'Fed Information Effect' "

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 113(3), 664–700 (2023); DOI 10.1257/aer.20201220.

Private forecast revisions often move the wrong way after monetary-policy surprises, which has been read as a Fed information effect. The authors show the results are also consistent with a Fed response to news, in which the Fed and forecasters react to the same incoming data, and support that channel with regressions controlling for news, their own survey of forecasters and market responses to FOMC announcements.

Version record

Tim A. Kroencke, Maik Schmeling and Andreas Schrimpf, "The FOMC Risk Shift"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 120, 21–39 (2021); DOI 10.1016/j.jmoneco.2021.02.003.

Extracts risk shifts, a component of monetary-policy news uncorrelated with risk-free-rate surprises, from high-frequency risky-asset prices. Risk shifts account for most stock moves around FOMC announcements and coincide with significant fund flows, and price pressure amplifies the response, so central-bank information effects are harder to identify than thought.

Version record

Francesca Brusa, Pavel Savor and Mungo Wilson, "One Central Bank to Rule Them All"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 24(2), 263–304 (2020); DOI 10.1093/rof/rfz015.

Global stock markets earn high returns around FOMC meetings but not around announcements by other central banks, even in their own markets and for domestically focused stocks. Surprises, currency moves, economy size and policy activism do not explain the difference; uncertainty falls worldwide only after FOMC announcements, consistent with a unique global role for the Fed.

Version record

Philippe Mueller, Alireza Tahbaz-Salehi and Andrea Vedolin, "Exchange Rates and Monetary Policy Uncertainty"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 72(3), 1213–1252 (2017); DOI 10.1111/jofi.12499.

A strategy short the U.S. dollar and long other currencies earns significantly larger excess returns on scheduled FOMC announcement days. The returns are larger for currencies with higher interest differentials against the United States, rise with monetary-policy uncertainty and during easing, and are interpreted as compensation for policy uncertainty borne by constrained intermediaries.

Version record

13.3.2 Real-time data, nowcasting and macro signals

Domenico Giannone, Lucrezia Reichlin and David Small, "Nowcasting: The real-time informational content of macroeconomic data"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 55(4), 665–676 (2008); DOI 10.1016/j.jmoneco.2008.05.010.

Dynamic factor nowcasting of GDP from mixed-frequency data releases with ragged edges.

Version record

Marta Bańbura, Domenico Giannone, Michele Modugno and Lucrezia Reichlin, "Now-casting and the real-time data flow"

Difficulty: Advanced **Type:** Research paper

Status/version: Handbook of Economic Forecasting, 195–237 (2013); DOI 10.1016/b978-0-444-53683-9.00004-9.

Handbook chapter on news decomposition and real-time data vintages in nowcasting models.

Version record

James H. Stock and Mark W. Watson, "Macroeconomic Forecasting Using Diffusion Indexes"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 20(2), 147–162 (2002); DOI 10.1198/073500102317351921.

Summarises 215 predictors with a few principal-component indexes in an approximate dynamic factor model and forecasts eight monthly U.S. macro series 6, 12 and 24 months ahead in simulated real time, 1970–1998. The diffusion-index forecasts beat univariate autoregressions, small vector autoregressions and leading-indicator models.

Version record

S. Borağan Aruoba, Francis X. Diebold and Chiara Scotti, "Real-Time Measurement of Business Conditions"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 27(4), 417–427 (2009); DOI 10.1198/jbes.2009.07205.

A dynamic factor framework with exact filtering for measuring economic activity at high frequency, potentially in real time, from stock and flow data observed at mixed frequencies. Illustrated with a prototype empirical example and a calibrated simulation.

Version record

Elena Andreou, Eric Ghysels and Andros Kourtellos, "Should Macroeconomic Forecasters Use Daily Financial Data and How?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 31(2), 240–251 (2013); DOI 10.1080/07350015.2013.767199.

Regression-based mixed-data-sampling (MIDAS) forecasts, combined across models, use daily financial data to predict quarterly real activity, and a small set of daily financial factors is extracted from about 1,000 daily series. The paper measures the value of daily financial information for nowcasting and forecasting real GDP growth with real-time updates.

Version record

Chiara Scotti, "Surprise and uncertainty indexes: Real-time aggregation of real-activity macro-surprises"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 82, 1–19 (2016); DOI 10.1016/j.jmoneco.2016.06.002.

Daily real-activity surprise and uncertainty indexes for the United States, the euro area, the United Kingdom, Canada and Japan, built as weighted averages of release surprises or squared surprises with weights from an Aruoba–Diebold–Scotti business-conditions index. The surprise index keeps the asset-price effects of the underlying releases; uncertainty tied only to real activity has a milder macro effect than uncertainty that also involves the financial sector.

Version record

Alessandro Beber, Michael W. Brandt and Maurizio Luisi, "Distilling the macroeconomic news flow"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 117(3), 489–507 (2015); DOI 10.1016/j.jfineco.2015.05.005.

A cross-sectional technique extracts daily factors from macro releases published at different times and frequencies, giving real-time measures of inflation, output, employment and macro sentiment and of economists' disagreement about them. The measures forecast changes in economic conditions more promptly and accurately than other real-time approaches.

Version record

Leif Anders Thorsrud, "Words are the New Numbers: A Newsy Coincident Index of the Business Cycle"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 38(2), 393–409 (2020); DOI 10.1080/07350015.2018.1506344.

A daily business-cycle index built from quarterly GDP growth and the topics of a daily business newspaper, using a time-varying dynamic factor model with sparse loadings. It classifies business-cycle phases almost perfectly, competes with forecast combinations and expert judgement in out-of-sample nowcasts, and predicts later GDP revisions.

Version record

Domenico Giannone, Michele Lenza and Giorgio E. Primiceri, "Economic Predictions With Big Data: The Illusion of Sparsity"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 89(5), 2409–2437 (2021); DOI 10.3982/ecta17842.

Compares sparse and dense predictive models in macroeconomics, microeconomics and finance with a prior that allows both variable selection and shrinkage. The posterior rarely concentrates on one sparse model; it spreads over many models that often include many predictors.

Version record

Philippe Goulet Coulombe, Maxime Leroux, Dalibor Stevanovic and Stéphane Surprenant, "How is machine learning useful for macroeconomic forecasting?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 37(5), 920–964 (2022); DOI 10.1002/jae.2910.

Isolates which features of machine learning improve macro forecasts: nonlinearity, regularisation, cross-validation and the loss function. Nonlinearity is the main source of gains, which are largest in periods of high uncertainty, financial stress and housing busts; the standard factor model remains the best regulariser and K-fold cross-validation the best practice.

Version record

Magnus Dahlquist and Henrik Hasseltoft, "Economic momentum and currency returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 136(1), 152–167 (2020); DOI 10.1016/j.jfineco.2019.09.002.

Past trends in fundamentals tied to economic activity and inflation predict currency returns. Going long currencies with strong economic momentum and short those with weak momentum earns an annualised Sharpe ratio of 0.70 with significant alpha against carry, momentum and value, and subsumes the alpha of carry trades.

Version record

Kyle Jurado, Sydney C. Ludvigson and Serena Ng, "Measuring Uncertainty"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 105(3), 1177–1216 (2015); DOI 10.1257/aer.20131193.

Econometric estimates of time-varying macroeconomic uncertainty from a data-rich environment. Large uncertainty episodes are rarer than popular proxies suggest but larger, more persistent and more correlated with real activity, so much of the variation in the proxies is not driven by uncertainty.

Version record

Dario Caldara and Matteo Iacoviello, "Measuring Geopolitical Risk"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 112(4), 1194–1225 (2022); DOI 10.1257/aer.20191823.

A news-based geopolitical risk (GPR) index that spikes around the two world wars, the Korean War, the Cuban Missile Crisis and 9/11. Higher geopolitical risk foreshadows lower investment and employment and higher disaster probability and downside risk, through both threats and realised events; industry- and firm-level versions are also provided.

Version record

13.4 Exchange-rate determination and forecasting

Why exchange rates are hard to forecast, what forward rates fail to predict, and models that link currencies to fundamentals, order flow and financial intermediaries. Order-flow evidence from dealer data is in Section 13.9.

Charles Engel, "Exchange Rates and Interest Parity"

Difficulty: Advanced **Type:** Book chapter

Status/version: Handbook of International Economics, 453–522 (2014); DOI 10.1016/b978-0-444-54314-1.00008-2.

Survey of theory and evidence on exchange-rate determination: monetary models under uncovered interest parity and rational expectations, then departures from them through foreign-exchange risk premia, private information, near-rational expectations and peso problems.

Version record

Charles Engel, "The forward discount anomaly and the risk premium: A survey of recent evidence"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Empirical Finance, 3(2), 123–192 (1996); DOI 10.1016/0927-5398(95)00016-x.

Survey of the forward-premium anomaly since Hodrick (1987): the change in the exchange rate is generally negatively related to the forward discount. Reviews forecast-error properties, the links between uncovered and real interest parity, and tests of consumption-CAPM, latent-variable and portfolio-balance risk-premium models; learning, peso problems, survey expectations and bubbles are not covered.

Version record

Lars Peter Hansen and Robert J. Hodrick, *Forward Exchange Rates as Optimal Predictors of Future Spot Rates: An Econometric Analysis* (1980); Eugene F. Fama, *Forward and Spot Exchange Rates* (1984)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected DOI-linked version; print 1980-10.

Forward-rate bias and uncovered-interest-parity tests; statistical rejection alone does not establish implementable carry profits.

Version record

Barbara Rossi, "Exchange Rate Predictability"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Literature, 51(4), 1063–1119 (2013); DOI 10.1257/jel.51.4.1063.

Critical review and updated empirical analysis of exchange-rate forecasting after the Meese–Rogoff puzzle. Whether exchange rates are predictable depends on the predictor, horizon, sample, model and evaluation method; predictability appears most with Taylor-rule or net-foreign-asset predictors, linear models and few estimated parameters, and the toughest benchmark is the random walk without drift.

Version record

Charles Engel and Kenneth D. West, "Exchange Rates and Fundamentals"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 113(3), 485–517 (2005); DOI 10.1086/429137.

In a rational-expectations present-value model the exchange rate behaves almost like a random walk when fundamentals are I(1) and the discount factor is near one, which helps explain why fundamentals barely predict exchange-rate changes. The data show the link the models imply: exchange rates help predict fundamentals.

Version record

Tanya Molodtsova and David H. Papell, "Out-of-sample exchange rate predictability with Taylor rule fundamentals"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 77(2), 167–180 (2009); DOI 10.1016/j.jinteco.2008.11.001.

Models with Taylor-rule fundamentals give short-horizon out-of-sample predictability for 11 of 12 currencies against the U.S. dollar over the post-Bretton Woods float, strongest with heterogeneous coefficients and interest-rate smoothing, and much stronger than interest-rate, purchasing-power-parity or monetary models.

Version record

Pasquale Della Corte, Lucio Sarno and Ilias Tsiakas, "An Economic Evaluation of Empirical Exchange Rate Models"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(9), 3491–3530 (2009); DOI 10.1093/rfs/hhn058.

Evaluates the short-horizon predictive power of fundamentals and forward premia for monthly exchange-rate returns with volatility timing, Bayesian estimation and model averaging, and measures it in economic terms. A risk-averse investor would pay a high fee to switch from a random-walk strategy to one conditioning on the forward premium with stochastic volatility, and combined forecasts give large gains that survive reasonably high transaction costs.

Version record

Charles Engel and Steve Pak Yeung Wu, "Forecasting the U.S. Dollar in the 21st Century"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 141, 103715 (2023); DOI 10.1016/j.jinteco.2023.103715.

The level of the dollar exchange rate appears to forecast dollar returns against major currencies after 2000 at 12- to 60-month horizons under conventional asymptotic tests, but correcting for small-sample bias leaves little evidence against a random walk. Persistent predictors create near-spurious long-horizon correlation, and measures of global risk add nothing to the forecasting power of the level.

Version record

Alan M. Taylor and Mark P. Taylor, "The Purchasing Power Parity Debate"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 18(4), 135–158 (2004); DOI 10.1257/0895330042632744.

History of the purchasing-power-parity debate from its interwar revival and its use as a short-run condition after Bretton Woods through two decades of empirical attack. Larger datasets and nonlinear econometrics since the mid-1990s have narrowed the gap between theory and data, and long-run PPP now has its strongest support in more than thirty years.

Version record

Xavier Gabaix and Matteo Maggiori, "International Liquidity and Exchange Rate Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 130(3), 1369–1420 (2015); DOI 10.1093/qje/qjv016.

Capital flows drive exchange rates through the balance sheets of financiers who absorb international imbalances in asset demand and change their required compensation for currency risk, moving both the level and the volatility of exchange rates. The theory helps explain the disconnect between exchange rates and macro fundamentals and implies that exchange rates seldom act as shock absorbers.

Version record

Oleg Itskhoki and Dmitry Mukhin, "Exchange Rate Disconnect in General Equilibrium"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 129(8), 2183–2232 (2021); DOI 10.1086/714447.

A dynamic general-equilibrium model with home bias and financial-market shocks that accounts for the Meese–Rogoff, Backus–Smith, purchasing-power-parity and uncovered-interest-parity puzzles. Financial shocks produce volatile, near-martingale exchange rates with realistic comovement with macro variables without spoiling the fit of business-cycle moments.

Version record

Robin Greenwood, Samuel Hanson, Jeremy C. Stein and Adi Sunderam, "A Quantity-Driven Theory of Term Premia and Exchange Rates"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 138(4), 2327–2389 (2023); DOI 10.1093/qje/qjad024.

Specialised bond investors absorb supply shocks to long-term bonds in two currencies. Because long bonds and currencies are both exposed to short-rate surprises, a supply shift in one currency moves the exchange rate and term premia in both; the model matches the comovement of exchange rates and term premia and the currency effects of quantitative easing, and an extension links spot rates to post-2008 covered-interest-parity deviations.

Version record

Semyon Malamud, Andreas Schrimpf and Yuan Zhang, "An Intermediation-Based Model of Exchange Rates"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(8), 2386–2433 (2025); DOI 10.1093/rfs/hhaf037.

A continuous-time general-equilibrium model in which global intermediaries bargain with households and earn rents from providing access to foreign claims. Their market power tilts state prices, breaks monetary neutrality and distorts international asset prices even with zero net positions, reproducing deviations from uncovered and covered interest parity in closed form.

Version record

Charles Engel, "Exchange Rates, Interest Rates, and the Risk Premium"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 106(2), 436–474 (2016); DOI 10.1257/aer.20121365.

Documents two puzzles with opposite implications for the foreign-exchange risk premium: high-interest currencies earn high expected short-term returns (the uncovered-interest-parity puzzle), yet currencies of high real-interest countries are stronger than the expected path of real interest differentials implies. Shows that existing models do not account for both and discusses a model that might.

Version record

Kenneth A. Froot and Tarun Ramadorai, "Currency Returns, Intrinsic Value, and Institutional-Investor Flows"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 60(3), 1535–1566 (2005); DOI 10.1111/j.1540-6261.2005.00769.x.

Splits currency returns into permanent intrinsic-value shocks and transitory expected-return shocks using institutional-investor flow data. Flows relate to the transitory part and short-horizon returns; intrinsic value moves with expected cumulative interest differentials and explains long-horizon values, which helps explain why exchange-rate models fit poorly at short horizons.

Version record

Nelson Camanho, Harald Hau and Hélène Rey, "Global Portfolio Rebalancing and Exchange Rates"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 35(11), 5228–5274 (2022); DOI 10.1093/rfs/hhac023.

Fund-level evidence that equity funds rebalance after excess foreign returns, generating capital flows that move currencies, with an equilibrium model in which incompletely traded FX risk partly segments equity markets. Rebalancing is stronger when FX volatility is high, and a granular instrumental-variable estimate gives a positive currency supply elasticity.

Version record

Andrew Lilley, Matteo Maggiori, Brent Neiman and Jesse Schreger, "Exchange Rate Reconnect"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 104(4), 845–855 (2022); DOI 10.1162/rest_a_00978.

Common proxies for global risk appetite, which did not co-move with exchange rates before 2007, explain a significant share of currency moves in sample since then, and U.S. purchases of foreign bonds tracked both in 2007–2012. Read as evidence that the dollar's safe-haven role grew after the global financial crisis; the fit is in sample, not a forecasting result.

Version record

Dagfinn Rime, Lucio Sarno and Elvira Sojli, "Exchange rate forecasting, order flow and macroeconomic information"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 80(1), 72–88 (2010); DOI 10.1016/j.jinteco.2009.03.005.

One year of live Reuters order flow for three major exchange rates relates closely to current and expected macroeconomic fundamentals and predicts daily exchange-rate changes out of sample, judged by Sharpe ratios and utility-based performance fees. Short sample; use it to see how order flow can carry dispersed macro information.

Version record

Richard K. Lyons, *The Microstructure Approach to Exchange Rates*

Difficulty: Advanced **Type:** Book / monograph

Status/version: The MIT Press (2001); DOI 10.7551/mitpress/4602.001.0001.

Monograph that brings microstructure tools to exchange-rate economics, treating exchange rates from the trading room: information is not all public, participants differ, and trading organisation matters. Combines theory and empirical work on order flow and dealer behaviour to explain puzzling exchange-rate behaviour.

Version record

Lucio Sarno and Mark P. Taylor, *The Economics of Exchange Rates*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press (2003); DOI 10.1017/cbo9780511754159.

Graduate text on exchange-rate theory and evidence: parity conditions, forward-rate bias, purchasing-power parity, monetary and portfolio-balance models, intervention, microstructure and currency crises, with appraisal of the empirical literature to the early 2000s.

Version record

Geert Bekaert and Robert J. Hodrick, *International Financial Management*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press (3rd edition, 2017); DOI 10.1017/9781316282274.

Third-edition textbook combining theory, data analysis and cases on foreign-exchange markets, parity conditions, currency risk management, international portfolio choice and international capital budgeting. A structured route into the market conventions and hedging decisions the specialist pathways assume.

Version record

13.5 Currency risk premia and factors

Carry, momentum, value and dollar factors, the risk-based and equilibrium explanations offered for them, and the theory linking exchange rates to stochastic discount factors. Covered-interest-parity and funding evidence is in Section 13.6; option-implied and crash-risk evidence is in Section 13.7.

Ryan Niladri Banerjee, Lena Boneva, Gabor Pinter, and Vladyslav Sushko, Monetary Policy Transmission to Exchange Rates: The Role of Currency Carry Trades

Difficulty: Research **Type:** Report / research note

Status/version: BIS Bulletin 124, published 6 May 2026; institutional policy analysis with eight-page PDF and online appendix.

Shows that exchange-rate responses to a monetary-policy shock depend on pre-carry positioning: large shorts in funding currencies can unwind and amplify appreciation after tightening. Positions and policy surprises are imperfectly observed, so the result should not be converted into a deterministic FX timing rule or assumed to hold uniformly across currencies.

Extended note. Version record

Hanno Lustig, Nikolai Roussanov and Adrien Verdelhan, "Common Risk Factors in Currency Markets"

Difficulty: Research **Type:** Research paper

Status/version: 2011

Establishes dollar and carry factors in global currency returns. It provides the empirical architecture for analysing FX strategies as compensated global risks rather than collections of independent bilateral trades.

Applications: FX forwards; carry, factor investing. Version record

Markus Brunnermeier, Stefan Nagel and Lasse Pedersen, "Carry Trades and Currency Crashes"

Difficulty: Research **Type:** Research paper

Status/version: 2009

Connects currency carry returns to crash risk, funding constraints and speculative positioning. It is indispensable for stress-testing carry portfolios whose normal-period Sharpe ratios conceal negatively skewed liquidation episodes.

Applications: FX; carry, macro, funding liquidity. Version record

Mai C. Dao, Pierre-Olivier Gourinchas, and Oleg Itskhoki, Breaking Parity: Equilibrium Exchange Rates and Currency Premia

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 34443, issued November 2025, DOI 10.3386/w34443; also in the IMF working-paper series, with no journal publication asserted.

Builds an empirical account of covered and uncovered currency premia, spot and forward rates, and dealer-bank balance-sheet constraints, using currency positions to explain time variation. Dealer net futures positions are an explanatory state variable in the paper, not proof of a freely observable or profitable trading rule; institutions must account for position-reporting lag, hedging versus speculation, execution cost, and funding basis.

Extended note. Version record

Craig Burnside, Martin Eichenbaum, Isaac Kleshchelski and Sergio Rebelo, *Do Peso Problems Explain the Returns to the Carry Trade?* (2011 journal version)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2011. Version distinctions remain in the annotation.

Examines carry returns and peso-problem explanations.

Version record

Lukas Menkhoff, Lucio Sarno, Maik Schmeling and Andreas Schrimpf, "Carry Trades and Global Foreign Exchange Volatility"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 67(2), 681–718 (2012); DOI 10.1111/j.1540-6261.2012.01728.x.

Global FX volatility innovations price carry-trade portfolios; high-yield currencies lose when volatility spikes.

Version record

Lukas Menkhoff, Lucio Sarno, Maik Schmeling and Andreas Schrimpf, "Currency momentum strategies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 106(3), 660–684 (2012); DOI 10.1016/j.jfineco.2012.06.009.

Cross-sectional currency momentum is large but concentrated in high-cost, minor currencies.

Version record

Lukas Menkhoff, Lucio Sarno, Maik Schmeling and Andreas Schrimpf, "Currency Value"
Difficulty: Research **Type:** Research paper
Status/version: Review of Financial Studies, 30(2), 416–441 (2017); DOI 10.1093/rfs/hhw067.
Real-exchange-rate value signals adjusted for fundamentals predict currency returns.
Version record

Hanno Lustig, Nikolai Roussanov and Adrien Verdelhan, "Countercyclical currency risk premia"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 111(3), 527–553 (2014); DOI 10.1016/j.jfineco.2013.12.005.
Dollar carry trade: US interest-rate differential predicts dollar returns countercyclically.
Version record

Zhengyang Jiang, Arvind Krishnamurthy and Hanno Lustig, "Foreign Safe Asset Demand and the Dollar Exchange Rate"
Difficulty: Research **Type:** Research paper
Status/version: The Journal of Finance, 76(3), 1049–1089 (2021); DOI 10.1111/jofi.13003.
Treasury convenience yield explains dollar movements; links safe-asset demand to FX.
Version record

Lukas Kremens, Ian W. R. Martin and Liliana Varela, "Long-Horizon Exchange Rate Expectations"
Difficulty: Research **Type:** Research paper
Status/version: The Journal of Finance, 80(6), 3695–3724 (2025); DOI 10.1111/jofi.13504.
Survey exchange-rate expectations of professionals forecast two-year currency appreciation in and out of sample and relate to risk-neutral covariance, real exchange rates and rates.
Version record

Zhengyang Jiang, Robert J. Richmond and Tony Zhang, "Understanding the strength of the dollar"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 168, 104052 (2025); DOI 10.1016/j.jfineco.2025.104052.
Analyses the sources of the US dollar's strength.
Extended note. Version record

Felipe Aldunate, Zhi Da, Borja Larrain and Clemens Sialm, "Pension fund flows, exchange rates, and covered interest rate parity"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 170, 104075 (2025); DOI 10.1016/j.jfineco.2025.104075.
Links pension-fund flows to exchange-rate movements and covered-interest-parity deviations.
Extended note. Version record

Ai Jun Hou, Lucio Sarno and Xiaoxia Ye, "The trade imbalance network and currency returns"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 172, 104112 (2025); DOI 10.1016/j.jfineco.2025.104112.
Adds network centrality of intermediaries' balance sheets to Gabaix–Maggiori; a centrality-based currency factor prices currency returns.
Version record

Alex Cheema-Fox, Edward S. Cuipa and Robin Greenwood, "Flight to Safety or Flight from Carry?"
Difficulty: Advanced **Type:** Research paper
Status/version: Financial Analysts Journal, 1–19 (2026); DOI 10.1080/0015198x.2026.2728601.
Custodian FX-forward and sovereign-bond flows classify crises as flights to safety or flights from carry; CHF, JPY and USD roles diverge, and the 2024 carry unwind centred on the yen.
Version record

Po-Hsuan Hsu, Mark P. Taylor, Zigan Wang and Yan Li, "On the profitability of influential carry-trade strategies: Data-snooping bias and post-publication performance"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Empirical Finance, 83, 101640 (2025); DOI 10.1016/j.jempfin.2025.101640.
Data-snooping-adjusted tests of influential carry-trade strategies.
Extended note. Version record

Tarek A. Hassan and Tony Zhang, "The Economics of Currency Risk"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Economics, 13(1), 281–307 (2021); DOI 10.1146/annurev-economics-092220-103354.

Review of currency and country risk and their macroeconomic consequences. Countries with safer currencies have persistently lower interest rates and required returns on capital and so accumulate more capital; the survey links currency risk premia to capital allocation, exchange-rate stabilisation, trade-deficit sustainability and cross-border spillovers.

Version record

Hanno Lustig and Adrien Verdelhan, "The Cross Section of Foreign Currency Risk Premia and Consumption Growth Risk"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 97(1), 89–117 (2007); DOI 10.1257/aer.97.1.89.

Sorts currencies into interest-rate portfolios: high-interest currencies depreciate when U.S. consumption growth is low and low-interest currencies appreciate, so the carry excess returns compensate for aggregate consumption growth risk. Read with Burnside's comment and the authors' 2011 reply.

Version record

Craig Burnside, "The Cross Section of Foreign Currency Risk Premia and Consumption Growth Risk: Comment"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 101(7), 3456–3476 (2011); DOI 10.1257/aer.101.7.3456.

The stochastic discount factor of the Lustig–Verdelhan benchmark model is approximately uncorrelated with the currency portfolio returns it is meant to price, so one cannot reject that the model explains none of the cross-section of expected returns. Concludes that the forward-premium puzzle remains a puzzle.

Version record

David K. Backus, Silverio Foresi and Chris I. Telmer, "Affine Term Structure Models and the Forward Premium Anomaly"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 56(1), 279–304 (2001); DOI 10.1111/0022-1082.00325.

Characterises the forward-premium anomaly inside affine term-structure models. Matching it requires state variables with asymmetric effects on state prices in different currencies or nominal rates that can be negative with positive probability, and both routes have quantitative shortcomings. The standard starting point for no-arbitrage models of currency risk premia.

Version record

Adrien Verdelhan, "The Share of Systematic Variation in Bilateral Exchange Rates"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 73(1), 375–418 (2018); DOI 10.1111/jofi.12587.

Sorting currencies by their betas on the dollar gives a new cross-section of currency excess returns priced by a dollar slope factor that is orthogonal to the carry factor. The dollar and carry factors together explain 18% to 80% of monthly bilateral exchange-rate changes, implying at least two global shocks in models of exchange rates.

Version record

Tarek A. Hassan and Rui C. Mano, "Forward and Spot Exchange Rates in a Multi-Currency World"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 134(1), 397–450 (2019); DOI 10.1093/qje/qjy026.

Decomposes uncovered-interest-parity violations into cross-currency, between and cross-time components. The forward-premium puzzle and the dollar trade are driven almost entirely by the cross-time component, the carry trade largely by the cross-currency one, so the anomalies are distinct and impose different restrictions on models of currency returns.

Version record

Hanno Lustig, Andreas Stathopoulos and Adrien Verdelhan, "The Term Structure of Currency Carry Trade Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 109(12), 4142–4177 (2019); DOI 10.1257/aer.20180098.

At a fixed investment horizon, carry-trade returns fall as the maturity of the foreign bonds rises, because local term premia offset currency risk premia; foreign bond returns in dollars become less predictable with maturity. Leading no-arbitrage models miss this downward term structure, and the paper derives a preference-free condition they must satisfy.

Version record

Kent Daniel, Robert J. Hodrick and Zhongjin Lu, "The Carry Trade: Risks and Drawdowns"

Difficulty: Research **Type:** Research paper

Status/version: Critical Finance Review, 6(2), 211–262 (2017); DOI 10.1561/104.00000051.

Dollar-neutral G10 carry trades earn positive average returns but are strongly negatively skewed, load on risk factors and show downside risk, whereas a diversified dollar-carry portfolio has a higher Sharpe ratio, little skewness and no downside risk. Daily drawdown distributions point to time-varying autocorrelation behind negative skewness at longer horizons.

Version record

Emmanuel Farhi and Xavier Gabaix, "Rare Disasters and Exchange Rates"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 131(1), 1–52 (2016); DOI 10.1093/qje/qjv040.

Exchange-rate model in which the probability of world disasters and each country's exposure to them vary over time. It accounts for excess volatility and disconnect, the forward-premium puzzle and carry returns, and stock–currency comovement, and predicts links between exchange rates and disaster risk measured in currency options.

Version record

Adrien Verdelhan, "A Habit-Based Explanation of the Exchange Rate Risk Premium"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 65(1), 123–146 (2010); DOI 10.1111/j.1540-6261.2009.01525.x.

Two-country model with external habits in which countercyclical risk aversion and procyclical real rates make expected currency excess returns rise with interest-rate differentials, reproducing the uncovered-interest-parity puzzle.

Version record

Riccardo Colacito and Mariano M. Croce, "Risks for the Long Run and the Real Exchange Rate"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 119(1), 153–181 (2011); DOI 10.1086/659238.

Equilibrium model with cross-country correlated long-run growth risk and Epstein–Zin preferences, estimated on U.S. and U.K. data. It reconciles smooth real exchange rates with volatile marginal utilities and explains highly correlated stock markets despite weakly correlated fundamentals.

Version record

Robert J. Richmond, "Trade Network Centrality and Currency Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 74(3), 1315–1361 (2019); DOI 10.1111/jofi.12755.

Countries more central in the global trade network have lower interest rates and currency risk premia. In a general-equilibrium model their consumption is more exposed to global shocks, so their currencies appreciate in bad times; the data confirm that central countries' consumption growth covaries more with world consumption.

Version record

Mirela Sandulescu, Fabio Trojani and Andrea Vedolin, "Model-Free International Stochastic Discount Factors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 76(2), 935–976 (2021); DOI 10.1111/jofi.12970.

Model-free framework relating international stochastic discount factors, exchange rates and market structure. In incomplete markets a stochastic wedge separates the exchange rate from the ratio of SDFs; market segmentation weakens the link and lets models match international returns with moderate SDF volatility and correlation.

Version record

Michael W. Brandt, John H. Cochrane and Pedro Santa-Clara, "International risk sharing is better than you think, or exchange rates are too smooth"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 53(4), 671–698 (2006); DOI 10.1016/j.jmoneco.2005.02.004.

Exchange rates equal the ratio of domestic to foreign marginal-utility growth, which equity premia imply varies by at least 50% a year, while exchange rates vary by about 10%. Marginal utility growth must therefore be highly correlated across countries: risk sharing is much better than consumption data suggest, or exchange rates are too smooth.

Version record

Robert Ready, Nikolai Roussanov and Colin Ward, "Commodity Trade and the Carry Trade: A Tale of Two Countries"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 72(6), 2629–2684 (2017); DOI 10.1111/jofi.12546.

Commodity-exporting countries have persistently high interest rates and finished-goods exporters low ones. In a general-equilibrium trade model commodity producers are insulated from global productivity shocks that final-goods producers absorb, so commodity-currency risk premia rise with productivity differentials and trade frictions, as the data confirm.

Version record

Angelo Ranaldo and Paul Söderlind, "Safe Haven Currencies"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 14(3), 385–407 (2010); DOI 10.1093/rof/rfq007.

High-frequency evidence for 1993–2008 that the Swiss franc and yen appreciate against the dollar when U.S. stocks fall and bond prices and FX volatility rise. The safe-haven behaviour appears from hours to days, is non-linear in volatility and strongest in crises, notably for the yen in 2008.

Version record

13.6 Covered interest parity, the cross-currency basis and dollar funding

How covered-interest-parity deviations are measured, why the cross-currency basis has persisted since 2008, and how hedging demand, dealer balance sheets, regulation and central-bank swap lines set the price of synthetic dollar funding. Hedging demand from investors is also covered in Section 13.8.

Wenxin Du, Alexander Tepper and Adrien Verdelhan, Deviations from Covered Interest Rate Parity

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance 73(3), 915–957, 2018.

Documents persistent covered-interest-parity deviations and their relationship to quarter-end bank balance-sheet constraints. It provides an empirical foundation for interpreting cross-currency basis as more than a mechanical curve-fitting residual. Reproduce contract dates, collateral, credit exposure and balance-sheet costs before calling a spread arbitrage; the paper's findings do not imply that every investor can finance and scale the same trade.

Extended note. Version record

Alyssa Anderson, Wenxin Du and Bernd Schlusche, "Arbitrage Capital of Global Banks"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(5), 2591–2638 (2025); DOI 10.1111/jofi.13478.

The 2016 US money-market-fund reform cut global banks' unsecured wholesale funding; banks responded mainly by cutting CIP and other arbitrage positions rather than lending.

Version record

Pēteris Kloks, Edouard Mattille and Angelo Ranaldo, "Segmented Dollar Funding"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 184, 104348 (2026); DOI 10.1016/j.jfineco.2026.104348.

CIP deviations reflect constrained access of non-US agents to wholesale dollar funding and a premium for unencumbered synthetic dollars, not only limits to arbitrage.

Version record

Wenxin Du and Jesse Schreger, "CIP deviations, the dollar, and frictions in international capital markets"

Difficulty: Advanced **Type:** Book chapter

Status/version: Handbook of International Economics: International Macroeconomics, Volume 6, 147–197 (2022); DOI 10.1016/bs.hesint.2022.03.001.

Handbook survey of the breakdown of covered interest parity in the global financial crisis and its persistence over the following decade: how to measure the deviations, what drives them, and what they imply for global financial markets.

Version record

Q. Farooq Akram, Dagfinn Rime and Lucio Sarno, "Arbitrage in the foreign exchange market: Turning on the microscope"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 76(2), 237–253 (2008); DOI 10.1016/j.jinteco.2008.07.004.

Tick data on executable quotes for three major currency and money markets show numerous short-lived covered-interest-arbitrage opportunities of economically significant size. They last long enough to exploit but too briefly to appear in lower-frequency data, which is why pre-2008 studies found parity holding.

Version record

Naohiko Baba and Frank Packer, "Interpreting deviations from covered interest parity during the financial market turmoil of 2007–08"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 33(11), 1953–1962 (2009); DOI 10.1016/j.jbankfin.2009.05.007.

Sharp, persistent euro–dollar CIP deviations in 2007–08 are associated with differences in counterparty risk between European and U.S. institutions, and ECB dollar auctions funded by Federal Reserve swap lines eased the FX swap market dislocation.

Version record

Claudio Borio, Robert N. McCauley, Patrick McGuire and Vladyslav Sushko, Covered interest parity lost: understanding the cross-currency basis

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Quarterly Review, September 2016 (published 18 September 2016); special feature.

Explains the post-crisis cross-currency basis by the interaction of hedging demand from banks, institutional investors and firms, which is largely insensitive to the basis, with tighter limits to arbitrage from risk management and balance-sheet costs: hedging demand opens the basis and costly balance sheet space keeps it from closing. The sign of each currency's basis lines up with net hedging positions.

Version record

Dagfinn Rime, Andreas Schrimpf and Olav Syrstad, "Covered Interest Parity Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 35(11), 5185–5227 (2022); DOI 10.1093/rfs/hhac026.

Measured with marginal funding costs and risk-free investments, CIP holds fairly well for most participants; only a few high-rated banks have arbitrage opportunities. Dealers offset lower-rated banks' FX-swap demand for dollars by inducing arbitrage flows from high-rated banks, but the arbitrage does not scale because funding costs rise with position size.

Version record

Stefan Avdjiev, Wenxin Du, Catherine Koch and Hyun Song Shin, "The Dollar, Bank Leverage, and Deviations from Covered Interest Parity"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review: Insights, 1(2), 193–208 (2019); DOI 10.1257/aeri.20180322.

A stronger dollar goes with wider CIP deviations and contracting cross-border dollar bank lending. The authors read the dollar as a barometer of risk-taking capacity in global capital markets.

Version record

Victoria Ivashina, David S. Scharfstein and Jeremy C. Stein, "Dollar Funding and the Lending Behavior of Global Banks"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 130(3), 1241–1281 (2015); DOI 10.1093/qje/qjv017.

Non-U.S. banks fund dollar lending with flighty wholesale dollars; when their credit quality falls they swap euros into dollars, which breaks CIP when arbitrage capital is limited and makes synthetic dollars expensive. In the 2011 euro-area crisis, as U.S. money funds withdrew, euro-area banks cut dollar lending relative to euro lending.

Version record

Puriya Abbassi and Falk Bräuning, "Demand Effects in the FX Forward Market: Micro Evidence from Banks' Dollar Hedging"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 34(9), 4177–4215 (2021); DOI 10.1093/rfs/hhaa123.

Contract-level supervisory data show that dollar forward sales by non-U.S. banks that start at a quarter-end and mature soon after trade at higher prices and volumes, driven by banks hedging large on-balance-sheet dollar assets to cut capital charges on open FX positions. Demand from banks' FX-exposure management moves the basis.

Version record

Gino Cenedese, Pasquale Della Corte and Tianyu Wang, "Currency Mispricing and Dealer Balance Sheets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 76(6), 2763–2803 (2021); DOI 10.1111/jofi.13079.

Trade-repository data with counterparty identities and the U.K. leverage-ratio framework as a shock show that affected dealers charge about 20 basis points a year more for synthetic dollar funding than unaffected dealers, after controlling for client demand; some clients shift to unaffected dealers they already traded with.

Version record

Patrick Augustin, Mikhail Chernov, Lukas Schmid and Dongho Song, "The Term Structure of Covered Interest Rate Parity Violations"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 79(3), 2077–2114 (2024); DOI 10.1111/jofi.13336.

Values currency derivatives with an SDF inferred from interest-rate swaps, which embeds risk-based intermediary constraints; the remaining wedge measures non-risk-based constraints. The wedge is absent at short horizons but accounts for 40% of long-term CIP violations and correlates with the shadow cost of intermediary capital.

Version record

Gordon Y. Liao, "Credit migration and covered interest rate parity"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 138(2), 504–525 (2020); DOI 10.1016/j.jfineco.2020.06.002.

The same corporate credit risk is priced differently in bonds of different currencies, and this credit-spread deviation moves with CIP violations across currencies and over time. In a segmented-markets model, hedged investment and debt issuance bundle the two violations, so limits to arbitrage spill from one market to the other.

Version record

Wenxin Du, Benjamin Hébert and Amy Wang Huber, "Are Intermediary Constraints Priced?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 36(4), 1464–1507 (2023); DOI 10.1093/rfs/hhac050.

A forward CIP trading strategy that bets on CIP violations shrinking identifies the price of the risk that intermediary constraints tighten. Its returns are highest for carry-trade currency pairs, it contributes substantially to SDF volatility, correlates with other near-arbitrages and intermediary wealth, and is priced consistently across asset classes.

Version record

Saleem Bahaj and Ricardo Reis, "Central Bank Swap Lines: Evidence on the Effects of the Lender of Last Resort"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 89(4), 1654–1693 (2022); DOI 10.1093/restud/rdab074.

Swap lines let a source central bank lend its currency to foreign banks through their own central bank. Variation in swap-line terms, access and opening days shows that they cap CIP deviations, lower ex post bank borrowing costs and raise ex ante inflows into privately issued source-currency assets.

Version record

Claudio Borio, Robert N. McCauley and Patrick McGuire, Dollar debt in FX swaps and forwards: huge, missing and growing

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Quarterly Review, December 2022 (published 5 December 2022); special feature.

Uses the 2022 Triennial Survey to estimate dollar payment obligations from FX swaps, forwards and currency swaps that standard debt statistics miss: about $26 trillion for non-banks outside the United States, double their on-balance-sheet dollar debt, and about $39 trillion for non-U.S. banks. Swap-line decisions in 2008 and 2020 were made with little information on who owed this debt.

Version record

Olav Syrstad and Ganesh Viswanath-Natraj, "Price-setting in the foreign exchange swap market: Evidence from order flow"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 146(1), 119–142 (2022); DOI 10.1016/j.jfineco.2022.07.004.

Inter-dealer transaction data show that the price impact of FX swap order flow rose from under one basis point before 2008 to about five after, but only when funding-cost dispersion is high and around quarter-ends. Swap lines reduce order flow into dollars, while at quarter-ends and after policy announcements dealers move prices to curb flow.

Version record

Eugenio M. Cerutti, Maurice Obstfeld and Haonan Zhou, "Covered interest parity deviations: Macrofinancial determinants"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 130, 103447 (2021); DOI 10.1016/j.jinteco.2021.103447.

For 2010–2018, FX liquidity and intermediaries' risk-taking capacity are strongly correlated with the cross-currency basis, and the link between broad dollar strength and the basis runs through a common safe-haven factor. Post-crisis monetary policy and bank regulation, including year-end effects, also matter.

Version record

13.7 FX volatility, option-implied risk and crash risk

What currency option prices reveal about skew, volatility and correlation risk premia and crash risk, and whether option-implied measures forecast currency returns. Option quotation conventions and smile construction are in Section 13.9.

Pasquale Della Corte, Tarun Ramadorai and Lucio Sarno, "Volatility risk premia and exchange rate predictability"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 120(1), 21–40 (2016); DOI 10.1016/j.jfineco.2016.02.015.

Currency option volatility risk premia predict exchange-rate returns.

Version record

Junye Li, Lucio Sarno and Gabriele Zinna, "Skewness Risk Premia and the Cross Section of Currency Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(4), 1565–1603 (2026); DOI 10.1017/s0022109025102123.

Currencies with high option-implied skewness risk premia outperform those with low SRP, robust to asset-pricing tests.

Version record

Peter Carr and Liuren Wu, "Stochastic skew in currency options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 86(1), 213–247 (2007); DOI 10.1016/j.jfineco.2006.03.010.

Over-the-counter options on two major currency pairs show risk-neutral return distributions whose asymmetry varies strongly over time and often changes sign. A class of stochastic-skew models fits this behaviour and beat jump-diffusion stochastic-volatility models in and out of sample.

Version record

Gurdip Bakshi, Peter Carr and Liuren Wu, "Stochastic risk premiums, stochastic skewness in currency options, and stochastic discount factors in international economies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 87(1), 132–156 (2008); DOI 10.1016/j.jfineco.2006.12.001.

International SDF models estimated on returns and options for a triangle of three currency pairs. The Japanese risk premium is on average larger than the U.S. or U.K. one, the global premium is more persistent and volatile than country premiums, and only downside jumps are priced.

Version record

Pasquale Della Corte, Lucio Sarno and Ilias Tsiakas, "Spot and forward volatility in foreign exchange"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 100(3), 496–513 (2011); DOI 10.1016/j.jfineco.2011.01.007.

Tests whether forward implied volatility, the delivery price of a forward volatility agreement, predicts future spot implied volatility. It is a biased predictor that overstates future moves, and trading on the bias earns high economic value, indicating predictable volatility term premiums in currencies.

Version record

Juan M. Londono and Hao Zhou, "Variance risk premiums and the forward premium puzzle"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 124(2), 415–440 (2017); DOI 10.1016/j.jfineco.2017.02.002.

The currency variance risk premium predicts currency appreciation at a six-month horizon and the stock variance risk premium at one month; the two premia are weakly correlated, and the currency one does not predict stock returns. A consumption-based model with separate local and global uncertainty, the latter unpriced in local stock markets, rationalises this.

Version record

Jakub W. Jurek, "Crash-neutral currency carry trades"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 113(3), 325–347 (2014); DOI 10.1016/j.jfineco.2014.05.004.

G10 carry trades in 1990–2012 earn Sharpe ratios comparable to equity factors. Hedging them with out-of-the-money FX options shows the returns are not a peso problem: crash-risk premia account for at most a third of the carry excess return.

Version record

Emmanuel Farhi, Samuel P. Fraiberger, Xavier Gabaix, Romain Ranciere and Adrien Verdelhan, "Crash Risk in Currency Markets"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 15062, June 2009; DOI 10.3386/w15062. Working-paper record; no journal version established in this check.

Since late 2008 out-of-the-money currency options have priced large depreciations of high-interest and appreciations of low-interest currencies. A structural model with Gaussian and disaster risk, estimated on spot, forward and option data, yields a real-time index of world disaster-risk premia; disaster risk accounts for more than a third of currency risk premia in advanced countries.

Extended note. Version record

Mikhail Chernov, Jeremy Graveline and Irina Zviadadze, "Crash Risk in Currency Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 53(1), 137–170 (2018); DOI 10.1017/s0022109017000801.

Empirical model of bilateral exchange rates with stochastic variance and jumps in both the rate and its variance. Jump probabilities in the rate rise with interest rates, rate jumps coincide with announcements, jumps make up about a quarter of currency risk, and option prices suggest jump risk is priced.

Version record

Ricardo J. Caballero and Joseph B. Doyle, "Carry Trade and Systemic Risk: Why are FX Options so Cheap?"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 18644, December 2012; DOI 10.3386/w18644. Working-paper record; no journal version established in this check.

Carry returns are highly correlated with a short-VIX-futures roll-down strategy that does at least as well, so their systemic-risk-adjusted returns are low; hedging carry with FX options, however, leaves large returns not explained by systemic risk, because the option portfolio is cheap systemic insurance. A counterpoint to crash-premium explanations.

Extended note. Version record

Zhenzhen Fan, Juan M. Londono and Xiao Xiao, "Equity tail risk and currency risk premiums"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 143(1), 484–503 (2022); DOI 10.1016/j.jfineco.2021.05.020.

An option-based equity tail-risk factor is priced in the cross-section of currencies: currencies more exposed to it hedge tail risk and earn lower premiums. The global component of the factor has a negative price in carry and momentum portfolios and in other asset classes.

Version record

Philippe Mueller, Andreas Stathopoulos and Andrea Vedolin, "International correlation risk"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 126(2), 270–299 (2017); DOI 10.1016/j.jfineco.2016.09.012.

Cross-sectional dispersion of conditional FX correlations is countercyclical, and currencies that do badly when dispersion is high earn higher excess returns. Option-implied correlation risk premia point the other way, which a no-arbitrage model with unspanned FX correlation risk reconciles.

Version record

Lukas Kremens and Ian Martin, "The Quanto Theory of Exchange Rates"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 109(3), 810–843 (2019); DOI 10.1257/aer.20180019.

An identity links expected exchange-rate appreciation to a risk-neutral covariance measurable from quanto index contracts. The quanto variable predicts currency appreciation and currency excess returns in panel regressions and, out of sample, beats uncovered interest parity, purchasing-power parity and a random walk for dollar-neutral appreciation.

Version record

Alessandro Beber, Francis Breedon and Andrea Buraschi, "Differences in beliefs and currency risk premiums"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 98(3), 415–438 (2010); DOI 10.1016/j.jfineco.2010.07.001.

A proxy for disagreement built from about 50 forecasters' exchange-rate forecasts over ten years affects currency-option implied volatility beyond fundamental volatility, and also the shape of the smile, volatility risk premia and future currency returns.

Version record

13.8 Currency hedging for international portfolios

How much of a foreign portfolio's currency exposure to hedge, over which horizon and at what cost, and how hedging demand feeds back into exchange rates and the cross-currency basis. Basis and funding evidence is in Section 13.6.

Fischer Black, "Universal Hedging: Optimizing Currency Risk and Reward in International Equity Portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal, 45(4), 16–22 (1989); DOI 10.2469/faj.v45.n4.16.

Derives a universal currency-hedge ratio for international equity investors who share common views, in liquid markets without barriers to international investing. It needs three inputs: the cross-country averages of the world market portfolio's expected return and volatility, and average exchange-rate volatility. The rules: hedge foreign equity, hedge less than 100%, and hedge equally across countries, wherever the investor lives.

Version record

Kenneth Froot, "Currency Hedging over Long Horizons"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 4355, May 1993; DOI 10.3386/w4355. Working-paper record; no journal version established in this check.

Re-examines the advice to hedge the currency exposure of international investments fully. Hedging reduces return variance at short horizons but not at horizons of several years, where full hedging can raise variance; variance-minimising long-horizon hedge ratios are low and make little difference to variance.

Extended note. Version record

John Y. Campbell, Karine Serfaty-de Medeiros and Luis M. Viceira, "Global Currency Hedging"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 65(1), 87–121 (2010); DOI 10.1111/j.1540-6261.2009.01524.x.

Over 1975–2005 the U.S. dollar (especially against the Canadian dollar), the euro and the Swiss franc moved against world equity markets, making them attractive to risk-minimising global equity investors despite low average returns. For a global bond investor the risk-minimising strategy is close to a full hedge with a modest long dollar position, and there is little evidence for adjusting hedges to interest differentials.

Version record

Tim A. Kroencke, Felix Schindler and Andreas Schrimpf, "International Diversification Benefits with Foreign Exchange Investment Styles"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 18(5), 1847–1883 (2014); DOI 10.1093/rof/rft047.

Managing the currency component of international investments with carry, FX momentum and FX value styles gives economically large diversification benefits beyond the hedging benefit of removing currency risk. The results hold after transaction costs and in a real-time out-of-sample experiment; adding a composite FX style portfolio to global stocks and bonds raises the out-of-sample Sharpe ratio from 0.64 to 1.05 without worsening skewness.

Version record

Wei Opie and Steven J. Riddiough, "Global currency hedging with common risk factors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 136(3), 780–805 (2020); DOI 10.1016/j.jfineco.2019.12.001.

Hedges the currency exposure of international equity and bond portfolios dynamically, using time-series predictability of currency returns that arises from a forecastable component in global factor returns. The hedge beats leading alternatives on many performance measures, and a stand-alone currency portfolio exploiting the predictability diversifies global portfolios better than carry, value or momentum.

Version record

Luis M. Viceira and Sally Shen, "Optimal Currency Exposure When Interest Parity Fails"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(11), 13–31 (2026); DOI 10.3905/jpm.2026.049.

Optimal currency exposure for US and Canadian equity and bond investors across developed and emerging markets, 1975–2023. The dollar and, less so, the euro appreciate when global equities fall, so risk-minimising investors should hold them and hedge commodity and emerging-market currencies; covered-interest-parity hedging costs do not change this, but persistent uncovered-interest-parity deviations give emerging-market currencies positive expected returns that can attenuate or reverse those hedges.

Version record

Pedro Castro, Carl Hamill, John Harber, Campbell R. Harvey and Otto Van Hemert, "The Best Strategies for FX Hedging"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 51(9), 37–78 (2025); DOI 10.3905/jpm.2025.1.740.

Decades of dynamic FX-hedging evidence across currency pairs: conditioning hedge ratios on carry, trend and value improves portfolio outcomes relative to static hedging, linking hedging policy to currency premia.

Version record

Wei Opie and Steven J. Riddiough, "The End of Crisis Insurance? Reassessing the US Dollar and the Case for Currency Hedging"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(11), 32–56 (2026); DOI 10.3905/jpm.2026.055.

Argues that the dollar's crisis insurance for non-US investors may weaken as shocks increasingly originate in the United States and the dollar's global role erodes. A scenario analysis re-estimates minimum-variance hedge ratios with crisis-state correlations that remove that insurance; the changes are economically large, especially for commodity currencies.

Version record

Gordon Y. Liao and Tony Zhang, "The Hedging Channel of Exchange Rate Determination"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(1), 1–38 (2025); DOI 10.1093/rfs/hhae072.

A currency-hedging channel: in financial distress investors raise currency hedging in proportion to their net foreign-asset exposure, and with constrained intermediaries this links slowly adjusting external imbalances to volatile spot and forward exchange rates. Evidence comes from conditional and unconditional moments of exchange rates, option prices and countries' use of Federal Reserve swap lines.

Version record

Clemens Sialm and Qifei Zhu, "Currency Management by International Fixed-Income Mutual Funds"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 79(6), 4037–4081 (2024); DOI 10.1111/jofi.13381.

New data on currency derivatives held by U.S. international fixed-income funds: 90% use forwards but hedge on average only 18% of their currency exposure. Positions reflect risk management, return enhancement through currency momentum and carry, and strategic motives tied to past performance and clientele; funds that hedge have lower return variability without lower abnormal returns.

Version record

13.9 FX options, fixes, execution and settlement

The FX market combines bilateral credit, dealer internalisation, aggregators and venue-specific execution rules. Read these institutional sources with the mathematical execution models and current counterparty disclosures. Subject context and qualifications.

Global Foreign Exchange Committee, FX Global Code; revision and disclosure context

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: FX Global Code updated December 2024; GFXC revision announcement 24 January 2025; checked 4 October 2026.

Principles for wholesale FX execution, information sharing, risk management, confirmation and settlement. The 2024 review specifically strengthens settlement-risk practices and transparency around execution and use of client-generated data. Read liquidity-provider and platform disclosures alongside the code when analysing last look, internalisation and algorithmic execution. Voluntary adherence and a statement of commitment do not establish the realised execution quality of a venue or replace applicable law.

Version record

ISDA and EMTA, 2026 FX Definitions: publication and implementation; Update InfoHub

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Version 1.0 published 3 March 2026; industry implementation scheduled 22 November 2027.

Adds the published successor framework for FX documentation while retaining the distinction between publication and implementation. The definitions consolidate earlier supplements and EMTA non-deliverable terms, update disruption/fallback and calendar-adjustment provisions, and align calculation-agent standards with the 2021 rates definitions. The public implementation hub lists a roadmap, annotated confirmations and July 2026 comparisons. Read the precise definitions incorporated by each confirmation; a future industry transition does not automatically rewrite an existing trade. Public summaries establish the release and transition plan; access to the complete MyLibrary book and contract-specific legal analysis remains separate.

Version record

BIS Markets Committee, FX Execution Algorithms and Market Functioning

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Markets Committee Papers 13, published 30 October 2020; historical survey and study-group report, checked 4 October 2026.

Study-group account of algorithm use in fragmented FX markets, including trade-offs in execution risk, internalisation, monitoring and market functioning. Connect scheduling and venue access to the benchmark, information leakage and who bears price risk during execution. Useful institutional context for extending equity execution models to dealer and aggregator markets. Its survey and market structure are dated; assess current providers and conditions separately rather than importing its market shares.

Version record

Iain J. Clark, Foreign Exchange Option Pricing: A Practitioner's Guide

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2011 copyright; Wiley online edition 2 January 2012.

Builds an FX-options progression from quotation conventions and vanilla smiles to barriers, numerical pricing and richer dynamics. Long-dated products also require coherent rate and collateral assumptions. It is a durable technical foundation, but its historical conventions and market examples must be checked against the selected currency pair, current confirmation and modern discounting setup.

Extended note. Version record

Dimitri Reiswich and Uwe Wystup, FX Volatility Smile Construction; Author working paper (2010 revision)

Difficulty: Advanced **Type:** Research paper

Status/version: Wilmott, issue 60, 58–69, 2012; linked author working-paper revision March 2010.

Explains the conversion from FX delta quotes, risk reversals and butterflies to a strike-based volatility smile. This is the practical bridge between a vendor quote sheet and an option-pricing engine: reproduce quote conventions and round-trip prices before fitting a surface. Pair this construction with the catalogue's static-arbitrage checks; matching the supplied quotes does not by itself guarantee coherent interpolation across strikes or maturities.

Extended note. Version record

Wenxin Du and Amy W. Huber, Dollar Asset Holdings and Hedging around the Globe; Earlier NBER working paper

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, advance article; 27 July 2026, corrected/typeset 7 August 2026.

Uses institutional filings to distinguish foreign demand for dollar-denominated securities from the currency exposure investors retain after hedging. It connects heterogeneous hedging decisions to constrained intermediary supply and hedging costs. Especially useful for currency overlays, cross-currency basis and portfolio flow research across mutual funds, pensions and insurers. Data coverage, aggregation and reporting delays must be respected when translating the paper into a real-time signal.

Extended note. Version record

Bank for International Settlements, OTC foreign exchange turnover in April 2025: Triennial Central Bank Survey; BIS derivatives tables

Difficulty: Intro **Type:** Mixed resources

Status/version: 2025 survey; September 2025 release, April 2025 observation month.

Supplies the global instrument, currency, counterparty and location context missing from a purely developed-market FX reading route. Use the tables to compare spot, forwards, swaps and options, and to understand which currencies trade principally offshore. Turnover is activity rather than outstanding risk or available executable depth. Survey comparisons also depend on the exceptional conditions of the observation month.

Extended note. Version record

Philip Wooldridge, Renminbi propels the growth of EME currency trading

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Quarterly Review, December 2025.

Extends the FX route to emerging-market currencies and the differences between onshore and offshore trading. The survey interpretation helps explain why an NDF, a deliverable forward and a domestic futures position are not interchangeable exposures, and why renminbi activity can dominate aggregate EME growth. This is market-structure evidence, not an exhaustive statement of current eligibility rules for each currency or jurisdiction.

Extended note. Version record

Martin D. D. Evans and Richard K. Lyons, "Order Flow and Exchange Rate Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 110(1), 170–180 (2002); DOI 10.1086/324391.

Interdealer order flow explains a large share of daily exchange-rate changes, where macro models fail.

Version record

Michael R. King, Carol L. Osler and Dagfinn Rime, "The market microstructure approach to foreign exchange: Looking back and looking forward"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of International Money and Finance, 38, 95–119 (2013); DOI 10.1016/j.jimonfin.2013.05.004.

Survey of FX market structure, order flow, dealer behaviour and electronic trading.

Version record

Alain P. Chaboud, Benjamin Chiquoine, Erik Hjalmarsson and Clara Vega, "Rise of the Machines: Algorithmic Trading in the Foreign Exchange Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(5), 2045–2084 (2014); DOI 10.1111/jofi.12186.

EBS data show algorithmic trading improves price efficiency and reduces triangular-arbitrage opportunities.

Version record

Martin D. D. Evans, "Forex trading and the WMR Fix"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 87, 233–247 (2018); DOI 10.1016/j.jbankfin.2017.09.017.

Price behaviour around the London 4pm fix and evidence relevant to benchmark manipulation cases.

Version record

Uwe Wystup, *FX Options and Structured Products*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2017); DOI 10.1002/9781119192183.

Second edition (Wiley 2017): FX conventions, smile construction, barriers, accumulators and structured FX products.

Version record

Wenqian Huang, Angelo Ranaldo, Andreas Schrimpf and Fabricius Somogyi, "Constrained liquidity provision in currency markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 167, 104028 (2025); DOI 10.1016/j.jfineco.2025.104028.

When FX dealers' intermediation capacity is constrained, liquidity costs rise disproportionately relative to dealer volume, weakening the usual volume–liquidity relation.

Version record

Craig Burnside, Mario Cerrato and Zhekai Zhang, "Foreign Exchange Order Flow as a Risk Factor"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(5), 2555–2582 (2025); DOI 10.1017/s0022109024000796.

An order-flow-based currency factor prices carry and momentum currency portfolios.

Version record

Ingomar Krohn, Philippe Mueller and Paul Whelan, "Foreign Exchange Fixings and Returns around the Clock"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 79(1), 541–578 (2024); DOI 10.1111/jofi.13306.

The U.S. dollar appreciates before FX fixes and depreciates afterwards, tracing a W-shaped return pattern around the clock. Reversals across the nine most traded currencies over 21 years are pervasive and large; natural experiments show that a published reference rate sets their timing, consistent with dealers bearing inventory risk when they supply dollars at the fixes.

Version record

Mark B. Garman and Steven W. Kohlhagen, "Foreign currency option values"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 2(3), 231–237 (1983); DOI 10.1016/s0261-5606(83)80001-1.

Derives the closed-form value of a European option on foreign currency by treating the foreign interest rate like a continuous dividend yield in the Black–Scholes framework, the model behind FX option quotation conventions. Annotation not checked against the text; see the evidence note.

Version record

Frédéric Bossens, Grégory Rayée, Nikos S. Skantzos and Griselda Deelstra, "Vanna-Volga methods applied to FX derivatives: from theory to market practice"

Difficulty: Research **Type:** Research paper

Status/version: International Journal of Theoretical and Applied Finance, 13(08), 1293–1324 (2010); DOI 10.1142/s0219024910006212.

Explains the vanna–volga adjustments dealers use to price first-generation FX exotics, rescaling the correction to Black–Scholes by survival probability or expected first-exit time, with a summary of the market data conventions involved. Compares results with indicative market prices and richer models and calibrates to one-touch prices.

Version record

Antonio Castagna, *FX Options and Smile Risk*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley; Crossref print date 2 January 2012, online 9 October 2015; DOI 10.1002/9781119207085. First-edition date not established in this check.

Practitioner book covering FX market conventions, pricing models for FX options, dynamic hedging and volatility trading, construction of the volatility surface, vanilla, barrier and other exotic options, risk-management tools and correlation. Description based on the registered chapter list; contents not inspected.

Version record

Financial Stability Board, Foreign Exchange Benchmarks: Final Report

Difficulty: Advanced **Type:** Report / research note

Status/version: Financial Stability Board report, 30 September 2014.

Reform recommendations prompted by the 2013 concerns about trading around FX benchmark fixes, chiefly the WM/Reuters 4pm London fix: the calculation methodology, IOSCO's review of the WM fixes, central-bank reference rates, infrastructure for executing fix trades, and market conduct around the fixes.

Version record

Michael Melvin and John Prins, "Equity hedging and exchange rates at the London 4p.m. fix"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 22, 50–72 (2015); DOI 10.1016/j.finmar.2014.11.001.

Passive currency hedges of global equity portfolios are typically traded at the London 4pm fix, where trading intensity and volatility peak. Hedging flows predict temporary exchange-rate moves at the fix, especially at month-ends when hedges are rebalanced; the impact is strong but transitory.

Version record

Takatoshi Ito and Masahiro Yamada, "Puzzles in the Tokyo fixing in the forex market: Order imbalances and Bank pricing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 109, 214–234 (2017); DOI 10.1016/j.jinteco.2017.09.005.

At the Tokyo 10am fix price spikes are more frequent than in London, customer orders predictably favour buying foreign currency, and volume concentrates in dollar–yen. Before 2008 banks set the fixing above the highest traded price in the window; the gap disappeared after 2008, though the fix still sits above average traded prices.

Version record

Loriano Mancini, Angelo Ranaldo and Jan Wrampelmeyer, "Liquidity in the Foreign Exchange Market: Measurement, Commonality, and Risk Premiums"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 68(5), 1805–1841 (2013); DOI 10.1111/jofi.12053.

First systematic study of FX liquidity: liquidity differs widely across currency pairs, illiquidity costs are substantial and liquidity co-moves across currencies and with equity and bond markets. Funding currencies insure against liquidity risk and investment currencies are exposed, and a liquidity factor strongly affected carry returns in 2007–2009.

Version record

Nina Karnaukh, Angelo Ranaldo and Paul Söderlind, "Understanding FX Liquidity"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 28(11), 3073–3108 (2015); DOI 10.1093/rfs/hhv029.

Shows that spot FX liquidity can be measured accurately from daily, readily available data over two decades and many currencies. Liquidity falls with funding constraints and global risk, more so for developed and riskier currencies, and co-moves more strongly in distress.

Version record

Marc Glowka and Thomas Nilsson, FX settlement risk: an unsettled issue

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Quarterly Review, December 2022 (published 5 December 2022).

Triennial Survey estimates that $2.2 trillion of daily FX turnover in April 2022 was exposed to settlement risk, up from about $1.9 trillion in 2019, because payment-versus-payment arrangements are unavailable, unsuitable or too costly for some trades. Explains pre-settlement netting, PvP and on-us settlement as the mitigants.

Version record

Committee on Payments and Market Infrastructures, Facilitating increased adoption of payment versus payment (PvP): final report

Difficulty: Advanced **Type:** Report / research note

Status/version: CPMI Papers, final report, 27 March 2023; G20 cross-border payments programme.

Examines why some FX trades still settle without payment versus payment and reviews existing and proposed PvP arrangements. Existing arrangements cover much of the market, but some segments remain exposed; adoption faces weak incentives, technical difficulties connecting to real-time gross settlement systems and legal differences across jurisdictions.

Version record

13.10 Emerging-market currencies, exchange-rate regimes and intervention

De facto exchange-rate regimes, pegs, floors and target zones, currency crises, foreign-exchange intervention and capital-flow measures, non-deliverable and offshore markets, and the dollar's role in the global financial cycle. Capital flows are in Section 13.15, local-currency government bonds in Section 13.16 and country market-access rules in Section 13.17.

Ethan Ilzetzki, Carmen M. Reinhart and Kenneth S. Rogoff, "Exchange Arrangements Entering the Twenty-First Century: Which Anchor will Hold?"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 134(2), 599–646 (2019); DOI 10.1093/qje/qjy033.

De facto classification of anchor currencies, exchange-rate arrangements and a new measure of FX restrictions for 194 countries and territories, 1946–2016. Regimes with limited flexibility remain the majority, the dollar is by far the dominant anchor while the euro's role has stalled, and reserve accumulation since 2002 partly reflects the wish to stabilise exchange rates.

Version record

Ethan Ilzetzki, Carmen M. Reinhart and Kenneth S. Rogoff, "Rethinking exchange rate regimes"

Difficulty: Advanced **Type:** Book chapter

Status/version: Handbook of International Economics: International Macroeconomics, Volume 6, 91–145 (2022); DOI 10.1016/bs.hesint.2022.02.010.

Handbook chapter using an updated classification algorithm and database to trace the global exchange-rate system, including parallel rates, capital controls and reserves: the dollar's growing centrality as anchor, the substitution of reserves for capital controls, the modern Triffin dilemma, falling volatility among core currencies and the rise of China.

Version record

Guillermo A. Calvo and Carmen M. Reinhart, "Fear of Floating"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 117(2), 379–408 (2002); DOI 10.1162/003355302753650274.

Exchange-rate, reserve and interest-rate behaviour shows that most countries that say they float do not: an epidemic of fear of floating, contrary to the view that countries were moving to hard pegs or free floats.

Version record

Paul R. Krugman, "Target Zones and Exchange Rate Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 106(3), 669–682 (1991); DOI 10.2307/2937922.

Model of a target-zone regime in which expected policy defence at the edges shapes the exchange rate even inside the band, where no intervention occurs. The analysis is formally similar to option pricing and investment under uncertainty.

Version record

Urban J. Jermann, "Financial Markets' Views about the Euro–Swiss Franc Floor"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 49(2–3), 553–565 (2017); DOI 10.1111/jmcb.12389.

Model of exchange rates under a policy floor, used to price options and estimated on euro–Swiss franc spot and option data. The implied three-month survival probability of the floor was often below 75% in its first months and reached 95% by August 2014, months before the floor was abandoned.

Version record

Paul Krugman, "A Model of Balance-of-Payments Crises"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 11(3), 311–325 (1979); DOI 10.2307/1991793.

The first-generation currency-crisis model: when domestic credit growth is inconsistent with a fixed exchange rate, speculators exhaust reserves in a sudden attack once the shadow floating rate reaches the peg. Annotation not checked against the text; see the evidence note.

Version record

Maurice Obstfeld, "Models of currency crises with self-fulfilling features"

Difficulty: Research **Type:** Research paper

Status/version: European Economic Review, 40(3–5), 1037–1047 (1996); DOI 10.1016/0014-2921(95)00111-5.

The government's cost of resisting speculation shapes speculators' incentives and can produce multiple equilibria, so high unemployment can trigger a crisis with self-fulfilling features. Reviews other self-reinforcing mechanisms and supporting econometric evidence.

Version record

Graciela L. Kaminsky and Carmen M. Reinhart, "The Twin Crises: The Causes of Banking and Balance-of-Payments Problems"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 89(3), 473–500 (1999); DOI 10.1257/aer.89.3.473.

Banking problems typically precede currency crises, the currency crisis deepens the banking crisis, and financial liberalisation often precedes both. Crises arrive as the economy enters recession after a credit- and capital-inflow-fuelled boom with an overvalued currency.

Version record

Jeffrey A. Frankel and Andrew K. Rose, "Currency crashes in emerging markets: An empirical treatment"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 41(3–4), 351–366 (1996); DOI 10.1016/s0022-1996(96)01441-9.

Annual panel of more than 100 developing countries, 1971–1992. Currency crashes are more likely when output growth is low, domestic credit growth high and foreign interest rates high; a low ratio of FDI to debt is consistently associated with crashes.

Version record

Lucio Sarno and Mark P. Taylor, "Official Intervention in the Foreign Exchange Market: Is It Effective and, If So, How Does It Work?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Economic Literature, 39(3), 839–868 (2001); DOI 10.1257/jel.39.3.839.

Review of theory and evidence concluding that, unlike the 1980s consensus, intervention can be effective, particularly as a signal of policy when announced and concerted. Notes the puzzle that much intervention is secret and proposes a coordination channel.

Version record

Marcel Fratzscher, Oliver Gloede, Lukas Menkhoff, Lucio Sarno and Tobias Stöhr, "When Is Foreign Exchange Intervention Effective? Evidence from 33 Countries"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics, 11(1), 132–156 (2019); DOI 10.1257/mac.20150317.

Daily intervention data for 33 countries, 1995–2011: intervention is widely used and succeeds more than 80% of the time under some criteria, smoothing exchange rates and stabilising narrow-band regimes. Moving the level in flexible regimes needs large, public interventions supported by communication.

Version record

Lukas Menkhoff, "Foreign Exchange Intervention in Emerging Markets: A Survey of Empirical Studies"

Difficulty: Research **Type:** Research paper

Status/version: World Economy, 36(9), 1187–1208 (2013); DOI 10.1111/twec.12027.

Survey showing that emerging-market interventions differ from advanced-economy ones: they are regular and backed by large reserves, partial non-sterilisation, information advantages and capital controls, and often move the level and volatility of exchange rates.

Version record

Gustavo Adler, Kyun Suk Chang, Rui C. Mano and Yuting Shao, "Foreign Exchange Intervention: A Data Set of Official Data and Estimates"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 57(5), 1241–1273 (2025); DOI 10.1111/jmcb.13137.

Monthly and quarterly FX-intervention data since 2000: published official data for about 40 countries and estimates for 122, covering spot and derivative operations, adjusted for valuation changes, income flows and other foreign-currency positions, with a sterilised/unsterilised classification. Distinguish reported from estimated series in use.

Version record

Silvia Miranda-Agrippino and Hélène Rey, "U.S. Monetary Policy and the Global Financial Cycle"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 87(6), 2754–2776 (2020); DOI 10.1093/restud/rdaa019.

A U.S. monetary tightening lowers a single global factor in risky asset prices, deleverages global intermediaries, reduces domestic credit worldwide, shrinks international credit flows and tightens foreign financial conditions. Countries with floating exchange rates face similar spillovers.

Version record

Hélène Rey, "Dilemma not Trilemma: The Global Financial Cycle and Monetary Policy Independence"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 21162, May 2015; DOI 10.3386/w21162. Version of the 2013 Jackson Hole symposium paper; working-paper record.

A global financial cycle in capital flows, asset prices and credit co-moves with the VIX and is partly driven by centre-country monetary policy through global bank leverage. With free capital mobility it constrains national monetary policy whatever the exchange-rate regime, turning the trilemma into a dilemma.

Extended note. Version record

Valentina Bruno and Hyun Song Shin, "Cross-Border Banking and Global Liquidity"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 82(2), 535–564 (2015); DOI 10.1093/restud/rdu042.

Model of global and local banks in which the bank leverage cycle transmits financial conditions across borders through banking capital flows, and local currency appreciation goes with higher bank leverage. A 46-country panel supports the predictions.

Version record

Stefan Avdjiev, Valentina Bruno, Catherine Koch and Hyun Song Shin, "The Dollar Exchange Rate as a Global Risk Factor: Evidence from Investment"

Difficulty: Research **Type:** Research paper

Status/version: IMF Economic Review, 67(1), 151–173 (2019); DOI 10.1057/s41308-019-00074-4.

Country- and firm-level evidence that a stronger dollar goes with slower growth of cross-border dollar bank flows and lower real investment in emerging markets, a financial channel that works against the standard trade channel.

Version record

Kristin Forbes, Marcel Fratzscher and Roland Straub, "Capital-flow management measures: What are they good for?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 96, S76–S97 (2015); DOI 10.1016/j.jinteco.2014.11.004.

Propensity-score matching on a new weekly database of capital controls and macroprudential measures for 60 countries, 2009–2011. Macroprudential measures reduce some indicators of financial fragility, but most capital-flow measures do not significantly affect exchange rates, flows, rate differentials, inflation or volatility.

Version record

Mai Chi Dao and Pierre-Olivier Gourinchas, "Covered interest parity in emerging markets: Measurement and drivers"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 163, 104231 (2026); DOI 10.1016/j.jinteco.2026.104231.

Builds a purified CIP basis for eight major emerging-market currencies from supranational bonds issued in local currency and dollars, free of local credit risk and segmentation, over twenty years. The basis tracks the supply of and demand for dollar forwards: global dollar funding costs, intermediary balance-sheet capacity and safe-asset demand interact with currency-specific hedging and funding needs.

Version record

Robert McCauley, Chang Shu and Guonan Ma, Non-deliverable forwards: 2013 and beyond

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Quarterly Review, March 2014 (published 9 March 2014); special feature.

Explains how NDFs settle the difference between the contracted and fixing rate in dollars, letting non-residents hedge or take currency exposure despite onshore restrictions. Uses the 2013 Triennial Survey to size NDF markets, compares NDF and onshore deliverable-forward pricing and price discovery under stress, and considers how NDF markets fade after liberalisation.

Version record

Yin-Wong Cheung and Dagfinn Rime, "The offshore renminbi exchange rate: Microstructure and links to the onshore market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 49, 170–189 (2014); DOI 10.1016/j.jimonfin.2014.05.012.

Microstructure data show that the offshore CNH rate responds to its own order flow and limit-order imbalance, increasingly leads the onshore rate and predicts the official central parity; CNH order flow also moves the onshore rate. The links vary over time.

Version record

Gita Gopinath, Emine Boz, Camila Casas, Federico J. Díez, Pierre-Olivier Gourinchas and Mikkel Plagborg-Møller, "Dominant Currency Paradigm"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 110(3), 677–719 (2020); DOI 10.1257/aer.20171201.

Trade prices are set in a dominant currency, mostly the dollar. Bilateral price and volume data for over 2,500 country pairs show that the dollar exchange rate, not the bilateral rate, dominates pass-through and trade elasticities, and a 1% dollar appreciation predicts a 0.6% fall in trade between other countries within a year.

Version record

Matteo Maggiori, Brent Neiman and Jesse Schreger, "International Currencies and Capital Allocation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 128(6), 2019–2066 (2020); DOI 10.1086/705688.

Security-level holdings show investors strongly biased toward their own currencies, so countries hold most foreign debt issued in their currency; only large firms borrow abroad in foreign currency. The United States is the exception, with large foreign holdings of dollar bonds, and the dollar share of cross-border holdings surged after 2008.

Version record

Zhengyang Jiang, Arvind Krishnamurthy and Hanno Lustig, "Dollar Safety and the Global Financial Cycle"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 91(5), 2878–2915 (2024); DOI 10.1093/restud/rdad108.

Model of the global financial cycle driven by international demand for safe dollar assets. It matches dollar borrowing and currency mismatch, the U.S. external balance sheet, the exorbitant privilege, U.S. monetary spillovers and the dollar as a global risk factor: the global financial cycle is a dollar cycle.

Version record

13.11 International equity integration, diversification and home bias

Whether equity markets are priced globally or locally: international asset-pricing models with investment barriers and segmentation, measures of integration and comovement, contagion tests, long-run international return histories and their survivorship, and explanations of home bias. Country factor evidence is in Section 13.12; emerging and frontier markets are in Section 13.14.

Geert Bekaert and Campbell R. Harvey, "Time-Varying World Market Integration"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 50(2), 403 (1995); DOI 10.2307/2329414.

Regime-switching model of emerging-market integration: expected returns move between local and global pricing as markets liberalise.

Version record

Bruno H. Solnik, "Why Not Diversify Internationally Rather Than Domestically?"

Difficulty: Intro **Type:** Research paper

Status/version: Financial Analysts Journal, 30(4), 48–54 (1974); DOI 10.2469/faj.v30.n4.48.

The original practitioner case for international diversification: because national markets are far from perfectly correlated, adding foreign stocks reduces portfolio risk well below what domestic diversification alone achieves. Read it as the baseline that later work on rising correlations, tail dependence and home bias qualifies.

Version record

Michael Adler and Bernard Dumas, "International Portfolio Choice and Corporation Finance: A Synthesis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 38(3), 925–984 (1983); DOI 10.1111/j.1540-6261.1983.tb02511.x.

Synthesis of international portfolio choice and asset pricing when purchasing-power parity fails. Investors of different nationalities hold a common world portfolio plus hedges against their own inflation and exchange-rate risk, so expected returns carry currency risk premia; the paper draws out the implications for corporate exposure and financing.

Version record

René M. Stulz, "On the Effects of Barriers to International Investment"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 36(4), 923–934 (1981); DOI 10.1111/j.1540-6261.1981.tb04893.x.

Models a cost of holding foreign assets proportional to the position. Foreign assets with betas above a threshold plot on one of two security market lines, while some low-beta foreign assets are not held at all even if their expected return rises slightly. A tractable benchmark for pricing under partial segmentation.

Version record

Vihang Errunza and Etienne Losq, "International Asset Pricing under Mild Segmentation: Theory and Test"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 40(1), 105–124 (1985); DOI 10.1111/j.1540-6261.1985.tb04939.x.

When some investors cannot hold a subset of securities, those securities earn a super risk premium and standard two-fund separation breaks down. Tests on a database that includes developing-country markets give tentative support. The model behind later partial-segmentation factor models.

Version record

G. Andrew Karolyi and René M. Stulz, "Are financial assets priced locally or globally?"

Difficulty: Advanced **Type:** Book chapter

Status/version: Handbook of the Economics of Finance, Volume 1B: Financial Markets and Asset Pricing, Elsevier, 975–1020 (2003); DOI 10.1016/s1574-0102(03)01025-2.

Handbook survey of international asset pricing. Country risk premia depend on covariance with the world market, with some evidence for exchange-rate risk, but mean–variance models fail to explain portfolio holdings, equity flows and time-varying cross-country correlations. Home bias strengthens local influences on prices; flows and correlations strengthen global ones.

Version record

Kenneth R. French and James M. Poterba, "Investor Diversification and International Equity Markets"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 3609, January 1991; DOI 10.3386/w3609.

Estimates that Japanese, US and British investors held about 98, 94 and 82 per cent of their equity portfolios at home, and backs out the expected-return gaps that would justify these holdings: several hundred basis points in favour of the home market. Concludes the bias reflects investor choices more than institutional constraints.

Extended note. Version record

Nicolas Coeurdacier and Hélène Rey, "Home Bias in Open Economy Financial Macroeconomics"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Literature, 51(1), 63–115 (2013); DOI 10.1257/jel.51.1.63.

Survey of explanations for home bias in open-economy models: hedging of real-exchange-rate and non-tradable income risk, trade costs such as transaction costs and tax differences, and information frictions and behavioural biases. Adds evidence on cross-border holdings of equities, bonds and bank loans and on fund-level equity holdings.

Version record

Jun-Koo Kang and René M. Stulz, "Why is there a home bias? An analysis of foreign portfolio equity ownership in Japan"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 46(1), 3–28 (1997); DOI 10.1016/s0304-405x(97)00023-8.

Foreign investors in Japan in 1975–1991 overweighted large, manufacturing, low-leverage firms with good accounting performance and low idiosyncratic risk rather than holding the market portfolio, yet did not underperform it. A model with size-related information asymmetries and deadweight costs fits these holdings.

Version record

Alan G. Ahearne, William L. Griever and Francis E. Warnock, "Information costs and home bias: an analysis of U.S. holdings of foreign equities"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 62(2), 313–336 (2004); DOI 10.1016/s0022-1996(03)00015-1.

US holdings of a country's equities rise with the share of its market listed on US exchanges, after controlling for capital controls, trade links, transaction costs and past returns. The authors attribute this to the disclosure and accounting reconciliation a US listing requires, which lowers investors' information costs.

Version record

Stijn Van Nieuwerburgh and Laura Veldkamp, "Information Immobility and the Home Bias Puzzle"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 64(3), 1187–1215 (2009); DOI 10.1111/j.1540-6261.2009.01462.x.

Investors with a small home information advantage choose what to learn before investing, and prefer to deepen what others do not know rather than learn what foreigners know. Learning therefore amplifies the asymmetry and can reproduce local and industry bias and observed patterns of outperformance.

Version record

Magnus Dahlquist, Lee Pinkowitz, René M. Stulz and Rohan Williamson, "Corporate Governance and the Home Bias"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 38(1), 87 (2003); DOI 10.2307/4126765.

Because controlling shareholders hold much of the largest firms in most countries, the portfolio available to outside investors differs sharply from the world market portfolio. US investors' country weights track this float-adjusted world portfolio, and market-capitalisation weights add no explanatory power. Benchmark home bias against available shares.

Version record

Steven L. Heston and K. Geert Rouwenhorst, "Does industrial structure explain the benefits of international diversification?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 36(1), 3–27 (1994); DOI 10.1016/0304-405x(94)90028-0.

Decomposes European stock returns into country and industry effects with a dummy-variable regression and finds that country effects dominate: the low correlation of national indices is not explained by differences in industrial composition, so diversifying across countries mattered more than across industries in the sample.

Version record

John M. Griffin and G. Andrew Karolyi, "Another look at the role of the industrial structure of markets for international diversification strategies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 50(3), 351–373 (1998); DOI 10.1016/s0304-405x(98)00041-5.

Uses the Dow Jones World Stock Index for 25 countries and 66 industries to decompose country and industry sources of return variation. Confirms that industrial composition explains little of the variation in country index returns, but finds the split between country and industry factors differs across industry indices.

Version record

Geert Bekaert, Robert J. Hodrick and Xiaoyan Zhang, "International Stock Return Comovements"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 64(6), 2591–2626 (2009); DOI 10.1111/j.1540-6261.2009.01512.x.

Risk-based factor models fit the covariance of country–industry and country–style portfolios better than the Heston–Rouwenhorst model. Correlations show no upward trend outside Europe, the rise of industry over country factors was short-lived, and large growth stocks are more correlated across countries than small value stocks.

Version record

Kuntara Pukthuanthong and Richard Roll, "Global market integration: An alternative measure and its application"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 94(2), 214–232 (2009); DOI 10.1016/j.jfineco.2008.12.004.

Shows that correlation is a poor integration measure, since perfectly integrated markets can be weakly correlated, and proposes the explanatory power of a multi-factor global model instead. On that measure integration rose markedly for most countries over three decades, even where index correlations did not.

Version record

Peter Christoffersen, Vihang Errunza, Kris Jacobs and Hugues Langlois, "Is the Potential for International Diversification Disappearing? A Dynamic Copula Approach"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 25(12), 3711–3751 (2012); DOI 10.1093/rfs/hhs104.

A dynamic asymmetric copula for large cross-sections shows correlations and tail dependence rising in developed and emerging markets, with both still lower in emerging markets. Diversification benefits measured with higher moments have fallen sharply for developed markets; emerging markets still diversify, especially in large downturns.

Version record

Kristin J. Forbes and Roberto Rigobon, "No Contagion, Only Interdependence: Measuring Stock Market Comovements"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 57(5), 2223–2261 (2002); DOI 10.1111/0022-1082.00494.

Correlation coefficients rise mechanically with market volatility, which biases crisis tests toward finding contagion. After adjusting for heteroskedasticity, correlations did not rise during the 1997 Asian crisis, the 1994 Mexican devaluation or the 1987 crash; markets are highly interdependent in all periods instead.

Version record

Kee-Hong Bae, G. Andrew Karolyi and René M. Stulz, "A New Approach to Measuring Financial Contagion"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 16(3), 717–763 (2003); DOI 10.1093/rfs/hhg012.

Measures contagion as the coincidence of extreme daily return shocks across countries within and across regions, modelled with a multinomial logit. In 1990s emerging-market data contagion is predictable from regional interest rates, exchange-rate changes and conditional volatility; evidence that it is stronger for negative extremes is mixed.

Version record

Geert Bekaert, Campbell R. Harvey, Christian T. Lundblad and Stephan Siegel, "What Segments Equity Markets?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 24(12), 3841–3890 (2011); DOI 10.1093/rfs/hhr082.

Proposes a valuation-based measure of segmentation. Segmentation has fallen in many countries but remains significant in emerging markets; it is explained locally by capital-flow regulation, political risk and stock-market development and globally by the US corporate credit spread.

Version record

Philippe Jorion and William N. Goetzmann, "Global Stock Markets in the Twentieth Century"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Finance, 54(3), 953–980 (1999); DOI 10.1111/0022-1082.00133.

Capital-appreciation indices for 39 markets from the 1920s show US equities had the highest real return in 1921–1996, 4.3 per cent a year against a median of 0.8 per cent elsewhere. US-based estimates of the equity premium suffer from survivorship; the US experience is the exception rather than the rule.

Version record

William N. Goetzmann, Lingfeng Li and K. Geert Rouwenhorst, "Long-Term Global Market Correlations"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business, 78(1), 1–38 (2005); DOI 10.1086/426518.

Over 150 years, correlations among major equity markets vary widely and peak in periods of integration such as the late nineteenth and late twentieth centuries. Diversification benefits split into lower average correlation and a larger opportunity set; recent benefits come mainly from marginal markets rather than low correlations among major ones.

Version record

Aizhan Anarkulova, Scott Cederburg and Michael S. O'Doherty, "Stocks for the long run? Evidence from a broad sample of developed markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 143(1), 409–433 (2022); DOI 10.1016/j.jfineco.2021.06.040.

Uses 39 developed countries from 1841 to 2019 to limit survivor and easy-data bias, and bootstraps long-horizon outcomes. A diversified investor with a 30-year horizon has about a 12 per cent chance of losing to inflation, contradicting the advice that stocks are safe over long holding periods.

Version record

Gikas A. Hardouvelis, Dimitrios Malliaropulos and Richard Priestley, "EMU and European Stock Market Integration"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business, 79(1), 365–392 (2006); DOI 10.1086/497414.

A conditional asset-pricing model with a time-varying degree of integration, intra-European currency risk and time-varying risk prices. Integration tracks forward interest differentials with Germany, rises sharply from 1995 and is nearly complete by mid-1998; the estimated cost-of-capital saving in 1992–1998 is about 2 per cent.

Version record

13.12 Global equity factors, country selection and return predictability

Value, momentum, profitability, investment, liquidity and volatility premia outside the United States; global, local and partially segmented factor models; country-level momentum, reversal and value; and cross-country lead–lag predictability. US factor evidence is in Section 10.1.

Kewei Hou, G. Andrew Karolyi and Bong-Chan Kho, "What Factors Drive Global Stock Returns?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 24(8), 2527–2574 (2011); DOI 10.1093/rfs/hhr013.

Cash-flow-to-price and momentum factors explain international return variation across 49 countries.

Version record

Eugene F. Fama and Kenneth R. French, "Size, value, and momentum in international stock returns"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 105(3), 457–472 (2012); DOI 10.1016/j.jfineco.2012.05.011.

Regional factor evidence for North America, Europe, Japan and Asia Pacific; global versus local factor models.

Version record

Eugene F. Fama and Kenneth R. French, "Value versus Growth: The International Evidence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 53(6), 1975–1999 (1998); DOI 10.1111/0022-1082.00080.

Value stocks beat growth stocks in twelve of thirteen major markets in 1975–1995, by 7.68 per cent a year in global portfolios. An international CAPM cannot explain the premium; a two-factor model with a relative-distress factor captures it.

Version record

K. Geert Rouwenhorst, "International Momentum Strategies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 53(1), 267–284 (1998); DOI 10.1111/0022-1082.95722.

In twelve European countries in 1980–1995, past medium-term winners beat losers by more than 1 per cent a month after risk adjustment. Continuation lasts about a year, is stronger in small firms but not limited to them, and correlates with US momentum, suggesting a common factor.

Version record

John M. Griffin, "Are the Fama and French Factors Global or Country Specific?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 15(3), 783–803 (2002); DOI 10.1093/rfs/15.3.783.

Country-specific three-factor models explain more time-series variation and generally price portfolios and stocks better than a world model, and adding foreign factors worsens in- and out-of-sample pricing. Cost-of-capital and performance work should use local factors.

Version record

John M. Griffin, Xiuqing Ji and J. Spencer Martin, "Momentum Investing and Business Cycle Risk: Evidence from Pole to Pole"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 58(6), 2515–2547 (2003); DOI 10.1046/j.1540-6261.2003.00614.x.

Neither unconditional Chen–Roll–Ross macroeconomic factors nor conditional forecasting models explain momentum internationally. Momentum profits are large in good and bad economic states and reverse over one to five years, which is hard to reconcile with risk-based explanations.

Version record

Andy C. W. Chui, Sheridan Titman and K. C. John Wei, "Individualism and Momentum around the World"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 65(1), 361–392 (2010); DOI 10.1111/j.1540-6261.2009.01532.x.

Momentum profits, trading volume and volatility are larger in countries with higher Hofstede individualism scores, a proxy for overconfidence and self-attribution bias. The relation survives controls for analyst dispersion, transaction costs, size, volatility and market familiarity to foreigners.

Version record

Eugene F. Fama and Kenneth R. French, "International tests of a five-factor asset pricing model"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 123(3), 441–463 (2017); DOI 10.1016/j.jfineco.2016.11.004.

Average returns in North America, Europe and Asia Pacific rise with book-to-market and profitability and fall with investment; in Japan only the value relation is strong. The five-factor model absorbs most patterns but, as in the US, fails for small stocks that invest heavily despite low profitability.

Version record

G. Andrew Karolyi and Ying Wu, "A New Partial-Segmentation Approach to Modeling International Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 53(2), 507–546 (2018); DOI 10.1017/s0022109017001016.

Builds size, value and momentum factors in a partial-segmentation framework that accounts for which stocks foreign investors can access. For more than 37,000 stocks in 46 developed and emerging countries the model has lower pricing errors and rejection rates than purely segmented or purely integrated models.

Version record

Anthony J. Richards, "Winner-Loser Reversals in National Stock Market Indices: Can They be Explained?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 52(5), 2129 (1997); DOI 10.2307/2329478.

Winner–loser reversals in 16 national indices are not explained by risk: loser countries are not more volatile, more exposed to world or other factors, or worse in adverse states. Reversals are larger in small markets but not confined to them, leaving the anomaly unresolved.

Version record

Ronald Balvers, Yangru Wu and Erik Gilliland, "Mean Reversion across National Stock Markets and Parametric Contrarian Investment Strategies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 55(2), 745–772 (2000); DOI 10.1111/0022-1082.00225.

National index data for 18 countries in 1969–1996 show strong mean reversion in relative index prices, with a half-life of three to three and a half years. Parametric contrarian strategies that exploit the estimated reversion beat buy-and-hold and standard contrarian strategies.

Version record

David E. Rapach, Jack K. Strauss and Guofu Zhou, "International Stock Return Predictability: What Is the Role of the United States?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 68(4), 1633–1662 (2013); DOI 10.1111/jofi.12041.

Lagged US returns predict monthly returns in many non-US industrialised markets, while lagged non-US returns barely predict US returns. A news-diffusion model attributes this to gradual incorporation of US return shocks into prices abroad.

Version record

Yakov Amihud, Allaudeen Hameed, Wenjin Kang and Huiping Zhang, "The illiquidity premium: International evidence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 117(2), 350–368 (2015); DOI 10.1016/j.jfineco.2015.04.005.

Across 45 countries the illiquid-minus-liquid return premium is positive and significant after controlling for other pricing factors, in both portfolio and Fama–MacBeth estimates. Illiquidity premia co-move across countries beyond common return and liquidity factors, more so in globally integrated markets.

Version record

G. Andrew Karolyi, Kuan-Hui Lee and Mathijs A. van Dijk, "Understanding commonality in liquidity around the world"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 105(1), 82–112 (2012); DOI 10.1016/j.jfineco.2011.12.008.

In 40 developed and emerging markets, commonality in liquidity and turnover follows the same patterns as commonality in returns: it is higher where investor protection is weaker and information more opaque, rises with volatility, market declines and interest rates, and falls with capital-market openness. Consistent with funding-constraint theories of liquidity.

Version record

Kuan-Hui Lee, "The world price of liquidity risk"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 99(1), 136–161 (2011); DOI 10.1016/j.jfineco.2010.08.003.

Tests the Acharya–Pedersen liquidity-adjusted CAPM globally: the covariances of a security's liquidity with local market liquidity and with local and global returns are priced separately from market risk. The US drives global liquidity risk, and its pricing varies with geographic, economic and political conditions.

Version record

Andrew Ang, Robert J. Hodrick, Yuhang Xing and Xiaoyan Zhang, "High idiosyncratic volatility and low returns: International and further U.S. evidence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 91(1), 1–23 (2009); DOI 10.1016/j.jfineco.2007.12.005.

Across 23 developed markets, stocks with high past idiosyncratic volatility earn 1.31 per cent a month less than low-volatility stocks after world market, size and value controls. The effect holds in each G7 country and co-moves across countries, pointing to a broad, hard-to-diversify source.

Version record

Louis K. C. Chan, Yasushi Hamao and Josef Lakonishok, "Fundamentals and Stock Returns in Japan"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 46(5), 1739–1764 (1991); DOI 10.1111/j.1540-6261.1991.tb04642.x.

In Japanese stocks from 1971 to 1988, including both Tokyo Stock Exchange sections and delisted firms, earnings yield, size, book-to-market and cash-flow yield relate to expected returns, with book-to-market and cash-flow yield the strongest. Early out-of-US evidence for value.

Version record

13.13 Cross-listings, dual listings and asynchronous trading

Why firms list abroad and what cross-listing does to value, liquidity and price discovery; ADR and dual-listed-company parity and the limits to arbitrage; return and volatility transmission across non-overlapping trading hours; and stale prices in international funds and country ETFs.

G. Andrew Karolyi, "The World of Cross-Listings and Cross-Listings of the World: Challenging Conventional Wisdom"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Finance, 10(1), 99–152 (2006); DOI 10.1007/s10679-006-6980-8.

Survey of why firms list abroad. The conventional view, a lower cost of capital through wider investor access, is challenged by evidence that agency conflicts, disclosure and transparency, and corporate governance shape the listing decision and its effects. Identifies open questions.

Version record

Stephen R. Foerster and G. Andrew Karolyi, "The Effects of Market Segmentation and Investor Recognition on Asset Prices: Evidence from Foreign Stocks Listing in the United States"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 54(3), 981–1013 (1999); DOI 10.1111/0022-1082.00134.

Non-US firms listing ADRs on US exchanges earn cumulative abnormal returns of 19 per cent in the year before listing and 1.2 per cent in the listing week, then lose 14 per cent in the following year. The pattern relates to shareholder-base expansion and capital raised, supporting segmentation and investor-recognition explanations.

Version record

Craig Doidge, G. Andrew Karolyi and René M. Stulz, "Why are foreign firms listed in the U.S. worth more?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 71(2), 205–238 (2004); DOI 10.1016/s0304-405x(03)00183-1.

At the end of 1997 foreign firms listed in the US had Tobin's q 16.5 per cent above unlisted firms from the same country, and 37 per cent above for exchange listings. A bonding explanation: a US listing limits controlling shareholders' private benefits, so firms with better-aligned controllers and growth opportunities list, and their growth is valued more.

Version record

Michael Halling, Marco Pagano, Otto Randl and Josef Zechner, "Where Is the Market? Evidence from Cross-Listings in the United States"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 21(2), 725–761 (2008); DOI 10.1093/rfs/hhm066.

The US share of trading in cross-listed stocks is larger for firms from countries near the US with low financial development and weak insider-trading protection. Domestic turnover rises after listing for developed-market firms but not for emerging-market firms, and falls where insider-trading enforcement is poor.

Version record

Cheol S. Eun and Sanjiv Sabherwal, "Cross-Border Listings and Price Discovery: Evidence from U.S.-Listed Canadian Stocks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 58(2), 549–575 (2003); DOI 10.1111/1540-6261.00537.

Toronto and US prices of interlisted Canadian stocks are cointegrated and adjust to each other. The US share of price discovery ranges from 0.2 to 98.2 per cent, averaging 38.1 per cent, rising with the US share of trading and of informative trades and falling with relative US bid–ask spreads.

Version record

Joachim Grammig, Michael Melvin and Christian Schlag, "Internationally cross-listed stock prices during overlapping trading hours: price discovery and exchange rate effects"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 12(1), 139–164 (2005); DOI 10.1016/j.jempfin.2003.10.004.

For firms trading simultaneously in the US and at home, most price discovery usually occurs in the home market, with significant exceptions; the US share rises with the relative liquidity of US trading.

Version record

Louis Gagnon and G. Andrew Karolyi, "Multi-market trading and arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 97(1), 53–80 (2010); DOI 10.1016/j.jfineco.2010.03.005.

Compares intraday prices of ADRs and other US cross-listed shares with synchronous, currency-adjusted home prices for 506 stocks from 35 countries. Parity deviations average 4.9 basis points but are volatile with large extremes, and they rise with holding-cost proxies even after controlling for transaction costs and foreign-investment restrictions.

Version record

Leonard Rosenthal and Colin Young, "The seemingly anomalous price behavior of Royal Dutch/Shell and Unilever N.V./PLC"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 26(1), 123–141 (1990); DOI 10.1016/0304-405x(90)90015-r.

Documents large and persistent deviations from the theoretical price ratios of the Royal Dutch/Shell and Unilever twin shares, whose division of cash flows is fixed by charter, that taxes, currency and transaction costs do not explain. The canonical evidence that claims on identical cash flows can trade at different prices in different markets.

Version record

Kenneth A. Froot and Emil M. Dabora, "How are stock prices affected by the location of trade?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 53(2), 189–216 (1999); DOI 10.1016/s0304-405x(99)00020-3.

Siamese-twin companies pool their cash flows, yet each twin's relative price moves with the market where it trades most. Dividend policy, voting rights, currency and ex-dividend timing do not explain this; tax-induced investor heterogeneity explains some of it, leaving country-specific sentiment or segmentation as candidates.

Version record

Abe de Jong, Leonard Rosenthal and Mathijs A. van Dijk, "The Risk and Return of Arbitrage in Dual-Listed Companies"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 13(3), 495–520 (2009); DOI 10.1093/rof/rfn031.

Simple trading rules on 12 dual-listed companies in 1980–2002 earn abnormal returns of up to almost 10 per cent a year after risk, transaction costs and margin requirements. Uncertain convergence horizons and volatile deviations give the trades high idiosyncratic risk and frequent large losses, which limits arbitrage.

Version record

Paolo Pasquariello, "Financial Market Dislocations"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 27(6), 1868–1914 (2014); DOI 10.1093/rfs/hhu007.

Builds a model-free monthly dislocation index from abnormal violations of ADR parity, triangular currency parity and covered interest parity. Assets that perform poorly when dislocations are high earn significant risk premia, consistent with high marginal utility when arbitrage capital is constrained.

Version record

Yasushi Hamao, Ronald W. Masulis and Victor Ng, "Correlations in Price Changes and Volatility across International Stock Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 3(2), 281–307 (1990); DOI 10.1093/rfs/3.2.281.

ARCH-family models on daily opening and closing index prices for Tokyo, London and New York find volatility spillovers from New York to Tokyo, London to Tokyo and New York to London, but not in other directions, before October 1987.

Version record

Wen-Ling Lin, Robert F. Engle and Takatoshi Ito, "Do Bulls and Bears Move Across Borders? International Transmission of Stock Returns and Volatility"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 7(3), 507–538 (1994); DOI 10.1093/rfs/7.3.507.

Intraday data split daytime and overnight returns: Tokyo daytime returns correlate with New York overnight returns and vice versa, so information revealed in one market's trading hours moves the other. A signal-extraction model with GARCH isolates the global factor.

Version record

Patrick Burns, Robert F. Engle and Joseph J. Mezrich, "Correlations and Volatilities of Asynchronous Data"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Derivatives, 5(4), 7–18 (1998); DOI 10.3905/jod.1998.408000.

Different closing times make returns look predictable and understate correlations, distorting portfolio values, value at risk and hedges. Synchronises prices by estimating the value of closed markets from open ones, then fits GARCH volatilities and correlations to G7 indices.

Version record

William N. Goetzmann, Zoran Ivković and K. Geert Rouwenhorst, "Day Trading International Mutual Funds: Evidence and Policy Solutions"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 36(3), 287 (2001); DOI 10.2307/2676284.

Net asset values struck from stale foreign closing prices let speculators profit from a simple day-trading rule in 391 US international funds, at the expense of other shareholders. Proposes fair-value pricing that corrects NAVs for stale prices instead of monitoring traders and charging redemption fees.

Version record

Ariel Levy and Offer Lieberman, "Overreaction of country ETFs to US market returns: Intraday vs. daily horizons and the role of synchronized trading"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 37(5), 1412–1421 (2013); DOI 10.1016/j.jbankfin.2012.03.024.

Country ETFs follow their NAV returns while the underlying market is open, but when it is closed the S&P 500 dominates their prices, more than it moves the underlying foreign indices: an overreaction to US returns during non-synchronous hours.

Version record

Antti Petajisto, "Inefficiencies in the Pricing of Exchange-Traded Funds"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 73(1), 24–54 (2017); DOI 10.2469/faj.v73.n1.7.

ETF prices deviate from NAVs within a band of about 200 basis points, more for funds holding international or illiquid securities. Using groups of similar ETFs to correct for stale NAVs leaves a band of about 100 basis points, and trading the deviations earns abnormal returns before costs.

Version record

13.14 Emerging and frontier equity markets

Stock-market liberalisation and its effects on prices, the cost of capital and growth; liquidity, price informativeness and efficiency tests; foreign investors' trading; investor protection; and factor evidence in emerging and frontier markets. Index-provider classification documents and the China subsection are in Section 13.17.

Geert Bekaert and Campbell R. Harvey, "Foreign Speculators and Emerging Equity Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 55(2), 565–613 (2000); DOI 10.1111/0022-1082.00220.

Liberalisation lowers the cost of capital by roughly 5–75 bp and raises correlation with world markets without much higher volatility.

Version record

K. Geert Rouwenhorst, "Local Return Factors and Turnover in Emerging Stock Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 54(4), 1439–1464 (1999); DOI 10.1111/0022-1082.00151.

Size, value and momentum premia appear in 20 emerging markets with low correlation to developed-market factors.

Version record

Geert Bekaert, Campbell R. Harvey and Christian Lundblad, "Liquidity and Expected Returns: Lessons from Emerging Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 20(6), 1783–1831 (2007); DOI 10.1093/rfs/hhm030.

Zero-return-day liquidity measures predict emerging-market returns; liquidity is a priced local factor where segmentation persists.

Version record

John M. Griffin, Patrick J. Kelly and Federico Nardari, "Do Market Efficiency Measures Yield Correct Inferences? A Comparison of Developed and Emerging Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 23(8), 3225–3277 (2010); DOI 10.1093/rfs/hhq044.

Standard efficiency measures can misrank developed and emerging markets; a methodological caution for EM anomaly work.

Version record

Geert Bekaert and Campbell R. Harvey, "Emerging markets finance"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Empirical Finance, 10(1–2), 3–55 (2003); DOI 10.1016/s0927-5398(02)00054-3.

Survey of emerging-market finance that starts from integration and segmentation, distinguishes liberalisation from integration, and covers effects on the real economy, contagion, corporate finance, microstructure and stock selection. Adds results on political risk and liberalisation, the volatility of capital flows and the performance of emerging-market investments.

Version record

Campbell R. Harvey, "Predictable Risk and Returns in Emerging Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 8(3), 773–816 (1995); DOI 10.1093/rfs/8.3.773.

Emerging equity markets have high expected returns and volatility and low correlations with developed markets, so they cut world portfolio risk. Integrated global pricing models fail to explain their average returns, and their returns are more predictable from local information than developed-market returns.

Version record

Peter Blair Henry, "Stock Market Liberalization, Economic Reform, and Emerging Market Equity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 55(2), 529–564 (2000); DOI 10.1111/0022-1082.00219.

Aggregate equity indices earn abnormal real dollar returns of 3.3 per cent a month over the eight months before a country's first stock-market liberalisation, consistent with a lower cost of equity from risk sharing with foreign investors. The analysis separates liberalisation from concurrent economic reforms.

Version record

E. Han Kim and Vijay Singal, "Stock Market Openings: Experience of Emerging Economies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business, 73(1), 25–66 (2000); DOI 10.1086/209631.

Around market openings, stock returns rise without higher volatility and markets become more efficient by random-walk tests. There is no evidence of higher inflation or currency appreciation; inflation and the volatility of inflation and exchange rates tend to fall.

Version record

Geert Bekaert, Campbell R. Harvey and Christian Lundblad, "Does financial liberalization spur growth?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 77(1), 3–55 (2005); DOI 10.1016/j.jfineco.2004.05.007.

Equity-market liberalisations raise annual real growth by about one percentage point on average, robust to liberalisation definitions, the world business cycle and growth opportunities. Capital-account liberalisation also matters but does not subsume the effect, and the response is largest where institutions are strong.

Version record

Geert Bekaert, Campbell R. Harvey, Christian Lundblad and Stephan Siegel, "Global Growth Opportunities and Market Integration"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 62(3), 1081–1137 (2007); DOI 10.1111/j.1540-6261.2007.01231.x.

Interacts a country's industry mix with global price–earnings ratios to measure exogenous growth opportunities, which predict GDP and investment growth most strongly where capital accounts, equity markets and banking systems are open. Uses the link between local and global PE ratios to test integration.

Version record

Randall Morck, Bernard Yeung and Wayne Yu, "The information content of stock markets: why do emerging markets have synchronous stock price movements?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 58(1–2), 215–260 (2000); DOI 10.1016/s0304-405x(00)00071-4.

Stock prices move together more in low-income economies, not because of market size and only partly because fundamentals are more correlated. Weak protection of property rights explains much of the difference, consistent with impeded informed trading and greater noise-trader risk.

Version record

Hyuk Choe, Bong-Chan Kho and René M. Stulz, "Do foreign investors destabilize stock markets? The Korean experience in 1997"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 54(2), 227–264 (1999); DOI 10.1016/s0304-405x(99)00037-9.

Trade data for Korea from November 1996 to the end of 1997 show positive feedback trading and herding by foreign investors before the crisis, weakening during it. The market absorbed large foreign sales quickly and without subsequent negative abnormal returns, so there is no evidence that foreigners destabilised it.

Version record

Anthony Richards, "Big Fish in Small Ponds: The Trading Behavior and Price Impact of Foreign Investors in Asian Emerging Equity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 40(1), 1–27 (2005); DOI 10.1017/s0022109000001721.

Daily aggregate foreign trading in six Asian emerging markets shows positive feedback to global as well as domestic returns, which looks behavioural or informational rather than rebalancing, and price impacts much larger than earlier estimates.

Version record

Dave Berger, Kuntara Pukthuanthong and J. Jimmy Yang, "International diversification with frontier markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 101(1), 227–242 (2011); DOI 10.1016/j.jfineco.2011.02.009.

Principal-component measures show frontier markets have low integration with the world market and, unlike developed and emerging markets, no upward trend, with structural breaks that vary by country. Their low integration offers significant diversification benefits.

Version record

Wilma de Groot, Juan Pang and Laurens Swinkels, "The cross-section of stock returns in frontier emerging markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 19(5), 796–818 (2012); DOI 10.1016/j.jempfin.2012.08.007.

A survivorship-free sample of more than 1,400 stocks in 24 frontier markets in 1997–2008 shows significant value and momentum effects and a local size effect, robust to conservative transaction costs and not explained by global risk factors.

Version record

Nusret Cakici, Frank J. Fabozzi and Sinan Tan, "Size, value, and momentum in emerging market stock returns"

Difficulty: Research **Type:** Research paper

Status/version: Emerging Markets Review, 16, 46–65 (2013); DOI 10.1016/j.ememar.2013.03.001.

Stock-level data for 18 emerging markets in 1990–2011 show value effects everywhere and momentum everywhere except Eastern Europe. Local factors price size–value and size–momentum portfolios better than US or global developed-market factors, pointing to segmentation.

Version record

Matthias X. Hanauer and Jochim G. Lauterbach, "The cross-section of emerging market stock returns"

Difficulty: Research **Type:** Research paper

Status/version: Emerging Markets Review, 38, 265–286 (2019); DOI 10.1016/j.ememar.2018.11.009.

In 28 emerging markets over 21 years, the Fama–French five-factor definitions are less robust than alternatives, while cash-flow-to-price, gross profitability, composite equity issuance and momentum are pervasive. There is no positive risk–return relation, and forecasts built on the alternative factors predict better out of sample.

Version record

Utpal Bhattacharya and Hazem Daouk, "The World Price of Insider Trading"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 57(1), 75–108 (2002); DOI 10.1111/1540-6261.00416.

Of 103 countries with stock markets, 87 had insider-trading laws but only 38 had ever prosecuted. A country's cost of equity does not change when a law is introduced but falls significantly after the first prosecution: enforcement, not legislation, matters.

Version record

Rafael La Porta, Florencio Lopez-de-Silanes, Andrei Shleifer and Robert W. Vishny, "Law and Finance"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 106(6), 1113–1155 (1998); DOI 10.1086/250042.

Codes shareholder and creditor protections and the quality of their enforcement in 49 countries: common-law countries protect investors most, French civil-law countries least, and German and Scandinavian civil-law countries lie between. Ownership is more concentrated where protection is weaker.

Version record

13.15 International capital flows and benchmark-driven investors

Push and pull drivers of capital flows, gross-flow episodes, daily custodian and mutual-fund flow evidence, benchmark and index-rebalancing effects, the spillovers of US monetary policy to emerging markets, and nationality-based restatements of cross-border holdings. The global financial cycle and the dollar are in Section 13.10.

Robin Koepke, "What Drives Capital Flows to Emerging Markets? A Survey of the Empirical Literature"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Surveys, 33(2), 516–540 (2019); DOI 10.1111/joes.12273.

Survey organised by push and pull drivers and by type of flow. Global risk aversion and external interest rates matter most for portfolio debt and equity flows and less for bank flows; domestic growth, asset returns and country risk matter for all three, most for bank flows.

Version record

Kristin J. Forbes and Francis E. Warnock, "Capital flow waves: Surges, stops, flight, and retrenchment"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 88(2), 235–251 (2012); DOI 10.1016/j.jinteco.2012.03.006.

Identifies extreme episodes from quarterly gross inflows and outflows, separating foreigners from residents: surges, stops, flight and retrenchment. Global risk is the main determinant; contagion through trade and bank exposures matters for stops and retrenchment, domestic factors matter less, and capital controls do little.

Version record

Marcel Fratzscher, "Capital flows, push versus pull factors and the global financial crisis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 88(2), 341–356 (2012); DOI 10.1016/j.jinteco.2012.05.003.

A factor model on high-frequency portfolio flows to 50 economies finds common shocks drove flows in the 2008 crisis, while country-specific factors dominated in 2009–2010, especially for emerging markets. Differences in exposure are explained by institutions, country risk and macroeconomic fundamentals.

Version record

Eugenio Cerutti, Stijn Claessens and Damien Puy, "Push factors and capital flows to emerging markets: why knowing your lender matters more than fundamentals"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 119, 133–149 (2019); DOI 10.1016/j.jinteco.2019.04.006.

Gross inflows to 34 emerging markets co-move, but only bank and portfolio bond and equity flows do. Push factors explain the common component, and a country's sensitivity depends on its foreign investor base and market liquidity, notably reliance on international funds and global banks, more than on institutional fundamentals.

Version record

Kenneth A. Froot, Paul G. J. O'Connell and Mark S. Seasholes, "The portfolio flows of international investors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 59(2), 151–193 (2001); DOI 10.1016/s0304-405x(00)00084-2.

State Street custody data on daily flows into 46 countries in 1994–1998 show regional factors, strong persistence and trend-following. Inflows forecast emerging-market but not developed-market returns, local prices rise with inflows, and transitory inflows predict lower future returns.

Version record

John M. Griffin, Federico Nardari and René M. Stulz, "Are Daily Cross-Border Equity Flows Pushed or Pulled?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 86(3), 641–657 (2004); DOI 10.1162/0034653041811725.

In a model with investment barriers and more extrapolative foreign investors, high world or local returns draw net equity inflows into small countries. Daily flow data for nine emerging markets confirm that flows respond to host-country and foreign market returns, although the effects fade quickly.

Version record

Graciela Kaminsky, Richard K. Lyons and Sergio L. Schmukler, "Managers, investors, and crises: mutual fund strategies in emerging markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 64(1), 113–134 (2004); DOI 10.1016/s0022-1996(03)00075-8.

Portfolio-level data on US mutual funds in emerging markets show momentum trading, buying winners and selling losers, and contagion trading, selling in one country when prices fall in another. Separating managers from fund investors, contemporaneous momentum is stronger in crises and stronger for investors than for managers.

Version record

Chotibhak Jotikasthira, Christian Lundblad and Tarun Ramadorai, "Asset Fire Sales and Purchases and the International Transmission of Funding Shocks"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 67(6), 2015–2050 (2012); DOI 10.1111/j.1540-6261.2012.01780.x.

Investor flows to funds domiciled in developed markets force changes in their emerging-market allocations, and these fire sales and purchases move emerging-market prices, correlations and betas. Push effects from domicile countries and spillovers between markets owned by the same funds carry the shocks.

Version record

Claudio Raddatz and Sergio L. Schmukler, "On the international transmission of shocks: Micro-evidence from mutual fund portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 88(2), 357–374 (2012); DOI 10.1016/j.jinteco.2012.05.006.

Micro data on equity and bond mutual funds show that both investor injections and redemptions and managers' changes in country weights and cash drive the volatility of fund investment. Both respond pro-cyclically to returns and crises, so mutual-fund capital does not stabilise recipient countries.

Version record

Claudio Raddatz, Sergio L. Schmukler and Tomás Williams, "International asset allocations and capital flows: The benchmark effect"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 108, 413–430 (2017); DOI 10.1016/j.jinteco.2017.06.007.

Monthly benchmark compositions and fund holdings for 1996–2012 show benchmarks explain about 70 per cent of country allocations, with significant effects even for active funds. Pre-announced benchmark changes move allocations when implemented rather than announced, affecting capital flows directly and through contagion.

Version record

Serkan Arslanalp, Dimitris Drakopoulos, Rohit Goel and Robin Koepke, "Benchmark-Driven Investments in Emerging Market Bond Markets: Taking Stock"

Difficulty: Advanced **Type:** Research paper

Status/version: IMF Working Paper 20/192, September 2020; DOI 10.5089/9781513557588.001.

Explains how the main emerging-market local-currency bond indices are built and how they shape fund behaviour. Assets benchmarked to them rose about fivefold from the mid-2000s to around $300 billion; index membership brings inflows but also outflow risk driven by external factors, concentrated in some countries by the way indices are constructed.

Extended note. Version record

Harald Hau, Massimo Massa and Joel Peress, "Do Demand Curves for Currencies Slope Down? Evidence from the MSCI Global Index Change"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 23(4), 1681–1717 (2010); DOI 10.1093/rfs/hhp095.

The 2001–2002 redefinition of the MSCI world index forced large exogenous index-fund flows across countries. Currencies of countries whose index weight rose appreciated on announcement, supporting downward-sloping currency demand and showing that uninformative shocks spread from equities to currencies.

Version record

Anusha Chari, Karlye Dilts Stedman and Christian Lundblad, "Taper Tantrums: Quantitative Easing, Its Aftermath, and Emerging Market Capital Flows"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 34(3), 1445–1508 (2021); DOI 10.1093/rfs/hhaa044.

US monetary policy shocks identified from Treasury futures move expected short rates and term premia and have sizeable effects on US holdings of emerging-market assets, mostly through valuation rather than flows, more for equities than bonds, and asymmetrically: flows matter more in the tapering period.

Version record

Barry Eichengreen and Poonam Gupta, "Tapering talk: The impact of expectations of reduced Federal Reserve security purchases on emerging markets"

Difficulty: Research **Type:** Research paper

Status/version: Emerging Markets Review, 25, 1–15 (2015); DOI 10.1016/j.ememar.2015.07.002.

Uses exchange rates, reserves and equity prices from April to August 2013 to see which emerging markets the Federal Reserve's tapering talk hit. Countries that let real appreciation and current-account deficits build during quantitative easing suffered most; better fundamentals did not insulate, and larger financial markets came under more pressure.

Version record

Antonio Coppola, Matteo Maggiori, Brent Neiman and Jesse Schreger, "Redrawing the Map of Global Capital Flows: The Role of Cross-Border Financing and Tax Havens"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 136(3), 1499–1556 (2021); DOI 10.1093/qje/qjab014.

Links securities issued by tax-haven subsidiaries to their ultimate parents and restates bilateral portfolio positions by nationality. Developed-country investment in large emerging markets is far larger than residency statistics show: US national accounts understate the position in Chinese firms by nearly $600 billion. The currency composition of external liabilities also changes.

Version record

Ralph S. J. Koijen and Motohiro Yogo, "Exchange Rates and Asset Prices in a Global Demand System"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 27342, June 2020; DOI 10.3386/w27342.

Estimates a demand system for financial assets across 36 countries from international holdings data. Macro and policy variables (short rates, debt quantities and reserves) account for 55, 57 and 69 per cent of the variation in exchange rates, long-term yields and stock prices; the estimated convenience yield is 2.15 per cent on US long-term debt and 1.70 per cent on US equity.

Extended note. Version record

Zhengyang Jiang, Robert J. Richmond and Tony Zhang, "A Portfolio Approach to Global Imbalances"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 79(3), 2025–2076 (2024); DOI 10.1111/jofi.13333.

A portfolio-based framework attributes the decline of the US net foreign asset position to global savings gluts and monetary policy, partly offset by shifts in investor demand. US exorbitant privilege declined after 2010 as foreign demand for US equity rose, and the US can issue much more debt than other countries for a given yield increase.

Version record

13.16 Emerging-market local-currency government bonds

Original sin and its partial redemption, the development and investor base of local-currency government bond markets, the effects of foreign holdings and index rebalancing on yields, liquidity and volatility, and the currency and duration risks behind flows. Sovereign default and local-currency credit risk are in Section 11.6; China's bond market is in Section 13.17.2.

Bernardus Doornik, Jon Frost, Rafael Guerra, Alexandre Tombini and Christian Upper, Towards Liquid and Resilient Government Debt Markets in EMEs

Difficulty: Research **Type:** Report / research note

Status/version: BIS Quarterly Review, March 2024, pp. 59–72; technical and measurement annexes; institutional research.

Links investor composition and hedge-market depth to bond bid–ask spreads and their recovery after shocks, using local projections and a panel of 16 EMEs through 2023. Provides a common empirical framework for the India, China, ASEAN, Latin America and South Africa paths without assuming identical market structures. Re-estimate with current instrument and investor definitions before using historical resilience as a liquidity budget.

Extended note. Version record

Iván Baumann Fonay, Inter-American Development Bank, The Effects of Foreign Investors' Holdings on the Local Currency Sovereign Bond Markets in Latin America

Difficulty: Research **Type:** Report / research note

Status/version: IDB technical note, May 2022; country panel includes Argentina, Brazil, Chile, Colombia, Mexico, Peru and Uruguay.

Tests whether foreign ownership of local-currency sovereign bonds stabilises or destabilises markets, finding the sign can depend on currency-depreciation expectations. Use its specification as a hypothesis for the Banxico holdings panel and Brazilian rates, with FX expectations and global-dollar conditions as controls. Follow instrument-level holdings. Decompose duration, foreign share, investor type and hedged versus unhedged returns; Latin American currencies and access rules are not interchangeable.

Extended note. Version record

Ricardo Hausmann and Ugo Panizza, "On the determinants of Original Sin: an empirical investigation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 22(7), 957–990 (2003); DOI 10.1016/j.jimonfin.2003.09.006.

Documents how few countries borrow abroad in their own currency. Development, institutional quality, monetary credibility and fiscal solvency are only weakly related to original sin; only economic size is robust. Monetary credibility and capital controls go with the ability to borrow at home long-term in local currency.

Version record

Stijn Claessens, Daniela Klingebiel and Sergio L. Schmukler, "Government Bonds in Domestic and Foreign Currency: the Role of Institutional and Macroeconomic Factors"

Difficulty: Research **Type:** Research paper

Status/version: Review of International Economics, 15(2), 370–413 (2007); DOI 10.1111/j.1467-9396.2007.00682.x.

Panel evidence that investor bases matter: deeper domestic financial systems go with larger local-currency government bond markets and less foreign-currency debt, foreign demand with more foreign-currency bonds, and less flexible exchange-rate regimes with more foreign-currency issuance. Inflation, fiscal burden, legal origin and capital-account openness also matter.

Version record

John D. Burger and Francis E. Warnock, "Foreign participation in local currency bond markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Economics, 16(3), 291–304 (2007); DOI 10.1016/j.rfe.2006.09.001.

Analyses foreign participation in the local-currency bond markets of more than 40 countries. US investors avoid markets with high variance and negative skewness, which a three-moment CAPM traces to diversifiable idiosyncratic risk; countries can attract foreign participation by reducing macroeconomic instability.

Version record

John D. Burger, Francis E. Warnock and Veronica Cacdac Warnock, "Emerging Local Currency Bond Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 68(4), 73–93 (2012); DOI 10.2469/faj.v68.n4.4.

Local-currency bond markets in emerging economies grew substantially over the 2000s, supported by macroeconomic stability and creditor rights, and gave dollar-based investors attractive returns. US holdings rose sharply, especially in countries with investor-friendly institutions and policies.

Version record

Barry Eichengreen, Ricardo Hausmann and Ugo Panizza, "Yet it Endures: The Persistence of Original Sin"

Difficulty: Research **Type:** Research paper

Status/version: Open Economies Review, 34(1), 1–42 (2023); DOI 10.1007/s11079-022-09704-3.

International original sin persists. Middle-income countries that built local-currency sovereign markets and attracted foreigners still hedge currency exposure through local pension funds and other residents, moving the mismatch within the economy rather than removing it; others limit external borrowing. Argues for an international initiative on currency risk.

Version record

Mert Onen, Hyun Song Shin and Goetz von Peter, "Overcoming original sin: Shedding new light on uneven progress"

Difficulty: Research **Type:** Research paper

Status/version: Economic Policy, 40(122), 575–620 (2025); DOI 10.1093/epolic/eiaf002.

A new dataset on sovereign bond markets shows major emerging markets reducing original sin by issuing in local currency and attracting foreign holders, driven by country factors as well as global push factors. Progress is slow and uneven, stronger for sovereigns than other issuers, and relies on investors whose currency sensitivity can make flows volatile.

Version record

Blaise Gadanecz, Ken Miyajima and Chang Shu, "Emerging market local currency sovereign bond yields: The role of exchange rate risk"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Economics & Finance, 57, 371–401 (2018); DOI 10.1016/j.iref.2018.02.004.

Exchange-rate volatility and expected depreciation raise emerging-market local-currency sovereign yields, robust to endogeneity and other factors. The volatility effect strengthened after the May 2013 taper announcement and is weaker where foreign ownership is higher, the capital account more open and the exchange rate more flexible.

Version record

Carol Bertaut, Valentina Bruno and Hyun Song Shin, "Original Sin Redux: Role of Duration Risk"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33816, May 2025; DOI 10.3386/w33816.

Uses comprehensive data on US investor flows into emerging-market local-currency government bonds. Long-term borrowing limits rollover risk but magnifies valuation changes that interact with currency moves, and strategic complementarities among mutual-fund investors can amplify market stress.

Extended note. Version record

Shanaka J. Peiris, "Foreign Participation in Emerging Markets' Local Currency Bond Markets"

Difficulty: Research **Type:** Research paper

Status/version: IMF Working Paper 10/88, April 2010; DOI 10.5089/9781451982602.001.

A panel of ten emerging markets in 2000–2009 shows that greater foreign participation in domestic government bond markets significantly lowers long-term local-currency yields without necessarily raising yield volatility, and sometimes dampens it.

Extended note. Version record

Christian Ebeke and Yinqiu Lu, "Emerging market local currency bond yields and foreign holdings – A fortune or misfortune?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 59, 203–219 (2015); DOI 10.1016/j.jimonfin.2015.07.006.

In 12 emerging markets after Lehman, foreign holdings of local-currency government bonds lowered yields but somewhat raised yield volatility, identified with financial remoteness from offshore centres as an instrument. Volatility effects are larger where fiscal and external positions are weak; a Poland case study complements the panel.

Version record

Ken Miyajima, M. S. Mohanty and Tracy Chan, "Emerging market local currency bonds: Diversification and stability"

Difficulty: Research **Type:** Research paper

Status/version: Emerging Markets Review, 22, 126–139 (2015); DOI 10.1016/j.ememar.2014.09.006.

Studies local-currency government bond yields since 2000 in three phases up to and after May 2013. Domestic factors anchor yields, raising their diversification value, and yields became more resilient to global risk aversion, but they remained sensitive to very low US Treasury yields and vulnerable to their rise.

Version record

Serkan Arslanalp and Takahiro Tsuda, "Tracking Global Demand for Emerging Market Sovereign Debt"

Difficulty: Advanced **Type:** Research paper

Status/version: IMF Working Paper 14/39, March 2014; DOI 10.5089/9781484326541.001; with an online dataset.

Builds quarterly estimates from 2004 of foreign holdings of emerging-market government debt in local and hard currency, by investor type, with an online dataset. About half a trillion dollars flowed in during 2010–12, mostly from foreign asset managers, and foreign central-bank holdings are concentrated in a few countries. Shows how investor-base data can gauge flow risk.

Extended note. Version record

Lorenzo Pandolfi and Tomas Williams, "Capital flows and sovereign debt markets: Evidence from index rebalancings"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 132(2), 384–403 (2019); DOI 10.1016/j.jfineco.2018.10.008.

Constructs flows implied by mechanical rebalancing of the largest emerging-market local-currency government bond index as an informationless shock. Implied inflows raise bond returns and market depth after rebalancing, and larger inflows (outflows) bring currency appreciation (depreciation).

Version record

Xiang Fang, Bryan Hardy and Karen K. Lewis, "Who Holds Sovereign Debt and Why It Matters"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 30087, May 2022; DOI 10.3386/w30087.

A dataset of sovereign debt holdings by foreign and domestic banks, non-bank private investors and official investors for 95 countries over twenty years shows private non-banks absorb disproportionately more supply and respond most to yields. For emerging markets a 10 per cent rise in debt raises costs by 6.7 per cent, or 9 per cent without non-bank investors.

Extended note. Version record

13.17 Regional institutions and market access

Subject context and qualifications. Subject context and qualifications.

13.17.1 Regional institutions and market access

Noritaka Fukuma, Tomiyuki Kitamura, Kohei Maehashi, Naoki Matsuda, Keita Takemura and Kota Watanabe, The Impact of Quantitative and Qualitative Easing and Yield Curve Control on the Functioning of the Japanese Government Bond Market

Difficulty: Research **Type:** Research paper

Status/version: Bank of Japan Working Paper 24-E-9, 30 August 2024.

Uses issue-level JGB panels to separate purchase-flow effects from the effects of central-bank holdings and fixed-rate operations on volume, bid–ask spreads and yield-curve distortions. Useful for studying scarcity, cash–futures basis and the limits of smooth-curve relative value in Japan. Treat them as evidence about a historical intervention, not a forecast of current policy or an assumption that all bond purchases affect liquidity in the same direction.

Extended note. Version record

Joji Ide, Tomoaki Masuda and Masato Takahashi, Developments in the Functioning of the JGB Markets from the Perspective of the Usage of the Securities Lending Facility

Difficulty: Advanced **Type:** Report / research note

Status/version: Bank of Japan Review 2025-E-3, 21 March 2025.

Adds securities lending and issue scarcity to the Japanese rates route. Useful for forming instrument-level research questions about specialness and settlement. Facility use is an imperfect proxy for the whole market: eligibility, fees and operational changes affect observed demand, and historical usage is not a live quote for obtaining a particular issue.

Extended note. Version record

Hong Kong Exchanges and Clearing, Stock Connect: Information Books for Investors and Market Participants; Market participant book (6 July 2026)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Both selected PDFs version-dated 6 July 2026.

Provides an operational entry to cross-border mainland equity research: eligible securities, trading calendars, order handling, quota arrangements, settlement and investor obligations. Read the investor book first and the participant book before building a tradable universe or execution simulation. Preserve historical eligibility and corporate actions; current information books do not reconstruct past access rules or replace exchange notices.

Extended note. Version record

European Securities and Markets Authority, Trends, Risks and Vulnerabilities: Risk Monitor No. 2, 2026; Statistical annex

Difficulty: Advanced **Type:** Report / research note

Status/version: 10 September 2026; read with the companion statistical annex.

Supplies a dated cross-market view of European securities, investment funds, infrastructures and financial innovation. Use the statistical annex to identify regional data definitions and to compare liquidity, credit, valuation and fund vulnerabilities across market segments. It complements asset-specific models by showing where supervisory aggregates reveal concentrations and transmission channels. Treat the report as surveillance and hypothesis formation rather than an alpha model or trading rulebook.

Extended note. Version record

Sanjay Malhotra, Reserve Bank of India, Indian financial markets: resilience and resurgence

Difficulty: Intro **Type:** Practitioner / commentary

Status/version: Keynote at FIMMDA–PDAI, 1 May 2026; BIS-hosted record dated 5 May 2026.

A current regional orientation to India's money, government-securities and derivatives markets, including investor access, benchmark issuance and product development. Use it to identify research questions about sovereign and state-development-loan curves, local interest-rate hedging and the interaction of domestic and offshore markets. It is not a comprehensive permission list or contract specification; a practical study still needs dated RBI directions, exchange or CCIL rules, and instrument-level data.

Extended note. Version record

Asian Development Bank, Bond Connect and Recent China Inter-Bank Bond Market Developments; Hong Kong Monetary Authority, The Hong Kong Bond Market in 2025

Difficulty: Advanced **Type:** Mixed resources

Status/version: ADB July 2026 brief and HKMA March 2026 bulletin; primary institutional route and data.

Maps Northbound Bond Connect from offshore order to onshore CIBM bond ownership, then connects 2025 offshore and cross-boundary RMB repo, onshore-bond margin eligibility and expanded Swap Connect interest-rate hedges. Read after equity Stock Connect to avoid transferring its ownership rules to bonds. Verify current investor admission, custodian/nominee rights, repo haircut, CNY–CNH basis, Swap Connect tenor/quota and settlement calendars.

Extended note. Version record

Juliana Robbert, Vladyslav Sushko and Frank Westermann, Shifting Forces behind RMB Internationalization: Evidence from the 2025 Triennial Survey

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1345, 29 April 2026; working paper and partially restricted underlying banking data.

Uses disaggregated FX-turnover, banking and debt statistics to distinguish local from cross-border RMB trading and spot from swaps. Finds banking links and investor-access policy important for turnover, with Hong Kong and Singapore FX-swap patterns unlike a simple trade-flow story. This informs where a foreign investor may actually hedge a China bond or equity position. Measure the actual hedge's tenor, deliverability, fixing, margin and capital controls separately.

Extended note. Version record

Reserve Bank of India, Fully Accessible Route: Exclusion of New 14- and 30-Year Securities; Sovereign Green Bond Trading and Settlement Scheme

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: RBI directions, 29 July and 29 August 2024; binding notices for their specified dates.

Shows why "Indian G-sec" is not an eligibility class: new 14- and 30-year central-government issues were removed from FAR, while green bonds have a separate IFSC trade/settlement scheme with FAR and non-FAR paths. Build an ISIN-level admissibility panel before estimating passive-index demand or a non-resident carry strategy. Confirm FPI registration, limits for non-FAR holdings, tax, currency hedge permissions and custodian settlement for the actual investor.

Extended note. Version record

Reserve Bank of India Working Group, Report on Comprehensive Review of Trading and Settlement Timings of Markets Regulated by the Reserve Bank; Clearing Corporation of India, Daily Data, MTM Prices

Difficulty: Advanced **Type:** Mixed resources

Status/version: RBI report, 2 May 2025, chapters 2–3; CCIL live trade and model-price routes.

The RBI maps G-sec/repo, FX and interest-rate-derivative trading and the sequence of primary auctions, secondary clearing and cash settlement; it documents non-resident participation and intraday funding frictions. CCIL's trades and settlement series let a researcher distinguish executed prices from modelled end-day marks based on a Nelson–Siegel–Svensson curve. Read after eligibility. An NDS-OM quote is not necessarily a completed and funded transaction.

Extended note. Version record

Reserve Bank of India, Master Direction: Risk Management and Inter-Bank Dealings

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Original 5 July 2016; consolidation updated 1 October 2026. Full text of Part A, Section I, 2.1–2.4 and Part C inspected alongside the circular sequence below.

Separates user class, purpose, product and dealer eligibility. Paragraph 2.2 permits INR NDDCs through eligible AD Category-I banks (or nonresident parents) operating an IBU and AD Category-III standalone primary dealers; resident settlement is in INR, nonresident settlement in INR or foreign currency. Prerequisite: distinguish these permissions from temporary restrictions. Product, counterparty, underlying exposure, settlement and the remaining related-party restrictions must all fit the trade.

Extended note. Version record

Reserve Bank of India, A.P. (DIR) 24, A.P. (DIR) 03 (revised) and A.P. (DIR) 07

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Original RBI circulars: 27 March, 1 April and 20 April 2026; full original texts inspected and archived. Prerequisite: dealer positions versus customer exposures.

27 March: ADs' end-of-business-day NOP-INR in the onshore deliverable market capped at USD 100 million; compliance no later than 10 April. Deliverable hedges remained subject to the stated genuine-hedge/no-offsetting-NDDC condition. Related parties follow Ind AS 24/IAS 24 or equivalent standards. Use dated regimes for historical execution studies; neither the cap nor April user restriction is a universal notional cap or ban on all interbank NDDCs.

Extended note. Version record

Reserve Bank of India, A.P. (DIR) 13, A.P. (DIR) 16; swap schemes S-56 and S-57

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Circulars issued 8 and 23 June 2026; full original texts inspected.

8 June permits AD Category-I banks to exclude swap positions from eligible FCNR(B), ECB and OFCB funding for the March cap. 23 June partially modifies this: banks shall exclude positions from the related hedged transactions both for March-cap compliance and net overnight open-position computation under the Master Direction. No wider permission follows from a consolidation's omission.

Extended note. Version record

Reserve Bank of India, Annual Report 2022–23, Economic Review, Box II.5.1: Drivers of Term Premia and Risk Premia in Financial Markets

Difficulty: Research **Type:** Report / research note

Status/version: Official staff empirical box, published 30 May 2023; sample November 2012–September 2022.

Relates Indian government-curve slope and corporate credit spreads to liquidity and macro variables using forward stepwise selection. Pair the local hypotheses with the catalogue's affine term-structure and validation readings, then rebuild variables from CCIL prices and transactions and dated RBI releases. Its in-sample selection and historical sample do not establish causal drivers or current predictability. Distinguish matched-tenor credit spread, stale marks and executable returns.

Extended note. Version record

Asian Development Bank ASEAN+3 Bond Market Forum, Bond Market Guides for Indonesia and Thailand; ADB, Local Currency Bond Markets, Foreign Investor Participation, and Capital Flow Volatility in Emerging Asia

Difficulty: Advanced **Type:** Mixed resources

Status/version: Country guide updates dated 15 December 2021; ADB empirical study using 1999–2020 data.

The guides map separate issuance, trading, settlement, custody, taxes and investor eligibility in Indonesia and Thailand; the empirical paper asks whether foreign participation reduces or amplifies volatile flows and argues for deeper FX hedging, particularly during stress. Use both to compare investable IDR and THB bond returns after custody, withholding and hedging. Read with versioned macro data. Never assume an ASEAN-wide rule gives uniform settlement or FX deliverability.

Extended note. Version record

Bank Indonesia, Regulation 6/2023 on Bank Indonesia Rupiah Securities (SRBI); SRBI ownership data; sample auction result

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Live central-bank regulation and data; 24 April 2026 auction illustration.

Defines SRBI as discounted, negotiable short-term rupiah central-bank securities held in BI-SSSS; nonresidents may own them in the secondary market. Continue from the bond market guides. The auction example is not a guaranteed secondary execution price, and ownership releases are delayed and may be revised. Confirm current issue rules, eligible broker/custodian, day count, withholding, IDR FX hedge and repatriation before comparing SRBI carry with sovereign bills.

Extended note. Version record

ASEAN Finance Ministers and Central Bank Governors, 12th AFMGM Joint Statement; Bank Negara Malaysia, The Significance of the Malaysian Government Bond Market

Difficulty: Advanced **Type:** Mixed resources

Status/version: ASEAN statement, 10 April 2025; BNM speech, 3 June 2025.

ASEAN endorsed model guidelines for a Local Currency Transaction Framework; Malaysia describes a dynamic hedging programme for non-resident ringgit bond investors. Together they show how regional payment/currency ambitions meet a country-specific hedge permission and domestic government-bond market. Pair with country settlement guides. The BNM speech is policy commentary; check the current Malaysia FX notices, eligible investor class, underlying exposure and custodian before executing a ringgit forward.

Extended note. Version record

B3, One-Day Interbank Deposit Futures (DI1); B3 clearing; Banco Central do Brasil and CVM, Joint Resolution 13/2024

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Live exchange rule and central-bank/CVM nonresident rule effective 1 January 2025.

DI1 prices the compounded DI rate on a 252-business-day convention, with BRL100,000 notional at expiry, daily variation margin and cash settlement via B3 Clearinghouse. The 2025 joint rule regulates nonresident registration, flows and accounts, replacing older 4,373-era guidance still displayed on some B3 pages. Check participant, representative/custody requirements, margin, settlement marks and holiday calendar under the current rule; the BCB English translation is informational and Portuguese text governs.

Extended note. Version record

Banco de México, Government Securities by Sector; S.D. Indeval, Clearing and Settlement; MexDer, Currency Futures

Difficulty: Advanced **Type:** Mixed resources

Status/version: Official live statistical, depository and exchange routes consulted 4 October 2026.

Banxico separates Cetes, fixed-rate Bonos, floating Bondes and inflation-linked Udibonos, with non-resident holdings. Indeval's DALÍ matches securities and cash for intraday delivery-versus-payment, with two-minute cycles and potential partial settlement. MexDer lists

MXN/USD futures, including a USD10,000 contract; use its March 2026 terms to model the separate FX hedge. Pair with Brazil only for comparison. Banxico holdings are not executable prices. Preserve series, settlement, fixing and release vintages.

Extended note. Version record

Adrian Alter, Khushboo Khandelwal, Thibault Lemaire, Hamza Mighri, Can Sever and Luc Tucker, Navigating the Evolving Landscape of External Financing in Sub-Saharan Africa

Difficulty: Research **Type:** Research paper

Status/version: IMF Working Paper 2025/139, 4 July 2025; open working paper.

Compares Eurobond, syndicated-loan and domestic local-currency channels; sections III–V study dollar/global-rate drivers of spreads and the role of non-resident investors in local debt. Read with macro data, then country debt offices. Domestic-bond records and FX hedges are uneven across countries, so a pooled regional regression is not a tradable security list.

Extended note. Version record

Johannesburg Stock Exchange, Bond Futures, Currency Derivatives; South African Reserve Bank, Currency and Exchanges Manual for Authorised Dealers

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Exchange product pages and SARB June 2025 manual.

The JSE provides deliverable bond futures and listed currency futures/options, letting an analyst separate ZAR duration from FX risk and margin funding. Read after Africa's funding channels. The JSE's public summary is not the full contract rulebook; identify the bond, conversion factor, delivery option, clearing membership and cash margin, and verify current SARB amendments with a dealer.

Extended note. Version record

Central Bank of Kenya, government-bond settlement guidelines; Clearstream, Kenya link announcement and market link guide

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: CBK July 2023; Clearstream 25 June 2026, link effective 29 June. Selected full operational text inspected.

Connects DhowCSD securities records and KEPSS cash settlement to the new indirect omnibus link through Standard Chartered Kenya. Eligible government bonds, bills and infrastructure bonds can settle domestically against KES; internal settlement permits other eligible currencies. Prerequisite: DvP, nominee custody and prefunding. Read current account/cash deadlines and legal-opinion scope; this route does not establish access to every Kenyan security or another African market.

Extended note. Version record

Central Bank of Kenya, September 2026 bond reopening prospectus, yield-curve methodology and Kenya FX Code

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: September 2026 transaction; March 2023 code; living data page observed 4 October. Selected official terms/methodology inspected.

Follow FXD3/2019/015 and SDB1/2011/030 from coupon and accrued interest to DhowCSD settlement and collateral treatment. CBK distinguishes auction-based bill rates, secondary bond yields and interpolated missing tenors. The FX Code addresses conduct, execution and confirmation/settlement in the wholesale market. Prerequisite: bond clean/dirty prices and currency cash flows. The September auction is a dated example, not a live offer or a collateral-funding guarantee.

Extended note. Version record

Central Bank of Kenya, Government Securities Yield Curve Shifts, Implications on Lending Rates and the Economy

Difficulty: Research **Type:** Research paper

Status/version: *CBK Discussion Papers*, May 2011, pp. 12–16; selected full text inspected. Retained historical analysis, not a 2026 publication.

Interprets 2009–2010 curve shifts with inflation measurement changes, monetary conditions and borrowing needs. Read alongside current instrument/data definitions. Prerequisite: level, slope and curvature versus individual-bond duration. Use to formulate regime-aware curve and funding questions before estimating a Kenyan term-structure model. Downloaded curves still require a reproducible vintage, missing-tenor and transaction-cost treatment.

Extended note. Version record

Debt Management Office Nigeria, FGN bond access guide; 2024 domestic USD bond offer and institutional subscription form

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living guide and August 2024 offer; official indexed terms inspected, not complete trust deeds.

The naira-bond guide connects dealer-submitted auctions to electronic securities registration. The separate USD offer specifies semiannual coupon, bullet redemption, eligible investor classes and bank-transfer funding; its institutional form identifies depository-account choices. Prerequisite: separate denomination, issuer credit, custody and source of subscription funds. Read the selected issue's trust deed and custodian terms before a trade; no current executed nonresident custody agreement was established here.

Extended note. Version record

FMDQ Exchange, OTC FX Futures Market Operational Standards and cleared naira-settled NDF documentation

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Standards dated 16 August 2018 on the PDF; current product-page record observed 4 October 2026. Selected full definitions/eligibility provisions and the 16 October 2023 architecture note inspected; original bulletin text available as indexed primary excerpts.

Separates dealer/client eligibility, underlying transaction evidence, CCI, collateral margin and final naira settlement. Despite the historical futures name, the defined contract is a non-deliverable forward: its payoff does not deliver the underlying USD. Prerequisite: spot funding, forward fixing, margin liquidity and capital-importation documentation. Current contract specifications, CBN amendments, bank documentation and executable quotes remain necessary; a CCI or hedge payoff does not guarantee dollar availability.

Extended note. Version record

FMDQ Exchange, MB-33: Rollover of OTC FX Futures Contracts for Foreign Portfolio Investments; cleared USD/NGN NDF rollover FAQ

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: MB-33 dated 25 June 2019; the FMDQ FAQ uses the later cleared NDF terminology and was inspected 4 October 2026. Historical bulletin plus undated living explanation; current complete amendment chain unverified.

For an eligible foreign portfolio investment, MB-33's rollover designation requires the new cleared USD/NGN NDF to match the original amount and tenor and keeps combined tenor within 12 months; only original tenors of one to six months can qualify. FMDQ's FAQ says a seven to 12 month new hedge is not prohibited merely by this rollover rule, but it is not a qualifying rollover and still needs a valid CCI and the operative standards. Model original maturity, new booking, clearing fees, margin and actual dollar repatriation separately. Read with the existing 2023 architecture and MB-46 suspension of underlying-transaction substitution; the latter is a different control. The indexed bulletin and FAQ do not establish reinstatement, executable quotes or all current CBN conditions.

Extended note. Version record

International Monetary Fund, Nigeria: Selected Issues; full report

Difficulty: Research **Type:** Report / research note

Status/version: Country Report 26/126, prepared 7 May, published 9 June 2026. Selected transmission analysis and Annex III inspected in full text.

Connects policy rates, interbank funding, bill yields and exchange-rate pass-through. Distinguishes pre-/post-unification regimes, asymmetric rate adjustments and a Bayesian VAR that separates external oil/rate shocks from domestic responses. Prerequisite: cointegration, identification restrictions and shrinkage. Use to challenge constant hedge/funding parameters in the preceding Nigeria instrument path. Publication and preparation dates differ; do not read staff policy recommendations as enacted market rules.

Extended note. Version record

International Monetary Fund, GCC: Enhancing Resilience to Global Shocks, Policy Challenges, bond-market discussion

Difficulty: Advanced **Type:** Report / research note

Status/version: IMF Policy Paper 2025/043, December 2025; published regional analysis.

Documents the distinction between fast-growing international USD/EUR issues and still uneven local bond/repo markets; foreign-currency issues were 85% of GCC outstanding bonds in 2024 excluding short-term bills. It also distinguishes the dollar pegs of most members from Kuwait's basket peg. Pegs reduce observed spot variability but leave liquidity, convertibility, forward basis and tail regime risks; the policy paper's data are historical.

Extended note. Version record

Saudi National Debt Management Center, Local Saudi Sukuk Issuance Program; Saudi Exchange, Sukuk and Bonds; 2026 Edaa access notice

Difficulty: Advanced **Type:** Mixed resources

Status/version: Issuer tranche data, exchange market route and 6 January 2026 depository notice.

NDMC identifies SAR onshore government sukuk by ISIN, settlement date and award yield, separate from international USD/EUR funding in its FY2026 debt plan. Saudi Exchange matches listed debt electronically for T+2 settlement in SAR. Read after GCC currency context. Auction awards are not executable secondary quotes. Verify sukuk terms, Edaa custody, repo eligibility, SAR funding/FX hedge and the particular ISIN's trading and settlement status.

Extended note. Version record

Central Bank of the UAE, Domestic Market Operations; M-Bills tender rules; EIBOR history

Difficulty: Advanced **Type:** Mixed resources

Status/version: Official living operations and rate pages; publisher-linked tender rules and 2026 auction records.

CBUAE M-Bills locate the short AED yield curve, while EIBOR is a bank-loan fixing. The tender rules distinguish an appointed dealer's primary-auction access from an eligible investor's underlying holding; the 22 June 2026 auction specifies ISINs and T+2 settlement. Compare bills, EIBOR and USD funding to estimate basis and liquidity risk under the peg. The AED peg does not make unsecured EIBOR and secured government-like bills interchangeable.

Extended note. Version record

MSCI, 2026 Global Market Accessibility Review

Difficulty: Advanced **Type:** Report / research note

Status/version: MSCI report, June 2026; data as of 31 May 2026; updated annually in June.

Assesses every equity market in the MSCI indices against five criteria: openness to foreign ownership, ease of capital inflows and outflows, efficiency of the operational framework, availability of investment instruments and stability of the institutional framework. The 18 measures, based on feedback from investors, brokers, custodians, exchanges and regulators, include investor qualification, foreign ownership limits and room, registration, settlement, custody, transferability and stock lending; the aggregate assessment feeds Developed, Emerging or Frontier classification.

Version record

FTSE Russell (LSEG), FTSE Russell announces results of September 2026 annual country classification review for equities and fixed income

Difficulty: Advanced **Type:** Report / research note

Status/version: LSEG press release, 6 October 2026; next update April 2027.

Results of FTSE's annual review: Greece moved from Advanced Emerging to Developed, Vietnam from Frontier to Secondary Emerging and Nigeria from Unclassified to Frontier, effective 21 September 2026. Egypt left the watch list for possible demotion, Oman joined it for possible promotion to Secondary Emerging, and fixed-income classification was unchanged. Read with FTSE's Quality of Markets criteria.

Version record

European Parliament and Council of the European Union, Regulation (EU) 2025/2075 amending Regulation (EU) No 909/2014 as regards a shorter settlement cycle in the Union

Difficulty: Advanced **Type:** Report / research note

Status/version: Regulation (EU) 2025/2075; in force 3 November 2025; applies from 11 October 2027.

Requires the intended settlement date of securities transactions executed on EU trading venues to be no later than the first business day after trading (T+1) from 11 October 2027. Privately negotiated trades executed on a venue, bilateral trades reported to one, first transactions in newly recorded securities and certain securities-financing transactions are excluded; the Commission and ESMA monitor fails and readiness during the transition.

Version record

Council of the European Union, Council Directive (EU) 2025/50 of 10 December 2024 on faster and safer relief of excess withholding taxes (FASTER)

Difficulty: Advanced **Type:** Report / research note

Status/version: Council Directive (EU) 2025/50 of 10 December 2024; transposition by 31 December 2028; national rules apply from 1 January 2030.

Introduces a common EU digital tax residence certificate and two fast-track procedures alongside the standard refund: relief at source, where the treaty rate applies at payment, and quick refund of overpaid tax within 50 days. Certified financial intermediaries, in a national register that large EU intermediaries must join, report dividend and interest payments so tax administrations can check eligibility and detect abuse.

Version record

13.17.2 China's equity and bond markets

Jianan Liu, Robert F. Stambaugh and Yu Yuan, "Size and value in China"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 134(1), 48–69 (2019); DOI 10.1016/j.jfineco.2019.03.008.

China-specific three- and four-factor models excluding shell-value small firms; the standard China factor benchmark.

Version record

Jennifer N. Carpenter, Fangzhou Lu and Robert F. Whitelaw, "The real value of China's stock market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 139(3), 679–696 (2021); DOI 10.1016/j.jfineco.2020.08.012.

Chinese stock prices have become as informative about future profits as US prices; relevant to market-access decisions.

Version record

Franklin Allen, Jun (QJ) Qian, Chenyu Shan and Julie Lei Zhu, "Dissecting the Long-Term Performance of the Chinese Stock Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 79(2), 993–1054 (2024); DOI 10.1111/jofi.13312.

A-share returns trailed other markets in 2000–2018 despite growth, concentrated in large firms and linked to listing, delisting and governance.

Version record

Grace Xing Hu, Jun Pan and Jiang Wang, "Chinese Capital Market: An Empirical Overview"

Difficulty: Advanced **Type:** Research paper

Status/version: Critical Finance Review, 10(2), 125–206 (2021); DOI 10.1561/104.00000097.

Data-rich overview of Chinese equity and bond markets: size, structure, returns, valuation and trading behaviour.

Version record

Jennifer N. Carpenter and Robert F. Whitelaw, "The Development of China's Stock Market and Stakes for the Global Economy"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 9(1), 233–257 (2017); DOI 10.1146/annurev-financial-110716-032333.

Review of research on China's stock market: stages of privatisation, selection biases in listing, IPO underpricing, the A-share premium over foreign shares, firm-specific information in prices, behavioural effects and cross-sectional return patterns. Warns against over-applying research paradigms from other markets.

Version record

Zheng (Michael) Song and Wei Xiong, "Risks in China's Financial System"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 10(1), 261–286 (2018); DOI 10.1146/annurev-financial-110716-032402.

Reviews commonly perceived risks in China's financial system and roots them in politico-economic institutions in which state and non-state sectors coexist and finance funds government policy. Argues a near-term crisis is unlikely, but that misallocation lowers growth and builds risk over time.

Version record

Kalok Chan, Albert J. Menkveld and Zhishu Yang, "Information Asymmetry and Asset Prices: Evidence from the China Foreign Share Discount"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 63(1), 159–196 (2008); DOI 10.1111/j.1540-6261.2008.01313.x.

Microstructure measures of information asymmetry in the A- and B-share markets, price impact and the adverse-selection component of spreads, explain 44 and 46 per cent of cross-sectional variation in B-share discounts and dominate the other controls.

Version record

Ting Chen, Zhenyu Gao, Jibao He, Wenxi Jiang and Wei Xiong, "Daily price limits and destructive market behavior"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 208(1), 249–264 (2019); DOI 10.1016/j.jeconom.2018.09.014.

Shenzhen account-level data show large investors buying stocks on the day they hit the 10 per cent upper limit and selling the next day, with their limit-day buying predicting stronger long-run reversal. Special-treatment stocks, which face 5 per cent limits, give a before-and-after comparison supporting a causal role for the limits.

Version record

Jiangze Bian, Zhiguo He, Kelly Shue and Hao Zhou, "Leverage-Induced Fire Sales and Stock Market Crashes"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 25040, September 2018; DOI 10.3386/w25040.

Account-level data on brokerage- and shadow-financed margin accounts in the 2015 crash show heavy selling as leverage nears its limit. Stocks held by such accounts suffer selling pressure and abnormal declines that later reverse; unregulated shadow-financed accounts on fintech platforms contributed more despite smaller holdings.

Extended note. Version record

Yi Huang, Jianjun Miao and Pengfei Wang, "Saving China's Stock Market?"

Difficulty: Research **Type:** Research paper

Status/version: IMF Economic Review, 67(2), 349–394 (2019); DOI 10.1057/s41308-019-00079-z.

Values the government share purchases made between the mid-June 2015 crash and the September recovery: the rescued firms gained RMB 5,697–6,635 billion, about 10 per cent of 2014 GDP, through higher demand, lower default probabilities and better liquidity. A benefit estimate; read it alongside evidence on the costs of intervention.

Version record

Sheridan Titman, Chishen Wei and Bin Zhao, "Corporate actions and the manipulation of retail investors in China: An analysis of stock splits"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 145(3), 762–787 (2022); DOI 10.1016/j.jfineco.2021.09.018.

Identifies suspicious firms that use stock splits to inflate share prices, which rise and then fall below pre-split levels. Shanghai account data show small retail investors buying, more sophisticated investors buying before announcements and selling afterwards, and insiders selling blocks and pledging shares around the splits.

Version record

Charles M. Jones, Donghui Shi, Xiaoyan Zhang and Xinran Zhang, "Retail Trading and Return Predictability in China"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(1), 68–104 (2025); DOI 10.1017/s0022109024000085.

Splits Chinese retail investors into five account-size groups. Small accounts mispredict returns, show daily momentum, fail to process public news and display overconfidence and gambling preferences; large accounts predict returns and trade as contrarians, but trading costs offset their stock-selection skill.

Version record

Charles M. C. Lee, Yuanyu Qu and Tao Shen, "Going public in China: Reverse mergers versus IPOs"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Corporate Finance, 58, 92–111 (2019); DOI 10.1016/j.jcorpfin.2019.04.003.

Chinese firms that list by reverse merger are larger, more profitable and less politically connected before listing than IPO firms, and perform better afterwards without long-run underperformance, the opposite of developed-market evidence. Restrictive and possibly biased IPO approval keeps good firms out, which explains the shell value of small listed firms.

Version record

Chang Ma, John Rogers and Sili Zhou, "The Effect of the China Connect"

Difficulty: Research **Type:** Research paper

Status/version: Finance and Economics Discussion Series 2019-087, Board of Governors of the Federal Reserve System, December 2019; DOI 10.17016/FEDS.2019.087.

Difference-in-differences evidence that firms admitted to the Shanghai and Shenzhen–Hong Kong Stock Connect cut investment more after contractionary US monetary shocks, especially firms whose returns covary more with the world market, while on average enjoying lower financing costs, higher investment and higher profitability.

Extended note. Version record

Marlene Amstad, Guofeng Sun and Wei Xiong (eds), *The Handbook of China's Financial System*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press (2020); DOI 10.1515/9780691205847.

Edited handbook with chapters on banking and its regulation, the monetary-policy framework and instruments, interest-rate liberalisation, the bond and interbank markets, real estate, infrastructure financing, RMB internationalisation, capital-account liberalisation, the stock market, corporate governance, accounting, investment funds, venture capital, pensions and fintech. Foreword by Darrell Duffie.

Version record

Marlene Amstad and Zhiguo He, "Chinese Bond Markets and Interbank Market"

Difficulty: Advanced **Type:** Book chapter

Status/version: The Handbook of China's Financial System, Princeton University Press, 105–148 (2020); DOI 10.1515/9780691205847-007.

Overview of China's two largely segmented bond markets, the over-the-counter interbank market and the exchange market: their instruments, links to the banking system and multi-layered regulators. Covers ratings and rating agencies, defaults since 2014, the tightening of shadow banking from late 2017 and the forces behind internationalisation.

Version record

Zhe Geng and Jun Pan, "The SOE Premium and Government Support in China's Credit Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 79(5), 3041–3103 (2024); DOI 10.1111/jofi.13380.

A structural default model with credit risk, liquidity and bailout shows improving price discovery and a widening gap between state-owned and other issuers. As liquidity worsened, expected government support made SOE bonds more valuable, while non-SOEs became more sensitive to credit quality and their performance deteriorated.

Version record

Yi Ding, Wei Xiong and Jinfan Zhang, "Issuance overpricing of China's corporate debt securities"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 144(1), 328–346 (2022); DOI 10.1016/j.jfineco.2021.06.010.

Chinese corporate debt is issued overpriced, unlike the underpricing seen in Western markets; average overpricing fell from 7.44 to 2.41 basis points after underwriter rebates were banned in October 2017. Underwriters competing for future mandates drive overpricing through rebates and self-purchases.

Version record

Chapter 14

Commodities and energy

Storage, location, delivery and operating constraints shape commodity valuation. Power, freight, weather and carbon contracts require their own exposure and settlement definitions. Separate sections cover the theory of storage, hedging, options and real options, financialisation, commodity allocation and index products, crude oil, oil supply and demand shocks and forecasting, oil shocks and the macroeconomy, oil and asset markets, natural gas markets, electricity market design, short-term power markets, contract design and delivery, price discovery, trading firms and margin stress, spot price assessment and benchmarks, spatial arbitrage, price cycles, long-run prices, price stabilisation and trade policy, carbon allowance prices and derivatives, carbon pass-through and leakage, and voluntary carbon credits.

14.1 Storage, futures curves and physical options

Physical delivery, non-storability, location, weather and operating constraints can dominate a purely financial factor model. These specialist readings complement the broad commodity and trend-following entries. Subject context and qualifications.

14.1.1 Storage, curves and physical optionality

Robert S. Pindyck, The Dynamics of Commodity Spot and Futures Markets: A Primer

Difficulty: Advanced **Type:** Research paper

Status/version: The Energy Journal 22(3), 1–29, July 2001; DOI 10.5547/ISSN0195-6574-EJ-Vol22-No3-1.

Starts from linked cash-commodity and storage markets to explain how production, inventories, volatility, spot prices, and futures curves jointly evolve, with petroleum examples. Its oil and refined-product evidence is historical, and a storage relation cannot be transferred unchanged to nonstorable electricity, location-constrained gas, or contract-specific delivery markets.

Extended note. Version record

Bryan R. Routledge, Duane J. Seppi, and Chester S. Spatt, Equilibrium Forward Curves for Commodities; author-hosted original article PDF

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 55(3), 1297–1338, June 2000; DOI 10.1111/0022-1082.00248. A Wiley migration field says online 17 December 2002, inconsistent with the original 2000 issue; the journal issue and original PDF establish 2000 publication.

Models storable-commodity forward curves with nonnegative inventories, giving the spot commodity a timing option whose value changes with inventories and transitory supply–demand shocks. The two-factor calibration uses crude oil; contract delivery, storage capacity, and regional basis have to be re-estimated for other physical markets, and the model should not be used unchanged for electricity.

Extended note. Version record

Hélyette Geman, Commodities and Commodity Derivatives: Modeling and Pricing for Agriculturals, Metals and Energy

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, March 2005 hardcover; ISBN 9780470012185. The 2009 e-book is a different format date.

Connects market institutions and physical supply with price-volume risk, spot and forward-curve models, and real-option valuation of energy assets. Use its chapters to route specialists into deeper empirical and operational papers, not as a source of live parameters: exchange rules, delivery geography, energy transitions, and storage economics have changed materially since the print edition.

Extended note. Version record

Rajna Gibson and Eduardo S. Schwartz, Stochastic Convenience Yield and the Pricing of Oil Contingent Claims

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 45(3), 959–976, July 1990

Treats oil spot price and instantaneous convenience yield as two stochastic state variables, then uses futures observations to estimate parameters and value contingent claims. The distinction helps an energy book avoid reducing every curve movement to spot-price mean reversion. The latent convenience yield is an inferred model state, not a directly observed storage fee.

Extended note. Version record

Eduardo S. Schwartz, The Stochastic Behavior of Commodity Prices: Implications for Valuation and Hedging

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 52(3), 923–973, July 1997

Compares one-, two-, and three-factor commodity price models with mean reversion, stochastic convenience yield, and stochastic interest rates, estimating them from copper, oil, and gold futures via the Kalman filter. This supplies a disciplined model-selection bridge

from a curve fit to hedging a production or storage asset. Long-dated real-asset values are particularly sensitive to state dynamics that the historical futures panel cannot identify precisely.

Extended note. Version record

Angus Deaton and Guy Laroque, On the Behaviour of Commodity Prices

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies 59(1), 1–23, January 1992; publisher abstract verified, PDF requires access.

Uses a competitive storage model across thirteen commodities to explain why inventories that cannot fall below zero create asymmetric price spikes and quieter persistence in normal states. It is the stochastic economics behind non-linear inventory stress rather than a curve-trading recipe. Its annual historical data and common stylized storage process should be supplemented with contract timing, harvest, transport, and policy-specific evidence.

Extended note. Version record

Hélyette Geman and Vu-Nhat Nguyen, Soybean Inventory and Forward Curve Dynamics

Difficulty: Research **Type:** Research paper

Status/version: Management Science 51(7), 1076–1091, July 2005; online 1 July 2005.

Reconstructs soybean inventories from the United States, Brazil, and Argentina, finds volatility rising with inverse inventories, and uses scarcity as a forward-curve state variable. Its 25-year CBOT sample predates major shifts in Chinese demand and South American supply; a present-day hedge model needs synchronized crop-year definitions, inventory revisions, basis by origin, and futures delivery economics.

Extended note. Version record

Hélyette Geman and William O. Smith, Theory of Storage, Inventory and Volatility in the LME Base Metals

Difficulty: Research **Type:** Research paper

Status/version: Resources Policy 38(1), 18–28, March 2013

Tests inventory, curve slope, and volatility relationships for aluminium, copper, lead, nickel, tin, and zinc, with Shanghai warehouse stocks strengthening the results. It turns the broad storage theory into a metal-specific monitoring design: track accessible inventory across regions and compare spot with nearby futures. The sample ends before later market-structure changes, so its relationships need current contract and inventory validation.

Extended note. Version record

Nicola Secomandi, Optimal Commodity Trading with a Capacitated Storage Asset

Difficulty: Advanced **Type:** Research paper

Status/version: Management Science 56(3), 449–467, March 2010; first online July 2009.

Models a merchant storage asset with both space and injection or withdrawal limits, yielding state-dependent buy, hold, and sell regions. It translates a gas forward view into feasible physical actions and clarifies why an unconstrained spread-option value overstates what a storage operator can capture. Reoptimization value is not automatically an attainable trading profit.

Extended note. Version record

Nicola Secomandi and Mulan X. Wang, A Computational Approach to the Real Option Management of Network Contracts for Natural Gas Pipeline Transport Capacity

Difficulty: Advanced **Type:** Research paper

Status/version: Manufacturing & Service Operations Management 14(3), 441–454, 2012; publisher online date 4 May 2012. Publisher abstract and secondary page range cross-checked.

Treats contracted pipeline capacity on a network as a real option, rather than a frictionless spread between gas hubs. The modeled contract portfolio still simplifies operational contingencies; a live implementation must reconcile tariff schedules, balancing rules, interruptibility, fuel retention, and timing of pipeline capacity auctions.

Extended note. Version record

International Energy Agency, Gas Market Report, Q3-2026; Report assumptions and summary

Difficulty: Advanced **Type:** Report / research note

Status/version: IEA quarterly fuel report, published 7 July 2026; reviews first-half 2026 developments and forecasts the full year. Released report, not a future announcement; assumptions conditional as of issue date.

Provides a dated global and regional map of gas demand, LNG supply, shipping disruption, and prices after a major 2026 shock. Its full-year projections depend on explicitly stated reopening and facility-restoration assumptions; treat them as conditional scenarios, preserve the report vintage, and do not substitute aggregate regional outlooks for contract-level LNG optionality.

Extended note. Version record

U.S. Energy Information Administration, Weekly Natural Gas Storage Report; EIA storage, price and LNG market supplement

Difficulty: Advanced **Type:** Report / research note

Status/version: Official weekly data series, continuously updated; inspected a September 24, 2026 release for the week ending September 18. Cite observation/release dates in any downstream use, not the access date as the data vintage.

Reports Lower 48 working-gas stocks, weekly changes, regional partitions, and comparisons with the prior year and five-year average. A research pipeline must keep the Thursday release timestamp distinct from the prior Friday measurement date, retain historical revisions and sampling uncertainty, and avoid treating regional aggregate stocks as immediately deliverable at a particular hub or pipeline node.

Extended note. Version record

USDA World Agricultural Outlook Board; USDA Economic Research Service, World Agricultural Supply and Demand Estimates (WASDE) and USDA Outlook Process; USDA outlook production process

Difficulty: Advanced **Type:** Report / research note

Status/version: Official monthly series; 11 September 2026 latest released issue at cutoff. October issue scheduled for 9 October; ERS methodology page updated 2025.

Provides dated US and world crop supply-and-use forecasts for grains, oilseeds, cotton, and other commodities, with monthly archives in machine-readable formats. The USDA outlook-process page explains how interagency committees combine surveys, trade data, weather, models, and expert judgment. Archive the exact release and any correction: production estimates and stock-to-use ratios can change after publication, and global balances do not resolve origin, quality, or delivery-basis risk.

Extended note. Version record

CME Group, CBOT Rulebook Chapter 10: Corn Futures

Difficulty: Advanced **Type:** Contract rulebook

Status/version: Current official CBOT chapter inspected 4 October 2026; undated PDF, so use an archived rulebook version for a trade date.

Specifies the 5,000-bushel Corn futures unit, delivery months, grades, delivery procedure and location differentials. Its distinct location tables through December 2027 and from March 2028 show why a crop-balance signal cannot be mapped to a generic spot or continuous futures series without contract-month, quality and delivery-basis controls.

Extended note. Version record

London Metal Exchange, LME Copper Contract Specifications and Warehouse and Stock Reports; LME warehouse and warrant reports; LME warrant mechanics

Difficulty: Intro **Type:** Contract / specification / guidance

Status/version: Official live specifications and stock reports checked 4 October 2026; 2026 reform decisions have staggered implementation, including copper certificate-of-analysis rules from 2028.

Defines the physically settled Grade A copper future, 25-tonne lot, prompt-date structure, approved brands, and warrant-based delivery; the companion reports separate live and cancelled warrants, arrivals, withdrawals, queue times, and off-warrant stock. Several 2026 warehouse reforms have future start dates, so proposed or decided changes must not be backfilled into older observations.

Extended note. Version record

Craig Pirrong, *Commodity Price Dynamics: A Structural Approach*

Difficulty: Research **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2011); DOI 10.1017/cbo9781139018142.

Structural storage models solved numerically, confronted with forward curves, volatility and correlations across commodities.

Version record

Robert A. Ritz, "Price discrimination and limits to arbitrage: An analysis of global LNG markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Economics, 45, 324–332 (2014); DOI 10.1016/j.eneco.2014.07.013.

Destination clauses, shipping constraints and market power allow persistent regional LNG price gaps; the economics of the free-destination option.

Version record

Xiaoyi Mu, "Weather, storage, and natural gas price dynamics: Fundamentals and volatility"

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Economics, 29(1), 46–63 (2007); DOI 10.1016/j.eneco.2006.04.003.

Weather shocks and storage surprises drive Henry Hub futures returns and volatility.

Version record

Alexander Boogert and Cyriel de Jong, "Gas Storage Valuation Using a Monte Carlo Method"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Derivatives, 15(3), 81–98 (2008); DOI 10.3905/jod.2008.702507.

Least-squares Monte Carlo valuation of gas storage with injection and withdrawal constraints; the practitioner baseline for storage optionality.

Version record

René Carmona and Michael Ludkovski, "Valuation of energy storage: an optimal switching approach"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 10(4), 359–374 (2010); DOI 10.1080/14697680902946514.

Storage valuation as an optimal-switching problem solved by regression Monte Carlo, applicable to gas storage and batteries.

Version record

Claude B. Erb and Campbell R. Harvey, "The Golden Dilemma"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 69(4), 10–42 (2013); DOI 10.2469/faj.v69.n4.1.

Critically examines popular claims about gold, such as its being an inflation hedge. Investors face a dilemma: the real gold price is historically very high and may mean-revert, but if large emerging markets add to their gold holdings it may rise further.

Version record

Dirk G. Baur and Brian M. Lucey, "Is Gold a Hedge or a Safe Haven? An Analysis of Stocks, Bonds and Gold"

Difficulty: Research **Type:** Research paper

Status/version: Financial Review, 45(2), 217–229 (2010); DOI 10.1111/j.1540-6288.2010.00244.x.

Distinguishes a hedge (uncorrelated with stocks or bonds on average) from a safe haven (uncorrelated in a crash). For the US, UK and Germany, gold hedges stocks on average and is a safe haven in extreme stock-market declines, but the safe-haven property is short-lived.

Version record

Robert J. Barro and Sanjay Misra, "Gold Returns"

Difficulty: Research **Type:** Research paper

Status/version: Economic Journal, 126(594), 1293–1317 (2016); DOI 10.1111/ecoj.12274.

From 1836 to 2011 gold's real price rose 1.1 per cent a year with 13.1 per cent volatility and negligible covariance with consumption growth, so it does not hedge macroeconomic declines. A rare-disaster model where gold provides services fits, with most expected return an unobserved dividend yield.

Version record

Brian D. Wright, "The Economics of Grain Price Volatility"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Economic Perspectives and Policy, 33(1), 32–58 (2011); DOI 10.1093/aepp/ppq033.

Review of the 2007–08 and later grain price spikes. They are less unusual than often claimed, and the balance of consumption, supply and stocks explains them as well as it explained earlier episodes; standard storage tools can guide policy better than many costly proposals.

Version record

Michael J. Roberts and Wolfram Schlenker, "Identifying Supply and Demand Elasticities of Agricultural Commodities: Implications for the US Ethanol Mandate"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 103(6), 2265–2295 (2013); DOI 10.1257/aer.103.6.2265.

Uses past yield shocks, which shift inventories and futures prices, to identify supply elasticities of storable crops. Applied to the 2009 Renewable Fuel Standard, prices of corn, rice, soybeans and wheat rise about 20 per cent if a third of ethanol feedstock is recycled, with a 95 per cent interval of 14–35 per cent.

Version record

European Commission, Directorate-General for Energy, Gas storage: Regulation (EU) 2022/1032 and its extension to the end of 2027 by Regulation (EU) 2025/1733

Difficulty: Advanced **Type:** Report / research note

Status/version: European Commission policy page; Regulation (EU) 2025/1733 published 10 September 2025; rules apply to the end of 2027.

Commission summary of the EU gas-storage rules. Regulation (EU) 2022/1032, adopted in June 2022 after the 2021–22 price spike and Russia's invasion of Ukraine, set filling targets to the end of 2025; Regulation (EU) 2025/1733 extends them to the end of 2027. The 90 per cent target can now be met at any time between 1 October and 1 December, trajectories are indicative, and countries may deviate in difficult market conditions. Storage typically supplies 25–30 per cent of EU winter gas use.

Version record

International Energy Agency, Global Critical Minerals Outlook 2026

Difficulty: Advanced **Type:** Report / research note

Status/version: IEA report, published 16 July 2026; licence CC BY 4.0.

The IEA's annual outlook for energy minerals. The 2026 edition reports that aluminium, copper and tin prices rose by a third between January 2025 and April 2026, with copper at record highs; lithium more than doubled and cobalt rose about 130 per cent after export quotas in the Democratic Republic of the Congo. Excluding rare earths, the top refiner's average share rose to 72 per cent in 2025, copper and lithium deficits persist to 2035, and Chinese export controls have opened wide price gaps for gallium, germanium and heavy rare earths.

Version record

Christopher L. Gilbert, "Warehouse Load-Out Queues and Aluminum Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Commodity Markets, 28, article 100243 (2022); DOI 10.1016/j.jcomm.2022.100243.

After the Global Financial Crisis, restrictive load-out rules at two LME exchange warehouses created queues of more than two years. The queues raised the physical premiums paid over the exchange price, and the exchange price did not fall to offset them, so transaction prices rose; the episode raises questions about mutual exchanges and UK oversight.

Version record

Giovanni Barone-Adesi, Hélyette Geman and John Theal, "On the Lease Rate, Convenience Yield and Speculative Effects in the Gold Futures Market"

Difficulty: Research **Type:** Research paper

Status/version: International Journal of Financial Engineering and Risk Management, 1(3), 282 (2014); DOI 10.1504/ijferm.2014.058766.

Using GOFO and lease rates, gold's convenience yield is better measured by the lease rate than by the interest-adjusted spread. Lease rates fall as COMEX inventories rise, and futures prices have increasingly exceeded forward prices, more so with speculative pressure and maturity.

Version record

Guoming Lai, François Margot and Nicola Secomandi, "An Approximate Dynamic Programming Approach to Benchmark Practice-Based Heuristics for Natural Gas Storage Valuation"

Difficulty: Research **Type:** Research paper

Status/version: Operations Research, 58(3), 564–582 (2010); DOI 10.1287/opre.1090.0768.

Develops an approximate dynamic programming method that, with Monte Carlo simulation, gives lower and upper bounds on the value of a gas storage facility, and uses them to benchmark practitioners' heuristics. The heuristics are fast but well below the tight upper bound unless reoptimised periodically within the simulation, when they become nearly optimal; the approach extends to storing metals, oil and refined products.

Version record

14.1.2 Commodity risk premia and factor strategies

Gary Gorton, Fumio Hayashi and K. Geert Rouwenhorst, "The Fundamentals of Commodity Futures Returns"

Difficulty: Research **Type:** Research paper

Status/version: 2013

Links commodity-futures risk premia to physical inventories and the futures basis; finds no evidence that participant positions predict those premia. Distinguish inventory-based explanations from hedging-pressure claims. *Review of Finance* 17(1), 35–105 (2013; online 2012).

Applications: Commodities; carry, curve, relative value. Version record

Ming Gu, Wenjin Kang, Dong Lou, and Ke Tang, Relative Basis and the Expected Returns of Commodity Futures; FMG Discussion Paper 942 PDF

Difficulty: Research **Type:** Research paper

Status/version: London School of Economics Financial Markets Group Discussion Paper 942, publication date 14 November 2025; PDF path dated January 2026. Selected discussion-paper title/version; no journal publication asserted.

Differences a nearby-contract basis from a more distant basis to reduce persistent storage and financing components and better isolate the transient convenience-yield component associated with inventories. An institutional test needs synchronized contract and inventory availability, roll and delivery rules, distant-tenor liquidity, trading costs, and out-of-sample selection controls before treating the relative basis as investable alpha.

Extended note. Version record

Gary Gorton and K. Geert Rouwenhorst, *Facts and Fantasies about Commodity Futures*, *Financial Analysts Journal* (2006; 2004 working paper)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2006, 2004. Version distinctions remain in the annotation.

Historical fully collateralised futures returns, diversification and business-cycle/inflation relationships; sample correlations are not constant hedge ratios.

Version record

Marta Szymanowska, Frans de Roon, Theo Nijman and Rob van den Goorbergh, "An Anatomy of Commodity Futures Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 69(1), 453–482 (2014); DOI 10.1111/jofi.12096.

Separates spot and term premia in commodity futures by basis sorts.

Version record

Devraj Basu and Joëlle Miffre, "Capturing the risk premium of commodity futures: The role of hedging pressure"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 37(7), 2652–2664 (2013); DOI 10.1016/j.jbankfin.2013.02.031.

CFTC hedging-pressure signals sort commodity futures returns.

Version record

Gurdip Bakshi, Xiaohui Gao and Alberto G. Rossi, "Understanding the Sources of Risk Underlying the Cross Section of Commodity Returns"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 65(2), 619–641 (2019); DOI 10.1287/mnsc.2017.2840.

Average, carry and momentum factors price commodity futures portfolios.

Version record

Claude B. Erb and Campbell R. Harvey, "The Strategic and Tactical Value of Commodity Futures"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 62(2), 69–97 (2006); DOI 10.2469/faj.v62.n2.4084.

Roll yield, diversification and the weak case for passive commodity index returns.

Version record

Athanasios Sakkas, "Risk premia in commodity markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 190, 107760 (2026); DOI 10.1016/j.jbankfin.2026.107760.

Risk premia in commodity markets.

Extended note. Version record

Yiyan Qian, Ying Jiang and Xiaoquan Liu, "Factor Momentum in Commodity Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 45(11), 1934–1969 (2025); DOI 10.1002/fut.70022.

Commodity factors show one-month factor momentum (US and UK, 1985–2022); time-series-efficient factors raise Sharpe ratios, but transaction costs erode the gains.

Version record

David Hirshleifer, "Hedging Pressure and Futures Price Movements in a General Equilibrium Model"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 58(2), 411–428 (1990); DOI 10.2307/2938209.

General-equilibrium model with two goods, stochastic output and sequential information. With costless trading, the partial-equilibrium link between hedging pressure and risk premia fails; it is restored when fixed costs deter consumers more than producers from trading futures, so producer hedging pressure moves premia.

Version record

Hendrik Bessembinder, "Systematic Risk, Hedging Pressure, and Risk Premiums in Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 5(4), 637–667 (1992); DOI 10.1093/rfs/5.4.637.

Tests whether futures and asset markets price risk uniformly. Complete integration is not rejected, but returns in currency and agricultural futures vary with hedgers' net positions after controlling for systematic risk, implying some segmentation and supporting hedging pressure as a source of futures premiums.

Version record

Frans A. de Roon, Theo E. Nijman and Chris Veld, "Hedging Pressure Effects in Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 55(3), 1437–1456 (2000); DOI 10.1111/0022-1082.00253.

In 20 futures markets in four groups, both own-market and within-group cross-market hedging pressure affect futures returns after controlling for systematic risk and price pressure. Hedging pressure also helps explain returns on the underlying assets.

Version record

Viral V. Acharya, Lars A. Lochstoer and Tarun Ramadorai, "Limits to Arbitrage and Hedging: Evidence from Commodity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 109(2), 441–465 (2013); DOI 10.1016/j.jfineco.2013.03.003.

Producers averse to cash-flow risk hedge with futures against risk-constrained speculators, so higher hedging demand raises hedging costs, lowers inventories and spot prices. In oil and gas in 1980–2006, producers' default risk forecasts hedging demand, futures premia and spot prices, more so when speculative capital is scarce.

Version record

Wenjin Kang, K. Geert Rouwenhorst and Ke Tang, "A Tale of Two Premiums: The Role of Hedgers and Speculators in Commodity Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 75(1), 377–417 (2020); DOI 10.1111/jofi.12845.

Short-term position changes reflect the liquidity demands of non-commercial traders; long-term variation reflects the hedging demands of commercials. The two components predict futures returns with opposite signs, and the liquidity premium commercials earn largely offsets the insurance premium they pay.

Version record

Harrison Hong and Motohiro Yogo, "What Does Futures Market Interest Tell Us about the Macroeconomy and Asset Prices?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 105(3), 473–490 (2012); DOI 10.1016/j.jfineco.2012.04.005.

With hedging demand and limited risk-bearing capacity, open interest can carry more information than futures prices. Commodity open interest is pro-cyclical and predicts commodity returns, bond returns and the short rate beyond known predictors, and to a lesser degree currency and stock returns.

Version record

Erkko Etula, "Broker-Dealer Risk Appetite and Commodity Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 11(3), 486–521 (2013); DOI 10.1093/jjfinec/nbs024.

Broker-dealers take the other side of producers' and consumers' hedges, so their risk-bearing capacity prices commodity risk. The price of aggregate commodity risk falls as broker-dealer leverage rises, and broker-dealer risk appetite forecasts commodity, especially energy, returns in and out of sample.

Version record

Michaël Dewally, Louis H. Ederington and Chitru S. Fernando, "Determinants of Trader Profits in Commodity Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 26(10), 2648–2683 (2013); DOI 10.1093/rfs/hht048.

Proprietary energy futures position data show that hedgers lose on average and speculators, especially hedge funds, gain. Traders positioned against likely hedgers earn more, and profits on long positions fall with inventories and rise with volatility, consistent with risk-premium, hedging-pressure and storage theories; commodity momentum may largely reflect hedging pressure.

Version record

James D. Hamilton and Jing Cynthia Wu, "Risk Premia in Crude Oil Futures Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 42, 9–37 (2014); DOI 10.1016/j.jimonfin.2013.08.003.

If producers or financial investors hedge with futures, the arbitrageurs on the other side earn a premium, which gives futures prices an affine factor structure. New estimation methods for unbalanced contract data show significant changes in oil futures risk premia since 2005, with smaller average compensation to long positions, consistent with index investing gaining weight relative to commercial hedging.

Version record

Geetesh Bhardwaj, Gary B. Gorton and K. Geert Rouwenhorst, "Facts and Fantasies about Commodity Futures Ten Years Later"

Difficulty: Advanced **Type:** Research paper

Status/version: NBER Working Paper 21243, June 2015; DOI 10.3386/w21243.

Revisits Gorton–Rouwenhorst (2006), which found that collateralised commodity futures matched US equities' return and Sharpe ratio with negative correlation to stocks and bonds. Out of sample, average premiums and the cross-sectional link between returns and the basis are not significantly different; correlations rose only temporarily during the crisis.

Extended note. Version record

Ari Levine, Yao Hua Ooi, Matthew Richardson and Caroline Sasseville, "Commodities for the Long Run"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 74(2), 55–68 (2018); DOI 10.2469/faj.v74.n2.4.

Daily futures prices back to 1877 show positive average returns on commodity indices. Premiums come from both carry and spot returns, but returns across inflation and growth states vary mostly with spot prices, even conditioning on backwardation or contango; the evidence supports commodities in stock–bond portfolios.

Version record

Claude B. Erb and Campbell R. Harvey, "Conquering Misperceptions about Commodity Futures Investing"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 72(4), 26–35 (2016); DOI 10.2469/faj.v72.n4.3.

The poor returns of long-only commodity futures over the previous decade came from weak "income returns" (roll and collateral), not falling prices. The historical link with inflation runs through income returns rather than price returns, and the asset class is small relative to investor expectations.

Version record

Katherine Dusak, "Futures Trading and Investor Returns: An Investigation of Commodity Market Risk Premiums"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 81(6), 1387–1406 (1973); DOI 10.1086/260133.

Recasts the Keynes–Hicks debate on whether speculators earn a premium within the CAPM: the required premium depends on systematic risk, not price variability. Wheat, corn and soybean futures in 1952–67 had betas and average returns close to zero.

Version record

K. Geert Rouwenhorst and Ke Tang, "Commodity Investing"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 4(1), 447–467 (2012); DOI 10.1146/annurev-financial-110311-101716.

Investor-oriented review. Re-estimating early tests on recent data, support for normal backwardation as the source of the commodity premium is weak and the evidence fits storage better. Money managers trade on momentum, producers are net short contrarians, and index traders and swap dealers show little response to past returns.

Version record

Joëlle Miffre and Georgios Rallis, "Momentum Strategies in Commodity Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 31(6), 1863–1886 (2007); DOI 10.1016/j.jbankfin.2006.12.005.

Contrarian strategies fail in commodity futures, but 13 momentum strategies earn about 9.4 per cent a year. Winners tend to be backwardated and losers contangoed, and the returns have low correlation with traditional asset classes.

Version record

Ana-Maria Fuertes, Joëlle Miffre and Georgios Rallis, "Tactical Allocation in Commodity Futures Markets: Combining Momentum and Term Structure Signals"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 34(10), 2530–2548 (2010); DOI 10.1016/j.jbankfin.2010.04.009.

Momentum and term-structure strategies earn annual alphas of about 10 and 13 per cent separately; a double sort on both earns about 21 per cent. The result survives liquidity, data-mining and transaction-cost checks and different risk adjustments.

Version record

Joëlle Miffre, "Long-Short Commodity Investing: A Review of the Literature"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Commodity Markets, 1(1), 3–13 (2016); DOI 10.1016/j.jcomm.2016.01.001.

Reviews long–short commodity futures strategies based on roll yields, inventories and hedging pressure, which follow from storage theory and the hedging-pressure hypothesis, and on past performance, risk, value, skewness, liquidity and inflation betas, together with attempts to refine or combine signals. Long–short positions outperform long-only exposure.

Version record

Fan Yang, "Investment Shocks and the Commodity Basis Spread"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 110(1), 164–184 (2013); DOI 10.1016/j.jfineco.2013.04.012.

A slope factor, on which low-basis futures load more than high-basis ones, and a level factor explain most of the average returns of basis-sorted portfolios. The slope factor correlates with investment-specific technology shocks, and a production model with this link reproduces the cross-section.

Version record

Charoula Daskalaki, Alexandros Kostakis and George Skiadopoulos, "Are There Common Factors in Individual Commodity Futures Returns?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 40, 346–363 (2014); DOI 10.1016/j.jbankfin.2013.11.034.

Tests equity-market factor models, commodity-theory models and principal-component models on individual commodity futures. None succeeds, implying commodity markets are segmented from equities and heterogeneous among themselves.

Version record

Martijn Boons and Melissa Porras Prado, "Basis-Momentum"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 74(1), 239–279 (2019); DOI 10.1111/jofi.12738.

Basis-momentum, related to the slope and curvature of the futures curve, predicts commodity spot and term premiums better than benchmark characteristics in time series and cross section, and its factor is priced. It reflects futures supply–demand imbalances when speculators' and intermediaries' capacity is impaired, not storage or hedging pressure.

Version record

Adrian Fernandez-Perez, Bart Frijns, Ana-Maria Fuertes and Joëlle Miffre, "The Skewness of Commodity Futures Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 86, 143–158 (2018); DOI 10.1016/j.jbankfin.2017.06.015.

Buying the commodity futures with the most negative skewness and shorting those with the most positive earns significant excess returns after controlling for known risk factors, and a tradeable skewness factor helps explain the cross-section. The authors attribute the effect to skewness preference under cumulative prospect theory and to selective hedging.

Version record

Robert J. Bianchi, Michael E. Drew and John Hua Fan, "Combining Momentum with Reversal in Commodity Futures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 59, 423–444 (2015); DOI 10.1016/j.jbankfin.2015.07.006.

Single-sort momentum earns about 11 per cent a year, and its profits reverse 12 to 30 months after formation. A double sort on momentum and reversal earns about 20 per cent a year, robust to seasonality and sample changes, unexplained by standard risk factors, the term structure, volatility, sentiment or costs, and related to global funding liquidity.

Version record

John Hua Fan and Tingxi Zhang, "The Untold Story of Commodity Futures in China"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 40(4), 671–706 (2020); DOI 10.1002/fut.22087.

Commodity futures risk premia in China, where retail traders dominate and entry is restricted. Term-structure and momentum premia persist, while hedging-pressure, skewness, volatility and liquidity premia are distorted by time-varying margins and strict position limits; the premia relate to liquidity, anchoring and regulation-induced limits to arbitrage.

Version record

Colin A. Carter, Gordon C. Rausser and Andrew Schmitz, "Efficient Asset Portfolios and the Theory of Normal Backwardation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 91(2), 319–331 (1983); DOI 10.1086/261148.

Re-estimates Dusak's CAPM test of commodity futures with two corrections: speculators may be net long or net short, and their diversified portfolio holds futures as well as stocks. Wheat, corn and soybean contracts then show significant positive systematic risk and a non-market return that depends on speculators' net position, with similar results for cotton and live cattle, supporting a generalised theory of normal backwardation.

Version record

Eric C. Chang, "Returns to Speculators and the Theory of Normal Backwardation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 40(1), 193–208 (1985); DOI 10.2307/2328055.

Applies a nonparametric test to the returns of speculators in wheat, corn and soybean futures and finds support for normal backwardation, with risk premia to speculators more prominent in later years. Large wheat speculators show some forecasting ability, and futures prices are not unbiased forecasts of later spot prices.

Version record

David Hirshleifer, "Residual Risk, Trading Costs, and Commodity Futures Risk Premia"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 1(2), 173–193 (1988); DOI 10.1093/rfs/1.2.173.

When speculators face a fixed cost of entering a futures market that producers use to hedge revenue, the futures risk premium departs from the perfect-market prediction. The deviation grows with the square root of the trading cost and with the residual standard deviation of returns, is unrelated to covariance with producers' non-marketable wealth, and depends on the coefficient of variation of their revenue, so even a minor commodity can carry a large residual-risk premium.

Version record

Hendrik Bessembinder and Kalok Chan, "Time-varying risk premia and forecastable returns in futures markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 32(2), 169–193 (1992); DOI 10.1016/0304-405x(92)90017-r.

Treasury bill yields, equity dividend yields and the junk-bond premium, which forecast stock and bond returns, also forecast returns in agricultural, metals and currency futures. The pattern fits a two-latent-variable equilibrium model, but the latent variables differ from those that price size-sorted equity portfolios, suggesting that futures carry different priced risks from equities.

Version record

Ivar Ekeland, Delphine Lautier and Bertrand Villeneuve, "Hedging pressure and speculation in commodity markets"

Difficulty: Research **Type:** Research paper

Status/version: Economic Theory, 68(1), 83–123 (2019; online 2018); DOI 10.1007/s00199-018-1115-y.

Micro-founded equilibrium model of the physical and futures markets of a commodity that nests the hedging-pressure and storage theories and produces a range of equilibrium price relations. Comparative statics identify who gains and who loses from financialisation, which clarifies regulatory debates about speculators' influence on prices.

Version record

Chris Brooks, Marcel Prokopczuk and Yingying Wu, "Commodity futures prices: More evidence on forecast power, risk premia and the theory of storage"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Review of Economics and Finance, 53(1), 73–85 (2013); DOI 10.1016/j.qref.2013.01.003.

Re-tests the theory of storage and the risk-premium view of futures prices for stability over time. Seasonality in the basis is stronger than earlier studies found, supporting storage theory, and the basis forecasts later price changes, while evidence for a risk premium is inconclusive. Structural breaks mostly shift intercepts, not slopes.

Version record

Andrew C. Szakmary, Qian Shen and Subhash C. Sharma, "Trend-following trading strategies in commodity futures: A re-examination"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 34(2), 409–426 (2010); DOI 10.1016/j.jbankfin.2009.08.004.

Dual moving-average crossover and channel rules applied to 28 commodity futures over 48 years of monthly data earn positive mean excess returns net of transaction costs in at least 22 markets. Pooled across markets the returns are highly significant, persist in most subperiods and survive data-mining adjustments, reconciling divergent earlier evidence on momentum and trend following.

Version record

Raphael Paschke, Marcel Prokopczuk and Chardin Wese Simen, "Curve momentum"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 113, article 105718 (2020); DOI 10.1016/j.jbankfin.2019.105718.

A momentum strategy that trades within each commodity's futures curve rather than across commodities. The diversified curve-momentum strategy has an annualised Sharpe ratio of 1.28, rising in recent years, and keeps a positive alpha after standard risk factors and transaction costs.

Version record

Adrian Fernandez-Perez, Ana-Maria Fuertes and Joëlle Miffre, "Is idiosyncratic volatility priced in commodity futures markets?"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Financial Analysis, 46, 219–226 (2016); DOI 10.1016/j.irfa.2016.06.002.

With pricing models that ignore backwardation and contango, the idiosyncratic volatility of 27 commodity futures appears negatively priced. Once the phases of backwardation and contango enter the model it is not priced, and portfolios that mimic idiosyncratic volatility look less profitable.

Version record

Peter Christoffersen, Asger Lunde and Kasper V. Olesen, "Factor Structure in Commodity Futures Return and Volatility"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 54(3), 1083–1115 (2019; online 2018); DOI 10.1017/s0022109018000765.

After financialisation, daily commodity volatility has a much stronger factor structure than commodity returns. The common volatility factor relates to stock-market volatility and the business cycle. Realised commodity betas to stocks were high in 2008–10 and have since fallen back near zero, so volatility shows more integration with equities than returns do.

Version record

Fabian Hollstein, Marcel Prokopczuk and Björn Tharann, "Anomalies in Commodity Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Finance, 11(4), article 2150017 (2021); DOI 10.1142/s2010139221500178.

Tests a large set of equity-market anomaly variables on commodity futures. Jump risk, momentum, skewness and volatility-of-volatility carry sizeable premia; downside beta, idiosyncratic volatility and MAX are not priced. Premia are much weaker with annual than with monthly rebalancing.

Version record

14.1.3 Energy and power price modelling

Hendrik Bessembinder and Michael L. Lemmon, Equilibrium Pricing and Optimal Hedging in Electricity Forward Markets

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance 57(3), 1347–1382, June 2002; DOI 10.1111/1540-6261.00463. Wiley migration displays a later online date; cite original issue date.

Builds equilibrium forward pricing and hedging around electricity's limited storability, volatile spot prices, and demand-dependent skew rather than a conventional inventory carry equation. The analysis is based on historical US power markets; current renewable penetration, nodal congestion, weather forecasts, caps, and local market design can change the premium and hedge effectiveness.

Extended note. Version record

Jonty Flottmann, Paul Simshauser, Phillip Wild, and Neda Todorova, The Forward Market Dilemma in Energy-Only Electricity Markets

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Economics 148, article 108676, August 2025; DOI 10.1016/j.eneco.2025.108676.

Models how off-market contracts for difference supporting renewable generation can withdraw hedge supply from an energy-only electricity forward market, leaving retailers more exposed to volatile spot prices. The analysis is grounded in Australia's National Electricity Market; its forward-shortage mechanism and proposed recycling design should be assessed against each jurisdiction's contract and capacity arrangements.

Extended note. Version record

International Energy Agency, Electricity 2026; IEA 2026 prices chapter; IEA 2026 flexibility chapter

Difficulty: Advanced **Type:** Research paper

Status/version: IEA Electricity 2026 published 6 February 2026; 2025 historical review and conditional 2026–2030 outlook, checked 4 October 2026.

Provides a dated cross-region map of electricity demand, supply, grids, flexibility, reliability, and wholesale prices, including battery deployment and negative-price episodes. The 2026–2030 paths are conditional forecasts and aggregate regional views, not trade recommendations or nodal price forecasts; data vintages and country-specific market rules must be retained when adapting them to a portfolio.

Extended note. Version record

Bartosz Uniejewski and Florian Ziel, The Role of Probabilistic Load and Renewable Prediction in Enhancing Day-Ahead Electricity Price Forecasts

Difficulty: Research **Type:** Research paper

Status/version: Renewable Energy 269, article 125844, 1 August 2026; DOI 10.1016/j.renene.2026.125844.

Embeds probabilistic load, wind, and solar forecasts into parsimonious and LASSO day-ahead electricity-price models, finding that extreme forecast quantiles matter and improve error metrics over point-input benchmarks in German EPEX data from 2015–2023. Forecast accuracy is not trading profitability: publication-time availability, bidding-zone changes, intraday imbalance charges, transaction costs, and weather-regime shifts need separate point-in-time tests.

Extended note. Version record

Les Clewlow and Chris Strickland, ***Energy Derivatives: Pricing and Risk Management***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Lacima Publications, 2000 (ISBN 9780953488919).

Spot and forward-curve models, mean reversion and spikes, swing and spread options for energy (Lacima 2000).

Version record

Alexander Eydeland and Krzysztof Wolyniec, ***Energy and Power Risk Management: New Developments in Modeling, Pricing, and Hedging***

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, 2003 (ISBN 9780471104001).

Hybrid power-price models, tolling and generation asset valuation, and risk management for power portfolios.

Version record

Fred Espen Benth, Jūratė Šaltytė Benth and Steen Koekebakker, ***Stochastic Modelling of Electricity and Related Markets***

Difficulty: Research **Type:** Book / monograph

Status/version: WORLD SCIENTIFIC (2008); DOI 10.1142/6811.

Lévy-based spot models, forward dynamics with delivery periods and pricing of electricity derivatives.

Version record

Rafał Weron, "Electricity price forecasting: A review of the state-of-the-art with a look into the future"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Forecasting, 30(4), 1030–1081 (2014); DOI 10.1016/j.ijforecast.2014.08.008.

Comprehensive review of statistical, structural and computational-intelligence electricity price forecasting.

Version record

Jesus Lago, Grzegorz Marcjasz, Bart De Schutter and Rafał Weron, "Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Energy, 293, 116983 (2021); DOI 10.1016/j.apenergy.2021.116983.

Open benchmark datasets and LEAR/DNN baselines with best-practice evaluation (epftoolbox).

Version record

Francis A. Longstaff and Ashley W. Wang, "Electricity Forward Prices: A High-Frequency Empirical Analysis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 59(4), 1877–1900 (2004); DOI 10.1111/j.1540-6261.2004.00682.x.

Hourly spot and day-ahead prices in the PJM market show significant forward risk premia that vary through the day and are related to the volatility of unexpected demand, spot prices and revenues, consistent with pricing by risk-averse agents.

Version record

Fred Espen Benth, Álvaro Cartea and Rüdiger Kiesel, "Pricing Forward Contracts in Power Markets by the Certainty Equivalence Principle: Explaining the Sign of the Market Risk Premium"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 32(10), 2006–2021 (2008); DOI 10.1016/j.jbankfin.2007.12.022.

Links the market risk premium in power forwards to buyers' and sellers' risk aversion and relative market power through a certainty-equivalence argument. Applied to the German market, the premium has a term structure explained by both.

Version record

Álvaro Cartea and Marcelo G. Figueroa, "Pricing in Electricity Markets: A Mean Reverting Jump Diffusion Model with Seasonality"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 12(4), 313–335 (2005); DOI 10.1080/13504860500117503.

Mean-reverting jump-diffusion model of electricity spot prices with seasonality and a closed-form forward price. Calibrated to England and Wales data, it produces month-, quarter- and season-ahead forward surfaces.

Version record

Jakub Nowotarski and Rafał Weron, "Recent advances in electricity price forecasting: A review of probabilistic forecasting"

Difficulty: Advanced **Type:** Research paper

Status/version: Renewable and Sustainable Energy Reviews, 81, 1548–1568 (2018); DOI 10.1016/j.rser.2017.05.234.

Tutorial review of probabilistic electricity price forecasting, with guidelines on methods, scoring rules and tests organised around maximising sharpness subject to reliability. It updates Weron's 2014 review of point forecasting for interval and density forecasts.

Version record

Chung-Li Tseng and Graydon Barz, "Short-Term Generation Asset Valuation: A Real Options Approach"

Difficulty: Research **Type:** Research paper

Status/version: Operations Research, 50(2), 297–310 (2002); DOI 10.1287/opre.50.2.297.429.

Values a power plant over a short horizon as a real option with unit-commitment constraints: hourly run decisions, commitment lead times, minimum up- and down-times and start-up and shut-down costs, with electricity and fuel prices following Itô processes. Combining forward Monte Carlo simulation with backward dynamic programming, it shows that ignoring physical constraints can significantly overvalue a plant, a caution for simple spark-spread option valuations.

Version record

Florian Ziel and Rafał Weron, "Day-ahead electricity price forecasting with high-dimensional structures: Univariate vs. multivariate modeling frameworks"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 70, 396–420 (2018); DOI 10.1016/j.eneco.2017.12.016.

Compares univariate and multivariate lasso-type models for day-ahead price forecasting on 12 datasets. Multivariate models have a small overall edge but do not dominate across markets, seasons or hours, and a simple average of the two structures improves accuracy; the variable-selection results guide model design.

Version record

Joanna Janczura and Rafal Weron, "An empirical comparison of alternate regime-switching models for electricity spot prices"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 32(5), 1059–1073 (2010); DOI 10.1016/j.eneco.2010.05.008.

Calibrates and tests Markov regime-switching models of spot price spikes with attention to goodness of fit. The best structure is an independent-spike three-regime model with time-varying transition probabilities, heteroscedastic base-regime dynamics and shifted spike distributions, which captures seasonal spike intensity, consecutive spikes and drops, and the inverse leverage effect of power prices.

Version record

Audun Botterud, Tarjei Kristiansen and Marija D. Ilic, "The relationship between spot and futures prices in the Nord Pool electricity market"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 32(5), 967–978 (2010); DOI 10.1016/j.eneco.2009.11.009.

Eleven years of Nord Pool spot and futures prices show futures above spot on average: the convenience yield and the realised return on long futures are negative, unlike most commodities, and vary with season, hydro reservoir levels, inflow and demand. The authors attribute this to differences in risk preferences and short-term flexibility between buyers and sellers.

Version record

14.2 Theory of storage and commodity term-structure models

Why commodities are stored when futures prices lie below spot prices: convenience yield, the supply-of-storage curve, competitive storage with stock-outs, inventories and volatility, and one-, two-, three- and N-factor, seasonal and jump models of spot prices, convenience yields and forward curves. Contract and delivery evidence for individual commodities is in Section 14.1.

Nicholas Kaldor, "Speculation and Economic Stability"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 7(1), 1–27 (1939); DOI 10.2307/2967593.

Classic analysis of when speculation stabilises prices. Its section on stocks introduces the convenience yield: holders of inventories accept a return below interest and carrying costs because stocks on hand let them meet unexpected demand, so spot can exceed futures while stocks are positive.

Version record

Holbrook Working, "Theory of the Inverse Carrying Charge in Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Farm Economics, 30(1), 1–28 (1948); DOI 10.2307/1232678.

Explains why futures prices for later delivery can stand below spot prices (an inverse carrying charge) while stocks are still carried. The intertemporal spread is a price of storage that depends on the size of stocks: positive and close to full carrying cost when stocks are large, negative when they are scarce. The basis for the supply-of-storage curve.

Version record

Michael J. Brennan, "The Supply of Storage"

Difficulty: Research **Type:** Book chapter

Status/version: American Economic Review 48(1), 50–72 (1958); reprint (DOI 10.1007/978-1-349-02693-7_5) in B. A. Goss and B. S. Yamey (eds), The Economics of Futures Trading, Palgrave Macmillan, 100–107 (1976).

Asks why stocks are carried when the futures price lies below the spot price, and answers with a supply-of-storage function: the marginal outlay on storage plus a risk premium, net of the marginal convenience yield, rises with the level of stocks and equals the intertemporal price spread. Estimated for several US agricultural commodities in the original article.

Version record

Eugene F. Fama and Kenneth R. French, "Commodity Futures Prices: Some Evidence on Forecast Power, Premiums, and the Theory of Storage"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business, 60(1), 55–73 (1987); DOI 10.1086/296385.

Tests two views of futures prices on 21 commodities. The basis varies with interest rates and with seasonal convenience yields, as the theory of storage predicts; splitting the futures price into an expected premium and a forecast of the spot price finds forecast power for 10 commodities and time-varying premiums for five.

Version record

Jeffrey C. Williams and Brian D. Wright, *Storage and Commodity Markets*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press (1991); DOI 10.1017/cbo9780511571855.

Theory of how the ability to store a surplus shapes commodity prices and production, built on competitive storage with a non-negativity constraint on stocks. Covers how much a country should store against harvest risk, storage versus trade as a buffer, raw versus processed stocks, and government price bands, buffer stocks and strategic reserves.

Version record

Angus Deaton and Guy Laroque, "Competitive Storage and Commodity Price Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 104(5), 896–923 (1996); DOI 10.1086/262046.

Speculators who buy cheap and sell dear can smooth prices and create serial dependence even when harvests are independent. Estimating the competitive-storage model shows that speculation raises autocorrelation substantially for weakly autocorrelated prices, but not to the high levels observed in commodity prices.

Version record

Victor K. Ng and Stephen Craig Pirrong, "Fundamentals and Volatility: Storage, Spreads, and the Dynamics of Metals Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business, 67(2), 203–230 (1994); DOI 10.1086/296630.

Tests the storage-theory predictions that inventories and demand drive both the spot–forward spread and the variances and correlations of spot and forward prices. For four industrial metals and one precious metal in 1986–92, the relations between spreads and volatility match the theory: fundamentals govern metals price dynamics.

Version record

Robert S. Pindyck, "Inventories and the Short-Run Dynamics of Commodity Prices"

Difficulty: Research **Type:** Research paper

Status/version: RAND Journal of Economics, 25(1), 141–159 (1994); DOI 10.2307/2555858.

Estimates a structural model of production, sales and storage for copper, heating oil and lumber, with marginal convenience yield measured from futures prices and convex in inventories. Inventories smooth production when prices are low or normal; when prices are temporarily high they mainly serve scheduling and stock-out avoidance.

Version record

Eduardo Schwartz and James E. Smith, "Short-Term Variations and Long-Term Dynamics in Commodity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 46(7), 893–911 (2000); DOI 10.1287/mnsc.46.7.893.12034.

Two-factor model in which short-term deviations mean-revert while the equilibrium price level is itself uncertain. Long-dated futures identify the equilibrium level and the slope identifies short-term deviations; the model is equivalent to Gibson–Schwartz stochastic convenience yield, is estimated on oil futures by Kalman filter, and is applied to oil-linked assets.

Version record

Jaime Casassus and Pierre Collin-Dufresne, "Stochastic Convenience Yield Implied from Commodity Futures and Interest Rates"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 60(5), 2283–2331 (2005); DOI 10.1111/j.1540-6261.2005.00799.x.

Three-factor affine model of spot prices, convenience yields and interest rates that nests earlier specifications and lets convenience yield depend on spot prices and rates, with time-varying risk premia. Crude oil and copper show spot-level dependence in convenience yields (risk-neutral mean reversion); silver, gold and copper show time-varying premia (physical mean reversion).

Version record

Robert H. Litzenberger and Nir Rabinowitz, "Backwardation in Oil Futures Markets: Theory and Empirical Evidence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 50(5), 1517–1545 (1995); DOI 10.1111/j.1540-6261.1995.tb05187.x.

Treats oil wells as call options to explain strong backwardation, which Hotelling's certainty model rules out. Production occurs only when discounted futures prices lie below spot, falls with price risk, and backwardation emerges when risk is high; US production is inversely, and backwardation directly, related to implied volatility.

Version record

Leonid Kogan, Dmitry Livdan and Amir Yaron, "Oil Futures Prices in a Production Economy with Investment Constraints"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 64(3), 1345–1375 (2009); DOI 10.1111/j.1540-6261.2009.01466.x.

Documents a V-shaped relation between the volatility of oil futures prices and the slope of the forward curve that storage models cannot produce. An equilibrium model with irreversible, capacity-constrained investment makes supply elasticity time-varying, and the estimated model reproduces the V-shape.

Version record

Carlo Cafiero, Eugenio S. A. Bobenrieth H., Juan R. A. Bobenrieth H. and Brian D. Wright, "The Empirical Relevance of the Competitive Storage Model"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 162(1), 44–54 (2011); DOI 10.1016/j.jeconom.2009.10.008.

Answers Deaton and Laroque's critique that competitive storage cannot produce the high serial correlation of commodity prices. With a much finer grid for the equilibrium price function, the model fits coffee, copper, jute, maize, palm oil, sugar and tin with a constant marginal storage cost; demand is inelastic, stock-outs are infrequent except for sugar, and the implied price autocorrelation matches the data.

Version record

Christophe Gouel and Nicolas Legrand, "Estimating the Competitive Storage Model with Trending Commodity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 32(4), 744–763 (2017); DOI 10.1002/jae.2553.

Estimates the storage model jointly with a linear or spline trend in prices. Models with a trend are always preferred: they give more plausible storage costs and demand elasticities, imply the occasional stock-outs that trendless models miss, and match observed price moments more closely.

Version record

Lazaros Symeonidis, Marcel Prokopczuk, Chris Brooks and Emese Lazar, "Futures Basis, Inventory and Commodity Price Volatility: An Empirical Analysis"

Difficulty: Research **Type:** Research paper

Status/version: Economic Modelling, 29(6), 2651–2663 (2012); DOI 10.1016/j.econmod.2012.07.016.

Tests the theory of storage with physical inventories for 21 commodities over 1993–2011. Low inventories go with backwardation and high inventories with contango, and price volatility falls as inventories rise for most commodities, more strongly when the curve is backwardated.

Version record

Hendrik Bessembinder, Jay F. Coughenour, Paul J. Seguin and Margaret Monroe Smoller, "Mean Reversion in Equilibrium Asset Prices: Evidence from the Futures Term Structure"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 50(1), 361–375 (1995); DOI 10.1111/j.1540-6261.1995.tb05178.x.

Uses the futures term structure to test whether investors expect spot prices to mean-revert. They do in every market studied: about 44 per cent of a typical spot oil shock is expected to reverse within eight months, reversion is large for agricultural commodities, smaller but significant for metals and weak for financial assets.

Version record

Gonzalo Cortazar and Lorenzo Naranjo, "An N-Factor Gaussian Model of Oil Futures Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 26(3), 243–268 (2006); DOI 10.1002/fut.20198.

Estimates an N-factor Gaussian model of oil futures by a Kalman filter that uses every daily transaction rather than fixed maturities. On 1992–2001 data three factors fit the term structure of prices but four are needed for the term structure of volatility; the model also fits copper and out-of-sample oil prices for 2002–04.

Version record

Carsten Sørensen, "Modeling Seasonality in Agricultural Commodity Futures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 22(5), 393–426 (2002); DOI 10.1002/fut.10017.

Models log agricultural prices as a deterministic seasonal component plus a non-stationary and a stationary factor, estimated by Kalman filter on weekly CBOT corn, soybean and wheat futures for 1972–97. The filtered convenience yields fall as inventories rise, as storage theory predicts.

Version record

Svetlana Borovkova and Hélyette Geman, "Seasonal and Stochastic Effects in Commodity Forward Curves"

Difficulty: Research **Type:** Research paper

Status/version: Review of Derivatives Research, 9(2), 167–186 (2006); DOI 10.1007/s11147-007-9008-4.

Forward-curve model for seasonal commodities such as natural gas, power and crops. The first factor is the average forward price, which is observable and free of seasonality; the second is a stochastic convenience-yield-like factor with a deterministic seasonal premium. Estimation procedures are applied to several energy markets.

Version record

Leif Andersen, "Markov Models for Commodity Futures: Theory and Practice"

Difficulty: Advanced **Type:** Research paper

Status/version: Quantitative Finance, 10(8), 831–854 (2010); DOI 10.1080/14697680903493599.

Practitioner framework for low-dimensional Markov models of the whole futures curve, covering jump-diffusions with stochastic volatility and regime switching. A worked US natural gas model captures seasonality in futures prices, implied volatilities and their smile, and correlations across maturities, with closed-form European options.

Version record

Gonzalo Cortazar, Cristobal Millard, Hector Ortega and Eduardo S. Schwartz, "Commodity Price Forecasts, Futures Prices, and Pricing Models"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 65(9), 4141–4155 (2019); DOI 10.1287/mnsc.2018.3035.

Commodity pricing models fit futures curves but not the distribution of future spot prices. Calibrating them to analysts' oil price forecasts as well as futures gives reasonable expected spot curves; adding forecasts changes long-maturity estimates substantially but leaves the short and medium end almost unchanged.

Version record

Kristian R. Miltersen, "Commodity Price Modelling That Matches Current Observables: A New Approach"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 3(1), 51–58 (2003); DOI 10.1080/713666159.

Spot-price and convenience-yield model whose volatility and mean-reversion parameters are deterministic functions of time, so that it matches the current term structures of futures prices and of futures volatilities and the intertemporal pattern of volatility.

Version record

John Crosby, "A Multi-Factor Jump-Diffusion Model for Commodities"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 8(2), 181–200 (2008); DOI 10.1080/14697680701253021.

Arbitrage-free multi-factor jump-diffusion model consistent with any initial futures curve, with mean reversion, stochastic convenience yields and stochastic interest rates. Letting long-dated futures jump less than short-dated ones greatly improves the fit to crude oil options, which have fast semi-analytic prices.

Version record

Raphael Paschke and Marcel Prokopczuk, "Commodity Derivatives Valuation with Autoregressive and Moving Average Components in the Price Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 34(11), 2742–2752 (2010); DOI 10.1016/j.jbankfin.2010.05.010.

Continuous-time factor model with autoregressive and moving-average components, motivated by the convenience yield's time-series dynamics, with closed-form futures and option prices. For crude oil it prices nearby futures better in and out of sample and improves long-horizon prices using the short end of the curve.

Version record

Peng Liu and Ke Tang, "The Stochastic Behavior of Commodity Prices with Heteroskedasticity in the Convenience Yield"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 18(2), 211–224 (2011); DOI 10.1016/j.jempfin.2010.12.003.

Shows that the volatility of the convenience yield of industrial commodities rises with its level, and proposes an affine three-factor model (spot price, interest rate, convenience yield) that captures this, the positive link between spot volatility and the convenience yield, and an upward-sloping implied-volatility smile.

Version record

Martin J. Nielsen and Eduardo S. Schwartz, "Theory of Storage and the Pricing of Commodity Claims"

Difficulty: Research **Type:** Research paper

Status/version: Review of Derivatives Research, 7(1), 5–24 (2004); DOI 10.1023/b:redr.0000017026.28316.c8.

Generalises the Gibson–Schwartz two-factor model so that spot volatility depends on the forward spread, as storage theory suggests. Estimated by Kalman filter on copper, the spread–volatility link holds, but the simpler model prices forwards almost as well; option prices differ materially between the two.

Version record

14.3 Commodity hedging and corporate risk management

Minimum-variance, cross- and time-varying hedge ratios; rolling hedges of long-dated commitments and the Metallgesellschaft case; and evidence on which producers and consumers hedge, how they do it and whether it adds value.

Holbrook Working, "Futures Trading and Hedging"

Difficulty: Research **Type:** Book chapter

Status/version: American Economic Review 43(3), 314–343 (1953); reprint (DOI 10.1007/978-1-349-02693-7_3) in B. A. Goss and B. S. Yamey (eds), The Economics of Futures Trading, Palgrave Macmillan, 68–82 (1976).

Argues that the textbook picture of hedging as pure risk avoidance is misleading. Merchants and processors hedge mainly to profit from expected changes in the spot–futures basis and to make storage and pricing decisions on the basis, which makes hedging a form of arbitrage. The standard source for the "basis trading" view of commercial hedging.

Version record

Leland L. Johnson, "The Theory of Hedging and Speculation in Commodity Futures"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 27(3), 139–151 (1960); DOI 10.2307/2296076.

Applies portfolio theory to a holder of commodity inventory who can sell futures. The optimal hedge is the position that minimises the variance of the combined cash and futures position, generally less than one-for-one, with any remainder reflecting a speculative view: the origin of the minimum-variance hedge ratio.

Version record

Louis H. Ederington, "The Hedging Performance of the New Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 34(1), 157–170 (1979); DOI 10.1111/j.1540-6261.1979.tb02077.x.

Applies portfolio hedging theory to the then-new GNMA and Treasury bill futures. The risk-minimising hedge ratio is the slope from regressing spot on futures price changes, and hedging effectiveness is the regression R-squared, the percentage reduction in variance. The template for estimating hedge ratios in any futures market.

Version record

Ronald W. Anderson and Jean-Pierre Danthine, "Cross Hedging"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 89(6), 1182–1196 (1981); DOI 10.1086/261028.

Theory of hedging with futures contracts whose underlying differs from the exposure. Mean–variance decision rules show how optimal cash and futures positions depend on price expectations, production possibilities and the number of futures markets available.

Version record

Richard T. Baillie and Robert J. Myers, "Bivariate GARCH Estimation of the Optimal Commodity Futures Hedge"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 6(2), 109–124 (1991); DOI 10.1002/jae.3950060202.

For six commodities, cash and futures prices follow martingales with near-integrated GARCH errors. The optimal hedge ratio, computed from a bivariate GARCH as conditional covariance over conditional futures variance, varies over time and appears non-stationary, so a constant hedge ratio is inappropriate.

Version record

Donald Lien and Y. K. Tse, "Some Recent Developments in Futures Hedging"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Surveys, 16(3), 357–396 (2002); DOI 10.1111/1467-6419.00172.

Survey of futures hedging: hedge strategies derived from expected-utility maximisation, the minimum-variance criterion as its simplification, alternative objectives, and the more sophisticated econometric methods for estimating optimal hedge ratios that followed advances in financial time-series modelling.

Version record

Kenneth A. Froot, David S. Scharfstein and Jeremy C. Stein, "Risk Management: Coordinating Corporate Investment and Financing Policies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 48(5), 1629–1658 (1993); DOI 10.1111/j.1540-6261.1993.tb05123.x.

If external finance costs more than internal funds, hedging adds value by keeping enough internal cash to fund good investments. The paper shows how hedging strategies should depend on shocks to investment and financing opportunities, and discusses exchange-rate hedging and nonlinear instruments such as options.

Version record

Christopher L. Culp and Merton H. Miller, "Metallgesellschaft and the Economics of Synthetic Storage"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Applied Corporate Finance, 7(4), 62–76 (1995); DOI 10.1111/j.1745-6622.1995.tb00263.x.

Defends the economic logic of Metallgesellschaft's 1993 programme: long-dated fixed-price supply contracts hedged by rolling a stack of short-dated energy futures, which creates synthetic storage. The authors argue the losses came from the parent's decision to unwind the hedge when cash calls mounted, not from the strategy itself.

Version record

Antonio S. Mello and John E. Parsons, "Maturity Structure of a Hedge Matters: Lessons from the Metallgesellschaft Debacle"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Applied Corporate Finance, 8(1), 106–121 (1995); DOI 10.1111/j.1745-6622.1995.tb00279.x.

Critique of the Metallgesellschaft stack-and-roll. A one-for-one rolling hedge of long-dated commitments is not a minimum-variance hedge, carries large basis and rollover risk when the curve moves from backwardation to contango, and creates funding needs that a financially constrained firm cannot meet: the maturity structure of a hedge matters.

Version record

Anthony Neuberger, "Hedging Long-Term Exposures with Multiple Short-Term Futures Contracts"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 12(3), 429–459 (1999); DOI 10.1093/rfs/12.3.429.

How to hedge a long-term supply commitment with short-maturity futures that must be rolled. The optimal strategy uses contracts of several maturities, is independent of risk aversion and, under some conditions, of beliefs about futures returns. It is compared with term-structure-model approaches and tested on oil futures.

Version record

Peter Tufano, "Who Manages Risk? An Empirical Examination of Risk Management Practices in the Gold Mining Industry"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 51(4), 1097–1137 (1996); DOI 10.1111/j.1540-6261.1996.tb04064.x.

New data on North American gold miners' hedging give little support to value-maximisation theories of risk management. Firms whose managers hold more options hedge less and those whose managers hold more stock hedge more, and hedging falls with CFO tenure, pointing to managerial risk aversion and preferences.

Version record

G. David Haushalter, "Financing Policy, Basis Risk, and Corporate Hedging: Evidence from Oil and Gas Producers"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 55(1), 107–152 (2000); DOI 10.1111/0022-1082.00202.

Oil and gas producers in 1992–94 hedge more when they are more levered, consistent with financing costs. Larger firms, and firms whose production is in regions whose prices correlate closely with exchange benchmarks, are more likely to hedge, pointing to scale economies and basis risk.

Version record

Yanbo Jin and Philippe Jorion, "Firm Value and Hedging: Evidence from U.S. Oil and Gas Producers"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 61(2), 893–919 (2006); DOI 10.1111/j.1540-6261.2006.00858.x.

Hand-collected hedging and reserve data for 119 US oil and gas producers in 1998–2001. Hedging reduces the sensitivity of share prices to oil and gas prices, but, contrary to earlier studies, it does not appear to raise market value in this industry.

Version record

David A. Carter, Daniel A. Rogers and Betty J. Simkins, "Does Hedging Affect Firm Value? Evidence from the US Airline Industry"

Difficulty: Research **Type:** Research paper

Status/version: Financial Management, 35(1), 53–86 (2006); DOI 10.1111/j.1755-053x.2006.tb00131.x.

US airlines in 1992–2003 fit the Froot–Scharfstein–Stein framework: investment opportunities rise with jet fuel costs while cash flow falls. Jet fuel hedging is associated with a value premium possibly as large as 10 per cent, mostly through its interaction with capital investment.

Version record

Tim R. Adam and Chitru S. Fernando, "Hedging, Speculation, and Shareholder Value"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 81(2), 283–309 (2006); DOI 10.1016/j.jfineco.2005.03.014.

Gold miners consistently realised economically significant cash-flow gains from their derivatives, with no offsetting change in systematic risk, contradicting the assumption that derivatives have zero NPV. Selective hedging is common, but its gains are small at best.

Version record

Gregory W. Brown, Peter R. Crabb and David Haushalter, "Are Firms Successful at Selective Hedging?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business, 79(6), 2925–2949 (2006); DOI 10.1086/508004.

Hedge ratios of 44 gold miners vary a lot over time and little of the variation is explained by theory; firms cut hedges as prices move against them. Survey evidence shows managers try to time gold prices. There is some timing success, but no sign that selective hedging improves operating or financial performance.

Version record

14.4 Commodity options, spread options and real options

Options on commodity futures, spread, average-price and swing options, unspanned volatility and variance risk premia, and real options in natural-resource investment. General option-pricing foundations are in Chapter 7.

Fischer Black, "The Pricing of Commodity Contracts"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 3(1–2), 167–179 (1976); DOI 10.1016/0304-405x(76)90024-6.

Derives prices for forward contracts, futures and commodity options in a continuous-time setting and shows that a futures contract has zero value at inception and is marked to market daily. Its formula for European options on futures, Black-76, remains the market quoting convention for commodity and interest-rate options.

Version record

William Margrabe, "The Value of an Option to Exchange One Asset for Another"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 33(1), 177–186 (1978); DOI 10.1111/j.1540-6261.1978.tb03397.x.

Closed-form value of a European option to exchange one risky asset for another when both follow lognormal processes: the price depends on the volatility of the ratio of the two prices, and interest rates drop out. The starting point for spread options with zero strike, such as the spark and crack spreads, and for Kirk's approximation.

Version record

René Carmona and Valdo Durrleman, "Pricing and Hedging Spread Options"

Difficulty: Advanced **Type:** Research paper

Status/version: SIAM Review, 45(4), 627–685 (2003); DOI 10.1137/s0036144503424798.

Survey of the theory and numerics of spread options, with energy examples: their uses for speculation and risk management, models for spot and forward-curve dynamics, closed-form approximations, numerical pricing and hedging algorithms, and the valuation of physical assets as real options on spreads.

Version record

Kristian R. Miltersen and Eduardo S. Schwartz, "Pricing of Options on Commodity Futures with Stochastic Term Structures of Convenience Yields and Interest Rates"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 33(1), 33–59 (1998); DOI 10.2307/2331377.

Heath–Jarrow–Morton-style model of commodity prices with stochastic term structures of both interest rates and convenience yields. Gives closed-form prices for options on futures under Gaussian assumptions and shows how they differ from options on forwards, with Gibson–Schwartz as a special case.

Version record

Anders B. Trolle and Eduardo S. Schwartz, "Unspanned Stochastic Volatility and the Pricing of Commodity Derivatives"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 22(11), 4423–4461 (2009); DOI 10.1093/rfs/hhp036.

Tractable model with unspanned stochastic volatility, quasi-analytical option prices and affine futures-curve dynamics, estimated on NYMEX crude oil with about 45,000 futures and 233,000 option prices over 4,082 days. There is strong evidence of two volatility factors that futures prices largely fail to span.

Version record

Anders B. Trolle and Eduardo S. Schwartz, "Variance Risk Premia in Energy Commodities"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Derivatives, 17(3), 15–32 (2010); DOI 10.3905/jod.2010.17.3.015.

Model-free variance risk premia in crude oil and natural gas over 11 years are significantly negative and hard to explain with systematic or commodity-specific factors. Natural gas variance swaps pay off like calls, crude oil swaps more like puts; Sharpe ratios from selling energy variance are sizeable but below those for the S&P 500.

Version record

Marcel Prokopczuk, Lazaros Symeonidis and Chardin Wese Simen, "Variance Risk in Commodity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 81, 136–149 (2017); DOI 10.1016/j.jbankfin.2017.05.003.

Synthetic variance swaps in commodity markets have significantly negative realised and expected payoffs in most markets. Payoffs share common components and comove with equity variance-swap payoffs, but are distinct from futures returns: commodity variance risk is not spanned by futures.

Version record

Robert McDonald and Daniel Siegel, "The Value of Waiting to Invest"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 101(4), 707–727 (1986); DOI 10.2307/1884175.

Optimal timing of an irreversible investment when both its value and its cost follow continuous-time stochastic processes. The option to invest has an explicit value, the same analysis covers scrapping, and for reasonable parameters it is optimal to wait until benefits are about twice costs.

Version record

Alberto Moel and Peter Tufano, "When Are Real Options Exercised? An Empirical Study of Mine Closings"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 15(1), 35–64 (2002); DOI 10.1093/rfs/15.1.35.

Annual opening and closing decisions for 285 North American gold mines in 1988–97 follow real-options predictions: they depend on the level and volatility of gold prices, interest rates, costs and reserves, with strong hysteresis. Closings also depend on the profitability of the firm's other mines and businesses.

Version record

A. G. Z. Kemna and A. C. F. Vorst, "A Pricing Method for Options Based on Average Asset Values"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 14(1), 113–129 (1990); DOI 10.1016/0378-4266(90)90039-5.

Prices Asian options on the arithmetic average of an asset price. The geometric-average option has a closed form and serves as a control variate in Monte Carlo pricing of the arithmetic-average option, which has none. Average-price settlement is standard for commodity swaps and options.

Version record

Patrick Jaillet, Ehud I. Ronn and Stathis Tompaidis, "Valuation of Commodity-Based Swing Options"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 50(7), 909–921 (2004); DOI 10.1287/mnsc.1040.0240.

Values swing (take-or-pay) options in gas and power contracts, which let the holder vary delivered volumes repeatedly within daily and periodic limits. A one-factor mean-reverting price process with seasonality is calibrated to forward prices and volatilities, and a numerical scheme values the option for natural gas.

Version record

M. A. H. Dempster, Elena Medova and Ke Tang, "Long Term Spread Option Valuation and Hedging"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 32(12), 2530–2540 (2008); DOI 10.1016/j.jbankfin.2008.04.004.

Models the spread between two cointegrated commodity prices directly with a two-factor model, avoiding the hard-to-estimate correlation between two price processes, and derives pricing and hedging formulas for options on spot and futures spreads. Applied to the heating oil–WTI crack spread and the Brent–WTI location spread.

Version record

14.5 Financialisation, speculation and index investing

Whether index investors and speculators changed commodity prices, volatility and correlations: the 2008 Senate testimony and CFTC survey that framed the debate, surveys, CFTC position data and their limits, storage-based and structural tests in oil, flow-based identification, models of financialisation, and its information and real effects. Commodity risk premia and factor strategies are in Section 14.1.

Ing-Haw Cheng and Wei Xiong, "Financialization of Commodity Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 6(1), 419–441 (2014); DOI 10.1146/annurev-financial-110613-034432.

Review of index-investor effects on commodity prices, risk sharing and information discovery.

Version record

Ke Tang and Wei Xiong, "Index Investment and the Financialization of Commodities"

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal, 68(6), 54–74 (2012); DOI 10.2469/faj.v68.n6.5.

Correlations of indexed commodities with oil rose after 2004 relative to off-index commodities.

Version record

Lutz Kilian and Daniel P. Murphy, The Role of Inventories and Speculative Trading in the Global Market for Crude Oil

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics 29(3), 454–478, 2014; first online 10 April 2013.

Adds inventories to a structural oil-market model so shocks to speculative demand can be separated from flow demand and supply. This is valuable for scenario attribution and for resisting a single-cause futures-position narrative. Its identification and historical inventory data do not prove that the same shares hold in a later oil shock, especially under sanctions or inventory-reporting gaps.

Extended note. Version record

Scott H. Irwin and Dwight R. Sanders, "Index Funds, Financialization, and Commodity Futures Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Economic Perspectives and Policy, 33(1), 1–31 (2011); DOI 10.1093/aepp/ppq032.

Review of whether index funds drove the 2007–08 commodity price spike. Studies finding an effect use data and methods that limit confidence in them, while others find no systematic link between index positions and futures prices; the lack of a direct empirical link casts doubt on the bubble view.

Version record

Scott H. Irwin and Dwight R. Sanders, "Testing the Masters Hypothesis in Commodity Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 34(1), 256–269 (2012); DOI 10.1016/j.eneco.2011.10.008.

Tests the claim that long-only index investment drove the 2007–08 spike. Index estimates based on agricultural data contain large errors in energy markets relative to the CFTC's Index Investment Data. Fama–MacBeth tests on 19 markets find very little effect of index positions on returns or volatility, and Granger tests on two energy index funds find no causal links.

Version record

James D. Hamilton and Jing Cynthia Wu, "Effects of Index-Fund Investing on Commodity Futures Prices"

Difficulty: Research **Type:** Research paper

Status/version: International Economic Review, 56(1), 187–205 (2015); DOI 10.1111/iere.12099.

If index-fund purchases move futures prices, index investors' notional positions should predict excess returns. CFTC index-trader positions do not predict returns in agricultural contracts; there is some predictability for oil in 2006–09, but it breaks down out of sample.

Version record

Brian J. Henderson, Neil D. Pearson and Li Wang, "New Evidence on the Financialization of Commodity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 28(5), 1285–1311 (2015); DOI 10.1093/rfs/hhu091.

Issuers of commodity-linked notes hedge their liabilities in futures, so flows into and out of the notes pass to futures markets without information about prices. These flows still move futures prices, consistent with non-informational financial investment affecting commodity prices.

Version record

Kenneth J. Singleton, "Investor Flows and the 2008 Boom/Bust in Oil Prices"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 60(2), 300–318 (2014); DOI 10.1287/mnsc.2013.1756.

Informational frictions and speculation can push oil prices from fundamentals. Investor flows, especially intermediate-term growth in index positions and managed-money spread positions, had significant effects on futures prices after controlling for stock returns, intermediary balance sheets, open interest, the basis and past returns.

Version record

Bahattin Büyükşahin and Jeffrey H. Harris, "Do Speculators Drive Crude Oil Futures Prices?"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 32(2), 167–202 (2011); DOI 10.5547/issn0195-6574-ej-vol32-no2-7.

Uses unique CFTC data on positions by trader type in crude oil futures for 2000–08. Granger-causality tests at daily and multi-day horizons find little evidence that hedge-fund or other non-commercial position changes lead prices; price changes precede position changes.

Version record

Celso Brunetti, Bahattin Büyükşahin and Jeffrey H. Harris, "Speculators, Prices, and Market Volatility"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 51(5), 1545–1574 (2016); DOI 10.1017/s0022109016000569.

CFTC data for 2005–09 identifying trader categories show little evidence that speculators destabilise markets. Hedge-fund position changes are negatively related to volatility in corn, crude oil and natural gas, and swap-dealer activity is largely unrelated to it, consistent with hedge funds supplying liquidity.

Version record

Ing-Haw Cheng, Andrei Kirilenko and Wei Xiong, "Convective Risk Flows in Commodity Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 19(5), 1733–1781 (2015); DOI 10.1093/rof/rfu043.

Around the 2008 crisis, financial traders cut net long commodity positions as the VIX rose, while hedgers reduced their net shorts as prices fell. This "convective risk flow" shifted risk back to hedgers and shows that financial traders' own distress moves commodity prices.

Version record

Michael Sockin and Wei Xiong, "Informational Frictions and Commodity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 70(5), 2063–2098 (2015); DOI 10.1111/jofi.12261.

Commodity prices aggregate dispersed information about the global economy and guide producers whose output is complementary. Noise from supply shocks or futures trading can therefore feed back into commodity demand and spot prices, which undercuts empirical work that treats shocks as publicly observed.

Version record

Suleyman Basak and Anna Pavlova, "A Model of Financialization of Commodities"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 71(4), 1511–1556 (2016); DOI 10.1111/jofi.12408.

Institutional investors who care about performance relative to a commodity index raise futures prices, volatilities and correlations, more for index than non-index commodities, and raise equity–commodity correlations. Spot prices and inventories also respond, and shocks to one index commodity spill over to all storable ones.

Version record

Bassam Fattouh, Lutz Kilian and Lavan Mahadeva, "The Role of Speculation in Oil Markets: What Have We Learned So Far?"

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Journal, 34(3), 7–33 (2013); DOI 10.5547/01956574.34.3.2.

Surveys six strands of evidence on whether financial speculation drove the 2003–08 oil price surge. The evidence does not support an important role for speculation; comovement of spot and futures prices reflects common fundamentals rather than financialisation.

Version record

Jonathan Brogaard, Matthew C. Ringgenberg and David Sovich, "The Economic Impact of Index Investing"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 32(9), 3461–3499 (2019); DOI 10.1093/rfs/hhy129.

After commodity index investing grew around 2004, firms that use index commodities made worse production decisions, earned 40 per cent lower profits and had 6 per cent higher costs. Index investing appears to distort the price signal firms learn from.

Version record

Itay Goldstein and Liyan Yang, "Commodity Financialization and Information Transmission"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 77(5), 2613–2667 (2022); DOI 10.1111/jofi.13165.

Model separating financial speculators from financial hedgers in commodity futures. It derives their effects on price informativeness, the futures price bias, comovement with other markets and the predictive power of financial trading, linked to spot prices and production; a dynamic version matches the financialisation period.

Version record

U.S. Commodity Futures Trading Commission, Commitments of Traders: explanatory notes and Disaggregated COT explanatory notes

Difficulty: Intro **Type:** Report / research note

Status/version: CFTC data documentation; weekly reports of Tuesday positions released on Fridays; pages retrieved 13 October 2026.

Documentation for the weekly Commitments of Traders reports, which break down each Tuesday's open interest for markets in which 20 or more traders hold reportable positions; reportable positions usually cover 70–90 per cent of open interest. The legacy report splits traders into commercial (hedgers, by Form 40 statement) and non-commercial. The disaggregated report, published since 2009, separates producer/merchant/processor/user, swap dealers, managed money and other reportables; classification is by trader, not by trade.

Version record

Christopher R. Knittel and Robert S. Pindyck, "The Simple Economics of Commodity Price Speculation"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics, 8(2), 85–110 (2016); DOI 10.1257/mac.20140033.

Oil rose from under $40 a barrel in 2004 to $145 in July 2008, fell to $30 and recovered to $110 in 2011. A simple model of supply and demand in the cash and storage markets checks whether speculation is consistent with data on production, inventory changes and convenience yields; it had little if any effect on oil prices. The approach applies to other storable commodities.

Version record

Lutz Kilian and Thomas K. Lee, "Quantifying the speculative component in the real price of oil: The role of global oil inventories"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 42, 71–87 (2014); DOI 10.1016/j.jimonfin.2013.08.005.

Statistical tools for attributing dollars of the real oil price to speculative demand for oil stocks, with two proxies for global crude inventories. Speculative demand raised the price by $5–14 in mid-2008 but shows no effect between early 2003 and early 2008; the 2011 Libyan crisis added $3–13 and the 2012 tensions with Iran $0–9.

Version record

Luciana Juvenal and Ivan Petrella, "Speculation in the Oil Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 30(4), 621–649 (2015; online 2014); DOI 10.1002/jae.2388.

Identifies oil shocks with a dynamic factor model on a large dataset, because a small VAR is not informationally sufficient. Global demand shocks explain most oil-price variation and speculative shocks are the second most important driver; speculation contributed significantly to the 2004–08 rise and the collapse that followed, while comovement with other commodities mainly reflects global demand.

Version record

Bahattin Büyükşahin and Michel A. Robe, "Speculators, commodities and cross-market linkages"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 42, 38–70 (2014); DOI 10.1016/j.jimonfin.2013.08.004.

Non-public CFTC position data for 17 US commodity futures markets show that the correlation between investable commodity and equity index returns rises with speculators' participation, especially hedge funds active in both equity and commodity futures. Commodity swap dealers, including index traders, show no such link, and the effect weakens in periods of financial stress.

Version record

Zeno Adams and Thorsten Glück, "Financialization in commodity markets: A passing trend or the new normal?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 60, 93–111 (2015); DOI 10.1016/j.jbankfin.2015.07.008.

Argues that large inflows changed the dependence between commodities and stocks, and that distressed selling in 2007–09 alone cannot explain the size and persistence of the higher comovement. Commodities have become an investment style for institutions, so the authors expect spillovers to stay high.

Version record

Yannick Le Pen and Benoît Sévi, "Futures Trading and the Excess Co-movement of Commodity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 22(1), 381–418 (2018; online 2017); DOI 10.1093/rof/rfx039.

Revisits Pindyck–Rotemberg excess comovement with a large approximate factor model: nine factors from 184 real and nominal series across developed and emerging economies, 1993–2013, for eight unrelated commodities. Excess comovement varies over time and is particularly high after 2007, and speculative intensity, correlated across futures markets, helps drive it.

Version record

Steven D. Baker, "The Financialization of Storable Commodities"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 67(1), 471–499 (2021); DOI 10.1287/mnsc.2019.3445.

Dynamic equilibrium model of spot and futures prices with storage and heterogeneous risk-averse traders, calibrated to crude oil. Financialisation lowers the futures risk premium and raises the correlation between open interest and the spot price, but leaves the long-run mean spot price unchanged, and speculative storage generally dampens its effect on spot volatility.

Version record

Scott Main, Scott H. Irwin, Dwight R. Sanders and Aaron Smith, "Financialization and the returns to commodity investments"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Commodity Markets, 10, 22–28 (2018); DOI 10.1016/j.jcomm.2018.05.004.

Asks whether financialisation lowered the risk premiums available to long-only commodity investors. Energy futures premiums generally fell after 2004, but premiums rose in all but one non-energy market, and the average unconditional return to individual commodity futures is about zero both before and after.

Version record

Valentina G. Bruno, Bahattin Büyükşahin and Michel A. Robe, "The Financialization of Food?"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 99(1), 243–264 (2017; online 2016); DOI 10.1093/ajae/aaw059.

Documents the growth of financial institutions' activity in agricultural futures since 1995 and estimates a structural VAR for weekly grain, livestock and equity returns over 1995–2015. World business-cycle shocks have a large and lasting effect on agriculture–equity comovement; shocks to speculative intensity have shorter-lived effects that are not significant in every specification.

Version record

Michael W. Masters, Testimony before the U.S. Senate Committee on Homeland Security and Governmental Affairs, 20 May 2008

Difficulty: Intro **Type:** Report / research note

Status/version: Hearing of 20 May 2008, *Financial Speculation in Commodity Markets*, printed in S. Hrg. 110-705, *High Price of Commodities—2008*.

The statement behind the Masters hypothesis. Institutional "index speculators" had raised assets in commodity index strategies from $13 billion to $260 billion in five years, bought futures without regard to price, and were in Masters's view a primary cause of the 2003–08 rise in food and energy prices; he compared their growth in petroleum futures demand with China's growth in oil demand. Read it as the claim that later tests examine, not as evidence.

Version record

U.S. Commodity Futures Trading Commission, Staff Report on Commodity Swap Dealers & Index Traders with Commission Recommendations

Difficulty: Advanced **Type:** Report / research note

Status/version: CFTC staff report with Commission recommendations, September 2008 (press release 5542-08, 11 September 2008).

Results of a June 2008 special call to 32 swap dealers and index funds. Index investment on and off exchange was about $200 billion at the end of June 2008, $161 billion of it in CFTC-regulated markets. In crude oil, index notional value rose with prices while index positions in futures equivalents fell between December 2007 and June 2008. Staff note that the commercial and non-commercial labels classify traders, not trades, and recommend better reporting of swap-dealer and index activity; a commissioner's dissent is appended.

Version record

14.6 Commodity allocation, inflation hedging and index products

Whether commodities belong in a multi-asset portfolio: diversification evidence in and out of sample, inflation hedging, commodity factor premia in the strategic mix, index rules and enhanced indices, futures-based ETFs, the price effects of index and ETF trading, and the USO case of April 2020. Risk premia and factor strategies are in Section 14.1; financialisation is in Section 14.5.

Zvi Bodie, "Investment Strategy in an Inflationary Environment"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 701 (June 1981); DOI 10.3386/w0701.

Mean-variance study of how an investor who cares about real returns should divide funds among stocks, bonds, bills and commodity futures. Bills anchor low-risk portfolios (the minimum-risk portfolio has a mean real return of zero and a standard deviation of about 1 per cent), stocks have the highest mean and risk, and commodity futures are the only asset positively correlated with inflation, so adding them reaches any target real return with less risk. Published in 1982 in *The Changing Roles of Debt and Equity in Financing U.S. Capital Formation*, 47–64.

Version record

Robert J. Greer, "The Nature of Commodity Index Returns"

Difficulty: Intro **Type:** Research paper

Status/version: Journal of Alternative Investments, 3(1), 45–52 (2000); DOI 10.3905/jai.2000.318924.

Practitioner case for an unleveraged long-only commodity index: historically, returns comparable to equities in size and volatility, with more positive skew and negative correlation with stock and bond returns. Asks whether there are fundamental reasons to expect the pattern to continue, which would let the asset class extend a portfolio's efficient frontier.

Version record

Gerald R. Jensen, Robert R. Johnson and Jeffrey M. Mercer, "Efficient use of commodity futures in diversified portfolios"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 20(5), 489–506 (2000); DOI 10.1002/(sici)1096-9934(200005)20:5<489::aid-fut5>3.0.co;2-a.

Over 1973–97, mean-variance optimisation across stocks, bonds, bills, real estate and commodity futures gives commodities substantial weight. The benefit depends on monetary conditions: commodities enter efficient portfolios and raise returns at all risk levels when policy is restrictive, and have little or no weight when it is expansive.

Version record

C. Sherman Cheung and Peter Miu, "Diversification benefits of commodity futures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Financial Markets, Institutions and Money, 20(5), 451–474 (2010); DOI 10.1016/j.intfin.2010.06.003.

Tests whether the diversification benefits of adding commodity futures are statistically significant, whether they extend to investors in a resource-based economy such as Canada, whether they survive regime-switching correlations in bear markets, and which investors should hold commodities. Concludes that the benefit is far more complex than the literature usually assumes.

Version record

Charoula Daskalaki and George Skiadopoulos, "Should investors include commodities in their portfolios after all? New evidence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 35(10), 2606–2626 (2011); DOI 10.1016/j.jbankfin.2011.02.022.

In-sample mean-variance and non-mean-variance spanning tests find that commodities help only investors who care about higher moments, and the gains vanish out of sample for optimal portfolios that account for higher moments. The result holds across performance measures, utility functions, datasets, transaction costs and subperiods, except the 2005–08 boom.

Version record

Julia Belousova and Gregor Dorfleitner, "On the diversification benefits of commodities from the perspective of euro investors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 36(9), 2455–2472 (2012); DOI 10.1016/j.jbankfin.2012.05.003.

Euro-investor view of the diversification contribution of individual commodities, separating its sources with statistical tests. Industrial metals, agriculturals and livestock mainly reduce risk; energy and precious metals reduce risk and add return. Exposure can be adjusted between bull and bear markets, and the gains hold for futures as well as physical commodities.

Version record

Wolfgang Bessler and Dominik Wolff, "Do commodities add value in multi-asset portfolios? An out-of-sample analysis for different investment strategies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 60, 1–20 (2015); DOI 10.1016/j.jbankfin.2015.06.021.

Adds commodities to stock–bond portfolios under equal and strategic weights, risk parity, minimum variance, reward-to-risk timing, mean-variance and Black–Litterman. Out-of-sample gains are much smaller than in-sample studies and spanning tests suggest; broad indices, industrial and precious metals and energy help in most strategies, agriculture and livestock hardly at all.

Version record

Xin Gao and Federico Nardari, "Do Commodities Add Economic Value in Asset Allocation? New Evidence from Time-Varying Moments"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 53(1), 365–393 (2018); DOI 10.1017/s002210901700103x.

Out-of-sample test of commodities in multi-asset strategies that exploit predictable means, variances and correlations. With predictability, commodities add value even without short sales or high leverage: a moderately risk-averse investor rebalancing quarterly would pay up to 108 basis points a year after costs (155 for a non-mean-variance investor) to add them.

Version record

Nikolay Gospodinov and Serena Ng, "Commodity Prices, Convenience Yields, and Inflation"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 95(1), 206–219 (2013); DOI 10.1162/rest_a_00242.

The first two principal components of 23 commodity convenience yields predict inflation beyond the unemployment gap and oil prices, out of sample, across horizons and across the G7. Convenience yields also explain commodity prices and carry information from futures markets about future economic conditions; a bootstrap handles inference with estimated components.

Version record

Shaun Roache and Alexander Attie, "Inflation Hedging for Long-Term Investors"

Difficulty: Research **Type:** Research paper

Status/version: IMF Working Paper 09/90 (April 2009); DOI 10.5089/9781451872378.001.

Uses a vector error-correction model to assess how traditional asset classes hedge inflation over different horizons within a diversified portfolio. Commodities hedge inflation surprises in the short run but may not over longer horizons, and tactical allocation after inflation surprises could raise returns.

Version record

Georgios Bampinas and Theodore Panagiotidis, "Are gold and silver a hedge against inflation? A two century perspective"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Financial Analysis, 41, 267–276 (2015); DOI 10.1016/j.irfa.2015.02.007.

Time-invariant and time-varying cointegration tests on UK and US data from 1791 to 2010. Gold fully hedges headline, expected and core CPI in the long run, more strongly when the relationship may vary over time and more in the US than in the UK; silver does not hedge US consumer prices, with some time-varying evidence for the UK.

Version record

David Blitz and Wilma de Groot, "Strategic Allocation to Commodity Factor Premiums"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Alternative Investments, 17(2), 103–115 (2014); DOI 10.3905/jai.2014.17.2.103.

Confirms sizeable momentum, carry and low-volatility premiums in commodity futures and argues they belong in the strategic asset allocation. A diversified portfolio of the factor premiums has much better risk-adjusted performance than the long-only commodity market and adds value to a stock–bond portfolio, while the market portfolio itself deserves little or no strategic weight.

Version record

Georgios Rallis, Joëlle Miffre and Ana-Maria Fuertes, "Strategic and Tactical Roles of Enhanced Commodity Indices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 33(10), 965–992 (2013; online 2012); DOI 10.1002/fut.21571.

Compares the S&P GSCI and the Dow Jones–UBS Commodity Index with enhanced versions that choose contract maturity or use momentum and term-structure signals. Maturity-enhanced indices earn alphas of 2.77–5.49 per cent a year after liquidity risk, momentum and term-structure versions about 2.10 per cent, and all keep the diversification and inflation-hedging properties of the traditional indices.

Version record

Hans R. Stoll and Robert E. Whaley, "Commodity Index Investing: Speculation or Diversification?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Alternative Investments, 14(1), 50–60 (2011); DOI 10.3905/jai.2011.14.1.050.

Responds to congressional findings that index investment rather than fundamentals caused the 2007–08 commodity price spikes. Using CFTC Commitments of Traders data, the authors argue that passive long-only index investment is not speculation and that index flows, from rolling existing positions or opening new ones, have little effect on futures prices.

Version record

Scott H. Irwin, Philip Garcia, Darrel L. Good and Eugene L. Kunda, "Spreads and Non-Convergence in Chicago Board of Trade Corn, Soybean, and Wheat Futures: Are Index Funds to Blame?"

Difficulty: Research **Type:** Research paper

Status/version: Applied Economic Perspectives and Policy, 33(1), 116–142 (2011); DOI 10.1093/aepp/ppr001.

Examines whether index funds caused the wide calendar spreads and failures of convergence between cash and futures prices in CBOT corn, soybeans and wheat. Statistical tests find no evidence that index rolls, or large index positions in a crowded market, widened the spreads.

Version record

Ilan Guedj, Guohua Li and Craig McCann, "Futures-Based Commodity ETFs"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Index Investing, 2(1), 14–24 (2011); DOI 10.3905/jii.2011.2.1.014.

Returns on futures-based commodity ETFs have deviated widely from changes in the prices of their underlying commodities. For crude-oil ETFs the deviation is serially correlated and largely predictable from the term structure of oil futures, so only investors who monitor the curve should use them for diversification or hedging.

Version record

Geetesh Bhardwaj, Gary B. Gorton and K. Geert Rouwenhorst, "Investor Interest and the Returns to Commodity Investing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Portfolio Management, 42(3), 44–55 (2016); DOI 10.3905/jpm.2016.42.3.044.

Monthly commodity futures returns in the decade after investor inflows began in 2004 were similar to their long-run means. Correlations among commodities and with equities rose temporarily around the financial crisis and then returned to normal, and the variation tracks the business cycle rather than financialisation.

Version record

Karamfil Todorov, "When Passive Funds Affect Prices: Evidence from Volatility and Commodity ETFs"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 28(3), 831–863 (2024; online 2023); DOI 10.1093/rof/rfad038.

Measures ETF price impact in VIX and commodity futures, the most ETF-dominated markets, using the specifics of futures contracts, and isolates a component of VIX futures prices strongly related to ETF rebalancing. ETF demand splits into leverage, calendar and flow rebalancing; leverage rebalancing matters most, amplifies price changes and makes trading against ETFs risky.

Version record

Bloomberg Index Services Limited, The Bloomberg Commodity Index Methodology

Difficulty: Advanced **Type:** Report / research note

Status/version: Index methodology published as the February 2026 edition, with 2026 target weights; no revision date is printed in the text.

Rules of the Bloomberg Commodity Index (BCOM). Weights combine two-thirds futures-market liquidity and one-third dollar-weighted production, averaged over five years, subject to caps of 15 per cent per commodity, 25 per cent for a commodity with its derivatives and 33 per cent per sector, and a 2 per cent floor; the index is reweighted annually. Positions roll from the sixth to the tenth business day of the month, 20 per cent a day. Twenty-three commodities are in the 2026 index. The risk section explains roll yield, spot return and the drag from contango.

Version record

U.S. Securities and Exchange Commission, In the Matter of United States Commodity Funds LLC and United States Oil Fund, LP: Order Instituting Cease-and-Desist Proceedings

Difficulty: Advanced **Type:** Report / research note

Status/version: SEC administrative order, Securities Act Release No. 11006, 8 November 2021 (press release 2021-229); settled without admitting or denying the findings.

USO aimed to track daily changes in the front-month WTI futures price. In April 2020, as the May contract settled below zero, it took record inflows, exhausted its registered shares and moved positions further down the curve. On 22 April its sole futures broker limited new positions; USO did not fully disclose the limit until 21 May. The SEC found a violation of Securities Act section 17(a)(3), and USO and its general partner agreed to a $2.5 million penalty, offset by up to $1.25 million paid in a parallel CFTC case.

Version record

14.7 Crude oil, refined products and price benchmarks

Crude benchmarks, futures contracts and price reporting agencies; quality and location differentials, the WTI–Brent spread and Cushing storage; the negative WTI settlement of April 2020; refining margins, crack spreads and retail pass-through; and OPEC behaviour.

U.S. Energy Information Administration, Weekly Petroleum Status Report

Difficulty: Advanced **Type:** Report / research note

Status/version: Official weekly petroleum data series, continuously updated; inspected September 16, 2026 release for week ending September 11. Dynamic front page; retain the chosen issue and table files when using historical data.

Supplies dated measures of crude and petroleum-product stocks, refinery activity, imports, exports, and related balances. It supports an oil or crack-spread desk in distinguishing refinery bottlenecks and product-specific scarcity from a crude-only signal. National stocks are only a starting point for location-specific deliverability or quality-adjusted refinery economics.

Extended note. Version record

U.S. Energy Information Administration, What Drives Petroleum Product Prices: Prices and Crack Spreads; Refinery Capacity Report

Difficulty: Advanced **Type:** Report / research note

Status/version: Official live explainer plus report released 26 June 2026 with 1 January 2026 capacity observations.

Defines gasoline and distillate cracks with regional grade and seasonal specifications, then supplies refinery unit capacity to test whether a spread could be processed. Separate product-specific margins from a generic crude-oil factor or 3:2:1 shortcut. Read after oil curves. A crack is a gross price spread, not net refinery profit. Align crude slate, yields, unit downtime, blend rules, pipeline/export routes and release vintages; capacity is not run-rate throughput.

Version record

Bassam Fattouh, "The Dynamics of Crude Oil Price Differentials"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 32(2), 334–342 (2010); DOI 10.1016/j.eneco.2009.06.007.

Standard unit-root tests suggest that differentials between crude grades are stationary, but two-regime threshold autoregressions show that adjustment toward equilibrium happens only beyond a threshold. The oil market is one great pool, yet not integrated at every moment: arbitrage between grades is costly and risky, and temporary breakdowns in benchmarks can decouple prices.

Version record

Bahattin Büyükşahin, Thomas K. Lee, James T. Moser and Michel A. Robe, "Physical Markets, Paper Markets and the WTI-Brent Spread"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 34(3), 129–152 (2013); DOI 10.5547/01956574.34.3.7.

From autumn 2008 WTI periodically traded at unprecedented discounts to Brent, which did not appear between Brent and grades comparable to WTI. Calendar and WTI–Brent spreads track storage conditions at Cushing, and the paper tests whether index traders' positions help predict the spread after controlling for fundamentals.

Version record

Adrian Fernandez-Perez, Ana-Maria Fuertes and Joëlle Miffre, "The Negative Pricing of the May 2020 WTI Contract"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 44(1), 119–142 (2023); DOI 10.5547/01956574.44.1.afer.

The super-contango of early 2020 drew cash-and-carry traders long the May 2020 WTI contract, short later ones and into Cushing storage. That filled storage, and the crash on 20 April came from long holders without booked storage reversing, aggravated by a liquidity squeeze, margin calls and trade-at-settlement. Index trackers did not trigger it.

Version record

Michael S. Haigh and Matthew T. Holt, "Crack Spread Hedging: Accounting for Time-Varying Volatility Spillovers in the Energy Futures Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 17(3), 269–289 (2002); DOI 10.1002/jae.628.

Hedges a trader's crack-spread exposure with crude oil, heating oil and gasoline futures. A multivariate GARCH that captures volatility spillovers and time-to-maturity effects cuts uncertainty significantly more than constant hedge ratios, even after trading costs.

Version record

Severin Borenstein, A. Colin Cameron and Richard Gilbert, "Do Gasoline Prices Respond Asymmetrically to Crude Oil Price Changes?"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 112(1), 305–339 (1997); DOI 10.1162/003355397555118.

Retail gasoline prices rise faster after crude increases than they fall after decreases ("rockets and feathers"). Asymmetry appears in spot gasoline's response to crude, possibly from inventory adjustment, and in retail's response to wholesale, possibly from short-run retail market power.

Version record

Pedro A. Almoguera, Christopher C. Douglas and Ana María Herrera, "Testing for the Cartel in OPEC: Non-cooperative Collusion or Just Non-cooperative?"

Difficulty: Research **Type:** Research paper

Status/version: Oxford Review of Economic Policy, 27(1), 144–168 (2011); DOI 10.1093/oxrep/grr007.

A switching-regression model of OPEC facing a competitive fringe in 1974–2004 finds both collusive and non-cooperative episodes. OPEC raised prices through collusion in some periods but was not systematically effective; on average its behaviour is best described as Cournot competition.

Version record

ICE Futures Europe, Brent Crude Futures: contract specification

Difficulty: Intro **Type:** Report / research note

Status/version: ICE product page; live contract specification, retrieved 13 October 2026.

Specification of the ICE Brent Crude futures contract: 1,000 barrels priced in US dollars and cents; trading ends on the last business day of the second month before the contract month; daily settlement is the weighted average price over a two-minute window from 19:28 London time. The contract is deliverable through exchange for physical, with an option to cash settle against the ICE Brent Index published the next day.

Version record

Board of the International Organization of Securities Commissions (IOSCO), Principles for Oil Price Reporting Agencies: Final Report (FR06/12)

Difficulty: Advanced **Type:** Report / research note

Status/version: IOSCO Final Report FR06/12, 5 October 2012.

IOSCO's response to the G20's request for better oversight of oil price reporting agencies, prepared with the IEA, IEF and OPEC. Agencies should publish their assessment methodologies, including the criteria for judgement, minimum transaction-data thresholds, treatment of thin markets and dependence on a few submitters, consult on methodology changes, and manage data quality, conflicts of interest, complaints and audit. The principles target assessments referenced in oil derivatives.

Version record

U.S. Commodity Futures Trading Commission staff (Division of Market Oversight and Office of the Chief Economist), Interim Staff Report: Trading in NYMEX WTI Crude Oil Futures Contract Leading up to, on, and around April 20, 2020

Difficulty: Advanced **Type:** Report / research note

Status/version: CFTC interim staff report, published 23 November 2020 (press release 8315-20).

Staff account of the May 2020 WTI contract, which fell from $17.73 to settle at minus $37.63 a barrel on 20 April 2020, the first negative price in the contract's 37 years. It describes fundamental factors (an oversupplied market, the COVID-19 demand collapse, concerns about storage at Cushing) and technical ones (unusually high open interest entering the session, liquidity that thinned well before 20 April, and an exceptionally fast fall before the 14:30 settlement despite exchange controls). Root-cause analysis was out of scope.

Version record

14.8 Oil supply and demand shocks and oil-price forecasting

Identification of oil supply and demand shocks with structural VARs, sign restrictions, elasticity bounds and external instruments; measures of global real activity; and forecasting the real price of oil in real time with futures, product spreads, VARs and forecast combinations. Macroeconomic effects are in Section 14.9 and asset-price effects in Section 14.10.

Lutz Kilian, "Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 99(3), 1053–1069 (2009); DOI 10.1257/aer.99.3.1053.

Structural VAR separating supply, aggregate demand and precautionary demand shocks in oil.

Version record

Christiane Baumeister and Lutz Kilian, "Forty Years of Oil Price Fluctuations: Why the Price of Oil May Still Surprise Us"

Difficulty: Intro/Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 30(1), 139–160 (2016); DOI 10.1257/jep.30.1.139.

Accessible history of major oil-price episodes and what drove them.

Version record

Robert B. Barsky and Lutz Kilian, "Oil and the Macroeconomy since the 1970s"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 18(4), 115–134 (2004); DOI 10.1257/0895330042632708.

Questions the view that oil price shocks cause recessions and stagflation. Major oil price moves are not exogenous to the US economy, there are good reasons to expect reverse causality from macroeconomic conditions to oil prices, and oil shocks are not needed to explain 1970s stagflation.

Version record

Christiane Baumeister and James D. Hamilton, "Structural Interpretation of Vector Autoregressions with Incomplete Identification: Revisiting the Role of Oil Supply and Demand Shocks"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 109(5), 1873–1910 (2019); DOI 10.1257/aer.20151569.

Treats identifying assumptions in structural VARs as Bayesian priors that can be relaxed. Applied to oil, supply disruptions explain more of historical price moves and inventory demand less than earlier estimates; supply shocks lower world activity after a lag, while demand shocks do not.

Version record

Diego R. Känzig, "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 111(4), 1092–1125 (2021); DOI 10.1257/aer.20190964.

Identifies oil supply news shocks from high-frequency oil futures moves around OPEC announcements. Negative news raises oil prices at once, gradually lowers production and raises inventories; US activity falls, prices and inflation expectations rise and the dollar depreciates.

Version record

Dario Caldara, Michele Cavallo and Matteo Iacoviello, "Oil Price Elasticities and Oil Price Fluctuations"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 103, 1–20 (2019); DOI 10.1016/j.jmoneco.2018.08.004.

Shows that structural VARs tie the short-run supply and demand elasticities of oil together, so plausible restrictions on one can imply implausible values for the other. Identifying elasticities from a survey of studies, supply shocks explain about half of oil price fluctuations and global demand shocks about a third.

Version record

Ron Alquist and Lutz Kilian, "What Do We Learn from the Price of Crude Oil Futures?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 25(4), 539–573 (2010); DOI 10.1002/jae.1159.

Oil futures prices forecast the spot price less accurately than a no-change forecast, because the futures spread varies with the marginal convenience yield. Greater uncertainty about supply shortfalls lowers the spread and raises precautionary demand and the spot price, so the negative spread indicates precautionary-demand shocks.

Version record

Ron Alquist, Lutz Kilian and Robert J. Vigfusson, "Forecasting the Price of Oil"

Difficulty: Advanced **Type:** Book chapter

Status/version: Handbook of Economic Forecasting, Volume 2A, Elsevier, 427–507 (2013); DOI 10.1016/b978-0-444-53683-9.00008-6.

Handbook guide to forecasting oil prices: choice of sample and price series, predictability of real and nominal prices from macroeconomic aggregates, out-of-sample comparison with no-change forecasts, the value of futures and survey forecasts, scenario-based sensitivity analysis and risk measures for forecasts.

Version record

Christiane Baumeister and Lutz Kilian, "Real-Time Forecasts of the Real Price of Oil"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 30(2), 326–336 (2012); DOI 10.1080/07350015.2011.648859.

Builds monthly data vintages for 1991–2010 with backcasting and nowcasting. Real-time forecasts can beat the no-change forecast up to a year ahead, with mean-squared error cut by up to a quarter at one and three months; VARs in oil-market variables beat futures-based, AR and ARMA forecasts at short horizons.

Version record

Lutz Kilian and Xiaoqing Zhou, "Modeling Fluctuations in the Global Demand for Commodities"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 88, 54–78 (2018); DOI 10.1016/j.jimonfin.2018.07.001.

Sets criteria a measure of global real activity must meet for modelling industrial commodity prices, compares the measures in use and explains why some popular ones are unsuitable. The commodity boom of the 2000s appears largely transitory.

Version record

Lutz Kilian, "Oil Price Shocks: Causes and Consequences"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Resource Economics, 6(1), 133–154 (2014); DOI 10.1146/annurev-resource-083013-114701.

Review of a decade of research: the real price of oil is endogenous to economic fundamentals, oil price shocks do not happen ceteris paribus, and studying their transmission requires structural models that identify the underlying supply and demand shocks.

Version record

Christiane Baumeister and Lutz Kilian, "Understanding the Decline in the Price of Oil since June 2014"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Association of Environmental and Resource Economists, 3(1), 131–158 (2016); DOI 10.1086/684160.

More than half of the $49 fall in Brent between June and December 2014 was predictable in June: $11 from earlier negative demand shocks and $16 from positive production shocks. The rest came from a July shock to price expectations that cut inventory demand ($9) and a December demand shock ($13).

Version record

Yu-Chin Chen, Kenneth Rogoff and Barbara Rossi, "Can Exchange Rates Forecast Commodity Prices?"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 125(3), 1145–1194 (2010); DOI 10.1162/qjec.2010.125.3.1145.

Exchange rates of commodity exporters predict world commodity prices in and out of sample against several benchmarks; the reverse relation is weaker. Exchange rates are forward-looking while commodity prices react more to short-term imbalances.

Version record

Lutz Kilian, "The Economic Effects of Energy Price Shocks"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Literature, 46(4), 871–909 (2008); DOI 10.1257/jel.46.4.871.

Survey of energy-price shocks: what they are and where they come from, how responsive energy demand is, how consumer spending responds, how the shocks affect US output, inflation and stock prices, why price increases seem to cause recessions while decreases do not cause expansions, and why the 2000s price surge had not caused a recession. Stresses that energy prices are endogenous and that demand and supply shocks must be separated.

Version record

James D. Hamilton, "Historical oil shocks"

Difficulty: Intro **Type:** Book chapter

Status/version: *Routledge Handbook of Major Events in Economic History*, Routledge (2013); DOI 10.4324/9780203067871.ch21.

History of the oil industry organised around large price changes, from nineteenth-century parallels to the Suez crisis of 1956–57, the 1973–74 embargo, the Iranian revolution, the Iran–Iraq war, the 1990–91 Gulf war and the 2007–08 spike, with the economic downturns that followed each major postwar shock.

Version record

Lutz Kilian and Clara Vega, "Do Energy Prices Respond to U.S. Macroeconomic News? A Test of the Hypothesis of Predetermined Energy Prices"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 93(2), 660–671 (2011); DOI 10.1162/rest_a_00086.

Regresses daily energy-price changes on news from US macroeconomic data releases to test whether energy prices are predetermined with respect to US aggregates. There is no compelling evidence of feedback at daily or monthly horizons, which supports the common VAR assumption that monthly energy-price innovations do not respond within the month to US macroeconomic aggregates.

Version record

Lutz Kilian and Daniel P. Murphy, "Why Agnostic Sign Restrictions Are Not Enough: Understanding the Dynamics of Oil Market VAR Models"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the European Economic Association, 10(5), 1166–1188 (2012); DOI 10.1111/j.1542-4774.2012.01080.x.

Sign restrictions alone cannot pin down the response of the real oil price to oil demand and supply shocks, and the assumption that all admissible models are equally likely is routinely violated, so posterior median responses mislead. Adding bounds on the short-run oil supply elasticity and on the impact response of real activity narrows the admissible models to a few similar ones, broadly matching earlier exclusion-restriction results.

Version record

Christiane Baumeister and Gert Peersman, "Time-Varying Effects of Oil Supply Shocks on the US Economy"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics, 5(4), 1–28 (2013); DOI 10.1257/mac.5.4.1.

Time-varying Bayesian VARs show that the short-run price elasticity of oil demand fell after the mid-1980s, so a given supply shortfall moves prices more, while a given price rise has smaller output effects. Supply shocks explain a smaller share of real oil-price variability than before and demand shocks a larger one; the overall effect of supply disruptions on the US economy has been modest.

Version record

Lutz Kilian, "Understanding the estimation of oil demand and oil supply elasticities"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 107, article 105844 (2022); DOI 10.1016/j.eneco.2022.105844.

Reviews methods for estimating oil supply and demand elasticities and building them into structural VARs, and points out overlooked econometric problems that reconcile apparently conflicting recent results. Reaffirms that the one-month oil supply elasticity is close to zero, so oil demand shocks are the dominant driver of the real oil price.

Version record

James D. Hamilton, "Measuring global economic activity"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 36(3), 293–303 (2021; online 2019); DOI 10.1002/jae.2740.

The shipping-cost index of world real activity used in many oil studies depends on a normalisation with substantive consequences. Comparing alternative measures on their treatment of trends, coherence with world output and correlation with commodity prices, Hamilton concludes that measures built from world industrial production are better indicators.

Version record

Lutz Kilian, "Measuring global real economic activity: Do recent critiques hold up to scrutiny?"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 178, 106–110 (2019); DOI 10.1016/j.econlet.2019.03.001.

Reply to Hamilton: the problem he documented in the Kilian (2009) index came from a coding error that logged nominal freight rates twice. The corrected index differs only slightly from the original, his concerns no longer hold, and the main results of Kilian (2009) and related studies are unchanged.

Version record

Ana María Herrera and Sandeep Kumar Rangaraju, "The effect of oil supply shocks on US economic activity: What have we learned?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 35(2), 141–159 (2020; online 2019); DOI 10.1002/jae.2735.

Most of the wide variation in estimated responses of the real oil price and US GDP to oil supply disruptions comes from identification assumptions and model specification. Models that allow a large short-run supply elasticity imply larger price responses and deeper, longer GDP contractions; conditioning on microeconomic elasticity estimates narrows the differences. Also examines lag length, pre-1973 data, real-activity measures and median versus modal responses.

Version record

Christiane Baumeister and James D. Hamilton, "Advances in Using Vector Autoregressions to Estimate Structural Magnitudes"

Difficulty: Research **Type:** Research paper

Status/version: Econometric Theory, 40(3), 472–510 (2024; online 2022); DOI 10.1017/s026646662200055x.

Survey of drawing structural conclusions from VARs from a unified view of prior information: traditional identification as a claim to exact prior knowledge, Bayesian inference to admit that the knowledge is imperfect, concerns about how results from sign-restricted VARs are reported, and a correction for a common error in estimating structural elasticities.

Version record

Christiane Baumeister and Lutz Kilian, "What Central Bankers Need to Know about Forecasting Oil Prices"

Difficulty: Research **Type:** Research paper

Status/version: International Economic Review, 55(3), 869–889 (2014); DOI 10.1111/iere.12074.

Addresses the econometric issues in the quarterly oil-price forecasts that central banks use. Quarterly forecasts of the real oil price from suitably designed VARs estimated on monthly data are the most accurate real-time forecasts among many methods, including averaged monthly futures-based forecasts, no-change forecasts and regressions on quarterly data.

Version record

Christiane Baumeister and Lutz Kilian, "Forecasting the Real Price of Oil in a Changing World: A Forecast Combination Approach"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Business & Economic Statistics, 33(3), 338–351 (2015); DOI 10.1080/07350015.2014.949342.

Combines six real-time econometric forecasting models of the real oil price, weighted by recent accuracy. Over the previous 20 years the combination would have beaten the no-change forecast at horizons up to 18 months, with MSPE reductions up to 12 per cent and directional accuracy up to 72 per cent; EIA forecasts were less accurate than no-change and lowered the accuracy of the combination.

Version record

Christiane Baumeister, Lutz Kilian and Xiaoqing Zhou, "Are Product Spreads Useful for Forecasting Oil Prices? An Empirical Evaluation of the Verleger Hypothesis"

Difficulty: Research **Type:** Research paper

Status/version: Macroeconomic Dynamics, 22(3), 562–580 (2018; online 2017); DOI 10.1017/s1365100516000237.

Tests the industry view, the Verleger hypothesis, that refined-product spreads predict crude prices. Some product-spread models forecast the real oil price out of sample even at one to two years; the best, a time-varying parameter model of gasoline and heating-oil spreads, cuts MSPE by up to 20 per cent at two years, complementing fundamentals-based models that work best at short horizons.

Version record

Christiane Baumeister, Dimitris Korobilis and Thomas K. Lee, "Energy Markets and Global Economic Conditions"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 104(4), 828–844 (2022); DOI 10.1162/rest_a_00977.

Evaluates indicators of global economic activity and other fundamentals for forecasting real oil prices and global petroleum consumption. World industrial production is among the most useful single indicators, but combining sources does better; the result is a new index of global economic conditions, applied to the early-2020 contraction and to shale producers' price risks.

Version record

Christoph Funk, "Forecasting the real price of oil – Time-variation and forecast combination"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 76, 288–302 (2018); DOI 10.1016/j.eneco.2018.04.016.

Out-of-sample comparison of seven real oil-price forecasting models, some extended with real-time Brent measures and alternative activity indices. Forecasts beat no-change for horizons up to 24 months, with MSPE gains up to 25 per cent; recursive MSPE ratios expose each model's weak periods, and a combination of four models is more accurate and more stable than the best single models over the medium run.

Version record

Anthony Garratt, Shaun P. Vahey and Yunyi Zhang, "Real-time forecast combinations for the oil price"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 34(3), 456–462 (2019; online 2018); DOI 10.1002/jae.2673.

Replication of Baumeister–Kilian (2015) that broadly reproduces its findings, extends the evaluation to Brent and to December 2017, and fills missing futures prices with a real-time Nelson–Siegel factor model. Combined forecasts work as well for Brent, and the futures-based model improves accuracy at longer horizons.

Version record

Reinhard Ellwanger and Stephen Snudden, "Forecasts of the real price of oil revisited: Do they beat the random walk?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 154, article 106962 (2023); DOI 10.1016/j.jbankfin.2023.106962.

The usual no-change benchmark uses the monthly average oil price. An end-of-month no-change forecast, the random walk in daily prices, is significantly more accurate up to a year ahead, and one month ahead every existing model that beats the monthly-average benchmark does worse than it, which questions the value of existing forecasting approaches.

Version record

U.S. Energy Information Administration, Short-Term Energy Outlook, October 2026

Difficulty: Intro **Type:** Report / research note

Status/version: Monthly official forecast; October 2026 edition released 6 October 2026 (forecast completed 1 October; next release 10 November 2026).

Monthly EIA forecasts of US and global energy prices, supply and demand through the end of the following year. The October 2026 edition assumes Middle East oil flows stay constrained through the fourth quarter but rise as convoys through the Strait of Hormuz, bypass routes and ship-to-ship transfers increase exports. It raises its Brent forecast to $105 a barrel for the fourth quarter of 2026 and $84 for 2027, citing attacks on Saudi Arabia's East–West pipeline and tight diesel markets. Archived vintages give a judgemental benchmark for model forecasts.

Version record

14.9 Oil shocks, inflation and monetary policy

How oil-price shocks affect output, consumption, inflation and inflation expectations; the breakdown, nonlinearity and asymmetry debates; and the role of monetary policy, from the 1970s to the 2021–22 inflation surge. Identification and forecasting are in Section 14.8.

James D. Hamilton, "Oil and the Macroeconomy since World War II"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 91(2), 228–248 (1983); DOI 10.1086/261140.

All but one postwar US recession was preceded, with a lag of about three quarters, by a sharp rise in crude oil prices. The correlation is statistically significant and not spurious even for 1948–72, suggesting oil shocks contributed to some recessions.

Version record

James D. Hamilton, "What Is an Oil Shock?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 113(2), 363–398 (2003); DOI 10.1016/s0304-4076(02)00207-5.

Flexible nonlinear estimates show oil price increases matter far more for GDP than decreases, and increases that only reverse earlier falls predict little; hence the "net oil price increase" measure. Instrumenting with exogenous supply disruptions suggests oil shocks work by disrupting spending on key sectors.

Version record

James D. Hamilton, "Causes and Consequences of the Oil Shock of 2007–08"

Difficulty: Research **Type:** Research paper

Status/version: Brookings Papers on Economic Activity, 2009(1), 215–261 (2009); DOI 10.1353/eca.0.0047.

Unlike earlier shocks caused by supply disruptions, the 2007–08 run-up came from strong demand meeting stagnant world production. The consequences were similar, through consumption and car purchases, and without them 2007Q4–2008Q3 would probably not have been a recession.

Version record

Mark A. Hooker, "What happened to the oil price-macroeconomy relationship?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 38(2), 195–213 (1996); DOI 10.1016/s0304-3932(96)01281-0.

Oil prices no longer Granger-cause many US macroeconomic indicators after 1973. Sample instability, endogenous oil prices and asymmetric or nonlinear specifications do not explain the breakdown; the OPEC price increases had significant effects, the price declines of the 1980s smaller and less clear ones.

Version record

James D. Hamilton, "This is what happened to the oil price-macroeconomy relationship"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 38(2), 215–220 (1996); DOI 10.1016/s0304-3932(96)01282-2.

Reply to Hooker: many quarterly oil-price increases after 1985 only reversed larger falls in the previous quarter. Measured as the net increase over the previous year, recent oil-price rises remain consistent with the historical correlation between oil shocks and recessions; the measure became known as the net oil price increase.

Version record

Ben S. Bernanke, Mark Gertler and Mark Watson, "Systematic Monetary Policy and the Effects of Oil Price Shocks"

Difficulty: Research **Type:** Research paper

Status/version: Brookings Papers on Economic Activity, 1997(1), 91–157 (1997); DOI 10.2307/2534702.

Adds term-structure assumptions to an identified VAR so the economy can be simulated under alternative systematic monetary policies, and applies the method mainly to oil-price shocks. Shutting off the Federal Reserve's response eliminates most of the output effect of an oil shock after the first eight to ten months, so an important part of the effect comes from the resulting monetary tightening rather than from oil prices themselves; with twelve lags instead of seven the result holds only at short horizons. Published with comments by Christopher Sims and Benjamin Friedman.

Version record

James D. Hamilton and Ana María Herrera, "Oil Shocks and Aggregate Macroeconomic Behavior: The Role of Monetary Policy: A Comment"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit, and Banking, 36(2), 265–286 (2004); DOI 10.1353/mcb.2004.0012.

Challenges Bernanke–Gertler–Watson on two grounds: the Federal Reserve probably could not have held the funds rate 900 basis points below its 1974 level, and their small oil-shock effects come from a shorter lag length than other studies use. With the longer lags, even the aggressive policy they propose would not have averted a downturn.

Version record

Olivier J. Blanchard and Jordi Galí, "The Macroeconomic Effects of Oil Price Shocks: Why Are the 2000s so Different from the 1970s?"

Difficulty: Research **Type:** Book chapter

Status/version: *International Dimensions of Monetary Policy*, University of Chicago Press (2010), 373–421; DOI 10.7208/chicago/9780226278872.003.0008.

Compares the stagflation that accompanied the oil shocks of the 1970s with the mild effects of similar price rises in the 2000s. VAR estimates attribute only part of the 1970s episodes to oil shocks, pointing to other shocks at the same time, and the authors weigh three reasons for the smaller recent effects: a lower oil share in production and consumption, more flexible labour markets and more credible monetary policy.

Version record

Paul Edelstein and Lutz Kilian, "How sensitive are consumer expenditures to retail energy prices?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 56(6), 766–779 (2009); DOI 10.1016/j.jmoneco.2009.06.001.

Quantifies how energy-price shocks reduce US consumer spending through discretionary income, precautionary saving and the operating cost of energy-using durables. Linear models fit the symmetric responses of 1979, when energy prices rose sharply, and 1986, when they fell; energy shocks have mattered for consumption growth but were not the dominant factor.

Version record

Lutz Kilian and Robert J. Vigfusson, "Are the responses of the U.S. economy asymmetric in energy price increases and decreases?"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Economics, 2(3), 419–453 (2011); DOI 10.3982/qe99.

Censored oil-price VARs suggest that GDP falls after oil-price increases but does not rise after decreases; these estimates are invalid. In a structural model that nests symmetric and asymmetric cases, correctly computed responses are about the same size in both directions, consistent with formal tests of symmetry.

Version record

Lutz Kilian and Logan T. Lewis, "Does the Fed Respond to Oil Price Shocks?"

Difficulty: Research **Type:** Research paper

Status/version: Economic Journal, 121(555), 1047–1072 (2011); DOI 10.1111/j.1468-0297.2011.02437.x.

Re-examines the view that the Federal Reserve's responses to oil-driven inflation caused large fluctuations. There is no credible evidence that monetary policy responses to oil shocks did so in the 1970s and 1980s or later; the authors argue for reaction functions that treat the oil price as endogenous and let the response depend on the cause of the shock.

Version record

Olivier Coibion and Yuriy Gorodnichenko, "Is the Phillips Curve Alive and Well after All? Inflation Expectations and the Missing Disinflation"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics, 7(1), 197–232 (2015); DOI 10.1257/mac.20130306.

Explains the missing disinflation of the Great Recession within a standard Phillips curve: if firms' inflation expectations track households', the rise in household expectations from 2009 to 2011 accounts for it, and that rise is explained by the increase in oil prices over the period.

Version record

Lutz Kilian and Xiaoqing Zhou, "Oil prices, gasoline prices, and inflation expectations"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 37(5), 867–881 (2022); DOI 10.1002/jae.2911.

Dynamic structural models quantify how gasoline-price shocks move households' one-year inflation expectations: on average they explain 42 per cent of the variation, and almost all of the rise from early 2009 to early 2013. They do not, however, explain why household expectations improved the fit of the Phillips curve in 2009–13.

Version record

Luca Gagliardone and Mark Gertler, "Oil Prices, Monetary Policy, and Inflation Surges"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics, 18(4), 468–501 (2026); DOI 10.1257/mac.20240295.

Quantitative New Keynesian model of the post-2021 rise and fall in inflation, estimated with identified oil and monetary shocks and then used to recover shocks without targeting inflation. Both oil-price shocks and accommodative monetary policy played important roles in the rise. Circulated as NBER Working Paper 31263 (2023).

Version record

Benjamin Wong, "Do Inflation Expectations Propagate the Inflationary Impact of Real Oil Price Shocks? Evidence from the Michigan Survey"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 47(8), 1673–1689 (2015); DOI 10.1111/jmcb.12288.

Michigan Survey inflation expectations respond to real oil-price shocks but play only a minimal role in passing them through to inflation, and after the 1990s possibly none.

Version record

John Elder and Apostolos Serletis, "Oil Price Uncertainty"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 42(6), 1137–1159 (2010); DOI 10.1111/j.1538-4616.2010.00323.x.

Real-options theory predicts that uncertainty about oil prices depresses investment. Oil-price volatility has a significant negative effect on several measures of investment, durables consumption and output; allowing for it deepens the negative response of activity to a negative oil-price shock and damps the response to a positive one.

Version record

14.10 Oil prices and asset markets

How oil shocks move stock prices, industry and firm returns and commodity-currency exchange rates; oil risk factors and oil-volatility risk in the cross-section of stocks; and oil–equity correlation at the zero lower bound.

Lutz Kilian and Cheolbeom Park, "The Impact of Oil Price Shocks on the U.S. Stock Market"

Difficulty: Research **Type:** Research paper

Status/version: International Economic Review, 50(4), 1267–1287 (2009); DOI 10.1111/j.1468-2354.2009.00568.x.

US real stock returns respond very differently to oil price changes driven by demand and by supply shocks. Oil-market shocks together explain 22 per cent of the long-run variation in returns, and industry responses point to transmission through lower domestic final demand.

Version record

Gerben Driesprong, Ben Jacobsen and Benjamin Maat, "Striking Oil: Another Puzzle?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 89(2), 307–327 (2008); DOI 10.1016/j.jfineco.2007.07.008.

Oil price changes predict stock returns in developed and emerging markets: a rise in oil prices lowers future returns. Time-varying risk premia do not explain this, since oil price changes predict negative excess returns; the predictability strengthens when lags are introduced, consistent with investors reacting slowly to oil price information.

Version record

Robert C. Ready, "Oil Prices and the Stock Market"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 22(1), 155–176 (2018); DOI 10.1093/rof/rfw071.

Classifies oil price changes as demand-driven, through returns on oil producers orthogonal to VIX changes, or supply-driven. Demand shocks are positively and supply shocks negatively correlated with market returns and output, with industry patterns that match consumer and energy-input exposure.

Version record

Charles M. Jones and Gautam Kaul, "Oil and the Stock Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 51(2), 463–491 (1996); DOI 10.1111/j.1540-6261.1996.tb02691.x.

Tests whether stock-market reactions to oil shocks are justified by changes in current and future real cash flows or in expected returns. In the postwar period, US and Canadian stock-price reactions are fully accounted for by the effect on real cash flows; in the United Kingdom and Japan, oil shocks move stock prices more than cash flows or expected returns can justify.

Version record

I-Hsuan Ethan Chiang, W. Keener Hughen and Jacob S. Sagi, "Estimating Oil Risk Factors Using Information from Equity and Derivatives Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 70(2), 769–804 (2015); DOI 10.1111/jofi.12222.

Estimates four latent oil risk factors with simple economic interpretations from oil derivatives prices and oil-related equity returns. The factors fit well in and out of sample, carry significant risk premia and relate to macroeconomic variables and to characteristic- and industry-sorted portfolios; the average non-oil portfolio has about a sixth of the oil industry's sensitivity.

Version record

Peter Christoffersen and Xuhui (Nick) Pan, "Oil volatility risk and expected stock returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 95, 5–26 (2018); DOI 10.1016/j.jbankfin.2017.07.004.

After the financialisation of commodity futures in 2004–05, oil volatility strongly predicts stock-market returns and volatility, and exposure to oil-volatility risk drives the cross-section: stocks with low exposure earn 0.66 per cent a month more than those with high exposure, a risk premium of minus 0.60 per cent a month. The effect is tied to measures of funding-liquidity constraints.

Version record

Deepa D. Datta, Benjamin K. Johannsen, Hannah Kwon and Robert J. Vigfusson, "Oil, Equities, and the Zero Lower Bound"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Macroeconomics, 13(2), 214–253 (2021); DOI 10.1257/mac.20180488.

Oil and equity returns were more positively correlated from late 2008 to 2014, and both became more responsive to macroeconomic news. The evidence points to the zero lower bound on nominal rates, as a model with the bound predicts, and is clear empirical evidence that the bound changed the economic environment.

Version record

Paresh Kumar Narayan and Susan Sunila Sharma, "New evidence on oil price and firm returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 35(12), 3253–3262 (2011); DOI 10.1016/j.jbankfin.2011.05.010.

Oil-price effects on the returns of 560 NYSE-listed US firms differ by sector, appear with lags, depend on the regime in five of 14 sectors and vary with firm size.

Version record

Domenico Ferraro, Kenneth Rogoff and Barbara Rossi, "Can oil prices forecast exchange rates? An empirical analysis of the relationship between commodity prices and exchange rates"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 54, 116–141 (2015); DOI 10.1016/j.jimonfin.2015.03.001.

At a daily frequency, changes in the price of a country's main commodity export are robustly related to changes in its nominal exchange rate when contemporaneous price changes are used. With lagged commodity-price changes the predictive ability is fleeting and appears mainly after allowing for instabilities.

Version record

14.11 Natural gas markets, hub pricing and market integration

Hub price formation after open access and liberalisation, regional and international market integration, the link between gas and oil prices, shale-gas supply and the 2022 European supply shock, and the EU rules phasing out Russian gas. Storage valuation is in Section 14.1 and LNG contracts in Section 14.15.

Peter R. Hartley, Kenneth B. Medlock III and Jennifer E. Rosthal, "The Relationship of Natural Gas to Oil Prices"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 29(3), 47–66 (2008); DOI 10.5547/issn0195-6574-ej-vol29-no3-3.

The long-run link between US natural gas and crude oil prices is indirect, running through competition between gas and residual fuel oil, and depends on technology. Short-run departures are driven by inventories, weather, seasonality and supply shocks such as hurricanes.

Version record

Stephen P. A. Brown and Mine K. Yücel, "What Drives Natural Gas Prices?"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 29(2), 45–60 (2008); DOI 10.5547/issn0195-6574-ej-vol29-no2-3.

As fuel-switching capacity declined, US gas prices moved more independently of crude. An error-correction model shows that once weather, inventories and other factors are included, crude oil prices still shape gas prices, implying a range of prices at which gas and oil products substitute.

Version record

Péter Erdős, "Have Oil and Gas Prices Got Separated?"

Difficulty: Research **Type:** Research paper

Status/version: Energy Policy, 49, 707–718 (2012); DOI 10.1016/j.enpol.2012.07.022.

Vector error-correction models show that US natural gas prices decoupled from oil around 2009, with the shale boom, while UK gas prices stayed linked to oil. Before then, US gas prices adjusted to oil and to UK prices through transatlantic LNG arbitrage.

Version record

Paul L Joskow, "Natural Gas: From Shortages to Abundance in the United States"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review, 103(3), 338–343 (2013); DOI 10.1257/aer.103.3.338.

Short history of US wellhead price regulation, pipeline regulation and their reform, which ended the shortages and pipeline inefficiencies of the 1970s and 1980s and left a market platform able to absorb the unexpected shale-gas supply boom. Read it for the institutional sequence behind today's hub prices; its 2013 outlook for shale gas is dated.

Version record

Arthur De Vany and W. David Walls, "Pipeline Access and Market Integration in the Natural Gas Industry: Evidence from Cointegration Tests"

Difficulty: Research **Type:** Research paper

Status/version: The Energy Journal, 14(4), 1–19 (1993); DOI 10.5547/issn0195-6574-ej-vol14-no4-1.

Tests daily spot prices for 190 market pairs at 20 US producing fields and pipeline interconnections. Most field markets were not cointegrated in 1987; by 1991 more than 65 per cent were, which the authors read as evidence that FERC's open-access pipeline policy made gas markets competitive. Cointegration shows that prices move together, not that arbitrage is costless; pair it with basis and pipeline-capacity data.

Version record

John T. Cuddington and Zhongmin Wang, "Assessing the Degree of Spot Market Integration for U.S. Natural Gas: Evidence from Daily Price Data"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Regulatory Economics, 29(2), 195–210 (2006); DOI 10.1007/s11149-006-6035-2.

Uses daily spot prices at 76 US locations in 1993–97, after FERC's open-access reforms, to map the geographic extent of the gas market and how fast prices return to equilibrium after shocks. The East and Central regions form one highly integrated market; the West is more loosely integrated and quite segmented from it.

Version record

Catherine Hausman and Ryan Kellogg, "Welfare and Distributional Implications of Shale Gas"

Difficulty: Research **Type:** Research paper

Status/version: Brookings Papers on Economic Activity, Spring 2015, 71–139 (2016); DOI 10.1353/eca.2016.0001.

Estimates US gas supply and demand elasticities, the price fall attributable to shale gas and its welfare effects: gains for residential, commercial, industrial and power-sector consumers, losses for producers, and a net gain of $48 billion a year over 2007–13. It also tests claims of a gas-led manufacturing renaissance and of pass-through to retail gas, electricity and chemical prices, and flags environmental costs that remain unpriced.

Version record

David J. Ramberg and John E. Parsons, "The Weak Tie Between Natural Gas and Oil Prices"

Difficulty: Research **Type:** Research paper

Status/version: The Energy Journal, 33(2), 13–36 (2012); DOI 10.5547/01956574.33.2.2.

Reconciles evidence that crude oil and US natural gas prices are cointegrated with their apparent decoupling in the late 2000s. Gas prices carry large unexplained short-horizon volatility and the cointegrating relation shifts over time, so confidence intervals around any oil-linked gas price are wide at both short and long horizons.

Version record

Christian Growitsch, Rabindra Nepal and Marcus Stronzik, "Price Convergence and Information Efficiency in German Natural Gas Markets"

Difficulty: Research **Type:** Research paper

Status/version: German Economic Review, 16(1), 87–103 (2015); DOI 10.1111/geer.12034.

Studies Germany's 2007 move to an entry–exit network-access system with cointegration tests and a state-space model with time-varying coefficients on spot prices at the two main German hubs and the interlinked Dutch market. Prices converged reasonably well, and information efficiency between Germany and the Netherlands increased considerably.

Version record

Daan Hulshof, Jan-Pieter van der Maat and Machiel Mulder, "Market fundamentals, competition and natural-gas prices"

Difficulty: Research **Type:** Research paper

Status/version: Energy Policy, 94, 480–491 (2016); DOI 10.1016/j.enpol.2015.12.016.

Explains the day-ahead price at the Dutch hub over 2011–14. The oil price had only a small positive effect and changes in supply concentration none; storage availability, temperature and wind-power output mattered. Hub prices were set mainly by gas-market fundamentals, gas-to-gas competition, even while much gas still flowed under oil-indexed long-term contracts.

Version record

Anne Neumann, "Linking Natural Gas Markets - Is LNG Doing its Job?"

Difficulty: Research **Type:** Research paper

Status/version: The Energy Journal, 30 (Special Issue: World Natural Gas Markets and Trade), 187–200 (2009); DOI 10.5547/issn0195-6574-ej-vol30-nosi-12.

Applies a Kalman-filter convergence test to 2,059 pairs of daily spot prices in North America and Europe over 1999–2008. Prices on the two sides of the Atlantic converged as LNG trade grew, supporting the claim that LNG links previously segmented regional markets. Part of the Energy Modeling Forum's study of world gas markets and trade.

Version record

Raymond Li, Roselyne Joyeux and Ronald D. Ripple, "International Natural Gas Market Integration"

Difficulty: Research **Type:** Research paper

Status/version: The Energy Journal, 35(4), 159–180 (2014); DOI 10.5547/01956574.35.4.7.

Multivariate cointegration analysis of North American, European and Asian gas prices from January 1997 to May 2011. Japanese, Korean, Taiwanese and UK prices converge, but the North American price behaves differently, so there is no fully integrated world gas market. The Europe–Asia link appears to come from oil-indexed contract formulas rather than supply and demand arbitrage.

Version record

Benjamin Moll, Moritz Schularick and Georg Zachmann, "The Power of Substitution: The Great German Gas Debate in Retrospect"

Difficulty: Research **Type:** Research paper

Status/version: Brookings Papers on Economic Activity, Fall 2023, 395–481 (2023); DOI 10.1353/eca.2023.a935431.

How Germany adjusted after Russia cut gas supplies in summer 2022. Germany replaced much Russian gas with imports from other countries and cut gas use by about 20 per cent (industry 26, households 17); the economy had only a mild one-quarter contraction, and industrial output fell little outside energy-intensive sectors. Supply shocks cost far less when substitution elasticities are low but non-zero than when they are zero.

Version record

Oliver Ruhnau, Clemens Stiewe, Jarusch Muessel and Lion Hirth, "Natural gas savings in Germany during the 2022 energy crisis"

Difficulty: Research **Type:** Research paper

Status/version: Nature Energy, 8(6), 621–628 (2023); DOI 10.1038/s41560-023-01260-5.

Estimates the crisis response of German gas consumers with a regression controlling for the non-linear temperature–heating relation, seasonality and trends. Industry began cutting consumption in September 2021, small consumers only from March 2022; across sectors, consumption in the second half of 2022 was 23 per cent below the temperature-adjusted baseline. A demand-side input for any analysis of 2022 gas prices or storage.

Version record

Richard G. Newell, Brian C. Prest and Ashley B. Vissing, "Trophy Hunting versus Manufacturing Energy: The Price Responsiveness of Shale Gas"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Association of Environmental and Resource Economists, 6(2), 391–431 (2019); DOI 10.1086/701531.

Separates the drilling, completion and production responses of US gas supply to prices. Drilling responds by about 0.9 per cent per 1 per cent price change for conventional and shale gas alike, while production from existing wells and completion times barely respond. Because shale wells produce about three times as much gas, supply was about three times more price-responsive in 2010–15 than in 2000–05.

Version record

Rüdiger Bachmann, David Baqaee, Christian Bayer, Moritz Kuhn, Andreas Löschel, Benjamin Moll, Andreas Peichl, Karen Pittel and Moritz Schularick, "What if? The macroeconomic and distributional effects for Germany of a stop of energy imports from Russia"

Difficulty: Research **Type:** Research paper

Status/version: Economica, 91(364), 1157–1200 (2024); DOI 10.1111/ecca.12546.

The ex ante estimate, first made in spring 2022, of what a stop of Russian energy imports would cost Germany: a short-run output loss of 0.5–3 per cent of GDP, substantial but manageable because of substitution and reallocation along the production chain. Read it with Moll, Schularick and Zachmann (2023), which tests the estimate against the outcome.

Version record

Frank Asche, Petter Osmundsen and Maria Sandsmark, "The UK Market for Natural Gas, Oil and Electricity: Are the Prices Decoupled?"

Difficulty: Research **Type:** Research paper

Status/version: The Energy Journal, 27(2), 27–40 (2006); DOI 10.5547/issn0195-6574-ej-vol27-no2-2.

Uses the UK gas market of 1995–98, liberalised but not yet linked to the Continent by the Interconnector, as a natural experiment. Monthly gas, oil and electricity prices were highly integrated, suggesting that wholesale demand was for energy rather than a specific fuel, so gas had not decoupled from oil.

Version record

Gary W. Emery and Qingfeng (Wilson) Liu, "An analysis of the relationship between electricity and natural-gas futures prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 22(2), 95–122 (2002); DOI 10.1002/fut.2209.

NYMEX California–Oregon Border and Palo Verde electricity futures were cointegrated with natural-gas futures, with coefficients reflecting regional differences in electricity consumption. Trading rules based on the mean-reverting spread were profitable in and out of sample: an early empirical study of the futures spark spread.

Version record

European Commission, Directorate-General for Energy, REPowerEU gas regulation: Regulation (EU) 2026/261 phasing out Russian natural gas imports; Legislative Observatory summary, 2025/0180(COD)

Difficulty: Advanced **Type:** Report / research note

Status/version: Regulation (EU) 2026/261, published in the Official Journal on 2 February 2026 and in force the next day; Council adoption 26 January 2026.

Bans imports of Russian pipeline gas and LNG into the EU in steps. New contracts are banned six weeks after entry into force; contracts concluded before 17 June 2025 have a transition period, and the full ban applies to LNG from the start of 2027 and to pipeline gas from 30 September 2027, or 1 November 2027 for a member state whose storage-filling target is at risk. Importers must prove the country of production, gas entering from Russia or Belarus is presumed Russian, and maximum penalties for companies must reach at least 3.5 per cent of turnover, EUR 40 million or 300 per cent of the transaction value at TTF day-ahead prices. Member states had to file diversification plans by 1 March 2026.

Version record

14.12 Power contracts and storage services

Subject context and qualifications.

National Renewable Energy Laboratory, Annual Technology Baseline: Financial Cases and Methods

Difficulty: Advanced **Type:** Data / specification

Status/version: 2024b assumptions page, explicitly dated to its release; agency methodology.

Separates a fixed-price, take-or-pay renewable PPA and its debt/tax-equity financing from a quasi-merchant gas plant. Use its debt-service coverage and post-contract value assumptions to build a PPA cash-flow skeleton before valuing shape and curtailment. Follow power-forward pricing. Its representative US financing assumptions and June 2024 tax-law snapshot are scenarios, not executable deal terms. Reconcile actual contracted MWh, delivery node, basis, negative-price clauses, curtailment and counterparty credit.

Version record

Julio Pombo-Romero, Oliver Rúas-Barrosa and Carlos Vázquez, Assessing the Value and Risk of Renewable PPAs

Difficulty: Research **Type:** Research paper

Status/version: *Energy Economics* 139, 107861, November 2024.

Models the off-taker's option to default and the producer's expected loss, connecting a long-dated fixed-price PPA to collateral and guarantee sizing. It is a useful second step after building a joint generation–price payoff, because an attractive mark can raise counterparty credit exposure. It does not establish a universal guarantee haircut.

Extended note. Version record

Edoardo Del Bianco, Cannibalization Effect and Volumetric Risk in Power Purchase Agreements

Difficulty: Research **Type:** Research thesis

Status/version: Politecnico di Milano 2024/25 thesis; university repository abstract.

Models joint wind output and power prices for Danish and Spanish markets and tests the effect of hourly settlement on wind-PPA value. This explicitly addresses covariance between generated volume and the price received, which a flat block-price hedge misses. Read after the forward-premium and PPA entries. Replicate with timestamped weather forecasts, market prices, delivery shape and curtailment before use in another bidding zone.

Extended note. Version record

Guannan He, Soummya Kar, Javad Mohammadi, Panayiotis Moutis and Jay F. Whitacre, Power System Dispatch with Marginal Degradation Cost of Battery Storage

Difficulty: Research **Type:** Research paper

Status/version: *IEEE Transactions on Power Systems* 36(4), 3552–3562, July 2021; journal version. Public author preprint v3, 9 June 2020.

Derives a time- and health-dependent marginal cost of cycling for dispatch, linking short-run arbitrage choices to lifetime value. Use it to put wear into a battery bid or scenario optimiser, alongside state-of-charge and round-trip efficiency. Start with the storage-option model. The preprint's grid case studies do not identify a universal degradation tariff: cell chemistry, calendar aging, warranty throughput, replacement price and merchant revenues need project calibration.

Version record

Volkan Kumtepeli, Holger Hesse, Thomas Morstyn, Seyyed Mostafa Nosratabadi, Marko Aunedi and David A. Howey, Depreciation Cost is a Poor Proxy for Revenue Lost to Aging in Grid Storage Optimization

Difficulty: Research **Type:** Research paper

Status/version: ACC 2024 proceedings, pp. 701–706; accepted manuscript dated 15 March 2024; published conference version.

Shows that even an accurate physical aging model can make poor dispatch choices when its depreciation penalty misprices foregone future revenue. Use it to compare lifetime simulation with a rolling-horizon cycling penalty and to stress the shadow value of remaining capacity. Read after marginal degradation. The paper studies arbitrage and assumes usable price forecasts; its case-study profit differences should not be extrapolated to capacity or ancillary-service contracts without service-specific opportunity costs.

Version record

Murali Baggu, Adarsh Nagarajan, Dylan Cutler, Dan Olis, Thomas Bialek and Martha Symko-Davies, Coordinated Optimization of Multiservice Dispatch for Energy Storage Systems with Degradation Model for Utility Applications; author manuscript

Difficulty: Research **Type:** Research paper

Status/version: *IEEE Transactions on Sustainable Energy* 10(2), 886–894, April 2019; online 20 July 2018, DOI 10.1109/TSTE.2018.2853673.

Co-optimises distribution support and market participation while tracking cycling wear in a utility battery case. Its core lesson is an implementable resource budget: voltage support, energy dispatch and reserve availability compete for the same power, energy and cell life. Read after marginal wear cost. A trading model must add the local ISO's ancillary qualification, telemetry, dispatch performance and payment rules, plus actual feeder limits.

Extended note. Version record

Federal Energy Regulatory Commission, Energy Markets; Ancillary Services

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official living explainers, consulted 4 October 2026.

Defines day-ahead versus real-time settlement, locational marginal price components and reserve/regulation services. Use the component split to model nodal congestion and losses separately from system energy; reserve capacity and dispatch calls compete with a battery's energy arbitrage. Match the actual ISO's registration, co-optimisation, telemetry, performance penalties, charge rules and settlement calendar; do not add independent revenue streams that spend the same MWh or headroom twice.

Extended note. Version record

PJM Interconnection, Manual 06: Financial Transmission Rights

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: PJM Manual 06, revision 35, effective 24 September 2026; operative market manual at the review date.

Defines FTR obligations and options as financial claims on day-ahead nodal congestion-price differences, not physical transmission rights. Sections 6–9 cover long-term, annual and monthly auctions, the secondary market, credit rules, target allocations, congestion-credit settlement and simultaneous feasibility testing. After nodal-price foundations, calculate an hourly source-to-sink obligation payoff and option payoff, then reconcile auction price with realised congestion credits and any funding adjustment. Preserve product, path, MW, on/off-peak period and rule revision. The manual explains PJM procedures; tariff provisions, current credit requirements and FTR Center data control a live implementation.

Extended note. Version record

Monitoring Analytics, 2026 Quarterly State of the Market Report for PJM: Section 13, Financial Transmission and Auction Revenue Rights

Difficulty: Advanced **Type:** Report / research note

Status/version: Independent market monitor report for January–June 2026, posted 13 August 2026; observed outcomes and monitor recommendations, not PJM rules.

Supplies auction, ownership, congestion-revenue, target-allocation and FTR-profit data for a dated PJM market period. Section 13 reports a 2026/2027 annual auction with USD 3,755.8 million of net revenue and analyses the prior planning period's funding and settlement. Use its tables to compare auction-implied congestion value with realised day-ahead congestion and holder payments after reading Manual 06. The monitor also argues for changes to ARR/FTR design; distinguish those recommendations from current settlement rules. Results mix auction periods and planning-period outcomes, so retain denominators, dates, self-scheduled positions and revisions before inferring a tradable return.

Extended note. Version record

PJM Interconnection, Data Miner and public Data Directory; Data Directory

Difficulty: Advanced **Type:** Data / specification

Status/version: Official public-data and directory descriptions inspected 4 October 2026; feed and extraction vintages are separate.

Connects the transmission-rights manual to public nodal-price and auction data. The directory identifies day-ahead locational marginal prices and FTR auction bids; retain node definitions, time zone, daylight-saving treatment, congestion components and revisions when joining them. Distinguish target congestion credits, actual settlement funding and collateral from a simple difference in total energy prices. Automated queries require the appropriate account/API access, and redistribution terms must be respected. The catalogue identifies the data route; no historical feed download, auction reconstruction or settlement replication was executed in this expansion.

Version record

Shi-Jie Deng and Zhendong Xia, A Real Options Approach for Pricing Electricity Tolling Agreements; related author paper

Difficulty: Research **Type:** Contract / specification / guidance

Status/version: *International Journal of Information Technology & Decision Making* 5(3), 421–436, September 2006; publisher DOI and author prepublication research.

Values a buyer's right to call generation under plant operating and contractual limits using dynamic programming and simulation. It turns a spark-spread option into a tolling contract with exercisable output, a useful bridge from power/gas curves to project cash flow. Read after power forwards. Model heat rate, starts, minimum run, outages, emissions, capacity payments and fuel transport from the actual plant and agreement.

Extended note. Version record

M. Angeles Carnero, Stein-Erik Fleten, Ståle Størdal and Sjur Westgaard, "Ex ante and ex post risk premiums in electricity futures"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 25(11), 1717–1729 (2025); DOI 10.1080/14697688.2025.2551768.

Ex-ante and ex-post risk premiums in Nordic electricity futures using commercial forecasts (2013–2024), varying with scarcity and season.

Version record

Roxana Dumitrescu, Redouane Silvente and Peter Tankov, "Price Impact and Long-Term Profitability of Energy Storage"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 17(2), 281–322 (2026); DOI 10.1137/25m1752638.

Price impact of electricity storage and its long-run profitability in energy-transition scenarios, with equilibrium prices solved as coupled FBSDEs.

Version record

William W. Hogan, "Contract networks for electric power transmission"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Regulatory Economics, 4(3), 211–242 (1992); DOI 10.1007/bf00133621.

The original design of financial transmission rights as hedges against nodal congestion charges.

Version record

Akshaya Jha and Frank A. Wolak, "Can Forward Commodity Markets Improve Spot Market Performance? Evidence from Wholesale Electricity"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Economic Policy, 15(2), 292–330 (2023); DOI 10.1257/pol.20200234.

Introducing convergence (virtual) bidding in California shrank day-ahead/real-time price spreads and lowered costs.

Version record

Ignacia Mercadal, "Dynamic Competition and Arbitrage in Electricity Markets: The Role of Financial Players"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Microeconomics, 14(3), 665–699 (2022); DOI 10.1257/mic.20190276.

Entry of financial traders in MISO's virtual market reduced day-ahead/real-time premia and affected generator market power.

Version record

John R. Birge, Ali Hortaçsu, Ignacia Mercadal and J. Michael Pavlin, "Limits to arbitrage in electricity markets: A case study of MISO"

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Economics, 75, 518–533 (2018); DOI 10.1016/j.eneco.2018.08.024.

Documents persistent DA–RT spreads at MISO nodes and links them to FTR positions and arbitrage limits.

Version record

Ali Hortaçsu and Steven L. Puller, "Understanding strategic bidding in multi-unit auctions: a case study of the Texas electricity spot market"

Difficulty: Research **Type:** Research paper

Status/version: The RAND Journal of Economics, 39(1), 86–114 (2008); DOI 10.1111/j.0741-6261.2008.00005.x.

Large ERCOT firms bid close to optimal; smaller firms deviate, causing productive inefficiency.

Version record

Yumi Oum, Shmuel Oren and Shijie Deng, "Hedging quantity risks with standard power options in a competitive wholesale electricity market"

Difficulty: Research **Type:** Research paper

Status/version: Naval Research Logistics (NRL), 53(7), 697–712 (2006); DOI 10.1002/nav.20184.

Derives the optimal zero-cost hedge for a load-serving entity that sells at a fixed regulated price but faces correlated price and volume risk, and replicates its payoff with forwards and a portfolio of calls and puts. The volumetric-risk problem behind retail supply contracts and shaped PPAs.

Version record

14.13 Electricity market design, renewables and market power

Scarcity pricing, capacity mechanisms and resource adequacy; market power and the effect of forward contracts and vertical arrangements; reliability events and market-design reforms; and the market value of wind and solar, merit-order and cannibalisation effects and market design for high-renewable systems. Power contracts, storage services and transmission rights are in Section 14.12.

Paul L. Joskow, "Capacity Payments in Imperfect Electricity Markets: Need and Design"

Difficulty: Research **Type:** Research paper

Status/version: Utilities Policy, 16(3), 159–170 (2008); DOI 10.1016/j.jup.2007.10.003.

Imperfections in energy-only wholesale markets leave generators with net revenues too low to support the least-cost capacity mix and the reliability consumers want ("missing money"). Better scarcity pricing helps but is unlikely to close the gap, so forward capacity obligations with auctions are needed; the paper also diagnoses the flaws of early US capacity payments.

Version record

Paul Joskow and Jean Tirole, "Reliability and Competitive Electricity Markets"

Difficulty: Research **Type:** Research paper

Status/version: RAND Journal of Economics, 38(1), 60–84 (2007); DOI 10.1111/j.1756-2171.2007.tb00044.x.

Derives optimal prices and investment when price-insensitive retail consumers are served by load-serving entities that can ration according to real-time prices, then examines what competitive markets need to reach that optimum. Analyses wholesale price caps, capacity obligations and operating-reserve requirements against network collapse.

Version record

William W. Hogan, "Electricity Scarcity Pricing Through Operating Reserves"

Difficulty: Advanced **Type:** Research paper

Status/version: Economics of Energy & Environmental Policy, 2(2) (2013); DOI 10.5547/2160-5890.2.2.4.

Suppressed real-time prices weaken incentives for generation investment and demand bidding. An operating-reserve demand curve built from first principles would improve reliability and scarcity pricing within economic dispatch, and support long-term resource adequacy.

Version record

Peter Cramton, Axel Ockenfels and Steven Stoft, "Capacity Market Fundamentals"

Difficulty: Advanced **Type:** Research paper

Status/version: Economics of Energy & Environmental Policy, 2(2) (2013); DOI 10.5547/2160-5890.2.2.2.

Explains why inflexible, price-insensitive demand causes blackouts in scarcity, fails to set prices that attract efficient capacity and enables market power, and why capacity markets alongside energy markets can secure adequacy. Addresses practical design challenges and misunderstandings.

Version record

Peter Cramton, "Electricity Market Design"

Difficulty: Advanced **Type:** Research paper

Status/version: Oxford Review of Economic Policy, 33(4), 589–612 (2017); DOI 10.1093/oxrep/grx041.

How the best market designs meet short-run efficiency (best use of existing resources) and long-run efficiency (efficient investment): day-ahead and real-time markets with security-constrained dispatch and locational marginal prices, and forward contracting for risk management and bidding incentives, adapted to intermittent renewables and storage.

Version record

Severin Borenstein, James B. Bushnell and Frank A. Wolak, "Measuring Market Inefficiencies in California's Restructured Wholesale Electricity Market"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 92(5), 1376–1405 (2002); DOI 10.1257/000282802762024557.

Decomposes wholesale payments into production costs, competitive rents and market power for June 1998 to October 2000. Summer 2000 spending rose to $8.98 billion from $2.04 billion a year earlier; 21 per cent of the rise was production costs, 20 per cent competitive rents and 59 per cent market power.

Version record

Catherine D. Wolfram, "Measuring Duopoly Power in the British Electricity Spot Market"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 89(4), 805–826 (1999); DOI 10.1257/aer.89.4.805.

Price–cost markups in the British pool, from marginal-cost data and methods that need none, are positive but far below what oligopoly models predict for two dominant suppliers facing inelastic demand. Regulatory constraints, entry threats and financial contracts may explain the restraint.

Version record

James B. Bushnell, Erin T. Mansur and Celeste Saravia, "Vertical Arrangements, Market Structure, and Competition: An Analysis of Restructured US Electricity Markets"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 98(1), 237–266 (2008); DOI 10.1257/aer.98.1.237.

Wholesalers that are vertically integrated or hold long-term contracts have less incentive to raise prices when retail prices are fixed beforehand. Simulations for three markets show that without vertical arrangements, as in California, prices and production inefficiencies would have been much higher.

Version record

Blaise Allaz and Jean-Luc Vila, "Cournot Competition, Forward Markets and Efficiency"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Economic Theory, 59(1), 1–16 (1993); DOI 10.1006/jeth.1993.1001.

When Cournot producers can sell forward before the spot market, each has an incentive to sell forward to commit to higher output, which lowers spot prices and moves the outcome toward competition even without uncertainty. The standard argument that forward contracting mitigates market power in electricity.

Version record

Frank A. Wolak, "An Empirical Analysis of the Impact of Hedge Contracts on Bidding Behavior in a Competitive Electricity Market"

Difficulty: Research **Type:** Research paper

Status/version: International Economic Journal, 14(2), 1–39 (2000); DOI 10.1080/10168730000000017.

Model of expected-profit-maximising bidding that accounts for a generator's position in financial hedge contracts, applied to bid and contract data from the first months of Australia's National Electricity Market. Bidding is highly sensitive to contract cover, and hedge contracts are effective in mitigating market power.

Version record

Natalia Fabra, "A Primer on Capacity Mechanisms"

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Economics, 75, 323–335 (2018); DOI 10.1016/j.eneco.2018.08.003.

Simple model of investment and pricing incentives in electricity markets with market power. Unlike the competitive energy-only paradigm, scarcity prices then fail to induce efficient investment even for risk-neutral investors. Price caps combined with capacity payments separate the investment and market-power problems, and tying capacity payments to financial obligations (reliability options) curbs market power further if strike prices are near marginal cost.

Version record

Paul L. Joskow, "Challenges for Wholesale Electricity Markets with Intermittent Renewable Generation at Scale: The US Experience"

Difficulty: Advanced **Type:** Research paper

Status/version: Oxford Review of Economic Policy, 35(2), 291–331 (2019); DOI 10.1093/oxrep/grz001.

Uses CAISO to show how zero-marginal-cost wind and solar change supply patterns, variability and energy prices. Reforms to capacity markets and scarcity pricing are needed to keep traditional designs workable; more planning, technology-specific subsidies and mandated contracts would require deeper institutional change.

Version record

Chiara Lo Prete, Karen Palmer and Molly Robertson, "Time for a Market Upgrade? A Review of Wholesale Electricity Market Designs for the Future"

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Economics, 148, article 108640 (2025); DOI 10.1016/j.eneco.2025.108640.

Recent US scarcity events stemmed from a lack of available energy at critical times, not of capacity at peak. Reviews 11 proposed market designs against 10 criteria covering reliability, long-run and short-run efficiency, affordability and clean-energy goals, and identifies open questions such as load-serving entities' purchase obligations.

Version record

Frank Sensfuß, Mario Ragwitz and Massimo Genoese, "The Merit-Order Effect: A Detailed Analysis of the Price Effect of Renewable Electricity Generation on Spot Market Prices in Germany"

Difficulty: Research **Type:** Research paper

Status/version: Energy Policy, 36(8), 3086–3094 (2008); DOI 10.1016/j.enpol.2008.03.035.

Agent-based simulation of the German market shows that supported renewable generation lowers spot prices substantially. In 2006 the value of this merit-order effect exceeded the net support payments consumers made, a short-run transfer from generators to buyers.

Version record

Lion Hirth, "The Market Value of Variable Renewables"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 38, 218–236 (2013); DOI 10.1016/j.eneco.2013.02.004.

Because wind output shifts the merit order, its relative price falls as penetration rises. From a literature review, market regressions and a calibrated European model, wind's value falls from 110 per cent of the average price to 50–75 per cent at 30 per cent penetration, and solar reaches similar levels at about 15 per cent.

Version record

Paul L. Joskow, "Comparing the Costs of Intermittent and Dispatchable Electricity Generating Technologies"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review, 101(3), 238–241 (2011); DOI 10.1257/aer.101.3.238.

Levelised cost per MWh is the wrong metric for comparing intermittent wind and solar with dispatchable plant: it ignores when output is produced and what electricity is worth then, overvaluing intermittent technologies and probably wind relative to solar.

Version record

Gautam Gowrisankaran, Stanley S. Reynolds and Mario Samano, "Intermittency and the Value of Renewable Energy"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 124(4), 1187–1234 (2016); DOI 10.1086/686733.

Model of system-operator decisions estimated for southeastern Arizona. At 20 per cent solar, social cost is about $138 per MWh excluding carbon, of which intermittency accounts for $46 and unforecastable intermittency $6; with stated cost and carbon assumptions, 20 per cent solar would be welfare-neutral.

Version record

Joseph Cullen, "Measuring the Environmental Benefits of Wind-Generated Electricity"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Economic Policy, 5(4), 107–133 (2013); DOI 10.1257/pol.5.4.107.

Uses the randomness of wind output in Texas to estimate the emissions it offsets, accounting for dynamic production costs. Only at high social costs of pollution do the emissions offsets exceed the value of renewable subsidies.

Version record

James Bushnell and Kevin Novan, "Setting with the Sun: The Impacts of Renewable Energy on Conventional Generation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Association of Environmental and Resource Economists, 8(4), 759–796 (2021); DOI 10.1086/713249.

California's utility-scale solar build-out lowered average wholesale prices, but by cutting midday prices sharply while raising shoulder-hour prices. This favours flexible conventional generation and undermines baseload economics.

Version record

David Newbery, Michael G. Pollitt, Robert A. Ritz and Wadim Strielkowski, "Market Design for a High-Renewables European Electricity System"

Difficulty: Advanced **Type:** Research paper

Status/version: Renewable and Sustainable Energy Reviews, 91, 695–707 (2018); DOI 10.1016/j.rser.2018.04.025.

Policy recommendations for a European system dominated by intermittent renewables: the market failures to address, the evolution of EU support mechanisms and market impacts so far, and principles for interconnection, storage, support design, network tariffs, flexibility pricing, long-term contracts and risk management.

Version record

Natalia Fabra and Imelda, "Market Power and Price Exposure: Learning from Changes in Renewable Energy Regulation"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Economic Policy, 15(4), 323–358 (2023); DOI 10.1257/pol.20210221.

Whether renewables depress prices depends on their exposure to market prices: fixed-price payment curbs market power more when dominant firms own large renewable shares. Short-lived Spanish regulatory changes show that moving from full price exposure to fixed prices cut markups by 2–4 per cent.

Version record

Javier López Prol, Karl W. Steininger and David Zilberman, "The Cannibalization Effect of Wind and Solar in the California Wholesale Electricity Market"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 85, article 104552 (2020); DOI 10.1016/j.eneco.2019.104552.

Daily unit revenues and value factors of solar and wind in CAISO day-ahead data for 2013–17 fall with their own penetration. Wind penetration lowers solar's value factor while solar raises wind's at high penetration, and cannibalisation is stronger when demand is low.

Version record

Janina C. Ketterer, "The Impact of Wind Power Generation on the Electricity Price in Germany"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 44, 270–280 (2014); DOI 10.1016/j.eneco.2014.04.003.

A GARCH model of German wholesale prices shows that wind generation lowers the price level but raises its volatility. Volatility fell after a change in the marketing mechanism for renewable electricity, suggesting that regulation can ease the integration of renewables.

Version record

Federal Energy Regulatory Commission, North American Electric Reliability Corporation and Regional Entity staff, The February 2021 Cold Weather Outages in Texas and the South Central United States

Difficulty: Advanced **Type:** Report / research note

Status/version: FERC–NERC–Regional Entity staff report, November 2021.

Joint inquiry into the cold-weather event of 8–20 February 2021. 1,045 generating units with 192,818 MW of capacity had 4,124 outages, derates or failures to start; ERCOT averaged 34,000 MW unavailable for over two days and shed up to 20,000 MW of firm load, part of the largest controlled load shed in US history. Freezing (44.2 per cent) and fuel problems (31.4 per cent, mostly natural-gas supply) caused three-quarters of outages; the report recommends mandatory winterisation standards.

Version record

European Commission, Directorate-General for Energy, Electricity market design: the 2024 reform (Directive (EU) 2024/1711 and Regulation (EU) 2024/1747)

Difficulty: Advanced **Type:** Report / research note

Status/version: European Commission policy page; amending directive and regulation in force since 16 July 2024.

Commission summary of the reform of EU electricity market rules proposed in March 2023 after the 2022 energy crisis and in force since 16 July 2024. It aims to make bills less dependent on short-term prices through long-term contracts such as power purchase agreements,

to structure public support for new generation as two-way contracts for difference, and to strengthen REMIT enforcement, with a larger role for ACER in cross-border cases and an LNG price assessment and benchmark.

Version record

Frank A. Wolak, "Long-Term Resource Adequacy in Wholesale Electricity Markets with Significant Intermittent Renewables"

Difficulty: Research **Type:** Research paper

Status/version: Environmental and Energy Policy and the Economy, 3, 155–220 (2022); DOI 10.1086/717221.

Analyses the August 2020 California and February 2021 Texas supply shortfalls and argues that, with large intermittent renewable shares, similar shortfalls will recur unless long-term resource adequacy is determined differently. Outlines an approach that explicitly recognises the characteristics of different generation technologies and compares its properties with current mechanisms.

Version record

Mar Reguant, "Complementary Bidding Mechanisms and Startup Costs in Electricity Markets"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Economic Studies, 81(4), 1708–1742 (2014); DOI 10.1093/restud/rdu022.

Extends multi-unit auction estimation to bids that express cost complementarities over time. Start-up costs are substantial, limit how fast firms change output and amplify price fluctuations; static market-power estimates that ignore them overstate markup volatility and can even imply negative markups when demand is low.

Version record

Duncan S. Callaway, Meredith Fowlie and Gavin McCormick, "Location, Location, Location: The Variable Value of Renewable Energy and Demand-Side Efficiency Resources"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Association of Environmental and Resource Economists, 5(1), 39–75 (2018); DOI 10.1086/694179.

Values wind, solar and energy-efficiency resources across US regions by emissions displaced, operating costs avoided and capacity value, using hourly emissions and generation data, simulated output at thousands of sites, wholesale prices and technology costs. Displaced emissions vary widely across regions but little across resources within a region, so the net cost per tonne of CO2 avoided varies substantially, which prevailing incentives ignore.

Version record

Lion Hirth, Falko Ueckerdt and Ottmar Edenhofer, "Integration costs revisited – An economic framework for wind and solar variability"

Difficulty: Research **Type:** Research paper

Status/version: Renewable Energy, 74, 925–939 (2015); DOI 10.1016/j.renene.2014.08.065.

Defines the integration costs of wind and solar through their marginal economic value and splits them into components for temporal variability, forecast uncertainty and location. A review of over 100 studies puts them at EUR 25–35/MWh at a 30–40 per cent wind share, up to half of generation costs; reduced utilisation of thermal capital is the largest component.

Version record

Harrison Fell, Daniel T. Kaffine and Kevin Novan, "Emissions, Transmission, and the Environmental Value of Renewable Energy"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Economic Policy, 13(2), 241–272 (2021); DOI 10.1257/pol.20190258.

With hourly ERCOT and MISO data, relaxing transmission constraints between windy regions and demand centres conservatively raises the non-market value of wind by 30 per cent in Texas and 17 per cent in the midcontinent, mostly because wind then displaces fossil plants near populated areas.

Version record

14.14 Short-term power markets, balancing and flexibility

Sequential day-ahead, intraday and balancing markets; intraday price formation, forecasting and optimal trading; imbalance pricing; negative prices; and the arbitrage value and equilibrium effects of storage, including Europe's 2025 move to 15-minute day-ahead products and ERCOT's real-time co-optimisation. Transmission rights and storage services are in Section 14.12.

Koichiro Ito and Mar Reguant, "Sequential Markets, Market Power, and Arbitrage"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 106(7), 1921–1957 (2016); DOI 10.1257/aer.20141529.

Models strategic behaviour in sequential electricity markets when firms have market power and entry into arbitrage is restricted; the model predicts a systematic price premium between the markets. Iberian market microdata support the predictions, and a structural model shows that with market power full arbitrage need not raise welfare: it lowers consumer costs but increases deadweight loss.

Version record

Lion Hirth and Inka Ziegenhagen, "Balancing power and variable renewables: Three links"

Difficulty: Advanced **Type:** Research paper

Status/version: Renewable and Sustainable Energy Reviews, 50, 1035–1051 (2015); DOI 10.1016/j.rser.2015.04.180.

Reviews three links between variable renewables and balancing: forecast errors raise reserve needs, renewables can supply balancing services, and imbalance charges reward better forecasts. German data show a paradox: wind and solar capacity tripled after 2008 while balancing reserves fell 15 per cent and their cost 50 per cent.

Version record

Christopher Koch and Lion Hirth, "Short-term electricity trading for system balancing: An empirical analysis of the role of intraday trading in balancing Germany's electricity system"

Difficulty: Research **Type:** Research paper

Status/version: Renewable and Sustainable Energy Reviews, 113, 109275 (2019); DOI 10.1016/j.rser.2019.109275.

Updates the German balancing paradox: since 2011 wind and solar nearly doubled while reserve requirements and reserve use fell by half. Better intraday trading, now around the clock and in quarter-hour products, explains part of it; the shift to quarter-hourly products alone cut balancing energy by 17 per cent, and parties respond efficiently to imbalance charges.

Version record

Anselm Eicke, Oliver Ruhnau and Lion Hirth, "Electricity balancing as a market equilibrium: An instrument-based estimation of supply and demand for imbalance energy"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 102, 105455 (2021); DOI 10.1016/j.eneco.2021.105455.

Treats balancing as a market in which the imbalance price and system imbalance are set by supply (reserve activation) and demand (deliberate or accidental schedule deviations). With instruments from a theoretical model, German data show demand for imbalance energy falling by 2.2 MW per EUR 1/MWh rise in the imbalance price, although strategic deviations are prohibited, and imbalance-price shocks feed into later intraday prices.

Version record

Rüdiger Kiesel and Florentina Paraschiv, "Econometric analysis of 15-minute intraday electricity prices"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 64, 77–90 (2017); DOI 10.1016/j.eneco.2017.03.002.

Reduced-form econometric analysis of 15-minute German intraday prices with intraday-updated wind and solar forecasts. Prices respond asymmetrically to renewable forecast errors and trade volumes, with a threshold given by the share of expected demand covered by conventional capacity planned in the day-ahead market.

Version record

Shadi Goodarzi, H. Niles Perera and Derek Bunn, "The impact of renewable energy forecast errors on imbalance volumes and electricity spot prices"

Difficulty: Research **Type:** Research paper

Status/version: Energy Policy, 134, 110827 (2019); DOI 10.1016/j.enpol.2019.06.035.

Using quarter-hourly German data and least-squares, quantile and ARMA models, larger wind and solar forecast errors increase absolute imbalance volumes, which can pass into higher spot prices; wind errors matter more than solar errors. Supports incentives for system operators to improve and publish renewable forecasts.

Version record

Michał Narajewski and Florian Ziel, "Econometric modelling and forecasting of intraday electricity prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Commodity Markets, 19, 100107 (2020); DOI 10.1016/j.jcomm.2019.100107.

Models the ID3 price of the German continuous intraday market for hourly and quarter-hourly products with lasso and elastic-net estimation, and runs an out-of-sample very short-term forecasting study against benchmarks. The results bear on the efficiency of the continuous intraday market.

Version record

Christopher Kath and Florian Ziel, "The value of forecasts: Quantifying the economic gains of accurate quarter-hourly electricity price forecasts"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 76, 411–423 (2018); DOI 10.1016/j.eneco.2018.10.005.

Builds multivariate elastic-net forecasts for German quarter-hourly intraday continuous and intraday auction prices, using early day-ahead EXAA prices as inputs, and evaluates trading strategies built on them. Even simple strategies earn substantial value when combined with a decent forecast: score forecasts by their trading outcome as well as their error.

Version record

René Aïd, Pierre Gruet and Huyên Pham, "An optimal trading problem in intraday electricity markets"

Difficulty: Research **Type:** Research paper

Status/version: Mathematics and Financial Economics, 10(1), 49–85 (2016); DOI 10.1007/s11579-015-0150-8.

Solves a power producer's optimal intraday trading problem: minimise the imbalance cost from random residual demand (load minus renewable output) under linear price impact, with thermal generation as a second lever. Explicit approximate strategies are extended to jumps in demand forecasts and prices from wind-forecast errors and to delays in thermal production.

Version record

Enzo Fanone, Andrea Gamba and Marcel Prokopczuk, "The case of negative day-ahead electricity prices"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 35, 22–34 (2013); DOI 10.1016/j.eneco.2011.12.006.

Proposes and estimates a non-Gaussian process for hourly German day-ahead prices that generates extreme positive and negative spikes, motivated by the growth of renewables under the Renewable Energy Act, which made negative prices more likely.

Version record

Joachim Seel, Dev Millstein, Andrew Mills, Mark Bolinger and Ryan Wiser, "Plentiful electricity turns wholesale prices negative"

Difficulty: Research **Type:** Research paper

Status/version: Advances in Applied Energy, 4, 100073 (2021); DOI 10.1016/j.adapen.2021.100073.

In 2020 US average wholesale prices fell to $21/MWh and real-time hourly prices were negative in about 4 per cent of hours and nodes, but in over 25 per cent in clusters such as the Permian Basin in west Texas and Kansas and western Oklahoma in SPP. Negative prices come from local congestion or system-wide oversupply; in SPP all generator types contributed through limited ramping and self-scheduling, and production credits enable negative bids. Frequent negative prices show where storage, transmission or new load is valuable.

Version record

Ramteen Sioshansi, Paul Denholm, Thomas Jenkin and Jurgen Weiss, "Estimating the value of electricity storage in PJM: Arbitrage and some welfare effects"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 31(2), 269–277 (2009); DOI 10.1016/j.eneco.2008.10.005.

Values a price-taking storage device by arbitrage in PJM over 2002–07, showing how fuel prices, transmission constraints, round-trip efficiency, storage capacity and fuel mix drive its value. For larger storage, falling arbitrage revenue is offset by shifts in consumer and producer surplus and welfare gains.

Version record

R. Andrew Butters, Jackson Dorsey and Gautam Gowrisankaran, "Soaking up the Sun: Battery Investment, Renewable Energy, and Market Equilibrium"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 93(3), 891–927 (2025); DOI 10.3982/ecta20411.

Estimates the equilibrium effects of grid-scale batteries in California. The first storage unit breaks even without subsidy by 2024 once renewables reach 50 per cent, but returns fall quickly: the first 5,000 MWh cut wholesale prices by 5.6 per cent, an increase from 25,000 to 50,000 MWh by only 2.6 per cent. Batteries lower revenues for both dispatchable and renewable generators; without a mandate or subsidy adoption is negligible until 2030, while a 30 per cent capital subsidy reaches 5,000 MWh by 2024.

Version record

Fabian Ocker and Karl-Martin Ehrhart, "The 'German Paradox' in the balancing power markets"

Difficulty: Research **Type:** Research paper

Status/version: Renewable and Sustainable Energy Reviews, 67, 892–898 (2017); DOI 10.1016/j.rser.2016.09.040.

Responds to Hirth and Ziegenhagen's German paradox: balancing demand did not rise with renewables because of market-design changes and the efficiency savings of grid control cooperation. In the secondary balancing market, suppliers appear to coordinate on prices far above the competitive level, guided by previous auction prices.

Version record

Simon Hagemann, "Price determinants in the German intraday market for electricity: an empirical analysis"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Energy Markets, 8(2), 21–45 (2015); DOI 10.21314/jem.2015.128.

First analysis of hourly price determinants in the German intraday market (2010–11): plant outages, wind, solar and load forecast errors, cross-border supply and demand, the merit-order shape, ramping costs and strategic behaviour. Most act as expected, but only a small share of outages and solar errors is traded on the exchange, and effects vary over the day with liquidity.

Version record

René Aid, Andrea Cosso and Huyên Pham, "Equilibrium price in intraday electricity markets"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 32(2), 517–554 (2022); DOI 10.1111/mafi.12340.

Equilibrium model of intraday power trading in which agents balance customer-demand forecasts, with idiosyncratic and common weather noise, against production subject to random outages. Equilibrium prices and strategies are martingales; without generation uncertainty the price is a weighted average of forecast marginal costs consistent with Almgren–Chriss impact, and heterogeneity across agents is needed for the Samuelson volatility effect.

Version record

EPEX SPOT, 15-minute products in Market Coupling; go-live confirmation of 12 September 2025

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Exchange explainer page and Market Coupling Steering Committee press release of 12 September 2025; 15-minute day-ahead coupling live from trading day 30 September 2025.

Since trading day 30 September 2025 (delivery 1 October), the EU's single day-ahead coupling clears in 15-minute rather than hourly market time units in all bidding zones and across their borders, except Ireland at 30 minutes. Intraday auctions trade only 15-minute contracts, and continuous intraday markets offer 60-, 30- and 15-minute products. The change follows the EU requirement that TSOs settle imbalances every 15 minutes. EPEX matches 60- and 30-minute orders against quarter-hour products and publishes an hourly index as the average of the four quarter-hour prices, so any day-ahead series that crosses October 2025 changes resolution.

Version record

Potomac Economics, Independent Market Monitor for ERCOT, 2025 State of the Market Report for the ERCOT Electricity Markets

Difficulty: Advanced **Type:** Report / research note

Status/version: Independent market monitor's annual report, May 2026; covers 2025, including real-time co-optimisation from 5 December 2025.

The market monitor's review of ERCOT in 2025. Real-time co-optimisation, live from 5 December 2025, procures energy and ancillary services together in the real-time market, with a single model for storage resources; day-ahead ancillary-service awards became financial positions settled at real-time prices, and virtual ancillary-service offers were added. With new solar and storage, operating reserves rose: shortage pricing from the operating-reserve demand curve added only $0.02/MWh to real-time prices, summer price spikes were 40 per cent less frequent than in 2024, and the all-in price rose less than 14 per cent while gas rose 61 per cent. The monitor flags ancillary-service price-formation issues after the change and argues that ERCOT over-procures reserves.

Version record

14.15 Delivery rules and physical submarkets

Subject context and qualifications.

London Bullion Market Association, About Loco London; Good Delivery Rules

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living market standard; 2026 rule changes effective 1 January 2026.

Distinguishes a specific allocated bar from an unallocated metal entitlement, electronic net clearing from physical vault movement, and quoted Loco London price from delivery elsewhere. Use it before comparing bullion, gold futures or a gold-backed vehicle. Read after gold curve modelling. Good Delivery specifies eligible bars, not ownership or credit protection for every account. Verify custody agreement, refiner and bar status, location premiums, settlement route and current rule version.

Version record

International Energy Agency, Gas 2025: Executive Summary and LNG Contract Database Findings

Difficulty: Advanced **Type:** Data / specification

Status/version: IEA 2025 report, institutional analytical source; conditional contracts-in-force projection to 2030.

Documents the growing share of destination-free and hub-indexed or hybrid LNG contracting. Use it to separate oil-indexed purchase cost, hub resale, shipping and diversion optionality when valuing a cargo or term portfolio. Continue from 2026 gas context and the pipeline/storage models. IEA aggregates do not reveal a counterparty's take-or-pay, destination, cargo cancellation or shipping rights; projected shares are conditional, and individual contracts are generally private.

Version record

A. Baba, A. Creti and O. Massol, What Can Be Learned from the Free Destination Option in the LNG Imbroglio?

Difficulty: Research **Type:** Research paper

Status/version: *Energy Economics* 89, article 104764, 2020; university repository provides an author manuscript and abstract.

Values an LNG supplier's destination flexibility by simulating regional gas-price paths and choosing the best shipping decision for each path. The authors use a threshold vector autoregression tied to oil-volatility regimes, then a real-option valuation. Reproduce the decision with Asian, European and North American destination prices, freight, transit time and discharge capacity after the basic LNG contract reading. Separate the gross value of redirection from charter costs, contractual notices, export permissions and any foregone delivery commitment. The estimated market relationships describe an earlier sample and do not establish current cargo-option value or universal market integration.

Extended note. Version record

Corpus Christi Liquefaction and Cheniere Marketing International, LNG Sale and Purchase Agreement (FOB), Exhibit 10.46

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Executed 25 February 2026; SEC-filed affiliated-party FOB agreement; selected clauses inspected in full text.

A concrete contract for turning LNG price spreads into exercise rights and obligations. Section 3.3 permits resale and destination choice subject to export-authorisation conditions; section 5.7 permits scheduled-cargo cancellation by a specified notice deadline and a fee equal to the indexed FOB fixed component times scheduled quantity. Section 9.1 prices delivered LNG from 1.15 times Henry Hub plus that fixed component. Build a cargo decision tree across lifting, diversion and cancellation, adding freight and downstream commitments. This is an affiliated-party agreement; its economics and flexibility cannot be treated as representative third-party terms or observed arbitrage profit.

Extended note. Version record

ICE Futures U.S., Coffee "C" Futures Rules; USDA Foreign Agricultural Service, Coffee: World Markets and Trade route

Difficulty: Advanced **Type:** Research paper

Status/version: Exchange rulebook and official crop report route; living sources consulted 4 October 2026.

Coffee's tenderable origin, cup grade and defect limits make a physical Arabica basis different from soybean or wheat carry. Pair exchange-grade rules with USDA origin production, consumption and stocks to build crop-year and deliverability features. Read after soybean inventory dynamics. Rulebook terms can change by delivery month; report estimates are revised and are not certified exchange stocks. Preserve contract version, warehouse location, grading and report vintage.

Version record

Thorsten Martin, "Real Effects of Centralized Markets: Evidence from Steel Futures"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(7), 2140–2181 (2025); DOI 10.1093/rfs/hhaf011.

Staggered introduction of steel futures improved price transparency and hedging and reduced price dispersion in the physical market.

Version record

14.16 Futures contract design, delivery and manipulation

Why some futures contracts succeed; how much is delivered and what the grade, timing and location options embedded in delivery are worth; convergence failures in grain futures and storage-rate remedies; corners, squeezes and manipulation of the delivery process; and the principles, storage-infrastructure practices and position limits that govern physically delivered contracts.

Stephen Craig Pirrong, "Manipulation of the Commodity Futures Market Delivery Process"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Business, 66(3), 335 (1993); DOI 10.1086/296608.

Large longs can corner deliverable supply; delivery-market design (deliverable grades, locations) determines manipulability.

Version record

Praveen Kumar and Duane J. Seppi, "Futures Manipulation with "Cash Settlement""

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 47(4), 1485 (1992); DOI 10.2307/2328948.

Cash-settled futures remove corners but open manipulation of the settlement index; design trade-offs for settlement methods.

Version record

Scott H. Irwin, Philip Garcia, Darrel L. Good and Eugene L. Kunda, "Spreads and Non-Convergence in Chicago Board of Trade Corn, Soybean, and Wheat Futures: Are Index Funds to Blame?"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Economic Perspectives and Policy, 33(1), 116–142 (2011); DOI 10.1093/aepp/ppr001.

Non-convergence of grain futures and delivery points in 2005–2010 reflected storage economics and delivery-instrument pricing, not index funds.

Version record

Philip Garcia, Scott H. Irwin and Aaron Smith, "Futures Market Failure?"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 97(1), 40–64 (2015); DOI 10.1093/ajae/aau067.

Explains grain-futures non-convergence by the wedge between the price of storage and the fixed storage rate on delivery instruments.

Version record

Vitor M. O. Fernandes, Eugene L. Kunda and Michel A. Robe, "Commodity Futures Deliveries: Theory and Evidence From the US Corn Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 45(7), 844–876 (2025); DOI 10.1002/fut.22585.

How corn futures deliveries work (including shipping certificates) and when traders go to delivery: a delivery-value-equivalent versus basis explains delivery timing and size, 2011–2021.

Version record

Dennis W. Carlton, "Futures Markets: Their Purpose, Their History, Their Growth, Their Successes and Failures"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Futures Markets, 4(3), 237–271 (1984); DOI 10.1002/fut.3990040302.

Survey of why futures markets exist, how US futures trading grew from agricultural contracts to financial futures, and why many new contracts fail to attract trading. A starting point for asking what contract design a physical commodity needs.

Version record

Anne E. Peck and Jeffrey C. Williams, "Deliveries on Commodity Futures Contracts"

Difficulty: Advanced **Type:** Research paper

Status/version: Economic Record, 68(S1), 63–74 (1992); DOI 10.1111/j.1475-4932.1992.tb02296.x.

Deliveries are often thought negligible and large deliveries a sign of trouble, yet on five of the largest US physical-delivery futures markets they average about 10 per cent of maximum open interest in each delivery month. The timing and location options in contract terms are valuable, so measures such as hedging effectiveness depend on them.

Version record

Gerald D. Gay and Steven Manaster, "The Quality Option Implicit in Futures Contracts"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 13(3), 353–370 (1984); DOI 10.1016/0304-405x(84)90004-7.

Values the seller's right to choose which grade to deliver as an option to exchange one deliverable for another and applies it to grain futures: the futures price should track the cheapest grade to deliver rather than the par grade.

Version record

Jana Hranaiova, Robert A. Jarrow and William G. Tomek, "Estimating the Value of Delivery Options in Futures Contracts"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Research, 28(3), 363–383 (2005); DOI 10.1111/j.1475-6803.2005.00129.x.

Values the timing and location options in commodity futures jointly. Early delivery is always optimal with a timing option alone but not with both; the joint option is worth about 5 per cent of the average basis on the first day of the delivery month, far less than the timing option alone, so delivery options can widen deliverable supply with little effect on the basis.

Version record

Jana Hranaiova and William G. Tomek, "Role of Delivery Options in Basis Convergence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 22(8), 783–809 (2002); DOI 10.1002/fut.10028.

Estimates the joint timing and location options in CBOT corn futures for 1989–97. Their variation helps explain the basis on the first day of the delivery month and raises basis variability at maturity, but improves forecasts of convergence only occasionally.

Version record

Nicole M. Aulerich, Raymond P. H. Fishe and Jeffrey H. Harris, "Why Do Expiring Futures and Cash Prices Diverge for Grain Markets?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 31(6), 503–533 (2011); DOI 10.1002/fut.20483.

Explains grain non-convergence through the delivery process: the long holds a real option to exchange the delivery instrument for another futures contract, worth more as the relative volatility of cash and futures prices rises. An option model fitted to corn, soybeans and wheat for 2000–08 tracks recent non-convergence closely.

Version record

Scott H. Irwin, "Trilogy for Troubleshooting Convergence: Manipulation, Structural Imbalance, and Storage Rates"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Commodity Markets, 17, article 100083 (2020); DOI 10.1016/j.jcomm.2018.11.002.

Weighs three explanations of the 2005–10 non-convergence in CBOT corn, soybean and wheat and KCBOT wheat futures: manipulation, structural imbalance and low storage rates. Convergence failed because the market value of storage exceeded the storage rate paid to holders of the delivery instrument.

Version record

Alankrita Goswami, Michael K. Adjemian and Berna Karali, "The Impact of Futures Contract Storage Rate Policy on Convergence Expectations in Domestic Commodity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Food Policy, 111, article 102301 (2022); DOI 10.1016/j.foodpol.2022.102301.

Studies convergence in CBOT and KC wheat, corn and soybeans over 2006–20. The variable storage rate introduced for wheat reduced the size and expected duration of non-convergence; higher fixed storage rates for corn and soybeans show no such effect so far, and index-trader activity does not cause non-convergence.

Version record

Alankrita Goswami, Berna Karali and Michael K. Adjemian, "Hedging with Futures during Nonconvergence in Commodity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Commodity Markets, 32, article 100364 (2023); DOI 10.1016/j.jcomm.2023.100364.

Asks how short hedgers fared during the 2005–10 grain non-convergence. The minimum-variance hedge ratio often failed to minimise the variance of wheat producers' profits and weakened in low-carryover years; non-convergence impaired every hedge-ratio choice, most of all in wheat.

Version record

Kevin Guo and Tim Leung, "Understanding the Non-Convergence of Agricultural Futures via Stochastic Storage Costs and Timing Options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Commodity Markets, 6, 32–49 (2017); DOI 10.1016/j.jcomm.2017.04.001.

Models non-convergence as the value of the holder's timing options on a shipping certificate under stochastic spot prices and storage costs, solved as an optimal double-stopping problem, and calibrates the certificate premium to futures data from the non-convergence period.

Version record

Robert A. Jarrow, "Market Manipulation, Bubbles, Corners, and Short Squeezes"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 27(3), 311–336 (1992); DOI 10.2307/2331322.

Shows that a large trader whose trades move prices can, under reasonable conditions on the equilibrium price process, earn riskless profits from manipulation strategies, and gives sufficient conditions that rule such strategies out.

Version record

Stephen Craig Pirrong, "The Self-Regulation of Commodity Exchanges: The Case of Market Manipulation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Law and Economics, 38(1), 141–206 (1995); DOI 10.1086/467328.

Challenges the view that exchanges will police manipulation on their own: members ignore its effects on inframarginal traders and price informativeness, competition between exchanges is limited and collective action is hard. Ten exchanges took few, if any, measures against manipulation before laws prohibited it.

Version record

Craig Pirrong, "The Economics of Commodity Market Manipulation: A Survey"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Commodity Markets, 5, 1–17 (2017); DOI 10.1016/j.jcomm.2017.02.001.

Survey of market-power, trade-based and fraud-based manipulation in commodity futures: how each occurs and its effects on prices, quantities and welfare. Manipulation law and regulation have often been confused and ineffective because they have not absorbed these economics.

Version record

Franklin Allen, Lubomir Litov and Jianping Mei, "Large Investors, Price Manipulation, and Limits to Arbitrage: An Anatomy of Market Corners"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 10(4), 645–693 (2006); DOI 10.1007/s10679-006-9008-5.

Rational-expectations model of corners and hand-collected evidence on stock and commodity corners between 1863 and 1980. Large investors and insiders had the market power to manipulate prices; corners raised volatility, hurt other assets and made it risky to short against the mispricing.

Version record

John J. Merrick Jr., Narayan Y. Naik and Pradeep K. Yadav, "Strategic Trading Behavior and Price Distortion in a Manipulated Market: Anatomy of a Squeeze"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 77(1), 171–218 (2005); DOI 10.1016/j.jfineco.2004.10.004.

Uses trade and inventory reports filed with the regulator to dissect an attempted delivery squeeze in a London bond futures contract: how dealers and customers traded, how prices responded and how depth eroded. Recommends updating contract specifications more often, flagging possession-oriented trades and aligning delivery-failure penalties with cash and repo market conventions.

Version record

Jeffrey C. Williams, *Manipulation on Trial: Economic Analysis and the Hunt Silver Case*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press (1995); DOI 10.1017/cbo9780511521966.

The 1979–80 silver episode and the trial of the Hunt brothers, by one of its expert witnesses: how manipulation was defined, how statistical evidence, damages and intent were argued, and why the author places the failure on futures exchanges that lacked rules to limit large-scale trading in advance.

Version record

U.S. Commodity Futures Trading Commission, Position Limits for Derivatives: Final Rule (86 FR 3236)

Difficulty: Advanced **Type:** Report / research note

Status/version: Federal Register, vol. 86, 3236–3493, 14 January 2021; effective 15 March 2021.

The CFTC rule conforming speculative position limits to the Dodd-Frank amendments: new or amended federal spot-month limits for 25 physical commodity derivatives, amended single-month and all-months limits for most agricultural contracts, a revised definition of bona fide hedging with new enumerated hedges, a definition of economically equivalent swaps, and rules for exchange-set limits and exemptions. Compliance dates run to January 2023.

Version record

Board of the International Organization of Securities Commissions (IOSCO), Principles for the Regulation and Supervision of Commodity Derivatives Markets: Final Report (FR02/2023)

Difficulty: Advanced **Type:** Report / research note

Status/version: IOSCO Final Report FR02/2023, January 2023; updates the September 2011 principles (FR07/11).

Updated IOSCO principles for commodity derivatives markets: design and review of physically delivered contracts, data and market transparency, surveillance, disorderly markets, enforcement and information sharing, and price discovery. The update reflects new trading venues, the principles for price reporting agencies, direct electronic access and reliance on electronic data, and adds guidance for extraordinary disruptions.

Version record

Board of the International Organization of Securities Commissions (IOSCO), Commodity Storage and Delivery Infrastructures: Good or Sound Practices: Final Report (FR03/2019)

Difficulty: Advanced **Type:** Report / research note

Status/version: IOSCO Final Report FR03/2019, February 2019.

Good or sound practices for the warehouses and other storage infrastructures behind physically delivered contracts and for their oversight bodies, in five areas: oversight, transparency, fees and incentives, conflicts of interest and operations. It explains how rent discounts and load-out queues can disconnect futures from the cash market and lists remedies such as penalties for delayed delivery and rent caps.

Version record

14.17 Price discovery and fundamental information

Where commodity prices are discovered across cash and futures markets and across exchanges; how futures react to weather, USDA crop reports, EIA inventory reports and analyst forecasts; and what satellite and ship-tracking data add to official statistics.

Kenneth D. Garbade and William L. Silber, "Price Movements and Price Discovery in Futures and Cash Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Economics and Statistics, 65(2), 289 (1983); DOI 10.2307/1924495.

Classic model of futures–cash price discovery and the role of delivery arbitrage in linking the two markets.

Version record

Richard Roll, "Orange Juice and Weather"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 74(5), 861–880 (1984); Caltech repository copy of the JSTOR article.

Frozen orange juice concentrate is grown almost entirely around Orlando, so weather is a nearly exogenous fundamental. Futures returns predict errors in National Weather Service temperature forecasts, yet measured weather and other fundamentals explain only a small fraction of futures price variability.

Version record

Jacob Boudoukh, Matthew Richardson, YuQing (Jeff) Shen and Robert F. Whitelaw, "Do Asset Prices Reflect Fundamentals? Freshly Squeezed Evidence from the OJ Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 83(2), 397–412 (2007); DOI 10.1016/j.jfineco.2004.06.017.

Revisits Roll's orange-juice puzzle. In a nonlinear model where only temperatures near or below freezing matter, temperature explains about half of winter futures return variation; two-thirds of winter variability falls on the 4.5 per cent of days with freezing temperatures, and USDA forecasts and Brazilian supply news explain more.

Version record

Isabel Figuerola-Ferretti and Jesús Gonzalo, "Modelling and Measuring Price Discovery in Commodity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Econometrics, 158(1), 95–107 (2010); DOI 10.1016/j.jeconom.2010.03.013.

Equilibrium model of spot and futures prices with convenience yields and limited arbitrage, in which the long-run spot–futures relation can have a slope above or below one (backwardation or contango) and price-discovery weights follow from relative liquidity. For LME aluminium, copper, nickel, lead and zinc, most markets are backwardated and futures lead in the most liquid ones.

Version record

Joseph P. Janzen and Michael K. Adjemian, "Estimating the Location of World Wheat Price Discovery"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 99(5), 1188–1207 (2017); DOI 10.1093/ajae/aax046.

Uses high-frequency prices and information-leadership shares to locate world wheat price discovery among Chicago, Kansas City, Minneapolis and Paris futures. US markets remain dominant, but their share of new information fell from 91 to 75 per cent after the Russian and Ukrainian supply shocks of August 2010.

Version record

Olga Isengildina-Massa, Scott H. Irwin, Darrel L. Good and Jennifer K. Gomez, "The Impact of Situation and Outlook Information in Corn and Soybean Futures Markets: Evidence from WASDE Reports"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Agricultural and Applied Economics, 40(1), 89–103 (2008); DOI 10.1017/s1074070800027991.

Over 1985–2006, WASDE reports that include NASS crop production estimates make report-day return variance about seven times normal in corn and soybean futures; reports with only international or outlook information matter less, and the effect has grown over time.

Version record

Michael K. Adjemian, "Quantifying the WASDE Announcement Effect"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 94(1), 238–256 (2012); DOI 10.1093/ajae/aar131.

Quantifies the WASDE announcement effect for cotton, soybeans and hard winter wheat with a two-stage GLS model. The effect persists across contract months, is not limited to reports with NASS survey data, is larger when carryover stocks are low and is quickly absorbed into futures prices.

Version record

Georg V. Lehecka, Xiaoyang Wang and Philip Garcia, "Gone in Ten Minutes: Intraday Evidence of Announcement Effects in the Electronic Corn Futures Market"

Difficulty: Research **Type:** Research paper

Status/version: Applied Economic Perspectives and Policy, 36(3), 504–526 (2014); DOI 10.1093/aepp/ppu010.

Intraday CBOT corn futures for 2009–12 show that major USDA reports carry substantial news. The strongest reactions come right after the open and last about ten minutes, with little sign of systematic under- or overreaction; traders also adjust exposure in the session before a release.

Version record

Michael K. Adjemian and Scott H. Irwin, "USDA Announcement Effects in Real-Time"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 100(4), 1151–1171 (2018); DOI 10.1093/ajae/aay018.

After the CBOT dropped the morning trading halt around USDA reports in 2012, real-time trading on announcements produced volatility spikes that faded within minutes. Continuous markets found it harder to distinguish reports' newsworthiness but absorbed the shocks on almost the same time path.

Version record

Berna Karali, Olga Isengildina-Massa, Scott H. Irwin, Michael K. Adjemian and Robert Johansson, "Are USDA Reports Still News to Changing Crop Markets?"

Difficulty: Research **Type:** Research paper

Status/version: Food Policy, 84, 66–76 (2019); DOI 10.1016/j.foodpol.2019.02.005.

Measures the news in USDA crop reports as the gap between the USDA estimate and private expectations for corn, soybeans and wheat. Surprises have not shrunk despite competing private data, and price reactions to most reports have grown.

Version record

Marketa W. Halova, Alexander Kurov and Oleg Kucher, "Noisy Inventory Announcements and Energy Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 34(10), 911–933 (2014); DOI 10.1002/fut.21633.

Inventory surprises in oil and gas announcements are measured with error, which biases event-study estimates. With intraday futures data and identification through censoring, the price response is about twice the OLS estimate for petroleum commodities and about four times for natural gas.

Version record

Gerald D. Gay, Betty J. Simkins and Marian Turac, "Analyst Forecasts and Price Discovery in Futures Markets: The Case of Natural Gas Storage"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 29(5), 451–477 (2009); DOI 10.1002/fut.20368.

With high-frequency analyst forecasts of the weekly natural gas storage report, the market conditions its expectations on analysts beyond statistical models, looks through the consensus and weights individual analysts by their accuracy, more by long-run than recent accuracy.

Version record

Chen Gu and Alexander Kurov, "What Drives Informed Trading before Public Releases? Evidence from Natural Gas Inventory Announcements"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 38(9), 1079–1096 (2018); DOI 10.1002/fut.21926.

Natural gas futures drift before inventory announcements. The gap between the median forecast of historically accurate analysts and the consensus predicts surprises and part of the drift, pointing to superior forecasting rather than leakage; a simple strategy on the predictor had an annualised Sharpe ratio of 1.26.

Version record

Abhiroop Mukherjee, George Panayotov and Janghoon Shon, "Eye in the Sky: Private Satellites and Government Macro Data"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 141(1), 234–254 (2021); DOI 10.1016/j.jfineco.2021.03.002.

Uses cloud cover over key storage hubs, which blinds satellites, to test whether commercial satellite estimates substitute for government data. In the US crude oil market, satellite estimates of tank levels are now good enough that the EIA's weekly inventory announcement no longer surprises the market when the hubs are visible.

Version record

Roar Adland, Haiying Jia and Siri P. Strandenes, "Are AIS-Based Trade Volume Estimates Reliable? The Case of Crude Oil Exports"

Difficulty: Research **Type:** Research paper

Status/version: Maritime Policy & Management, 44(5), 657–665 (2017); DOI 10.1080/03088839.2017.1309470.

Compares crude oil export volumes built from AIS ship-position data with official customs data. Aggregates align well, but pipelines and transshipment create large differences across countries and periods, so cargo-level data must be structured and aggregated with care.

Version record

Hui Bu, "Effect of Inventory Announcements on Crude Oil Price Volatility"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 46, 485–494 (2014); DOI 10.1016/j.eneco.2014.05.015.

The information shock in the EIA weekly crude inventory report, rather than the inventory change itself, lowers crude oil futures returns on release days but does not change conditional variance; the effect weakens in rapid-growth periods and disappears in steep price falls.

Version record

Sultan Alturki and Alexander Kurov, "Market Inefficiencies Surrounding Energy Announcements"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 42(1), 172–188 (2022); DOI 10.1002/fut.22264.

Uses sequential energy inventory announcements to show that crude oil futures and stock markets incorporate inventory news inefficiently. A predictor of inventory surprises forecasts surprises and pre-announcement returns in and out of sample, and a combination forecast proxies market expectations.

Version record

Jian Yang, David A. Bessler and David J. Leatham, "Asset Storability and Price Discovery in Commodity Futures Markets: A New Look"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 21(3), 279–300 (2001); DOI 10.1002/1096-9934(200103)21:3<279::aid-fut5>3.0.co;2-l.

Cash and futures prices are cointegrated and futures predict cash prices for both storable and non-storable commodities once stochastic interest rates are allowed for; storability may change only the size of the bias in futures forecasts.

Version record

Isabel Figuerola-Ferretti and Christopher L. Gilbert, "Price Discovery in the Aluminum Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 25(10), 967–988 (2005); DOI 10.1002/fut.20173.

Measures the information content of four aluminium prices quoted since 1970: the US producer price, a trade-journal transactions price, and LME and Comex futures. Futures trading from 1978 made transactions prices more informative; LME prices soon led published prices, and the 1999 Comex contract, unlike the first, became as transparent as the LME.

Version record

14.18 Commodity trading firms, financing and margin stress

How physical trading firms finance inventories and use commodities as collateral, and how margin calls, leverage and concentration carried the 2022 commodity price shock into clearing and funding markets.

Craig Pirrong, "The Economics of Commodity Trading Firms"

Difficulty: Intro/Advanced **Type:** Report / research note

Status/version: White paper commissioned by Trafigura, 2014.

How physical commodity traders transform commodities in space, time and form, and manage price and credit risk (Trafigura-commissioned white paper, 2014).

Version record

World Bank, Using Commodities as Collateral for Finance: Commodity-Backed Finance

Difficulty: Advanced **Type:** Report / research note

Status/version: World Bank technical guide; official PDF, historical framework.

Distinguishes warehouse receipts, collateral management agreements and stock monitoring in inventory-backed lending. Use it to map a commodity trader's gross curve return into borrowing base, haircut, title, warehouse control and liquidation proceeds. Pair with deliverable crop rules and the storage models. The guide describes structures, not a current jurisdictional opinion; verify enforceability, perfection, inspection, insurance, fraud controls and margin calls locally.

Version record

Ke Tang and Haoxiang Zhu, "Commodities as Collateral"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 29(8), 2110–2160 (2016); DOI 10.1093/rfs/hhw029.

Under capital controls and collateral constraints, investors import commodities and pledge them as collateral for financing. Collateral demand raises commodity prices and makes the inventory–convenience yield relation less negative; deviations from covered interest parity proxy for that demand. Evidence from eight commodities in China and developed markets supports the model.

Version record

Randy Priem, "A Relaxation of European Derivatives Clearing Legislation as a Consequence of the 2021–2023 Energy Crisis"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of World Energy Law & Business, 17(4), 277–293 (2024); DOI 10.1093/jwelb/jwae010.

How the EU amended EMIR during the 2021–23 energy crisis so that central counterparties could accept unsecured bank guarantees, emission allowances and public debt as collateral, and the trade-off between keeping energy firms in central clearing and the macroprudential risks of weaker collateral.

Version record

Financial Stability Board, The Financial Stability Aspects of Commodities Markets

Difficulty: Advanced **Type:** Report / research note

Status/version: FSB report, 20 February 2023.

Maps the commodities ecosystem and its links to banks, clearing members and CCPs, and reviews how it absorbed the 2022 price shock. The vulnerabilities are concentration (a few large, sometimes highly leveraged trading firms and the banks and CCPs that serve them), widespread leverage and opacity of OTC exposures; some European activity moved from cleared to uncleared markets, and some firms cut hedges to reduce margin calls.

Version record

Basel Committee on Banking Supervision, Committee on Payments and Market Infrastructures and IOSCO, Margin Dynamics in Centrally Cleared Commodities Markets in 2022

Difficulty: Advanced **Type:** Report / research note

Status/version: BCBS–CPMI–IOSCO report, 24 May 2023.

Analyses central counterparties' margin models, their use of discretion and its effect on procyclicality, and other risk-management responses during the 2022 commodity turmoil, together with participants' preparedness to meet margin calls given the transparency and predictability of margin practices. It complements the FSB's commodities report and feeds the international work on initial-margin models.

Version record

Fernando Avalos, Wenqian Huang and Kevin Tracol, Margins and Liquidity in European Energy Markets in 2022 (BIS Bulletin No 77)

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Bulletin No 77, 13 September 2023.

European natural gas and electricity futures were deeply disrupted after the Russian invasion of Ukraine, partly through large swings in margins. The swings put heavy liquidity demands on participants, prompting liquidity facilities in some jurisdictions, and rising initial margins went with sharp falls in open interest, consistent with deleveraging.

Version record

14.19 Spot price assessment and benchmarks

How spot commodity prices are assessed and published: price reporting agencies and their oversight, auction benchmarks for precious metals, the design of fixings that traders can manipulate, the move from annual contract to index pricing, and contract pricing where spot markets are thin.

Adi Imsirovic, *Trading and Price Discovery for Crude Oils: Growth and Development of International Oil Markets*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Springer International Publishing (2021); DOI 10.1007/978-3-030-71718-6.

How crude oil prices are set. Most oil is never traded: its value is fixed by a few benchmark grades and baskets—Brent, WTI, Dubai and Oman—traded as physical and paper barrels in volumes many times world output, which separates price makers from price takers. Separate chapters cover the Brent, Dubai–Oman and WTI benchmarks; earlier ones trace oil pricing from the early US market through the eras of the majors and OPEC, and a late chapter draws lessons from the 2020 pandemic.

Version record

Board of the International Organization of Securities Commissions (IOSCO), Implementation of the Principles for Oil Price Reporting Agencies: Report (FR22/2015)

Difficulty: Advanced **Type:** Report / research note

Status/version: IOSCO Report FR22/2015, September 2015; second implementation review, following FR05/2014 (September 2014).

Second review, with the IEA, IEF and OPEC, of how Argus, ICIS, OPIS and Platts implemented the 2012 principles for oil price reporting agencies. External assurance reviews and stakeholder comment lead IOSCO to conclude that the principles are now part of the agencies' management and operations. An IEA–IEF–OPEC survey of physical market participants reports more confidence in assessments and more transparency about methodology changes, and no fall in the quality or quantity of data submitted, although a slight majority of those with earlier concerns said they had not been resolved.

Version record

Darrell Duffie and Piotr Dworczak, "Robust Benchmark Design"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 142(2), 775–802 (2021); DOI 10.1016/j.jfineco.2021.06.024.

Treats a benchmark fixing as an estimator of fair value computed from the trades of agents whose profits depend on it, so they have reason to manipulate it. Derives the optimal linear fixing when transaction weights are unidimensional and characterises the unique linear fixing robust to a form of collusion among traders, which gives a foundation for volume-weighted average prices and their analogues; which design is better depends on market characteristics.

Version record

Darrell Duffie, Piotr Dworczak and Haoxiang Zhu, "Benchmarks in Search Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 72(5), 1983–2044 (2017); DOI 10.1111/jofi.12525.

Role of benchmarks in over-the-counter markets with search frictions. A published benchmark can raise welfare by drawing more investors into trade, improving dealer–customer matching and cutting search costs. It narrows dealers' margins, yet dealers may still introduce one to raise participation, and low-cost dealers to gain market share; under some conditions announcing the benchmark is the welfare-maximising mechanism.

Version record

ICE Benchmark Administration (IBA), LBMA Gold and Silver Prices: Questionnaire for Usage Licence Holders—Feedback Statement

Difficulty: Advanced **Type:** Report / research note

Status/version: IBA feedback statement, 30 July 2024.

How the LBMA Gold and Silver Prices, the benchmarks for unallocated metal delivered in London, are set: electronic auctions at 10:30 and 15:00 London time for gold and 12:00 for silver, run in 30-second rounds at algorithm-set prices until the buy–sell imbalance is within 10,000 oz of gold or 500,000 oz of silver, with the residual shared equally among direct participants and the dollar price converted to sterling and euro benchmarks. After a 2024 survey of licence holders (13 responses), IBA kept the auction times, round length and currencies unchanged.

Version record

Andrew Caminschi and Richard Heaney, "Fixing a Leaky Fixing: Short-Term Market Reactions to the London PM Gold Price Fixing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 34(11), 1003–1039 (2014); DOI 10.1002/fut.21636.

Studies the London PM gold fixing, the conference-call benchmark that an electronic auction replaced in 2015, and its effect on COMEX gold futures and the GLD exchange-traded fund. Volume and volatility jump as soon as the fixing starts, well before its result is published; trades in its opening minutes predict the direction of the fix, in some cases more than 90 per cent of the time, and earn significant returns. Information was leaking from the fixing.

Version record

Martin Hauptfleisch, Tālis J. Putniņš and Brian Lucey, "Who Sets the Price of Gold? London or New York"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 36(6), 564–586 (2016); DOI 10.1002/fut.21775.

Price discovery between the London spot gold market and New York gold futures over 17 years of intraday data. Both contribute, but New York leads on average although it trades less than a tenth of London's volume. Shares vary within the day and across years with market structure, liquidity, daylight hours and macroeconomic announcements, and an upgrade of the New York trading platform reduced the noise in futures prices.

Version record

Linda Wårell, "The Effect of a Change in Pricing Regime on Iron Ore Prices"

Difficulty: Research **Type:** Research paper

Status/version: Resources Policy, 41, 16–22 (2014); DOI 10.1016/j.resourpol.2014.02.002.

Asks whether iron ore's move from annual benchmark contracts to spot-market pricing in 2008–09 raised volatility or changed the price level, with monthly data for 2003–12. With freight costs included, prices are not more volatile after the switch and the regime change has no significant effect on the price; iron ore prices, Chinese GDP growth and freight rates are cointegrated, and Chinese growth has the strongest short-run effect.

Version record

Yiqun Ma, "Iron Ore Spot Price Volatility and Change in Forward Pricing Mechanism"

Difficulty: Research **Type:** Research paper

Status/version: Resources Policy, 38(4), 621–627 (2013); DOI 10.1016/j.resourpol.2013.10.002.

Models the daily Platts IODEX iron ore spot price from October 2008 to September 2012 with a structural break at the change in the forward pricing mechanism and an EGARCH(1,1). The spot price is less volatile after quarterly pricing was introduced, and volatility responds asymmetrically to news.

Version record

R. Glenn Hubbard and Robert J. Weiner, "Efficient Contracting and Market Power: Evidence from the U.S. Natural Gas Industry"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Law and Economics, 34(1), 25–67 (1991); DOI 10.1086/467218.

Long-term contracts between gas producers and pipelines in US field markets in the 1950s, where few buyers and sellers and large up-front investments make bargaining bilateral. The initial price is modelled as the outcome of bargaining over a fixed payment and price-escalator clauses as a way to keep the contract responsive to changes in the value of gas; evidence on price determination and on most-favoured-nation clauses supports the model.

Version record

Jaqueline Taquiri, Thomas Lassourd and Andrew Viola (OECD and Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development), "Determining the Price of Minerals: A Transfer Pricing Framework for Lithium"

Difficulty: Advanced **Type:** Report / research note

Status/version: OECD and IISD/IGF practice note (August 2024); DOI 10.1787/a607ff0a-en.

Practice note for tax authorities pricing lithium carbonate, hydroxide and spodumene sold between related parties. Price reporting agencies build spot and contract assessments from transactions, bids and offers and market intelligence; spot assessments rest on few trades, contract prices mix old and new contracts, and reliable assessments exist mainly for China, Japan and Korea, which take over 90 per cent of traded volume. Explains how quality, incoterms, quotation periods and contract terms adjust an assessed price.

Version record

14.20 Spatial arbitrage and market integration

How spot prices in different places are tied together by transport, storage and information: tests of market integration with transfer costs, the law of one price under slow transport and costly information, and location differentials created by pipeline constraints, export bans and sanctions.

Andrew Coleman, "Storage, Slow Transport, and the Law of One Price: Theory with Evidence from Nineteenth-Century U.S. Corn Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 91(2), 332–350 (2009); DOI 10.1162/rest.91.2.332.

Rational-expectations model of arbitrage between regions when shipping takes time. Arbitrageurs let inventories at the importing centre run down to zero regularly, so local price spikes are a normal feature of competitive markets and pay for small losses at other times. Supported by Chicago and New York corn prices.

Version record

Paul L. Fackler and Barry K. Goodwin, "Spatial Price Analysis"

Difficulty: Advanced **Type:** Book chapter

Status/version: Handbook of Agricultural Economics, vol. 1 (Elsevier, 2001), chapter 17, 971–1024; DOI 10.1016/S1574-0072(01)10025-3.

Survey of spatial price analysis: the arbitrage condition that price differences between locations cannot exceed transfer costs, and the tools used to test market integration and price transmission, from correlation and regression tests through cointegration and error-correction models to threshold and switching-regime models that allow for unobserved transfer costs.

Version record

Bob Baulch, "Transfer Costs, Spatial Arbitrage, and Testing for Food Market Integration"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 79(2), 477–487 (1997); DOI 10.2307/1244145.

Argues that conventional tests, which ask whether prices in different places move together, are often misleading. Develops the parity-bounds model, which combines transfer-cost data with prices to judge how efficiently spatial arbitrage works; Monte Carlo experiments support it, and in Philippine rice markets it finds efficient arbitrage where other tests do not.

Version record

Christopher B. Barrett and Jau Rong Li, "Distinguishing between Equilibrium and Integration in Spatial Price Analysis"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 84(2), 292–307 (2002); DOI 10.1111/1467-8276.00298.

Mixture-distribution model estimated by maximum likelihood on prices, transfer costs and trade flows. It separates market integration (goods flow between markets) from competitive equilibrium (arbitrage profits are exhausted), yields probabilities of perfect integration, segmented equilibrium and segmented disequilibrium that can vary over time, and is applied to soybean meal trade among Pacific Rim economies.

Version record

David S. Jacks, Kevin H. O'Rourke and Jeffrey G. Williamson, "Commodity Price Volatility and World Market Integration since 1700"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 93(3), 800–813 (2011); DOI 10.1162/rest_a_00091.

Three centuries of commodity and manufactures prices. Commodity price volatility shows little trend since 1700 and has always exceeded that of manufactures; periods of world market integration under peace and open trade had lower commodity price volatility than periods of war or autarky.

Version record

Robert Jensen, "The Digital Provide: Information (Technology), Market Performance, and Welfare in the South Indian Fisheries Sector"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 122(3), 879–924 (2007); DOI 10.1162/qjec.122.3.879.

Mobile phone service reached Kerala's fishing markets between 1997 and 2001. Once fishermen and wholesalers adopted phones, price dispersion across beach markets fell sharply, waste disappeared and prices came close to the law of one price; consumer and producer welfare both rose.

Version record

Jenny C. Aker, "Information from Markets Near and Far: Mobile Phones and Agricultural Markets in Niger"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Applied Economics, 2(3), 46–59 (2010); DOI 10.1257/app.2.3.46.

The rollout of mobile phone service in Niger between 2001 and 2006 cut grain price dispersion across markets by 10 to 16 per cent, more for market pairs with higher transport costs.

Version record

Treb Allen, "Information Frictions in Trade"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 82(6), 2041–2083 (2014); DOI 10.3982/ecta10984.

Regional agricultural trade in the Philippines. Producers search sequentially, at a cost, for where to sell; adding this information friction to a competitive trade model reconciles it with observed trade flows and prices, and structural estimates attribute about half of regional price dispersion to information frictions.

Version record

Claudia Steinwender, "Real Effects of Information Frictions: When the States and the Kingdom Became United"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 108(3), 657–696 (2018); DOI 10.1257/aer.20150681.

The 1866 transatlantic telegraph as a natural experiment, with cotton prices, trade and news flows collected from historical newspapers. After the telegraph the mean and volatility of the transatlantic cotton price difference fell, while trade flows rose and became more volatile; a model in which exporters forecast foreign prices from the latest news puts the efficiency gain at about 8 per cent of export value.

Version record

Severin Borenstein and Ryan Kellogg, "The Incidence of an Oil Glut: Who Benefits from Cheap Crude Oil in the Midwest?"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 35(1), 15–34 (2014); DOI 10.5547/01956574.35.1.2.

From early 2011 Midwest and Canadian crude output exceeded pipeline capacity to the Gulf Coast, and WTI at Cushing fell well below internationally traded crude. The discount did not pass through to Midwest wholesale gasoline and diesel prices, because the marginal gallon was still imported from coastal locations, so new pipelines to the Gulf would not raise Midwest fuel prices.

Version record

Mark Agerton and Gregory B. Upton, "Decomposing Crude Price Differentials: Domestic Shipping Constraints or the Crude Oil Export Ban?"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 40(3), 155–172 (2019); DOI 10.5547/01956574.40.3.mage.

Why WTI traded at steep discounts to Brent in the decade before 2019. Scarce domestic pipeline capacity explains half to three quarters of the deviation of mid-continent crude from its long-run relation with Brent; there is no evidence that a mismatch between refinery configurations and domestic crude quality mattered, so the short-run effect of the US crude export ban may have been exaggerated.

Version record

Matthew E. Oliver, Charles F. Mason and David Finnoff, "Pipeline Congestion and Basis Differentials"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Regulatory Economics, 46(3), 261–291 (2014); DOI 10.1007/s11149-014-9256-9.

Network model of natural gas spot prices at two hubs, estimated on a Rocky Mountain pipeline route. As congestion rises the scarcity value of transport capacity drives a wedge between the two spot prices; access to storage mitigates the effect by shifting transport over time.

Version record

Lutz Kilian, David Rapson and Burkhard Schipper, "The Impact of the 2022 Oil Embargo and Price Cap on Russian Oil Prices"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 47(3), 1–39 (2026); DOI 10.1177/01956574251414076.

By March 2023 the oil embargo had forced Russia to accept a discount on Urals exports $32 a barrel wider than in January 2022. Nearly half is the higher cost of shipping crude the longer distance to India; a calibrated model of global oil supply and demand attributes the remaining $17 to India's greater bargaining power. The price cap had a negligible effect on export prices in March 2023 but may have eased fears of a wider ban on Western maritime services. A similar analysis covers the ESPO discount on exports to China.

Version record

14.21 Commodity price cycles, comovement and macro drivers

Surveys and current outlooks of primary commodity markets; the dating of booms and slumps; common factors in commodity prices and their links to global activity, real interest rates and the dollar; and forecasts of commodity spot prices.

Colin A. Carter, Gordon C. Rausser and Aaron Smith, "Commodity Booms and Busts"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Resource Economics, 3(1), 87–118 (2011); DOI 10.1146/annurev.resource.012809.104220.

Compares the broad commodity booms that peaked in 1974 and 2008. Both combined simultaneous supply and demand surprises with low inventories, magnified by macroeconomic shocks and policy responses; the authors do not expect such episodes to become more frequent.

Version record

Marian Radetzki and Linda Wårell, *A Handbook of Primary Commodities in the Global Economy*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, third edition (2020); DOI 10.1017/9781108886529.

Third edition of a guide to primary commodity markets, written after the 2014–15 price falls ended the strongest commodity boom since the Second World War. Chapters cover the geography of production and trade, trade-policy distortions, fossil fuels, price formation and price trends, booms, exchanges and speculation, depletion, supply security, producer cartels, public ownership and economies dependent on commodity exports, with updates on China's slowdown and US trade policy.

Version record

World Bank, Commodity Markets Outlook, April 2026

Difficulty: Intro **Type:** Report / research note

Status/version: World Bank Group report, April 2026; data cut-off 20 April 2026; published each April and October (CC BY 3.0 IGO).

Semi-annual review and price forecasts for 46 commodities. The April 2026 edition treats the war in the Middle East as a historic shock: in March the largest oil supply loss on record and the near-halt of shipping through the Strait of Hormuz took Brent from $72 to $118 a barrel. Assuming the acute disruption ends in May, average commodity prices rise 16 per cent in 2026 and energy prices 24 per cent, with Brent averaging $86, fertiliser prices soaring and base and precious metals at record highs; risks are tilted upward, with Brent at $95–115 if disruption persists. A special focus estimates oil-market responses to geopolitical supply shocks and their spillovers.

Version record

Paul Cashin, C. John McDermott and Alasdair Scott, "Booms and Slumps in World Commodity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Development Economics, 69(1), 277–296 (2002); DOI 10.1016/s0304-3878(02)00062-7.

Dates cycles in commodity prices. For most commodities slumps last longer than booms, and prices fall slightly further in a slump than they rebound in the next boom; cycles have no consistent shape, and the probability that a slump or boom ends does not depend on how long it has lasted.

Version record

Robert S. Pindyck and Julio J. Rotemberg, "The Excess Co-Movement of Commodity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Economic Journal, 100(403), 1173–1189 (1990); DOI 10.2307/2233966.

Prices of largely unrelated raw commodities move together persistently, far more than the common effects of past, current or expected inflation, industrial production, interest rates and exchange rates can explain. The authors read this as a rejection of the standard competitive model of commodity price formation with storage.

Version record

Partha Deb, Pravin K. Trivedi and Panayotis Varangis, "The Excess Co-Movement of Commodity Prices Reconsidered"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 11(3), 275–291 (1996); DOI 10.1002/(sici)1099-1255(199605)11:3<275::aid-jae392>3.0.co;2-3.

Re-tests excess comovement with univariate and multivariate GARCH(1,1) models and new score and likelihood-ratio tests, on monthly data for up to nine commodities over 1960–85 and 1974–92. Unlike earlier work, it finds only weak evidence of excess comovement.

Version record

Joseph P. Byrne, Giorgio Fazio and Norbert Fiess, "Primary Commodity Prices: Co-movements, Common Factors and Fundamentals"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Development Economics, 101, 16–26 (2013); DOI 10.1016/j.jdeveco.2012.09.002.

Nonstationary panel methods find a significant common factor in primary commodity prices. In a factor-augmented VAR, real interest rates and uncertainty are negatively related to the factor, as a simple asset-pricing model predicts, and the result survives controls for demand and supply shocks, both of which raise comovement.

Version record

Simona Delle Chiaie, Laurent Ferrara and Domenico Giannone, "Common Factors of Commodity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 37(3), 461–476 (2022); DOI 10.1002/jae.2887.

Splits a large cross-section of fuel and non-fuel commodity prices into a global factor, block-specific factors and idiosyncratic components. A single global factor summarises most of the fluctuations, tracks global economic activity, and has explained more of commodity price variation since the early 2000s.

Version record

Ron Alquist, Saroj Bhattarai and Olivier Coibion, "Commodity-Price Comovement and Global Economic Activity"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 112, 41–56 (2020); DOI 10.1016/j.jmoneco.2019.02.004.

Builds a macroeconomic model in which non-energy commodity prices are endogenous and uses the factor structure it implies to identify shocks from a cross-section of commodity prices since 1968. One factor captures shocks that reach commodity markets only through general-equilibrium forces; the data satisfy the model's restrictions, and commodity-related shocks have contributed only modestly to global economic fluctuations.

Version record

Jeffrey A. Frankel, "The Effect of Monetary Policy on Real Commodity Prices"

Difficulty: Advanced **Type:** Book chapter

Status/version: *Asset Prices and Monetary Policy*, University of Chicago Press (2008), 291–327; DOI 10.7208/chicago/9780226092126.003.0008.

Argues that monetary policy, through real interest rates, is an important and usually underappreciated determinant of the real prices of oil and other mineral and agricultural products. Analyses the channel first for a large country and then for a small one, and considers the reverse link: whether commodity prices should enter the price index used as a nominal anchor.

Version record

Jeffrey A. Frankel, "Effects of Speculation and Interest Rates in a "Carry Trade" Model of Commodity Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 42, 88–112 (2014); DOI 10.1016/j.jimonfin.2013.08.006.

Estimates a carry-trade model of oil and other storable commodity prices, using survey data on expected price changes and options data on perceived risk. Higher interest rates reduce inventory demand and so lower prices, while expected price gains raise inventory demand and current prices.

Version record

Q. Farooq Akram, "Commodity Prices, Interest Rates and the Dollar"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 31(6), 838–851 (2009); DOI 10.1016/j.eneco.2009.05.016.

Structural VARs on quarterly data for 1990–2007 for real prices of crude oil, food, metals and industrial raw materials. Commodity prices rise significantly when real interest rates fall, oil and industrial raw materials overshoot, a weaker dollar raises commodity prices, and interest-rate and dollar shocks account for substantial shares of commodity price fluctuations.

Version record

Yu-chin Chen and Kenneth Rogoff, "Commodity Currencies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 60(1), 133–160 (2003); DOI 10.1016/s0022-1996(02)00072-7.

For Australia, Canada and New Zealand, whose exports are heavily primary commodities, the world dollar price of those exports—largely exogenous to them—has a strong and stable effect on floating real exchange rates, especially for Australia and New Zealand, of the size standard models predict. A purchasing-power-parity puzzle remains in the residual.

Version record

Andrés Fernández, Stephanie Schmitt-Grohé and Martín Uribe, "World Shocks, World Prices, and Business Cycles: An Empirical Investigation"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 108, S2–S14 (2017); DOI 10.1016/j.jinteco.2017.01.001.

Models with a single world price such as the terms of trade find that world shocks explain under 10 per cent of domestic output movements. Letting several commodity prices transmit world disturbances, estimates on 138 countries for 1960–2015 attribute on average 33 per cent of output fluctuations to world shocks, twice that on post-2000 data.

Version record

Menzie D. Chinn and Olivier Coibion, "The Predictive Content of Commodity Futures"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 34(7), 607–636 (2014); DOI 10.1002/fut.21615.

Tests whether futures prices are unbiased and accurate predictors of later spot prices for energy, agricultural, precious and base metal commodities. Precious and base metals fail most unbiasedness tests and predict poorly, energy and agricultural futures do much better, liquidity does not explain the difference, and predictive content has declined broadly since the early 2000s.

Version record

Antonio Gargano and Allan Timmermann, "Forecasting Commodity Price Indexes Using Macroeconomic and Financial Predictors"

Difficulty: Research **Type:** Research paper

Status/version: International Journal of Forecasting, 30(3), 825–843 (2014); DOI 10.1016/j.ijforecast.2013.09.003.

Out-of-sample forecasts of commodity spot price indexes over 1947–2010. Commodity currencies have some predictive power at monthly and quarterly horizons, and industrial production growth and the investment–capital ratio at yearly horizons; multivariate methods that account for estimation error forecast best, and predictability is strongest in recessions.

Version record

John Baffes and Alain Kabundi, "Commodity Price Shocks: Order within Chaos?"

Difficulty: Research **Type:** Research paper

Status/version: Resources Policy, 83, article 103640 (2023); DOI 10.1016/j.resourpol.2023.103640.

Splits monthly prices of 27 commodities over 1970–2020 into transitory and permanent shocks with an ideal band-pass filter. The two contribute about equally overall, with permanent shocks dominating agriculture and transitory ones energy; the transitory component shows three medium-term cycles. Permanent components are non-linear where investment is irreversible (energy, metals, tree crops) and linear for annual crops, and prices subject to commodity agreements deviate persistently from trend.

Version record

14.22 Long-run commodity prices and resource scarcity

Tests of the Hotelling rule, the Prebisch–Singer hypothesis of declining relative commodity prices, long-run trends, supercycles and boom–bust episodes, and the supply responses and market power that shape long-run prices.

Margaret E. Slade and Henry Thille, "Whither Hotelling: Tests of the Theory of Exhaustible Resources"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Resource Economics, 1(1), 239–260 (2009); DOI 10.1146/annurev.resource.050708.144223.

Reviews empirical extensions and tests of the Hotelling model of optimal depletion of an exhaustible resource. Sets out the theory, compares descriptive and structural tests and their strengths and weaknesses, surveys studies of extraction under exhaustibility, and discusses the econometric pitfalls of testing the model.

Version record

Robert S. Pindyck, "The Long-Run Evolution of Energy Prices"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 20(2), 1–27 (1999); DOI 10.5547/issn0195-6574-ej-vol20-no2-1.

Uses up to 127 years of oil, coal and natural gas prices to ask how energy prices evolve and should be modelled, whether models of reversion to stochastically fluctuating trend lines help forecast 20 years or more ahead, and what this implies for investment decisions that depend on long-run prices.

Version record

Enzo R. Grilli and Maw Cheng Yang, "Primary Commodity Prices, Manufactured Goods Prices, and the Terms of Trade of Developing Countries: What the Long Run Shows"

Difficulty: Research **Type:** Research paper

Status/version: World Bank Economic Review, 2(1), 1–47 (1988); DOI 10.1093/wber/2.1.1.

Builds the Grilli–Yang commodity price index and finds that from 1900 to 1986 the prices of all primary commodities relative to manufactures fell by about 0.5 per cent a year, and of non-fuel commodities by 0.6 per cent, confirming the sign but not the size of the Prebisch trend; incomplete adjustment for quality in manufactures may exaggerate even this. The purchasing power of commodity exports and developing countries' income terms of trade nonetheless rose.

Version record

Paul Cashin and C. John McDermott, "The Long-Run Behavior of Commodity Prices: Small Trends and Big Variability"

Difficulty: Research **Type:** Research paper

Status/version: IMF Staff Papers, 49(2), 175–199 (2002); DOI 10.2307/3872481.

The Economist's industrial commodity price index for 1862–1999 shows a downward trend in real prices of about 1 per cent a year with little evidence of a break, and rising variability: larger price moves from the early 1900s and more frequent ones after Bretton Woods. The trend is small and dominated by variability, so of little practical policy relevance.

Version record

David I. Harvey, Neil M. Kellard, Jakob B. Madsen and Mark E. Wohar, "The Prebisch–Singer Hypothesis: Four Centuries of Evidence"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economics and Statistics, 92(2), 367–377 (2010); DOI 10.1162/rest.2010.12184.

Tests for trends in the relative prices of 25 primary commodities from the seventeenth to the twenty-first century, with trend tests robust to the order of integration. Eleven series have a significant downward trend over all or part of the sample, so a secular deterioration is relevant for a significant share of commodities in the very long run.

Version record

Bilge Erten and José Antonio Ocampo, "Super Cycles of Commodity Prices since the Mid-Nineteenth Century"

Difficulty: Research **Type:** Research paper

Status/version: World Development, 44, 14–30 (2013); DOI 10.1016/j.worlddev.2012.11.013.

Band-pass decomposition of real commodity prices finds four super cycles over 1865–2010, each 30 to 40 years long with amplitudes 20 to 40 per cent above or below trend. Non-oil super cycles follow world GDP and are demand-driven, while for oil causality runs the other way; each non-oil cycle has a lower mean than the last, supporting Prebisch–Singer, whereas real oil prices trend up.

Version record

David S. Jacks, "From Boom to Bust: A Typology of Real Commodity Prices in the Long Run"

Difficulty: Research **Type:** Research paper

Status/version: Cliometrica, 13(2), 201–220 (2019); DOI 10.1007/s11698-018-0173-5.

Real prices of 40 commodities from 1900 to 2015, split into long-run trends, medium-run cycles and short-run boom–bust episodes. Real commodity prices have risen modestly since 1950; cycles bring large, long-lived deviations from trend; and boom–bust episodes punctuate those cycles throughout history.

Version record

Martin Stuermer, "150 Years of Boom and Bust: What Drives Mineral Commodity Prices?"

Difficulty: Research **Type:** Research paper

Status/version: Macroeconomic Dynamics, 22(3), 702–717 (2018); DOI 10.1017/s136510051600050x.

New price and production data for copper, lead, tin and zinc from 1840 to 2014. Price fluctuations are driven mainly by demand rather than supply shocks; demand shocks move prices for up to 15 years and supply shocks for up to 5. Surges caused by rapid industrialisation recur, and prices return to declining or stable trends in the long run.

Version record

John Baffes and Alain Kabundi, "Do Supercycles Dominate Commodity Price Movements?"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 237, article 111646 (2024); DOI 10.1016/j.econlet.2024.111646.

Frequency-domain decomposition of 21 commodity prices from 1900 to 2022 into transitory, supercycle and permanent components. Four heterogeneous supercycles averaging 24 years appear, but they contribute only modestly to price variability; the permanent component contributes most, and different commodities fit Engel's law, Hotelling's rule or Schumpeterian cycles.

Version record

John Asker, Allan Collard-Wexler and Jan De Loecker, "(Mis)Allocation, Market Power, and Global Oil Extraction"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 109(4), 1568–1615 (2019); DOI 10.1257/aer.20171438.

Measures misallocation by comparing actual industry cost curves with undistorted supply curves, using field-level cost data. In global crude oil extraction over 1970–2014, misallocation is about $744 billion, of which 14 to 22 per cent is attributable to market power.

Version record

Richard G. Newell and Brian C. Prest, "The Unconventional Oil Supply Boom: Aggregate Price Response from Microdata"

Difficulty: Research **Type:** Research paper

Status/version: Energy Journal, 40(3), 1–30 (2019); DOI 10.5547/01956574.40.3.rnew.

Price responsiveness of US conventional and unconventional oil supply at the drilling, completion and production stages. Drilling is the main margin, with price elasticities of 1.3 for conventional and 1.6 for unconventional drilling; because shale wells produce about ten times more initially, the shale revolution made the supply response 13 times larger, yet a rise from $50 to $80 a barrel adds 0.6 million barrels a day after 6 months and 4.2 million after 5 years—far slower than a 30–90-day swing producer.

Version record

14.23 Price stabilisation, public stocks and trade policy

Buffer stocks, price bands and international commodity agreements analysed with the storage model; strategic oil reserves and coordinated stock releases; and export restrictions and insulating trade policy in the food price spikes of 2007–08 and 2022.

Christophe Gouel and Brian D. Wright, "Storage and the Economics of Commodity Price Stabilization"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Resource Economics, 18(1), 63–87 (2026); DOI 10.1146/annurev-resource-112624-121752.

Reassesses price stabilisation through the rational-expectations storage model. In frictionless markets private storage already smooths prices efficiently, so public intervention needs market or endowment failures, common in low-income agricultural markets. Optimal state-contingent policies are complex and crowd out private storage; narrow price bands or storage subsidies approximate them, subsidies more efficiently. Storage can support prices in open economies with high trade costs, export controls insulate domestic prices but are non-cooperative, and the redistribution involved far exceeds the efficiency gains.

Version record

Christophe Gouel, "Agricultural Price Instability: A Survey of Competing Explanations and Remedies"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Surveys, 26(1), 129–156 (2012); DOI 10.1111/j.1467-6419.2010.00634.x.

Contrasts cobweb models, in which expectation errors generate complex or chaotic dynamics, with rational-expectations models driven by real shocks; the evidence favours the latter but is not conclusive. In the rational-expectations benchmark public intervention cannot improve on the market, but market failures, especially in developing countries, may qualify this, and a sound stabilisation policy design is still lacking.

Version record

Brian D. Wright and Jeffrey C. Williams, "The Roles of Public and Private Storage in Managing Oil Import Disruptions"

Difficulty: Research **Type:** Research paper

Status/version: Bell Journal of Economics, 13(2), 341–353 (1982); DOI 10.2307/3003458.

A government storing oil against import disruptions must allow for private storage: public stocks are justified only by a distortion in the private market, such as a price ceiling the government cannot rule out. Public storage can then substantially offset the ceiling's harm, and how it should behave depends on tariff policy and other constraints and on private responses to expected public behaviour.

Version record

Lutz Kilian and Xiaoqing Zhou, "Does Drawing Down the US Strategic Petroleum Reserve Help Stabilize Oil Prices?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Applied Econometrics, 35(6), 673–691 (2020); DOI 10.1002/jae.2798.

Global oil-market models with storage demand and unanticipated changes in the SPR. Historical SPR interventions helped stabilise the oil price but modestly: releases after Iraq's 1990 invasion of Kuwait lowered the real price by about $2 a barrel after seven months.

Emergency drawdowns lower the real price, exchanges raise it in the long run, and the fiscal gains from the 2018 proposal to sell half the SPR slightly exceed the $16.6 billion projected.

Version record

International Energy Agency, IEA Member Countries to Carry Out Largest Ever Oil Stock Release amid Market Disruptions from Middle East Conflict

Difficulty: Intro **Type:** Report / research note

Status/version: IEA press release, 11 March 2026.

On 11 March 2026 the 32 IEA member countries agreed unanimously to make 400 million barrels of emergency stocks available, the agency's sixth collective action after 1991, 2005, 2011 and two in 2022. The war that began on 28 February had cut crude and product exports through the Strait of Hormuz, which carried about 20 million barrels a day in 2025, to under 10 per cent of pre-conflict levels. Members hold over 1.2 billion barrels of public emergency stocks and a further 600 million barrels of industry stocks held under government obligation.

Version record

Christopher L. Gilbert, "How to Understand High Food Prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Agricultural Economics, 61(2), 398–425 (2010); DOI 10.1111/j.1477-9552.2010.00248.x.

Agricultural price booms are better explained by common factors than by market-specific supply shocks. Granger-causality tests since 1971 point to demand growth, monetary expansion and exchange rates; there is little direct evidence for biofuel demand as the main cause of the 2007–08 rise, and index investment in agricultural futures is identified as the main channel through which macroeconomic and monetary factors raised food prices.

Version record

Derek Headey, "Rethinking the Global Food Crisis: The Role of Trade Shocks"

Difficulty: Research **Type:** Research paper

Status/version: Food Policy, 36(2), 136–146 (2011); DOI 10.1016/j.foodpol.2010.10.003.

Fundamentals moved food prices to a higher equilibrium before 2008, but prices overshot it at the June 2008 peak. Monthly data for Thailand (rice) and the United States (wheat, maize, soybeans) show large surges in export volumes before the price surges in all cases but soybeans; together with the rice export restrictions of India and Vietnam, this suggests trade shocks played a much larger role than thought.

Version record

Will Martin and Kym Anderson, "Export Restrictions and Price Insulation During Commodity Price Booms"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 94(2), 422–427 (2012); DOI 10.1093/ajae/aar105.

Variable trade barriers can stabilise one country's domestic prices, but when large shares of the market insulate, world price volatility rises and the policy fails to stabilise on average—a collective-action problem. Insulating behaviour explains about 45 per cent of the rise in world rice prices in 2006–08 and 30 per cent of that in wheat; industrial countries insulated much less than in 1972–74.

Version record

Paolo E. Giordani, Nadia Rocha and Michele Ruta, "Food Prices and the Multiplier Effect of Trade Policy"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Economics, 101, 102–122 (2016); DOI 10.1016/j.jinteco.2016.04.001.

With loss-averse citizens, governments use trade policy to shield domestic markets from large food price shocks, which makes trade policy complementary to world prices: exporters restrict and importers cut protection when prices rise, amplifying the shock. Monthly data on trade measures for 77 countries and 32 food products in 2008–11 show this multiplier effect.

Version record

Linde Götz, Thomas Glauben and Bernhard Brümmer, "Wheat Export Restrictions and Domestic Market Effects in Russia and Ukraine during the Food Crisis"

Difficulty: Research **Type:** Research paper

Status/version: Food Policy, 38, 214–226 (2013); DOI 10.1016/j.foodpol.2012.12.001.

Markov-switching vector error-correction models of wheat price transmission in Russia and Ukraine, with Germany and the US as controls, during the 2007–08 crisis. Export controls temporarily cut the integration of Russian and Ukrainian markets with world wheat markets, pushed growers' prices below equilibrium and raised instability, more so and for longer under Ukraine's quota than Russia's export tax, discouraging investment.

Version record

Joseph W. Glauber, David Laborde Debucquet and Abdullah Mamun, "From Bad to Worse: How Russia–Ukraine War-Related Export Restrictions Exacerbate Global Food Insecurity"

Difficulty: Advanced **Type:** Report / research note

Status/version: Chapter in *The Russia–Ukraine Conflict and Global Food Security*, International Food Policy Research Institute (IFPRI), 2023; DOI 10.2499/9780896294394_18.

Russia and Ukraine account for 12 per cent of calories traded worldwide. With food prices already high after COVID-19 supply disruptions and the 2021 droughts, the 2022 invasion led a number of countries to impose export restrictions, and the authors warn that growing shortages of grains and vegetable oils would push more countries to restrict trade.

Version record

Christopher L. Gilbert, "International Commodity Agreements: An Obituary Notice"

Difficulty: Advanced **Type:** Research paper

Status/version: World Development, 24(1), 1–19 (1996); DOI 10.1016/0305-750x(95)00121-r.

Reviews the international commodity agreements for products such as cocoa, coffee, natural rubber, sugar and tin, which tried to stabilise prices with buffer stocks or export quotas, and explains why their economic provisions had collapsed or lapsed by the mid-1990s.

Version record

Christophe Gouel and Sébastien Jean, "Optimal Food Price Stabilization in a Small Open Developing Country"

Difficulty: Research **Type:** Research paper

Status/version: World Bank Economic Review, 29(1), 72–101 (2015); DOI 10.1093/wber/lht018.

Optimal storage and trade policy for a small open developing country facing yield and world price shocks without insurance markets. Optimal trade policy subsidises imports and taxes exports, helping consumers at producers' expense, and import subsidies prevent spikes when stocks run out; storage policy alone hurts consumers because its benefits leak abroad, but combined with trade policy it stabilises domestic prices strongly.

Version record

14.24 Freight and weather

Subject context and qualifications.

Amir H. Alizadeh and Nikos K. Nomikos, Shipping Derivatives and Risk Management

Difficulty: Advanced **Type:** Book / monograph

Status/version: Palgrave Macmillan, 28 April 2009; publisher contents and subscription preview.

Builds shipping-market and risk-management foundations through route data, forward freight agreements, freight options, value at risk, and bunker-fuel exposure. It gives a commodity investor a coherent account of what a freight derivative hedges and why vessel class, route, voyage costs, and fuel can overwhelm an index-level signal. The publisher preview supports chapter scope, while full numerical methods would require the book.

Extended note. Version record

Manolis G. Kavussanos and Ilias D. Visvikis, Market Interactions in Returns and Volatilities between Spot and Forward Shipping Freight Markets

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance 28(8), 2015–2049, August 2004.

Uses over-the-counter forward freight agreement data to study information flow and volatility between route-specific forwards and physical spot freight. The results concern the routes and market design in its historical sample; current forward assessments, clearing, fuel adjustments, vessel mix, and voyage duration all need rechecking before using a reported lead-lag relation as a live trading signal.

Extended note. Version record

Baltic Exchange, Understanding FFAs: What Are FFAs and Why They Matter

Difficulty: Intro **Type:** Practitioner / commentary

Status/version: Baltic Exchange market explainer, 10 June 2026; official industry source, not peer-reviewed research.

States how forward freight agreements are cash-settled against Baltic spot freight indices, with no obligation to deliver a vessel, and identifies the OTC broker-to-clearing workflow and market participants. A cash-settled hedge still leaves route, vessel, fuel, port-delay, and contractual basis; the explainer does not document an individual clearinghouse's margin, collateral, or contract rulebook.

Extended note. Version record

Baltic Exchange, Guide to Market Benchmarks, version 8.8

Difficulty: Advanced **Type:** Benchmark methodology

Status/version: Official methodology version 8.8, September 2026, linked from the current Baltic methodology page; checked 4 October 2026.

Specifies route descriptions, broker/panellist submissions, assessment controls and index production used in physical freight and FFA settlement analysis. Benchmark assessments can involve expert judgment when fixtures are sparse, so an index is not a transparent

invoice for a named vessel, route deviation, fuel bill or port delay. Versioned route definitions and licensed underlying data matter for any backtest.

Extended note. Version record

UN Trade and Development (UNCTAD), Review of Maritime Transport 2025: Staying the Course in Turbulent Waters

Difficulty: Advanced **Type:** Report / research note

Status/version: UNCTAD/RMT/2025, released 24 September 2025; latest confirmed edition by 4 October 2026.

Maps trade volumes, fleet supply, freight rates, port performance, rerouting, and environmental-compliance costs in one dated physical-market reference. It is useful for stress scenarios linking commodity flows to voyage length, effective fleet capacity, bunker consumption, and port delays. Separate container, dry-bulk, tanker, and gas-shipping regimes rather than transferring a single volatility estimate across them.

Extended note. Version record

Stephen Jewson and Anders Brix, with contributions by Christine Ziehmann, Weather Derivative Valuation: The Meteorological, Statistical, Financial and Mathematical Foundations; Publisher chapter contents and online-date distinction

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, 2005 print edition; e-book online 22 September 2009.

Works from station data cleaning and climate trends through burn analysis, index and daily-temperature models, probabilistic forecasts, portfolio valuation, and non-temperature contracts. It is the broad foundation for separating meteorological uncertainty from market risk loading in a weather hedge. The identified edition is 2005, so current listed instruments and data feeds require fresh verification.

Extended note. Version record

Fred Espen Benth and Jūratė Šaltytė Benth, The Volatility of Temperature and Pricing of Weather Derivatives; Publisher article abstract and DOI

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 7(5), 553–561, 2007; full text subscription restricted.

Fits an Ornstein–Uhlenbeck temperature process with seasonal volatility to Stockholm daily observations and derives pricing dynamics for heating- and cooling-degree-day futures, cumulative-average-temperature futures, and related options. Stockholm's climate, exchange station, and sample period do not transfer mechanically to another location; trend, observation errors, microclimate mismatch, and unhedgeable temperature risk all matter when turning a model price into a hedge.

Extended note. Version record

CME Group, Weather Options Overview and Weather Futures and Options; CME HDD/CDD fact card

Difficulty: Intro **Type:** Course / learning resource

Status/version: Official CME educational contract overview, live page accessed 4 October 2026; related CME weather fact card is a separate undated/current PDF. Use current product rulebook for executable terms.

Defines exchange heating- and cooling-degree-day exposures and explains how a monthly contract cash-settles on the cumulative observed weather index. Hedge design must check the named weather station, observation source, base temperature, settlement calendar, contract multiplier, and available depth; liquidity and listed cities cannot be assumed from an educational example.

Extended note. Version record

CME Group, CME Rulebook Chapter 403: Monthly Temperature Futures

Difficulty: Advanced **Type:** Contract rulebook

Status/version: Current official Rule 403 PDF inspected 4 October 2026; document has no stated effective date, so check archived rules at trade date.

Defines listed monthly heating- and cooling-degree-day index calculation, named observation stations, the Speedwell Settlement Services processing role, local-standard-time day, contract multiplier and final-settlement timing. It makes settlement source and station identity explicit; historical NOAA GHCNd data are only a modeling proxy until reconciled to the official final index.

Extended note. Version record

NOAA National Centers for Environmental Information, 1991–2020 U.S. Climate Normals

Difficulty: Advanced **Type:** Data / specification

Status/version: Official data product first released May 2021; NCEI documents version 1.0.1 following a 2023 reissue for 23 sites. Product page accessed 4 October 2026.

Provides standardized station-level temperature and precipitation normal values, methodology, and bulk CSV access for a fixed 1991–2020 climate baseline. It supports weather-derivative station selection, seasonal comparison, and audit of trend adjustments. Keep exact station identifiers, version, missing-data treatment, and the actual daily settlement source alongside any historical hedge backtest, especially where stations move or measuring practices change.

Extended note. Version record

NOAA National Centers for Environmental Information, Global Historical Climatology Network–Daily (GHCNd)

Difficulty: Advanced **Type:** Data / documentation

Status/version: Living official daily station product, checked 4 October 2026; observations and source/quality revisions are separately dated.

Provides daily station maximum/minimum temperature and precipitation with station inventory and source/quality flags for constructing a weather-exposure proxy and checking missingness. NOAA archive-ready replacement can lag real-time feeds; a GHCNd reconstruction is not automatically the exact observation or settlement series used by a CME weather contract.

Extended note. Version record

Robin Greenwood and Samuel G. Hanson, "Waves in Ship Prices and Investment"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 130(1), 55–109 (2015); DOI 10.1093/qje/qju035.

Dry-bulk ship prices and investment overreact to demand shocks because owners neglect competitors' supply response; high prices predict low returns.

Version record

Myrto Kalouptsidi, "Time to Build and Fluctuations in Bulk Shipping"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 104(2), 564–608 (2014); DOI 10.1257/aer.104.2.564.

Structural model of shipbuilding with time-to-build: construction lags raise freight and ship-price volatility.

Version record

Manolis G. Kavussanos, Ilias D. Visvikis and David A. Menachof, "The Unbiasedness Hypothesis in the Freight Forward Market: Evidence from Cointegration Tests"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Derivatives Research, 7(3), 241–266 (2004); DOI 10.1007/s11147-004-4811-7.

Tests whether freight forward agreement prices are unbiased predictors of realised spot freight, route by route.

Version record

Sean D. Campbell and Francis X. Diebold, "Weather Forecasting for Weather Derivatives"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of the American Statistical Association, 100(469), 6–16 (2005); DOI 10.1198/016214504000001051.

Time-series models of daily temperature (trend, seasonality, seasonal volatility) for pricing HDD/CDD contracts.

Version record

Peter Alaton, Boualem Djehiche and David Stillberger, "On modelling and pricing weather derivatives"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Mathematical Finance, 9(1), 1–20 (2002); DOI 10.1080/13504860210132897.

Mean-reverting Ornstein–Uhlenbeck temperature model with seasonal volatility and pricing of degree-day options.

Version record

Francisco Pérez-González and Hayong Yun, "Risk Management and Firm Value: Evidence from Weather Derivatives"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 68(5), 2143–2176 (2013); DOI 10.1111/jofi.12061.

The 1997 introduction of weather derivatives raised the value of weather-exposed utilities that used them, a clean test of hedging and firm value.

Version record

Amiyatosh Purnanandam and Daniel Weagley, "Can Markets Discipline Government Agencies? Evidence from the Weather Derivatives Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 71(1), 303–334 (2016); DOI 10.1111/jofi.12366.

Weather-derivative prices reveal measurement errors at weather stations; market pressure improved data quality. A data-integrity lesson for index contracts.

Version record

Giulia Brancaccio, Myrto Kalouptsidi and Theodore Papageorgiou, "Geography, Transportation, and Endogenous Trade Costs"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 88(2), 657–691 (2020); DOI 10.3982/ecta15455.

Spatial model of world trade with an endogenous market for ocean shipping, estimated with vessel-movement and freight-rate data. Shipping attenuates comparative-advantage differences, creates network effects in trade costs and dampens shocks to trade flows; the model evaluates projects such as the Panama Canal and the Northwest Passage.

Version record

14.25 Emissions trading and carbon allowances

Subject context and qualifications.

Jonathan D. Rubin, A Model of Intertemporal Emission Trading, Banking, and Borrowing

Difficulty: Research **Type:** Research paper

Status/version: Journal of Environmental Economics and Management 31(3), 269–286, November 1996.

Derives permit-price and emissions paths when firms may bank and borrow allowances, and compares that setting with banking only. It provides the intertemporal economic logic behind an allowance's inventory value and explains why an ETS curve depends on future caps and compliance choices, not conventional commodity warehousing. The continuous-time least-cost benchmark abstracts from market power, uncertain enforcement, transaction costs, and political redesign.

Extended note. Version record

Jörn C. Richstein, Émile J. L. Chappin, and Laurens J. de Vries, The Market (In-)Stability Reserve for EU Carbon Emission Trading: Why It Might Fail and How to Improve It

Difficulty: Research **Type:** Research paper

Status/version: Utilities Policy 35, 1–18, August 2015; analyzes a proposed reserve design.

Uses an electricity and carbon-market simulation with endogenous investment to test backloading and a proposed Market Stability Reserve. It warns that a delayed, volume-triggered reserve can alter hedging supply and amplify allowance volatility under its assumptions. Its parameterization predates implemented MSR revisions, so it should not be presented as a forecast of the 2026 rule; current thresholds and invalidation law must be overlaid explicitly.

Extended note. Version record

European Commission, Directorate-General for Climate Action, 2025 Carbon Market Report: Report on the Functioning of the European Carbon Market in 2024; Commission report COM(2025) 735

Difficulty: Advanced **Type:** Report / research note

Status/version: European Commission report adopted 3 December 2025; evaluates 2024 functioning and developments through first-half 2025. Linked COM(2025) 735 is the underlying formal report.

Documents the operating EU ETS, emissions, auctioning, market oversight, and the addition of maritime transport using a clearly dated official record. It gives a carbon desk the policy and physical-emissions context needed before extrapolating a historical allowance price regression. The 2024 emissions reference period and the first-half 2025 update separately from its December 2025 publication date.

Extended note. Version record

European Commission, Directorate-General for Climate Action, Market Stability Reserve and 2025 Total Number of Allowances in Circulation; Dated 2026 TNAC communication

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official EU ETS rule page accessed 4 October 2026; 2025 TNAC communication published 29 May 2026 for the September 2026–August 2027 intake. A separate April 2026 invalidation amendment was only proposed, not treated as enacted.

Explains the rule-based link from circulating allowance surplus to auction withdrawals or releases, giving a concrete supply variable for modeling EU allowance curves. A trader should separately track banked private allowances, reserve holdings, invalidations, auction calendars, and live amendments. The Commission's April 2026 proposal to stop invalidation is a scenario until legally adopted; mixing proposed and operative rules would corrupt a point-in-time backtest.

Extended note. Version record

World Bank, State and Trends of Carbon Pricing 2026

Difficulty: Advanced **Type:** Report / research note

Status/version: World Bank 2026 report, DOI 10.1596/978-1-4648-2348-0; report states a 1 April 2026 data cutoff unless specified. Released report and PDF verified.

Surveys implemented and emerging emissions trading, carbon taxes, and crediting mechanisms with comparable global coverage and a visible data cutoff. It is the starting point for distinguishing a mandatory allowance, a tax price, and a voluntary or compliance credit before constructing cross-jurisdiction comparisons. Archive report edition and dashboard extract dates; do not infer that credits issued in different programs are fungible or deliverable into a given ETS.

Extended note. Version record

International Carbon Action Partnership, Emissions Trading Worldwide: ICAP Status Report 2026

Difficulty: Advanced **Type:** Report / research note

Status/version: ICAP annual status report, released 14 April 2026; full downloadable report available. Program-specific policy reference, not a peer-reviewed pricing study.

Catalogs active and planned emissions trading systems, comparing coverage, caps, allocation, price-control tools, and market development across jurisdictions. It prevents an apparently global carbon factor from conflating programs with different banking, offset, vintage, and participation rules. The report is a dated snapshot: 2026 launches and policy proposals require a later legal-status check before being treated as executable market infrastructure.

Extended note. Version record

W. David Montgomery, "Markets in Licenses and Efficient Pollution Control Programs"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Economic Theory, 5(3), 395–418 (1972); DOI 10.1016/0022-0531(72)90049-x.

Proves that a competitive market in tradable emission licences reaches any given pollution target at least total cost, whatever the initial allocation, and characterises licences defined by emissions and by ambient pollution. The theoretical foundation of cap-and-trade.

Version record

Richard Schmalensee and Robert N. Stavins, "Lessons Learned from Three Decades of Experience with Cap and Trade"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Environmental Economics and Policy, 11(1), 59–79 (2017); DOI 10.1093/reep/rew017.

Reviews the design and performance of seven major emissions trading programmes implemented over 30 years, draws lessons for future applications, and briefly covers several other proposed or implemented programmes.

Version record

A. Denny Ellerman, Claudio Marcantonini and Aleksandar Zaklan, "The European Union Emissions Trading System: Ten Years and Counting"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Environmental Economics and Policy, 10(1), 89–107 (2016); DOI 10.1093/reep/rev014.

Introduction to the EU ETS: its legislative development, the move from free allocation to auctioning and central rules, links to Kyoto and other policies, and its record on emissions, allowance prices and offsets, with the reform debate of the mid-2010s.

Version record

Grischa Perino, "New EU ETS Phase 4 Rules Temporarily Puncture Waterbed"

Difficulty: Research **Type:** Research paper

Status/version: Nature Climate Change, 8(4), 262–264 (2018); DOI 10.1038/s41558-018-0120-2.

Under the Phase 4 rules, the long-term cap of the EU ETS becomes a function of past and future market outcomes through the Market Stability Reserve. This temporarily punctures the waterbed, so overlapping policies that cut emissions can now lower total emissions, with retroactive effects on their abatement.

Version record

Simone Borghesi, Michael Pahle, Grischa Perino, Simon Quemin and Maximilian Willner, "The Market Stability Reserve in the EU Emissions Trading System: A Critical Review"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Resource Economics, 15(1), 131–152 (2023); DOI 10.1146/annurev-resource-111820-030145.

Critical review of research on the Market Stability Reserve, which adjusts allowance supply to the number in circulation, assessing it against its policy objectives with conceptual and quantitative evidence, and setting out policy implications and open questions.

Version record

Severin Borenstein, James Bushnell, Frank A. Wolak and Matthew Zaragoza-Watkins, "Expecting the Unexpected: Emissions Uncertainty and Environmental Market Design"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 109(11), 3953–3977 (2019); DOI 10.1257/aer.20161218.

Before California's cap-and-trade began, uncertainty in business-as-usual emissions and in abatement from other policies far exceeded the abatement a politically acceptable price could induce. The allowance price was therefore likely to be set by the administrative floor or ceiling, a feature probably shared by other markets.

Version record

Harrison Fell, Dallas Burtraw, Richard D. Morgenstern and Karen L. Palmer, "Soft and Hard Price Collars in a Cap-and-Trade System: A Comparative Analysis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Environmental Economics and Management, 64(2), 183–198 (2012); DOI 10.1016/j.jeem.2011.11.004.

Stochastic dynamic comparison of price collars when baseline emissions and costs are uncertain. Soft collars release a limited allowance reserve at the ceiling, hard collars an unlimited supply. A larger reserve lowers expected compliance costs with diminishing returns, so a modest soft collar gives most of the cost assurance.

Version record

Richard G Newell, William A Pizer and Daniel Raimi, "Carbon Markets 15 Years after Kyoto: Lessons Learned, New Challenges"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 27(1), 123–146 (2013); DOI 10.1257/jep.27.1.123.

Takes stock of carbon markets after eight years of trading: fewer free allowances, better handling of market-sensitive information and recognition that rule adjustments affect participants and market confidence. The emerging architecture is a set of separate trading

systems alongside other policies rather than Kyoto's integrated global market, which raises questions of linking and of comparing effort across jurisdictions.

Version record

Lawrence H. Goulder and Andrew R. Schein, "Carbon Taxes versus Cap and Trade: A Critical Review"

Difficulty: Advanced **Type:** Research paper

Status/version: Climate Change Economics, 4(3), 1350010 (2013); DOI 10.1142/s2010007813500103.

Compares a carbon tax, pure cap and trade and a hybrid with a price ceiling or floor. The instruments are equivalent along more dimensions than usually recognised and none dominates, but exogenous emissions pricing, through a tax or a hybrid, helps prevent price volatility, reduces expected policy errors under uncertainty, avoids problematic interactions with overlapping climate policies and limits wealth transfers to oil exporters.

Version record

Peter Cramton and Suzi Kerr, "Tradeable carbon permit auctions"

Difficulty: Research **Type:** Research paper

Status/version: Energy Policy, 30(4), 333–345 (2002); DOI 10.1016/s0301-4215(01)00100-8.

Argues for auctioning carbon permits rather than grandfathering them: permits required upstream at refineries, gas pipelines and coal processors, fully tradable and bankable, and sold in quarterly ascending-clock auctions for reliable price discovery. Auctions reduce tax distortions, allow flexible cost distribution, sharpen innovation incentives and avoid contentious arguments over allocating rents.

Version record

European Commission, Directorate-General for Climate Action, ETS2: buildings, road transport and additional sectors

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official programme page, last updated 2 September 2026; ETS2 fully operational from 2028.

The second EU emissions trading system covers CO2 from fuel combustion in buildings, road transport and small industry, regulating fuel suppliers upstream rather than households. All allowances are auctioned and the cap cuts covered emissions 42 per cent by 2030 against 2005. Monitoring began in 2025; the system becomes fully operational in 2028, with 30 per cent extra auction volume for 2028 and a dedicated market stability reserve that can release allowances if the price exceeds EUR 45 (2020 prices) in the first two years or rises too fast. From 2028, allowances are surrendered by 31 May of the following year; auction revenues help finance the Social Climate Fund.

Version record

14.26 Carbon allowance prices, derivatives and market participants

Allowance price drivers and structural breaks, stochastic and equilibrium models of allowance prices, futures, options, convenience yields and price discovery, and the role of banks and other financial participants. Cap-and-trade design is in Section 14.25.

Beat Hintermann, "Allowance Price Drivers in the First Phase of the EU ETS"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Environmental Economics and Management, 59(1), 43–56 (2010); DOI 10.1016/j.jeem.2009.07.002.

In Phase I of the EU ETS the allowance price rose above 30 euros and then collapsed to almost zero by mid-2007. A structural model of the allowance price, relaxed to allow delayed adjustment, shows that prices were not driven by marginal abatement costs at first, but that this was largely corrected after 2005 emissions were reported in April 2006.

Version record

Nicolas Koch, Sabine Fuss, Godefroy Grosjean and Ottmar Edenhofer, "Causes of the EU ETS Price Drop: Recession, CDM, Renewable Policies or a Bit of Everything?—New Evidence"

Difficulty: Research **Type:** Research paper

Status/version: Energy Policy, 73, 676–685 (2014); DOI 10.1016/j.enpol.2014.06.024.

EU allowance prices fell from almost 30 euros a tonne in mid-2008 to under 5 euros in mid-2013. Only economic activity and the growth of wind and solar output robustly explain price changes, with moderate interaction between the ETS and renewables policy; about 90 per cent of the variation in price changes is unexplained by abatement fundamentals, which contradicts the view that demand shocks were the main cause.

Version record

George Daskalakis, Dimitris Psychoyios and Raphael N. Markellos, "Modeling CO2 Emission Allowance Prices and Derivatives: Evidence from the European Trading Scheme"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 33(7), 1230–1241 (2009); DOI 10.1016/j.jbankfin.2009.01.001.

Studies Phase I allowance trading on Powernext, Nord Pool and the European Climate Exchange. The ban on banking allowances between phases has significant implications for futures pricing, and the paper develops a framework for pricing and hedging intra-phase and inter-phase futures and options on futures.

Version record

ICE Endex, EUA Futures Contract Specification

Difficulty: Intro **Type:** Contract / specification / guidance

Status/version: Official exchange product specification, live page accessed 4 October 2026; document current contract version and expiry at use. Not a dated research paper.

Defines a physically delivered European Union allowance futures contract, its eligible allowance, lot and delivery-account mechanics, and listed expiries. Cross-hedges against a company's actual compliance need must consider jurisdiction, surrender calendar, collateral, registry access, and the contract's current rulebook.

Extended note. Version record

Maximilian Fuchs, Johannes Stroebel and Julian Terstegge, "Carbon VIX: Carbon price uncertainty and decarbonization investments"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 183, 104328 (2026); DOI 10.1016/j.jfineco.2026.104328.

Carbon VIX: option-implied carbon-price uncertainty and its effect on decarbonisation investment.

Extended note. Version record

Emilie Alberola, Julien Chevallier and Benoît Chèze, "Price drivers and structural breaks in European carbon prices 2005–2007"

Difficulty: Research **Type:** Research paper

Status/version: Energy Policy, 36(2), 787–797 (2008); DOI 10.1016/j.enpol.2007.10.029.

Identifies two structural breaks in daily Phase I EUA prices, in April 2006 after verified 2005 emissions were disclosed and in October 2006 after the Commission announced stricter Phase II allocation, and estimates price fundamentals within each sub-period. Spot prices respond to energy prices and temperatures and also to production in the main sectors covered by the scheme.

Version record

Jan Seifert, Marliese Uhrig-Homburg and Michael Wagner, "Dynamic behavior of CO2 spot prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Environmental Economics and Management, 56(2), 180–194 (2008); DOI 10.1016/j.jeem.2008.03.003.

Builds a stochastic equilibrium model of the EU ETS to derive CO2 spot price dynamics for derivative valuation. Prices need not follow seasonal patterns, discounted prices should be martingales, and volatility should depend on time and the price level; a brief empirical check of market efficiency complements the model.

Version record

Marc S. Paolella and Luca Taschini, "An econometric analysis of emission allowance prices"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 32(10), 2022–2032 (2008); DOI 10.1016/j.jbankfin.2007.09.024.

Studies the distribution and forecastability of EU CO2 and US SO2 allowance returns and proposes a model for their stylized facts, which beats standard risk models in fit and in out-of-sample value-at-risk forecasts.

Version record

Eva Benz and Stefan Trück, "Modeling the price dynamics of CO2 emission allowances"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 31(1), 4–15 (2009); DOI 10.1016/j.eneco.2008.07.003.

Reviews the stylized facts of Phase I EUA spot returns and models them with Markov-switching and AR-GARCH specifications that capture distinct volatility phases, skewness and excess kurtosis; in- and out-of-sample forecasts support the models.

Version record

Stefan Trück and Rafał Weron, "Convenience Yields and Risk Premiums in the EU-ETS—Evidence from the Kyoto Commitment Period"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 36(6), 587–611 (2016); DOI 10.1002/fut.21780.

In the first Kyoto commitment period (2008–12), EUA futures moved from backwardation to contango with significantly negative convenience yields. Falling euro-area risk-free rates help explain deviations from cost of carry, convenience yields fall as surplus allowances and banking build up, and futures buyers pay a premium to hedge EUA volatility.

Version record

René Carmona, Max Fehr, Juri Hinz and Arnaud Porchet, "Market Design for Emission Trading Schemes"

Difficulty: Advanced **Type:** Research paper

Status/version: SIAM Review, 52(3), 403–452 (2010); DOI 10.1137/080722813.

Reviews quantitative models of emissions markets and builds an equilibrium model in which risk-neutral firms meet inelastic demand and trade permits to avoid a penalty at compliance. It proves existence of an equilibrium and uniqueness of allowance prices, reproduces Phase I EU ETS features including windfall profits, shows weaknesses of tax and subsidy alternatives and proposes a relative allocation scheme with smaller windfalls, illustrated with the Texas electricity market.

Version record

Steffen Hitzemann and Marliese Uhrig-Homburg, "Equilibrium Price Dynamics of Emission Permits"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 53(4), 1653–1678 (2018); DOI 10.1017/s0022109018000297.

Stochastic equilibrium model in which emission permits are highly non-linear contingent claims on economy-wide emissions, part investment asset and part consumption asset. It predicts the dynamics and volatility structure of permit prices, the forward curve and option prices, and evidence from existing markets matches the stylized facts.

Version record

Johanna Cludius and Regina Betz, "The Role of Banks in EU Emissions Trading"

Difficulty: Research **Type:** Research paper

Status/version: The Energy Journal, 41(2), 275–300 (2020); DOI 10.5547/01956574.41.2.jclu.

Combines interviews with transaction data from the first EU ETS trading period to study banks' role. Banks handle a large and rising share of transactions and provide trading services; large firms with trading experience deal with banks and exchanges, while smaller firms rely on brokers, especially to sell allowances. Rules on non-regulated participants should go together with market oversight.

Version record

Simon Quemin and Michael Pahle, "Financials threaten to undermine the functioning of emissions markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Nature Climate Change, 13(1), 22–31 (2023); DOI 10.1038/s41558-022-01560-w.

Perspective on financial participants in allowance markets: they perform necessary functions, but excessive speculation may undermine market functioning, and tighter caps attract more speculators and investors. The authors propose preliminary elements of a diagnostic toolbox for the scale and effects of speculation and apply it to the EU ETS.

Version record

Daniel Rittler, "Price discovery and volatility spillovers in the European Union emissions trading scheme: A high-frequency analysis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 36(3), 774–785 (2012); DOI 10.1016/j.jbankfin.2011.09.009.

High-frequency analysis of Phase II EUA spot and futures prices. Common-factor weights, Hasbrouck information shares and Granger tests show that the futures market leads long-run price discovery, increasingly so over time, and that volatility spills over mainly from futures to spot.

Version record

Marc Chesney and Luca Taschini, "The Endogenous Price Dynamics of Emission Allowances and an Application to CO2 Option Pricing"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 19(5), 447–475 (2012); DOI 10.1080/1350486x.2011.639948.

Derives permit price dynamics endogenously from the dynamic optimisation of firms with stochastic emissions, asymmetric information, banking and borrowing. The discounted permit price is a martingale, the model is solved numerically, and a closed-form formula prices European options.

Version record

European Securities and Markets Authority, Final Report: Emission allowances and associated derivatives (ESMA70-445-38)

Difficulty: Advanced **Type:** Report / research note

Status/version: ESMA final report, 28 March 2022; price data to 4 March 2022.

ESMA's analysis of EU carbon trading, requested by the Commission after the 2021 price rise. Using EMIR, MiFIR, position, auction and Union Registry data, it finds no major abnormality: prices and volatility followed fundamentals. Most secondary trading, EUR 57 billion a month in June–December 2021, is in derivatives, mainly the front December future; compliance entities hold long hedges against investment firms that are short and hold allowances in registry trading accounts, funds' positions are limited and high-frequency firms hold small net positions. It recommends position-management controls on venues trading EUA derivatives, better position reporting and a registry structure that identifies account holders.

Version record

14.27 Carbon cost pass-through, leakage and border adjustment

Pass-through of allowance costs to electricity prices, emissions leakage and how to measure it, free allocation and output-based relief, the effect of carbon pricing on regulated firms, and border carbon adjustment, including the EU CBAM.

European Commission, Directorate-General for Taxation and Customs Union, Carbon Border Adjustment Mechanism (CBAM): definitive regime and price of CBAM certificates

Difficulty: Advanced **Type:** Report / research note

Status/version: European Commission CBAM pages under Regulation (EU) 2023/956; definitive regime from 1 January 2026; retrieved 13 October 2026.

Since 1 January 2026, EU importers of more than 50 tonnes a year of cement, iron and steel, aluminium, fertilisers, electricity or hydrogen must be authorised CBAM declarants, declare embedded emissions and surrender certificates, net of carbon prices paid abroad. Certificate prices follow EU ETS auction clearing prices: quarterly averages in 2026 (EUR 75.36, 75.28 and 82.32 for the first three quarters) and weekly averages from 2027, with purchases on a common platform from February 2027.

Version record

Ralf Martin, Mirabelle Muûls and Ulrich J. Wagner, "The Impact of the European Union Emissions Trading Scheme on Regulated Firms: What Is the Evidence after Ten Years?"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Environmental Economics and Policy, 10(1), 129–148 (2016); DOI 10.1093/reep/rev016.

Reviews ex post evaluations of the EU ETS's effects on regulated industrial and power firms: carbon dioxide emissions, economic performance and competitiveness, and innovation, with the main gaps for future research.

Version record

Patrick Bayer and Michaël Aklin, "The European Union Emissions Trading System Reduced CO2 Emissions Despite Low Prices"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the National Academy of Sciences, 117(16), 8804–8812 (2020); DOI 10.1073/pnas.1918128117.

A carbon market can cut emissions despite low prices if it is a credible institution expected to tighten. Counterfactual sectoral emissions show the EU ETS saved about 1.2 billion tonnes of CO2 in 2008–16, about 3.8 per cent, with larger cuts in covered sectors.

Version record

Diego R. Känzig, "The Unequal Economic Consequences of Carbon Pricing"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 31221, May 2023; DOI 10.3386/w31221.

Identifies carbon policy shocks from high-frequency EU allowance futures moves around regulatory events. A tighter carbon policy raises energy prices, cuts emissions and spurs green innovation but lowers activity, hurting poorer households most through energy costs and larger income losses.

Extended note. Version record

Jos Sijm, Karsten Neuhoff and Yihsu Chen, "CO2 cost pass-through and windfall profits in the power sector"

Difficulty: Research **Type:** Research paper

Status/version: Climate Policy, 6(1), 49–72 (2006); DOI 10.1080/14693062.2006.9685588.

Estimates how much EU ETS allowance costs power producers passed into electricity prices: 40–100 per cent of CO2 costs, or EUR 3–18/MWh, depending on the carbon intensity of the marginal plant and market factors. With free allocation this produced substantial windfall profits, so the authors recommend auctioning allowances to the power sector.

Version record

Natalia Fabra and Mar Reguant, "Pass-Through of Emissions Costs in Electricity Markets"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 104(9), 2872–2899 (2014); DOI 10.1257/aer.104.9.2872.

Measures the pass-through of emissions costs to electricity prices with reduced-form and structural estimates based on optimal bidding and rich micro-level data. Pass-through is almost complete: with high-frequency auctions and highly inelastic demand, firms have weak incentives to adjust markups after a cost shock, and price adjustment is cheap.

Version record

Beat Hintermann, "Pass-Through of CO2 Emission Costs to Hourly Electricity Prices in Germany"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Association of Environmental and Resource Economists, 3(4), 857–891 (2016); DOI 10.1086/688486.

Identifies the pass-through of allowance costs to hourly German wholesale prices by building a fossil supply curve for every hour, finding the marginal generator and using its fuel and allowance costs. Pass-through is complete or nearly complete on average but varies along the load curve, which supports ending free allocation to German power generation.

Version record

Meredith L Fowlie, "Incomplete Environmental Regulation, Imperfect Competition, and Emissions Leakage"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Economic Policy, 1(2), 72–112 (2009); DOI 10.1257/pol.1.2.72.

Models emissions leakage when regulation covers only some producers in an imperfectly competitive industry and simulates CO2 regulation of California's electricity sector. Exempting out-of-state producers achieves about one-third of the emissions reductions of complete regulation at more than twice the cost per tonne.

Version record

Meredith Fowlie, Mar Reguant and Stephen P. Ryan, "Market-Based Emissions Regulation and Industry Dynamics"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 124(1), 249–302 (2016); DOI 10.1086/684484.

Dynamic structural model of the US cement industry under market-based emissions policies. Regulation worsens the distortion from domestic market power, and leakage through trade offsets domestic reductions, so fully internalising the externality can reduce welfare; designs that mitigate market power and leakage deliver gains when carbon damages are high.

Version record

Ralf Martin, Mirabelle Muûls, Laure B. de Preux and Ulrich J. Wagner, "Industry Compensation under Relocation Risk: A Firm-Level Analysis of the EU Emissions Trading Scheme"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 104(8), 2482–2508 (2014); DOI 10.1257/aer.104.8.2482.

Efficient compensation for relocation risk equalises firms' marginal relocation probabilities weighted by the damage of relocation. Applied with firm-level data to EU ETS free allocation for carbon-intensive and trade-exposed industries, current rules substantially overcompensate for leakage risk; efficient allocation would halve the aggregate risk of job loss without raising total compensation.

Version record

Helene Naegele and Aleksandar Zaklan, "Does the EU ETS cause carbon leakage in European manufacturing?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Environmental Economics and Management, 93, 125–147 (2019); DOI 10.1016/j.jeem.2018.11.004.

Reviews how leakage and pollution-haven effects are identified, then tests whether EU ETS costs caused leakage in European manufacturing using carbon and value embodied in trade from the Global Trade Analysis Project. Finds no evidence of carbon leakage.

Version record

Antoine Dechezleprêtre, Caterina Gennaioli, Ralf Martin, Mirabelle Muûls and Thomas Stoerk, "Searching for carbon leaks in multinational companies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Environmental Economics and Management, 112, 102601 (2022); DOI 10.1016/j.jeem.2021.102601.

Uses Carbon Disclosure Project data on the regional emissions of 1,122 multinationals, 261 of them regulated by the EU ETS, over 2007–14. Finds no displacement of emissions from Europe to the rest of the world, including countries with lax climate policies and within energy-intensive firms: modest carbon-price differences did not induce leakage.

Version record

Jonathan Colmer, Ralf Martin, Mirabelle Muûls and Ulrich J Wagner, "Does Pricing Carbon Mitigate Climate Change? Firm-Level Evidence from the European Union Emissions Trading System"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 92(3), 1625–1660 (2025); DOI 10.1093/restud/rdae055.

With administrative data, the EU ETS led regulated manufacturing firms to cut CO2 emissions by 14–16 per cent with no detectable contraction in economic activity and no outsourcing to unregulated firms or markets. Firms made targeted investments that lowered emissions intensity, consistent with a model in which carbon pricing triggers fixed-cost energy-saving investment.

Version record

Christoph Böhringer, Carolyn Fischer, Knut Einar Rosendahl and Thomas Fox Rutherford, "Potential impacts and challenges of border carbon adjustments"

Difficulty: Advanced **Type:** Research paper

Status/version: Nature Climate Change, 12(1), 22–29 (2022); DOI 10.1038/s41558-021-01250-z.

Review of border carbon adjustments, which apply the domestic carbon price to emissions embodied in traded goods, against five aims: leakage reduction, competitiveness, cost-effectiveness, equity and international cooperation. Legal and practical constraints on actual schemes can substantially reduce their viability.

Version record

Kimberly A. Clausing and Catherine Wolfram, "Carbon Border Adjustments, Climate Clubs, and Subsidy Races When Climate Policies Vary"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 37(3), 137–162 (2023); DOI 10.1257/jep.37.3.137.

Distinguishes two spillovers of uneven climate policy: free-riding by less ambitious countries, and the disadvantage of carbon-intensive producers in carbon-pricing jurisdictions against those in subsidising ones. Border adjustments and climate clubs respond, but no single policy handles both when approaches diverge; they could encourage upward harmonisation but risk subsidy races.

Version record

Frederic Branger, Philippe Quirion and Julien Chevallier, "Carbon Leakage and Competitiveness of Cement and Steel Industries Under the EU ETS: Much Ado About Nothing"

Difficulty: Research **Type:** Research paper

Status/version: The Energy Journal, 37(3), 109–136 (2016); DOI 10.5547/01956574.37.3.fbra.

Tests for competitiveness-driven operational leakage in EU cement and steel during EU ETS Phases I and II by relating net imports to domestic and foreign demand and the carbon price. Demand drives trade flows and the carbon price has no significant effect, so there is no evidence of leakage, at least in the short run.

Version record

Meredith L. Fowlie and Mar Reguant, "Mitigating Emissions Leakage in Incomplete Carbon Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the Association of Environmental and Resource Economists, 9(2), 307–343 (2022); DOI 10.1086/716765.

Derives industry-specific leakage-risk measures and calibrates output-based subsidies to them, using US energy-price variation as a proxy for a domestic carbon price. Without mitigation leakage is substantial; targeted output-based subsidies cut it sharply, while targeting by emissions intensity and trade exposure, as is common, delivers a small fraction of the benefit at significant cost.

Version record

14.28 Voluntary carbon credits and offset integrity

Project-based voluntary credits: integrity frameworks, baselines and additionality, over-crediting evidence, permanence, corporate demand and pricing, double counting under Article 6 of the Paris Agreement, and the regulation of credit derivatives. Renewable-energy, fuel and biodiversity claims are in Section 14.29.

Integrity Council for the Voluntary Carbon Market, Core Carbon Principles and Assessment Framework, Version 2; Current CCP program and methodology assessments

Difficulty: Intro **Type:** Contract / specification / guidance

Status/version: January 2024 Version 2 framework, official ICVCM PDF. Assessment-status table was updated 4 August 2026; distinguish static criteria from changing decisions.

Defines integrity criteria for carbon crediting programs and methodology categories, including additionality, permanence, robust quantification, and governance. It gives credit researchers a checklist for eligibility and claim quality before considering price spreads between projects or vintages. A desk must also inspect serial number, registry retirement, host-country authorization, reversal liability, and any buyer claim rules.

Extended note. Version record

Raphael Calel, Jonathan Colmer, Antoine Dechezleprêtre and Matthieu Glachant, "Do Carbon Offsets Offset Carbon?"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Applied Economics, 17(1), 1–40 (2025); DOI 10.1257/app.20230052.

New method for identifying wasted subsidies, applied to more than 1,000 Indian wind farms in the Clean Development Mechanism. At least 52 per cent of approved offsets went to projects that would very likely have been built anyway, and selling them to regulated polluters substantially raised global emissions.

Version record

Thales A. P. West, Sven Wunder, Erin O. Sills, Jan Börner, Sami W. Rifai, Alexandra N. Neidermeier, Gabriel P. Frey and Andreas Kontoleon, "Action Needed to Make Carbon Offsets from Forest Conservation Work for Climate Change Mitigation"

Difficulty: Research **Type:** Research paper

Status/version: Science, 381(6660), 873–877 (2023); DOI 10.1126/science.ade3535.

Synthetic-control estimates for 26 voluntary avoided-deforestation project sites in six countries: most did not significantly reduce deforestation, and those that did reduced it far less than claimed, because ex ante baselines exceeded observed counterfactuals. Baseline methodologies need urgent revision.

Version record

Benedict S. Probst, Malte Toetzke, Andreas Kontoleon, Laura Díaz Anadón, Jan C. Minx, Barbara K. Haya, Lambert Schneider, Philipp A. Trotter, Thales A. P. West, Annelise Gill-Wiehl and Volker H. Hoffmann, "Systematic Assessment of the Achieved Emission Reductions of Carbon Crediting Projects"

Difficulty: Research **Type:** Research paper

Status/version: Nature Communications, 15(1), article 9562 (2024); DOI 10.1038/s41467-024-53645-z.

Synthesises rigorous studies of 2,346 crediting projects covering nearly a billion tonnes of credits. Less than 16 per cent of credits represent real reductions: 11 per cent for cookstoves, 16 per cent for SF6 destruction, 25 per cent for avoided deforestation, 68 per cent for HFC-23, and none significant for wind and improved forest management.

Version record

Grayson Badgley, Jeremy Freeman, Joseph J. Hamman, Barbara Haya, Anna T. Trugman, William R. L. Anderegg and Danny Cullenward, "Systematic Over-crediting in California's Forest Carbon Offsets Program"

Difficulty: Research **Type:** Research paper

Status/version: Global Change Biology, 28(4), 1433–1445 (2022); DOI 10.1111/gcb.15943.

California's forest offset protocol credits projects against coarse regional carbon averages, inviting adverse selection. Project records and forest inventories show over-crediting of about 30 million tCO2e, 29 per cent of the credits analysed, worth about $410 million.

Version record

Tom Swinfield, Abby Williams, David Coomes, et al., Learning Lessons from Over-Crediting to Ensure Additionality in Forest Carbon Credits

Difficulty: Research **Type:** Research paper

Status/version: Nature Communications 17, article 3944, published 11 April 2026

Synthesizes six independent ex-post evaluations of 44 first-generation REDD+ projects and traces large certified-versus-estimated credit differences to reference-area selection and ex-ante baseline modeling. It gives a concrete causal diligence procedure for forest credits beyond accepting a registry's tonnage at face value. The sample and legacy methodologies should not be extrapolated mechanically to all current credits or removals, and uncertainty in counterfactual measurement remains material.

Extended note. Version record

Andrew Balmford, Srinivasan Keshav, Frank Venmans, et al., Realizing the Social Value of Impermanent Carbon Credits; Author-hosted abstract and PDF

Difficulty: Research **Type:** Research paper

Status/version: Nature Climate Change 13, 1172–1178, November 2023 issue; online 30 October 2023

Frames a nature-based credit's permanence as additional climate benefit through time and includes the social cost of possible future reversal in valuation. That turns a binary permanent-versus-temporary label into a time path requiring monitoring and conservative estimates. The framework estimates social benefit, not an exchange-traded clearing price; any financial hedge needs explicit legal responsibility for reversals, buffer-pool rules, discounting, and verification of project-specific survival assumptions.

Extended note. Version record

Matthias Rodemeier, "Willingness to Pay for Carbon Mitigation: Field Evidence from the Market for Carbon Offsets"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(1), 1–29 (2026); DOI 10.1093/rfs/hhaf054.

Field experiment: consumers respond to offset prices but are insensitive to offset impact, implying low willingness to pay for actual mitigation.

Version record

Lambert Schneider, Maosheng Duan, Robert Stavins, Kelley Kizzier, Derik Broekhoff, Frank Jotzo, Harald Winkler, Michael Lazarus, Andrew Howard and Christina Hood, "Double counting and the Paris Agreement rulebook"

Difficulty: Advanced **Type:** Research paper

Status/version: Science, 366(6462), 180–183 (2019); DOI 10.1126/science.aay8750.

Policy forum written before the December 2019 climate conference, when the Article 6 rules for international carbon markets were still missing from the Paris rulebook. Explains why double counting, using the same emission reduction toward more than one mitigation target, blocked agreement and threatens environmental integrity, and sets out the elements of a robust accounting outcome.

Version record

Antonio Bento, Ravi Kanbur and Benjamin Leard, "On the importance of baseline setting in carbon offsets markets"

Difficulty: Research **Type:** Research paper

Status/version: Climatic Change, 137(3-4), 625–637 (2016); DOI 10.1007/s10584-016-1685-2.

Models offsets within a cap-and-trade programme in which projects may be over- or under-credited. Stringent baselines let under-credited reductions balance the emissions effect of over-credited offsets without losing much offset supply; with high business-as-usual uncertainty or low carbon prices, baselines may need to sit below half of predicted business-as-usual emissions.

Version record

Vishal V. Agrawal, Soudipta Chakraborty and Şafak Yücel, "Additionality of Carbon Offsets: Project-Specific vs. Standardized Baselines"

Difficulty: Research **Type:** Research paper

Status/version: Manufacturing & Service Operations Management, articles in advance (2026); DOI 10.1287/msom.2025.0755.

Game between a registry and developers whose project revenue is private information, comparing project-specific and standardised baselines. Project-specific baselines can yield fewer non-additional offsets but smaller emissions reductions, so minimising non-additional credits is the wrong objective; registries may prefer project-specific baselines even when they produce more non-additional offsets, and their preference usually matches corporate buyers'.

Version record

Barbara K. Haya, Samuel Evans, Letty Brown, Jacob Bukoski, Van Butsic, Bodie Cabiyo, Rory Jacobson, Amber Kerr, Matthew Potts and Daniel L. Sanchez, "Comprehensive review of carbon quantification by improved forest management offset protocols"

Difficulty: Research **Type:** Research paper

Status/version: Frontiers in Forests and Global Change, 6, article 958879 (2023); DOI 10.3389/ffgc.2023.958879.

Reviews improved-forest-management offset protocols, 293 projects producing 11 per cent of voluntary registry credits, against the science on additionality, baselines, leakage, durability and forest carbon accounting. Protocols deviate in ways that risk significant over-crediting; more conservative baselines reduce but do not resolve the risk.

Version record

Annelise Gill-Wiehl, Daniel M. Kammen and Barbara K. Haya, "Pervasive over-crediting from cookstove offset methodologies"

Difficulty: Research **Type:** Research paper

Status/version: Nature Sustainability, 7(2), 191–202 (2024); DOI 10.1038/s41893-023-01259-6.

Compares five cookstove offset methodologies with the literature and the authors' own analysis. Assumptions on non-renewable biomass, fuel conversion, stove adoption and use, fuel consumption, stove stacking, rebound and emission factors are misaligned; the project sample is over-credited 9.2 times, and Gold Standard's metered methodology comes closest (1.5 times).

Version record

Gregory Trencher, Sascha Nick, Jordan Carlson and Matthew Johnson, "Demand for low-quality offsets by major companies undermines climate integrity of the voluntary carbon market"

Difficulty: Research **Type:** Research paper

Status/version: Nature Communications, 15(1), article 6863 (2024); DOI 10.1038/s41467-024-51151-w.

Examines the offsets retired by the 20 largest corporate buyers in the voluntary market over 2020–23 by project type, age, price and country. 87 per cent carry a high risk of not delivering real and additional reductions, mostly from forest-conservation and renewable-energy projects, and most fail industry norms on age and location: demand for cheap, low-quality credits sustains the integrity problem.

Version record

Florian Berg, Marco Ceccarelli, Florian Heeb, Alexey Ivashchenko, Roberto Rigobon and Remco C. J. Zwinkels, "The Market for Voluntary Carbon Offsets"

Difficulty: Research **Type:** Research paper

Status/version: SAFE Working Paper No. 462, Leibniz Institute for Financial Research SAFE (December 2025); DOI 10.2139/ssrn.5822702.

Uses a leading dealer's proprietary sales data to study voluntary-market prices, which range from a few cents to $100 a tonne and depend on project, buyer and trade characteristics rather than a common carbon value. Credits from the least reliable technologies with non-carbon co-benefits cost twice as much as trusted industrial ones; buyers in low-emission industries, rich countries and large firms pay premia, while heavy polluters do not; volume and relationship discounts are typical of OTC markets. The launch of voluntary-credit futures created premia for credits eligible for delivery into expiring contracts.

Version record

Commodity Futures Trading Commission, Commission Guidance Regarding the Listing of Voluntary Carbon Credit Derivative Contracts; withdrawal notice

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: 89 FR 83378, 15 October 2024; withdrawn as of 10 September 2025 (90 FR 44321, 15 September 2025).

Non-binding guidance for US exchanges listing voluntary-carbon-credit derivatives. Contract terms should name the eligible crediting programmes and consider credit quality (transparency, additionality, permanence and reversal risk, robust quantification), delivery (programme governance, registry tracking, no double counting) and the programme's validation and verification as inspection. The Commission withdrew it in September 2025, saying it added little to the existing core principles and singled out one class of contracts. Read it as a checklist of the quality terms a credit contract must specify, not as current regulation.

Version record

European Commission, Directorate-General for Climate Action, 2040 climate target

Difficulty: Intro **Type:** Report / research note

Status/version: Official policy page; amendment to the European Climate Law adopted in March 2026.

In March 2026 the EU amended its Climate Law to cut net greenhouse-gas emissions 90 per cent by 2040 against 1990: 85 per cent domestically and up to 5 per cent through high-quality international carbon credits, with a limited role for such credits in the second half of the 2030s and domestic permanent removals admitted to the EU ETS. A potential source of sovereign demand for Paris Agreement Article 6 credits; post-2030 sectoral legislation will set the rules.

Version record

14.29 Renewable-energy, fuel and biodiversity claims

Center for Resource Solutions / Green-e, Renewable Energy Standard for Canada and the United States; certification criteria

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official programme page identifies North America standard v4.6, updated 11 August 2026.

Introduces renewable-electricity attributes as claims separate from the physical MWh and from carbon offsets. Study eligible generation, certificate tracking, ownership, retirement and double counting, then distinguish voluntary purchases from a jurisdiction's compliance obligations. The programme links the North American standard, approved tracking systems and separate international market standards. Build a generator–registry–buyer–retirement ledger and test how geography and generation vintage change eligibility. This is a certification framework; neither programme approval nor electricity production alone establishes a certificate's executable price or transferable rights in every market.

Version record

Center for Resource Solutions / Green-e, Annual Verification Reports

Difficulty: Advanced **Type:** Data / specification

Status/version: Latest report listed on inspected index: 2025 Green-e Verification Report (2024 Data); inspected 4 October 2026.

Adds an empirical source for the volume, generation mix, geography and customers of certified renewable-electricity transactions. Keep report publication year separate from the observation year and distinguish certified programme activity from the full voluntary or compliance market. The archive supports comparisons across annual vintages; reconcile any changed scope or categories before treating growth as a price or demand signal. The inspected index establishes report identity and coverage, not a trade-level panel. Use actual registry records and executable quotes for settlement or return calculations rather than inventing prices from aggregate certificate volumes.

Version record

NEPOOL Generation Information System, Public Reports

Difficulty: Advanced **Type:** Data / specification

Status/version: Living registry report index inspected 4 October 2026; pin individual reporting periods.

Adds the registry reporting layer behind renewable-energy certificate accounting: issuance by period and fuel, aggregate transfers, system mix and residual mix. The public reports exclude account-holder-specific data, and generator-production reports mask participant identities. Reconcile the definitions and period boundaries before comparing these counts with Green-e programme volumes. Residual certificates, issuance and transfers are different states, not interchangeable measures of retirement or demand. The separate CPEC reports concern a different certificate category. Use these reports for aggregate reconciliation; a specific REC claim still requires its serial, ownership and retirement evidence.

Version record

U.S. Environmental Protection Agency, Renewable Identification Numbers (RINs) under the Renewable Fuel Standard Program

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official lifecycle and transaction reference, inspected 4 October 2026.

Explains the credit lifecycle linking renewable-fuel production, assigned credits, separation, transfers and retirement for compliance. An assigned RIN travels with fuel; a separated RIN can be traded independently. Build a ledger by fuel category, vintage, owner and transaction state, then reconcile generation and retirement with the applicable annual obligation. This distinguishes the compliance claim from both physical fuel exposure and an electricity REC. Current eligibility, banking and obligation rules require the actual regulation; this introductory source is not a complete legal or trading specification.

Version record

U.S. Environmental Protection Agency, RIN Trades and Price Information; public RFS data

Difficulty: Advanced **Type:** Data / specification

Status/version: Weekly volume-weighted separated-RIN price information and public RFS data; inspected 4 October 2026.

Provides a public empirical bridge from compliance-credit mechanics to observed transactions. The price chart uses historical weekly volume-weighted separated-RIN prices by transfer date; related EMTS reports cover generation, holdings, separation and retirement. Reconcile category, vintage, units, transfer date and release timing before comparing prices with physical fuel or policy changes. Aggregates are not an executable bid–ask history and can hide within-week dispersion and participant composition. Preserve downloaded vintages and document data gaps before calculating a hypothetical strategy's turnover, liquidity or net returns.

Version record

U.S. Environmental Protection Agency, Final Renewable Fuel Standards for 2026 and 2027

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final Set 2 rule; March 2026 regulatory impact analysis; official summary updated 9 July 2026.

Pairs the credit ledger with dated policy inputs for renewable-fuel obligations. Read the final volume requirements and linked regulatory impact analysis, retaining distinctions between base volumes, exemption reallocations, categories and vintages. The rule also changes programme eligibility, including removal of renewable electricity as a qualifying fuel. A scenario should recompute the obligation and available credit supply under stated rules, then separate that accounting effect from any assumed price response. The official summary and final-rule links establish status; they do not prove a particular credit is eligible or forecast future rule changes.

Version record

International Advisory Panel on Biodiversity Credits, Framework for High Integrity Biodiversity Credit Markets

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: October 2024 framework and current official explanatory page; inspected 4 October 2026.

Defines a project-finance path for biodiversity outcomes with measurement, additionality, durability, stewardship and governance. The framework distinguishes contributions, insetting and tightly constrained local compensation; biodiversity outcomes are not interchangeable global units. It does not support secondary markets at this stage. Analyse a project's baseline, delivery milestones, verification costs, benefit sharing and reversal exposure before discussing a cash-flow claim. This provides a framework for evaluating an emerging financing instrument, not evidence of a liquid asset class, a universal offset right or an established investable return index.

Version record

Biodiversity Credit Alliance, IAPB and World Economic Forum, High-level Principles to Guide the Biodiversity Credit Market; May 2025 white paper

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: May 2025 principles; prototype Assessment Matrix v1.0 guidance dated September 2025.

Turns broad biodiversity-market claims into questions about outcome integrity, equitable participation and governance. Compare a selected methodology against the principles, identifying which facts require field measurement, independent verification or enforceable project terms. The associated matrix is a prototype for learning and alignment assessment; it is not proof of certification or a valuation model. Pair with the IAPB framework and a specific project's evidence, then disclose unavailable baseline and cash-flow data. A carbon-credit label, conservation activity or promotional target alone does not establish a durable biodiversity outcome.

Version record

Caroline Flammer, Thomas Giroux and Geoffrey M. Heal, "Biodiversity finance"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 164, 103987 (2025); DOI 10.1016/j.jfineco.2024.103987.

Framework and evidence for biodiversity finance through private and blended capital structures, with outcomes of conservation finance projects.

Version record

Gabriel E. Lade, C.-Y. Cynthia Lin Lawell and Aaron Smith, "Policy Shocks and Market-Based Regulations: Evidence from the Renewable Fuel Standard"

Difficulty: Advanced **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 100(3), 707–731 (2018); DOI 10.1093/ajae/aax097.

Event study of RIN prices around EPA policy announcements; regulatory news moves RIN values sharply.

Version record

Christopher R. Knittel, Ben S. Meiselman and James H. Stock, "The Pass-Through of RIN Prices to Wholesale and Retail Fuels under the Renewable Fuel Standard"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of the Association of Environmental and Resource Economists, 4(4), 1081–1119 (2017); DOI 10.1086/692071.

RIN costs pass through quickly to wholesale fuel spreads and more slowly to retail prices.

Version record

Daan Hulshof, Catrinus Jepma and Machiel Mulder, "Performance of markets for European renewable energy certificates"

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Policy, 128, 697–710 (2019); DOI 10.1016/j.enpol.2019.01.051.

Assesses liquidity and price formation in the European guarantee-of-origin (renewable certificate) market.

Version record

Antonio M. Bento, Teevrat Garg and Daniel Kaffine, "Emissions reductions or green booms? General equilibrium effects of a renewable portfolio standard"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Environmental Economics and Management, 90, 78–100 (2018); DOI 10.1016/j.jeem.2018.05.006.

Renewable portfolio standards can raise total generation through REC subsidies, muting emissions effects.

Version record

Richard Schmalensee, "Evaluating Policies to Increase Electricity Generation from Renewable Energy"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Environmental Economics and Policy, 6(1), 45–64 (2012); DOI 10.1093/reep/rer020.

Reviews US and EU renewable-electricity policy. Who bears short-run subsidy costs depends on industry organisation; renewable portfolio standards may carry less long-run social risk than feed-in tariffs; the EU renewables programme will not minimise the cost of its goals; and US state portfolio standards, though market-based, cost more than necessary, as a detailed account of renewable-energy-credit markets and their shortcomings shows.

Version record

Michael Coulon, Javad Khazaei and Warren B. Powell, "SMART-SREC: A stochastic model of the New Jersey solar renewable energy certificate market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Environmental Economics and Management, 73, 13–31 (2015); DOI 10.1016/j.jeem.2015.05.004.

SMART-SREC, a stochastic model of solar renewable energy certificate prices that builds on carbon-pricing models and lets generation respond to prices. Calibrated and backtested on New Jersey's volatile market, it reproduces historical dynamics, and simulations show how regulatory parameters drive prices.

Version record

Scott H. Irwin, Kristen McCormack and James H. Stock, "The Price of Biodiesel RINs and Economic Fundamentals"

Difficulty: Research **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 102(3), 734–752 (2020); DOI 10.1002/ajae.12014.

Prices the D4 biomass-based-diesel RIN with option-pricing theory from two fundamentals: the biodiesel–diesel price spread and the status of the biodiesel blenders' tax credit. The fundamental price tracks actual D4 prices closely, so RIN volatility reflects the design of the Renewable Fuel Standard and the US fuel-supply system; the D4 price sets a ceiling on the ethanol (D6) RIN.

Version record

Stephen P Holland, Jonathan E Hughes and Christopher R Knittel, "Greenhouse Gas Reductions under Low Carbon Fuel Standards?"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Economic Policy, 1(1), 106–146 (2009); DOI 10.1257/pol.1.1.106.

A low carbon fuel standard limits the carbon intensity of fuels, which reduces high-carbon fuel output but raises low-carbon output and can increase emissions; it cannot be efficient. A simulated national standard on gasoline and ethanol cuts emissions at large cost, $307–2,272 per tonne of CO2 on average, against $60–868 for a cost-effective policy.

Version record

Sonia Yeh, Julie Witcover, Gabriel E. Lade and Daniel Sperling, "A review of low carbon fuel policies: Principles, program status and future directions"

Difficulty: Advanced **Type:** Research paper

Status/version: Energy Policy, 97, 220–234 (2016); DOI 10.1016/j.enpol.2016.07.029.

Reviews low carbon fuel standards: an intensity standard implicitly taxes emissions and subsidises output, so it is inferior to a carbon tax in a first-best world. Covers design features and programmes in California, the EU, British Columbia and Oregon; in California the alternative-fuel share rose 30 per cent in 2011–15 and credit prices exceeded $100 in the first half of 2016.

Version record

Gabriel E. Lade and C.-Y. Cynthia Lin Lawell, "The Design of Renewable Fuel Mandates and Cost Containment Mechanisms"

Difficulty: Research **Type:** Research paper

Status/version: Environmental and Resource Economics, 79(2), 213–247 (2021); DOI 10.1007/s10640-021-00558-w.

Compares a renewable share mandate and a carbon intensity standard, with and without a cost containment mechanism, in theory and in a numerical model of the US fuel market. An optimal mandate alone gains little over business as usual; cost containment limits both compliance costs and deadweight loss and raises efficiency substantially, especially for intensity standards.

Version record

Maria Gerveni, Todd Hubbs, Scott H. Irwin and James H. Stock, "The Biofuels Blueprint: Understanding the U.S. Renewable Fuel Standard"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Economic Perspectives and Policy, 48(3), 565–590 (2026); DOI 10.1002/aepp.70083.

Review of nearly two decades of the US Renewable Fuel Standard: nested mandates by life-cycle emissions, enforced through tradeable RINs; volumes of 20 billion gallons in 2022 against the statutory 36 billion because cellulosic biofuel never scaled; volatile RIN prices that follow economic fundamentals; and inflation-adjusted compliance costs estimated at $252.1 billion over 2011–25.

Version record

Gregory B. Upton and Brian F. Snyder, "Funding renewable energy: An analysis of renewable portfolio standards"

Difficulty: Research **Type:** Research paper

Status/version: Energy Economics, 66, 205–216 (2017); DOI 10.1016/j.eneco.2017.06.003.

Synthetic-control comparison of states with and without renewable portfolio standards: standard states saw higher electricity prices and lower demand, but no relative increase in renewable generation and only weak evidence of emissions reductions.

Version record

California Air Resources Board, Monthly LCFS Credit Transfer Activity Reports; LCFS regulation page

Difficulty: Intro **Type:** Data / specification

Status/version: Monthly official reports, posted on the second Tuesday of each month; August 2026 report posted 8 September 2026; amended regulation effective 1 July 2025.

Official monthly counts, volumes and average prices of California Low Carbon Fuel Standard credit transfers. The August 2026 report shows 3.19 million credits transferred at an average $77 (range $53–159), against $57 in the fourth quarter of 2025; calendar-year averages were $75 in 2023, $64 in 2024 and $60 in 2025, and 344 million credits have been transferred in total. Averages exclude zero- or near-zero-price transfers and include transfers pending confirmation; the regulation page links the amended rule in force since 1 July 2025.

Version record

Chapter 15

Private capital

Separate security underwriting from fund performance, commitment pacing and financing. Illiquidity, selection, contractual priority and access affect reported returns.

15.1 Private equity, venture capital and performance

Start with deal and fund economics, then cash-flow benchmarking, risk adjustment, manager selection and secondary transactions. IRR, TVPI, PME and alpha answer different questions; commitment pacing and access constraints belong alongside reported returns.

15.1.1 Fund performance, persistence and secondaries

Steven N. Kaplan and Antoinette Schoar, Private Equity Performance: Returns, Persistence, and Capital Flows

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance 60(4), 2005, pp. 1791–1823; version of record

The foundational LP cash-flow study introduces public-market-equivalent benchmarking alongside IRR and TVPI, then links fund returns, GP persistence, scale, and capital inflows. Reproduce its cash-flow timing and benchmark choices before comparing vintages or managers. Its 1980–2001 Venture Economics sample relies on voluntary reporting and largely realized early funds; the authors explicitly discuss potential selection bias and do not claim that a simple S&P 500 PME fully adjusts systematic risk.

Version record

Steven N. Kaplan and Per Strömberg, Leveraged Buyouts and Private Equity

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives 23(1), Winter 2009, pp. 121–146; published review

A compact economic map of buyout funds, transactions, capital structures, governance, and operating changes, with a comparison of the 1980s and later LBO waves. Use it before modelling a fund: the asset is a levered portfolio company inside a fee-bearing limited partnership, and debt capacity changes the equity payoff. Its historical synthesis predates the recent expansion of private credit and continuation vehicles, so pair it with current evidence for funding and exits.

Version record

Andrew Metrick and Ayako Yasuda, The Economics of Private Equity Funds

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Financial Studies 23(6), June 2010, pp. 2303–2341; published article

Models how management fees, carry, fund size, and reinvestment choices generate manager revenue from 238 funds raised in 1993–2006. It helps translate a gross deal return into GP and LP economics and explains why larger buyout funds can scale manager revenue differently from venture funds. Apply the contract model to actual partnership terms; the source investor and vintage window do not establish fee terms or economics for every modern strategy.

Version record

Robert S. Harris, Tim Jenkinson, and Steven N. Kaplan, Private Equity Performance: What Do We Know?

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance 69(5), October 2014, pp. 1851–1882; version of record

Uses Burgiss institutional LP cash flows for nearly 1,400 US buyout and VC funds and compares performance with public markets across indices and data vendors. It is the key replication companion to Kaplan–Schoar when deciding which denominator and reporting population to trust. The paper finds stronger historical buyout results than earlier databases, but vintage composition, VC regime shifts, fees, and public-market risk exposure all matter before turning sample averages into forward assumptions.

Version record

Arthur Korteweg and Morten Sørensen, Skill and Luck in Private Equity Performance; Version of record DOI

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics 124(3), 2017, pp. 535–562; institutional author record and accepted manuscript

Separates persistent GP skill from noisy fund outcomes with a model that uses successive fund histories, giving manager selection a more disciplined prior than ranking one vintage by IRR. Read alongside newer evidence that buyout persistence weakened as the industry matured. The inference depends on its hierarchical model, observed fund sample, and assumptions about common risk; a posterior skill estimate is not an investable promise, especially when an LP lacks access to the same managers.

Extended note. Version record

Will Gornall and Ilya A. Strebulaev, Squaring Venture Capital Valuations with Reality

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics 135(1), 2020, pp. 120–143; published article

Reconstructs the capital structures of 135 US unicorns from certificates of incorporation and values each security's distinct cash-flow rights. It shows why multiplying the latest preferred-share price by all shares can overstate company value, especially after protective terms are added. Use its waterfall logic for venture NAV and dilution analysis. The sample is late-stage US unicorns rather than all VC-backed firms, and conclusions depend on observed contractual rights and valuation-model inputs.

Version record

Robert S. Harris, Tim Jenkinson, Steven N. Kaplan, and Ruediger Stucke, Has Persistence Persisted in Private Equity? Evidence from Buyout and Venture Capital Funds

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Corporate Finance 81, August 2023, article 102361; published article

Revisits GP persistence with Burgiss cash flows and distinguishes a prior fund's final result from the interim result actually visible when its successor fundraises. That distinction is essential for an investable selection rule. The study finds durable VC persistence but weak buyout persistence, especially after 2000, when decisions use information available at fundraising. Its proprietary data and US-focused sample limit independent replication and direct extrapolation to emerging strategies or markets.

Version record

Arthur Korteweg, Stavros Panageas, and Anand Systla, Private Equity for Pension Plans? Evaluating Private Equity Performance from an Investor's Perspective; journal version

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics 184 (October 2026), 104336; NBER Working Paper 33194, November 2024 retained.

Evaluates pension PE using investor-specific stochastic discount factors, separating access to successful managers from superior selection and from compensation for bearing risk. It is a useful bridge from fund PME to a particular plan's opportunity cost and portfolio mix. The publisher version of record is linked alongside the 2024 preprint; this metadata update does not establish full-text equivalence.

Extended note. Version record

Taylor D. Nadauld, Berk A. Sensoy, Keith Vorkink, and Michael S. Weisbach, The Liquidity Cost of Private Equity Investments: Evidence from Secondary Market Transactions

Difficulty: Advanced **Type:** Research paper

Status/version: NBER Working Paper 22404, July 2016; Journal of Financial Economics published version 2018

Uses intermediary transactions in LP stakes to compare sale prices with NAV and the later market-adjusted outcomes of buyers and sellers. The observed trades are selected decisions to sell, often under specific liquidity conditions; their discounts should not be applied mechanically to all unsold fund interests or to current continuation-vehicle deals.

Extended note. Version record

Ege Y. Ercan, Steven N. Kaplan and Ilya A. Strebulaev, Interim Valuations, Predictability, and Outcomes in Private Equity

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33637; Stanford working-paper version dated 2024-05-20; no journal version established. Issued 2025-04.

Adds evidence on what interim private-equity marks reveal about ultimate cash flows and exit timing, complementing existing secondaries pricing and valuation-process entries.

Extended note. Version record

Rustam Abuzov, Will Gornall, Sophie Shive, Ilya A. Strebulaev, and Michael S. Weisbach, Selling to Yourself: Continuation Funds in Private Equity

Difficulty: Advanced **Type:** Research paper

Status/version: NBER Working Paper 34471, November 2025; working paper as of 4 October 2026

Combines a model with 472 hand-collected continuation funds to study why GPs transfer assets to new vehicles and why LPs sell or roll. The hand-collected sample captures disclosed transactions, not every continuation fund; its model does not by itself prove that transfers were fairly priced or that rolling outperforms selling.

Extended note. Version record

Institutional Limited Partners Association (ILPA), Continuation Funds: Considerations for Limited Partners and General Partners; ILPA status of 2026 draft

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: May 2023 published guidance; ILPA was reviewing updated draft guidance in 2026

Sets a due-diligence agenda for a GP-led sale: rationale, conflicts, valuation process, LPAC engagement, disclosure, choice to cash out or roll, and decision time. Use it to write a repeatable investment-committee checklist and insist that the rollover be evaluated as a new commitment with revised fees and carry. ILPA says these are general parameters for well-run processes, not universally applicable rules; its 2026 update remained draft at the cut-off.

Version record

Institutional Limited Partners Association (ILPA), ILPA Performance Template

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Template released 2025; version 1.1 on the publisher page; go-forward use for funds commencing from 1 January 2026

Standardizes the LP cash-flow and performance fields needed to reconcile gross and net return metrics at fund and portfolio level. The two template paths, granular and gross-up, help identify why reported gross IRRs differ when capital calls and transaction categories are mapped differently. It supplements governing agreements and accounting records; matching the template does not validate the underlying valuations or eliminate reporting bias.

Extended note. Version record

Reiner Braun, Nils Dorau, Tim Jenkinson, and Daniel Urban, Size, Returns, and Value: Do Private Equity Firms Allocate Capital According to Manager Skill?

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance 81(3), June 2026, pp. 1661–1700; version of record

Links PE deals to individual managers and measures value creation in dollars of NPV as well as relative returns. The paper finds that relative returns decline with scale while NPV can rise, and that firms direct more capital to skilled managers. NPV depends on discount-rate choices and deal attribution; observing internal allocation does not reveal what an outside LP can access.

Extended note. Version record

15.1.2 Performance measurement and public-market equivalents

Steven N. Kaplan and Berk A. Sensoy, "Private Equity Performance: A Survey"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 7(1), 597–614 (2015); DOI 10.1146/annurev-financial-111914-041858.

Survey of buyout and VC returns, PME methods, persistence and fund economics.

Version record

Arthur Korteweg, "Risk Adjustment in Private Equity Returns"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 11(1), 131–152 (2019); DOI 10.1146/annurev-financial-110118-123057.

Review of methods to estimate private-equity risk and alpha from cash flows, appraisals and deals.

Version record

Morten Sorensen and Ravi Jagannathan, "The Public Market Equivalent and Private Equity Performance"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 71(4), 43–50 (2015); DOI 10.2469/faj.v71.n4.4.

Shows PME is a valid SDF-based performance measure under log utility and explains its assumptions.

Version record

Oleg R. Gredil, Barry Griffiths and Rüdiger Stucke, "Benchmarking private equity: The Direct Alpha method"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Corporate Finance, 81, 102360 (2023); DOI 10.1016/j.jcorpfin.2023.102360.

Direct Alpha: an annualised excess return from compounded cash flows consistent with PME.

Version record

Andrew Ang, Bingxu Chen, William N. Goetzmann and Ludovic Phalippou, "Estimating Private Equity Returns from Limited Partner Cash Flows"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 73(4), 1751–1783 (2018); DOI 10.1111/jofi.12688.

Bayesian estimation of a time series of private-equity returns from fund cash flows.

Version record

Gregory W. Brown, Eric Ghysels and Oleg R. Gredil, "Nowcasting Net Asset Values: The Case of Private Equity"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 36(3), 945–986 (2023); DOI 10.1093/rfs/hhac045.

Mixed-frequency model that nowcasts fund NAVs from public-market data and reported marks.

Version record

Arpit Gupta and Stijn Van Nieuwerburgh, "Valuing Private Equity Investments Strip by Strip"
Difficulty: Research **Type:** Research paper
Status/version: The Journal of Finance, 76(6), 3255–3307 (2021); DOI 10.1111/jofi.13073.
Replicates fund cash flows with listed strips of equity and bond claims to measure risk-adjusted profits.
Version record

Erik Stafford, "Replicating Private Equity with Value Investing, Homemade Leverage, and Hold-to-Maturity Accounting"
Difficulty: Research **Type:** Research paper
Status/version: The Review of Financial Studies, 35(1), 299–342 (2021); DOI 10.1093/rfs/hhab020.
Levered small-value public portfolios with held-to-maturity accounting replicate buyout returns and smoothing.
Version record

Ludovic Phalippou, *Private Equity Laid Bare*
Difficulty: Intro/Advanced **Type:** Book / monograph
Status/version: Self-published (CreateSpace/Independently Published); editions 2017 onward; confirm edition before citing page references.
Critical guide to fund economics, fees, IRR manipulation, NAV and performance reporting; self-published editions updated regularly.
Version record

Ludovic Phalippou, Christian Rauch and Marc Umber, "Private equity portfolio company fees"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 129(3), 559–585 (2018); DOI 10.1016/j.jfineco.2018.05.010.
Monitoring and transaction fees charged to portfolio companies and their effect on LP returns.
Version record

Amit Goyal, Sunil Wahal and M. Deniz Yavuz, "Picking partners: Manager selection in private markets"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 178, 104250 (2026); DOI 10.1016/j.jfineco.2026.104250.
Evidence on how institutions select private-market managers and how selection relates to subsequent performance.
Extended note. Version record

Ludovic Phalippou, "The Tyranny of IRR"
Difficulty: Advanced **Type:** Research paper
Status/version: The Journal of Portfolio Management, 52(4), 66–84 (2026); DOI 10.3905/jpm.2026.1.809.
Since-inception IRR is not a rate of wealth growth and has inflated perceptions of private-equity outperformance (including the Yale-model case); argues for banning si-IRR in favour of horizon-based and market-equivalent measures.
Version record

15.1.3 Venture capital economics

John H. Cochrane, "The risk and return of venture capital"
Difficulty: Research **Type:** Research paper
Status/version: Journal of Financial Economics, 75(1), 3–52 (2005); DOI 10.1016/j.jfineco.2004.03.006.
Selection-bias-corrected estimates of VC risk and return from financing rounds.
Version record

Paul A. Gompers, Will Gornall, Steven N. Kaplan and Ilya A. Strebulaev, "How do venture capitalists make decisions?"
Difficulty: Advanced **Type:** Research paper
Status/version: Journal of Financial Economics, 135(1), 169–190 (2020); DOI 10.1016/j.jfineco.2019.06.011.
Survey of 885 VCs on sourcing, selection, valuation, contracting and value-add.
Version record

Steven N. Kaplan and Josh Lerner, "It Ain't Broke: The Past, Present, and Future of Venture Capital"
Difficulty: Advanced **Type:** Research paper
Status/version: Journal of Applied Corporate Finance, 22(2), 36–47 (2010); DOI 10.1111/j.1745-6622.2010.00272.x.
Long-run evidence on VC fundraising, returns and the industry's structural stability.
Version record

Arthur Korteweg and Morten Sorensen, "Risk and Return Characteristics of Venture Capital-Backed Entrepreneurial Companies"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 23(10), 3738–3772 (2010); DOI 10.1093/rfs/hhq050.

Dynamic selection model for VC-backed company returns, alphas and betas.

Version record

Michael Ewens, Charles M. Jones and Matthew Rhodes-Kropf, "The Price of Diversifiable Risk in Venture Capital and Private Equity"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 26(8), 1854–1889 (2013); DOI 10.1093/rfs/hht035.

Undiversified VC investors price idiosyncratic risk, affecting deal valuations.

Version record

Robert E. Hall and Susan E. Woodward, "The Burden of the Nondiversifiable Risk of Entrepreneurship"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 100(3), 1163–1194 (2010); DOI 10.1257/aer.100.3.1163.

Entrepreneurial payoffs are highly skewed; risk-averse founders value VC-backed ventures far below expected value.

Version record

Katja Kisseleva, Aksel Mjøs and David T. Robinson, "Evaluating Selection Bias in Early-Stage Investment Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(2), 841–871 (2026); DOI 10.1017/s0022109025101701.

Norwegian administrative data compare VC-backed with similar non-backed startups to quantify selection bias in early-stage returns and valuation premia.

Version record

Thomas Hellmann, Alexander Montag and Joacim Tåg, "Financing J-curves in venture capital"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, rfag037 (2026); DOI 10.1093/rof/rfag037.

Swedish start-up cash flows show deeper J-curves (larger early losses, higher later sales) when backed by US VCs, consistent with greater financing capacity.

Version record

15.2 Transactions and security design

Subject context and qualifications.

Joshua Rosenbaum and Joshua Pearl, *Investment Banking: Valuation, LBOs, M&A, and IPOs* (3rd ed.)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, 2022, ISBN 9781119867883; publisher lists an LBO valuation model download with the book.

Use its LBO analysis after the Kaplan–Strömberg overview: bridge purchase enterprise value to seller equity, sources and uses, operating projection, debt schedule, cash sweep and exit proceeds. The downloadable templates may require book access; neither a template nor historical comparable multiples validates a live bid, financing commitment or realizable exit price.

Extended note. Version record

Mike Glossop and Bjoern Koertner, Leveraged Buyout Modeling: An Excel Application

Difficulty: Advanced **Type:** Book chapter

Status/version: Chapter 13 in *International Private Equity* by Eli Talmor and Florin Vasvari, Wiley, first online 12 April 2011; chapter DOI verified.

The chapter separates model construction from optimization and identifies purchase of seller equity, fees and existing-debt retirement as uses, against new debt, sponsor equity and target cash as sources. Work it after Rosenbaum–Pearl to challenge a model whose headline entry multiple balances but whose closing funds do not. Rebuild tax, covenants, financing fees and available target cash from transaction documents.

Extended note. Version record

Viral V. Acharya, Oliver F. Gottschalg, Moritz Hahn and Conor Kehoe, Corporate Governance and Value Creation: Evidence from Private Equity

Difficulty: Research **Type:** Research paper

Status/version: *Review of Financial Studies* 26(2), 368–402, February 2013; published article, online 2012.

Deal-level evidence separates abnormal performance from leverage and sector returns and associates stronger outcomes with sales and margin improvement during PE ownership. The authors observe transactions from large houses; GP background and observed improvement are associations, not a transferable causal estimate or a forward underwriting hurdle.

Extended note. Version record

Steven J. Davis, John C. Haltiwanger, Kyle Handley, Ben Lipsius, Josh Lerner and Javier Miranda, The (Heterogeneous) Economic Effects of Private Equity Buyouts

Difficulty: Research **Type:** Research paper

Status/version: *Management Science* 71(11), 9569–9587, November 2025; published online 25 March 2025, DOI 10.1287/mnsc.2021.03890.

Thousands of US buyouts and controls show that employment, productivity and reallocation vary with buyout type and credit conditions. The historical sample ends in 2013; operating outcomes are not an LP return decomposition. Use it to challenge a uniform margin-expansion forecast and examine worker, supplier and capital-spending effects rather than importing its average productivity change into a specific deal.

Extended note. Version record

Andrew Metrick and Ayako Yasuda, *Venture Capital and the Finance of Innovation* (3rd ed.)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, March 2021, ISBN 9781119494379; selected current publisher edition, with table of contents and chapter-one preview.

Use the venture-finance text to connect staged investment, ownership, security terms and exit payoffs before applying the unicorn valuation study. The publisher preview verifies edition and scope, not a particular chapter's calculations. Teaching assumptions and US venture conventions are a starting point; an actual certificate of incorporation, side letters and financing documents govern proceeds, anti-dilution and control.

Extended note. Version record

Paul A. Gompers, Optimal Investment, Monitoring, and the Staging of Venture Capital

Difficulty: Research **Type:** Research paper

Status/version: *Journal of Finance* 50(5), 1461–1489, December 1995; version of record.

Explains why staged capital lets an investor revise continuation decisions as information arrives. Useful for a reserve policy and downside financing tree. Venture organization and contracts have changed since the sample; staging is neither a guarantee of profitable abandonment nor evidence that a new round can be raised in a weak market.

Extended note. Version record

Steven N. Kaplan and Per Strömberg, Financial Contracting Theory Meets the Real World: An Empirical Analysis of Venture Capital Contracts

Difficulty: Research **Type:** Research paper

Status/version: *Review of Economic Studies* 70(2), 281–315, April 2003; publisher abstract, full text subject to access.

Documents how actual VC contracts separately allocate cash-flow, board, voting and liquidation rights and how they evolve over rounds. Read after Gompers and before the cap-table waterfall: percentage ownership alone does not determine control or proceeds. Use a rights matrix for each round, including vetoes and contingent board seats. Use signed company documents for seniority, conversion and participation.

Extended note. Version record

Y Combinator, Post-money SAFE forms and user guide

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living primary contract and conversion guide, consulted 4 October 2026; US and selected non-US forms differ.

The form and user guide give concrete conversion math for caps, discounts and pro-rata side letters. Work multiple SAFEs, a priced round and option-pool increase through a cap table, then apply actual preferred liquidation rights to the exit waterfall. Compare with security-level valuation. These are model forms, not the governing instrument of a portfolio company; amended terms, side letters, jurisdiction and tax may change the result.

Extended note. Version record

Adair Morse, Venture Debt

Difficulty: Research **Type:** Book chapter

Status/version: In *The Palgrave Encyclopedia of Private Equity*, first online 3 June 2024; publisher entry DOI 10.1007/978-3-030-38738-9_220-1.

Distinguishes patent-backed loans, leverage against investor support and bridge loans, explaining why a pre-profit startup can sometimes borrow despite limited ordinary collateral. Useful for a growth company's financing-choice tree. It is a six-page reference synthesis, not a default-rate table or a valuation of a specified facility; lender security, cash dominion, IP collateral, covenants and the next-equity-round scenario must come from actual documents.

Extended note. Version record

Ulf Axelson, Per Strömberg and Michael S. Weisbach, "Why Are Buyouts Levered? The Financial Structure of Private Equity Funds"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 64(4), 1549–1582 (2009); DOI 10.1111/j.1540-6261.2009.01473.x.

Theory of PE fund structure: committed equity plus deal-level debt mitigates agency problems and explains leverage procyclicality.

Version record

Ulf Axelson, Tim Jenkinson, Per Strömberg and Michael S. Weisbach, "Borrow Cheap, Buy High? The Determinants of Leverage and Pricing in Buyouts"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 68(6), 2223–2267 (2013); DOI 10.1111/jofi.12082.

Buyout leverage tracks credit-market conditions rather than firm fundamentals, and higher leverage goes with higher prices and lower returns.

Version record

Shourun Guo, Edith S. Hotchkiss and Weihong Song, "Do Buyouts (Still) Create Value?"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 66(2), 479–517 (2011); DOI 10.1111/j.1540-6261.2010.01640.x.

Post-2000 LBO returns come from leverage, industry performance and operating gains, but operating improvements are smaller than in the 1980s.

Version record

Paul Gompers, Steven N. Kaplan and Vladimir Mukharlyamov, "What do private equity firms say they do?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 121(3), 449–476 (2016); DOI 10.1016/j.jfineco.2016.06.003.

Survey of PE firms on deal sourcing, valuation (IRR and multiples, rarely DCF/CAPM), capital structure and value creation.

Version record

Steven N. Kaplan, Berk A. Sensoy and Per Strömberg, "Should Investors Bet on the Jockey or the Horse? Evidence from the Evolution of Firms from Early Business Plans to Public Companies"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 64(1), 75–115 (2009); DOI 10.1111/j.1540-6261.2008.01429.x.

Business lines are far more stable than management teams from business plan to IPO: bet on the horse.

Version record

Thomas Hellmann, "The Allocation of Control Rights in Venture Capital Contracts"

Difficulty: Research **Type:** Research paper

Status/version: The RAND Journal of Economics, 29(1), 57 (1998); DOI 10.2307/2555816.

Why entrepreneurs cede board control and CEO-replacement rights to VCs; the theory behind VC term sheets.

Version record

Victoria Ivashina and Anna Kovner, "The Private Equity Advantage: Leveraged Buyout Firms and Relationship Banking"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 24(7), 2462–2498 (2011); DOI 10.1093/rfs/hhr024.

Repeat PE sponsors obtain cheaper, looser loan terms from relationship banks, an advantage in LBO financing.

Version record

Cem Demiroglu and Christopher M. James, "The role of private equity group reputation in LBO financing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 96(2), 306–330 (2010); DOI 10.1016/j.jfineco.2010.02.001.

Reputable PE sponsors borrow at lower spreads with looser covenants and more institutional loans.

Version record

15.3 Fund commitments, financing and liquidity

Subject context and qualifications.

Dean Takahashi and Seth Alexander, Illiquid Alternative Asset Fund Modeling

Difficulty: Advanced **Type:** Research paper

Status/version: *Journal of Portfolio Management* 28(2), 90–100, Winter 2002; journal DOI verified, subscription access may apply.

Provides a fund-life projection of contributions, NAV and distributions under alternative commitments and growth assumptions. Start with it to turn target private weight into a vintage schedule, then add stochastic calls, exits and market conditions with Luxenberg et al. and Robinson–Sensoy. Use as a transparent base case for commitments and unfunded exposure. Backtest vintages and retain cash to meet calls when distributions stop.

Extended note. Version record

Eric Sager Luxenberg, Stephen Boyd, Misha van Beek, Wen Cao and Mykel Kochenderfer, Strategic Asset Allocation with Illiquid Alternatives

Difficulty: Research **Type:** Research paper

Status/version: ACM ICAIF 2022, 8 pages, DOI 10.1145/3533271.3561769; author-hosted paper.

Models uncertain returns, calls and distributions as a stochastic system and selects liquid holdings and new commitments in successive periods. Use its state and control design to test overcommitment and liquidity policy. Its experiments assume no external liabilities or income and do not model secondary sales; pensions and endowments must add benefits, spending, collateral and sale haircuts before treating the policy as feasible.

Extended note. Version record

David T. Robinson and Berk A. Sensoy, Cyclicality, Performance Measurement, and Cash Flow Liquidity in Private Equity

Difficulty: Research **Type:** Research paper

Status/version: *Journal of Financial Economics* 122(3), 521–543, 2016; NBER working-paper page links the published version.

Quarterly buyout and VC cash flows show that distributions respond more to public valuations than calls do, leaving net flows weak when liquid markets fall. Pair with the deterministic and stochastic pacing models to impose joint public-market and private-distribution shocks, rather than independently sampling draws and exits. The 1984–2010 sample and fund-age effects inform scenario structure, not current manager-level forecast coefficients; most cash-flow variation is idiosyncratic.

Extended note. Version record

Institutional Limited Partners Association, Guidance on Enhancing Transparency Around Subscription Lines of Credit

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: ILPA final guidance, June 2020; free institutional PDF.

The worked example holds an investment and exit constant while delaying LP capital calls through a subscription line, showing how IRR can rise while TVPI falls after financing cost. ILPA guidance is voluntary and illustrative; governing LPA, bank agreement and actual transaction-level cash flows determine LP exposure and preferred-return timing.

Extended note. Version record

Golub Capital BDC 4, Inc. / PNC Bank, Sixth Amendment and marked credit agreement

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Executed 15 March 2024; original agreement 8 July 2022. EDGAR full-text definitions and Sections 5, 7–10 inspected.

Connects investor exclusions, side-letter excuses, unfunded commitments and capital-call collateral to borrowing availability, overadvance cure and lender remedies. Calls can be required to repay obligations after the investment period; an eligible-investor default can shrink capacity precisely when liquidity is scarce. Read insertions/deletions in the amended annex, governing documents and separate fee letter; selected historical clauses do not establish current capacity, every investor's enforceability or net LP returns.

Extended note. Version record

Institutional Limited Partners Association, Capital Call & Distribution Template (v. 2.0)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Released 11 September 2025 with suggested guidance; publisher template and guidance, implementation timing is fund-specific.

Provides transaction categories and changes in unfunded commitment for LP notices, making the input to a pacing model auditable. Use with the performance template, but keep notices, fund NAV, facility draws and estimated future calls as distinct records. Template adoption is not universal; it supplements notices and the LPA and does not forecast exits or assure that reported NAV can be sold.

Extended note. Version record

Institutional Limited Partners Association, NAV-Based Facilities Guidance & Roadmap

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: ILPA five-part guidance, July 2024; public roadmap and PDF.

Distinguishes financing secured on fund NAV from a subscription line secured by uncalled commitments. The roadmap covers structures, LP concerns, process and disclosures. Read beside secondaries and continuation-fund economics: a debt-funded distribution is liquidity today with a senior claim on later exits. This is general LP/GP guidance, not an LPA amendment or a legal ranking opinion; access to facility terms and portfolio marks is needed for underwriting.

Extended note. Version record

HarbourVest Global Private Equity, Annual Report and Accounts 2025

Difficulty: Advanced **Type:** Report / research note

Status/version: Year ended 31 January 2025; full balance-sheet-management discussion and credit-facility note inspected. Selected historical issuer case, not a claim to the latest report.

The USD 1.2 billion 2024 facility is secured by investments and cash, with LTV tests incorporating fund-level borrowing and portfolio diversity. The report reconciles calls, distributions, operating outflows, debt draws and a minimum-draw requirement. Prerequisite: look-through leverage and the distinction between investment NAV and spendable cash. The Guernsey-listed fund-of-funds structure differs from a single buyout fund's NAV loan. Fund/vehicle returns and shareholder price returns remain distinct.

Extended note. Version record

HSEC Holdings / JPMorgan Chase, Revolving Credit Agreement

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Execution version dated 24 June 2026, Exhibit 10.3 to the June-quarter 10-Q. Selected definitions, Sections 2.05, 7.10–7.12, 8.03 and signatures inspected in full text.

Connects eligible investment values, concentration exclusions and collateral-account cash to a borrowing base. Initial LTV is 25%; reaching 30% activates mandatory prepayment or a conditional, lender-approved cure plan targeting 25%. The numerator includes interest, fees and specified hedge exposure. Prerequisite: fund NAV, look-through concentration and liquidity for unfunded commitments. Uninspected schedules, fee arrangements, later amendments and live portfolio marks still prevent a complete current-capacity reconstruction.

Extended note. Version record

Institutional Limited Partners Association, Continuation Fund Disclosure Template; ILPA/Coller mock multi-asset and single-asset disclosures

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Template 27 January 2026; examples announced 9 April 2026. Selected full-text sections 2.9, 3.2–3.4 and 4 inspected.

The multi-asset example distinguishes selling, rolling through an SPV and selling/reinvesting; selling can include deferred consideration, while rolling can require additional unfunded capital for investments, fees and expenses. It contrasts fees on original cost with a stepped-up base, carry/clawback and expense allocation. The dated 24 June 2026 guidance resource remains labelled draft; an undated/staging page is insufficient evidence of final adoption.

Extended note. Version record

Odyssey Investment Partners, Project Lightning election/transaction materials, in the Golden Gate Transit–Amalgamated Retirement Plan's 21 November 2025 packet

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Selected citation year(s): 2025. Version distinctions remain in the annotation.

Contrasts selling 70%, seeking a full sale and retaining exposure in the original fund; elections face cutbacks. Shows expense/carry deductions, retained unfunded obligations and different economics for new continuation investors. Prerequisite: LP commitments, distributions, carry and NAV versus realised proceeds; compare the preceding ILPA mock disclosures. The referenced new-fund LPA, all schedules, closing settlement and later realised outcomes remain unverified.

Applications: PDF pp. 56–81 and selected 8 October agreement clauses/signature counterparts inspected; public pension attachment..
Extended note. Version record

Jamil Baz, Lloyd Han and Marc-Antoine Loo, "Three Models of the Liquidity Premium"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 51(9), 209–219 (2025); DOI 10.3905/jpm.2025.1.700.

Three complementary models of the illiquidity premium: forgone liquid opportunities, the (possibly negative) value of return smoothing, and insurance against running out of accessible capital, combined into an overall estimate.

Version record

Wen Cao and Misha van Beek, "A Latent Factor Cash Flow Model for Alternative Investment Funds"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 81(3), 60–75 (2025); DOI 10.1080/0015198x.2025.2489923.

Extends Takahashi–Alexander to a latent-factor model of private-fund calls, distributions and growth linked to macro factors, supporting stress tests, commitment pacing and total-portfolio risk.

Version record

15.4 Private credit and workouts

Subject context and qualifications. Read the private-credit path first. Underwrite the borrower cash flow, collateral and each lender's contractual claim through amendment, default and enforcement, not just a quoted spread.

Sergey Chernenko, Isil Erel, and Robert Prilmeier, Why Do Firms Borrow Directly from Nonbanks?

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Financial Studies 35(11), November 2022, pp. 4902–4947; published article

Hand-collects 2010–2015 middle-market credit agreements to compare direct nonbank lending with bank lending in borrower risk, price, and nonprice terms. It identifies regulatory constraints as an important channel for highly levered or unprofitable firms moving to nonbanks. Build a loan-comparison panel that controls for borrower quality before attributing spread differences to an illiquidity premium. The historical agreements and regulatory episode do not describe today's entire private-credit market, including asset-based finance.

Version record

Joern Block, Young Soo Jang, Steven N. Kaplan, and Anna Schulze, A Survey of Private Debt Funds; Earlier working paper

Difficulty: Advanced **Type:** Research paper

Status/version: The Review of Corporate Finance Studies 13(2), May 2024, pp. 335–383; published version supersedes NBER WP 30868

Surveys US and European private-debt managers controlling more than $390 billion to map sourcing, sponsor relationships, underwriting, leverage, covenant use, and monitoring. It is a strong qualitative prior for designing a lender due-diligence questionnaire and for distinguishing cash-flow lending from a bank term loan. Survey responses describe managers' stated practice rather than independently verified loan outcomes; compare claims with credit agreements, borrower financials, and realized losses before using them in return forecasts.

Version record

Isil Erel, Thomas Flanagan, and Michael S. Weisbach, Risk-Adjusting the Returns to Private Debt Funds

Difficulty: Advanced **Type:** Research paper

Status/version: NBER Working Paper 32278, March 2024; working-paper status verified

Uses private-debt fund cash flows to build a replicating risk benchmark with debt and equity exposures. It finds that typical net abnormal returns are statistically insignificant when both risks are included, while gross returns or debt-only benchmarks tell a more favourable story. Its fund sample and replicating-portfolio assumptions matter: private loans are not frictionlessly tradable replicas of the benchmark securities.

Extended note. Version record

David T. Robinson and Melanie Wallskog, The Growth of Private Lending and Retail Access to Alternative Investments

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 34617, revised May 2026; earlier circulation under 'Why is Private Lending So Popular?' dated 2025-12-30; no journal version established. Issued 2026-01.

Provides investor-level economic evidence on listed private-lending vehicles and fee incidence, filling a gap between broad private-credit vehicle descriptions and direct-lending asset-side studies.

Extended note. Version record

Young Soo Jang, Dasol Kim, Amir Sufi, and Xiangyu Chen, The Lending Technology of Direct Lenders in Private Credit

Difficulty: Advanced **Type:** Research paper

Status/version: NBER Working Paper 34500, issued November 2025; revised May 2026; current NBER landing-page author list

Uses secured-borrowing filings for US private middle-market firms to compare direct lenders, banks, and finance companies. The revised paper emphasizes enterprise and continuation value, PE-backed intangible-heavy borrowers, and industry specialization; use those variables in a loan-level underwriting and collateral taxonomy. It also tempers a pure bank-regulation explanation for direct-lending growth. The dataset is US secured borrowing, so it does not establish behavior for unsecured loans or non-US jurisdictions.

Version record

Gregor Matvos, Tomasz Piskorski, and Amit Seru, Private Credit, Balance Sheets and Financial Stability

Difficulty: Advanced **Type:** Research paper

Status/version: NBER Working Paper 34991, March 2026; revised August 2026

Examines fund- and asset-level data on private-credit financing, fund maturity, diversification, and loss absorption. Its August revision finds substantial equity funding and limited maturity transformation relative to deposit banks, while identifying remaining concerns around liquidity provision, valuations, retail structures, and interconnected investors. Read it against the IMF risk taxonomy rather than assuming every private-credit vehicle has the same fragility. Fund averages may hide strategy-specific leverage, concentrated BDCs, or correlated draws during stress.

Version record

U.S. Securities and Exchange Commission, Investor Bulletin: Interval Funds

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC investor-education bulletin dated 25 September 2020; explanatory staff material, not binding rule text.

Explains the liquidity mechanics of an interval-fund wrapper around illiquid assets. Repurchase offers occur at disclosed intervals; the fund generally offers to buy 5–25% of shares, and oversubscription can lead to pro rata fulfilment. Shareholders accept before the later NAV pricing date, so requested liquidity, paid cash and realised sale price need separate dates. Use the bulletin to stress a private-credit or real-asset allocation with simultaneous investor requests, limited liquid assets and valuation lags. Read the chosen fund's prospectus and filings for its actual interval, fees, proration and suspension terms; this bulletin is education, not the governing contract.

Extended note. Version record

I. Aldasoro, S. Doerr and K. Todorov, Retail Investors in Private Credit

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Bulletin 106, 9 July 2025; published policy research on retail private-credit vehicles.

Places the growth of retail-access vehicles around private credit in a wider liquidity and price-discovery setting. The bulletin focuses on business development companies and private-credit ETFs, and notes that secondary prices can diverge from reported NAV during stress. Read it beside interval-fund mechanics to distinguish a traded share-price discount from a fund's scheduled, capacity-limited repurchase at NAV. Trace a loan portfolio through the vehicle's valuation, financing and exit channels before comparing quoted performance. The

bulletin's ETF observations are not evidence that interval-fund shares trade continuously or that every private-credit loan can be sold quickly.

Extended note. Version record

U.S. Securities and Exchange Commission, Interval Fund Modernization and Multiple Share Classes for Registered Closed-End Funds and BDCs

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Proposed rule, Release 33-11444, issued 30 September 2026; proposal open for comment at the review date, not adopted law.

The SEC proposes changes to rule 23c-3, including a regular monthly repurchase option, altered notice and liquidity-management requirements, and related closed-end-fund share-class disclosures. The release describes current 3-, 6- and 12-month intervals alongside proposed alternatives, making it a dated way to compare existing and possible future fund cash timing. Build two scenarios for a private-credit interval fund: current governing terms and proposed terms only if adopted. Check subsequent rulemaking before updating a live model. Its economic analysis and examples are regulatory arguments, not evidence that monthly liquidity has already become generally available.

Extended note. Version record

International Monetary Fund, Monetary and Capital Markets Department, The Rise and Risks of Private Credit

Difficulty: Advanced **Type:** Report / research note

Status/version: Global Financial Stability Report, April 2024, chapter 2; final institutional report

Maps borrower fragility, fund and portfolio leverage, stale valuations, semiliquid vehicles, and links to banks, insurers, and pensions. Its definitions are useful for building a complete risk inventory before comparing funds or allocating capital. The chapter says aggregate risks then appeared contained but data gaps were substantial; treat that as a dated assessment, not a standing safety guarantee. It focuses on performing corporate private credit rather than real-estate, infrastructure, or distressed debt.

Version record

Fang Cai and Sharjil Haque, Private Credit: Characteristics and Risks

Difficulty: Advanced **Type:** Report / research note

Status/version: Federal Reserve FEDS Notes, 23 February 2024; corrected 26 February 2024

Builds a loan-level portrait of private-credit terms, borrower sectors, pricing, maturity, and interest coverage using PitchBook and other sources. It helps turn a broad private-credit allocation into measurable exposures: floating-rate burden, sponsor backing, collateral tangibility, and potential fund-level calls. Its market-size and spread observations are snapshots through 2023; the note itself warns that opaque coverage constrains risk inference. Use its definitions and figures, but refresh the data before making current-cycle claims.

Version record

Fernando Avalos, Sebastian Doerr, and Gabor Pinter, The Global Drivers of Private Credit

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Quarterly Review, March 2025; published feature

Relates international direct-lending growth to bank efficiency, regulation, policy rates, and narrowing bank versus BDC funding-cost differences. It also highlights industry concentration inside apparently diversified funds. Use the paper to separate loan-demand, bank-supply, and nonbank-funding hypotheses in a market entry study. Its cross-country loan panel ends in 2019 and the listed BDC cost-of-capital comparison is a US proxy, so it should not stand in for every private fund's present funding terms.

Version record

Iñaki Aldasoro and Sebastian Doerr, Collateralized Lending in Private Credit

Difficulty: Advanced **Type:** Research paper

Status/version: BIS Working Paper 1267, May 2025; working paper

Uses US loan-level evidence to show that secured direct loans are large and increasingly important, and that security is more likely when lender-borrower information frictions are stronger. The collateral channel links local house prices with secured credit supply. The house-price instrument is suggestive evidence, and a security interest alone does not determine recovery without seniority and enforceability.

Extended note. Version record

International Private Equity and Venture Capital Valuation (IPEV) Board, International Private Equity and Venture Capital Valuation Guidelines

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: December 2025 edition, published 11 December 2025; current final guidelines at cut-off

Sets a fair-value discipline for private-capital investments that can be applied to illiquid private-credit marks: use market-participant assumptions, calibrate models to transactions, refresh evidence, and document judgment. It is a control framework for valuation committees facing stale prices, covenant amendments, or changed borrower risk. Treat its estimates as modelled exit prices, not cash recoveries or tradable quotes. Pair the guidance with IFRS 13 or the relevant accounting framework and loan-specific credit analysis.

Version record

Puriya Abbassi, Iñaki Aldasoro, and Sebastian Doerr, Financing the Digital Economy: The Role of Private Credit

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Quarterly Review, 14 September 2026; published feature

Studies the post-2020 rise of direct lending to technology firms with intangible assets and recurring revenue, examining the association between local private-credit supply, firm entry, and employment. The sector and time window are unusually specific; a correlation between local credit capacity and firm growth should not be carried over to every borrower, especially if technology disruption changes renewal cash flows.

Extended note. Version record

S&P Global Ratings, Private Credit Payment Defaults Rose in 2023 as Weaker Borrowers Struggled to Service Debt

Difficulty: Advanced **Type:** Report / research note

Status/version: Credit Trends report, 26 June 2024; published ratings analysis

Tracks payment defaults among S&P credit-estimated middle-market issuers and describes out-of-court amendments, PIK, exchanges, and lender exits. Its careful distinction between payment and selective default is useful when reconciling a manager's zero-default claim with economic impairment. The covered universe is middle-market CLO collateral with credit estimates, not all private credit, and observed exchange prices are not universal ultimate recoveries. Record the denominator, default definition, seniority, and workout path before comparing losses.

Version record

Simeon Brown, Jessica Ann Rissmiller, Michael Graziano and Heather Sonnenberg, Unitranche Harmony: A 101 on the Agreement Among Lenders

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: American Bar Association, 2025 fall meeting program overview; public summary, not a model contract.

Explains why first-out and last-out lenders under one borrower credit agreement and one lien need a separate agreement among lenders to allocate cash, voting, standstills, remedies and bankruptcy rights. Read after direct-lending basics and before assuming "unitranche" means a homogeneous senior claim. The overview describes negotiable structures; it cannot establish recovery priority or enforceability in a particular jurisdiction.

Extended note. Version record

Office of the Comptroller of the Currency, *Asset-Based Lending*, Comptroller's Handbook

Difficulty: Advanced **Type:** Book / monograph

Status/version: January 2017 booklet, current OCC page consulted 4 October 2026; bank-supervision guidance.

Works from receivables and inventory eligibility, advance rates, concentration, dilution, field exams and cash dominion to availability and collateral control. Use it to underwrite ABL or specialty-finance exposure separately from enterprise-value or EBITDA-based direct lending. OCC guidance applies to supervised banks, not automatically every private lender; actual lien perfection, collateral audits and liquidation values remain deal-specific.

Extended note. Version record

S&P Global Ratings, Pre-Tariff U.S. Middle-Market Collateralized Loan Obligation Rally Is Unlikely To Last

Difficulty: Advanced **Type:** Research paper

Status/version: 22 May 2025 rating-agency research; public article, credit-estimate universe.

Separates conventional payment defaults from selective defaults involving PIK or amendment and reports wide maintenance-covenant cushions in its covered middle-market issuers. Monitor economic impairment and accrual quality even when contractual payment default is avoided. The article's middle-market CLO credit estimates are selected observations and its default taxonomy is agency-specific; neither is a population default rate for all private credit or a forecast recovery for a loan.

Extended note. Version record

U.S. Well Services lenders, Agreement Among Lenders

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Executed 28 February 2022; EDGAR full text, priority, enforcement, buyout and insolvency provisions inspected; New York governing law.

Separates a borrower's common facility from lenders' internal allocation of proceeds, voting, enforcement and purchase rights. The buyout provisions include defined transferred obligations and retained interests, so paying par principal alone need not acquire every claim. Prerequisite: secured-loan waterfalls and contingent enforcement control. Related credit documents, amendments and court orders matter; the agreement is evidence of negotiated rights, not a universal lender waterfall or realised-loss dataset.

Extended note. Version record

TEAM, Inc. / Eclipse Business Capital, Amendment No. 3 and amended Credit Agreement

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Executed 16 June 2023; original 11 February 2022 agreement. Selected full-text collateral definitions and reporting provisions inspected.

Combines a revolving borrowing base with machinery/equipment and real-estate facilities, reserves and collateral/debt reporting. Monthly reporting accelerates to weekly during the defined increased-reporting period. Prerequisite: distinguish eligible receiv-

ables/inventory, appraised collateral and lender reserves from book asset totals. Eligibility, cash controls and appraisal terms require the operative agreement, and recovery cannot be inferred from the advance rate.

Extended note. Version record

U.S. Court of Appeals for the Fifth Circuit, In re Serta Simmons Bedding, L.L.C.

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Opinion originally filed 31 December 2024, revised 14 February 2025; selected full opinion inspected. Historical judicial decision, not a claim that all subsequent proceedings are concluded.

Tests the 2020 uptier against the specific 2016 credit agreement's pro-rata and open-market-purchase provisions and addresses bankruptcy-plan indemnity. Prerequisite: consent thresholds, purchase exceptions and the difference between priority and an indemnity claim. Read beside the negotiated lender agreement: a contractual label does not determine judicial treatment. Select the revised opinion rather than citing only the widely discussed December 2024 decision date.

Extended note. Version record

Jose L. Fillat, Leo Shen and J. Christina Wang (Federal Reserve Bank of Boston), "BDC lending, PIK usage and borrower credit quality (Current Policy Perspectives 26-6)"

Difficulty: Advanced **Type:** Report / research note

Status/version: Federal Reserve Bank of Boston Current Policy Perspectives 26-6, 5 August 2026.

SEC filings from 168 BDCs (about 890,000 company-quarter loans) show rising payment-in-kind usage since 2022 alongside pricing and creditworthiness changes; confirm the exact title on the Boston Fed page.

Version record

Author as listed on arXiv (Zou), "Private Credit Markets: Theory, Evidence, and Emerging Frontiers"

Difficulty: Advanced **Type:** Research paper

Status/version: arXiv 2603.14491, March 2026.

Literature survey of direct lending economics, risk-adjusted returns and fees; argues the premium over syndicated loans is largely absorbed by fund fees. Unrefereed survey.

Extended note. Version record

15.5 Trade finance, leasing, royalties and legal claims

Office of the Comptroller of the Currency, *Trade Finance and Services*, Comptroller's Handbook

Difficulty: Advanced **Type:** Book / monograph

Status/version: October 2018 booklet, current OCC page consulted 4 October 2026; US bank-supervision guidance.

Maps documentary letters of credit, guarantees, acceptances, open-account financing, supply-chain finance and prepayments to their different obligors, documents and settlement steps. Read before comparing a short-dated trade receivable with a corporate loan: importer, issuing bank, exporter and collateral can fail at different points. The supervisory booklet is not the actual UCP undertaking, receivable purchase agreement or legal opinion; program controls and jurisdiction determine enforceability and loss.

Extended note. Version record

International Chamber of Commerce, ICC Trade Register: The Global Benchmark for Trade and Supply Chain Finance

Difficulty: Advanced **Type:** Report / research note

Status/version: 22 October 2025 product and methodology page; detailed default and LGD tables require paid access.

The multibank register separates trade-finance products and regions and offers default and loss data for comparison with a proposed portfolio. Use after the OCC instrument taxonomy; a letter of credit, open-account receivable and export loan should not share one default assumption. The public page verifies scope and access tiers, not the full rates. Do not enter a numerical loss estimate without the licensed table and its denominator.

Extended note. Version record

Pol Antràs and C. Fritz Foley, Poultry in Motion: A Study of International Trade Finance Practices; author-hosted published text

Difficulty: Research **Type:** Practitioner / commentary

Status/version: *Journal of Political Economy* 123(4), 853–901, August 2015; online 30 June.

Models importer nonpayment against exporter nondelivery and compares cash in advance, open account and bank-intermediated terms. Evidence comes from one US food exporter's transactions across countries in 1996–2009. Prerequisite: working-capital financing, contracting and selection in observational data. Use to challenge a uniform trade-finance default assumption: counterparty, relationship and enforcement affect the chosen contract. No net investor-return estimate is imported.

Extended note. Version record

Majesco / HSBC Bank USA, Receivables Purchase Agreement, Schedule and Standard Terms, Exhibit 10.1

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Signed agreement dated 13 January 2017; SEC-filed full agreement, schedule, standard terms and signatures inspected. Historical U.S. services receivables case.

An actual factoring contract with a USD 10 million limit, 90% stated advance percentage, 40% customer concentration percentage, 2% margin and 0.2% arrangement and annual review fees. The schedule expressly selects advances and non-notification, and declines HSBC credit-risk assumption. Build invoice-level eligibility, reserve, advance, interest, customer payment and recourse branches, including what happens if a receivable is disputed or overdue. The lender may reduce advance percentage or reserves, so the stated maximum is not guaranteed liquidity. This is domestic services factoring, not cross-border documentary credit or a representative trade-fund net return; historical LIBOR pricing needs replacement before a current model.

Extended note. Version record

Office of the Comptroller of the Currency, *Lease Financing*, Comptroller's Handbook

Difficulty: Advanced **Type:** Book / monograph

Status/version: Revised September 2023 booklet, current OCC page; US bank-supervision guidance.

Separates lessee credit, contracted rentals, tax and accounting treatment, maintenance, off-lease time and unguaranteed residual value. Stress tenant default and asset residual together; an aircraft engine or specialized machine is not interchangeable with generic equipment. The handbook is for supervised banks and does not provide current aircraft appraisal values, legal title, registry status or insurance for a particular asset.

Extended note. Version record

Alessandro Gavazza, Asset Liquidity and Financial Contracts: Evidence from Aircraft Leases

Difficulty: Research **Type:** Research paper

Status/version: *Journal of Financial Economics* 95(1), 62–84, January 2010; author-hosted published text.

Relates aircraft marketability to lease type, maturity and rental markups. Combines an April 2003 fleet cross-section and aircraft histories with appraised model/vintage prices and operating-lease rates for 1967–2003. Prerequisite: lease versus secured-loan economics, fixed effects and appraisal measurement. Inform remarketing and residual-value scenarios, then compare the following actual lease. The historical commercial data are not a verified freely reusable loan tape or a causal guarantee for today's aircraft.

Extended note. Version record

Bank of Utah (owner trustee) / Global Crossing Airlines, Aircraft Operating Lease Agreement, A321-231 MSN 1153

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Agreement dated 29 April 2024, SEC exhibit filed 2025.

Exhibit H links reserve payments to flight hours/cycles, engine severity and escalation, with reimbursement for qualifying work subject to conditions. Return condition, deposits and maintenance balances alter cash flows independently of headline rent. Prerequisite: aircraft utilisation, maintenance events and residual value. Trace rent, reserves, reimbursement, downtime and redelivery separately; retained reserves are not simply rental profit. Read with Cape Town rights.

Extended note. Version record

Andrea L. Eisfeldt and Adriano A. Rampini, Leasing, Ability to Repossess, and Debt Capacity; Census CES-07-19

Difficulty: Research **Type:** Research paper

Status/version: *Review of Financial Studies* 22(4), 1621–1657, April 2009; online 29 March 2008.

Models a repossession advantage against ownership/control agency costs. Census manufacturing data distinguish machinery/equipment from structures, while the Compustat merge links leasing to financial constraints. Prerequisite: secured debt, true versus finance leases and selection in firm-level data. Inform residual-value and workout scenarios without assuming identical recovery across equipment classes or jurisdictions. Full journal/preprint equivalence, current equipment loss tapes and a representative investable net-return panel remain open.

Extended note. Version record

CHG-MERIDIAN U.S. Finance / CDF2 Holdings, Master Equipment Lease and Schedule 001, Exhibit 10.3

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Signed execution version dated 18 October 2011; SEC exhibit contains master lease, rider, one executed digital-cinema equipment schedule and signature counterparts.

A contract-level equipment case with 618 scheduled units, a 66-month initial term, monthly rent of USD 216,300 for the first 15 months and USD 482,040 for months 16–66, plus a six-month mandatory renewal at the initial-term rate. The master lease allocates loss, return, default and residual risk; casualty damages include remaining rent, recovery costs and estimated residual, while disposal proceeds have a stated order. Model the lessor's contracted receipts, funding, lessee/sublessee performance, loss and end-of-term option separately. The schedule's rent is not realised cash collection or investor IRR, and linked sale-leaseback, senior-credit and multiparty records need reconciliation.

Extended note. Version record

International Civil Aviation Organization, Cape Town Convention and Aircraft Protocol portal

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living ICAO treaty and implementation portal; reviewed 4 October 2026.

Explains the cross-border legal problem of obtaining and enforcing interests in mobile airframes, engines and helicopters. Read with lease cash-flow underwriting: recovery after airline failure depends on registered interests, priority, jurisdiction, deregistration and repossession, not only appraised aircraft value. The portal establishes treaty purpose and texts, not that a given lender's interest is perfected or that repossession will be immediate in a particular country.

Extended note. Version record

Peter Tschmuck, Dennis Collopy and Christian Handke, IP Finance in the Music Industry

Difficulty: Research **Type:** Research paper

Status/version: WIPO Economic Research Working Paper 98/2026, 2026; free 137-page primary report.

Distinguishes composition, master, performance and synchronization rights, and studies catalogue sales, royalty marketplaces and music-backed securitization. Work from audited revenue by right, platform, territory and vintage to after-fee cash flow, then model rights term, decay, recoupment and licensing upside separately. Confirm title chain, writer and publisher shares, collecting-society data, audit rights, concentration and copyright law before treating a catalogue multiple as investable.

Extended note. Version record

Royalty Pharma, 2025 Form 10-K; 30 June 2026 Form 10-Q

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Audited 2025 statements and unaudited June 2026 update; selected full-text royalty, funding, valuation and accounting notes inspected.

Separates marketed-product royalties from development funding, contingent milestones and loan-like claims. Probability-adjusted cash-flow valuation and commitment models depend on clinical/regulatory outcomes, duration and counterparty terms. Prerequisite: product sales forecasts, patent/licence rights, discounting and financial-statement cash reconciliation. Bridge asset receipts to acquisition outlays, losses, funding and corporate costs before discussing investor returns. Drug-rights evidence does not cover film, trademarks or all patents.

Extended note. Version record

PTC Therapeutics / Royalty Pharma, 2023 amended royalty purchase agreement; signed 2025 Amendment No. 2; June 2026 filing

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Base agreement signed 18 October 2023, amended 17 June 2024 and 29 December 2025; 2025 amendment full text and June 2026 issuer accounting inspected.

The executed chain shows why a royalty slice must be dated. The 2023 agreement split Evrysdi-related rights around a USD 1.3 billion assigned-payment cap and included seller put options; the signed December 2025 amendment transferred the remaining interest for USD 240 million upfront plus at most USD 60 million in contingent purchase-price payments. After that closing Royalty Pharma holds 100% of the specified royalty, which the June 2026 10-Q confirms alongside USD 2.1 billion total funding. The agreement governs payment instructions, reports, reductions and enforcement; a licence payer's credit nonpayment is not automatically a seller true-up. Reconstruct contract receipts and accounting liability separately, without treating upfront purchase price as fund profit or this one drug as broad IP returns.

Extended note. Version record

U.S. Government Accountability Office, Third-Party Litigation Financing: Market Characteristics, Data, and Trends

Difficulty: Advanced **Type:** Report / research note

Status/version: GAO-23-105210, dated 20 December 2022 and publicly released 19 January 2023; free federal report.

Distinguishes commercial case or law-firm finance from consumer advances and maps contingent payment, disclosure, control, regulation and data gaps. Model expected litigation proceeds by legal stage, duration, expenses, settlement, appeal and collection, including a zero-recovery branch. Counsel must review privilege, assignment, priority, funder control and local law for a real claim; legal outcome is not a normal traded-credit payoff.

Extended note. Version record

Ronen Avraham and Anthony Sebok, An Empirical Investigation of Third Party Consumer Litigant Funding; journal full text

Difficulty: Research **Type:** Research paper

Status/version: *Cornell Law Review* 104(5), 1133 ff., 2019; selected full-text introduction and haircut analysis inspected.

Distinguishes requested and approved funding, contractually owed amounts, compounding, fees, settlement delay, negotiated haircuts and collections. Prerequisite: nonrecourse claims and the difference between a quoted charge and money actually received. Read after GAO's separation of consumer and commercial funding. Publication of aggregate evidence does not make the case-level data public; the authors' disclosed industry relationships should accompany use of the findings.

Extended note. Version record

Alessandro Gavazza, "Leasing and Secondary Markets: Theory and Evidence from Commercial Aircraft"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 119(2), 325–377 (2011); DOI 10.1086/660730.

Operating leases and secondary-market trading of aircraft: leasing reduces transaction costs and improves allocation.

Version record

Efraim Benmelech and Nittai K. Bergman, "Liquidation Values and the Credibility of Financial Contract Renegotiation: Evidence from U.S. Airlines"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 123(4), 1635–1677 (2008); DOI 10.1162/qjec.2008.123.4.1635.

Airlines renegotiate aircraft lease payments down when fleet liquidation values are low; collateral redeployability sets bargaining power.

Version record

Efraim Benmelech and Nittai K. Bergman, "Collateral pricing"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 91(3), 339–360 (2009); DOI 10.1016/j.jfineco.2008.03.003.

Airline secured-debt spreads fall with collateral redeployability; evidence for EETC and aircraft ABS pricing.

Version record

Todd C. Pulvino, "Do Asset Fire Sales Exist? An Empirical Investigation of Commercial Aircraft Transactions"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 53(3), 939–978 (1998); DOI 10.1111/0022-1082.00040.

Financially constrained airlines sell aircraft at a 14% discount, more in industry downturns: fire-sale discounts for aircraft collateral.

Version record

Mitchell A. Petersen and Raghuram G. Rajan, "Trade Credit: Theories and Evidence"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 10(3), 661–691 (1997); DOI 10.1093/rfs/10.3.661.

Who extends and uses trade credit and why: financing advantages of suppliers and price discrimination.

Version record

Leora Klapper, "The role of factoring for financing small and medium enterprises"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Banking & Finance, 30(11), 3111–3130 (2006); DOI 10.1016/j.jbankfin.2006.05.001.

Cross-country evidence that factoring (receivables purchase) substitutes for weak creditor rights because the receivable, not the borrower, carries the credit risk.

Version record

Justin Murfin and Ken Njoroge, "The Implicit Costs of Trade Credit Borrowing by Large Firms"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 28(1), 112–145 (2015); DOI 10.1093/rfs/hhu051.

Large buyers stretch payables to small suppliers, who then cut investment; the economics behind supply-chain finance programs.

Version record

Jean-Noël Barrot, "Trade Credit and Industry Dynamics: Evidence from Trucking Firms"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 71(5), 1975–2016 (2016); DOI 10.1111/jofi.12371.

A French cap on payment terms boosted entry of liquidity-constrained trucking firms; payment terms act as a barrier to entry.

Version record

Niklas Amberg, Tor Jacobson, Erik von Schedvin and Robert Townsend, "Curbing Shocks to Corporate Liquidity: The Role of Trade Credit"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 129(1), 182–242 (2021); DOI 10.1086/711403.

Swedish firms absorb liquidity shocks by adjusting trade credit with suppliers and customers.

Version record

Jose-Maria Fernandez, Roger M. Stein and Andrew W. Lo, "Commercializing biomedical research through securitization techniques"

Difficulty: Advanced **Type:** Research paper

Status/version: Nature Biotechnology, 30(10), 964–975 (2012); DOI 10.1038/nbt.2374.

Proposes a megafund financed with tranched debt and equity over many drug-development projects; the financial logic behind royalty and pharma-asset securitisation.

Version record

David S. Abrams and Daniel L. Chen, "A Market for Justice: A First Empirical Look at Third Party Litigation Funding"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper 2404483 (2014); DOI 10.2139/ssrn.2404483. Working-paper record; no journal version established in this check.

Data from an Australian litigation funder: case selection, returns and duration of funded claims.

Extended note.

Version record

Chapter 16

Property, infrastructure and natural-resource rights

Property, projects and natural-resource rights generate different cash flows. Appraisal evidence, operating constraints and legal rights should remain visible in valuation.

16.1 Property valuation and investment

Subject context and qualifications.

David M. Geltner, Norman G. Miller, Alex Van De Minne, Piet Eichholtz, Thies Lindenthal and Lily Shen, Commercial Real Estate Analysis for Investment, Finance, and Development

Difficulty: Advanced **Type:** Book / monograph

Status/version: Fourth edition, Routledge, January 2026, 440 pages; paperback ISBN 9781041076391; publisher record and 2026 catalogue

The fully revised edition connects property-market assumptions, valuation, development, mortgage finance, leverage and institutional returns. Work print chapter 4 on discounted cash flow and chapter 10 on leases, then the publisher-listed online chapters 16–18 on development and 19–21 on mortgage finance. Online chapters need the publisher's companion resources; exact exercises and pagination require the edition itself. The title and six-author list differ from the 2014 third edition below.

Extended note. Version record

David M. Geltner, Norman G. Miller, Jim Clayton, and Piet Eichholtz, Commercial Real Estate: Analysis and Investments; Edition bibliographic record

Difficulty: Advanced **Type:** Book / monograph

Status/version: Third edition, OnCourse Learning, 2014, ISBN 9781133108825; MIT author profile confirms edition

The earlier property-level finance text remains useful for lease cash flows, net operating income, cap rates, discounted cash flows, development options, debt and institutional return measures, and for tracing older academic references. Use the revised fourth edition for current course sequence and expanded development and finance material. Keep third-edition page and chapter numbers attached to the 2014 work; bibliographic catalogues mix package and student editions.

Version record

David M. Geltner, Estimating Market Values from Appraised Values without Assuming an Efficient Market

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Real Estate Research 8(3), 1993, pp. 325–346; published article

Develops a structural way to infer underlying property-market returns from appraisal-based index returns while allowing real return predictability. It explicitly handles individual appraisal smoothing, temporal aggregation, and staggered reappraisals. The correction is estimated from an appraisal process; it cannot reveal a unique true price path when the underlying data are sparse or regimes change.

Extended note. Version record

National Council of Real Estate Investment Fiduciaries (NCREIF), NCREIF Property Index (NPI): Index Methodology and FAQs

Difficulty: Advanced **Type:** Data / specification

Status/version: Living primary methodology page; reviewed 4 October 2026; 2Q2026 latest public return shown

Defines the quarterly, value-weighted, unleveraged return series for institutionally held US operating commercial properties, including the treatment of acquisitions, sales, appraisal values, and property types. Use it to determine exactly what a private-property benchmark measures before calculating active return or covariance. Because members contribute properties and the composition changes, NPI is a fiduciary-held sample rather than a census of US real estate. Its appraisal frequency also smooths and delays measured capital returns.

Version record

National Council of Real Estate Investment Fiduciaries (NCREIF), NCREIF Transaction Based Index (NTBI): Methodology and FAQs

Difficulty: Advanced **Type:** Data / specification

Status/version: Living primary methodology page; reviewed 4 October 2026

Uses prices on properties sold from the NPI pool to adjust appraisals and estimate a transaction-oriented return series. Read alongside NPI to see why appraisal and deal-price histories tell different short-horizon risk stories. Transaction selection, changing asset quality, and sparse sales require explicit treatment before interpreting a quarterly movement as the whole market.

Extended note. Version record

National Council of Real Estate Investment Fiduciaries (NCREIF), NCREIF Fund Index – Open End Diversified Core Equity (NFI-ODCE)

Difficulty: Advanced **Type:** Data / specification

Status/version: Living primary index methodology; reviewed 4 October 2026

Measures gross-of-fee, time-weighted returns for open-end US core real-estate funds, with net-of-fee and equal-weight versions available. It is the vehicle-level counterpart to the property-level NPI, making leverage, cash balances, fund costs, and structure visible in comparisons. Use matching gross/net and leverage definitions when grading a manager. The index covers a defined group of diversified core funds, not value-add or closed-end vehicles, and periodic redemptions do not guarantee immediate cash liquidity.

Version record

FTSE Russell and Nareit, FTSE Nareit US Real Estate Index Series Ground Rules

Difficulty: Advanced **Type:** Data / specification

Status/version: Version 5.5, August 2025; current publisher-hosted rules found at cut-off

Specifies eligibility, size, liquidity, free-float, sector, and return variants for listed US REIT benchmarks. It lets a researcher construct a public comparator without silently mixing all REITs, equity REITs, mortgage REITs, or the Composite. The All REITs and Composite universes apply different screens and weighting rules. Listed securities reprice daily and embed corporate leverage and management, so match asset exposure and vehicle economics when comparing their returns with appraised private properties.

Version record

MSCI Real Assets, RCA Commercial Property Price Indexes (RCA CPPI); Published methodology notes and revision policy

Difficulty: Advanced **Type:** Data / specification

Status/version: September 24, 2026 US update; recurring index report with current methodology notes

Provides transaction-based repeat-sales price indexes by property type and metro, a complementary price signal to appraisal-based NPI returns. Use the published notes to track the data cut-off, property mix, and backward revisions before joining the monthly series to fund reports or stress tests. Pin the report vintage when reproducing historical signals.

Extended note. Version record

Arpit Gupta, Vrinda Mittal, and Stijn Van Nieuwerburgh, Work from Home and the Office Real Estate Apocalypse

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review 116(2), February 2026, pp. 674–709; version of record

Links remote work to office rents, occupancy, lease revenues, and discount rates, then revalues New York and US office assets. Its separate cash-flow and discount-rate channels are a useful scenario template for sector concentration and debt coverage. Its office-specific calibration, tenant behavior, and financing environment are not direct forecasts for industrial, residential, or every city.

Extended note. Version record

Peter Shepard (MSCI), The MSCI Private Real Estate Factor Model

Difficulty: Advanced **Type:** Research paper

Status/version: MSCI research paper, 17 December 2025; publisher preview verified

Presents a global risk-factor framework with Bayesian desmoothing, regional and property-type factors, and separate income versus appreciation components. It is a practical bridge from sparse appraisal observations to cross-asset risk estimates in a total portfolio. Desmoothing is model dependent, and the page notes that its private-real-estate series is unlevered while public comparators include leverage.

Extended note. Version record

Karl E. Case and Robert J. Shiller, "The Efficiency of the Market for Single-Family Homes"

Difficulty: Advanced **Type:** Research paper

Status/version: NBER Working Paper 2506 (1988); American Economic Review 79(1), 125–137 (1989).

Repeat-sales indices show house-price changes are positively autocorrelated, unlike stocks.

Extended note. Version record

Andrew Ang, Neil Nabar and Samuel J. Wald, "Searching for a Common Factor in Public and Private Real Estate Returns"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 39(5), 120–133 (2013); DOI 10.3905/jpm.2013.39.5.120.

Estimates a common latent real-estate factor across REIT and appraisal-based returns with smoothing correction.

Version record

Joseph L. Pagliari Jr., Kevin A. Scherer and Richard T. Monopoli, "Public Versus Private Real Estate Equities: A More Refined, Long-Term Comparison"

Difficulty: Advanced **Type:** Research paper

Status/version: Real Estate Economics, 33(1), 147–187 (2005); DOI 10.1111/j.1080-8620.2005.00115.x.

After leverage and sector adjustments, REIT and private core real estate returns are similar over long horizons.

Version record

Stijn Van Nieuwerburgh, "Why Are REITs Currently So Expensive?"

Difficulty: Advanced **Type:** Research paper

Status/version: Real Estate Economics, 47(1), 18–65 (2019); DOI 10.1111/1540-6229.12238.

REIT valuations explained by low expected returns rather than high expected growth.

Version record

David Geltner, "Smoothing in Appraisal-Based Returns"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Real Estate Finance and Economics, 4(3), 327–345 (1991); DOI 10.1007/bf00161933.

Shows appraisal-based property indices are smoothed and lagged and derives the unsmoothing filter used to estimate true volatility and correlations.

Version record

Jeffrey D. Fisher, David M. Geltner and R. Brian Webb, "Value Indices of Commercial Real Estate: A Comparison of Index Construction Methods"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Real Estate Finance and Economics, 9(2), 137–164 (1994); DOI 10.1007/bf01099972.

Compares appraisal, hedonic and repeat-sales commercial property indices and their biases.

Version record

David C. Ling and Andy Naranjo, "Returns and Information Transmission Dynamics in Public and Private Real Estate Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Real Estate Economics, 43(1), 163–208 (2015); DOI 10.1111/1540-6229.12069.

REIT returns lead private property returns; public and private real estate converge over longer horizons.

Version record

16.2 Infrastructure and project finance

Subject context and qualifications. Underwrite the lease, plant, contract or regulated concession first; then map that cash flow through construction, financing, refinancing and the fund vehicle. A property or infrastructure index cannot replace those layers.

Stefano Gatti, Project Finance in Theory and Practice: Designing, Structuring, and Financing Private and Public Projects (4th ed.)

Difficulty: Advanced **Type:** Book / monograph

Status/version: Elsevier fourth edition, published 24 May 2023; publisher lists this as the latest edition at the 4 October 2026 cut-off.

A practical foundation for valuing a ring-fenced infrastructure project from contracted and forecast cash flows rather than applying corporate multiples mechanically. Government guarantees and country risk matter to the funding package. The cases teach deal design, but their assumptions must be replaced with current project contracts, tariffs, capital costs, and jurisdiction-specific law before underwriting an actual asset.

Extended note. Version record

Aleksandar Andonov; Roman Kräussl; Joshua Rauh, Institutional Investors and Infrastructure Investing

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies 34(8), 3880–3934, 2021; first online 20 April 2021.

Tests the claim that infrastructure funds supply stable, countercyclical cash flows. This is essential when setting an allocation benchmark or liquidity model: project assets and fund wrappers create different exposures. The sample concerns mainly equity positions and historic fund structures, so it does not establish the same result for directly held project debt.

Extended note. Version record

Asian Development Bank; EBRD; Global Infrastructure Hub; IDB; IsDB; OECD; UNESCAP; World Bank Group, PPP Reference Guide, Version 3.0

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Joint institutional guide, published 27 April 2017; version 3.0.

A structured map of how a public-private partnership is conceived, appraised, procured, contracted, and monitored. Use it before modeling a concession: determine who carries demand, construction, operating, political, and residual-value risk, and which government commitments support the cash flows. The guide explicitly describes itself as a synthesis and bibliography rather than a step-by-step manual; deal terms and public-law constraints must be checked in the relevant sector and jurisdiction.

Extended note. Version record

Ofgem, RIIO-3 Final Determinations; Finance Annex

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Ofgem final determination issued 4 December 2025 for the 1 April 2026–31 March 2031 control period; official finance annex and public models.

Adds regulated UK energy networks to the infrastructure path alongside concession and project-finance cases. The final determination and finance annex specify the regulatory asset value, cost-of-debt and equity allowances, depreciation, inflation indexation and revenue profiling used to set permitted network cash flows. Reconstruct a selected network company's allowed-revenue bridge with the public financial model, then stress capital expenditure, inflation, incentive performance and financing costs. Keep the regulatory asset value distinct from enterprise value and the allowance distinct from distributable cash. Verify licence modifications, errata and model version before applying 2026–2031 parameters to a company or security.

Extended note. Version record

World Bank Group, Private Participation in Infrastructure Database: Methodology; PPI data access

Difficulty: Advanced **Type:** Data / specification

Status/version: Living World Bank Group methodology page, consulted at the 4 October 2026 cut-off; page retains some stale coverage-year text.

Read this before using PPI financial-close counts or investment commitments as an infrastructure opportunity set. It provides project type, sponsors, debt providers, public support, revenue source, and status for cross-country analysis. Public-source coverage misses smaller deals and can misstate commitments, so it is unsuitable as a complete universe or a return series.

Extended note. Version record

EDHEC Scientific Infra & Private Assets Research Institute, Private Equity & Infrastructure Fund Alpha Tracking Using a Private Market Equivalent

Difficulty: Advanced **Type:** Report / research note

Status/version: Private Markets 2.0 Use Case Series, September 2024; 26-page institute PDF.

A worked implementation of Direct Alpha against privateMetrics and infraMetrics benchmarks, separating a fund's strategic allocation tilt from asset selection. It is useful when LPs need a cash-flow-matched manager scorecard and want to know whether a reported excess return reflects sector or project exposure. Its private-market benchmark is model-built and institute-provided, so the attribution depends on taxonomy, benchmark coverage, beta estimation, and the completeness of fund cash flows.

Extended note. Version record

World Bank Group, Infrastructure Monitor 2024: Global Trends in Private Investment in Infrastructure; Official report record

Difficulty: Advanced **Type:** Report / research note

Status/version: 2024 report, ©2025 World Bank Group; official resource page dated 1 January 2025.

Use Chapter 3 to stress-test assumptions about infrastructure equity, project debt default, and recovery rather than treating all assets as low risk. It also documents differences by sector, geography, and instrument. These are pooled historical benchmarks compiled from several vendors, with varying coverage and selection; they cannot replace transaction-level covenants, offtake quality, or downside modeling.

Extended note. Version record

Evan Clark, The Exit Test: What a Decade of Realisations Reveals About Valuation Accuracy

Difficulty: Advanced **Type:** Research paper

Status/version: EDHEC Infra & Private Assets research paper, June 2026; publisher summary verified.

Compares realized exit prices with the last unaffected carrying values for 333 private-equity and infrastructure-equity exits from 29 UK-listed investment trusts during 2015–2026. The reported median uplift is about 23% for infrastructure, challenging the assumption that periodic NAVs are unbiased transaction-price proxies. Exits are selected transactions and the time from mark to sale varies; neither uplift nor the underlying appraisal error should be extrapolated mechanically to unsold holdings.

Extended note. Version record

MSCI, Global Quarterly Property Fund Index: Q1 2026 Results

Difficulty: Advanced **Type:** Data / specification

Status/version: Publisher research page, 8 June 2026; sample of open-ended funds in North America, Europe and Asia Pacific.

Shows that property-sector and regional performance diverged even as broad open-end returns recovered: retail led the main sectors, while office had its first positive capital growth since 2022Q2. Use the breakdown to challenge a single generic cap-rate or rental-growth assumption across office, retail, industrial and residential. Read after property-level NPI and fund-level ODCE. The observation is Q1 2026, not a forecast.

Extended note. Version record

Board of Governors of the Federal Reserve System, Financial Stability Report: Overview and Asset Valuations, May 2026; full report

Difficulty: Advanced **Type:** Report / research note

Status/version: Official stability report dated 8 May 2026; market conditions through 23 April 2026.

Identifies substantial upcoming US commercial-real-estate refinancing needs even as transaction-price indexes stabilised. Use its credit and valuation context to run asset-level debt-yield, loan-to-value and debt-service tests at maturity, with alternative lender extensions or forced-sale outcomes. Read after mortgage and asset valuation. Attach the current rent roll, debt agreement, hedge maturity, appraised value and lender terms to each scenario.

Extended note. Version record

Keith Crouch, Wendy Hu and Jorge Velazquez (MSCI), The Rise and Concentration Risk of Data Centers in Private Markets

Difficulty: Advanced **Type:** Report / research note

Status/version: MSCI research note, 16 April 2026; proprietary 584-holding analysis to Q3 2025.

Maps data-center holdings across closed-end infrastructure, property and private-equity funds; seemingly distinct mandates can own the same companies or depend on the same compute-demand driver. Pair with infrastructure fund evidence. Its constructed historical index and cross-fund overlaps depend on proprietary coverage and cannot validate a particular data-center deal's power or lease terms.

Extended note. Version record

Global Infrastructure Hub and Turner & Townsend, PPP Contract Management Reference Tool; full guide

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: GI Hub original reference tool, based on 250 PPP projects; World Bank-hosted copy, historical guidance.

Turns a PPP financial close into an operating-control plan: construction completion and handover, performance reporting, payment deductions, change control, dispute records and refinancing-gain sharing. Use its "routine contract management" and refinancing material to model actual concession distributions after availability penalties and lifecycle maintenance. Governing concession, lender, direct-agreement and step-in provisions control the case; retain the procurement jurisdiction and contract version.

Extended note. Version record

Wakefield Council, Waste Management PFI contract repository; Schedule 4: Payment Mechanism

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Council describes a signed 25-year PFI contract beginning services in 2013; its public contract repository dated 6 December 2022 supplies the 72-page payment schedule. Selected schedule clauses inspected.

An actual PPP payment mechanism: annual unitary charge combines tonnage-based base payment, landfill and compensation amounts with performance, reporting, non-acceptance, recycling and third-party-income adjustments. Section 3.3 caps some deductions but expressly excludes specified diversion and non-acceptance deductions; section 14 converts the forecast charge and adjustments into monthly payments, and section 15 indexes selected components. Use forecast versus actual tonnage, service performance, indexation and third-party income to stress project cash available for debt service. The public project agreement PDF was too large for this retrieval and the schedule contains redactions; the governing agreement, amendment history, actual invoices and fund distributions are not certified from the schedule alone.

Extended note. Version record

Eduardo Engel, Ronald Fischer and Alexander Galetovic, *The Economics of Public-Private Partnerships: A Basic Guide*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Cambridge University Press, edition 1 (2014); DOI 10.1017/cbo9781139565615.

Economics of PPP contracts: risk allocation, demand risk, renegotiation and fiscal accounting; essential for availability-payment and concession assets.

Version record

16.3 Timberland and farmland

Subject context and qualifications.

Michael G. McIntosh and Daowei Zhang, Faustmann Formula and Its Use in Forest Asset Valuation: A Review and a Suggestion

Difficulty: Advanced **Type:** Research paper

Status/version: Forest Policy and Economics 160 (March 2024), article 103158; doi:10.1016/j.forpol.2024.103158.

Re-centers the Faustmann model on valuing forest land and stands as renewable assets rather than treating rotation age as an isolated forestry rule. Its survey connects discounted harvest revenues, regeneration costs, repeated rotations, and land expectation value, providing the deterministic base case for a timberland DCF. The framework abstracts from uncertain prices, fire, disease, carbon payments, and operational constraints unless those cash flows and risks are modeled explicitly.

Extended note. Version record

Randall Morck, Eduardo Schwartz, and David Stangeland, The Valuation of Forestry Resources under Stochastic Prices and Inventories

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis 24(4), December 1989, pp. 473–487; doi:10.2307/2330980.

Treats a forest as a contingent claim whose value changes with timber prices, biological growth, and inventory. The authors solve the harvest-timing decision under stochastic conditions and include the option to halt timber production temporarily, showing why a fixed cutting schedule can misvalue operational flexibility. The publisher abstract establishes the model, but calibration choices and the practical costs of delaying harvest need separate review for a particular forest.

Extended note. Version record

Oscar R. Burt, Econometric Modeling of the Capitalization Formula for Farmland Prices

Difficulty: Advanced **Type:** Research paper

Status/version: American Journal of Agricultural Economics 68(1), February 1986, pp. 10–26; doi:10.2307/1241645.

An empirical capitalization model linking farmland prices to net crop-share rents received by landlords. Burt estimates a lagged response in which rent changes feed into land values over a prolonged damped cycle, rather than requiring price to equal current rent divided by a constant cap rate. The paper's implicit 4% equilibrium tax-free capitalization rate is historical and sample-specific, not a current underwriting assumption.

Extended note. Version record

National Council of Real Estate Investment Fiduciaries (NCREIF), NCREIF Farmland Property Index: Overview and Methodology

Difficulty: Advanced **Type:** Data / specification

Status/version: Living index methodology and quarterly return page; base 1990 Q4.

A benchmark for income-producing U.S. farmland held by tax-exempt institutional investors. Its value-weighted quarterly return splits income and appreciation and reports returns as unlevered, making it useful for separating farm cash yield from mark changes in an allocation model. Property type, crop mix, region, and lease structure still need separate underwriting.

Extended note. Version record

National Council of Real Estate Investment Fiduciaries (NCREIF), NCREIF Timberland Property Index: Overview and Methodology

Difficulty: Advanced **Type:** Data / specification

Status/version: Living index methodology and quarterly return page.

The institutional benchmark for U.S. private timberland return analysis, reported quarterly on an unlevered property basis. Use its national and regional series to compare biological growth and harvest income with land-value changes and with liquid timber REITs. Consequently, the apparent volatility and correlations may reflect appraisal timing; the contributor-owned sample is not a census of timberland transactions.

Extended note. Version record

USDA Economic Research Service; USDA National Agricultural Statistics Service, Farmland Value (2026 update) and Land Values, 2026 Summary; NASS Quick Stats for state-level values and rents

Difficulty: Advanced **Type:** Data / specification

Status/version: ERS page updated 23 September 2026; underlying NASS Land Values, 2026 Summary dated July 2026.

Use the regional cropland, pasture, and cash-rent tables as external anchors for a U.S. farmland purchase model. The 2026 page reports a $4,500 national average per acre and shows that land use and regional farm economics produce substantial dispersion; it also discusses inflation-adjusted trends and factors such as soil quality and urban proximity. Estimate a farm's net rent, water rights, taxes, and crop risk separately.

Extended note. Version record

USDA Forest Service, The Forest Inventory and Analysis Database User Guide (NFI); Download user-guide PDF

Difficulty: Advanced **Type:** Data / specification

Status/version: User guide published 25 August 2025; host page last updated 28 May 2026.

The official schema, codes, and definitions for the Forest Inventory and Analysis database across U.S. ownerships. It supports an empirical timberland model by linking plot, tree, species, area, and inventory concepts to estimates of standing stock, growth, mortality, and potential harvest. The data describe forest resources, not land title, merchantable sale prices, transport cost, discount rates, or investor cash flows; those inputs require separate market and property diligence.

Extended note. Version record

Srijana Baral; Bin Mei, The Time-Varying Link of Public Timber REITs with Private Timberland, Real Estate, and Financial Assets

Difficulty: Advanced **Type:** Research paper

Status/version: Forest Science 68(5–6), October 2022, pp. 464–472; published online 22 August 2022; doi:10.1093/forsci/fxac031.

Compares liquid public timber vehicles with private timberland, commercial property, stocks, and bonds using a state-space volatility decomposition. The authors unsmooth the NCREIF Timberland Index and find public timber REIT risk is strongly linked to large-cap equities while private timberland contributes little to their measured volatility. The estimates cover U.S. quarterly returns through 2019 and depend on the chosen unsmoothing and factor-orthogonalization methods.

Extended note. Version record

Bin Mei, "Timberland investments in the United States: a review and prospects"

Difficulty: Advanced **Type:** Research paper

Status/version: Forest Policy and Economics, 109, 101998 (2019); DOI 10.1016/j.forpol.2019.101998.

Review of US institutional timberland investing: ownership history, return drivers, indices, risk and diversification.

Version record

Wolfram Schlenker, W. Michael Hanemann and Anthony C. Fisher, "Will U.S. Agriculture Really Benefit from Global Warming? Accounting for Irrigation in the Hedonic Approach"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review, 95(1), 395–406 (2005); DOI 10.1257/0002828053828455.

Hedonic farmland-value regressions that separate dryland and irrigated land show climate damages to non-irrigated farmland value.

Version record

Andrew J. Plantinga, Ruben N. Lubowski and Robert N. Stavins, "The effects of potential land development on agricultural land prices"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Urban Economics, 52(3), 561–581 (2002); DOI 10.1016/s0094-1190(02)00503-x.

Development option value accounts for about 10% of US farmland value on average and much more near cities.

Version record

Barry Falk, "Formally Testing the Present-Value Model of Farmland Prices"

Difficulty: Advanced **Type:** Research paper

Status/version: American Journal of Agricultural Economics, 73(1), 1–10 (1991); DOI 10.2307/1242877.

Iowa farmland prices are too volatile relative to rents for a constant-discount-rate present-value model.

Version record

16.4 Mineral and water rights

U.S. Bureau of Land Management, General Leasing: Federal Oil and Gas

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living federal agency page, reviewed 4 October 2026; current lease terms must be checked at sale date.

Describes federal mineral lease issuance, rentals, royalties, development approvals and assignment. Read after the farmland/timber and energy paths to distinguish surface ownership, mineral ownership, operator working interest and a passive royalty interest. The federal framework is not a template for state, tribal or private minerals; the web page's rate examples may refer to older leases and should never override signed terms.

Extended note. Version record

Government of Alberta, Petroleum and Natural Gas Tenure, Royalty Overview and Modernized Royalty Framework C* Calculator

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living official explanations inspected 4 October 2026; MRF applies to specified post-1 January 2017 wells/early opt-ins.

Connects Crown tenure and rental/royalty obligations to a revenue threshold based on a proxy drilling/completion allowance. The initial 5% royalty changes after cumulative revenue reaches C*; later rates depend on resource, reference price and production. Prerequisite: production decline, netback and government-versus-private royalty interests. Title, Indigenous rights, abandonment liabilities and the operative tenure still require asset-specific evidence.

Extended note. Version record

Petrinex, Alberta Public Data and Conventional Volumetrics learning aid; Alberta Energy Regulator, public-data catalogue

Difficulty: Not specified **Type:** Data / specification

Status/version: September 2022 learning aid; portal observed 4 October 2026. Documentation and downloaded January 2022 XML header/initial records inspected; no full-history reconciliation.

Provides monthly conventional volumetrics, well/facility links, operators and product/activity codes, with CSV/XML downloads and documented API patterns. Pair production and dispositions with price, ownership and royalty inputs from the preceding entry. Prerequisite: identifiers, units, gross versus allocated volumes and vintage control. Administrative coverage is broader than selected royalty-company transactions but excludes confidential/experimental information. The portal flags potentially stale business-associate licence-eligibility fields and directs users to AER ST104.

Extended note. Version record

Western Australia, Mining (Royalties) Regulations 2025; RevenueWA, Mineral Royalties in WA; DMPE, TENGRAPH/Mineral Titles route

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official consolidation 00-d0-00, current from 2 April 2026; selected valuation/rate/reporting provisions and commencement table inspected. Guidance updated 1 October 2026.

Separates mining tenure from State royalties and privately contracted royalty interests. Statutory cash obligations depend on mineral, processing stage, valuation basis and allowed deductions, with special mineral rules and State Agreements. Prerequisite: production, realised price, processing and title. The 2025 regime's substantive commencement was 1 September; January/April 2026 amendments must also be read. This selected WA path does not establish rights throughout Australia.

Extended note. Version record

Western Australian Department of Mines, Petroleum and Exploration, Mineral and Petroleum Statistics Digest 2024–2025; full report

Difficulty: Not specified **Type:** Report / research note

Status/version: Published 11 May 2026; 2024–25 observations. Selected tables and methodology/data-availability passages inspected.

Links commodity quantities, sales values, prices and State receipts, with documented major-commodity, economic-indicator and spatial/regional workbook families. Read after the preceding royalty rules; prerequisite: distinguish ore, concentrate and metal units, financial/calendar periods and price versus volume changes. Workbook families are documented, but their complete downloadable contents and a representative private-royalty return panel were not validated here.

Extended note. Version record

Australian Competition and Consumer Commission, Murray–Darling Basin Water Markets Inquiry: Final Report

Difficulty: Advanced **Type:** Report / research note

Status/version: Published 26 March 2021, corrected electronic version 10 May 2021; public Australian inquiry.

Uses trade records from 2012 and stakeholder evidence to examine entitlement and allocation markets, trade constraints, intermediaries and market information. Value a right by reliability, seasonal allocations, storage and delivery, transfer restrictions and agricultural demand rather than assuming water is one fungible spot commodity. The inquiry is Basin-specific and its recommendations are not proof of later rule adoption. The corrected version for citations.

Extended note. Version record

Victorian Water Register, Allocation trading and Water share trading

Difficulty: Advanced **Type:** Data / specification

Status/version: Living export interfaces inspected 4 October 2026; year selections include 2009–2010 through 2026–2027.

Supplies transaction-record export paths for allocation trades and water-share transfers by region and year, with reliability filters for shares. Distinguish seasonal allocation from ownership of the entitlement. Allocation exports include trade-reason fields; pre-27 August 2020 and paper applications can lack those reasons or deal dates. Share exports include recorded, rejected and expired applications, with applicant-reported prices; bundled land transactions and non-monetary transfers complicate return inference. Filter status, commercial purpose, zone and reliability before comparison. These Victorian records add a transaction-data path to the Australian inquiry and Chilean material; they are not a national or all-cost return index.

Version record

Chile Dirección General de Aguas, registered water rights and regional/national files; water legislation register

Difficulty: Advanced **Type:** Data / specification

Status/version: Registry page cut-off 4 September 2026, observed 4 October; current official page and legislative listings inspected. National file retrieval was unavailable here.

Documents original rights, reported transfers and approved changes of abstraction point/exercise, with historical regional spreadsheets and a monthly national consolidation. Read the 2022 Water Code reform with subsequent registration/enforcement amendments, including laws listed in 2025. Prerequisite: title, permitted use, hydrological supply and abstraction location. No current statutory deadline is inferred from an old summary.

Extended note. Version record

Eric C. Edwards, Oscar Cristi, Gonzalo Edwards and Gary D. Libecap, An Illiquid Market in the Desert; NBER version and publisher's online appendix

Difficulty: Research **Type:** Research paper

Status/version: *Environment and Development Economics* 23(6), 615–634, December 2018; NBER January 2016, revised February 2018. Indexed primary chronology and full appendix inspected.

Compares restricted agricultural-to-mining transfers with desalination and pumping alternatives. The appendix exposes energy, distance, efficiency, electricity-price and discount-rate assumptions, with transaction regressions and sensitivity cases. Prerequisite: hydrology, location-specific delivery cost and counterfactual welfare analysis. Use the worked assumptions to distinguish a transferable right from the cost of delivered water. Journal full-text equivalence and raw microdata access remain unverified.

Extended note. Version record

Harold Hotelling, "The Economics of Exhaustible Resources"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Political Economy, 39(2), 137–175 (1931); DOI 10.1086/254195.

Resource rents rise at the rate of interest in competitive extraction; the benchmark for valuing reserves and royalties.

Version record

Michael J. Brennan and Eduardo S. Schwartz, "Evaluating Natural Resource Investments"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Business, 58(2), 135 (1985); DOI 10.1086/296288.

Real-options valuation of a mine with options to open, close and abandon under stochastic commodity prices.

Version record

James L. Paddock, Daniel R. Siegel and James L. Smith, "Option Valuation of Claims on Real Assets: The Case of Offshore Petroleum Leases"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 103(3), 479 (1988); DOI 10.2307/1885541.

Values undeveloped offshore oil leases as options and compares with auction bids; the classic real-options application to resource rights.

Version record

Ryan Kellogg, "The Effect of Uncertainty on Investment: Evidence from Texas Oil Drilling"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 104(6), 1698–1734 (2014); DOI 10.1257/aer.104.6.1698.

Texas drillers delay investment when expected oil-price volatility rises, as real-options theory predicts.

Version record

Soren T. Anderson, Ryan Kellogg and Stephen W. Salant, "Hotelling under Pressure"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 126(3), 984–1026 (2018); DOI 10.1086/697203.

Oil production from existing wells does not respond to prices; drilling does. Reformulates Hotelling around the drilling decision.

Version record

R. Quentin Grafton, Gary Libecap, Samuel McGlennon, Clay Landry and Bob O'Brien, "An Integrated Assessment of Water Markets: A Cross-Country Comparison"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Environmental Economics and Policy, 5(2), 219–239 (2011); DOI 10.1093/reep/rer002.

Compares water markets in Australia, Chile, China, South Africa and the western US on institutional and market performance.

Version record

Avinash K. Dixit and Robert S. Pindyck, *Investment under Uncertainty*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Princeton University Press, 1994; JSTOR edition DOI 10.2307/j.ctt7sncv.

The real-options text: irreversible investment, timing and option value, with resource-extraction and entry/exit applications.

Version record

16.5 Collectibles

Jianping Mei and Michael Moses, Art as an Investment and the Underperformance of Masterpieces

Difficulty: Research **Type:** Research paper

Status/version: *American Economic Review* 92(5), 1656–1668, December 2002; publisher article.

Builds repeat-sale evidence and tests whether expensive works outperform. Read as a method for separating repeat-sale price changes from broad claims that unique objects earn an illiquidity premium. Add insurance, conservation, storage, tax and auction costs to any actual holding-period return. A repeat-sales index is not a purchasable portfolio; compare the counterevidence in Etro–Stepanova before applying a masterpiece premium.

Extended note. Version record

Federico Etro and Elena Stepanova, Art Return Rates from Old Master Paintings to Contemporary Art

Difficulty: Research **Type:** Research paper

Status/version: *Journal of Economic Behavior & Organization* 181, 94–116, January 2021; published article with author-institution proof.

Uses worldwide Christie's and Sotheby's repeat sales from 2000–2018 and controls buyer premiums, resale rights, guarantees, bought-in lots and news such as attribution. It reports sector differences and no uniform masterpiece effect, refining the older Mei–Moses inference. Use only for a documented auction-return benchmark. The sample still omits most private sales and ongoing ownership costs; lot-level authenticity, condition, provenance and the bid/ask wedge determine an investable collectible's realized outcome.

Version record

Elroy Dimson and Christophe Spaenjers, Ex Post: The Investment Performance of Collectible Stamps; 2009 working paper

Difficulty: Research **Type:** Research paper

Status/version: *Journal of Financial Economics* 100(2), 443–458, May 2011. Published identity verified; working-paper data and cost discussion inspected.

Studies British stamp catalogue prices over 1900–2008, with smoothing and nonsynchronous-market adjustments. Prerequisite: historical price-index construction and the distinction between dealer/catalogue quotations and executable transactions. Pair with the auction-selection critique for art. Use to examine how quoted-price smoothing hides volatility, not as an investable return forecast. The accessible working paper is retained separately from the journal version; full version equivalence and a public raw-price panel were not established.

Extended note. Version record

Elroy Dimson, Kuntara Pukthuanthong and Blair Vorsatz, Emotional Yields of Collectibles; Cambridge manuscript

Difficulty: Research **Type:** Research paper

Status/version: *Financial Analysts Journal* 82(2), 19–45; online 16 April 2026. Publisher chronology and February 8 manuscript's data/method sections inspected; full version equivalence unchecked.

Uses factor-mimicking portfolios to estimate noncash ownership benefits across 30 series in 13 categories, including coins, wine, classic cars, violins, furniture and rugs. Prerequisite: factor exposure matching, nonsynchronous prices and index construction. The data discussion distinguishes auction selection, quality changes, look-ahead selection and smoothing. No complete public raw panel or representative investor returns after ownership costs and fees were verified.

Extended note. Version record

William N. Goetzmann, "Accounting for Taste: Art and the Financial Markets Over Three Centuries"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review 83(5), 1370–1376 (December 1993); bibliographic record via RePEc.

Repeat-sales art index over three centuries; art returns correlate with wealth and are volatile.

Version record

Arthur Korteweg, Roman Kräussl and Patrick Verwijmeren, "Does it Pay to Invest in Art? A Selection-Corrected Returns Perspective"

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies, 29(4), 1007–1038 (2016); DOI 10.1093/rfs/hhv062.

Correcting for sale selection lowers painting index returns from 8.7% to 6.3% and the Sharpe ratio from 0.27 to 0.11.

Version record

Luc Renneboog and Christophe Spaenjers, "Buying Beauty: On Prices and Returns in the Art Market"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 59(1), 36–53 (2013); DOI 10.1287/mnsc.1120.1580.

Hedonic art price index from over a million auction sales; art underperforms equities with sentiment-linked prices.

Version record

Elroy Dimson, Peter L. Rousseau and Christophe Spaenjers, "The price of wine"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 118(2), 431–449 (2015); DOI 10.1016/j.jfineco.2015.08.005.

Long-run fine-wine returns with storage, ageing and insurance costs; separates consumption and investment value.

Version record

William J. Baumol, "Unnatural Value: Or Art Investment as Floating Crap Game"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Arts Management and Law, 15(3), 47–60 (1985); DOI 10.1080/07335113.1985.9942162.

Art prices have no fundamental anchor, so returns behave like a random walk with low average returns; the starting point of art-investment economics.

Version record

Benjamin J. Burton and Joyce P. Jacobsen, "Measuring Returns on Investments in Collectibles"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 13(4), 193–212 (1999); DOI 10.1257/jep.13.4.193.

Review of repeat-sales and hedonic methods for collectibles returns and their biases; collectibles underperform equities with high volatility.

Version record

Benjamin R. Mandel, "Art as an Investment and Conspicuous Consumption Good"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review, 99(4), 1653–1663 (2009); DOI 10.1257/aer.99.4.1653.

Consumption (emotional) dividends explain low financial returns and high volatility of art.

Version record

William N. Goetzmann, Luc Renneboog and Christophe Spaenjers, "Art and Money"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review, 101(3), 222–226 (2011); DOI 10.1257/aer.101.3.222.

Top income and equity-market wealth drive art prices over two centuries.

Version record

Julien Pénasse and Luc Renneboog, "Speculative Trading and Bubbles: Evidence from the Art Market"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 68(7), 4939–4963 (2022); DOI 10.1287/mnsc.2021.4088.

Short holding-period resales spike in art booms and predict lower subsequent returns, consistent with speculative bubbles.

Version record

Stefano Lovo and Christophe Spaenjers, "A Model of Trading in the Art Market"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 108(3), 744–774 (2018); DOI 10.1257/aer.20160522.

Auction-trading model with emotional dividends and liquidity shocks explains art-price dynamics and selection in repeat sales.

Version record

Amy Whitaker and Roman Kräussl, "Fractional Equity, Blockchain, and the Future of Creative Work"

Difficulty: Advanced **Type:** Research paper

Status/version: Management Science, 66(10), 4594–4611 (2020); DOI 10.1287/mnsc.2020.3633.

Fractional ownership of art and creative work via tokens: design, artist-equity and valuation issues for tokenised collectibles.

Version record

Chapter 17

Insurance-linked investments

Connect insured losses and demographic risks to triggers, payout timing, model uncertainty and capital release. Index and portfolio exposure need not coincide.

17.1 Catastrophe, mortality and longevity

Subject context and qualifications.

Kenneth A. Froot, The Market for Catastrophe Risk: A Clinical Examination

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 8110, February 2001, DOI 10.3386/w8110; NBER records published Journal of Financial Economics 60(2–3), 529–571, May 2001. Link retains accessible working-paper version and journal-status distinction.

Examines why insurers historically purchased relatively little protection for the largest catastrophe layers despite strong risk-management reasons to transfer them. Clinical transactions point to capital supply restrictions and reinsurer market power as plausible explanations for high premiums relative to expected loss. The empirical market and catastrophe models are early-2000s; current ILS capacity, collateral, attachment structures, and peril correlations require contemporary measurement.

Extended note. Version record

J. David Cummins and Mary A. Weiss, Convergence of Insurance and Financial Markets: Hybrid and Securitized Risk-Transfer Solutions

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Risk and Insurance 76(3), 493–545, September 2009; first online 19 August 2009

Surveys the boundary between reinsurance and capital markets, including catastrophe bonds, industry-loss warranties, and hybrid contracts, with evidence on pricing and returns. Its transaction data describe an earlier market and should not anchor current spread assumptions. Modelled expected loss, collateral structure, trapped capital after an event, and sponsor-specific contract wording must be reconciled before comparing an ILS coupon with a conventional bond yield.

Extended note. Version record

Alexander Braun, Pricing in the Primary Market for Cat Bonds: New Empirical Evidence

Difficulty: Research **Type:** Research paper

Status/version: Journal of Risk and Insurance 83(4), 811–847, December 2016; first online 24 February 2015.

Uses a near-complete set of 1997–2012 issuance tranches to estimate how expected loss, territory, sponsor, reinsurance cycle, and comparable corporate spreads explain primary cat-bond spreads. The reported out-of-sample comparison is a useful benchmark for separating modeled event loss from capital-market and issuance conditions. An investor should rerun calibration on current perils, model versions, attachments, and rate environments before using the historical coefficients.

Extended note. Version record

Simone Beer and Alexander Braun, Market-Consistent Valuation of Natural Catastrophe Risk

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance 134, 106350, January 2022; indexed

Extracts implied catastrophe intensities from secondary cat-bond quotes and relates them to time to maturity and modeled first-loss probability. It connects deal-level catastrophe modeling with an observable market-consistent valuation input, helping an investor mark less liquid protection between issuance and maturity. Thin trading and model changes should be treated explicitly when using it for daily valuation.

Extended note. Version record

Tse-Ling Teh and Christopher Woolnough, A Better Trigger: Indices for Insurance

Difficulty: Research **Type:** Research paper

Status/version: Journal of Risk and Insurance 86(4), 861–885, December 2019; first online 28 February 2018

Shows that a high correlation between an index and insured loss can still produce a poor trigger, because missed-loss and unnecessary-payment basis risk affect utility differently. The framework ranks binary and multivalue indices under risk aversion and is directly relevant to parametric catastrophe bonds, weather cover, and crop protection. Utility assumptions and estimated joint tails limit how confidently a ranking transfers across insureds.

Extended note. Version record

World Bank, World Bank Catastrophe Bonds Provide Mexico with $595 Million Financial Protection against Hurricanes and Earthquakes

Difficulty: Advanced **Type:** Report / research note

Status/version: World Bank Treasury case study dated 27 February 2025 about four bonds issued April–May 2024; official three-page PDF. The 2024 deal date and 2025 publication date are distinct.

Documents four sovereign catastrophe-bond classes with earthquake and hurricane triggers, showing how physical event measures, attachment probability, expected loss, coupon risk margin, and payout timing appear in an actual risk-transfer structure. The revised earthquake grid and earlier hurricane data address distinct types of basis and settlement delay. The PDF summary and table disagree on one attachment-probability range, so quote tranche statistics from the table only.

Extended note. Version record

Swiss Re Alternative Capital Partners, ILS Market Insights: July 2026

Difficulty: Advanced **Type:** Report / research note

Status/version: Swiss Re market report page dated 23 July 2026, reviewing the first half of 2026. Released institutional market commentary, not independent academic research.

Updates issuance, sponsor mix, spread conditions, secondary activity, and Swiss Re catastrophe-bond index context after the older academic samples. It is useful for checking whether a historical expected-loss-to-spread relationship still resembles the current primary market and for forming capacity scenarios. Issuance and spread figures describe deals actually placed, not the loss-adjusted return an investor will realize after future catastrophe events.

Extended note. Version record

O. Kolesnikov, A. Markov, D. Smagulov and S. Solovjovs, Cyber Loss Distribution Fitting: A General Framework towards Cyber Bonds and Their Pricing Models

Difficulty: Research **Type:** Research paper

Status/version: *International Journal of Mathematics and Mathematical Sciences*, article 7689828, first published 31 December 2022; open-access journal article.

Builds a cyber-bond pricing example from observed cyber-event losses, modelling time between events with an exponential distribution and loss size with a log-normal distribution. The article then simulates note prices, yields and coupon structures. Use it to construct a frequency–severity loss engine and compare attachment probability, expected loss and risk margin with natural-catastrophe bond methods. Its fitted parameters come from publicly observed historical events; reporting bias, changing exposure, shared service providers and correlated attacks can defeat an independent-event assumption. Treat the output as a transparent modelling exercise, then stress systemic aggregation and trigger wording before applying it to an issued bond.

Extended note. Version record

Conyers, PoleStar Re Ltd. US$300 Million Series 2026-1 Note Offering and Listing

Difficulty: Advanced **Type:** Report / research note

Status/version: Transaction-counsel announcement dated December 2025 for a US$300 million Series 2026-1 cyber catastrophe note offering and listing.

Provides primary transaction evidence that an insurer transferred cyber risk through a principal-at-risk note rather than only a hypothetical model. Conyers says it advised the PoleStar special-purpose insurer on a US$300 million Series 2026-1 offering linked to Beazley and assisted with Bermuda Stock Exchange listing and closing. Pair the announcement with cyber-loss modelling, then seek offering documents to identify covered cyber events, trigger, attachment, exhaustion, collateral, coupon and maturity. The brief counsel notice establishes size, structure and completed transaction work; it does not reveal expected loss, spread, investor return or the full risk-transfer contract.

Extended note. Version record

Andrew J. G. Cairns, David Blake, and Kevin Dowd, A Two-Factor Model for Stochastic Mortality with Parameter Uncertainty: Theory and Calibration; Author-hosted manuscript

Difficulty: Research **Type:** Research paper

Status/version: Journal of Risk and Insurance 73(4), 687–718, December 2006; first online 29 November 2006

Models post-60 UK mortality with one factor shifting all ages and another more strongly affecting higher ages, then considers cohort-dependent longevity-bond pricing with parameter uncertainty. It shows why long-horizon annuity or pension risk cannot be reduced to a single average life-expectancy shock. Re-estimate age, cohort, population, trend-break, and parameter uncertainty for the liability being hedged; the risk premium is not directly observable.

Extended note. Version record

Human Mortality Database, Overview, country data and methods

Difficulty: Advanced **Type:** Data / specification

Status/version: Official overview and data portal inspected 4 October 2026; country and download vintages must be pinned.

Supplies national death counts, exposures, mortality rates and life tables for a reproducible longevity exercise. Its harmonised methods support comparison across selected countries with sufficiently complete registration; coverage is not representative of every insured or pension population. Separate period tables from cohort survival assumptions and document age, sex, exposure construction and revisions. Full database access requires registration and acceptance of the user agreement. Use a permitted country extract or an explicitly synthetic example, then test population basis risk rather than treating national mortality as an insurer's realised experience.

Version record

Office for National Statistics, National life tables: UK; 2022–2024 release

Difficulty: Advanced **Type:** Data / specification

Status/version: Released 10 December 2025; observation window 2022–2024; public XLSX link listed.

Adds a public starting dataset for longevity work alongside the registration-based Human Mortality Database. National tables pool three consecutive years and report period life expectancy by age and sex. Use the UK workbook and its definitions to construct a period survival curve, then state the improvement assumptions required for a cohort projection. Population mortality does not represent an insured or pension population without a basis adjustment. Preserve observation years, release date and downloaded vintage separately. The release and download listing were checked; the workbook cells were not audited in this reading-list review.

Version record

David Blake, Andrew Cairns, Kevin Dowd, and Richard MacMinn, Longevity Bonds: Financial Engineering, Valuation, and Hedging

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Risk and Insurance 73(4), 647–672, December 2006; first online 29 November 2006.

Explores alternative longevity-bond designs, how their cash flows respond to longevity shocks, and the challenges of valuation and hedging in an incomplete market. It highlights reference-population choice and the scarcity of ultra-long fixed-income instruments, both essential when matching pension liabilities. Model value does not eliminate basis risk between a broad mortality index and a specific insured cohort.

Extended note. Version record

Guy D. Coughlan, Marwa Khalaf-Allah, Yijing Ye, Sumit Kumar, Andrew J. G. Cairns, David Blake, and Kevin Dowd, Longevity Hedging 101: A Framework for Longevity Basis Risk Analysis and Hedge Effectiveness; Author-hosted working manuscript

Difficulty: Research **Type:** Research paper

Status/version: North American Actuarial Journal 15(2), 150–176, 2011; publisher migration date is 2012.

Gives a framework for calibrating an index-based longevity hedge when a pension or annuity portfolio's mortality differs from the reference population. This moves the mortality model from an elegant population curve to an actual liability hedge. Stable historical relationships may break under selection, migration, medical changes, or small-cohort noise; hedge effectiveness also depends on tenor, collateral, and the instrument's sparse market liquidity.

Extended note. Version record

Ronald D. Lee and Lawrence R. Carter, "Modeling and Forecasting U.S. Mortality"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of the American Statistical Association, 87(419), 659 (1992); DOI 10.2307/2290201.

Log-bilinear mortality model with a single time index; the benchmark for longevity forecasting and hedging.

Version record

Shaun S. Wang, "A Class of Distortion Operators for Pricing Financial and Insurance Risks"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Risk and Insurance, 67(1), 15 (2000); DOI 10.2307/253675.

Wang transform for pricing insurance and financial risks; used in cat-bond pricing.

Version record

Morton Lane and Olivier Mahul, "Catastrophe Risk Pricing: An Empirical Analysis"

Difficulty: Advanced **Type:** Report / research note

Status/version: World Bank Policy Research Working Paper 4765, November 2008.

Empirical cat-bond spread model linking spreads to expected loss and market cycle (World Bank policy research working paper).

Extended note. Version record

Kirsten Mitchell-Wallace, Matthew Jones, John Hillier and Matthew Foote (eds.), *Natural Catastrophe Risk Management and Modelling: A Practitioner's Guide*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Wiley, edition 1 (2017); DOI 10.1002/9781118906057.

Hazard, vulnerability, financial modules and model uncertainty in catastrophe models, with portfolio-management use.

Version record

Ralph S. J. Koijen and Motohiro Yogo, "The Cost of Financial Frictions for Life Insurers"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 105(1), 445–475 (2015); DOI 10.1257/aer.20121036.

Life insurers sold annuities below reserve value in 2008–09 to relieve statutory capital constraints.

Version record

Ralph S. J. Koijen and Motohiro Yogo, "Shadow Insurance"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 84(3), 1265–1287 (2016); DOI 10.3982/ecta12401.

Reinsurance to affiliated captives reduces reported capital requirements; risk implications for life insurers.

Version record

Edward W. Frees, *Constructing Insurable Risk Portfolios*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Chapman & Hall/CRC, 8 April 2025 (listing date).

CRC (April 2025) treatment of portfolio construction for insurable risks: retention, reinsurance, risk measures and optimisation.

Extended note. Version record

Tuomas Tomunen, "Failure to Share Natural Disaster Risk"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 39(3), 661–701 (2026); DOI 10.1093/rfs/hhaf055.

Intermediary risk exposures explain 71% of security-level variation in catastrophe-bond expected returns; evidence that intermediary constraints price cat risk.

Version record

17.2 Loss aggregation, contracts and capital release

The same expected loss can produce different cash timing, reinstatement and collateral requirements. Separate the insured exposure, reinsurance contract, special-purpose vehicle and investor note before modelling an ILS payoff.

National Association of Insurance Commissioners, Catastrophe Models (Property); Catastrophe Model Primer, "Key Metrics and Outputs"

Difficulty: Advanced **Type:** Mixed resources

Status/version: NAIC living topic page and 2025 primer; regulator educational material.

Defines occurrence exceedance probability from the largest event in a year and aggregate exceedance probability from all events in a year. Build separate event and annual-loss tables before pricing an occurrence layer, an annual aggregate layer or a reinstating cover; the same annual average loss does not determine either tail. Verify those in the final deal documents and retain model version and exposure date.

Extended note. Version record

Haibo Liu, Robust Indifference Valuation of Catastrophe Bonds

Difficulty: Research **Type:** Research paper

Status/version: *Insurance: Mathematics and Economics* 122, 1–10, May 2025; version of record abstract.

Prices a catastrophe bond when event intensity and severity distributions are ambiguous to a risk-averse investor. The robust bid–ask spread widens with uncertainty aversion, giving a disciplined sensitivity test around a single vendor's expected-loss estimate. Follow market-consistent valuation. The publisher abstract supports the model result; a full calibration was not inspected. Ambiguity sets, peril correlations, exposure growth and secondary-market liquidity require independent estimates, and the model is not a quoted spread rule.

Version record

Haibo Liu, Yuhao Liu, Qihe Tang and Jinxia Zhu, Pricing Catastrophe Risk during Transitions of Physical and Economic Environments; authors' numerical code

Difficulty: Research **Type:** Research paper

Status/version: *European Journal of Operational Research* 331(2), 615–628, June 2026; version of record.

Models event intensity, risk-free discounting and floating coupon rates jointly as physical and economic regimes shift. Its hypothetical hurricane bond shows how a fixed catastrophe hazard or independent interest-rate scenario can miss state-dependent note value; replicate the authors' code as a method study before substituting locally calibrated hazards. The GitHub repository is mutable; pin a commit and match exposure, event definition and trigger to actual note documents.

Extended note. Version record

Swiss Re Capital Markets, Topanga Re Series 2025-1 for Farmers Insurance Group

Difficulty: Advanced **Type:** Report / research note

Status/version: Issuer-side transaction release, 18 December 2025; USD 400 million two-class placement.

The Class A notes' drop-down lets collateralised per-occurrence cover become reinstatement protection in specified third-event scenarios. Trace the event sequence through attachment, exhaustion, drop-down and note principal to see why a simple annual loss threshold is insufficient. Read after occurrence and aggregate metrics. The exact trigger, covered territory, event count, fees and principal return require offering documents; do not generalise the mechanism to all cat bonds.

Extended note. Version record

Enstar Group, Second Property ILS Transaction; Enstar–Artex Exit Arrangement

Difficulty: Advanced **Type:** Report / research note

Status/version: Issuer releases, 5 November 2024 completed transaction and 15 April 2026 strategic arrangement.

Shows two different paths for releasing investor capital after underwriting ends: a completed transfer to a rated balance sheet that eliminated collateral requirements, and prospective or retrospective exit products intended to shorten trapped-capital tails. Press releases are transaction-party descriptions; the 2026 arrangement is a product offering, not evidence that all offered exits have closed. Price counterparty substitution, residual loss creep, legal finality and release conditions from transaction documents.

Extended note. Version record

Slide Insurance / White Rock Insurance (SAC), account T104, Aggregate Catastrophe Excess of Loss Reinsurance Contract

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Signed 19–20 November 2024; SEC exhibit filed 2025. Articles 28, 32–33 and signatures inspected in full text.

At expiration, possible claims retain collateral based on paid losses, outstanding reserves, IBNR and loss-adjustment expenses, adjusted by loss-age buffers, retention, limit and prior payments. Quarterly/new-estimate recalculation governs release above 102% of obligations; a 67-month reporting period ends in commutation. Prerequisite: distinguish coverage expiry, ultimate loss and cash release. It is collateralised reinsurance, not evidence that every cat bond or quota-share sidecar uses these terms.

Extended note. Version record

Aon Reinsurance Solutions, Reinsurance Market Dynamics, April 2026

Difficulty: Advanced **Type:** Report / research note

Status/version: Broker market report, April 2026; institutional but commercial source.

Reports more than USD 5 billion of sidecar-capacity growth in 2025, largely casualty and multiline structures. Use it to distinguish a quota-share sidecar's proportional premium, claims and float from a fixed-principal, event-triggered catastrophe bond. Read after securitised risk-transfer architecture. The broker's capacity estimate is not a cash-flow dataset or an independent return series. Contract terms, claims reporting, investment of float, recapture and sponsor leverage determine an actual sidecar's return and liquidity.

Version record

RenaissanceRe, 2025 Form 10-K, variable-interest-entity note: Upsilon RFO

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Fiscal year ended 31 December 2025; full note inspected, filed 2026.

Links prior underwriting years' cedant collateral releases to subsequent returns of capital to segregated-account investors. It also distinguishes consolidated Upsilon accounts from an unconsolidated account and separates the sponsor's participation from third-party capital. Returned capital includes principal and is not automatically profit or IRR; sponsor management/performance fees and investors' cash dates are needed. Surviving issuer accounts do not form a representative ILS net-return panel.

Extended note. Version record

AllianceBernstein L.P., June 2026 Form 10-Q, commitments note on RGA and FCA Re sidecars

Difficulty: Advanced **Type:** Report / research note

Status/version: Unaudited issuer 10-Q filed 31 July 2026, observations at 30 June 2026; selected commitments note inspected in full text.

Investor-side disclosure separates a sidecar commitment from contributed capital: AllianceBernstein states USD 100 million committed to an RGA-sponsored vehicle, with only USD 0.1 million funded by 30 June 2026, and a separate USD 100 million FCA Re commitment with none funded at that date. The latter targets Asian life and annuity liabilities. Map commitment, capital calls, invested assets and future claims before modelling an investor's earned return; a manager's separate asset-management mandate is also not a sidecar distribution. This 10-Q does not supply either sidecar's executed investor agreement, actual loss/runoff allocation or after-fee returns.

Extended note. Version record

Malene Kallestrup-Lamb and Nicolai Søgaard Laursen, Longevity Hedge Effectiveness Using Socioeconomic Indices; author-university open copy

Difficulty: Research **Type:** Research paper

Status/version: *Insurance: Mathematics and Economics* 114, 242–251, January 2024; published open manuscript.

Compares full-population and socioeconomic-group mortality indices for hedging annuity liabilities, measuring both risk reduction and the premium or notional cost. It sharpens the older generic basis-risk framework: better demographic matching can improve hedge effectiveness, but a narrower index may trade less deeply. Read after longevity basis-risk analysis. Validate sex, age, socioeconomic, region and data-release lags before selecting an index.

Extended note. Version record

David Landriault, Bin Li, Hong Li and Yuanyuan Zhang, Contract Structure and Risk Aversion in Longevity Risk Transfers; university open copy

Difficulty: Research **Type:** Research paper

Status/version: *Insurance: Mathematics and Economics* 128, article 103251, May 2026; peer-reviewed open-access article.

Compares long-dated pre-committed and short-dated rolled longevity swaps when buyer and seller have different risk aversion. Use it to price term, renewal and indemnity-versus-index design jointly. Continue from longevity securities and demographic basis. The

equilibrium result rests on the model's Stackelberg bargaining, risk preferences and ambiguity set; it is a design framework, not evidence that an actual short swap can always be rolled at the assumed price.

Extended note. Version record

UK Prudential Regulation Authority, SS18/16: Longevity Risk Transfers; DP2/25: Alternative Life Capital

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SS18/16 current 15 November 2024, effective 31 December 2024; DP2/25 November 2025 discussion, not final policy.

The live supervisory statement frames genuine risk transfer, counterparty and residual basis risk for longevity swaps. The discussion paper contrasts uncapped, long-duration longevity reinsurance with contractually bounded, time-limited capital-market structures, including possible life sidecars and insurance SPVs. DP2/25 solicits views and cannot be cited as an enacted capital rule; match survivor data, cohort, index, term, collateral and recapture rights to the actual transaction.

Extended note. Version record

Darius Lakdawalla and George Zanjani, "Catastrophe Bonds, Reinsurance, and the Optimal Collateralization of Risk Transfer"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Risk and Insurance, 79(2), 449–476 (2012); DOI 10.1111/j.1539-6975.2011.01425.x.

Why fully collateralised cat bonds and sidecars coexist with traditional reinsurance: collateral reduces counterparty risk at a cost.

Version record

Marc Gürtler, Martin Hibbeln and Christine Winkelvos, "The Impact of the Financial Crisis and Natural Catastrophes on CAT Bonds"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Risk and Insurance, 83(3), 579–612 (2016); DOI 10.1111/jori.12057.

Cat-bond spreads rose in the financial crisis and after major catastrophes, showing non-zero beta to financial stress.

Version record

Shan Ge, "How Do Financial Constraints Affect Product Pricing? Evidence from Weather and Life Insurance Premiums"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 77(1), 449–503 (2022); DOI 10.1111/jofi.13093.

Insurers hit by weather losses raise life-insurance prices, showing internal capital constraints transmit across lines.

Version record

Chapter 18

Digital assets

Separate protocol design, spot trading, derivatives, financing, tokenisation and security. Recent mechanism research sits beside the market and contract foundations it extends; crypto spot markets and tokenised securities and real-world assets have their own sections. Event contracts and prediction markets, including on-chain venues, have their own chapter (Chapter 19).

18.1 Protocols, market structure and risk

Digital-market applications share protocol and market-structure foundations. Research outlets are consolidated in Section 23.8. Cross-references: Easley et al., Flash Boys 2.0, Crypto Market Structure 3.0, Common Risk Factors in Cryptocurrency, and Gauntlet model evaluation.

18.1.1 Foundations and protocol texts

Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, *Bitcoin and Cryptocurrency Technologies* (Princeton University Press, 2016)

Difficulty: Intro **Type:** Book / monograph

Status/version: 2016 published book; free author-hosted pre-publication draft is a separate version.

A rigorous computer-science foundation for consensus, transactions, mining, anonymity and protocol design. It is necessary because digital-asset risk cannot be understood solely from prices and order books; settlement and protocol mechanics are part of the asset.

Applications: Bitcoin, blockchains; protocol and operational risk. Version record

Andreas M. Antonopoulos and David A. Harding, Mastering Bitcoin: Programming the Open Blockchain

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Third edition, first print, December 2023. Fixed third_edition_print_1 tag; development branch may differ.

Adds transaction-level programming to the existing conceptual Bitcoin text. Read the transaction walkthrough, keys, transactions, fees, networking and consensus selectively; trace UTXOs, change outputs and fee units before modelling exchange transfers.

Extended note. Version record

Roger Wattenhofer, *Blockchain Science: Distributed Ledger Technology*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2019

A distributed-systems treatment of consensus, fault tolerance, Proof-of-Work and the mechanisms underlying blockchains. Connects network and adversary assumptions to settlement and protocol behaviour, complementing Narayanan's introduction. The author's ETH course lists the 2019 Inverted Forest Publishing edition. Useful preparation for on-chain mechanism design and protocol risk analysis.

Applications: Blockchains; consensus, settlement, protocol design. Version record

Kang Li et al., *Security Analysis and Formal Verification on Blockchain and its Applications*

Difficulty: Research **Type:** Book / monograph

Status/version: 2025

Surveys blockchain security and formal verification of smart contracts and consensus, including vulnerabilities, tools and research challenges. Published in *Foundations and Trends in Privacy and Security*, 8(1), 1–121, DOI 10.1561/3300000044. For researchers with formal-methods or computer-science preparation evaluating DeFi contract correctness and protocol assumptions.

Applications: DeFi, smart contracts; security, formal verification. Version record

Andreas M. Antonopoulos, Gavin Wood, Carlo Parisi, Alessandro Mazza and Niccolò Pozzolini, Mastering Ethereum: Implementing Smart Contracts; Publisher edition record; Ethereum transaction reference; Solidity documentation

Difficulty: Intro/Advanced **Type:** Book / monograph with documentation

Status/version: Second edition, October 2025 (publisher record). Selected second-edition online book plus living transaction/compiler documentation, inspected 12 October 2026.

Supplies the missing implementation bridge from blockchain concepts to EVM trading: nonces, calldata, gas, token allowances, receipts, oracles and contract state. Read transactions and smart contracts before the DeFi, EVM, consensus and scaling chapters. Use the linked language reference for the compiler actually studied.

Extended note. Version record

Satoshi Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System"

Difficulty: Intro **Type:** Research paper

Status/version: 2008

Original proof-of-work ledger design, settlement and issuance. Establishes the security and settlement assumptions behind on-chain data, even for researchers whose strategies trade derivatives or centralised order books.

Applications: Bitcoin; protocol, settlement.

Version record

Kose John, Maureen O'Hara and Fahad Saleh, "Bitcoin and Beyond"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 14(1), 95–115 (2022); DOI 10.1146/annurev-financial-111620-011240.

Annual Review of blockchain economics: consensus, mining, stablecoins, DeFi and regulation.

Version record

Campbell R. Harvey, Ashwin Ramachandran and Joey Santoro, *DeFi and the Future of Finance*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Wiley, 2021 (ISBN 9781119836018).

Introduction to decentralised exchanges, lending, derivatives and tokenisation, with risk taxonomy.

Version record

Eric Budish, "Trust at Scale: The Economic Limits of Cryptocurrencies and Blockchains"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 140(1), 1–62 (2025); DOI 10.1093/qje/qjae033.

Security cost of Nakamoto consensus must scale with the value it protects, limiting economic uses of anonymous decentralised trust; journal version of the 2018 NBER paper.

Version record

18.1.2 DeFi surveys, financial stability and regulation

Start with a survey and a systematisation, then the official assessments. Regulators describe the same mechanisms in terms of leverage, liquidity mismatch, governance and responsible persons.

Sam M. Werner, Daniel Perez, Lewis Gudgeon, Ariah Klages-Mundt, Dominik Harz and William J. Knottenbelt, "SoK: Decentralized Finance (DeFi)"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the 4th ACM Conference on Advances in Financial Technologies, 30–46 (2022); DOI 10.1145/3558535.3559780.

Taxonomy of DeFi primitives, protocols and technical and economic security risks.

Version record

Kose John, Leonid Kogan and Fahad Saleh, "Smart Contracts and Decentralized Finance"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 15(1), 523–542 (2023); DOI 10.1146/annurev-financial-110921-022806.

Survey of smart contracts for finance. Smart contracts can solve commitment problems, but they cannot see information outside the chain without an oracle, are hard to join to legal enforcement, and cost more to run without a trusted intermediary. Closes with the main DeFi applications: token issuance, decentralised exchanges and protocols for loanable funds, with the institutional detail and literature for each.

Version record

Fabian Schär, "Decentralized Finance: On Blockchain- and Smart Contract-Based Financial Markets"

Difficulty: Intro **Type:** Research paper

Status/version: Federal Reserve Bank of St. Louis Review, 103(2), 153–174 (2021); DOI 10.20955/r.103.153-74.

Early, widely cited map of DeFi on Ethereum. Proposes a layered framework (settlement, asset, protocol, application and aggregation layers) and places token standards, decentralised exchanges, debt markets, derivatives and on-chain asset management in it, with their opportunities and risks. A short first reading before the systematisations.

Version record

Raphael Auer, Bernhard Haslhofer, Stefan Kitzler, Pietro Saggese and Friedhelm Victor, "The technology of decentralized finance (DeFi)"

Difficulty: Advanced **Type:** Research paper

Status/version: Digital Finance, 6(1), 55–95 (2023); DOI 10.1007/s42521-023-00088-8.

Technical tour of DeFi through a three-layer stack reference model: settlement, applications and interfaces. Describes decentralised exchanges, lending, derivatives and aggregators, and shows how composability lets protocols build on each other's functions. Ends with sources of systemic risk and a research agenda. Journal version of BIS Working Paper 1066 (2023).

Version record

Jonathan Chiu, Charles M. Kahn and Thorsten V. Koeppl, "Grasping decentralized finance through the lens of economic theory"

Difficulty: Advanced **Type:** Research paper

Status/version: Canadian Journal of Economics/Revue canadienne d'économique, 55(4), 1702–1728 (2022); DOI 10.1111/caje.12627.

A stylised model of collateralised lending used to weigh DeFi's value proposition. Smart contracts that enforce loan terms and hold collateral can lower intermediation costs and widen access; the limits are volatile crypto collateral and settlement stablecoins, incomplete contracts and the lack of a reliable oracle. A short economist's frame for the lending and oracle readings.

Version record

Agostino Capponi, Garud Iyengar and Jay Sethuraman, *Decentralized Finance: Protocols, Risks, and Governance*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Foundations and Trends in Privacy and Security, 5(3), 144–188 (2023); DOI 10.1561/3300000036.

Monograph on how smart contracts implement token exchange and lending, the pros and cons of governance tokens, and the operational risks DeFi adds for users, sorted into five groups: consensus, protocol, oracle, front-running and systemic risk. Useful as a single longer text between the surveys and the specialist papers.

Version record

Francesca Carapella, Edward Dumas, Jacob Gerszten, Nathan Swem and Larry Wall, "Decentralized Finance (DeFi): Transformative Potential & Associated Risks"

Difficulty: Intro **Type:** Report / research note

Status/version: Federal Reserve Board, Finance and Economics Discussion Series 2022-057 (August 2022); DOI 10.17016/FEDS.2022.057.

Federal Reserve staff overview of DeFi payments, lending, trading, capital raising and insurance, and of the financial-stability risks they may pose while largely outside the prudential perimeter. A plain-language complement to the BIS and FSB reports.

Version record

Sirio Aramonte, Wenqian Huang and Andreas Schrimpf, "DeFi risks and the decentralisation illusion"

Difficulty: Intro **Type:** Report / research note

Status/version: BIS Quarterly Review, December 2021; published 6 December 2021.

BIS special feature arguing that DeFi has a 'decentralisation illusion': governance has to be centralised somewhere, and consensus mechanisms tend to concentrate power, so those governance structures are where policy can engage. Its vulnerabilities are high leverage, liquidity mismatches, built-in interconnectedness and no shock absorbers such as banks, which makes stablecoin runs a key channel.

Version record

Financial Stability Board, The Financial Stability Risks of Decentralised Finance

Difficulty: Advanced **Type:** Report / research note

Status/version: FSB report, 16 February 2023.

FSB assessment of DeFi vulnerabilities: operational fragilities (governance, congested chains, oracles, cross-chain bridges, coding errors), liquidity and maturity mismatches in stablecoins and lending protocols, leverage and collateral chains with automatic liquidation and no circuit breakers, and interconnectedness with centralised platforms. Links to traditional finance were judged limited in 2022, and the report sets out what monitoring data would be needed if they grow.

Version record

International Organization of Securities Commissions, Final Report with Policy Recommendations for Decentralized Finance (DeFi)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: IOSCO Final Report FR/14/2023, December 2023.

IOSCO's nine policy recommendations for DeFi (FR/14/2023): analyse the economic substance of each arrangement, identify the responsible persons, apply common regulatory outcomes, and require conflicts, material risks (including operational and technology risks) and disclosures to be addressed, with enforcement, cross-border cooperation and monitoring of links to traditional markets. Complements the November 2023 crypto-asset recommendations.

Version record

18.1.3 Market structure, arbitrage and MEV

Igor Makarov and Antoinette Schoar, "Trading and Arbitrage in Cryptocurrency Markets"; LSE discussion paper

Difficulty: Research **Type:** Research paper

Status/version: 2020

The 2018 discussion paper (2020 journal article) documents large, persistent cross-exchange price differences and relates them to capital controls, settlement frictions and segmented investor demand. It remains the foundational empirical paper for crypto cross-venue arbitrage and market integration.

Applications: Crypto spot; arbitrage, fragmentation. Version record

Fei Wu, Danning Sui, Thomas Thiery and Mallesh Pai, Measuring CEX-DEX Extracted Value and Searcher Profitability: The Darkest of the MEV Dark Forest; Full arXiv version

Difficulty: Research **Type:** Empirical research paper

Status/version: 7th Conference on Advances in Financial Technologies (AFT 2025), LIPIcs 354, Article 26; published 6 October 2025. Published conference version; arXiv v3 dated 3 August 2025.

Measures Ethereum CEX-DEX arbitrage with transaction-identification heuristics and a revenue estimator that does not observe searchers' actual CEX trades. Study the August 2023-March 2025 window, searcher concentration, builder integration and estimated extracted value, then reconstruct which costs and off-chain hedge outcomes remain invisible. Treat its dollar estimate as method-dependent extracted value, not audited net P&L. Pair the paper with actual local receive times, executable CEX quotes, hedge fills, builder bids and inventory restoration before testing whether an apparent opportunity was attainable.

Extended note. Version record

Philip Daian et al., "Flash Boys 2.0: Frontrunning, Transaction Reordering, and Consensus Instability in Decentralized Exchanges"

Difficulty: Research **Type:** Research paper

Status/version: 2020

The 2019 preprint (2020 conference version) introduces miner/maximal extractable value through decentralised-exchange arbitrage and transaction reordering. It is indispensable for understanding why on-chain execution has endogenous ordering risk that does not appear in a conventional central-limit-order-book model. Priority gas auctions connect transaction-ordering competition to consensus incentives.

Applications: Ethereum, DEXs; MEV, arbitrage, execution. Version record

Lioba Heimbach, Vabuk Pahari and Eric Schertenleib, Non-Atomic Arbitrage in Decentralized Finance; Author preprint, arXiv v3; Author replication repository

Difficulty: Research **Type:** Empirical research paper

Status/version: 2024 IEEE Symposium on Security and Privacy. Published proceedings paper; arXiv v3 revised 8 April 2024.

Identifies likely non-atomic arbitrage involving Ethereum DEX trades and a second venue or domain, extending MEV analysis beyond self-contained on-chain cycles. Inspect its classification rules, sample ending 31 October 2023 and reported searcher concentration; compare inferred arbitrage volume with observable settlement traces. A DEX transaction alone does not show the external hedge, its price or execution costs. For a tradability study, add synchronized venue states, failed-leg accounting, inventory funding and bridge or withdrawal delays before turning observed price gaps into candidate profits.

Extended note. Version record

Bartosz Bieganowski and Robert Ślepaczuk, "Explainable Patterns in Cryptocurrency Microstructure"; SSRN posting, 19 February 2026

Difficulty: Research **Type:** Research paper

Status/version: Preprint v1, 31 January 2026

Reports stable SHAP structures for order-book and trade features across several perpetual-futures markets and explicitly compares maker and taker implementations. This is a useful 2026 preprint on portable microstructure representations, but exchange concentration and flash-crash sensitivity warrant replication. [40]

Applications: Crypto perpetuals; LOB alpha, market making. Version record

David Easley, Maureen O'Hara, Songshan Yang and Zhibai Zhang, "Microstructure and market dynamics in crypto markets"; April 2024 draft; 2025 SSRN version

Difficulty: Research **Type:** Research paper

Status/version: 2026 journal article; drafts 2024–2025

Predicts changes in volatility and return-distribution measures for five cryptocurrencies using liquidity and information metrics; the journal version compares futures-market dynamics and toxicity. Findings remain sample-, horizon- and execution-dependent. *Journal of Financial Markets*, article 101071; publisher preview corroborates the displayed Easley–O'Hara–Yang–Zhang order. Indexed publisher byline/abstract/introduction inspected; issue assignment and exact online/acceptance dates remain unverified. Read after microstructure and classification foundations; predictive labels are not net trading returns.

Applications: Crypto; microstructure, price dynamics. Version record

Robert Miller / Flashbots, "MEV and the Limits of Scaling"

Difficulty: Research **Type:** Practitioner / research commentary

Status/version: Flashbots Writings, 16 June 2025; practitioner article.

Practitioner thesis that MEV competition and transaction spam can constrain scaling and worsen user experience. Use it to study auction design, private order flow and builder/relay/searcher incentives; the market-design thesis is an attributed argument. The selected evidence uses OP-Stack traces and an explicit spam-classification heuristic; results and proposed auction benefits are not independently reproduced.

Applications: Blockchains; MEV, market design. Version record

Flashbots, *Flashbots Auction: Overview and Documentation*

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living documentation

Distinguishes historical PoW `mev-geth`/`mev-relay` from PoS MEV-Boost proposer–builder separation. Read the overview's trust assumptions: private submission does not establish complete privacy or permissionlessness. Pin protocol era, builder/relay and latency, censorship and inclusion assumptions.

Applications: Ethereum; on-chain execution. Version record

CoW Protocol contributors, CoW Protocol: Solver Engine and Solver Competition Rules; Competition rules

Difficulty: Advanced **Type:** Protocol / execution specification

Status/version: Living documentation; no fixed publication date asserted. Solver architecture, baseline and competition rules inspected 12 October 2026.

Introduces user-intent batch execution as a distinct intermediation problem. Separate the engine's matching/routing solution from the driver's simulation, scoring and settlement; enforce user limits and inventory/funding feasibility before evaluating solver revenue.

Extended note. Version record

Uniswap Labs contributors, UniswapX Overview

Difficulty: Advanced **Type:** Protocol / execution specification

Status/version: Living documentation; no fixed publication date asserted. Current overview inspected 12 October 2026; mechanism varies by chain.

Studies signed swap outputs, competitive filling and inventory versus external liquidity. Auction parameters and output amounts constrain the fill; settlement timing affects the executable terms. Read as an intent-filling protocol, separate from LP positions in Uniswap v3/v4.

Extended note. Version record

Fei Wu, Thomas Thiery, Stefanos Leonardos and Carmine Ventre, "Strategic Bidding Wars in On-chain Auctions"

Difficulty: Research **Type:** Research paper

Status/version: 2023; revised 2024

Game-theoretic modelling and simulations of MEV-Boost bidding, relay connectivity, latency and access to opportunities. The ICBC 2024 proceedings version, pp. 503–511, was online 21 August 2024; the linked preprint was revised 17 March. University publication record and accepted-manuscript route verified; simulated bidding performance is not a deployed-market result.

Applications: Ethereum; MEV-Boost auctions. Version record

Paradigm, "Crypto Market Structure 3.0"

Difficulty: Advanced **Type:** Research paper

Status/version: 2020

Practitioner synthesis of fragmentation, global 24/7 trading and the convergence of centralised and decentralised markets. Use as historical context for venue selection and settlement boundaries, rather than a current venue specification. Arjun Balaji extquotesingle s 2020 practitioner essay treats its named firms as historical examples.

Applications: Crypto; CeFi/DeFi market structure. Version record

Agostino Capponi, Ruizhe Jia and Kanye Ye Wang, "Maximal extractable value and allocative inefficiencies in public blockchains"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 172, 104132 (2025); DOI 10.1016/j.jfineco.2025.104132.

Public settlement lets validators and bots front-run systematically, which misallocates block space. Private transaction pools can reduce the inefficiency and raise welfare, but validators are reluctant to give up front-running rents, so adoption is partial and front-running persists. Flashbots data support this: validators earn more, users more exposed to front-running use the private pool more, and attackers' cost-to-revenue ratio in private pools converges to one.

Extended note. Version record

Daniele Maria Di Nosse, Federico Gatta, Fabrizio Lillo and Sebastian Jaimungal, "Deviations from tradition: stylized facts in the era of DeFi"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 26(8), 1231–1252 (2026); DOI 10.1080/14697688.2026.2678231.

Stylised facts on 24 major DEX pools with concentrated liquidity differ from limit-order-book markets.

Version record

18.1.4 Block building, transaction fees and MEV measurement

Measurement of front-running and private order flow, proposer-builder separation and builder concentration, transaction fee mechanisms, countermeasures and MEV on rollups.

Kaihua Qin, Liyi Zhou and Arthur Gervais, "Quantifying Blockchain Extractable Value: How dark is the forest?"

Difficulty: Research **Type:** Research paper

Status/version: 2022 IEEE Symposium on Security and Privacy (SP), 198–214 (2022); DOI 10.1109/sp46214.2022.9833734.

Measures extractable value from sandwich attacks, liquidations and DEX arbitrage on Ethereum: about USD 540m over 32 months, shared among about 11,000 addresses, with single opportunities worth hundreds of block rewards. Gives a generalised front-running algorithm and argues that centralised relays for these transactions worsen consensus-layer attacks by rewarding forks.

Version record

Christof Ferreira Torres, Ramiro Camino and Radu State, "Frontrunner Jones and the Raiders of the Dark Forest: An Empirical Study of Frontrunning on the Ethereum Blockchain"

Difficulty: Research **Type:** Research paper

Status/version: 30th USENIX Security Symposium (USENIX Security 21), 1343–1359 (August 2021); preprint arXiv:2102.03347.

Defines and measures three kinds of front-running on Ethereum: displacement, insertion (sandwiching) and suppression. Scanning more than 11 million blocks finds almost 200,000 attacks with about USD 18m of attacker profit, evidence that public-mempool ordering was a routine cost before private order flow.

Version record

Ben Weintraub, Christof Ferreira Torres, Cristina Nita-Rotaru and Radu State, "A Flash(bot) in the Pan: Measuring Maximal Extractable Value in Private Pools"

Difficulty: Research **Type:** Research paper

Status/version: ACM Internet Measurement Conference (IMC 2022), 458–471 (2022); DOI 10.1145/3517745.3561448.

Tests whether Flashbots met its stated goals in the proof-of-work era. Flashbots miners held over 99.9% of hash power; large miners earned more than twice their earlier revenue while others' share shrank; more than 90% of Flashbots blocks came from two miners; and 13.2% of MEV went through other private pools. Private order flow moved extraction rather than removing it.

Version record

Lioba Heimbach, Lucianna Kiffer, Christof Ferreira Torres and Roger Wattenhofer, "Ethereum's Proposer-Builder Separation: Promises and Realities"

Difficulty: Research **Type:** Research paper

Status/version: ACM Internet Measurement Conference (IMC 2023), 406–420 (2023); DOI 10.1145/3618257.3624824.

Measures proposer-builder separation after the Merge. Builders and relays are highly concentrated; PBS gives validators access to optimised blocks but tends to increase rather than reduce censorship; and relays do not always deliver the promised value or the filtering they advertise. Read before assuming that a block auction pays the proposer the full bid.

Version record

Tivas Gupta, Mallesh M. Pai and Max Resnick, "The Centralizing Effects of Private Order Flow on Proposer-Builder Separation"

Difficulty: Research **Type:** Research paper

Status/version: 5th Conference on Advances in Financial Technologies (AFT 2023), LIPIcs 282, 20:1–20:15 (2023); DOI 10.4230/LIPIcs.AFT.2023.20.

Models why private order flow sold through order-flow auctions favours builders that also run searchers: private flow complements top-of-block (CEX-DEX) opportunities, so integrated builders bid more for it and win more blocks. Post-Merge data pairing PBS outcomes with 12-second Binance volatility show integrated builders win more often when volatility is high.

Version record

Burak Öz, Danning Sui, Thomas Thiery and Florian Matthes, "Who Wins Ethereum Block Building Auctions and Why?"

Difficulty: Research **Type:** Research paper

Status/version: 6th Conference on Advances in Financial Technologies (AFT 2024), LIPIcs 316, 22:1–22:25 (2024); DOI 10.4230/LIPIcs.AFT.2024.22.

Six months of MEV-Boost auctions (October 2023 to March 2024), when three builders produced 80% of blocks. Market share rises with the diversity of order flow; profit rises with exclusive flow from integrated searchers and exclusive deals. Features such as exclusive signals, non-atomic arbitrage and Telegram-bot flow drive both, a chicken-and-egg barrier to entry.

Version record

Maryam Bahrani, Pranav Garimidi and Tim Roughgarden, "Centralization in Block-Building and Proposer-Builder Separation"

Difficulty: Research **Type:** Research paper

Status/version: Financial Cryptography and Data Security, Lecture Notes in Computer Science, 331–349 (2025); DOI 10.1007/978-3-031-78676-1_19.

Models whether differences in skill among block producers must centralise staking and whether PBS undoes that. Quantifies how heterogeneous rewards concentrate stake in equilibrium and over time as rewards are restaked, and how far PBS narrows reward differences across proposers depending on builder competition. Theory to set against the measurement papers.

Version record

Tim Roughgarden, "Transaction Fee Mechanism Design"

Difficulty: Research **Type:** Research paper

Status/version: Journal of the ACM, 71(4), 1–25 (2024); DOI 10.1145/3674143.

Mechanism-design analysis of blockchain transaction fee mechanisms. Introduces incentive-compatibility notions for profit-maximising block producers (MMIC) and against off-chain collusion (OCA-proofness), and shows that EIP-1559's base fee, burn and variable block size satisfy both and are dominant-strategy incentive compatible except during demand spikes. Journal version of the EC 2021 paper.

Version record

Yulin Liu, Yuxuan Lu, Kartik Nayak, Fan Zhang, Luyao Zhang and Yinhong Zhao, "Empirical Analysis of EIP-1559: Transaction Fees, Waiting Times, and Consensus Security"

Difficulty: Research **Type:** Research paper

Status/version: ACM Conference on Computer and Communications Security (CCS 2022), 2099–2113 (2022); DOI 10.1145/3548606.3559341.

Causal study of EIP-1559 using chain, mempool and exchange data. It reduced differences in gas prices within blocks and users' waiting times but had little effect on fee levels or consensus security; waiting times rise when ETH is volatile. Pairs with Roughgarden's theory.

Version record

Sen Yang, Fan Zhang, Ken Huang, Xi Chen, Youwei Yang and Feng Zhu, "SoK: MEV Countermeasures"

Difficulty: Advanced **Type:** Research paper

Status/version: ACM CCS Workshop on Decentralized Finance and Security (DeFi 2024), 21–30 (2024); DOI 10.1145/3689931.3694911.

Systematises 32 proposed MEV countermeasures in four technical directions and checks each against four security problems MEV causes. A map of the design space (private flow, fair ordering, encryption, auctions) and of the gap between academic proposals and what practitioners use.

Version record

Matheus Venturyne Xavier Ferreira and David C. Parkes, "Credible Decentralized Exchange Design via Verifiable Sequencing Rules"

Difficulty: Research **Type:** Research paper

Status/version: ACM Symposium on Theory of Computing (STOC 2023), 723–736 (2023); DOI 10.1145/3564246.3585233.

Proposes verifiable sequencing rules: block producers still choose a block's contents but commit to an ordering rule that anyone can check in polynomial time, so a deviation yields a proof of non-compliance. Shows what execution guarantees such rules can give DEX users against front-running.

Version record

Christof Ferreira Torres, Albin Mamuti, Ben Weintraub, Cristina Nita-Rotaru and Shweta Shinde, "Rolling in the Shadows: Analyzing the Extraction of MEV Across Layer-2 Rollups"

Difficulty: Research **Type:** Research paper

Status/version: ACM Conference on Computer and Communications Security (CCS 2024), 2591–2605 (2024); DOI 10.1145/3658644.3690259.

Nearly three years of MEV on Ethereum and on the Arbitrum, Optimism and zkSync rollups. Rollups have no public mempool but MEV is widespread, with volume comparable to Ethereum and lower costs and profits. No sandwiching was found on the rollups studied, but cross-layer sandwich attacks via bridge transactions were feasible, worth an estimated USD 2m.

Version record

18.1.5 Returns, factors and allocation

Yukun Liu, Aleh Tsyvinski and Xi Wu, "Common Risk Factors in Cryptocurrency"; NBER working paper

Difficulty: Research **Type:** Research paper

Status/version: 2022; NBER 2019

Empirical crypto factor framework for cross-sectional modelling, portfolio risk and performance attribution. Read as a distinct paper from the Borri–Shakhnov entry; test systematic exposures instead of assuming cryptocurrencies are mutually uncorrelated. Market, size and momentum exposures support attribution; Yale record.

Applications: Crypto; factors, portfolio risk.

Version record

Yukun Liu and Aleh Tsyvinski, Risks and Returns of Cryptocurrency; NBER working paper 24877

Difficulty: Research **Type:** Empirical research paper

Status/version: Review of Financial Studies 34(6), 2689-2727 (June 2021 issue); first published online 26 September 2020. Published journal article; compare with August 2018 NBER working paper when reproducing tables.

Studies whether early Bitcoin, Ripple and Ethereum returns load on traditional asset factors or crypto-specific variables, and tests time-series momentum and investor-attention proxies. Read it as a historical research design, then rebuild signals with point-in-time asset eligibility, exchange coverage, volume, financing and trading costs. Separate explanatory exposures from forecasts available before an order could be placed. The paper's early sample and reported predictability should not be transferred to current token universes without post-publication tests and a clear market, size and liquidity benchmark.

Extended note.

Version record

ARK Invest, "On-Chain Data: A Framework to Evaluate Bitcoin"

Difficulty: Advanced **Type:** Research paper

Status/version: 2021

Source-named framework developed with Glassnode data: network health, holder behaviour and valuation proxies. Organises on-chain state variables and investment-committee discussion; validate timestamp availability and incremental predictive value before treating a proxy as a valuation signal.

Applications: Bitcoin; on-chain analytics. Version record

Klaus Grobys and Syed Jawad Hussain Shahzad, "Cryptocurrency Momentum: Is It an Illusion?" (SSRN: written 22 September, posted 30 November 2023); journal version

Difficulty: Advanced **Type:** Research paper

Status/version: Selected citation year(s): 2023. Version distinctions remain in the annotation.

The journal abstract examines realised-variance tails of six crypto momentum strategies, expanding the earlier abstract's single-portfolio description. The authors question variance-based performance measures under sample- and tail-model-dependent tests. *International Journal of Finance & Economics* 31(2), 2180–2193 (April 2026; online 13 August 2025). Its existence is established, but complete preprint revision history and full-text equivalence remain unverified.

Applications: distributional statistics and tail risk. Extended note. Version record

Marja Ekström, *Bitcoin trading performance evaluation: A comparative study of momentum and mean reversion strategies from 2020 to 2025* (2025); Gian Pietro Enzo Bellocca, *Exploiting the momentum effect in the cryptocurrency market: A machine learning-based trading system* (2019–2020)

Difficulty: Not specified **Type:** Research thesis

Status/version: Selected citation year(s): 2020, 2025, 2019. Version distinctions remain in the annotation.

Two distinct educational theses: Ekström compares momentum and Bollinger-band reversal; Bellocca studies momentum with an ML classifier. Strategy evidence is sample- and implementation-dependent.

Version record

Kasper Johansson and Stephen Boyd, "Simple and Effective Portfolio Construction with Crypto Assets" (January 2025)

Difficulty: Intermediate **Type:** Research paper

Status/version: January 2025 TechREG Chronicle article.

The article contrasts risk allocation with a 90/10 traditional/crypto mix dynamically diluted with cash to target ex-ante risk (DD90/10). CVaR optimisation appears in the literature review; DD90/10 is not a CVaR optimiser. Treat the historical backtest as an illustration, not prospective performance.

Applications: risk allocation and volatility targeting. Version record

Wilshire Indexes, "Traditional Asset Allocation with Digital Assets"

Difficulty: Intermediate **Type:** Research paper

Status/version: Edition/revision and publication date not established for the selected resource(s).

Practitioner research on digital assets in traditional portfolios; assess allocation assumptions and sample dependence alongside Johansson–Boyd.

Applications: asset allocation. Version record

Christian Fieberg, Gerrit Liedtke, Thorsten Poddig, Thomas Walker and Adam Zaremba, "A Trend Factor for the Cross Section of Cryptocurrency Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(7), 3116–3153 (2025); DOI 10.1017/s0022109024000747.

CTREND: an ML-aggregated price and volume trend factor across 3,000+ coins that predicts returns beyond known crypto factors.

Version record

Mykola Babiak and Daniele Bianchi, "Mispricing and Risk Compensation in Cryptocurrency Returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(5), 2106–2132 (2026); DOI 10.1017/s0022109025102329.

IPCA decomposes crypto returns into mispricing and risk-compensation components operating through distinct mechanisms.

Version record

Ran Duchin, David H. Solomon, Jun Tu and Xi Wang, "The cryptocurrency elephant in the room"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, rfaf037 (2025); DOI 10.1093/rof/rfaf037.

Bayesian portfolio choice makes zero crypto weights hard to justify: only extremely pessimistic priors avoid buying; optimal weights are small (1–5%) and certainty-equivalent gains rival international diversification.

Version record

Weijia Xue and Chunyang Zhou, "Predicting cryptocurrency returns with vision transformers: A look-ahead bias-free approach"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 190, 107750 (2026); DOI 10.1016/j.jbankfin.2026.107750.

Vision-transformer return prediction for cryptocurrencies with an explicit look-ahead-bias-free design.

Extended note. Version record

Siddharth M. Bhambhwani, Stefanos Delikouras and George M. Korniotis, "Blockchain characteristics and cryptocurrency returns"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Financial Markets, Institutions and Money, 86, 101788 (2023); DOI 10.1016/j.intfin.2023.101788.

Network size (adoption and use) and computing power (resources spent securing transactions) move with cryptocurrency prices, and factors built from them explain expected returns at least as well as market, size and momentum factors. A fundamentals-based alternative to the purely return-based crypto factors.

Version record

Dominik Metelski and Janusz Sobieraj, "Decentralized Finance (DeFi) Projects: A Study of Key Performance Indicators in Terms of DeFi Protocols' Valuations"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Financial Studies, 10(4), 108 (2022); DOI 10.3390/ijfs10040108.

Relates the valuations of 30 DeFi protocols (exchanges, lending and asset management) to total value locked, protocol and total revenue, volume and token inflation, using Granger tests and fixed-effects panels. Valuations depend partly on these measures; a starting point for protocol-fundamentals signals, not a tested trading rule.

Version record

Lin William Cong, Ke Tang, Yanxin Wang and Xi Zhao, "Inclusion and Democratization Through Web3 and DeFi? Initial Evidence from the Ethereum Ecosystem"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 30949 (February 2023); DOI 10.3386/w30949.

Large-scale description of Ethereum's network: mining income and ether ownership are concentrated in exchanges and a few nodes, activity moved from peer-to-peer transfers to DApps and DeFi, and fees, gas-price swings and failed transactions weigh most on small and new users. Suggestive evidence that fee burning and airdrops redistribute toward inclusion. Useful for wallet-level on-chain features.

Version record

18.1.6 CFMM and AMM theory

Jason Milionis, Ciamac C. Moallemi and Tim Roughgarden, "A Myersonian Framework for Optimal Liquidity Provision in Automated Market Makers"; preprint

Difficulty: Research **Type:** Research paper

Status/version: 2024; preprint 2023

Frames optimal AMM liquidity provision using mechanism-design ideas and explicitly studies informed flow and fee choice. It is a major theoretical reference for comparing passive LP returns with market-making economics in centralised venues. [37]

Applications: DEXs; AMMs, liquidity provision. Version record

Conor McMenamin and Vanesa Daza, "An AMM minimizing user-level extractable value and loss-versus-rebalancing" (V0LVER)

Difficulty: Research **Type:** Research paper

Status/version: 2023

Combines an encrypted mempool, liquidity allocation before decryption and block-producer incentives to limit extractable value and LP losses. Version 2, 15 February 2023. Evaluate cryptographic and incentive assumptions alongside implementation costs. [38]

Applications: DEXs; AMM design, MEV. Version record

Mengqian Zhang, Sen Yang and Fan Zhang, "RediSwap: MEV Redistribution Mechanism for CFMMs"; preprint; evaluation code/data

Difficulty: Research **Type:** Research paper

Status/version: 2025; preprint 2024

DeFi '25 workshop paper on redistributing MEV to users and liquidity providers; first preprint October 2024. Experimental viability depends on gas, routing, solver concentration and manipulation resistance as well as theoretical surplus. [39]

Applications: DEXs; MEV redistribution, liquidity. Version record

Guillermo Angeris and Tarun Chitra, "Improved Price Oracles: Constant Function Market Makers"

Difficulty: Research **Type:** Research paper

Status/version: 2020

Theoretical CFMM reference for arbitrage incentives, price-oracle behaviour and reserve properties, including the constant-product family. Connect invariant curves to liquidity, slippage and LP risk. The paper has two authors; the source's "and others" is removed.

Applications: DEXs; AMMs, oracles, LP risk. Version record

Guillermo Angeris, Akshay Agrawal, Alex Evans, Tarun Chitra, Stephen Boyd, Constant Function Market Makers: Multi-Asset Trades via Convex Optimization; Published handbook contents; Author manuscript

Difficulty: Research **Type:** Research paper

Status/version: Handbook on Blockchain, Springer, 2022, pp. 415-444. Author manuscript and arXiv preprint first posted 26 July 2021.

Formulates utility-maximizing trades against general CFMMs as convex optimization problems and demonstrates reliable methods for multi-asset pools. It is a mathematical bridge from one-pool swap formulas to portfolio-level trading; actual token arithmetic, fees, gas, route availability, and state changes still require implementation checks.

Extended note. Version record

Guillermo Angeris, Alex Evans, Tarun Chitra, Stephen Boyd, Optimal Routing for Constant Function Market Makers; Author manuscript and code; Preprint

Difficulty: Research **Type:** Research paper

Status/version: ACM EC 2022, pp. 115-128; DOI 10.1145/3490486.3538336. Author manuscript and 2022 arXiv preprint retained alongside conference identity.

Optimizes trade execution across a network of CFMMs. With no fixed pool-activation costs the route problem is convex; with fixed costs it becomes mixed-integer convex. The formulation also covers arbitrage detection. Model the transaction cost, gas, timing, and state availability of the actual chain separately.

Extended note. Version record

Theo Diamandis, Max Resnick, Tarun Chitra and Guillermo Angeris, An Efficient Algorithm for Optimal Routing Through Constant Function Market Makers; Financial Cryptography 2023 proceedings DOI; Conference preproceedings manuscript

Difficulty: Advanced **Type:** Research paper

Status/version: 2023 preprint; presented at Financial Cryptography 2023. arXiv v1, 9 February 2023; compare the proceedings edition when using numerical results.

Develops a decomposition algorithm for choosing trades across a network of constant-function market makers, including aggregate representations of complex pools such as Uniswap v3. Work through the optimization objective, pool constraints and dual variables before comparing it with the earlier general routing theory. Reproduce a small multi-pool route using exact swap arithmetic and executable token amounts. Report gas, fee tiers, stale pool state and failed or delayed settlement separately: faster numerical optimization does not establish a profitable executable route.

Extended note. Version record

Hayden Adams et al., "Uniswap v3 Core"

Difficulty: Advanced **Type:** Research paper

Status/version: Uniswap v3 Core whitepaper dated March 2021.

Concentrated-liquidity design for positions restricted to price ranges. Study how range boundaries and price paths change LP exposure, fee earning, execution and arbitrage; connect the option-like economic exposure to explicit inventory and rebalancing assumptions.

Applications: DEXs; concentrated liquidity. Version record

Zhou Fan et al., "Strategic Liquidity Provision in Uniswap v3"

Difficulty: Research **Type:** Research paper

Status/version: 2021 preprint; AFT 2023

Dynamic concentrated-liquidity allocation balancing fees, out-of-range exposure and repositioning gas costs; results depend on price-process assumptions.

Applications: DEXs; LP allocation. Version record

18.1.7 Perpetuals, options and crypto rates

Will Gornall, Martin Rinaldi and Yizhou Xiao, "Perpetual Futures and Basis Risk: Evidence from Cryptocurrency"; 19 May 2025 manuscript; earlier-title draft

Difficulty: Research **Type:** Research paper

Status/version: Selected citation year(s): 2025. Version distinctions remain in the annotation.

Capital-constrained arbitrage, funding and basis dislocations; financial-stability claims are conditional on the studied markets. SSRN search metadata reports posting 14 January and revision 19 May 2025, with Date Written 27 November 2024; an earlier-title PDF is dated 28 November 2024. The May 2025 manuscript is preliminary. The 2026 ASSA programme lists it, and the author site lists it In Progress; neither establishes journal publication.

Applications: derivatives and constrained arbitrage. Version record

Maik Schmeling, Andreas Schrimpf, Karamfil Todorov, Crypto carry; Official revised PDF

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1087, 4 April 2023. Selected official PDF revised October 2025.

Documents crypto spot-futures carry across crypto and traditional venues, its large and time-varying basis, market segmentation, convenience-yield drivers, constrained arbitrage capital, and the relation between carry and crash risk. Treat the sample estimates as historical evidence; cash-and-carry remains exposed to margin spikes, liquidation, basis moves, and capital access.

Extended note. Version record

Damien Ackerer, Julien Hugonnier, Urban Jermann, Perpetual Futures Pricing; Author preprint

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance 36(3), 481-499, 2026. Published article; arXiv v2 revised 4 September 2024.

Derives no-arbitrage prices for linear, inverse, and quanto perpetual futures in discrete and continuous time, interprets the price through a random-time expectation, and identifies funding specifications that anchor the future to spot. Apply the cash-flow assumptions to the actual contract and account for funding clamps, costs, margin, and collateral before treating the result as an executable bound.

Extended note. Version record

Songrun He, Asaf Manela, Omri Ross, Victor von Wachter, Fundamentals of Perpetual Futures

Difficulty: Research **Type:** Research preprint

Status/version: Preprint first posted 13 December 2022. arXiv:2212.06888 v7, 17 September 2026; read the corrected version.

Develops frictionless perpetual-futures values and transaction-cost bounds, then studies observed crypto price deviations and a funding-aware trading construction. The empirical result complements pure pricing theory; rebuild all economics from the corrected current version, accessible funding and borrow, fees, margin, inventory, and liquidation paths.

Extended note. Version record

Philippe Bergault, Sébastien Bieber, Olivier Guéant, Wenkai Zhang, Cryptocurrencies and Interest Rates: Inferring Yield Curves in a Bondless Market

Difficulty: Research **Type:** Research preprint

Status/version: Preprint first posted 4 September 2025. arXiv:2509.03964 v2, 17 December 2025.

Constructs currency-specific crypto yield curves from derivatives when conventional bond markets and fixed-rate lending are sparse. It gives a framework for linking futures, option parity, discounting, and financing assumptions. An implied curve is a model estimate, not proof that an account can borrow or lend at those rates.

Extended note. Version record

Hyperliquid contributors, Hyperliquid HyperCore: Order Book, Funding, Price Indices and HLP; Order-book matching and block ordering; Funding calculation; Robust price indices; HLP protocol vault

Difficulty: Advanced **Type:** Software / protocol documentation

Status/version: Living documentation; no fixed publication date stated. Official Hyperliquid docs inspected 12 October 2026; verify instrument rules and vault configuration for the study period.

Use the order-book and funding pages to reconstruct HyperCore's matching priority, batch ordering, margin checks, oracle-based funding and liquidation price inputs. Its robust-price page distinguishes oracle, mark and executable book prices; the protocol-vault page describes HLP's market-making, liquidation and other strategy exposures. This gives a transparent, crypto-native perpetual venue case for comparing funding, basis and market microstructure, while documenting where the risk engine's marks diverge from immediately executable prices.

Extended note. Version record

GMX contributors, GMX GM and GLV Pools: Liquidity, Pricing and Risk Mechanics; GM/GLV contract architecture and execution; GM/GLV token valuation inputs

Difficulty: Advanced **Type:** Software / protocol documentation

Status/version: Living documentation; no fixed publication date stated. Official GMX docs inspected 12 October 2026; pool parameters and supported markets are mutable.

Model GM as a share of one market's pool and GLV as a share of a vault spanning selected GM markets. Reconstruct token value from backing assets, borrowing fees and capped pending trader PnL, then distinguish pool borrowing charges from trader funding transfers. Reserve factors, open-interest caps, impact and queued execution shape both trader cost and LP liquidity. This is a protocol-native counterparty and inventory case for crypto derivatives, beyond exchange basis and funding-rate studies.

Extended note. Version record

Julia Kończal, "Pricing options on the cryptocurrency futures contracts" (17 June 2025, v1)

Difficulty: Intermediate/Advanced **Type:** Research paper

Status/version: Selected citation year(s): 2025. Version distinctions remain in the annotation.

Sample-specific calibration comparison of diffusion, jump and stochastic-volatility option models. The 17 June 2025 v1 is preserved; the selected current preprint is v2, 13 August 2026. Repository metadata were checked; no journal publication or full technical equivalence between revisions is asserted.

Applications: option theory and calibration. Version record

Carol Alexander, Ding Chen, Arben Imeraj, Inverse and Quanto Inverse Options in a Black-Scholes World

Difficulty: Research **Type:** Research preprint

Status/version: Preprint first posted 26 July 2021. arXiv:2107.12041 v5, 17 October 2022; earlier v4 marked withdrawn.

Sets out inverse, direct, and quanto option payoffs, numeraire choices, pricing, hedging, and sources of market incompleteness in a Black-Scholes framework. Its market inventory and venue claims are historical and now stale; use it for payoff and currency-risk mathematics, then verify live contract terms separately.

Extended note. Version record

Vladimir Lucic, Artur Sepp, Valuation and hedging of cryptocurrency inverse options; Author manuscript

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance 24(7), 851-869, 2024; online 12 July 2024. Published article; separately linked SSRN manuscript.

Derives inverse-option valuation and adjusted delta under a forward numeraire, then tests delta-hedged Deribit strategies using separate coin and USD P&L ledgers. The reported negative volatility premium and strategy results depend on the historical sample and implementation assumptions; rebuild costs, funding, collateral, and instrument conventions.

Extended note. Version record

Carol Alexander, Jun Deng, Jianfen Feng, Huning Wan, Net Buying Pressure and the Information in Bitcoin Option Trades; Preprint

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets 63, article 100764, March 2023. Published article with separately identified 2021 arXiv preprint.

Uses tick-level Deribit option data to separate volatility-driven demand from directional information across moneyness, maturity, and time of day, and studies how market makers manage order imbalance as volume grows. This is a signal and inventory study, not a plug-in quoting rule; test trade signing, timestamp alignment, spread, and fill-conditioned outcomes.

Extended note. Version record

Carol Alexander, Arben Imeraj, The Bitcoin VIX and Its Variance Risk Premium; Author publication record; Working paper

Difficulty: Research **Type:** Research paper

Status/version: Journal of Alternative Investments 23(4), 84-109, Spring 2021. 2019 working-paper posting; conflicting revision snippets do not establish a revision date.

Constructs a term structure of Bitcoin implied-volatility indices from Deribit options and compares it with realized variance to study crypto variance risk premia. The sample covers March 2019-March 2020, and the VIX-style replication assumes no jumps; reproduce with current data and compare jump-aware alternatives before inferring a tradable premium.

Extended note. Version record

Jovanka Lili Matic, Natalie Packham and Wolfgang Karl Härdle, "Hedging cryptocurrency options"

Difficulty: Research **Type:** Research paper

Status/version: Review of Derivatives Research, 26(1), 91–133 (2023); DOI 10.1007/s11147-023-09194-6.

Compares delta, delta-gamma and minimum-variance hedges of bitcoin options under jump and stochastic-volatility models; jump-aware models hedge better.

Version record

18.1.8 Leverage, liquidations and crypto lenders

Exchange leverage and perpetual design, volatility transmission from leveraged venues, liquidation cascades and auto-deleveraging, the 2022 failures of centralised lenders, and US rules for crypto collateral at futures commission merchants.

Kyle Soska, Jin-Dong Dong, Alex Khodaverdian, Ariel Zetlin-Jones, Bryan Routledge and Nicolas Christin, "Towards Understanding Cryptocurrency Derivatives:A Case Study of BitMEX"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the Web Conference 2021 (WWW '21), 45–57 (2021); DOI 10.1145/3442381.3450059.

Case study of BitMEX, the venue that set the template for settled and perpetual crypto futures with up to 100x leverage. Traces the evolution of its products and uses on-chain forensics, public liquidation events and the venue's chat room to describe its traders, from wealthy automated strategies to small, short-horizon risky positions. Discusses how derivative trading produced sharp moves in spot markets.

Version record

Riccardo De Blasis and Alexander Webb, "Arbitrage, contract design, and market structure in Bitcoin futures markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 42(3), 492–524 (2022); DOI 10.1002/fut.22305.

Compares bitcoin quarterly and perpetual futures. Perpetuals, first proposed by Shiller (1993), show several intraday U-shapes, seasonal and opening effects although they never close; there is suggestive spillover between perpetual and quarterly contracts, and quarterly futures offer cash-and-carry opportunities mainly during market dislocations.

Version record

Carol Alexander, Daniel F. Heck and Andreas Kaeck, "The Role of Binance in Bitcoin Volatility Transmission"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 29(1), 1–32 (2022); DOI 10.1080/1350486x.2022.2125885.

High-frequency realised-volatility spillovers across bitcoin spot and derivative pairs against the US dollar and tether. The tether-margined perpetual on Binance is the main source of volatility, transmitting strongly to every other instrument and receiving little; traders react more during US hours, and interconnectedness is higher while Western stock markets are open. Argues that regulators should look beyond fiat spot venues to unregulated tether-margined derivatives.

Version record

Warodom Werapun, Tanakorn Karode, Jakapan Suaboot, Tanwa Arpornthip and Esther Sangiamkul, "Exploring risk and return profiles of funding rate arbitrage on CEX and DEX"

Difficulty: Advanced **Type:** Research paper

Status/version: Blockchain: Research and Applications, 7(4), 100354 (2026); DOI 10.1016/j.bcra.2025.100354.

Risk and return of funding-rate arbitrage (long spot, short perpetual or the reverse) on centralised venues such as Binance and BitMEX and on decentralised ones such as ApolloX and Drift. Reports returns of up to 115.9% over six months with small losses and no correlation with holding the coins; treat these as backtests whose fees, borrow, margin and venue-failure terms must be rebuilt before calling the yield realisable.

Version record

Tarun Chitra, "Autodeleveraging: Impossibilities and Optimization"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2512.01112, v3 (16 February 2026); DOI 10.48550/arXiv.2512.01112.

First formal model of auto-deleveraging (ADL), the last-resort loss socialisation used by perpetual venues when liquidations cannot keep them solvent. Proves a trilemma: no policy delivers solvency, exchange revenue and fairness to traders together, and moral hazard grows with participation. Applies the model to Hyperliquid on 10 October 2025, when ADL closed USD 2.1 billion of positions in twelve minutes, and estimates the excess haircut on profitable traders.

Version record

OECD, "Lessons from the crypto winter"

Difficulty: Intro **Type:** Report / research note

Status/version: OECD Business and Finance Policy Papers (December 2022); DOI 10.1787/199edf4f-en.

OECD account of the 2022 failures: Terra, the centralised lenders (Celsius, Voyager, BlockFi), Three Arrows Capital and, in an update, FTX. Identifies high interconnectedness, leverage built through DeFi composability and market concentration as the channels, and the disproportionate losses of retail participants; notes that much of the evidence comes from industry sources.

Version record

Shoba Pillay (Examiner), Final Report of Shoba Pillay, Examiner, In re Celsius Network LLC

Difficulty: Advanced **Type:** Report / research note

Status/version: US Bankruptcy Court, S.D.N.Y., Case No. 22-10964 (MG), Doc. 1956, filed 31 January 2023.

Court-appointed examiner's report on a centralised crypto lender. Documents that the business sold to customers was not the business run: the CEL token's price was propped up with customer assets, coin-by-coin liabilities were not reconciled, customer BTC and ETH funded CEL buybacks until shortfalls were covered with customer stablecoins, leaving a hole of about USD 1 billion, and pledged coins had to be unwound when withdrawals surged in May and June 2022. A primary source on CeFi leverage and customer-asset risk.

Version record

Commodity Futures Trading Commission, Acting Chairman Pham Announces Launch of Digital Assets Pilot Program for Tokenized Collateral in Derivatives Markets (Release 9146-25)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: CFTC press release 9146-25, 8 December 2025, with CFTC Staff Letters 25-39 and 25-40.

CFTC staff actions of 8 December 2025: a no-action pilot (Staff Letter 25-40) letting futures commission merchants accept bitcoin, ether and USDC as customer margin for an initial three months with weekly reporting by account class; guidance on tokenised collateral such as Treasuries and money market fund shares (25-39), covering enforceability, custody, haircuts and operational risk; and withdrawal of the 2020 advisory that kept virtual currencies out of segregation.

Version record

Boon Chuan Lim, "Two-Regime Liquidity Recovery After a Perpetual Futures Liquidation Cascade: Evidence from Hyperliquid and the October 10, 2025 Event"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2026); DOI 10.2139/ssrn.6636998.

Minute-level study of Hyperliquid order books for five perpetuals around the 10 October 2025 liquidation cascade. Quoted spreads spiked (median 9.4 and 95th percentile 87.5 basis points at 21:00 UTC) and returned to near pre-event levels within weeks, but inside-quote

depth kept shrinking, to 66–79% of pre-event levels after three weeks: prices recover faster than market makers' inventory commitment. Working paper; one venue and one event.

Version record

18.1.9 Crypto option markets: implied volatility, smiles and risk premia

Option-pricing models fitted to bitcoin options, implied-volatility indices and smiles, smile-adjusted hedging, variance and illiquidity premia, and the 2024 approval of options on spot bitcoin ETPs.

Ai Jun Hou, Weining Wang, Cathy Y. H. Chen and Wolfgang Karl Härdle, "Pricing Cryptocurrency Options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 18(2), 250–279 (2020); DOI 10.1093/jjfinec/nbaa006.

Prices bitcoin and CRIX options with a stochastic-volatility model with correlated jumps (SVCJ) and compares it with a co-jump model. Jumps and co-jumps shape the implied-volatility curve, and a sizeable share of price jumps are contemporaneously anti-correlated with volatility jumps. An early reference model for crypto option pricing.

Version record

Noshaba Zulfiqar and Saqib Gulzar, "Implied volatility estimation of bitcoin options and the stylized facts of option pricing"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Innovation, 7(1), 67 (2021); DOI 10.1186/s40854-021-00280-y.

Estimates implied volatilities of 14-day Deribit bitcoin options by Newton–Raphson and bisection and compares them with Black–Scholes benchmarks. Finds a smile with a forward skew, which places bitcoin options with commodity rather than equity-index options.

Version record

Fabian Woebbeking, "Cryptocurrency volatility markets"

Difficulty: Research **Type:** Research paper

Status/version: Digital Finance, 3(3-4), 273–298 (2021); DOI 10.1007/s42521-021-00037-3.

Builds a cryptocurrency volatility index (CVX) from intraday option prices, with two methods for the thin and uneven quote environment, over a sample that includes the COVID-19 crash. The two series are cointegrated, and the error-correction term serves as an indicator of implied tail risk. Crypto volatility is often disconnected from VIX and gold volatility but shares common shocks. A template for building an index from thin quotes.

Version record

Carol Alexander and Arben Imeraj, "Delta hedging bitcoin options with a smile"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 23(5), 799–817 (2023); DOI 10.1080/14697688.2023.2181205.

Tests smile-implied and smile-adjusted deltas for dynamic hedging of bitcoin options on hourly data. Bitcoin implied-volatility curves behave differently from equity-index curves. In some periods smile-implied deltas beat the Black–Scholes delta, by more than 30% in hedging-error variance for out-of-the-money puts, especially when the perpetual rather than a calendar future is the hedge, because its basis risk is much smaller.

Version record

Christina Atanasova, Terrel Miao, Ignacio Segarra and Frederick Willeboordse, "Aggregate illiquidity and crypto option returns"

Difficulty: Research **Type:** Research paper

Status/version: Finance Research Letters, 85, 108003 (2025); DOI 10.1016/j.frl.2025.108003.

Measures aggregate gamma imbalance, the hedging pressure on bitcoin option market makers, from Deribit transactions. As the imbalance turns more negative, option spreads widen (a one-standard-deviation fall raises out-of-the-money call and put spreads by 0.07 and 0.16 percentage points), and illiquidity is a priced factor in bitcoin option returns. Highlights only in the registered abstract.

Version record

Caio Almeida, Maria Grith, Ratmir Miftachov and Zijin Wang, "Risk Premia in the Bitcoin Market"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2410.15195, v2 (1 August 2025).

Compares bitcoin and S&P 500 risk premia using options and realised returns. Bitcoin carries a much larger variance risk premium; moderately negative returns explain only a third of its equity premium. Clustering option-implied risk-neutral densities gives two regimes: in calm periods more premium comes from positive returns and the variance premium is higher. Preprint.

Version record

Julian Winkel and Wolfgang Karl Härdle, "Pricing Kernels and Risk Premia implied in Bitcoin Options"

Difficulty: Research **Type:** Research paper

Status/version: Risks, 11(5), 85 (2023); DOI 10.3390/risks11050085.

Estimates bitcoin pricing kernels from Deribit options as the ratio of risk-neutral to physical densities, with bootstrap bands. Kernels are U-shaped for short maturities and W-shaped for long ones; investors pay a large premium to insure against short-term moves, less at longer horizons.

Version record

Melanie Cao and Batur Celik, "Valuation of bitcoin options"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 41(7), 1007–1026 (2021); DOI 10.1002/fut.22214.

Equilibrium valuation of bitcoin options that treats bitcoin as a foreign currency in a small open economy with exogenous money supply and aggregate dividend. Diffusive and jump risk in both raise bitcoin prices; closed-form option formulas nest Merton's model, call values rise with money-supply growth, and every risk adds a premium over Black–Scholes.

Version record

Tak Kuen Siu and Robert J. Elliott, "Bitcoin option pricing with a SETAR-GARCH model"

Difficulty: Research **Type:** Research paper

Status/version: The European Journal of Finance, 27(6), 564–595 (2021); DOI 10.1080/1351847x.2020.1828962.

Prices bitcoin options with a SETAR model for regime switching combined with Heston–Nandi GARCH, using conditional Esscher and variance-dependent pricing kernels. A model-choice reading; compare its fitted smiles with executable quotes before using them.

Version record

Alessio Brini and Jimmie Lenz, "Pricing cryptocurrency options with machine learning regression for handling market volatility"

Difficulty: Research **Type:** Research paper

Status/version: Economic Modelling, 136, 106752 (2024); DOI 10.1016/j.econmod.2024.106752.

Prices cryptocurrency options with regression-tree models fed with high-frequency volatility estimators and compares them with Black–Scholes and Heston. Crypto options prove harder to price than equity options, which the authors read as inefficiency, and the data-driven models adapt better to features such as an inverted leverage effect.

Version record

Pierre J. Venter and Eben Maré, "GARCH Generated Volatility Indices of Bitcoin and CRIX"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Risk and Financial Management, 13(6), 121 (2020); DOI 10.3390/jrfm13060121.

Tests a symmetric GARCH option-pricing model on BTCUSD options and uses it to generate 30-, 60- and 90-day implied-volatility indices for bitcoin and the CRIX index, with their term structure. An early model-based alternative to a VIX-style index.

Version record

Akanksha Jalan, Roman Matkovskyy and Saqib Aziz, "The Bitcoin options market: A first look at pricing and risk"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Economics, 53(17), 2026–2041 (2021); DOI 10.1080/00036846.2020.1854671.

First look at Deribit bitcoin options on 27 January 2020: compares near-the-money premiums with Black–Scholes–Merton and Heston–Nandi GARCH values and computes the Greeks against those of commodity options. Finds slight over- and under-pricing by strike. A single-day snapshot, useful for method rather than evidence.

Version record

US Securities and Exchange Commission, Order Granting Accelerated Approval of a Proposed Rule Change . . . To Permit the Listing and Trading of Options on the iShares Bitcoin Trust

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Release No. 34-101128 (SR-ISE-2024-03), 20 September 2024; 89 FR 78942–78947.

SEC order of 20 September 2024 approving Nasdaq ISE's rule to list options on the iShares Bitcoin Trust, the first US options on a spot bitcoin ETP. The options are physically settled and American-style, with position and exercise limits fixed at 25,000 contracts, which the exchange called the lowest in the options industry. NYSE American and Cboe orders for other bitcoin ETP options followed in October 2024.

Version record

18.2 Crypto spot markets

Spot trading of bitcoin and other tokens on centralised exchanges, in futures and through exchange-traded products: price discovery and arbitrage, exchange market quality and manipulation, valuation models, return dynamics, investors and access, and exchange failures and regulation. Derivatives on these assets are in Perpetuals, options and crypto rates; cross-sectional factor research is in Returns, factors and allocation.

18.2.1 Price discovery, arbitrage and reference rates

Which venues lead the price, why cross-exchange arbitrage is limited by settlement latency and counterparty risk, how futures and spot ETFs changed price discovery, and how reference rates are fixed from constituent-exchange trades.

José Almeida and Tiago Cruz Gonçalves, "Cryptocurrency market microstructure: a systematic literature review"

Difficulty: Intermediate **Type:** Research paper

Status/version: Annals of Operations Research, 332(1–3), 1035–1068 (2024); DOI 10.1007/s10479-023-05627-5.

Systematic review of 138 cryptocurrency market-microstructure papers in ABS-listed journals indexed by Web of Science, mapped by bibliographic coupling into research strands on efficiency, liquidity and information, with trends and open questions.

Applications: intermediate to graduate. Version record

Nicola Borri and Kirill Shakhnov, "The Cross-Section of Cryptocurrency Returns"

Difficulty: Research **Type:** Research paper

Status/version: Review of Asset Pricing Studies, 12(3), 667–705 (2022); DOI 10.1093/rapstu/raac007.

Bitcoin trades at different prices on exchanges in different countries and against different currencies. After transaction costs and trading limits, the more expensive pairs are riskier: they depreciate more in bad times for crypto investors, when aggregate liquidity and sentiment are low. A risk-based account of cross-market premia, not a cross-token factor study; compare the distinct Liu–Tsyvinski–Wu paper. [36]

Applications: Bitcoin; exchange segmentation, liquidity risk. Version record

Igor Makarov and Antoinette Schoar, "Price Discovery in Cryptocurrency Markets"

Difficulty: Research **Type:** Research paper

Status/version: AEA Papers and Proceedings, 109, 97–99 (2019); DOI 10.1257/pandp.20191020.

Which exchanges drive the bitcoin price? Price formation is heterogeneous across exchanges and over time. When markets are integrated, shocks on every exchange contribute to price discovery; when they segment, exchanges trading at large premia to the US price, where investors are most exuberant, contribute much less. A short companion to the authors' 2020 study of cross-exchange arbitrage.

Version record

Morten Brandvold, Peter Molnár, Kristian Vagstad and Ole Christian Andreas Valstad, "Price discovery on Bitcoin exchanges"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Financial Markets, Institutions and Money, 36, 18–35 (2015); DOI 10.1016/j.intfin.2015.02.010.

Early information-share study of bitcoin exchanges using public high-frequency trade data. Mt.Gox and BTC-e were the price leaders, and each exchange's share of price discovery changed substantially over time, so leadership has to be re-estimated rather than assumed.

Version record

Thomas Dimpfl and Franziska J. Peter, "Nothing but noise? Price discovery across cryptocurrency exchanges"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 54, 100584 (2021); DOI 10.1016/j.finmar.2020.100584.

Crypto exchanges carry far more microstructure noise than the NYSE or Nasdaq, which biases conventional information shares. Using Putniņš' information leadership share, which adjusts for noise, the paper finds Bitfinex leading price discovery. A methods reading: compare noise levels before ranking venues.

Version record

Nikolaus Hautsch, Christoph Scheuch and Stefan Voigt, "Building trust takes time: limits to arbitrage for blockchain-based assets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 28(4), 1345–1381 (2024); DOI 10.1093/rof/rfae004.

Blockchain settlement is slow, so cross-exchange arbitrageurs bear price risk while coins move; settling off-chain instead exposes them to exchange default. With Bitcoin network and order-book data from 16 exchanges, price differences widen when settlement latency is high, asset flows chase arbitrage opportunities, and spreads are smaller between exchanges with low default risk. Trusted intermediation, not the blockchain, is what lowers arbitrage costs. Replication data and code are public.

Version record

Sinan Krückeberg and Peter Scholz, "Decentralized Efficiency? Arbitrage in Bitcoin Markets"

Difficulty: Research **Type:** Research paper

Status/version: Financial Analysts Journal, 76(3), 135–152 (2020); DOI 10.1080/0015198x.2020.1733902.

Tick data from February 2013 to April 2018 show large, patterned arbitrage spreads between bitcoin markets: price extremes cluster, and spreads widen in the early UTC hours, when new exchanges enter and after hacks. Net exploitable arbitrage profits were at least USD 380 million in 2017 and again in the first quarter of 2018, and measured inefficiency rose over the sample.

Version record

Tommy Crépellière, Matthias Pelster and Stefan Zeisberger, "Arbitrage in the market for cryptocurrencies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 64, 100817 (2023); DOI 10.1016/j.finmar.2023.100817.

Confirms cross-exchange arbitrage opportunities but shows that their size fell sharply from April 2018; after costs, the remaining price differences are barely exploitable. Informed trading is correlated with the reduction. Read with Krückeberg–Scholz to see why pre-2018 results do not carry forward.

Version record

Patrick Augustin, Alexey Rubtsov and Donghwa Shin, "The Impact of Derivatives on Spot Markets: Evidence from the Introduction of Bitcoin Futures Contracts"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 69(11), 6752–6776 (2023); DOI 10.1287/mnsc.2023.4900.

The December 2017 launch of BTC–USD futures as a natural experiment: relative to bitcoin pairs in other currencies without futures, BTC–USD prices became more synchronised across exchanges, with fewer arbitrage opportunities and lower volatility, and price efficiency, market quality and liquidity improved. Futures let traders get round short-sale constraints, block-confirmation risk and segmentation.

Version record

Dirk G. Baur and Thomas Dimpfl, "Price discovery in bitcoin spot or futures?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 39(7), 803–817 (2019); DOI 10.1002/fut.22004.

Hasbrouck and Gonzalo–Granger information shares for bitcoin spot against the CBOE and CME futures launched in December 2017: in the early futures period the spot price leads, which the authors attribute to the spot market's larger volume and round-the-clock global trading. Compare with Aleti–Mizrach and Hu–Hou–Oxley, who reach the opposite conclusion with later or time-varying samples.

Version record

Carol Alexander, Jaehyuk Choi, Heungju Park and Sungbin Sohn, "BitMEX bitcoin derivatives: Price discovery, informational efficiency, and hedging effectiveness"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 40(1), 23–43 (2020); DOI 10.1002/fut.22050.

Minute data from BitMEX, then the largest unregulated bitcoin derivatives exchange: its derivatives lead prices on the major spot exchanges, with spreads, inter-exchange spreads and relative volumes driving price discovery. BitMEX contracts were informationally more efficient than spot prices and effective spot hedges.

Version record

Saketh Aleti and Bruce Mizrach, "Bitcoin spot and futures market microstructure"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 41(2), 194–225 (2021); DOI 10.1002/fut.22163.

Microstructure of BTC/USD on the CME and four spot exchanges that feed its settlement price: spot median trades are under USD 1,300 against over USD 18,000 on the CME, spreads average about three basis points, and trades over USD 1 million move prices by less than 1 per cent. Many executions trade through better quotes, at an estimated cost of USD 36 million. In this sample the CME leads price discovery, and bitcoin leads ether.

Version record

Yang Hu, Yang Greg Hou and Les Oxley, "What role do futures markets play in Bitcoin pricing? Causality, cointegration and price discovery from a time-varying perspective?"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Financial Analysis, 72, 101569 (2020); DOI 10.1016/j.irfa.2020.101569.

Revisits the conflicting spot–futures results with time-varying Granger causality, time-varying cointegration and rolling information shares on daily data. Bitcoin futures prices Granger-cause spot prices and dominate price discovery once time variation is allowed for.

Version record

Takahiro Hattori and Ryo Ishida, "The relationship between arbitrage in futures and spot markets and Bitcoin price movements: Evidence from the Bitcoin markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 41(1), 105–114 (2021); DOI 10.1002/fut.22171.

Reconstructs the actual cash-and-carry arbitrage condition between bitcoin spot and CBOE futures from intraday data. Arbitrage profits are rare in normal markets but large during bitcoin crashes, when the basis is hardest to trade.

Version record

Azhar Mohamad, "Do Bitcoin ETFs Lead Price Discovery Following their Introduction in the Bitcoin Market?"

Difficulty: Research **Type:** Research paper

Status/version: Computational Economics, 66(1), 947–969 (2025); DOI 10.1007/s10614-025-10998-x.

Price discovery between US spot bitcoin ETFs and spot bitcoin from their 11 January 2024 launch to October 2024, using 5-minute information shares, component shares and information leadership shares. By the information leadership share, the most traded ETFs (IBIT, FBTC, GBTC) lead spot bitcoin about 85 per cent of the time. ETF prices trade only in US hours, so read the result for those hours only.

Version record

CF Benchmarks, CME CF Cryptocurrency Reference Rates: Methodology Guide; Constituent Exchanges List

Difficulty: Advanced **Type:** Benchmark methodology

Status/version: Methodology version 17.4 and constituent-exchange list version 13.7, both dated 24 August 2026; living documents.

How the CME CF Bitcoin Reference Rate and its sister rates are fixed. All relevant spot trades on approved constituent exchanges during a TWAP window are pooled and split into equal time partitions; the rate is the equally weighted average of each partition's volume-weighted median price. The guide covers erroneous-data filters, restatement thresholds, market-failure events and the New York and Asia–Pacific variants; since July 2022 trades quoted in an 'accepted asset' (a fully reserved stablecoin, currently USDC) can count as dollar trades. CF Benchmarks' index page states that the BRR settles the bitcoin futures listed by CME Group and Crypto Facilities and prices the NAV of several exchange-traded products; check which rate and variant a contract or fund document names before using it. CF Benchmarks is authorised by the UK FCA as a benchmark administrator.

Version record

18.2.2 Exchange liquidity, fake volume and manipulation

Spreads, depth and their measurement on crypto exchanges; wash trading and inflated volume; price manipulation, stablecoin-funded buying and pump-and-dump schemes.

Peter Reinhard Hansen, Chan Kim and Wade Kimbrough, "Periodicity in Cryptocurrency Volatility and Liquidity"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 22(1), 224–251 (2024); DOI 10.1093/jjfinec/nbac034.

Day-of-week, hour-of-day and within-hour patterns in volatility and volume for bitcoin and ether on Coinbase Pro, Binance and Uniswap V2. The patterns have grown stronger over time, plausibly through algorithmic trading and futures funding times; price formation happens mainly on the centralised exchanges, with sluggish adjustment on the DEX. Useful for execution scheduling, volatility forecasting and comparing CEX and DEX market quality. [35]

Applications: Crypto spot and DEX; liquidity, volatility.

Version record

Lin William Cong, Xi Li, Ke Tang and Yang Yang, "Crypto Wash Trading"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 69(11), 6427–6454 (2023); DOI 10.1287/mnsc.2021.02709.

Statistical tests (Benford, trade-size clustering) show pervasive wash trading on unregulated exchanges.

Version record

John M. Griffin and Amin Shams, "Is Bitcoin Really Untethered?"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 75(4), 1913–1964 (2020); DOI 10.1111/jofi.12903.

Blockchain flow analysis linking Tether issuance to Bitcoin price support in 2017–18.

Version record

Tao Li, Donghwa Shin and Baolian Wang, "Cryptocurrency Pump-and-Dump Schemes"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 60(8), 3622–3659 (2025); DOI 10.1017/s0022109025000201.

Documents cryptocurrency pump-and-dump schemes with pre-event run-ups and reversals implying wealth transfers to insiders.

Version record

Jakob Albers, Mihai Cucuringu, Sam Howison and Alexander Y. Shestopaloff, "The good, the bad, and latency: exploratory trading on Bybit and Binance"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 25(6), 919–947 (2025); DOI 10.1080/14697688.2025.2515933.

Millions of live market orders on Bybit and Binance measure execution versus order-book snapshots, quantifying latency effects.

Version record

Anne H. Dyhrberg, Sean Foley and Jiri Svec, "How investible is Bitcoin? Analyzing the liquidity and transaction costs of Bitcoin markets"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 171, 140–143 (2018); DOI 10.1016/j.econlet.2018.07.032.

Intraday trades and quotes on three US exchanges: although trading is continuous, activity, volatility and the lowest spreads coincide with US trading hours, consistent with retail and non-algorithmic trading. Quoted and effective spreads were below those on major equity exchanges, so bitcoin was highly investible for retail-size trades.

Version record

Alexander Brauneis, Roland Mestel, Ryan Riordan and Erik Theissen, "How to measure the liquidity of cryptocurrency markets?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 124, 106041 (2021); DOI 10.1016/j.jbankfin.2020.106041.

Benchmarks low-frequency liquidity measures against high-frequency liquidity on crypto venues. Corwin–Schultz and Abdi–Ranaldo spread estimators track time variation best across frequencies, venues and coins; Kyle–Obizhaeva and Amihud measures do better for liquidity levels and for ranking venues. No single measure dominates, but several low-frequency proxies are usable.

Version record

Alexander Brauneis, Roland Mestel and Erik Theissen, "What drives the liquidity of cryptocurrencies? A long-term analysis"

Difficulty: Research **Type:** Research paper

Status/version: Finance Research Letters, 39, 101537 (2021); DOI 10.1016/j.frl.2020.101537.

Four years of hourly Abdi–Ranaldo spreads for four cryptocurrencies on four large venues. Return volatility, dollar volume and the number of trades explain liquidity; general financial-market variables have no explanatory power.

Version record

Neil Gandal, J. T. Hamrick, Tyler Moore and Tali Oberman, "Price manipulation in the Bitcoin ecosystem"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 95, 86–96 (2018); DOI 10.1016/j.jmoneco.2017.12.004.

Identifies suspicious trading on the Mt.Gox exchange, where about 600,000 bitcoins valued at USD 188 million were fraudulently acquired. On days with suspicious trades the USD–BTC rate rose by about 4 per cent on average, against a slight fall on other days, and the activity likely caused the late-2013 spike from about USD 150 to over USD 1,000 in two months.

Version record

Arash Aloosh and Jiasun Li, "Direct Evidence of Bitcoin Wash Trading"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 70(12), 8875–8921 (2024); DOI 10.1287/mnsc.2021.01448.

Uses internal trading records of a major bitcoin exchange (Mt.Gox) leaked by hackers to identify wash trades directly, confirming the fake-volume allegation that earlier work could only estimate. The direct evidence is then used to grade indirect detectors: Benford's law, trade-size clustering, lognormality and structural-break tests work; power-law tail tests give the opposite answer. Data files are published with the article.

Version record

Dan Amiram, Evgeny Lyandres and Daniel Rabetti, "Trading Volume Manipulation and Competition Among Centralized Crypto Exchanges"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 71(10), 8604–8622 (2025); DOI 10.1287/mnsc.2021.02903.

Centralised crypto exchanges inflate reported volume more when they face stronger static and dynamic competition. Inflating exchanges gain in the short run but are punished in the long run as reputation suffers. A model of why volume manipulation persists and of the evidence for detecting it.

Version record

Anirudh Dhawan and Tālis J. Putniņš, "A New Wolf in Town? Pump-and-Dump Manipulation in Cryptocurrency Markets"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 27(3), 935–975 (2023); DOI 10.1093/rof/rfac051.

Why do traders join announced crypto pump-and-dump schemes with negative expected returns? A simple model shows overconfidence and gambling preferences can explain participation, and 355 pumps on Binance and Yobit over six months support both. Pumps distort prices by 65 per cent on average, generate abnormal volume in the millions of dollars and transfer wealth between participants.

Version record

J. T. Hamrick, Farhang Rouhi, Arghya Mukherjee, Amir Feder, Neil Gandal, Tyler Moore and Marie Vasek, "An examination of the cryptocurrency pump-and-dump ecosystem"

Difficulty: Research **Type:** Research paper

Status/version: Information Processing & Management, 58(4), 102506 (2021); DOI 10.1016/j.ipm.2021.102506.

Joins every relevant Telegram and Discord pump channel to document thousands of pumps on hundreds of coins. Pumps raise prices modestly in the short run, the effect diminished over time, the most transparent pumps were the most successful, and activity was concentrated in a few channels.

Version record

Guénolé Le Pennec, Ingo Fiedler and Lennart Ante, "Wash trading at cryptocurrency exchanges"

Difficulty: Research **Type:** Research paper

Status/version: Finance Research Letters, 43, 101982 (2021); DOI 10.1016/j.frl.2021.101982.

Predicts each exchange's true volume from web traffic and administered user funds, calibrated on exchanges that report accurately. Exchanges with evidence of wash trading overstate volume by a factor of 25 to 50; those with mixed evidence by 1.25 to 33.

Version record

18.2.3 Valuation, adoption and mining economics

Monetary and equilibrium models of bitcoin and token prices, network effects and adoption, transaction fees and the economics of mining.

Bruno Biais, Christophe Bisière, Matthieu Bouvard, Catherine Casamatta and Albert J. Menkveld, "Equilibrium Bitcoin Pricing"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 78(2), 967–1014 (2023); DOI 10.1111/jofi.13206.

Overlapping-generations model where Bitcoin value derives from transactional benefits, with estimated fundamentals.

Version record

Hanna Halaburda, Guillaume Haeringer, Joshua Gans and Neil Gandal, "The Microeconomics of Cryptocurrencies"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Literature, 60(3), 971–1013 (2022); DOI 10.1257/jel.20201593.

Survey of the microeconomics of cryptocurrencies: what drives their supply, demand, trading price and competition with one another, drawing on economics, law and computer science. The starting point for the valuation and market-structure pathways.

Version record

Igor Makarov and Antoinette Schoar, "Cryptocurrencies and Decentralized Finance (DeFi)"

Difficulty: Advanced **Type:** Research paper

Status/version: Brookings Papers on Economic Activity, 2022(1), 141–215 (2022); DOI 10.1353/eca.2022.0014.

Overview of cryptocurrencies and decentralised finance against the traditional intermediated system. DeFi may cut transaction costs, but rents accumulate at several layers because competition is constrained; the permissionless, pseudonymous design makes tax, anti-money-laundering and anti-fraud enforcement hard. Proposes ways to regulate DeFi that keep most of the benefits of the blockchain architecture. Published with comments and discussion.

Version record

David Yermack, "Is Bitcoin a Real Currency? An Economic Appraisal"

Difficulty: Advanced **Type:** Book chapter

Status/version: In David K. C. Lee (ed.), *Handbook of Digital Currency* (Elsevier, 2015), 31–43; DOI 10.1016/b978-0-12-802117-0.00002-3.

Asks whether bitcoin is money. It fails as a medium of exchange (scant merchant use), as a store of value (volatility far above major currencies) and as a unit of account (prices need many decimal places); its exchange rate is uncorrelated with currencies and gold, so it cannot be hedged, and it lacks deposit insurance and credit use. Bitcoin behaves more like a speculative investment than a currency.

Version record

Rodney Garratt and Neil Wallace, "Bitcoin 1, Bitcoin 2, . . . : An Experiment in Privately Issued Outside Monies"

Difficulty: Research **Type:** Research paper

Status/version: Economic Inquiry, 56(3), 1887–1897 (2018); DOI 10.1111/ecin.12569.

In a monetary model, the value of bitcoin rests on self-fulfilling beliefs that are hard to pin down, whether bitcoin is the only money or competes with clones or with fiat currency. Some of the indeterminacy disappears if the token pays interest.

Version record

Linda Schilling and Harald Uhlig, "Some simple bitcoin economics"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Monetary Economics, 106, 16–26 (2019); DOI 10.1016/j.jmoneco.2019.07.002.

An endowment economy with dollars, supplied to hit an inflation target, and bitcoin, whose supply grows deterministically. In the simplest case the fundamental pricing equation makes the bitcoin price a martingale; with mutual impatience there is no speculation, so volatility does not rule out medium-of-exchange use. Block rewards are financed by a dollar tax, not by bitcoin holders.

Version record

Emiliano S. Pagnotta, "Decentralizing Money: Bitcoin Prices and Blockchain Security"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 35(2), 866–907 (2022); DOI 10.1093/rfs/hhaa149.

Prices bitcoin jointly with blockchain security. Users value holdings for transactions and resale and price the risk of systemic attacks; miners supply security for block rewards. The same technology supports multiple equilibria with very different price and security levels, the price–security feedback amplifies fundamental shocks into booms and busts, and the fixed monetary policy can cause welfare losses.

Version record

Michael Sockin and Wei Xiong, "A Model of Cryptocurrencies"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 69(11), 6684–6707 (2023); DOI 10.1287/mnsc.2023.4756.

Models cryptocurrencies as utility tokens on a decentralised platform. Network effects and the token price's non-neutrality can make the token market break down; resale helps young platforms by drawing on user optimism but worsens fragility when speculators crowd out users. Elastic issuance reduces the fragility and strategic miner attacks increase it.

Version record

Gur Huberman, Jacob D. Leshno and Ciamac Moallemi, "Monopoly without a Monopolist: An Economic Analysis of the Bitcoin Payment System"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Economic Studies, 88(6), 3011–3040 (2021); DOI 10.1093/restud/rdab014.

Models the Bitcoin payment system: free entry of miners means no one can set fees, which are instead determined in a market for transaction priority. Derives closed-form fees and waiting times, compares them with a profit-maximising monopoly payment system, and suggests protocol changes. The appendix explains Bitcoin's design for economists.

Version record

Julien Prat and Benjamin Walter, "An Equilibrium Model of the Market for Bitcoin Mining"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 129(8), 2415–2452 (2021); DOI 10.1086/714445.

A structural model in which the bitcoin price drives the network's computing power. Free entry caps mining revenues; calibrated, the model forecasts hash-rate growth out of sample, so mining capacity follows price rather than supporting it.

Version record

Wilko Bolt and Maarten R. C. van Oordt, "On the Value of Virtual Currencies"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Money, Credit and Banking, 52(4), 835–862 (2020); DOI 10.1111/jmcb.12619.

The exchange rate of a virtual currency depends on the current value of transactions settled in it, on forward-looking investors' purchases and on the factors driving future adoption by consumers and merchants. As the currency becomes established its exchange rate becomes less sensitive to shocks to speculators' beliefs, so volatility need not prevent wider use.

Version record

Neil Gandal and Hanna Halaburda, "Can We Predict the Winner in a Market with Network Effects? Competition in Cryptocurrency Market"

Difficulty: Research **Type:** Research paper

Status/version: Games, 7(3), 16 (2016); DOI 10.3390/g7030016.

Network effects in early competition between cryptocurrencies. For a long period other coins appreciated faster than bitcoin, consistent with their use as financial assets rather than winner-take-all dynamics; towards the end of the sample bitcoin rose against the dollar while others fell, consistent with strong network effects.

Version record

Lin William Cong, Zhiguo He and Jiasun Li, "Decentralized Mining in Centralized Pools"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 34(3), 1191–1235 (2021); DOI 10.1093/rfs/hhaa040.

Mining pools need not centralise proof-of-work: because miners diversify across pools and pool managers set fees, larger pools charge more, attract proportionately fewer miners and grow more slowly. Pools nonetheless escalate the mining arms race and raise energy use. Bitcoin mining data support the predictions.

Version record

Bank for International Settlements, "Cryptocurrencies: looking beyond the hype" (Annual Economic Report 2018, Chapter V)

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Annual Economic Report 2018, Chapter V; published 24 June 2018.

The central banks' case against cryptocurrencies as money. Good money needs trust in the stability of its value and an elastic supply that scales with the economy; decentralised consensus requires honest majorities of computing power, verification of the full history by every user and a supply fixed by protocol. Transactions are slow, costly and prone to congestion, consensus is fragile and energy-intensive, and payments are never fully final. Distributed ledgers may still be useful elsewhere.

Version record

18.2.4 Return dynamics, bubbles and cross-asset links

Stylised facts, efficiency tests, bubble dating, volatility and jump models, monetary-policy effects, and correlation, hedge and safe-haven evidence against other assets.

Albert S. Hu, Christine A. Parlour and Uday Rajan, "Cryptocurrencies: Stylized facts on a new investible instrument"

Difficulty: Research **Type:** Research paper

Status/version: Financial Management, 48(4), 1049–1068 (2019); DOI 10.1111/fima.12300.

Stylised facts on the returns of 222 coins and 64 initial coin offerings: summary statistics and common variation in secondary-market returns. All other coins' returns are strongly correlated with bitcoin's.

Version record

Andrew Urquhart, "The inefficiency of Bitcoin"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 148, 80–82 (2016); DOI 10.1016/j.econlet.2016.09.019.

A battery of efficiency tests rejects weak-form efficiency for bitcoin returns over the full sample, but some tests accept it in the later sub-period, so bitcoin may be moving towards efficiency.

Version record

Saralees Nadarajah and Jeffrey Chu, "On the inefficiency of Bitcoin"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 150, 6–9 (2017); DOI 10.1016/j.econlet.2016.10.033.

Reply to Urquhart (2016): eight tests show that a simple power transformation of bitcoin returns, which loses no information, satisfies the efficient-market hypothesis. A reminder that efficiency verdicts depend on the test and the transformation.

Version record

Eng-Tuck Cheah and John Fry, "Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 130, 32–36 (2015); DOI 10.1016/j.econlet.2015.02.029.

Econometric modelling of bitcoin prices that finds speculative bubbles and evidence that the fundamental price of bitcoin is zero. An early, much-cited bubble result to read against the equilibrium models in the valuation pathway.

Version record

Peter C. B. Phillips, Shuping Shi and Jun Yu, "Testing for Multiple Bubbles: Historical Episodes of Exuberance and Collapse in the S&P 500"

Difficulty: Research **Type:** Research paper

Status/version: International Economic Review, 56(4), 1043–1078 (2015); DOI 10.1111/iere.12132.

The generalised sup ADF test: a recursive, flexible-window right-tailed unit-root test that dates the start and end of multiple bubbles in real time and performs well in long samples with several episodes. Applied here to the S&P 500; it is the standard tool for date-stamping crypto bubbles.

Version record

Paraskevi Katsiampa, "Volatility estimation for Bitcoin: A comparison of GARCH models"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 158, 3–6 (2017); DOI 10.1016/j.econlet.2017.06.023.

Compares conditional-variance models for bitcoin by goodness of fit: the AR-CGARCH, with separate short- and long-run variance components, fits best.

Version record

Olivier Scaillet, Adrien Treccani and Christopher Trevisan, "High-Frequency Jump Analysis of the Bitcoin Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 18(2), 209–232 (2020); DOI 10.1093/jjfinec/nby013.

Uses the leaked Mt.Gox database (June 2011 to November 2013), with trader identifiers at tick level, to study jumps. Jumps are frequent and cluster; order-flow imbalance, a preponderance of aggressive traders and widening spreads predict them, and they have a persistent price effect and short-run effects on activity and illiquidity.

Version record

Christian Conrad, Anessa Custovic and Eric Ghysels, "Long- and Short-Term Cryptocurrency Volatility Components: A GARCH-MIDAS Analysis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Risk and Financial Management, 11(2), 23 (2018); DOI 10.3390/jrfm11020023.

GARCH-MIDAS decomposition of cryptocurrency volatility into long- and short-run parts. S&P 500 realised volatility lowers long-run bitcoin volatility, an unusual co-movement; the S&P 500 variance risk premium and the Baltic Dry Index raise it, linking bitcoin volatility to global activity.

Version record

Dirk G. Baur, KiHoon Hong and Adrian D. Lee, "Bitcoin: Medium of exchange or speculative assets?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Financial Markets, Institutions and Money, 54, 177–189 (2018); DOI 10.1016/j.intfin.2017.12.004.

Bitcoin returns are uncorrelated with stocks, bonds and commodities in normal and turbulent times, and transaction data on bitcoin accounts show coins used mainly for speculative investment rather than as a medium of exchange.

Version record

Elie Bouri, Peter Molnár, Georges Azzi, David Roubaud and Lars Ivar Hagfors, "On the hedge and safe haven properties of Bitcoin: Is it really more than a diversifier?"

Difficulty: Research **Type:** Research paper

Status/version: Finance Research Letters, 20, 192–198 (2017); DOI 10.1016/j.frl.2016.09.025.

Dynamic conditional correlations of bitcoin with world equity indices, bonds, oil, gold, commodities and the dollar, 2011–2015. Bitcoin is a poor hedge and useful mainly for diversification; it was a strong safe haven only against weekly extreme falls in Asian stocks, and its properties vary by horizon.

Version record

Shaen Corbet, Andrew Meegan, Charles Larkin, Brian Lucey and Larisa Yarovaya, "Exploring the dynamic relationships between cryptocurrencies and other financial assets"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 165, 28–34 (2018); DOI 10.1016/j.econlet.2018.01.004.

Time- and frequency-domain links between three cryptocurrencies and other financial assets: the coins were relatively isolated, offering diversification for short-horizon investors, with linkages varying with external shocks. Crossref showed no correction or retraction notice for this article on 14 October 2026; two other crypto papers by overlapping authors were retracted in 2025–26 over editorial conflicts.

Version record

Anne Haubo Dyhrberg, "Bitcoin, gold and the dollar – A GARCH volatility analysis"

Difficulty: Research **Type:** Research paper

Status/version: Finance Research Letters, 16, 85–92 (2016); DOI 10.1016/j.frl.2015.10.008.

GARCH models find bitcoin's volatility behaviour similar in some respects to gold and the dollar, placing it between a pure medium of exchange and a pure store of value. Early-sample evidence that later studies (Baur–Hong–Lee, Bouri et al.) qualify.

Version record

Marc Gronwald, "Is Bitcoin a Commodity? On price jumps, demand shocks, and certainty of supply"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 97, 86–92 (2019); DOI 10.1016/j.jimonfin.2019.06.006.

Compares bitcoin with exhaustible commodities. Jump and Student-t GARCH models fit best and extreme moves matter more for bitcoin than for oil or gold; with supply fixed and certain, bitcoin price movements can be read as demand shocks.

Version record

Ladislav Kristoufek, "What Are the Main Drivers of the Bitcoin Price? Evidence from Wavelet Coherence Analysis"

Difficulty: Research **Type:** Research paper

Status/version: PLOS ONE, 10(4), e0123923 (2015); DOI 10.1371/journal.pone.0123923.

Wavelet coherence of bitcoin prices with fundamental, speculative and technical drivers and with the Chinese market, separating short- and long-run links. Bitcoin shows properties of both a standard financial asset and a speculative one.

Version record

Daniele Bianchi, "Cryptocurrencies as an Asset Class? An Empirical Assessment"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Alternative Investments, 23(2), 162–179 (2020); DOI 10.3905/jai.2020.1.105.

Cryptocurrency returns have only a mild correlation with commodities and precious metals and no volatility spillovers; past returns drive trading volume, consistent with sentiment; term-structure and inflation-expectation variables do not explain market activity.

Version record

Nicola Borri, "Conditional tail-risk in cryptocurrency markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 50, 1–19 (2019); DOI 10.1016/j.jempfin.2018.11.002.

CoVaR estimates for bitcoin, ether, XRP and litecoin: each is highly exposed to tail risk within crypto markets but not to equity or gold tail risk. Portfolios of coins cut idiosyncratic risk, but once liquidity is accounted for the optimal crypto share of an investor's portfolio is small.

Version record

Emmanouil Platanakis and Andrew Urquhart, "Should investors include Bitcoin in their portfolios? A portfolio theory approach"

Difficulty: Research **Type:** Research paper

Status/version: The British Accounting Review, 52(4), 100837 (2020); DOI 10.1016/j.bar.2019.100837.

Out-of-sample portfolio tests to June 2018 across eight allocation rules: adding bitcoin to a stock–bond portfolio raised risk-adjusted returns substantially, robust to costs, estimation windows and short-sale constraints. A sample-specific result; compare with Borri's liquidity-adjusted allocation.

Version record

Sören Karau, "Monetary policy and Bitcoin"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 137, 102880 (2023); DOI 10.1016/j.jimonfin.2023.102880.

High-frequency evidence that bitcoin did not respond to US monetary-policy announcements until late 2020, after which it reacts like other risky assets. A structural VAR shows that historically US tightening raised bitcoin prices; cross-currency price spreads and blockchain data link this to demand from East Asian economies with capital controls, particularly China.

Version record

Lin William Cong, G. Andrew Karolyi, Ke Tang and Weiyi Zhao, "Crypto Value, Factor Pricing, and Market Segmentation"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, advance online publication (5 June 2026); DOI 10.1287/mnsc.2024.05875.

In the largest crypto-asset sample to date, a value effect based on active addresses relative to market capitalisation. A four-factor model (market, size, momentum and this value factor) prices the cross-section better than existing benchmarks; the premium plausibly compensates on-chain activity risk. A functional classification of tokens reveals significant, time-varying segmentation across categories. Advance online publication; volume and pages not yet assigned.

Version record

Abdullah Mohammad Ghazi Al khatib and Bayan Mohamad Alshaib, "From contagion to stabilization: Spot Bitcoin ETFs and the regime shift in crypto-equity integration"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Economics & Finance, 111, 105664 (2026); DOI 10.1016/j.iref.2026.105664.

Daily data from January 2016 to May 2026 test whether the January 2024 spot ETF approvals raised or lowered crypto–equity contagion. Contagion appears in the 2020 liquidity crisis; slope-equality tests point to a stabilisation phase after 2024, with bitcoin leading equities mainly in crashes. IBIT volume rises with volatility, so the authors do not attribute the change causally to ETF flows.

Version record

Robert J. R. Elliott and Andrew Urquhart, "Bitcoin Research: What Are the New Frontiers?"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Review, 60(4), 1121–1127 (2025); DOI 10.1111/fire.70015.

Short agenda piece after sixteen years of Bitcoin: briefly reviews the literature and lists open research questions on institutional and country adoption, criminality, and bitcoin's relationship with stablecoins.

Version record

18.2.5 Investors, spot ETFs and bank custody

Who buys crypto and why, how retail trading responds to prices and income, the approval and early evidence of US spot ETFs, and the capital rules for banks that hold or custody cryptoassets.

Andrew M. Hornback and Robert E. Whaley, "Spot Bitcoin ETFs: The Struggle Was Worth It"

Difficulty: Advanced **Type:** Research paper

Status/version: Financial Analysts Journal, 81(2), 39–50 (2025); DOI 10.1080/0015198x.2025.2458054.

First-year performance of US spot bitcoin ETFs relative to bitcoin futures and futures-based ETFs, and the regulatory history behind their approval.

Version record

Shimon Kogan, Igor Makarov, Marina Niessner and Antoinette Schoar, "Are cryptos different? Evidence from retail trading"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 159, 103897 (2024); DOI 10.1016/j.jfineco.2024.103897.

eToro account data: the same retail traders are contrarian in stocks and gold but follow momentum in cryptocurrencies. Investor characteristics, composition, inattention, fees and a taste for lottery-like assets do not explain the difference; the authors conjecture that traders believe price rises raise the chance of widespread adoption, so they extrapolate.

Version record

Raphael Auer, Giulio Cornelli, Sebastian Doerr, Jon Frost and Leonardo Gambacorta, "Crypto Trading and Bitcoin Prices: Evidence from a New Database of Retail Adoption"

Difficulty: Research **Type:** Research paper

Status/version: IMF Economic Review, 74(2), 425–460 (2026); DOI 10.1057/s41308-025-00275-0.

A public daily database of crypto exchange app use in 95 countries, 2015–2022. Rising bitcoin prices are followed by entry of new users, particularly more risk-seeking ones, while larger holders sell as prices rise, likely at retail users' expense. An exogenous price fall during the January 2022 unrest in Kazakhstan confirms the pattern. Earlier version: BIS Working Paper 1049.

Version record

Anantha Divakaruni and Peter Zimmerman, "Uncovering Retail Trading in Bitcoin: The Impact of COVID-19 Stimulus Checks"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 70(4), 2066–2085 (2024); DOI 10.1287/mnsc.2023.4790.

Treats the April 2020 US stimulus payments as a household wealth shock: bitcoin buy trades of exactly USD 1,200, the modal payment, rose sharply, with similar effects in other countries that paid stimulus. The payments raised USD–BTC buy volume by 3.8 per cent and the price by 0.6 per cent, and bitcoin demand is far less price-elastic than stock demand.

Version record

Andreas Hackethal, Tobin Hanspal, Dominique M. Lammer and Kevin Rink, "The Characteristics and Portfolio Behavior of Bitcoin Investors: Evidence from Indirect Cryptocurrency Investments"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 26(4), 855–898 (2022); DOI 10.1093/rof/rfab034.

Administrative brokerage data on investors who buy cryptocurrency exposure through structured retail products. They trade actively, show investment biases, chase high-sentiment stocks and technical-analysis heuristics, log in and trade more after their first crypto purchase, and tilt towards riskier securities afterwards.

Version record

Matthias Pelster, Bastian Breitmayer and Tim Hasso, "Are cryptocurrency traders pioneers or just risk-seekers? Evidence from brokerage accounts"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 182, 98–100 (2019); DOI 10.1016/j.econlet.2019.06.013.

Brokerage data around investors' first cryptocurrency trade: they simultaneously trade stocks more often, with more leverage and more risk, especially when crypto volatility is low, consistent with excitement-seeking rather than diversification.

Version record

Tim Hasso, Matthias Pelster and Bastian Breitmayer, "Who trades cryptocurrencies, how do they trade it, and how do they perform? Evidence from brokerage accounts"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Behavioral and Experimental Finance, 23, 64–74 (2019); DOI 10.1016/j.jbef.2019.04.009.

Demographics, trading patterns and performance in 465,926 brokerage accounts: crypto trading spread across ages and genders, but men trade crypto more often and more speculatively and earn lower returns. Crossref records a 2021 erratum to the article.

Version record

Darren Aiello, Scott R. Baker, Tetyana Balyuk, Marco Di Maggio, Mark J. Johnson and Jason D. Kotter, "Who Invests in Crypto? Wealth, Financial Constraints, and Risk Attitudes"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 31856 (November 2023); DOI 10.3386/w31856.

Transaction data for millions of US households: apart from high-income early adopters, crypto investors resemble the general population, with most dollars from high-income households as in equities. High past crypto returns and income shocks raise crypto investment, and so do household inflation expectations; for most households crypto is treated like a traditional asset.

Version record

Michael Weber, Bernardo Candia, Olivier Coibion and Yuriy Gorodnichenko, "Do You Even Crypto, Bro? Cryptocurrencies in Household Finance"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 31284 (May 2023); DOI 10.3386/w31284.

Repeated large surveys of US households: crypto holders are younger, more often white, male and libertarian, expect much higher returns, see crypto as safer and as an inflation hedge, and spend bitcoin gains on durables. Randomised information on past returns raises desired and actual crypto holdings.

Version record

Mieszko Mazur and Efstathios Polyzos, "Spot Bitcoin ETFs: The Effect of Fund Flows on Bitcoin Price Formation"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Alternative Investments, 27(4), 110–123 (2025); DOI 10.3905/jai.2025.1.239.

Early evidence on US spot bitcoin ETFs: daily inflows above USD 500 million (about five times new mining supply), net flows that track the bitcoin price with an R-squared of 95 per cent, most price changes outside ETF trading hours, abnormal ETF volume after price rises, and inflows that mirror gold-ETF outflows. Descriptive; the flow–price relation is not identified causally.

Version record

Vassilios Babalos, Elie Bouri and Rangan Gupta, "Does the introduction of US spot Bitcoin ETFs affect spot returns and volatility of major cryptocurrencies?"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Review of Economics and Finance, 102, 102006 (2025); DOI 10.1016/j.qref.2025.102006.

Event study of the January 2024 spot ETF launch with GARCH models: positive returns around the event for bitcoin, ether and litecoin, lower volatility for bitcoin and XRP afterwards, and wavelet evidence of volatility spillovers from the Grayscale fund to bitcoin futures and, less, to spot.

Version record

U.S. Securities and Exchange Commission, Order Granting Accelerated Approval of Proposed Rule Changes . . . to List and Trade Bitcoin-Based Commodity-Based Trust Shares and Trust Units (Release 34-99306)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC order, 10 January 2024; NYSE Arca, Nasdaq and Cboe BZX filings for eleven spot bitcoin products.

The order that let US exchanges list spot bitcoin ETPs, after the D.C. Circuit vacated the SEC's disapproval of the Grayscale conversion in August 2023. The test is whether exchanges can prevent fraud and manipulation: the SEC accepts the exchanges' surveillance-sharing with the CME through the Intermarket Surveillance Group because its own analysis of Coinbase and Kraken BTC/USD returns against the front CME future, March 2021 to October 2023, finds correlations of at least 98.4 per cent hourly, 94.2 per cent at five minutes and 76.9 per cent at one minute, consistently over rolling windows. It criticises correlation estimates based on non-stationary price levels, and the approved products create and redeem only in cash.

Version record

U.S. Securities and Exchange Commission, Order Granting Accelerated Approval . . . to Adopt Generic Listing Standards for Commodity-Based Trust Shares (Release 34-103995)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC order of 17 September 2025, 90 FR 45414 (22 September 2025); Nasdaq's amendments approved 30 July 2026 (91 FR 48204).

Lets Nasdaq, Cboe BZX and NYSE Arca list spot-commodity trusts, including crypto ETPs, without a separate rule filing for each product. Every commodity held must trade on an Intermarket Surveillance Group market, or underlie a futures contract listed on a designated contract market for at least six months with a surveillance-sharing agreement, or (initially) be at least 40 per cent of the net assets of an existing ETF. The six-month futures test is the route by which tokens beyond bitcoin and ether qualify.

Version record

Basel Committee on Banking Supervision, Prudential Treatment of Cryptoasset Exposures (SCO60); Cryptoasset standard amendments (July 2024); February 2026 review update

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Standard of 16 December 2022 (Basel Framework chapter SCO60); targeted amendments of 17 July 2024, to be implemented by 1 January 2026; expedited targeted review under way in 2026.

The global bank capital standard for cryptoasset exposures, covering tokenised traditional assets, stablecoins and unbacked cryptoassets such as bitcoin. The 2024 amendments tightened the conditions under which stablecoins receive the preferential Group 1b treatment and clarified other parts of the standard. In February 2026 the Committee reported progress on an expedited review of targeted elements, prompted by market developments, with an update promised later in 2026. Sets the cost to banks of custody, trading and lending in crypto.

Version record

18.2.6 Exchange failures and market regulation

Exchange closures, hacks and the FTX collapse; proofs of solvency; multifunction intermediaries; market reactions to regulation; and the international and US frameworks for crypto trading platforms.

Tyler Moore and Nicolas Christin, "Beware the Middleman: Empirical Analysis of Bitcoin-Exchange Risk"

Difficulty: Research **Type:** Research paper

Status/version: Financial Cryptography and Data Security (FC 2013), Lecture Notes in Computer Science 7859 (Springer, 2013), 25–33; DOI 10.1007/978-3-642-39884-1_3.

Of 40 bitcoin exchanges opened in three years, 18 had closed, often wiping out customer balances. A proportional-hazards model shows that low-volume exchanges were more likely to close, and a logistic model that popular exchanges were more likely to be breached. The first systematic study of exchange counterparty risk.

Version record

Tyler Moore, Nicolas Christin and Janos Szurdi, "Revisiting the Risks of Bitcoin Currency Exchange Closure"

Difficulty: Research **Type:** Research paper

Status/version: ACM Transactions on Internet Technology, 18(4), 1–18 (2018); DOI 10.1145/3155808.

Extends the 2013 study to 80 exchanges opened between 2010 and 2015: 38 closed, sometimes wiping out customer balances, and 25 suffered breaches, 15 of which later closed. A breach raises the odds of closure in that quarter thirteenfold; each doubling of volume cuts the odds of closure by 12 per cent.

Version record

Gaby G. Dagher, Benedikt Bünz, Joseph Bonneau, Jeremy Clark and Dan Boneh, "Provisions: Privacy-preserving Proofs of Solvency for Bitcoin Exchanges"

Difficulty: Research **Type:** Research paper

Status/version: ACM CCS 2015 conference proceedings, 720–731; DOI 10.1145/2810103.2813674.

Exchanges hold customers' coins like banks, and several have lost them. A proof of solvency shows cryptographically that reserves cover every customer balance; Provisions does so without revealing the exchange's addresses, total holdings or liabilities, or customer data, adds a defence against exchanges colluding to share reserves, and is practical for millions of customers. Shows what a proof of reserves without liabilities cannot establish.

Version record

John J. Ray III, First Interim Report of John J. Ray III to the Independent Directors on Control Failures at the FTX Exchanges

Difficulty: Advanced **Type:** Report / research note

Status/version: Filed 9 April 2023 in In re FTX Trading Ltd., Case No. 22-11068 (JTD), U.S. Bankruptcy Court for the District of Delaware (Docket 1242).

The debtors' own account of how the FTX exchanges failed: no effective management or governance, no appropriate accounting systems, extensive commingling of assets, and private keys to crypto assets stored in plain text or single-signature form in the cloud. Code changes exempted Alameda Research, the affiliated trading firm, from the auto-liquidation applied to other customers and let it trade and withdraw regardless of its balance; its borrowing limit stood at USD 65 billion at the petition date. A cyberattack drained about USD 432 million on the day of the bankruptcy. A primary source for what a multifunction intermediary can hide.

Version record

David Vidal-Tomás, Antonio Briola and Tomaso Aste, "FTX's downfall and Binance's consolidation: The fragility of centralised digital finance"

Difficulty: Research **Type:** Research paper

Status/version: Physica A: Statistical Mechanics and its Applications, 625, 129044 (2023); DOI 10.1016/j.physa.2023.129044.

On-chain data show that FTX relied heavily on leveraging and misusing its own token, FTT. Network models of 199 coins at hourly frequency and tick data around November 2022 identify the Terra–Luna collapse as the event that drained FTX's liquidity, and Binance's tweets as an accelerant; the collapse left crypto more centralised.

Version record

Luca Galati, Alexander Webb and Robert I. Webb, "Financial contagion in cryptocurrency exchanges: Evidence from the FTT collapse"

Difficulty: Research **Type:** Research paper

Status/version: Finance Research Letters, 67, 105747 (2024); DOI 10.1016/j.frl.2024.105747.

Contagion across major exchanges during the collapse of FTX and its token FTT, and information cascades while FTX withdrawals were frozen: abnormal returns indicate a flight from less to more established crypto assets.

Version record

Brian D. Feinstein and Kevin Werbach, "The Impact of Cryptocurrency Regulation on Trading Markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Regulation, 7(1), 48–99 (2021); DOI 10.1093/jfr/fjab003.

Original data on crypto regulations worldwide and trading activity on exchanges around regulatory announcements. Across many models and categories of action (licensing regimes, anti-money-laundering and anti-fraud enforcement) the results are almost entirely null: no systematic evidence that regulation drives traders out of or into a jurisdiction.

Version record

Ahmad Chokor and Elise Alfieri, "Long and short-term impacts of regulation in the cryptocurrency market"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Review of Economics and Finance, 81, 157–173 (2021); DOI 10.1016/j.qref.2021.05.005.

Event study of regulatory news, 2015–2019: events that raise the probability of regulation bring negative abnormal returns, smaller for illiquid coins and coins with more information asymmetry; risk-adjusted returns positive before the events are insignificant afterwards.

Version record

Raphael Auer and Stijn Claessens, "Regulating cryptocurrencies: assessing market reactions" (BIS Quarterly Review, September 2018)

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Quarterly Review, September 2018; published 23 September 2018.

Event study of crypto prices, volumes and user bases around regulatory news. General bans and treatment under securities law have the largest negative effects, followed by anti-money-laundering news and limits on links to regulated markets; news of tailored legal frameworks for cryptocurrencies and ICOs brings strong gains. Crypto markets rely on regulated institutions and are segmented by jurisdiction, so national regulation reaches them. Compare with Feinstein–Werbach's null results on trading volumes.

Version record

Board of the International Organization of Securities Commissions (IOSCO), Policy Recommendations for Crypto and Digital Asset Markets: Final Report (FR11/2023); Thematic Review of their implementation (FR/13/25)

Difficulty: Advanced **Type:** Report / research note

Status/version: IOSCO Final Report FR11/2023, 16 November 2023; thematic review FR/13/25, October 2025.

Eighteen recommendations applying securities-regulation outcomes to crypto-asset service providers across the life cycle: governance and disclosure of conflicts, order handling and trade disclosure (separating intermediaries from market operators), admission to trading, fraud, market abuse and surveillance, cross-border cooperation, custody (segregation, reconciliation, independent assurance), operational risk and retail distribution. The 2025 thematic review finds significant progress in the jurisdictions reviewed, with frameworks still developing.

Version record

Financial Stability Board, The Financial Stability Implications of Multifunction Crypto-asset Intermediaries

Difficulty: Advanced **Type:** Report / research note

Status/version: FSB report, 28 November 2023.

Multifunction crypto-asset intermediaries combine a trading platform with brokerage, proprietary trading, lending, custody, token issuance and sometimes their own stablecoin, combinations restricted or separated in traditional finance. They earn mainly trading fees, plus reserve income and proprietary returns, often through offshore and opaque group structures. Vulnerabilities are leverage, liquidity mismatch, operational fragility and interconnection, amplified by the combinations of functions and by conflicts of interest; the report sets out data gaps and monitoring elements. The report names Binance, Bitfinex and Coinbase, and FTX before its failure, as examples.

Version record

U.S. Securities and Exchange Commission and Commodity Futures Trading Commission staff, Joint Statement on Trading of Certain Spot Crypto Asset Products; CFTC, First-Ever Listed Spot Crypto Trading on U.S. Regulated Exchanges (Release 9145-25)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC press release 2025-110 and joint staff statement, 2 September 2025; CFTC press release 9145-25, 4 December 2025.

The US route to exchange-listed spot crypto trading without new legislation. SEC Trading and Markets staff and CFTC market-oversight and clearing staff state that SEC- and CFTC-registered exchanges are not prohibited from facilitating trading in certain spot crypto products, as part of the SEC's Project Crypto and the CFTC's Crypto Sprint. In December 2025 the CFTC announced that listed spot crypto

products would begin trading on CFTC-registered futures exchanges, citing the statutory requirement that leveraged retail commodity trading take place on such exchanges. The joint statement is a staff view, not a Commission rule.

Version record

United States Congress, H.R. 3633, Digital Asset Market Clarity Act (CLARITY Act)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Passed the House 17 July 2025 (294–134); reported by the Senate Banking Committee with a substitute 1 June 2026; cloture on the motion to proceed failed 49–50 on 15 September 2026; a motion to reconsider was entered. Not enacted.

The proposed US market-structure law for crypto. As passed by the House it gives the CFTC primary jurisdiction over digital commodities (assets whose value relies on a blockchain), including digital-commodity exchanges, brokers and dealers; tokens qualify for exchange trading if their blockchain is mature or decentralised or the issuer files reports; exchanges must monitor trading, keep records and segregate customer assets; and the SEC keeps jurisdiction over such trading on alternative trading systems and securities exchanges. Read as a stalled bill, not current law.

Version record

18.3 Crypto market making

Liquidity provision in crypto markets: order-book microstructure, fees and impact on centralised venues; quoting and execution strategies, learning agents and their backtests; token market makers, listings and enforcement; and liquidity provision on decentralised exchanges, including just-in-time liquidity and fee-tier fragmentation. AMM design and LP returns are in Section 18.6; venue documentation and replay tools are in Sections 18.1 and 21.1.

18.3.1 Order-book microstructure, fees and impact

J. Donier and J. Bonart, "A Million Metaorder Analysis of Market Impact on the Bitcoin"

Difficulty: Research **Type:** Research paper

Status/version: Market Microstructure and Liquidity, 1(2), 1550008 (2015); DOI 10.1142/S2382626615500082.

Reconstructs more than a million metaorders from complete Mt.Gox bitcoin data and confirms the square-root law of market impact over four decades of size, although statistical arbitrage and market making were nearly absent. Impact follows the square root along the whole execution path, and the uninformed part of the flow decays almost completely. A clean benchmark for the impact a crypto market maker's hedges and inventory unwinds will face.

Version record

Jonathan Donier and Jean-Philippe Bouchaud, "Why Do Markets Crash? Bitcoin Data Offers Unprecedented Insights"

Difficulty: Research **Type:** Research paper

Status/version: PLOS ONE, 10(10), e0139356 (2015); DOI 10.1371/journal.pone.0139356.

Uses bitcoin order-book data to show that crashes are conditioned by the liquidity available in the book. Proposes a liquidity measure, built from impact theory and public data, that can serve as an early warning of instability; read it as the market-quality view of when quoting depth disappears.

Version record

Alexander Brauneis, Roland Mestel, Ryan Riordan and Erik Theissen, "The anatomy of a fee change — evidence from cryptocurrency markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 67, 152–167 (2022); DOI 10.1016/j.jempfin.2022.03.003.

Difference-in-differences study of a leading BTC–USD platform that introduced maker fees while cutting taker fees. With a negligible tick size, the Colliard–Foucault model predicts wider quoted spreads, and they widen; the taker-fee cut more than compensates, while depth and the number of trades fall and trade size rises. The cleanest evidence on how maker-taker pricing shapes crypto quoting.

Version record

Luca Galati, "Exchange market share, market makers, and murky behavior: The impact of no-fee trading on cryptocurrency market quality"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 165, 107222 (2024); DOI 10.1016/j.jbankfin.2024.107222.

Binance's July 2022 removal of maker–taker fees on bitcoin pairs as a natural experiment. Zero fees raised willingness to trade and Binance's market share, but market makers widened spreads and thinned depth, so liquidity fell and customers' total transaction costs rose; liquidity providers gained at takers' expense. Read with Brauneis et al. for the opposite fee change.

Version record

Daniele Bianchi, Mykola Babiak and Alexander Dickerson, "Trading volume and liquidity provision in cryptocurrency markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 142, 106547 (2022); DOI 10.1016/j.jbankfin.2022.106547.

Measures the return to liquidity provision with a short-term reversal strategy across USD crypto pairs on several centralised exchanges (2017–2022). Returns concentrate in smaller, more volatile and less active pairs, where adverse selection is likely higher, and the interaction of past returns with volume shocks predicts returns, consistent with inventory and adverse-selection theories.

Version record

Alla A. Petukhina, Raphael C. G. Reule and Wolfgang Karl Härdle, "Rise of the machines? Intraday high-frequency trading patterns of cryptocurrencies"

Difficulty: Advanced **Type:** Research paper

Status/version: The European Journal of Finance, 27(1-2), 8–30 (2021); DOI 10.1080/1351847x.2020.1789684.

Describes intraday high-frequency patterns in crypto trading on European venues: returns, volumes, volatility periodicity and correlations with the CRIX index, and evidence of intraday momentum. Useful background on the time-of-day profile a quoting engine must handle.

Version record

Eduard Silantyev, "Order flow analysis of cryptocurrency markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Digital Finance, 1(1-4), 191–218 (2019); DOI 10.1007/s42521-019-00007-w.

Order-flow analysis of the BitMEX XBTUSD perpetual: trade-flow imbalance explains contemporaneous price changes better than aggregate order-flow imbalance, and over long enough intervals price changes are close to linear in order-flow imbalance. Thin depth and slow book updates set crypto microstructure apart from mature markets.

Version record

18.3.2 Quoting strategies, learning agents and backtests

Matthias Schnaubelt, "Deep reinforcement learning for the optimal placement of cryptocurrency limit orders"

Difficulty: Research **Type:** Research paper

Status/version: European Journal of Operational Research, 296(3), 993–1006 (2022); DOI 10.1016/j.ejor.2021.04.050.

Large-scale deep reinforcement learning for limit-order placement on crypto exchanges, with a purpose-built reward, hand-crafted state features and a virtual limit-order exchange, trained on 18 months of data (300 million trades, 3.5 million book states). Proximal policy optimisation learns placement rules known from established markets, becoming more aggressive when trade and order imbalances signal lower fill probability.

Version record

Jonathan Sadighian, "Deep Reinforcement Learning in Cryptocurrency Market Making"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 1911.08647 (20 November 2019); DOI 10.48550/arXiv.1911.08647.

Early deep reinforcement learning framework for crypto market making: two policy-gradient agents observe limit-order-book and order-flow arrival features, and two reward functions are compared on daily and average trade returns. A template for the agent, environment and reward choices; its backtests assume fills that a queue-aware simulator should test.

Version record

Jiafa He, Cong Zheng and Can Yang, "Integrating Tick-level Data and Periodical Signal for High-frequency Market Making"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2306.17179 (19 June 2023); DOI 10.48550/arXiv.2306.17179.

Deep reinforcement learning market maker that combines tick-level order-book data with periodic prediction signals, tested in simulation and on crypto market data, where it reports better profitability and risk control than benchmark methods. Preprint; check its fill and latency assumptions before comparing results.

Version record

Thomas Günter Fischer, Christopher Krauss and Alexander Deinert, "Statistical Arbitrage in Cryptocurrency Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Risk and Financial Management, 12(1), 31 (2019); DOI 10.3390/jrfm12010031.

Random-forest statistical arbitrage on minute data for 40 coins: buy the three predicted outperformers and short the three predicted underperformers over 120 minutes. Out of sample (June–September 2018) it earns 7.1 basis points a day after 15 basis points per half-turn, but the authors stress volume limits to scaling the intraday strategy.

Version record

Álvaro Cartea, Fayçal Drissi and Marcello Monga, "Execution and Statistical Arbitrage with Signals in Multiple Automated Market Makers"

Difficulty: Research **Type:** Research paper

Status/version: IEEE International Conference on Distributed Computing Systems Workshops (ICDCSW 2023), 37–42 (2023); DOI 10.1109/ICDCSW60045.2023.00012.

Closed-form optimal strategy for a liquidity taker who executes large orders and statistical arbitrage across a basket of co-moving crypto pairs, using signals from AMM pools and limit-order-book venues. Uniswap v3 pools and Binance data show co-movement and lead–lag effects between venues; the cross-venue counterpart to single-venue quoting models.

Version record

18.3.3 Token market makers, listings and enforcement

Hugo Benedetti and Ehsan Nikbakht, "Returns and network growth of digital tokens after cross-listings"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Corporate Finance, 66, 101853 (2021); DOI 10.1016/j.jcorpfin.2020.101853.

Studies 3,625 tokens on 108 marketplaces around a token's first cross-listing: prices, volume, network growth and on-chain activity all rise, with a 16% market-adjusted return over two weeks. Listings that reduce segmentation or improve information earn more. Background for why issuers pay for listings and liquidity, and for the incentives market makers are hired into.

Version record

US Securities and Exchange Commission, SEC Charges Three So-Called Market Makers and Nine Individuals in Crackdown on Manipulation of Crypto Assets Offered and Sold as Securities (Press Release 2024-166)

Difficulty: Advanced **Type:** Report / research note

Status/version: SEC press release 2024-166, 9 October 2024.

SEC fraud charges (with a parallel criminal case) against three so-called market makers, ZM Quant, Gotbit and CLS Global, and nine individuals. Promoters allegedly hired them for "market-manipulation-as-a-service": self-trading and other trading with no economic purpose, run by bots generating billions of dollars of artificial volume a day, including in a token created by the FBI. The primary record of what fake liquidity provision looks like.

Version record

US Securities and Exchange Commission, Tai Mo Shan to Pay $123 Million for Negligently Misleading Investors About Stability of Terra USD (Press Release 2024-212)

Difficulty: Advanced **Type:** Report / research note

Status/version: SEC press release 2024-212, 20 December 2024, with the settled administrative order.

SEC settlement (about USD 123 million) with Tai Mo Shan, a Jump Crypto subsidiary. When UST lost its peg in May 2021, Terraform vested the firm's option to buy LUNA at a discount in exchange for buying UST; the firm bought more than USD 20 million of UST, which the SEC found misled the market into thinking the algorithm had restored the peg. A case study of how option-based compensation for a market maker can turn support trading into deception.

Version record

18.3.4 DEX liquidity provision: JIT liquidity, fee tiers and LP strategies

Alfred Lehar, Christine Parlour and Marius Zoican, "Fragmentation and optimal liquidity supply on decentralized exchanges"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2307.13772, v7 (21 May 2024); DOI 10.48550/arXiv.2307.13772.

How liquidity providers choose between high- and low-fee Uniswap pools when gas is a fixed cost. High-fee pools hold 58% of liquidity but execute 21% of volume; large LPs dominate low-fee pools and reposition often against informed flow, while small LPs sit in high-fee pools, accepting fewer fills to limit adverse selection and management costs. Fragmentation draws in more LPs than a single fee tier.

Version record

Agostino Capponi, Ruizhe Jia and Brian Zhu, "The Paradox Of Just-in-Time Liquidity in Decentralized Exchanges: More Providers Can Sometimes Mean Less Liquidity"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2311.18164, v2 (16 February 2024); DOI 10.48550/arXiv.2311.18164.

Game-theoretic model of just-in-time (JIT) liquidity: a JIT LP sees pending swaps in the mempool, adds liquidity just before an order and removes it just after, so it serves only uninformed flow and avoids the adverse selection passive LPs bear. When volume is not elastic to depth, JIT crowds out passive LPs and can reduce liquidity; a two-tier fee split or Cournot competition among JIT LPs mitigates this.

Version record

Xin Wan and Austin Adams, "Just-in-time Liquidity on the Uniswap Protocol"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper (2023); DOI 10.2139/ssrn.4382303.

Uniswap Labs study of JIT liquidity on the Uniswap protocol: JIT is rare in practice, and when it happens it improves the execution of the trader it serves. Derives and validates a model of the price improvement, bounded above by twice the pool's fee rate. Working paper by protocol researchers; read it against the more critical JIT papers.

Version record

Xihan Xiong, Zhipeng Wang, William Knottenbelt and Michael Huth, "Demystifying Just-in-Time (JIT) Liquidity Attacks on Uniswap V3"

Difficulty: Research **Type:** Research paper

Status/version: IEEE BRAINS 2023 conference, 1–8 (2023); DOI 10.1109/BRAINS59668.2023.10316935.

Measures 36,671 JIT liquidity events on Uniswap v3 over 20 months, earning 7,498 ETH in total. A few bots dominate (one took 92% of profit), JIT positions average 269 times the swap size, return on investment is about 0.007%, and existing LPs' shares are diluted by 85% on average.

Version record

Bruno Llacer Trotti, Weizhao Tang, Rachid El-Azouzi, Giulia Fanti and Daniel Sadoc Menasché, "Strategic Analysis of Just-In-Time Liquidity Provision in Concentrated Liquidity Market Makers"

Difficulty: Research **Type:** Research paper

Status/version: 7th Conference on Advances in Financial Technologies (AFT 2025), LIPIcs 354, 8:1–8:22 (2025); DOI 10.4230/LIPIcs.AFT.2025.8.

First transaction-level model of JIT provision in concentrated-liquidity AMMs: characterises price impact and fee allocation and proves an optimal JIT strategy exists. Fitted to real pools, observed JIT LPs ignore price impact and could earn up to 69% more; strategic JIT reduces slippage for traders but erodes passive LP profits by up to 44% per trade on average.

Version record

Basile Caparros, Amit Chaudhary and Olga Klein, "Blockchain scaling and liquidity concentration on decentralized exchanges"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2306.17742, v4 (18 March 2024); DOI 10.48550/arXiv.2306.17742.

Uses the entry of Arbitrum and Polygon, where gas is cheaper, as an instrument for how often LPs reposition. More frequent and more precise repositioning concentrates liquidity around the market price, which lowers slippage for small trades: gas costs are a direct input to an LP's quoting frequency.

Version record

Brian Zhu, Dingyue Liu, Xin Wan, Gordon Liao, Ciamac C. Moallemi and Brad Bachu, "What Drives Liquidity on Decentralized Exchanges? Evidence from the Uniswap Protocol"

Difficulty: Research **Type:** Research paper

Status/version: Financial Cryptography and Data Security: FC 2025 International Workshops, Lecture Notes in Computer Science, 78–93 (2026); DOI 10.1007/978-3-032-00492-5_6.

Predictors of depth on Uniswap at platform, chain, pair and pool level, with a "v2 counterfactual spread" that splits a factor's effect into total value locked and concentration. Gas prices, returns and volume share act through concentration; internalisation by private market makers on aggregators changes TVL but not overall depth; volatility, fee revenue and markouts act through both.

Version record

Stefan Loesch, Nate Hindman, Mark B Richardson and Nicholas Welch, "Impermanent Loss in Uniswap v3"

Difficulty: Advanced **Type:** Research preprint

Status/version: arXiv preprint 2111.09192 (17 November 2021); DOI 10.48550/arXiv.2111.09192.

Treats impermanent loss as the gamma of the LP's self-financing strategy and measures it for 17 Uniswap v3 pools (43% of TVL): fees of USD 199.3 million against impermanent loss of USD 260.1 million, so LPs in aggregate would have done USD 60.8 million better by holding. Concentrated ranges are leveraged liquidity: more fees and more loss.

Version record

Robin Fritsch, "Concentrated Liquidity in Automated Market Makers"

Difficulty: Advanced **Type:** Research paper

Status/version: ACM CCS Workshop on Decentralized Finance and Security (DeFi 2021), 15–20 (2021); DOI 10.1145/3464967.3488590.

Compares average LP fee returns before and after concentrated liquidity and backtests basic concentrated-liquidity strategies on historical trades, showing which range rules work best for which pairs and conditions. A simple first benchmark for active LP management.

Version record

18.4 Crypto trading strategies

Directional and relative-value strategies on crypto assets, excluding market making: cross-sectional factors and anomalies, trend following, technical rules and calendar effects, statistical arbitrage and pairs trading, arbitrage and DeFi carry, attention, sentiment and event-driven signals, and machine-learning prediction and portfolio allocation. Crypto factor models and on-chain signals are in Section 18.1; perpetual carry in Section 18.1; market making in Section 18.3.

18.4.1 Cross-sectional factors and anomalies

Adam Zaremba, Mehmet Huseyin Bilgin, Huaigang Long, Aleksander Mercik and Jan J. Szczygielski, "Up or down? Short-term reversal, momentum, and liquidity effects in cryptocurrency markets"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Financial Analysis, 78, 101908 (2021); DOI 10.1016/j.irfa.2021.101908.

Shows on more than 3,600 coins that yesterday's return predicts today's in the cross-section: last-day losers beat last-day winners, and the effect survives a battery of controls. The reversal comes from illiquidity, so it fades with liquidity, and the largest, most tradeable coins show daily momentum instead. Reconciles earlier conflicting results on return persistence.

Version record

Victoria Dobrynskaya, "Cryptocurrency Momentum and Reversal"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Alternative Investments, 26(1), 65–76 (2023); DOI 10.3905/jai.2023.1.189.

Diversified cross-sectional momentum and reversal strategies on the 2,000 largest coins (2014–2020), with sorting and holding periods from a week to two years. Momentum holds up to two to four weeks and reverses after about a month, much faster than in equities; the reversal, driven by past losers, is large and not explained by standard crypto factors.

Version record

Klaus Grobys and Niranjan Sapkota, "Cryptocurrencies and momentum"

Difficulty: Advanced **Type:** Research paper

Status/version: Economics Letters, 180, 6–10 (2019); DOI 10.1016/j.econlet.2019.03.028.

Counter-evidence: on 143 coins over 2014–2018, the standard monthly cross-sectional momentum strategy earns no significant payoff. Read with Dobrynskaya, whose shorter horizons and wider sample find momentum, to see how much horizon and universe choices drive the answer.

Version record

Christian Fieberg, Gerrit Liedtke, Daniel Metko and Adam Zaremba, "Cryptocurrency factor momentum"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 23(12), 1853–1869 (2023); DOI 10.1080/14697688.2023.2269999.

Replicates 34 crypto anomalies on more than 3,900 coins (2014–2022) and finds factor momentum: past winning factors keep beating losers, at a size similar to equities. The autocorrelation comes mainly from size and volatility anomalies, and unlike in stocks it originates in price momentum that passes to the factor level.

Version record

Dehua Shen, Andrew Urquhart and Pengfei Wang, "A three-factor pricing model for cryptocurrencies"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 34, 101248 (2020); DOI 10.1016/j.frl.2019.07.021.

Market, size and reversal factors priced on more than 1,700 coins (2013–2019). Small coins earn more and reversal returns grow from large to small coins; the three-factor model beats a crypto CAPM and is robust to factor construction. A simple alternative baseline to Liu, Tsyvinski and Wu.

Version record

Panagiotis Tzouvanas, Renatas Kizys and Bayasgalan Tsend-Ayush, "Momentum trading in cryptocurrencies: Short-term returns and diversification benefits"

Difficulty: Advanced **Type:** Research paper

Status/version: Economics Letters, 191, 108728 (2020); DOI 10.1016/j.econlet.2019.108728.

J/K momentum portfolios on twelve coins over three years: momentum is significant at short horizons and disappears at longer ones. Dynamic correlations show crypto momentum portfolios behave unlike traditional assets and can diversify, hedge or act as a safe haven for them. Small universe; read as exploratory.

Version record

Nicola Borri, Yukun Liu, Aleh Tsyvinski and Xi Wu, "Cryptocurrency as an Investable Asset Class: Coming of Age"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics (2026); DOI 10.1146/annurev-financial-111524-010949.

Survey that organises crypto's empirical regularities into seven stylised facts and reads them through empirical asset pricing. Risk-adjusted performance has been broadly comparable with traditional assets and a few factors summarise the cross-section, but jumps are frequent and large and blockchain information moves prices. Also covers data-quality problems and regulatory change. The best single starting point for crypto strategy research.

Version record

Klaus Grobys and Juha Junttila, "Speculation and lottery-like demand in cryptocurrency markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Financial Markets, Institutions and Money, 71, 101289 (2021); DOI 10.1016/j.intfin.2021.101289.

First test of lottery-like demand in crypto: a weekly MAX measure from the previous week's daily returns, with tests robust to unknown dependence. Coins in the lowest MAX quintile beat the highest by more than 1.5% a week in raw and risk-adjusted terms, robust to bitcoin risk and microstructure controls. The crypto version of the equity MAX anomaly.

Version record

Klaus Grobys, James W. Kolari, Davide Sandretto, Syed Jawad H. Shahzad and Janne Äijö, "Cryptocurrency momentum has (not) its moments"

Difficulty: Research **Type:** Research paper

Status/version: Financial Markets and Portfolio Management, 39(4), 443–476 (2025); DOI 10.1007/s11408-025-00474-9.

Studies the tails of crypto momentum on large-cap coins: momentum suffers severe crashes, and a single coin can make portfolio returns insignificant. Volatility-managed momentum, as in equities, mitigates the crashes; momentum appears to be a large-cap phenomenon.

Version record

Arben Kita and Yue Zhang, "Volatility $\neq$ Risk: When Timing Alpha in Crypto Markets Reflects Mispricing"

Difficulty: Research **Type:** Research paper

Status/version: Financial Review, advance online publication (2026); DOI 10.1111/fire.70080.

Volatility timing on S&P crypto indices (2017–2023) earns significant alpha, but only under loose monetary policy and high uncertainty. Realised volatility reflects noise-driven speculative flows rather than risk, most strongly in small coins, so volatility works as a signal of time-varying inefficiency.

Version record

Manisha Yadav, "Intraday lottery demands in cryptocurrency market"

Difficulty: Research **Type:** Research paper

Status/version: Studies in Economics and Finance, 42(4), 799–835 (2025); DOI 10.1108/sef-07-2024-0461.

Hourly MAX measure from 5-minute returns on the 100 most liquid coins, with idiosyncratic volatility and skewness: a one-standard-deviation rise in MAX lowers next-hour returns by 0.043%, and MAX dominates other lottery proxies. An intraday extension of Grobys and Junttila.

Version record

18.4.2 Trend following, technical rules and calendar effects

Robert Hudson and Andrew Urquhart, "Technical trading and cryptocurrencies"

Difficulty: Research **Type:** Research paper

Status/version: Annals of Operations Research, 297(1-2), 191–220 (2021); DOI 10.1007/s10479-019-03357-1.

Tests almost 15,000 technical rules from five classes on two bitcoin markets and three other coins, with multiple-testing corrections against data snooping. Rules show significant predictability and break-even costs well above crypto trading costs, and cut drawdowns relative to buy-and-hold; but bitcoin shows no out-of-sample predictability, while the other coins still do.

Version record

Andrew Detzel, Hong Liu, Jack Strauss, Guofu Zhou and Yingzi Zhu, "Learning and predictability via technical analysis: Evidence from bitcoin and stocks with hard-to-value fundamentals"

Difficulty: Research **Type:** Research paper

Status/version: Financial Management, 50(1), 107–137 (2021); DOI 10.1111/fima.12310.

Equilibrium model in which technical analysis arises from rational learning about hard-to-value fundamentals. Ratios of price to moving averages forecast daily bitcoin returns in and out of sample and give significant alpha over buy-and-hold; similar results hold for small, young and little-covered stocks and for NASDAQ stocks in the dot-com era. The theory reading for crypto trend following.

Version record

Dirk F. Gerritsen, Elie Bouri, Ehsan Ramezanifar and David Roubaud, "The profitability of technical trading rules in the Bitcoin market"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 34, 101263 (2020); DOI 10.1016/j.frl.2019.08.011.

Seven trend-following indicators on daily bitcoin prices (July 2010–January 2019): trading-range breakout rules carry significant forecasting power and beat buy-and-hold on a bootstrapped Sharpe ratio, mainly in strongly trending markets.

Version record

Klaus Grobys, Shaker Ahmed and Niranjan Sapkota, "Technical trading rules in the cryptocurrency market"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 32, 101396 (2020); DOI 10.1016/j.frl.2019.101396.

Simple moving-average rules on the eleven most-traded coins (2016–2018): a 20-day variable moving average works, earning 8.76% a year above the market return when bitcoin is excluded. Short sample covering one boom and bust; a replication exercise more than a strategy.

Version record

Krzysztof Kosc, Paweł Sakowski and Robert Ślepaczuk, "Momentum and contrarian effects on the cryptocurrency market"

Difficulty: Advanced **Type:** Research paper

Status/version: Physica A: Statistical Mechanics and its Applications, 523, 691–701 (2019); DOI 10.1016/j.physa.2019.02.057.

Momentum and contrarian portfolios on the 100 largest coins by market value with volume filters, varying rebalancing period, ranking window, portfolio size and transaction costs. The short-term contrarian strategy clearly dominates momentum and all benchmarks, with very high information ratios; check how much survives realistic costs and liquidity.

Version record

Guglielmo Maria Caporale and Alex Plastun, "The day of the week effect in the cryptocurrency market"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 31 (2019); DOI 10.1016/j.frl.2018.11.012.

Day-of-the-week tests (t-tests, ANOVA, Kruskal–Wallis, dummy regressions) and trading simulations for bitcoin, Litecoin, Ripple and Dash. Only bitcoin shows an anomaly, higher Monday returns; the simulated profits are mostly not significantly different from random trading, so the authors stop short of rejecting efficiency.

Version record

Lars Kaiser, "Seasonality in cryptocurrencies"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 31 (2019); DOI 10.1016/j.frl.2018.11.007.

Tests equity calendar effects on ten coins: no robust seasonality in returns, but trading volume, volatility and spreads are lower in January, at weekends and in summer. Shows how a single sell-off (January 2018) can flip monthly seasonality results.

Version record

Dehua Shen, Andrew Urquhart and Pengfei Wang, "Bitcoin intraday time series momentum"

Difficulty: Research **Type:** Research paper

Status/version: Financial Review, 57(2), 319–344 (2022); DOI 10.1111/fire.12290.

Intraday time-series momentum in bitcoin, using volume to define the "session" in a 24-hour market: the first half-hour predicts the last half-hour return, most strongly in the highest-volume or most volatile sessions. Momentum timing adds economic value, especially in downturns, and the effect comes from liquidity provision rather than late-informed trading.

Version record

Zhuzhu Wen, Elie Bouri, Yahua Xu and Yang Zhao, "Intraday return predictability in the cryptocurrency markets: Momentum, reversal, or both"

Difficulty: Research **Type:** Research paper

Status/version: The North American Journal of Economics and Finance, 62, 101733 (2022); DOI 10.1016/j.najef.2022.101733.

High-frequency bitcoin data (2013–2020) show both intraday momentum and intraday reversal, with patterns that change around large jumps, FOMC announcements, liquidity levels and COVID-19; ether, litecoin and XRP show similar predictability. Timing strategies beat always-long and buy-and-hold. Momentum fits late-informed investors; the reversal, unique to crypto, fits overreaction.

Version record

18.4.3 Statistical arbitrage and pairs trading

Tim Leung and Hung Nguyen, "Constructing cointegrated cryptocurrency portfolios for statistical arbitrage"

Difficulty: Advanced **Type:** Research paper

Status/version: Studies in Economics and Finance, 36(4), 581–599 (2019); DOI 10.1108/sef-08-2018-0264.

Builds mean-reverting portfolios of bitcoin, ether, bitcoin cash and litecoin with Johansen and Engle–Granger tests, then compares trading rules with and without stop-losses. The strategies are profitable under several configurations; a clear worked example of cointegration trading in crypto.

Version record

Masood Tadi and Irina Kortchemski, "Evaluation of dynamic cointegration-based pairs trading strategy in the cryptocurrency market"

Difficulty: Research **Type:** Research paper

Status/version: Studies in Economics and Finance, 38(5), 1054–1075 (2021); DOI 10.1108/sef-12-2020-0497.

Dynamic cointegration pairs trading on BitMEX with Engle–Granger, KSS and Johansen tests, an Ornstein–Uhlenbeck half-life for pair selection and an optimised look-back window. Accounting for microstructure limits, it beats buy-and-hold with low drawdown, and using many coins to build the spread raises risk-adjusted returns.

Version record

Miroslav Fil and Ladislav Kristoufek, "Pairs Trading in Cryptocurrency Markets"

Difficulty: Research **Type:** Research paper

Status/version: IEEE Access, 8, 172644–172651 (2020); DOI 10.1109/access.2020.3024619.

Distance and cointegration pairs trading on 26 liquid Binance coins at 5-minute, hourly and daily frequencies. Daily strategies underperform (the distance method returns -0.07% a month), while 5-minute trading earns 11.61% a month; results are very sensitive to parameters, transaction costs and execution windows. Evidence of intraday mean reversion missing from daily data.

Version record

Po-Chang Ko, Ping-Chen Lin, Hoang-Thu Do, Yuan-Heng Kuo, Linh My Mai and You-Fu Huang, "Pairs trading in cryptocurrency markets: A comparative study of statistical methods"

Difficulty: Advanced **Type:** Research paper

Status/version: Investment Analysts Journal, 53(2), 102–119 (2024); DOI 10.1080/10293523.2023.2268386.

Compares six pair-selection methods (cointegration, correlation, distance, fluctuation behaviour, Hurst exponent, stochastic differential residual) on 30 Binance coins at 1-, 5- and 60-minute frequencies over three months in 2022. The distance method does best at every frequency; the three-month window makes this a method comparison rather than evidence of a durable edge.

Version record

18.4.4 Arbitrage and DeFi carry

Matthias Muck, Thomas Schmidl and Julian Wolf, "Wish or reality? On the exploitability of triangular arbitrage in cryptocurrency markets"

Difficulty: Research **Type:** Research paper

Status/version: Finance Research Letters, 73, 106508 (2025); DOI 10.1016/j.frl.2024.106508.

Triangular arbitrage among bitcoin, litecoin and the dollar on Binance with high-frequency data: 4,879 apparent opportunities, all eliminated by transaction costs and order-book depth. Counting opportunities is not a measure of inefficiency.

Version record

Pietro Saggese, Alessandro Belmonte, Nicola Dimitri, Angelo Facchini and Rainer Böhme, "Arbitrageurs in the Bitcoin ecosystem: Evidence from user-level trading patterns in the Mt. Gox exchange platform"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Economic Behavior & Organization, 213, 251–270 (2023); DOI 10.1016/j.jebo.2023.07.025.

Uses the leaked Mt.Gox trade history with user identifiers to find 440 triangular arbitrageurs. Unsophisticated users lose money after costs, while a few expert users earn a significant premium by reacting within seconds to exogenous moves in official exchange rates; arbitrage is concentrated in a small number of fast, skilled traders.

Version record

Nicola Borri and Kirill Shakhnov, "Cryptomarket discounts"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 139, 102963 (2023); DOI 10.1016/j.jimonfin.2023.102963.

Documents persistent bitcoin price discounts across exchange–currency pairs, with a half-life of about one day and a fat-tailed, right-skewed distribution (standard deviation 3.9%). Discounts vary more where capital controls are tighter, through segmentation and local supply and demand shocks linked to mining and attention.

Version record

Amit Chaudhary, Roman Kozhan and Ganesh Viswanath-Natraj, "Interest Rate Parity in Decentralized Finance"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2023); DOI 10.2139/ssrn.4437706.

Shows DeFi lending protocols are used for leveraged long and short positions, and that the borrowing-rate differential is significantly but weakly related to the ETH/USDT perpetual premium. The markets are segmented: wide no-arbitrage bounds from trading costs, gas and price impact make arbitrage between them ineffective. Working paper.

Version record

Patrick Augustin, Roy Chen-Zhang and Donghwa Shin, "Reaching for Yield in Decentralized Financial Markets"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2022); DOI 10.2139/ssrn.4063228.

Uses blockchain data on yield farms that compete with salient headline rates. Investors chase high yields, and the farms with the highest headline rates have the most negative risk-adjusted returns; randomised changes to how risk is displayed reduce yield chasing and improve performance, consistent with salience-driven reaching for yield. Working paper.

Version record

Simon Cousaert, Jiahua Xu and Toshiko Matsui, "SoK: Yield Aggregators in DeFi"

Difficulty: Advanced **Type:** Research paper

Status/version: IEEE International Conference on Blockchain and Cryptocurrency (ICBC 2022), 1–14 (2022); DOI 10.1109/ICBC54727.2022.9805523.

Systematisation of yield farming: where the yield comes from, three example strategies with simulated performance, and a comparison of four aggregators (Idle, Pickle, Harvest, Yearn) with on-chain vault data, plus their benefits and risks. A map of DeFi carry strategies before measuring their returns.

Version record

Kyoung Jin Choi, Alfred Lehar and Ryan Stauffer, "Bitcoin Microstructure and the Kimchi Premium"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2018; author version April 2019); DOI 10.2139/ssrn.3189051.

Bitcoin traded on average 4.73% higher in Korea than in the US between January 2016 and February 2018, peaking at 54.48% in January 2018. Capital controls create and amplify frictions: the premium rises with transaction costs, blockchain confirmation time and volatility, is higher in countries with less financial freedom, and falls with volume. The classic case of an arbitrage that capital controls keep open. Working paper.

Version record

18.4.5 Attention, sentiment and event-driven signals

Ladislav Kristoufek, "BitCoin meets Google Trends and Wikipedia: Quantifying the relationship between phenomena of the Internet era"

Difficulty: Advanced **Type:** Research paper

Status/version: Scientific Reports, 3(1), 3415 (2013); DOI 10.1038/srep03415.

Early evidence that bitcoin prices and Google Trends and Wikipedia search interest are linked, with an asymmetry: more interest pushes prices further up when they are above trend and further down when below. The origin of attention-based crypto signals.

Version record

Andrew Urquhart, "What causes the attention of Bitcoin?"

Difficulty: Advanced **Type:** Research paper

Status/version: Economics Letters, 166, 40–44 (2018); DOI 10.1016/j.econlet.2018.02.017.

Asks what drives attention to bitcoin and finds that realised volatility and volume predict next-day attention. Read before treating attention as a cause: it responds to market activity as well as moving it.

Version record

Dehua Shen, Andrew Urquhart and Pengfei Wang, "Does twitter predict Bitcoin?"

Difficulty: Advanced **Type:** Research paper

Status/version: Economics Letters, 174, 118–122 (2019); DOI 10.1016/j.econlet.2018.11.007.

Uses the number of tweets about bitcoin as an attention measure from better-informed investors. Tweet counts drive next-day trading volume and realised volatility in linear and nonlinear Granger tests, which makes them more useful for volatility and liquidity forecasts than for direction.

Version record

Olivier Kraaijeveld and Johannes De Smedt, "The predictive power of public Twitter sentiment for forecasting cryptocurrency prices"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of International Financial Markets, Institutions and Money, 65, 101188 (2020); DOI 10.1016/j.intfin.2020.101188.

Lexicon-based Twitter sentiment for the nine largest coins: sentiment Granger-causes returns for bitcoin, bitcoin cash and litecoin, and a bullishness ratio for EOS and TRON. Estimates that at least 1–14% of the tweets came from bots, a warning for any social-media signal.

Version record

Dominique Guégan and Thomas Renault, "Does investor sentiment on social media provide robust information for Bitcoin returns predictability?"

Difficulty: Research **Type:** Research paper

Status/version: Finance Research Letters, 38, 101494 (2021); DOI 10.1016/j.frl.2020.101494.

About a million StockTwits messages and intraday bitcoin returns: sentiment predicts returns at horizons up to 15 minutes, mostly around the 2017 bubble, and not at lower frequencies. The effect is too small to trade profitably, a useful counterweight to sentiment backtests.

Version record

Lennart Ante, "How Elon Musk's Twitter activity moves cryptocurrency markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Technological Forecasting and Social Change, 186, 122112 (2023); DOI 10.1016/j.techfore.2022.122112.

Event study of Elon Musk's crypto-related Twitter activity and bitcoin and Dogecoin prices and volume at minute-to-hour horizons. In the 2021 working-paper version (six events), the two non-reactive events (adding #bitcoin to his profile and tweeting "One word: Doge") produced cumulative abnormal returns of about 6% in 30 minutes for bitcoin and 8% in five minutes for Dogecoin, with large volume spikes.

Version record

Christos A. Makridis, Michael Fröwis, Kiran Sridhar and Rainer Böhme, "The rise of decentralized cryptocurrency exchanges: Evaluating the role of airdrops and governance tokens"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Corporate Finance, 79, 102358 (2023); DOI 10.1016/j.jcorpfin.2023.102358.

Documents the growth of decentralised exchanges and how their volume and price dynamics differ from centralised ones, then studies airdrops and governance tokens as ways to grow users and value. Preliminary, non-causal evidence that both work, especially for DEXs; two event studies suggest growth reflects security benefits, not only speculation.

Version record

Darcy W.E. Allen, Chris Berg and Aaron M. Lane, "Why airdrop cryptocurrency tokens?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Business Research, 163, 113945 (2023); DOI 10.1016/j.jbusres.2023.113945.

Case studies of high-profile airdrops (2014–2022) explaining why projects give tokens away: marketing to attract and keep users, and decentralising ownership and control for community, regulatory and security reasons. Background for airdrop-event strategies.

Version record

Johnnatan Messias, Aviv Yaish and Benjamin Livshits, "Airdrops: Giving Money Away Is Harder Than It Seems"

Difficulty: Advanced **Type:** Research preprint

Status/version: arXiv preprint 2312.02752, v4 (21 July 2025); DOI 10.48550/arXiv.2312.02752.

On-chain analysis of six large airdrops: a substantial share of tokens is sold off quickly by airdrop farmers, which explains post-airdrop selling pressure. Sets out the airdrop design space and guidelines. Preprint.

Version record

Lennart Ante, Ingo Fiedler and Elias Strehle, "The influence of stablecoin issuances on cryptocurrency markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 41, 101867 (2021); DOI 10.1016/j.frl.2020.101867.

Event study of 565 issuances of seven stablecoins (April 2019–March 2020): crypto markets fall in the week before an issuance and earn positive abnormal returns in the 24 hours around it, though not for every coin, and issuance size does not matter. Stablecoin minting as an observable on-chain event.

Version record

Andrew Morin, Tyler Moore and Eric Olson, "Breaking the stablecoin buck: The attenuating impact of security breach shocks at cryptocurrency exchanges"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Research, advance online publication (2026); DOI 10.1111/jfir.70069.

Event study of security breaches at crypto exchanges: breaches briefly break Tether's peg, which quickly returns to par, but do not move bitcoin's price, and the effects shrink over time, consistent with a maturing market. Evidence that hack-driven trades have weakened.

Version record

Jiahua Xu and Benjamin Livshits, "The Anatomy of a Cryptocurrency Pump-and-Dump Scheme"

Difficulty: Research **Type:** Research paper

Status/version: 28th USENIX Security Symposium (USENIX Security 19), 1609–1625 (2019).

Studies 412 pump-and-dumps organised on Telegram (June 2018–February 2019) and builds a model that predicts which coins on an exchange will be pumped. A strategy that buys predicted targets before the announcement earns up to 60% on small positions over two and a half months; the capacity is tiny and the strategy front-runs a manipulation.

Version record

Massimo La Morgia, Alessandro Mei, Francesco Sassi and Julinda Stefa, "The Doge of Wall Street: Analysis and Detection of Pump and Dump Cryptocurrency Manipulations"

Difficulty: Advanced **Type:** Research paper

Status/version: ACM Transactions on Internet Technology, 23(1), 1–28 (2023); DOI 10.1145/3561300.

Three years of monitoring Telegram and Discord pump groups (about 900 events) and a machine-learning detector that flags a pump within 25 seconds of its start (F1 score 94.5%), plus an analysis of "crowd pumps" after GameStop. Detection, not a trading endorsement.

Version record

18.4.6 Machine-learning prediction and portfolio allocation

Nusret Cakici, Syed Jawad Hussain Shahzad, Barbara Będowska-Sójka and Adam Zaremba, "Machine learning and the cross-section of cryptocurrency returns"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Financial Analysis, 94, 103244 (2024); DOI 10.1016/j.irfa.2024.103244.

Machine-learning models for cross-sectional crypto returns: all earn large gains, but complexity adds little; a few simple characteristics (price, past alpha, illiquidity, momentum) drive predictability. Abnormal returns come from the long leg and persist, and most strategies survive costs, but the alpha is concentrated in small, illiquid, volatile coins that are hard to trade.

Version record

Patrick Jaquart, David Dann and Christof Weinhardt, "Short-term bitcoin market prediction via machine learning"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance and Data Science, 7, 45–66 (2021); DOI 10.1016/j.jfds.2021.03.001.

Machine-learning prediction of bitcoin over 1- to 60-minute horizons with technical, blockchain, sentiment and cross-asset features; recurrent networks and gradient boosting do best and technical features matter most. A long–short strategy earns up to 39% a month before costs but loses money after them because holding periods are so short.

Version record

Erdinc Akyildirim, Ahmet Goncu and Ahmet Sensoy, "Prediction of cryptocurrency returns using machine learning"

Difficulty: Research **Type:** Research paper

Status/version: Annals of Operations Research, 297(1-2), 3–36 (2021); DOI 10.1007/s10479-020-03575-y.

Classifies the direction of the twelve most liquid coins at daily and minute frequencies with SVMs, logistic regression, neural networks and random forests using past prices and technical indicators. Accuracy averages about 55–65%, with SVMs most consistent; accuracy is not profit, so pair it with a cost-aware backtest.

Version record

Jing-Zhi Huang, William Huang and Jun Ni, "Predicting bitcoin returns using high-dimensional technical indicators"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance and Data Science, 5(3), 140–155 (2019); DOI 10.1016/j.jfds.2018.10.001.

Classification-tree model on 124 bitcoin technical indicators with strong out-of-sample power for narrow ranges of daily returns. An early high-dimensional technical-signal study; check the number of indicators against the multiple-testing corrections of the technical-trading pathway.

Version record

Fan Fang, Carmine Ventre, Michail Basios, Leslie Kanthan, David Martinez-Rego, Fan Wu and Lingbo Li, "Cryptocurrency trading: a comprehensive survey"

Difficulty: Intro **Type:** Research paper

Status/version: Financial Innovation, 8, article 13 (2022); DOI 10.1186/s40854-021-00321-6.

Survey of 146 papers on crypto trading: trading systems, bubbles and extreme conditions, volatility and return prediction, portfolio construction, technical trading and more, with a summary of datasets, open problems and directions. The map for the strategy pathways.

Version record

Simon Trimborn, Mingyang Li and Wolfgang Karl Härdle, "Investing with Cryptocurrencies—a Liquidity Constrained Investment Approach*"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Econometrics, 18(2), 280–306 (2020); DOI 10.1093/jjfinec/nbz016.

Liquidity-bounded risk-return optimisation (LIBRO) that adds crypto to stock, bond and commodity portfolios only in amounts that can actually be traded for the intended investment size. In and out of sample, liquidity-constrained crypto weights improve the risk–return trade-off.

Version record

Alexander Brauneis and Roland Mestel, "Cryptocurrency-portfolios in a mean-variance framework"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 28, 259–264 (2019); DOI 10.1016/j.frl.2018.05.008.

Mean–variance portfolios of the 500 largest coins (2015–2017), out of sample with costs: combining coins widens low-risk choices, but the equally weighted portfolio beats single coins and more than 75% of optimised portfolios on Sharpe ratio and certainty equivalent.

Version record

Weiyi Liu, "Portfolio diversification across cryptocurrencies"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 29, 200–205 (2019); DOI 10.1016/j.frl.2018.07.010.

Compares common allocation models across ten coins out of sample: diversification across coins helps, the maximum-utility model wins on utility, but none consistently beats 1/N on Sharpe ratio.

Version record

18.5 On-chain data and infrastructure

Yassine Elmandjra and David Puell, "On-Chain Data: A New Framework to Evaluate Bitcoin"; companion ARK white paper

Difficulty: Intermediate **Type:** Practitioner / commentary

Status/version: Edition/revision and publication date not established for the selected resource(s).

Network health, investor behaviour and valuation measures such as MVRV and Puell Multiple. The article is dated 11 January 2021; the separate white paper is dated 30 December 2021.

Applications: blockchain basics and data science. Version record

Harry Kalodner, Malte Möser, Kevin Lee, Steven Goldfeder, Martin Plattner, Alishah Chator and Arvind Narayanan, BlockSci: Design and applications of a blockchain analysis platform; Proceedings paper

Difficulty: Research **Type:** Research paper

Status/version: 29th USENIX Security Symposium, August 2020, pp. 2721-2738. Published proceedings paper; earlier 2017 preprint has a different author list.

Adds a concrete UTXO research architecture: chain parsing, in-memory analysis and address-clustering heuristics. The paper's supported transaction-graph model excludes Ethereum. Use it to inspect how raw transfers become analytical observations before treating address activity as economic flows.

Extended note. Version record

Igor Makarov and Antoinette Schoar, Blockchain Analysis of the Bitcoin Market

Difficulty: Research **Type:** Research manuscript

Status/version: NBER Working Paper 29396, October 2021; selected author manuscript dated 10 May 2026. Author-hosted 2026 revision; abstract reports 2009-2025 data. Journal status not established in this inspection.

Uses address-to-entity analysis to study transaction activity, miners and ownership concentration. Adds a measurement-focused study to the catalogue's exchange-price arbitrage paper by the same authors; the two papers answer different questions.

Extended note. Version record

Jaime Castillo León and Alfred Lehar, "What data have told us about decentralized finance"; DOI record

Difficulty: Not specified **Type:** Research paper

Status/version: Selected DOI-linked version; print 2026-01.

Empirical survey of DEXs, lending, governance, liquidations and composability. 2025 online publication; *Journal of Corporate Finance* 96, 102916 (2026).

Applications: empirical DeFi. Version record

Dune documentation; The Graph subgraphs; Glassnode Studio (documentation)

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Dune SQL/API queries and The Graph subgraphs offer distinct on-chain data access. Preserve query logic, metric definitions, entity heuristics and revision dates; continuing provider research is in Section 23.8.

Version record

Kaiko, "The Crypto Data Problem" (source: 2023)

Difficulty: Intro **Type:** Practitioner / commentary

Status/version: Kaiko article dated 9 November 2023.

Provider account of fragmented continuous-market data and reference-price construction; historical venue counts are not current coverage claims.

Applications: data architecture context. Version record

CryptoCompare; Nansen API documentation; Amberdata API documentation; Awesome-Crypto-Trading; Chainlink 2.0 whitepaper (2021)

Difficulty: Not specified **Type:** Software / documentation

Status/version: Selected citation year(s): 2021. Version distinctions remain in the annotation.

Data/API resources and an oracle-design white paper; coverage, label revisions and access terms are product-specific. CCXT is an exchange interface rather than a historical database; deployed Chainlink feed specifications remain separate from the 2.0 design. For protocol-risk methods, see Gauntlet and DeFiSafety.

Version record

Binance spot REST / WebSocket; Deribit APIs; selected BitMEX technical post; Coinbase engineering blog; OKX V5 API; OKX checksum change; Bitfinex books; Bitfinex raw books; Bitfinex checksum

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s). Selected companion documents inspected 12 October 2026; earlier source verification retained separately.

Exchange and engineering documentation for data ingestion and execution. The named backtesting engines have separate entries in Section 21.1; fees, funding and fill models differ by venue. The added OKX and Bitfinex readings distinguish venue-specific snapshots, updates and integrity checks. OKX's affected JSON book channels retain checksum as zero after 23 June 2026; use seqId/prevSeqId continuity checks. Bitfinex documents its own checksum procedure.

Extended note. Version record

Deribit FZE, *Exchange Rulebook* and Funding Specifications; Inverse Options; Linear USDC Options; Portfolio Margin

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Rulebook version 2.0 applicable 13 August 2026; page updated 15 September 2026. Funding page updated 27 July 2026. Selected companion documents inspected 12 October 2026; earlier source verification retained separately.

A dated venue case for perpetual futures, margin and default handling. Read rulebook sections 9–11 for maintenance margin, discretionary liquidation, insurance-fund use, socialised losses and settlement. Pair them with the funding formula: an eight-hour quoted rate is accrued continuously, with a premium damper and instrument-specific caps. Separate equity credit from withdrawable cash and reconstruct the applicable margin mode, index, collateral currency and funding path. An estimated liquidation price is not a guaranteed close-out price. Current terms describe the selected vintage; historical simulations require contemporaneous rules and market observations. Added option specifications separate coin-settled inverse payoffs from linear USDC payoffs and account-level portfolio-margin scenarios. The inverse specification states that expiries from 1 August 2026 use an intermediate inverse future before base-currency cash settlement.

Extended note. Version record

Sarah Meiklejohn et al., "A fistful of bitcoins: characterizing payments among men with no names"

Difficulty: Advanced **Type:** Research paper

Status/version: Proceedings of the 2013 conference on Internet measurement conference, 127–140 (2013); DOI 10.1145/2504730.2504747.

Address clustering heuristics that link Bitcoin addresses to entities: the basis of most on-chain entity analytics.

Version record

Dorit Ron and Adi Shamir, "Quantitative Analysis of the Full Bitcoin Transaction Graph"

Difficulty: Advanced **Type:** Research paper

Status/version: Lecture Notes in Computer Science, 6–24 (2013); DOI 10.1007/978-3-642-39884-1_2.

Early full-graph analysis of Bitcoin: dormant coins, large-entity behaviour and transaction structure.

Version record

Sean Foley, Jonathan R. Karlsen and Tālis J. Putniņš, "Sex, Drugs, and Bitcoin: How Much Illegal Activity Is Financed through Cryptocurrencies?"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 32(5), 1798–1853 (2019); DOI 10.1093/rfs/hhz015.

Network clustering of on-chain users estimates that about a quarter of Bitcoin users were involved in illegal activity in 2009–2017.

Version record

Friedhelm Victor and Andrea Marie Weintraud, "Detecting and Quantifying Wash Trading on Decentralized Cryptocurrency Exchanges"

Difficulty: Advanced **Type:** Research paper

Status/version: Proceedings of the Web Conference 2021, 23–32 (2021); DOI 10.1145/3442381.3449824.

On-chain detection of wash trading on IDEX and EtherDelta, using trade-graph cycles.

Version record

Andrea Barbon and Angelo Ranaldo, "On the quality of cryptocurrency markets: Centralized versus decentralized exchanges"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, mnsc.2024.07703 (2026); DOI 10.1287/mnsc.2024.07703.

Compares price efficiency and transaction costs on centralised and decentralised exchanges; DEX trading costs are higher, largely because of gas fees.

Version record

18.6 AMMs, lending and liquidity design

Read the CFMM and Uniswap v3 foundations first. These works separate passive liquidity exposure, informed-flow losses, fee design and programmable execution; mechanism-design claims remain conditional on the specified protocol and participants. Lending protocols, liquidations and the oracles they depend on follow in their own subsection.

18.6.1 AMM design and liquidity provision

Guillermo Angeris, Hsien-Tang Kao, Rei Chiang, Charlie Noyes and Tarun Chitra, *An analysis of Uniswap markets* (2019 preprint); Uniswap v2 Core specification

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2019. Version distinctions remain in the annotation. Selected companion documents inspected 12 October 2026; earlier source verification retained separately.

The Uniswap analysis concerns arbitrage, fees and reference-price deviations. The published article is dated 5 April 2021, DOI 10.21428/58320208.c9738e64; retain the 2019 preprint separately. See the separate CFMM paper for invariant pricing and theoretical reserve bounds; those bounds do not guarantee protection from exploits. Pair the 2019 analysis with the separate March 2020 Uniswap v2 Core specification by Hayden Adams, Noah Zinsmeister and Dan Robinson for reserve updates, fees, price-oracle accumulation and flash-swap repayment.

Extended note. Version record

Jason Milionis, Ciamac C. Moallemi, Tim Roughgarden and Anthony Lee Zhang, Automated Market Making and Loss-Versus-Rebalancing; author publication and version list

Difficulty: Research **Type:** Research paper

Status/version: Working paper; first posted 11 August 2022; v6 posted 20 August 2026

Decomposes liquidity-provider returns into market exposure and fees net of losses to arbitrageurs. Loss-versus-rebalancing compares an AMM with a dynamically rebalanced benchmark, so it answers a different question from impermanent loss relative to holding. Use the decomposition to separate inventory beta, adverse selection and fees when evaluating LP strategies. Continuous-time assumptions, discrete blocks, transaction fees and hedging costs matter; a pool-specific variance decomposition does not establish a portable net return.

Version record

Jason Milionis, Ciamac C. Moallemi, Tim Roughgarden, Automated Market Making and Arbitrage Profits in the Presence of Fees

Difficulty: Research **Type:** Research preprint

Status/version: Preprint first posted 24 May 2023. arXiv:2305.14604 v2, 23 July 2025.

Extends loss-versus-rebalancing analysis to include trading fees and discrete Poisson block times, deriving a closed-form expected arbitrage-profit rate. It explains when fees limit LP adverse selection; the arrival model is stylized and needs chain-specific timing and fee calibration.

Extended note. Version record

Hayden Adams et al., Uniswap v4 Core; core repository

Difficulty: Advanced **Type:** Protocol paper / software

Status/version: Technical white paper, August 2024 version

Extends concentrated liquidity with hooks, dynamic fees, a singleton contract, flash accounting and native ETH. Study the action callbacks and custom accounting to understand which parts of execution, fee allocation and oracle behaviour belong to the core and which belong to a hook. Permissionless hook creation is not a guarantee that an individual hook has safe economics or the same withdrawal and upgrade assumptions as the core.

Extended note. Version record

Austin Adams, Ciamac C. Moallemi, Sara Reynolds and Dan Robinson, am-AMM: An Auction-Managed Automated Market Maker; author proceedings record

Difficulty: Research **Type:** Research paper

Status/version: FC 2025, LNCS 15751, 93–108; preprint v4, 12 February 2025; first posted 5 March 2024

Auctions temporary pool-management rights so the manager sets swap fees and captures part of arbitrage value while paying rent to liquidity providers. Connects fee optimisation and informed-flow losses in a single economic mechanism. Read its equilibrium assumptions, manager incentives and withdrawal rules before comparing it with fixed-fee pools. The paper focuses on constant-product AMMs; extending the result to concentrated liquidity, bidder concentration and live implementation costs requires additional work.

Version record

Leonardo Baggiani, Martin Herdegen, Leandro Sánchez-Betancourt, Optimal Dynamic Fees in Automated Market Makers

Difficulty: Research **Type:** Research preprint

Status/version: Preprint first posted 3 June 2025. arXiv:2506.02869 v3, 30 July 2026.

Models fee choice as a control problem for a CFMM and identifies different fee regimes: raise fees to deter arbitrage when prices are misaligned, or lower fees to attract noise flow. The inventory- and external-price-sensitive rules are hypotheses for dynamic-fee design, not evidence that a specific hook earns more after implementation costs or changes in flow.

Extended note. Version record

Andreas Lehar and Christine A. Parlour, "Decentralized Exchange: The Uniswap Automated Market Maker"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 80(1), 321–374 (2025); DOI 10.1111/jofi.13405.

Characterises equilibrium liquidity in Uniswap pools and compares the AMM with a centralised limit order book using all 95.8 million Uniswap interactions. Long-lived arbitrage opportunities are absent, and the paper gives conditions under which the AMM dominates a limit order market.

Version record

Agostino Capponi and Ruizhe Jia, "Liquidity Provision on Blockchain-Based Decentralized Exchanges"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(10), 3040–3085 (2025); DOI 10.1093/rfs/hhaf046.

Liquidity providers face adverse selection from arbitrageurs and competition for block space; characterises when decentralised exchanges sustain liquidity. Published under this title in RFS (2025), developed from the earlier adoption-of-DEXs working paper.

Version record

Andreas Park, "The Conceptual Flaws of Decentralized Automated Market Making"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 69(11), 6731–6751 (2023); DOI 10.1287/mnsc.2021.02802.

Constant-function AMMs create front-running and stale-price exploitation; proposes improvements.

Version record

Agostino Capponi, Ruizhe Jia and Shihao Yu, "Price Discovery on Decentralized Exchanges"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag002 (2026); DOI 10.1093/rfs/hhag002.

High-priority-fee DEX trades are more informative and contribute more to price discovery; informed traders persistently bid higher fees.

Version record

Tian Chen, Jun Deng, Jing Nie, Bin Zou and Qi Fu, "Liquidity provision and its information content in decentralized markets"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Markets, 101073 (2026); DOI 10.1016/j.finmar.2026.101073.

Information content of liquidity provision on decentralised exchanges.

Extended note. Version record

Alif Aqsha, Philippe Bergault and Leandro Sánchez-Betancourt, "Equilibrium Reward for Liquidity Providers in Automated Market Makers"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, mafi.70030 (2026); DOI 10.1111/mafi.70030.

Equilibrium contract between an AMM venue and strategic liquidity providers as a leader–follower stochastic game.

Version record

Christoph Schlegel, Mateusz Kwaśnicki and Akaki Mamageishvili, "Axioms for Constant Function Market Makers"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, mafi.70041 (2026); DOI 10.1111/mafi.70041.

Axiomatic characterisation of constant-product market makers and the LMSR prediction-market maker.

Version record

Mnacho Echenim, Emmanuel Gobet and Anne-Claire Maurice, "Uniswap V3: Impermanent Loss Modeling and Swap Fees Asymptotic Analysis"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 17(1), 1–40 (2026); DOI 10.1137/23m1606149.

Rigorous treatment of Uniswap v3 mechanics, impermanent loss and asymptotics of swap fees without the constant-liquidity simplification.

Version record

Bastien Baude, Damien Challet and Ioane Muni Toke, "Optimal Execution on Uniswap v2/v3 under Transient Price Impact"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 1–34 (2026); DOI 10.1080/1350486x.2026.2721676.

Optimal liquidation on Uniswap v2/v3 with transient impact: closed form for v2, dynamic programming for multi-layer v3 liquidity.

Version record

Carol Alexander, Xi Chen, Jun Deng and Qi Fu, "Price Discovery and Efficiency in Uniswap Liquidity Pools"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 45(8), 1023–1048 (2025); DOI 10.1002/fut.22593.

Minute data show Uniswap v3 pools approaching centralised exchanges in price discovery, though informed traders move to CEXs in volatile periods.

Version record

Jiahua Xu, Krzysztof Paruch, Simon Cousaert and Yebo Feng, "SoK: Decentralized Exchanges (DEX) with Automated Market Maker (AMM) Protocols"

Difficulty: Advanced **Type:** Research paper

Status/version: ACM Computing Surveys, 55(11), 1–50 (2023); DOI 10.1145/3570639.

Systematisation of AMM-based exchanges: a general state-space framework, then the conservation, slippage and divergence-loss functions of the main protocols, their security and privacy issues, and a literature review linking DeFi to market microstructure. A reference text for comparing pool designs.

Version record

Vijay Mohan, "Automated market makers and decentralized exchanges: a DeFi primer"

Difficulty: Intro **Type:** Research paper

Status/version: Financial Innovation, 8(1), 20 (2022); DOI 10.1186/s40854-021-00314-5.

Primer that treats an AMM as a black box turning tokens into prices through an exchange function, and compares constant-product, constant-mean, constant-sum, hybrid and dynamic market makers and concentrated liquidity geometrically. A gentle entry before the CFMM papers.

Version record

Joel Hasbrouck, Thomas J. Rivera and Fahad Saleh, "The Need for Fees at a DEX: How Increases in Fees Can Increase DEX Trading Volume"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 72(10), 8309–8324 (2026); DOI 10.1287/mnsc.2023.00726.

Model of trading and liquidity provision at a DEX in which a higher fee can raise volume: fees attract LP inventory, deeper pools lower price impact, and the lower impact can more than offset the fee, so the marginal cost of trading falls. Explains why fee tiers are not a simple price for trading.

Version record

Joel Hasbrouck, Thomas J. Rivera and Fahad Saleh, "An Economic Model of a Decentralized Exchange with Concentrated Liquidity"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 72(5), 3666–3683 (2026); DOI 10.1287/mnsc.2024.04510.

Economic model of concentrated liquidity (Uniswap v3 and v4). Providing liquidity to a risky/risk-free pool resembles a covered call whose option is sold at intrinsic rather than market value, so LPs give up the time premium for fees; equilibrium liquidity falls as that premium rises. Gives an expression for equilibrium liquidity for empirical work.

Version record

Jun Aoyagi and Yuki Ito, "Coexisting Exchange Platforms: Limit Order Books and Automated Market Makers"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy Microeconomics, 3(3), 611–648 (2025); DOI 10.1086/732831.

Model of an AMM coexisting with a limit order book. Changes in AMM liquidity provoke disproportionate responses from informed and noise traders, spilling over positively to order-book liquidity, and AMM liquidity supply has a hump-shaped response to volatility. A bridge between DEX design and centralised microstructure.

Version record

Álvaro Cartea, Fayçal Drissi and Marcello Monga, "Predictable Losses of Liquidity Provision in Constant Function Markets and Concentrated Liquidity Markets"

Difficulty: Research **Type:** Research paper

Status/version: Applied Mathematical Finance, 30(2), 69–93 (2023); DOI 10.1080/1350486x.2023.2277957.

Defines predictable loss: the gap between an LP's pool holdings, without fees, and a self-financing portfolio that replicates them. It splits into a convexity cost from liquidity taking and an opportunity cost of locked assets. Uniswap v3 and Binance data show liquidity provision was loss-making at the time. A model-free complement to loss-versus-rebalancing.

Version record

Álvaro Cartea, Fayçal Drissi and Marcello Monga, "Decentralized Finance and Automated Market Making: Predictable Loss and Optimal Liquidity Provision"

Difficulty: Research **Type:** Research paper

Status/version: SIAM Journal on Financial Mathematics, 15(3), 931–959 (2024); DOI 10.1137/23m1602103.

Continuous-time wealth of strategic LPs in concentrated-liquidity pools: fee income, pool holdings and rebalancing costs. Derives a closed-form optimal range width driven by pool profitability net of gas, predictable loss and concentration risk, and how to skew it when the rate drifts. Uniswap v3 data show LPs on average lost money; the strategy beats historical LP performance out of sample.

Version record

Lioba Heimbach, Eric Schertenleib and Roger Wattenhofer, "Risks and Returns of Uniswap V3 Liquidity Providers"

Difficulty: Research **Type:** Research paper

Status/version: ACM Conference on Advances in Financial Technologies (AFT 2022), 89–101 (2022); DOI 10.1145/3558535.3559772.

Model and data on the choices Uniswap v3 LPs face. Performance varies widely: simple strategies in low-volatility pools earn modest returns, and large returns require more risk and active management, so concentrated liquidity favours sophisticated providers over retail.

Version record

18.6.2 Lending, liquidations and oracles

Interest-rate rules, liquidation mechanics and evidence, leverage and fragility, and the price oracles that trigger liquidations.

Maker Team, *The Dai Stablecoin System* (historical 2017 whitepaper); Compound Finance documentation (Comet specification)

Difficulty: Not specified **Type:** Software / documentation

Status/version: Maker whitepaper dated December 2017; Compound documentation is living and version-dependent.

The 2017 Maker whitepaper describes original single-collateral Dai, CDPs, stability fees and liquidation, with multi-collateral Dai then prospective. It is historical design evidence, not current Sky/Maker deployment evidence. Compound III/Comet specification discusses

utilisation-linked rates, collateral factors and liquidation; check selected deployed version and parameters. Aave and Gauntlet have main entries.

Version record

Kaihua Qin, Liyi Zhou, Pablo Gamito, Philipp Jovanovic and Arthur Gervais, An Empirical Study of DeFi Liquidations: Incentives, Risks, and Instabilities; Author preprint, arXiv v2; UCL accepted manuscript

Difficulty: Research **Type:** Empirical research paper

Status/version: ACM Internet Measurement Conference (IMC 2021), 336-350. Published proceedings paper; arXiv v2 revised 1 October 2021.

Compares historical Aave, Compound, MakerDAO and dYdX liquidation mechanisms, measuring liquidator incentives, borrower collateral discounts and protocol instability. Use it to separate the trigger rule from auction or fixed-spread execution and to construct an event ledger for eligibility, gas, bidding, received collateral and realized resale. The study reports that some designs favor liquidators at borrowers' expense; do not apply its historical parameter values to current deployments. Pin chain, block, contract revision, oracle update and market liquidity before estimating a present liquidation opportunity.

Extended note. Version record

Morpho Association / Morpho contributors, Morpho Blue: Variable-Rate Markets and Adaptive Curve IRM; Interest rate model documentation; Morpho Blue core contracts

Difficulty: Advanced **Type:** Software / protocol documentation

Status/version: Living documentation; no fixed publication date stated. Official docs and core-contract repository inspected 12 October 2026; version and deployment are chain-specific.

Treat each market as a separately specified lending instrument: collateral and loan assets, liquidation LTV, oracle and immutable interest-rate model. Follow utilization through AdaptiveCurveIRM to distinguish borrow cost, supplier yield and withdrawal liquidity; connect liquidation thresholds and incentives to collateral-market risk. This supplies a crypto-native rates and credit case beyond broad DeFi surveys. Read the mechanism first, then inspect the market parameters and oracle configuration at the study block.

Extended note. Version record

Chainlink Labs, Smart Value Recapture (SVR) Feeds; SVR feed addresses; Flashbots MEV-Share documentation

Difficulty: Advanced **Type:** Software / protocol documentation

Status/version: Living documentation; no fixed publication date stated. Official docs inspected 12 October 2026; feed variants, chains and address availability are deployment-specific.

Trace the dual path for oracle reports: a public standard aggregator and a private route that auctions the right to backrun a liquidation, followed by fallback if the private path fails or times out. The documentation also describes the outcome when no searcher bid wins. This makes oracle extractable value and liquidation ordering concrete beside transaction-auction material. Treat it as an auction design and integration specification; an interface or announcement alone does not establish a feed deployment or realized recapture.

Extended note. Version record

Charles Bertucci, Louis Bertucci, Mathis Gontier Delaunay, Olivier Guéant and Matthieu Lesbre, "Agents' Behavior and Interest Rate Model Optimization in DeFi Lending"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 36(2), 374–396 (2026); DOI 10.1111/mafi.70002.

DeFi lending rates are set by rule rather than by market interaction, so the interest-rate model matters. Under perfect information, optimal control with a state constraint yields a rate curve shaped like those of popular markets; PID-controller rate policies work with fewer assumptions. Ethereum data drive a simulation of agents' behaviour and the design trade-offs.

Version record

Lewis Gudgeon, Sam Werner, Daniel Perez and William J. Knottenbelt, "DeFi Protocols for Loanable Funds: Interest Rates, Liquidity and Market Efficiency"

Difficulty: Research **Type:** Research paper

Status/version: ACM Conference on Advances in Financial Technologies (AFT 2020), 92–112 (2020); DOI 10.1145/3419614.3423254.

Coins 'protocols for loanable funds' and reviews how Compound, Aave and dYdX set interest rates from utilisation. Examines how these rules behaved with liquidity since launch, tests uncovered interest parity within a protocol, and models rate dependence across protocols with a vector error-correction model.

Version record

Daniel Perez, Sam M. Werner, Jiahua Xu and Benjamin Livshits, "Liquidations: DeFi on a Knife-Edge"

Difficulty: Research **Type:** Research paper

Status/version: Financial Cryptography and Data Security, Lecture Notes in Computer Science, 457–476 (2021); DOI 10.1007/978-3-662-64331-0_24.

First detailed study of liquidations on Compound, to September 2020. A 3% price move could make over USD 10m liquidatable; liquidators became fast, with more than 70% of liquidatable positions liquidated immediately; and misplaced confidence in non-custodial stablecoins raised users' liquidation risk.

Version record

Kanis Saengchote, "Decentralized lending and its users: Insights from Compound"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of International Financial Markets, Institutions and Money, 87, 101807 (2023); DOI 10.1016/j.intfin.2023.101807.

Who uses Compound and why: on-chain data show short loans (31 days on average) and many borrowers funding leveraged yield-farming strategies. Concentration and interconnection create systemic risk that traditional risk management struggles to address.

Version record

Ariah Klages-Mundt and Andreea Minca, "(In)Stability for the Blockchain: Deleveraging Spirals and Stablecoin Attacks"

Difficulty: Research **Type:** Research paper

Status/version: Cryptoeconomic Systems (2021); DOI 10.21428/58320208.e46b7b81.

Model of crypto-collateralised stablecoins showing deleveraging spirals: liquidations cause illiquidity in crises and deepen collateral drawdowns. Predicted before Dai's 'Black Thursday' in March 2020, when mempool congestion let liquidation auctions clear near zero at a cost of about USD 8m. Also describes profitable attacks around liquidations.

Version record

Alfred Lehar and Christine A. Parlour, "Systemic fragility in decentralised markets"

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1062 (December 2022).

Collateral liquidations on Compound and Aave, distinguishing liquidators who repay from inventory and those who use flash loans and must sell the collateral at once. The sales have temporary and permanent price impact on nine DEXs, as large block trades do in equities, and change return distributions: evidence of systemic fragility in decentralised markets. BIS Working Paper 1062.

Version record

Lioba Heimbach and Wenqian Huang, "DeFi leverage"

Difficulty: Research **Type:** Research paper

Status/version: BIS Working Paper 1171 (March 2024).

Wallet-level leverage (assets over equity) on major lending platforms: typically 1.4 to 1.9, higher for the largest and most active users, and driven by loan-to-value limits, borrowing costs, crypto prices and sentiment. Higher leverage raises the share of debt near liquidation, and highly leveraged borrowers shift toward volatile collateral as liquidation nears. BIS Working Paper 1171.

Version record

Jonathan Chiu and Furkan Berk Danışman, "DeFi Lending: Returns, Leverage, and Liquidation Risk"

Difficulty: Advanced **Type:** Report / research note

Status/version: Bank of Canada Staff Analytical Paper 2026-13 (April 2026); DOI 10.34989/sap-2026-13.

Transaction-level study of Aave V3: protocol earnings are concentrated in a few tokens, many users build recursive leverage despite overcollateralisation, and liquidations come in concentrated waves with limited effects on wider markets. Concludes that governed DeFi lending is operationally viable but limited by capital efficiency, liquidation risk and fragility within crypto. Bank of Canada staff analytical paper.

Version record

Shayan Eskandari, Mehdi Salehi, Wanyun Catherine Gu and Jeremy Clark, "SoK: Oracles from the Ground Truth to Market Manipulation"

Difficulty: Advanced **Type:** Research paper

Status/version: ACM Conference on Advances in Financial Technologies (AFT 2021), 127–141 (2021); DOI 10.1145/3479722.3480994.

Systematisation of blockchain oracles: the design choices from data source to on-chain contract, their trust models, attacks on them and mitigations. The reference for naming which oracle a lending market or derivative depends on and how it could be manipulated.

Version record

Torgin Mackinga, Tejaswi Nadahalli and Roger Wattenhofer, "TWAP Oracle Attacks: Easier Done than Said?"

Difficulty: Research **Type:** Research paper

Status/version: 2022 IEEE International Conference on Blockchain and Cryptocurrency (ICBC), 1–8 (2022); DOI 10.1109/icbc54727.2022.9805499.

Challenges the assumption that manipulating an arithmetic time-weighted average price oracle is expensive and grows linearly with the window. A single-block attack is cheaper than the known multi-block attack, and a proposer who controls two consecutive blocks can manipulate the oracle orders of magnitude more cheaply.

Version record

Xun Deng, Sidi Mohamed Beillahi, Cyrus Minwalla, Han Du, Andreas Veneris and Fan Long, "Safeguarding DeFi Smart Contracts against Oracle Deviations"

Difficulty: Research **Type:** Research paper

Status/version: IEEE/ACM International Conference on Software Engineering (ICSE 2024), 1–12 (2024); DOI 10.1145/3597503.3639225.

OVer, a tool that symbolically analyses how DeFi contracts behave under skewed oracle inputs and uses an SMT solver to find safe parameters and generate guard statements. Most of the ten benchmark protocols had parameters too weak for large oracle deviations.

Version record

Giulio Caldarelli and Joshua Ellul, "The Blockchain Oracle Problem in Decentralized Finance—A Multivocal Approach"

Difficulty: Intro **Type:** Research paper

Status/version: Applied Sciences, 11(16), 7572 (2021); DOI 10.3390/app11167572.

Multivocal review (papers, white papers and practitioner posts) of the oracle problem in DeFi: how platforms choose and manage oracles, how poor choices led to hacks, and why standardisation and economic incentives are needed. A non-technical complement to Eskandari et al.

Version record

18.7 Staking, liquid claims and token economics

Distinguish native ETH, a validator deposit, a liquid-staking claim, a restaked position, and a protocol governance token. Their rewards, exit rights, loss waterfalls and price exposures are different even when all are quoted in ETH.

Lin William Cong, Zhiheng He and Ke Tang, The Tokenomics of Staking

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 33640, April 2025; working paper at the cut-off.

Models allocation between transactional tokens and staked tokens; staking ratio affects productivity, adoption, reward rate and token price. Its empirical tests separate cross-sectional staking rewards from time-series responses and help frame staking yield as an endogenous network outcome, not a bond coupon. Read after the crypto risk-factor study. Preserve working-paper status and estimate issuance and fee changes separately.

Extended note. Version record

Ethereum Foundation, Staking Withdrawals; EIP-7002: Execution-Layer-Triggerable Withdrawals

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living protocol explainer consulted 4 October 2026; EIP final core standard, Pectra activated May 2025.

Separates partial reward sweeps, validator exit, execution-layer withdrawal credentials and the withdrawal queue. Pectra's compounding validator type can accrue effective balance to 2048 ETH, changing cash-flow timing from the earlier 32 ETH design; EIP-7002 lets a withdrawal address initiate exits or partial withdrawals. Read after proof-of-stake mechanics. Check credential type, active balance, exit capacity, activation and custodial control at the measurement block.

Extended note. Version record

Ethereum Foundation, Proof-of-Stake Rewards and Penalties

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living protocol documentation consulted 4 October 2026.

Separates routine missed-attestation penalties, slashable consensus faults and a correlation penalty that grows when many validators are slashed together. Use correlated operator/client failure scenarios instead of treating a published staking rate as a risk-free cash yield. Read with exit mechanics. Documentation is descriptive and protocol parameters can change. A fund or custody wrapper adds smart-contract, operational, oracle and liquidity risks absent from the native validator penalty schedule.

Version record

Lido DAO, Lido Tokens Integration Guide; Accounting Oracle

Difficulty: Advanced **Type:** Software / documentation

Status/version: Live protocol documentation consulted 4 October 2026.

Distinguishes rebasing stETH from non-rebasing wstETH, and secondary-market sale from asynchronous protocol withdrawal. A withdrawal request locks tokens, mints a transferable NFT and becomes claimable after oracle finalisation; significant losses can reduce the rate and bunker mode can delay finalisation. Read after native withdrawals. Stress stETH/ETH market discount, oracle and contract risk, queue depth, validator exits and the exact claimable-rate rule at the queried contract version.

Extended note. Version record

Pendle contributors, Pendle V2: Yield Tokenization, AMM and Order Book; Pendle V2 order book; Standardized Yield contracts

Difficulty: Advanced **Type:** Software / protocol documentation

Status/version: Living documentation; no fixed publication date stated. Pendle V2 official docs inspected 12 October 2026; pool parameters and supported assets can change.

Follow the conversion chain from a yield-bearing asset to Standardized Yield (SY), Principal Token (PT) and Yield Token (YT), then read PT price as a maturity-specific yield quote. Pendle's PT/SY curve tightens toward expiry, supports YT trades through flash swaps, and charges maturity-sensitive fees; its order book adds implied-APY limit orders. The route connects fixed-versus-floating yield views to pool mechanics, redemption units and market liquidity rather than treating token yields as one rate.

Extended note. Version record

Allan Niemerg, Dan Robinson and Lev Livnev, YieldSpace: An Automated Liquidity Provider for Fixed Yield Tokens

Difficulty: Advanced **Type:** Research paper

Status/version: August 2020 working draft, revision 1 (as stated on the paper). Primary PDF inspected 12 October 2026; retain its working-draft status.

Read the constant power-sum invariant as a curve designed for fixed-yield tokens rather than spot assets. The paper incorporates time to maturity and reserve ratio so a pool quotes an interest rate, not a constant spot price, and derives the invariant from desired reserve properties. Its fyToken example exposes why generic constant-product curves can become capital-inefficient near maturity. Work through the derivation before comparing the design with contemporary term-yield AMMs such as Pendle V2.

Extended note. Version record

Eigen Labs, Slashing Goes Live on Mainnet; ELIP-002: Slashing

Difficulty: Advanced **Type:** Software / documentation

Status/version: 17 April 2025 deployment release and maintained protocol proposal; primary protocol sources.

Operator sets and actively validated services may impose additional, service-specific slash conditions on delegated stake; the protocol's withdrawal delay was set to 14 days at slashing launch to permit accounting. Map each position's operator, service allocation, reward and slashing exposure rather than adding advertised yields without contingent losses. Read after liquid-staking claims. A liquid-restaking token adds its own redemption, oracle and secondary-market basis.

Extended note. Version record

Uniswap Labs, UNI Token; Protocol Fees

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living protocol documentation consulted 4 October 2026; burn mechanism operating since December 2025.

Traces protocol fees through collection and external exchange for UNI that is burned, separately from liquidity-provider fees and LP ownership. Read after Uniswap v4 mechanics. Deployment and fee settings differ by version, pool and chain and can be changed by governance. The documentation's activation examples may lag governance execution; use on-chain events and the effective proposal at the valuation block, not total swap volume as automatic tokenholder revenue.

Extended note. Version record

Lin William Cong, Ye Li and Neng Wang, Tokenomics: Dynamic Adoption and Valuation; NBER working paper

Difficulty: Research **Type:** Research paper

Status/version: Review of Financial Studies 34(3), 1105-1155, March 2021; first online 11 August 2020; NBER WP 27222, May 2020. Published article and separately identified working-paper access.

Develops a transactional-demand and adoption model for token valuation. Use it to distinguish platform utility from an enforceable claim on protocol fees before transferring an equity cash-flow model to a token.

Extended note. Version record

Kose John, Thomas J. Rivera and Fahad Saleh, "Proof-of-Work versus Proof-of-Stake: A Comparative Economic Analysis"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 38(7), 1955–2004 (2025); DOI 10.1093/rfs/hhaf013.

Compares equilibrium security of proof-of-work and proof-of-stake; with realistic parameters PoS is more secure than equivalent PoW.

Version record

Fahad Saleh, "Blockchain without Waste: Proof-of-Stake"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 34(3), 1156–1190 (2021); DOI 10.1093/rfs/hhaa075.

Shows proof-of-stake can achieve consensus because stakeholders bear the cost of forks through the coin value; the economic foundation of staking.

Version record

Lin William Cong, Ye Li and Neng Wang, "Token-based platform finance"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 144(3), 972–991 (2022); DOI 10.1016/j.jfineco.2021.10.002.

Platforms finance themselves by issuing tokens; dynamic model of token supply, platform investment and agency conflicts.

Version record

Krzysztof Gogol, Yaron Velner, Benjamin Kraner and Claudio J. Tessone, "SoK: Liquid Staking Tokens (LSTs) and Emerging Trends in Restaking"

Difficulty: Advanced **Type:** Research paper

Status/version: arXiv preprint 2404.00644, v3 (26 December 2024); DOI 10.48550/arXiv.2404.00644.

Systematisation of liquid staking: a framework for the technical and economic models of LST protocols (operator selection, reward distribution, slashing), evidence that design and market conditions affect LST prices, and a review of restaking and liquid restaking tokens and their risks. arXiv preprint.

Version record

Stefan Scharnowski and Hossein Jahanshahloo, "The Economics of Liquid Staking Derivatives: Basis Determinants and Price Discovery"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Futures Markets, 45(2), 91–117 (2024); DOI 10.1002/fut.22556.

The liquid staking basis, the price gap between a liquid staking token and its underlying, varies widely. It widens when the protocol's yield is low relative to direct staking, when crypto is volatile and when secondary liquidity is thin, and narrows with investor attention and positive sentiment. Liquid staking tokens contribute a growing share of price discovery.

Version record

Xihan Xiong, Zhipeng Wang and Qin Wang, "Exploring the Market Dynamics of Liquid Staking Derivatives (LSDs)"

Difficulty: Research **Type:** Research paper

Status/version: 2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC), 334–342 (2024); DOI 10.1109/icbc59979.2024.10634417.

Measures the gap between primary and secondary prices of liquid staking tokens, the arbitrage it allows and its barriers, and LP returns in LST pools: 66% of liquidity positions returned less than holding the tokens.

Version record

Dominic Grandjean, Lioba Heimbach and Roger Wattenhofer, "Ethereum Proof-of-Stake Consensus Layer: Participation and Decentralization"

Difficulty: Research **Type:** Research paper

Status/version: Financial Cryptography and Data Security. FC 2024 International Workshops, Lecture Notes in Computer Science, 253–280 (2024); DOI 10.1007/978-3-031-69231-4_17.

Longitudinal measurement of Ethereum's beacon chain. Every dip in participation traces to network upgrades, problems in major consensus clients or operators running many validators; staking power is about as decentralised as proof-of-work mining was, with liquid staking services having a large effect. Useful for correlated slashing and outage scenarios.

Version record

Caspar Schwarz-Schilling, Joachim Neu, Barnabé Monnot, Aditya Asgaonkar, Ertem Nusret Tas and David Tse, "Three Attacks on Proof-of-Stake Ethereum"

Difficulty: Research **Type:** Research paper

Status/version: Financial Cryptography and Data Security, Lecture Notes in Computer Science, 560–576 (2022); DOI 10.1007/978-3-031-18283-9_28.

Refines two known attacks on proof-of-stake Ethereum (short-range reorganisations for profit and network-delay attacks that stall consensus) and combines them into a third that lets a small staker cause long-range reorganisations. Motivates the fork-choice changes and why finality, not inclusion, ends settlement risk.

Version record

Naveen Durvasula and Tim Roughgarden, "Robust Restaking Networks"

Difficulty: Research **Type:** Research paper

Status/version: 16th Innovations in Theoretical Computer Science Conference (ITCS 2025), LIPIcs 325, 48:1–48:21 (2025); DOI 10.4230/LIPIcs.ITCS.2025.48.

Security of restaking networks in which validators secure several services. Robustness depends on the buffer between attack costs and profits: if costs exceed profits by 10%, a sudden loss of 0.1% of stake cannot cascade beyond 1.1%. Gives tight bounds on worst-case losses, local versions for single services and a bound on cascade length.

Version record

18.8 Tokenisation and stablecoin funding

Connect these readings to credit, collateral and settlement risk. The token representation, the underlying legal claim, the redemption arrangement and the source of settlement money can have different risk profiles.

Iñaki Aldasoro, Giulio Cornelli, Jon Frost, Priscilla Koo Wilkens, Ulf Lewrick and Vatsala Shreeti, Tokenisation of Government Bonds: Assessment and Roadmap

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Bulletin 107, published 10 July 2025; eight-page bulletin with an online appendix.

Examines tokenised sovereign and other government-related bond issuance, trading frictions and the infrastructure needed for broader adoption. Connect programmable settlement to collateral mobility, delivery-versus-payment and interoperability rather than treating tokenisation as a new credit exposure by itself. The early issuance sample and comparisons do not identify a universal causal liquidity gain; distinguish native legal claims from wrappers and off-chain redemption arrangements.

Version record

Alexandra Born, Johanne Evrard, Claudia Lambert, Wagner Eduardo Schuster and Anna Tskhakaya, Tokenised Bonds: Assessing Efficiency and Liquidity in a Nascent Market

Difficulty: Research **Type:** Institutional research article

Status/version: ECB Macroprudential Bulletin 33 research article, April 2026; institutional article, not a peer-reviewed journal article. Issued 2026-04.

A carefully bounded early empirical comparison of tokenised versus conventional corporate bonds. It usefully tempers tokenisation claims by separating tentative quoted-liquidity and funding-cost associations from unchanged underwriting fees.

Extended note. Version record

Bank for International Settlements, Anchoring Trust in Money: Innovation Beyond Stablecoins

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Annual Economic Report 2026 Chapter III, published 23 June 2026; checked 4 October 2026.

Studies stablecoins, tokenised money and distributed ledgers through monetary singleness, elasticity, interoperability and integrity. Its reserve-composition and redemption scenarios connect digital-asset funding to bank deposits, money markets and cross-border capital flows. Useful for defining depeg and funding stresses around an on-chain strategy. Treat the chapter as BIS policy analysis with stylised scenarios; issuer-specific asset quality, legal redemption rights and intraday market liquidity still need direct evidence.

Version record

Yiming Ma, Yao Zeng and Anthony Lee Zhang, Stablecoin Runs and the Centralization of Arbitrage; author manuscript

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, hhag072; online 3 September 2026, corrected and typeset 11 September 2026.

Connects access to primary redemption with secondary-market peg arbitrage and run incentives. Concentrated arbitrage can support price stability while making sales less costly for investors and amplifying withdrawals. Read the empirical mechanism beside issuer eligibility, reserve liquidity and redemption terms, then stress the mint–trade–redeem chain when a major arbitrageur exits. The journal page and abstract establish the published identity and central result; the linked author manuscript provides a separate accessible research version. They do not establish current issuer exposures or guarantee a stable peg.

Version record

Circle, USDC Terms; Circle Mint tokenization and redemption lifecycle; USDC reserve and issuance disclosures; Circle Mint native-chain support

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Updated 12 December 2025; outside-EEA terms inspected 4 October 2026. Selected companion documents inspected 12 October 2026; earlier source verification retained separately.

A concrete issuer-contract case for separating token ownership from direct redemption access. The terms distinguish Circle Mint account holders from other USDC holders, describe reserve arrangements and identify eligibility and service conditions. Map which entity holds the claim, who can redeem directly, which intermediary a secondary holder needs, and where settlement can be delayed. These terms expressly apply outside the EEA; use the separately referenced European documents for that jurisdiction. Contract language and issuer representations do not independently verify reserve assets or their liquidation value. Pair the legal terms with Circle's fiat-in/mint and burn/fiat-out lifecycle, weekly reserve and issuance disclosures, and chain-specific Mint support. Identify the exact native token contract: a bridged token does not inherit direct Circle Mint redemption.

Extended note. Version record

United States Congress, "Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS) Act, Public Law 119-27"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Enacted 18 July 2025 as Public Law 119-27 (S. 1582); implementing regulations pending or in progress.

Federal framework for payment-stablecoin issuers: one-for-one permitted reserves, redemption, disclosure and supervision. Effective on the earlier of 18 months after enactment or 120 days after final regulations; implementing rules determine practical terms.

Extended note. Version record

European Parliament and Council, "Regulation (EU) 2023/1114 on markets in crypto-assets (MiCA)"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official Journal L 150, 9 June 2023; phased application 2024; Level 2 measures separate.

EU regime for crypto-asset issuers, asset-referenced and e-money tokens, and crypto-asset service providers; stablecoin provisions applied from 30 June 2024, CASP rules from 30 December 2024 with transitional periods.

Extended note. Version record

Gary B. Gorton and Jeffery Y. Zhang, "Taming Wildcat Stablecoins"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper first posted July 2021; later law-review publication not established in this check.

Compares stablecoins with nineteenth-century private banknotes and argues for public-money or bank-like regulation.

Extended note. Version record

Bank for International Settlements, "The rise of tokenised money market funds (BIS Bulletin No 115)"

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Bulletin No 115, 26 November 2025.

Tokenised MMFs grew to roughly USD 9 billion and serve as DeFi collateral; risks mirror and may amplify conventional MMF liquidity mismatch.

Version record

Bank for International Settlements, "The next-generation monetary and financial system (Annual Economic Report 2025, Chapter III)"

Difficulty: Advanced **Type:** Report / research note

Status/version: BIS Annual Economic Report, 24 June 2025.

Proposes a tokenised unified ledger with central bank reserves, commercial bank money and government bonds; assesses stablecoins against singleness, elasticity and integrity.

Version record

Raphael Auer, Cyril Monnet and Hyun Song Shin, "Distributed ledgers and the governance of money"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 167, 104026 (2025); DOI 10.1016/j.jfineco.2025.104026.

Theory of how distributed ledgers govern money and payments, comparing permissionless and permissioned designs; central-bank perspective on tokenised money.

Extended note. Version record

18.8.1 Stablecoin pegs, runs and reserve assets

Peg arbitrage and its limits, the design theory of collateralised and algorithmic stablecoins, run evidence from Terra, the 2021 algorithmic runs and the March 2023 USDC depeg, market-implied devaluation risk, and the international standards for stablecoin arrangements.

Richard K. Lyons and Ganesh Viswanath-Natraj, "What keeps stablecoins stable?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of International Money and Finance, 131, 102777 (2023); DOI 10.1016/j.jimonfin.2022.102777.

Arbitrage, issuer redemption and demand shocks determine stablecoin peg deviations.

Version record

Gary B. Gorton, Elizabeth C. Klee, Chase P. Ross, Sharon Y. Ross and Alexandros P. Vardoulakis, "Leverage and Stablecoin Pegs"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial and Quantitative Analysis, 61(1), 99–136 (2026); DOI 10.1017/s0022109025000134.

Stablecoin holders are compensated for run risk by lending coins to leveraged crypto speculators, linking peg stability to crypto leverage.

Version record

Jiageng Liu, Igor Makarov and Antoinette Schoar, "Anatomy of a Run: The Terra Luna Crash"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 184, 104359 (2026); DOI 10.1016/j.jfineco.2026.104359.

On-chain analysis of the 2022 Terra/UST run: who ran first, how the algorithmic peg failed and how losses were distributed; published in the Journal of Financial Economics (2026).

Version record

Alexander Hammerl, "Optimal control of reserve asset portfolios for stablecoins"

Difficulty: Research **Type:** Research paper

Status/version: Quantitative Finance, 1–16 (2026); DOI 10.1080/14697688.2026.2671176.

Continuous-time control of stablecoin reserve allocation between cash and bills and mint/burn fees to keep the peg credible.

Version record

Tirupam Goel, Ulf Lewrick and Isha Agarwal, "Making stablecoins stable(r): can regulation help?"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, rfag040 (2026); DOI 10.1093/rof/rfag040.

Model of a stablecoin issuer facing persistent redemption shocks; usable capital and liquidity buffers reduce default and fire-sale spillovers through asymmetric channels.

Version record

Adrien d'Avernas, Vincent Maurin and Quentin Vandeweyer, "Can Stablecoins Be Stable?"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, advance online publication (2026); DOI 10.1287/mnsc.2024.06992.

General model of a stablecoin protocol that earns seigniorage from issuance, used to compare designs by their ability to hold the peg. Algorithmic backing can be rationalised but is more fragile than collateral; even full collateral does not make promises to repurchase coins when demand falls credible, so immutable contracts or decentralised commitment improve stability. The design theory behind the peg evidence in this pathway.

Version record

Bryan Routledge and Ariel Zetlin-Jones, "Currency stability using blockchain technology"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Economic Dynamics and Control, 142, 104155 (2022); DOI 10.1016/j.jedc.2021.104155.

Treats a partly backed stablecoin as a currency peg exposed to self-fulfilling speculative attacks. A conversion rate that adjusts dynamically to traders' redemption demand can eliminate the attack equilibrium while keeping much of the exchange-rate stability, but it requires commitment by the party that manages the reserve. Read with the classical currency-crisis model it adapts.

Version record

Kenechukwu Anadu, Pablo Azar, Marco Cipriani, Thomas M. Eisenbach, Catherine Huang, Mattia Landoni, Gabriele La Spada, Marco Macchiavelli, Antoine Malfroy-Camine and J. Christina Wang, "Runs and Flights to Safety: Are Stablecoins the New Money Market Funds?"

Difficulty: Advanced **Type:** Report / research note

Status/version: Federal Reserve Bank of New York Staff Reports 1073 (September 2023); DOI 10.59576/sr.1073.

Compares investor behaviour in the stablecoin runs of 2022 and 2023 with the money market fund runs of 2008 and 2020. Stablecoin holders also fly to safety, with net flows from riskier to safer coins in stress, but runs hit different stablecoin types in 2022 and 2023, whereas fund runs struck only prime funds. Flows stay mostly within a blockchain, and redemptions accelerate below an estimated break-the-buck threshold of USD 0.99. Federal Reserve Bank of New York Staff Report 1073.

Version record

Barry Eichengreen, My T. Nguyen and Ganesh Viswanath-Natraj, "Stablecoin devaluation risk"

Difficulty: Research **Type:** Research paper

Status/version: The European Journal of Finance, 31(11), 1469–1496 (2025); DOI 10.1080/1351847x.2025.2505757.

Builds market-based measures of Tether devaluation risk from spot and futures prices, treating reliance on centralised custodians like a pegged exchange-rate regime. Conditional on full default the implied devaluation probability averages about 60 basis points a year, above 200 during the Terra-Luna crash; a 5% partial devaluation is priced at roughly 12 percentage points a year. Volatility and transaction velocity drive the risk, and deviations from covered interest parity show stablecoin markets segmented from traditional ones by leverage and arbitrage costs.

Version record

Klaus Grobys, Juha Junttila, James W. Kolari and Niranjan Sapkota, "On the stability of stablecoins"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Empirical Finance, 64, 207–223 (2021); DOI 10.1016/j.jempfin.2021.09.002.

Studies the volatility processes of stablecoins with power-law methods. Bitcoin volatility behaves well statistically, but stablecoin volatilities are unstable and respond contemporaneously to bitcoin volatility; lagged bitcoin volatility Granger-causes stablecoin volatility, not the reverse.

Version record

Lai T. Hoang and Dirk G. Baur, "How stable are stablecoins?"

Difficulty: Research **Type:** Research paper

Status/version: The European Journal of Finance, 30(16), 1984–2000 (2024); DOI 10.1080/1351847x.2021.1949369.

Proposes tests of absolute and relative stability and finds excess price variation in stablecoins in high-frequency data. Stablecoin returns, volatility and volume co-move with bitcoin, and a quasi-natural experiment indicates that stablecoins raise bitcoin trading volume.

Version record

Roman Kozhan and Ganesh Viswanath-Natraj, "Decentralized Stablecoins and Collateral Risk"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2021); DOI 10.2139/ssrn.3866975.

Uses MakerDAO's DAI to study the trade-off between decentralisation and peg stability when collateral is volatile and held on-chain. Peg deviations come from collateral risk and limits to arbitrage; the Peg Stability Module anchors the peg through par conversion into fiat-backed coins at the cost of more reliance on centralised assets. Working paper.

Version record

Harald Uhlig, "A Luna-tic Stablecoin Crash"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 30256 (July 2022); DOI 10.3386/w30256.

Theory of the May 2022 UST collapse, in which the algorithmic stablecoin and its LUNA backing lost more than USD 50 billion within a week. Lets the market possibly return to normal at any moment, so the crash unfolds gradually until convertibility is suspended, and fits the model to the data by quantitative interpretation. NBER Working Paper 30256.

Version record

Antonio Briola, David Vidal-Tomás, Yuanrong Wang and Tomaso Aste, "Anatomy of a Stablecoin's failure: The Terra-Luna case"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 51, 103358 (2023); DOI 10.1016/j.frl.2022.103358.

Event reconstruction of the Terra failure from social-media news and hourly and transaction data for bitcoin, LUNA and UST. Stresses the project's dependence on the Anchor protocol's yield, identifies the trigger events and tracks network dependence among 61 large cryptocurrencies during the fall; finds no herding by the cross-sectional absolute deviation measure.

Version record

Austin Adams and Markus Ibert, "Runs on Algorithmic Stablecoins: Evidence from Iron, Titan, and Steel"

Difficulty: Intro **Type:** Report / research note

Status/version: FEDS Notes, Board of Governors of the Federal Reserve System, 2 June 2022; DOI 10.17016/2380-7172.3121.

Federal Reserve note on three runs on algorithmic stablecoins in 2021 (Iron, Titan and Steel), written weeks after Terra. Attributes IRON's June 2021 collapse on Polygon to a flaw in its no-arbitrage mechanism: TITAN backing was valued at a ten-minute average price, so after TITAN's crash users could mint IRON cheaply and sell it below the peg. The same code on Binance's chain (Steel) held its peg, and larger accounts ran before smaller ones.

Version record

Gordon Y. Liao and John Caramichael, "Stablecoins: Growth Potential and Impact on Banking"

Difficulty: Advanced **Type:** Report / research note

Status/version: Federal Reserve Board, IFDP 1334 (January 2022); DOI 10.17016/IFDP.2022.1334.

Federal Reserve discussion paper on stablecoin use cases and their effects on banks. The effect on credit depends on where inflows come from and how reserves are held: a two-tiered system in which reserves sit as bank deposits keeps credit creation, while a narrow-bank model can disintermediate banks but may give the most stable peg. Safe, liquid backing could make dollar stablecoins a haven in crypto distress.

Version record

Nico Oefele, Dirk G. Baur and Lee A. Smales, "Flight-to-quality—Money market mutual funds and stablecoins during the March 2023 banking crisis"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 234, 111464 (2024); DOI 10.1016/j.econlet.2023.111464.

Short comparison of flows in the March 2023 banking crisis, when USDC lost its peg over reserves held at Silicon Valley Bank: money market funds took inflows while stablecoins saw outflows, and within both classes money moved from riskier to safer products. Reserve assets and issuer size mattered.

Version record

Jon Frost, Hyun Song Shin and Peter Wierts, "An Early Stablecoin? The Bank of Amsterdam and the Governance of Money"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper (2020), also BIS Working Paper 902; DOI 10.2139/ssrn.3732132.

History of the Bank of Amsterdam (1609–1820), which began as a 'stablecoin' issuing deposits backed by coin and settling payments across its books. Monthly balance sheets show how lending and the shocks of the 1780s, without fiscal support, moved confidence to a run equilibrium as the deposit premium (agio) turned negative. Also BIS Working Paper 902.

Version record

Financial Stability Board, High-level Recommendations for the Regulation, Supervision and Oversight of Global Stablecoin Arrangements: Final Report

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: FSB final report, 17 July 2023.

Ten FSB recommendations for global stablecoin arrangements: regulatory powers, oversight by function, cross-border cooperation, governance, risk management, data, recovery and resolution, disclosure and conformance before operation. Recommendation 9 sets the stablecoin-specific core: a robust legal claim on the issuer or reserve, timely redemption at par into fiat for single-currency coins, and an effective stabilisation mechanism with prudential requirements.

Version record

Committee on Payments and Market Infrastructures and International Organization of Securities Commissions, Application of the Principles for Financial Market Infrastructures to Stablecoin Arrangements

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: CPMI–IOSCO final guidance, 13 July 2022.

CPMI–IOSCO guidance that a systemically important stablecoin arrangement performing a transfer function should observe the PFMI. Addresses what sets these arrangements apart: a settlement asset that is neither central-bank nor commercial-bank money, interdependent functions, decentralised operation or governance and DLT, with focus on governance, risk management, settlement finality and money settlement. Read with the PFMI themselves.

Version record

18.9 Tokenised securities and real-world assets

Tokenised versions of existing securities and funds, and tokens that create new fractional claims on ventures, property and unique assets. Stablecoins, tokenised bonds and tokenised money market funds are in Tokenisation and stablecoin funding; fractional shares traded at brokers are in Fractional shares, odd lots and contract size.

18.9.1 Tokenised securities: legal form, venues and settlement

What a tokenised security is in law, how it settles and trades, and the 2024–2026 official reports, pilots, exchange rules and exemptions that let tokenised stocks, bonds and funds enter regulated markets.

Jonathan Chiu and Thorsten V. Koeppl, "Blockchain-Based Settlement for Asset Trading"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 32(5), 1716–1753 (2019); DOI 10.1093/rfs/hhy122.

Can securities settle on a blockchain, and what is gained? Faster and more flexible settlement, provided settlement fails from participants forking the chain to undo losses are ruled out. Under proof of work the chain must limit block size and frequency to generate fees that pay for mining; even so, net gains for the US corporate bond market are estimated at 1–4 basis points.

Version record

Financial Stability Board, The Financial Stability Implications of Tokenisation

Difficulty: Advanced **Type:** Report / research note

Status/version: FSB report, 22 October 2024.

Defines tokenisation as issuing or representing assets as tokens on new technologies such as distributed ledgers, and reviews adoption, benefits and obstacles. Adoption is very low; claimed gains in efficiency, cost and transparency are largely unproven, may be achievable without tokenisation and involve trade-offs; scaling is held back by uncertain demand, lack of interoperability, scarce settlement assets and legal differences. Vulnerabilities are liquidity and maturity mismatch, leverage, asset price and quality, interconnectedness and operational fragility; not yet material for financial stability.

Version record

Board of the International Organization of Securities Commissions (IOSCO), Tokenization of Financial Assets: Final Report (FR/17/25)

Difficulty: Advanced **Type:** Report / research note

Status/version: IOSCO Final Report FR/17/25, November 2025 (press release 11 November 2025).

IOSCO's survey of tokenisation across the securities life cycle. Tokenisation promises fractionalisation, programmability, composability and atomic settlement, but fractionalisation is also achievable through securitisation and other intermediation. Adoption is growing but nascent, led by fixed income and money market funds; issuance has changed more than distribution and secondary trading, which still run on conventional infrastructure; given a choice, participants still prefer traditional settlement. Interoperability and credible settlement assets are the binding constraints, and IOSCO's crypto and DeFi recommendations are to be applied.

Version record

U.S. Securities and Exchange Commission, Divisions of Corporation Finance, Investment Management and Trading and Markets, Statement on Tokenized Securities

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC staff statement, 28 January 2026; not a rule or Commission statement.

A taxonomy of tokenised securities. Issuer-sponsored tokens either make the blockchain part of the master securityholder file, so that a token transfer is the security transfer, or use tokens only to instruct off-chain record changes. Third-party tokens of another issuer's security are either custodial (a security entitlement to shares held by the third party) or synthetic (the third party's own security, such as a linked note or security-based swap, giving economic exposure only). Format does not change the application of the securities laws, and third-party holders bear the sponsor's insolvency risk.

Version record

U.S. Securities and Exchange Commission, Division of Trading and Markets, No-Action Letter to The Depository Trust Company on the DTCC Tokenization Services

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC staff no-action letter, 11 December 2025; withdrawn three years after DTC launches the pilot.

Staff relief for a three-year DTC pilot in which participants can hold security entitlements to DTC-held securities as tokens in registered blockchain wallets instead of only on DTC's ledger. Eligible securities are Russell 1000 stocks, US Treasuries and ETFs on major indices; DTC debits a participant's account, credits a digital omnibus account and mints the token, and burns it on de-tokenisation. During the pilot tokens carry no collateral or settlement value in DTC's risk management.

Version record

U.S. Securities and Exchange Commission, Order Approving a Proposed Rule Change . . . to Enable the Trading of Securities on the Exchange in Tokenized Form (Release 34-105047)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC order of 18 March 2026 approving SR-NASDAQ-2025-072, as modified by Amendment No. 2.

Lets participants in the DTC tokenisation pilot trade tokenised versions of eligible stocks and ETFs on Nasdaq on the same order book and with the same priority as the traditional shares, but only where the token is fungible with the traditional security, shares its CUSIP and symbol and carries the same rights. Eligibility follows the DTC pilot: Russell 1000 stocks and ETFs on major indices. Trades clear and settle through DTC in token form. Tokenisation here changes the settlement record, not the market.

Version record

U.S. Securities and Exchange Commission, Order Granting Temporary Conditional Exemptive Relief . . . for the Use of Certain Distributed Ledger Trading Venues for Tokenized NMS Stocks (Release 34-106402)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: SEC order of 17 September 2026, 91 FR 60168 (22 September 2026); expires five years after publication; request for comment.

Exempts 'Tokenized Securities Venues', which match permissioned participants through automated-market-maker liquidity pools, from the definition of an exchange, and certain liquidity providers in those pools from the definition of a dealer. Covers NMS stocks tokenised by the issuer or by an unaffiliated third party, but not synthetic tokens. Conditions include smart contracts that are auditable, public and deployed on a public permissionless ledger, disclosure to participants, and halting trading when the underlying stock is halted. An interim measure while the SEC considers rulemaking.

Version record

European Securities and Markets Authority, Report on the Functioning and Review of the DLT Pilot Regime (ESMA75-117376770-460)

Difficulty: Advanced **Type:** Report / research note

Status/version: ESMA report under Article 14 of Regulation (EU) 2022/858, 25 June 2025.

ESMA's review of the EU's DLT Pilot Regime, which since March 2023 has let market infrastructures trade and settle tokenised securities on distributed ledgers with exemptions from existing rules. Uptake has been limited: three authorised infrastructures (CSD Prague, 21X and 360X) and minimal live trading, though the regime encouraged experiments in DLT trading, settlement and compliance. Obstacles are interoperability and access to central-bank money; ESMA recommends recalibrating the thresholds that cap participation and settling the regime's long-term status.

Version record

Joachim Lebrun, Tony Malghem, Kevin Thizy, Luc Falempin and Adam Boudjemaa, ERC-3643: T-REX – Token for Regulated EXchanges

Difficulty: Advanced **Type:** Software / protocol documentation

Status/version: Ethereum Request for Comments, created 9 July 2021; status Final.

A permissioned token standard for tokenised securities: an ERC-20-compatible token that checks every transfer against an on-chain identity registry, a registry of trusted claim issuers and a modular compliance contract, so only eligible investors can hold it. Agents can pause the token and freeze holders' balances. The standard shows how transfer restrictions, investor eligibility and corporate-action controls are coded into the token rather than enforced by an intermediary.

Version record

18.9.2 Token offerings and tokenised real assets

Initial coin and security token offerings, the design of tokens against equity, and fractional, tokenised ownership of real estate, art and other unique assets.

Sabrina T. Howell, Marina Niessner and David Yermack, "Initial Coin Offerings: Financing Growth with Cryptocurrency Token Sales"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 33(9), 3925–3974 (2020); DOI 10.1093/rfs/hhz131.

More than 1,500 initial coin offerings raising USD 12.9 billion: real success (higher employment, survival) is associated with disclosure, credible commitment and quality signals, and an instrumental-variables analysis finds that listing the token on an exchange causes higher future employment, so token liquidity has real effects.

Version record

Hugo Benedetti and Leonard Kostovetsky, "Digital Tulips? Returns to investors in initial coin offerings"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Corporate Finance, 66, 101786 (2021); DOI 10.1016/j.jcorpfin.2020.101786.

2,390 completed ICOs raising USD 12 billion: average returns of 179 per cent from the ICO price to the first day's opening price over an average 16 days, and buy-and-hold abnormal returns of 48 per cent over the first 30 trading days. Relates underpricing to issue characteristics and prices to Twitter activity.

Version record

Jingxing (Rowena) Gan, Gerry Tsoukalas and Serguei Netessine, "Initial Coin Offerings, Speculation, and Asset Tokenization"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 67(2), 914–931 (2021); DOI 10.1287/mnsc.2020.3796.

Token offerings as asset tokenisation: firms sell tokens backed by future inventory to finance growth. Designs optimal floating and pricing for utility tokens and for equity tokens (security token offerings) under cash diversion and demand uncertainty. Unregulated ICOs cause agency costs, underproduction and lost value, fading as margins and demand improve; equity tokens align incentives better even without regulation.

Version record

Katya Malinova and Andreas Park, "Tokenomics: When Tokens Beat Equity"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 69(11), 6568–6583 (2023); DOI 10.1287/mnsc.2023.4882.

Generic revenue-based token contracts are inferior to equity and cause over- or underproduction. An optimally designed token, combining an output presale with incremental revenue sharing, replicates equity and debt payoffs, and with entrepreneurial moral hazard tokens can finance a strictly larger set of ventures than equity.

Version record

Sebastian Gryglewicz, Simon Mayer and Erwan Morellec, "Optimal financing with tokens"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 142(3), 1038–1067 (2021); DOI 10.1016/j.jfineco.2021.05.004.

A start-up financing a digital platform with tokens faces agency conflicts with outsiders. Token financing beats equity unless the platform expects strong cash flows, needs a lot of financing or faces severe agency conflicts; security features that grant cash-flow rights spur adoption but dilute developers' incentives, so their optimal level falls with agency problems.

Version record

Andrew Baum, "Tokenization—The Future of Real Estate Investment?"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 47(10), 41–61 (2021); DOI 10.3905/jpm.2021.1.260.

Journal version of the Oxford Future of Real Estate Initiative report on tokenisation. Tokens could open lumpy, illiquid property to small investors, but single-asset tokenisation needs demand for fractional units and comfort with blockchains, and in many land markets an intermediate vehicle, which adds cost, because land cannot be owned directly by many. Assets already held in funds are the likelier candidates; mass tokenisation of single commercial buildings is some way off.

Version record

Laurens Swinkels, "Empirical evidence on the ownership and liquidity of real estate tokens"

Difficulty: Research **Type:** Research paper

Status/version: Financial Innovation, 9(1), 45 (2023); DOI 10.1186/s40854-022-00427-5.

58 tokenised US rental homes, mostly in Detroit (RealT): an average of 254 owners per property; investors with more than USD 5,000 diversify across properties and cities; ownership changes about once a year, more often for tokens on decentralised exchanges; token prices move with the local house price index, so the tokens deliver residential price exposure.

Version record

Julia Kreppmeier, Ralf Laschinger, Bertram I. Steininger and Gregor Dorfleitner, "Real estate security token offerings and the secondary market: Driven by crypto hype or fundamentals?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Banking & Finance, 154, 106940 (2023); DOI 10.1016/j.jbankfin.2023.106940.

173 US real estate tokens, 2019–2021, and 238,433 blockchain transactions: tokens give many small investors fractional property ownership with low entry barriers, but holders' token portfolios are poorly diversified. Offering success, secondary trading and daily capital flows depend on property characteristics and also on crypto-market transaction costs and sentiment.

Version record

Roman Kräussl and Alessandro Tugnetti, "Non-Fungible Tokens (NFTs): A Review of Pricing Determinants, Applications and Opportunities"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Surveys, 38(2), 555–574 (2024); DOI 10.1111/joes.12597.

Survey of the NFT market: investor perceptions, the financial and econometric pricing models used in the literature and their predictive performance, and a framework for NFT price formation and the determinants of value. The entry point for unique tokenised assets, including fractionalised NFTs.

Version record

18.10 Security, exploits and governance

Security practice and regional sources come first; exploits, cross-chain bridges, layer-two protocols and DAO governance follow. Chinese-language resources provide regional industry and academic context; primary and secondary evidence are distinguished below.

18.10.1 Security practice, audits and regional sources

Optional practitioner context on mechanisms and derivatives complements the research outlets in Section 23.8.

Olympix, "A Comprehensive Analysis of Smart Contract Security Approaches: Auditing, Mutation Testing, Formal Verification, and Fuzzing" (source: 2024)

Difficulty: Intro **Type:** Practitioner / commentary

Status/version: Selected citation year(s): 2024. Version distinctions remain in the annotation.

Comparison of auditing, mutation testing, formal verification and fuzzing. Proofs cover specified properties under assumptions; neither an audit nor a proof eliminates all economic risk.

Applications: basic programming.

Version record

OpenZeppelin Ethernaut; Damn Vulnerable DeFi; Trail of Bits, ConsenSys Diligence and CertiK audit reports

Difficulty: Not specified **Type:** Report / research note

Status/version: Edition/revision and publication date not established for the selected resource(s).

Security exercise collections and audit archives for interpreting protocol failures. Solidity/EVM knowledge is required; individual reports are specific to contract versions and audit scope.

Applications: applied security and audit interpretation.

Version record

DeFiSafety, *Protocol Review Process*

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Edition/revision and publication date not established for the selected resource(s).

Published criteria for code transparency, documentation, testing, audits, admin controls and oracles. Process-quality scores complement code audits; they do not establish protocol safety or measure expected financial loss.

Applications: process-quality methodology.

Version record

Huobi / HTX Research (火币研究院); SMARTChain and the *Huobi Blockchain Asset Assessment Report*

Difficulty: Not specified **Type:** Report / research note

Status/version: Historical industry lead; 2017. Original report not retrieved.

Historical blockchain-asset assessment framework described in a 2017 news report and a 2019 academic analysis. These secondary accounts do not establish the complete original report.

Version record

2024/25年度中国量化投资白皮书 **(China Quant Investment Whitepaper)**

Difficulty: Not specified **Type:** Report / research note

Status/version: Industry context; 2024/25 edition. Crypto relevance unverified.

The producing partner's catalogue lists the 2024/25 edition. A Sina-hosted release article names participants including Kuangbang Technology (宽邦科技) and Huatai Securities (华泰证券); the catalogue is maintained by 金融阶. Useful for Chinese quant-industry

strategies, technology and risk context. Annual publication and direct or indirect crypto coverage remain unconfirmed; treat this as adjacent industry background.

Version record

Zhang Tianyi, Tsinghua thesis on Bitcoin price drivers using system dynamics (2024); Zhejiang smart-contract verification research

Difficulty: Not specified **Type:** Mixed resources

Status/version: Selected citation year(s): 2024. Version distinctions remain in the annotation.

Identified university research on Bitcoin price drivers and smart-contract verification; the entries are distinct thesis and technical-research resources.

Applications: Academic discovery category; evaluate individual papers and theses.. Version record

***Uncommon Core*; Hasu and MEV Senpai, "Interview with a Searcher" (19 July 2021)**

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Selected citation year(s): 2021. Version distinctions remain in the annotation.

Historical practitioner interview and transcript on MEV searchers, private relays and transaction bundles. The archive also includes Uncommon Core 2.0; distinguish episode dates and protocol versions when interpreting claims. See Flashbots for technical research.

Version record

Deribit Insights

Difficulty: Research **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Options education, crypto derivatives analytics, volatility discussion and exchange-related technical material. Connect pricing and surface concepts to crypto market conventions and flow; identify contributor, model assumptions and observation date. Exchange and partner publications are useful practitioner evidence, with their commercial context retained.

Applications: options and volatility.. Version record

18.10.2 Exploits, bridges, layer two and governance

Attack taxonomies and the limits of automated tools, cross-chain communication and bridge failures, layer-two and rollup designs, crypto-enabled crime and the concentration of DAO voting power.

Liyi Zhou, Xihan Xiong, Jens Ernstberger, Stefanos Chaliasos, Zhipeng Wang, Ye Wang, Kaihua Qin, Roger Wattenhofer, Dawn Song and Arthur Gervais, "SoK: Decentralized Finance (DeFi) Attacks"

Difficulty: Research **Type:** Research paper

Status/version: IEEE Symposium on Security and Privacy 2023.

Systematisation of DeFi incidents by layer (network, consensus, protocol, application) with loss data.

Version record

Kaihua Qin, Liyi Zhou, Benjamin Livshits and Arthur Gervais, "Attacking the DeFi Ecosystem with Flash Loans for Fun and Profit"

Difficulty: Research **Type:** Research paper

Status/version: Lecture Notes in Computer Science, 3–32 (2021); DOI 10.1007/978-3-662-64322-8_1.

Flash loans exist only within one atomic transaction, which raises arbitrage revenue and enables attacks. Analyses two February 2020 attacks with returns above 500,000% and frames finding attack parameters as an optimisation over chain and DeFi state; optimised versions of the two attacks would have earned USD 0.83m and 1.1m.

Version record

Nicola Atzei, Massimo Bartoletti and Tiziana Cimoli, "A Survey of Attacks on Ethereum Smart Contracts (SoK)"

Difficulty: Advanced **Type:** Research paper

Status/version: Principles of Security and Trust, Lecture Notes in Computer Science, 164–186 (2017); DOI 10.1007/978-3-662-54455-6_8.

Early taxonomy of programming pitfalls in Ethereum smart contracts and the attacks that exploit them (reentrancy as in the DAO attack, and others), with examples. The starting point for the later DeFi attack systematisations.

Version record

Stefanos Chaliasos, Marcos Antonios Charalambous, Liyi Zhou, Rafaila Galanopoulou, Arthur Gervais, Dimitris Mitropoulos and Benjamin Livshits, "Smart Contract and DeFi Security Tools: Do They Meet the Needs of Practitioners?"

Difficulty: Research **Type:** Research paper

Status/version: IEEE/ACM International Conference on Software Engineering (ICSE 2024), 1–13 (2024); DOI 10.1145/3597503.3623302.

Tests five automated security tools against 127 high-impact attacks (USD 2.3bn of losses) and surveys 49 DeFi developers and auditors. The tools could have prevented only 8% of the attacks (USD 149m), all reentrancy; logic bugs and protocol-level flaws, which practitioners rank as main threats, go undetected. A reason not to treat a clean tool scan as safety.

Version record

Alexei Zamyatin, Mustafa Al-Bassam, Dionysis Zindros, Eleftherios Kokoris-Kogias, Pedro Moreno-Sanchez, Aggelos Kiayias and William J. Knottenbelt, "SoK: Communication Across Distributed Ledgers"

Difficulty: Research **Type:** Research paper

Status/version: Financial Cryptography and Data Security, Lecture Notes in Computer Science, 3–36 (2021); DOI 10.1007/978-3-662-64331-0_1.

Formal treatment of cross-chain communication. Shows that it is impossible without a trusted third party, contrary to common belief, then gives a framework for evaluating bridge and interoperability protocols by use case, trust model and security assumptions. The theory behind bridge risk.

Version record

Lewis Gudgeon, Pedro Moreno-Sanchez, Stefanie Roos, Patrick McCorry and Arthur Gervais, "SoK: Layer-Two Blockchain Protocols"

Difficulty: Research **Type:** Research paper

Status/version: Financial Cryptography and Data Security, Lecture Notes in Computer Science, 201–226 (2020); DOI 10.1007/978-3-030-51280-4_12.

Systematisation of layer-two protocols up to 2019: payment and state channels, commit-chains and refereed delegation, with their synchronisation, routing, privacy and security. Layer two settles off-chain and uses the base chain only to resolve disputes; read before treating a rollup balance as final.

Version record

Louis Tremblay Thibault, Tom Sarry and Abdelhakim Senhaji Hafid, "Blockchain Scaling Using Rollups: A Comprehensive Survey"

Difficulty: Advanced **Type:** Research paper

Status/version: IEEE Access, 10, 93039–93054 (2022); DOI 10.1109/access.2022.3200051.

Survey of rollups as a scaling solution: how optimistic and zero-knowledge rollups operate, the main implementations with their features and limits, and a performance comparison.

Version record

Mengya Zhang, Xiaokuan Zhang, Yinqian Zhang and Zhiqiang Lin, "Security of Cross-chain Bridges: Attack Surfaces, Defenses, and Open Problems"

Difficulty: Research **Type:** Research paper

Status/version: International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2024), 298–316 (2024); DOI 10.1145/3678890.3678894.

Systematic study of cross-chain bridge security: bridge features, verification and communication models, 12 potential attack vectors and a taxonomy of attacks over three years in 10 categories with Solidity examples, plus defences and open problems.

Version record

Jakob Svennevik Notland, Jingyue Li, Mariusz Nowostawski and Peter Halland Haro, "SoK: cross-chain bridging architectural design flaws and mitigations"

Difficulty: Research **Type:** Research paper

Status/version: Blockchain: Research and Applications, 7(1), 100315 (2026); DOI 10.1016/j.bcra.2025.100315.

Multivocal review of 64 bridges, 31 exploits and four known bugs (2021–2023). Identifies 13 architectural components, links them to eight design flaws and proposes 11 measures to reduce impact. Bridge hacks cost USD 1.5–2bn in 2022 alone.

Version record

Lin William Cong, Campbell R. Harvey, Daniel Rabetti and Zong-Yu Wu, "An Anatomy of Crypto-Enabled Cybercrimes"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 71(4), 3622–3633 (2025); DOI 10.1287/mnsc.2023.03691.

Anatomy of crypto-enabled cybercrime from on- and off-chain, proprietary and hand-collected data, including ransom negotiations and Russian-language dark-web forums. Ransomware gangs run like firms with revenue models and reputations; blockchain transparency supports forensics, so blanket bans on crypto use may backfire.

Version record

Robin Fritsch, Marino Müller and Roger Wattenhofer, "Analyzing voting power in decentralized governance: Who controls DAOs?"

Difficulty: Research **Type:** Research paper

Status/version: Blockchain: Research and Applications, 5(3), 100208 (2024); DOI 10.1016/j.bcra.2024.100208.

Token holders, delegates, proposals and votes in the Compound, Uniswap and ENS DAOs. Most voting power sits with a few addresses, but they rarely vote against the outcome the wider community chooses.

Version record

Tom Barbereau, Reilly Smethurst, Orestis Papageorgiou, Johannes Sedlmeir and Gilbert Fridgen, "Decentralised Finance's timocratic governance: The distribution and exercise of tokenised voting rights"

Difficulty: Advanced **Type:** Research paper

Status/version: Technology in Society, 73, 102251 (2023); DOI 10.1016/j.techsoc.2023.102251.

Multiple-case study of the voting-rights tokens of the nine projects in Bloomberg's first Galaxy DeFi index, using on-chain distribution, trading, staking and delegation data. Voting rights are highly concentrated and rarely used: minority rule rather than democracy.

Version record

Jungsuk Han, Jongsub Lee and Tao Li, "Decentralized Autonomous Organization Governance"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, advance online publication (2026); DOI 10.1287/mnsc.2024.07033.

Model of token-based DAO voting with strategic token trading and a conflict between a large holder (a whale) and many small ones. Ownership concentration slows platform growth, less so with larger platforms, illiquid tokens and long-term incentives; voting data on major DAOs from 2020 to 2024 support the predictions.

Version record

Jungsuk Han, Jongsub Lee and Tao Li, "A review of DAO governance: Recent literature and emerging trends"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Corporate Finance, 91, 102734 (2025); DOI 10.1016/j.jcorpfin.2025.102734.

Review comparing DAO proposal and voting mechanisms with corporate governance, describing new voting models, the whale agency problem, investor activism and contested votes, and the literature on optimal DAO design.

Version record

Chapter 19

Event contracts and prediction markets

Event contracts pay a fixed amount if a stated event happens. The chapter runs from what an event price reveals, through the wagering-market literature on biases and through decision markets, to venues, outcome tokens, resolution, integrity and regulation, and finally to trading: automated and order-book market making, event derivatives and combos, and trading strategies. Exchange-listed, on-chain and bookmaker markets appear side by side: the economics of an event contract does not depend on its settlement layer, but its data, resolution and custody do.

19.1 Event prices, beliefs and forecasts

Subject context and qualifications. Four subsections: what an event price can reveal; the forecast record; the wagering-market literature on the favourite–longshot bias, bookmakers and exchanges; and decision, corporate and scientific markets.

19.1.1 Price interpretation and theory

Justin Wolfers and Eric Zitzewitz, "Prediction Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 18(2), 107–126 (2004); DOI 10.1257/0895330041371321. NBER Working Paper 10504 (2004) is the earlier version.

Provides the canonical overview of information aggregation, contract design and empirical forecast accuracy. It also explains when prices can reveal probabilities, moments or conditional beliefs and where causal interpretation remains unavailable. [41]

Applications: Prediction markets; forecasting, market design. Version record

Charles Manski, "Interpreting the Predictions of Prediction Markets"

Difficulty: Research **Type:** Research paper

Status/version: Economics Letters, 91(3), 425–429 (2006); DOI 10.1016/j.econlet.2006.01.004. NBER Working Paper 10359 (2004) is the earlier version.

Shows that a binary-contract price need not equal the average subjective probability except under restrictive assumptions. This is an essential counterweight to treating every event-contract price as a calibrated physical probability. [42]

Applications: Binary contracts; probability interpretation. Version record

Justin Wolfers and Eric Zitzewitz, "Interpreting Prediction Market Prices as Probabilities"

Difficulty: Research **Type:** Research paper

Status/version: NBER Working Paper 12200, May 2006; DOI 10.3386/w12200.

Replies to Manski by giving sufficient conditions under which a binary-contract price equals traders' mean belief, and argues that across a broad class of models prices stay close to mean beliefs. Read the two papers together: the probability reading rests on risk preferences and the dispersion of beliefs, and the remaining bias can be material.

Extended note. Version record

Erik Snowberg, Justin Wolfers and Eric Zitzewitz, "Prediction Markets for Economic Forecasting"

Difficulty: Advanced **Type:** Book chapter

Status/version: Handbook of Economic Forecasting (Elsevier, 2013), 657–687; DOI 10.1016/B978-0-444-53683-9.00011-6.

Summarises prediction-market research for economic forecasters. Markets quickly incorporate new information, are largely efficient and hard to manipulate, and generally show lower errors than professional forecasters and polls; the chapter also shows how prices can uncover or test the economic model behind a forecast.

Version record

Michael Ostrovsky, "Information Aggregation in Dynamic Markets With Strategic Traders"

Difficulty: Research **Type:** Research paper

Status/version: Econometrica, 80(6), 2595–2647 (2012); DOI 10.3982/ECTA8479.

Asks when repeated trading by strategic, privately informed traders reveals their pooled information. Prices converge to the full-information value in every equilibrium for securities the paper calls separable, but not in general, which bears on how combinatorial or conditional event contracts are specified.

Version record

Marco Ottaviani and Peter Norman Sørensen, "Price Reaction to Information with Heterogeneous Beliefs and Wealth Effects: Underreaction, Momentum, and Reversal"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 105(1), 1–34 (2015); DOI 10.1257/aer.20120881.

In a binary market with heterogeneous prior beliefs, the equilibrium price underreacts to information when investment is bounded, or when traders become less risk averse as wealth rises; underreaction grows with belief dispersion. Dynamically it produces momentum and then reversal, a benchmark for slow-moving event prices and the favourite–longshot pattern.

Version record

Erik Snowberg and Justin Wolfers, "Explaining the Favorite–Long Shot Bias: Is it Risk-Love or Misperceptions?"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 118(4), 723–746 (2010); DOI 10.1086/655844.

Uses the prices of win and compound (exotic) horse-race bets to separate two explanations of overpriced long shots. The evidence favours misperception of small probabilities over a taste for risk, so a price near zero should not be read as an unbiased probability.

Version record

Xue-Zhong He and Nicolas Treich, "Prediction market prices under risk aversion and heterogeneous beliefs"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Mathematical Economics, 70, 105–114 (2017); DOI 10.1016/j.jmateco.2017.02.005.

Equilibrium prices of state claims equal traders' mean beliefs if and only if the common utility function is logarithmic. A necessary and sufficient condition tells when prices sit systematically above or below mean beliefs, giving a rational, risk-aversion explanation of the favourite–longshot bias. The general-utility companion to Manski and to Wolfers and Zitzewitz.

Version record

Ned Augenblick and Matthew Rabin, "Belief Movement, Uncertainty Reduction, and Rational Updating"

Difficulty: Research **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 136(2), 933–985 (2021); DOI 10.1093/qje/qjaa043.

A Bayesian's beliefs must on average move only as much as uncertainty falls. The paper turns this into measures of belief movement and uncertainty reduction that are equal in expectation, and into statistical tests of Bayesian updating, then applies them to individual, algorithmic and market beliefs. Use it to ask whether an event-price path moves too much, or too little, to be a rational probability.

Version record

19.1.2 Forecast accuracy, calibration and event studies

Joyce E. Berg, Forrest D. Nelson and Thomas A. Rietz, "Prediction Market Accuracy in the Long Run"

Difficulty: Research **Type:** Research paper

Status/version: International Journal of Forecasting, 24(2), 285–300 (2008); DOI 10.1016/j.ijforecast.2008.03.007.

Compares Iowa Electronic Markets vote-share forecasts with 964 polls across the five U.S. presidential elections from 1988 to 2004. The market was closer to the eventual result 74% of the time and beat the polls in every election more than 100 days ahead; read with Erikson and Wlezien, who dispute how the polls are scored.

Version record

Robert S. Erikson and Christopher Wlezien, "Are Political Markets Really Superior to Polls as Election Predictors?"

Difficulty: Research **Type:** Research paper

Status/version: Public Opinion Quarterly, 72(2), 190–215 (2008); DOI 10.1093/poq/nfn010.

Argues that raw polls are the wrong benchmark because they measure current preference, not the election-day outcome. Once poll leads are properly discounted, poll-based forecasts outperform the Iowa vote-share markets, and poll-based win projections beat the winner-take-all markets.

Version record

David Rothschild, "Forecasting Elections: Comparing Prediction Markets, Polls, and Their Biases"

Difficulty: Research **Type:** Research paper

Status/version: Public Opinion Quarterly, 73(5), 895–916 (2009); DOI 10.1093/poq/nfp082.

Corrects market prices for the favourite–longshot bias and polls for their known biases before comparing state-level forecasts of the 2008 U.S. elections. The method of debiasing both sources before scoring them carries over directly to current venue comparisons.

Version record

Lionel Page and Robert T. Clemen, "Do Prediction Markets Produce Well-Calibrated Probability Forecasts?"

Difficulty: Research **Type:** Research paper

Status/version: Economic Journal, 123(568), 491–513 (2013); DOI 10.1111/j.1468-0297.2012.02561.x.

Tests calibration across a large set of exchange-traded event contracts. Prices are well calibrated for events close to resolution but show a favourite–longshot bias for distant ones, consistent with the cost of tying up capital; adjust for horizon before reading long-dated prices as probabilities.

Version record

Pavel Atanasov, Phillip Rescober, Eric Stone, Samuel A. Swift, Emile Servan-Schreiber, Philip Tetlock, Lyle Ungar and Barbara Mellers, "Distilling the Wisdom of Crowds: Prediction Markets vs. Prediction Polls"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 63(3), 691–706 (2017); DOI 10.1287/mnsc.2015.2374.

A randomised two-season test in a geopolitical forecasting tournament: more than 2,400 forecasters on 261 events. Market prices beat the simple mean of poll forecasts, but team polls beat markets once forecasts were aggregated with temporal decay, performance weighting and recalibration.

Version record

Anthony M. Diercks, Jared Dean Katz and Jonathan H. Wright, "Kalshi and the Rise of Macro Markets"; Federal Reserve FEDS version

Difficulty: Research **Type:** Research paper

Status/version: 2026 working paper: NBER w34702 and Federal Reserve Board FEDS 2026-010 (February 2026)

Evaluates regulated event-contract forecasts for macroeconomic releases against surveys and conventional market-implied expectations. It positions prediction markets as high-frequency distributional data sources for systematic macro, not only as standalone trading venues. [45]

Applications: Kalshi, macro events; nowcasting, event trading. Version record

Mikhail Chernov, Vadim Elenev and Dongho Song, "The Comovement of Voter Preferences: Insights from U.S. Presidential Election Prediction Markets Beyond Polls"; journal version

Difficulty: Research **Type:** Research paper

Status/version: 2026 journal online; 2025 NBER

Combines polls, fundamentals and prediction prices in a multivariate state-level forecasting framework. Its key quantitative insight is that correlated outcomes materially affect aggregate event probabilities and cannot be handled by independent state bets. Journal of Econometrics article 106192, online 3 February 2026; publisher lists a corrected proof without an assigned volume. The 2024-election setting limits transport to other events. [44]

Applications: Political prediction markets; factor and correlation models. Version record

Erik Snowberg, Justin Wolfers and Eric Zitzewitz, "Partisan Impacts on the Economy: Evidence from Prediction Markets and Close Elections"

Difficulty: Research **Type:** Research paper

Status/version: Quarterly Journal of Economics, 122(2), 807–829 (2007); DOI 10.1162/qjec.122.2.807.

Scales election-night asset-price moves by the simultaneous change in a prediction-market probability to estimate how a Republican victory affected equity prices and bond yields. The template for event studies that use an event-contract price as the measure of news.

Version record

Author as listed on arXiv (Le), "Decomposing Crowd Wisdom: Domain-Specific Calibration Dynamics in Prediction Markets"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2602.19520, first posted February 2026; revised August 2026.

Calibration of Kalshi and Polymarket prices by domain, horizon and trade size across hundreds of millions of trades; politics markets appear underconfident. Associational preprint; version figures differ.

Extended note. Version record

Karen Croxson and J. James Reade, "Information and Efficiency: Goal Arrival in Soccer Betting"

Difficulty: Research **Type:** Research paper

Status/version: Economic Journal, 124(575), 62–91 (2014); DOI 10.1111/ecoj.12033.

Uses goals scored just before half-time, when play stops but exchange betting continues, to test how quickly prices absorb news. Prices update swiftly and fully: a clean test of semi-strong efficiency and of how fast in-play event contracts reprice.

Version record

Paul W. Rhode and Koleman S. Strumpf, "Historical Presidential Betting Markets"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economic Perspectives, 18(2), 127–142 (2004); DOI 10.1257/0895330041371277.

Large, organised markets in U.S. presidential elections from 1868 to 1940 forecast results well before scientific polling, and stayed fairly efficient despite limited information and active attempts to manipulate odds. They faded as polls arrived and other gambling spread. The historical baseline for today's election contracts.

Version record

Felix Reichenbach and Martin Walther, "When Can Prediction Market Prices Be Trusted? Price Informativeness, Calibration, and Bias on Polymarket"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2026); DOI 10.2139/ssrn.7552098.

1.5 billion trades across 779,000 Polymarket markets: prices are highly informative and improve as markets mature, with more trading linked to better forecasts early on. Accuracy differs by category (weather most precise, mentions and geopolitics least). No market-wide longshot bias; low-probability outcomes are underpriced early and reverse late, and default outcomes stay overpriced. In sports, prices match professional bookmakers. Working paper.

Version record

Seungju Lee, Yebon Lee and Jaewook Lee, "Underreaction, salience, and hot-hand beliefs in sports prediction markets: Evidence from Kalshi NBA and NFL event contracts"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Behavioral and Experimental Finance, 52, 101259 (2026); DOI 10.1016/j.jbef.2026.101259.

41.8 million Kalshi trades across 1,496 NBA and NFL games, matched to second-by-second win-probability benchmarks. Prices absorb scoring news gradually and incompletely, a robust underreaction; salient highlight-type events get a weaker short-run response; there is no hot-hand extrapolation. Deviations exceed fee-and-spread hurdles, yet a mean-reversion rule earns nothing net of costs: slow, but not exploitable.

Version record

19.1.3 Wagering markets, bookmakers and the favourite–longshot bias

Richard H. Thaler and William T. Ziemba, "Anomalies: Parimutuel Betting Markets: Racetracks and Lotteries"

Difficulty: Intro **Type:** Research paper

Status/version: Journal of Economic Perspectives, 2(2), 161–174 (1988); DOI 10.1257/jep.2.2.161.

The classic survey of racetrack and lottery anomalies. Wagering markets suit efficiency tests because every bet has a known terminal value, yet they show the favourite–longshot bias, profitable place-and-show overlays and lotto numbers that are systematically over- and under-chosen. The starting point for treating event contracts as short-lived assets with known terminal value.

Version record

Mukhtar M. Ali, "Probability and Utility Estimates for Racetrack Bettors"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Political Economy, 85(4), 803–815 (1977); DOI 10.1086/260600.

20,247 harness races: subjective (odds-implied) probabilities exaggerate longshots and depress favourites. Under simplifying assumptions the paper backs out a representative bettor's utility of wealth, who is a risk lover and takes more risk as capital dwindles. The original risk-preference explanation of the favourite–longshot bias.

Version record

Bruno Jullien and Bernard Salanié, "Estimating Preferences under Risk: The Case of Racetrack Bettors"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 108(3), 503–530 (2000); DOI 10.1086/262127.

Estimates bettors' risk preferences from 34,443 British horse races with a multinomial model that goes beyond expected utility. Rank-dependent utility fits no better than expected utility, while cumulative prospect theory explains more; the preferred estimates show Friedman–Savage-style local risk aversion. Read with Snowberg and Wolfers on risk-love against misperception.

Version record

Donald B. Hausch, William T. Ziemba and Mark Rubinstein, "Efficiency of the Market for Racetrack Betting"

Difficulty: Advanced **Type:** Research paper

Status/version: Management Science, 27(12), 1435–1452 (1981); DOI 10.1287/mnsc.27.12.1435.

Win-pool inefficiencies (underbet favourites, overbet longshots) are too small to beat the track take. A technical system compares model probabilities of every finishing order with the amounts bet in the place and show pools, and finds bets with positive expected value, a weak-form inefficiency. It also sizes bets with growth-optimal reasoning, an early link from mispricing to staking.

Version record

Linda M. Woodland and Bill M. Woodland, "Market Efficiency and the Favorite–Longshot Bias: The Baseball Betting Market"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 49(1), 269–279 (1994); DOI 10.1111/j.1540-6261.1994.tb04429.x.

Weak-form tests in the legal market for Major League Baseball bets: the favourite–longshot bias appears in reverse, with favourites overbet. The deviations are too small to trade profitably once commissions are paid. Shows that the direction of the bias depends on the market.

Version record

Steven D. Levitt, "Why are Gambling Markets Organised so Differently from Financial Markets?"

Difficulty: Advanced **Type:** Research paper

Status/version: The Economic Journal, 114(495), 223–246 (2004); DOI 10.1111/j.1468-0297.2004.00207.x.

Why sports bookmakers do not act as market makers: they post a price, adjust it rarely and take large positions on outcomes. With a unique data set, Levitt shows this earns them higher profits, because they predict games better than bettors and set prices away from market clearing to exploit bettor biases. The benchmark for dealer-style event pricing.

Version record

Leighton Vaughan Williams, "Information Efficiency in Betting Markets: a Survey"

Difficulty: Advanced **Type:** Research paper

Status/version: Bulletin of Economic Research, 51(1), 1–39 (1999); DOI 10.1111/1467-8586.00069.

Surveys tests of weak, semi-strong and strong-form information efficiency in betting markets, which offer clean tests because every bet has a definite terminal value. A map of the pre-exchange literature on biases, insider information and the speed of price adjustment.

Version record

Michael A. Smith, David Paton and Leighton Vaughan Williams, "Market Efficiency in Person-to-Person Betting"

Difficulty: Advanced **Type:** Research paper

Status/version: Economica, 73(292), 673–689 (2006); DOI 10.1111/j.1468-0335.2006.00518.x.

Matched UK horse-racing data from betting exchanges and traditional bookmakers: person-to-person exchanges show significantly smaller market biases, consistent with lower transaction costs. An information-based model explains the favourite–longshot bias better than risk-preference models. The first exchange-versus-dealer comparison to read.

Version record

Egon Franck, Erwin Verbeek and Stephan Nüesch, "Prediction Accuracy of Different Market Structures—Bookmakers versus a Betting Exchange"

Difficulty: Advanced **Type:** Research paper

Status/version: International Journal of Forecasting, 26(3), 448–459 (2010); DOI 10.1016/j.ijforecast.2010.01.004.

Compares forecast accuracy across market structures on 5,478 football matches in the five largest European leagues over three seasons. The betting exchange predicts better than bookmakers, and simply betting where bookmakers' odds imply lower probabilities than the exchange earns above-average and sometimes positive returns.

Version record

Tadgh Hegarty and Karl Whelan, "Forecasting soccer matches with betting odds: A tale of two markets"

Difficulty: Research **Type:** Research paper

Status/version: International Journal of Forecasting, 41(2), 803–820 (2025); DOI 10.1016/j.ijforecast.2024.06.013.

The traditional home/draw/away market for European football shows inefficiencies and a strong favourite–longshot bias. The Asian handicap market for the same matches gives efficient forecasts once its odds are mapped into probabilities with the paper's new method. Market structure, not the event, decides calibration.

Version record

Ramon Casadesus-Masanell and Neil Campbell, "Platform competition: Betfair and the UK market for sports betting"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Economics & Management Strategy, 28(1), 29–40 (2019); DOI 10.1111/jems.12310.

Industrial-organisation case study of the UK betting industry: Betfair, an underfunded second mover, drew punters faster than the first exchange, Flutter, and competed aggressively with bookmakers while developing a complementary relationship with them. Useful for thinking about exchange entry, platform design and incumbent dealers in today's event-contract market.

Version record

Guy Elaad, J. James Reade and Carl Singleton, "Information, prices and efficiency in an online betting market"

Difficulty: Advanced **Type:** Research paper

Status/version: Finance Research Letters, 35, 101291 (2020); DOI 10.1016/j.frl.2019.09.006.

Odds from 51 online bookmakers on over 16,000 English football matches since 2010, tested with forecast-efficiency regressions. Odds are generally unbiased against outcomes, with no favourite–longshot or outcome-type bias, but individual bookmakers are inefficient: their odds do not fully use the information in competitors' odds.

Version record

Philip W. S. Newall and Dominic Cortis, "Are Sports Bettors Biased toward Longshots, Favorites, or Both? A Literature Review"

Difficulty: Intro **Type:** Research paper

Status/version: Risks, 9(1), 22 (2021); DOI 10.3390/risks9010022.

Literature review: early horse-racing studies suggested longshot bias dominates, but favourite bias appears in many sports markets. Two-outcome markets, where the favourite's probability exceeds one half, tend to show favourite bias; multi-outcome markets show longshot bias. Restricted odds ranges explain why single studies find only one. A quick orientation before testing binary event contracts.

Version record

Marco Ottaviani and Peter Norman Sørensen, "Surprised by the Parimutuel Odds?"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Review, 99(5), 2129–2134 (2009); DOI 10.1257/aer.99.5.2129.

An information explanation of the favourite–longshot bias in parimutuel markets: bettors take simultaneous positions on private information and cannot condition on the final odds. A high final market probability therefore signals favourable news, so favourites are underestimated and longshots overestimated. Short paper; read before the AEJ follow-up.

Version record

Marco Ottaviani and Peter Norman Sørensen, "Noise, Information, and the Favorite–Longshot Bias in Parimutuel Predictions"

Difficulty: Research **Type:** Research paper

Status/version: American Economic Journal: Microeconomics, 2(1), 58–85 (2010); DOI 10.1257/mic.2.1.58.

Derives testable predictions for the direction and size of the bias from the ratio of private information to noise. That ratio links the bias to observables: the number of bettors and outcomes, the amount of private information, recreational participation, the divisibility of bets, ex post noise and asymmetries across outcomes.

Version record

Andrew Meyer and Sean Hundtofte, "The Longshot Bias Is a Context Effect"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 69(11), 6954–6968 (2023); DOI 10.1287/mnsc.2023.4684.

A context-effect explanation: gambles shown by their payoffs are easier to compare on payoff than on probability, so contrasts make longshots attractive. Historical odds fit the account, and laboratory experiments show the bias disappears when bets are judged in isolation or probabilities are easy to compare. Relevant to how venues display event contracts.

Version record

Karl Whelan, "Risk Aversion and Favourite–Longshot Bias in a Competitive Fixed-Odds Betting Market"

Difficulty: Research **Type:** Research paper

Status/version: Economica, 91(361), 188–209 (2024); DOI 10.1111/ecca.12500.

Parimutuel explanations cannot explain a favourite–longshot bias in competitive fixed-odds markets. Here disagreement among gamblers and risk aversion on the part of bookmakers produce a pattern resembling the evidence. The model for dealer-quoted event prices where the quote setter carries the risk.

Version record

19.1.4 Decision, corporate and scientific markets

Bo Cowgill and Eric Zitzewitz, "Corporate Prediction Markets: Evidence from Google, Ford, and Firm X"

Difficulty: Research **Type:** Research paper

Status/version: Review of Economic Studies, 82(4), 1309–1341 (2015); DOI 10.1093/restud/rdv014.

Studies internal prediction markets at three firms. Despite thin trading and documented biases, including optimism, the markets compared well with available alternative forecasts; a reference for event prices in small, closed populations of traders.

Version record

Kenneth J. Arrow, Robert Forsythe, Michael Gorham, Robert Hahn, Robin Hanson, John O. Ledyard, Saul Levmore, Robert Litan, Paul Milgrom, Forrest D. Nelson, George R. Neumann, Marco Ottaviani, Thomas C. Schelling, Robert J. Shiller, Vernon L. Smith, Erik Snowberg, Cass R. Sunstein, Paul C. Tetlock, Philip E. Tetlock, Hal R. Varian, Justin Wolfers and Eric Zitzewitz, "The Promise of Prediction Markets"

Difficulty: Intro **Type:** Report / research note

Status/version: Science, 320(5878), 877–878 (2008); DOI 10.1126/science.1157679.

A two-page policy statement by 22 economists, among them Arrow, Milgrom, Shiller, Smith and Wolfers: the ability of groups of people to make predictions is a potent research tool and should be freed of unnecessary government restrictions. Read it for the case made in 2008 for small-stakes, research and corporate markets, before exchange-listed event contracts existed in the United States.

Version record

Robin Hanson, "Shall We Vote on Values, But Bet on Beliefs?"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Political Philosophy, 21(2), 151–178 (2013); DOI 10.1111/jopp.12008.

Proposes "futarchy": vote on values, bet on beliefs. Elected representatives would define a measure of national welfare, and speculative markets would price that measure conditional on a policy being adopted and on it not being adopted, through called-off trades; a policy would become law when markets clearly estimate it raises welfare. An engineering sketch that answers twenty-five objections rather than a formal model. The source of conditional and decision markets.

Version record

Yiling Chen, Ian Kash, Mike Ruberry and Victor Shnayder, "Decision Markets with Good Incentives"

Difficulty: Research **Type:** Research paper

Status/version: Internet and Network Economics (WINE 2011), Lecture Notes in Computer Science 7090, 72–83 (2011); DOI 10.1007/978-3-642-25510-6_7.

When a market's forecast drives a decision, only the chosen action's prediction can be scored, so experts may mislead the decision maker. The paper characterises decision markets that keep myopic incentive compatibility: the decision maker must risk taking every action, and reducing that risk raises worst-case loss. It links strictly proper decision markets to sets of prediction markets.

Version record

Martin Spann and Bernd Skiera, "Internet-Based Virtual Stock Markets for Business Forecasting"

Difficulty: Advanced **Type:** Research paper

Status/version: Management Science, 49(10), 1310–1326 (2003); DOI 10.1287/mnsc.49.10.1310.17314.

Internet-based virtual stock markets for business forecasting: design choices, three real-world applications, accuracy against expert predictions and a validity test for the forecasts. An early template for internal markets on sales or product outcomes.

Version record

Benjamin J. Gillen, Charles R. Plott and Matthew Shum, "A Pari-Mutuel-Like Mechanism for Information Aggregation: A Field Test inside Intel"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Political Economy, 125(4), 1075–1099 (2017); DOI 10.1086/692714.

A pari-mutuel-like information aggregation mechanism, developed in the laboratory, run for years inside Intel to forecast sales as probability distributions. Its predictive distributions are very accurate, especially at short horizons and in direct channels, and often beat Intel's internal forecast. Field evidence that mechanism design carries from the lab into a firm.

Version record

Emile Servan-Schreiber, Justin Wolfers, David M. Pennock and Brian Galebach, "Prediction Markets: Does Money Matter?"

Difficulty: Advanced **Type:** Research paper

Status/version: Electronic Markets, 14(3), 243–251 (2004); DOI 10.1080/1019678042000245254.

Real money against play money: TradeSports (real money) and NewsFutures (play money) prices for the 2003–04 NFL season both forecast well and beat individual humans, and the play-money market did as well as the real-money one. The authors suggest real money motivates information discovery while play money may aggregate better. Relevant when a venue or a firm chooses whether to pay out cash.

Version record

Sharad Goel, Daniel M. Reeves, Duncan J. Watts and David M. Pennock, "Prediction without markets"

Difficulty: Advanced **Type:** Research paper

Status/version: ACM Conference on Electronic Commerce (EC 2010), 357–366 (2010); DOI 10.1145/1807342.1807400.

Counter-evidence on the size of the market advantage: across thousands of sports and film events, prediction markets beat polls and simple statistical models only slightly on squared error, calibration and discrimination, and two or three parameters capture nearly all predictive power. Weigh the cost of running a market against modest benefits.

Version record

Anna Dreber, Thomas Pfeiffer, Johan Almenberg, Siri Isaksson, Brad Wilson, Yiling Chen, Brian A. Nosek and Magnus Johannesson, "Using prediction markets to estimate the reproducibility of scientific research"

Difficulty: Advanced **Type:** Research paper

Status/version: Proceedings of the National Academy of Sciences, 112(50), 15343–15347 (2015); DOI 10.1073/pnas.1516179112.

Prediction markets on whether 44 published psychology studies would replicate in the Reproducibility Project predicted the replication outcomes well and beat a survey of individual forecasts. A scientific use of event contracts and a clean test of aggregation, since outcomes were determined independently of the market.

Version record

Lionel Page and Christoph Siemroth, "An experimental analysis of information acquisition in prediction markets"

Difficulty: Research **Type:** Research paper

Status/version: Games and Economic Behavior, 101, 354–378 (2017); DOI 10.1016/j.geb.2015.11.002.

Laboratory asset markets in which traders can buy information: those with larger endowments, inconclusive information, lower risk aversion and less market experience buy more. Traders overacquire information, so informed traders lose money net of its cost, and the losses persist. More acquired information brings prices closer to fundamentals, a new explanation of prediction-market accuracy.

Version record

19.2 Event-contract venues, settlement and regulation

The spot market for event contracts: how outcomes are resolved and paid, how claims are held as fractional or tokenised positions, how the trade record is abused, and how venues are regulated.

19.2.1 Resolution, disputes and settlement data

Polymarket, How Are Markets Disputed?

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official help article dated 5 April 2026; living documentation inspected 4 October 2026.

Supplies the dispute lifecycle missing from a purely probabilistic treatment of event contracts: proposal, challenge, bond, discussion, voting and final resolution. The page describes a two-hour challenge window and outcomes including a premature proposal and an unknown/50–50 result. Reconstruct an individual market's rules, oracle dependency, dispute costs and capital lock-up before interpreting its quote as a probability. The help article describes a general process; save the particular contract wording and protocol version because exceptions and changed rules affect both payoff and timing.

Version record

Polymarket, Market Data, Data API v1 to v2 migration and Resolution; Polymarket market making; Polymarket positions and tokens; Polymarket negative-risk markets

Difficulty: Advanced **Type:** Data / specification

Status/version: Living documentation inspected 4 October 2026; listed legacy Data API routes scheduled to retire 24 October 2026, excluding the accounting-snapshot endpoint. Selected companion documents inspected 12 October 2026; earlier source verification retained separately.

Maps events to markets and outcome-token identifiers, then connects prices and order books to the correct token. The v2 Data API uses cursor pagination, snake-case fields and explicit position states; its announced v1 retirement is still future at this catalogue date, and the v1 accounting-snapshot endpoint is expressly excluded. Preserve original contract rules and resolution context. A closed market need not be resolved, and a redeemable position is not cash already received. Current resolution documentation describes CTF redemption through a collateral adapter into pUSD; record token, chain, transaction and later conversion separately. Retain historical schemas for older samples. For maker studies, join each outcome ID to its CTF or Protocol V2 position ledger. Account for complete-set split/merge, negative-risk conversion, quote cancellation and resolution separately; spread P&L and maker incentives are different cash flows.

Extended note. Version record

Kalshi, Market FAQs

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official help article dated 12 April 2026; living documentation inspected 4 October 2026.

Distinguishes trading close, official determination and payment to the cash balance. Official-source delays and contract-specific determination times can keep capital committed after an event appears resolved. The page also explains exceptional nonbinary settlements and how individual leg values enter a combination payout. Compare this venue-administered process with the oracle-dispute path in Polymarket; use actual market rules and official source data for the exercise. Illustrative settlement times are descriptive, not guaranteed deadlines, and a binary forecast does not capture every contractual payoff.

Version record

Kalshi, Historical Market Data

Difficulty: Advanced **Type:** Data / specification

Status/version: Living API documentation inspected 4 October 2026; publication/revision date not stated.

Provides a concrete archive path for settled markets, candles and trades. Query the historical cutoff separately for each data type, then join live and historical endpoints using the documented timestamp and stable identifier. User fills, orders and positions remain account-scoped. Event positions can straddle tiers during migration, so deduplicate rather than assuming an atomic snapshot. The settlements endpoint is live-only: historical position P&L is not an equivalent cash-settlement record. Use public market records for forecast and microstructure exercises; do not infer a wallet payment time from market close.

Version record

Austin K. Williams and Jack Peterson, "Decentralized Common Knowledge Oracles"

Difficulty: Research **Type:** Research paper

Status/version: Ledger, 4 (2019); DOI 10.5195/ledger.2019.166.

Defines three mechanisms for getting common-knowledge, after-the-fact truths onto a blockchain, as repeated games between queriers and a pool of reporters who are paid fees and can be punished later. Under a stated economic condition the mechanisms are individually rational and incentive compatible and decide the true outcome; reporters could make the oracle lie but are paid not to. The theory behind Augur-style reporting.

Version record

UMA, How Does UMA Work? (Optimistic Oracle and Data Verification Mechanism)

Difficulty: Advanced **Type:** Software / protocol documentation

Status/version: Living protocol documentation (GitBook), inspected 14 October 2026.

Requests carry resolution instructions, a proposer reward, a bond and a minimum challenge period. A bonded proposer posts an answer; anyone may dispute it during the challenge window, and disputes escalate to token-holder voting in the Data Verification Mechanism. The page states that the optimistic step resolves about 99.8% of requests. Read with Polymarket's dispute guide and resolution page.

Version record

Polymarket, Resolution

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living developer documentation, inspected 14 October 2026.

Two resolution sources: the UMA Optimistic Oracle for prediction markets and a Chainlink time-weighted price for up–down markets. Every market has written rules naming the resolution source, end date and edge cases, and the page warns that the rules, not the title, define the payoff. Winning tokens redeem for $1 and losing tokens become worthless.

Version record

Kalshi, Market Settlement; Market Settlement

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living API documentation, inspected 14 October 2026; settlement floors marked as not yet live.

Settlement pays $1 per net contract to the winning side, shortly after expiry but with timing that depends on market type, source data and manual review; settlement fees are zero for simple yes/no outcomes. Settlement bounds describe a planned floor for scalar markets that are partly determined early (a shutdown market paying $0.10 a day, for example): raising the floor releases collateral to YES holders before settlement. The page states the feature is not live.

Version record

19.2.2 Outcome tokens, fractional contracts and cross-venue identity

Jack Peterson, Joseph Krug, Micah Zoltu, Austin K. Williams and Stephanie Alexander, "Augur: A Decentralized Oracle and Prediction Market Platform"

Difficulty: Advanced **Type:** Research preprint

Status/version: White paper, arXiv 1501.01042, v3 (30 November 2020); DOI 10.48550/arXiv.1501.01042.

Augur white paper: a trustless oracle and prediction-market platform in which holders of the Reputation token stake on observed outcomes for settlement fees. Disputes escalate through growing bonds; past a threshold Reputation forks into one version per outcome, and only the version matching reality should keep value. Read for how a token can underwrite settlement, and why that is costly and slow.

Version record

Jonas Gebele and Florian Matthes, "Semantic Non-Fungibility and Violations of the Law of One Price in Prediction Markets"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2601.01706, v1 (5 January 2026); DOI 10.48550/arXiv.2601.01706.

The same event listed on different venues is not the same asset: without a shared event identity, liquidity does not pool and arbitrage is costly or unenforceable. A semantic alignment of descriptions, resolution rules and timing builds a human-validated cross-platform data set of over 100,000 events on ten venues (2018–2025). About 6% of events are listed on more than one venue, and equivalent markets show persistent execution-aware price gaps of 2–4% on average.

Version record

Gnosis, Conditional Tokens Contracts (documentation, version 1.0.3)

Difficulty: Advanced **Type:** Software / protocol documentation

Status/version: Developer documentation, version 1.0.3 (copyright 2018), inspected 14 October 2026.

The conditional-token framework: an oracle-defined condition with several outcomes, outcome collections and positions held as ERC-1155 balances, with operations to split collateral into a full set of positions, merge a full set back into collateral, and redeem after the oracle reports. Conditions can be nested, which is how combinatorial markets are built without thin separate order books. Polymarket's outcome tokens use this standard.

Version record

Polymarket, How Positions Work; Negative Risk Markets; Combinatorial Positions

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living developer documentation, inspected 14 October 2026.

Outcome positions are ERC-1155 balances held either in the Conditional Tokens contract or in the newer Protocol V2 position manager, with separate identifiers for each. Splitting turns one pUSD into a YES and a NO token, merging reverses it, and redemption pays winners after resolution. In negative-risk events, where only one outcome can win, a NO share in one market converts into a YES share in every other. Combinatorial positions are fully collateralised YES/NO pairs on a conjunction of legs. The tokenised claim is a fraction of a dollar of collateral.

Version record

Kalshi, Fixed-Point Representation

Difficulty: Advanced **Type:** Data / specification

Status/version: API documentation last updated 20 August 2026; inspected 14 October 2026.

Prices are fixed-point dollar strings with up to four decimals and contract quantities fixed-point strings with up to two, so fractional contracts and sub-cent ticks exist. Each market publishes its own price grid; some use a 0.1-cent tick near 0 and 1 and a 1-cent tick in between. Integer-cent fields cannot represent sub-cent prices. Data built from older cent fields will misstate prices and sizes on these markets.

Version record

19.2.3 Insider trading, manipulation and market integrity

Robin Hanson and Ryan Oprea, "A Manipulator Can Aid Prediction Market Accuracy"

Difficulty: Research **Type:** Research paper

Status/version: Economica, 76(302), 304–314 (2009); DOI 10.1111/j.1468-0335.2008.00734.x.

Adds a manipulator with a preferred price to a Kyle-style model. When others are uncertain about the target, the mean target does not move prices, and more manipulation can raise accuracy by increasing the return to informed trading; a theoretical benchmark to set against field evidence.

Version record

Itzhak Rasooly and Roberto Rozzi, "How manipulable are prediction markets?"

Difficulty: Research **Type:** Research paper

Status/version: arXiv 2503.03312, posted 5 March 2025.

A field experiment that randomly moved prices in 817 markets on Manifold, a play-money platform, and tracked them hourly. The shocks faded but remained visible 60 days later; markets with more traders, more volume or an external probability benchmark were harder to move. Transfer to real-money venues is not tested here.

Extended note. Version record

Allen Sirolly, Hongyao Ma, Yash Kanoria and Rajiv Sethi, "Network-Based Detection of Wash Trading"

Difficulty: Research **Type:** Research paper

Status/version: Working paper (Columbia Business School); first circulated November 2025; version dated 17 January 2026.

Detects wash trading as nearly closed clusters of accounts that trade mostly with each other, scored iteratively over the counterparty network. On Polymarket, wash-like activity rose from July 2024 to nearly 60% of volume in December 2024, subsided after April 2025 and returned to about 20% in early October 2025; screen volume before treating it as liquidity or interest.

Extended note. Version record

Hyun Song Shin, "Optimal Betting Odds Against Insider Traders"

Difficulty: Research **Type:** Research paper

Status/version: The Economic Journal, 101(408), 1179–1185 (1991); DOI 10.2307/2234434.

A bookmaker facing some better-informed gamblers optimally sets odds by a square-root rule: the ratio of posted prices equals the square root of the ratio of winning probabilities. Odds therefore understate favourites and exaggerate longshots, the observed bias, as a defence against insiders. The model behind Shin's method for removing the bookmaker margin.

Version record

Hyun Song Shin, "Measuring the Incidence of Insider Trading in a Market for State-Contingent Claims"

Difficulty: Research **Type:** Research paper

Status/version: The Economic Journal, 103(420), 1141–1153 (1993); DOI 10.2307/2234240.

Insider traders open a spread between buying and selling prices as intermediaries recoup losses from other traders, so the spread measures insider trading. Solving for equilibrium prices when bookmakers compete for horse-race bets gives the spread as a function of the incidence of insider trading, which is then estimated from UK data. A template for inferring informed flow from quotes.

Version record

Colin F. Camerer, "Can Asset Markets Be Manipulated? A Field Experiment With Racetrack Betting"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Political Economy, 106(3), 457–482 (1998); DOI 10.1086/250018.

A field experiment in manipulation: $500 and $1,000 win bets were placed and then cancelled at racetracks. The bets visibly moved odds relative to matched control horses and drew slightly more money to the horse, but the net effect was close to zero: some bettors inferred information and others did not, and their reactions roughly cancelled.

Version record

Jan Hansen, Carsten Schmidt and Martin Strobel, "Manipulation in political stock markets – preconditions and evidence"

Difficulty: Advanced **Type:** Research paper

Status/version: Applied Economics Letters, 11(7), 459–463 (2004); DOI 10.1080/1350485042000191700.

From a behavioural model, sets out preconditions under which political stock markets can be manipulated, then reports evidence of manipulation in two such markets during the 1999 Berlin state election. Short; a counterpoint to claims that thin political markets are a reliable substitute for polls.

Version record

Cary Deck, Shengle Lin and David Porter, "Affecting policy by manipulating prediction markets: Experimental evidence"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Economic Behavior & Organization, 85, 48–62 (2013); DOI 10.1016/j.jebo.2012.10.017.

Laboratory counter-evidence to robustness results: single-minded, well-funded manipulators can destroy a prediction market's ability to aggregate information and mislead forecasters who rely on it. The manipulators mainly move traded prices, so outstanding bids and asks stay informative. Read with Hanson and Oprea and with Rasooly and Rozzi.

Version record

Justin Wolfers, "Point Shaving: Corruption in NCAA Basketball"

Difficulty: Advanced **Type:** Research paper

Status/version: American Economic Review, 96(2), 279–283 (2006); DOI 10.1257/000282806777211757.

Forensic economics applied to college basketball: point-spread betting gives gamblers and players asymmetric, nonlinear payoffs that reward "point shaving", winning by less than the spread. The distribution of outcomes around the spread suggests the practice may be widespread. The model for how a contract's payoff design creates incentives to fix the underlying event.

Version record

Karl Whelan, "How Does Inside Information Affect Sports Betting Odds?"

Difficulty: Research **Type:** Research paper

Status/version: Scottish Journal of Political Economy, 72(5) (2025); DOI 10.1111/sjpe.70017.

Generalises Shin's model of bookmakers facing insiders. Disagreement among uninformed bettors, not insiders, generates the favourite–longshot bias; insiders lower odds but need not worsen the bias, and with realistic beliefs markets collapse if the insider share rises above low levels. Corrects the common reading of the Shin margin as an insider measure.

Version record

Joshua Mitts and Moran Ofir, "From Iran to Taylor Swift: Informed Trading in Prediction Markets"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2026); DOI 10.2139/ssrn.6426778.

Legal and empirical study of informed trading on prediction markets. Case studies on Polymarket and Kalshi (from the February 2026 U.S.–Israel strike on Iran to pre-announcement trading on a celebrity engagement) and a screen of all Polymarket markets from February 2024 to February 2026 over 210,000 suspicious wallet–market pairs: flagged traders win 69.9% of the time, with about $143 million in anomalous profit. Securities-fraud theories and CFTC Rule 180.1 fit poorly; the authors argue for registration and surveillance duties. Working paper; the screen may over- or under-include.

Version record

Siyang Liu, "Wisdom of the Crowd or Wisdom of the Insider? Insider Trading on Prediction Markets"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2026); DOI 10.2139/ssrn.6678718.

Uses Polymarket's on-chain transparency: realised profits concentrate in insiders and market makers; in markets flagged for insider trading, market makers lose much more per dollar than in matched clean markets (the Kyle–Glosten adverse-selection channel); insiders operate Sybil clusters of accounts that send winnings to one withdrawal address. The rent from information is paid largely by liquidity providers. Working paper.

Version record

19.2.4 Regulation and litigation

Congressional Research Service, "CFTC Issues Proposed Rule Regarding Prediction Markets (Legal Sidebar LSB11441)"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: CRS Legal Sidebar, 2026; underlying CFTC proposal 10 June 2026 with 45-day comment period; final rule not established in this check.

Summary of the CFTC June 2026 proposal on when event contracts are contrary to the public interest, sports-contract limits, election contracts and the circuit split over state gaming preemption.

Extended note. Version record

Karl E. Schneider and Rena S. Miller, Congressional Research Service, "Prediction Markets: Policy Issues for Congress" (In Focus IF13187)

Difficulty: Advanced **Type:** Report / research note

Status/version: CRS In Focus IF13187, version 1, 20 March 2026.

A three-page map of the current market: event contracts as binary-payoff instruments, CFTC-regulated exchanges beside an offshore venue that blocks U.S. users, sports as the largest category by volume, and open questions on state gaming law, regulatory arbitrage and insider trading.

Extended note. Version record

U.S. Commodity Futures Trading Commission, proposed rule on event contracts in the swap definition (Release 9310-26); interim final rule excluding casino-style gambling (Release 9309-26)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Press releases dated 9 October 2026; 30-day comment periods from Federal Register publication.

The proposed rule would expressly include event contracts, among them sports, political, cultural and weather contracts, in the definition of swap; the interim final rule codifies the exclusion of casino-style gambling such as sportsbook wagers and casino games, effective on publication. Read with the June 2026 proposal in the CRS legal sidebar; the swap-definition proposal is not final at this catalogue date.

Extended note. Version record

U.S. Commodity Futures Trading Commission, 17 CFR 40.11, Review of event contracts based upon certain excluded commodities

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Electronic Code of Federal Regulations, current through 9 October 2026; inspected 14 October 2026.

The rule under which a designated contract market may not list event contracts on excluded commodities that involve, relate to or reference terrorism, assassination, war, gaming or activity unlawful under federal or state law, and under which the CFTC may review a self-certified contract. Read with the court opinions on sports contracts and with the CFTC's October 2026 proposals, which would change the surrounding definitions.

Version record

U.S. Commodity Futures Trading Commission, CFTC Orders Event-Based Binary Options Markets Operator to Pay $1.4 Million Penalty (Release 8478-22)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: CFTC press release 8478-22, 3 January 2022, announcing a settled order.

Settled charges against Blockratize, Inc. (Polymarket) for offering off-exchange event-based binary options without designation as a contract market or registration as a swap execution facility. Polymarket had run more than 900 event markets on smart contracts since June 2020; the order found the contracts were swaps, imposed a $1.4 million penalty and required non-compliant markets to be wound down. The precedent that an on-chain venue is still a derivatives venue.

Version record

U.S. Court of Appeals for the Third Circuit, KalshiEX, LLC v. Flaherty, No. 25-1922 (precedential opinion)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Opinion decided 6 April 2026 (Porter, J., for the court; panel Chagares, C.J., Porter and Roth, JJ.); reported at 172 F.4th 220 per a later court filing.

Over a dissent, affirms a preliminary injunction barring New Jersey from enforcing its gambling laws against Kalshi's sports-event contracts: Kalshi showed a reasonable chance that the Commodity Exchange Act preempts state law for contracts on a CFTC-designated market. The first appellate ruling for Kalshi; New Jersey has petitioned the Supreme Court (No. 26-299). Read with the Sixth and Ninth Circuit opinions, which go the other way.

Version record

U.S. Court of Appeals for the Sixth Circuit, KalshiEX LLC v. Schuler; KalshiEX LLC v. Orgel, Nos. 26-3196/5235 (opinion 26a0272p.06)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Opinion decided and filed 25 September 2026 (Gibbons, J., for Clay, Gibbons and Bloomekatz, JJ.); recommended for publication.

Holds that Kalshi has not shown its sports-event contracts are "swaps" within the CFTC's exclusive jurisdiction, and alternatively that the Commodity Exchange Act would not preempt Ohio's or Tennessee's gambling laws even if they were. Affirms the Ohio denial of an injunction and vacates the Tennessee injunction. With the Ninth Circuit, it creates the split with the Third.

Version record

U.S. Court of Appeals for the Ninth Circuit, KalshiEX, LLC v. Assad, No. 25-7516 (opinion)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Opinion filed 28 August 2026 (R. Nelson, J.; Lee, J., concurring); for publication.

Affirms the dissolution of Kalshi's preliminary injunction against Nevada gaming regulators: Kalshi did not show a likelihood that the Commodity Exchange Act preempts state gaming law as applied to its sports-event contracts. The court staff summary sets out the swap definition, the special rule on gaming contracts and Rule 40.11.

Version record

19.3 Event-contract trading: market making, derivatives and strategies

Who supplies liquidity and how it is priced, from scoring-rule market makers to order books and request-for-quote auctions; contracts built on event prices (binary options, parimutuel auctions, combos and belief-volatility models); and the evidence on trading strategies, arbitrage and stake sizing.

19.3.1 Automated market makers and market design

Robin Hanson, "Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation"

Difficulty: Research **Type:** Research paper

Status/version: 2007

Introduces the logarithmic market scoring rule widely used in automated prediction markets. The paper connects bounded-loss liquidity provision, information aggregation and combinatorial contract design. [43]

Applications: Prediction markets; automated market making.

Version record

Robin Hanson, "Combinatorial Information Market Design"

Difficulty: Research **Type:** Research paper

Status/version: Information Systems Frontiers, 5(1), 107–119 (2003); DOI 10.1023/A:1022058209073.

Introduces market scoring rules: each trader updates the forecast of a proper scoring rule, so one automated market maker can price combinations of many events where order books would be too thin. The design origin of LMSR and later cost-function market makers.

Version record

Yiling Chen and David M. Pennock, "A Utility Framework for Bounded-Loss Market Makers"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the 23rd Conference on Uncertainty in Artificial Intelligence (UAI 2007); arXiv 1206.5252 (posted June 2012).

Derives automated market makers from a utility function: HARA-utility market makers are equivalent to scoring-rule market makers, with LMSR corresponding to exponential utility. It maps the trade-off between liquidity and worst-case loss.

Version record

Jacob Abernethy, Yiling Chen and Jennifer Wortman Vaughan, "Efficient Market Making via Convex Optimization, and a Connection to Online Learning"

Difficulty: Research **Type:** Research paper

Status/version: ACM Transactions on Economics and Computation, 1(2), 1–39 (2013); DOI 10.1145/2465769.2465777.

Proves that any market meeting a few natural conditions prices securities through a convex cost function built by conjugate duality, and makes large or infinite outcome spaces tractable by optimising over a convex hull. It also links market making to online learning.

Version record

Abraham Othman, David M. Pennock, Daniel M. Reeves and Tuomas Sandholm, "A Practical Liquidity-Sensitive Automated Market Maker"

Difficulty: Research **Type:** Research paper

Status/version: ACM Transactions on Economics and Computation, 1(3), 1–25 (2013); DOI 10.1145/2509413.2509414.

Addresses two practical failures of LMSR-style market makers: prices move by the same amount whatever the volume, and the maker typically runs at a deficit. The liquidity-sensitive design deepens with volume, keeps loss bounded and can book a profit, at the cost of prices that need not sum to one.

Version record

Christian Kroer, Miroslav Dudík, Sébastien Lahaie and Sivaraman Balakrishnan, "Arbitrage-Free Combinatorial Market Making via Integer Programming"

Difficulty: Research **Type:** Research paper

Status/version: Proceedings of the 2016 ACM Conference on Economics and Computation (EC '16), 161–178; DOI 10.1145/2940716.2940767.

Keeps the prices of logically related securities free of arbitrage by projecting them, with an integer-programming oracle, onto the set of coherent prices. The same coherence problem arises when related outcome contracts on a venue must sum to one.

Version record

Ciamac C. Moallemi, Dan Robinson and Brian Zhu, Uniform-Loss Automated Market Making for Prediction Markets; AFT 2026 proceedings volume

Difficulty: Research **Type:** Research paper

Status/version: Full preprint: 19 July 2026, v2 23 July; extended abstract in AFT 2026, LIPIcs 395, article 14, published 2 October 2026

Uses loss-versus-rebalancing to design prediction-market AMMs whose instantaneous subsidy loss is proportional to pool value across price states. Extends static win-martingale pricing to time-varying liquidity and target expected loss schedules. Read after scoring-rule market makers to distinguish a bound on total loss from control over where and when losses occur. Results are conditional on the assumed resolution process and regularity; oracle disputes, jumps, fees and strategic participants require separate modelling.

Version record

Hamed Amini, Maxim Bichuch and Zachary Feinstein, "Decentralized Prediction Markets and Sports Books"

Difficulty: Research **Type:** Research paper

Status/version: Mathematical Finance, 36(3), 465–480 (2026); DOI 10.1111/mafi.70017.

Liquidity-based AMM structure for decentralised prediction markets and sports books.

Version record

David M. Pennock, "A dynamic pari-mutuel market for hedging, wagering, and information aggregation"

Difficulty: Research **Type:** Research paper

Status/version: ACM Conference on Electronic Commerce (EC 2004), 170–179 (2004); DOI 10.1145/988772.988799.

The dynamic pari-mutuel market: a hybrid of a pari-mutuel pool and a continuous double auction. Like a pool it offers unlimited buy-in liquidity at zero risk to the operator, which pays out exactly what it takes in; like an order book it updates prices continuously and lets traders sell before resolution. Closed-form price functions for several variants. The mechanism behind early exchange-run prediction markets.

Version record

Yiling Chen and David M. Pennock, "Designing Markets for Prediction"

Difficulty: Advanced **Type:** Research paper

Status/version: AI Magazine, 31(4), 42–52 (2010); DOI 10.1609/aimag.v31i4.2313.

Survey of prediction mechanisms, from markets to peer-prediction systems, organised around the design process: objectives such as information elicitation, liquidity, bounded loss and expressiveness, and the properties that make a mechanism good. A map before the cost-function papers.

Version record

Stanko Dimitrov and Rahul Sami, "Non-myopic strategies in prediction markets"

Difficulty: Research **Type:** Research paper

Status/version: ACM Conference on Electronic Commerce (EC 2008), 200–209 (2008); DOI 10.1145/1386790.1386823.

Market scoring rules are only myopically strategyproof: a trader may profit by misleading others and correcting later. With two partially informed traders, the myopic strategy is generically not an equilibrium under LMSR; results extend to more traders and signals. A discounted scoring rule reduces bluffing, and prices converge in any equilibrium at a bounded rate.

Version record

Jianjun Gao, Zizhuo Wang, Weiping Wu and Dian Yu, "Price Interpretability of Prediction Markets: A Convergence Analysis"

Difficulty: Research **Type:** Research paper

Status/version: Operations Research, 73(1), 157–177 (2025); DOI 10.1287/opre.2022.0417.

A multivariate-utility mechanism that nests existing automated market makers, with convergence results when risk-averse traders with heterogeneous beliefs trade repeatedly with the market maker. Analytical and numerical results describe the limiting price, and an efficient approximation shows how beliefs shape it. Connects the price-interpretation question to AMM design.

Version record

Aseem Brahma, Mithun Chakraborty, Sanmay Das, Allen Lavoie and Malik Magdon-Ismail, "A bayesian market maker"

Difficulty: Research **Type:** Research paper

Status/version: ACM Conference on Electronic Commerce (EC 2012), 215–232 (2012); DOI 10.1145/2229012.2229031.

A Bayesian market maker for binary markets that learns from the information in trades. By giving up a bounded-loss guarantee it achieves lower expected loss at equal liquidity and converges quickly after jumps in beliefs, unlike LMSR and liquidity-sensitive variants. Agent-based and human-subject experiments show a trade-off between adapting to shocks and converging in equilibrium.

Version record

19.3.2 Order books, market makers and trader returns

David Rothschild and Rajiv Sethi, "Trading Strategies and Market Microstructure: Evidence from a Prediction Market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Prediction Markets, 10(1), 1–29 (2016); DOI 10.5750/jpm.v10i1.1179.

Uses account-level data for 6,300 traders in Intrade's 2012 presidential-winner market to classify strategies, from fleeting arbitrage to large one-directional positions. It presents evidence suggestive of manipulation by a single large trader, a case study in market ecology and price interpretation.

Version record

Philipp D. Dubach, "The Anatomy of a Decentralized Prediction Market: Microstructure Evidence from the Polymarket Order Book"

Difficulty: Research **Type:** Research paper

Status/version: 2026 working paper

Preprint first posted 27 April 2026; version 2, 14 May 2026. Constructs a tick-level Polymarket order-book archive joined to on-chain fills and documents spreads, depth, maker concentration and trade-direction measurement error. The finding that public-feed direction inference can be unreliable is directly relevant to anyone backtesting order-flow signals on the venue. [46]

Applications: Polymarket; microstructure, liquidity provision. Version record

Hunter Ng, Lin Peng, Yubo Tao and Dexin Zhou, "Price Discovery and Trading in Modern Prediction Markets"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 5331995 (2025); DOI 10.2139/ssrn.5331995.

Uses common 2024 U.S. presidential-election contracts on Polymarket, Kalshi, PredictIt and Robinhood. More liquid markets beat polls, yet cross-platform prices differed materially; Polymarket led Kalshi in price discovery when liquidity and activity were high, and order imbalance from large trades predicted subsequent returns.

Extended note. Version record

Constantin Bürgi, Wanying Deng and Karl Whelan, "Makers and Takers: The Economics of the Kalshi Prediction Market"

Difficulty: Research **Type:** Research paper

Status/version: CESifo Working Paper 12122 (2026); DOI 10.65864/s9kc4p0b7t. Earlier versions circulated as UCD and CEPR discussion papers in 2025.

Transaction-level evidence on more than 300,000 Kalshi contracts. Prices grow more accurate towards close but show a favourite–longshot bias: low-priced contracts win too rarely and high-priced ones earn small positive returns. A model of quote-driven trading predicts different bias patterns for makers who post offers and takers who accept them, which the data support.

Extended note. Version record

Roberto Gómez-Cram, Yunhan Guo, Theis Ingerslev Jensen and Howard Kung, "Prediction Market Accuracy: Crowd Wisdom or Informed Minority?"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper 6617059 (2026); DOI 10.2139/ssrn.6617059.

Using the full transaction record of a large platform, identified as Polymarket in a Yale summary of the paper, finds that about 3% of accounts generate most price discovery, with trades that predict later prices and outcomes. The majority supplies volume and funds the informed minority rather than producing accuracy.

Extended note. Version record

Robert Bartlett and Maureen O'Hara, "Adverse Selection in Prediction Markets: Evidence from Kalshi"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2026); DOI 10.2139/ssrn.6615739.

41.6 million Kalshi trades. Kyle's lambda and a Glosten–Harris decomposition show more informed price impact in single-name than in broad-based markets, yet effective spreads are only modestly wider and makers earn twice as much per contract. The resolution: takers overbet YES in markets that mostly settle NO, a behavioural surplus that cross-subsidises adverse selection. A VPIN-style toxicity measure predicts maker losses only in single-name markets. Working paper.

Version record

Pat Akey, Vincent Grégoire, Nicolas Harvie and Charles Martineau, "Who Wins and Who Loses In Prediction Markets? Evidence from Polymarket"

Difficulty: Research **Type:** Research paper

Status/version: SSRN working paper (2026); DOI 10.2139/ssrn.6443103.

Gains and losses of 2.4 million Polymarket users (2022–2026; 588 million trades, $67 billion volume). The top 1% capture 76.5% of all gains, yet prices are well calibrated. Winners trade with limit orders at prices favourable relative to outcomes; losses go with liquidity taking, and for about one loser in five the cost of taking liquidity alone turns profit into loss. Longshot buying marks losers but matters less after controlling for activity; performance persists only weakly. Working paper.

Version record

John R. Birge, Yifan Feng, N. Bora Keskin and Adam Schultz, "Dynamic Learning and Market Making in Spread Betting Markets with Informed Bettors"

Difficulty: Research **Type:** Research paper

Status/version: Operations Research, 69(6), 1746–1766 (2021); DOI 10.1287/opre.2021.2109.

The bookmaker's spread-setting problem as dynamic learning under information asymmetry. Bayesian and other popular learning-and-pricing policies learn well but can be manipulated by strategic informed bettors; the proposed inertial policy still gathers information from the market while protecting the market maker.

Version record

Polymarket, Market Making; Maker Rebates Program

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living developer documentation, inspected 14 October 2026.

Venue guidance for market makers: account set-up, two-sided quoting around a fair value, and the equivalence of buying NO at 0.48 with selling YES at 0.52. The rebate page explains that a share of taker fees (15–25% by category) is paid daily to makers in proportion to filled liquidity, weighted by the fee curve. Use it to compute a maker's net spread, not just the quoted one.

Version record

Polymarket, Fees

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living developer documentation, inspected 14 October 2026.

Takers pay a fee equal to shares times the fee rate times p(1 - p), with rates of 0.04–0.07 by category; makers pay nothing, and geopolitical markets are fee-free. The fee in dollars is symmetric around 50 cents and vanishes near 0 and 1, so cost per dollar staked is highest for cheap longshots. Needed before any return-by-price-bin calculation.

Version record

Kalshi, Orderbook Responses; Request for Quote (RFQ)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living API documentation, inspected 14 October 2026.

The order book returns only YES bids and NO bids, because a bid for YES at price p is an offer of NO at 1 - p; asks must be derived. The RFQ system lets a requester solicit private, full-size quotes from makers on any market, including combo (multivariate) markets, with accept-then-confirm execution and shorter timing windows for combos. The two routes by which liquidity reaches a taker.

Version record

19.3.3 Event derivatives, combos and belief volatility

Refet S. Gürkaynak and Justin Wolfers, "Macroeconomic Derivatives: An Initial Analysis of Market-Based Macro Forecasts, Uncertainty, and Risk"; NBER working paper

Difficulty: Research **Type:** Research paper

Status/version: NBER International Seminar on Macroeconomics 2005, 11–64; DOI 10.1086/653967. Working paper: NBER 11929 (January 2006).

Studies the auction-based "economic derivatives on U.S. data releases launched in 2002. Market-based expectations are similar to survey forecasts but somewhat better at predicting market responses to data surprises; the implied distributions are well calibrated and measure uncertainty, and risk aversion drives little wedge between prices and probabilities.

Version record

Justin Wolfers, "Event Derivatives"

Difficulty: Advanced **Type:** Book chapter

Status/version: Chapter in R. W. Kolb and J. A. Overdahl (eds.), *Financial Derivatives: Pricing and Risk Management*, Wiley, 157–176 (2009); DOI 10.1002/9781118266403.ch12.

Textbook chapter on event derivatives: types of prediction contracts, applications and evidence, accuracy, arbitrage, manipulation, market design and how to draw inferences from prices, in the Kolb–Overdahl derivatives text. Read to place event contracts in the derivatives toolkit rather than in gambling.

Version record

Jeffrey Lange and Nicholas Economides, "A Parimutuel Market Microstructure for Contingent Claims"

Difficulty: Research **Type:** Research paper

Status/version: European Financial Management, 11(1), 25–49 (2005); DOI 10.1111/j.1354-7798.2005.00274.x.

Parimutuel trading of contingent claims as a call auction: payouts are funded by the premiums paid, relative prices equal relative amounts wagered, and state prices are unique. The microstructure allows notional trades, limit orders and bundles of states. Investment banks used it for auctions of claims on U.S. nonfarm payrolls and euro-area inflation, the "economic derivatives" studied by Gürkaynak and Wolfers.

Version record

Nikolay Archak and Panagiotis G. Ipeirotis, "Modeling volatility in prediction markets"

Difficulty: Research **Type:** Research paper

Status/version: ACM Conference on Electronic Commerce (EC 2009), 275–284 (2009); DOI 10.1145/1566374.1566414.

Models each side's ability to win as an Itô diffusion. If the market is efficient, the contract price is also a diffusion, and its instantaneous volatility depends only on the current price and time to expiry; the result extends to more than two parties. The basis for pricing options on event contracts and for volatility forecasts.

Version record

Shaw Dalen, "Toward Black–Scholes for Prediction Markets: A Unified Kernel and Market Maker's Handbook"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2510.15205, v2 (6 April 2026); DOI 10.48550/arXiv.2510.15205.

Proposes a logit jump-diffusion with risk-neutral drift, so the traded probability is a martingale, and belief volatility, jump intensity and cross-event dependence become quotable risk factors. Calibration filters microstructure noise and separates jumps by expectation–maximisation; variance, correlation, corridor and first-passage contracts are defined on top. Single-author preprint; its forecast gains over diffusion-only baselines are self-reported.

Version record

Seungju Lee, Yebon Lee and Jaewook Lee, "Cryptocurrency prediction markets through the derivatives lens: Evidence from Kalshi and Polymarket"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Financial Analysis, 119, 105433 (2026); DOI 10.1016/j.irfa.2026.105433.

Treats crypto event contracts as cash-or-nothing digital options. On 113,338 bitcoin and ether contracts on Kalshi and Polymarket (September 2025–February 2026) it extracts implied volatility surfaces, risk-neutral densities and variance risk premia. Risk-neutral skewness is near zero with a symmetric smile; implied variance exceeds realised; Polymarket's premium is about 16 times Kalshi's, consistent with its forward-start contract design; both venues are underconfident.

Version record

Aditya Goel, "Measuring macroeconomic surprise magnitude with prediction markets: Market-implied dispersion from Kalshi"

Difficulty: Research **Type:** Research paper

Status/version: International Review of Economics & Finance, 110, 105577 (2026); DOI 10.1016/j.iref.2026.105577.

Recovers the market-implied interquartile range of CPI, payrolls and GDP outcomes from Kalshi strike ladders just before release. Implied width predicts the size of the headline-CPI surprise (N = 45) beyond standard uncertainty proxies and the Bloomberg survey dispersion; the implied median predicts direction only weakly and skew not at all. Intervals are too narrow (10–90 bands cover two-thirds) and the signal is concentrated in 2022–23. Small sample.

Version record

Niusha Moshrefi, "Prices, Probabilities, and Parlays: Systematic Bias in Sports Prediction Markets"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2607.14430, v1 (15 July 2026); DOI 10.48550/arXiv.2607.14430.

23 million Kalshi moneyline trades. Calibration varies with time to expiry: near perfect mid-life, step-like in the last ten minutes, consistent with insurance demand from holders of losing positions. Cross-game parlays are overpriced relative to the product of contemporaneous leg prices, increasingly so with more legs, even when legs are well calibrated: a markup at the parlay-pricing stage. Preprint.

Version record

Polymarket, How Combos Work; Market Makers (Combos)

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Living developer documentation, inspected 14 October 2026.

Combos join two or more outcomes into one YES/NO pair that pays only if every leg pays: a parlay. Prices come from a request-for-quote auction: makers have 400 ms to submit signed quotes, the requester 10 seconds to accept the best, and makers may get a one-second last look. The leg dependence and the quote are the maker's to price, which is where parlay markups arise.

Version record

19.3.4 Trading strategies, arbitrage and stake sizing

Tobias J. Moskowitz, "Asset Pricing and Sports Betting"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Finance, 76(6), 3153–3209 (2021); DOI 10.1111/jofi.13082.

Sports contracts carry no systematic risk and their terminal values do not depend on betting, so mispricing can be detected without a joint-hypothesis problem. It finds momentum consistent with delayed overreaction, with returns too small to overcome transaction costs; relevant now that sports is the largest exchange-listed event-contract category.

Version record

Oriol Saguillo, Vahid Ghafouri, Lucianna Kiffer and Guillermo Suarez-Tangil, "Unravelling the Probabilistic Forest: Arbitrage in Prediction Markets"; arXiv preprint

Difficulty: Research **Type:** Research paper

Status/version: 7th Conference on Advances in Financial Technologies (AFT 2025), LIPIcs vol. 354, article 27; DOI 10.4230/LIPIcs.AFT.2025.27.

Measures two kinds of arbitrage on Polymarket: within one multi-outcome market whose prices should sum to one, and across logically dependent markets found with text embeddings and a language model. From on-chain order data it estimates about $40 million of realised arbitrage profit over the study period.

Version record

John L. Kelly, Jr., "A New Interpretation of Information Rate"

Difficulty: Advanced **Type:** Research paper

Status/version: Bell System Technical Journal, 35(4), 917–926 (1956); DOI 10.1002/j.1538-7305.1956.tb03809.x.

A gambler with private information about a chance event, betting at fair odds, can grow capital exponentially at a maximum rate equal to the information rate of the channel; the result extends to arbitrary odds. The origin of log-optimal, growth-maximising stake sizing for event bets.

Version record

Matej Uhrín, Gustav Šourek, Ondřej Hubáček and Filip Železný, "Optimal sports betting strategies in practice: an experimental review"

Difficulty: Advanced **Type:** Research paper

Status/version: IMA Journal of Management Mathematics, 32(4), 465–489 (2021); DOI 10.1093/imaman/dpaa029.

Experimental review of betting-portfolio strategies built on modern portfolio theory and the Kelly criterion, with the risk-control modifications used in practice. A unified evaluation on horse racing, basketball and football shows those controls are needed, and an adaptive fractional-Kelly rule works well across settings.

Version record

Ondřej Hubáček, Gustav Šourek and Filip Železný, "Exploiting sports-betting market using machine learning"

Difficulty: Research **Type:** Research paper

Status/version: International Journal of Forecasting, 35(2), 783–796 (2019), with a 2021 erratum; DOI 10.1016/j.ijforecast.2019.01.001.

A machine-learning betting system that also penalises correlation with the bookmaker's implied probabilities, so the model finds where the odds are wrong rather than merely predicting well. Convolutional networks use player statistics, and portfolio theory allocates stakes; the method gives positive cumulative profit on NBA seasons 2007–2014. An erratum was published in 2021 (DOI 10.1016/j.ijforecast.2021.01.011).

Version record

Lisandro Kaunitz, Shenjun Zhong and Javier Kreiner, "Beating the Bookies with Their Own Numbers—and How the Online Sports Betting Market Is Rigged"

Difficulty: Advanced **Type:** Research preprint

Status/version: arXiv preprint 1710.02824, v2 (11 November 2017); DOI 10.48550/arXiv.1710.02824.

Bets where one bookmaker's odds stray from the consensus implied by all bookmakers' odds, instead of building a forecasting model. Profitable in a ten-year simulation on closing odds, a six-month minute-level simulation and five months of real-money betting, until bookmakers restricted the accounts. Code and data are public. Evidence that dealer venues limit winners, which exchanges do not.

Version record

Jonas Gebele, Timm Mutzel and Florian Matthes, "Executable Arbitrage and Market Efficiency in Prediction Markets"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2608.00666, v1 (1 August 2026); DOI 10.48550/arXiv.2608.00666.

Separates payoff-space no-arbitrage (from terminal payoffs) from protocol-executable no-arbitrage (from the position conversions available before settlement). In Polymarket's negative-risk markets the adapter converts NO into YES but not the reverse; depth-aware reconstruction finds about $1.12 million of realised arbitrage, mostly through converter strategies, and violations concentrate on the unsupported side. Pre-settlement conversion strengthens enforcement.

Version record

Guang Cheng, Jiaxin Yang and Haoxuan Zou, "Arbitrage Analysis in Polymarket NBA Markets"

Difficulty: Advanced **Type:** Research preprint

Status/version: arXiv preprint 2605.00864, v1 (22 April 2026); DOI 10.48550/arXiv.2605.00864.

75 million order-book snapshots across 173 Polymarket NBA games. Single-market arbitrage is very rare (7 executable episodes, median 3.6 seconds); combinatorial opportunities are more frequent (290 episodes, late in games) with a median return of 101 basis points, but 76.9% are limited to about 15 shares. Executable mispricings exist but are capacity-bound.

Version record

Marcos Cardozo and José Ignacio Rivero-Wildemauwe, "The Favorite-Longshot Bias in Prediction Markets: Evidence from Polymarket"

Difficulty: Research **Type:** Research preprint

Status/version: arXiv preprint 2609.12878, v1 (11 September 2026); DOI 10.48550/arXiv.2609.12878.

588 million Polymarket trades: purchases below 10 cents lose 19.3 cents per dollar, purchases at 90 cents or more earn 0.83 cents. Grouping matters: longshots lose 6.3 cents with contracts weighted equally but gain 4.1 cents once related contracts are grouped by parent event. The bias holds in crypto and politics, not sports; habitual longshot buyers earn returns similar to others. Preprint.

Version record

Joyce E. Berg and Thomas A. Rietz, "Longshots, overconfidence and efficiency on the Iowa Electronic Market"

Difficulty: Research **Type:** Research paper

Status/version: International Journal of Forecasting, 35(1), 271–287 (2019); DOI 10.1016/j.ijforecast.2018.03.004.

Iowa Electronic Markets binary contracts tested against the longshot bias and the overconfidence bias, scoring probabilities against realisation frequencies rather than single outcomes. No longshot bias appears and short-horizon prices are efficient; overconfidence may bias intermediate horizons. Sharpe ratios of static strategies and returns of dynamic ones gauge whether the inefficiencies are economically meaningful.

Version record

Part IV

Data, systems and research resources

Chapter 20

Data specifications, identifiers and sources

Use identified data specifications and versioned identifiers to reconstruct observations. Product availability, revisions and rights to use data are distinct from economic coverage.

20.1 Identifiers, disclosures and trade lifecycles

Read the source specifications before building a return panel or event replay. Observation time, publication time, revisions, identifiers, corporate actions and trade lifecycle events determine which research claims the data can support.

US Securities and Exchange Commission, EDGAR Application Programming Interfaces (APIs)

Difficulty: Advanced **Type:** Software / documentation

Status/version: Documentation, 6 June 2024; updated 8 April 2025; inspected 4 October 2026

Official submissions-history and XBRL company-concept, company-facts and frames interfaces, with nightly bulk archives. Study taxonomy, units, fiscal periods and filing identity before using accounting or text features. A frame selects the last-filed fact closest to a calendar period: a current download is therefore not an as-of historical panel. Preserve accession, dissemination time, original filing and amendments; custom-tag and entity-level coverage also limit comparisons.

Version record

FINRA, TRACE Historic Data Information and Enhanced Historical File Layouts; historical layout and reversal semantics

Difficulty: Advanced **Type:** Data / specification

Status/version: Living product documentation; inspected 4 October 2026

Transaction-level corporate, agency, Treasury and securitised-product data with product-specific layouts, dissemination lags and licensing. Enhanced files can reveal uncapped volume and counterparty categories absent from public feeds, so distinguish research access from information available to a trading strategy. Reconstruct cancellations, corrections and reversals using the applicable vintage specification; trade execution time and report time serve different purposes. Matched dealer legs and changing coverage can distort apparent liquidity if treated as independent trades.

Version record

CRSP, CRSP US Stock and Indexes Databases: Summary of CIZ Differences to Legacy Files

Difficulty: Advanced **Type:** Data / specification

Status/version: CIZ migration documentation; inspected 4 October 2026

Explains the move to monthly returns compounded from daily returns, new flags, table restructuring and metadata crosswalks. Use it when reconciling historical factor replications, delistings and corporate-action treatment across CRSP vintages. A schema migration can change return construction without any strategy change; retain the original release and mapping rules. Read with the French library's documented 2025 FIZ-to-CIZ transition rather than splicing series silently.

Version record

Global Legal Entity Identifier Foundation (GLEIF), Golden Copy and Delta Files

Difficulty: Advanced **Type:** Data / specification

Status/version: Living LEI reference-data documentation; inspected 4 October 2026

Defines entity records, direct and ultimate accounting-consolidation relationships, reporting exceptions and incremental revisions. Useful for counterparty aggregation, issuer joins and ownership-based alternative-data research. Entity identity is different from security identity, and a reporting exception is not proof of no parent. Retain dated files and both relationship validity and observation times; today's corrected ownership graph can leak later knowledge into historical exposures.

Version record

Federal Reserve Bank of New York, Additional Information about Reference Rates Administered by the New York Fed; SOFR averages and index; 2024 methodology change

Difficulty: Advanced **Type:** Data / specification

Status/version: Living rate methodology; inspected 4 October 2026

Primary methodology for SOFR, compounded averages and the SOFR Index, including publication, revision and value-date conventions. The overnight rate is published after its value date, whereas the index and averages carry the compounding end date. Check accrual periods, holidays, rounding and information availability: an overnight observation or backward-looking average is not a forward-looking term rate or a complete derivative coupon convention.

Extended note. Version record

FINOS, with ISDA, ICMA and ISLA, Common Domain Model: Overview and Event Model; documentation and event model

Difficulty: Advanced **Type:** Software / documentation

Status/version: Documentation displays version 7.0.0; inspected 4 October 2026

Machine-readable product, event, process, agreement and reference-data models connect derivatives, repo, securities lending and collateral workflows. Pin the CDM release and conventions. The documentation distinguishes the complete Java executable distribution from language distributions that may supply data types without executable functions; a shared schema does not itself establish correct valuation or legal enforceability.

Extended note. Version record

U.S. Securities and Exchange Commission and CAT NMS, LLC, "Consolidated Audit Trail (CAT) NMS Plan and reporting technical specifications"

Difficulty: Advanced **Type:** Data / specification

Status/version: Living specifications; plan amendments and cost-allocation litigation ongoing.

Order-lifecycle reporting across US equities and options; regulator-facing data, but its event model clarifies order, route and execution semantics.

Version record

Object Management Group and Bloomberg, "Financial Instrument Global Identifier (FIGI) and OpenFIGI API"

Difficulty: Advanced **Type:** Data / specification

Status/version: Living standard and API documentation.

Open instrument identifier with share-class and venue-level granularity; useful for mapping vendor identifiers.

Version record

International Organization for Standardization, "ISO 20022 Universal financial industry message scheme"

Difficulty: Advanced **Type:** Data / specification

Status/version: Living message catalogue; payments migration milestones differ by market infrastructure.

Message standard for payments, securities settlement and reporting; relevant to settlement-event data and CDM mapping.

Version record

U.S. Securities and Exchange Commission, "Form 13F and Form N-PORT structured data sets"

Difficulty: Advanced **Type:** Data / specification

Status/version: Living SEC data sets; N-PORT public with 60-day lag for quarter-end filings.

Quarterly institutional holdings and monthly fund portfolio holdings; filing lags and confidential-treatment rules govern point-in-time availability.

Version record

20.2 Data sources and research tools

Subject context and qualifications. Subject context and qualifications.

ALFRED/FRED; Bloomberg ECO; IMF, World Bank and BIS databases; CFTC *Commitments of Traders*

Difficulty: Not specified **Type:** Data / specification

Status/version: Edition/revision and publication date not established for the selected resource(s).

Macro vintages, economic releases and positioning data. ALFRED provides archival vintages; a default FRED download is not automatically point-in-time. Commercial market-data sources appear in the data directory.

Version record

Karsten Müller, Chenzi Xu, Mohamed Lehbib and Ziliang Chen, The Global Macro Database: A New International Macroeconomic Dataset; Dataset paper

Difficulty: Advanced **Type:** Data / specification

Status/version: NBER Working Paper 33714, April 2025; selected dataset release 2026_09, 30 September 2026.

Provides harmonised annual macroeconomic series across countries and long historical periods. The September 2026 release documents corrections to exchange rates, currency units and government deficits, illustrating why versioned data are essential for global macro research. Use it for regional comparisons and long-run context, keeping historical observations separate from forecasts and source reconstructions. Review academic/non-commercial licensing and any separate commercial arrangement before institutional use.

Extended note. Version record

LOBSTER

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Reconstructed Nasdaq books for microstructure research. The output specification defines seconds after midnight (precision varies by period), prices scaled by 10,000 and direction as the resting order's side. Hidden-execution IDs change across December 2010. Pin sample dates and depth; hidden liquidity and feed latency still require care.

Applications: US equities; LOB research; Commercial academic data. Version record

Kenneth French Data Library

Difficulty: Not specified **Type:** Data / specification

Status/version: Edition/revision and publication date not established for the selected resource(s).

Factor returns and sorted portfolios for benchmark replication and attribution: market, size, value, momentum, profitability and investment, among others. Record the download vintage and construction conventions; the library documents the 2025 switch from CRSP FIZ to CIZ and supplies historical archives.

Applications: Academic data. Version record

Nasdaq Data Link (formerly Quandl); Yahoo Finance; Alpha Vantage; Bloomberg and Refinitiv

Difficulty: Not specified **Type:** Data / specification

Status/version: Edition/revision and publication date not established for the selected resource(s).

Data-discovery routes ranging from quick prototypes to institutional vendor feeds. The source's WIKI Prices reference is a historical dataset lead, not a current live feed or a description of Wikipedia-sourced prices. Assess adjustments, delistings, point-in-time histories, licences and revisions for every dataset; an institutional vendor name alone does not guarantee suitability.

Applications: Data sources / discovery. Version record

Michael W. McCracken and Serena Ng, "FRED-MD: A Monthly Database for Macroeconomic Research"

Difficulty: Advanced **Type:** Data / specification

Status/version: Journal of Business & Economic Statistics, 34(4), 574–589 (2016); DOI 10.1080/07350015.2015.1086655.

Monthly vintage database of 100+ US macro series with transformation codes; standard input for factor and nowcasting models.

Version record

Òscar Jordà, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick and Alan M. Taylor, "The Rate of Return on Everything, 1870–2015"

Difficulty: Advanced **Type:** Research paper

Status/version: The Quarterly Journal of Economics, 134(3), 1225–1298 (2019); DOI 10.1093/qje/qjz012.

Long-run returns on equities, housing, bonds and bills for 16 countries, with the public Macrohistory database behind them.

Version record

Dean Croushore and Tom Stark, "A real-time data set for macroeconomists"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Econometrics, 105(1), 111–130 (2001); DOI 10.1016/s0304-4076(01)00072-0.

The Philadelphia Fed real-time data set of macro vintages, essential for honest backtests of macro signals.

Version record

Michael W. McCracken and Serena Ng, "FRED-QD: A Quarterly Database for Macroeconomic Research"

Difficulty: Advanced **Type:** Data / specification

Status/version: Review, 103(1) (2021); DOI 10.20955/r.103.1-44.

Quarterly companion to FRED-MD with about 250 series and standard transformations.

Version record

Robert J. Shiller, "Online Data: US Stock Markets 1871–Present and CAPE Ratio"

Difficulty: Advanced **Type:** Data / specification

Status/version: Updated dataset maintained by the author; check the vintage used.

Monthly S&P composite prices, dividends, earnings and CAPE since 1871; the standard long-history equity valuation dataset.

Version record

Federal Reserve Bank of Philadelphia, "Survey of Professional Forecasters"

Difficulty: Advanced **Type:** Data / specification

Status/version: Quarterly survey; ongoing.

Quarterly survey of macro forecasts since 1968, used for expectations, forecast-error and nowcasting benchmarks.

Version record

20.3 Data biases, cleaning and vendor documentation

Delisting returns, history rewriting, survivorship and reporting errors change backtests before any model is fitted. Record vendor vintages and cleaning rules with every result.

Tyler Shumway, "The Delisting Bias in CRSP Data"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 52(1), 327–340 (1997); DOI 10.1111/j.1540-6261.1997.tb03818.x.

Missing delisting returns, especially performance delistings, bias anomaly returns; prescribes replacement values.

Version record

Alexander Ljungqvist, Christopher Malloy and Felicia Marston, "Rewriting History"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 64(4), 1935–1960 (2009); DOI 10.1111/j.1540-6261.2009.01484.x.

Documents retroactive changes to I/B/E/S analyst recommendation histories that alter backtest results; a canonical point-in-time warning.

Version record

Jens Dick-Nielsen, "How to Clean Enhanced TRACE Data"

Difficulty: Advanced **Type:** Research paper

Status/version: SSRN working paper 2337908 (2013); DOI 10.2139/ssrn.2337908. Working-paper record; no journal version established in this check.

Step-by-step filters for cancellations, corrections, reversals and agency duplicates in Enhanced TRACE.

Extended note. Version record

Edwin J. Elton, Martin J. Gruber and Christopher R. Blake, "Survivor Bias and Mutual Fund Performance"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 9(4), 1097–1120 (1996); DOI 10.1093/rfs/9.4.1097.

Quantifies survivorship bias in fund databases and its effect on performance persistence tests.

Version record

William Fung and David A. Hsieh, "Performance Characteristics of Hedge Funds and Commodity Funds: Natural versus Spurious Biases"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Financial and Quantitative Analysis, 35(3), 291 (2000); DOI 10.2307/2676205.

Survivorship, instant-history and selection biases in hedge-fund and CTA databases.

Version record

Hendrik Bessembinder, Kathleen M. Kahle, William F. Maxwell and Danielle Xu, "Measuring Abnormal Bond Performance"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 22(10), 4219–4258 (2009); DOI 10.1093/rfs/hhn105.

Compares bond event-study methods and shows daily TRACE data improve power over monthly data.

Version record

Wharton Research Data Services, "WRDS platform and dataset documentation (CRSP, Compustat, TAQ, OptionMetrics, TRACE)"

Difficulty: Advanced **Type:** Data / specification

Status/version: Living documentation; institutional subscription required.

Access layer and documentation for the main academic financial datasets; variable definitions, vintages and point-in-time availability must be checked dataset by dataset.

Version record

OptionMetrics, "IvyDB US Reference Manual"

Difficulty: Advanced **Type:** Data / specification

Status/version: Living vendor documentation; version varies by subscription.

Definitions of option prices, implied volatilities, Greeks, volatility surfaces and dividend projections; methodology changes affect historical comparisons.

Version record

Pat Akey, Adriana Z. Robertson and Mikhail Simutin, "Noisy factors? The retroactive impact of methodological changes on the Fama–French factors"

Difficulty: Research **Type:** Research paper

Status/version: Review of Finance, 30(5), 1437–1485 (2026); DOI 10.1093/rof/rfag002.

Fama–French factor returns differ depending on download date, mainly because of construction changes rather than data revisions, with material effects on fund-performance and asset-pricing results. Version-pin your factor data.

Version record

Stephen J. Brown, William Goetzmann, Roger G. Ibbotson and Stephen A. Ross, "Survivorship Bias in Performance Studies"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 5(4), 553–580 (1992); DOI 10.1093/rfs/5.4.553.

Survivorship alone can create apparent performance persistence; the classic warning for manager and strategy databases.

Version record

Mark M. Carhart, Jennifer N. Carpenter, Anthony W. Lynch and David K. Musto, "Mutual Fund Survivorship"

Difficulty: Advanced **Type:** Research paper

Status/version: Review of Financial Studies, 15(5), 1439–1463 (2002); DOI 10.1093/rfs/15.5.1439.

Quantifies survivorship bias in mutual fund returns and its effect on persistence tests.

Version record

Richard B. Evans, "Mutual Fund Incubation"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 65(4), 1581–1611 (2010); DOI 10.1111/j.1540-6261.2010.01579.x.

Funds incubated privately and launched after good performance create upward-biased backfilled returns.

Version record

Edwin J. Elton, Martin J. Gruber and Christopher R. Blake, "A First Look at the Accuracy of the CRSP Mutual Fund Database and a Comparison of the CRSP and Morningstar Mutual Fund Databases"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 56(6), 2415–2430 (2001); DOI 10.1111/0022-1082.00410.

Omission bias in the CRSP mutual fund database, especially for small funds; compare vendors before relying on one.

Version record

Ozgur S. Ince and R. Burt Porter, "Individual Equity Return Data from Thomson Datastream: Handle with Care!"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Research, 29(4), 463–479 (2006); DOI 10.1111/j.1475-6803.2006.00189.x.

Datastream coverage, rounding and padded-price errors and the screens needed for international equity research.

Version record

S. P. Kothari, Jay Shanken and Richard G. Sloan, "Another Look at the Cross-Section of Expected Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 50(1), 185 (1995); DOI 10.2307/2329243.

Compustat selection (backfill) bias inflates book-to-market effects; beta looks better with annual returns.

Version record

Rolf W. Banz and William J. Breen, "Sample-Dependent Results Using Accounting and Market Data: Some Evidence"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Finance, 41(4), 779 (1986); DOI 10.2307/2328228.

Look-ahead and ex-post selection bias in Compustat-based studies change measured earnings-yield and size effects.

Version record

William Beaver, Maureen McNichols and Richard Price, "Delisting returns and their effect on accounting-based market anomalies"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Accounting and Economics, 43(2-3), 341–368 (2007); DOI 10.1016/j.jacceco.2006.12.002.

Missing delisting returns bias accounting anomalies; shows how to impute them.

Version record

Juhani T. Linnainmaa and Michael R. Roberts, "The History of the Cross-Section of Stock Returns"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Financial Studies, 31(7), 2606–2649 (2018); DOI 10.1093/rfs/hhy030.

Hand-collected pre-1963 accounting data show most anomalies are weaker out of sample, consistent with data snooping.

Version record

Andrew J. Patton, Tarun Ramadorai and Michael Streatfield, "Change You Can Believe In? Hedge Fund Data Revisions"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 70(3), 963–999 (2015); DOI 10.1111/jofi.12240.

Hedge funds revise previously reported returns in commercial databases, usually downwards; revisers subsequently underperform.

Version record

Nicolas P. B. Bollen and Veronika K. Pool, "Do Hedge Fund Managers Misreport Returns? Evidence from the Pooled Distribution"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 64(5), 2257–2288 (2009); DOI 10.1111/j.1540-6261.2009.01500.x.

A discontinuity around zero in reported hedge-fund returns suggests managers avoid reporting small losses.

Version record

Chapter 21

Software and research infrastructure

Software provides reference implementations and infrastructure, not automatic validation. Papers and their implementation repositories retain distinct identities and maintenance conditions.

21.1 Libraries, platforms and simulation

Subject context and qualifications.

21.1.1 Pricing, rates and risk libraries

QuantLib / repository

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

The core open-source C++ library for curves, instruments, calendars, pricing engines and risk calculations. Select release 1.43, 14 July 2026: its release notes include inflation-index interpolation deprecations, additional overnight-rate/calendar support and cross-currency swaps. Useful for reference implementations, with independent market-data, convention and numerical validation; the release was inspected, not installed or tested. [47]

Applications: Rates, credit, equity/FX derivatives, fixed income. Version record

OpenGamma Strata

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

A production-orientated Java library for market conventions, curve calibration, pricing and risk. It is particularly useful for rates and multi-curve systems where conventions and lifecycle handling are as important as model equations.

Applications: Rates, FX, credit, commodities. Version record

finmath-lib

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

A Java library with Monte Carlo, term-structure and stochastic-calculus components. It is useful for transparent numerical research and model validation, especially in rates and XVA-related projects.

Applications: Rates, derivatives, Monte Carlo. Version record

rateslib

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

A Python library for curves, rates instruments, calendars and automatic differentiation. The current official documentation accompanies source-available non-commercial and paid commercial licensing; it is not an unrestricted open-source library. Use selected-version examples for reference pricing, subject to independent convention/numerical checks and the applicable licence.

Applications: Rates, FX, fixed income. Version record

gs-quant

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Goldman Sachs' open Python toolkit for market data abstractions, risk, derivatives analytics and portfolio workflows. It provides insight into production-style interfaces, though some functionality naturally depends on proprietary services or data.

Applications: Derivatives, risk, portfolios. Version record

Acadia / Open Source Risk Engine contributors, "Open Source Risk Engine (ORE)"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

QuantLib-based exposure simulation, xVA, SIMM and market-risk analytics.

Version record

21.1.2 Portfolio and risk engines

CVXPY

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

The standard Python modelling language for convex optimisation. It is invaluable for portfolio, execution and calibration prototypes, but researchers must understand solver tolerances, duals, scaling and whether the formulation is genuinely convex.

Applications: Portfolio, risk, execution, calibration. Version record

PyPortfolioOpt

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Implements conventional efficient-frontier, Black–Litterman and hierarchical allocation methods. It is useful for baselines and teaching internal users, not a replacement for a desk's constrained, cost-aware and tax-/liquidity-aware optimiser.

Applications: Multi-asset portfolio construction. Version record

Riskfolio-Lib

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Provides a broad set of convex risk measures, clustering and portfolio formulations. It is valuable for rapid comparison of allocation objectives, subject to independent checking of estimators and solver behaviour.

Applications: Multi-asset risk and allocation. Version record

Enzo Busseti, Stephen Boyd et al., "Cvxportfolio"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

Multi-period portfolio optimisation and backtesting with transaction and holding costs, implementing Boyd et al. (2017).

Version record

skfolio contributors, "skfolio"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

scikit-learn-compatible portfolio optimisation with cross-validation, risk measures, HRP and distributionally robust models.

Version record

21.1.3 Statistical and ML libraries

Microsoft Qlib

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

An integrated AI-oriented research platform covering data ingestion, technical-factor generation, forecasting (including LSTM templates), model and hyperparameter experiments, backtests, performance evaluation, portfolio construction, online learning, concept drift and RL execution. The Microsoft Research paper describes the platform design. It now also connects to Microsoft's RD-Agent work on automated factor and model research; institutional users should replace public demonstration data and independently validate its backtesting assumptions. [48]

Applications: Equities; ML, alpha research, execution. Version record

RD-Agent

Difficulty: Not specified **Type:** Software / documentation

Status/version: 2024–2026

A multi-agent framework for automating data-centric research and factor/model optimisation. It is relevant to quant research productivity, but autonomous experimentation substantially increases multiple-testing, provenance and governance risks. [48]

Applications: Automated factor research, model generation. Version record

DeepLOB code

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

A reference implementation for reproducing the influential architecture. It should be treated as a model benchmark rather than a production trading stack, with evaluation rebuilt around point-in-time event sequencing and costs.

Applications: LOB prediction; code. Version record

TimesFM

Difficulty: Not specified **Type:** Software / documentation

Status/version: Version 2.5, 2025–2026

A pretrained time-series foundation model with current support for long contexts, quantile forecasts, covariates and fine-tuning examples. Its Apache-licensed open repository is suitable for benchmarking, but Google states that the open version is not an officially supported product. [23]

Applications: General time-series and risk forecasts. Version record

Chronos

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Amazon's open implementation of pretrained token-based time-series models. It is useful for probabilistic baseline comparisons and transfer learning across sparse financial series.

Applications: Forecasting, probabilistic scenarios. Version record

iTransformer

Difficulty: Not specified **Type:** Software / documentation

Status/version: 2024

Official implementation of variate-token attention for multivariate time series. It is a compact research baseline for yield curves, volatility surfaces and asset panels. [52]

Applications: Multivariate forecasting. Version record

MASTER

Difficulty: Not specified **Type:** Software / documentation

Status/version: 2024

Official code for market-guided cross-sectional stock forecasting. It is useful for reproducing recent transformer results and examining market-state conditioning within Qlib-style datasets. [53]

Applications: Equity ranking and alpha. Version record

DoubleML

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Implements double/debiased ML with sample splitting and orthogonal scores. It is the preferred open-source starting point for high-dimensional causal estimation where the identification design has already been established.

Applications: Causal ML, policy evaluation. Version record

EconML

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Provides heterogeneous treatment estimators, orthogonal learners and policy tools. It is appropriate for client-level or liquidity-state treatment effects, but not a substitute for randomisation or valid instruments.

Applications: Causal ML, pricing and experimentation. Version record

grf

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

The reference implementation of generalized random forests. It is particularly useful for heterogeneous treatment, instrumental-variable and local parameter estimation.

Applications: Causal inference. Version record

mlfinlab

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Implements financial-ML procedures including dollar/volume bars, event labelling, sampling and bootstrapping, fractional differentiation, entropy features, purged cross-validation, bet sizing and backtest diagnostics such as deflated Sharpe and probability of overfitting. Check which modules are available in the chosen release. Review the selected version and access rights; original papers remain the methodological authority.

Applications: Financial ML and validation; Commercial; public issue repository. Extended note. Version record

pandas; TA-Lib; pandas_market_calendars

Difficulty: Research **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Data transforms, technical indicators, trading sessions, holidays and date alignment. The pandas 3.0.0 release notes (21 January 2026) document new default string types, Copy-on-Write and datetime-resolution inference; earlier examples can change behavior during migration. Calendar versions, time zones, daylight saving and early closes affect features and execution. Indicators and rolling windows must respect availability time.

Version record

NLTK, spaCy, BERT / transformer models

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Text processing, entity recognition and representation learning for news, filings and social data. Record model/training vintages, languages and document timestamps; evaluate dictionary baselines and downstream net returns. Explanations or sentiment labels are not causal identification.

Applications: NLP methods / tools.

Version record

AI4Finance Foundation

Difficulty: Research **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Research projects and examples for financial ML and reinforcement learning. The source's generic reference to deep learning on limit-order books does not identify a specific implementation; use the DeepLOB entry for a concrete order-book benchmark.

Version record

Skipper Seabold, Josef Perktold et al., "statsmodels"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

Regression, time-series (ARIMA, VAR, state-space), robust covariance and statistical tests in Python.

Version record

Kevin Sheppard et al., "arch: ARCH models, bootstrap and multiple comparison in Python"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

GARCH-family volatility models, stationary and circular block bootstraps, SPA and model-confidence-set tests.

Version record

Kevin Sheppard et al., "linearmodels"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

Panel regressions with clustered errors, Fama–MacBeth, IV/GMM and linear factor-model estimation.

Version record

scikit-learn developers, "scikit-learn"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

Standard Python machine-learning toolkit; TimeSeriesSplit and pipelines help prevent leakage when used correctly.

Version record

Microsoft and contributors, "LightGBM (with XGBoost as alternative)"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

Fast gradient-boosted trees for large tabular panels; standard baseline in return-prediction competitions.

Version record

PyTorch Foundation; Google JAX team, "PyTorch and JAX"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source projects; JAX at https://github.com/jax-ml/jax.

Deep-learning frameworks with automatic differentiation; JAX also suits differentiable pricing and Monte Carlo.

Version record

PyMC developers; Stan Development Team, "PyMC and Stan"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source projects; Stan at https://mc-stan.org/.

Probabilistic programming for Bayesian hierarchical, stochastic-volatility and shrinkage models.

Version record

Polars contributors, "Polars"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

Columnar DataFrame engine with lazy query optimisation and as-of joins for large point-in-time panels.

Version record

Nixtla, "statsforecast, neuralforecast and related forecasting libraries"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source projects.

Fast statistical baselines (ETS, ARIMA, Theta) and neural forecasters with cross-validation utilities.

Version record

Optuna contributors, "Optuna"

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active open-source project.

Hyperparameter search with trial logging; records the number of configurations tried for multiple-testing adjustment.

Version record

21.1.4 Backtesting, trading and simulation platforms

QuantConnect LEAN (platform)

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

A C# event-driven engine with Python support, multi-asset backtesting and live brokerage integrations. Research notebooks, fee/slippage/fill models and brokerage buying-power rules support prototypes and accounting tests. Data, alternative-data subscriptions and Quantpedia integrations have separate access conditions; the engine does not come with every historical dataset. Institutional HFT and OTC workflows need additional venue and lifecycle models.

Applications: Equities, options, futures, FX, crypto. Version record

NautilusTrader

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active, 2026 release line

A Rust-native, deterministic event-driven engine designed for research-to-live parity, nanosecond timestamps and multi-venue trading. It is one of the strongest public foundations for crypto, futures and multi-asset event-driven systems, but adapter and fill-model fidelity still need venue-by-venue validation. [49]

Applications: Crypto, futures, FX, equities, options, betting. Version record

HftBacktest contributors, HftBacktest: Order Fill

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living documentation; no fixed publication date asserted. Selected order-fill documentation inspected 12 October 2026; pin software and documentation release.

A focused execution-model reading on full versus partial fills and queue assumptions in market-data replay. The replay cannot alter recorded depth or trades, so a simulated taker order can appear to consume liquidity that remains in the subsequent data.

Extended note. Version record

ABIDES-JPMC

Difficulty: Not specified **Type:** Software / documentation

Status/version: Archived public fork; 2 June 2025

A multi-agent discrete-event simulator with explicit messaging and latency models, divided into core, markets and Gym components. It is valuable for market-design and RL experiments where counterfactual participant interaction is required, not for claiming that a synthetic market exactly reproduces a live venue. [50]

Applications: Equity-style LOBs; market design, RL, execution. Version record

JAX-LOB

Difficulty: Not specified **Type:** Software / documentation

Status/version: 2023

A GPU-accelerated limit-order-book simulator and RL environment designed for large-scale policy training. Its speed is attractive for execution and market-making research; validation should focus on queue semantics, calibration and the sim-to-real gap. [51]

Applications: LOB execution, market making, RL. Version record

Alphalens

Difficulty: Not specified **Type:** Software / documentation

Status/version: Analysis / historical code

Cross-sectional factor analysis: forward-return alignment, rank IC, quantile returns, spreads, turnover and tear sheets. Check sector/group treatment, missingness and execution lag. Useful for evaluating single factors and combined value/momentum forecasts; diagnostics do not themselves constitute an executable strategy.

Version record

Backtrader

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Python event-driven strategy experiments and intraday prototypes. Validate data granularity, fill timing, multiple feeds and broker assumptions. Aggregating tick data into bars can erase queue and latency effects; use a venue-level simulator for those questions.

Applications: Supplementary backtester. Version record

Zipline; Zipline-reloaded

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Historical Quantopian backtester and a separate community continuation. Pipeline-style factor research and integration with factor/performance analytics make these useful companions to the lecture archive and Jansen. Pin bundles, calendars and corporate-action handling. Live order routing requires additional infrastructure.

Applications: Backtesting / community code. Version record

PyFolio, QuantStats, Empyrical

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Portfolio tear sheets and performance statistics, including Sharpe, drawdown, exposure and benchmark comparisons. Reconcile gross/net returns, cash flows, risk-free rates, sampling intervals and annualisation; use factor regressions for unintended beta and style exposure. Legacy examples may need dependency migration.

Applications: Performance analytics. Version record

quantstrat (R); Marketcetera

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Strategy-framework and trading-platform leads named in the HFT section. Evaluate the selected project's event handling, data adapters and execution facilities; neither the name nor intraday support establishes microsecond simulation accuracy. LOBSTER, ABIDES and JAX-LOB address different data/simulation needs.

Applications: Supplementary infrastructure. Version record

KX kdb+ / q

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Time-series database and language for storing and querying tick data. Learn time-based joins, aggregation and event sequencing. Consult current licensing and platform documentation; the source's “free 32-bit version” is an historical access claim, not a current entitlement.

Applications: Tick-data engineering. Version record

21.1.5 Crypto and on-chain tooling

CCXT

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Normalises REST and WebSocket interfaces across crypto exchanges, with bindings in multiple programming languages for market-data ingestion and order submission. Useful for multi-exchange research and lower-frequency connectivity; exchange coverage and endpoint capabilities must be checked individually. See also the Upbit integration guide. Native venue semantics remain necessary for low-latency work.

Applications: Crypto CEX data and trading. Version record

Cryptofeed

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Provides normalised streaming trades, order books and tickers through Python callbacks, handling WebSocket connections across crypto venues. It is valuable for cross-exchange data infrastructure, provided sequence gaps, snapshots, timestamp domains and exchange-specific update rules are monitored explicitly.

Applications: Crypto microstructure. Version record

Ethereum ETL

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Extracts blocks, transactions, token transfers, contracts and related on-chain entities into CSV and database-oriented pipelines; see the official documentation for export and streaming workflows. A useful foundation for on-chain factors and protocol monitoring, although chain reorganisations, traces and entity attribution require additional handling.

Applications: Ethereum and EVM on-chain data. Version record

Foundry contributors, Foundry: Anvil, forking and transaction tracing

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living documentation; no fixed publication date asserted. Anvil overview inspected 12 October 2026; pin release and fork block for an exercise.

Adds EVM execution replay to the catalogue's extraction tools. Anvil supports a local node, fixed-block forks, controlled mining, saved state and transaction traces. Use it to verify exact pool arithmetic, gas and reverts against a specified pre-state.

Extended note. Version record

Freqtrade

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Supplementary Python framework for developing and backtesting exchange-connected crypto strategies, with parameter optimisation and machine-learning workflows. Useful for learning and intermediate strategy prototypes. Its simulation assumptions and retail deployment model require independent scrutiny before institutional use; it is not evidence of HFT execution fidelity.

Applications: Crypto strategy prototyping and backtesting; Supplementary. Version record

Hummingbot

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

Provides connectors and strategy infrastructure for centralised and decentralised crypto market making. It is suitable for prototyping venue interactions; serious institutional deployment requires independent risk, key management, latency and reconciliation systems.

Applications: Crypto market making and arbitrage. Version record

Flashbots repositories and specifications

Difficulty: Not specified **Type:** Software / documentation

Status/version: Active

The primary open technical ecosystem for Ethereum block building, relay infrastructure, MEV research and transaction-ordering mechanisms. These repositories are indispensable for protocol-aware execution research but evolve too rapidly to be treated as stable venue specifications.

Applications: Ethereum MEV and block markets. Version record

CCXT and native cryptocurrency exchange APIs

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

See CCXT and the core digital-asset infrastructure entries.

Applications: Covered in infrastructure. Version record

21.2 Systems and performance engineering

Protocol and data-system references support research infrastructure. Performance diagnosis, BPF tracing and C++ concurrency are optional specialist readings for HFT, execution and infrastructure work; select them for the systems being studied.

FIX Trading Community, *Online Specification*

Difficulty: Advanced **Type:** Software / documentation

Status/version: FIX Latest is a living versioned specification; exact current extension-pack cut was not established because the cited online page did not load on 4 October 2026.

Versioned normative FIX message semantics are foundations for order lifecycle and production debugging. The 2019 FIX Trading Community note distinguishes the evolving FIX Latest application specification from session-layer and encoding standards; use only the protocol/extension-pack and counterparty rules of engagement actually deployed. The current cited specification page did not load during this check, so its current EP cannot be asserted.

Version record

KX, *Technical White Papers: kdb+ and q*

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living archive

Production time-series patterns for tick capture, partitioning, compression, throughput profiling and analytics. Extends the kdb+/q platform mention with the specific source-recommended white-paper archive.

Version record

KX, *PyKX Documentation*

Difficulty: Advanced **Type:** Software / documentation

Status/version: Living documentation

Python-to-q/kdb+ bridge covering embedded q, IPC and analytics. Study deployment modes and constraints when connecting Python research to an kdb+ estate, including where computation and data transfer occur.

Version record

Brendan Gregg, *Systems Performance: Enterprise and the Cloud*, 2nd ed.

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2020

Performance diagnosis, profiling, observability and bottleneck analysis. Apply the methods to trading-system tail latency, dropped packets, garbage-collection pauses and capacity limits; investigate distributions and contention rather than relying only on average throughput.

Version record

Brendan Gregg, *BPF Performance Tools*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Kernel and application instrumentation for root-cause analysis of latency and throughput regressions. Complements systems-level observability with detailed tracing; match tools and overhead to the operating system and environment being measured.

Applications: specialist tracing. Version record

Titus Winters, Tom Manshreck and Hyrum Wright, *Software Engineering at Google*

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2020

Engineering practices for long-lived code: testing, maintainability, reliability and coordinated change. Especially relevant to quant organisations where exploratory code becomes shared infrastructure and production behaviour must remain understandable over time.

Version record

Anthony Williams, *C++ Concurrency in Action*, 2nd ed.

Difficulty: Advanced **Type:** Book / monograph

Status/version: 2nd edition

The C++ memory model, concurrency primitives and thread coordination for market-data handlers, execution systems and performance-critical libraries. Use to reason about races, ordering and hard-to-reproduce concurrency failures.

Version record

Martin Kleppmann and Chris Riccomini, *Designing Data-Intensive Applications*, 2nd ed.

Difficulty: Advanced **Type:** Book / monograph

Status/version: Second edition, February 2026.

Storage/streaming, consistency, replication and failure modes in data systems; publisher identifies the second edition as February 2026.

Version record

Ulrich Drepper, "What Every Programmer Should Know About Memory"

Difficulty: Advanced **Type:** Report / research note

Status/version: Red Hat technical paper, 21 November 2007.

Caches, NUMA, TLBs and memory bandwidth; background for low-latency market-data and order-handling code.

Version record

Sourav Ghosh, *Building Low Latency Applications with C++*

Difficulty: Advanced **Type:** Book / monograph

Status/version: Packt, July 2023 (ISBN 9781837639359).

Builds an exchange, market-data publisher and trading client in C++ with latency measurement (Packt 2023).

Version record

21.3 Data architecture and experiment lineage

These references cover point-in-time data, reproducible experiments, model review and simulation. Select technical documentation according to the research and operational requirements. These are compact examples for readers evaluating telemetry, service deployment or credential handling. Choose tools according to scale, operational boundaries and controls. Existing Systems Performance remains the broader performance-diagnosis reading.

Martin Fowler, *Bitemporal History* (2021)

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Published 7 April 2021; page lists March 2021 drafting/review dates.

Explains actual/valid time alongside record/transaction time and retroactive changes. Apply the distinction to release dates, corrections, late observations and historical as-of queries; timestamps alone do not preserve overwritten versions.

Version record

Apache Kafka; Apache Airflow

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Optional documentation for event streaming and scheduled workflow orchestration, respectively. Airflow users should distinguish the current release notes from historical 2.x examples; the stable page lists 3.3.2 (17 September 2026), while 3.3.1 describes pandas 3 serialization compatibility. Release inspection does not validate any installed stack; major-version APIs and deployment assumptions are version-sensitive. Relevant to continuous collection, data builds, replay and backfills where scale or operational requirements justify these tools.

Version record

Apache Parquet; Apache Iceberg

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Columnar file storage and analytical table metadata, snapshots and schema/partition evolution. Table snapshots do not themselves establish when financial information became available to a strategy.

Version record

DuckDB; QuestDB; PostgreSQL / hypertables; Tiger Data table-modelling guide

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Local analytical SQL and optional online time-series query/storage references. The linked single-versus-multiple-hypertable guidance is product-specific; evaluate online serving separately from the historical archive and retain financial availability/revision semantics.

Version record

Protocol Buffers; gRPC; OpenAPI

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Typed cross-language messages, remote procedure calls and HTTP API descriptions/client generation.

Version record

Apache Arrow; Arrow Flight

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Columnar in-memory interchange and its data-transfer RPC. Consider these when a real cross-language or service boundary needs them; the source treats Arrow Flight as an optional performance path.

Version record

Great Expectations / GX

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Documentation for versioned checks on data tables and events. Attach check outcomes to dataset-building and promotion evidence; the original documentation URL is retained without asserting that its navigation or version is current.

Version record

MLflow; 3.16.0 release notes

Difficulty: Not specified **Type:** Software / documentation

Status/version: Selected release 3.16.0, 3 September 2026; living documentation inspected 4 October 2026.

Experiment tracking, artefacts, model versions and registry lineage. Useful for retaining attempted variants and their inputs; registry status does not replace independent model validation or a promotion decision.

Version record

Feast; Feast point-in-time joins

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

References for feature reuse and historical-versus-online access. Feature-store joins still need the financial release, availability and revision rules of the underlying data. Consider a feature store when shared serving needs justify it.

Version record

Bailey–López de Prado, Deflated Sharpe Ratio; Marcos López de Prado, *The 10 Reasons Most Machine Learning Funds Fail* (version 27 January 2018; first version 25 December 2017)

Difficulty: Not specified **Type:** Mixed resources

Status/version: Selected citation year(s): 2018, 2017. Version distinctions remain in the annotation.

Read with the financial-ML validation material for selection bias and time-series evaluation. The 2018-version article discusses research-process failure modes; such controls do not certify future returns.

Version record

Zipline; Backtrader; VectorBT

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Simulation-engine references spanning event-driven and vectorised research. Zipline and Backtrader already have canonical entries; VectorBT is the additional vectorised reference. Compare them using explicit accounting and timing assumptions; the original Quantopian Zipline is not a complete live-trading deployment.

Version record

OpenTelemetry; OpenTelemetry Collector; Prometheus; Alertmanager; Grafana; Grafana Loki

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Optional telemetry examples spanning traces, metrics, logs, dashboards and alert routing. OpenTelemetry and its Collector describe instrumentation/export paths; Prometheus/Alertmanager and Grafana/Loki are separate metric, alert and log components. The grouped references are not a mandatory vendor stack.

Version record

Docker / Docker Compose; Kubernetes; HashiCorp Vault

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Optional container, orchestration and secret-management references. Evaluate orchestration and credential rotation only where the deployment needs them.

Version record

21.4 Research-to-production practice

Related software and technical documentation: Sections 21.1, 21.2 and 21.3.

Martin Zinkevich, *Rules of Machine Learning: Best Practices for ML Engineering*

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Edition/revision and publication date not established for the selected resource(s).

Practitioner engineering guidance on metrics, simple baselines, feature consistency and training/serving skew; financial labels also require time-aware evaluation.

Applications: Applied engineering guide; 43 rules. Version record

Chip Huyen, *Designing Machine Learning Systems* (2022), and "Design a machine learning system"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2022. Version distinctions remain in the annotation.

Applied ML system design across data, modelling, serving and maintenance. The author notes complement the distinct 2022 book.

Applications: Applied book and author notes. Version record

Alexandra Posoldova, "Machine learning pipelines: From research to production"

Difficulty: Not specified **Type:** Research paper

Status/version: , 2020

Data processing and engineering considerations when transferring an ML prototype into a service.

Version record

Lester Solbakken, "From research to production: scaling a state-of-the-art machine learning system"

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Vespa engineering case study, 2020

Engineering case study of the latency, cost, retrieval and model-compression trade-offs in deploying a question-answering system.

Version record

Microsoft Team Data Science Process (TDSP); CRISP-DM; KDD

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Historical process documentation connecting business criteria, data acquisition, modelling and deployment; related CRISP-DM and KDD frameworks are conceptual context. The Microsoft repository was archived on 12 October 2023 and is read-only; retain it as a process reference, not a maintained implementation.

Applications: Enterprise process and comparison leads. Version record

D. Sculley et al., "Machine Learning: The High Interest Credit Card of Technical Debt" (2014); Eric Breck et al., "The ML Test Score" (2017)

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2014, 2017. Version distinctions remain in the annotation.

The 2014 technical-debt paper is distinct from Hidden Technical Debt (2015). Breck et al. offer a production-readiness rubric covering tests and monitoring.

Version record

Rishi K. Narang, Inside the Black Box: The Simple Truth About Quantitative Trading

Difficulty: Intro **Type:** Book / monograph

Status/version: Edition/revision and publication date not established for the selected resource(s).

Accessible overview of quant-fund idea generation, research, backtesting, risk, portfolio calibration and technology. Use it to connect data cleaning, version control and research-to-live consistency with the trading process rather than treating a fitted signal as a complete system.

Version record

Keith Fitschen, Building Reliable Trading Systems: Tradable Strategies That Perform as They Backtest and Meet Your Risk-Reward Goals (2013)

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2013. Version distinctions remain in the annotation.

Practitioner treatment of strategy specification, testing, implementation and sizing; Keith Fitschen is the identified author.

Applications: Supplementary / practitioner.

Version record

Emmanuel Raj, *Engineering MLOps: Rapidly build, test, and manage production-ready machine learning life cycles at scale*, first edition, Packt (April 2021)

Difficulty: Not specified **Type:** Book / monograph

Status/version: Selected citation year(s): 2021. Version distinctions remain in the annotation.

Optional engineering reference for deployment, CI/CD, monitoring and traceability. Azure-based examples are version-sensitive; operational reliability does not establish a financial model's predictive validity.

Version record

Sculley et al. (2015), "Hidden Technical Debt in Machine Learning Systems"

Difficulty: Not specified **Type:** Research paper

Status/version: Selected citation year(s): 2015. Version distinctions remain in the annotation.

Entanglement, data dependencies, feedback loops and technical debt. The 2015 NeurIPS paper is distinct from the 2014 workshop paper listed in Section 5.1.

Version record

ONNX documentation

Difficulty: Not specified **Type:** Software / documentation

Status/version: Edition/revision and publication date not established for the selected resource(s).

Graph representation, operators and model checking for exchanging trained models. The versioning specification distinguishes IR, per-domain operator sets and model versions from the library release. Record all applicable versions and verify backend support; execution and optimisation depend on the chosen runtime.

Applications: model interchange.

Version record

Yves Hilpisch, *Python for Finance: Mastering Data-Driven Finance*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: O'Reilly, second edition, 2018 (ISBN 9781492024330).

Second edition (O'Reilly 2018) on pandas, numerical methods, simulation, valuation and algorithmic trading in Python.

Version record

Wes McKinney, *Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter*

Difficulty: Intro **Type:** Book / monograph

Status/version: O'Reilly, third edition, 2022; open-access online edition.

Third edition (2022) by pandas' creator, with free online edition; time-series indexing and grouping chapters are directly relevant.

Version record

Chris Kelliher, *Quantitative Finance with Case Studies in Python*

Difficulty: Intro/Advanced **Type:** Book / monograph

Status/version: Chapman & Hall/CRC, second edition (retitled), December 2025 (ISBN 9781032868004); first edition 2022 titled Quantitative Finance with Python.

Broad practitioner text on stochastic models, numerical methods, calibration, rates, inflation, MBS, portfolio construction and ML, with Python code; the 2025 second edition adds data science and LLM material.

Version record

Greg Wilson, Jennifer Bryan, Karen Cranston, Justin Kitzes, Lex Nederbragt and Tracy K. Teal, "Good enough practices in scientific computing"

Difficulty: Intro **Type:** Research paper

Status/version: PLOS Computational Biology, 13(6), e1005510 (2017); DOI 10.1371/journal.pcbi.1005510.

Minimal practices for data management, code organisation, collaboration and project structure for research teams.

Version record

Chapter 22

Governance and institutional controls

Read model, data and dependency governance alongside clearing and settlement obligations. A dated standard or implementation report has a different role from a research result.

22.1 Model and risk-data governance

22.1.1 Model risk and AI governance

Supervisory Guidance on Model Risk Management (SR 11-7, 2011); SR 26-2 (17 April 2026)

Difficulty: Not specified **Type:** Contract / specification / guidance

Status/version: Selected citation year(s): 2011, 2026. Version distinctions remain in the annotation.

Banking model-lifecycle, validation and governance references. The 2011 guidance is historical: SR 26-2 explicitly superseded SR 11-7 on 17 April 2026. Applying these ideas to a research organisation does not establish supervisory applicability.

Version record

Basel Committee on Banking Supervision, Principles for Effective Risk Data Aggregation and Risk Reporting (BCBS 239)

Difficulty: Advanced **Type:** Report / research note

Status/version: BCBS final guidelines published 9 January 2013; BIS catalogue marks this version current at check on 4 October 2026.

Governance, architecture, accuracy, completeness, timeliness and adaptability of risk aggregation explain why desk-level research must reconcile to enterprise exposures. Use the principles to examine identifiers, lineage, manual adjustments and stress-time reporting across books and legal entities. They are a banking supervisory reference with a stated scope, not a universal software specification. Read with the later implementation evidence to distinguish design principles from demonstrated institutional capability.

Version record

Basel Committee on Banking Supervision, Implementation of the Principles for Effective Risk Data Aggregation and Risk Reporting (BCBS 239 Principles)

Difficulty: Advanced **Type:** Report / research note

Status/version: BCBS Newsletter, 6 January 2026

Recent supervisory and industry observations on data lineage, governance, cross-border aggregation, emerging technology and compensating controls. Useful for assessing whether new platforms and AI tools preserve reliable risk information during organisational or system changes. This is an informational newsletter and explicitly does not introduce new supervisory guidance or expectations. Pair it with the 2013 principles instead of interpreting a new data architecture as completed risk-data remediation.

Version record

Financial Stability Board, The Financial Stability Implications of Artificial Intelligence

Difficulty: Advanced **Type:** Report / research note

Status/version: 14 November 2024; policy report

Connects AI adoption to correlated behaviour, third-party concentration, cyber vulnerabilities, data quality and model governance. Read when evaluating shared foundation models or automated research across strategies: individually useful tools can create common dependencies and crowded responses. The report supplies a system-wide risk framework, not evidence that a particular model generates alpha or that correlations necessarily cause a crisis.

Version record

Financial Stability Board, Monitoring Adoption of Artificial Intelligence and Related Vulnerabilities in the Financial Sector

Difficulty: Advanced **Type:** Report / research note

Status/version: 10 October 2025; report to the G20

Extends the 2024 report with monitoring approaches, candidate indicators, data gaps and a case study of AI service-provider dependencies. Use it to frame concentration inventories spanning model, cloud, data and operational suppliers. Proxy indicators require interpretation and do not substitute for testing model behaviour, recovery or portfolio exposures. The contribution is monitoring design and institutional context, distinct from a model-performance benchmark.

Version record

Prudential Regulation Authority (Bank of England), "SS1/23: Model risk management principles for banks"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Published 17 May 2023; effective 17 May 2024.

Five principles covering model identification, governance, development, independent validation and risk mitigants; effective 17 May 2024.

Extended note. Version record

European Parliament and Council, "Regulation (EU) 2024/1689 (Artificial Intelligence Act)"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official Journal 12 July 2024; staggered application 2025–2027; later amendments possible.

Risk-based AI regulation; creditworthiness and insurance pricing uses are high-risk; general-purpose AI obligations phase in from 2025.

Extended note. Version record

National Institute of Standards and Technology, "Artificial Intelligence Risk Management Framework (AI RMF 1.0)"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: NIST AI 100-1, January 2023; generative-AI profile July 2024.

Govern, map, measure and manage functions for AI risk, with a generative-AI profile (NIST AI 600-1, July 2024).

Extended note. Version record

European Central Bank Banking Supervision, "ECB guide to internal models"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Revised guide, 2024 (latest revision to be confirmed).

Supervisory expectations for internal market, credit and counterparty-risk models, including validation and model change.

Extended note. Version record

Joseph Simonian, "An Evidence-Based Framework for Model Governance"

Difficulty: Advanced **Type:** Research paper

Status/version: The Journal of Portfolio Management, 52(7), 98–112 (2026); DOI 10.3905/jpm.2026.1.836.

An evidence-based model-governance framework that replaces one-off pass/fail validation with continuous scoring of empirical success and consistency (building on Belnap's four-valued logic), with escalation and retirement rules.

Version record

22.1.2 Algorithmic-trading controls and incidents

European Commission, "Commission Delegated Regulation (EU) 2017/589 (MiFID II RTS 6) on algorithmic trading organisational requirements"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official Journal 31 March 2017; applied from 3 January 2018.

Testing, kill functionality, pre- and post-trade controls, real-time monitoring and annual self-assessment for algorithmic trading firms.

Extended note. Version record

U.S. Securities and Exchange Commission, "Rule 15c3-5: Risk Management Controls for Brokers or Dealers with Market Access"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Release 34-63241, November 2010; compliance from 2011.

Market access rule requiring pre-trade credit, capital and erroneous-order controls under the broker-dealer's direct control.

Extended note. Version record

U.S. Securities and Exchange Commission, "In the Matter of Knight Capital Americas LLC (Administrative Proceeding 3-15570)"

Difficulty: Advanced **Type:** Report / research note

Status/version: SEC order dated 16 October 2013.

Order describing the 1 August 2012 deployment failure that generated about USD 460 million of losses in 45 minutes; a case study in release management and market-access controls.

Version record

Financial Conduct Authority, "Algorithmic Trading Compliance in Wholesale Markets"

Difficulty: Advanced **Type:** Report / research note

Status/version: FCA multi-firm review, February 2018.

Supervisory review of algorithm development, testing, governance and market-abuse surveillance.

Version record

Prudential Regulation Authority (Bank of England), "SS5/18: Algorithmic trading"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Published June 2018 with PS12/18; PRA Rulebook version dated 4 October 2025.

PRA expectations on governance, algorithm approval, testing and deployment, inventories and risk controls, covering unregulated instruments such as spot FX; effective 30 June 2018.

Extended note. Version record

FIA, "Best Practices for Automated Trading Risk Controls and System Safeguards"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: FIA white paper, July 2024.

Industry best practices for pre-trade limits, kill switches, cancel-on-disconnect, exchange volatility controls and post-trade reconciliation (July 2024).

Extended note. Version record

Andrei Kirilenko, Albert S. Kyle, Mehrdad Samadi and Tugkan Tuzun, "The Flash Crash: High-Frequency Trading in an Electronic Market"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 72(3), 967–998 (2017); DOI 10.1111/jofi.12498.

Audit-trail data from E-mini futures on 6 May 2010: HFTs did not cause the crash but amplified volatility by trading in the direction of price moves.

Version record

U.S. Commodity Futures Trading Commission and U.S. Securities and Exchange Commission, "Findings Regarding the Market Events of May 6, 2010"

Difficulty: Advanced **Type:** Report / research note

Status/version: Joint CFTC–SEC staff report, 30 September 2010.

Official reconstruction of the Flash Crash: a large automated sell program, liquidity withdrawal and stub quotes.

Version record

U.S. Department of the Treasury, Federal Reserve, FRBNY, SEC and CFTC, "Joint Staff Report: The U.S. Treasury Market on October 15, 2014"

Difficulty: Advanced **Type:** Report / research note

Status/version: Joint staff report, July 2015.

Reconstruction of the Treasury flash rally: automated principal trading firms dominated volume and depth thinned before the move.

Version record

Albert J. Menkveld and Bart Zhou Yueshen, "The Flash Crash: A Cautionary Tale About Highly Fragmented Markets"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 65(10), 4470–4488 (2019); DOI 10.1287/mnsc.2018.3040.

During the Flash Crash, a large seller's algorithm broke the E-mini–SPY arbitrage link, so liquidity in fragmented markets evaporated.

Version record

22.1.3 Operational resilience

Basel Committee on Banking Supervision, "Principles for Operational Resilience"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: BCBS d516, March 2021.

Governance, mapping of critical operations, third-party dependency, incident management and ICT resilience.

Extended note. Version record

European Parliament and Council, "Regulation (EU) 2022/2554 on digital operational resilience for the financial sector (DORA)"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official Journal 27 December 2022; application from 17 January 2025.

ICT risk management, incident reporting, resilience testing and third-party oversight; applies from 17 January 2025.

Extended note. Version record

Prudential Regulation Authority (Bank of England), "SS1/21: Operational resilience: Impact tolerances for important business services"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: First published 29 March 2021.

Identify important business services, set impact tolerances and test the ability to remain within them; full compliance expected by 31 March 2025.

Extended note. Version record

Financial Stability Board, "Effective Practices for Cyber Incident Response and Recovery"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: FSB final report, 19 October 2020.

Forty-six effective practices across governance, preparation, analysis, mitigation, restoration and communication (October 2020).

Extended note. Version record

Basel Committee on Banking Supervision, "Revisions to the Principles for the Sound Management of Operational Risk"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: BCBS, March 2021.

Updated operational-risk management principles covering governance, change management, ICT and business continuity; read with the operational-resilience principles.

Extended note. Version record

CPMI and IOSCO, "Guidance on cyber resilience for financial market infrastructures"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: CPMI–IOSCO, June 2016.

Cyber-resilience expectations for FMIs: governance, identification, protection, detection, response and recovery, testing.

Extended note. Version record

Financial Stability Board, "Enhancing Third-Party Risk Management and Oversight: A toolkit for financial institutions and financial authorities"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: FSB, 4 December 2023.

Toolkit for managing critical third-party and cloud dependencies, including concentration and substitutability.

Extended note. Version record

U.S. Securities and Exchange Commission, "Regulation Systems Compliance and Integrity (Regulation SCI), Release No. 34-73639"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final rule, 79 FR 72252 (5 December 2014); later amendments not reviewed here.

Requirements for exchanges, clearing agencies and other SCI entities on system capacity, integrity, resilience, testing and incident notification.

Extended note. Version record

Thomas M. Eisenbach, Anna Kovner and Michael Junho Lee, "Cyber risk and the U.S. financial system: A pre-mortem analysis"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 145(3), 802–826 (2022); DOI 10.1016/j.jfineco.2021.10.007.

Simulates a cyberattack that impairs a large bank's payments: liquidity spillovers through Fedwire could affect a large share of the system.

Version record

Shinichi Kamiya, Jun-Koo Kang, Jungmin Kim, Andreas Milidonis and René M. Stulz, "Risk management, firm reputation, and the impact of successful cyberattacks on target firms"

Difficulty: Advanced **Type:** Research paper

Status/version: Journal of Financial Economics, 139(3), 719–749 (2021); DOI 10.1016/j.jfineco.2019.05.019.

Successful cyberattacks with personal-data loss cause significant shareholder losses and lead firms to change risk management.

Version record

Anil K. Kashyap and Anne Wetherilt, "Some Principles for Regulating Cyber Risk"

Difficulty: Advanced **Type:** Research paper

Status/version: AEA Papers and Proceedings, 109, 482–487 (2019); DOI 10.1257/pandp.20191058.

Short principles-based framework for cyber regulation focused on outcomes, recovery and system-wide dependencies.

Version record

Matteo Crosignani, Marco Macchiavelli and André F. Silva, "Pirates without borders: The propagation of cyberattacks through firms' supply chains"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 147(2), 432–448 (2023); DOI 10.1016/j.jfineco.2022.12.002.

The NotPetya attack propagated through supply chains, hurting customers of affected firms; third-party dependency risk in practice.

Version record

22.2 Clearing, settlement and institutional dependencies

These are specific institutional references for model oversight, aggregation, funding and market infrastructure. Read them with the model-risk guidance and distinguish general principles, dated policy analysis and jurisdiction-specific implementation material.

CPSS and IOSCO Technical Committee, Principles for Financial Market Infrastructures

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Final CPSS-IOSCO Principles for financial market infrastructures published 16 April 2012; checked 4 October 2026.

Framework for payment systems, central securities depositories, settlement systems, central counterparties and trade repositories. Prioritise credit and liquidity risk, collateral, margin, settlement finality, default procedures and operational resilience when studying how a strategy survives participant failure. Clearing transforms and mutualises exposures; it does not remove funding needs or every tail loss. Pair the general principles with the actual CCP rulebook and account structure for a concrete strategy.

Version record

US Securities and Exchange Commission, Treasury Clearing Implementation

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: Official implementation hub; inspected 4 October 2026

Primary route to the US Treasury central-clearing rule, access arrangements and implementation guidance. The hub states extended compliance dates of 31 December 2026 for eligible cash transactions and 30 June 2027 for eligible repo transactions. These are eligibility-dependent future milestones as of this catalogue date, not a statement that every Treasury transaction is already centrally cleared.

Extended note. Version record

US Securities and Exchange Commission, Division of Examinations, Shortening the Securities Transaction Settlement Cycle

Difficulty: Advanced **Type:** Report / research note

Status/version: 2024 risk alert; US T+1 transition on 28 May 2024

Concrete reference for the shortened settlement cycle and operational readiness around allocation, confirmation, affirmation and recordkeeping. For global equity portfolios, study how trade-date cash, FX funding, securities lending and holiday calendars interact with settlement. Reduced calendar exposure does not make settlement instantaneous or eliminate fails. Use the jurisdiction and instrument scope of the actual rule rather than applying T+1 to every asset or market.

Version record

Oliver Wyman (for the London Metal Exchange), "Independent Review of Events in the Nickel Market in March 2022"

Difficulty: Advanced **Type:** Report / research note

Status/version: Final report and LME response published 10 January 2023; governance matters left to FCA and Bank of England reviews.

Review of the March 2022 nickel squeeze, OTC exposure opacity, margin calls and trade cancellation at a clearing house.

Version record

BCBS, CPMI and IOSCO, "Review of margining practices"

Difficulty: Advanced **Type:** Report / research note

Status/version: September 2022 (d537); later policy reports 2024–2025.

Analysis of March 2020 margin calls, procyclicality, transparency and liquidity preparedness, with follow-up policy work.

Version record

Albert J. Menkveld, "Crowded Positions: An Overlooked Systemic Risk for Central Clearing Parties"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Asset Pricing Studies, 7(2), 209–242 (2017); DOI 10.1093/rapstu/rax016.

CCP exposure measure based on tail risk of crowded member portfolios; crowding raises exposure by about a third on turbulent days.

Version record

CPMI and IOSCO, "Resilience of central counterparties (CCPs): Further guidance on the PFMI"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: CPMI–IOSCO, July 2017 (d163).

Detailed guidance on CCP governance, stress testing, margin, liquidity risk and recovery planning (July 2017).

Extended note. Version record

Financial Stability Board, "Guidance on Financial Resources to Support CCP Resolution and on the Treatment of CCP Equity in Resolution"

Difficulty: Advanced **Type:** Contract / specification / guidance

Status/version: FSB, 16 November 2020.

Resolution resources and loss allocation for CCPs (November 2020).

Extended note. Version record

Behnoud Alizadeh and Marius Zoican, "Preferred Clearing and Post-Trade Costs"

Difficulty: Research **Type:** Research paper

Status/version: The Review of Asset Pricing Studies, raag008 (2026); DOI 10.1093/rapstu/raag008.

Fee competition between incumbent and entrant CCPs under interoperability versus preferred clearing; network effects favour incumbents.

Version record

Rama Cont and Thomas Kokholm, "Central clearing of OTC derivatives: bilateral vs multilateral netting"

Difficulty: Research **Type:** Research paper

Status/version: Statistics & Risk Modeling, 31(1), 3–22 (2014); DOI 10.1515/strm-2013-1161.

Netting gains from a CCP depend on asset-class correlations and the number of participants; complements Duffie–Zhu.

Version record

Yee Cheng Loon and Zhaodong (Ken) Zhong, "The impact of central clearing on counterparty risk, liquidity, and trading: Evidence from the credit default swap market"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 112(1), 91–115 (2014); DOI 10.1016/j.jfineco.2013.12.001.

Voluntary CDS clearing lowered counterparty risk and improved liquidity for reference entities that became cleared.

Version record

Albert J. Menkveld and Guillaume Vuillemey, "The Economics of Central Clearing"

Difficulty: Advanced **Type:** Research paper

Status/version: Annual Review of Financial Economics, 13(1), 153–178 (2021); DOI 10.1146/annurev-financial-100520-100321.

Review of CCP economics: netting, margin, default waterfalls, incentives and systemic risk.

Version record

Bruno Biais, Florian Heider and Marie Hoerova, "Risk-Sharing or Risk-Taking? Counterparty Risk, Incentives, and Margins"

Difficulty: Research **Type:** Research paper

Status/version: The Journal of Finance, 71(4), 1669–1698 (2016); DOI 10.1111/jofi.12396.

Margins mitigate the risk-taking incentives created by derivatives hedging; the theory behind margin design.

Version record

Darrell Duffie, Martin Scheicher and Guillaume Vuillemey, "Central clearing and collateral demand"

Difficulty: Research **Type:** Research paper

Status/version: Journal of Financial Economics, 116(2), 237–256 (2015); DOI 10.1016/j.jfineco.2014.12.006.

Bilateral CDS position data estimate how clearing mandates and margin rules change collateral demand.

Version record

Mark Paddrik, Sriram Rajan and H. Peyton Young, "Contagion in Derivatives Markets"

Difficulty: Research **Type:** Research paper

Status/version: Management Science, 66(8), 3603–3616 (2020); DOI 10.1287/mnsc.2019.3354.

Variation-margin payment shortfalls propagate through CDS networks; stress-testing payment liquidity rather than solvency.

Version record

Chapter 23

Course and research-outlet directories

These compact directories support discovery and continuing study. Courses and outlets are resources in their own right; follow specific claims to the underlying work.

23.1 Probability, statistics and finance courses

Peking University HSBC Business School, *Research Methodology (QF)*

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Peng Xianhua; Module 2, 2023–2024

Historical quantitative-finance thesis-methods syllabus covering literature review and research proposals; examples include factors, pricing and portfolio selection.

Version record

Harvard Stat 110, Joe Blitzstein

Difficulty: Intro **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Introductory probability. Textbook, handouts, exercises and lecture videos; the hub also links Stat110x.

Version record

MIT 18.05: Introduction to Probability and Statistics; MIT 6.041: Probabilistic Systems Analysis and Applied Probability

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Two distinct introductory courses: 18.05 offers readings, problems and interactive material; 6.041 offers videos, recitations, assignments and exams.

Version record

MIT 18.650: Statistics for Applications; MIT 18.655: Mathematical Statistics

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

18.650 (Philippe Rigollet, Fall 2016) includes videos; 18.655 (Peter Kempthorne, Spring 2016) offers graduate notes, assignments and exams on decision theory, inference and asymptotics.

Version record

NPTEL: Statistical Inference, Somesh Kumar, IIT Kharagpur

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Undergraduate/postgraduate estimation and testing, with worked examples; official course catalogue and syllabus, distinct from the unavailable 2020 cohort archive.

Version record

NPTEL: Measure Theoretic Probability 1 and Measure Theoretic Probability 2, Suprio Bhar, IIT Kanpur

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Measure construction, integration and convergence in Part 1; limit laws, characteristic functions and asymptotic applications in Part 2. The syllabuses do not promise martingale coverage.

Version record

Çağın Ararat, Bilkent IE 523: Probabilistic Analysis; lecture playlist

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Graduate measure-theoretic probability. The instructor identifies the videos as Fall 2017; the separate Fall 2016 syllabus lists conditioning, convergence, LLN and CLT, with Resnick and Çınlar among its texts. Playlist count and completeness are unverified.

Version record

Todd Kemp, UC San Diego Math 280: Probability Theory and Stochastic Processes

Difficulty: Advanced **Type:** Course / learning resource

Status/version: Instructor's 2020–2021 full-year graduate lecture archive; page accessible 4 October 2026.

The instructor's course page links 135 topical videos and accompanying before/after slides (about 57 hours) across measure construction, integration, limit theorems, conditioning, Markov processes, martingales and Brownian motion. It offers a concrete recorded graduate probability sequence for quantitative-finance foundations. This is an independent UC San Diego resource, not an identification of Jem Corcoran's unverified 2024 *A Probability Space* series.

Version record

MIT 18.675: Theory of Probability

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Nike Sun's Fall 2019 graduate probability course, renumbered from 18.175. The Spring 2014 OCW archive is a separate notes/problem-set resource.

Version record

UCCS Mathematics Video Course Archive

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Archive of MATH 4810 Mathematical Statistics I, MATH 4820 Mathematical Statistics II, and MATH 4830 Linear Statistical Models.

Version record

Krishna Jagannathan, IIT Madras EE5110: Probability Foundations; NPTEL Probability Foundations for Electrical Engineers

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Measure-theoretic probability course and official syllabus; distinct from the IIT Kanpur courses. The original playlist attribution remains unconfirmed.

Version record

MIT 18.676: Stochastic Calculus, Spring 2020

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Selected citation year(s): 2020. Version distinctions remain in the annotation.

Graduate course, formerly 18.176; 18.675 is the prerequisite. The instructor page does not establish an OCW video archive.

Version record

Columbia, Introduction to Financial Engineering and Risk Management

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Applied finance companion connecting probability with financial-engineering examples.

Version record

Robert Almgren / NYU time-series notes; Equilibrium of Nothing outline and flashcards

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Historical notes on ARMA, state-space methods and volatility. The 2008–09 NYU catalogue identifies Almgren–Reider's Financial Econometrics and Statistical Arbitrage; the third-party mirror is not authenticated in full.

Version record

SUNY / Coursera, Practical Time Series Analysis, Tural Sadigov and William Thistleton

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Applied time-series course; access and cohort terms are provider-specific.

Version record

QuantInsti ARIMA tutorial; QuantStart ARIMA+GARCH example

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Practitioner time-series examples; historical trading claims need independent reconstruction and cost-aware evaluation.

Version record

MIT 14.32/14.320: Econometric Data Science, Spring 2022

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Selected citation year(s): 2022. Version distinctions remain in the annotation.

Regression, experiments and causal research designs; identified historical syllabus.

Version record

Will Kurt, Count Bayesie measure-theory introduction; QuantStart quant-study guide; Kshitij Anand stochastic-calculus post

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Optional explanatory and orientation resources. These are distinct from the formal textbooks and course syllabuses.

Version record

John H. Cochrane (University of Chicago), "Asset Pricing online course (Coursera, two parts)"

Difficulty: Advanced **Type:** Course / learning resource

Status/version: Recorded course; Coursera availability varies.

Video lectures following Cochrane's Asset Pricing, with problem sets; the most direct companion to the theory text.

Version record

MIT OpenCourseWare, "18.S096 Topics in Mathematics with Applications in Finance (Fall 2013)"

Difficulty: Intermediate **Type:** Course / learning resource

Status/version: Archived OCW course, Fall 2013.

Lectures by MIT faculty and practitioners on linear algebra, statistics, time series, stochastic calculus and portfolio theory.

Version record

Stephen Boyd (Stanford), "EE364a: Convex Optimization I"

Difficulty: Advanced **Type:** Course / learning resource

Status/version: Recorded lectures; current course materials on the Stanford course site.

Lecture videos and slides accompanying Boyd–Vandenberghe; portfolio and finance examples throughout.

Version record

Martin Haugh (Columbia/Imperial), "Course notes in financial engineering, Monte Carlo, risk management and machine learning"

Difficulty: Advanced **Type:** Course / learning resource

Status/version: Author-hosted notes; individual notes dated separately.

Concise, rigorous lecture notes on term-structure models, Monte Carlo, risk measures and portfolio optimisation.

Version record

23.2 Blockchain and security courses

Use these as structured bridges between finance and blockchain engineering. Introductory MOOCs supply prerequisites; they do not replace graduate stochastic calculus, microstructure or a protocol security review. Course availability and syllabuses can change.

Princeton / Coursera: *Bitcoin and Cryptocurrency Technologies*

Difficulty: Intro **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Companion to Narayanan et al. in Section 18.1. The author site provides lectures, quizzes and three Java programming assignments following the textbook. Covers cryptographic transactions, Bitcoin design, consensus, mining, privacy and security. Useful for finance researchers who need to understand what blockchain data and settlement events mean.

Applications: technical blockchain foundation..

Version record

Duke / Campbell Harvey: *Decentralized Finance (DeFi): The Future of Finance*

Difficulty: Intro **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Four-course sequence: Infrastructure, Primitives, Deep Dive, and Opportunities and Risks. Introduces blockchains, smart contracts, oracles and stablecoins, then exchanges, lending, derivatives and protocol risks. Use it to connect financial intermediation to DeFi mechanics. The provider labels the specialization beginner level.

Applications: DeFi financial architecture..

Version record

University at Buffalo: Blockchain Specialization

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

The sequence covers blockchain basics, smart contracts, decentralised applications and platforms. Course 2, *Smart Contracts*, develops Solidity contract design, coding, deployment and testing; DApp development continues in the broader sequence. Provides hands-on preparation for reading DeFi code and building on-chain research prototypes. Production protocols need additional security and economic validation.

Applications: contract-programming preparation..

Version record

University of Nicosia: *Introduction to Digital Currencies* and *Introduction to DeFi* (historical MOOCs); Fall 2022 university course record

Difficulty: Intro **Type:** Course / learning resource

Status/version: Historical named MOOCs; the original 2022–2023 brochure is unavailable for direct recheck.

The university's Fall 2022 course page names both MOOCs and identifies George Giaglis as their instructor. An earlier catalogue pass used the university's 2022–2023 brochure URL for DeFi syllabus topics and a 12-week duration, but that PDF was unavailable at the 4 October 2026 audit; those edition-specific details are retained only as historical audit claims. For a separately documented current route through digital-asset and DeFi foundations, see the following University of Nicosia MOOC.

Version record

University of Nicosia, *Introduction to Blockchain and Digital Assets* (current MOOC)

Difficulty: Intro **Type:** Course / learning resource

Status/version: University page updated 2 October 2026; next cohort advertised to start 8 February 2027.

The university advertises a free eight-week introductory track covering Bitcoin, Ethereum, wallets, DeFi, prediction markets, tokenization and stablecoins, with recorded lectures, weekly Q&A and optional exercises. A paid advanced track extends through weeks 9–12 to NFTs, policy and scaling. Its concrete current syllabus supplies a route into on-chain market mechanics; it is a separately titled course, not a verified revision of the older Digital Currencies or DeFi MOOCs.

Version record

OpenZeppelin: Ethernaut / source

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Solidity and EVM security puzzles that require identifying and exploiting deliberately vulnerable contracts. Use the exercises to understand failure modes before evaluating smart-contract-dependent strategies; complement them with the formal-verification monograph in Section 18.1.

Version record

Dan Boneh et al. (Stanford), "CS251: Cryptocurrencies and Blockchain Technologies"

Difficulty: Advanced **Type:** Course / learning resource

Status/version: Course site updated by term; check current offering.

Rigorous course on consensus, Bitcoin, Ethereum, DeFi, scaling and privacy with public lecture slides.

Version record

Dawn Song, Arthur Gervais and Christine Parlour (UC Berkeley), "Decentralized Finance MOOC"

Difficulty: Intro/Advanced **Type:** Course / learning resource

Status/version: Fall 2022 offering; recordings available.

Lecture series on DeFi primitives, AMMs, lending, MEV and security from researchers active in the field.

Version record

23.3 Research courses and communities

Retained for structured practice and idea discovery. Provider catalogues establish course identity; fees, access, schedules and individual modules should be checked for the selected cohort. Community and competition results require independent time-ordered validation.

Google Cloud / NYIF: Machine Learning for Trading Specialization (Coursera)

Difficulty: Intro **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

A three-course route into Python, financial ML and trading applications. Use technical, factor, news and reinforcement-learning exercises as practice in specifying targets, backtesting and evaluating forecasts. Independently enforce point-in-time validation; course participation does not certify a strategy.

Version record

Georgia Tech CS 7646: Machine Learning for Trading

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Edition/revision and publication date not established for the selected resource(s).

Academic course associated with Tucker Balch's original material, with trading simulation and ML assignments such as decision trees and reinforcement learning. Use the current university page for syllabus and access; older Udacity material mentioned by the source is historical.

Applications: Academic / applied course.

Version record

Quantopian lecture notebooks

Difficulty: Not specified **Type:** Course / learning resource

Status/version: Archived educational material

Historical exercises on statistics, factors, portfolio optimisation, market impact, backtesting and event studies. Reproduce the combined value/momentum example and compare single-factor versus combined IC, turnover and net performance. Replace retired platform services and pin compatible packages/data.

Version record

QuantInsti EPAT; QuantInsti blog

Difficulty: Not specified **Type:** Practitioner / commentary

Status/version: Edition/revision and publication date not established for the selected resource(s).

Commercial professional-training lead on systematic strategies, ideation, backtesting and implementation. Free tutorials include regime detection and event-driven topics. Evaluate the actual curriculum and exercises; certification is distinct from empirical validation.

Version record

Numerai community; Kaggle Two Sigma: Using News to Predict Stock Movements

Difficulty: Not specified **Type:** Community / competition

Status/version: Community / historical competition

Practical work on anonymised/tabular data, news, feature construction, ensembles, leakage detection and efficient inference. Public solutions and leaderboard performance can suggest baselines but are affected by competition design and selection; rerun the full selection process on independent temporal holdouts.

Version record

Stanford University, "CS229: Machine Learning (lecture notes and videos)"

Difficulty: Intro/Advanced **Type:** Course / learning resource

Status/version: Course site updated by term.

Standard graduate-level ML course notes covering supervised learning, kernels, trees, unsupervised learning and RL.

Version record

23.4 Managers and strategy research

Subject context and qualifications.

Two Sigma insights; Jane Street technical blog

Difficulty: Research **Type:** Practitioner / commentary

Status/version: Edition/revision and publication date not established for the selected resource(s).

Practitioner perspectives on data analysis, modelling, news/NLP, software and research tooling. Read for problem formulation and implementation lessons; no assumption that posts disclose deployable proprietary trading strategies.

Applications: Industry feeds.

Version record

Quantpedia

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Strategy-discovery catalogue linking academic ideas such as PEAD, analyst revisions, factors and event trades to implementation summaries. Read the original papers, verify data and include realistic costs. The PDF cites a LuxAlgo guide, not a primary Quantpedia study; access and any LEAN integration are service-specific.

Applications: Supplementary discovery.

Version record

AQR Capital Management – Research

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Factors, portfolio construction, risk premia, alternatives, macro-finance and implementation. Climate Risk Pricing (30 December 2025); Active Extension (24 November 2025). Climate-risk examples are model-based research, with assumptions rather than guaranteed investment implications.

Version record

Man Group – Man Institute / Man AHL insights

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Systematic macro, ML in production, execution, smart order routing and research commentary. The Rise of Machine Learning at Man AHL; the article reports trading ML-based systems since early 2014. Companion: An Introduction to Machine Learning, retained from the bibliography. Provider accounts of live ML use are not independently verified performance evidence.

Version record

Two Sigma – Factor Lens

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Multi-asset factors, risk decomposition and portfolio analytics. Introducing the Two Sigma Factor Lens (2018); Revisiting the Two Sigma Factor Lens (Robert Xie, December 2023). The December 2023 PDF and June 2022 web article are different versions.

Version record

Robeco – Quantitative research

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Factor investing, anomalies and transaction-cost minimisation in quantitative processes. Active Quant unboxed (2025); the Our ground-breaking papers hub linked from the research portal.

Version record

Research Affiliates – Publications

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Factor investing, smart-beta construction and valuation-aware factor timing. Finding Smart Beta in the Factor Zoo; smart-beta research hub.

Version record

Barclays – Quantitative Portfolio Strategy / QIS insights

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Systematic strategies, risk budgeting, cross-asset quant research and defensive structuring. Demystifying defensive quantitative investment strategies (August 2025). Public summaries and client-gated research have different access.

Version record

BNP Paribas – QIS Lab

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Systematic strategies, factor-investing foundations, trend, dispersion and asset allocation. Demystifying algorithmic trend following (23 January 2025). The public article introduces Following trends, one brick at a time; the research paper is offered to institutional clients via Brio.

Version record

Goldman Sachs – Systematic Trading Strategies / gs-quant

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Systematic portfolios, signals, portfolio/risk tooling and analytics workflows. gs-quant platform overview; open-source Python toolkit. The GSAM link is historical; some gs-quant analytics depend on proprietary data/services.

Version record

23.5 Exchanges, regulators and central banks

NYSE, Cboe, SEC DERA, Federal Reserve, New York Fed and CFTC research portals

Difficulty: Research **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

These institutional sources document tick regimes, options structure, payment for order flow, Treasury liquidity, dealer behaviour and futures market quality using data unavailable to most academics. They should be monitored continuously because venue rules and participant composition can invalidate historical execution models. Individual profiles and named examples are in Section 23.5.

Applications: Equities, options, rates, futures; market structure; Ongoing. Version record

New York Stock Exchange – Research Insights

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Market microstructure, liquidity, tick/lot sizing, trading hours and market quality. Q2 2025 Extended Hours Trading Review (31 July 2025 in the research index); also see the data-insights archive.

Version record

Cboe Global Markets – Market structure docs & insights

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Equity and options microstructure, execution quality and market-structure policy. Options Market Structure: A Half Century of Innovation (December 2024).

Version record

U.S. Securities and Exchange Commission – DERA Working Papers

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Market structure, routing/payment for order flow, liquidity and empirical market quality. How Does Payment for Order Flow Influence Markets? Evidence from Robinhood Crypto Token Introductions (January 2025 draft), already annotated in the microstructure readings. Staff working papers are preliminary research, not SEC policy determinations.

Version record

Federal Reserve Board – FEDS / Market Microstructure field

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Liquidity, Treasuries, principal trading firms, microstructure invariance and macro-finance. Microstructure Invariance in U.S. Stock Market Trades, Kyle, Obizhaeva and Tuzun (FEDS 2016-034); Market Microstructure field index. FEDS papers are work in progress. Also: The Relationship Between Market Depth and Liquidity Fragility in the Treasury Market.

Version record

Federal Reserve Bank of New York – Staff Reports

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Treasury microstructure, liquidity crises, dealer intermediation and market plumbing. The Microstructure of a U.S. Treasury ECN: The BrokerTec Platform, already in the fixed-income readings.

Version record

Commodity Futures Trading Commission – Research Papers

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Futures/swaps market structure, liquidity, trader behaviour, HFT and speed studies. Determinants of Commodity Market Liquidity: positions and microstructure-noise measures. A separate CFTC anticipatory-trading paper addresses a distinct market-structure question.

Version record

Bank for International Settlements – Working Papers

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

FX microstructure, macro-finance, bank/intermediary constraints and market functioning. The foreign exchange market (BIS Working Paper 1094).

Version record

European Central Bank – Working papers

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Monetary and market liquidity, banking and market functioning for expert readers. Liquidity dependencies in the euro area (Working Paper 3056, 2025).

Version record

23.6 Index and risk providers

MSCI – Research & Insights

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Factor models, risk modelling, index research and risk decomposition. A Multidimensional View of Risk: integrating time-series and factor models. Also read Factor Indexing Through the Decades for an index-construction perspective. Commercial risk/index provider; product and estimation conventions matter.

Version record

S&P Dow Jones Indices – Research / Indexology / SPIVA

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Index methodology, factor research and active-versus-passive measurement. SPIVA overview, About SPIVA and the Indexology blog (series).

Version record

FTSE Russell – Research portal

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Factors, ESG, multi-asset research and digital-asset taxonomy/indexing.

Version record

Qontigo – STOXX / Applied Research & Whitepapers

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Factor-lens analytics, thematic index risk, portfolio efficiency and risk models. Tech-oriented Thematic Indices through a Factor Lens (references an Axioma model). Qontigo/STOXX labels preserve historical naming rather than assert current corporate structure.

Version record

SimCorp – Axioma factor risk models

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Fundamental versus statistical risk models, factor libraries and multi-asset look-through.

Version record

23.7 Academic and execution research

Quantitative Finance Stack Exchange; Quantivity; Alpha Architect

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Practitioner questions, technical discussions and research leads; verify claims against the cited methods and original papers.

Applications: Supplementary discovery. Version record

Oxford–Man Institute of Quantitative Finance – Academic lab

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

ML/AI in finance, limit order books and data-driven models. Selected publications, including DeepLOB and BDLOB; the individual readings appear in the microstructure references and Section 3.1.

Version record

arXiv – q-fin

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

New research in quantitative finance. Categories include trading and microstructure (TR), statistical finance (ST), portfolio management (PM) and computational finance (CP). Recent q-fin submissions. Preprints are preliminary and may be revised.

Version record

SSRN – Financial Economics Network

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Finance working papers and early-stage dissemination, including asset pricing, banking and market microstructure. FEN overview and FEN eJournals. Working-paper versions and access differ by author.

Version record

National Bureau of Economic Research – Working Paper Series

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Asset pricing, macro-finance, financial markets and systemic risk. New This Week and newsletter subscriptions. Working papers are not automatically peer-reviewed journal articles.

Version record

Quantitative Brokers – Whitepapers

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Execution algorithms, regime modelling, reinforcement learning and futures/rates execution. Reinforcement Learning For Trade Execution (March 2023).

Version record

The Microstructure Exchange, "The Microstructure Exchange online seminar series"

Difficulty: Research **Type:** Research outlet

Status/version: Living seminar series.

Online seminar series for market-microstructure research with recorded talks and a public schedule.

Version record

23.8 Digital-asset research outlets

Coin Metrics – Insights / State of the Network

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Crypto market structure, on-chain and market data, and systemic narratives supported by metrics. State of the Network; Crypto Trends To Watch in 2026 (December 2025 issue). Newsletter archive; network activity, flows, supply and realised-capitalisation metrics require their published definitions.

Version record

Kaiko – Kaiko Research

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Liquidity, fragmentation, exchange dynamics and cross-venue microstructure. Weekly Data Debrief; Liquidity Lowdown: Asset Ranking, an example of liquidity-component charts.

Version record

Glassnode – Glassnode Insights

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

On-chain metrics, market regimes and Bitcoin/Ethereum/DeFi. The Week On-chain; subscription/access terms vary. Entity heuristics and historical revisions affect on-chain metrics.

Version record

Flashbots – Writings

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

MEV, block building, auctions, mechanism design and market structure. Migrating to BuilderNet (December 2024); also see the Flashbots homepage.

Version record

Paradigm – Writing

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Market design, MEV, protocol economics and trading mechanics. See Crypto Market Structure 3.0.

Version record

a16z crypto – Research

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Market structure, MEV, mechanism design and policy.

Version record

Gauntlet – Applied Research

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

DeFi risk modelling, incentive design and simulation-based stress tests. How to Measure Risk Models (Part I) (December 2023). Model evaluation covers strategic borrower/lender/liquidator incentives, not only code vulnerabilities.

Version record

Delphi Digital – Reports

Difficulty: Not specified **Type:** Research outlet

Status/version: Edition/revision and publication date not established for the selected resource(s).

Macro and crypto, systematic-style research and subscription strategy reports. Subscription-gated; interpretive research is secondary evidence.

Version record

23.9 Numbered source-link directory

Numbered links support the superscript citations in this catalogue; resource entries also link directly to primary or official destinations.

[1] https://www.aqr.com/insights/research/journal-article/value-and-momentum-everywhere
[2] https://academic.oup.com/rfs/article/33/5/2223/5758276
[3] https://academic.oup.com/princeton-scholarship-online/book/42452/chapter/356713759
[4] https://onlinelibrary.wiley.com/doi/book/10.1002/9781119835622
[5] https://link.springer.com/book/10.1007/978-3-540-34604-3
[6] https://www.gsb.stanford.edu/faculty-research/books/credit-risk
[7] https://www.nber.org/papers/w33320
[8] https://www.nber.org/papers/w33434
[9] https://www.vitalsource.com/sg/products/arbitrage-theory-in-continuous-time-tomas-bjork-v9780192592453
[10] https://onlinelibrary.wiley.com/doi/book/10.1002/9781119202073
[11] https://arxiv.org/abs/1410.3394
[12] https://epubs.siam.org/doi/10.1137/1.9781611977783.ch1
[13] https://epubs.siam.org/doi/10.1137/24M1636952
[14] https://www.tandfonline.com/doi/abs/10.1080/14697688.2019.1571683
[15] https://econpapers.repec.org/paper/arxpapers/2207.00932.htm
[16] https://www.tandfonline.com/doi/full/10.1080/14697688.2022.2136037
[17] https://papers.ssrn.com/sol3/Delivery.cfm/4484004.pdf?abstractid=4484004
[18] https://arxiv.org/abs/2504.20349
[19] https://arxiv.org/abs/2310.06625
[20] https://arxiv.org/abs/2403.07815
[21] https://research.google/blog/a-decoder-only-foundation-model-for-time-series-forecasting/
[22] https://papers.ssrn.com/sol3/Delivery.cfm/5022829.pdf?abstractid=5022829&mirid=1
[23] https://github.com/google-research/timesfm
[24] https://arxiv.org/abs/1809.09441
[25] https://ojs.aaai.org/index.php/AAAI/article/view/27767
[26] https://digitalcommons.njit.edu/fac_pubs/1334/
[27] https://www.microsoft.com/en-us/research/publication/learning-multiple-stock-trading-patterns-with-temporal-routing-adaptor-and-optimal-transport/?locale=fr-ca
[28] https://arxiv.org/abs/2106.10024
[29] https://econpapers.repec.org/paper/arxpapers/2207.07467.htm
[30] https://research.ibm.com/publications/neural-progressive-hedging-enforcing-constraints-in-reinforcement-learning-with-stochastic-programming
[31] https://www.nber.org/papers/w34054
[32] https://arxiv.org/abs/2506.22611
[33] https://arxiv.org/abs/2507.19003
[34] https://arxiv.org/abs/2605.27113
[35] https://academic.oup.com/jfec/article-abstract/22/1/224/6759403
[36] https://academic.oup.com/raps/article-abstract/12/3/667/6543623
[37] https://arxiv.org/abs/2303.00208
[38] https://arxiv.org/abs/2301.13599
[39] https://arxiv.org/abs/2410.18434
[40] https://arxiv.org/abs/2602.00776
[41] https://www.nber.org/papers/w10504
[42] https://www.nber.org/papers/w10359
[43] https://www.ubplj.org/index.php/jpm/article/view/417
[44] https://www.nber.org/papers/w33339
[45] https://www.nber.org/papers/w34702
[46] https://arxiv.org/abs/2604.24366
[47] https://www.quantlib.org/git.shtml
[48] https://github.com/microsoft/qlib
[49] https://nautilustrader.io/
[50] https://github.com/jpmorganchase/abides-jpmc-public
[51] https://github.com/KangOxford/jax-lob
[52] https://github.com/thuml/iTransformer
[53] https://github.com/SJTU-DMTai/MASTER